

R&S®CMW290/500

LTE UE Firmware Applications

User Manual



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Version 41

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Make ideas real



This user manual describes the following options for R&S®CMW290 and R&S®CMW500:

- R&S®CMW-KM500/-KM550 (LTE FDD/TDD R8, TX measurement, uplink)
- R&S®CMW-KM012 (TX measurement, multi-evaluation list mode)
- R&S®CMW-KS500/-KS550 (LTE FDD/TDD R8, SISO, basic signaling)
- R&S®CMW-KS510 (LTE R8, SISO, advanced signaling)
- R&S®CMW-KS520 (LTE MIMO 2x2, generic signaling)
- R&S®CMW-KS590 (LTE MTC, generic signaling)

And the following options for R&S®CMW500 only:

- R&S®CMW-KM502/-KM552 (LTE FDD/TDD R10 UL CA, TX measurement, uplink)
- R&S®CMW-KM504/-KM554 (LTE FDD/TDD R12 UL 3-4 CA, TX measurement, uplink)
- R&S®CMW-KM570 (LTE R14 C-V2X PC5, measurement)
- R&S®CMW-KS170 (alerting network support, generic signaling)
- R&S®CMW-KS502/-KS552 (LTE FDD/TDD R10, CA, basic signaling)
- R&S®CMW-KS504/-KS554 (LTE FDD/TDD R12, basic signaling)
- R&S®CMW-KS505/-KS555 (LTE FDD/TDD R15, basic signaling)
- R&S®CMW-KS512 (LTE R10, CA, advanced signaling)
- R&S®CMW-KS514 (LTE R12, advanced signaling)
- R&S®CMW-KS515 (LTE R15, advanced signaling)
- R&S®CMW-KS521 (LTE MIMO 4x2, generic signaling)
- R&S®CMW-KS522 (LTE MIMO 8x2, generic signaling)
- R&S®CMW-KS525 (LTE user-defined bands, generic signaling)
- R&S®CMW-KS540 (LTE MIMO 4x4, DL, generic signaling)
- R&S®CMW-KS541 (LTE MIMO 8x4, DL, generic signaling)
- R&S®CMW-KE100 (basic fading support: AWGN generator)
- R&S®CMW-KE500 (LTE fading profiles TS 36.521, excerpts)
- R&S®CMW-KE501 (LTE fading profiles MIMO 4x2 from TS 36.521 B2.3)
- R&S®CMW-KE502 (LTE fading profiles MIMO 8x2 from TS 36.521 B2)
- R&S®CMW-KE504 (LTE fading profiles MIMO 2x4 and 4x4 from TS 36.521 and SCME)
- R&S®CMW-KE505 (LTE fading profiles MIMO 8x4 from TS 36.521)

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The following abbreviations are used throughout this manual: R&S®CMW is abbreviated as R&S CMW.

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1 Preface

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1.1 Documentation overview

This section provides an overview of the R&S CMW user documentation. Most documents are available in PDF format on the customer web.

The customer web is a section of GLORIS, the global Rohde & Schwarz information system, see <https://gloris.rohde-schwarz.com>. A registration is required.

1.1.1 Getting started manual / quick start guide

Depending on your instrument model, there is either a getting started manual or a quick start guide.

The document introduces the R&S CMW and describes how to set up and start working with the product.

A PDF file is available on the customer web.

1.1.2 User manual and help

Separate user manuals are provided for the base unit and the firmware applications:

- The base unit manual describes the base software and common features of the firmware applications. It also describes basic principles for manual operation and remote control. The manual includes the contents of the getting started manual or quick start guide.
- Additional documents, like this one, describe the firmware applications, including the graphical user interface of each application and the remote control commands.

All user manuals are available on the customer web.

The contents of the user manuals are also available as help in the R&S CMW. The help offers quick, context-sensitive access to the information needed for operation and programming. If you perform a software update, the help is also updated.

1.1.3 Features & functions

The features & functions documents list the available software options and the related features, settings and measurement results. The documents are available on the customer web.

1.1.4 Safety instructions

The safety instructions are available in many languages.

The printed document is delivered with the instrument.

1.1.5 Specifications and product brochure

The specifications document, also called "data sheet", contains the technical specifications of the R&S CMW. It also lists the options and their order numbers and optional accessories.

The brochure provides an overview of the instrument and deals with the specific characteristics.

The documents are available on the customer web.

1.1.6 Calibration certificate

The document is available on <https://gloris.rohde-schwarz.com/calcert>. You need the device ID of your instrument, which you can find on a label on the rear panel.

1.1.7 Release notes and open-source acknowledgment (OSA)

The release notes list new features, improvements and known issues per software version and installation package.

The software uses several valuable open source software packages. The open source acknowledgment provides verbatim license texts of the used open source software.

The documents are available on the customer web.

1.1.8 Application notes, application cards, white papers, etc.

These documents deal with special applications or background information on particular topics.

The documents are available on the customer web.

1.2 How to read the documentation

For each firmware application, there are the following sections.

- [General description](#)..... 15
- [GUI reference](#)..... 15
- [Programming examples](#)..... 15
- [Command reference](#)..... 15

1.2.1 General description

This section provides a general description of the firmware application, independent of a specific operation mode (manual or remote control). Background information is given as far as it is directly related to configurable parameters.

1.2.2 GUI reference

The GUI reference describes the manual operation via the graphical user interface.

The description of a group of settings usually starts with a screenshot presenting the preset values of the parameters. Sometimes, preset values are modified to enable hidden settings.

Below the screenshot, the shown parameters are described. For each parameter, a link to the related command in the command reference is provided. Ranges for numeric parameters and reset values are given there.

For measurement results, links to the remote commands for result query are provided.

1.2.3 Programming examples

The programming examples show how to control and configure the firmware application via a remote control program and how to retrieve measurement results.

The examples consist of comprehensive command sequences. You can check just a single command of a sequence to get an example for the syntax of this single command. But you can also consider an entire sequence showing the commands in the context of a command script, under consideration of dependencies and required orders of the commands.

The command sequences are written with the intention to list most commands of the firmware application. They do not show the fastest way for a given configuration task. The fastest way uses as many reset values as possible and omits the corresponding commands.

The examples are linked from the command reference.

1.2.4 Command reference

The command reference provides information on the remote commands of the firmware application. The commands are grouped according to their function.

Each command description indicates the syntax of the command header and of the parameters. For input parameters, the allowed ranges, reset values and default units are listed.

Most commands have a command form and a query form. Exceptions are marked by "Setting only", "Query only" or "Event".

Links to the programming examples and to the GUI reference are provided. The first software version supporting the command is indicated.

2 LTE signaling

The LTE signaling application emulates an E-UTRAN cell and communicates with the UE under test. The UE can synchronize to the downlink signal and attach to the PS domain. A connection can be set up (3GPP-compliant RMC or user-defined channel).

Two basic signaling options are available: R&S CMW-KS500 for R8 FDD signals and R&S CMW-KS550 for R8 TDD signals. At least one of these basic options is required to use the signaling application. Both basic options support only configurations with a single antenna (SISO). For additional options, see [Section 2.2.1, "Overview of options"](#), on page 18.

Tests can be performed using the LTE multi-evaluation measurement, the PRACH measurement or the SRS measurement (all included in the options R&S CMW-KM500 and R&S CMW-KM550).

Data transfer tests can be performed using the data application unit (DAU, option R&S CMW-B450x and R&S CMW-KM050). The DAU also provides an IMS server for voice over IMS and SMS over IMS. Audio tests can be performed using the audio board plus a speech codec (R&S CMW-B400B and R&S CMW-B405A).

The LTE signaling application provides additional measurements. For details, refer to:

- [Section 2.3.1, "Combined signal path measurements"](#), on page 99
- [Section 2.2.25, "Relative power control tests"](#), on page 96
- [Section 2.3.2, "LTE IP-based data tests"](#), on page 103
- [Section 3, "LTE BLER measurement"](#), on page 921
- [Section 4, "LTE RLC throughput measurement"](#), on page 974

2.1 What's new

This documentation describes version 4.0.250 and higher of the LTE signaling application. Compared to version 4.0.230, it provides the following new features and changes:

- Configurable ATTACH ACCEPT message:
 - [EPS Attach Result](#)
 - [Additional Update Result](#)

2.2 General description

The following sections describe how to use the R&S CMW for LTE signaling tests and provide background information.

- [Overview of options](#).....18
- [Scenarios](#).....19
- [Test setups](#).....27
- [Initiating signaling tests](#).....31

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• CQI channels.....	86
• Semi-persistent scheduling (SPS).....	90
• Operating bands.....	91
• Trigger signals.....	95
• Relative power control tests.....	96

2.2.1 Overview of options

Two basic signaling options are available: R&S CMW-KS500 for R8 FDD signals and R&S CMW-KS550 for R8 TDD signals. At least one of these basic options is required to use the signaling application. Both basic options support only configurations with a single antenna (SISO).

The basic functionality can be enhanced via the following options:

- R&S CMW-KS502 adds support of DL carrier aggregation for FDD signals.
R&S CMW-KS552 adds support of DL carrier aggregation for TDD signals.
- R&S CMW-KS504 adds R12 features for FDD signals.
R&S CMW-KS554 adds R12 features for TDD signals.
- R&S CMW-KS505 adds R15 features for FDD signals.
R&S CMW-KS555 adds R15 features for TDD signals.
- R&S CMW-KS510 provides advanced parameter settings for R8.
R&S CMW-KS512 provides advanced parameter settings for R10.
R&S CMW-KS514 provides advanced parameter settings for R12.
R&S CMW-KS515 provides advanced parameter settings for R15.
- R&S CMW-KS520 adds support of DL MIMO 2x2 and SIMO 1x2.
R&S CMW-KS521 is required for DL MIMO 4x2.
R&S CMW-KS522 is required for DL MIMO 8x2.
- R&S CMW-KS540 adds support of DL MIMO 2x4 and 4x4.
R&S CMW-KS541 is required for DL MIMO 8x4.
- R&S CMW-KS525 adds a user-defined operating band and special bands.
- R&S CMW-KS590 is required for machine-type communication (MTC).

- R&S CMW-KE100 enables basic internal fading.
R&S CMW-KE500 is required for SISO and MIMO 2x2 fading.
R&S CMW-KE501 is required for MIMO 4x2 fading.
R&S CMW-KE502 is required for MIMO 8x2 fading.
R&S CMW-KE504 is required for MIMO 2x4 and 4x4 fading.
R&S CMW-KE505 is required for MIMO 8x4 fading.
- R&S CMW-KS170 is required for cell broadcast services for alerting networks

The dependencies between the options are shown in the following figures.

Examples:

- R&S CMW-KS521 needs R&S CMW-KS520 which needs R&S CMW-KS500 or R&S CMW-KS550.
- R&S CMW-KS515 needs for FDD: R&S CMW-KS514 + KS512 + KS502 + KS505 + KS504 + KS500

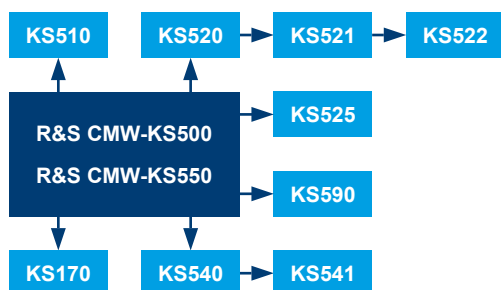


Figure 2-1: Option dependencies part 1

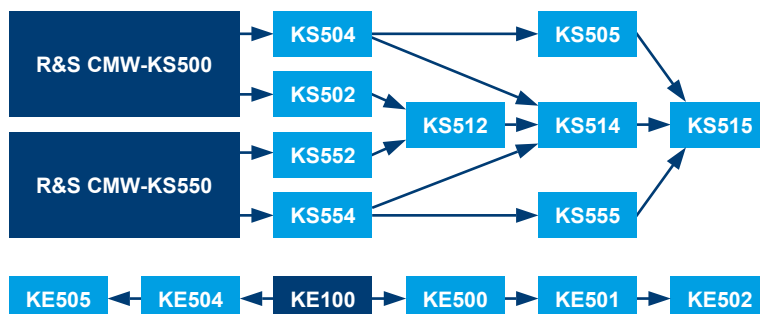


Figure 2-2: Option dependencies part 2

2.2.2 Scenarios

The scenario selection is the first setting to be done for test configuration.

The LTE signaling application provides different scenarios for tests with or without MIMO, fading and carrier aggregation (CA). The number of component carriers (CC) is also selected via the scenario (one PCC + n SCC). Selecting a scenario hides/shows parts of the GUI as required by the scenario.

• Scenario names	20
• Prerequisites for scenarios	20
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2.2.2.1 Scenario names

Most scenario names follow the scheme "<n>CC - <PCC> <SCC1> <SCC2> ...".

<n> is the number of DL component carriers. <PCC>, <SCC1>, and so on, state the antenna configuration for the DL component carriers: SISO 1x1, MIMO nx2 or MIMO nx4.

The number of UL component carriers is configurable. It is limited by the number of DL component carriers and by the absolute limit 4. So you can configure *1 to min(<n>, 4)* UL component carriers.

The string "- Fading -" in the scenario name indicates fading support.

Table 2-1: Examples

Name	Meaning
"1CC - 1x1"	1 carrier, SISO
"1CC - nx2"	1 carrier, MIMO nx2
"2CC - Fading - nx2 nx2"	2 carriers, both MIMO nx2, with fading
"3CC - nx4 nx2 nx2"	3 carriers, PCC MIMO nx4, SCCs MIMO nx2
"4CC - nx2 nx4 nx2 nx2"	4 carriers, PCC nx2, SCC1 nx4, SCC2 nx2, SCC3 nx2
"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx2 nx2"	8 carriers, PCC to SCC5 nx4, SCC6/7 nx2

2.2.2.2 Prerequisites for scenarios

Which scenarios are offered for selection, depends on the available software and hardware options. The dependencies are described in this section.

General hints:

- **Single-CMW setups**

In a single CMW, you can equip up to 4 TX/RX and up to 3 SUA. If these resources are sufficient for a scenario, you can execute the scenario on a single CMW. But you can also use the scenario with a CMWflexx setup.

- **CMWflexx setups**

For scenarios needing more hardware, you need a CMWflexx setup.

The instruments are controlled by an R&S CMWC. The test software including the LTE signaling application is installed and executed on the R&S CMWC.

The options can be distributed over several instruments. But the signaling units and TX/RX modules for each carrier must be in the same instrument. Example: For one carrier, you cannot use a signaling unit of one instrument plus the TX/RX modules of another instrument.

Coprocessing SUAs do not belong to a specific carrier and can be located in any instrument.

The SW licenses are pooled in the R&S CMWC, so it does not matter where they are installed.

- **Co-location**

Some scenarios support co-location. That means you can use the same TX module for two carriers, if you put the carriers into the same band within a bandwidth of 80 MHz. If the required number of resources differs without and with co-location, both numbers are listed in column "TX" of the following table.

- **SUA**

Each SUA board provides two SUA partitions. The GUI and the remote commands select SUA partitions, not SUA boards. In one CMW, you have up to six SUA partitions.

For fading with the SUA, you need additional SUA resources.

All options listed in the following tables are R&S CMW-... options. Example: KS510 means R&S CMW-KS510. Some of the listed options require other options, see [Section 2.2.1, "Overview of options"](#), on page 18.

Scenarios without fading

The following table lists all non-fading scenarios. Column "SUA" indicates the number of SUA partitions not the number of SUA boards.

Table 2-2: Scenarios without fading, required hardware and software options

	TX ¹⁾	SUA ²⁾	Software options
"1CC - 1x1"	1/-	1/-	-
"1CC - nx2"	2/-	1/-	KS520
"1CC - nx4"	4/-	1/-	KS521 or KS540
"2CC - 1x1 1x1"	2/1	2/-	KS502/552
"2CC - nx2 nx2"	4/2	2/-	KS502/552, KS520
"2CC - nx4 nx2"	6/4	2/-	KS502/552, KS520, KS540
"2CC - nx2 nx4"	6/4	2/-	KS502/552, KS520, KS540
"2CC - nx4 nx4"	8/4	2/-	KS502/552, KS540
"3CC - 1x1 1x1 1x1"	3/2	3/-	KS512
"3CC - nx2 1x1 1x1"	4/3	3/-	KS512, KS520
"3CC - 1x1 nx2 1x1"	4/3	3/-	KS512, KS520
"3CC - nx2 nx2 nx2"	6/4	3/5	KS512, KS520
"3CC - nx4 nx2 nx2"	8/6	3/5	KS512, KS520, KS540
"3CC - nx2 nx4 nx2"	8/6	3/5	KS512, KS520, KS540
"3CC - nx4 nx4 nx2"	10/6	4/6	KS512, KS520, KS540
"3CC - nx2 nx4 nx4"	10/6	4/6	KS512, KS520, KS540
"3CC - nx4 nx4 nx4"	12/8	-/5	KS512, KS540
"4CC - 1x1 1x1 1x1 1x1"	4/2	4/6	KS512

	TX¹⁾	SUA²⁾	Software options
"4CC - nx2 nx2 nx2 nx2"	8/4	4/6	KS512, KS520
"4CC - nx4 nx2 nx2 nx2"	10/6	5/7	KS512, KS520, KS540
"4CC - nx2 nx4 nx2 nx2"	10/6	5/7	KS512, KS520, KS540
"4CC - nx4 nx4 nx2 nx2"	12/6	-/6	KS512, KS520, KS540
"4CC - nx2 nx4 nx4 nx2"	12/6	-/6	KS512, KS520, KS540
"4CC - nx4 nx4 nx4 nx2"	14/8	-/6	KS512, KS520, KS540
"4CC - nx2 nx4 nx4 nx4"	14/8	-/6	KS512, KS520, KS540
"4CC - nx4 nx4 nx4 nx4"	16/8	-/6	KS512, KS540
"5CC - 1x1 1x1 1x1 1x1 1x1" ³⁾	5/3	6/8	KS512
"5CC - nx2 nx2 nx2 nx2 nx2"	10/6	6/8	KS512, KS520
"5CC - nx4 nx2 nx2 nx2 nx2"	12/8	-/7	KS512, KS520, KS540
"5CC - nx2 nx4 nx2 nx2 nx2"	12/8	-/7	KS512, KS520, KS540
"5CC - nx4 nx4 nx2 nx2 nx2"	14/8	-/7	KS512, KS520, KS540
"5CC - nx2 nx4 nx4 nx2 nx2"	14/8	-/7	KS512, KS520, KS540
"5CC - nx4 nx4 nx4 nx2 nx2"	16/10	-/7	KS512, KS520, KS540
"5CC - nx2 nx4 nx4 nx4 nx2"	16/10	-/7	KS512, KS520, KS540
"5CC - nx4 nx4 nx4 nx4 nx2"	-/10	-/7	KS512, KS520, KS540
"5CC - nx2 nx4 nx4 nx4 nx4"	-/10	-/7	KS512, KS520, KS540
"5CC - nx4 nx4 nx4 nx4 nx4"	-/12	-/7	KS512, KS540
"6CC - 1x1 1x1 1x1 1x1 1x1 1x1"	6/3	7/9	KS514
"6CC - nx2 nx2 nx2 nx2 nx2 nx2"	12/6	-/8	KS514, KS520
"6CC - nx4 nx2 nx2 nx2 nx2 nx2"	14/8	-/8	KS514, KS520, KS540
"6CC - nx2 nx4 nx2 nx2 nx2 nx2"	14/8	-/8	KS514, KS520, KS540
"6CC - nx4 nx4 nx2 nx2 nx2 nx2"	16/8	-/8	KS514, KS520, KS540
"6CC - nx2 nx4 nx4 nx2 nx2 nx2"	16/8	-/8	KS514, KS520, KS540
"6CC - nx4 nx4 nx4 nx2 nx2 nx2"	-/10	-/8	KS514, KS520, KS540
"6CC - nx2 nx4 nx4 nx4 nx2 nx2"	-/10	-/8	KS514, KS520, KS540
"6CC - nx4 nx4 nx4 nx4 nx2 nx2"	-/10	-/8	KS514, KS520, KS540
"6CC - nx2 nx4 nx4 nx4 nx4 nx2"	-/10	-/8	KS514, KS520, KS540
"6CC - nx4 nx4 nx4 nx4 nx4 nx2"	-/12	-/8	KS515, KS520, KS540
"6CC - nx2 nx4 nx4 nx4 nx4 nx4"	-/12	-/8	KS515, KS520, KS540
"6CC - nx4 nx4 nx4 nx4 nx4 nx4"	-/12	-/8	KS515, KS540
"7CC - 1x1 1x1 1x1 1x1 1x1 1x1 1x1"	7/4	-/9	KS514

	TX¹⁾	SUA²⁾	Software options
"7CC - nx2 nx2 nx2 nx2 nx2 nx2 nx2"	14/8	-/9	KS514, KS520
"7CC - nx4 nx2 nx2 nx2 nx2 nx2 nx2"	16/10	-/9	KS514, KS520, KS540
"7CC - nx2 nx4 nx2 nx2 nx2 nx2 nx2"	16/10	-/9	KS514, KS520, KS540
"7CC - nx4 nx4 nx2 nx2 nx2 nx2 nx2"	-/10	-/9	KS514, KS520, KS540
"7CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2"	-/10	-/9	KS514, KS520, KS540
"7CC - nx4 nx4 nx4 nx2 nx2 nx2 nx2"	-/12	-/9	KS514, KS520, KS540
"7CC - nx2 nx4 nx4 nx4 nx2 nx2 nx2"	-/12	-/9	KS514, KS520, KS540
"7CC - nx4 nx4 nx4 nx4 nx2 nx2 nx2"	-/12	-/9	KS515, KS520, KS540
"7CC - nx2 nx4 nx4 nx4 nx4 nx2 nx2"	-/12	-/9	KS515, KS520, KS540
"7CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2"	-/14	-/9	KS515, KS520, KS540
"7CC - nx2 nx4 nx4 nx4 nx4 nx4 nx2"	-/14	-/9	KS515, KS520, KS540
"7CC - nx4 nx4 nx4 nx4 nx4 nx4 nx2"	-/14	-/9	KS515, KS520, KS540
"7CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4"	-/14	-/9	KS515, KS520, KS540
"7CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4"	-/16	-/9	KS515, KS540
"8CC - 1x1 1x1 1x1 1x1 1x1 1x1 1x1"	8/4	-/10	KS514
"8CC - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2"	16/8	-/10	KS514, KS520
"8CC - nx4 nx2 nx2 nx2 nx2 nx2 nx2 nx2"	-/10	-/10	KS514, KS520, KS540
"8CC - nx2 nx4 nx2 nx2 nx2 nx2 nx2 nx2"	-/10	-/10	KS514, KS520, KS540
"8CC - nx4 nx4 nx2 nx2 nx2 nx2 nx2 nx2"	-/10	-/10	KS514, KS520, KS540
"8CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2 nx2"	-/10	-/10	KS514, KS520, KS540
"8CC - nx4 nx4 nx4 nx2 nx2 nx2 nx2 nx2"	-/12	-/10	KS515, KS520, KS540
"8CC - nx2 nx4 nx4 nx4 nx2 nx2 nx2 nx2"	-/12	-/10	KS515, KS520, KS540
"8CC - nx4 nx4 nx4 nx4 nx2 nx2 nx2 nx2"	-/12	-/10	KS515, KS520, KS540
"8CC - nx2 nx4 nx4 nx4 nx4 nx2 nx2 nx2"	-/12	-/10	KS515, KS520, KS540
"8CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2 nx2"	-/14	-/10	KS515, KS520, KS540
"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx2 nx2"	-/14	-/10	KS515, KS520, KS540
"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx2 nx2"	-/14	-/10	KS515, KS520, KS540
"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4 nx2"	-/14	-/10	KS515, KS520, KS540
"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4 nx2"	-/16	-/10	KS515, KS520, KS540
"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4 nx4"	-/16	-/10	KS515, KS520, KS540

	TX ¹⁾	SUA ²⁾	Software options
"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4 nx4"	-/16	-/10	KS515, KS540
¹⁾ Required TX modules. "x/y" means "without co-location" / "with co-location". The number of required RX modules <n> is not listed. It equals the number of carriers (scenario "<n>CC..."). Exception: 4 RX required for "5CC - 1x1 1x1 1x1 1x1 1x1" in a single-CMW setup. ²⁾ Number of required SUA partitions. "x/y" means "without coprocessing" / "with coprocessing". Coprocessing can be not possible "x/-", optional to reach the maximum throughput "x/y", or mandatory "-/y". In general, each carrier needs one SUA partition. Exceptions: • If sum = nx10, the PCC needs two SUA partitions (considered in the table). • If sum < nx12 and CC = 5 or 6, the PCC needs two SUA partitions (considered in the table). • If sum < nx12 and CC ≥ 7, coprocessing is mandatory (two SUA partitions more, is considered). • If sum ≥ nx12, coprocessing is mandatory (two SUA partitions more, considered in the table). • If an uplink uses 64-QAM or 256-QAM, the carrier needs two SUA partitions. Possibly, you need more SUA partitions than listed in the table. ³⁾ If you run the scenario in a single-CMW setup, you cannot use the UL of SCC4.			

For further reference:

- [Section 2.2.5, "Carrier aggregation"](#), on page 32 (also for co-location)
- ["Scenario"](#) on page 140 (scenario selection)
- [Section 2.2.2.3, "Throughput, SUA usage and coprocessing"](#), on page 25 (assigning optional SUA resources)

Scenarios with internal fading

The following table lists all scenarios with internal fading. For SUA partitions, you need the sum of both SUA columns.

Table 2-3: Scenarios with internal fading, required hardware and software options

	TX ¹⁾	SUA for CC ²⁾	SUA fading ²⁾	Software options (plus KE100)
"1CC - Fading - 1x1"	1/-	1/-	1	KS510, KE500
"1CC - Fading - nx2"	2/-	1/-	1	KS510, KS520, KE500
"1CC - Fading - nx4"	4/-	1/-	1	KS510, KS540, KE504
"2CC - Fading - 1x1 1x1"	2/1	2/-	2	KS512, KE500
"2CC - Fading - nx2 nx2"	4/2	2/-	2	KS512, KS520, KE500
"2CC - Fading - nx4 nx2"	6/4	2/-	2	KS512, KS520, KS540, KE500, KE504
"2CC - Fading - nx4 nx4"	8/4	2/-	2	KS512, KS540, KE504
"3CC - Fading - nx2 nx2 nx2"	6/4	3/5	3	KS512, KS520, KE500
"3CC - Fading - nx4 nx2 nx2"	8/6	3/5	3	KS512, KS520, KS540, KE500, KE504
"3CC - Fading - nx4 nx4 nx2"	10/6	4/6	3	KS512, KS520, KS540, KE500, KE504
"3CC - Fading - nx2 nx4 nx4"	10/6	4/6	3	KS512, KS520, KS540, KE500, KE504
"4CC - Fading - nx2 nx2 nx2 nx2"	8/4	4/6	4	KS512, KS520, KE500
"4CC - Fading - nx4 nx4 nx4 nx4"	16/8	-/6	4	KS512, KS540, KE504
"5CC - Fading - nx2 nx2 nx2 nx2 nx2"	10/6	6/8	5	KS512, KS520, KE500

	TX ¹⁾	SUA for CC ²⁾	SUA fading ²⁾	Software options (plus KE100)
"5CC - Fading - nx4 nx4 nx4 nx2 nx2"	16/10	-/7	5	KS512, KS520, KS540, KE500, KE504
"5CC - Fading - nx2 nx4 nx4 nx4 nx2"	16/10	-/7	5	KS512, KS520, KS540, KE500, KE504
"6CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2"	12/6	-/8	6	KS514, KS520, KE500
"6CC - Fading - nx4 nx4 nx2 nx2 nx2 nx2"	16/8	-/8	6	KS514, KS520, KS540, KE500, KE504
"6CC - Fading - nx2 nx4 nx4 nx2 nx2 nx2"	16/8	-/8	6	KS514, KS520, KS540, KE500, KE504
"7CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2 nx2"	14/8	-/9	7	KS514, KS520, KE500
"7CC - Fading - nx4 nx2 nx2 nx2 nx2 nx2 nx2"	16/10	-/9	7	KS514, KS520, KS540, KE500, KE504
"7CC - Fading - nx2 nx4 nx2 nx2 nx2 nx2 nx2"	16/10	-/9	7	KS514, KS520, KS540, KE500, KE504
"8CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2"	16/8	-/10	8	KS514, KS520, KE500
¹⁾ Required TX modules. "x/y" means "without co-location" / "with co-location". The number of required RX modules is ≤ the number of required TX modules. ²⁾ Number of SUA partitions required for the unfaded carriers ("SUA for CC") plus number of SUA partitions required for fading ("SUA fading"). You need the sum of both columns. For "SUA for CC", "x/y" means "without coprocessing" / "with coprocessing". Coprocessing can be not possible "x/-", optional to reach the maximum throughput "x/y", or mandatory "-/y". In general, you need one SUA partition for each unfaded carrier plus one SUA partition for fading of the carrier. Exceptions that are considered in the table: • If sum = nx10, the unfaded PCC needs two SUA partitions. • If sum < nx12 and CC = 5 or 6, the unfaded PCC needs two SUA partitions. • If sum < nx12 and CC ≥ 7, coprocessing is mandatory (two SUA partitions more). • If sum ≥ nx12, coprocessing is mandatory (two SUA partitions more). Exceptions that are not considered in the table. Possibly, you need more partitions than listed in the table. • If a carrier uses MIMO 8x2, you need two partitions for the carrier and two partitions for fading. • If an uplink uses 64-QAM or 256-QAM, the unfaded carrier needs two partitions. • For SCME profiles plus MIMO 4x2 or higher, fading of the carrier needs two partitions.				

For further reference:

- [Section 2.2.6, "Internal fading"](#), on page 36

2.2.2.3 Throughput, SUA usage and coprocessing

The number of SUA partitions required by a scenario is listed in [Section 2.2.2.2, "Prerequisites for scenarios"](#), on page 20.

For some scenarios, you can assign additional SUA resources to increase the reachable maximum throughput:

- If the scenario requires only one SUA partition for the PCC, you can nevertheless assign two SUA partitions.
- If coprocessing is optional for the scenario, you can enable coprocessing (that means, assign an additional SUA).

Assigning additional SUA partitions is relevant in particular for maximum IP throughput tests with the data application unit (DAU).

The following table lists maximum IP throughputs reachable in the data application mode, depending on: the SUA configuration, the number of uplink component carriers and the protocol.

Table 2-4: Examples for maximum IP throughput

SUA resources	UL CC	Protocol	Throughput / (Mbit/s)		No of layers x modulation (examples)
			DL	UL	
SUA for PCC: one partition	1, 2	UDP, TCP	600	100	6 x 256-QAM (DL) or 8 x 64-QAM (DL) 2 x 16-QAM (UL)
SUA for PCC: two partitions	1, 2, 3	UDP, TCP	1000	150	10 x 256-QAM (DL) 2 x 64-QAM (UL) or 3 x 16-QAM (UL)
	3, 4	UDP, TCP	600	225	6 x 256-QAM (DL) or 8 x 64-QAM (DL) 3 x 64-QAM (UL)
Coproprocessing active	1	UDP	2000	50	20 x 256-QAM (DL) 1 x 16-QAM (UL)
	2, 3, 4	UDP	1600	300	16 x 256-QAM (DL) 4 x 64-QAM (UL)
	2, 3, 4	TCP	1600	150	16 x 256-QAM (DL) 2 x 64-QAM (UL) or 3 x 16-QAM (UL)
	3, 4	TCP	1500	225	15 x 256-QAM (DL) or 20 x 64-QAM (DL) 3 x 64-QAM (UL)

Examples for selected scenarios

Assume that you have 8 DL layers, scenario "4CC - nx2 nx2 nx2 nx2".

If you assign one SUA partition to the PCC, you can reach throughputs of 600 Mbit/s (DL) and 100 Mbit/s (UL), though the physical layer supports 800 Mbit/s (DL). If you want to increase the DL throughput or the UL throughput, you need two SUA partitions for the PCC.

Table 2-5: IP throughput with 8 layers

SUA resources	UL CC	Protocol	Throughput / (Mbit/s)		No of layers x modulation (examples)
			DL	UL	
SUA for PCC: one partition	2	UDP, TCP	600	100	8 x 64-QAM (DL) 2 x 16-QAM (UL)
SUA for PCC: two partitions	1, 2, 3	UDP, TCP	800	150	8 x 256-QAM (DL) 3 x 16-QAM (UL) or 2 x 64-QAM (UL)

Assume a scenario with 10 DL layers, for example: "3CC - nx4 nx4 nx2", "4CC - nx4 nx2 nx2 nx2", "5CC - nx2 nx2 nx2 nx2 nx2".

Without coprocessing, you can reach throughputs of 1000 Mbit/s (DL) and 150 Mbit/s (UL). You already have the maximum DL throughput that is possible with 10 layers. If you want to increase the UL throughput, you must enable coprocessing.

Table 2-6: IP throughput with 10 layers

SUA resources	UL CC	Protocol	Throughput / (Mbit/s)		No of layers x modulation (examples)
			DL	UL	
SUA for PCC: two partitions	1, 2, 3	UDP, TCP	1000	150	10 x 256-QAM (DL) 2 x 64-QAM (UL) or 3 x 16-QAM (UL)
Coprocessing active	3, 4	UDP, TCP	1000	225	10 x 256-QAM (DL) 3 x 64-QAM (UL)

Troubleshooting

To reach the maximum IP throughput, ensure the following:

- Use the latest base software and the latest DAU software.
- For TCP tests, configure the maximum bit rate in the QoS settings of the DAU, as described in the DAU documentation.
- In the RX measurement, check that you have 0% BLER. If you see a higher BLER:
 - Check your RF cabling for damages and for low-quality RF cables.
 - Ensure that the RS EPRE setting is high enough.

2.2.3 Test setups

The following sections provide an overview of typical test setups for the individual scenarios.

- [Test setup for scenario 1CC - 1x1](#).....27
- [Test setup for scenario 1CC - nx2](#).....28
- [Test setup for CA scenarios \(single CMW\)](#)..... 28
- [Test setup for CA scenarios \(CMWflexx\)](#)..... 30
- [Test setup for internal fading scenarios](#).....31

2.2.3.1 Test setup for scenario 1CC - 1x1

The basic test setup for a "1CC - 1x1" scenario uses a bidirectional RF connection between the tester and the tested UE. It carries both the downlink (DL) and the uplink (UL) signal:

- The R&S CMW transmits the DL signal to which the UE can synchronize to perform an attach. The DL signal transfers signaling messages and user data to the UE.
- The UE transmits a UL signal that the R&S CMW can receive, decode and analyze.

For this setup, the UE is connected to one of the bidirectional RF COM connectors on the front panel of the R&S CMW. No additional cabling and no external trigger are needed. The input level ranges of all RF COM connectors are identical.

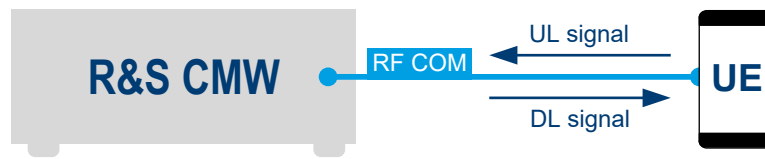


Figure 2-3: Test setup for 1CC - 1x1

2.2.3.2 Test setup for scenario 1CC - nx2

The scenario "1CC - nx2" is used for setups with two UE RX antennas (SIMO 1x2, MIMO nx2 and beamforming). A test setup for this scenario involves one UL and two DL signals at the UE side. Typically, one bidirectional connection carries one UL and one DL signal. An additional connection carries the second DL signal. The two DL signals must be transmitted via different TX modules and different RF connectors. So the instrument must support at least two TX paths. You can use, for example, RF 1 COM for the bidirectional connection and RF 3 COM for the second DL connection.

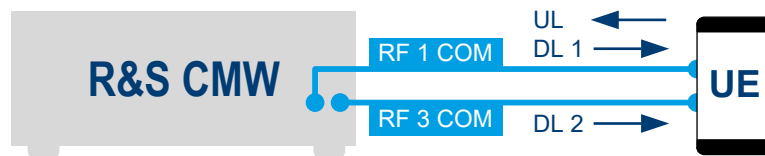


Figure 2-4: Test setup for 1CC - nx2

Using several signaling applications in parallel

Depending on the installed hardware and software options, you can run several signaling application instances in parallel.

Example: Assume an instrument with one signaling unit advanced (SUA, two partitions), one signaling unit universal (SUU), four TX paths and four RX paths.

This instrument allows you to run for example:

- Two LTE signaling instances using the "1CC - nx2" scenario, or
- Two LTE signaling instances using the "1CC - 1x1" scenario, plus one instance of an SUU signaling application (e.g. GSM / CDMA2000)
- One LTE signaling instance using the "1CC - nx2" scenario, plus one instance of an SUA signaling application (e.g. WCDMA)

The LTE and WCDMA signaling applications both use the SUA. An SUA partition can only be used by one instance at a time.

2.2.3.3 Test setup for CA scenarios (single CMW)

The carrier aggregation (CA) scenarios allow you to establish a Release-10 connection with several component carriers. A carrier can be a primary component carrier (PCC) or a secondary component carrier (SCC).

Independent of the scenario, there is always one UL PCC and one DL PCC. For a CA scenario, there is at least one additional DL SCC. You can also enable additional UL SCCs.

Depending on the scenario, a DL carrier can have one output path (SISO) or two output paths (for example MIMO nx2). In total, up to four output paths per carrier are supported. As always for MIMO, the output paths of a carrier must use different TX modules and different RF connectors.

Over all carriers, all input paths must use different RX modules and different RF connectors. With co-location, you can let two carriers share TX modules. Without co-location, all output paths must use different TX modules. See also [Section 2.2.5.1, "RF paths and co-location"](#), on page 33.

With an advanced frontend, you can route two output paths to the same connector. This feature allows you to choose the optimum test setup depending on the number of antenna connectors of your UE. There is no need for external combiners.

The following example setup is suitable for CA with four DL and a UE with only two antenna connectors. There are two PCC DL (MIMO), two SCC DL (one MIMO SCC or two SISO SCCs), one PCC UL and one SCC UL.

RF 1 COM is used for the uplinks, the first PCC DL and the first SCC DL. RF 3 COM is used for the second PCC DL and the second SCC DL.

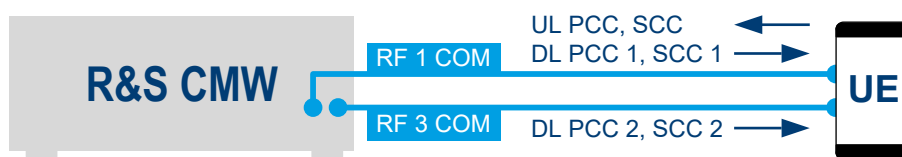


Figure 2-5: Test setup for CA, four DL, two UL, two UE antenna connectors

For a UE with four antenna connectors, the following test setup could be used instead. Each DL signal is now routed to a different connector.

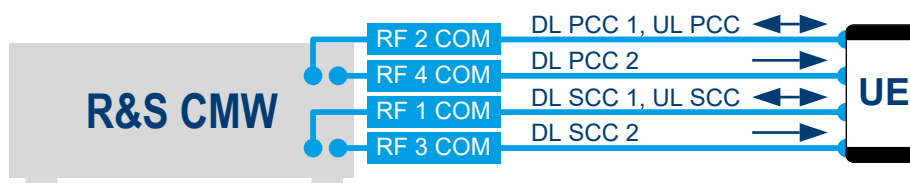


Figure 2-6: Test setup for CA, four DL, two UL, four UE antenna connectors

For CA with one UL and two DL, you could, for example, use only RF 1 COM (UL, PCC DL and SCC DL). Or you could use RF 1 COM (UL and PCC DL) and RF 3 COM (SCC DL).

2.2.3.4 Test setup for CA scenarios (CMWflexx)

Some carrier aggregation scenarios require a CMWflexx setup, comprising one R&S CMWC plus several R&S CMW.



The instruments are connected to each other via their rear panels. This rear panel cabling is identical for all CMWflexx setups. For details, refer to the R&S CMWC documentation.

The required RF cabling depends on the scenario and on the RF antenna configuration of the UE.

The following RF cabling example is suitable for DL CA with four carriers plus MIMO nx2 (eight DL, no co-location) and a UE with only two antenna connectors. One antenna is used for the first DL of each carrier and for the uplinks (UL CA with two carriers). The second antenna is used for the second DL of each carrier.

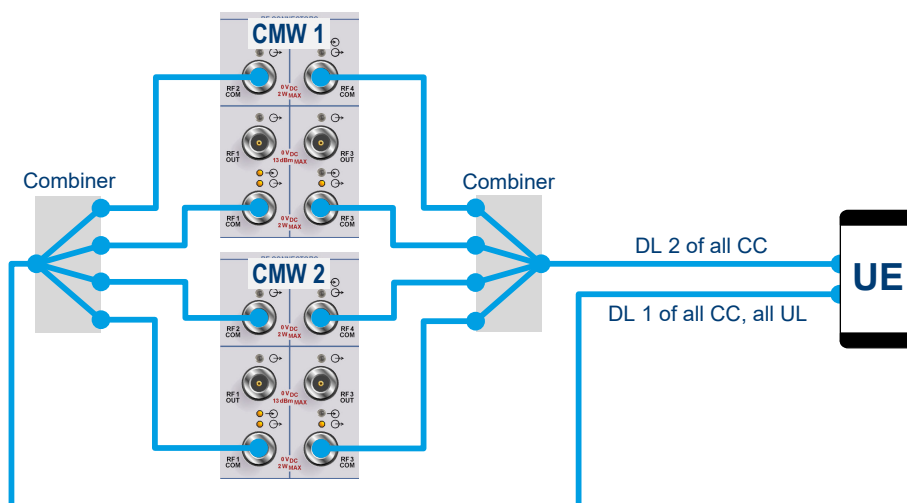


Figure 2-7: CMWflexx setup, four CC, MIMO, two UE antenna connectors

As an external combiner, you can for example use the R&S CMW-Z24. It provides two 4-way combiners. For the given example, you would connect the R&S CMW-Z24 as follows:

- Connect the first UE antenna (UL + first DL) to the port AC.
- Connect the second UE antenna (second DL) to the port BC.
- Connect RF 1 COM and RF 2 COM of both R&S CMW to the ports A1 to A4, in any order.
- Connect RF 3 COM and RF 4 COM of both R&S CMW to the ports B1 to B4, in any order.

When configuring the RF settings, consider that the external combiner increases the external attenuation.

2.2.3.5 Test setup for internal fading scenarios

For internal fading scenarios, the test setup is the same as for the corresponding scenario without fading.

2.2.4 Initiating signaling tests

The required settings vary depending on the measurement to be performed. However, the following general procedure is applicable to most measurements.

1. Connect your UE to the R&S CMW (see [Section 2.2.3, "Test setups"](#), on page 27).
2. Open the LTE signaling application.
3. Configure the signaling application according to your test case.
You can modify the most important settings directly in the main view. The "Config" hotkey provides access to the configuration dialog box.
4. To turn on the DL signal, press [ON | OFF]. Alternatively, right-click the "LTE Signaling" softkey. And click ON.
Wait until the "LTE Signaling" softkey indicates the "ON" state and the hour glass symbol has disappeared.
5. Switch on the UE.
The UE synchronizes to the DL signal and attaches. Note the connection states displayed in the main view.
6. If there is no RRC connection, set up a connection.
You can keep the RRC connection after the attach. See ["Keep RRC Connection"](#) on page 209.
You can set up a connection at the UE or use the hotkeys of the signaling application. See [Section 2.4.2, "Signaling control"](#), on page 130.
7. To switch to the measurement application, press the "LTE TX Meas" or "LTE RX Meas" softkey. The LTE RX measurements are part of the LTE signaling applica-

tion. The LTE TX measurements are available as option R&S CMW-KM500/-KM550.

8. Configure and start the measurement.

If the UE fails to attach

Check the following R&S CMW settings:

- The "Duplex Mode" must be supported by the UE.
- The "Frequency" of the generated DL signal must be within the frequency band supported by the UE.
- The "Downlink Power Levels", especially the "RS EPRE" must be sufficient so that the UE under test can receive the DL signal.
- The "Uplink Power Control" settings must be appropriate so that the UL signal is strong enough to be received and decoded by the R&S CMW.
- The UE capabilities must be in accordance with the security settings in the "Network" section of the configuration dialog.

The attach procedure can fail if authentication or security is disabled but the UE expects/requires an authentication or security procedure. It can also fail if authentication or security is enabled but not supported by the UE or the secret key does not match.

An appropriate USIM can be obtained from Rohde & Schwarz (R&S CMW-Z06, stock no. 1209.2010.02).

- Some UEs need a DL signal without padding bits for attach. You can disable downlink padding in the "Connection" section of the configuration dialog.

If the measurement does not start

If the measurement does not start at all while the signaling application generates a downlink signal, verify that the combined signal path scenario is selected in the measurement.

2.2.5 Carrier aggregation

Carrier aggregation (CA) is an LTE-Advanced Release 10 feature. It aggregates several component carriers to reach a higher bandwidth. Each single component carrier is compliant with Release 8. Release 11 and 12 enhance the basic carrier aggregation feature of Release 10. Release 13 adds enhanced carrier aggregation with more than five DL carriers.

The signaling application supports the aggregation of up to eight downlink carriers and of up to six uplink carriers. With eight 20-MHz carriers, you can reach a total bandwidth of 160 MHz.

The aggregated carriers can be located in the same operating band or in different operating bands. They can use a continuous spectrum or there can be gaps between the carriers. So all kinds of carrier aggregation are supported: intra-band contiguous CA, intra-band non-contiguous CA and inter-band CA.

Most settings can be configured independently per carrier, for example the carrier frequency, bandwidth, power, scheduling type and related channel settings.

According to 3GPP, a connection setup with carrier aggregation is a two-step procedure. In the first step, a Release 8 single carrier connection is set up. The used carrier is called primary component carrier (PCC). In the second step, one or more additional carriers are added, called secondary component carrier (SCC). The individual SCCs are distinguished via a number (SCC1, SCC2, ...).

Access procedures like cell search, cell selection and initial random access are only performed on the PCC, not on the SCCs. The SCCs are added, deleted or modified using an "RRC Connection Reconfiguration" message.

Software options for carrier aggregation

You need the following options:

- R&S CMW-KS502/552 (FDD/TDD) for 2 DL carriers
- R&S CMW-KS502/552 plus R&S CMW-KS512 for 2 UL carriers, for more than 2 DL carriers and for advanced carrier aggregation settings
- R&S CMW-KS504/554 plus R&S CMW-KS514 for 3 to 6 UL carriers and for more than 5 DL carriers
- R&S CMW-KS505/555 plus R&S CMW-KS515 for more than 20 layers (sum of DL carriers > nx20)
- For carrier aggregation with MIMO: MIMO software options as for MIMO without carrier aggregation

2.2.5.1 RF paths and co-location

For full carrier aggregation flexibility, you need one TX module per output path. For a list of required TX modules per scenario, see [Section 2.2.2, "Scenarios"](#), on page 19.

To reduce the number of required TX modules, you can let two DL carriers share TX modules. This feature is called co-location. It reduces the number of TX modules required due to carrier aggregation, but not the number of TX modules required due to MIMO. The following table gives some examples.

Table 2-7: Configuration examples

Scenario	Without co-location	With co-location	Comments
2 CC 4 RF out	4 TX PCC: TX 1.1, 1.2 SCC1: TX 1.3, 1.4	2 TX PCC: TX 1.1, 1.2 SCC1: TX 1.1, 1.2	Separate TX required for MIMO. Same TX shared by two CC.
3 CC 6 RF out	6 TX PCC: TX 1.1, 1.2 SCC1: TX 1.3, 1.4 SCC2: TX 2.1, 2.2	4 TX PCC: TX 1.1, 1.2 SCC1: TX 1.1, 1.2 SCC2: TX 1.3, 1.4	Two carriers co-located. No co-location partner for third carrier. Only one CMW instead of two CMW needed.
4 CC 16 RF out	16 TX PCC: TX 1.1, 1.2, 1.3, 1.4 SCC1: TX 2.1, 2.2, 2.3, 2.4 SCC2: TX 3.1, 3.2, 3.3, 3.4 SCC3: TX 4.1, 4.2, 4.3, 4.4	8 TX PCC: TX 1.1, 1.2, 1.3, 1.4 SCC1: TX 2.1, 2.2, 2.3, 2.4 SCC2: TX 1.1, 1.2, 1.3, 1.4 SCC3: TX 2.1, 2.2, 2.3, 2.4	Two pairs of co-located carriers. Only two CMW instead of four CMW needed.

Restrictions for co-located output paths:

- The RS EPRE of co-located carriers can differ by maximum 30 dB.
- The co-located channels must be located in the same band, within the 80-MHz bandwidth of a TX module. So the maximum carrier center frequency spacing equals 80 MHz minus half of the two cell bandwidths.
Example: 20-MHz channel + 10-MHz channel
Max spacing = 80 MHz - 10 MHz - 5 MHz = 65 MHz

The GUI takes care of these restrictions. With co-location, the allowed value ranges are adapted automatically.

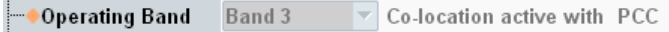
Configuring co-location

To co-locate two carriers, proceed as follows.

1. Select the signaling units for the two carriers. You need two different signaling units, located in the same instrument.
2. Select the same TX modules for both carriers.

Co-location is indicated at the GUI as follows:

- The icons on the component carrier tabs change.
Carriers with the same number in the icon are co-located to each other.

- The operating band setting of one carrier shows the coupling to the other carrier.


2.2.5.2 Setting up a connection with carrier aggregation

Proceed as follows:

1. Connect your UE to the R&S CMW (see [Section 2.2.3.3, "Test setup for CA scenarios \(single CMW\)"](#), on page 28).
2. Select a scenario with carrier aggregation.
3. Configure the SCC activation mode as desired. See ["SCC Activation Mode"](#) on page 142.
4. If you want to use uplink carrier aggregation, enable the uplinks of the desired SCCs. See ["Use UL"](#) on page 143.
5. If you want to align several uplinks for contiguous uplink carrier aggregation:
 - a) Configure the cell bandwidths of all relevant component carriers.
 - b) Specify the band and channel / center frequency of the component carrier with the lowest frequency.
 - c) Press the "Intraband Contiguous UL" > "Edit" button.
 - d) Select the component carriers that you want to align. Sort the entries from left to right, with increasing frequency.
 - e) Press "Apply".
The leftmost entry serves as primary carrier. The algorithm aligns the other selected component carriers to the primary carrier, so that the spacing between the carriers is correct, as defined in 3GPP TS 36.101, section 5.7.1A.
6. Configure the RF settings of all carriers. The channels of the carriers must not overlap.
7. Configure any other signaling application settings as desired, for example DL power, cell setup and connection settings.
If you change the physical cell ID or the E-UTRAN cell identifier, set a different value for each carrier.
If you want to add an SCC in packet-switched state "Attached", enable the connection setting "Keep RRC Connection".
8. To turn on the primary cell signal, press [ON | OFF]. Alternatively, right-click the "LTE Signaling" softkey. And click ON.
Wait until the "LTE Signaling" softkey indicates the "ON" state and the hour glass symbol has disappeared.
9. Switch on the UE.
The UE synchronizes to the DL signal and attaches. Note the packet-switched state in the main view and wait until it equals "Attached".
10. If the RRC state is "Idle", press the "Connect" hotkey to set up an RRC connection on the PCC.
Note the RRC state displayed in the main view. Wait until it equals "Connected".
11. Depending on the configured SCC activation mode, the SCCs are already active, or you can now activate them via hotkeys.
All hotkeys are related to a single SCC. On the "PCC" tab, the hotkeys affect the SCC1. On an SCC tab, the hotkeys affect the corresponding SCC.
 - Mode "Auto":

An automatism adds and activates the SCCs upon RRC connection establishment.

- Mode "Semiautomatic":
Press the "Activate MAC" hotkey to add and activate an SCC.
- Mode "Manual":
To add and activate an SCC, press the hotkey "SCC On", then "SCC add RRC", then "SCC activate MAC".

Note the SCC state displayed in the main view. After successful SCC activation, the state is "MAC Activated".

2.2.6 Internal fading

Testing under realistic air interface conditions is important in order to verify the receiver performance and the correct operation of the protocol stack implementation. For example, block error rates, throughput performance and correct operation of layer 1 procedures like hybrid automatic repeat request (HARQ) retransmission can be evaluated.

The SUA comes with a fading simulator and AWGN generator that can be selectively enabled. It manipulates the generated downlink I/Q data stream to emulate typical signal conditions at the receiver.

The following options are required to use the internal fading simulator in LTE:

- Option R&S CMW-KS510 for scenarios without carrier aggregation
Option R&S CMW-KS512 for scenarios with carrier aggregation
- Option R&S CMW-KE100 for any fading
- Option R&S CMW-KE500 for SISO and MIMO 2x2
Plus R&S CMW-KE501 for MIMO 4x2
Plus R&S CMW-KE502 for MIMO 8x2
- Option R&S CMW-KE504 for MIMO 2x4, MIMO 4x4 and for SCME profiles
Plus R&S CMW-KE505 for MIMO 8x4

2.2.6.1 Fading simulator

Multi-path fading is an effect which occurs in real world situations. A signal sent from the base station follows the direct line of sight and/or takes routes with reflections. The phase-shifted signals sum up at the receiving antenna. If the receiver is moving, there are also frequency-shifted signals.

The internal fading simulator supports multipath propagation conditions defined in annex B.2 of 3GPP TS 36.101.

A faded signal has a higher crest factor than an unfaded one. To avoid distortion, the baseband signal must be attenuated before entering the fading module. The necessary attenuation (insertion loss) depends on the selected fading profile.

In LTE signaling, the insertion loss at the baseband level can be calculated automatically or set manually. It is automatically compensated on the HF level. The compensa-

tion implies a shift of the allowed DL power range to the same extent, but in opposite direction.

2.2.6.2 AWGN generator

Additive white Gaussian noise (AWGN) is typically modeled in receiver tests, because it can lead to a decrease of throughput. The quality of the received signal is affected by the ratio of the signal power to the surrounding traffic noise level (signal to noise ratio). The modulated signals from neighbor cells simply appear as noise. This effect is simulated by adding AWGN to the signal.

The internal fading module supports AWGN insertion with configurable bandwidth and signal to noise ratio. Insertion loss at the baseband level is calculated and compensated automatically at the HF.

AWGN insertion via the signaling unit is disabled for fading scenarios (see ["AWGN"](#) on page 160).

2.2.7 Data tests and voice over LTE

The LTE signaling application provides a test mode, using only layer 1 and 2 of the protocol stack. And it provides a data application mode that supports also layer 3.

To use the data application mode, you need the following options in addition to the LTE signaling application:

- Data application unit (DAU, R&S CMW-B450x, mandatory)
- IPv4 enabler (R&S CMW-KA100, mandatory)
IPv6 enabler (R&S CMW-KA150, optional)
- DAU measurements (R&S CMW-KM050, optional)
- IMS server (R&S CMW-KAA20, optional)
- Installed DAU software package

If there is no space left for installation of a DAU in your instrument, you can install all these options in another R&S CMW and access the external DAU via LAN. For such a setup, you need also R&S CMW-KA120, both on the instrument hosting the external DAU and on the instrument where the LTE signaling application is running.

In data application mode, you can perform IP-based data tests using the DAU measurements or you can perform an RLC throughput measurement. You can also use the IMS server provided by the DAU to set up a voice over LTE (VoLTE) call.

For further reference:

- [Section 4, "LTE RLC throughput measurement"](#), on page 974
- [Section 2.3.2, "LTE IP-based data tests"](#), on page 103
This application sheet describes how to set up an LTE connection in data application mode and provides some examples for IP-based data tests.
- [Section 2.3.3, "VoLTE call setup and audio tests"](#), on page 107
This application sheet describes how to set up a VoLTE call and how to perform basic audio tests for the established call.

- Data application unit documentation
- ["External DAU"](#) on page 207

2.2.8 Audio tests and speech quality tests

You can connect the LTE signaling application to the speech codec of an installed audio board. This connection allows you, to feed an audio signal into an RF connection to the UE or to extract an audio signal from the RF connection for audio measurements. You can, for example, connect the R&S UPV to the audio board of the R&S CMW and perform a speech quality analysis.

To set up a voice over LTE connection, you need the DAU, see [Section 2.2.7, "Data tests and voice over LTE"](#), on page 37.

To connect the LTE signaling application to the speech codec, you need the following options in addition to the LTE signaling application:

- Audio board, R&S CMW-B400B
- Speech codec, R&S CMW-B405A
- Installed audio software package

For further reference:

- [Section 2.3.3, "VoLTE call setup and audio tests"](#), on page 107
This application sheet describes how to set up a VoLTE call and how to perform basic audio tests for the established call.
- "Audio Measurements" documentation

2.2.9 Connection states

An LTE connection is always a packet-switched connection. There can be several carriers in each direction (carrier aggregation).

A connection setup with carrier aggregation is a two-step procedure. In the first step, a Release 8 single carrier connection is set up. The used carrier is called primary component carrier (PCC). In the second step, the additional carriers are added, called secondary component carrier (SCC1, SCC2, ...).

The related connection states are described in the following sections. For a connection without carrier aggregation, only the packet-switched states are relevant.

2.2.9.1 Packet-switched states

The main connection states related to the primary packet-switched connection are described in the following table. They are related to the PCC downlink and uplink connection. The states are displayed in the main view as "Packet Switched" (PS) state.

PS state	Description
"(Cell) Off"	No downlink signal transmission
"(Cell) On"	The R&S CMW emulates an E-UTRAN cell, transmitting an LTE signal to which the UE can synchronize. After synchronization, the UE can initiate an attach towards the instrument.
"Attached"	Synchronization and attach have been performed. A default bearer has been established. Depending on parameter Keep RRC Connection , a radio resource control (RRC) connection is still established or has been released. To send or receive SMS messages, to start an inter-RAT handover or to establish an SCC connection, an established RRC connection is required.
"Connection Established"	A dedicated bearer has been set up. An RRC connection has been established (during attach or during connect). User data can be exchanged via shared channels. The R&S CMW can vary connection parameters, perform transmitter and receiver tests, or initiate a handover.

State transitions can be initiated by the instrument or by the UE. The following figure provides an overview.

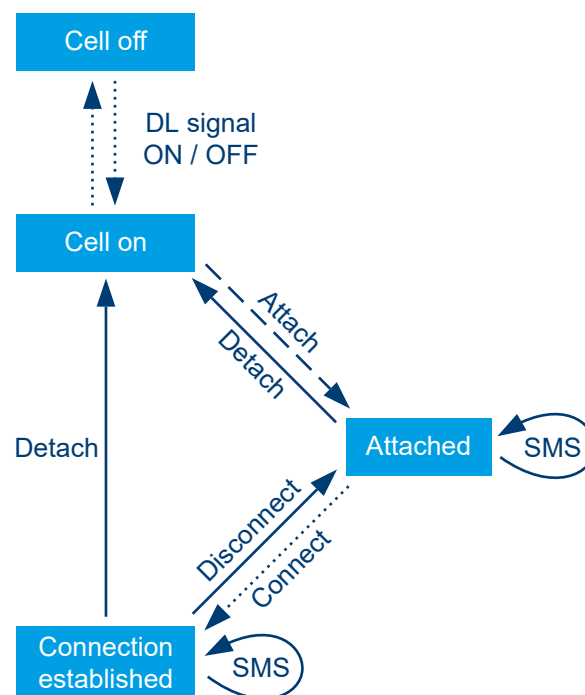


Figure 2-8: PS state transitions

Dotted line = Action initiated by instrument
 Dashed line = Action initiated by UE
 Solid line = Action initiated by UE or instrument

In addition to the main states shown in the table and the figure the instrument indicates the following transitory states:

- **"Signaling in Progress"**

For example, displayed during attach or when the channel changes for an established connection.

- **"Connection in Progress"**
Displayed during connection setup.
- **"Sending Message"**
Displayed while an SMS message is sent to the MS.
- **"Receiving Message"**
Displayed while an SMS message is received from the MS. To check the received message, see [Section 2.4.16.2, "Incoming SMS"](#), on page 288.
- **"Incoming Handover in Progress"**
Displayed while a handover is received (not for handover within signaling application).
- **"Outgoing Handover in Progress"**
Displayed while a handover to another signaling application / instrument is performed.
- **"Disconnect in Progress"**



Additional transitions and handover

The list of transitions in [Figure 2-8](#) is not complete. The "Off" state can be reached from any state by turning off the cell signal. Moreover, incidents like an alerting timeout or a loss of the radio link cause additional transitions.

A handover within the signaling application can be performed in the "Connection Established" state. An inter-RAT / inter-instrument redirection can also be performed in the "Attached" state (established RRC connection required).

2.2.9.2 SCC states

The state of each secondary component carrier (SCC) is displayed in the main view and described in the following table. An SCC can be used in downlink direction only, or in downlink plus uplink direction.

SCC state	Description
"(SCC) Off"	No downlink signal transmission via the SCC frequency
"(SCC) On"	An SCC downlink signal is transmitted, including synchronization signals, reference signal and system information. The UE is not yet informed about the SCC and ignores it. Prerequisite: Packet-switched state at least "Cell On"
"RRC Added"	An "RRC Connection Reconfiguration" message has been sent to the UE via the PCC. The UE has acknowledged this message. In this state, the UE is informed about the presence of the SCC and knows important SCC properties like the carrier frequency. Prerequisite: RRC connection established on the PCC (packet-switched state "Attached" or "Connection Established", depending on parameter Keep RRC Connection)
"MAC Activated"	The SCC has been activated via a MAC message, sent to the UE via the PCC. The UE has acknowledged this message. After SCC activation, the SCC scheduling is started. The scheduling information is sent to the UE via the PDCCH of the SCC (cross-carrier scheduling is not used).

You can set up the SCC automatically, semiautomatically or manually:

- Manual activation means that you initiate each state transition step separately.
- Semiautomatic activation means that you initiate the complete SCC setup. So one action initiates all transitions from "SCC Off" to "MAC Activated".
- Automatic activation means that the RRC connection establishment triggers also the setup of the SCC. Depending on "Keep RRC Connection", the SCC setup is done with the attach or later on.

The following figure shows the transitions between the SCC states.

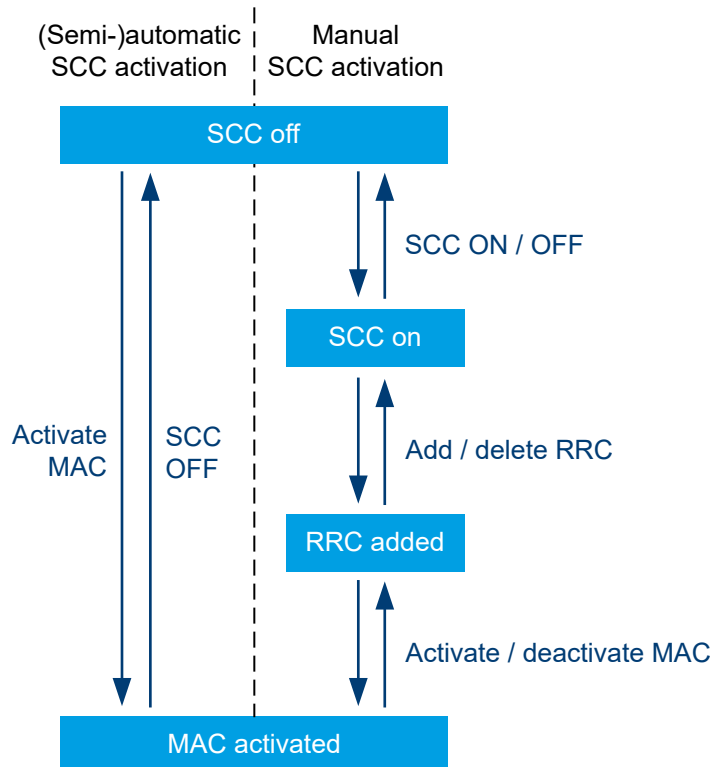


Figure 2-9: SCC state transitions

For selection of the mode, see ["SCC Activation Mode"](#) on page 142.

You can control the state per SCC. Or you can define SCC synchronization sets and control the state per set, see ["Multiple SCC Actions hotkey"](#) on page 132.



Additional transitions

The list of transitions in [Figure 2-9](#) is not complete. The "SCC Off" state can be reached at any time by releasing the RRC connection, for example by turning off the PCC cell signal. Incidents like a loss of the radio link also cause transitions.

2.2.10 Handover

The LTE signaling application supports a handover within the signaling application, to another signaling application or to an external instrument.

The following handover mechanisms are supported:

- **Handover:**
The R&S CMW performs an RRC connection reconfiguration. The reconfiguration message includes inter-RAT mobility control information.
This mechanism allows you to perform a connected-mode handover to a WCDMA or GSM signaling application. Option R&S CMW-KS510 is required.
After handover completion, the UE has been detached in the LTE signaling application. In the target signaling application, a new connection has been established.
- **Blind handover:**
The R&S CMW performs an RRC connection reconfiguration. The reconfiguration message includes intra-RAT mobility control information.
This mechanism is relevant for a handover within the signaling application. You can change the operating band and the channel, but not the cell bandwidth.
- **Redirection:**
The R&S CMW performs an RRC connection release with redirection information. This mechanism is relevant for a handover within the signaling application and for a handover to another signaling application or another instrument.
For a handover within the signaling application, it is possible to change the operating band, the channel and the cell bandwidth. A new connection with the changed parameters is established.
A handover to another signaling application or instrument results in a new registration of the UE at the handover destination. No new connection is set up.
- **Mobile-terminating circuit-switched fallback:**
The R&S CMW informs the UE via a CS service notification about an incoming mobile-terminating call. The UE answers with an extended service request with circuit-switched fallback (CSFB) response. Then the R&S CMW performs an RRC connection release with redirection information. The UE sends a paging response to the handover destination and a new CS connection is established by the target signaling application.
This mechanism is only supported for handover to another signaling application. Supported technologies are, for example, GSM, WCDMA and TD-SCDMA. The target signaling application can be at the same instrument or at another R&S CMW. Option R&S CMW-KS510 is required.

For a handover to another instrument, there is no communication at all between the two instruments and no cabling between the two instruments is required. The UE must be connected to both instruments, e.g. via an external combiner.

How to perform a handover

1. For a handover within the signaling application, configure the easy mode as desired. See ["Easy Mode"](#) on page 204.
2. In the LTE signaling application, establish a connection to the UE.
3. Press the hotkey "Inter/Intra-RAT". A configuration dialog box opens.

4. Configure the settings in the dialog:
 - a) Select the handover target - either the LTE signaling application or another signaling application or "No Connection" for an external instrument.
 - b) Select a handover mechanism.
 - c) Configure the destination parameters. For a handover to an external instrument, the parameters must reflect the actual configuration of the external instrument.
 - d) If you have selected another signaling application as target, the LTE signaling application automatically activates the target cell (switches on the downlink signal).
Wait until the cell icon  includes "RDY" to indicate that the handover target is ready to receive the handover.

See also ["Troubleshooting"](#) on page 43.

5. Press the button "Execute".

You can monitor the process in the "Event Log" area of the main view of the signaling applications.

If you want to reconfigure only one parameter of the LTE signaling application, you can change it directly, without using the "Inter/Intra-RAT" hotkey. Simply modify the channel or the operating band or the cell bandwidth during an established connection. The R&S CMW then initiates a redirection procedure to reconfigure the parameter.

Troubleshooting

If you get an error message, stating that blind handover is not supported, this message indicates that the handover source and target must use different RF paths. Otherwise, they cannot be active in parallel.

In that case, configure different RX/TX modules ("Converter" setting) for the handover source and target. Keep in mind that several RF paths can be used by the LTE signaling application, for example for MIMO or carrier aggregation.

Whether a blind handover is supported depends on the target RAT and on the used handover mechanism.

2.2.11 eMTC

To use the LTE signaling application for enhanced machine type communication (eMTC), you need option R&S CMW-KS590.

eMTC is designed for the Internet of things (IoT). The focus is on long battery life, low device cost and extended coverage. eMTC UEs are typically delay-tolerant and can be located in areas with poor coverage, for example in basements.

eMTC uses two modes of operation for coverage enhancement (CE), CE mode A and CE mode B. CE mode A is mandatory for UE category M1, supports full mobility and results in a small coverage enhancement. CE mode B is optional, supports only limited mobility and results in a much larger coverage enhancement.

To enhance the coverage, most physical channels support repeated transmissions. The allowed number of repetitions in CE mode B is much higher than in CE mode A.

Narrowbands

eMTC uses narrowband operation. A narrowband covers 1.4 MHz and contains six resource blocks. In Release 13, one eMTC connection uses only one narrowband at a time. But repeated transmissions can be combined with frequency hopping. So the same information is transmitted repeatedly in a changing narrowband.

The number and location of the narrowbands within the cell bandwidth is specified by 3GPP. There are as many narrowbands as fit into the bandwidth. Depending on the bandwidth, the center RB is not used. If there are more unused RBs, they are at the edge of the bandwidth. The narrowbands are numbered 0 to n, with increasing frequency. The following table shows the assignment of resource blocks to narrowbands.

Table 2-8: Narrowbands and resource blocks per cell BW

Cell BW / #RB	#NB	Narrowband positions	Unused RB
1.4 MHz / 6	1	NB 0 (RB 0 to 5)	-
3 MHz / 15	2	NB 0 (RB 1 to 6), NB 1 (RB 8 to 13)	0, 7, 14
5 MHz / 25	4	NB 0 (RB 0 to 5), NB 1 (RB 6 to 11), NB 2 (RB 13 to 18), NB 3 (RB 19 to 24)	12
10 MHz / 50	8	NB 0 (RB 1 to 6), ..., NB 3 (RB 19 to 24) NB 4 (RB 25 to 30), ..., NB 7 (RB 43 to 48)	0, 49
15 MHz / 75	12	NB 0 (RB 1 to 6), ..., NB 5 (RB 31 to 36), NB 6 (RB 38 to 43), ..., NB 11 (RB 68 to 73)	0, 37, 74
20 MHz / 100	16	NB 0 (RB 2 to 7), ..., NB 7 (RB 44 to 49), NB 8 (RB 50 to 55), ..., NB 15 (RB 92 to 97)	0, 1, 98, 99

In Release 14 and higher, one eMTC connection can use a bandwidth of up to 5 MHz in CE mode A. The connection uses up to four narrowbands in parallel.

Even with several used narrowbands, the uplink allocation is contiguous. Unused RBs between two used narrowbands are not allowed in the uplink. Such unused RBs are listed in the table for cell BW 3 MHz (#7), 5 MHz (#12) and 15 MHz (#37). If a narrowband below and above such an RB is used in the uplink, the upper narrowband moves down by one RB.

Example: 5-MHz cell BW, NB 1 and NB 2 are used. In that case, NB 2 contains the RBs 12 to 17 in the uplink, not 13 to 18.

References

For more eMTC background information and eMTC settings, see:

- [Table 2-30](#) (DL channels)
- ["eMTC"](#) on page 142 (enabling eMTC)
- [Section 2.4.11, "eMTC settings"](#), on page 193

The scheduling types for eMTC are described in the following sections.

2.2.12 Scheduling type RMC for eMTC

This scheduling type is only supported for CE mode A.

An RMC can be defined via the following set of parameters:

- Cell bandwidth
- Half-duplex operation or full-duplex operation
- Subframe pattern
- Number of allocated resource blocks (RB)
- Position of the first allocated RB within the lowest used narrowband
- Position of lowest used narrowband
- Modulation type
- Transport block size index (TBS index)

RMCs do not use repetitions or narrowband hopping.

The parameter combinations supported for eMTC are listed in the following sections.

In addition to the 3GPP-compliant combinations, the tables list also the configurable combinations without RB allocation ("Allocated RBs" = 0).

If you want to force the UE to use the PUCCH instead of the PUSCH, set the number of UL resource blocks to 0. Thus the UE is not allowed to use the PUSCH and must use the PUCCH to send ACK/NACKs.

• DL RMCs	45
• UL RMCs	47
• Subframe pattern	49

2.2.12.1 DL RMCs

The allocated RBs start at the lower end of a narrowband (start RB = #0). If more than 4 RBs are allocated, up to three next higher narrowbands are also used.

The narrowband selection mechanism depends on whether only one narrowband is allowed (max. eMTC BW 1.4 MHz) or up to four narrowbands are allowed (max. eMTC BW 5 MHz). Refer to the following tables.

Table 2-9: NB position values, maximum eMTC BW 1.4 MHz

Cell bandwidth	Available NBs	Configurable NB positions	Meaning	
			Mid	3GPP
1.4 MHz	NB0	3GPP	-	NB0
3 MHz	NB0, NB1	3GPP	-	NB1
5 MHz	NB0 to NB3	low, high, 3GPP	-	NB3
10 MHz	NB0 to NB7	low, mid, high, 3GPP	NB5	NB6

Cell bandwidth	Available NBs	Configurable NB positions	Meaning	
			Mid	3GPP
15 MHz	NB0 to NB11	3GPP	-	NB3
20 MHz	NB0 to NB15	low, mid, high, 3GPP	NB9	NB4
Meaning of the NB position values: • "low" selects NB0. "high" selects the highest NB within the cell bandwidth. • "mid" selects an NB as listed in the table (above channel center, to avoid PBCH and synch signals). • "3GPP" selects an NB as listed in the table (as in 3GPP TS 36.508, IE "mpdcch-Narrowband-r13").				

Table 2-10: NB position values, maximum eMTC BW 5 MHz

Cell bandwidth	Available NBs	Configurable NB positions	
		for QPSK	for 16-QAM
1.4 MHz	NB0	low	low
3 MHz	NB0, NB1	low	low
5 MHz	NB0 to NB3	low	low, high
10 MHz	NB0 to NB7	low, high	low, high
15 MHz	NB0 to NB11	low, high	low, high
20 MHz	NB0 to NB15	low, high	low, high
Meaning of the NB position values: • "low": The lowest used NB is NB0. • "high": The highest used NB is the highest NB in the cell bandwidth.			

Some RMCs are allowed for both maximum eMTC bandwidths (1.4 MHz and 5 MHz). And some RMCs are only allowed for maximum 5-MHz eMTC bandwidth. Refer to the following tables.

Table 2-11: DL RMCs, for all maximum eMTC bandwidths

Cell bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz to 5 MHz	0	QPSK	0
	2	16-QAM	14
	3	QPSK	5
	4	QPSK	4
10 MHz to 20 MHz	0	QPSK	0
	2	16-QAM	14
	3	QPSK	5
	4	QPSK	5

Table 2-12: DL RMCs, only for max. 5-MHz eMTC bandwidth

Cell bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	2	16-QAM	14
	4	QPSK	4
3 MHz	8	16-QAM	14
	8	QPSK	5
5 MHz to 20 MHz	15	16-QAM	14
	16	QPSK	5

2.2.12.2 UL RMCs

The position of the RBs within the cell bandwidth is configured via two parameters:

- The RB position selects the position within the narrowbands.
- The NB position selects the lowest used narrowband.

The allowed values have dependencies described in the following tables.

Table 2-13: RB position values

RB position	Allocated RBs	Meaning
low	0 to 6, 9, 12, 15, 18, 21, 24	start RB = first RB (#0) of selected NB
high	0 to 6	last RB = last RB (#5) of selected NB
	9, 15, 21	start RB = RB #3 of selected NB

Table 2-14: NB position values depending on cell bandwidth and number of RBs

Cell bandwidth	Available NBs	Configurable NB positions
1.4 MHz	NB0	low
3 MHz	NB0, NB1	low, high
5 MHz	NB0 to NB3	low, NB1, NB2, high
10 MHz	NB0 to NB7	low, NB1 to NB6, high
15 MHz	NB0 to NB11	low, NB1 to NB10, high
20 MHz	NB0 to NB15	low, NB1 to NB14, high
"low" selects NB0. "high" selects the highest NB within the cell bandwidth. If you allocate more than 6 RBs, they span over several NBs and the maximum allowed lowest NB position is reduced accordingly: • 9 or 12 RBs: highest NB not allowed • 15 or 18 RBs: highest two NBs not allowed • 21 or 24 RBs: highest three NBs not allowed		

The other configurable parameters are listed in the following tables.

Table 2-15: UL RMCs for max. 1.4-MHz eMTC bandwidth

Cell bandwidth	Allocated RBs	TBS index (QPSK)	TBS index (16-QAM)
1.4 MHz to 20 MHz	0	0	-
	1	5	14
	2	6	14
	3	6	14
	4	6	12
	5	5	10
	6	6	-

Table 2-16: UL RMCs for max. 5-MHz eMTC bandwidth

Cell bandwidth	Allocated RBs	TBS index (QPSK)	TBS index (16-QAM)
1.4 MHz	0	0	-
	1	5	14
	2	6	14
	3	6	14
	4	6	14
	5	5	14
	6	6	10
3 MHz	0	0	-
	1	5	14
	2	6	14
	3	6	14
	4	6	14
	5	5	14
	6	6	14
	9	5	14
	12	5	14
	12	6	10
5 MHz to 20 MHz	0	0	-
	1	5	14
	2	6	14
	3	6	14
	4	6	14
	5	5	14

Cell bandwidth	Allocated RBs	TBS index (QPSK)	TBS index (16-QAM)
	6	6	14
	9	5	14
	12	5	14
	15	5	14
	18	6	12
	21	6	11
	24	6	10

2.2.12.3 Subframe pattern

The subframe pattern defines which subframes of the radio frame are scheduled. The standard pattern has a periodicity of 10 subframes. Per 10 subframes, there are 3 DL subframes and 3 UL subframes.

For some tests, 3GPP defines subframe patterns deviating from the standard pattern.

You can select which pattern you want to use. For easy configuration, the patterns are identified via the section numbers of 3GPP TS 36.521 where they are defined.

Some patterns restrict the allowed values for the UL settings, as follows:

- "6.5.2.1EA.2-A" and "6.5.2.1EA.2-B" allow only:
 - NB position "low"
 - RB position "low"
 - 6 RB plus QPSK or 5 RB plus 16-QAM

2.2.13 User-defined channels for eMTC

For eMTC CE mode A, global user-defined channel configuration is available.

User-defined channels allow a more flexible RB allocation than RMCs. Repetitions and narrowband hopping are not used.

You can configure the following settings for user-defined channels:

- Cell bandwidth
- Full-duplex operation or half-duplex operation (see also [Subframe pattern](#))
- Number of used resource blocks (#RB), contiguous allocation
- Position of the lowest resource block in its narrowband (start RB)
- Narrowbands
- Modulation type
- Transport block size (TBS) index

The following tables provide an overview of allowed values for a maximum eMTC bandwidth of 1.4 MHz (one narrowband).

Table 2-17: eMTC user-defined channel settings, maximum eMTC BW 1.4 MHz

	Downlink	Uplink
Allocated RBs (#RB)	0 to 4	0 to 6
Start RB	0 to (4 - #RB)	0 to (6 - #RB)
Modulation	QPSK, 16-QAM	QPSK, 16-QAM
TBS index	0 to 9 (QPSK) 9 to 14 (16-QAM)	0 to 10 (QPSK) 10 to 14, 16 to 21 (16-QAM)

Table 2-18: NB position values, maximum eMTC BW 1.4 MHz

Cell BW	Available NBs	Configurable NB positions		Meaning	
		Uplink	Downlink	Mid	3GPP
1.4 MHz	NB0	low	3GPP	-	NB0
3 MHz	NB0, NB1	low, high	3GPP	-	NB1
5 MHz	NB0 to NB3	low, NB1, NB2, high	low, high, 3GPP	-	NB3
10 MHz	NB0 to NB7	low, NB1 to NB6, high	low, mid, high, 3GPP	NB5	NB6
15 MHz	NB0 to NB11	low, NB1 to NB10, high	3GPP	-	NB3
20 MHz	NB0 to NB15	low, NB1 to NB14, high	low, mid, high, 3GPP	NB9	NB4
Meaning of the NB position values: • "low" selects NB0. "high" selects the highest NB within the cell bandwidth. • "mid" selects an NB as listed in the table (above channel center, to avoid PBCH and synch signals). • "3GPP" selects an NB as listed in the table (as in 3GPP TS 36.508, IE "mpdcch-Narrowband-r13").					

For a maximum eMTC bandwidth of 5 MHz, the narrowband and RB configuration method differs for DL and UL:

- DL:
 - Select up to four narrowbands (the lowest narrowband m and any of the three next higher narrowbands $m+1$, $m+2$, $m+3$).
 - Define the number of RBs and the start RB for one narrowband. The same settings apply to all used narrowbands.
- UL:
 - Select the lowest narrowband.
 - Define the total number of RBs over all narrowbands and the start RB.

The following table provides an overview of the allowed values for a maximum eMTC bandwidth of 5 MHz.

Table 2-19: eMTC user-defined channel settings, maximum eMTC BW 5 MHz

	Downlink	Uplink	
Allocated RBs (#RB)	0 to 4	0 to 6	9, 12, 15, 18, 21, 24 ⁴⁾
Start RB	0 to (4 - #RB)	0 to (6 - #RB)	0, 3
Narrowband	low, high ¹⁾	0 to (n-1) ²⁾	0 to (n - ceiling(#RB+start RB) / 6)

	Downlink	Uplink
Additional narrowbands	lowest +1/+2/+3 ³⁾	selected automatically
Modulation	QPSK, 16-QAM	QPSK, 16-QAM
TBS index	0 to 9 (QPSK) 9 to 14 (16-QAM)	0 to 10 (QPSK) 10 to 14 (16-QAM)
¹⁾ "low" uses NB0 as lowest NB. "high" uses highest NB of cell BW as highest NB (for cell BW > 5 MHz). ²⁾ n is the number of narrowbands fitting into the cell BW, see Table 2-8 . ³⁾ 1st, 2nd and 3rd narrowband above the lowest narrowband. Only possible, if they fit into the cell BW. ⁴⁾ For 1.4-MHz cell BW max 6 RB. For 3-MHz cell BW max 12 RB.		

According to 3GPP, the maximum transport block size allowed for eMTC is 1000 bits. Some supported value combinations result in larger transport blocks.

Subframe pattern

The following subframe patterns are supported:

- Full duplex: Each subframe is scheduled for DL and UL.
- Half duplex: You can select between three subframe (SF) patterns. In the given order, the patterns provide higher downlink throughput but need more advanced UE capabilities.
 - Standard pattern:
No PDSCH HARQ-ACK bundling
Periodicity 10 SF, with 3 DL SF and 3 UL SF per 10 SF.
 - Bundling with 8 processes:
PDSCH HARQ-ACK bundling with 8 HARQ processes.
Periodicity 30 SF, with 16 DL SF and 6 UL SF per 30 SF.
The following UE capabilities are required: "ce-HARQ-AckBundling-r14", "ce-SchedulingEnhancement-r14".
 - Bundling with 10 processes:
PDSCH HARQ-ACK bundling with 10 HARQ processes.
Periodicity 34 SF, with 20 DL SF and 6 UL SF per 34 SF.
The following UE capabilities are required: "ce-HARQ-AckBundling-r14", "ce-SchedulingEnhancement-r14", "ce-PDSCH-TenProcesses-r14".

2.2.14 eMTC compact scheduling

This scheduling type is only supported for CE mode A.

"Compact" means, that the scheduling in the time domain is more compact than for other scheduling types. Use this scheduling type for maximum throughput.

Compact scheduling does not use repetitions or narrowband hopping.

You can configure the following settings for compact scheduling:

- Cell bandwidth
- Full-duplex operation or half-duplex operation (see also [Subframe pattern](#))

- Number of used resource blocks (#RB), contiguous allocation
- Position of the lowest resource block in its narrowband (start RB)
- More DL narrowbands (above the lowest automatically selected narrowband)
- Modulation type
- Transport block size (TBS) index

The values allowed for the RB allocation settings are listed in the following table, for a maximum eMTC bandwidth of 1.4 MHz (one narrowband).

Table 2-20: eMTC compact scheduling settings, maximum eMTC BW 1.4 MHz

	Downlink	Uplink
Allocated RBs (#RB)	0 to 4	0 to 6
Start RB	0 to (4 - #RB)	0 to (6 - #RB)
Modulation	QPSK, 16-QAM	QPSK, 16-QAM
TBS index	0 to 9 (QPSK) 9 to 14 (16-QAM)	0 to 10 (QPSK) 10 to 14, 16 to 21 (16-QAM)

For a maximum eMTC bandwidth of 5 MHz, the narrowband and RB configuration method differs for DL and UL:

- DL:
 - Enable up to three additional narrowbands above the automatically selected lowest narrowband m (next higher narrowbands $m+1$, $m+2$, $m+3$).
 - Define the number of RBs and the start RB for one narrowband. The same settings apply to all used narrowbands.
- UL:
 - Define the total number of RBs over all narrowbands and the start RB.

The following table provides an overview of the allowed values for a maximum eMTC bandwidth of 5 MHz.

Table 2-21: eMTC compact scheduling settings, maximum eMTC BW 5 MHz

	Downlink	Uplink	
Allocated RBs (#RB)	0 to 4	0 to 6	9, 12, 15, 18, 21, 24 ²⁾
Start RB	0 to (4 - #RB)	0 to (6 - #RB)	0, 3
Additional narrowbands	lowest +1/+2/+3 ¹⁾	selected automatically	
Modulation	QPSK, 16-QAM	QPSK, 16-QAM	
TBS index	0 to 9 (QPSK) 9 to 14 (16-QAM)	0 to 10 (QPSK) 10 to 14 (16-QAM)	

¹⁾ 1st, 2nd and 3rd narrowband above the lowest narrowband. Only possible, if they fit into the cell BW.

²⁾ For 1.4-MHz cell BW max 6 RB. For 3-MHz cell BW max 12 RB.

According to 3GPP, the maximum transport block size allowed for eMTC is 1000 bits. Some supported value combinations result in larger transport blocks.

Subframe pattern

The scheduling in the time domain is more compact than for other scheduling types. The following subframe patterns are supported:

- Full duplex: Each subframe is scheduled for DL and UL.
- Half duplex: You can select between three subframe (SF) patterns, each with a periodicity of 40 ms. In the given order, the patterns provide higher downlink throughput but need more advanced UE capabilities.
 - Standard pattern:
No PDSCH HARQ-ACK bundling
3 DL SF and 3 UL SF per 8 SF.
 - Bundling with 8 processes:
PDSCH HARQ-ACK bundling with 8 HARQ processes.
4 DL SF and 2 UL SF per 8 SF.
The following UE capabilities are required: "ce-HARQ-AckBundling-r14", "ce-SchedulingEnhancement-r14".
 - Bundling with 10 processes:
PDSCH HARQ-ACK bundling with 10 HARQ processes.
6 DL SF and 2 UL SF per 10 SF.
The following UE capabilities are required: "ce-HARQ-AckBundling-r14", "ce-SchedulingEnhancement-r14", "ce-PDSCH-TenProcesses-r14".

2.2.15 eMTC auto mode

eMTC uses cross-subframe scheduling. If you want to use repetitions and narrowband hopping, you must consider many aspects to configure a meaningful scheduling and to fit all channels into the narrowband.

To facilitate the scheduling configuration for eMTC, the signaling application provides an auto mode. The auto mode takes care of all these aspects. It uses the configured repetitions and the configured hopping and creates a scheduling without collisions.

The auto mode uses half-duplex. The scheduling contains alternating DL and UL phases:

- DL phase: MPDCCH, then PDSCH, then PUCCH
The DL phase includes the DL information (MPDCCH), the DL data (PDSCH) and the acknowledgments of the DL data (PUCCH).
- UL phase: MPDCCH, then PUSCH
The UL phase includes the scheduling (MPDCCH) and the UL data (PUSCH).

As for other scheduling types, you can configure the resource block allocation for the eMTC auto mode. The following settings are available:

- Number of used resource blocks (#RB), contiguous allocation
- Position of the lowest resource block in its narrowband (start RB)
- Narrowbands used for the first transmission in the first DL phase, before hopping
- Modulation type
- Transport block size (TBS) index

The following table provides an overview of the allowed values for a maximum eMTC bandwidth of 1.4 MHz (one narrowband), depending on the CE mode.

Table 2-22: eMTC auto mode settings, maximum eMTC BW 1.4 MHz

	CE mode A		CE mode B	
	Downlink	Uplink	Downlink	Uplink
Allocated RBs (#RB)	0 to 6	0 to 6	0, 4, 6	0, 1, 2
Start RB	0 to (6 - #RB)	0 to (6 - #RB)	0	0 to 5 (#RB=0, 1) 0, 2 (#RB = 2)
Narrowband ¹⁾	0 to (n-1)	0 to (n-1)	0 to (n-1)	0 to (n-1)
Modulation	QPSK, 16-QAM	QPSK, 16-QAM	QPSK	QPSK
TBS index	0 to 9 (QPSK) 9 to 14 (16-QAM)	0 to 10 (QPSK) 10 to 14, 16 to 21 (16-QAM)	0 to 9	0 to 10

¹⁾ n is the number of narrowbands fitting into the cell bandwidth, see [Table 2-8](#).

For a maximum eMTC bandwidth of 5 MHz, the narrowband and RB configuration method differs for DL and UL:

- DL:
 - Select up to four narrowbands (the lowest narrowband m and any of the three next higher narrowbands m+1, m+2, m+3).
 - Define the number of RBs and the start RB for one narrowband. The same settings apply to all used narrowbands.
- UL:
 - Select the lowest narrowband.
 - Define the total number of RBs over all narrowbands and the start RB.

The following table provides an overview of the allowed values for a maximum eMTC bandwidth of 5 MHz.

Table 2-23: eMTC auto mode settings, maximum eMTC BW 5 MHz, CE mode A

	Downlink	Uplink	
Allocated RBs (#RB)	0 to 6	0 to 6	9, 12, 15, 18, 21, 24 ³⁾
Start RB	0 to (6 - #RB)	0 to (6 - #RB)	0, 3
Lowest narrowband ¹⁾	0 to (n-1)	0 to (n-1)	0 to (n – ceiling(#RB+start RB) / 6)
Additional narrowbands	lowest +1/+2/+3 ²⁾	selected automatically	
Modulation	QPSK, 16-QAM	QPSK, 16-QAM	
TBS index	0 to 9 (QPSK) 9 to 14 (16-QAM)	0 to 10 (QPSK) 10 to 14 (16-QAM)	

¹⁾ n is the number of narrowbands fitting into the cell bandwidth, see [Table 2-8](#).

²⁾ 1st, 2nd and 3rd narrowband above the lowest narrowband. Only possible, if they fit into the cell BW.

³⁾ For 1.4-MHz cell BW max 6 RB. For 3-MHz cell BW max 12 RB.

According to 3GPP, the maximum transport block size allowed for eMTC is 1000 bits. Some supported value combinations result in larger transport blocks.

2.2.16 Licensed-assisted access (LAA)

LAA is a carrier aggregation feature using unlicensed spectrum for the SCC DL. The same spectrum can be used by WLAN.

To use LAA in the LTE signaling application, you need option R&S CMW-KS514.

Within one scenario, you can mix SCCs in LAA operation mode and SCCs in normal operation mode. To see in which mode an SCC is, check the "Frame Structure" setting:

- "Type 1" = normal FDD operation
- "Type 2" = normal TDD operation
- "Type 3" = LAA operation mode

To put an SCC into LAA operation mode, select the duplex mode TDD and an LAA band (46 or 49). The frame structure setting changes to "Type 3". If you select the user-defined TDD band, you can set "Type 3" manually.

Restrictions

3GPP defines many restrictions for LAA.

The following applies to SCCs in LAA operation mode in the LTE signaling application:

- Only SCC DL allowed (no SCC UL, R13 restriction)
- Only normal CP (no extended CP)
- Only 20-MHz cell bandwidth in the current software version (3GPP allows also 10-MHz bandwidth)
- Transmission modes TM 1 to TM 4, TM 8, TM 9

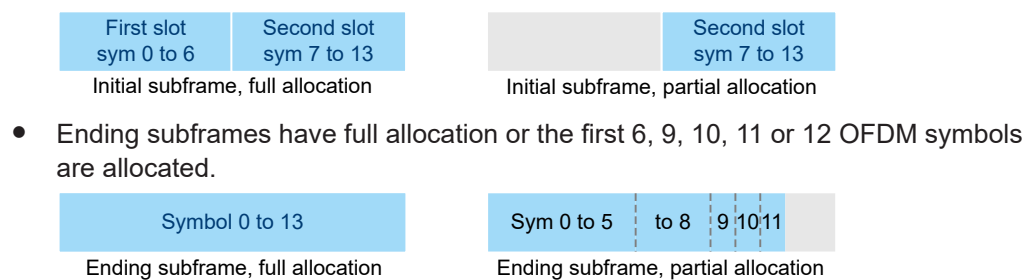
Frame structure and partial allocation

LAA allows downlink transmissions in any subframe. There can be partially allocated subframes at the beginning and at the end of a transmission burst. Between two downlink bursts, there is no signal transmission at all. Only discovery reference signals (DRS) can be transmitted between bursts, with a minimum periodicity of 40 ms. There is no PBCH on frame structure type 3 cells.

The first subframe of a burst is also called initial subframe. The last subframe of a burst is also called ending subframe. For initial subframes with partial allocation, you find the term PIPS in the user interface. For ending subframes with partial allocation, you find the term PEPS.

The following rules apply for partial/full allocation:

- Each burst contains at least one subframe with full allocation.
So a burst with only one subframe has full allocation. In a burst with two subframes, only one subframe can have partial allocation.
- Initial subframes have full allocation or the second half is allocated.



The following figure shows some examples of possible LAA bursts, relative to the PCC subframe positions.

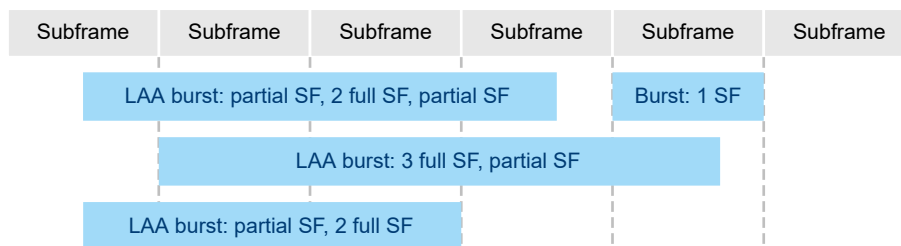


Figure 2-10: Burst examples with full / partial allocation

The subframe allocation is defined via the scheduling settings. The scheduling types supported for LAA are described in the following sections.

- [User-defined channels for LAA](#)..... 56
- [Scheduling type RMC for LAA](#)..... 58
- [CQI channels for LAA](#)..... 59

2.2.16.1 User-defined channels for LAA

For LAA, global user-defined channel configuration is available. TTI-based configuration is not supported. Option R&S CMW-KS510 is required.

Two burst types with different sets of settings must be distinguished: fixed transmission bursts and random transmission bursts. They are described in the following sections.

Random transmission bursts

Random transmission bursts are requested by 3GPP TS 36.101, annex B.8. The scheduling has a random component, so that the duration of the individual bursts and of the pause between subsequent bursts is not predictable.

The following settings influence the random procedure:

- Possible number of subframes in a burst
You can select up to 10 values from the set {1, 2, 3, 4, 5, 6, 7, 8, 9, 10}.
- Possible number of allocated OFDM symbols in the ending subframe
You can allow up to 6 values, from the set {6, 9, 10, 11, 12, 14}.
- Partial allocation allowed for initial subframe or not
- Probability for burst transmission, no transmission means muting for the burst duration

- Number and position of resource blocks (RB)
The same rules as without LAA apply, see [Section 2.2.20.1, "Number and position of RBs \(DL\)"](#), on page 83.
- Modulation type and transport block size index, configurable per subframe type (full allocation, initial partial allocation, ending partial allocation with n symbols)

The allowed transport block size indices for initial partial allocation differ from the allowed values for full allocation / ending partial allocation. The allowed transport block size indices depend also on the modulation type and on whether 256-QAM is used or 1024-QAM is used.

"256-QAM used" means that 256-QAM is configured for an allowed subframe type and 1024-QAM is not configured. If initial partial subframes are disabled, their modulation setting is ignored for the decision. If a certain OFDM symbol number for the ending subframe is disabled, the corresponding modulation setting is also ignored. "1024-QAM used" means that 1024-QAM is configured for an allowed subframe type.

The dependencies are listed in the following tables.

Table 2-24: Range of transport block size index, for first SF with partial allocation

	QPSK	16-QAM	64-QAM	256-QAM	1024-QAM
max 64-QAM used	0 to 4	5 to 9	9 to 26	n.a.	n.a.
256-QAM used (no 1024)	0, 2, 4	6, 8	10 to 14	15 to 25, 27 to 33	n.a.
1024-QAM used	0, 2, 4	6, 8	11, 13	15, 16, 18, 20 to 25, 27 to 33	34 to 37

Table 2-25: Range of transport block size index, for full SF and last SF

	QPSK	16-QAM	64-QAM	256-QAM	1024-QAM
max 64-QAM used	0 to 9	9 to 15	15 to 26	n.a.	n.a.
256-QAM used (no 1024)	0, 2, 4, 6, 8	10 to 15	16 to 24	25, 27 to 33	n.a.
1024-QAM used	0, 2, 4, 6, 8	11, 13, 15	16, 18, 20 to 24	25, 27 to 33	34 to 37

For DL 256-QAM, option R&S CMW-KS554 is required. For DL 1024-QAM, option R&S CMW-KS555 is required.

Fixed transmission bursts

With this burst type, you have full control over the burst configuration, including partial allocation and full allocation. The defined burst is transmitted with a configurable periodicity.

The following settings configure the burst sequence:

- Number of subframes per burst
- Periodicity of bursts (a burst starts every n^{th} subframe)
- Partial allocation or full allocation for the first subframe of the burst
- Number of allocated OFDM symbols in the last subframe of the burst
The first 6, 9, 10, 11, 12 or 14 symbols are allocated.
- Number and position of resource blocks (RB)

The same rules as without LAA apply, see [Section 2.2.20.1, "Number and position of RBs \(DL\)"](#), on page 83.

- Modulation type and transport block size index, configurable per subframe type (full allocation, initial partial allocation, ending partial allocation)

The allowed transport block size indices for initial partial allocation differ from the allowed values for full allocation / ending partial allocation. The allowed transport block size indices depend also on the modulation type and on whether 256-QAM or 1024-QAM is used.

"256-QAM used" means that 256-QAM is configured for at least one subframe of the burst and 1024-QAM is not configured. "1024-QAM used" means that 1024-QAM is configured for at least one subframe of the burst.

For the dependencies, see [Table 2-24](#).

For DL 256-QAM, option R&S CMW-KS554 is required. For DL 1024-QAM, option R&S CMW-KS555 is required.

2.2.16.2 Scheduling type RMC for LAA

For an RMC, you can define the following settings:

- Burst type
- Partial subframe configuration for random bursts
- Modulation type

Two burst types are supported:

- Fixed bursts: Full allocation for the subframes 0 and 3 to 9 of each radio frame. Subframes 1 and 2 are not used.
- Random bursts: The subframe pattern is determined according to 3GPP TS 36.521, section B.8.

For random bursts, you can configure partial subframes as follows:

- Full allocation only
- Partial allocation allowed for initial subframes
- Partial allocation allowed for ending subframes
- Partial allocation allowed for initial and ending subframes

With allowed partial allocation, the rules defined in 3GPP TS 36.521 section B.8 determine whether partial allocation is used for a subframe or not. For ending subframes, S_2 is {6, 9, 12, 14}.

The other parameters can be combined for LAA as listed in the following tables. The configurable TBSI values apply to full allocation. The TBSI values for partial allocation are set automatically.

Table 2-26: DL RMCs for LAA, fixed bursts

Bandwidth	Allocated RBs	Modulation	TBS index
20 MHz	100	QPSK	5
		64-QAM	24

Bandwidth	Allocated RBs	Modulation	TBS index
		256-QAM	32
		1024-QAM	35

Table 2-27: DL RMCs for LAA, random bursts

Bandwidth	Allocated RBs	Modulation	TBS index
20 MHz	100	16-QAM	12
		64-QAM	20

2.2.16.3 CQI channels for LAA

The following scheduling types are also supported for CQI tests with LAA. Option R&S CMW-KS510/-KS512 is required as without LAA:

- Follow wideband CQI
- Follow wideband PMI
- Follow wideband PMI-RI
- Follow wideband CQI-RI
- Follow wideband CQI-PMI-RI

For these scheduling types, you can configure the same DL settings as without LAA. For details, see [Section 2.2.21, "CQI channels"](#), on page 86.

In addition, you can configure the following settings:

- Number of subframes per burst
- Periodicity of bursts (a burst starts every n^{th} subframe)

The following additional characteristics apply:

- Fixed transmission bursts only, no random transmission bursts
- Only full allocation, even in the first and last subframe of the burst

2.2.17 Physical DL channels and signals

This section provides an overview of the LTE downlink radio resources, physical channels and physical signals.

See also [Section 2.2.23, "Operating bands"](#), on page 91.

- [Resources in time and frequency domain](#).....60
- [Physical channel overview](#).....63
- [Physical signal overview](#).....64

2.2.17.1 Resources in time and frequency domain

The DL radio resources in an LTE system are divided into time-frequency units called resource elements. In the time domain, a resource element corresponds to one OFDM symbol. In the frequency domain, it corresponds to one subcarrier (see next figure).

For the mapping of physical channels to resources, the resource elements are grouped into resource blocks (RB). Each RB consists of 12 consecutive subcarriers (180 kHz) and six or seven consecutive OFDM symbols (0.5 ms).

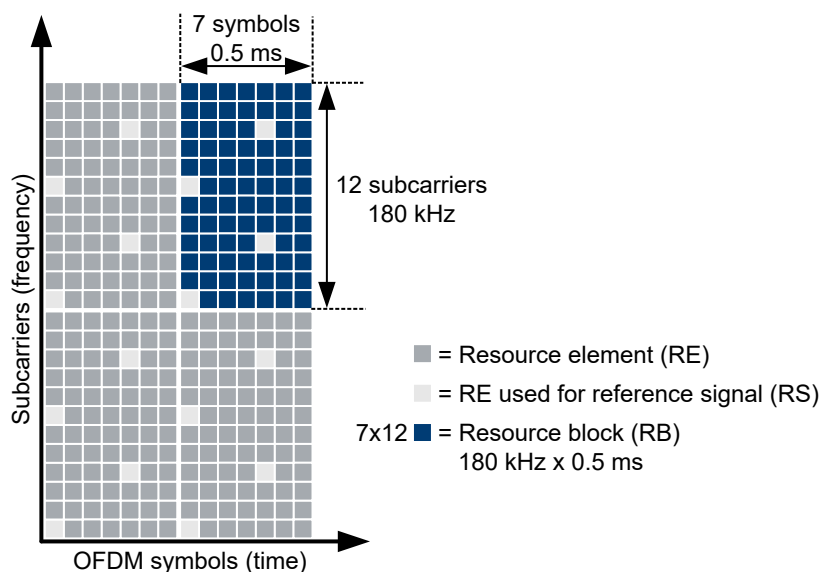


Figure 2-11: Resource element grid (seven symbols per RB, one TX antenna)

The positions of resource elements carrying reference signals (pilots) are standardized and depend on the number of transmit antennas. The preceding figure applies to single-antenna configurations. If more than one transmit antenna is used, each antenna uses different resource elements for reference signals. These resource elements are reserved for one antenna and not used at all by the other antennas. The following figure shows the resource element grid for a two transmit antenna configuration, used for MIMO 2x2.

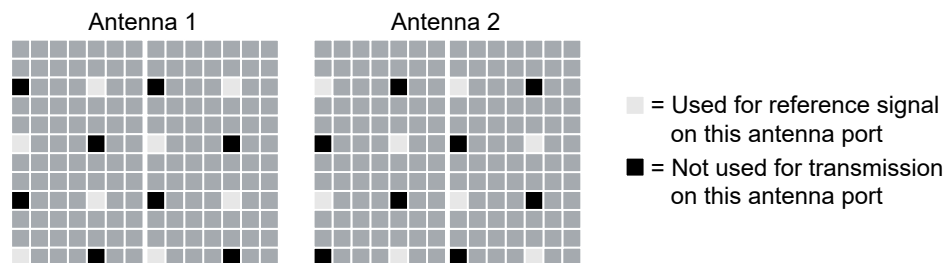


Figure 2-12: Resource element grids for two TX antennas

The smallest resource unit that can be assigned to a UE consists of two resource blocks (180 kHz, 1 ms). The assignment of resources to a UE can vary in time and frequency domain (see next figure).

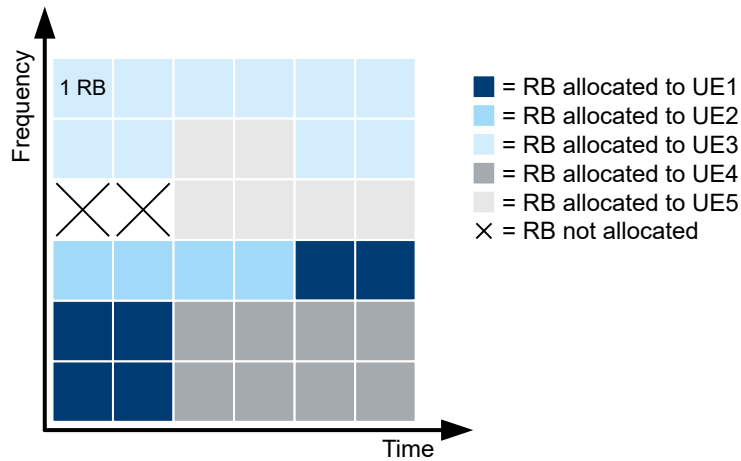


Figure 2-13: Assignment of resource blocks (RB) to UEs

In the time domain, the additional units radio frame, subframe and slot (containing the OFDM symbols) are defined. A guard time called cyclic prefix (CP) is added to each OFDM symbol. Depending on the duration of the guard time, it is either called normal CP or extended CP and the slot contains either seven or six OFDM symbols.

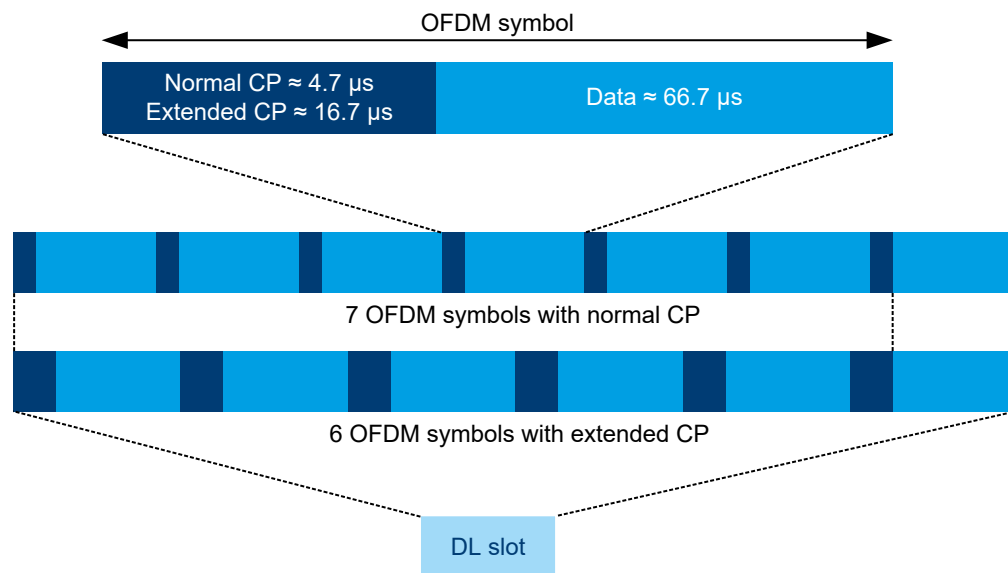


Figure 2-14: LTE DL slot structure

The radio frame structure depends on the duplex mode. An FDD DL radio frame contains 20 DL slots, grouped into 10 subframes.

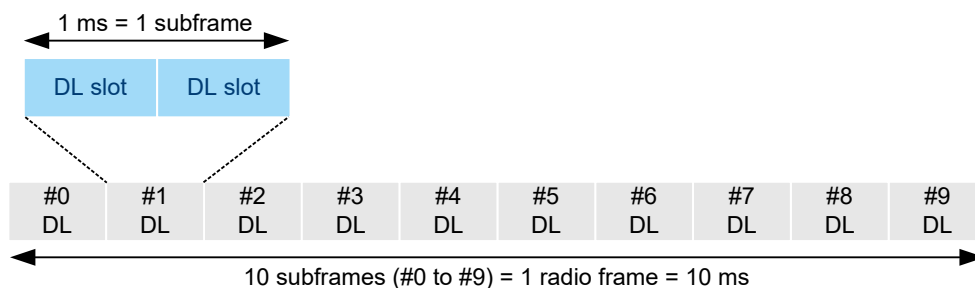


Figure 2-15: LTE DL frame structure for FDD

A TDD radio frame is also divided into 10 subframes. But for TDD, three subframe types are defined: DL subframe (two DL slots), UL subframe (two UL slots) and special subframe. A special subframe contains the fields DwPTS, GP and UpPTS. 3GPP defines several possible special subframe configurations, resulting in different lengths of these fields, see 3GPP TS 36.211, table 4.2-1. The total length of a special subframe equals 1 ms (same length for all subframe types).

The type of subframe number 0, 1, 2 and 5 is fixed. The type of the other subframes depends on the used UL-DL configuration, see [Table 2-28](#).

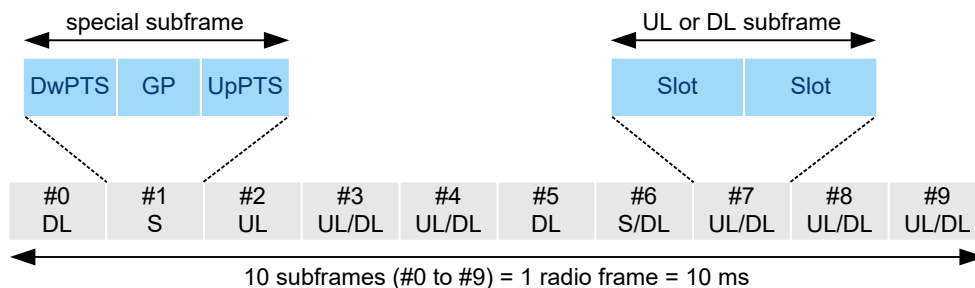


Figure 2-16: LTE frame structure for TDD

Table 2-28: UL-DL configuration 0 to 6

Configuration	Subframe number									
	0	1	2	3	4	5	6	7	8	9
0	D	S	U	U	U	D	S	U	U	U
1	D	S	U	U	D	D	S	U	U	D
2	D	S	U	D	D	D	S	U	D	D
3	D	S	U	U	U	D	D	D	D	D
4	D	S	U	U	D	D	D	D	D	D
5	D	S	U	D	D	D	D	D	D	D
6	D	S	U	U	U	D	S	U	U	D

D = downlink, U = uplink, S = special subframe

For eMTC narrowbands, see ["Narrowbands"](#) on page 44.

2.2.17.2 Physical channel overview

A downlink physical channel corresponds to a set of resource elements carrying information originating from higher layers. Physical channels can be either broadcast channels or shared channels. Dedicated channels are not used for LTE.

Broadcast channels carry messages that are not directed at a particular UE; they are point-to-multipoint channels. Shared channels are shared by several UEs. At a given time, a shared channel is assigned to one UE only, but the assignment can change within a few timeslots.

An overview of the physical channels of the generated downlink signal is given in the following table.

Table 2-29: DL channel overview, no eMTC

Physical DL channel	Purpose / type / modulation scheme
Physical broadcast channel (PBCH)	Provides physical layer information of the cell to be read during cell search, e.g. system bandwidth, number of transmit antennas, reference signal transmit power Broadcast; QPSK
Physical control format indicator channel (PCFICH)	Carries information about PDCCH format (number of OFDM symbols in the beginning of a subframe used for PDCCH) Broadcast; QPSK
Physical hybrid ARQ indicator channel (PHICH)	Carries ACK/NACK indicators for uplink packets Shared; QPSK
Physical downlink control channel (PDCCH)	Carries UE-specific control information, like scheduling information or UL power control commands Shared; QPSK
Physical downlink shared channel (PDSCH)	Carries user data Shared; QPSK, 16-QAM, 64-QAM, 256-QAM

Table 2-30: DL channel overview eMTC

Physical DL channel	Purpose / type / modulation scheme
Physical broadcast channel (PBCH)	Provides physical layer information of the cell to be read during cell search, e.g. system bandwidth, number of transmit antennas, reference signal transmit power, MPDCCH format, hopping on/off Broadcast; QPSK
MTC physical downlink control channel (MPDCCH)	Carries UE-specific control information, like scheduling information or UL power control commands Shared; QPSK
Physical downlink shared channel (PDSCH)	Carries user data Shared; QPSK, 16-QAM

2.2.17.3 Physical signal overview

A downlink physical signal corresponds to a set of resource elements carrying information originating from the physical layer. Two types of downlink physical signals are available: Reference signals and synchronization signals.

Table 2-31: Signal overview

Physical DL signal	Purpose / transmission modes / antenna ports
Cell-specific reference signal (C-RS)	For demodulation and channel quality feedback calculation Available for all transmission modes Up to TM 7 port 0, TM 8/9 ports 0 and 1
UE-specific reference signal (UE-RS)	For demodulation of signals with beamforming (BF), TM 7 to 9 TM 7 port 5 TM 8/9 single-layer BF port 8 TM 8/9 dual-layer BF ports 7 and 8 TM 9 four-layer BF ports 7 to 10
Channel state information reference signal (CSI-RS)	For channel quality feedback calculation, TM 9 Ports configurable: 15 / 15-16 / 15-18 / 15-22
Primary synchronization signal (PSS) Secondary synchronization signal (SSS)	Acquisition of cell timing and cell identity during cell search

For more details about reference signals, see 3GPP TS 36.211, section 6.10.

2.2.18 MIMO and beamforming

Multiple input multiple output (MIMO) systems are essential for the LTE downlink to achieve the specified throughput and spectral efficiency requirements. The term MIMO refers to the use of multiple antennas at transmitter side (input to radio channel) and receiver side (output of radio channel). As a special MIMO technique, beamforming improves the transmission to users at the cell edges via directional downlink transmission.

Different gains can be achieved via MIMO depending on the used transmission scheme. The LTE signaling application supports the following transmission schemes:

- **Spatial multiplexing**

Spatial multiplexing uses at least two transmit antennas and two receive antennas. Different data streams are transmitted simultaneously on the same resource blocks. The data streams can belong to one single user (SU-MIMO) or to different users (MU-MIMO). While SU-MIMO increases the data rate of one user, MU-MIMO increases the overall capacity.

Spatial multiplexing requires precoding at the transmitter side. Two types of spatial multiplexing are defined, using different mechanisms for selection of the precoding matrix. For closed loop (CL) spatial multiplexing, the UE estimates the radio channel, determines the optimum matrix and proposes this precoding matrix by reporting a PMI value to the eNodeB. With open loop (OL) spatial multiplexing, the eNodeB selects a matrix without UE feedback.

The LTE signaling application supports open-loop and closed-loop spatial multiplexing for a single user.

- **Transmit diversity**

Transmit diversity uses several antennas transmitting essentially the same stream of data. This mechanism increases the signal to noise ratio at the receiver side and thus the robustness of data transmission, especially in fading scenarios.

- **Receive diversity (SIMO)**

Receive diversity uses several antennas at the receiver side to receive the same data stream. Like transmit diversity, this mechanism increases the signal to noise ratio.

- **Beamforming**

Beamforming uses multiple antennas. It controls the direction of the wavefront by weighting the magnitude and phase of the individual antenna signals. Beamforming makes it possible, to provide better coverage to specific areas along the cell edges. The UE-specific reference signals and the data transmission use the same antenna weightings and antenna ports (see [Table 2-31](#)). For the cell-specific reference signals and the CSI reference signals, there is no beamforming.

For transmission schemes without beamforming, the number of used antennas is denoted as "TxR", with T transmit antennas and R receive antennas. The LTE signaling application supports the following antenna configurations for the LTE downlink:

- SISO 1x1
- SIMO 1x2 (requires R&S CMW-KS520)
- MIMO 2x2 (requires R&S CMW-KS520)
- MIMO 4x2 (requires R&S CMW-KS521)
- MIMO 8x2 (requires R&S CMW-KS522)
- MIMO 2x4 (requires R&S CMW-KS540)
- MIMO 4x4 (requires R&S CMW-KS540)
- MIMO 8x4 (requires R&S CMW-KS541)

For MIMO configuration, you must set the parameters "Scenario" and "Transmission Mode" (see ["Scenario"](#) on page 140 and [Section 2.4.12.2, "General MIMO settings"](#), on page 210).

For some transmission modes, several DCI formats, antenna configurations, precoding matrix indices or transmission schemes are supported, so that you must also configure the corresponding settings. The following table provides an overview of the possible parameter combinations and the resulting transmission scheme.

For more information about the transmission modes and DCI formats, see 3GPP TS 36.213, section 7.1.

Table 2-32: Transmission scheme overview

Scenario	TM ¹⁾	DCI ⁸⁾	Antennas/ beamform.	PMI ¹⁾²⁾	Transmission scheme
"<n>CC...1x1..." 1x1 carrier	1	1, 1A	1x1	-	SISO
	7	1	1x1	-	Single-layer beamforming (port 5)
"<n>CC...nx2....."	1	1, 1A	1x2	-	SIMO

Scenario	TM ¹⁾	DCI ⁸⁾	Antennas/ beamform.	PMI ¹⁾²⁾	Transmission scheme
nx2 carrier	2	1, 1A	2x2, 4x2	-	Transmit diversity
	3	1A ⁴⁾	2x2, 4x2	-	Transmit diversity
		2A			OL spatial multiplexing
	4	2	2x2	0, 1	CL spatial multiplexing ³⁾
			4x2	0 to 15	
	6	1B	2x2	0 to 3	CL spatial multiplexing, single layer
			4x2	0 to 15	
	7	1	1x2	-	Single-layer beamforming (port 5)
	8	2B	1x2, 2x2	0 to 3	Single-layer beamforming (port 8)
				1, 2	Dual-layer beamforming (ports 7, 8)
	9	2C	2x2	0, 1, 2, 3, random ⁵⁾	Single-layer beamforming (port 8)
				0, 1, 2, random ⁶⁾	Dual-layer beamforming (ports 7, 8)
			4x2	0 to 15	Single-layer beamforming (port 8)
			8x2	2x 0 to 15	Dual-layer beamforming (ports 7, 8)
"1CC...4x2"	2	1, 1A	4x2	-	Transmit diversity
	3	1A ⁴⁾	4x2	-	Transmit diversity
		2A	4x2	-	OL spatial multiplexing
	4	2	4x2	0 to 15	CL spatial multiplexing ³⁾
	6	1B	4x2	0 to 15	CL spatial multiplexing, single layer
	9	2C	4x2	0 to 15	Single-layer beamforming (port 8) Dual-layer beamforming (ports 7, 8)
"<n>CC...nx4....." nx4 carrier	2	1, 1A	2x4, 4x4	-	Transmit diversity
	3	1A ⁴⁾	2x4, 4x4	-	Transmit diversity
		2A			OL spatial multiplexing
	4	2	2x4	0, 1	CL spatial multiplexing ³⁾
			4x4	0 to 15	
	6	1B	2x4	0 to 3	CL spatial multiplexing, single layer
			4x4	0 to 15	
	9	2C	2x4	0, 1, 2, 3, random ^{5), 7)}	Single-layer beamforming (port 8)
				0, 1, 2, random ^{6), 7)}	Dual-layer beamforming (ports 7, 8)

Scenario	TM ¹⁾	DCI ⁸⁾	Antennas/ beamform.	PMI ¹⁾²⁾	Transmission scheme
			4x4	0 to 15	Single-layer beamforming (port 8) Dual-layer beamforming (ports 7, 8) Four-layer beamforming (ports 7-10)
			8x4	2x 0 to 15	Single-layer beamforming (port 8) Dual-layer beamforming (ports 7, 8)
				0 to 3 and 0 to 7	Four-layer beamforming (ports 7-10)

¹⁾ TM = transmission mode, PMI = precoding matrix indicator

²⁾ Depending on the scheduling type, the used PMI value is configured statically or it follows the PMI value reported by the UE

³⁾ Depending on the scheduling type, the transmission scheme is configured statically or it follows the RI value reported by the UE. In the latter case, the scheme can change to transmit diversity.

⁴⁾ For scheduling type "Follow WB CQI-RI", only DCI format 2A is supported.

⁵⁾ For the "Follow..." scheduling types, random is not allowed.

⁶⁾ For the "Follow..." scheduling types, 0 and random are not allowed.

⁷⁾ For scenarios without fading, random is not allowed.

⁸⁾ For eMTC, the DCI format is 6-1A for CE mode A and 6-1B for CE mode B.

2.2.18.1 Radio channel coefficients for MIMO

A MIMO 2x2 configuration uses two transmit antennas (TX) and two receive antennas (RX). An ideal radio channel would provide isolated connections between pairs of antennas, so that each RX antenna receives only the signal of one TX antenna.

In practice, an ideal radio channel is not possible. A real radio channel couples the sent signals, so that the RX antennas always receive a combination of the TX antenna signals. The following figure shows a radio channel for MIMO 2x2, assuming that there is only one signal path between each TX and RX antenna.

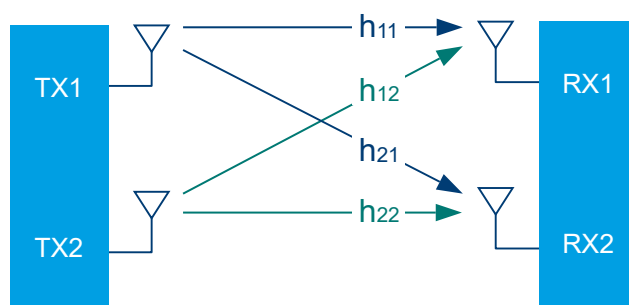


Figure 2-17: Radio channel for MIMO 2x2, TM 2 to 6

The channel amplitude and phase responses for the individual transmission paths can be characterized by the complex coefficients h_{11} to h_{22} .

The indices for TM 2 to 6 specify the receiver first (h_{ab} for receiver a / transmitter b). For other transmission modes, the indices specify the transmitter first (h_{ab} for transmitter a / receiver b).

In a test setup with the R&S CMW and a UE, the radio channel is typically located within the instrument. Each RF output connector is connected directly to one UE antenna connector.

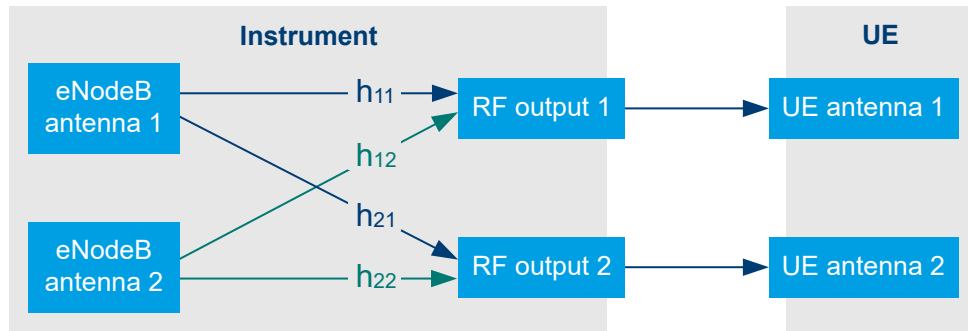


Figure 2-18: "Radio channel" in a test setup, MIMO 2x2, TM 2 to 6

For fading scenarios, the coefficients are determined dynamically. For scenarios without fading, you can configure static channel coefficients. See for example [Section 2.4.12.3, "MIMO channel matrix TM 2 to TM 6"](#), on page 212.

A MIMO 4x2 configuration uses four TX and two RX antennas. So the setup is as follows:

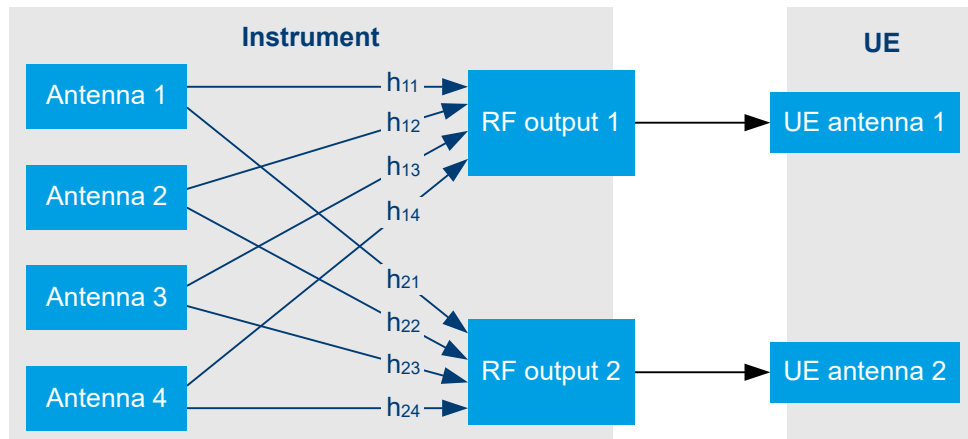


Figure 2-19: "Radio channel" in a test setup, MIMO 4x2, TM 2 to 6

The principles described here for MIMO 2x2 and MIMO 4x2 apply also to the other MIMO modes. A "MIMO <a> x " configuration uses <a> TX antennas, RF output paths and RX antennas.

2.2.18.2 Beamforming matrix

For transmission schemes with beamforming, beamforming is applied to antenna ports with UE-specific signals. For cell-specific signals, there is no beamforming.

The used ports depend on the transmission scheme and on the number of beamforming layers. In general, the antenna ports 0 and 1 can be used by a transmission scheme for cell-specific signals. The antenna ports 5, 7 and higher can be used for UE-specific signals.

The coefficients of a beamforming matrix are similar to the coefficients of a channel matrix. They are complex coefficients, have a magnitude and a phase and define an amplitude response and a phase response.

For TM 8, there is a beamforming matrix and a channel matrix. They are independent and can have different coefficients. Without fading, you can configure both matrices.

The following example visualizes the matrices for TM 8 with dual-layer beamforming.

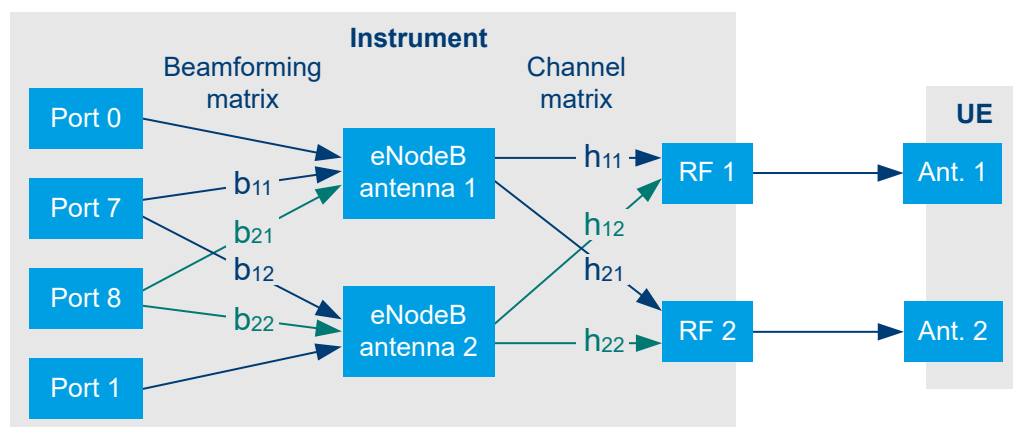


Figure 2-20: Beamforming and channel matrices, 2x2, transmission mode 8

For TM 7, there is only a beamforming matrix. The following example shows a 1x2 matrix for TM 7 with single-layer beamforming. Port 0 is in this example mapped to only one RF output. You can also map it to both RF outputs.

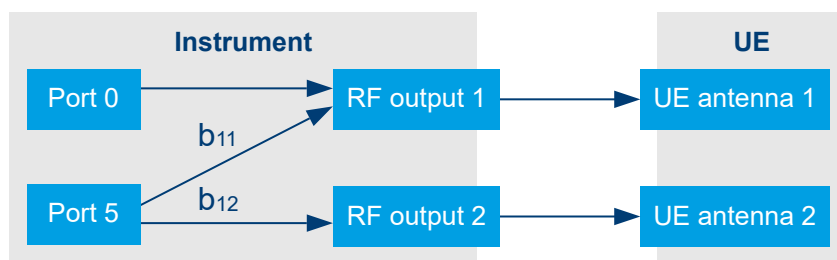


Figure 2-21: Beamforming matrix, 1x2, transmission mode 7

For transmission mode 9, you do not configure a beamforming matrix. Instead, you define the beamforming via the number of TX antennas and one or two precoding matrix indices.

2.2.19 Scheduling type RMC

Reference measurement channels (RMC) as defined in 3GPP TS 36.521 are required for various transmitter and receiver conformance tests.

An RMC can be defined via the following set of parameters:

- Cell bandwidth
- Number of allocated resource blocks (RB)
- Position of the first allocated RB
- Modulation type
- Transport block size index (TBS index)

The parameter combinations supported by the R&S CMW are listed in the following sections. Many other parameters are indirectly determined by these parameters. Refer to 3GPP TS 36.521 for detailed tables. Some of these parameters are also displayed at the GUI for information.

In addition to the 3GPP-compliant combinations, the tables list also the configurable combinations without RB allocation ("Allocated RBs" = 0).

If you want to force the UE to use the PUCCH instead of the PUSCH, set the number of UL resource blocks to 0. Thus the UE is not allowed to use the PUSCH and must use the PUCCH to send ACK/NACKs.



The following sections apply without eMTC and LAA. See also:

- [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45
- [Section 2.2.16.2, "Scheduling type RMC for LAA"](#), on page 58

• UL RMCs, contiguous allocation	70
• UL RMCs, multi-cluster	74
• DL RMCs, one TX antenna (TM 1)	75
• DL RMCs, multiple TX antennas (TM 2 to 6)	77
• DL RMCs, transmission mode 7	79
• DL RMCs, transmission mode 8	80
• DL RMCs, transmission mode 9	81

2.2.19.1 UL RMCs, contiguous allocation

This section lists the supported parameter sets for UL RMCs with contiguous RB allocation (resource allocation type 0).

For allowed RB positions, see ["Position of first RB, contiguous allocation"](#) on page 72.

For 64-QAM and 256-QAM, option R&S CMW-KS504/-KS554 is required.

Table 2-33: UL RMCs for FDD and TDD, contiguous

Bandwidth	Allocated RBs	"TBS Idx"			
		QPSK	16-QAM	64-QAM	256-QAM
1.4 MHz to 20 MHz	0	0	-	-	-
	1	5	19	25	32
	2	6	19	25	32
	3	6	19	25	32

Bandwidth	Allocated RBs	"TBS Idx"			
		QPSK	16-QAM	64-QAM	256-QAM
	4	6	19	25	32
	5	5	19	25	32
	6	6	19	25	32
3 MHz to 20 MHz	8	6	19	25	32
	9	5	19	25	32
	10	5	19	25	32
	12	6	19	25	32
	15	6 for 3 MHz, else 5	14	25	32
5 MHz to 20 MHz	16	5	14	25	32
	18	6	14	25	32
	20	5	11	25	32
	24	6	11	25	32
	25	5	11	25	32
10 MHz to 20 MHz	27	6	10	25	32
	30	5	19	25	32
	32	5	19	25	32
	36	6	19	25	32
	40	6	19	25	32
	45	5	19	25	32
	48	5	19	25	32
	50	6	19	25	32
15 MHz to 20 MHz	54	5	19	25	32
	60	4	18	25	32
	64	4	18	25	32
	72	4	14	25	32
	75	3	14	25	32
20 MHz	80	3	14	25	32
	81	3	14	25	32
	90	2	12	23	32
	96	2	12	25	32
	100	2	11	25	32

Position of first RB, contiguous allocation

Two start positions are always allowed:

- Low: lower end of cell bandwidth, starting with first RB
- High: upper end of cell bandwidth, up to last RB

If the number of allocated RBs equals one, an additional position is allowed:

- Middle: RB located in the middle of the cell bandwidth
The position is calculated as the smallest integer number $\geq N_{RB} / 2$, with N_{RB} being the number of RBs in a full RB allocation.

Additional special start positions are allowed if certain conditions are fulfilled. The special start positions are listed in the following tables. Column "Conditions" lists the operating band and the network signaled value for which the special start positions are supported. Column "3GPP tables" indicates in which table of 3GPP TS 36.521 the start positions are defined.

Table 2-34: Special start positions, UL RMCs for FDD, contiguous

Conditions	3GPP tables	Bandwidth	Alloc. RBs	Modulation	Position first RB
Band 1 NS_05	6.2.4.4.1-3 6.6.3.3.4.1-1	15 MHz	1	QPSK	8, 66
			16	QPSK	8, 51
			30	QPSK	8, 37
			48	QPSK	8, 19
		20 MHz	1	QPSK	24, 75
			18	QPSK	24, 58
			24	QPSK	24, 52
			48	QPSK	24, 28
Band 13 NS_07	6.2.4.4.1-5 6.6.2.2.4.1-3 6.6.3.3.4.1-2	10 MHz	2	QPSK	48
			6	QPSK	13, 43
			12	QPSK	13, 19
			16	QPSK, 16-QAM	19
			20	QPSK	13
			30	QPSK, 16-QAM	19
			36	16-QAM	13
Band 26 NS_12	6.2.4.4.1-10	1.4 MHz	1, 5	QPSK	1
		3 MHz	4, 10	QPSK	4
		5 MHz	8, 15	QPSK	7
		10 MHz	18	QPSK	16
		15 MHz	30	QPSK	31
Band 26 NS_13	6.2.4.4.1-11	5 MHz	15	QPSK	7

Conditions	3GPP tables	Bandwidth	Alloc. RBs	Modulation	Position first RB
Band 26 NS_14	6.2.4.4.1-12	10 MHz	25	QPSK	1
		15 MHz	50	QPSK	15
Band 26 NS_15	6.2.4.4.1-13	3 MHz	6	QPSK	7
			12	QPSK	1
		5 MHz	6	QPSK	14
			16	QPSK	9
		10 MHz	1	QPSK	10, 39
			20	QPSK	3, 25
			36	QPSK	1
		15 MHz	18	QPSK	36, 44
			25	QPSK	1
			60	QPSK	2
Band 27 NS_16	6.2.4.4.1-14	3 MHz	12	QPSK	1
		5 MHz	12, 18, 20	QPSK	2
		10 MHz	1	QPSK	10
			27, 32	QPSK	15
			40	QPSK	1
Band 20	7.3.3-2	15 MHz	20	QPSK	11
		20 MHz	20	QPSK	16
Band 31	7.3.3-2	3 MHz	5	QPSK	9
		5 MHz	5	QPSK	10
Band 12	7.3A.1.3-0b	1.4 MHz	2	QPSK	middle
		3 MHz	5	QPSK	middle
		5 MHz	8	QPSK	middle
		10 MHz	16	QPSK	middle
Band 17	7.3A.1.3-0b	5 MHz	8	QPSK	middle
		10 MHz	16	QPSK	middle
Band 1 CA_NS_01	6.2.4A.1.4.1-1	15 MHz	45	QPSK	7
	6.6.3.3A.1.4.1-1	20 MHz	64	QPSK	24
Band 1 CA_NS_02	6.2.4A.1.4.1-2	15 MHz	1	QPSK	54
	6.6.3.3A.1.4.1-2	20 MHz	1	QPSK	21, 83
Band 1 CA_NS_03	6.2.4A.1.4.1-3	15 MHz	1	QPSK	21, 44
	6.6.3.3A.1.4.1-3	20 MHz	1	QPSK	63, 70

Conditions	3GPP tables	Bandwidth	Alloc. RBs	Modulation	Position first RB
Band 2	7.xA.4.4.1-1 7.x.xA.4.4.1-1	20 MHz	16	QPSK	57
Band 3	7.xA.4.4.1-1 7.x.xA.4.4.1-1	20 MHz	16	QPSK	50
Band 25	7.xA.4.4.1-1 7.x.xA.4.4.1-1	10 MHz	10	QPSK	33
		20 MHz	12	QPSK	62

Table 2-35: Special start positions, UL RMCs for TDD, contiguous

Conditions	3GPP tables	Bandwidth	Alloc. RBs	Modulation	Position first RB
Band 41 NS_04	6.2.4.4.1-2	10 MHz	24	QPSK, 16-QAM	13
			36	QPSK	13
			12	QPSK	37
			1	QPSK	49
		15 MHz	36	QPSK, 16-QAM	19
			50	QPSK	19
			18	QPSK	56
			1	QPSK	74
		20 MHz	50	QPSK, 16-QAM	25
			75	QPSK	25
			25	QPSK	75
			1	QPSK	99
Band 41 CA_NS_04	6.2.4A.1.4.1-4 6.6.2.2A.1.4.1-1	15 MHz	10	QPSK	20
			20	16-QAM	15
		20 MHz	10	QPSK	20, 25
			15	16-QAM	40
			20	16-QAM	30
Band 39 CA_NS_07	6.2.4A.1.4.1-7 6.6.3.3A.1.4.1-6	15 MHz	40	QPSK	20
		20 MHz	1	QPSK	1
			10, 30	QPSK	30
			40	QPSK	20

2.2.19.2 UL RMCs, multi-cluster

The following table lists the supported parameter sets for UL RMCs with multi-cluster RB allocation (resource allocation type 1).

The allocated resource blocks are divided into two clusters. Within each cluster, the RB allocation is contiguous. The number of allocated resource blocks and the position of the first allocated resource block are specified per cluster.

Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

For 64-QAM, option R&S CMW-KS504/-KS554 is required.

Table 2-36: UL RMCs for FDD and TDD, multi-cluster

Bandwidth	Alloc. RBs Cluster 1	First RB Cluster 1	Alloc. RBs Cluster 2	First RB Cluster 2	"TBS Idx"		
					16-QAM	64-QAM	256-QAM
5 MHz	2	0	1	24	11	25	32
			18	6	11	25	32
	18	0	2	22	11	25	32
10 MHz	3	0	2	48	11	25	32
			42	6	11	25	32
	42	0	3	45	11	25	32
15 MHz	4	0	60	12	11	25	32
	8	0	7	68	11	25	32
	60	0	4	68	11	25	32
20 MHz	4	0	4	96	11	25	32
			92	8	11	25	32
	92	0	4	96	11	25	32

2.2.19.3 DL RMCs, one TX antenna (TM 1)

The following table lists the supported parameter sets for downlink RMCs when only one TX antenna is used in the downlink.

The resource blocks allocated in the DL are contiguous. The position of the first allocated RB is indicated as follows:

- L: lower end of cell bandwidth, starting with first RB
- H: upper end of cell bandwidth, up to last RB
- 5, 10, 23, 35, 48: special start positions for 3GPP TS 36.521, table C.2-2.

For 256-QAM, option R&S CMW-KS504/-KS554 is required. For 1024-QAM, option R&S CMW-KS505/-KS555 is required.

Table 2-37: DL RMCs, single TX antenna

Bandwidth	Allocated RBs	Position first RB	Modulation	TBS index
1.4 MHz	0	L, H	QPSK	0
	3	L, H	QPSK	1
	6	L, H	QPSK	4

Bandwidth	Allocated RBs	Position first RB	Modulation	TBS index
			16-QAM	12
			64-QAM	21
			256-QAM	29
			1024-QAM	34
3 MHz	0	L, H	QPSK	0
	1	L, H	16-QAM	13
	4	L, H	QPSK	5
	6	L, H, 5	16-QAM	12
	15	L, H	QPSK	5
			64-QAM	23
			256-QAM	31
			1024-QAM	34
5 MHz	0	L, H	QPSK	0
	6	L, H, 10	16-QAM	12
	8	L, H	QPSK	5
	16	L, H	64-QAM	25*
	18	L, H	64-QAM	23
	25	L, H	QPSK	5
			16-QAM	13
			64-QAM	23
			256-QAM	30
			1024-QAM	34
10 MHz	0	L, H	QPSK	0
	1	L, H	16-QAM	14
	6	L, H, 23	16-QAM	12
	16	L, H	QPSK	5
			64-QAM	25*
	17	L, H	64-QAM	24
	50	L, H	QPSK	5
			16-QAM	14
			64-QAM	24
			256-QAM	32
			1024-QAM	35

Bandwidth	Allocated RBs	Position first RB	Modulation	TBS index
15 MHz	0	L, H	QPSK	0
	6	L, H, 35	16-QAM	12
	16	L, H	64-QAM	25*
	17	L, H	64-QAM	24
	25	L, H	QPSK	5
	75	L, H	QPSK	5
			64-QAM	25
			256-QAM	32
			1024-QAM	35
20 MHz	0	L, H	QPSK	0
	1	L, H	16-QAM	14
	6	L, H, 48	16-QAM	12
	16	L, H	64-QAM	25*
	17	L, H	64-QAM	24
	30	L, H	QPSK	5
	83	L, H	64-QAM	24
	100	L, H	QPSK	5
			64-QAM	24
			256-QAM	32
			1024-QAM	35

*) The entries with a * in the last column have been removed from 3GPP TS 36.521, but are still supported for FDD for backward compatibility reasons. They are not supported for TDD.

2.2.19.4 DL RMCs, multiple TX antennas (TM 2 to 6)

For multiple TX antennas (e.g. MIMO 2x2), 3GPP has not yet specified DL RMCs for all cell bandwidths. For that reason, the R&S CMW supports also some RMCs only specified for single-antenna configurations. All supported parameter combinations are listed in the following tables.

The allocated resource blocks are contiguous and at the low or high end of the cell bandwidth.

Table 2-38: DL RMCs for FDD, multiple TX antennas

Bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	0	QPSK	0
	6	QPSK	4
3 MHz	0	QPSK	0
	15	QPSK	5
5 MHz	0	QPSK	0
	25	QPSK	5
		16-QAM	12
10 MHz	0	QPSK	0
	40	16-QAM	13
	50	QPSK	5
		16-QAM	13
		64-QAM	18
15 MHz	0	QPSK	0
	75	QPSK	5
20 MHz	0	QPSK	0
	100	16-QAM	13

Some TDD RMCs defined by 3GPP have the same cell bandwidth, number of RBs, modulation type and TBS index. The additional parameter "Version" is introduced to distinguish between these RMCs. The RMCs are defined in 3GPP TS 36.521, table A.3.4.2.1-1.

Table 2-39: DL RMCs for TDD, multiple TX antennas

Bandwidth	Allocated RBs	Modulation	TBS index	Version
1.4 MHz	0	QPSK	0	-
	6	QPSK	4	-
3 MHz	0	QPSK	0	-
	15	QPSK	5	-
5 MHz	0	QPSK	0	-
	25	16-QAM	12	-
10 MHz	0	QPSK	0	-
	40	16-QAM	13	-
	50	QPSK	5	-
		16-QAM	13	0: R.11 1: R.11-1

Bandwidth	Allocated RBs	Modulation	TBS index	Version
		64-QAM	18	-
15 MHz	0	QPSK	0	-
	75	QPSK	5	-
20 MHz	0	QPSK	0	-
	100	16-QAM	13	0: R.30 1: R.30-1

2.2.19.5 DL RMCs, transmission mode 7

The following tables list the supported parameter sets for downlink RMCs when transmission mode 7 is used.

The TDD RMCs for 5-MHz and 10-MHz bandwidth are defined in 3GPP TS 36.521, table A.3.4.3.1-1. For the other bandwidths and FDD, 3GPP has not yet specified DL RMCs for TM 7, but the listed parameter combinations are supported.

The listed allocated resource blocks are contiguous and at the low or high end of the cell bandwidth. In subframe 0, a non-contiguous RB allocation is applied (not indicated in the tables).

Table 2-40: DL RMCs for FDD, TM 7

Bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	0	16-QAM	0
	6	16-QAM	13
3 MHz	0	16-QAM	0
	15	16-QAM	13
5 MHz	0	16-QAM	0
	25	16-QAM	12
10 MHz	0	16-QAM	0
	50	16-QAM	13
15 MHz	0	16-QAM	0
	75	16-QAM	13
20 MHz	0	16-QAM	0
	100	16-QAM	13

Table 2-41: DL RMCs for TDD, TM 7

Bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	0	16-QAM	0
	6	16-QAM	13

Bandwidth	Allocated RBs	Modulation	TBS index
3 MHz	0	16-QAM	0
	15	16-QAM	13
5 MHz	0	16-QAM	0
	25	16-QAM	12
10 MHz	0	QPSK	0
	1	16-QAM	13
	18	64-QAM	23
	50	QPSK	5
		16-QAM	13
		64-QAM	23
15 MHz	0	16-QAM	0
	75	16-QAM	13
20 MHz	0	16-QAM	0
	100	16-QAM	13

2.2.19.6 DL RMCs, transmission mode 8

The following tables list the supported parameter sets for downlink RMCs when transmission mode 8 is used.

The TDD RMCs for 5-MHz and 10-MHz bandwidth are defined in 3GPP TS 36.521, table A.3.4.3.2-1. For the other bandwidths and FDD, 3GPP has not yet specified DL RMCs for TM 8, but the listed parameter combinations are supported.

The listed allocated resource blocks are contiguous and at the low or high end of the cell bandwidth. In FDD subframe 0 and TDD subframe 0, 1 and 6, a non-contiguous RB allocation is applied (not indicated in the tables).

Table 2-42: DL RMCs for FDD, TM 8

Bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	0	16-QAM	0
	6	16-QAM	12
3 MHz	0	16-QAM	0
	15	16-QAM	12
5 MHz	0	16-QAM	0
	25	16-QAM	12
10 MHz	0	16-QAM	0
	50	16-QAM	12
15 MHz	0	16-QAM	0

Bandwidth	Allocated RBs	Modulation	TBS index
	75	16-QAM	12
20 MHz	0	16-QAM	0
	100	16-QAM	12

Table 2-43: DL RMCs for TDD, TM 8

Bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	0	16-QAM	0
	6	16-QAM	12
3 MHz	0	16-QAM	0
	15	16-QAM	12
5 MHz	0	16-QAM	0
	25	16-QAM	12
10 MHz	0	QPSK	0
	18	64-QAM	22
	50	QPSK	4
		16-QAM	12
		64-QAM	17
			22
15 MHz	0	16-QAM	0
	75	16-QAM	12
20 MHz	0	16-QAM	0
	100	16-QAM	12

2.2.19.7 DL RMCs, transmission mode 9

The following tables list the supported parameter sets for downlink RMCs when transmission mode 9 is used.

The RMCs for 10-MHz bandwidth are defined in 3GPP TS 36.521, table A.3.3.3.1-1, A.3.3.3.2-1, A.3.4.3.3-1, A.3.4.3.4-1 and A.3.4.3.5-1. For the other bandwidths, 3GPP has not yet specified DL RMCs for TM 9, but the listed parameter combinations are supported.

The listed allocated resource blocks are contiguous and at the low or high end of the cell bandwidth. In FDD subframe 0 and TDD subframe 0, 1 and 6, a non-contiguous RB allocation is applied (not indicated in the tables).

Table 2-44: DL RMCs for FDD, TM 9

Bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	0	16-QAM	0
	6	16-QAM	12
3 MHz	0	16-QAM	0
	15	16-QAM	12
5 MHz	0	16-QAM	0
	25	16-QAM	12
10 MHz	0	16-QAM	0
	50	QPSK	4
			7
		16-QAM	12
15 MHz	0	16-QAM	0
	75	16-QAM	12
20 MHz	0	16-QAM	0
	100	16-QAM	12

Table 2-45: DL RMCs for TDD, TM 9

Bandwidth	Allocated RBs	Modulation	TBS index
1.4 MHz	0	16-QAM	0
	6	16-QAM	12
3 MHz	0	16-QAM	0
	15	16-QAM	12
5 MHz	0	16-QAM	0
	25	16-QAM	12
10 MHz	0	QPSK	0
	50	QPSK	4
			7
		16-QAM	12
15 MHz	0	16-QAM	0
	75	16-QAM	12
20 MHz	0	16-QAM	0
	100	16-QAM	12

2.2.20 User-defined channels

The following applies to user-defined channels without eMTC and LAA. See also:

- [Section 2.2.13, "User-defined channels for eMTC"](#), on page 49
- [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56

In addition to RMCs required by conformance tests, the signaling application supports flexibly configurable user-defined channels (option R&S CMW-KS510 required).

User-defined channels are specified via a set of parameters:

- Cell bandwidth
- Number and position of resource blocks (RB)
- Modulation type
- Transport block size index

The supported modulation types are independent of the allocated RBs. For each modulation type, a range of transport block size indices is available.

Two modes are available. One mode allows you to configure UL and DL channels globally, for all subframes. The other mode supports individual settings for each subframe (TTI) of a radio frame (all parameters except cell bandwidth configurable per TTI).



Force PUCCH

If you want to force the UE to use the PUCCH instead of the PUSCH, set the number of resource blocks to 0. Thus the UE is not allowed to use the PUSCH and must use the PUCCH to send ACK/NACKs.

2.2.20.1 Number and position of RBs (DL)

In the downlink, the used resource allocation type depends on the DCI format:

- Allocation type 2 for DCI formats 1A, 1B
- Allocation type 0 for DCI formats 1, 2, 2A, 2B and 2C

The allowed DCI formats depend on the selected transmission mode, see [Table 2-32](#).

Resource allocation type 2

With resource allocation type 2, the allocated resource blocks are contiguous. You can configure the number of RBs and the position of the first RB.

The number of allocated DL RBs is restricted by an upper limit. Any number of RBs between 0 and this upper limit can be allocated. The maximum number of RBs depends on the cell bandwidth and is listed in [Table 2-46](#). The position of the first allocated RB is freely selectable within the cell bandwidth. Thus allowed positions are 0 to <Maximum no of RBs> - 1.

Resource allocation type 0

With resource allocation type 0, the maximum number of RBs is divided into resource block groups (RBG) with a certain size (see 3GPP TS 36.213, section 7.1.6.1). The RBG size depends on the maximum number of RBs and thus on the cell bandwidth. If the maximum number of RBs is not a multiple of the RBG size, the last RBG is smaller than the other RBGs.

With resource allocation type 0, you can only allocate entire RBGs. You can freely select the allocated RBGs (multi-cluster allocation - only for global configuration, not TTI-based). Or you can allocate a contiguous sequence of RBGs by selecting the start RB and the number of allocated RBs.

The following tables provide an overview of the resource block groups.

Table 2-46: RBG parameters

Channel bandwidth	Maximum no. of RBs	RBG size	Size of last RBG	No. of RBGs	Allowed start RB
1.4 MHz	6	1	1	6	0, 1, 2, 3, 4, 5
3 MHz	15	2	1	8	0, 2, 4, 6, 8, 10, 12, 14
5 MHz	25	2	1	13	0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24
10 MHz	50	3	2	17	0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48
15 MHz	75	4	3	19	0, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72
20 MHz	100	4	4	25	0, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 88, 92, 96

Table 2-47: Example: Allowed contiguous allocations, BW 3 MHz

Start RB	0	2	4	6	8	10	12	14
No of RBs	2, 4, ..., 14, 15	2, 4, ..., 12, 13	2, 4, ..., 10, 11	2, 4, ..., 8, 9	2, 4, 6, 7	2, 4, 5	2, 3	1

2.2.20.2 Number and position of RBs (UL)

In the uplink, the total number of allocated RBs is restricted by the following formula, specified in 3GPP TS 36.211. For multi-cluster RB allocation, it applies to the sum of both clusters:

$$N_{RB} = 2^i \times 3^j \times 5^k, \text{ with } i, j, k \text{ being non-negative integer values}$$

Example: For a cell bandwidth of 3 MHz, the following values are allowed in the uplink: 0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15. Not allowed are 7, 11, 13 and 14.

The other rules depend on the selected resource allocation type as described in the following.

Contiguous RB allocation, allocation type 0

This resource allocation type is defined in 3GPP TS 36.213 section 8.1.1.

The position of the first allocated RB is freely selectable within the cell bandwidth. The number of resource blocks is restricted by the maximum number of RBs, depending on the cell bandwidth, in the same way as for the downlink.

You can also allocate no resource blocks at all ($N_{RB} = 0$). As a result, the UE is not allowed to use the PUSCH and must use the PUCCH instead, e.g. to send ACK/NACK messages.

Resource allocation type 0 is supported for global and TTI-based channel configuration.

Multi-cluster RB allocation, allocation type 1

This resource allocation type is defined in 3GPP TS 36.213 section 8.1.2.

The allocated resource blocks are divided into two clusters. Within each cluster, the RB allocation is contiguous. The number of allocated resource blocks and the position of the first allocated resource block are configured per cluster.

Only entire resource block groups (RBG) can be allocated. This restriction affects both the number of RBs that can be allocated per cluster and the allowed positions for the first RB of each cluster. For details, see [Table 2-46](#).

There must be at least one unused RBG between the two clusters. Allocating no RBs to a cluster is not allowed.

Resource allocation type 1 is only supported for global channel configuration. With carrier aggregation, option R&S CMW-KS512 is required.

2.2.20.3 Supported transport block size indices

The ranges allowed for transport block size indices depend on several settings:

- The UL and DL ranges depend on the modulation type.
- The DL ranges depend also on whether at least one DL TTI uses 256-QAM or 1024-QAM.
- The UL ranges for global configuration depend also on whether 64-QAM and 256-QAM are allowed in the UL.
- The UL ranges for TTI-based configuration depend also on whether at least one UL TTI uses a TBS index greater than 26. Possible combinations are 64-QAM plus index 27/28/29 and 256-QAM plus index 30/31/32.

The allowed TBS index ranges are listed in the following tables.

Table 2-48: Range of UL TBS index

	QPSK	16-QAM	64-QAM	256-QAM
UL 64/256-QAM not allowed	0 to 10	10 to 26	n.a.	n.a.
Global config, 64/256-QAM allowed	0 to 10	10 to 19	19 to 29	30 to 32
TTI-based, TBSI > 26 used in a TTI	0, 2, 4, 6, 8, 10	11 to 14, 16 to 19	20 to 25, 27 to 29	30 to 32

Table 2-49: Range of DL TBS index

	QPSK	16-QAM	64-QAM	256-QAM	1024-QAM
max 64-QAM used	0 to 9	9 to 15	15 to 26	n.a.	n.a.
256-QAM used (no 1024)	0, 2, 4, 6, 8	10 to 15	16 to 24	25, 27 to 33	n.a.
1024-QAM used	0, 2, 4, 6, 8	11, 13, 15	16, 18, 20 to 24	25, 27 to 33	34 to 37

For DL 256-QAM, UL 256-QAM and UL 64-QAM, option R&S CMW-KS504/-KS554 is required.

For DL 1024-QAM, option R&S CMW-KS505/-KS555 is required.

2.2.21 CQI channels

The CQI reporting tests specified in section 9 of 3GPP TS 36.521 require downlink signals with a specific CQI index value. For some tests, the CQI index value must be static (for example reported median CQI value plus or minus one). For other tests, the CQI index value must follow the wideband CQI value reported by the UE.

The LTE signaling application provides the following scheduling types for CQI tests:

- Fixed CQI
- Follow wideband CQI
- Follow wideband PMI
- Follow wideband PMI-RI
- Follow wideband CQI-RI
- Follow wideband CQI-PMI-RI

The downlink channel for the individual scheduling types is described in the following sections.

The uplink channel is identical to a user-defined TTI-based uplink channel, see [Section 2.2.20, "User-defined channels"](#), on page 83.

2.2.21.1 Fixed CQI channels

This scheduling type is supported for the transmission modes 1 to 9. Option R&S CMW-KS510 is required.

The downlink channel is configured via a set of parameters:

- Cell bandwidth
- Number of allocated resource blocks (RB)
- Position of the first allocated RB
- CQI index
- 256-QAM usage
- 1024-QAM usage

All parameters except the cell bandwidth and the QAM usages can be configured individually for each downlink subframe (TTI) of a radio frame.

The allocated resource blocks are contiguous and the position of the first RB can be configured. The configuration rules are the same as for TTI-based user-defined downlink channels, see [Section 2.2.20.1, "Number and position of RBs \(DL\)"](#), on page 83.

The CQI index determines the channel code rate. The code rate is defined as the number of downlink information bits (including CRC bits) divided by the number of physical channel bits on the PDSCH. The relation is defined in 3GPP TS 36.213, table 7.2.3-1 (without 256-QAM), table 7.2.3-2 (with 256-QAM) and table 7.2.3-4 (with 1024-QAM).

Table 2-50: CQI indices with max 64-QAM

CQI index	Modulation scheme	Target code rate
1	QPSK	0.07617
2		0.11719
3		0.18848
4		0.30078
5		0.43848
6		0.58789
7	16-QAM	0.36914
8		0.47852
9		0.60156
10	64-QAM	0.45508
11		0.55371
12		0.65039
13		0.75391
14		0.85254
15		0.92578

Table 2-51: CQI indices with 256-QAM (no 1024-QAM)

CQI index	Modulation scheme	Target code rate
1	QPSK	0.07617
2		0.18848
3		0.43848
4	16-QAM	0.36914
5		0.47852
6		0.60156
7	64-QAM	0.45508
8		0.55371
9		0.65039
10		0.75391

CQI index	Modulation scheme	Target code rate
11	256-QAM	0.85254
12		0.69434
13		0.77832
14		0.86426
15		0.92578

Table 2-52: CQI indices with 1024-QAM

CQI index	Modulation scheme	Target code rate
1	QPSK	0.07617
2		0.18848
3		0.43848
4	16-QAM	0.36914
5		0.60156
6	64-QAM	0.55371
7		0.65039
8		0.75391
9		0.85254
10	256-QAM	0.69434
11		0.77832
12		0.86426
13		0.92578
14	1024-QAM	0.83301
15		0.92578

The modulation scheme is selected automatically according to the table. Indirectly, the table also determines the transport block size. The transport block size is selected automatically so that the resulting effective channel code rate approaches the listed target code rate as far as possible. Thus the transport block size depends on the CQI index and on the number of allocated RBs.

2.2.21.2 Follow WB CQI channels

This scheduling type is supported for the transmission modes 1 to 9. Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

The downlink signal reacts to the reported CQI index value. That means, the MCS index value used for the downlink depends on the previous wideband CQI value reported by the UE.

The dependency between CQI and MCS is configured via a mapping table. The table assigns an MCS index value to each possible reported wideband CQI index value.

The reported PMI and RI values have no effect on the downlink signal.

In addition to the mapping table, you can configure the following downlink settings:

- Cell bandwidth
- Number and position of resource blocks (RB)
- 256-QAM usage
- 1024-QAM usage

All downlink settings are configured globally, not per TTI. The rules for configuration of the allocated RBs are the same as for user-defined downlink channels (global configuration, not TTI-based), see [Section 2.2.20.1, "Number and position of RBs \(DL\)"](#), on page 83.

Not scheduled subframes

By default, some subframes are not scheduled, as defined in 3GPP TS 36.521, table C3.2-4:

- FDD: Subframes 0 and 5 are not scheduled.
- TDD: Subframes 0, 5, 6 and special subframes are not scheduled.

You can override this behavior and individually enable also these subframes.

2.2.21.3 Follow WB PMI and PMI-RI channels

The scheduling type "Follow WB PMI" is supported for the transmission modes 4, 6, 8 and 9. The scheduling type "Follow WB PMI-RI" is supported for the transmission modes 4, 8 and 9. Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

The PMI value used for the downlink signal follows the reported PMI value. The reported CQI value has no effect on the downlink signal.

For scheduling type "Follow WB PMI-RI", the reported rank indicator (RI) value determines the transmission scheme:

- Transmission mode 4:
RI=1: transmit diversity
RI=2: spatial multiplexing
- Transmission mode 8:
RI=1: single-layer beamforming (port 8)
RI=2: dual-layer beamforming (ports 7 and 8)
- Transmission mode 9:
RI=1: single-layer beamforming (port 8)
RI=2: dual-layer beamforming (ports 7 and 8)
RI=4: four-layer beamforming (ports 7 to 10)

The downlink signal settings are similar to the settings for user-defined downlink channels (global configuration, not TTI-based), see [Section 2.2.20, "User-defined channels"](#), on page 83.

Not scheduled subframes

By default, some subframes are not scheduled, as defined in 3GPP TS 36.521, table C3.2-3:

- FDD: Subframe 5 is not scheduled.
- TDD: Subframe 5 and special subframes are not scheduled.

You can override this behavior and individually enable also these subframes.

2.2.21.4 Follow WB CQI-RI and CQI-PMI-RI channels

The scheduling type "Follow WB CQI-RI" is supported for the transmission mode 3. Scheduling type "Follow WB CQI-PMI-RI" is supported for the transmission modes 4, 8 and 9. Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

The downlink signal reacts to the reported values.

The reported CQI index value determines the MCS index value used for the downlink. The dependency between CQI and MCS is configured via a mapping table, as for "Follow wideband CQI". The resource block settings are also similar to the "Follow wideband CQI" settings.

The reported rank indicator (RI) value determines the used transmission scheme, in the same way as for "Follow WB PMI-RI".

For scheduling type "Follow WB CQI-PMI-RI", the used PMI value follows the reported PMI value.

Not scheduled subframes

By default, some subframes are not scheduled, as defined in 3GPP TS 36.521, table C3.2-4:

- FDD: Subframes 0 and 5 are not scheduled.
- TDD: Subframes 0, 5, 6 and special subframes are not scheduled.

You can override this behavior and individually enable also these subframes.

2.2.22 Semi-persistent scheduling (SPS)

Applications like voice over LTE require constant but few radio resources. Voice data is sent in small packets that are transmitted in regular intervals.

For such applications, semi-persistent scheduling (SPS) can be used instead of dynamic scheduling. SPS reduces the control channel overhead on the PDCCH, so that a cell can handle more voice calls in parallel. With SPS, the UE gets a semi-persistent grant. The grant allows the UE to use every n^{th} subframe until further notice.

The LTE signaling application supports SPS in transmission mode 1 and 2, uplink and downlink. Option R&S CMW-KS510 is required.

The configuration of the resource block allocation is similar to the configuration for a user-defined channel (global configuration, not TTI-based), see [Section 2.2.20, "User-defined channels"](#), on page 83. The rules for the allowed number of RBs and position

of the first RB are identical. Exceptions: For SPS, setting the number of RBs to 0 is not allowed. Multi-cluster allocation is not supported.

In contrast to user-defined channels, the highest modulation scheme supported for SPS is 16-QAM. The maximum transport block size index is 14, see following table.

Table 2-53: Range of transport block size index depending on modulation type

Modulation type	QPSK	16-QAM	64-QAM/256-QAM
Indices for uplink channels	0 to 10	10 to 14	not supported
Indices for downlink channels	0 to 9	9 to 14	not supported

The configured resource block allocation is granted every n^{th} subframe, where n is configurable separately for UL and DL.

If the UE does not use the granted uplink resources for transmission, the allocation expires after some unused subframes (implicit release). In that case, a new grant is sent automatically to the UE.

SPS is only supported for the PCC, not for SCCs. HARQ cannot be combined with SPS. Connected DRX is supported, but without UL dynamic scheduling.

2.2.23 Operating bands

The carrier frequencies for LTE signals are defined in 3GPP TS 36.101. Each operating band contains carrier frequencies identified by channel numbers (EARFCN, E-UTRA absolute radio frequency channel number). The assignment between channel numbers N and carrier center frequencies F is defined as:

$$N = 10 \times (F - F_{\text{Offset}}) / \text{MHz} + N_{\text{Offset}}$$

The tables in this section provide an overview of all supported FDD and TDD operating bands. For each band, they list the channel numbers N and the carrier center frequencies F . The offsets F_{Offset} and N_{Offset} equal the lower boundaries of the listed ranges (example: channel 18000 to 18599 means $N_{\text{Offset}} = 18000$).

The table for FDD uplink signals lists also the separation between uplink carrier frequency and downlink carrier frequency (frequency pair for one UE in FDD mode).

In addition to the 3GPP defined bands listed in the tables, a user-defined operating band is supported (option R&S CMW-KS525 required). It uses a 100-kHz channel raster like the 3GPP defined bands. The frequency range and the channel number range are configurable for uplink and downlink. For configuration, see ["Band Definition"](#) on page 149.

For RF frequencies above 3.3 GHz, you need the generic option R&S CMW-KB036 "3.3 to 6 GHz enabling". Bands above 3.3 GHz are, for example, band 22, 42, 43, 46, 48, 250, 252 and 255.

Table 2-54: Operating bands for FDD uplink signals

Band	Channel N _{UL}	F _{UL} / MHz	(F _{DL} -F _{UL}) / MHz
1	18000 to 18599	1920 to 1979.9	190
2	18600 to 19199	1850 to 1909.9	80
3	19200 to 19949	1710 to 1784.9	95
4	19950 to 20399	1710 to 1754.9	400
5	20400 to 20649	824 to 848.9	45
6	20650 to 20749	830 to 839.9	45
7	20750 to 21449	2500 to 2569.9	120
8	21450 to 21799	880 to 914.9	45
9	21800 to 22149	1749.9 to 1784.8	95
10	22150 to 22749	1710 to 1769.9	400
11	22750 to 22949	1427.9 to 1447.8	48
12	23010 to 23179	699 to 715.9	30
13	23180 to 23279	777 to 786.9	-31
14	23280 to 23379	788 to 797.9	-30
15	23380 to 23579	1900 to 1919.9	700
16	23580 to 23729	2010 to 2024.9	575
17	23730 to 23849	704 to 715.9	30
18	23850 to 23999	815 to 829.9	45
19	24000 to 24149	830 to 844.9	45
20	24150 to 24449	832 to 861.9	-41
21	24450 to 24599	1447.9 to 1462.8	48
22	24600 to 25399	3410 to 3489.9	100
23	25500 to 25699	2000 to 2019.9	180
24	25700 to 26039	1626.5 to 1660.4	-101.5
25	26040 to 26689	1850 to 1914.9	80
26	26690 to 27039	814 to 848.9	45
27	27040 to 27209	807 to 823.9	45
28	27210 to 27659	703 to 747.9	55
30	27660 to 27759	2305 to 2314.9	45
31	27760 to 27809	452.5 to 457.4	10
65	131072 to 131971	1920 to 2009.9	190
66	131972 to 132671	1710 to 1779.9	400
68	132672 to 132971	698 to 727.9	55

Band	Channel N _{UL}	F _{UL} / MHz	(F _{DL} -F _{UL}) / MHz
70 ¹⁾	132972 to 133121	1695 to 1709.9	300
	132972 to 133071	1700 to 1709.9	295
71	133122 to 133471	663 to 697.9	-46
72	133472 to 133521	451 to 455.9	10
73	133522 to 133571	450 to 454.9	10
74	133572 to 134001	1427 to 1469.9	48
85	134002 to 134181	698 to 715.9	30
87	134182 to 134231	410 to 414.9	10
88	134231 to 134280	412 to 416.9	10
106	134292 to 134341	896 to 900.9	39
¹⁾ There are two possible UL/DL separations, with different channel and frequency ranges. The separation is configurable.			

Table 2-55: Operating bands for FDD downlink signals

Band	Channel N _{DL}	F _{DL} / MHz
1	0 to 599	2110 to 2169.9
2	600 to 1199	1930 to 1989.9
3	1200 to 1949	1805 to 1879.9
4	1950 to 2399	2110 to 2154.9
5	2400 to 2649	869 to 893.9
6	2650 to 2749	875 to 884.9
7	2750 to 3449	2620 to 2689.9
8	3450 to 3799	925 to 959.9
9	3800 to 4149	1844.9 to 1879.8
10	4150 to 4749	2110 to 2169.9
11	4750 to 4949	1475.9 to 1495.8
12	5010 to 5179	729 to 745.9
13	5180 to 5279	746 to 755.9
14	5280 to 5379	758 to 767.9
15	5380 to 5579	2600 to 2619.9
16	5580 to 5729	2585 to 2599.9
17	5730 to 5849	734 to 745.9
18	5850 to 5999	860 to 874.9
19	6000 to 6149	875 to 889.9

Band	Channel N _{DL}	F _{DL} / MHz
20	6150 to 6449	791 to 820.9
21	6450 to 6599	1495.9 to 1510.8
22	6600 to 7399	3510 to 3589.9
23	7500 to 7699	2180 to 2199.9
24	7700 to 8039	1525 to 1558.9
25	8040 to 8689	1930 to 1994.9
26	8690 to 9039	859 to 893.9
27	9040 to 9209	852 to 868.9
28	9210 to 9659	758 to 802.9
29 ¹⁾	9660 to 9769	717 to 727.9
30	9770 to 9869	2350 to 2359.9
31	9870 to 9919	462.5 to 467.4
32 ¹⁾	9920 to 10359	1452 to 1495.9
65	65536 to 66435	2110 to 2199.9
66	66436 to 67135 67136 to 67335 ¹⁾	2110 to 2179.9 2180 to 2199.9 ¹⁾
67 ¹⁾	67336 to 67535	738 to 757.9
68	67536 to 67835	753 to 782.9
69 ¹⁾	67836 to 68335	2570 to 2619.9
70	68336 to 68435 68436 to 68485 ³⁾ 68486 to 68585 ¹⁾	1995 to 2004.9 2005 to 2009.9 ³⁾ 2010 to 2019.9 ¹⁾
71	68586 to 68935	617 to 651.9
72	68936 to 68985	461 to 465.9
73	68986 to 69035	460 to 464.9
74	69036 to 69465	1475 to 1517.9
75 ¹⁾	69466 to 70315	1432 to 1516.9
76 ¹⁾	70316 to 70365	1427 to 1431.9
85	70366 to 70545	728 to 745.9
87	70546 to 70595	420 to 424.9
88	70596 to 70645	422 to 426.9
106	70656 to 70705	935 to 939.9
252 ²⁾	255242 to 256046	5159.8 to 5240.2

Band	Channel N _{DL}	F _{DL} / MHz
255 ²⁾	261092 to 261896	5744.8 to 5825.2
¹⁾ SCC DL only, not for PCC or UL ²⁾ SCC DL only, not for PCC or UL, R&S CMW-KS525 required ³⁾ SCC DL only; for PCC/SCC with UL if F _{DL} - F _{UL} = 300 MHz		

Table 2-56: Operating bands for TDD signals

Band	Channel N	F / MHz
33	36000 to 36199	1900 to 1919.9
34	36200 to 36349	2010 to 2024.9
35	36350 to 36949	1850 to 1909.9
36	36950 to 37549	1930 to 1989.9
37	37550 to 37749	1910 to 1929.9
38	37750 to 38249	2570 to 2619.9
39	38250 to 38649	1880 to 1919.9
40	38650 to 39649	2300 to 2399.9
41	39650 to 41589	2496 to 2689.9
42	41590 to 43589	3400 to 3599.9
43	43590 to 45589	3600 to 3799.9
44	45590 to 46589	703 to 802.9
45	46590 to 46789	1447 to 1466.9
46 ²⁾	46790 to 54539	5150 to 5924.9
48	55240 to 56739	3550 to 3699.9
49 ²⁾	56740 to 58239	3550 to 3699.9
50	58240 to 59089	1432 to 1516.9
51	59090 to 59139	1427 to 1431.9
52	59140 to 60139	3300 to 3399.9
53	60140 to 60254	2483.5 to 2494.9
250 ¹⁾	253644 to 255143	3550 to 3699.9
¹⁾ Option R&S CMW-KS525 required ²⁾ SCC DL with LAA only, R&S CMW-KS514 required		

2.2.24 Trigger signals

The LTE signaling application provides trigger signals that can be used by other R&S CMW applications to synchronize to the generated LTE downlink signal.

To address the trigger signals in remote commands, use the following strings, with <i> replaced by the instance number of the signaling application and <n> replaced by the SCC number:

- "LTE Sig<i>: FrameTrigger" (for PCC)
- "LTE Sig<i>: FrameTrigger SCC<n>"
- "LTE Sig<i>: PRACH Trigger"
- "LTE Sig<i>: TPC Trigger" (for PCC)
- "LTE Sig<i>: TPC Trigger SCC<n>"

Frame trigger

Trigger event at the beginning of each uplink radio frame. If an SCC uplink is configured, there are separate trigger signals for the PCC and the SCC.

This signal can be used to trigger LTE multi-evaluation measurements (option R&S CMW-KM500 / -KM550).

PRACH trigger

Trigger event for each received PRACH preamble. When a preamble has been successfully received and detected by the LTE signaling application, a trigger signal is generated. It is aligned to the beginning of the fourth subframe after the subframe in which the preamble has started. The following figure illustrates this timing for preamble format 3.

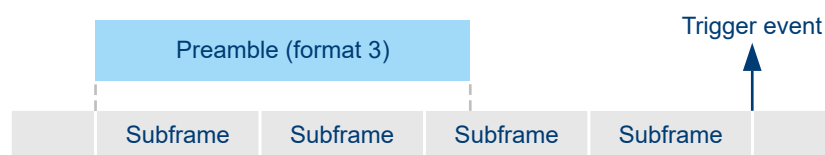


Figure 2-22: PRACH trigger timing

The PRACH trigger signal can be used to trigger LTE "PRACH" measurements (option R&S CMW-KM500 / -KM550). When the measurement receives a trigger event, it evaluates already measured data to provide results for the preamble.

TPC trigger

Trigger event generated when a single TPC pattern is executed ("Single Pattern" or "User-defined Single Pattern" or "Flexible UL Power"). If an SCC uplink is configured, there are separate trigger signals for the PCC and the SCC.

This signal can be used to trigger a single shot power measurement using the LTE multi-evaluation measurement (option R&S CMW-KM500 / -KM550).

2.2.25 Relative power control tests

Uplink power control tests with relative commands are specified in 3GPP TS 36.521, section 6.3.5.2 "Power Control Relative power tolerance". The first two subtests require special TPC power control patterns combined with a dynamic resource block allocation.

tion. Such patterns are provided by the signaling application and are described in this section.

The basic test procedure specified by 3GPP for both subtests is as follows:

1. Command the UE to the initial target power, depending on the carrier frequency.
2. Configure the initial RB allocation, depending on the duplex mode and the cell bandwidth.
3. Send the TPC pattern.
4. Measure the resulting power steps. In parallel, change the resource block allocation.

For subtest 1, the test has to be performed three times, with the ramping up TPC patterns A, B and C. All three patterns consist of 1-dB commands over 40 active uplink subframes. Within each pattern, there is one RB allocation change. The change depends on the cell bandwidth. The position of the change within the pattern differs for pattern A, B and C.

Subtest 2 is similar, with ramping down TPC patterns instead of ramping up patterns.

The following figure shows the ramping up TPC pattern A for FDD as an example. The RB allocation change is located after the first radio frame (10 subframes).

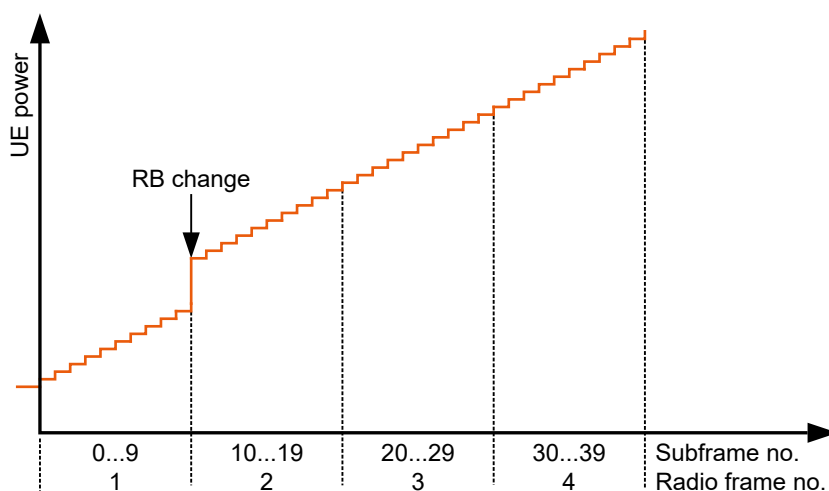


Figure 2-23: Ramping up pattern A for FDD

The 3GPP specification allows us to interrupt the power ramping. The interruptions must be whole numbers of radio frames without power change (0-dB commands). The R&S CMW inserts such interruptions to reconfigure the input path according to the changing expected nominal power.

Thus the measured UE power diagram with TPC pattern A for FDD looks typically as follows.

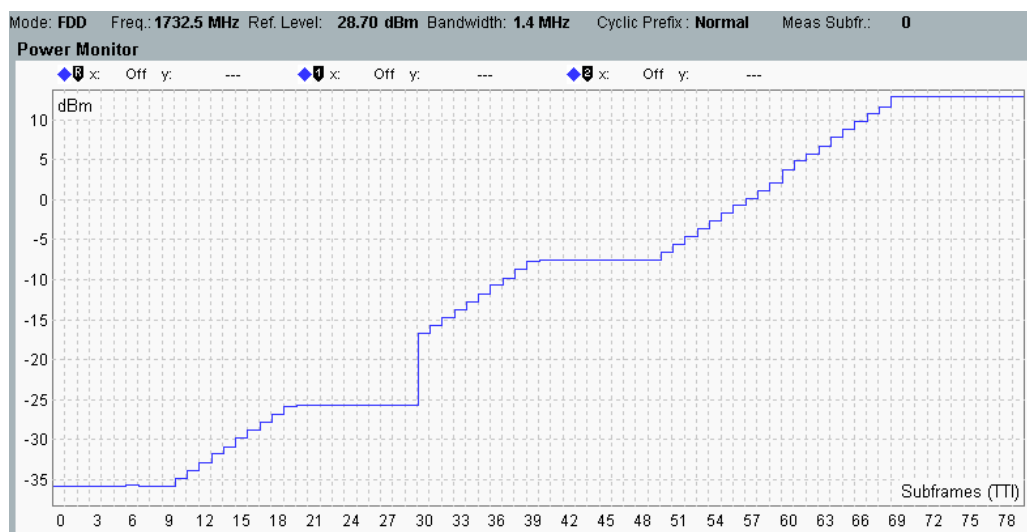


Figure 2-24: Power measurement with ramping up pattern A for FDD

Explanation of the diagram:

- Frame 1: constant initial target power
- Frame 2: ramping up
- Frame 3: constant power for input path configuration
- End of frame 3: change of RB allocation
In this example, the cell bandwidth is 1.4 MHz. Thus the allocation changes from 1 RB to 6 RB (8.78-dB power step expected).
- Frame 4: ramping up
- Frame 5: constant power for input path configuration
- Frame 6 and 7: ramping up

To perform a relative power control test

You need the LTE signaling application and the LTE measurement application (option R&S CMW-KM500/550, LTE FDD/TDD R8, TX measurement).

1. Set up an RMC connection with the desired duplex mode and cell bandwidth.
Note that the test procedure adapts the expected nominal power automatically. You do not need to set a specific value.
2. Switch to the LTE multi-evaluation measurement:
In the signaling application, press the "LTE TX Meas" softkey.
The measurement application is opened and the combined signal path scenario is selected.
3. In the measurement, select the TPC trigger signal provided by the signaling application:
Press the softkey "Trigger" followed by the hotkey "Trigger Source".
Select "LTE Sig<n>: TPC Trigger".
4. Ensure that the number of measured subframes is sufficient (e.g. 80):

Press the softkey "Multi Evaluation" followed by the hotkey "Measurement Subframes" and set "No. of Subframes".

5. Start the measurement. Ignore an indicated trigger timeout.
6. Configure and execute the TPC pattern:
 - a) Press the "Signaling Parameter" softkey followed by the "TPC" hotkey.
 - b) Set "Active TPC Setup" to "3GPP Relative Power Control".
 - c) Set "3GPP..." > "Pattern" to the desired ramping pattern (A, B or C up or down).
 - d) Press "Execute".

The TPC pattern is executed. The initial RB allocation and the RB change are configured automatically. The expected nominal power is also configured automatically. After completion of the TPC pattern, the RB allocation is reset to the allocation used before TPC pattern execution.

To check the measurement results, open the "Power Monitor" view.

7. To repeat the measurement with another pattern:
 - a) Select the pattern.
 - b) Press the "Execute" button again.

2.3 Application sheets

Application sheets describe short application examples for select issues and provide related background information. The following application sheets are related to the LTE signaling application.

- [Combined signal path measurements](#).....99
- [LTE IP-based data tests](#)..... 103
- [VoLTE call setup and audio tests](#)..... 107

2.3.1 Combined signal path measurements

This application sheet describes how to establish a connection to an LTE user equipment (UE) and perform TX measurements on the received uplink signal using the LTE multi-evaluation measurement. For SRS and PRACH measurements, see [Section 2.3.1.5, "SRS and PRACH measurements"](#), on page 103.



Sequencer tool R&S CMWrun

The automated test capabilities of R&S CMWrun make many measurement tasks easier. Option R&S CMW-KT055 provides configurable LTE test modules and test plans for R&S CMWrun.

2.3.1.1 Options and equipment required

An LTE combined signal path measurement requires the following equipment:

- R&S CMW with software version $\geq$ V1.0.15.20. The latest software version is recommended.
This application sheet describes software version V3.2.10.
- For FDD tests:
 - Option R&S CMW-KS500, LTE FDD R8, basic signaling
 - Option R&S CMW-KM500, LTE FDD R8, TX measurement
- For TDD tests:
 - Option R&S CMW-KS550, LTE TDD R8, basic signaling
 - Option R&S CMW-KM550, LTE TDD R8, TX measurement

The following sections describe a combined signal path measurement for FDD. A combined signal path measurement for TDD is performed in the same way, using the R&S CMW-Kx550 options instead of the R&S CMW-Kx500 options.

2.3.1.2 Setting up a connection

An established connection to the UE is a prerequisite for many signaling tests, including the combined signal path measurement described in this application sheet. The following example uses a single downlink stream. For MIMO tests, additional cabling and related configuration of routing settings are required.

To set up a connection, proceed as follows:

1. Reset your R&S CMW to ensure a definite instrument state.
2. Connect your UE to the RF 1 COM connector.
3. Open the LTE signaling application, e.g. from the task bar (press "TASKS" to open the task bar).
If the application is not present in the task bar, enable it in the "Generator/Signaling Controller" dialog (press "SIGNAL GEN" to open the dialog).
4. In the main view of the signaling application, adjust the RF settings to the capabilities of your UE.

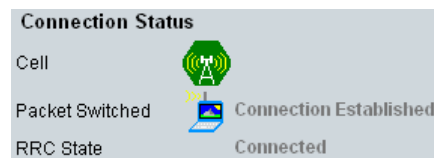
Configure especially the duplex mode (FDD / TDD), the band, the channel and the cell bandwidth. The "RS EPRE" must be sufficient so that the UE under test can receive the DL signal.

Operating Band	Band 1	FDD
	Downlink	Uplink
Channel	300 Ch	18300 Ch
Frequency	2140.0 MHz	1950.0 MHz
Cell Bandwidth	10.0 MHz	10.0 MHz
RS EPRE	-85.0 dBm/15kHz	
Full Cell BW Pow.	-57.2 dBm	

5. Press the "Config" hotkey to open the configuration dialog.
6. In section "RF Settings", select a bidirectional RF connector for input and output. This example uses RF 1 COM.
If necessary, also adjust the "External Attenuation" settings.
7. Close the configuration dialog.
8. To turn on the DL signal, press [ON | OFF]. And wait until the "LTE Signaling" soft-key indicates the "ON" state and the hour glass symbol has disappeared.
9. Switch on the UE.
The UE synchronizes to the DL signal and attaches. A default bearer is established.
Note the connection states displayed in the main view and wait until the attach procedure is complete.
By default, the RRC connection is kept after the attach is complete.



10. The default bearer and the established RRC connection are sufficient for many tests.
If you want to set up a dedicated bearer, press the "Connect" hotkey.
Note the connection states displayed in the main view and wait until the connection has been established.



Attach failure

If the attach procedure fails, check whether the UE capabilities are in accordance with the security settings in the "Network" section of the configuration dialog.

If your UE needs a DL signal without padding bits for attach, disable downlink padding in the "Connection" section of the configuration dialog.

You can check the power settings in the configuration dialog sections "Downlink Power Levels" and "Uplink Power Control".

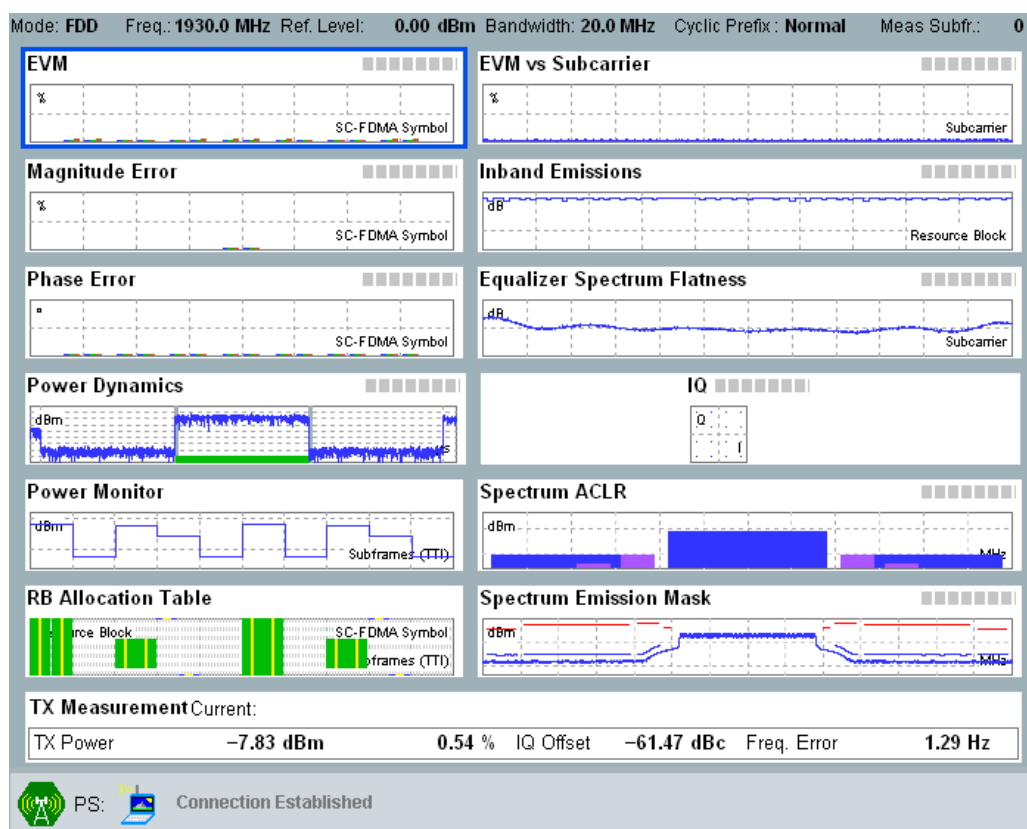
2.3.1.3 Analyzing the UL signal from the UE

While an established connection is available, the UL signals from the UE can be monitored using the LTE multi-evaluation measurement (included in option R&S CMW-KM500).

To ensure compatible measurement settings, the measurement must be coupled to the LTE signaling application. For this purpose, the measurement provides a combined signal path scenario. With this scenario, the measurement application uses the most important settings of the signaling application, for example the RF settings.

Proceed as follows:

1. Use the "LTE TX Meas" softkey to switch to the multi-evaluation measurement.
The measurement application is opened and the combined signal path scenario is selected automatically. Furthermore the frame trigger signal provided by the LTE signaling application is selected as trigger source.
2. Press [ON | OFF] to start the measurement.
The main view provides an overview of the measurement results.



To enlarge a diagram, perform one of the following actions:

- Double-click it using a connected mouse.
- Select it by turning the rotary knob and open it by pressing the rotary knob.

2.3.1.4 Possible extensions

While the connection is established, you can vary LTE signaling settings and observe the behavior of the UE under test using the LTE multi-evaluation measurement. You

can, for example, configure the signaling application to send transmit power control (TPC) commands to the UE and observe the resulting uplink power in the measurement dialogs. Or you can modify the channel configuration for the uplink and observe the result in the "RB Allocation Table".

2.3.1.5 SRS and PRACH measurements

In addition to the LTE multi-evaluation measurement, option R&S CMW-KM500 provides also an SRS measurement and a PRACH measurement.

The proceeding for a combined signal path SRS measurement is similar to the proceeding for a multi-evaluation measurement. Before switching on the downlink signal, you must enable SRS in section "Physical Cell Setup" of the signaling application. As a result, the UE will send an SRS signal after connection setup. The other steps are similar.

For PRACH measurements, you do not establish a connection. Configure the PRACH settings in section "Physical Cell Setup" of the signaling application. Activate "No Response to Preambles" and set "Power Ramping Step" to 0 dB.

When you have configured all required settings, switch on the downlink signal and the UE. The UE sends random access preambles which are not answered by the instrument. Use the PRACH measurement to measure these preambles.

2.3.2 LTE IP-based data tests

This application sheet provides examples for testing LTE data transfer over IP, using the LTE signaling application and the data application unit (DAU). It describes how to prepare the measurements, how to measure the network latency using ping and how to measure the downlink throughput for data transfer via UDP/IP.

2.3.2.1 Options and equipment required

The described tests require the following equipment:

- R&S CMW with software version $\geq$ V2.0.20. The latest software version is recommended.
This application sheet describes software version V3.2.60.
- One of the options:
 - R&S CMW-KS500, LTE FDD R8, basic signaling
 - R&S CMW-KS550, LTE TDD R8, basic signaling
- Option R&S CMW-KM050, data application measurements (requires hardware R&S CMW-B450x, data application unit)
- Option R&S CMW-KA100, enabling of IP-Data interface for IPv4

2.3.2.2 Preparing the tests

For test preparation, you need to configure the signaling application, set up a connection and configure the measurement.

Proceed as follows:

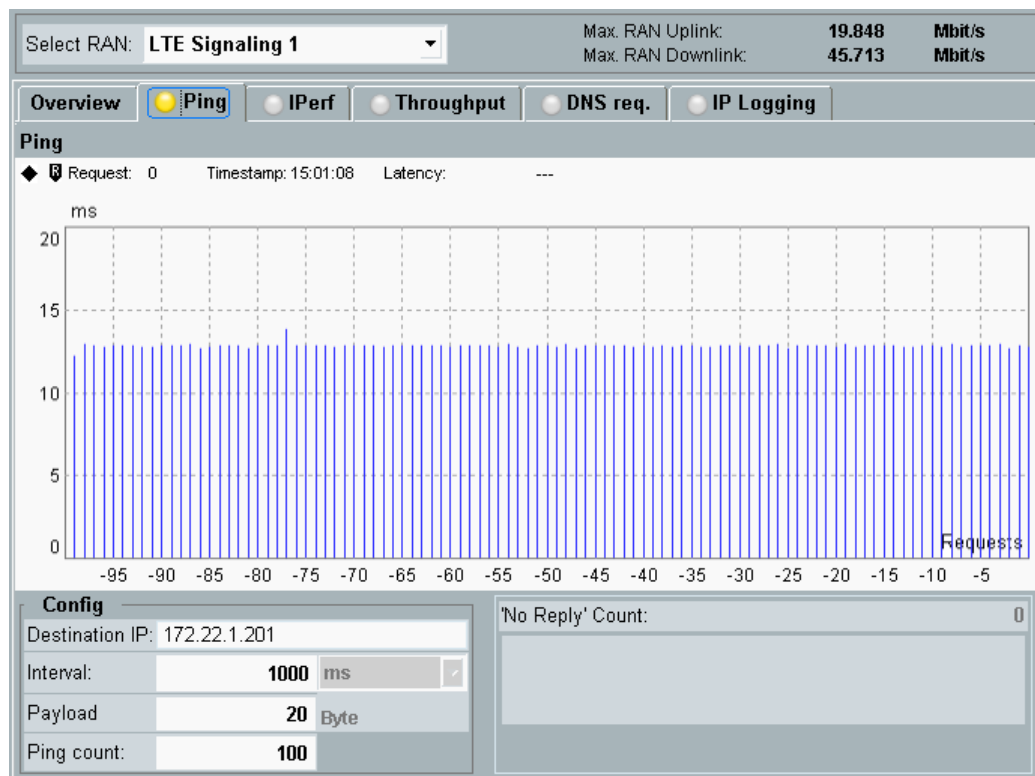
1. Set up a connection.
See [Section 2.3.1.2, "Setting up a connection"](#), on page 100.
For IP-based data tests, insert the following step after opening the configuration dialog:
 - a) In section "Connection", set parameter "Connection Type" to "Data Application".
This step enables the support of protocol stack layer 3 in the signaling application.
2. In the main view of the signaling application, note the IPv4 address assigned to the UE during attach. You need this information in the following sections.
If you have connected your UE to a PC with Windows operating system, you can also retrieve the IP address information via the PC. Open a command shell at the PC and type `IPconfig`.
3. Open the "Data Application Measurements", e.g. from the task bar (press "TASKS" to open the task bar).
If the application is not present in the task bar, enable it in the "Measurement Controller" dialog (press "MEASURE" to open the dialog).
4. At the top of the GUI, select the LTE signaling application (parameter "Select RAN").
The expected maximum throughput resulting from the current signaling settings is displayed on the right for information.

2.3.2.3 Measuring the latency with ping

After completed test preparation, you can ping your UE and evaluate the reported round-trip latency.

Proceed as follows:

1. Select the "Ping" tab.
2. For parameter "Destination IP", enter the IP address of the UE.
3. Press [ON | OFF].
The measurement starts and the "Ping" softkey indicates the "RUN" state.
The graph shows the measured round-trip latency for each executed ping request.

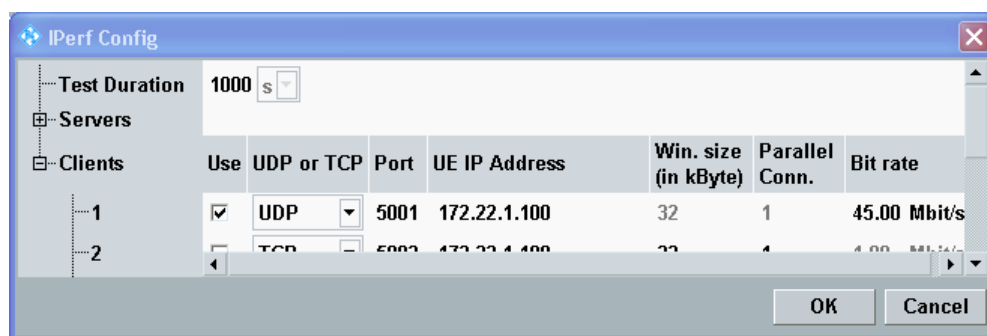


2.3.2.4 Measuring the throughput with iperf

The following steps describe how to measure the downlink throughput for data transfer via UDP/IP. As a prerequisite, the iperf tool must be installed on the UE, or on a PC connected to the UE. A compatible program version can be downloaded from the web pages provided by the DAU. Refer to the DAU documentation for details.

Proceed as follows:

1. Select the "IPerf" tab.
2. Press "Config..." to open the configuration dialog.
3. In section "Servers", disable all entries (column "Use").
4. In section "Clients" configure the first entry:
 - a) Enable the entry.
 - b) Select "UDP".
 - c) Note the port. You need it in a later step.
 - d) Enter the IP address of the UE (see [Preparing the tests](#)).
 - e) Configure a bit rate compatible to the expected maximum downlink bit rate displayed at the top of the iperf view as "Max. Downlink".



5. Press OK to close the configuration dialog.

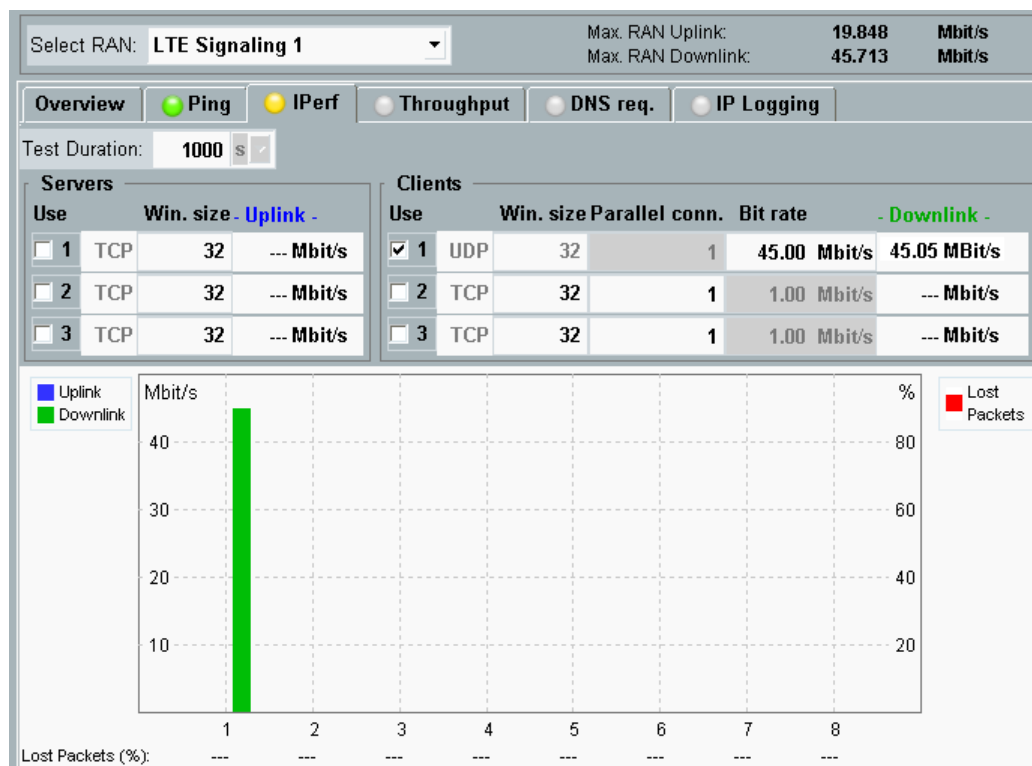
6. Press [ON | OFF].

The application starts and the "Iperf" softkey indicates the "RUN" state.

The R&S CMW sends data packets to the UE, using the downlink bit rate indicated in the last column and the graphic.

7. Configure the iperf tool at UE side so that it listens to the UDP port configured for the used client entry.

8. Start the iperf tool at UE side. Compare the bit rates received at the UE with the sent bit rates at the R&S CMW.



2.3.2.5 Possible extensions

While the connection is established, you can vary LTE signaling settings and "Data Application Measurements" settings and observe the impact on the measurement results. You can, for example, modify the downlink power levels, the downlink channel configuration and the sent bit rate. You can also set up a MIMO configuration (option required, see below) and perform LTE E2E data tests with MIMO.

The applications can be enhanced, for example, by the following options:

- Option R&S CMW-KS510 provides advanced parameter settings for LTE signaling
- Option R&S CMW-KS520 adds support of MIMO 2x2 in the downlink
- Option R&S CMW-KS521 adds support of MIMO 4x2 in the downlink
- Option R&S CMW-KS522 adds support of MIMO 8x2 in the downlink
- Option R&S CMW-KS540 adds support of MIMO 4x4 in the downlink
- Option R&S CMW-KS541 adds support of MIMO 8x4 in the downlink
- Option R&S CMW-KA150 adds support of IPv6
- Option R&S CMW-KAA20 provides an IMS server

2.3.3 VoLTE call setup and audio tests

This application sheet describes how to set up a voice over LTE (VoLTE) call, using the LTE signaling application and the IMS server of the data application unit (DAU).

For the established VoLTE call, basic audio tests are performed with the R&S CMW audio board. For speech quality tests with a connected R&S UPV, the R&S CMW configuration and the test setup are described.

2.3.3.1 Required equipment

The following equipment is required to set up a voice over LTE call and to perform audio tests with the R&S CMW:

- R&S CMW
- Hardware options:
 - Signaling unit, R&S CMW-B300x or R&S CMW-B500x
 - Audio board, R&S CMW-B400B
Speech codec, R&S CMW-B405A
 - Option carrier, R&S CMW-B660x
Ethernet switch, R&S CMW-B661x
Data application unit, R&S CMW-B450x
- RF cable, for connection of a UE antenna port to an RF COM port
- Audio cables between the UE microphone / speaker and the AF OUT / IN ports
Alternatives at UE side:
 - UE headset jack or UE microphone/speaker jacks
 - Microphone at UE speaker plus speaker at UE microphone

Use a high-quality microphone and speaker designed for that purpose, for example an artificial head.

- Software options:
 - LTE R8, SISO, basic signaling, R&S CMW-KS500 (FDD) or -KS550 (TDD)
 - IPv4 enabler, R&S CMW-KA100, for IPv6 also R&S CMW-KA150
 - IMS basic service, R&S CMW-KAA20
- Software and minimum software versions:
 - Base software $\geq$ V3.2.40
 - LTE signaling $\geq$ V3.2.70
 - Audio measurements $\geq$ V3.2.10
 - Data application software $\geq$ V3.2.30

The described step-by-step procedures apply to the base software V3.5.110, LTE V3.5.50, DAU V3.5.50, audio V3.5.20.

For a functional test of the audio transmission, a handset is required (option R&S CMW-Z50, only for [Section 2.3.3.3, "Checking the audio transmission"](#), on page 113).

To perform a speech quality analysis, the following additional equipment is required (only for [Section 2.3.3.5, "Testing the speech quality with an R&S UPV"](#), on page 116):

- Audio analyzer R&S UPV with relevant software options
- Audio cables for the following connections:
 - R&S UPV, generator output – UE, microphone input
 - UE, speaker output – R&S UPV, analyzer input
 - R&S UPV, generator output – R&S CMW, AF IN
 - R&S CMW, AF OUT – R&S UPV, analyzer input

For details, refer to the R&S UPV documentation.

2.3.3.2 Setting up a VoLTE call

The following sections describe in detail how to set up a voice over LTE call. We use the default scenario, a standard SISO LTE cell. And we use the default RF connector, RF 1 COM.

The following sections are based on each other. Execute them in the given order:

- Connecting the UE and initializing the R&S CMW
- Configuring and starting the IMS server of the R&S CMW
- Configuring and switching on the LTE cell
- Attaching the UE to the cell and registering it to the IMS server
- Initiating a voice over LTE call

Connecting the UE and initializing the R&S CMW

The instrument is already running. The UE is switched off.

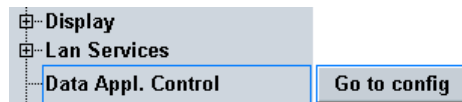
1. Connect your UE to the RF 1 COM port of the instrument.

2. Press [RESET] and perform a global preset.
3. Enable the applications:
 - a) Press [SIGNAL GEN].
The "Generator/Signaling Controller" dialog box opens.
 - b) Enable "LTE Signaling 1".
 - c) Press [MEASURE].
The "Measurement Controller" dialog box opens.
 - d) Enable "Audio Measurements 1".
 - e) Enable "Data Appl. Measurement 1".
If this entry is not available, you can skip this step. The presence of the entry depends on the presence of option R&S CMW-KM050.

The task bar at the bottom contains now hotkeys for access to the LTE signaling application, the audio measurements and - optionally - to the data application measurements. To show or hide the task bar, press [TASKS].

Configuring and starting the IMS server

1. Open the "Data Application Control" dialog box.
 - a) Alternative 1, with R&S CMW-KM050:
On the task bar, press "Data Meas 1". The "Data Application Measurement" view opens.
Press the "Configure Services" softkey.
 - b) Alternative 2, without R&S CMW-KM050:
Press [SETUP]. The "Setup" dialog box opens.
In the "System" section > "Data Appl. Control", press the "Go to config" button.



The "Data Application Control" dialog box opens.

2. Select the "IMS" tab.
3. Press the "Subscriber" hotkey.
A dialog box opens. Configure "Subscriber 1" as follows:
 - a) Enter the private user ID of your mobile.
 - b) Configure the authentication settings compatible to your mobile.

- c) Enter all public user IDs of your mobile.

- d) Press "Apply" and then "Close".

4. Press the "Virtual Subscriber" hotkey.

A dialog box opens. Configure "Virtual 1" as follows:

- Set the "Signaling Type" compatible to your mobile (call setup with or without SIP preconditions).
- Enable the "Dedicated Bearer" checkbox.
- Configure the "Audio" and "Video" codec settings compatible to your mobile.
- Set "Media Endpoint" to "Audioboard".

- e) Press "Apply" and then "Close".

5. Press [ON | OFF].

The IMS service is started. Wait until the "IMS" softkey displays the state "ON".



Configuring and switching on the LTE cell

- On the task bar, press "LTE Signaling 1".
The "LTE Signaling" view opens.
- Adjust the RF settings to the capabilities of your UE.

Configure especially the duplex mode (FDD / TDD), the band, the channel number and the cell bandwidth. The "RS EPRE" must be sufficient so that the UE can receive the DL signal.

Operating Band	Band 1	FDD
	Downlink	Uplink
Channel	300 Ch	18300 Ch
Frequency	2140.0 MHz	1950.0 MHz
Cell Bandwidth	10.0 MHz	10.0 MHz
RS EPRE	-85.0 dBm/15kHz	
Full Cell BW Pow.	-57.2 dBm	

3. Press the "Config" hotkey.
The LTE signaling configuration dialog box opens.
4. In section "Network" > "NAS Signaling":
 - a) Enable "EPS Network Feature Support".
 - b) Set "IMS Voice Over PS Sess. Ind." to "Supported".

Time	
NAS Signaling	
Send DNS PCO	☒
Attach Reject Cause	☐ PLMN not allowe
TAU Reject Cause	☐ PLMN not allowe
EPS Network Feature Support	☒
IMS Voice Over PS Sess. Ind.	Supported
Emergency Bearer Serv. Ind.	Not Supported
Loc. Service Indicator in EPC	Not Supported
Loc. Service Indicator in CS	Not Supported
Synchronization	

5. Configure the remaining "Network" settings compatible to your UE, especially the sections "NAS Signaling", "Identity" and "Security Settings".
6. In section "Connection", set "Connection Type" to "Data Application".
7. Disable "Use 'Activate Testmode' Message"

Connection Type	Data Application
Testmode	
Use "Activate Testmode" Message	☐

8. Close the dialog box.
9. Press [ON | OFF].
The LTE cell signal is switched on. Note the displayed connection states and wait until the process is complete.

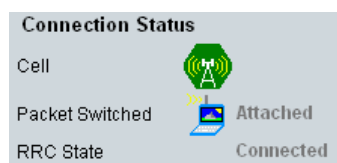


Attaching and registering the UE

1. Switch on the UE.

The UE synchronizes to the LTE cell signal and attaches. A default bearer is established.

Note the displayed connection states and wait until the attach procedure is complete and the RRC connection is established.

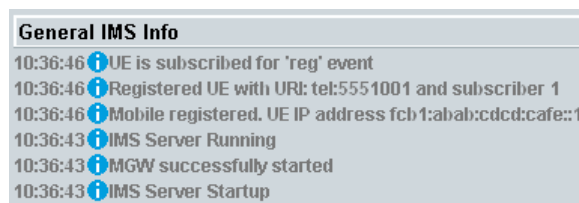


2. Open the "Data Application Control" dialog box.
 - a) Task bar > "Data Meas 1" > "Configure Services" softkey, or
 - b) [SETUP] > "Go to config" button

The "Data Application Control" dialog box opens.

3. After attaching to the LTE cell, the UE registers to the IMS server.

Check the successful registration in the "General IMS Info" area on the IMS tab.



Initiating a mobile-originating VoLTE call

Perform the following steps or set up a mobile-terminating call, see ["Initiating a mobile-terminating VoLTE call"](#) on page 113.

- Dial an arbitrary number (public user ID) at the UE.

The call is accepted by virtual subscriber 1 of the IMS server.

A new line is added to the "Events" area. After call setup completion, the status in that line equals "Established".

Events				
Timestamps	Source	Destination	Event	Status
16:09:03	S1	VS1		Established

In the LTE signaling application, a dedicated bearer has been set up.

Initiating a mobile-terminating VoLTE call

Perform the following steps or set up a mobile-originating call, see ["Initiating a mobile-originating VoLTE call"](#) on page 112.

1. Press the "Virtual Subscriber" hotkey.
A dialog box opens.
2. Press the  button.
The mobile-terminating call settings are displayed.
3. Configure the following call settings:
 - a) Select the "Destination" that you want to call.
 - b) Select the "Call Type".

Calling	
Destination	[1] LTE1 - subscriber 1 ▾
Call Type	Audio ▾

All other call settings are copies of already configured settings.

4. Press the "Call" button.
The dialog box is closed.
The mobile is alerted.
A new line is added to the "Events" area. The status in that line equals "Ringing".
5. Accept the call at the mobile.
The status in the "Events" area changes to "Established".

Events				
Timestamps	Source	Destination	Event	Status
16:09:03	VS1	S1		Established

In the LTE signaling application, a dedicated bearer has been set up.

2.3.3.3 Checking the audio transmission

This section describes a simple functional check of the audio transmission in uplink and downlink direction. You need the handset R&S CMW-Z50 for this check.

1. Connect the handset to AF 1 IN and AF 1 OUT.
2. Open the "Audio Measurement 1" view:
On the task bar, press "Audio Measurement 1".
3. At the top, select the scenario "External Analog Speech Analysis".
4. For "Controlled by", select "DAU IMS Server".
5. Press the "Input Level" hotkey. Enter 0.05 V.
6. Press the "Output Level" hotkey. Enter 1 V.
7. To check the downlink direction:

- a) Speak into the handset microphone.
- b) Listen to the UE speaker.

The speech must be audible at the UE speaker after a small delay.

- 8. To check the uplink direction:

- a) Speak into the UE microphone.
- b) Listen to the handset speaker.

The speech must be audible at the handset speaker after a small delay.

- 9. Disconnect the handset.

2.3.3.4 Performing basic audio tests

This section describes how to perform audio tests for an established VoLTE connection. As audio test signal, a single tone is used. The tests analyze mainly the harmonic distortions and noise of an audio signal.

The tests are performed in two steps. The first step is a speaker test, the second step a microphone test. You can perform speaker- and microphone tests with a stand-alone R&S CMW. No additional test instruments are required.

Performing a speaker test

The digital generator of the audio board feeds a 1000-Hz audio tone to the speech encoder. The tone is transmitted to the UE via the already established VoLTE connection. The UE demodulates and decodes the signal and feeds the resulting audio signal to its speaker.

The analog measurement of the audio board analyzes the audio signal of the speaker.

If the UE has a speaker jack, headphones jack or headset jack, connect it directly to the AF 1 IN port. Alternatively, you can use a microphone and connect it to AF 1 IN. You could, for example, use the microphone of an artificial head, designed for this purpose.

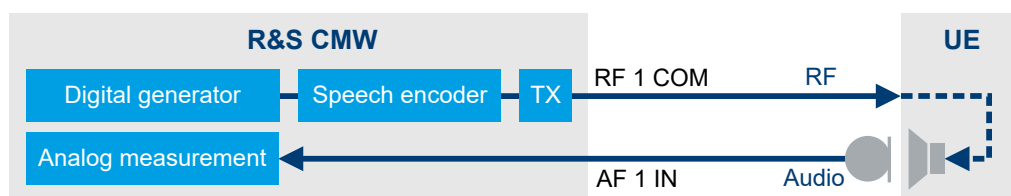


Figure 2-25: Test setup for a speaker test

To prepare and execute the speaker test, proceed as follows:

1. Open the "Audio Measurement 1" view:
On the task bar, press "Audio Measurement 1".
2. At the top, select the scenario "Microphone- and Speakertest".
3. Connect the UE speaker to AF 1 IN.

4. Select the "Digital Generator" softkey and press [ON | OFF].
The generator is started.
5. Wait until the softkey indicates the "ON" state.
A 1000-Hz tone is generated, fed to the speech encoder and transmitted to the UE via the VoLTE connection.
6. Select the "Analog Meas" softkey and press [ON | OFF].
The measurement is started.
7. Wait until the softkey indicates the "RUN" state.
8. Evaluate the measurement results, for example the total harmonic distortion (THD) and the signal to noise ratio (SNR) of the audio signal.

Analog Measurement				
	CUR	AVG	EXT	STD
THD [%]	0.1121	0.1095	0.1371	0.0037
THD [dB]	-59.01	-59.21	-57.26	0.28
THD + N [%]	0.4402	0.4465	0.4639	0.0051
SINAD [dB]	47.13	47.00	46.67	0.10
SNR [dB]	47.42	47.27	46.97	0.11
DC Level [V]	0.000	0.000	0.000	0.000
Frequency [Hz]	1000.0	1003.3	1095.5	12.3
Weighted Level, RMS [V]	0.0872	0.0873	0.0880	0.0002
Bandpass Level, RMS [V]	0.0872	0.0873	0.0880	0.0002

9. Stop the generator and the measurement:
 - a) Select the "Analog Meas" softkey and press [ON | OFF].
 - b) Select the "Digital Generator" softkey and press [ON | OFF].

Performing a microphone test

In the following test, the analog generator of the audio board feeds a 1000-Hz audio tone to the microphone of the UE.

The UE encodes the audio signal and modulates the RF signal. It transmits the RF signal via the already established VoLTE connection to the R&S CMW.

The R&S CMW demodulates and decodes the signal. The digital audio measurement of the audio board analyzes the resulting audio signal.

If the UE has a microphone jack or a headset jack, connect it directly to the AF 1 OUT port. Alternatively, you can use a speaker and connect it to AF 1 OUT. You could, for example, use the speaker of an artificial head.

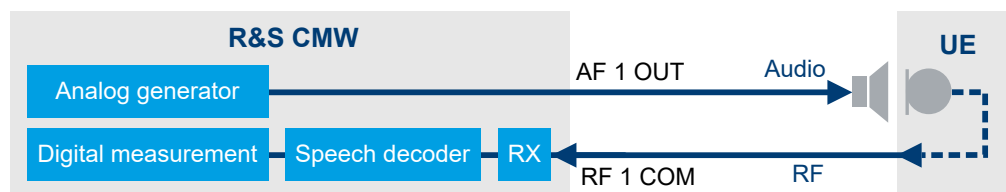


Figure 2-26: Test setup for a microphone test

To prepare and execute the microphone test, proceed as follows:

1. Connect the UE microphone to AF 1 OUT.
2. Select the "Digital Meas" tab.
3. Press the "Analog Generator" softkey.
4. Press the "Level" hotkey. Enter a value compatible to the microphone input of your UE, for example 100 mV.
5. Press [ON | OFF].
The generator is started.
6. Wait until the softkey indicates the "ON" state.
A 1000-Hz tone is generated and fed to the UE microphone.
7. Select the "Digital Meas" softkey and press [ON | OFF].
The measurement is started.
8. Wait until the softkey indicates the "RUN" state.
9. Evaluate the measurement results, for example the total harmonic distortion (THD) and the signal to noise ratio (SNR) of the audio signal.

Digital Measurement				
	CUR	AVG	EXT	STD
THD [%]	0.2841	0.2956	25.6570	0.0588
THD [dB]	-50.92	-50.75	-11.82	1.76
THD + N [%]	2.5354	2.7075	24.9839	0.1183
SINAD [dB]	31.92	31.36	12.05	0.38
SNR [dB]	31.66	31.08	11.42	0.39
Weighted Level, RMS [FS]	0.0610	0.0595	1.0000	0.0024
Bandpass Level, RMS [FS]	0.0602	0.0586	0.9887	0.0025

10. Stop the generator and the measurement:
 - a) Select the "Analog Meas" softkey and press [ON | OFF].
 - b) Select the "Digital Generator" softkey and press [ON | OFF].

2.3.3.5 Testing the speech quality with an R&S UPV

To measure the speech quality of a VoLTE connection, add an audio analyzer to your test setup. The following example uses an R&S UPV.

The test setup in the following figure is similar to the combination of the microphone test setup and the speaker test setup. But the audio generators and measurements are now located in the R&S UPV, not in the R&S CMW. The RF connection / VoLTE connection is the same as for the preceding tests.

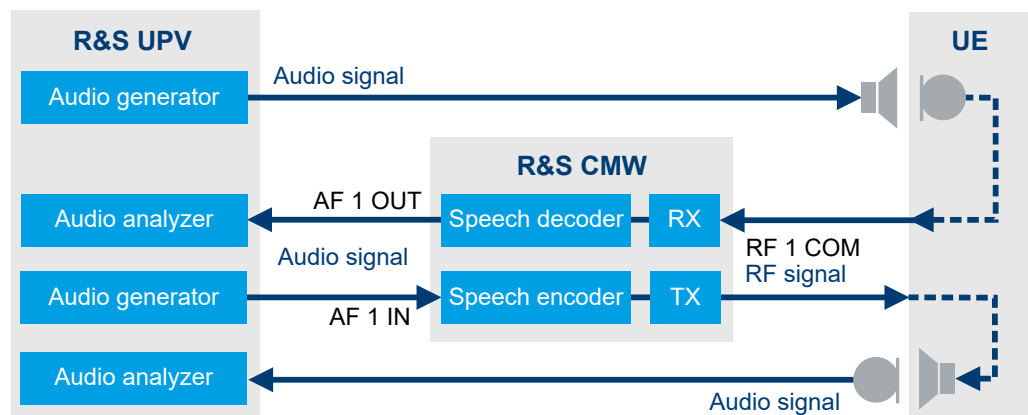


Figure 2-27: Test setup for external speech analysis

A generator of the R&S UPV feeds an audio signal to the microphone of the UE. The UE encodes the audio signal and modulates the RF uplink signal. It transmits the RF signal via the VoLTE connection to the R&S CMW. The R&S CMW demodulates and decodes the signal. It routes the resulting audio signal via the AF 1 OUT port to an audio analyzer of the R&S UPV.

A generator of the R&S UPV feeds an audio signal via the AF 1 IN port to the speech encoder of the audio board. The test signal is transmitted via the VoLTE connection to the UE. The UE demodulates and decodes the signal and feeds the resulting audio signal to its speaker. An analyzer of the R&S UPV measures the audio signal of the speaker.

If the UE has microphone and speaker jacks or a headset jack, connect the UE via an audio cable to the R&S UPV. Alternatively, you can position a microphone at the UE speaker and a speaker at the UE microphone and connect them to the R&S UPV. Use a high-quality microphone and speaker designed for that purpose, for example an artificial head.

Performing a speech quality test

The following sequence describes the configuration steps required at the R&S CMW. The R&S UPV configuration is out of the scope of this document.

1. Set up the VoLTE connection as described in the previous sections.
2. Set up the two audio connections between the R&S UPV and the UE and the two audio connections between the R&S UPV and the R&S CMW.
3. Configure the audio settings of the R&S CMW:
 - a) At the top of the "Audio Measurement 1" view, select the scenario "External Analog Speech Analysis".
 - b) Press the "Config" hotkey to open the audio configuration dialog box.
 - c) Configure the input level compatible to the R&S UPV generator output.
 - d) Configure the output level compatible to the R&S UPV analyzer input.
Check whether a calibration procedure of the R&S UPV requires specific level settings.

4. Configure the R&S UPV. Calibrate it, if necessary.
5. Start the R&S UPV generators and analyzers.

For more information about speech quality tests with an R&S UPV, refer to the following documents:

- Application note "1MA204", "Voice over LTE (VoLTE) Speech Quality Measurements"
- Application note "1GA62", "Test Automation Tool for POLQA® and PESQ® Speech Quality Tests"

These documents are available for download at <http://www.rohde-schwarz.com>.

The R&S CMWrun provides test modules for automated speech quality tests with the R&S CMW and the R&S UPV.

2.3.3.6 Possible extensions

While you perform audio tests or speech quality tests, you can vary connection parameters and study the effect on the audio quality.

With option R&S CMW-KM050, you can use the data application unit to add network impairments to the downlink VoLTE connection. Thus you can simulate challenges of real IP networks and test the effect on the speech quality.

Configurable network impairments are for example:

- Packet delay, static or with jitter distribution
- Packet loss and packet corruption
- Reordering and duplication of packets

2.4 GUI reference

The following sections provide detailed reference information on the graphical user interface (GUI) of the LTE signaling application.

Many of the signaling settings are available in a subset of connection states only. Temporarily unavailable parameters are grayed out in the configuration dialogs. Hotkeys appear and disappear dynamically, depending on the connection state.



The screenshots in this section show the GUI with all available options installed. Depending on the installed options, some parameters are not configurable (display the default value) or are not visible at all. The required options are stated.

The GUI reference is structured as follows.

- [Main view](#)..... 119
- [Signaling control](#).....130
- [Using the shortcut softkeys](#)..... 137
- [General settings](#)..... 138

• RF settings.....	144
• Internal fading.....	152
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• TTI-based channel configuration.....	265
• CQI reporting.....	278
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• Messaging (SMS) parameters.....	284
• Messaging (CBS) parameters.....	289
• Shortcut configuration.....	293
• Message monitoring settings.....	294
• Debug configuration.....	294
• Annex: UE capabilities.....	296

2.4.1 Main view

The main view of the signaling application shows status information on the left and the most important settings on the right. All settings in this view can also be accessed via the configuration dialog.

Figure 2-28: LTE signaling main view

For the hotkeys of the "LTE Signaling" softkey, see [Section 2.4.2, "Signaling control"](#), on page 130.

For the shortcut softkeys, see [Section 2.4.3, "Using the shortcut softkeys"](#), on page 137.

For descriptions of the individual areas of the main view, refer to the following sections.

• Connection status	120
• Event log	121
• UE measurement report	121
• UE capabilities	124
• UE info	125
• Settings	127

2.4.1.1 Connection status

Access: LTE signaling main view > "Connection Status" area.

The area displays the following information.

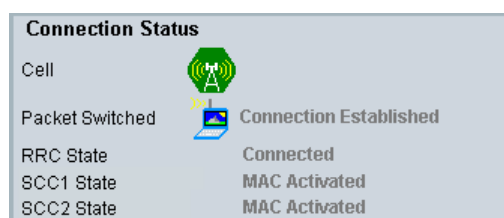


Figure 2-29: Connection status area of the main view

For background information, see [Section 2.2.9, "Connection states"](#), on page 38.

For control of the states, see [Section 2.4.2, "Signaling control"](#), on page 130.

Cell

Displays the current state (or state transition) of the downlink signal generator.

Remote command:

`SOURce:LTE:SIGN<i>:CELL:STATe`

Packet Switched

Displays the current state or state transition of the packet-switched network domain.

Remote command:

`FETCh:LTE:SIGN<i>:PSWitched:STATe?`

RRC State

Indicates whether a PCC RRC connection is established ("Connected") or not ("Idle").

Remote command:

`SENSe:LTE:SIGN<i>:RRCState?`

SCC<n> State

Displays the current state of the secondary component carrier (SCC) number <n>. Only visible for carrier aggregation scenarios.

Remote command:

```
FETCh:LTE:SIGN<i>:SCC<c>:STATe?
```

2.4.1.2 Event log

Access: LTE signaling main view > "Event Log" area.

The area reports events and errors like PS connection state changes, RRC connection establishment/release, SCC state changes and authentication failure.

The button on the right clears the displayed event entries.



Figure 2-30: Event log area of the main view

Event log entries

Each entry consists of a timestamp, an icon indicating the category of the event and a short text describing the event.

Meaning of the category icons: [i] [w] [e] = information, warning and error

Remote command:

```
SENSe:LTE:SIGN<i>:ELOG:LAST?
```

```
SENSe:LTE:SIGN<i>:ELOG:ALL?
```

```
CLEAn:LTE:SIGN<i>:ELOG
```

2.4.1.3 UE measurement report

Access: LTE signaling main view > below the "Event Log" area, select "UE Measurement Report".

The displayed information is retrieved from measurement reports provided by the connected UE. The individual report values are defined in 3GPP TS 36.133.

To enable/disable measurement reports, use the checkbox.

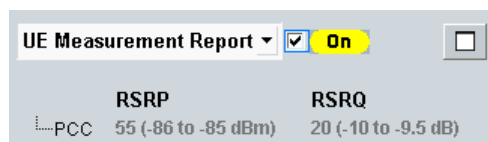


Figure 2-31: Measurement report in the main view

The minimized default presentation shows all measurement report values for the serving LTE cell.

To show also neighbor cell measurement results, press the button on the right. It maximizes the area vertically.

To maximize the area also horizontally, press the button again.

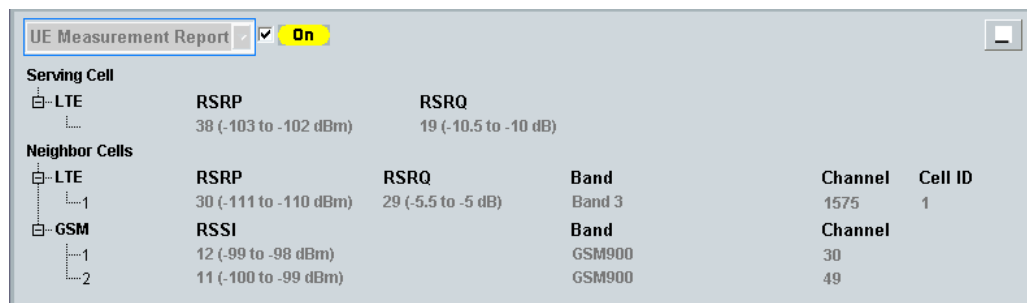


Figure 2-32: Maximized report area

Neighbor cell measurements are by default disabled and can be enabled separately for each neighbor cell, see [Section 2.4.10.1, "Neighbor cell settings"](#), on page 179. Option R&S CMW-KS510 is required.

The measurement results displayed in the maximized report area are described in the following. Configured neighbor cell settings are also displayed (band, channel, ...).

RSRP for serving LTE cell.....	122
RSRQ for serving LTE cell.....	123
RSRP and RSRQ for LTE neighbor cells.....	123
RSSI for GSM.....	123
RSCP and EcNO for WCDMA.....	123
pilot Pn Phase and pilot Strength for CDMA2000 and 1xEV-DO.....	123
RSCP for TD-SCDMA.....	124
RSSI Measurements for LAA.....	124
Channel Occupancy for LAA.....	124

RSRP for serving LTE cell

The reference signal received power (RSRP) denotes the average power of the resource elements carrying cell-specific reference signals.

The measurement report displays the reported dimensionless value and in brackets the corresponding measured value range. For carrier aggregation scenarios, the information is displayed per carrier.

Remote command:

```
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRP?
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRP:RANGE?
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCEL1?
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCEL1:RANGE?
```

Plus corresponding PCC commands.

RSRQ for serving LTE cell

The reference signal received quality (RSRQ) is calculated as $RSRQ = N \times RSRP / (E-UTRA \text{ carrier RSSI})$. N is the number of resource blocks in the measurement bandwidth. The "E-UTRA carrier RSSI" denotes the average of the total received power (including interferers, and so on), observed in OFDM symbols containing reference symbols for antenna port 0.

The measurement report displays the reported dimensionless value and in brackets the corresponding measured value range. For carrier aggregation scenarios, the information is displayed per carrier.

Remote command:

```
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRQ?
```

```
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRQ:RANGe?
```

```
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCELL?
```

```
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCELL:RANGe?
```

Plus corresponding PCC commands.

RSRP and RSRQ for LTE neighbor cells

The RSRP and RSRQ reported for LTE neighbor cells are displayed in the same way as the values for the serving cell.

Remote command:

```
SENSe:LTE:SIGN<i>:UEReport:NCELL:LTE:CELL<no>?
```

```
SENSe:LTE:SIGN<i>:UEReport:NCELL:LTE:CELL<no>:RANGe?
```

RSSI for GSM

The received signal strength indicator (RSSI) denotes the received wideband power within the GSM channel bandwidth, measured on a GSM BCCH carrier.

The measurement report displays the reported dimensionless value and in brackets the corresponding measured value range.

Remote command:

```
SENSe:LTE:SIGN<i>:UEReport:NCELL:GSM:CELL<no>?
```

```
SENSe:LTE:SIGN<i>:UEReport:NCELL:GSM:CELL<no>:RANGe?
```

RSCP and EcNO for WCDMA

The received signal code power (RSCP) denotes the received power on one code, measured on the primary CPICH of the neighbor WCDMA cell.

The Ec/No denotes the ratio of the received energy per PN chip for the primary CPICH to the total received power spectral density in the WCDMA band.

The measurement report displays the reported dimensionless values and in brackets the corresponding measured value range.

Remote command:

```
SENSe:LTE:SIGN<i>:UEReport:NCELL:WCDMa:CELL<no>?
```

```
SENSe:LTE:SIGN<i>:UEReport:NCELL:WCDMa:CELL<no>:RANGe?
```

pilot Pn Phase and pilot Strength for CDMA2000 and 1xEV-DO

The pilot PN phase indicates the arrival time of a pilot, measured relative to the time reference of the UE in units of PN chips. For details refer to 3GPP2 C.S0005, section 2.6.6.2.4.

The pilot strength denotes the ratio of the received pilot energy per chip to the total received power spectral density in the signal bandwidth of the forward channel.

For reporting, the ratio is converted into a dB value, multiplied with -2 and truncated, so that a positive integer value is reported (0 to 63):

*Reported pilot strength value = INT (-2 * 10 log₁₀(pilot energy / total power))*

For more details refer to 3GPP2 C.S0005, section 2.6.6.2.2 and section 2.7.2.3.2.5.

Remote command:

`SENSe:LTE:SIGN<i>:UEReport:NCELL:CDMA:CELL<no>?`

`SENSe:LTE:SIGN<i>:UEReport:NCELL:EVDO:CELL<no>?`

RSCP for TD-SCDMA

The received signal code power (RSCP) denotes the received power, measured on the P-CCPCH of the neighbor TD-SCDMA cell.

The measurement report displays the reported dimensionless values and in brackets the corresponding measured value range.

Remote command:

`SENSe:LTE:SIGN<i>:UEReport:NCELL:TDSCdma:CELL<no>?`

`SENSe:LTE:SIGN<i>:UEReport:NCELL:TDSCdma:CELL<no>:RANGE?`

RSSI Measurements for LAA

The received signal strength indicator (RSSI) denotes the received power within an SCC DL channel. The measurement is only relevant for frame structure type 3 (LAA).

The measurement report displays the reported dimensionless value. For the mapping to dBm values, see 3GPP TS 36.133, table 9.1.18.5.1-1.

Remote command:

`SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RRESult?`

Channel Occupancy for LAA

The channel occupancy is the percentage of samples, where the measured RSSI value is greater than the configured threshold, see "[Channel Occupancy Threshold](#)" on page 284.

The measurement is only relevant for frame structure type 3 (LAA).

Remote command:

`SENSe:LTE:SIGN<i>:UEReport:SCC<c>:COCC?`

2.4.1.4 UE capabilities

Access: LTE signaling main view > below the "Event Log" area, select "UE Capabilities".



Figure 2-33: UE capabilities in the main view

The displayed information comprises the following extracts from the UE capability report:

- General UE capability information
- RF UE capabilities

To maximize the UE capabilities area and display all capability information, press the button on the right.

The UE capabilities characterize the radio access capabilities of the UE. This information is received from the UE during registration. The radio access capabilities are specified in 3GPP TS 36.331.

For a description of all capability information, see [Section 2.4.21, "Annex: UE capabilities"](#), on page 296.

2.4.1.5 UE info

Access: LTE signaling main view > below the "Event Log" area, select "UE Info".

The area displays UE-related information for an attached UE.

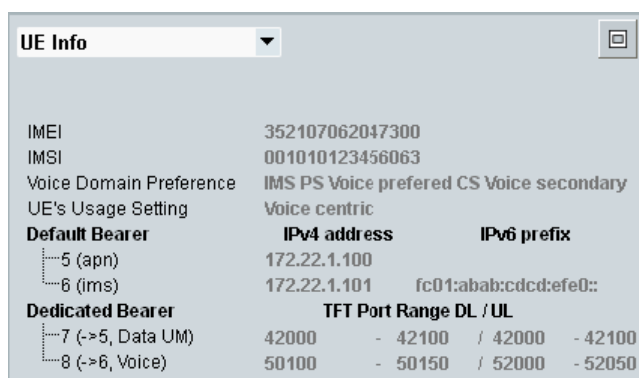


Figure 2-34: UE info area of the main view

IMEI	126
IMSI	126
Voice Domain Preference	126
UE's Usage Setting	126
Default Bearer	126
Dedicated Bearer	127

IMEI

International mobile equipment identity (IMEI) received from the UE. The IMEI is only known if NAS security is enabled.

Remote command:

`SENSe:LTE:SIGN<i>:UESinfo:IMEI?`

IMSI

International mobile subscriber identity (IMSI) received from the UE.

Remote command:

`SENSe:LTE:SIGN<i>:UESinfo:IMSI?`

Voice Domain Preference

The voice domain preference received from the UE indicates whether the UE uses the CS domain or the IMS for voice calls.

CS domain means that a circuit-switched fallback is performed and the voice call is established via GERAN or UTRAN.

IMS means that the voice call is established as voice over IMS call, via E-UTRAN.

The following preference values are possible:

- "CS voice only"
- "IMS PS voice only"
- "CS voice preferred, IMS PS voice as secondary"
- "IMS PS voice preferred, CS voice as secondary"

Remote command:

`SENSe:LTE:SIGN<i>:UESinfo:VDPReference?`

UE's Usage Setting

The usage setting received from the UE determines the behavior of the UE, if voice services are not possible in the current E-UTRAN cell.

Example: The UE has the voice domain preference "IMS PS Voice only", but the cell does not offer IMS services.

"Voice centric" The UE leaves the cell to ensure the support of voice services. It disables the E-UTRAN capability and performs a reselection to GERAN or UTRAN.

"Data centric" The UE stays in the cell even if voice services are not possible.

Remote command:

`SENSe:LTE:SIGN<i>:UESinfo:UEUSage?`

Default Bearer

List of all established default bearers, one line per bearer.

Each line contains the following information:

- Default bearer ID, composed as follows: *<EPS default bearer ID> (<APN>)*
Example: "5 (cmw500.rohde.schwarz.com)" means default bearer 5, with access point name "cmw500.rohde.schwarz.com"
- IPv4 address assigned to the UE for this bearer
- IPv6 prefix assigned to the UE for this bearer

Remote command:

```
SENSe:LTE:SIGN<i>:UESInfo:UEADdress:IPV<n>?
```

Dedicated Bearer

List of all established dedicated bearers, one line per bearer.

Each line contains the following information:

- Dedicated bearer ID, composed as follows:
 <EPS dedicated bearer ID> (-> <EPS default bearer ID>, <profile>)
 Example: "6 (->5, Voice)" means dedicated bearer 6, mapped to default bearer 5, using dedicated bearer profile "Voice"
 For calls set up by the IMS, the <profile> indicates the call type ("IMS Voice" or "IMS Video") and the initial digits of the call ID.
- TFT port range assigned to the dedicated bearer downlink / uplink
 This information is not displayed for calls setup by the IMS.

Remote command:

```
SENSe:LTE:SIGN<i>:UESInfo:UEADdress:DEDBearer?
```

```
SENSe:LTE:SIGN<i>:UESInfo:UEADdress:DEDBearer:SEParate?
```

2.4.1.6 Settings

Access: LTE signaling main view > tabs on the right.

The main view provides the most important settings for fast access.

For carrier aggregation scenarios, the settings for the individual component carriers are provided on separate tabs.

Figure 2-35: Settings in the main view, PCC and SCC1

Upper Part.....	128
L Swap button.....	128
Lower Part.....	129
L Copy button.....	129

Upper Part

The upper part contains the following settings:

- Duplex mode
See [Section 2.4.4, "General settings"](#), on page 138
- RF frequency settings
See [Section 2.4.5.2, "RF frequency"](#), on page 147
- Cell bandwidths
See [Section 2.4.9, "Physical cell setup"](#), on page 170
- Most important DL power parameters (RS EPRE, full cell BW power)
See [Section 2.4.7, "Downlink power levels"](#), on page 157
- Most important UL power parameters
See [Section 2.4.8, "Uplink power control"](#), on page 161
- Intra-band contiguous SCC configuration
See ["Intraband Contiguous UL"](#) on page 143

Swap button ← Upper Part

The "Swap" button is only available on SCC tabs. It exchanges PCC settings with the settings of one SCC.

The following settings are exchanged:

- Operating band, channel number and frequency

- Cell bandwidth
- Active scheduling with all related settings (RB configuration, modulation etc.)
Only the active scheduling settings for the active cell bandwidth are written to the other carrier. The settings related to inactive scheduling types or inactive cell bandwidths are not swapped.

Example 1 – operating band and channel number, with SCC UL:

- Before swap
 - PCC: band 3, channel DL 1500 / UL 19500
 - SCC1: band 20, channel DL 6300 / UL 24300
- After swap
 - PCC: band 20, channel DL 6300 / UL 24300
 - SCC1: band 3, channel DL 1500 / UL 19500

Example 2 – cell BW, scheduling type and #RB, no SCC UL:

- Before swap
 - PCC: 10 MHz, RMC, DL 50 RB, UL 40 RB
 - SCC1: 5 MHz, user-defined channel, 22 RB
 - PCC inactive settings for 5 MHz, user-defined channel: DL 15 RB, UL 18 RB
- After swap
 - PCC: 5 MHz, user-defined channel, DL 22 RB, UL 18 RB
 - SCC1: 10 MHz, RMC, 50 RB
 - PCC inactive settings for 10 MHz, RMC: DL 50 RB, UL 40 RB
 - SCC1 inactive settings for 5 MHz, user-defined channel: 22 RB

If you perform a swap for an established connection, all connection states after the swap equal the states before the swap. Either redirection or blind handover is used for the swap, see ["Operating Band Change, Frequency Change"](#) on page 228.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SEXecute
```

Lower Part

The contents of the lower part depend on the scheduling type. Refer to:

- [Section 2.4.12.11, "RMC settings \(no eMTC\)"](#), on page 240
- [Section 2.4.12.12, "RMC settings for eMTC"](#), on page 243
- [Section 2.4.12.13, "User-defined channels \(no LAA, no eMTC\)"](#), on page 246
- [Section 2.4.12.14, "User-defined channels for eMTC"](#), on page 249
- [Section 2.4.12.17, "User-defined TTI-based channels"](#), on page 258
- [Section 2.4.12.18, "CQI channel configuration"](#), on page 259
- [Section 2.4.12.19, "SPS configuration"](#), on page 260
- [Section 2.4.12.20, "eMTC auto mode settings"](#), on page 262
- [Section 2.4.12.21, "eMTC compact scheduling settings"](#), on page 263

If a transmission scheme using several data streams is active, the parameter "MIMO DL Stream" allows you to switch between the settings of the downlink streams. The parameter is disabled if all streams use the same set of settings.

Copy button ← Lower Part

The "Copy" button is only available on SCC tabs. It copies PCC settings to an SCC.

The following settings are copied:

- Cell bandwidth
- Active scheduling with all related settings (RB configuration, modulation etc.)

The copy operation is only possible, if the PCC settings and the SCC settings are compatible. Examples for incompatibilities:

- The active scenario is not symmetric for the carriers. Example: The scenario allows MIMO for the PCC but not for the SCC.
- The PCC scheduling type is not supported for the SCC (for example SPS).
- The resource allocation type is two for the PCC and zero for the SCC.
- The PCC scheduling type equals RMC. The PCC uses transmission mode 1 and the SCC uses another transmission mode (or vice versa).

Remote command:

CONFigure:LTE:SIGN<i>:CONNectio:n:SCC<c>:CEXecute

2.4.2 Signaling control

The connection states are controlled via the [ON | OFF] key, via hotkeys and via the UE.

The available hotkeys depend on the current connection state. All possible hotkeys of the "LTE Signaling" softkey are described in the following.

For background information, see [Section 2.2.9, "Connection states"](#), on page 38.

[ON OFF] key / LTE Signaling softkey.....	130
Connection control hotkeys.....	131
Multiple SCC Actions hotkey.....	132
Connect hotkey.....	133
L Connect button.....	133
L Disconnect button.....	134
Inter/Intra-RAT hotkey.....	134
L Target.....	135
L Mobility Mode.....	136
L Operating Band Change, Frequency Change.....	136
L Connection Type.....	136
L Destination Parameters.....	136
L Execute.....	137

[ON | OFF] key / LTE Signaling softkey

To turn the DL signal transmission on or off, select the "LTE Signaling" softkey and press [ON | OFF]. Alternatively, right-click the softkey.

The current state is shown on the softkey. A yellow hour glass symbol indicates that the state is changing.



The state "RDY" means that the signaling application is ready to receive an inter-RAT handover from another signaling application (e.g. from WCDMA). This state is initiated by the application acting as source of the handover.

Remote command:

`SOURCE:LTE:SIGN<i>:CELL:STATE`

`SOURCE:LTE:SIGN<i>:CELL:STATE:ALL?`

Connection control hotkeys

Any interaction with a UE requires an LTE downlink signal (cell). If the signal is available (state ON, no hour glass), connection control hotkeys can be present in the hotkey bar.

The available hotkeys depend on the current packet-switched state, RRC state, SCC state, connection type and selected component carrier tab.

For SCC activation, the available hotkeys depend also on:

- The SCC activation mode, see ["SCC Activation Mode"](#) on page 142
- Whether SCCs are grouped into a synchronization set or not, see ["Multiple SCC Actions hotkey"](#) on page 132

To access hotkeys for a specific component carrier, select the corresponding tab.

Example: For hotkey "SCC2 On", select the tab "SCC2". Via the "PCC" tab, you can access hotkeys for the PCC and for the SCC1.

The possible hotkeys of the "LTE Signaling" softkey are listed in the following table.

Hotkey	Description
"Connect"	See "Connect hotkey" on page 133.
"Disconnect"	Release a dedicated bearer connection in test mode.
"Detach"	Send a detach request to the UE and detach the UE (independent of UE reaction to the request).
"Send SMS"	Send an SMS message to the UE.
"Inter/Intra-RAT"	See "Inter/Intra-RAT hotkey" on page 134.
"SCC<n> On" "SyncSet A/B On"	Switch on the DL signal of SCC<n> or of all SCCs being member of the synchronization set A/B.
"SCC<n> Off" "SyncSet A/B Off"	Switch off the DL signal of SCC<n> or of all SCCs being member of the synchronization set A/B.
"SCC<n> add RRC" "SyncSet A/B add RRC"	Add an RRC connection for SCC<n> or for all SCCs of the synchronization set A/B.
"SCC<n> delete RRC" "SyncSet A/B delete RRC"	Delete the RRC connection for SCC<n> or for all SCCs of the synchronization set A/B.
"SCC<n> activate MAC" "SyncSet A/B activate MAC"	Activate MAC for SCC<n> or for all SCCs of the synchronization set A/B.
"SCC<n> deactivate MAC" "SyncSet A/B deactiv. MAC"	Deactivate MAC for SCC<n> or for all SCCs of the synchronization set A/B.

Remote command:

`CALL:LTE:SIGN<i>:PSWitched:ACTion`

`FETCh:LTE:SIGN<i>:PSWitched:STATE?`

`CALL:LTE:SIGN<i>:SCC<c>:ACTion`

```

FETCh:LTE:SIGN<i>:SCC<c>:STATe?
CALL:LTE:SIGN<i>:A:ACTioN
FETCh:LTE:SIGN<i>:A:STATe?
CALL:LTE:SIGN<i>:B:ACTioN
FETCh:LTE:SIGN<i>:B:STATe?

```

Multiple SCC Actions hotkey

The hotkey is displayed for carrier aggregation scenarios with more than one SCC. It opens a dialog box for configuration of SCC synchronization sets. The dialog box allows you also to trigger state transitions.

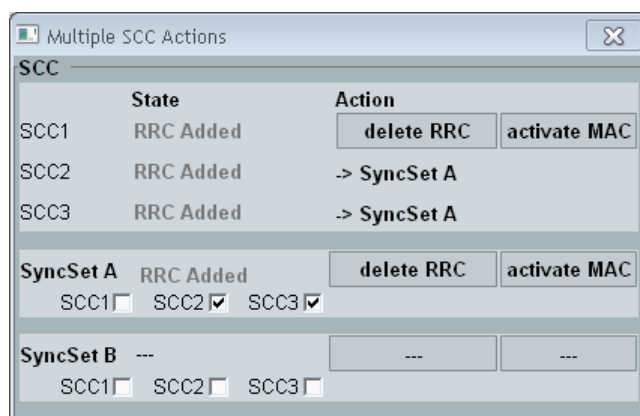


Figure 2-36: Dialog box for three SCC (four CC)

The upper part lists the individual SCCs. The lower part lists the synchronization sets A and B.

Via the checkboxes, you can assign SCCs to synchronization sets. An SCC can belong to one of the two sets or it can belong to no set at all. A set and an SCC to be added to the set must have the same state.

The column "Action" contains buttons for SCC state transitions. These buttons are an alternative to the corresponding hotkeys, see ["Connection control hotkeys"](#) on page 131.

There are state transition buttons for synchronization sets containing at least one SCC. And there are buttons for SCCs that do not belong to a set. For SCCs that belong to a set, the column "Action" indicates the set.

If you change the state of a synchronization set, a single message is sent to the UE, ordering the transition for all SCCs of the set. This rule applies to all SCC activation modes.

Remote command:

```

CONFigure:LTE:SIGN<i>:A:SCC<c>:ENABle
CONFigure:LTE:SIGN<i>:B:SCC<c>:ENABle
CALL:LTE:SIGN<i>:SCC<c>:ACTioN
FETCh:LTE:SIGN<i>:SCC<c>:STATe?
CALL:LTE:SIGN<i>:A:ACTioN
FETCh:LTE:SIGN<i>:A:STATe?

```


CALL:LTE:SIGN<i>:B:ACTion

FETCh:LTE:SIGN<i>:B:STATe?

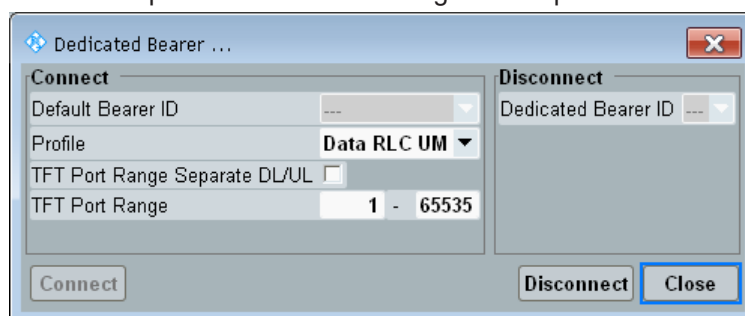
Connect hotkey

The UE must attach to make the "Connect" hotkey available.

The effect of the hotkey depends on the parameter [Connection Type](#):

- "Testmode": The hotkey sets up a single dedicated bearer with predefined default settings.
- "Data Application": The hotkey opens a dialog box for configuration, connection and release of multiple dedicated bearers.

Option R&S CMW-KS510 is required. Without this option, a single dedicated bearer with predefined default settings is set up.



The dialog box is described in the following.

Connect button ← Connect hotkey

Use the "Connect" area of the dialog box to configure a dedicated bearer. Then press the "Connect" button to establish the bearer with the configured settings.

"Default Bearer ID" selects the default bearer, to which the dedicated bearer is to be mapped. For more details about the ID, see ["Default Bearer"](#) on page 126.

"Profile" selects a dedicated bearer profile. The profiles contain optimum bearer settings for voice calls, video calls and data connections with high throughput in RLC acknowledged or unacknowledged mode. The most important bearer settings are listed in [Table 2-57](#).

"TFT Port Range" selects the destination ports for which traffic is to be routed to the dedicated bearer. If you set up several dedicated bearers, the port ranges of the bearers must not overlap.

By default, the configured TFT port range is used for the uplink and for the downlink. If you want to configure different ranges for the uplink and the downlink, enable "TFT Port Range Separate DL/UL".

In total, up to eight bearers can be established in parallel (default plus dedicated bearers).

Table 2-57: Bearer settings depending on the Profile

	Voice	Video	Data AM	Data UM
QCI	1	2	6	6
PDCP "discardTimer"	50ms	50ms	infinity	infinity
"pdcp-SN-size" (RLC UM)	len7bits	len7bits	-	len12bits

	Voice	Video	Data AM	Data UM
RLC mode (RLC AM/UM)	UM	UM	AM	UM
"sn-FieldLength" (RLC UM)	size5	size5	-	size10
"t-Reordering" (RLC UM, AM)	10ms	10ms	20ms	20ms
"t-StatusProhibit" (RLC AM)	-	-	5ms	-
"t-PollRetransmit" (RLC AM)	-	-	60ms	-
"pollPDU" (RLC AM)	-	-	p32	-
"pollByte" (RLC AM)	-	-	250kB	-
"maxRetxThreshold" (RLC AM)	-	-	t16	-

Remote command:

```
CATalog:LTE:SIGN<i>:CONNection:DEFBearer?
PREPare:LTE:SIGN<i>:CONNection:DEDBearer
PREPare:LTE:SIGN<i>:CONNection:DEDBearer:SEParate
CALL:LTE:SIGN<i>:PSWitched:ACTion
```

Disconnect button ← Connect hotkey

Use the "Disconnect" area of the dialog box to release an established dedicated bearer.

Select the bearer via the "Dedicated Bearer ID" list. Then press the "Disconnect" button.

For more details about the ID, see ["Dedicated Bearer"](#) on page 127.

Remote command:

```
CATalog:LTE:SIGN<i>:CONNection:DEDBearer?
CONFIgure:LTE:SIGN<i>:CONNection:DEDBearer
CALL:LTE:SIGN<i>:PSWitched:ACTion
```

Inter/Intra-RAT hotkey

The hotkey opens a dialog box for selection and configuration of the handover destination and initiation of the handover.

The LTE signaling application supports a handover within the signaling application, e.g. to another operating band, and a handover to another instrument or to another signaling application.

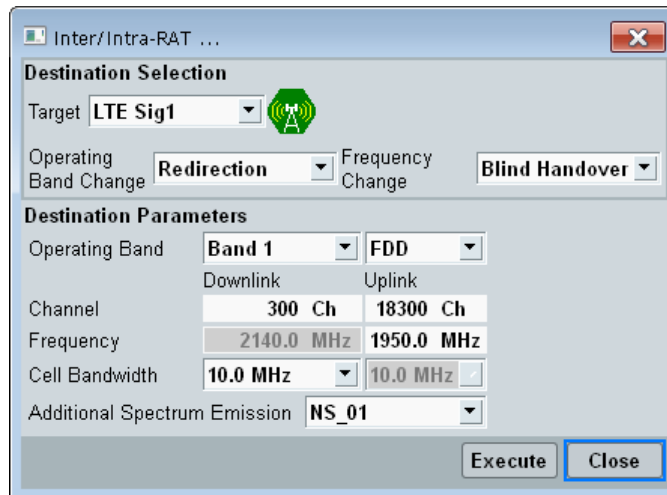


Figure 2-37: Intra-RAT handover

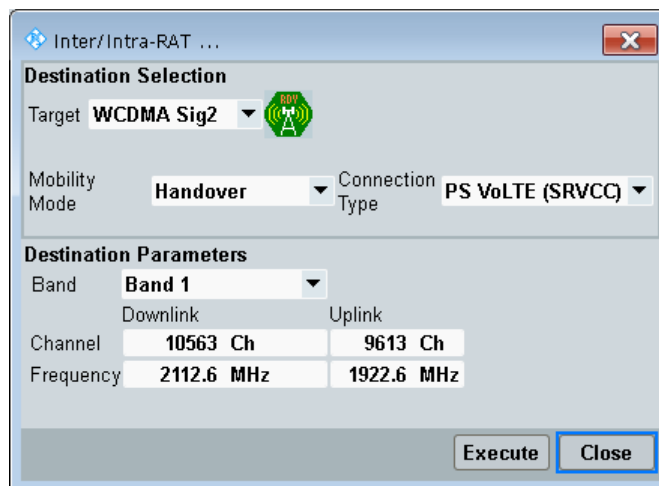


Figure 2-38: Handover of VoLTE call to WCDMA

For a detailed step-by-step description of a handover, see ["How to perform a hand-over"](#) on page 42.

Target ← Inter/Intra-RAT hotkey

This parameter selects the handover destination. For a handover to another instrument, select "No Connection" as target.

The cell icon indicates the cell state of the currently selected destination. When you select another signaling application, e.g. "WCDMA Sig1", the destination cell is switched on automatically and the target cell state changes to RDY (ready for the handover).

Remote command:

```
PREPare:LTE:SIGN<i>:HANDover:CATalog:DESTination?
PREPare:LTE:SIGN<i>:HANDover:DESTination
PREPare:LTE:SIGN<i>:HANDover:EXTernal:DESTination
```

Mobility Mode ← Inter/Intra-RAT hotkey

Selects the mechanism to be used for a handover to another LTE signaling instance or to another signaling application, for example to GSM or WCDMA.

Circuit-switched fallback requires option R&S CMW-KS510.

For a description of the mobility modes, see [Section 2.2.10, "Handover"](#), on page 42.

Remote command:

```
PREPare:LTE:SIGN<i>:HANDover:MMODE
```

Operating Band Change, Frequency Change ← Inter/Intra-RAT hotkey

These settings are displayed for a handover within the signaling application.

The setting "Operating Band Change" selects the mechanism to be used for changing the band, without changing the cell bandwidth.

The setting "Frequency Change" selects the mechanism to be used for changing the frequency, without changing the band and the cell bandwidth.

If the cell bandwidth is changed, redirection is used independent of these settings.

For a description of the mobility modes, see [Section 2.2.10, "Handover"](#), on page 42.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:OBCHange
```

```
CONFigure:LTE:SIGN<i>:CONNection:FCHange
```

Connection Type ← Inter/Intra-RAT hotkey

This setting is only relevant for the handover of a VoLTE call to another signaling application.

It selects the call type to be set up at the destination. Option R&S CMW-KS510 is required.

"PS VoLTE (SRVCC)"	A voice call is established. The VoLTE call is handed over seamlessly via SRVCC.
--------------------	--

"PS Data (End2End)"	An E2E packet data connection via the DAU is established.
---------------------	---

Remote command:

```
PREPare:LTE:SIGN<i>:HANDover:CTYPE
```

Destination Parameters ← Inter/Intra-RAT hotkey

The "Destination Parameters" display current settings of the selected signaling application target, typically operating band and channels. You can modify these settings before starting the handover.

For a handover to another instrument, the "External Destination Parameters" are displayed instead (radio access technology and typically operating band and channel). Configure them according to the actual configuration of the other instrument. There is no communication between the two instruments, so the settings at both instruments must match.

The commands listed here are relevant for handover within the LTE signaling application and for handover to another instrument.

For a handover to another signaling application in the same instrument, use the commands provided by the signaling application target. There are no special handover commands for this purpose.

Remote command:

```
PREPare:LTE:SIGN<i>:HANDover
PREPare:LTE:SIGN<i>:HANDover:ENHanced
PREPare:LTE:SIGN<i>:HANDover:EXTernal:CDMA
PREPare:LTE:SIGN<i>:HANDover:EXTernal:EVDO
PREPare:LTE:SIGN<i>:HANDover:EXTernal:GSM
PREPare:LTE:SIGN<i>:HANDover:EXTernal:LTE
PREPare:LTE:SIGN<i>:HANDover:EXTernal:TDSCdma
PREPare:LTE:SIGN<i>:HANDover:EXTernal:WCDMa
```

Execute ← Inter/Intra-RAT hotkey

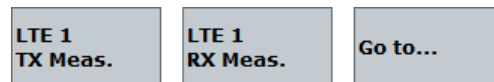
To initiate a handover with the configured settings, press the "Execute" button.

Remote command:

```
CALL:LTE:SIGN<i>:PSWitched:ACTion
```

2.4.3 Using the shortcut softkeys

The LTE signaling main view provides three configurable shortcut softkeys.



You can use the shortcut softkeys for fast access to other measurements, running in parallel to the signaling application.

If you use an LTE TX measurement in parallel to the signaling application, use always a shortcut softkey to switch to the measurement. The softkey accesses the TX measurement and it configures some measurement settings for parallel usage:

- A suitable trigger signal provided by the signaling application is selected as trigger source for the TX measurement.
- The combined signal path scenario is activated in the TX measurement.
The RF input path settings of the signaling application are used for the TX measurement.
Some measurement control settings are automatically configured compatible with the signaling settings.

If the softkey label indicates a measurement name, this measurement has been assigned to the softkey as fixed target, see [Section 2.4.18, "Shortcut configuration"](#), on page 293.

If the softkey label equals "Go to...", the softkey opens a dialog box with a list of all available measurements.

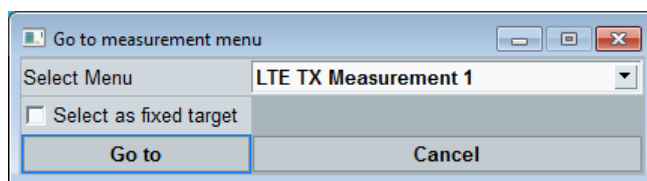


Figure 2-39: Dialog box opened by Go to... softkey

Select Menu

Selects the target measurement you want to switch to.

Select as fixed target

Sets the selected measurement as fixed target of the softkey. The softkey label indicates the measurement name and switches directly to the selected target without opening the dialog box.

When the dialog box has been disabled, you can still change the target measurement or re-enable the dialog box using the configuration menu, see [Section 2.4.18, "Shortcut configuration"](#), on page 293.

Go to / Cancel

Press the "Go to" button to switch to the selected measurement or "Cancel" to abort.

2.4.4 General settings

Access: "Config" hotkey > top of the tabs.

Configure the general settings according to your test scenario, before configuring any other settings.

The general settings reflect basic decisions: FDD or TDD signal, use MIMO / fading / carrier aggregation / eMTC or not.

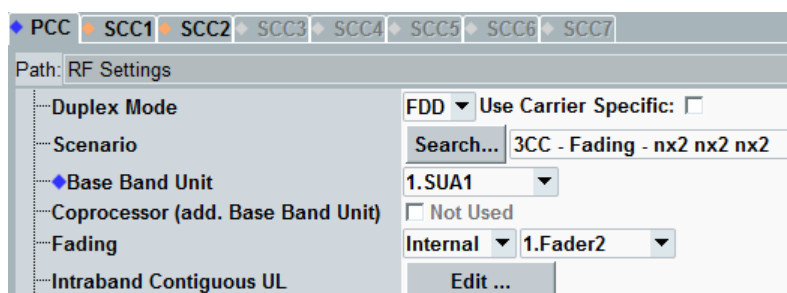


Figure 2-40: Top of dialog for PCC, with three CC and fading

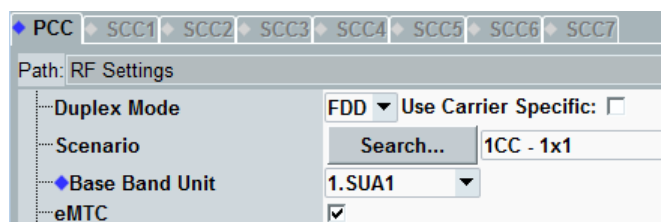


Figure 2-41: Top of dialog with eMTC

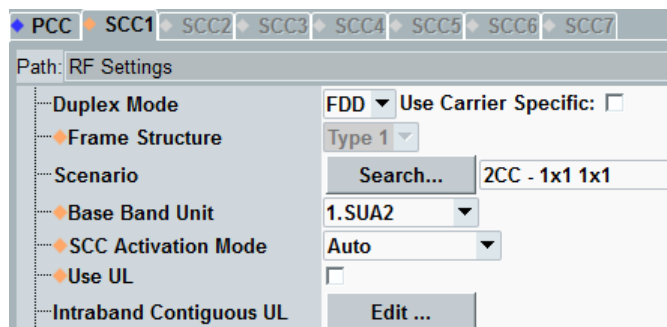


Figure 2-42: Top of dialog for SCC, with two CC, no fading

The configuration dialog contains one tab per component carrier:

- "PCC" tab: This tab is always active, independent of the selected scenario. The settings are either PCC-specific (marked by ) , or they apply to the PCC and to the SCCs (if SCC tabs are active).
- "SCC<n>" tab: The tabs are active or inactive depending on the selected scenario. The settings are either SCC-specific (marked by ) , or they apply to the PCC and to the SCC. In the latter case, changing the setting on one tab changes it also on the other tab. Most settings marked by  are configurable per SCC. So different values can be set on the individual SCC tabs.

If a setting is individually configurable for PCC and SCC, two remote control commands are listed for the setting: one [: PCC] command and one : SCC command. If the SCC command has a suffix (: SCC<c>), the setting is individually configurable per SCC.

The SCC tab contains a subset of the settings present on the PCC tab. For that reason, most screenshots in the following sections show the settings on the PCC tab. The general section on the SCC tab contains additional parameters that are not present on the PCC tab.

The general settings are described in the following.

Duplex Mode.....	140
Scenario.....	140
L Search button.....	140
Base Band Unit.....	141
Coprocessor (add. Base Band Unit).....	141
eMTC.....	142
Fading.....	142
Frame Structure.....	142

SCC Activation Mode.....	142
Use UL.....	143
Intraband Contiguous UL.....	143

Duplex Mode

Selects the duplex mode of the LTE signal: FDD or TDD.

If you enable "Use Carrier Specific", you can configure the duplex mode per component carrier.

Option R&S CMW-KS500 required for FDD, R&S CMW-KS550 required for TDD.

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:DMODE
```

```
CONFigure:LTE:SIGN<i>:SCC<c>:DMODE
```

```
CONFigure:LTE:SIGN<i>[:PCC]:DMODE:UCSPecific
```

Scenario

Different test scenarios require different sets of parameters. Selecting a scenario hides/shows parts of the GUI as required by the scenario.

The active scenario determines, for example, whether MIMO and fading are used or not. It determines also the number of component carriers.

Scenario names:

- "<n>CC" indicates the number of DL component carriers. So <n> greater 1 means DL carrier aggregation with <n> component carriers.
- Fading support is indicated by "- Fading -" in the scenario name.
- For each component carrier, an "axb" information is stated. The order is: PCC, SCC1, SCC2, and so on. 1x1 means SISO, nx2 means MIMO nx2, and so on.

The individual scenarios are only offered for selection, if the required software and hardware options are available, see [Section 2.2.2, "Scenarios"](#), on page 19.

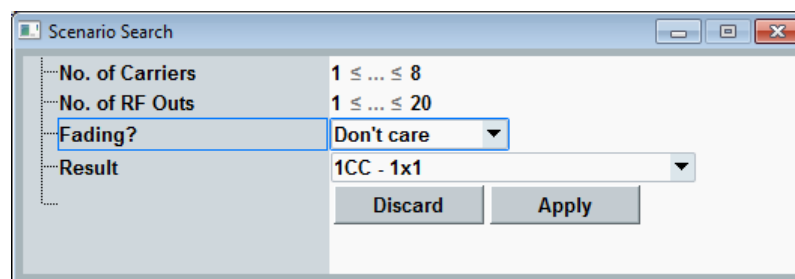
Remote command:

For each scenario, there is at least one `ROUTE:LTE:SIGN<i>:SCENario:... com-`mand. For fading scenarios, there are several commands for the individual fading types.

All scenario commands are listed in [Section 2.6.8.1, "Scenario selection and signal routing"](#), on page 451.

Search button ← Scenario

The button opens a dialog box for search of a scenario via filter criteria.



You can specify a minimum and maximum value for the number of component carriers and RF output paths. And you can specify whether you want scenarios with fading, without fading, or both.

The result list shows only the scenarios that match your filter criteria and for which you have the required software and hardware options. The result "Undefined" means that there are no such scenarios.

Select a scenario from the list and click "Apply" to set this scenario. Or click "Discard" to close the box without changing the scenario.

Base Band Unit

Selects the signaling unit to be used for a carrier. This setting is especially useful for CMWflexx setups, to control which carrier is available at which instrument.

A prefix "1." to "4." selects the CMW in a CMWflexx setup.

You select one or two SUA partitions. The offered values depend on the selected scenario. If you want to use 64-QAM or 256-QAM in the UL of a carrier, select two SUA partitions for that carrier. Otherwise, you cannot configure 64-QAM/256-QAM.

Some examples:

- "SUA2" selects SUA partition 2 in a single-CMW setup.
- "3.SUA1" selects SUA partition 1 in CMW number 3 in a CMWflexx setup.
- "1.SUA3&4" selects SUA partitions 3 and 4 in CMW number 1.

Remote command:

Configured via the scenario selection command, see [Section 2.6.8.1, "Scenario selection and signal routing"](#), on page 451.

Coprocessor (add. Base Band Unit)

Configures an additional SUA (two SUA partitions) as coprocessor.

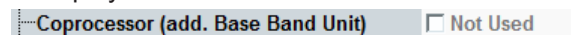
There are three possible behaviors, depending on the scenario:

- The scenario does not need coprocessing. The setting is hidden.
- Coprocessing is optional. The scenario works without coprocessing. But depending on the settings, you need a coprocessor to reach the maximum throughput, especially for data application tests with the DAU.
- Coprocessing is mandatory. The scenario does not work without coprocessing.

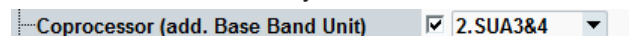
Configuration:

- Optional coprocessing:

The checkbox enables/disables coprocessing. With disabled checkbox, "Not Used" is displayed.

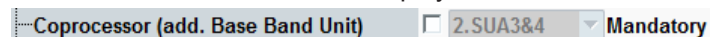
 Coprocessor (add. Base Band Unit) ☐ Not Used

With enabled checkbox, you can select a SUA.

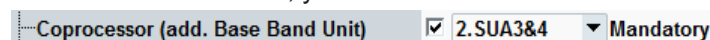
 Coprocessor (add. Base Band Unit) ☒ 2.SUA3&4 ▼

- Mandatory coprocessing:

The checkbox enables/disables the SUA selection. With disabled checkbox, the used default SUA selection is displayed.

 Coprocessor (add. Base Band Unit) ☐ 2.SUA3&4 ▼ Mandatory

With enabled checkbox, you can select a SUA.

 Coprocessor (add. Base Band Unit) ☒ 2.SUA3&4 ▼ Mandatory

To check which behavior applies to which scenario, refer to the tables in [Section 2.2.2, "Scenarios"](#), on page 19.

Remote command:

Configured via the scenario selection command, see [Section 2.6.8.1, "Scenario selection and signal routing"](#), on page 451.

eMTC

Enables enhanced machine type communication (eMTC).

eMTC is only supported for FDD, for the scenario "1CC - 1x1". Option R&S CMW-KS590 is required.

For a short introduction, see [Section 2.2.11, "eMTC"](#), on page 43.

For eMTC-specific settings, see [Section 2.4.11, "eMTC settings"](#), on page 193.

Remote command:

`CONFfigure:LTE:SIGN<i>[:PCC]:EMTC:ENABLe`

Fading

This parameter is displayed for the fading scenarios. Only internal fading is supported.

The fader setting selects an SUA partition. The numbering is reversed, so that the lowest fader number selects the highest SUA partition. This reverse numbering allows you to select the same number for the fader and the baseband unit, because they address different SUA partitions.

Example: Baseband unit "2.SUA4" selects SUA partition 4 in CMW 2. Fader "2.Fader4" selects SUA partition 1 in CMW 2. So they can be combined.

The fader selection offers only values compatible with the baseband setting. If there is only one possible value, the selection is grayed out, displaying this value.

Remote command:

Configured via the scenario selection command, see [Section 2.6.8.1, "Scenario selection and signal routing"](#), on page 451.

Frame Structure

This setting is only displayed on the SCC tabs.

It shows the frame structure:

- "Type 1" = normal FDD operation (requires R&S CMW-KS500)
- "Type 2" = normal TDD operation (requires R&S CMW-KS550)
- "Type 3" = LAA operation mode (requires R&S CMW-KS514)

The value is set automatically depending on the configured duplex mode and operating band. For "Type 3", select TDD and an LAA operating band.

For the TDD user-defined band, the setting is configurable.

Remote command:

`CONFfigure:LTE:SIGN<i>:SCC<c>:FSTRucture`

SCC Activation Mode

This setting is only displayed on the SCC tabs. It configures the activation mode of the SCCs for carrier aggregation scenarios. The setting applies to all SCCs (same value on all SCC tabs).

Option R&S CMW-KS512 is required for the modes "Manual" and "Semiautomatic".

See also [Section 2.2.9.2, "SCC states"](#), on page 40.

"Auto"	The SCC is activated automatically, so that the state "MAC Activated" is reached. If the PCC RRC connection is kept after the attach (Keep RRC Connection), the SCC activation is done at UE attach. If the PCC RRC connection is not kept after the attach, the SCC activation is done with the next RRC connection establishment.
"Semiautomatic"	Initiate the SCC activation manually. As a result, all state transitions required to reach the state "MAC Activated" are performed. This mode is useful if you want to establish first a PCC connection and add the SCC connection later on after some tests.
"Manual"	Initiate each state transition step separately. There are three steps from "SCC Off" via "SCC On" and "RRC Added" to "MAC Activated". This mode is useful for debugging of the UE. From each state, you can go back to the previous state or continue to the next state. For example from state "RRC Added", you can go to "MAC Activated" or to "SCC On".

Remote command:

`CONFfigure:LTE:SIGN<i>:SCC:AMODE`

Use UL

This setting is only displayed on the SCC tabs. It activates the uplink for the SCC.

You can activate the uplink of maximum five SCCs, resulting in uplink carrier aggregation with six carriers (PCC plus five SCC).

SCC uplink is not allowed for frame structure type 3.

Option R&S CMW-KS512 is required for UL carrier aggregation. R&S CMW-KS514 is required for aggregation of more than two UL.

Remote command:

`CONFfigure:LTE:SIGN<i>:SCC<c>:UUL`

Intraband Contiguous UL

This button opens a dialog box for alignment of component carrier uplinks according to the rules for intraband contiguous uplink carrier aggregation. Enable the desired SCC uplinks before opening the dialog box.

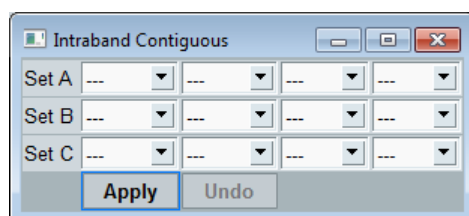


Figure 2-43: Dialog box for 6CC scenario

The dialog box contains one, two or three rows ("Set A", "Set B", "Set C"), depending on the selected scenario.

Each line defines a set of up to four component carriers with contiguous allocation. For a scenario with four carriers, for example, you can assign two carriers to one set and two carriers to the other set. Or you can assign two, three or four carriers to one set and no carrier to the other set.

The leftmost setting of a set selects the primary carrier. The set uses the operating band of the primary carrier. And the primary carrier keeps its configured frequency.

When you press "Apply", all other carriers of the set are automatically aligned to the primary carrier. Their frequency settings are grayed out and coupled to the primary carrier frequency. You can modify the frequency and band of the primary carrier. The other carriers follow the primary carrier, maintaining the contiguous allocation (as far as possible - selecting a too small band, for example, results in an error message).

If possible, the carriers are aligned in the spectrum in the order selected in the dialog box. So the frequency increases from the leftmost entry to the rightmost entry. Carriers that do not fit into the band to the right of the primary carrier are positioned to the left of the primary carrier.

Remote command:

`CONFigure:LTE:SIGN<i>:CAGGregation:SET`

2.4.5 RF settings

Access: "Config" hotkey > "RF Settings" node.

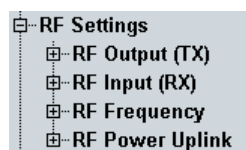


Figure 2-44: RF settings (1CC - 1x1)

The settings for the RF input and output paths are described in the following sections.

- [RF signal routing](#)..... 144
- [RF frequency](#)..... 147
- [Expected uplink power](#)..... 151

2.4.5.1 RF signal routing

Access: "Config" hotkey > "RF Settings" node > "RF Output (TX)" / "RF Input (RX)" nodes.

The two nodes configure the RF input path and the RF output paths. Most parameters are configurable per path.

The number of configurable input and output paths depends mainly on the selected scenario.

For carrier aggregation scenarios, you can configure the settings per component carrier (except "External Delay Compensation").

Depending on your hardware configuration, there are dependencies between some parameters. Configure the settings in the following order to access all possible values:

1. Select the signaling unit. See ["Base Band Unit"](#) on page 141.
In a CMWflexx setup, this action selects also the instrument.
2. Select the RF connector.
3. Select the RF converter.

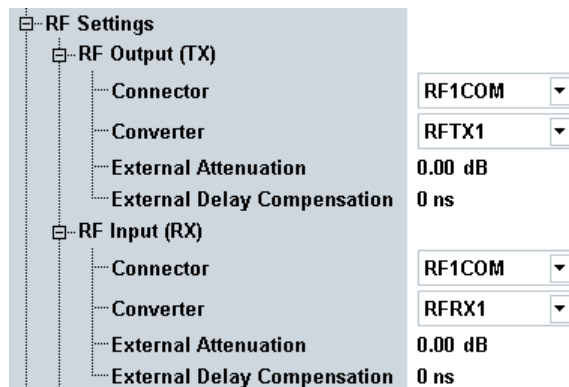


Figure 2-45: RF path settings (1CC - 1x1)

RF Output (TX).....	145
L Connector, Converter.....	145
L External Attenuation.....	145
L External Delay Compensation.....	146
RF Input (RX).....	146
L Connector, Converter.....	146
L External Attenuation.....	146
L External Delay Compensation.....	146

RF Output (TX)

The following parameters configure the RF output paths of the R&S CMW.

Connector, Converter ← RF Output (TX)

Selects the output paths for the generated RF signals. That means the output connector and the TX module for each downlink path.

Remote command:

Configured via the scenario selection command, see [Section 2.6.8.1, "Scenario selection and signal routing"](#), on page 451.

External Attenuation ← RF Output (TX)

Defines the value of an external attenuation (or gain, if the value is negative) in the output path. With an external attenuation of x dB, the power of the generated signal is increased by x dB. The actual generated levels are equal to the displayed values plus the external attenuation.

If a correction table for frequency-dependent attenuation is active for the chosen connector, the table name and a button are displayed. Press the button to display the table entries.

You can configure the external attenuation per output path.

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:OUTPut<n>
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:OUTPut<n>
```

External Delay Compensation ← RF Output (TX)

Defines the value of an external time delay in the output path. A delay can, for example, be caused by a long optical fiber cable or by an additional instrument in the output path.

As a result, the downlink signal is sent earlier, so that the downlink signal arrives at the UE without delay.

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings:EDC:OUTPut
```

RF Input (RX)

The following parameters configure the RF input path of the R&S CMW.

Connector, Converter ← RF Input (RX)

Selects the input path for the measured RF signal, i.e. the input connector and the RX module to be used.

Remote command:

Configured via the scenario selection command, see [Section 2.6.8.1, "Scenario selection and signal routing"](#), on page 451.

For configuration of SCC uplinks, see `CONFigure:LTE:SIGN<i>:SCC<c>:UUL`.

External Attenuation ← RF Input (RX)

Defines the value of an external attenuation (or gain, if the value is negative) in the input path. The power readings are corrected by the external attenuation value.

The external attenuation value is also used in the calculation of the maximum configurable input power value.

If a correction table for frequency-dependent attenuation is active for the chosen connector, then the table name and a button are displayed. Press the button to display the table entries.

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:INPut
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:INPut
```

External Delay Compensation ← RF Input (RX)

Defines the value of an external time delay in the input path, for example caused by a long optical fiber cable.

The signaling application uses this information to compensate for the delay and to synchronize the uplink and the downlink despite the delay.

Remote command:

CONFigure:LTE:SIGN<i>:RFSettings:EDC:INPut

2.4.5.2 RF frequency

Access: "Config" hotkey > "RF Settings" node > "RF Frequency" node.

This node configures the operating band and the channel/frequency for uplink and downlink.

The uplink settings configure the center frequency of the RF analyzer. The downlink settings configure the carrier center frequency of the generated LTE signal.

If option R&S CMW-KS525 is available, a user-defined band can be specified and used.

For carrier aggregation scenarios, you can configure the settings per component carrier.

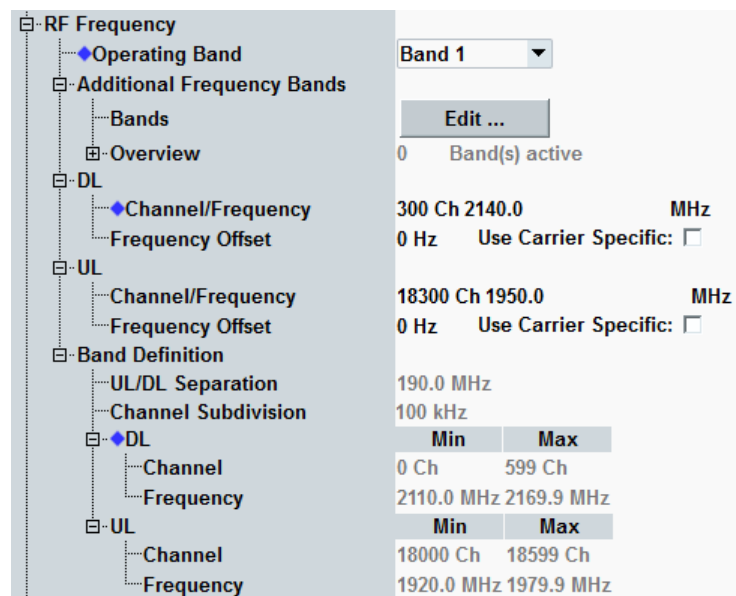


Figure 2-46: Frequency settings

Operating Band, Channel, Frequency.....	148
Additional Frequency Bands.....	148
Frequency Offset.....	149
Band Definition.....	149
L UL/DL Separation.....	149
L Channel Subdivision.....	150
L Band Indicator.....	150
L DL Channel, Frequency.....	150
L UL Channel, Frequency.....	150

Operating Band, Channel, Frequency

To specify the DL and UL frequencies, select an operating band first, then enter a valid channel number or frequency for UL or DL. The related frequency or channel number and the parameters for the other direction are calculated automatically.

For bands with configurable UL/DL separation like band 70, set also ["UL/DL Separation"](#) on page 149.

For carrier aggregation, select different DL channels for the carriers. The channels must not overlap and can be located in the same band or in different bands. For co-location, see [Section 2.2.5.1, "RF paths and co-location"](#), on page 33.

The relation between operating band, carrier frequencies and channel numbers is defined by 3GPP (see [Section 2.2.23, "Operating bands"](#), on page 91). You can define an additional band if option R&S CMW-KS525 is available, see ["Band Definition"](#) on page 149.

You can change the operating band and the channels in all main connection states. In the state "Connection Established", you can either change one parameter directly or you can perform a handover to reconfigure several parameters, see [Section 2.2.10, "Handover"](#), on page 42.

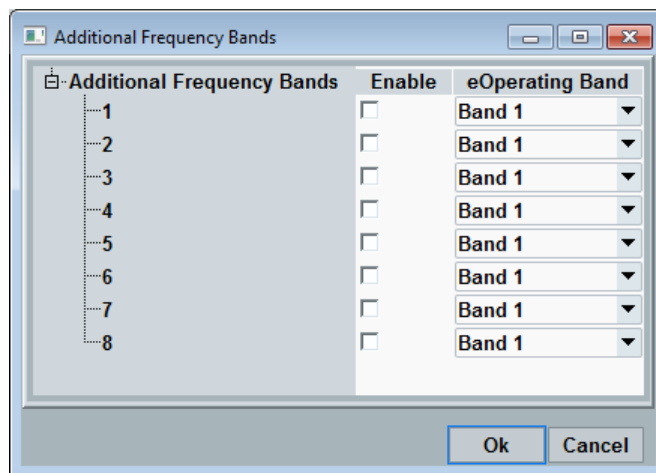
If you have configured intraband contiguous uplink carrier aggregation, the settings of all aligned component carriers are grayed out. Only the settings of the primary carrier are configurable. See also ["Intraband Contiguous UL"](#) on page 143.

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:BAND
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:DL
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:UL
CONFigure:LTE:SIGN<i>:SCC<c>:BAND
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:DL
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:UL
CONFigure:LTE:SIGN<i>:RFSettings:ALL:BWCHannel
```

Additional Frequency Bands

A cell that supports the feature multiple frequency band indicator (MFBI) can belong to several frequency bands. The additional bands are signaled to the UE in a list ("multi-BandInfoList"). To configure the list of additional bands, press "Edit". A dialog box opens.



Enable the entries to be added to the "multiBandInfoList" and select a different band for each enabled entry.

If no entry is enabled, no "multiBandInfoList" is signaled to the UE and only one band is supported by the cell.

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:AFBands:ALL
```

Frequency Offset

You can specify positive or negative offsets to be added to the carrier center frequencies. This setting is useful, for example, if the UE has a frequency offset relative to the channels defined by 3GPP.

To configure different PCC and SCC settings for carrier aggregation scenarios, enable "Use Carrier Specific".

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:DL
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:DL:UCSPecific
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL:UCSPecific
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:DL
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:UL
```

Band Definition

Displays the definition of the selected operating band. Operating band "User-Defined" can be edited.

Option R&S CMW-KS525 is required for user-defined bands.

UL/DL Separation ← Band Definition

$$F_{DL} - F_{UL}$$

This setting is configurable for user-defined bands, even for TDD. And it is configurable for band 70.

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:UDSeparation
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:UDSeparation
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDSeparation
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDSeparation
```

Channel Subdivision ← Band Definition

Displays the spacing between center frequencies of two adjacent channels, always 100 kHz - also for user-defined bands.

Band Indicator ← Band Definition

Frequency band indicator identifying a user-defined band in signaling messages. The value is sent to the UE via broadcast in system information block 1. It is also used in handover messages.

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:BINdicator
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:BINdicator
```

DL Channel, Frequency ← Band Definition

For user-defined PCC or SCC channels, you can edit the minimum DL channel number, the maximum DL channel number and the minimum DL center frequency.

The maximum DL center frequency is calculated automatically according to the "Channel Subdivision".

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:DL:
MINimum
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:DL:
MAXimum
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:DL:
MINimum
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:DL:
MAXimum?
```

Plus corresponding PCC commands.

UL Channel, Frequency ← Band Definition

For user-defined PCC channels, you can edit the minimum UL channel number and thus the relation between DL and UL channel numbers.

The other UL settings are calculated automatically according to the DL settings and the "UL/DL Separation".

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:UL:
MINimum
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:UL:
MAXimum?
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:UL:
MINimum?
```

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:UL:MAXimum?

Plus corresponding PCC commands.

2.4.5.3 Expected uplink power

Access: "Config" hotkey > "RF Settings" node > "RF Power Uplink" node.

This node configures the expected uplink power.

With UL carrier aggregation, you can configure the settings per component carrier.

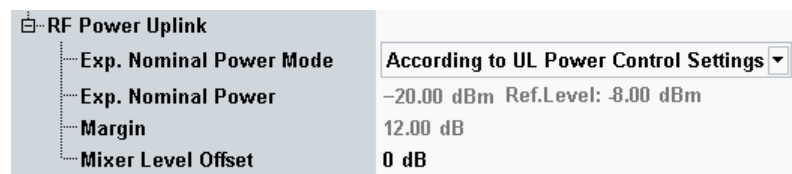


Figure 2-47: RF uplink power settings

Exp. Nominal Power..., Margin

These parameters configure the expected UL power. Two modes are available:

- "According to UL Power Control Settings"
The UL power is calculated automatically from the UL power control settings. The resulting expected nominal power is displayed for information. The displayed reference level is calculated as:
 $Reference\ Level = Expected\ Nominal\ Power + 12\ dB\ Margin$
For UL power control settings, see [Section 2.4.8, "Uplink power control"](#), on page 161.
- "Manual"
In manual mode, the expected nominal power and a margin can be defined manually. The displayed reference level is calculated as:
 $Reference\ Level = Expected\ Nominal\ Power + Margin$

The margin is used to account for the known variations (crest factor) of the RF input signal power.

Note: The actual input power at the connectors must be within the level range of the selected RF input connector; refer to the data sheet. If all power settings are configured correctly, the actual power equals the "Reference Level" minus the "External Attenuation (Input)" value.

The parameters can be changed in all main connection states including "Connection Established".

Remote command:

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPMODE
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPower
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UMARGIN

Plus corresponding PCC commands.

Mixer Level Offset

Varies the input level of the mixer in the analyzer path. A negative offset reduces the mixer input level. A positive offset increases the level. Optimize the mixer input level according to the properties of the uplink signal.

Mixer level offset	Advantages	Possible shortcomings
< 0 dB	Suppression of distortion (e.g. of the inter-modulation products generated in the mixer)	Lower dynamic range (due to smaller signal-to-noise ratio)
> 0 dB	High signal-to-noise ratio, higher dynamic range	Risk of intermodulation, smaller overdrive reserve

Remote command:

```
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:MLOffset
```

```
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:MLOffset
```

2.4.6 Internal fading

Access: "Config" hotkey > "Internal Fading" node.

This node is only visible, if a fading scenario is selected and the fading source is set to "Internal".

For general prerequisites/required options and background information, see [Section 2.2.6, "Internal fading"](#), on page 36.

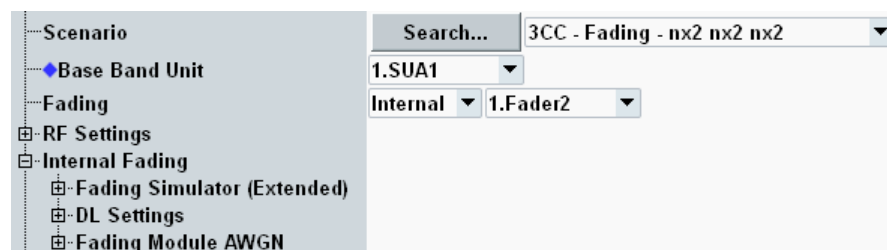


Figure 2-48: Internal fading settings

2.4.6.1 Fading simulator

Access: "Config" hotkey > "Internal Fading" node > "Fading Simulator (Extended)" node.

The node enables and sets up the fading simulator. For background information, see [Section 2.2.6.1, "Fading simulator"](#), on page 36.

Only extended fading is supported. Standard fading is no longer relevant.

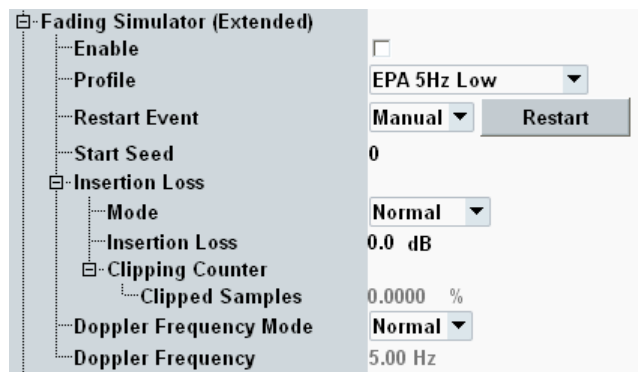


Figure 2-49: Fading simulator settings

For carrier aggregation scenarios, you can configure all settings per component carrier.

Enable	153
Profile	153
Restart Event	154
Start Seed	154
Insertion Loss	154
Doppler Frequency Mode, Doppler Frequency	154

Enable

Enables/disables the fading simulator.

Remote command:

`CONFfigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ENABle`

`CONFfigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ENABle`

Profile

Selects a propagation condition profile.

The following profiles are available:

- Delay profiles for low, medium and high delay spread environments:
 - "Extended pedestrian A" model (EPA), maximum Doppler frequency 5 Hz
 - "Extended vehicular A" model (EVA), maximum Doppler frequency 5 Hz, 70 Hz or 200 Hz
 - "Extended typical urban" model (ETU), maximum Doppler frequency 30 Hz, 70 Hz or 300 Hz

All profiles are available with low, medium and high correlation level, relevant for MIMO scenarios.

The delay profiles are defined in 3GPP TS 36.101, annex B.2.

- High-speed train (HST) scenario defined in 3GPP TS 36.101, annex B.3
- Multi-path profile used for CQI tests with parameter specifications of 3GPP TS 36.521-1, section 9.3 "CQI Reporting under fading conditions"
- Spatial channel model extension (SCME) profiles:
 - SCME urban micro-cell (UMi), UE speed 3 km/h or 30 km/h
 - SCME urban macro-cell (UMa), UE speed 3 km/h or 30 km/h

The profiles are defined in 3GPP TR 37.977. They are available for carriers with MIMO 2x2, 2x4, 4x2 and 4x4.

Option R&S CMW-KE504 is required. For carriers with MIMO 4x2, two SUA partitions are required for fading of the carrier with the SCME profiles.

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:PROFile
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:PROFile
```

Restart Event

There are the following modes.

- | | |
|----------|--|
| "Auto" | Fading starts automatically with the downlink signal. |
| "Manual" | Fading is started and restarted manually. A "Restart" button is displayed. |

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:REStart:MODE
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:REStart
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:REStart:MODE
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:REStart
```

Start Seed

Sets the start seed for the pseudo-random fading algorithm. This setting ensures reproducible fading conditions.

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:GLOBal:SEED
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:GLOBal:SEED
```

Insertion Loss

The insertion loss is the attenuation at the fader input. It can be calculated automatically or you can define it manually.

- | | |
|----------|--|
| "Normal" | The insertion loss is calculated automatically, based on the currently selected Profile . |
| "User" | Specify the insertion loss value. A lower insertion loss allows for a higher downlink power but can result in clipping.
You can use the displayed information "Clipped Samples" to find the lowest insertion loss value for which no clipping occurs. For scenarios with several output paths, the information is displayed per path. |

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:MODE
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:LOSS
SENSe:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:CSAMples<n>?
```

Plus corresponding PCC commands.

Doppler Frequency Mode, Doppler Frequency

The maximum Doppler frequency can be calculated automatically or you can define it manually.

- | | |
|----------|--|
| "Normal" | The maximum Doppler frequency is calculated automatically, based on the currently selected Profile . |
| "User" | Specify the maximum Doppler frequency. |

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:DSHift:MODE
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:DSHift
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:DSHift:MODE
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:DSHift
```

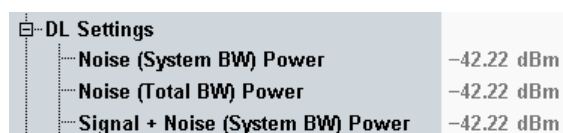
2.4.6.2 DL settings

Access: "Config" hotkey > "Internal Fading" node > "DL Settings" node.

The node displays noise power values, calculated from the downlink power, the cell bandwidth and the fading module AWGN settings.

For carrier aggregation scenarios, this information is available per component carrier.

The reference point for the power values is the RF output connector.



DL Settings	
Noise (System BW) Power	-42.22 dBm
Noise (Total BW) Power	-42.22 dBm
Signal + Noise (System BW) Power	-42.22 dBm

Figure 2-50: Noise information

Noise (System BW) Power

Displays the noise power on the downlink channel, i.e. within the cell bandwidth.

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:POWer:NOISe?
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:NOISe?
```

Noise (Total BW) Power

Displays the total noise power, within and outside of the cell bandwidth.

The total noise power is irrelevant for 3GPP test cases. They specify the SNR or the noise power within the cell bandwidth. The total noise power is only displayed for compatibility reasons with Rohde & Schwarz signal generators.

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:POWer:NOISe:TOTal?
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:NOISe:TOTal?
```

Signal + Noise (System BW) Power

Displays the total power (signal + noise) on the downlink channel, i.e. within the cell bandwidth.

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:POWer:SUM?
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:SUM?
```

2.4.6.3 Fading module AWGN

Access: "Config" hotkey > "Internal Fading" node > "Fading Module AWGN" node.

The node enables and configures the AWGN insertion via the fading module. For background information, see [Section 2.2.6.2, "AWGN generator"](#), on page 37.

Fading Module AWGN	
Enable	☐
Frequency Offset	0 Hz
Min. Noise/System BW Ratio	1.0
Noise Bandwidth	10.00 MHz
Effective Signal BW	9.00 MHz
Signal/Noise Ratio	0.00 dB

Figure 2-51: AWGN settings

For carrier aggregation scenarios, you can configure the settings per component carrier.

If you combine contiguous carrier aggregation with AWGN, we recommend the following AWGN settings:

- Enable AWGN for one carrier only.
- Configure the frequency offset to shift the noise center frequency to the center frequency of the aggregated bandwidth.
- Increase the "Min. Noise/System BW Ratio" so that the noise bandwidth equals the aggregated bandwidth.

These settings avoid undesired effects due to overlapping noise bandwidths of adjacent carriers.

Enable

Enables/disables AWGN insertion via the fading module.

Remote command:

```
CONFfigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:ENABle
CONFfigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:ENABle
```

Frequency Offset

Shifts the center frequency of the noise bandwidth relative to the carrier center frequency.

Remote command:

```
CONFfigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:FOFFset
CONFfigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:FOFFset
```

Min. Noise/System BW Ratio

Configures the minimum ratio between the noise bandwidth and the cell bandwidth.

Remote command:

```
CONFfigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:BWIDth:RATio
CONFfigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:BWIDth:RATio
```

Noise Bandwidth

Displays the actual noise bandwidth, resulting from the "Min. Noise/System BW Ratio" and the cell bandwidth.

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:BWIDth:NOISe?
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:BWIDth:NOISe?
```

Effective Signal BW

Displays the part of the cell bandwidth that is dedicated to the downlink subcarriers.

Signal/Noise Ratio

Specifies the signal to noise ratio.

Remote command:

```
CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:SNRatio
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:SNRatio
```

2.4.7 Downlink power levels

Access: "Config" hotkey > "Downlink Power Levels" node.

The node defines power levels of physical downlink channels and physical downlink signals.

The settings can be changed in all main connection states including "Connection Established".

For carrier aggregation scenarios, you can configure the settings per component carrier.

For background information, see also [Section 2.2.17, "Physical DL channels and signals"](#), on page 59.

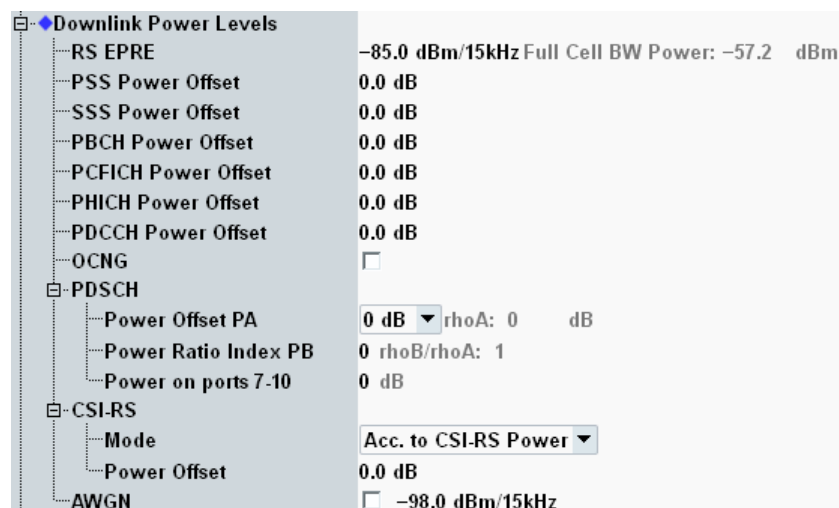


Figure 2-52: DL power levels

The reference point for all power settings in this section (except AWGN) is the simulated eNodeB antenna. For scenarios with several RF output paths per component carrier, there is a simulated radio channel between the eNodeB antennas and the RF out-

put connectors. The power at each single connector can be lower than the power at the eNodeB antenna.

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PSS Power Offset.....	158
SSS Power Offset.....	158
PBCH Power Offset.....	158
PCFICH Power Offset.....	159
PHICH Power Offset.....	159
PDCCH Power Offset.....	159
OCNG.....	159
PDSCH.....	159
CSI-RS.....	160
AWGN.....	160

RS EPRE

Defines the energy per resource element (EPRE) of the cell-specific reference signal (C-RS). The RS EPRE corresponds to the DL power averaged over all resource elements carrying cell-specific reference signals within one subcarrier (15 kHz).

The "Full Cell BW Power" resulting from the RS EPRE is also displayed. It is calculated assuming that the full DL cell bandwidth is used (all subcarriers) and all power offsets equal 0 dB.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:RSEPre:LEVel
SENSe:LTE:SIGN<i>:DL[:PCC]:FCPower?
CONFigure:LTE:SIGN<i>:DL:SCC<c>:RSEPre:LEVel
SENSe:LTE:SIGN<i>:DL:SCC<c>:FCPower?
```

PSS Power Offset

Power level of a primary synchronization signal (PSS) resource element relative to the RS EPRE.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:PSS:POFFset
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PSS:POFFset
```

SSS Power Offset

Power level of a secondary synchronization signal (SSS) resource element relative to the RS EPRE.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:SSS:POFFset
CONFigure:LTE:SIGN<i>:DL:SCC<c>:SSS:POFFset
```

PBCH Power Offset

Power level of a physical broadcast channel (PBCH) resource element relative to the RS EPRE.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:PBCH:POFFset
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PBCH:POFFset
```

PCFICH Power Offset

Power level of a physical control format indicator channel (PCFICH) resource element relative to the RS EPRE.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:PCFich:POFFset
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PCFich:POFFset
```

PHICH Power Offset

Power level of a physical hybrid ARQ indicator channel (PHICH) resource element relative to the RS EPRE.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:PHICH:POFFset
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PHICH:POFFset
```

PDCCH Power Offset

Power level of a physical downlink control channel (PDCCH) resource element relative to the RS EPRE.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:PDCCh:POFFset
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDCCh:POFFset
```

OCNG

Enables or disables the OFDMA channel noise generator (OCNG).

When the OCNG is enabled, it uses all not allocated resource blocks (RB) of the cell bandwidth, so that the full cell bandwidth is used. Example: if for a bandwidth of 10 MHz only 16 RBs are used by the RMC, the remaining 34 RBs are used by the OCNG.

The power level of the OCNG is chosen automatically so that the displayed "Full Cell BW Power" is reached. Thus the overall DL power is constant in each transmission time interval.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:OCNG
CONFigure:LTE:SIGN<i>:DL:SCC<c>:OCNG
```

PDSCH

These parameters define the power level of the physical downlink shared channel (PDSCH).

The relevant settings depend on the transmission mode:

- For transmission modes using an antenna port in the range 7 to 10, the configured antenna port power and the configured power ratio index are used. The power offset P_A is configurable, but it is only signaled to the UE. It is not used for downlink power calculation.
- For transmission modes not using these antenna ports, the configured power offset P_A and the configured power ratio index are used.

According to 3GPP TS 36.213, the power level of a PDSCH resource element relative to the RS EPRE is denoted by:

- ρ_A ("rhoA") if no RS resource elements are transmitted simultaneously on other subcarriers

- ρ_B ("rhoB") if RS resource elements are transmitted simultaneously on other sub-carriers

The power offset P_A and the power ratio index P_B are related to these ratios as follows:

- $P_A = \rho_A$
- $P_B = 0, 1, 2, 3$ corresponds to the following ρ_B/ρ_A values:
 - 1, 4/5, 3/5, 2/5 for single TX antenna configurations
 - 5/4, 1, 3/4, 1/2 for multiple TX antenna configurations

The displayed ratios "rhoA" and "rhoB" are calculated from the configured settings.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:PDSch:PA
CONFigure:LTE:SIGN<i>:DL[:PCC]:PDSch:RINdex
CONFigure:LTE:SIGN<i>:DL[:PCC]:POWer:PORTs
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDSch:PA
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDSch:RINdex
CONFigure:LTE:SIGN<i>:DL:SCC<c>:POWer:PORTs
```

CSI-RS

These parameters define the CSI-RS power offset for TM 9.

"Acc. to CSI-RS Power"	The used power offset matches the signaled value. The parameter "Power Offset" is ignored. For configuration of the signaled value, see "CSI-RS Power" on page 219.
"Manual"	The used power offset is configured via the parameter "Power Offset". The setting is independent from the signaled power offset.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:CSIRs:MODE
CONFigure:LTE:SIGN<i>:DL[:PCC]:CSIRs:POFFset
CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:MODE
CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:POFFset
```

AWGN

Total level of the additive white Gaussian noise (AWGN) interferer in dBm/15 kHz (the spectral density integrated across one subcarrier). If enabled, the AWGN signal is added to the DL LTE signal for the entire cell bandwidth.

For fading scenarios, this parameter is disabled, so that AWGN cannot be added by the signaling unit. Instead AWGN can be added by the fader (external or internal).

The reference point for this power setting is the RF output connector.

Option R&S CMW-KS510 required.

Remote command:

```
CONFigure:LTE:SIGN<i>:DL[:PCC]:AWGN
CONFigure:LTE:SIGN<i>:DL:SCC<c>:AWGN
```

2.4.8 Uplink power control

The settings in this section define parameters related to the control of the UE uplink power by the instrument.

2.4.8.1 General power control parameters

Access: "Config" hotkey > "Uplink Power Control" node.

At the highest level of the node, some general power control settings are available.

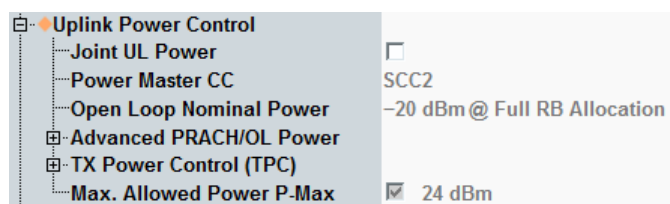


Figure 2-53: UL power control settings, highest level

Joint UL Power, Power Master CC

The parameter "Joint UL Power" is only available, if at least one SCC uplink is enabled.

Checkbox states:

- Enabled: The power of all uplinks is controlled via the PCC settings. All uplinks have the same power.
- Disabled: The power of at least one uplink can be controlled independently of the other uplinks.

Whether two uplinks can be controlled independently, depends for example on the distance of the uplink frequencies. If all uplinks must have the same power, the parameter "Joint UL Power" is enabled and grayed out. Otherwise, it is configurable.

If the UL power of a carrier cannot be controlled independently, the dependent uplink power control settings of that carrier are grayed out and show the primary carrier settings. The parameter "Power Master CC" displays which carrier is used as primary carrier.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:JUPower
```

```
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PMCC?
```

Max. allowed Power P-Max

Specifies the maximum allowed output power for the UE in the cell. The UE output power must not exceed this value. And it must not exceed the maximum power value resulting from the UE power class.

If the checkbox is enabled, the configured value is signaled to the UE within the system information. If it is disabled, the parameter has no effect.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:PMAX
```

```
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PMAX
```

```

CONFigure:LTE:SIGN<i>:UL:SETA:PMAX
CONFigure:LTE:SIGN<i>:UL:SETB:PMAX
CONFigure:LTE:SIGN<i>:UL:SETC:PMAX

```

2.4.8.2 PRACH and initial PUSCH power

Access: "Config" hotkey > "Uplink Power Control" node.

The following parameters configure the PRACH power and the initial PUSCH power.

By default, the basic settings apply (see [Figure 2-54](#)). Only the "Open Loop Nominal Power" is configurable. The advanced settings are not configurable and the grayed-out values indicate the used basic settings.

If you enable the advanced settings, the grayed-out values become configurable. Instead, the "Open Loop Nominal Power" parameter is disabled.

For carrier aggregation scenarios, you can configure the settings per uplink, if joint UL power control is disabled.

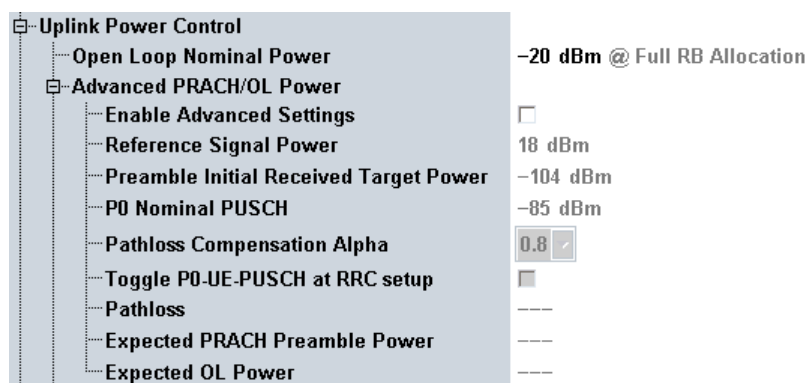


Figure 2-54: OL and PRACH power control

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P0 Nominal PUSCH.....	164
Pathloss Compensation Alpha.....	164
Toggle P0-UE-PUSCH at RRC Setup.....	165
Pathloss.....	165
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Expected OL Power.....	166

Open Loop Nominal Power

Defines a cell-specific nominal power value for full resource block allocation in the UL (entire cell bandwidth used). From this value, the cell-specific nominal power value related to one resource block is determined and sent to all UEs via broadcast.

The UE uses this value for calculation of the average power of an SC-FDMA symbol in which the PUSCH is transmitted. The calculation is described in 3GPP TS 36.213 (see $P_{O_NOMINAL_PUSCH}$).

This power control procedure is performed during connection setup. Afterwards, the power remains constant until changed by TPC commands.

The parameter is only relevant and configurable if the "Advanced PRACH/OL Power" settings are disabled.

Remote command:

```
CONFfigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:OLNPower
CONFfigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:OLNPower
CONFfigure:LTE:SIGN<i>:UL:SETA:PUSCh:OLNPower
CONFfigure:LTE:SIGN<i>:UL:SETB:PUSCh:OLNPower
CONFfigure:LTE:SIGN<i>:UL:SETC:PUSCh:OLNPower
```

Enable Advanced Settings

Enables configuration of the advanced parameters.

If disabled, the following parameters are grayed out and display the used basic settings.

Remote command:

```
CONFfigure:LTE:SIGN<i>:UL[:PCC]:APPower:EASettings
CONFfigure:LTE:SIGN<i>:UL:SCC<c>:APPower:EASettings
CONFfigure:LTE:SIGN<i>:UL:SETA:APPower:EASettings
CONFfigure:LTE:SIGN<i>:UL:SETB:APPower:EASettings
CONFfigure:LTE:SIGN<i>:UL:SETC:APPower:EASettings
```

Reference Signal Power

Specifies the parameter "referenceSignalPower", signaled to the UE as PDSCH configuration parameter (see 3GPP TS 36.331, section 6.3.2).

The value is used by the UE to determine the pathloss, see 3GPP TS 36.213 section 5.1.1.1.

Remote command:

```
CONFfigure:LTE:SIGN<i>:UL[:PCC]:APPower:RSPower:ADVanced
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:RSPower:BASic?
CONFfigure:LTE:SIGN<i>:UL:SCC<c>:APPower:RSPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:RSPower:BASic?
CONFfigure:LTE:SIGN<i>:UL:SETA:APPower:RSPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SETA:APPower:RSPower:BASic?
CONFfigure:LTE:SIGN<i>:UL:SETB:APPower:RSPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SETB:APPower:RSPower:BASic?
CONFfigure:LTE:SIGN<i>:UL:SETC:APPower:RSPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SETC:APPower:RSPower:BASic?
```

Preamble Initial Received Target Power

Specifies the parameter "preambleInitialReceivedTargetPower", signaled to the UE as common RACH parameter (see 3GPP TS 36.331, section 6.3.2).

In 3GPP TS 36.213, section 5.1.1.1 this parameter is called P_{O_PRE} .

The value is used by the UE, for example, to calculate the power of the first preamble.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PIRPower:ADVanced
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PIRPower:BASic?
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PIRPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PIRPower:BASic?
CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PIRPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PIRPower:BASic?
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PIRPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PIRPower:BASic?
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PIRPower:ADVanced
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PIRPower:BASic?
```

P0 Nominal PUSCH

Specifies the parameter "p0-NominalPUSCH", signaled to the UE as uplink power control parameter (see 3GPP TS 36.331, section 6.3.2).

In 3GPP TS 36.213, section 5.1.1.1 this parameter is called $P_{O_NOMINAL_PUSCH}$.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PNPusch:ADVanced
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PNPusch:BASic?
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PNPusch:ADVanced
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PNPusch:BASic?
CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PNPusch:ADVanced
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PNPusch:BASic?
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PNPusch:ADVanced
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PNPusch:BASic?
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PNPusch:ADVanced
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PNPusch:BASic?
```

Pathloss Compensation Alpha

Specifies the parameter "alpha", signaled to the UE as uplink power control parameter (see 3GPP TS 36.331, section 6.3.2).

In 3GPP TS 36.213, section 5.1.1.1 this parameter is called α .

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PCALpha:ADVanced
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PCALpha:BASic?
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PCALpha:ADVanced
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PCALpha:BASic?
CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PCALpha:ADVanced
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PCALpha:BASic?
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PCALpha:ADVanced
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PCALpha:BASic?
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PCALpha:ADVanced
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PCALpha:BASic?
```


Toggle P0-UE-PUSCH at RRC Setup

With enabled toggling, the following P0-UE-PUSCH values are set during RRC connection setup:

- RRC connection setup message (SRB1): P0-UE-PUSCH = 1 dB
- RRC connection reconfiguration message (SRB2+DRB): P0-UE-PUSCH = 0 dB

P0-UE-PUSCH toggling is required for some 3GPP conformance tests, for example 3GPP TS 36.521, section 6.3.5.1 "Power Control Absolute power tolerance", table 6.3.5.1.4.3-4.

The toggling initiates a "TPC accumulation reset" at the UE ($f_c(i)$ is set to 0, see 3GPP TS 36.213, section 5.1.1.1).

Note: Toggling deletes a PRACH ramp-up procedure. Disable toggling if a PRACH ramp-up procedure is required for decoding of the UE signal. Otherwise, the RRC connection setup fails.

Note: Use toggling only if the external attenuation is compensated accurately. Otherwise, do not use toggling, for example for over-the-air tests with fluctuating external attenuation.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:TPRRcsetup:ADVanced
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:TPRRcsetup:BASic?
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:TPRRcsetup:ADVanced
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:TPRRcsetup:BASic?
CONFigure:LTE:SIGN<i>:UL:SETA:APPower:TPRRcsetup:ADVanced
SENSe:LTE:SIGN<i>:UL:SETA:APPower:TPRRcsetup:BASic?
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:TPRRcsetup:ADVanced
SENSe:LTE:SIGN<i>:UL:SETB:APPower:TPRRcsetup:BASic?
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:TPRRcsetup:ADVanced
SENSe:LTE:SIGN<i>:UL:SETC:APPower:TPRRcsetup:BASic?
```

Pathloss

Displays the pathloss resulting from the "Reference Signal Power" on page 163 and the "RS EPRE" on page 158.

Remote command:

```
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PATHloss?
```

Expected PRACH Preamble Power

Displays the expected power of the first preamble.

The value depends on the "preambleInitialReceivedTargetPower" ("Preamble Initial Received Target Power" on page 163) and on the preamble format resulting from the configuration index ("Configuration Index" on page 176).

For details, see 3GPP TS 36.321 section 5.1.

Remote command:

```
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:EPPower?
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:EPPower?
SENSe:LTE:SIGN<i>:UL:SETA:APPower:EPPower?
SENSe:LTE:SIGN<i>:UL:SETB:APPower:EPPower?
SENSe:LTE:SIGN<i>:UL:SETC:APPower:EPPower?
```

Expected OL Power

Displays the expected initial PUSCH power. Most of the advanced power settings influence this value.

You can use the displayed values to verify your configuration: With a correct configuration, the expected PRACH power is not larger than the expected OL power. It is also not very much lower. Otherwise the PRACH cannot be detected and the attach procedure fails.

Remote command:

```
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:EOPower?
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:EOPower?
SENSe:LTE:SIGN<i>:UL:SETA:APPower:EOPower?
SENSe:LTE:SIGN<i>:UL:SETB:APPower:EOPower?
SENSe:LTE:SIGN<i>:UL:SETC:APPower:EOPower?
```

2.4.8.3 TX power control (TPC)

Access: "Config" hotkey > "Uplink Power Control" node > "TX Power Control (TPC)" node.

The node configures settings for UL power control via TPC commands.

For carrier aggregation scenarios, you can configure the settings per uplink, if joint UL power control is disabled.

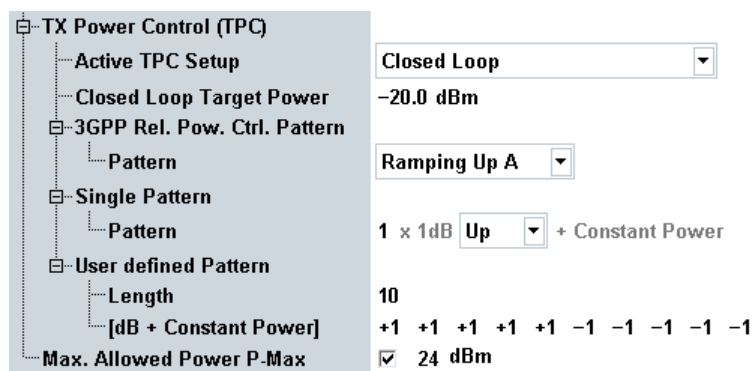


Figure 2-55: TPC power control

Active TPC Setup.....	167
Target Power PCC / SCC.....	168
3GPP Rel. Pow. Ctrl. Pattern.....	168

Closed Loop Target Power.....	169
Single Pattern.....	169
User-Defined Pattern.....	169

Active TPC Setup

Selects the TPC setup to be executed. All setups use relative (accumulative) TPC commands and are only executed when a connection has been established. During connection setup, 0-dB commands are sent.

The string "Adjusted" behind a node of settings indicates that the currently selected setup is further configured by this node.

- **"Min Power", "Max Power", "Constant Power"**

The UE is commanded to maximum power, to minimum power or is ordered to keep the UL power constant.

If one of these setups is selected, it is executed automatically without additional user action.

- **"3GPP Relative Power Control"**

Ramping up or down test pattern according to 3GPP TS 36.521, section 6.3.5.2.

Select the test pattern via "3GPP Rel. Pow. Ctrl. Pattern" and start the pattern via the "Execute" button.

For details about the test pattern, see [Relative power control tests](#).

This TPC setup is not allowed if "Joint UL Power" is enabled. It is also not allowed for the eMTC scheduling types "Compact Scheduling" and "Auto Mode".

- **"Single Pattern", "User-defined Single Pattern"**

A configurable TPC pattern is sent to the UE when the "Execute" button is pressed. Before and after execution of the pattern, the power is kept constant.

A single pattern contains only one type of TPC command while a user-defined single pattern allows you to combine different TPC commands.

- **"User-defined Continuous Pattern"**

A configurable user-defined TPC pattern is sent to the UE continuously. If this setup is selected, it is executed automatically without additional user action.

While this setup is selected, the relevant parameter for pattern configuration is marked by the string "Adjusted".

- **"Alternating +1, -1, ..."**

An alternating TPC pattern of plus and minus 1-dB commands is sent to the UE continuously. If this setup is selected, it is executed automatically without additional user action.

- **"Closed Loop"**

The UE is commanded to the configured target power. Plus 1-dB or minus 1-dB commands are sent until the difference between the measured UL power and the target power is less than 1 dB. Afterwards, the power is kept constant (0-dB commands).

This setup is executed automatically without additional user action. Even the measurement of the uplink power is performed automatically in the background. No additional measurement application is needed.

- **"Flexible UL Power"**

This setup is only available for uplink carrier aggregation, if "Joint UL Power" is grayed-out. The setup configures different PCC and SCC uplink powers for some seconds. The duration is long enough for a measurement, but short enough to pre-

vent a synchronization error. A synchronization error can nevertheless occur if you decrease the "Out of Sync" timeout value in the network settings.

The following parameters are related to this TPC setup.

TX Power Control (TPC)	
Active TPC Setup	Flexible UL Power Execute
Closed Loop Target Power	-30.0 dBm
Target Power PCC	-20.0 dBm
Target Power SCC1	-20.0 dBm

When you select this TPC setup, the UE is commanded to the closed loop target power. If you press the "Execute" button, the UE is commanded to the PCC and SCC target powers. After some seconds, it is commanded back to the closed loop target power.

For a combined signal path measurement while the two powers are different, configure your measurement as single-shot measurement with a small statistic count.

Use the TPC trigger (SCC) to start the measurement.

Depending on the current configuration, some values are grayed out. The reason is displayed in that case, for example active PUCCH scheduling (number of RBs = 0). TPC setups for which skipping a bit is no problem, are always available (for example max power or closed loop, in contrast to single or alternating pattern).

If you change the configuration so that the currently active setup is no longer allowed, the TPC setup is corrected automatically and a notice is displayed.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:SET
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:SET
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:SET
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:SET
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:SET
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:PEXecute
```

Target Power PCC / SCC

Defines the target powers for the TPC setup "Flexible UL Power".

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:TPOWer
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:TPOWer
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:TPOWer
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:TPOWer
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:TPOWer
```

3GPP Rel. Pow. Ctrl. Pattern

Selects the pattern to be executed for the TPC setup "3GPP Relative Power Control". The available up and down patterns are defined in 3GPP TS 36.521, section 6.3.5.2.

See also [Relative power control tests](#).

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:RPControl
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:RPControl
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:RPControl
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:RPControl
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:RPControl
```

Closed Loop Target Power

Defines the target power for the TPC setup "Closed Loop". The target power applies to PUCCH and PUSCH.

For carriers with a power primary carrier, the additional setting "Offset" is displayed. It shifts the target power relative to the target power of the power primary carrier. See also ["Joint UL Power, Power Master CC"](#) on page 161.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:CLTPower
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:CLTPower
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:CLTPower:OFFSet
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:CLTPower
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:CLTPower:OFFSet
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:CLTPower
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:CLTPower:OFFSet
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:CLTPower
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:CLTPower:OFFSet
```

Single Pattern

Defines a pattern for the TPC setup "Single Pattern". The pattern consists of up to 35 up (+1 dB) or down (-1 dB) commands.

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:SINGLE
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:SINGLE
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:SINGLE
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:SINGLE
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:SINGLE
```

User-Defined Pattern

Defines a pattern for the TPC setups "User-defined Single Pattern" and "User-defined Continuous Pattern". The pattern consists of up to 20 TPC commands (-1 / 0 / +1 / +3 dB).

Remote command:

```
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:UDPattern
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:UDPattern
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:UDPattern
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:UDPattern
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:UDPattern
```

2.4.9 Physical cell setup

Access: "Config" hotkey > "Physical Cell Setup" node.

The node defines physical layer attributes of the simulated cell.

For background information, see [Section 2.2.17, "Physical DL channels and signals"](#), on page 59.

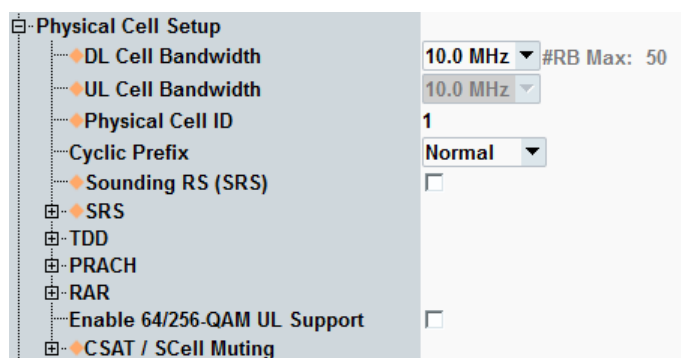


Figure 2-56: Cell parameters

• General cell setup settings	170
• SRS settings	171
• TDD settings	174
• PRACH settings	175
• RAR settings	177
• CSAT / SCell muting settings	177

2.4.9.1 General cell setup settings

Access: "Config" hotkey > "Physical Cell Setup" node.

This section describes the following settings at the highest level of the node.

DL / UL Cell Bandwidth	170
Physical Cell ID	171
Cyclic Prefix	171
Enable 64/256-QAM UL Support	171

DL / UL Cell Bandwidth

Define the DL and UL cell bandwidths, also called channel bandwidth by 3GPP. In the current release, the two values are identical. For frame structure type 3, only 20 MHz are allowed.

The resulting maximum number of DL resource blocks is indicated for information. It is smaller than the DL bandwidth divided by 180 kHz, because some space at the channel borders must not be occupied by resource blocks. 3GPP defines the relation as follows:

Cell bandwidth / MHz	1.4	3	5	10	15	20
Maximum no of RBs	6	15	25	50	75	100

You can change the cell bandwidth in all main connection states. In the state "Connection Established", you can either change it directly or you can perform a handover to reconfigure several parameters, see [Section 2.2.10, "Handover"](#), on page 42.

For carrier aggregation scenarios, you can configure different DL cell bandwidths for the carriers.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL
CONFigure:LTE:SIGN<i>:CELL:BANDwidth:SCC<c>:DL
CONFigure:LTE:SIGN<i>:RFSettings:ALL:BWChannel
```

Physical Cell ID

The cell ID is used for generation of the physical synchronization signals. During cell search, the UE determines the cell ID from the primary and secondary synchronization signal.

The physical cell ID can be set independent of the E-UTRAN cell identifier sent to the UE via broadcast (see ["E-UTRAN Cell Identifier"](#) on page 184).

If you use carrier aggregation, configure different values for the component carriers.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:PCID
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:PCID
```

Cyclic Prefix

Defines whether a normal or extended cyclic prefix (CP) is used.

An extended cyclic prefix cannot be combined with the transmission mode 8 or with frame structure type 3 (combination not foreseen by 3GPP).

Option R&S CMW-KS510 is required for extended CP.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL:CPRefix
```

Enable 64/256-QAM UL Support

Selects whether 64-QAM and 256-QAM are allowed in the uplink or not. This information is signaled to the UE in the IEs "enable64QAM..." and "enable256QAM...".

The setting affects also the channel configuration. If the checkbox is disabled, you cannot configure 64-QAM or 256-QAM for the uplink.

To enable the checkbox, you need option R&S CMW-KS504/-KS554 and two SUA partitions for the carrier, see ["Base Band Unit"](#) on page 141.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULSupport:QAM<m>:ENABLE
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULSupport:QAM<m>:ENABLE
```

2.4.9.2 SRS settings

Access: "Config" hotkey > "Physical Cell Setup" node > "SRS" node.

The following cell settings are related to sounding RS (SRS). You can enable SRS and configure SRS parameters to be signaled to the UE.

Uplink SRS signals can be measured using the "LTE SRS" measurement included in option R&S CMW-KM500/-KM550.

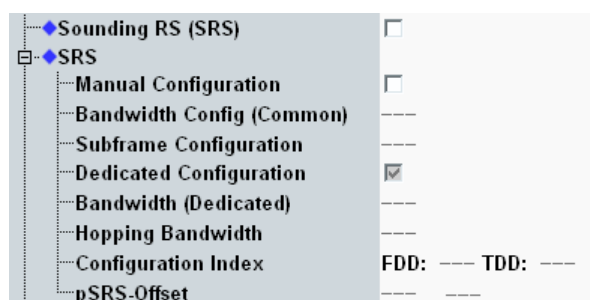


Figure 2-57: SRS settings

Sounding RS (SRS).....	172
SRS.....	172
L Manual Configuration.....	172
L Bandwidth Config (Common).....	173
L Subframe Configuration.....	173
L Dedicated Configuration.....	173
L Bandwidth (Dedicated).....	173
L Hopping Bandwidth.....	173
L Configuration Index.....	174
L pSRS-Offset.....	174

Sounding RS (SRS)

Enables support of SRS. The information that the cell supports SRS is then broadcasted to the UE. During connection setup, the UE is requested to send a sounding reference signal.

For carrier aggregation scenarios, you can configure this setting per component carrier.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:ENABle
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:ENABle
```

SRS

This node configures SRS parameters signaled to the UE. Their values can be set automatically or manually.

For carrier aggregation scenarios, you can configure the settings per component carrier.

Option R&S CMW-KS510 is required (except for eMTC).

Manual Configuration ← SRS

With disabled "Manual Configuration", the following values are used:

- "srs-BandwidthConfig": 7
- "srs-SubframeConfig": 3 for FDD, 0 for TDD
- "srs-Bandwidth": 3
- "srs-HoppingBandwidth": 3
- "srs-ConfigIndex": 7 for FDD, 0 for TDD
- "pSRS-Offset": 3

With enabled "Manual Configuration", the signaled values are configured manually via the following parameters.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:MCENable
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:MCENable
```

Bandwidth Config (Common) ← SRS

Configures the parameter "srs-BandwidthConfig" manually.

This cell-specific value influences the mapping to physical resources. The meaning of the individual values is specified in 3GPP TS 36.211, section 5.5.3.2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:BWConfig
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:BWConfig
```

Subframe Configuration ← SRS

Configures the parameter "srs-SubframeConfig" manually.

The value determines the cell-specific SRS timing. The meaning of the individual values is specified in 3GPP TS 36.211, section 5.5.3.3.

At time instances with cell-specific SRS but without UE-specific SRS, the UE sends a "blank SRS symbol" (and another UE sends an SRS symbol).

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SFCConfig
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SFCConfig
```

Dedicated Configuration ← SRS

Selects whether the UE-specific SRS parameters are signaled to the UE or not.

The UE-specific parameters are also called dedicated parameters (dedicated to one UE). They are configured via the remaining settings of this node.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:DConfig
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:DConfig
```

Bandwidth (Dedicated) ← SRS

Configures the parameter "srs-Bandwidth" manually.

This UE-specific value influences the mapping to physical resources. The meaning of the individual values is specified in 3GPP TS 36.211, section 5.5.3.2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:DBANDwidth
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:DBANDwidth
```

Hopping Bandwidth ← SRS

Configures the parameter "srs-HoppingBandwidth" manually.

The value determines the frequency hopping for periodic SRS transmission. The meaning of the individual values is specified in 3GPP TS 36.211, section 5.5.3.2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:HBANdwidth
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:HBANdwidth
```

Configuration Index ← SRS

Configures the parameter "srs-ConfigIndex" manually.

The value determines the UE-specific SRS timing. The meaning of the individual values is specified in 3GPP TS 36.213, section 8.2.

At time instances with configured UE-specific SRS, the UE sends an SRS symbol. Other UEs in the cell send a "blank SRS symbol".

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SCINdex:FDD
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SCINdex:TDD
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SCINdex:FDD
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SCINdex:TDD
```

pSRS-Offset ← SRS

Configures the parameter "pSRS-Offset" manually (child of field "UplinkPowerControl-Dedicated"). The UE translates the signaled value into a dB value. The dB value corresponding to the configured signaled value is displayed for information.

The setting influences the SRS power as described in 3GPP TS 36.213 section 5.1.3.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:POFFset
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:POFFset
```

2.4.9.3 TDD settings

Access: "Config" hotkey > "Physical Cell Setup" node > "TDD" node.

The node configures TDD-specific settings, not relevant for FDD.

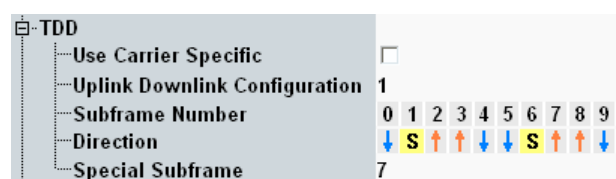


Figure 2-58: TDD settings

Use Carrier Specific.....	174
Uplink Downlink Configuration.....	175
Special Subframe.....	175

Use Carrier Specific

Selects whether the UL/DL configuration and the special subframe configuration can be configured individually per component carrier or whether the same settings apply to all carriers.

Option R&S CMW-KS512 is required for carrier-specific configuration.

Remote command:

CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific

Uplink Downlink Configuration

Selects the UL-DL configuration of a TDD signal, see [Table 2-28](#). Each configuration defines a combination of uplink subframes, downlink subframes and special subframes within a radio frame.

If duplex mode TDD is active, the radio frame structure is displayed for information (lines "Subframe Number" and "Direction").

If a ☹ is displayed for a special subframe, it indicates that there is no PDSCH transmission in the DwPTS. This state depends, for example, on the special subframe configuration and on the cyclic prefix.

Option R&S CMW-KS510 and R&S CMW-KS550 are required.

Remote command:

CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULDL

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULDL

Special Subframe

Configuration of the special subframes of a TDD signal, as defined in 3GPP TS 36.211, section 4, "Frame Structure". Each configuration defines the inner structure of a special subframe, i.e. the lengths of DwPTS, GP and UpPTS.

Value 8 and 9 can only be used with normal cyclic prefix.

Option R&S CMW-KS550 is required. R&S CMW-KS512 is required for value 7 combined with extended cyclic prefix and for value 9.

Remote command:

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SSUBframe

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SSUBframe

2.4.9.4 PRACH settings

Access: "Config" hotkey > "Physical Cell Setup" node > "PRACH" node.

The node configures settings related to the random access procedure.

Setting	Value
No Response to Preambles	Off
# ignored Preambles	1
Power Ramping Step	2 dB
Configuration Index	FDD: 12 TDD: 12
Frequency Offset	0
Logical Root Sequ.Idx	123
Zero Corr. Zone Conf.	9

Figure 2-59: PRACH settings

No Response to Preambles, # Ignored Preambles.....	176
Power Ramping Step.....	176
Configuration Index.....	176

Frequency Offset.....	176
Logical Root Sequ.Idx.....	177
Zero Corr. Zone Conf.....	177

No Response to Preambles, # Ignored Preambles

By default the signaling application responds to received preambles, so that the UE can perform an attach. Alternatively, received preambles can be ignored (no response), so that the UE continues to send preambles and PRACH measurements can be performed (included in R&S CMW-KM500/-KM550).

"Off"	The signaling application responds to received preambles.
"On"	Any received preambles are ignored by the signaling application (no response).
"# Ignored Preambles"	The number of preambles to be ignored by the signaling application is configured via parameter "# Ignored Preambles". Subsequent preambles are answered. This value can only be selected if parameter "Power Ramping Step" is set to 0 dB.

Remote command:

CONFigure:LTE:SIGN<i>:CELL:PRACH:NRPPreambles

CONFigure:LTE:SIGN<i>:CELL:PRACH:NIPRach

Power Ramping Step

Specifies the transmit power difference between two consecutive preambles. This value is broadcasted to the UE.

If you set the step size to 0 dB, the preamble power is kept constant by the UE. Thus you can use a constant expected nominal power setting during a PRACH measurement.

Remote command:

CONFigure:LTE:SIGN<i>:CELL:PRACH:PRSTep

Configuration Index

Sets the PRACH configuration index. It defines the preamble format and other PRACH signal properties, e.g. which resources in the time domain are allowed for transmission of preambles. This value is broadcasted to the UE.

There are separate settings for FDD and TDD (if both R&S CMW-KS500 and R&S CMW-KS550 are available). The values allowed for TDD depend on the UL-DL configuration as listed in the command description.

For details, see 3GPP TS 36.211, section 5.7.1 "Time and frequency structure".

If you enable eMTC, the FDD configuration index is set to 3.

Remote command:

CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:FDD

CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:TDD

Frequency Offset

The frequency offset is used by the UE to calculate the location of the preamble resource blocks within the cell bandwidth. The value is broadcasted to the UE.

For details, see "prach-FrequencyOffset" in 3GPP TS 36.211, section 5.7.1 "Time and frequency structure".

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:PRACH:PFOffset`

Logical Root Sequ.Idx

Specifies the logical root sequence index to be used by the UE for generation of the preamble sequence. The value is broadcasted to the UE as RACH_ROOT_SEQUENCE.

For details see 3GPP TS 36.211, section 5.7.2 "Preamble sequence generation".

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:PRACH:LRSIndex`

Zero Corr. Zone Conf.

The zero correlation zone config determines which N_{CS} value of an N_{CS} set has to be used by the UE for generation of the preamble sequence. The value is broadcasted to the UE.

For details, see "zeroCorrelationZoneConfig" in 3GPP TS 36.211, section 5.7.2 "Preamble sequence generation".

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:PRACH:ZCZConfig`

2.4.9.5 RAR settings

Access: "Config" hotkey > "Physical Cell Setup" node > "RAR" node.

The node configures settings related to the random access response (RAR) message sent to the UE after reception of a preamble.

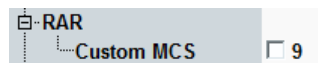


Figure 2-60: RAR settings

Custom MCS

Specifies the MCS index value for RAR messages. With disabled checkbox, the MCS index is selected automatically. With enabled checkbox, the entered MCS index is used.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:RAR:CMCS:ENABLE`

`CONFigure:LTE:SIGN<i>:CELL:RAR:CMCS`

2.4.9.6 CSAT / SCell muting settings

Access: "Config" hotkey > "SCC" tab > "Physical Cell Setup" node > "CSAT / SCell Muting" node.

Carrier-sensing adaptive transmission (CSAT) is defined by the LTE-U Forum for the SCC DL in unlicensed bands. With CSAT, the DL transmission is adapted depending on how busy the channel is. Furthermore, regular signal OFF periods allow others to use the channel.

In an OFF period, the SCC DL (SCell) transmits nothing, not even reference or synchronization signals. In an ON period, it transmits reference signals and user data or the MIB or an LTE-U discovery signal (LDS).

To simulate an SCC DL signal with CSAT, the signaling application supports LDS transmission and SCell muting with alternating ON and OFF periods. You can configure the ON and OFF duration and the periodicity of the LDS transmission. Option R&S CMW-KS525 is required.

You can use the feature for any bands, not only for unlicensed bands. The supported ON and OFF durations are more flexible than defined by the LTE-U Forum. The MIB is transmitted with a fixed periodicity of 160 ms.

CSAT and SCell muting cannot be combined with frame structure type 3.

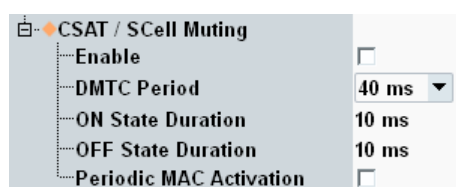


Figure 2-61: CSAT settings

Enable.....	178
DMTC Period.....	178
ON State Duration / OFF State Duration.....	178
Periodic MAC Activation.....	179

Enable

Enables CSAT and SCell muting.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CSAT:ENABle`

DMTC Period

The signaling application transmits an LTE-U discovery signal (LDS) with the configured periodicity. The information is signaled to the UE in the R12 DRS DMTC IE.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CSAT:DMTCperiod`

ON State Duration / OFF State Duration

The SCC DL signal is alternating ON and OFF. The settings define the duration of the ON and OFF periods.

The actual ON state duration can be greater than configured, if an LDS or MIB transmission occurs in a subframe immediately after the configured ON period.

To ensure a DL signal during ON periods, you can for example use MAC padding, see ["Downlink MAC Padding"](#) on page 210.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:ONSDuration
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:OFFSduration
```

Periodic MAC Activation

Enable the checkbox if you want to activate / deactivate the MAC for each ON / OFF period. With disabled checkbox, the MAC is kept enabled.

With enabled checkbox, the signaling application sends an activate message eight subframes before the start of each ON period. It sends a deactivate message eight subframes before the start of each OFF period.

Remote command:

```
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:PMAC
```

2.4.10 Network settings

Access: "Config" hotkey > "Network" node.

The node configures parameters of the simulated radio network.

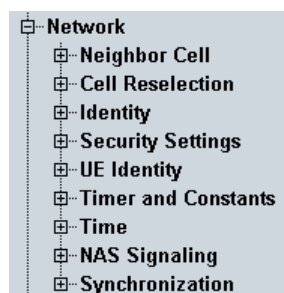


Figure 2-62: Network settings

For parameter descriptions, refer to the following sections.

• Neighbor cell settings	179
• Cell reselection	182
• Identity settings	183
• Security settings	184
• UE identity	186
• Timer and constants	187
• Time	188
• NAS signaling settings	189
• Synchronization	192

2.4.10.1 Neighbor cell settings

Access: "Config" hotkey > "Network" node > "Neighbor Cell" node.

The node defines neighbor cell information to be broadcasted to the UE. For each radio access technology, you can define a reselection threshold and several neighbor

cell entries. The signaling messages for broadcast of neighbor cell information are defined in 3GPP TS 36.331.

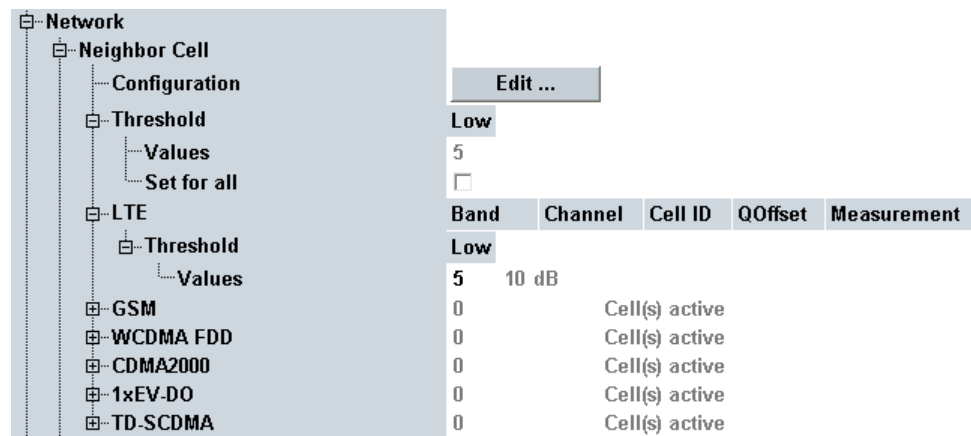


Figure 2-63: Neighbor cell settings

To configure the neighbor cell entries, press the "Edit" button. The configuration dialog contains one tab per technology.

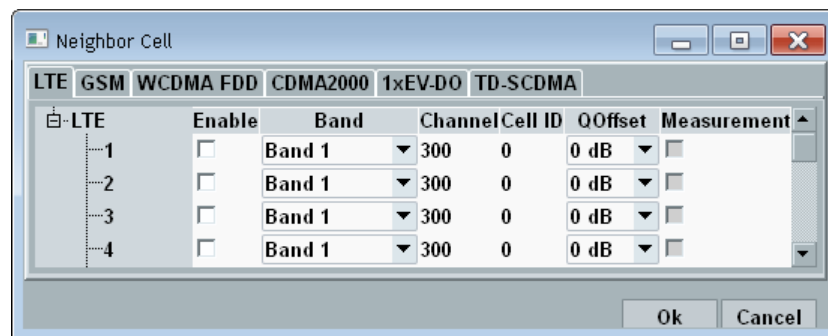


Figure 2-64: Neighbor cell list configuration dialog

Only the enabled entries are broadcasted.

The column "Measurement" is only configurable for enabled entries. It specifies whether measurement reports for the neighbor cell are requested from the UE. Received reports are displayed in the main view, see [Section 2.4.1.3, "UE measurement report"](#), on page 121. Option R&S CMW-KS510 is required for neighbor cell measurements.

The individual neighbor cell settings are described in the following.

Threshold.....	181
LTE.....	181
GSM.....	181
WCDMA FDD.....	181
CDMA2000, 1xEV-DO.....	182
TD-SCDMA.....	182

Threshold

The configured "Low" reselection threshold value is written into the system information block element "threshX-Low" defined in 3GPP TS 36.331. It corresponds to the parameter "Thresh_{X, LowP}" in 3GPP TS 36.304. The resulting threshold value in dB is displayed for information.

You can define an individual threshold per technology or a common threshold applicable to all technologies. To apply common thresholds, enable "Threshold > Set for all". To apply the individual thresholds, disable the parameter.

Remote command:

```
CONFigure:LTE:SIGN<i>:NCELL:ALL:THResholds:LOW
CONFigure:LTE:SIGN<i>:NCELL:LTE:THResholds:LOW
CONFigure:LTE:SIGN<i>:NCELL:GSM:THResholds:LOW
CONFigure:LTE:SIGN<i>:NCELL:WCDma:THResholds:LOW
CONFigure:LTE:SIGN<i>:NCELL:CDMA:THResholds:LOW
CONFigure:LTE:SIGN<i>:NCELL:EVDO:THResholds:LOW
CONFigure:LTE:SIGN<i>:NCELL:TDSCdma:THResholds:LOW
```

LTE

For an LTE (E-UTRA) neighbor cell entry, you can specify the operating band and downlink channel number, the physical layer cell ID and the "QOffset".

Parameter "QOffset" corresponds to the value "q-OffsetCell" in 3GPP TS 36.331, which equals "Qoffset_{s,n}" in 3GPP TS 36.304. It is used by the UE when evaluating candidates for cell reselection or triggering conditions for measurement reporting.

If the channel number of an entry and the currently used channel number are identical, the parameters are written into system information block 4 (intra-frequency cell reselection). Otherwise they are written into system information block 5 (inter-frequency cell reselection).

Reselection thresholds defined for LTE are only relevant for system information block 5.

The list supports up to 16 active neighbor cell entries. The active entries can use up to five different channel numbers. Active entries with the same channel number must have different cell IDs.

Remote command:

```
CONFigure:LTE:SIGN<i>:NCELL:LTE:CELL<n>
```

GSM

For a GSM (GERAN) neighbor cell entry, you can specify the operating band and the channel number used for the broadcast control channel (BCCH).

This information and the GSM reselection thresholds are written into system information block 7.

Remote command:

```
CONFigure:LTE:SIGN<i>:NCELL:GSM:CELL<n>
```

WCDMA FDD

For a WCDMA (UTRA FDD) neighbor cell entry, you can specify the operating band, the downlink channel number and the primary scrambling code of the cell.

The channel number and the WCDMA reselection thresholds are written into system information block 6.

Remote command:

```
CONFigure:LTE:SIGN<i>:NCELL:WCDMa:CELL<n>
```

CDMA2000, 1xEV-DO

For a CDMA2000 (1xRTT) or 1xEV-DO (HRPD) neighbor cell entry, you can specify the band class, the channel number and the physical cell ID which identifies the PN offset.

This information and the reselection thresholds, are written into system information block 8 - the CDMA2000 parameters into element "parameters1XRTT", the 1xEV-DO parameters into element "parametersHRPD".

Remote command:

```
CONFigure:LTE:SIGN<i>:NCELL:CDMA:CELL<n>
```

```
CONFigure:LTE:SIGN<i>:NCELL:EVDO:CELL<n>
```

TD-SCDMA

For a TD-SCDMA (UTRA TDD) neighbor cell entry, you can specify the operating band, the channel number and the cell parameter ID.

The channel number and the TD-SCDMA reselection thresholds are written into system information block 6.

Remote command:

```
CONFigure:LTE:SIGN<i>:NCELL:TDSCdma:CELL<n>
```

2.4.10.2 Cell reselection

Access: "Config" hotkey > "Network" node > "Cell Reselection" node.

The node defines cell reselection information to be transmitted in the system information blocks SIB1 and SIB3. For detailed information, refer to 3GPP TS 36.304 and 3GPP TS 36.331.

The settings are only visible if R&S CMW-KS510 is available.

Cell Reselection	
S IntraSearch	☒ 32 dB
S NonIntraSearch	☒ 32 dB
ThreshServingLow	16 dB
Q rxlevmin	-132 dBm

Figure 2-65: Settings for cell reselection

S IntraSearch.....	182
S NonIntraSearch.....	183
ThreshServingLow.....	183
Q rxlevmin.....	183

S IntraSearch

Threshold $S_{\text{IntraSearch}}$ for intra-frequency measurements.

The threshold is configured in dB. System information block 3 contains the configured value divided by 2. If the checkbox is disabled, the information element is omitted in SIB 3.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:RESelection:SEARCh:INTRasearch`

S NonIntraSearch

Threshold $S_{\text{nonIntraSearch}}$ for inter-frequency and inter-RAT measurements.

The threshold is configured in dB. System information block 3 contains the configured value divided by 2. If the checkbox is disabled, the information element is omitted in SIB 3.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:RESelection:SEARCh:NINTrasearch`

ThreshServingLow

The parameter "Thresh_{Serving,Low}" is used by the UE for reselection towards a lower priority RAT/ frequency.

The threshold is configured in dB. System information block 3 contains the configured value divided by 2.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:RESelection:TSLow`

Q rxlevmin

Minimum required received RSRP level in the cell in dBm (Q_{rxlevmin}).

The level is configured in dBm. System information block 1 contains the configured value divided by 2.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:RESelection:QUALity:RXLevmin`

2.4.10.3 Identity settings

Access: "Config" hotkey > "Network" node > "Identity" node.

The node configures identities of the simulated radio network. The values are transferred to the UE under test via broadcast.

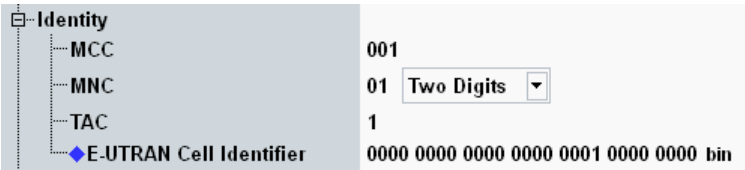


Figure 2-66: Network identity settings

MCC.....	184
MNC.....	184
TAC.....	184
E-UTRAN Cell Identifier.....	184

MCC
 Specifies the three-digit mobile country code (MCC).
 Remote command:
 CONFIGure:LTE:SIGN<i>:CELL:MCC

MNC
 Specifies the mobile network code (MNC). A two or three-digit MNC can be set.
 Remote command:
 CONFIGure:LTE:SIGN<i>:CELL:MNC
 CONFIGure:LTE:SIGN<i>:CELL:MNC:DIGits

TAC
 Specifies the tracking area code (TAC).
 Remote command:
 CONFIGure:LTE:SIGN<i>:CELL:TAC

E-UTRAN Cell Identifier
 Specifies the E-UTRAN cell identifier, unique within a PLMN. It is sent to the UE via broadcast and can be set independent of the physical cell ID (see "Physical Cell ID" on page 171).
 If you use carrier aggregation, configure different values for the carriers.
 Remote command:
 CONFIGure:LTE:SIGN<i>:CELL[:PCC]:CID:EUTRan
 CONFIGure:LTE:SIGN<i>:CELL:SCC<c>:CID:EUTRan

2.4.10.4 Security settings

Access: "Config" hotkey > "Network" node > "Security Settings" node.

The node configures parameters related to the authentication procedure and other security procedures.

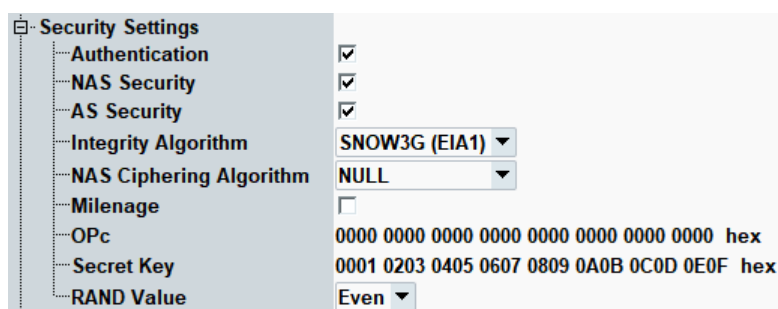


Figure 2-67: Security settings

Authentication.....	185
NAS Security.....	185
AS Security.....	185
Integrity Algorithm.....	185
NAS Ciphering Algorithm.....	186
Milenage.....	186
OPc.....	186
Secret Key.....	186
RAND Value.....	186

Authentication

Enables or disables authentication, to be performed during the attach procedure. Authentication requires a test USIM. An appropriate 3GPP USIM can be obtained from Rohde & Schwarz (R&S CMW-Z06, stock no. 1209.2010.02).

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:AUTHenticat`

NAS Security

Enables or disables non-access stratum (NAS) security. With enabled NAS security, integrity protection and ciphering are used for NAS signaling. This setting is only relevant if authentication is enabled.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:NAS`

AS Security

Enables or disables access stratum (AS) security. With enabled AS security, integrity protection is used for RRC signaling. This setting is only relevant if authentication is enabled.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:AS`

Integrity Algorithm

Selects an algorithm for integrity protection. NULL means that integrity protection is disabled. Use this setting for UEs which do not support the SNOW3G (EIA1) algorithm.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:IALgorithm`

NAS Ciphering Algorithm

Selects an algorithm for ciphering of NAS signaling. NULL means that ciphering is disabled.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:NCAlgorithm`

Milenage

Enable this parameter to use a USIM with MILENAGE algorithm set.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:MILenage`

OPc

The key OP_c is used for authentication and integrity check procedures with the MILENAGE algorithm set (parameter "Milenage" enabled). The value is entered as 32-digit hexadecimal number.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:OPC`

Secret Key

The secret key K is used for the authentication procedure (including a possible integrity check). The value is entered as 32-digit hexadecimal number.

The integrity check fails unless the secret key is equal to the value stored on the test USIM of the UE. The test USIM R&S CMW-Z06 is compatible with the default setting of this parameter.

If authentication is switched off, the secret key is ignored.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:SKEY`

RAND Value

The random number RAND is used for the authentication procedure (including a possible integrity check). This parameter selects whether an odd or even RAND value is used.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:SECurity:RVALue`

2.4.10.5 UE identity

Access: "Config" hotkey > "Network" node > "UE Identity" node.



Figure 2-68: UE identity settings

Default IMSI

International mobile subscriber identity (IMSI)

14 to 16 digits are allowed.

Remote command:
CONFigure:LTE:SIGN<i>:CELL:UEIDentity:IMSI

2.4.10.6 Timer and constants

Access: "Config" hotkey > "Network" node > "Timer and Constants" node.

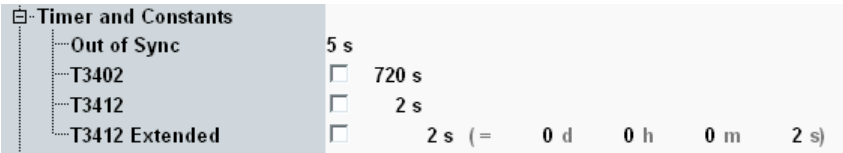


Figure 2-69: Timer settings

Out of Sync.....	187
T3402.....	187
T3412.....	187
T3412 Extended.....	187

Out of Sync

The "Out of Sync" timer specifies the time after which the instrument, having waited for a signal from the connected UE, releases the connection. The timer is started when an uplink grant message is sent to the UE.

Remote command:
CONFigure:LTE:SIGN<i>:CELL:TOUT:OSYNch

T3402

Timer "T3402" controls reattempts for failed attach procedures and failed tracking area update procedures (after five attempts).

Remote command:
CONFigure:LTE:SIGN<i>:CELL:TOUT:T<no>

T3412

Timer "T3412" controls the initiation of a periodic tracking area update by the UE. If the timer is disabled, no periodic tracking area update is required.

Remote command:
CONFigure:LTE:SIGN<i>:CELL:TOUT:T<no>

T3412 Extended

The extended T3412 setting allows much longer timeout values than the T3412 setting. If the UE supports extended values for T3412, it uses this setting instead of the T3412 setting.

Configure the timer in seconds. The setting is displayed for information in days, hours, minutes and seconds.

Remote command:
CONFigure:LTE:SIGN<i>:CELL:TOUT:TEXT<no>

2.4.10.7 Time

Access: "Config" hotkey > "Network" node > "Time" node.

The node allows you to send configurable date and time information to the UE. Thus you can update the date and time displayed by the mobile. In a real network, this service is typically used to send the current local time to the UE.

The node is only visible if R&S CMW-KS510 is available.

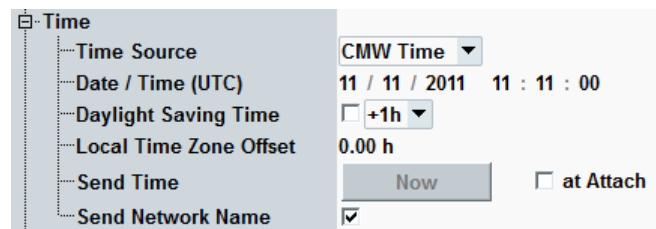


Figure 2-70: Time settings

Time Source.....	188
Date / Time (UTC).....	188
Daylight Saving Time.....	188
Local Time Zone Offset.....	189
Send Time.....	189
Send Network Name.....	189

Time Source

This parameter selects the date and time source that delivers the UTC date, the UTC time, the current daylight saving time offset and the time zone offset.

Option R&S CMW-KS510 is required.

"CMW Time" Selects the current CMW (Windows) date and time as source.

"Date / Time" Selects the other settings in the "Time" node as source.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:TIME:TSource`

Date / Time (UTC)

Defines the UTC date and time to be used if "Time Source" is set to "Date / Time".

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:TIME:DATE`

`CONFigure:LTE:SIGN<i>:CELL:TIME:TIME`

Daylight Saving Time

Specifies a daylight saving time (DST) offset to be used if "Time Source" is set to "Date / Time".

You can disable DST or enable it with an offset of +1 hour or +2 hours.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:TIME:DSTime`

Local Time Zone Offset

Defines the time zone offset to be used if "Time Source" is set to "Date / Time".

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:TIME:LTZoffset`

Send Time

Press "Now" to send the date and time information to the UE. This action is only possible if an RRC connection has been established.

"at Attach" selects whether the date and time information is sent to the UE during the attach procedure or not.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:TIME:SNOW`

`CONFigure:LTE:SIGN<i>:CELL:TIME:SATTach`

Send Network Name

Enables sending the network name together with the date and time information.

The network name is fixed. With disabled checkbox, no network name is sent. With enabled checkbox, the full and short network names are sent.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:TIME:SNName`

2.4.10.8 NAS signaling settings

Access: "Config" hotkey > "Network" node > "NAS Signaling" node.

The node configures settings related to NAS signaling messages, to be sent to the UE.

Figure 2-71: NAS signaling settings

Send DNS PCO.....	190
Attach Reject Cause.....	190
Attach Accept Cause.....	190
EPS Attach Result.....	191
Additional Update Result.....	191
TAU Reject Cause.....	191
EPS Network Feature Support.....	191
IMS Voice Over PS Session Indicator.....	191
Emergency Bearer Services Indicator.....	191
Location Service Indicator in EPC.....	192
Location Service Indicator in CS.....	192

Send DNS PCO

Enables or disables sending of a DNS IP address to the UE.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNECTION:SDNSpco`

Attach Reject Cause

If the checkbox is enabled, the application rejects attach requests from the UE. The reject message contains the selected reject cause.

The rejection causes are defined in 3GPP TS 24.301, section 9.9.3.9. The purpose of rejecting UE requests is to test the reaction of the UE: Does it repeat the request at all and if so, in which time intervals?

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:RCause:ATTach`

Attach Accept Cause

Selects whether the displayed cause is added to ATTACH ACCEPT messages or not.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:ACause:ATTach`

EPS Attach Result

If the checkbox is enabled, the application returns the configured "EPS attach result" IE in ATTACH ACCEPT messages. If the checkbox is disabled, the IE is configured automatically.

The IE is defined in 3GPP TS 24.301, section 9.9.3.10.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:EAResult`

Additional Update Result

If the checkbox is enabled, the application sends the configured value as "additional update result" IE in ATTACH ACCEPT messages.

The IE is defined in 3GPP TS 24.301, section 9.9.3.0A.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:AUResult`

TAU Reject Cause

If the checkbox is enabled, the application rejects tracking area update (TAU) requests from the UE. The reject message contains the selected reject cause.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:RCause:TAU`

EPS Network Feature Support

Enables or disables sending of the information element "EPS Network Feature Support" to the UE in the ATTACH ACCEPT message.

The information element indicates the support of specific features by the network (see 3GPP TS 24.301, section 9.9.3.12A). The individual feature flags are configured via the following parameters.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:NAS:EPSNetwork`

IMS Voice Over PS Session Indicator

Configures the first bit of the information element "EPS Network Feature Support".

The flag indicates whether voice over LTE (VoLTE) is supported, or a circuit-switched fallback (CSFB) is required for voice calls.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:NAS:IMSVops`

Emergency Bearer Services Indicator

Configures the second bit of the information element "EPS Network Feature Support".

The flag indicates whether emergency bearer services are supported. Such bearers have a higher priority than ordinary bearers.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:NAS:EMCBs`

Location Service Indicator in EPC

Configures the third bit of the information element "EPS Network Feature Support".

The flag indicates whether location services are supported by the LTE network (evolved packet core network).

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:NAS:EPCLcs`

Location Service Indicator in CS

Configures the fourth and fifth bit of the information element "EPS Network Feature Support".

The flag indicates whether location services are supported by the CS domain or not or no information is available.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL:NAS:CSLCs`

2.4.10.9 Synchronization

Access: "Config" hotkey > "Network" node > "Synchronization" node.

The node configures the synchronization to other signaling applications and the synchronization of PCC and SCC.

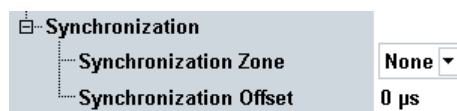


Figure 2-72: Synchronization settings

Synchronization Zone

Select the same synchronization zone in all signaling applications that you want to synchronize. "None" means that the application is not synchronized to other signaling applications.

The PCC and SCC of an LTE signaling application are always synchronized to each other, independent of this setting.

Synchronizing signaling applications means synchronizing the used system time. This feature is useful, for example, for evaluation of message logs, because the time stamps in the logs are synchronized.

Synchronizing two LTE signaling applications means also synchronizing the used system frame numbers.

Remote command:

`CONFigure:LTE:SIGN<i>:CELL[:PCC]:SYNC:ZONE`

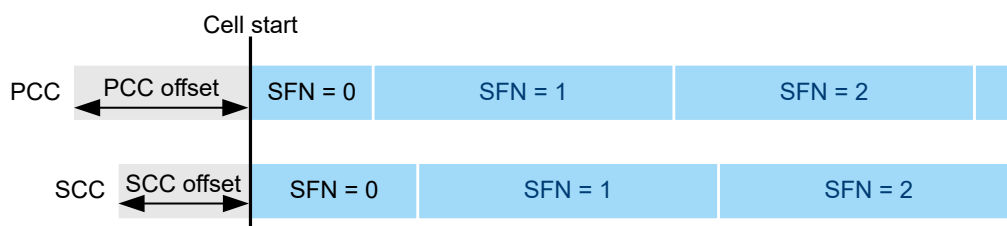
Synchronization Offset

Configures the timing offset at cell start, relative to the time zone.

Without offset, the cell signal starts with system frame number 0 and a system time according to the time zone. With an offset, the cell starts with system frame number 0 plus the offset and a system time according to the time zone plus the offset.

Option R&S CMW-KS510 is required for the PCC setting. Option R&S CMW-KS512 is required for the SCC setting.

Setting different offsets for component carriers is required for example for performance requirement tests with carrier aggregation, see 3GPP TS 36.521 section "8.2.1.1.1_A".



Remote command:

```
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SYNC:OFFSet
```

```
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SYNC:OFFSet
```

2.4.11 eMTC settings

Access: "Config" hotkey > "eMTC" node.

The node is only visible if enhanced machine type communication (eMTC) is enabled, see "eMTC" on page 142.

Option R&S CMW-KS590 is required for eMTC.

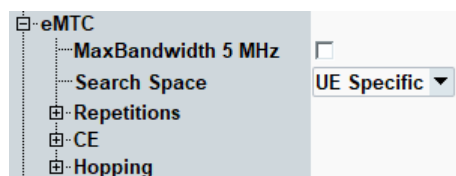


Figure 2-73: eMTC settings

For parameter descriptions, refer to the following sections.

- [General settings](#)..... 193
- [Repetition settings](#)..... 194
- [CE settings](#)..... 198
- [Hopping settings](#)..... 201

2.4.11.1 General settings

Access: "Config" hotkey > "eMTC" node.

This section describes the settings at the highest level of the "eMTC" node.

Max Bandwidth 5 MHz

Selects whether the maximum bandwidth for an eMTC connection is enhanced from 1.4 MHz to 5 MHz. This feature is only supported in CE mode A.

The enhanced bandwidth is signaled to the UE in the field "ce-PDSCH-PUSCH-Max-Bandwidth".

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MB<number>`

Search Space

Selects where the signaling application puts the MPDCCH.

"Common" MPDCCH in common search space
This selection is only supported for the scheduling type "eMTC Auto Mode".

"UE Specific" MPDCCH in UE-specific search space

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:SSpace`

2.4.11.2 Repetition settings

Access: "Config" hotkey > "eMTC" node > "Repetitions" node.

eMTC allows the repetition of control message and data transfers to enhance the coverage. The parameters in this section configure repetition settings for the individual channel types.

The settings are only available for the scheduling type "eMTC Auto Mode". The other scheduling types do not use repetitions.

Channel Type	Parameter	Value
MPDCCH	Max Repetition Paging	1
	Max Repetition	1
	Repetition Level	1
PDSCH	Max Repetition CE Mode A	Not configured
	Repetition CE Mode A	1
	Max Repetition CE Mode B	Not configured
	Repetition CE Mode B	4
PUCCH	Repetitions CE Mode A	1
	Repetitions CE Mode B	4
PUSCH	Max Repetition CE Mode A	Not configured
	Repetition CE Mode A	1
	Max Repetition CE Mode B	Not configured
	Repetition CE Mode B	4

Figure 2-74: Repetition settings

MPDCCH.....	195
L Max Repetition Paging.....	195
L Max Repetition.....	195
L Repetition Level.....	195
PDSCH.....	196
L Max Repetitions CE Mode A.....	196
L Repetition CE Mode A.....	196
L Max Repetitions CE Mode B.....	196
L Repetition CE Mode B.....	196
PUCCH.....	197
L Repetitions CE Mode A.....	197
L Repetitions CE Mode B.....	197
PUSCH.....	197
L Max Repetitions CE Mode A.....	197
L Repetition CE Mode A.....	197
L Max Repetitions CE Mode B.....	198
L Repetition CE Mode B.....	198

MPDCCH

The following settings configure the MPDCCH repetitions.

Max Repetition Paging ← MPDCCH

Maximum number of MPDCCH repetitions for paging.

The setting is signaled to the UE via the field "mpdcch-NumRepetition-Paging".

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:MRPaging`

Max Repetition ← MPDCCH

Maximum number of MPDCCH repetitions.

The setting is signaled to the UE via the field "mpdcch-NumRepetition-r13".

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:MREPetitions`

Repetition Level ← MPDCCH

Repetition level for MPDCCH repetitions.

The allowed values depend on the setting "Max Repetition" ($r_{\max}$), as listed in the following table.

Table 2-58: Allowed repetition levels, depending on $r_{\max}$

$r_{\max}$	Repetition level
1	1
2	1, 2
4	1, 2, 3
≥ 8	1, 2, 3, 4

The meaning of the repetition level depends on the setting "[Max Bandwidth 5 MHz](#)" on page 194, see 3GPP TS 36.213, section 9.1.5 "MPDCCH assignment procedure".

Remote command:

[CONFigure:LTE:SIGN<i>\[:PCC\]:EMTC:MPDCch:RLEVel](#)

PDSCH

The following settings configure the PDSCH repetitions.

Max Repetitions CE Mode A ← PDSCH

Maximum number of PDSCH repetitions for CE mode A.

The setting is signaled to the UE via the field "pdsch-maxNumRepetitionCEmodeA". "Not configured" means that the field is omitted.

Remote command:

[CONFigure:LTE:SIGN<i>\[:PCC\]:EMTC:PDSCh:A:MRCE](#)

Repetition CE Mode A ← PDSCH

Number of PDSCH repetitions for CE mode A (repetition level).

The allowed values depend on the configured maximum repetitions, as listed in the following table.

Table 2-59: Repetition values depending on max repetitions, mode A

Max repetitions	Repetitions
Not configured	1, 2, 4, 8
16	1, 4, 8, 16
32	1, 4, 16, 32

Remote command:

[CONFigure:LTE:SIGN<i>\[:PCC\]:EMTC:PDSCh:A:CERepetition](#)

Max Repetitions CE Mode B ← PDSCH

Maximum number of PDSCH repetitions for CE mode B.

The setting is signaled to the UE via the field "pdsch-maxNumRepetitionCEmodeB". "Not configured" means that the field is omitted.

Remote command:

[CONFigure:LTE:SIGN<i>\[:PCC\]:EMTC:PDSCh:B:MRCE](#)

Repetition CE Mode B ← PDSCH

Number of PDSCH repetitions for CE mode B (repetition level).

The allowed values depend on the configured maximum repetitions, as listed in the following table.

Table 2-60: Repetition values depending on max repetitions, mode B

Max repetitions	Repetitions
Not configured	4, 8, 16, 32, 64, 128, 256, 512
192	1, 4, 8, 16, 32, 64, 128, 192

Max repetitions	Repetitions
256	4, 8, 16, 32, 64, 128, 192, 256
384	4, 16, 32, 64, 128, 192, 256, 384
512	4, 16, 64, 128, 192, 256, 384, 512
768	8, 32, 128, 192, 256, 384, 512, 768
1024	4, 8, 16, 64, 128, 256, 512, 1024
1536	4, 16, 64, 256, 512, 768, 1024, 1536
2048	4, 16, 64, 128, 256, 512, 1024, 2048

Remote command:

`CONFfigure:LTE:SIGN<i>[:PCC]:EMTC:PDSCh:B:CERepetition`

PUCCH

The following settings configure the PUCCH repetitions.

The settings are sent to the UE via the children of the field "pucch-NumRepetitionCE-r13".

Repetitions CE Mode A ← PUCCH

Number of PUCCH repetitions for CE mode A.

Remote command:

`CONFfigure:LTE:SIGN<i>[:PCC]:EMTC:PUCCh:A:CERepetition`

Repetitions CE Mode B ← PUCCH

Number of PUCCH repetitions for CE mode B.

Remote command:

`CONFfigure:LTE:SIGN<i>[:PCC]:EMTC:PUCCh:B:CERepetition`

PUSCH

The following settings configure the PUSCH repetitions.

Max Repetitions CE Mode A ← PUSCH

Maximum number of PUSCH repetitions for CE mode A.

The setting is signaled to the UE via the field "pusch-maxNumRepetitionCEmodeA".

Remote command:

`CONFfigure:LTE:SIGN<i>[:PCC]:EMTC:PUSCh:A:MRCE`

Repetition CE Mode A ← PUSCH

Number of PUSCH repetitions for CE mode A (repetition level).

The allowed values depend on the configured maximum repetitions as for the PDSCH, see [Table 2-59](#).

Remote command:

`CONFfigure:LTE:SIGN<i>[:PCC]:EMTC:PUSCh:A:CERepetition`

Max Repetitions CE Mode B ← PUSCH

Maximum number of PUSCH repetitions for CE mode B.

The setting is signaled to the UE via the field "pusch-maxNumRepetitionCEmodeB".

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSCh:B:MRCE`

Repetition CE Mode B ← PUSCH

Number of PUSCH repetitions for CE mode B (repetition level).

The allowed values depend on the configured maximum repetitions, as listed in the following table.

Table 2-61: Repetition values depending on max repetitions, mode B

Max repetitions	Repetitions
Not configured	4, 8, 16, 32, 64, 128, 256, 512
192	1, 4, 8, 16, 32, 64, 128, 192
256	4, 8, 16, 32, 64, 128, 192, 256
384	4, 16, 32, 64, 128, 192, 256, 384
512	4, 16, 64, 128, 192, 256, 384, 512
768	8, 32, 128, 192, 256, 384, 512, 768
1024	4, 8, 16, 64, 128, 256, 512, 1024
1536	4, 16, 64, 256, 512, 768, 1024
2048	4, 16, 64, 128, 256, 512, 1024

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSCh:B:CERepetition`

2.4.11.3 CE settings

Access: "Config" hotkey > "eMTC" node > "CE" node.

The node configures the coverage enhancement (CE) levels and modes.

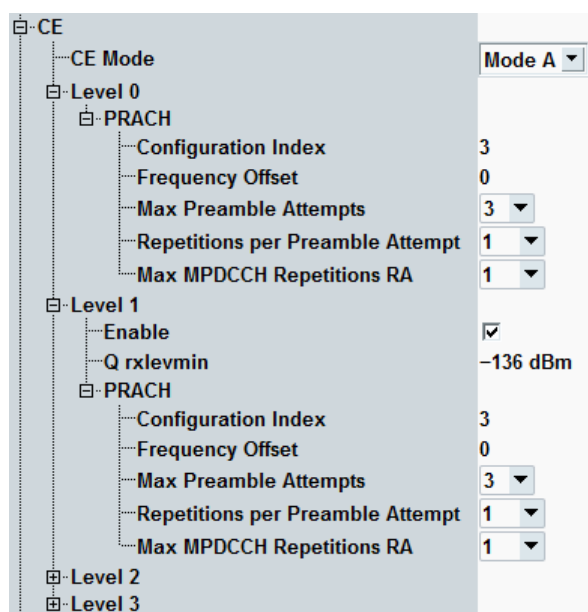


Figure 2-75: CE settings

CE Mode.....	199
Level 0 to Level 3.....	199
Enable.....	199
Q rxlevmin.....	200
PRACH.....	200
L Configuration Index.....	200
L Frequency Offset.....	200
L Max Preamble Attempts.....	200
L Repetitions per Preamble Attempt.....	201
L Max MPDCCH Repetitions RA.....	201

CE Mode

Selects the coverage enhancement mode. The information is signaled to the UE during the random access procedure, via the field "ce-Mode-r13".

Before the UE receives this message, it can use a different CE mode for the random access procedure, depending on the other settings in this section.

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:MODE
```

Level 0 to Level 3

CE levels supported by the eNodeB for PRACH. The UE selects one of these levels, see "Q rxlevmin".

A higher CE level means a higher coverage enhancement (for example more repetitions / higher power). If the UE selects CE level 0 or 1, it uses CE mode A. With CE level 2 and 3, it uses CE mode B.

Enable

Selects whether the eNodeB supports the CE level. Level 0 is always supported.

If you disable a level, the higher levels are disabled automatically.

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:ENABLe
```

Q rxlevmin

Threshold for selection of a CE level by the UE. The UE compares the measured RSRP to the configured "Q rxlevmin" values and selects the CE level accordingly.

With increasing CE level, the "Q rxlevmin" setting must decrease. CE level 0 has no "Q rxlevmin" setting. It can be used by the UE for all RSRP levels larger than the "Q rxlevmin" of CE level 1.

The "Q rxlevmin" is configured in dBm. The system information field "q-RxLevMinCE-r13" contains the configured value divided by two.

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:QRXLevmin
```

PRACH

This node configures settings for the random access procedure with coverage enhancement.

All settings are configurable per PRACH CE level. All settings for all CE levels are signaled to the UE via the field "PRACH-ParametersListCE-r13".

Configuration Index ← PRACH

Sets the PRACH configuration index. The configuration index defines the preamble format and other PRACH signal properties. The setting is signaled to the UE via the field "prach-ConfigIndex-r13".

For details, see 3GPP TS 36.211, section 5.7.1 "Time and frequency structure".

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:CINdex
```

Frequency Offset ← PRACH

The frequency offset is used by the UE to calculate the location of the preamble resource blocks within the cell bandwidth. The setting is signaled to the UE via the field "prach-FreqOffset-r13".

For details, see "prach-FrequencyOffset" in 3GPP TS 36.211, section 5.7.1 "Time and frequency structure".

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:FOFFset
```

Max Preamble Attempts ← PRACH

Specifies the maximum number of preamble transmission attempts within one CE level. The UE increases the power for each attempt (as long as the maximum power is not yet reached).

After the maximum number of attempts, the UE can increase the CE level.

The setting is signaled to the UE via the field "maxNumPreambleAttemptCE-r13".

Remote command:

```
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MPATtempts
```

Repetitions per Preamble Attempt ← PRACH

Specifies the number of repetitions per preamble transmission attempt (repetitions with same preamble and power).
The setting is signaled to the UE via the field "numRepetitionPerPreambleAttempt-r13".

Remote command:
`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:RPATtempt`

Max MPDCCH Repetitions RA ← PRACH

Specifies the maximum number of MPDCCH repetitions for the random access response.
The setting is signaled to the UE via the field "mpdcch-NumRepetition-RA-r13".

Remote command:
`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MMRRepetitio`

2.4.11.4 Hopping settings

Access: "Config" hotkey > "eMTC" node > "Hopping" node.
The node configures frequency hopping between narrowbands for the scheduling type "eMTC Auto Mode". The other scheduling types do not use repetitions.
For some transmissions, frequency hopping is mandatory and configured automatically as specified by 3GPP. The settings in this section are not relevant for mandatory frequency hopping.
The settings apply to all transmissions with optional frequency hopping. You can configure hopping separately for the DL and the UL.

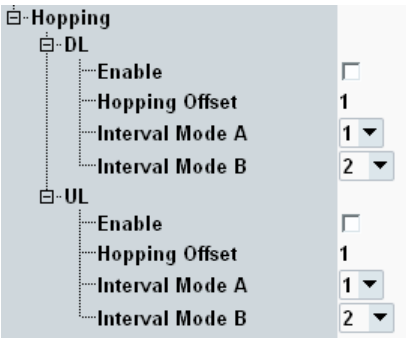


Figure 2-76: Frequency hopping settings

Enable.....	201
Hopping Offset.....	202
Interval Mode A.....	202
Interval Mode B.....	202

Enable
Enables or disables frequency hopping.

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:DL:ENABLE`

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:UL:ENABLE`

Hopping Offset

Specifies the size of one frequency hop. The unit is narrowbands.

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:DL:HOFFset`

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:UL:HOFFset`

Interval Mode A

Specifies the time interval between two hops for CE mode A. The unit is subframes.

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:DL:A:INTerval`

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:UL:A:INTerval`

Interval Mode B

Specifies the time interval between two hops for CE mode B. The unit is subframes.

Remote command:

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:DL:B:INTerval`

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:UL:B:INTerval`

2.4.12 Connection configuration

The settings in this section define parameters for the connection, for example the resource configuration to be allocated to the UE. Most parameters can be reconfigured for an established connection.

For parameter descriptions, refer to the following sections.

• Miscellaneous connection settings part 1	203
• General MIMO settings	210
• MIMO channel matrix TM 2 to TM 6	212
• MIMO beamforming TM 7	214
• MIMO beamforming TM 8	215
• MIMO TM 9	218
• Miscellaneous connection settings part 2	221
• ROHC settings	233
• Connected DRX settings	234
• Capability request settings	238
• RMC settings (no eMTC)	240
• RMC settings for eMTC	243
• User-defined channels (no LAA, no eMTC)	246
• User-defined channels for eMTC	249
• User-defined ch. for LAA, fixed bursts	251
• User-defined ch. for LAA, random bursts	254
• User-defined TTI-based channels	258
• CQI channel configuration	259

• SPS configuration.....	260
• eMTC auto mode settings.....	262
• eMTC compact scheduling settings.....	263

2.4.12.1 Miscellaneous connection settings part 1

Access: "Config" hotkey > "Connection" node.

This section describes the following settings.

Setting	Value
Easy Mode	☐
Group Hopping	☐
UE Category	3 Use Reported (if available): ☒
UE Category 0 allowed	☐
PSM allowed	☐
IMS Emergency Support	☒
eCall over IMS Support	☐
Default Paging Cycle	#64
PCCH-Config nB	T/2
Additional Spectrum Emission	NS_01
UE Meas. Filter Coefficient	FC4
Connection Type	Testmode
External DAU	
Testmode	
Use "Activate Testmode" Message	☒
Default Bearer	
Testmode Dedicated Bearer	
RLC Mode	Unacknowledged
SIB Reconfiguration	SIB Paging
Keep RRC Connection	☒ Inactivity Timer 5 s
Downlink MAC Padding	☒
Downlink MAC Error Insertion	0 %

Figure 2-77: Connection configuration settings part 1

Easy Mode.....	204
L Band/Frequency/BW Change.....	204
Group Hopping.....	204
UE Category.....	205
UE Category 0 Allowed.....	205
PSM Allowed.....	205
IMS Emergency Support.....	205
eCall Over IMS Support.....	205
Default Paging Cycle.....	206
PCCH-Config nB.....	206
Additional Spectrum Emission.....	206
Add. Spectrum Emission Scell.....	206
UE Meas. Filter Coefficient.....	206
Connection Type.....	207
External DAU.....	207
L Use external DAU.....	207
L Network Segment, Subnet Node ID.....	207

Test Mode > Use "Activate Testmode" Message.....	207
Default Bearer.....	208
L RLC Mode.....	208
L IP Version.....	208
L APN.....	208
L QCI.....	208
Testmode Dedicated Bearer.....	208
L RLC Mode.....	209
SIB Reconfiguration.....	209
Keep RRC Connection.....	209
Downlink MAC Padding.....	210
Downlink MAC Error Insertion.....	210

Easy Mode

The easy mode is available for a redirection or a blind handover within the signaling application.



With disabled easy mode, such an action can fail. For example, because the power level or the number of scheduled resource blocks is too low for a reliable exchange of the required signaling messages.

With enabled easy mode, critical settings are set to optimum values directly before the redirection or handover, to eliminate possible failure causes. After completion of the procedure, the settings are restored to the previous values. The final settings are the same with or without easy mode.

With enabled easy mode, the following intermediate settings are configured:

- Full UL and DL resource block allocation in all subframes
- Suitable UL and DL power level
- Modulation type QPSK
- TBS index 5
- DCI format 1A
- No AWGN, no error insertion

Band/Frequency/BW Change ← Easy Mode

Specifies whether the easy mode is used if the band or the frequency or the cell bandwidth is changed.

Disable the easy mode if you want to test whether a redirection or a blind handover is successful under certain circumstances.

Enable the easy mode if you want to change settings reliably via a redirection or a blind handover.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:EASY:BFBW`

Group Hopping

Enables or disables group hopping, specified in 3GPP TS 36.211.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:GHOPping`

UE Category

The UE category to be used by the R&S CMW can be set manually. Alternatively, it can be set automatically according to the UE category reported by the UE in the capability report.

"Manual"	If you want to set the category of your UE manually, select it here. A value not reflecting the capabilities of the UE results in problems when trying to reach high data rates.
"Use Reported (if available)"	If no reported value is available, the manually configured value is used. If a reported value is available, it is displayed for information. If the checkbox is enabled, the reported value is used. Otherwise the manually configured value is used. The reported value is the "ue-Category", not the "ue-CategoryDL" or "UE Category UL".

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:UECategory:MANual:ENHanced
CONFigure:LTE:SIGN<i>:CONNection:UECategory:REPorted:ENHanced
```

UE Category 0 Allowed

Specifies whether category 0 UEs are allowed to access the cell. This information is sent to the UE via broadcast in system information block 1.

Enabling the checkbox is relevant for machine-type communication (MTC). Option R&S CMW-KS590 is required.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:UECategory:CZAllowed
```

PSM Allowed

Specifies whether a UE request for power-saving mode (PSM) is accepted or rejected. See also 3GPP TS 24.301, section 5.3.11 "Power saving mode".

Enabling the checkbox is relevant for machine-type communication (MTC). Option R&S CMW-KS590 is required.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:PSMallowed
```

IMS Emergency Support

Adds the optional field "ims-EmergencySupport-r9" to the system information block 1.

The field indicates that the cell supports IMS emergency bearer services for UEs in limited service mode.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:IEmergency
```

eCall Over IMS Support

Adds the optional field "eCallOverIMS-Support-r14" to the system information block 1.

The field indicates that the cell supports eCall over IMS services for UEs.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:EOISupport
```

Default Paging Cycle

Selects the cell-specific default paging cycle (32, 64, 128 or 256 radio frames). The value is signaled to the UE as "defaultPagingCycle" and used by the UE as input for the calculation of paging radio frame and subframe positions.

The R&S CMW considers the setting when sending paging messages.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:DPCycle`

PCCH-Config nB

Configures the field "nB" in the "PCCH-Config" in system information block 2.

The value is used by the UE as input for the calculation of paging radio frame and paging occasion according to 3GPP TS 36.304, section 7.

The values T/64 to T/256 are only allowed for eMTC. And they are only allowed, if the default paging cycle has the same or a greater value. Example: T/64 needs cycle $\geq$ #64.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:PCNB`

Additional Spectrum Emission

The selected value is signaled to the UE within a system information block. It determines which additional ACLR and spectrum emission requirements have to be met and whether additional maximum power reduction (A-MPR) is allowed. For details, see 3GPP TS 36.521-1.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:ASEMission`

Add. Spectrum Emission Scell

This setting is only displayed on an SCC tab with enabled uplink. The selected value is signaled to the UE and determines additional ACLR and spectrum emission requirements for the SCC uplink.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:ASEMission:CAGGregation`

UE Meas. Filter Coefficient

Some RRC messages to be sent to the UE for conformance tests contain an information element "filterCoefficient" = "fc4" or "fc8". This parameter selects the value to be sent.

The relevant value depends on the conformance test to be performed, see 3GPP TS 36.521 and 3GPP TS 36.508.

The "filterCoefficient" is used for uplink power control. Do not confuse it with "filterCoefficientRSRP" and "filterCoefficientRSRQ" for UE measurement reports, see ["Filter Coefficient RSRP/RSRQ"](#) on page 282.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:FCEfficient`

Connection Type

Configures the connection type to be applied at the R&S CMW.

- | | |
|--------------------|--|
| "Testmode" | The test mode uses only layer 1 and 2 of the protocol stack. Layer 3 is not used. This mode is suitable for any signaling tests not involving the data application unit (DAU). |
| "Data Application" | The data application mode supports also layer 3, required for IP-based services. Select this mode for data application measurements with the DAU. See also Section 2.2.7, "Data tests and voice over LTE" , on page 37 and Section 2.3.2, "LTE IP-based data tests" , on page 103.
This value requires an installed DAU or an external DAU. |

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:CTYPe`

External DAU

Usually, the DAU is installed on the same R&S CMW as the LTE signaling application. If all expansion slots of this R&S CMW are occupied by other hardware options, a DAU installed on another R&S CMW can be used instead (external DAU).

Using an external DAU is only possible if the instrument where the LTE signaling application is running fulfills the following prerequisites:

- No DAU is installed
- Option R&S CMW-KA120 is available
- An Ethernet switch is installed (option R&S CMW-B660x plus -B661x)



Use external DAU ← External DAU

Enable the checkbox if you want to use an external DAU.

The two instruments must be connected via LAN. For details, refer to the DAU documentation.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:EDAU:ENABLE`

Network Segment, Subnet Node ID ← External DAU

Select the values of the instrument where the external DAU is installed. You can check the values in the "Setup" dialog > "Misc" > "IP Subnet Config".

The two instruments must use the same network segment and must have different node IDs.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:EDAU:NSEGment`

`CONFigure:LTE:SIGN<i>:CONNection:EDAU:NID`

Test Mode > Use "Activate Testmode" Message

When enabled, an "ACTIVATE TEST MODE" message is sent to the UE. No loop mode is requested. Test modes are specified in 3GPP TS 36.509 and 36.508.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:TMode`

Default Bearer

The default bearer settings are applied to all default bearers.

For connection type "Testmode", the RLC mode is configurable and the other default bearer settings are fixed. For connection type "Data Application", the default bearer settings are configurable.



RLC Mode ← Default Bearer

Selects the RLC mode for downlink transmissions: unacknowledged mode (UM) or acknowledged mode (AM), see 3GPP TS 36.322.

Option R&S CMW-KS510 is required.

"UM" There are no ARQ retransmissions on the RLC layer.

"AM" There are ARQ retransmissions on the RLC layer.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:RLCMode`

IP Version ← Default Bearer

Allowed IP versions. With "IPv4 only", for example, IPv6 is not used, even if requested by the UE.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:IPVersion`

APN ← Default Bearer

Default access point name (APN), used if no APN is provided by the UE.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:APN`

QCI ← Default Bearer

Quality-of-service class identifier signaled to the UE. The values are specified in 3GPP TS 23.203.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:QCI`

Testmode Dedicated Bearer

The settings are applied to all dedicated bearers in connection type "Testmode".

RLC Mode ← Testmode Dedicated Bearer

Selects the RLC mode for downlink transmissions: unacknowledged mode (UM) or acknowledged mode (AM), see 3GPP TS 36.322.

"UM" There are no ARQ retransmissions on the RLC layer.

"AM" There are ARQ retransmissions on the RLC layer.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNectio:n:TDBearer:RLCMode`

SIB Reconfiguration

Selects a method for information of an attached UE about changes in the system information, resulting from modified parameters.

The UE is, for example, informed about changes of the following parameters: maximum allowed uplink power P-Max, neighbor cell information, PUSCH open loop nominal power, SRS enable/disable, PRACH configuration index and default paging cycle.

"RRC Reconfiguration" An RRC reconfiguration message is sent to the UE, containing the system information values in a mobility control information element. This method is only used if an RRC connection is established. Without RRC connection, SIB paging is used.

"SIB Paging" The UE is paged. This action triggers the UE to evaluate the broadcasted system information.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNectio:n:SIBReconfig`

Keep RRC Connection

Selects whether the RRC connection is kept or released after attach.

"Enabled" The RRC connection established during attach is kept when the attach procedure is completed. For a subsequent connection setup, the already established RRC connection is used. So the connection setup is faster. When the dedicated bearer is released via a disconnect, the RRC connection is still kept.

"Disabled"

- For "Connection Type" = "Testmode", the RRC connection established during attach is released immediately when the attach procedure is completed.
- For "Connection Type" = "Data Application", there is an additional setting "Inactivity Timer". The RRC connection established during attach is released when the attach procedure is completed and there has been no activity on the connection (no traffic) for the configured time. The inactivity timer requires option R&S CMW-KS510. Without this option, the RRC connection is released immediately when the attach procedure is completed.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNectio:n:KRRC`

`CONFigure:LTE:SIGN<i>:CONNectio:n:RITimer`

Downlink MAC Padding

Activates or deactivates downlink padding at the MAC layer.

When no data (signaling or user data) is available from higher layers, an allocated channel can be filled with padding bits (DL padding on). This scenario is foreseen in many conformance tests.

Switching off DL padding is useful, for example, if the UE has problems to attach because the DL signal contains padding bits.

If UL dynamic scheduling is enabled, MAC padding is disabled and grayed out.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:DLPadding`

Downlink MAC Error Insertion

Configures the rate of transport block errors to be inserted into the downlink data. This setting is useful for BLER measurements.

The parameter can be changed in all main connection states including "Connection Established".

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:DLEinsertion`

2.4.12.2 General MIMO settings

Access: "Config" hotkey > "Connection" node > "MIMO Settings" node.

This section describes general MIMO connection settings applicable to most transmission modes.

For background information, see [Section 2.2.18, "MIMO and beamforming"](#), on page 64.

For carrier aggregation scenarios, you can configure all MIMO settings per component carrier.

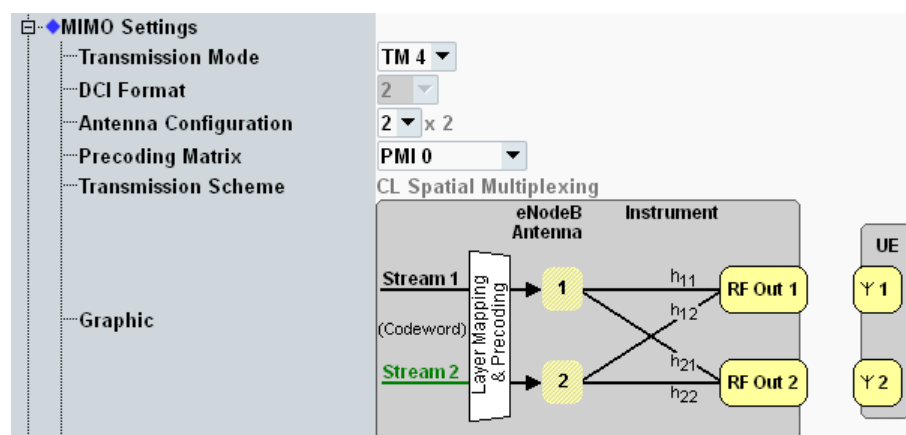


Figure 2-78: General MIMO settings, TM 4

Transmission Mode.....	211
DCI Format.....	211
Antenna Configuration.....	211
Precoding Matrix.....	211
Transmission Scheme.....	212

Transmission Mode

Selects the LTE transmission mode. The available values depend on the active scenario, see [Table 2-32](#).

For eMTC, only TM 1 is supported. For frame structure type 3, the supported transmission modes are 1 to 4, 8 and 9.

In the state "Connection Established", the parameter can be changed within the following groups: TM 1/7, TM 2/3/4/6/8/9.

The following combinations are not foreseen by 3GPP and are not allowed:

- TM 7 plus normal cyclic prefix, plus "number of PDCCH symbols" = 4
- TM 8 plus extended cyclic prefix

Remote command:

```
CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:TRANsmission
```

```
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:TRANsmission
```

DCI Format

Selects the downlink control information (DCI) format. The available values depend on the transmission mode. For most transmission modes, the DCI format is fixed.

For scheduling type "Follow WB CQI-RI", only DCI format 2A is supported. Changing the format to 1A modifies the scheduling type.

For 256-QAM and 1024-QAM, DCI format 1A is not supported.

Remote command:

```
CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:DCIFormat
```

```
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:DCIFormat
```

Antenna Configuration

Selects the number of downlink TX antennas. The allowed values depend on the selected scenario and transmission mode. The setting is available for transmission mode 1 to 6.

"2" 2x2 carrier (R&S CMW-KS520) or 2x4 carrier (R&S CMW-KS540)

"4" 4x2 carrier (R&S CMW-KS521) or 4x4 carrier (R&S CMW-KS540)

The number of downlink RX antennas depends on the selected scenario and is displayed for information.

Remote command:

```
CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:NENBantennas
```

```
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:NENBantennas
```

Precoding Matrix

Selects the precoding matrix for transmission mode 4, 6 and 9. For TM 8, see "[Precoding Matrix](#)" on page 217.

Depending on the scheduling type, the downlink signal follows the reported PMI value. In that case, the configured PMI value is used as initial value until a PMI report has been received.

The meaning of the PMI values is defined in 3GPP TS 36.213, section 7.2.4.

With the setting "random PMI", the PMI value is selected randomly as defined in 3GPP TS 36.521, annex B.4.1 and B.4.2. The setting is available for transmission mode 9 with two TX antennas if no "Follow..." scheduling type is active.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PMATrix
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PMATrix
```

Transmission Scheme

Displays the PDSCH transmission scheme resulting from the settings.

The displayed value does not reflect transmission scheme changes due to reported rank indicator values. TM 4 combined with follow RI and RI=1 results in transmit diversity.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection[:PCC]:TSCHEME?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:TSCHEME?
```

2.4.12.3 MIMO channel matrix TM 2 to TM 6

Access: "Config" hotkey > "Connection" node > "MIMO Settings" node > "Transmission Mode" = "TM 2" to "TM 6" > "Channel Matrix" node.

The channel matrix settings are displayed for transmission mode 2 to 6, if supported by the scenario. They are hidden for fading scenarios and for scenarios with RF output before the radio channel.

The matrix defines the radio channel coefficients, see [Section 2.2.18.1, "Radio channel coefficients for MIMO"](#), on page 67.

Each element of the matrix is a complex number, defined via its phase and its magnitude.

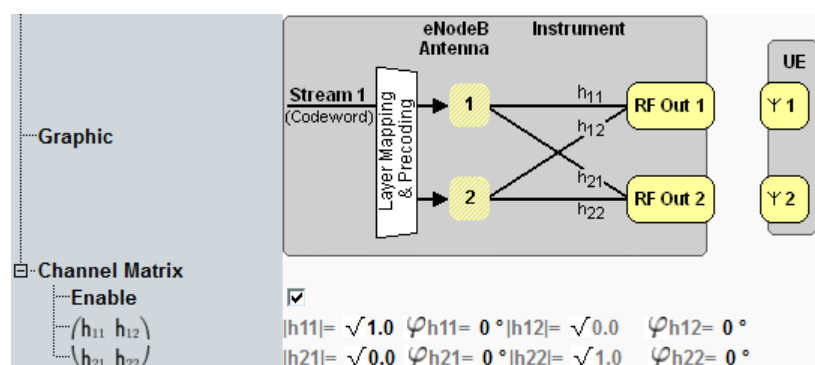


Figure 2-79: MIMO 2x2 channel matrix

The size of the matrix depends on the number of antennas.

2x2 channel matrix

You can configure the phase of each number and the square of the magnitude of h_{11} and h_{21} . The other magnitudes are adapted automatically.

If the matrix is disabled, there is no coupling between the signals of the eNodeB antennas. So each UE antenna receives only the signal of one eNodeB antenna.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:ENABle
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:ENABle
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel
```

2x4 channel matrix

The matrix is not configurable and displayed for information.

4x2 channel matrix

You can configure the phase and the square of the magnitude of all coefficients. The sum of all magnitude squares within each line of the matrix must equal 1.

The sum is checked automatically. A correct sum is indicated by , a wrong sum by , followed by the sum. Configure all values of a line so that the sum equals 1, then press the "Apply" button.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:MIMO<Mimo>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:MIMO<Mimo>
```

4x4 channel matrix

Via the following parameters, you can disable the channel matrix or select a predefined matrix or define your own matrix.

Enable ← 4x4 channel matrix

If the channel matrix is disabled, there is no coupling between the signals of the eNodeB antennas. So each UE antenna receives only the signal of one eNodeB antenna.

The other channel matrix settings are irrelevant in that case.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:ENABle:
MIMO<Mimo>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:ENABle:
MIMO<Mimo>
```

Matrix Selection ← 4x4 channel matrix

Selects a predefined matrix or the user-defined matrix.

If you select a predefined matrix, the resulting matrix coefficients are displayed for information.

"User Defined"	You can configure the phase and the square of the magnitude of all coefficients. The sum of all magnitude squares within each line of the matrix must equal 1. The sum is checked automatically. A correct sum is indicated by 😊, a wrong sum by 😞, followed by the sum. Configure all values of a line so that the sum equals 1, then press the "Apply" button of that line.
"3GPP Channel Matrix"	The square of the magnitude is 0.25 for all coefficients. The used phases are 0°, 90°, 180° and 270°.
"Hadamard Matrix"	The square of the magnitude is 0.25 for all coefficients. The used phases are 0° and 180°.
"Identity Matrix"	The square of the magnitude is 1 for the main diagonal and 0 for all other coefficients. The phase is 0° for all coefficients.

Remote command:

```

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:MSElection:
MIMO<Mimo>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:MSElection:
MIMO<Mimo>
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:MIMO<Mimo>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:MIMO<Mimo>

```

2.4.12.4 MIMO beamforming TM 7

Access: "Config" hotkey > "Connection" node > "MIMO Settings" node > "Transmission Mode" = "TM 7" > "Beamforming Model" node.

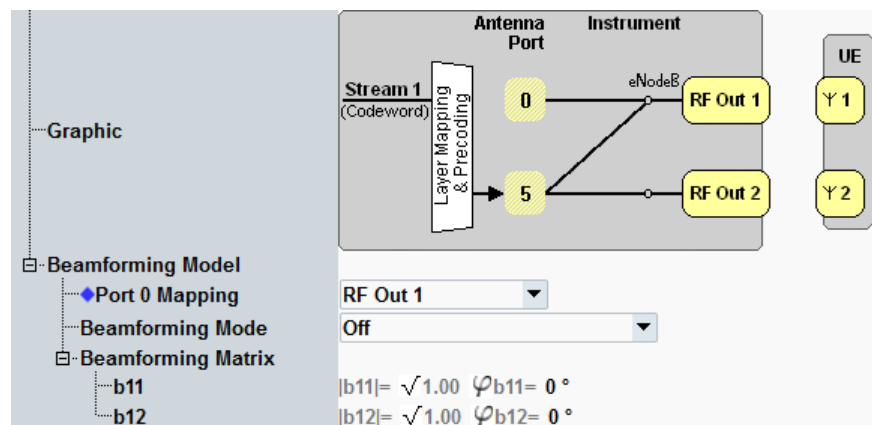


Figure 2-80: Beamforming settings for TM 7

Port 0 Mapping.....	214
Beamforming Mode.....	215
Beamforming Matrix.....	215

Port 0 Mapping

Selects the mapping of antenna port 0 to the RF output paths. This setting is available for TM 7 in scenarios with two RF output paths, without fading.

You can map port 0 to the first RF output path only. Or you can map it to both output paths.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PZERo:MAPPing
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PZERo:MAPPing
```

Beamforming Mode

Enables or disables beamforming.

"Off"	Beamforming is disabled.
"On"	Beamforming is enabled. The configured beamforming matrix is used.
"TS 36.521 Beamforming Mode"	Beamforming is enabled. The beamforming matrix is selected randomly as defined in 3GPP TS 36.521, annex B.4.1 and B.4.2. Not allowed for 1x1 beamforming matrices and for "Follow..." scheduling types.
"Precoding Matrix"	Not allowed for TM 7

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MODE
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MODE
```

Beamforming Matrix

This node defines the beamforming matrix for the beamforming mode "On".

The matrix characterizes the mapping of the UE-specific antenna port 5 to the transmit antennas, see [Section 2.2.18.2, "Beamforming matrix"](#), on page 68.

Each element of the matrix is a complex number, defined via its phase and its magnitude. The size of the matrix is dynamic:

- 1x1 matrix (b_{11}) for scenarios with one RF output path
You can configure the phase of the coefficient.
- 1x2 matrix (b_{11} and b_{21}) for scenarios with two RF output paths.
You can configure the phase of both coefficients.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MATRix
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MATRix
```

2.4.12.5 MIMO beamforming TM 8

Access: "Config" hotkey > "Connection" node > "MIMO Settings" node > "Transmission Mode" = "TM 8" > "Beamforming Model" node.

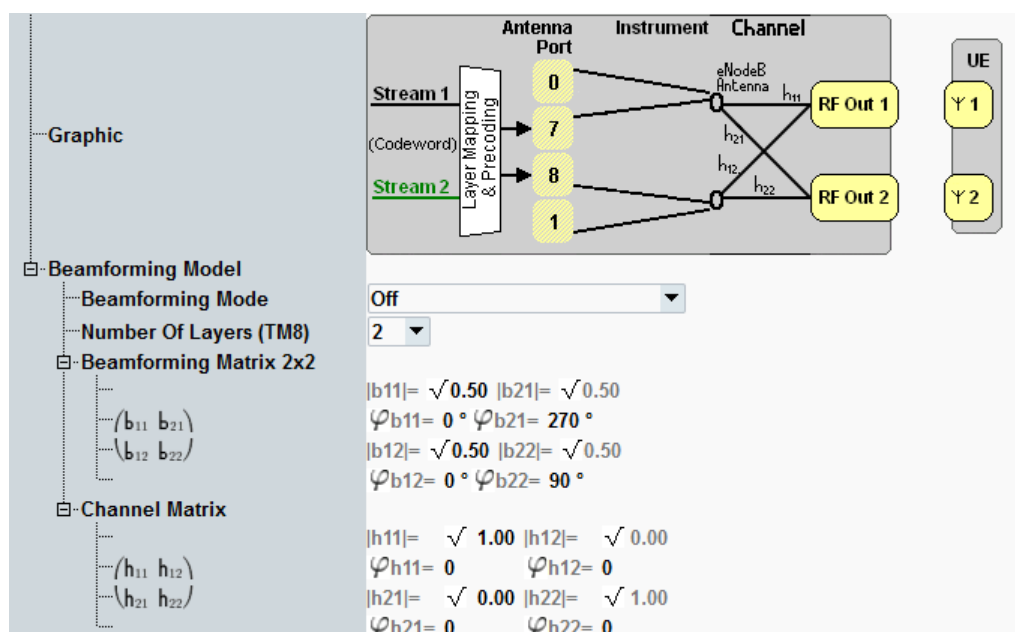


Figure 2-81: Beamforming settings for TM 8, 2 layers

Beamforming Mode.....	216
Number of Layers.....	217
Precoding Matrix.....	217
Beamforming Matrix.....	217
Channel Matrix.....	217

Beamforming Mode

Enables or disables beamforming.

"Off"	Beamforming is disabled. Not allowed for TM 8 plus "Follow..." scheduling types.
"On"	Beamforming is enabled. The configured beamforming matrix is used. Not allowed for TM 8 plus "Follow..." scheduling types
"TS 36.521 Beamforming Mode"	Beamforming is enabled. The beamforming matrix is selected randomly as defined in 3GPP TS 36.521, annex B.4.1 and B.4.2. Not allowed for 1x1 beamforming matrices and for "Follow..." scheduling types.
"Precoding Matrix"	Beamforming is enabled. A precoding matrix is used as beamforming matrix. The matrix is selected via the parameter "Precoding Matrix" on page 217. For "Follow...PMI..." scheduling types, the beamforming matrix is the precoding matrix corresponding to the PMI reported by the UE.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MODE
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MODE
```

Number of Layers

You can select between one layer (single-layer beamforming) and two layers (dual-layer beamforming).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:NOLayers
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:NOLayers
```

Precoding Matrix

Selects a precoding matrix to be used as beamforming matrix. Only relevant for beamforming mode "Precoding Matrix".

The meaning of the PMI values is defined in 3GPP TS 36.213, section 7.2.4.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PMATrix
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PMATrix
```

Beamforming Matrix

This node defines the beamforming matrix for the beamforming mode "On".

The matrix characterizes the mapping of the UE-specific antenna ports 7 and 8 to the transmit antennas, see [Section 2.2.18.2, "Beamforming matrix"](#), on page 68.

Each element of the matrix is a complex number, defined via its phase and its magnitude. The size of the matrix is dynamic:

- 1x2 matrix (b_{11} and b_{21}) for single-layer beamforming.
You can configure the phase of both coefficients.
- 2x2 matrix (b_{11} to b_{22}) for dual-layer beamforming.
You can configure the phase of each coefficient and the square of the magnitude of b_{11} and b_{12} . The other magnitudes are adapted automatically.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MATrix
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MATrix
```

Channel Matrix

The channel matrix is configurable for scenarios without fading. It is not configurable for "TS 36.521 Beamforming Mode".

The matrix defines the radio channel coefficients, see [Section 2.2.18.1, "Radio channel coefficients for MIMO"](#), on page 67.

Each element of the matrix is a complex number, defined via its phase and its magnitude.

You can configure the phase of each number and the square of the magnitude of h_{11} and h_{21} . The other magnitudes are adapted automatically.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<8>:CHMatrix
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<8>:CHMatrix
```

2.4.12.6 MIMO TM 9

Access: "Config" hotkey > "Connection" node > "MIMO Settings" node > "Transmission Mode" = "TM 9" > "TM9 Parameters" node.

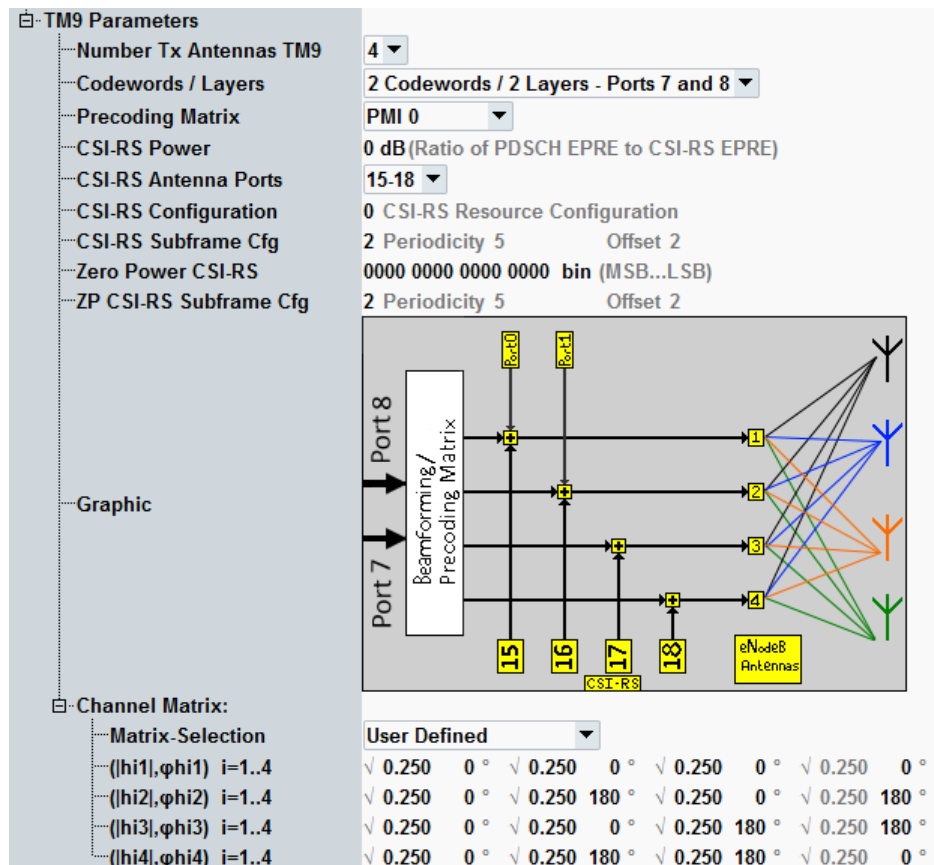


Figure 2-82: TM 9 settings, 4 TX antennas, 4 CSI-RS antenna ports

Number Tx Antennas TM 9	218
Codewords / Layers	219
Precoding Matrix 2	219
CSI-RS Power	219
CSI-RS Antenna Ports	219
CSI-RS Configuration	220
CSI-RS Subframe Cfg	220
Zero Power CSI-RS	220
ZP CSI-RS Subframe Cfg	220
Channel Matrix	220
Matrix Selection	221

Number Tx Antennas TM 9

Selects the number of downlink TX antennas. The allowed values and the required options depend on the selected scenario.

"2"	2x2 carrier (R&S CMW-KS520) or 2x4 carrier (R&S CMW-KS540)
"4"	4x2 carrier (R&S CMW-KS521) or 4x4 carrier (R&S CMW-KS540)

"8" 8x2 carrier (R&S CMW-KS522) or 8x4 carrier (R&S CMW-KS541)

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:NTXantennas
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:NTXantennas
```

Codewords / Layers

You can select between one and two layers. For MIMO 4x4 and 8x4, you can also select four layers.

The used ports are fixed and indicated.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CODewords
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CODewords
```

Precoding Matrix 2

For transmission mode 9 with 8 TX antennas, the precoding matrix is selected via two PMI values.

The parameter "Precoding Matrix" selects the first value, see ["Precoding Matrix"](#) on page 211. The additional parameter "Precoding Matrix 2" selects the second value.

The meaning of the PMI values is defined in 3GPP TS 36.213, section 7.2.4.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:PMATrix
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:PMATrix
```

CSI-RS Power

Value P_c to be signaled to the UE. P_c is the assumed ratio of the RS EPRE to the CSI-RS EPRE. To increase the CSI-RS power, decrease the setting.

The used CSI-RS power offset can be different from the signaled value. For configuration of the used CSI-RS power offset, see ["CSI-RS"](#) on page 160.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:POWer
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:POWer
```

CSI-RS Antenna Ports

Selects the antenna ports used for the CSI-RS. Select "None" to transmit no CSI-RS at all.

Besides "None", you can use 1, 2, 4 or 8 antenna ports, depending on other settings:

- 2 or 8 TX antennas: All values are allowed.
- 4 TX antennas: Up to 4 ports are allowed.
- If a scheduling type with follow PMI or follow RI or follow CQI is active, the number of CSI-RS antenna ports equals the number of TX antennas. You cannot configure this setting in that case.

The required options depend on the type of the carrier, as follows.

"15-16" nx2 carrier (R&S CMW-KS520) or nx4 carrier (R&S CMW-KS540)

"15-18" nx2 carrier (R&S CMW-KS521) or nx4 carrier (R&S CMW-KS540)

"15-22" nx2 carrier (R&S CMW-KS522) or nx4 carrier (R&S CMW-KS541)

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:APORTs
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:APORTs
```

CSI-RS Configuration

Selects the CSI reference signal configuration, influencing which resource elements are used for CSI-RS within a subframe.

The allowed values depend on the duplex mode, the cyclic prefix and the number of CSI-RS antenna ports.

The values are specified in 3GPP TS 36.211, table 6.10.5.2-1 and table 6.10.5.2-2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:RESource
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:RESource
```

CSI-RS Subframe Cfg

The CSI-RS subframe configuration ($I_{\text{CSI-RS}}$) determines in which subframes the CSI-RS is transmitted.

The selected value determines the periodicity ($T_{\text{CSI-RS}}$) and the subframe offset ($\Delta_{\text{CSI-RS}}$). Both values are displayed for information.

The mapping is defined in 3GPP TS 36.211, section 6.10.5.3.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:SUBFrame
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:SUBFrame
```

Zero Power CSI-RS

The 16-bit pattern is signaled to the UE and indicates for which CSI-RS configurations the UE must assume zero transmission power. In resource elements with zero transmission power, there is no downlink transmission at all - no CSI-RS and no PDSCH.

Bit value 1 means zero transmission power. For details, see 3GPP TS 36.211, section 6.10.5.2.

For FDD plus normal cyclic prefix, the last 6 bits are always 0. For FDD plus extended cyclic prefix, the last 8 bits are always 0.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:ZP:BITS
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:ZP:BITS
```

ZP CSI-RS Subframe Cfg

Selects the subframe configuration for zero power CSI-RS, similar as parameter "CSI-RS Subframe Cfg" for non-zero power CSI-RS.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:ZP:CSIRs:SUBFrame
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:ZP:CSIRs:SUBFrame
```

Channel Matrix

The matrix defines the radio channel coefficients, see [Section 2.2.18.1, "Radio channel coefficients for MIMO"](#), on page 67.

The size of the matrix depends on the number of TX antennas. Each element of the matrix is a complex number, defined via its phase and its magnitude.

For 2x4 and 8x4, the channel matrix is not configurable and displayed for information.

For 2x2, 4x2, 8x2 and 4x4, you can configure the phase of each element. And you can configure the square of the magnitude of all but one element in each matrix line. The magnitude of the last element is adapted automatically, so that the sum of all magnitude squares within each line of the matrix equals 1.

For 4x4, you can also select a predefined matrix instead of configuring your own matrix, see parameter "Matrix Selection".

The channel matrix settings are hidden for fading scenarios and for scenarios with RF output before the radio channel.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:TWO<line>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:
FOUR<line>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:
EIGHt<line>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:
MIMO<Mimo>:LINE<line>
```

Plus corresponding PCC commands.

Matrix Selection

Selects a predefined 4x4 matrix or the user-defined 4x4 matrix.

If you select a predefined matrix, the resulting matrix coefficients are displayed for information.

"User Defined"	The configured channel matrix is used.
"3GPP Chan- nel Matrix"	The square of the magnitude is 0.25 for all coefficients. The used phases are 0°, 90°, 180° and 270°.
"Hadamard Matrix"	The square of the magnitude is 0.25 for all coefficients. The used phases are 0° and 180°.
"Identity Matrix"	The square of the magnitude is 1 for the main diagonal and 0 for all other coefficients. The phase is 0° for all coefficients.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:
MIMO<Mimo>:MSElection
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:
MIMO<Mimo>:MSElection
```

2.4.12.7 Miscellaneous connection settings part 2

Access: "Config" hotkey > "Connection" node (lower part).

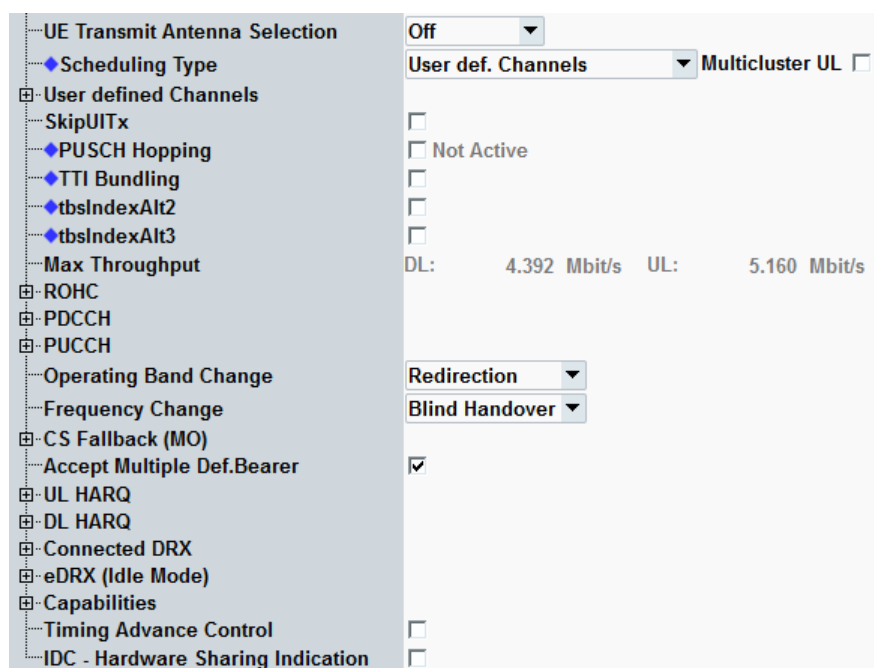


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UE Transmit Antenna Selection

Configures the parameter "ue-TransmitAntennaSelection" signaled to the UE, see 3GPP TS 36.213, section 8.7 "UE Transmit Antenna Selection".

You can disable UE antenna selection or allow open loop antenna selection. Closed loop antenna selection is not supported. For MIMO scenarios and for multi-cluster allocation, antenna selection is not allowed.

If you allow open loop antenna selection, the UE can change the used transmit antenna between TTIs. Connect the possible UE transmit antennas to the configured RF input connector via a combiner.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:UETSelection`

Scheduling Type

Selects the channel type to be scheduled for the UE.

For carrier aggregation scenarios, you can configure this setting per component carrier.

"RMC"	The instrument allocates a 3GPP-compliant reference measurement channel. For configuration, see Section 2.4.12.11, "RMC settings (no eMTC)", on page 240 and subsequent section.
"User-defined Channels"	3GPP-compliant RMCs allow only specific combinations of cell bandwidth, number of allocated RBs, RB position and modulation type. User-defined channels provide more flexibility concerning the allowed combinations. For configuration, see Section 2.4.12.13, "User-defined channels (no LAA, no eMTC)", on page 246 and subsequent sections. Option R&S CMW-KS510 is required (except for eMTC).
"User-defined TTI-Based"	Like "User-defined Channels", but settings can be configured individually per TTI (subframe). For configuration, see Section 2.4.13, "TTI-based channel configuration", on page 265. Option R&S CMW-KS510 is required.
"Fixed CQI"	Provides a downlink signal with CQI index and RB allocation configurable individually per TTI. The uplink signal is the same as for "User-defined TTI-Based". For configuration, see Section 2.4.13, "TTI-based channel configuration", on page 265. Option R&S CMW-KS510 is required.

"Follow ..."	The "Follow..." scheduling types provide a downlink signal configured according to the values reported by the UE. The scheduling types are available depending on the transmission mode: • Follow CQI (TM 1 to 9) • Follow PMI (TM 4, 6, 8, 9) • Follow CQI and RI (TM 3) • Follow CQI, PMI and RI (TM 4, 8, 9) • Follow PMI and RI (TM 4, 8, 9) The uplink signal is the same as for "User-defined TTI-Based". For configuration, see Section 2.4.13, "TTI-based channel configuration", on page 265. Selecting these scheduling types also enables CQI reporting. The "Follow..." scheduling types can only work correctly, if the UE sends the corresponding reports. For carrier aggregation, configure the CQI reporting settings, so that reports for the individual carriers arrive in different subframes. Otherwise, the "follow" mechanism does not work. See also Section 2.4.14, "CQI reporting", on page 278. Option R&S CMW-KS510/-KS512 is required (without CA/with CA).
"SPS"	Semi-persistent scheduling (PCC only). A configured RB allocation is granted to the UE every n^{th} subframe. Selecting this value disables HARQ and UL dynamic scheduling for connected DRX. For configuration, see Section 2.4.12.19, "SPS configuration", on page 260. Option R&S CMW-KS510 is required.
"eMTC Auto Mode"	Automatic consideration of repetitions, hopping, and so on, for comfortable eMTC scheduling configuration. To enable eMTC, see "eMTC" on page 142. For configuration of the auto mode, see Section 2.4.12.20, "eMTC auto mode settings", on page 262.
"eMTC Compact Scheduling"	Like "User-defined Channels", but with more compact scheduling in the time-domain, for maximum eMTC throughput. To enable eMTC, see "eMTC" on page 142. For configuration of the scheduling, see Section 2.4.12.21, "eMTC compact scheduling settings", on page 263.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:STYPe
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:STYPe
```

Multicluster UL

Enables multi-cluster allocation (resource allocation type 1) for RMC or user-defined channels. With disabled checkbox, contiguous allocation is used (resource allocation type 0).

Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:MCLuster:UL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:UL
```

Number of Layers

This setting is displayed for MIMO 4x4 with spatial multiplexing, TM 3 and TM 4.

You can select between two layers and four layers.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:NOLayers
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:NOLayers
```

SPS TwoIntervalsConfig TDD

Configures the parameter "twoIntervalsConfig", signaled to the UE for scheduling type SPS in TDD mode. The parameter is specified in 3GPP TS 36.321.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:TIConfig
```

SkipUITx

Allows the UE to skip granted uplink transmissions if no data is available in the UE buffer. Skipping a transmission means sending nothing (DTX).

The fields "skipUplinkTxSPS" and "skipUplinkTxDynamic" are set to true and signaled to the UE.

For SPS with an UL interval of less than 10 subframes, the setting is ignored and skipping is automatically allowed, see also "[Interval DL / UL](#)" on page 261.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SUITx
```

PUSCH Hopping

Enables inter-subframe PUSCH frequency hopping, type 2.

Depending on the RB allocation, PUSCH hopping is allowed or not. You can always enable the checkbox. The text to the right of the checkbox tells you whether PUSCH hopping is active (enabled and allowed) or not active (disabled or not allowed).

To provoke retransmissions and thus PUSCH hopping, configure also "[UL HARQ](#)" on page 229.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:HPUSch:ENABle
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:HPUSch:ENABle
SENSe:LTE:SIGN<i>:CONNection[:PCC]:HPUSch:ACTive?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:HPUSch:ACTive?
```

TTI Bundling

Enables or disables TTI bundling for the uplink.

With TTI bundling, the UE sends the same data with different redundancy versions in four subsequent TTIs. TTI bundling is faster than HARQ. So TTI bundling makes sense if the channel quality is bad and the application is delay-sensitive, for example voice over LTE.

The following restrictions apply for TTI bundling:

- FDD:
 - User-defined scheduling (global, not TTI-based) or SPS
- TDD:

- User-defined scheduling (global, not TTI-based)
- UL/DL configuration = 0, 1 or 6
- UL RB configuration:
 - Number of RB ≤ 3
 - Modulation type QPSK
- No UL carrier aggregation

Option R&S CMW-KS510 is required.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TTIBundling
```

tbsIndexAlt2

Enables or disables sending the parameter "tbsIndexAlt2-r14" to the UE. The parameter informs the UE that TBS index 33B is used instead of 33.

Option R&S CMW-KS514 is required. Not configurable for RMCs.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TIA<no>
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TIA<no>
```

tbsIndexAlt3

Enables or disables sending the parameter "tbs-IndexAlt3-r15" to the UE. The parameter informs the UE that TBS index 37A is used instead of 37.

Option R&S CMW-KS505/-KS555 is required. Not configurable for RMCs.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TIA<no>
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TIA<no>
```

Max Throughput

Displays the expected maximum throughput in Mbit/s (averaged over one frame). For the downlink, the value refers to the sum of all streams of a component carrier.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]?
```

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?
```

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>?
```

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:SCC<c>?
```

ROHC

See [Section 2.4.12.8, "ROHC settings"](#), on page 233.

PDCCH

Access: "Config" hotkey > "Connection" node > "PDCCH" node.



This node configures the used PDCCH configuration. For carrier aggregation scenarios, the settings can be configured per component carrier.

PDCCH Symbol Config, #PDCCH Symbols ← PDCCH

Configures and displays the number of OFDM symbols used for the PDCCH per normal subframe.

The available values depend on the cell bandwidth as follows:

- 1.4 MHz: four symbols
- 3 MHz and 5 MHz: two or three symbols
- 10 MHz, 15 MHz and 20 MHz: one, two, three symbols or "Auto"
"Auto" configures two or three symbols, depending on the scheduling type.

With enabled "Reduced PDCCH", the lowest allowed value is set automatically.

"#PDCCH Symbols" displays the number of OFDM symbols resulting from the settings.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:SYMBOL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:SYMBOL
SENSe:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:PSYMBOLs?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:PSYMBOLs?
```

Aggr. Level DL/UL Config, Aggreg. Level ... ← PDCCH

There are three aggregation levels, for different types of DCI messages: DCI messages for DL with C-RNTI, for UL with C-RNTI and for DL with SI-RNTI. The aggregation levels are displayed as numbers of control channel elements (CCE), see 3GPP TS 36.211, section 6.8.1.

"Aggr. Level DL/UL Config" configures the C-RNTI aggregation levels. Value <a>/ means <a> CCE for DL and CCE for UL. Which values are possible, depends also on the scheduling settings. If the selected value is not possible, a lower value is used automatically.

"Auto" configures the aggregation levels automatically, depending on the scheduling settings.

The "Aggreg. Level..." parameters display the used aggregation levels, resulting from the settings.

Prerequisites for the "Aggr. Level DL/UL Config" values:

- "Auto": always possible
- "1/1": cell BW 1.4 MHz
- "4/2": cell BW 1.4 MHz or 3 MHz
- "4/4": cell BW 3 MHz or 5 MHz
- "8/4": cell BW 5 MHz
- "8/8": cell BW 10 MHz or 20 MHz, normal CP, frame structure not type 2, 1 or 3 PDCCH symbols

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:ALEVEL
SENSe:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:ALEVEL?
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:ALEVEL
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:ALEVEL?
```

Reduced PDCCH ← PDCCH

Enable "Reduced PDCCH" to reduce the resources used for the PDCCH and to increase the resources available for the PDSCH. Thus you can influence the used coding rates and the PDSCH data rate.

This setting is a comfort function. You can achieve the same effect by setting low values for "PDCCH Symbol Config".

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:RPDCch
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:RPDCch
```

PUCCH

Access: "Config" hotkey > "Connection" node > "PUCCH" node.



This node selects the PUCCH format used for carrier aggregation scenarios. Without CA, the format is not configurable but selected depending on the transported message type, as defined by 3GPP.

"PUCCH format for CA" selects the desired format. "Act. PUCCH format for CA" displays the used format.

The configured format and the used format differ, if the configured format is incompatible with other settings. In that case, the required format is selected automatically and is displayed for information. Example: A PUCCH with the selected format cannot transport enough ACK/NACKs for the configured number of aggregated carriers.

Available formats:

- Format 1b with channel selection: Needs less space than the other formats and can transport fewer ACK/NACKs. Only used if configured.
- Format 3: Selected automatically if format 1a is not sufficient. Typically sufficient for up to 5 CC.
- Format 4: Can transport the most ACK/NACKs. Selected automatically if formats 1a and 3 are not sufficient.
- Format 5: Maximum payload size smaller than with format 4, but more efficient due to spreading (code division multiplexing of data of two UEs). Only used if configured.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PUCCh:FFCA
SENSe:LTE:SIGN<i>:CONNection[:PCC]:PUCCh:FFCA?
```

Operating Band Change, Frequency Change

Select the mechanism to be used for inter-band handover and inter-frequency handover within the LTE signaling application.

Either blind handover or redirection can be used, see [Section 2.2.10, "Handover"](#), on page 42.

The selected mechanism is also relevant for a swap of SCC and PCC settings during an established connection. If the cell bandwidth is changed by the swap, redirection is used. If the cell bandwidth is not changed but the band is changed, the setting "Operating Band Change" is used. If also the band is not changed but the frequency is changed, the setting "Frequency Change" is used.

If you enable eMTC, redirection is set automatically.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:OBCHange
```

```
CONFigure:LTE:SIGN<i>:CONNection:FCHange
```

CS Fallback (MO)

Access: "Config" hotkey > "Connection" node > "CS Fallback (MO)" node.

The following settings configure the feature circuit-switched fallback (CSFB) for mobile-originating calls.

A fallback for a mobile-originating voice call can be performed to a WCDMA cell, a GSM cell or a TD-SCDMA cell. If you select "None", fallback requests are rejected.

Select the technology, specify the target band and the channel. For GSM, you can also set the band indicator for distinction of GSM 1800 and GSM 1900 bands.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CSFB:DESTination
```

```
CONFigure:LTE:SIGN<i>:CONNection:CSFB:GSM
```

```
CONFigure:LTE:SIGN<i>:CONNection:CSFB:WCDMa
```

```
CONFigure:LTE:SIGN<i>:CONNection:CSFB:TDSCdma
```

Accept Multiple Default Bearer

Enables/disables accepting multiple default bearer requests.

- Disabled: Only the first default bearer request of a UE is accepted. Additional requests are rejected.
- Enabled: If the UE sends several default bearer requests, several bearers are established. For each bearer, a different IP address is assigned to the UE.

During the establishment of additional default bearers, do not modify any settings. So do not change parameters when the UE sends a default bearer request, until the event log indicates that the requested bearer has been established.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:AMDBearer
```

UL HARQ

Access: "Config" hotkey > "Connection" node > "UL HARQ" node.

The following parameters configure the HARQ procedure for uplink transmissions.

Option R&S CMW-KS510 is required. For eMTC, UL HARQ is disabled and the node is hidden.



UL HARQ ← UL HARQ

Enables or disables HARQ for uplink transmissions.

HARQ is disabled for TTI bundling and multi-cluster allocation. For the scheduling type SPS, HARQ is disabled and the node is hidden.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:ENABLE`

Number of HARQ Transmissions ← UL HARQ

Specifies the maximum number of uplink transmissions, including initial transmissions and retransmissions.

For TTI bundling, only the value 4 is supported.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:NHT`

DCI-0 / PHICH ← UL HARQ

Selects how the UE is informed about required UL retransmissions / successful UL transmissions.

There are two methods:

- "DCI-0": The PDCCH transports a DCI 0 and a new data indicator (NDI) bit that is toggled (success, no retransmission) or not toggled (fail, retransmission) relative to the previous transmission.
- "PHICH": The physical hybrid-ARQ indicator channel (PHICH), transports an ACK (success, no retransmission) or NACK (fail, retransmission).

In normal operation, a retransmission is only requested for failed UL transmissions. To test PUSCH hopping, you can request a retransmission for any UL transmission.

"DCI-0 only"	DCI-0, normal operation
"PHICH only"	PHICH, normal operation
"DCI-0 & PHICH"	DCI-0 and PHICH, normal operation
"PHICH NACK"	PHICH, retransmission for all transmissions
"PHICH NACK & DCI-0"	DCI-0 and PHICH, retransmission for all transmissions

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:DPHich`

maxHARQ-Tx

For the scheduling type SPS, this parameter is available instead of the node "UL HARQ". The value is signaled to the UE. UL HARQ is disabled.

For TTI bundling, only the value 4 is supported.

Option R&S CMW-KS510 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:MAXTx`

DL HARQ

Access: "Config" hotkey > "Connection" node > "DL HARQ" node.

The following parameters configure the HARQ procedure for downlink transmissions.

For scenarios without carrier aggregation, option R&S CMW-KS510 is required. For scenarios with carrier aggregation, R&S CMW-KS512 is required.

DL HARQ

DL HARQ

Number of HARQ Transmissions: 2

Redundancy Version Coding Sequence: TS 36.101 {0,1,2,3}/{0,0,1,2}

User Defined Sequence

Length: 4 { 0 0 0 0 }

DL HARQ ← DL HARQ

Enables or disables HARQ for downlink transmissions.

HARQ is disabled for scheduling type SPS.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:ENABLE`

Number of HARQ Transmissions ← DL HARQ

Specifies the maximum number of downlink transmissions, including initial transmissions and retransmissions.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:NHT`

Redundancy Version Coding Sequence, User Defined Sequence ← DL HARQ

The coding sequence defines the redundancy versions for repeated transmissions of a packet. The first value of the sequence is used for initial transmissions, the second value for the first retransmission, and so on.

If the sequence is shorter than the maximum allowed number of transmissions, it is used several times. Example: Four transmissions allowed and configured sequence {0, 2} results in the used sequence {0, 2, 0, 2}.

If the sequence is longer than the maximum allowed number of transmissions, the surplus values have no effect. Example: Two transmissions allowed and configured sequence {0, 1, 2, 3} results in the used sequence {0, 1}.

You can configure a user-defined sequence or use a sequence defined by 3GPP:

- **TS 36.101:** Depending on the modulation scheme, either the sequence {0, 1, 2, 3} or {0, 0, 1, 2} is used, as defined in 3GPP TS 36.101.
- **TS 36.104:** The sequence {0, 2, 3, 1} is used, as defined in 3GPP TS 36.104.
- **User-Defined:** The user-defined sequence is used. Define first the length of the sequence, then the sequence itself.

For details about the meaning of the individual redundancy versions, see "redundancy version number" in 3GPP TS 36.212.

Remote command:

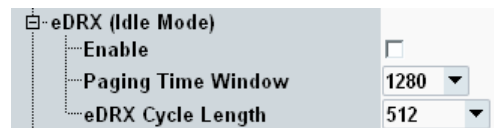
```
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:RVCSequence
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence:LENGth
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence
```

Connected DRX

See [Section 2.4.12.9, "Connected DRX settings"](#), on page 234.

eDRX

Access: "Config" hotkey > "Connection" node > "eDRX (Idle Mode)" node.



This node configures extended discontinuous reception (eDRX). eDRX has been specified for IoT devices for power saving.

For established connections, eDRX increases the maximum allowed duration of a long DRX cycle, see ["Long DRX Cycle"](#) on page 236.

For the idle mode, eDRX introduces discontinuous reception. The UE is informed about the eDRX cycle length and the paging time window size. Each eDRX cycle includes one paging time window during which the UE monitors the PDCCH for paging messages. The remaining time of the cycle, the UE does not monitor the PDCCH.

Option R&S CMW-KS590 is required.

Enabled ← eDRX

Enables or disables eDRX.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:ENABLE
```

Paging Time Window ← eDRX

Duration of one paging time window in idle mode, in subframes.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:PTWindow
```

eDRX Cycle Length ← eDRX

Duration of one extended DRX cycle in idle mode, in radio frames.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:CLENGTH
```

Capabilities

See [Section 2.4.12.10, "Capability request settings"](#), on page 238.

Timing Advance Control

Enables the correction of a changing UL frame timing via timing advance commands, sent to the UE.

A drift of the UL frame timing can cause a call drop after some time. Enabling timing advance control corrects such a drift via correction commands sent to the UE.

Remote command:
CONFigure:LTE:SIGN<i>:CONNection:TAControl

IDC - Hardware Sharing Indication
Enables sending the information element "idc-HardwareSharingIndication" to the UE when adding an LAA SCell.

Option R&S CMW-KS514 is required.
Remote command:
CONFigure:LTE:SIGN<i>:CONNection:IDCHsindic

2.4.12.8 ROHC settings

Access: "Config" hotkey > "Connection" node > "ROHC" node.
The node configures robust header compression (ROHC).
Header compression is relevant for the continuous transfer of small packets, for example for voice over LTE. Here, the uncompressed header is typically bigger than the payload. Header compression reduces the radio resources required for a voice call.
ROHC is specified in 3GPP TS 36.323 and the references stated therein.
Option R&S CMW-KS510 is required.

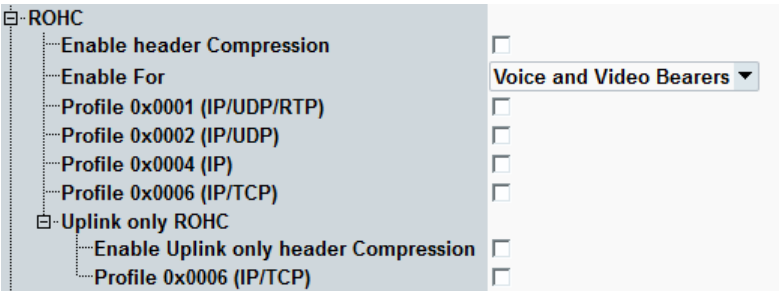


Figure 2-84: ROHC settings

Enable Header Compression.....	233
Enable for.....	234
Profile	234
Uplink Only ROHC.....	234
L Enable Uplink Only Header Compression.....	234
L Profile	234

Enable Header Compression
Enables or disables bidirectional header compression.

Remote command:
CONFigure:LTE:SIGN<i>:CONNection:ROHC:ENABLE

Enable for

Selects for which types of dedicated bearers ROHC is enabled. The setting applies to bidirectional header compression and to uplink-only header compression.

"Voice and Video Bearers" Dedicated bearers with the profile "Voice" or "Video".

"All Dedicated Bearers" All dedicated bearers, including voice, video and data connections.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:ROHC:EFOR`

Profile ...

Selects the header compression profiles allowed for bidirectional header compression.

If more than one enabled profile is compatible to the used protocols, the most specific profile is used. If no profile is compatible, no compression is performed.

Example - profile 2 (IP/UDP) and 4 (IP) enabled:

- IP/UDP/RTP traffic: profile 2 used
- IP/TDP/... traffic: profile 4 used

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:ROHC:PROFiles`

Uplink Only ROHC

This node configures uplink-only header compression.

You can enable bidirectional header compression or uplink-only header compression, not both in parallel.

Enable Uplink Only Header Compression ← Uplink Only ROHC

Enables or disables uplink-only header compression.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:ROHC:ULONly:ENABLE`

Profile ... ← Uplink Only ROHC

Selects the header compression profiles allowed for uplink-only header compression.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:ROHC:ULONly:PROFiles`

2.4.12.9 Connected DRX settings

Access: "Config" hotkey > "Connection" node > "Connected DRX" node.

The node configures settings for discontinuous reception (DRX) of the UE during an established connection. If DRX is enabled, the UE monitors the PDCCH discontinuously and thus reduces battery consumption. The related signaling parameters are configurable and specified in 3GPP TS 36.321, section 5.7.

There are two types of DRX cycles. Long DRX cycles are mandatory while short DRX cycles are optional. Both types have a fixed length and start with a fixed "On" time dur-

ing which the UE receiver is active. After the "On" time, there is an opportunity for DRX.

The following figure shows a typical example including short and long DRX cycles. The example starts with reception of a PDCCH during a long DRX cycle "On" time of two subframes. Due to this reception, the inactivity timer set to 40 subframes is started and the receiver stays on until the timer expires. In this example, no further PDCCH is received, so that the timer is not restarted.

After the timer is expired, four short DRX cycles are executed (short DRX cycle = 10 subframes, short cycle timer = 4 cycles). Finally, long DRX cycles are executed (long DRX cycle = 20 subframes).

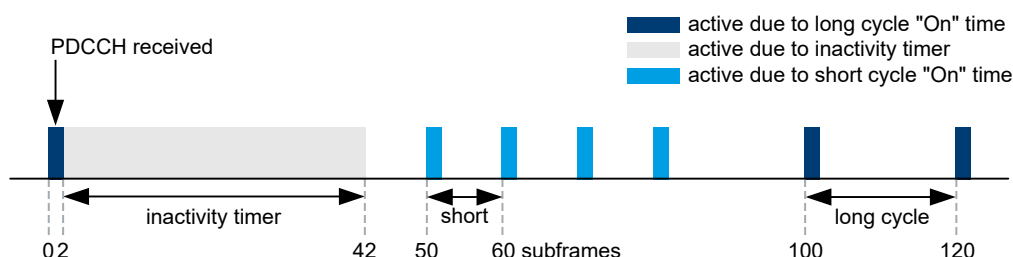


Figure 2-85: Example, short and long cycles after inactivity

Active MAC padding and uplink RB allocation cause PDCCH traffic and prevent DRX, see also "UL Dynamic Scheduling" on page 237.

The following settings are available.

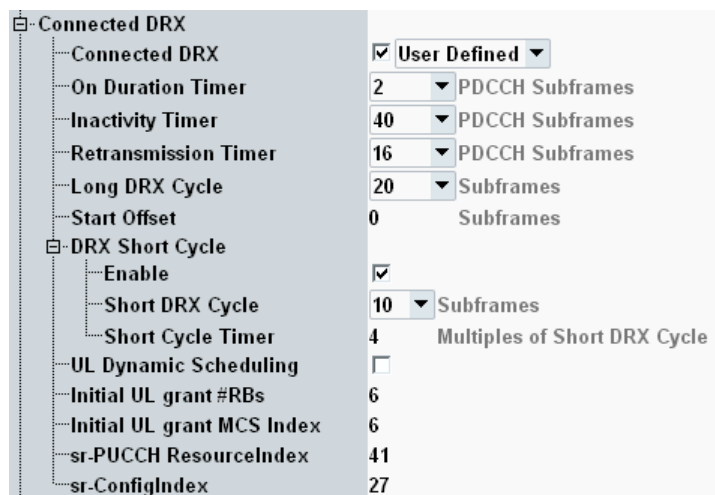


Figure 2-86: Connected DRX settings (values used in example)

The unit "PDCCH subframe" used in this context refers to subframes with PDCCH. For FDD, all subframes are PDCCH subframes. For TDD, only DL subframes and subframes with DwPTS are counted.

Connected DRX.....	236
On Duration Timer.....	236
Inactivity Timer.....	236

Retransmission Timer.....	236
Long DRX Cycle.....	236
Start Offset.....	237
DRX Short Cycle.....	237
L Enable.....	237
L Short DRX Cycle.....	237
L Short Cycle Timer.....	237
UL Dynamic Scheduling.....	237
Initial UL Grant #RBs / MCS Index.....	237
sr-PUCCH ResourceIndex.....	238
sr-ConfigIndex.....	238

Connected DRX

Enables or disables DRX and selects a set of DRX settings.

"DRX_S" Settings according to 3GPP TS 36.521-3, table H.3.6-1.

"DRX_L" Settings according to 3GPP TS 36.521-3, table H.3.6-2.

"User-Defined" You can configure all DRX settings.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:CDRX:ENABLe`

On Duration Timer

"On" time at the beginning of each short or long DRX cycle ("onDurationTimer").

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:CDRX:ODTimer`

Inactivity Timer

"On" time after reception and decoding of a PDCCH for this UE ("drx-InactivityTimer").

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:CDRX:ITIMer`

Retransmission Timer

"On" time if the UE expects a DL retransmission ("drx-RetransmissionTimer"). The timer expires early when the retransmission is received.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:CDRX:RTIMer`

Long DRX Cycle

Duration of one long DRX cycle ("longDRX-Cycle").

If short DRX cycles are enabled, the long DRX cycle duration is always divisible by the short DRX cycle duration. Enabling short DRX cycles or modifying the short DRX cycle duration modifies also the long DRX cycle duration, so that it is compatible.

The highest values 5120 and 10240 are only available if eDRX is enabled, see ["eDRX"](#) on page 232.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection:CDRX:LDCYcle`

Start Offset

Offset shifting all short and long DRX cycles ("drxStartOffset").

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SOFFset
```

DRX Short Cycle

The following settings configure optional short DRX cycles.

Enable ← DRX Short Cycle

Enables short DRX cycles.

For eMTC, short DRX cycles are disabled.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SCENable
```

Short DRX Cycle ← DRX Short Cycle

Duration of one short DRX cycle ("shortDRX-Cycle").

The long DRX cycle duration is always divisible by the short DRX cycle duration and adapted automatically to this setting (if short cycles are enabled).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SDCYcle
```

Short Cycle Timer ← DRX Short Cycle

Number of short DRX cycles to be processed ("drxShortCycleTimer"), for example after the inactivity timer has expired.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SCTimer
```

UL Dynamic Scheduling

With uplink dynamic scheduling, the UE gets uplink grants only upon request. Furthermore, MAC padding is disabled automatically.

Enable dynamic scheduling to force DRX, but allow the UE to get an UL grant upon request.

If you want to force DRX without allowing any uplink grants, disable dynamic scheduling, set the number of UL RBs to zero and disable MAC padding.

For scheduling type SPS, "UL Dynamic Scheduling" is disabled.

You can use uplink dynamic scheduling without enabling "Connected DRX".

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:UDScheduling
```

Initial UL Grant #RBs / MCS Index

The two settings are only relevant for UL dynamic scheduling. They configure the initial UL grant that the UE gets after a request. Subsequent UL grants are configured automatically, depending on how much data the UE still wants to send. For the first UL grant, this information is not available at the eNodeB.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:IUGNrb
CONFigure:LTE:SIGN<i>:CONNection:IUGMcsidx
```

sr-PUCCH ResourceIndex

Defines which PUCCH resource the UE must use for scheduling requests, see $n_{\text{PUCCH,SRI}}$ in 3GPP TS 36.213, section 10.1.5.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SRPRindex
```

sr-ConfigIndex

Defines how often and when the UE must send scheduling requests via the PUCCH. The effect of the individual SR configuration index values is specified in 3GPP TS 36.213, section 10.1.5.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SRCindex
```

2.4.12.10 Capability request settings

Access: "Config" hotkey > "Connection" node > "Capabilities" node.

The node defines which capability information is requested from the UE. That is, the settings configure information elements of the "ueCapabilityEnquiry" message.

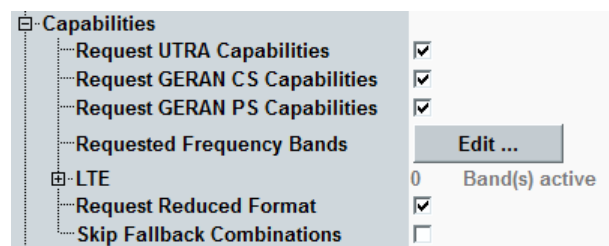


Figure 2-87: Capabilities settings

Request UTRA Capabilities.....	238
Request GERAN CS Capabilities.....	239
Request GERAN PS Capabilities.....	239
Requested Frequency Bands, LTE.....	239
Request Reduced Format.....	239
Skip Fallback Combinations.....	239

Request UTRA Capabilities

Selects whether UTRA capabilities are requested from the UE (entry "utra" in field "ue-CapabilityRequest" of "ueCapabilityEnquiry" message).

Remote command:

```
CONFigure:LTE:SIGN<i>:UECapability:RUTRa
```

Request GERAN CS Capabilities

Selects whether GERAN CS capabilities are requested from the UE (entry "geran-cs" in field "ue-CapabilityRequest" of "ueCapabilityEnquiry" message).

Remote command:

`CONFfigure:LTE:SIGN<i>:UECapability:RGCS`

Request GERAN PS Capabilities

Selects whether GERAN PS capabilities are requested from the UE (entry "geran-ps" in field "ue-CapabilityRequest" of "ueCapabilityEnquiry" message).

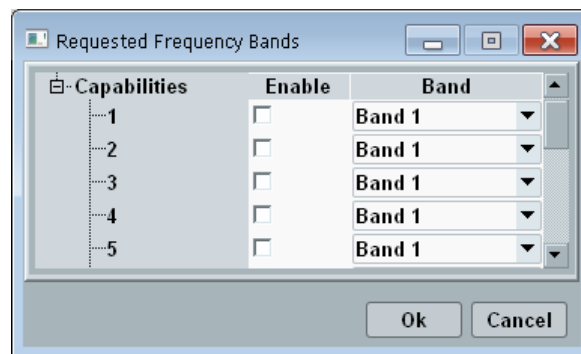
Remote command:

`CONFfigure:LTE:SIGN<i>:UECapability:RGPS`

Requested Frequency Bands, LTE

Configures the information element "requestedFrequencyBands" of the "ueCapabilityEnquiry" message. The element contains a list of LTE operating bands for which the UE is requested to report its capabilities.

To configure the list of bands, press the "Edit" button.



You can enable up to 16 entries with different bands. The element "requestedFrequencyBands" is filled with all enabled entries.

Remote command:

`CONFfigure:LTE:SIGN<i>:UECapability:RFBands:ALL`

Request Reduced Format

Adds the optional field "requestReducedFormat-r13" to the "ueCapabilityEnquiry" message.

Remote command:

`CONFfigure:LTE:SIGN<i>:UECapability:RRFormat`

Skip Fallback Combinations

Adds the optional field "requestSkipFallbackComb-r13" to the "ueCapabilityEnquiry" message.

Remote command:

`CONFfigure:LTE:SIGN<i>:UECapability:SFC`

2.4.12.11 RMC settings (no eMTC)

Access: "Config" hotkey > "Connection" node > "RMC" node.

The parameters in this section configure a 3GPP-conform reference measurement channel (RMC) for non-eMTC tests.

To display these settings, disable eMTC and select the scheduling type "RMC", see ["Scheduling Type"](#) on page 223.

To configure the RMC, proceed as follows (for UL and DL):

1. Select the cell bandwidth (see [Section 2.4.9, "Physical cell setup"](#), on page 170).
2. If you want to use UL multi-cluster, enable it.
3. For LAA, select the partial subframe configuration and the burst type.
4. Select the number and position of allocated resource blocks (RB).
For UL multi-cluster, configure first cluster 1, then cluster 2.
5. Select the modulation type.
If you want to use UL 64-QAM or UL 256-QAM, configure also ["Enable 64/256-QAM UL Support"](#) on page 171.
6. If the transport block size index is configurable, select it.

For valid parameter combinations and background information, see:

- [Section 2.2.19, "Scheduling type RMC"](#), on page 69
- [Section 2.2.16.2, "Scheduling type RMC for LAA"](#), on page 58

For carrier aggregation scenarios, you can configure the settings per component carrier.

	Stream 1	Stream 2
DL		
Use Stream 1 Settings		☒
# Resource Blocks	50	50
RB Position/Start RB	low Start RB: 0	low Start RB: 0
Modulation Type	QPSK	QPSK
Transport Block Size Ind...	5	5
Transport Block Size	4392	4392
Throughput	3.953 Mbit/s	3.953 Mbit/s
UL		
# Resource Blocks	50	
RB Position/Start RB	low Start RB: 0	
Modulation Type	QPSK	
Transport Block Size Ind...	6	
Transport Block Size	5160	
Throughput	5.160 Mbit/s	

Figure 2-88: RMC connection settings, MIMO

Partial Subframe Configuration	241
Use Stream 1 Settings	241
Burst Type	241

# Resource Blocks.....	241
Version.....	242
RB Position/Start RB.....	242
Modulation Type.....	242
Transport Block Size Index.....	242
Transport Block Size.....	242
Throughput.....	243

Partial Subframe Configuration

Only displayed for LAA with random transmission bursts.

Selects which subframes can have a partial allocation:

- None, all subframes of a burst have full allocation.
- Initial subframes, the first subframe of a burst can have a partial allocation.
- Ending subframes, the last subframe of a burst can have a partial allocation.
- Initial and ending subframes, the first and/or the last subframe can have a partial allocation.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:PSFConfig
```

Use Stream 1 Settings

This parameter is visible if the selected transmission scheme involves several downlink streams.

If you enable this parameter, all stream 1 settings are also applied to stream 2.

If you disable this parameter, you can configure some DL parameters individually per stream. Other parameters are configurable for stream 1 only and are applied automatically to stream 2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:DLEQual
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:DLEQual
```

Burst Type

Only displayed for LAA. Selects between fixed transmission bursts and random transmission bursts.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:TBURsts
```

Resource Blocks

Selects the number of allocated resource blocks. This parameter influences the allowed modulation types.

For multi-cluster allocation, this setting is configurable separately for cluster 1 and cluster 2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:UL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:MCLuster:UL
```

Plus corresponding PCC commands.

Version

Some TDD RMCs defined by 3GPP have the same cell bandwidth, number of RBs, modulation type and transport block size index. The parameter "Version" is introduced to distinguish between these RMCs.

For the meaning of the version values, see [Section 2.2.19.4, "DL RMCs, multiple TX antennas \(TM 2 to 6\)"](#), on page 77.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:VERSion:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:VERSion:DL<s>
```

RB Position/Start RB

Selects the position of the first allocated resource block.

For multi-cluster allocation, this setting is configurable separately for cluster 1 and cluster 2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:RBPosition:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:RBPosition:UL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:MCLuster:UL
```

Plus corresponding PCC commands.

Modulation Type

Selects the modulation type. This parameter influences the transport block size index.

256-QAM requires option R&S CMW-KS504/-KS554.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:DL<s>
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:UL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:UL
```

Transport Block Size Index

Displays or selects the transport block size index, depending on the modulation type.

For most RMCs, the TBS index can be determined from the other settings and is displayed for information.

Only for few RMCs, the combination of the other settings is ambiguous and you can also select a TBS index.

For LAA, the value applies to subframes with full allocation.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:DL<s>
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:UL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:UL
```

Transport Block Size

Displays the transport block size in bits.

Remote command:

n/a

Throughput

Displays the expected maximum throughput in Mbit/s (averaged over one frame). The value is calculated per downlink and uplink stream.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]:STReam<s>?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>:STReam<s>?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:SCC<c>?
```

2.4.12.12 RMC settings for eMTC

Access: "Config" hotkey > "Connection" node > "RMC" node.

The parameters in this section configure a 3GPP-conform reference measurement channel (RMC) for eMTC (CE mode A only).

To display these settings, enable eMTC and select the scheduling type "RMC", see ["Scheduling Type"](#) on page 223.

To configure the RMC, proceed as follows (for UL and DL):

1. Select the cell bandwidth. See [Section 2.4.9, "Physical cell setup"](#), on page 170.
2. Select the maximum eMTC bandwidth. See ["Max Bandwidth 5 MHz"](#) on page 194.
3. Configure half duplex or full duplex.
4. Select the subframe pattern.
5. Select the number of allocated resource blocks (RB).
6. Select the RB position within the narrowband (NB).
7. Select the NB position.
8. Select the modulation type.

For valid parameter combinations and background information, see:

- [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45

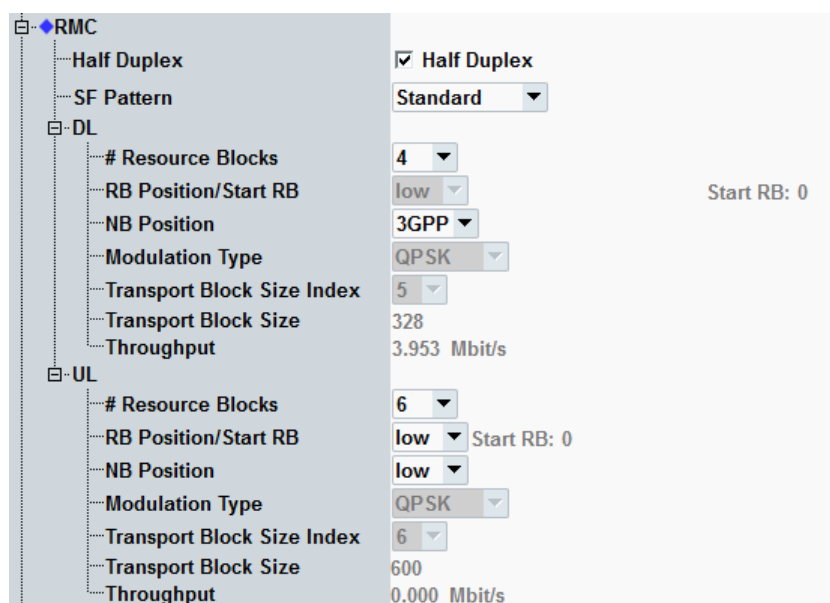


Figure 2-89: RMC connection settings, eMTC

Half Duplex.....	244
SF Pattern.....	244
# Resource Blocks.....	245
RB Position/Start RB.....	245
NB Position.....	245
Modulation Type.....	245
Transport Block Size Index.....	245
Transport Block Size.....	245
Throughput.....	246

Half Duplex

Selects between half-duplex operation and full-duplex operation.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:HDUPlex
```

SF Pattern

Determines the subframe pattern for eMTC RMCs.

The eMTC RMCs are defined in 3GPP TS 36.521 (keyword: UE category M1). This parameter selects according to which 3GPP TS 36.521 section the subframe pattern is configured.

"Standard"	The subframe pattern is defined in 3GPP TS 36.521 "Annex A", in the footnotes of the RMC definition tables.
"6.3.4EA"	The subframe pattern is defined in section 6.3.4EA "ON/OFF time mask for UE category M1".
"6.3.5EA.3"	The subframe pattern is defined in section 6.3.5EA.3 "Aggregate power control tolerance for UE category M1".

"6.5.2.1EA.2-A/B"

The subframe patterns are defined in section 6.5.2.1EA.2 "PUSCH-EVM with exclusion period for UE category M1".

Pattern A contains a subframe with high power followed by a subframe with low power.

Pattern B contains a subframe with low power followed by a subframe with high power.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:SFPattern
```

Resource Blocks

Selects the number of allocated resource blocks. This parameter influences the allowed modulation types and NB positions.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:UL
```

RB Position/Start RB

Selects the position of the first allocated resource block within the lowest used narrowband.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:RBPosition:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:RBPosition:UL
```

NB Position

Selects the lowest used narrowband, see also [Table 2-18](#).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:NBPosition:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:NBPosition:UL
```

Modulation Type

Selects the modulation type. This parameter influences the transport block size index.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:UL
```

Transport Block Size Index

Displays the transport block size index, depending on the other settings.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:UL
```

Transport Block Size

Displays the transport block size in bits.

Remote command:

n/a

Throughput

Displays the expected maximum throughput in Mbit/s (averaged over one frame).

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]?
```

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?
```

2.4.12.13 User-defined channels (no LAA, no eMTC)

Access: "Config" hotkey > "Connection" node > "User Defined Channels" node.

The parameters in this section configure user-defined channels for normal operation mode (no LAA, no eMTC). Option R&S CMW-KS510 is required.

To display these settings, disable eMTC and LAA and select the scheduling type "User-Defined Channels", see ["Scheduling Type"](#) on page 223.

To configure the channels, proceed as follows (for UL and DL):

1. Select the cell bandwidth (see [Section 2.4.9, "Physical cell setup"](#), on page 170).
2. If you want to use multi-cluster, enable it.
For UL, see [Section 2.4.12.7, "Miscellaneous connection settings part 2"](#), on page 221).
3. Configure the RB allocation.
For UL multi-cluster, configure first cluster 1, then cluster 2.
For DL multi-cluster, press the button "RB Clusters".
4. Select the modulation type.
If you want to use UL 64-QAM or UL 256-QAM, configure also ["Enable 64/256-QAM UL Support"](#) on page 171.
5. Select the transport block size index.

For valid parameter combinations and background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

For carrier aggregation scenarios, you can configure the settings per component carrier.

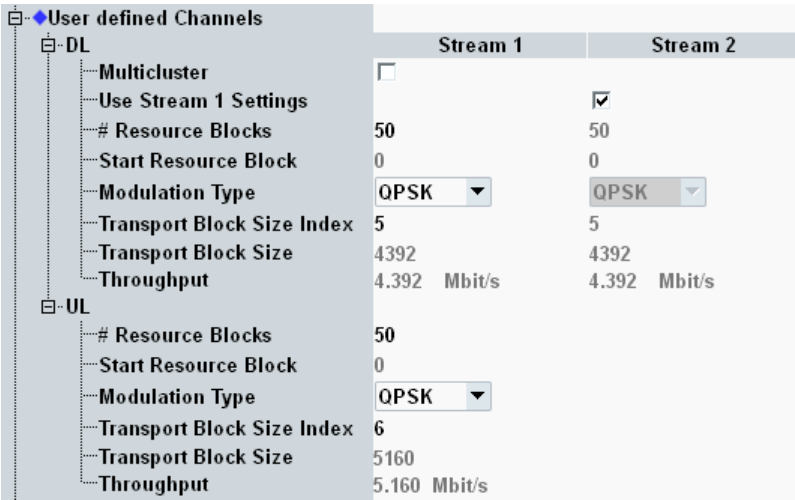
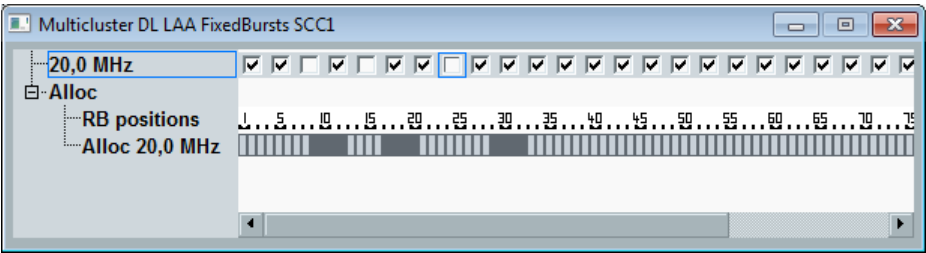


Figure 2-90: User-defined channel settings, MIMO

Multicluster.....	247
Use Stream 1 Settings.....	247
# Resource Blocks ... Transport Block Size.....	248
Throughput.....	248
Code Rate.....	248

Multicluster

You can enable DL multi-cluster allocation only for resource allocation type 0 (resource block groups). Option R&S CMW-KS510/-KS512 is required (without CA/with CA).
To use DL multi-cluster allocation, enable the checkbox. The button "RB Clusters" is displayed. Press the button to open the following configuration window.



In this dialog box, you can enable or disable each individual RBG via the checkboxes.
Remote command:
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:MCLuster:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:MCLuster:
DL<s>

Use Stream 1 Settings

This parameter is visible if the selected transmission scheme involves several downlink streams.
If you enable this parameter, all stream 1 settings are also applied to stream 2.

If you disable this parameter, you can configure some DL parameters individually per stream. Other parameters are configurable for stream 1 only and are applied automatically to stream 2.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:DLEQual
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:DLEQual
```

Resource Blocks ... Transport Block Size

"# Resource Blocks" selects the number of allocated resource blocks.

"Start Resource Block" specifies the number of the first allocated resource block. This parameter allows you to shift the allocated RBs within the cell bandwidth.

For UL multi-cluster allocation, the two resource block settings are configurable separately for cluster 1 and cluster 2. For DL multi-cluster allocation, see ["Multicluster"](#) on page 247.

"Modulation Type" selects the modulation type and influences the allowed transport block size indices. 256-QAM requires option R&S CMW-KS504/-KS554. 1024-QAM requires option R&S CMW-KS505/-KS555.

"Transport Block Size Index" selects the TBS index. The resulting "Transport Block Size" in bits is displayed

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:UL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:MCLuster:UL
Plus corresponding PCC commands.
```

Throughput

Displays the expected maximum throughput in Mbit/s (averaged over one frame). The value is calculated per downlink and uplink stream.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]:STReam<s>?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>:STReam<s>?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:SCC<c>?
```

Code Rate

Only displayed in the main view: Effective channel code rate, i.e. the number of information bits (including CRC bits) divided by the number of physical channel bits on PDSCH/PUSCH.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:DL<s>:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:UL:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:DL<s>:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:UL:CRATe:ALL?
```

2.4.12.14 User-defined channels for eMTC

Access: "Config" hotkey > "Connection" node > "User Defined Channels" node.

The parameters in this section configure user-defined channels for eMTC (CE mode A only).

To display these settings, enable eMTC and select the scheduling type "User-Defined Channels", see ["Scheduling Type"](#) on page 223.

To configure the channels, proceed as follows (for UL and DL):

1. Select the cell bandwidth. See [Section 2.4.9, "Physical cell setup"](#), on page 170.
2. Select the maximum eMTC bandwidth. See ["Max Bandwidth 5 MHz"](#) on page 194.
3. Configure half duplex or full duplex.
4. Select the SF pattern.
5. Select the number of resource blocks.
6. Select the start resource block, the narrowbands and the modulation type.
7. Select the transport block size index.

For valid parameter combinations and background information, see [Section 2.2.13, "User-defined channels for eMTC"](#), on page 49.

Figure 2-91: User-defined channel settings, eMTC

Half Duplex.....	250
SF Pattern.....	250
DL, # Resource Blocks ... Transport Block Size.....	250

UL, # Resource Blocks ... Transport Block Size.....	250
Throughput.....	251
Code Rate.....	251

Half Duplex

Selects between half-duplex operation and full-duplex operation.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:HDUPlex
```

SF Pattern

For half-duplex operation, you can select a subframe pattern. There is a standard pattern without PDSCH HARQ-ACK bundling. And there are two patterns with bundling and 8 or 10 HARQ processes.

Select a pattern with bundling to increase the downlink throughput. For pattern details and required UE capabilities, see "[Subframe pattern](#)" on page 51.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:SFPattern
```

DL, # Resource Blocks ... Transport Block Size

"# Resource Blocks" selects the number of resource blocks allocated per narrowband.

"Start Resource Block" specifies the number of the first allocated resource block within each narrowband.

"NB Position" selects the position of the narrowbands within the cell bandwidth.

"Additional NB" selects additional narrowbands for a maximum bandwidth of 5 MHz.

"Modulation Type" selects the modulation type and influences the allowed transport block size indices.

"Transport Block Size Index" selects the TBS index. The resulting "Transport Block Size" in bits is displayed.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:DL
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:
NBPosition:DL
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:DL:
ANB<Number>
```

UL, # Resource Blocks ... Transport Block Size

"# Resource Blocks" selects the total number of resource blocks, allocated contiguously over up to four used narrowbands.

"Start Resource Block" specifies the number of the first allocated resource block within the lowest used narrowband.

"NB Position" selects the lowest used narrowband.

"Modulation Type" selects the modulation type and influences the allowed transport block size indices.

"Transport Block Size Index" selects the TBS index. The resulting "Transport Block Size" in bits is displayed.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:UL
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:
NBPosition:UL
```

Throughput

Displays the expected maximum throughput in Mbit/s (averaged over one frame).

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]:STReam<s>?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?
```

Code Rate

Only displayed in the main view: Effective channel code rate, i.e. the number of information bits (including CRC bits) divided by the number of physical channel bits on PDSCH/PUSCH.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:DL<s>:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:UL:CRATe:ALL?
```

2.4.12.15 User-defined ch. for LAA, fixed bursts

Access: "Config" hotkey > "Connection" node > "User Defined Channels" node.

The parameters in this section configure user-defined channels for LAA, with a fixed burst configuration. Options R&S CMW-KS510 and -KS514 are required.

To display these settings, enable LAA and select the scheduling type "User-Defined Channels", see ["Frame Structure"](#) on page 142 and ["Scheduling Type"](#) on page 223.

To configure the channels, proceed in the following order:

1. Set the burst type to "Fixed transmission bursts".
2. If you want to use multi-cluster, enable it.
3. Configure the burst length.
4. Configure the first subframe (starting position) and the last subframe (OFDM symbols).
5. Select the modulation type.
6. Configure the remaining settings in any order.

For valid parameter combinations and background information, see ["Fixed transmission bursts"](#) on page 57.

You can configure the settings per SCC with LAA.

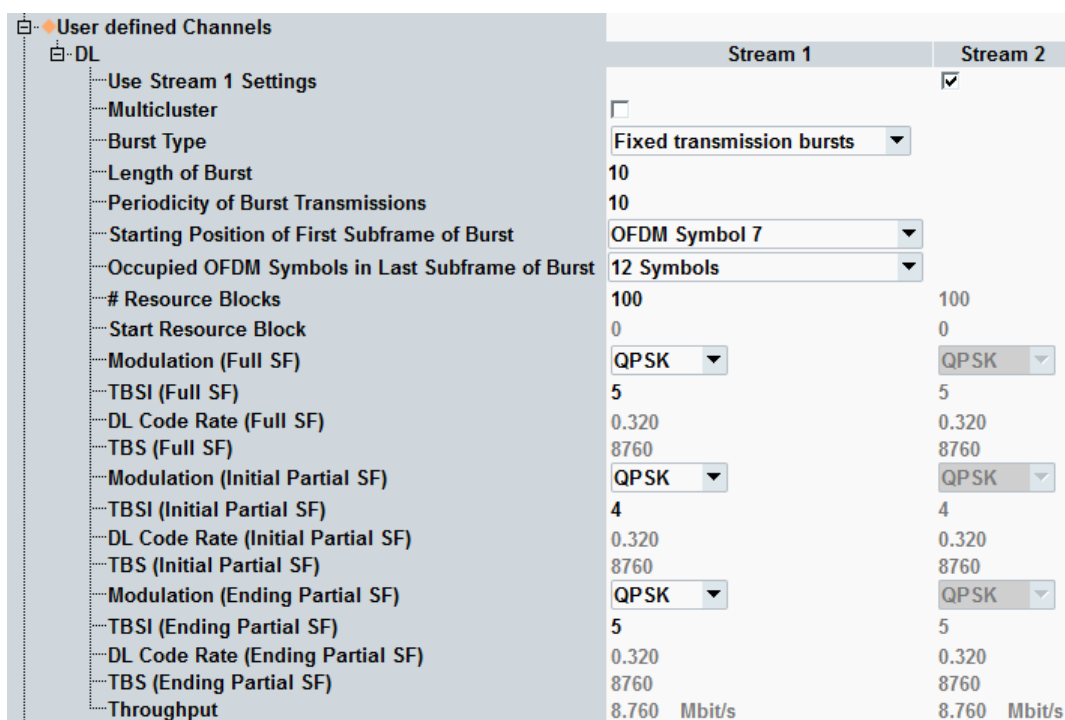


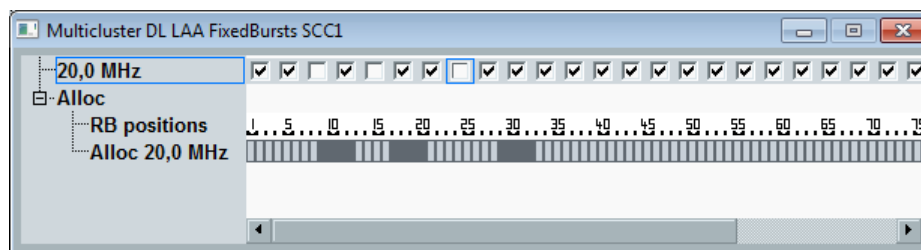
Figure 2-92: User-defined channel settings, LAA fixed bursts

Multicluster.....	252
Burst Type.....	253
Length of Burst.....	253
Periodicity of Burst Transmissions.....	253
Start Position of First Subframe of Burst.....	253
Occupied OFDM Symbols in Last Subframe of Burst.....	253
# Resource Blocks ... TBS (Ending Partial SF).....	253
DL Code Rate (...)	254
Throughput.....	254

Multicluster

Multi-cluster allocation is only possible for resource allocation type 0 (resource block groups). Option R&S CMW-KS512 is required.

To use DL multi-cluster allocation, enable the checkbox. The button "RB Clusters" is displayed. Press the button to open the following configuration window.



In this dialog box, you can enable or disable each individual RBG via the checkboxes.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst:
FSUBframes:MCLuster:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst:
PIPSubframes:MCLuster:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst:
PEPSubframes:MCLuster:DL<s>
```

Burst Type

This section describes "Fixed transmission bursts".

See also [Section 2.4.12.16, "User-defined ch. for LAA, random bursts"](#), on page 254.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:TBURsts
```

Length of Burst

Number of subframes per burst.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:BLENGTH
```

Periodicity of Burst Transmissions

Periodicity n for subsequent bursts (a burst starts every n^{th} subframe). The minimum value equals the burst length.

Example: Burst length = 3, Periodicity = 4 results in the sequence: (3 SF burst) (1 SF muting) (3 SF burst) and so on.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:PBTR
```

Start Position of First Subframe of Burst

Enables or disables partial allocation for the first subframe of the burst.

"OFDM Symbol 0" Full allocation, both slots allocated (OFDM symbol 0 to 13)

"OFDM Symbol 7" Partial allocation, second slot allocated (OFDM symbol 7 to 13)
Only allowed for burst length > 1.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:SPFSubframe
```

Occupied OFDM Symbols in Last Subframe of Burst

Number of allocated OFDM symbols at the beginning of the last subframe of the burst.

For burst length = 1, full allocation is mandatory (14 symbols).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:OSLSubframe
```

Resource Blocks ... TBS (Ending Partial SF)

- "# Resource Blocks" selects the number of allocated resource blocks.
"Start Resource Block" specifies the number of the first allocated resource block.

For DL multi-cluster allocation, see ["Multicluster"](#) on page 252.

- "Modulation (...)" selects the modulation type and influences the allowed TBSI values. 256-QAM requires option R&S CMW-KS504/-KS554. 1024-QAM requires option R&S CMW-KS505/-KS555.

"TBSI (...)" selects the TBS index. The resulting transport block size "TBS (...)" in bits is displayed.

The settings are configurable for:

- Subframes with full allocation "(Full SF)"
- First subframes with partial allocation "(Initial Partial SF)"
- Last subframes with partial allocation "(Ending Partial SF)"

The settings for partial allocation are only displayed if you have configured partial allocation.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:
FSUBframes:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:
PIPSubframes:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:
PEPSubframes:DL<s>
```

DL Code Rate (...)

Effective channel code rate, i.e. the number of information bits (including CRC bits) divided by the number of physical channel bits on PDSCH.

The information is displayed for all used subframe types (full allocation, first SF partial allocation, last SF partial allocation).

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:DL<s>:
FSUBframes:CRATe?
```

```
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:DL<s>:
PIPSubframes:CRATe?
```

```
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:DL<s>:
PEPSubframes:CRATe?
```

Throughput

Displays the expected average maximum throughput in Mbit/s. The value is calculated per downlink stream.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>:STream<s>?
```

2.4.12.16 User-defined ch. for LAA, random bursts

Access: "Config" hotkey > "Connection" node > "User Defined Channels" node.

The parameters in this section configure user-defined channels for LAA, with a random burst configuration. Options R&S CMW-KS510 and -KS514 are required.

To display these settings, enable LAA and select the scheduling type "User-Defined Channels", see ["Frame Structure"](#) on page 142 and ["Scheduling Type"](#) on page 223.

To configure the channels, proceed in the following order:

1. Set the burst type to "Random transmission bursts".
2. If you want to use multi-cluster, enable it.
3. Select the modulation types.
4. Configure the remaining settings in any order.

For valid parameter combinations and background information, see ["Random transmission bursts"](#) on page 56.

You can configure the settings per SCC with LAA.

	Stream 1	Stream 2
☐ Multicenter		☒
☐ Random transmission bursts		
8.760 Mbit/s		8.760 Mbit/s
100		
0		
Modulation (Full SF)	QPSK	QPSK
Modulation PIPS	QPSK	QPSK
Modulation (PEPS - 6 Symbols)	QPSK	QPSK
Modulation (PEPS - 9 Symbols)	QPSK	QPSK
Modulation (PEPS - 10 Symbols)	QPSK	QPSK
Modulation (PEPS - 11 Symbols)	QPSK	QPSK
Modulation (PEPS - 12 Symbols)	QPSK	QPSK
TBS Index (Full SF)	5	5
TBS Index PIPS	4	4
TBS Index (PEPS 6Symbols)	5	5
TBS Index (PEPS 9Symbols)	5	5
TBS Index (PEPS 10Symbols)	5	5
TBS Index (PEPS 11Symbols)	5	5
TBS Index (PEPS 12Symbols)	5	5

Figure 2-93: User-defined channel settings, LAA random bursts, part 1

TBS (Full SF)	8760	8760
TBS (PIPS)	8760	8760
TBS (PEPS 6Symbols)	8760	8760
TBS (PEPS 9Symbols)	8760	8760
TBS (PEPS 10Symbols)	8760	8760
TBS (PEPS 11Symbols)	8760	8760
TBS (PEPS 12Symbols)	8760	8760
Code Rate (Full SF)	0.000	0.000
Code Rate (PIPS)	0.000	0.000
Code Rate (PEPS 6 Symbols)	0.000	0.000
Code Rate (PEPS 9 Symbols)	0.000	0.000
Code Rate (PEPS 10 Symbols)	0.000	0.000
Code Rate (PEPS 11 Symbols)	0.000	0.000
Code Rate (PEPS 12 Symbols)	0.000	0.000
Burst Lengths	1 2 3 4 5 6 7 8 9 10	
Possible Burst Lengths	☒ ☐ ☒ ☐ ☒ ☐ ☒ ☐ ☐ ☐	
Last Subframes	6 9 10 11 12 14 (Number of Symbols)	
Possible last Subframe Of Burst	☐ ☐ ☐ ☐ ☐ ☒	
Initial Partial Subframes	☐	
Burst Transmission Probability	50	

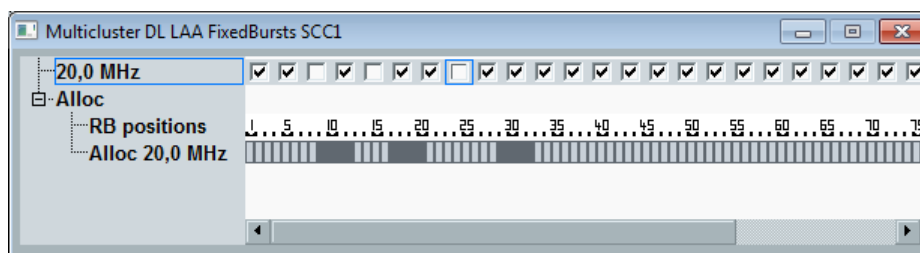
Figure 2-94: User-defined channel settings, LAA random bursts, part 2

Multiccluster	256
Burst Type	257
Throughput	257
#RB ... TBS (PEPS 12 Symbols)	257
Code Rate (...)	257
Possible Burst Lengths	258
Possible Last Subframe of Burst	258
Initial Partial Subframes	258
Burst Transmission Probability	258

Multiccluster

Multi-cluster allocation is only possible for resource allocation type 0 (resource block groups). Option R&S CMW-KS512 is required.

To use DL multi-cluster allocation, enable the checkbox. The button "RB Clusters" is displayed. Press the button to open the following configuration window.



In this dialog box, you can enable or disable each individual RBG via the checkboxes.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNECTION:SCC<c>:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNECTION:SCC<c>:UDChannels:LAA:RBURst:
FSUBframes:MCLuster:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
PIPSubframes:MCLuster:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
PEPSubframes:MCLuster:DL<s>
```

Burst Type

This section describes "Random transmission bursts".

See also [Section 2.4.12.15, "User-defined ch. for LAA, fixed bursts"](#), on page 251.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:TBURsts
```

Throughput

Displays the expected average maximum throughput in Mbit/s. The value is calculated per downlink stream.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHThroughput:DL:SCC<c>:STReam<s>?
```

#RB ... TBS (PEPS 12 Symbols)

- "#RB" selects the number of allocated resource blocks.
"Start RB" specifies the number of the first allocated resource block.
For DL multi-cluster allocation, see ["Multicluster"](#) on page 256.
- "Modulation (...)" selects the modulation type and influences the allowed TBS indices. 256-QAM requires option R&S CMW-KS504/-KS554.
"TBS Index (...)" selects the TBS index. The resulting transport block size "TBS (...)" in bits is displayed.
The settings are configurable for:
 - Subframes with full allocation "(Full SF)"
 - First subframes with partial allocation "(PIPS)"
 - Last subframes with n allocated symbols "(PEPS n Symbols)"

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
FSUBframes:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
PIPSubframes:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
PEPSubframes:DL<s>
```

Code Rate (...)

Effective channel code rate, i.e. the number of information bits (including CRC bits) divided by the number of physical channel bits on PDSCH.

The information is displayed per subframe type (full allocation, first SF partial allocation, last SF with n symbols).

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:DL<s>:
FSUBframes:CRATe?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:DL<s>:
PIPSubframes:CRATe?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:DL<s>:
PEPSubframes:CRATe?
```

Possible Burst Lengths

Selects allowed values for the number of subframes in a burst. For each burst, the random procedure selects one of the enabled values.

The number above each checkbox indicates the number of subframes enabled by the checkbox.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:BLENgtH
```

Possible Last Subframe of Burst

Selects allowed values for the number of allocated OFDM symbols in the last subframe of a burst. For each burst, the random procedure selects one of the enabled values.

The number above each checkbox indicates the number of symbols enabled by the checkbox.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:LSConfig
```

Initial Partial Subframes

Allows partial allocation for the first subframe of a burst.

If the checkbox is enabled, the probability for full allocation and partial allocation is equal. Partial allocation means that only the second slot is allocated (OFDM symbol 7 to 13).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:IPSubframe
```

Burst Transmission Probability

Probability for the decision, whether a burst is transmitted or whether muting is applied for the burst duration.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:TProBability
```

2.4.12.17 User-defined TTI-based channels

Parameters specific for the scheduling type "User-defined TTI-Based" can only be configured from the main view, not from the configuration tree.

Sched. **User def. TTI**

MIMO DL Stream ☒ 1 ☐ 2 Equal ☒

TTI **0** **Edit All** 64-QAM ☐

Downlink Uplink

# RB	50	50
Start RB	0	0
Mod / TBSI	QPSK 5	QPSK 6
Code Rate / TBS	0.320 4392	0.360 5160
Throughput	4.392 Mbit/s	5.160 Mbit/s
Throughput overall	8.784 Mbit/s	

Figure 2-95: Scheduling settings in the main view

In the main view, you can configure the settings of a selected subframe (TTI) directly. Alternatively, press the button "Edit All" to open a dialog box where you can edit the settings of all subframes.

If a transmission scheme using several data streams is active, the parameter "MIMO DL Stream" allows you to switch between the settings of the downlink streams. The parameter is disabled if all streams use the same set of settings.

For a parameter description, see [Section 2.4.13, "TTI-based channel configuration"](#), on page 265.

2.4.12.18 CQI channel configuration

Most parameters for the CQI scheduling types can only be configured from the main view, not from the configuration tree.

Sched. **CQI-PMI-RI** **Fixed CQI**

MIMO DL Stream ☒ 1 ☐ 2 Equal ☒

256-QAM ☐ TTI **0** **Edit All** 64/256-QAM ☐

1024-QAM ☐ Downlink Uplink

# RB	50	50
Start RB	0	0
Mod / CQI TBSI	QPSK 1	QPSK 6
Code Rate / TBS	0.320 4392	0.360 5160
Throughput	4.392 Mbit/s	5.160 Mbit/s
Throughput overall	8.784 Mbit/s	

Figure 2-96: CQI scheduling settings in the main view

In the main view, you can configure the settings of a selected subframe (TTI) directly. Alternatively, press the button "Edit All" to open a dialog box where you can edit the settings of all subframes.

For the "Follow..." CQI scheduling types, the downlink settings are identical for all subframes.

For some CQI scheduling types, the downlink settings are also contained in the configuration tree.

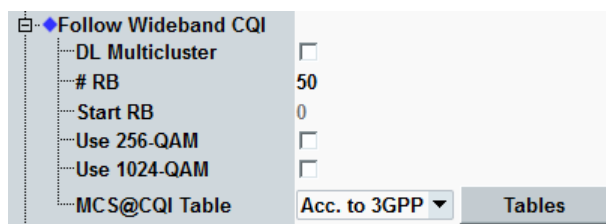


Figure 2-97: CQI scheduling settings in the configuration tree

For a description of the CQI settings, see [Section 2.4.13, "TTI-based channel configuration"](#), on page 265.

2.4.12.19 SPS configuration

Most parameters for semi-persistent scheduling can only be configured from the main view, not from the configuration tree.

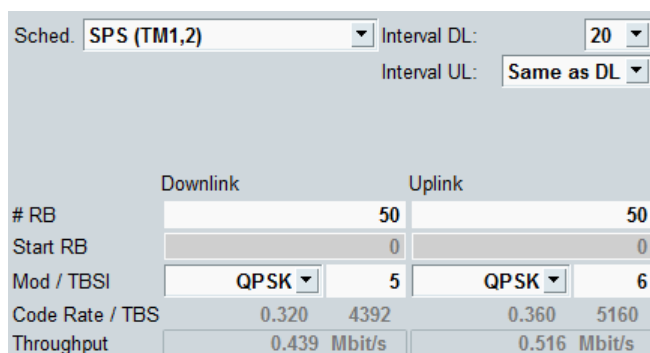


Figure 2-98: SPS settings in the main view

To configure SPS, proceed as follows:

1. Select the cell bandwidth. See [Section 2.4.9, "Physical cell setup"](#), on page 170.
2. Select the scheduling type SPS.
3. Select the subframe intervals.
4. Select the number of allocated resource blocks (RB), the position of the first RB and the modulation type.
5. Select the transport block size index.

For valid parameter combinations and background information, see [Section 2.2.22, "Semi-persistent scheduling \(SPS\)"](#), on page 90.

The parameters are described in the following. Option R&S CMW-KS510 is required.

Sched.....	261
Interval DL / UL.....	261
# RB ... TBSI / TBS.....	261
Code Rate.....	261
Throughput.....	261

Sched.

Selects the scheduling type, see "[Scheduling Type](#)" on page 223.

Interval DL / UL

Configures the subframe periodicity n for the downlink and the uplink. The UE is granted the configured RB allocation in every n^{th} subframe.

For TDD, only multiples of radio frames are allowed. For that reason, the selected value is internally rounded down to a multiple of 10. Example: 128 selected means 120.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:SINterval[:DL]
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:SINterval:UL
```

RB ... TBSI / TBS

"# RB" selects the number of allocated resource blocks.

"Start RB" specifies the number of the first allocated resource block. This parameter allows you to shift the allocated RBs within the cell bandwidth.

"Mod" selects the modulation type. The setting influences the allowed transport block size indices.

"TBSI" selects the transport block size index. "TBS" displays the resulting transport block size in bits.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:DL<s>
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:UL
```

Code Rate

Effective channel code rate, i.e. the number of information bits (including CRC bits) divided by the number of physical channel bits on PDSCH/PUSCH.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection[:PCC]:SPS:DL<s>:CRATe:ALL?
```

```
SENSe:LTE:SIGN<i>:CONNection[:PCC]:SPS:UL:CRATe:ALL?
```

Throughput

Expected maximum throughput in Mbit/s (averaged over one frame).

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]:STReam<s>?
```

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?
```

2.4.12.20 eMTC auto mode settings

Access: "Config" hotkey > "Connection" node > "Scheduling Type" = "eMTC Auto Mode" > "eMTC Auto Mode" node.

The node configures the resource block configuration, modulation type and transport block size for the eMTC auto mode (CE mode A or B). Option R&S CMW-KS590 is required.

To configure the settings, proceed as follows (for UL and DL):

1. Select the cell bandwidth. See [Section 2.4.9, "Physical cell setup"](#), on page 170.
2. Select the CE mode. See ["CE Mode"](#) on page 199.
3. Select the maximum eMTC bandwidth. See ["Max Bandwidth 5 MHz"](#) on page 194.
4. Select the number of resource blocks.
5. Select the start resource block, the narrowbands and the modulation type.
6. Select the transport block size index.

For valid parameter combinations and background information, see [Section 2.2.15, "eMTC auto mode"](#), on page 53.

Figure 2-99: eMTC auto mode settings

Half Duplex.....	262
DL, # Resource Blocks ... Transport Block Size.....	263
UL, # Resource Blocks ... Transport Block Size.....	263
Throughput.....	263

Half Duplex

Half-duplex operation is used in auto mode.

DL, # Resource Blocks ... Transport Block Size

"# Resource Blocks" selects the number of resource blocks allocated per narrowband.

"Start RB" specifies the number of the first allocated resource block within each narrowband.

"Narrow Band" selects the lowest used narrowband.

"Additional NB" selects additional narrowbands for a maximum bandwidth of 5 MHz. The numbers are relative to the selected "Narrow Band" (1st, 2nd and 3rd higher narrowband).

"Modulation Type" selects the modulation type and influences the allowed transport block size indices.

"Transport Block Size Index" selects the TBS index. The resulting "Transport Block Size" in bits is displayed.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:DL:ANB<Number>
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:DL:ALL
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:B:DL:ALL
```

UL, # Resource Blocks ... Transport Block Size

"# Resource Blocks" selects the total number of resource blocks, allocated contiguously over up to four used narrowbands.

"Start RB" specifies the number of the first allocated resource block within the lowest narrowband.

"Narrow Band" selects the lowest used narrowband.

"Modulation Type" selects the modulation type and influences the allowed transport block size indices.

"Transport Block Size Index" selects the TBS index. The resulting "Transport Block Size" in bits is displayed.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:UL:ALL
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:B:UL:ALL
```

Throughput

In the eMTC auto mode, no expected maximum throughput is calculated.

2.4.12.21 eMTC compact scheduling settings

Access: "Config" hotkey > "Connection" node > "Scheduling Type" = "eMTC Compact Scheduling" > "eMTC Compact Scheduling" node.

The node configures the resource block configuration, modulation type and transport block size for eMTC compact scheduling (CE mode A only). Option R&S CMW-KS590 is required.

To configure the settings, proceed as follows (for UL and DL):

1. Select the cell bandwidth. See [Section 2.4.9, "Physical cell setup"](#), on page 170.

2. Select the maximum eMTC bandwidth. See ["Max Bandwidth 5 MHz"](#) on page 194.
3. Select the duplex mode (half duplex or full duplex).
4. Select the SF pattern.
5. Select the number of resource blocks.
6. Select the start resource block, the narrowbands and the modulation type.
7. Select the transport block size index.

For valid parameter combinations and background information, see [Section 2.2.14, "eMTC compact scheduling"](#), on page 51.

Figure 2-100: eMTC compact scheduling

Half Duplex.....	264
SF Pattern.....	264
DL, # Resource Blocks ... Transport Block Size.....	265
UL, # Resource Blocks ... Transport Block Size.....	265
Throughput.....	265

Half Duplex

Selects between half-duplex operation and full-duplex operation.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNECTION[:PCC]:HDUPlex`

SF Pattern

For half-duplex operation, you can select a subframe pattern. There is a standard pattern without PDSCH HARQ-ACK bundling. And there are two patterns with bundling and 8 or 10 HARQ processes.

Select a pattern with bundling to increase the downlink throughput. For pattern details and required UE capabilities, see ["Subframe pattern"](#) on page 53.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNectio[n]:PCC]:CSCHeduling:SFPattern
```

DL, # Resource Blocks ... Transport Block Size

"# Resource Blocks" selects the number of resource blocks allocated per narrowband.

"Start Resource Block" specifies the number of the first allocated resource block within each narrowband.

"Additional NB" selects additional narrowbands for a maximum bandwidth of 5 MHz.

"Modulation Type" selects the modulation type and influences the allowed transport block size indices.

"Transport Block Size Index" selects the TBS index. The resulting "Transport Block Size" in bits is displayed.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNectio[n]:PCC]:CSCHeduling:A:DL:ALL
CONFigure:LTE:SIGN<i>:CONNectio[n]:PCC]:CSCHeduling:A:DL:
ANB<Number>
```

UL, # Resource Blocks ... Transport Block Size

"# Resource Blocks" selects the total number of resource blocks, allocated contiguously over up to four used narrowbands.

"Start Resource Block" specifies the number of the first allocated resource block within the lowest used narrowband.

"Modulation Type" selects the modulation type and influences the allowed transport block size indices.

"Transport Block Size Index" selects the TBS index. The resulting "Transport Block Size" in bits is displayed.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNectio[n]:PCC]:CSCHeduling:A:UL:ALL
```

Throughput

Displays the expected maximum throughput in Mbit/s (averaged over one frame). The value is calculated per downlink and uplink stream.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNectio[n]:ETHRoughput:DL[:PCC]:STReam<s>?
SENSe:LTE:SIGN<i>:CONNectio[n]:ETHRoughput:UL[:PCC]?
```

2.4.13 TTI-based channel configuration

This section describes the TTI-based channel configuration via a special dialog box.

Before you open the dialog box, select the duplex mode, the cell bandwidth and the scheduling type. Then open the dialog box via the button "Edit All" in the lower part of the main view. You cannot access the dialog box from the main configuration tree.

TTI-based channel configuration is available for the scheduling type "User-defined TTI-Based" and for all "CQI-PMI-RI" scheduling types ("Fixed CQI" and "Follow WB ...").

The dialog box contains several tabs, which are active or inactive, depending mainly on the scheduling type and the scenario.

The tabs contain settings at the top and a graphical presentation of the resulting resource block configuration at the bottom.

For valid parameter combinations and background information, refer to:

- [Section 2.2.20, "User-defined channels"](#), on page 83
- [Section 2.2.21, "CQI channels"](#), on page 86

2.4.13.1 DL stream tabs and UL tab

The DL stream tabs and the UL tab contain usually one column of settings for each subframe of a radio frame. For some transmission schemes, the DL stream tab contains only one column, applicable to all subframes.

For TDD, the UL tab presents only the UL subframes of a radio frame, while the DL tabs present both DL and special subframes. The special subframe settings configure the DwPTS field for data transfer in downlink direction. To modify the UL-DL configuration of the radio frame, see ["Use Carrier Specific"](#) on page 174. For UL-DL configuration 0, all UL subframes have the same settings.

The subframe number (0 to 9) and the subframe type are indicated in the column header. All columns must be configured top down (first "# RB", then "Start RB", then ...).

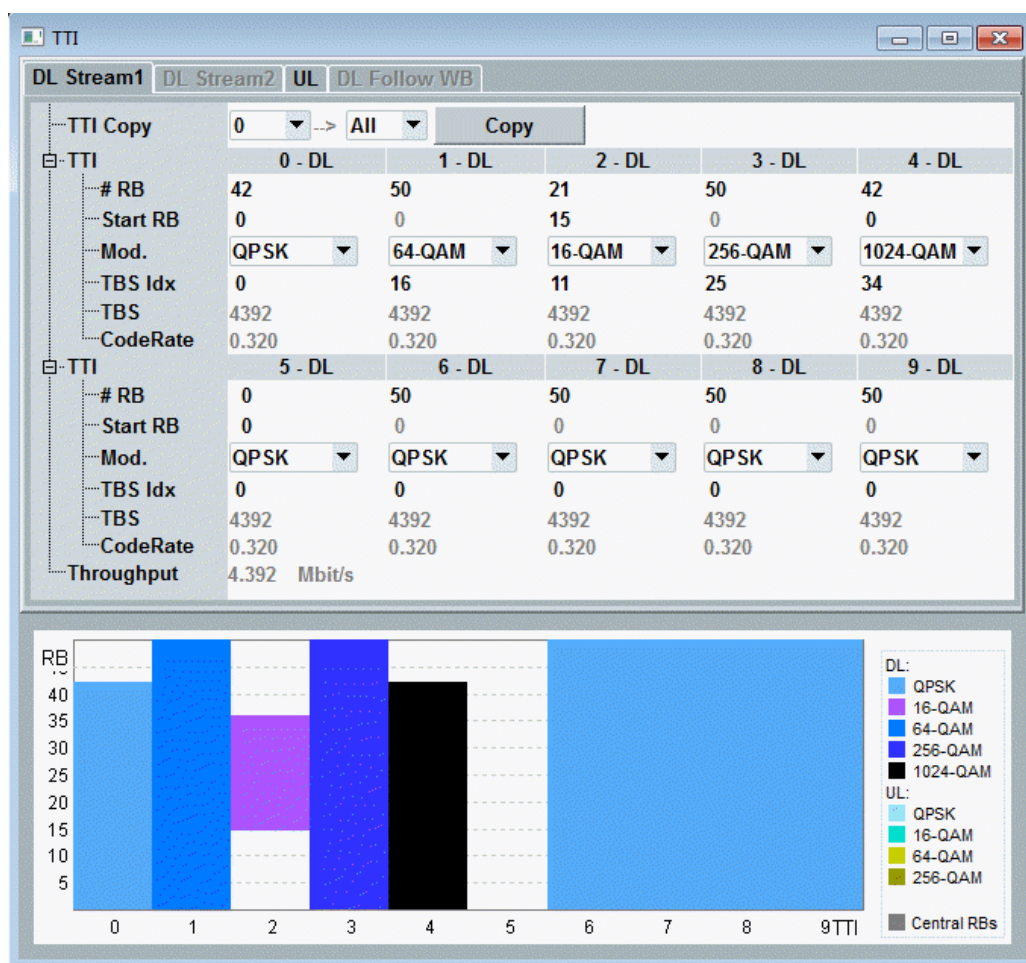


Figure 2-101: User-defined TTI-based configuration (PCC)

If the selected scenario involves several downlink streams, some DL parameters can be configured individually per stream. Other parameters are configurable for stream 1 only and are applied automatically to stream 2. If you enable parameter "Equal" in the main view, all stream 1 settings are also applied to stream 2 and the stream 2 tab is disabled.

For carrier aggregation scenarios, you can configure the settings per component carrier.

All parameters of the tabs are explained in the following.

TTI Copy.....	268
Use 256-QAM, Use 1024-QAM.....	268
#RB / Start RB.....	268
CQI Idx.....	268
Mod.....	268
TBS Idx.....	268
UL configuration commands.....	268
DL configuration commands.....	269
L User-Defined TTI-Based.....	269

L Fixed CQI.....	269
TBS.....	269
Code Rate.....	269
Throughput.....	269

TTI Copy

Allows you to copy the settings of a selected subframe to other subframes. The source and target subframes of the copy operation are identified via their number.

To perform a copy operation, select the number of the subframe to be copied, select the target subframes and press the "Copy" button.

Remote command:

n/a

Use 256-QAM, Use 1024-QAM

Selects a CQI definition table for "Fixed CQI". There is a table up to 64-QAM, a table with 256-QAM and a table with 1024-QAM, see [Section 2.2.21.1, "Fixed CQI channels"](#), on page 86.

256-QAM requires option R&S CMW-KS504/-KS554.

1024-QAM requires option R&S CMW-KS505/-KS555.

Remote command:

`CONFigure:LTE:SIGN<i>:CONNection[:PCC]:QAM<m>:DL`

`CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:QAM<m>:DL`

#RB / Start RB

Select the number of allocated resource blocks and specifies the number of the first allocated resource block.

CQI Idx.

Selects the CQI index for a downlink subframe. Only for scheduling type "Fixed CQI".

Mod.

Selects or displays the modulation type.

If you want to use UL 64-QAM or UL 256-QAM, configure also ["Enable 64/256-QAM UL Support"](#) on page 171.

TBS Idx

Selects the transport block size index.

The allowed values depend on whether you select 256-QAM or 1024-QAM for at least one TTI or not, see [Section 2.2.20.3, "Supported transport block size indices"](#), on page 85.

UL configuration commands

The following commands configure the uplink parameters "#RB", "Start RB", "Mod." and "TBS Idx".

The settings apply to all scheduling types with TTI-based uplink configuration.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL:ALL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL:ALL
```

DL configuration commands

For downlink configuration, there are different commands per scheduling type and for PCC and SCC.

User-Defined TTI-Based ← DL configuration commands

The following commands configure the parameters "#RB", "Start RB", "Mod." and "TBS Idx" for the scheduling type "User-Defined TTI-Based".

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>:ALL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>:ALL
```

Fixed CQI ← DL configuration commands

The following commands configure the parameters "#RB", "Start RB" and "CQI Idx" for the scheduling type "Fixed CQI".

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCTTibased:DL<s>
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCTTibased:DL<s>:ALL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCTTibased:DL<s>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCTTibased:DL<s>:ALL
```

TBS

Displays the transport block size in bits.

Code Rate

Displays the effective channel code rate, i.e. the number of information bits (including CRC bits) divided by the number of physical channel bits on PDSCH/PUSCH.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL:CRATe:ALL?
```

Throughput

Displays the expected maximum throughput in Mbit/s (averaged over one frame). The value is calculated per downlink and uplink stream.

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]:STReam<s>?
```

SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:SCC<c>?

SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>:STReam<s>?

2.4.13.2 DL follow WB tab - CQI variant

There are two variants of this tab. The CQI variant provides mapping tables and is active for the scheduling types "Follow WB CQI", "Follow WB CQI-RI" and "Follow WB CQI-PMI-RI".

Most settings are not configurable per TTI. They apply to all downlink subframes of the carrier. For carrier aggregation scenarios, you can configure the downlink settings per component carrier.

The expected maximum throughput that is displayed, for example, in the main view, is calculated using the maximum MCS index value of the mapping table.

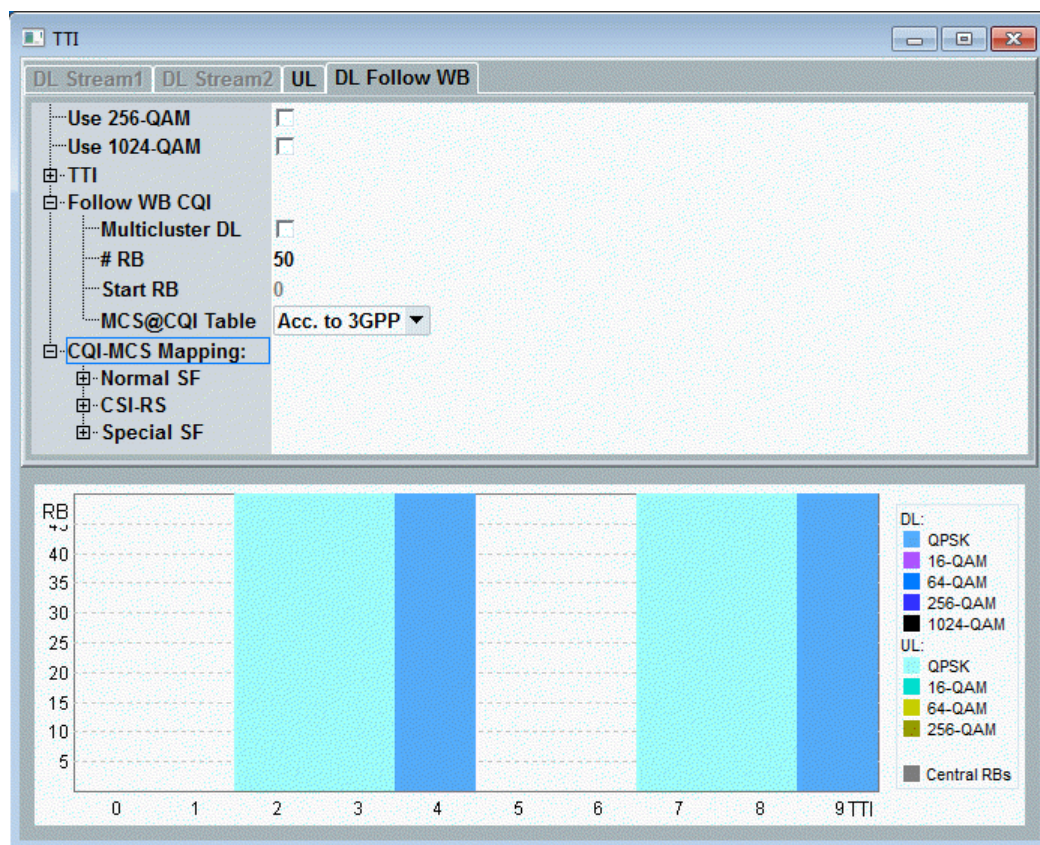


Figure 2-102: Follow WB CQI DL settings, TDD, UL-DL config 1

Use 256-QAM, Use 1024-QAM	271
Scheduled CQI	271
Length of Burst	271
Periodicity of Burst Transmissions	271
Multicluseter DL	271
#RB / Start RB	272
MCS@CQI Table	272

CQI-MCS Mapping.....	272
L Normal SF, CSI-RS, Special SF.....	273
L CW / Layers.....	273
L Commands for Follow WB CQI.....	274
L Commands for Follow WB CQI-RI.....	275
L Commands for Follow WB CQI-PMI-RI.....	275

Use 256-QAM, Use 1024-QAM

There are several CQI and MCS definition tables defined in 3GPP TS 36.213. There is one table for modulations up to 64-QAM, one table up to 256-QAM and one table up to 1024-QAM. The settings select which of the tables is used.

256-QAM requires option R&S CMW-KS504/-KS554.

1024-QAM requires option R&S CMW-KS505/-KS555.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:QAM<m>:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:QAM<m>:DL
```

Scheduled CQI

According to 3GPP TS 36.521, some subframes must be inactive (not scheduled), depending on the duplex mode and the scheduling type, see [Section 2.2.21, "CQI channels"](#), on page 86. The default behavior complies with the 3GPP definition.

The checked boxes indicate which subframes are scheduled. The configurable checkboxes allow you to activate additional DL subframes and special subframes.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:STTI
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:STTI
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:STTI
```

Plus corresponding PCC commands.

Length of Burst

Only displayed for LAA. Defines the number of subframes per burst.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:BLENgtH
```

Periodicity of Burst Transmissions

Only displayed for LAA. Defines the periodicity n for subsequent bursts (a burst starts every n^{th} subframe). The minimum value equals the burst length.

Example: Burst length = 3, Periodicity = 4 results in the sequence: (3 SF burst) (1 SF muting) (3 SF burst) and so on.

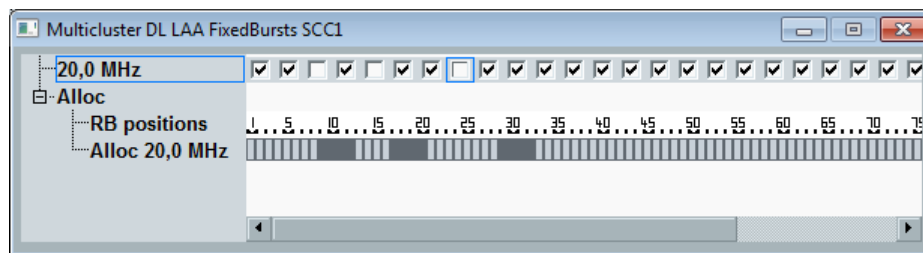
Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:PBTR
```

Multicenter DL

You can enable DL multi-cluster allocation only for resource allocation type 0 (resource block groups). Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

To use DL multi-cluster allocation, enable the checkbox. The button "RB Clusters" is displayed. Press the button to open the following configuration window.



In this dialog box, you can enable or disable each individual RBG via the checkboxes.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPri:MCLuster:DL
Plus corresponding PCC commands.
```

#RB / Start RB

Select the number of allocated resource blocks and the number of the first allocated resource block (without DL multi-cluster allocation).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPri:DL
Plus corresponding PCC commands.
```

MCS@CQI Table

Selects a table mode for MCS to CQI mapping.

- "Acc. to 3GPP" The mapping tables are defined automatically based on 3GPP specifications. The tables are displayed for information. The table contents are not configurable.
- "User-Defined" You can define the mapping tables manually (assign an MCS index value to each CQI index value).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPri:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:MCLuster:DL
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPri:MCLuster:DL
Plus corresponding PCC commands.
```

CQI-MCS Mapping

The MCS index value used for the downlink is dynamically determined via a mapping table. The table assigns an MCS index value to each possible reported CQI index value (1 to 15). The last wideband CQI value reported by the UE determines the applicable table column and thus the used MCS index.

CQI	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
MCS	0	0	2	3	5	7	10	12	14	17	19	21	23	24	25
Modulation	QPSK	QPSK	QPSK	QPSK	QPSK	QPSK	16-QAM	16-QAM	16-QAM	64-QAM	64-QAM	64-QAM	64-QAM	64-QAM	64-QAM
TBS Index	0	0	2	3	5	7	9	11	13	15	17	19	21	22	23
Code Rate	0.130	0.130	0.207	0.267	0.409	0.581	0.373	0.462	0.604	0.474	0.569	0.664	0.790	0.849	0.879

Figure 2-103: Mapping table, MCS to CQI

The table lists also the modulation type, TBS index and code rate resulting from the MCS index. The relations are defined in 3GPP TS 36.213, depending on the highest used modulation scheme. So the table contents depend on the settings "Use 256-QAM" and "Use 1024-QAM".

The displayed code rate applies to default DL subframes. If you enable additional subframes via **"Scheduled CQI"** on page 271, their code rate is slightly less than the displayed code rate.

Depending on settings like the duplex mode and the transmission scheme, there can be several mapping tables displayed in parallel, applicable in different situations. A table can, for example, apply to a certain subframe type and a certain mapping of code words to layers. The GUI elements distinguishing between these tables are described in the following.

Normal SF, CSI-RS, Special SF ← CQI-MCS Mapping

Nodes for up to three subframe types are displayed in parallel.

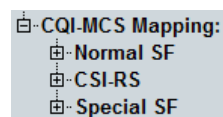


Figure 2-104: Subframe types

- "Special SF" Only displayed for TDD signals. Applies to special subframes.
- "CSI-RS" Only displayed for TM 9. Applies to subframes with CSI-RS.
- "Normal SF" Always displayed. Applies to all other subframes (no CSI-RS, no special subframe).

CW / Layers ← CQI-MCS Mapping

There can be several mapping tables per subframe type. The line "CW / Layers" indicates to which situation the table applies.

The modulation type, TBS index and code rate are only listed in the first table for a subframe type.

CQI-MCS Mapping:	
Normal SF	
CW / Layers	Each codeword mapped to 1 layer
CQI	1 2 3
MCS	0 0 2 3
Modulation	QPSK QPSK QPSK QPSK
TBS Index	0 0 2 3
Code Rate	0.130 0.130 0.207 0.2
Normal SF	
CW / Layers	Each codeword mapped to 2 layer
CQI	1 2 3 4 5 6
MCS	0 0 1 3 5 7
Normal SF	
Normal SF	

Figure 2-105: Four tables for normal subframes

The following "CW / Layers" texts are possible:

- "Any": There is only one table for the subframe type. The table is used independent of the number of code words and layers.
- "Each codeword mapped to 1 layer"
- "Each codeword mapped to 2 layers"
- "Each codeword mapped to 1 layer / 1 or 2 layers in total"
- "Each codeword mapped to 2 layers / 2 layers in total"
- "Each codeword mapped to 1 layer / 3 or 4 layers in total"
- "Each codeword mapped to 2 layers / 3 or 4 layers in total"

Which of these texts is relevant and displayed depends on many settings like the duplex mode, the scheduling type, the transmission mode and the number of antennas.

In the user-defined mode, there is only one mapping table per subframe type (text "Any").

Commands for Follow WB CQI ← CQI-MCS Mapping

The following commands apply to the scheduling type "Follow WB CQI".

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCS:ATable:LIST?
CONFIgure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:
UDEfined
CONFIgure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:
CSIRs:UDEfined
CONFIgure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:
SSUBframe:UDEfined
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:
DETermined?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:CSIRs:
DETermined?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:
SSUBframe:DETermined?
```

Plus corresponding PCC commands.

Commands for Follow WB CQI-RI ← CQI-MCS Mapping

The following commands apply to the scheduling type "Follow WB CQI-RI".

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCS:ATable:LIST?
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:
UDEfined
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:
SSUBframe:UDEfined
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:DETermined?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:SSUBframe:
DETermined?
```

Plus corresponding PCC commands.

Commands for Follow WB CQI-PMI-RI ← CQI-MCS Mapping

The following commands apply to the scheduling type "Follow WB CQI-PMI-RI".

Remote command:

```
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCS:ATable:LIST?
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:
UDEfined
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:CSIRs:
UDEfined
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:
SSUBframe:UDEfined
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:
DETermined?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:CSIRs:
DETermined?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:SSUBframe:
DETermined?
```

Plus corresponding PCC commands.

2.4.13.3 DL follow WB tab - PMI variant

There are two variants of this tab. The PMI variant is active for the scheduling types "Follow WB PMI" and "Follow WB PMI-RI".

Most settings are not configurable per TTI. They apply to all downlink subframes of the carrier. For carrier aggregation scenarios, you can configure the downlink settings per component carrier.

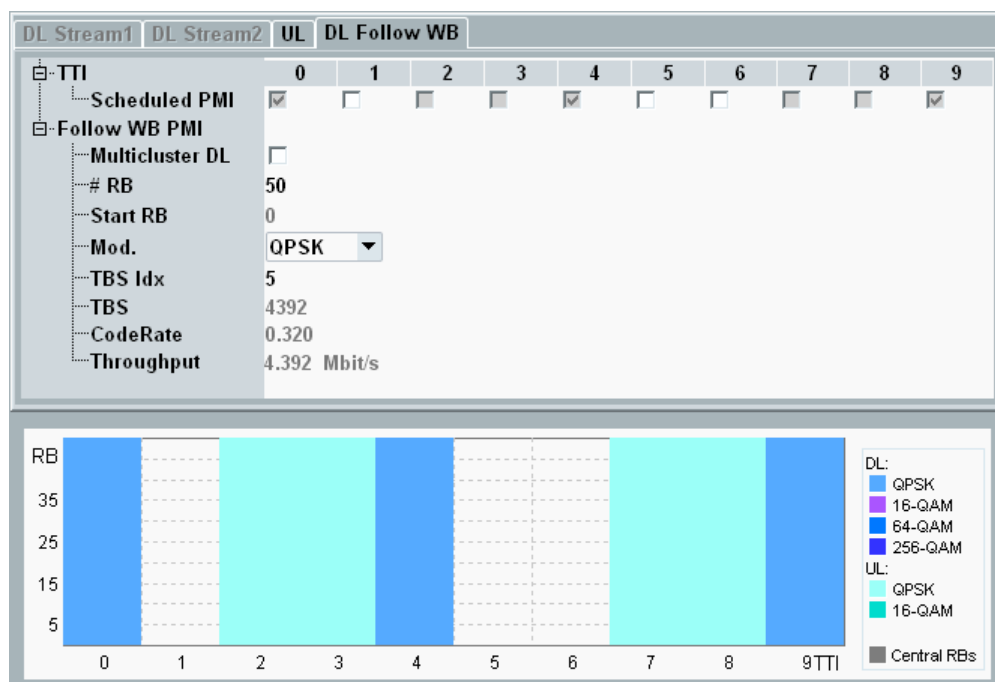


Figure 2-106: Follow WB PMI DL settings, TDD, UL-DL config 1

Scheduled CQI	276
Length of Burst	276
Periodicity of Burst Transmissions	277
Multicenter DL	277
#RB / Start RB	277
Mod. / TBS Idx	277
TBS	277
Code Rate	278
Throughput	278

Scheduled CQI

According to 3GPP TS 36.521, some subframes must be inactive (not scheduled), depending on the duplex mode and the scheduling type, see [Section 2.2.21, "CQI channels"](#), on page 86. The default behavior complies with the 3GPP definition.

The checked boxes indicate which subframes are scheduled. The configurable check-boxes allow you to activate additional DL subframes and special subframes.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPMI:DL:STTI
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL:STTI
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPMI:DL:STTI
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL:STTI
```

Length of Burst

Only displayed for LAA. Defines the number of subframes per burst.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:BLENght
```


Periodicity of Burst Transmissions

Only displayed for LAA. Defines the periodicity n for subsequent bursts (a burst starts every n^{th} subframe). The minimum value equals the burst length.

Example: Burst length = 3, Periodicity = 4 results in the sequence: (3 SF burst) (1 SF muting) (3 SF burst) and so on.

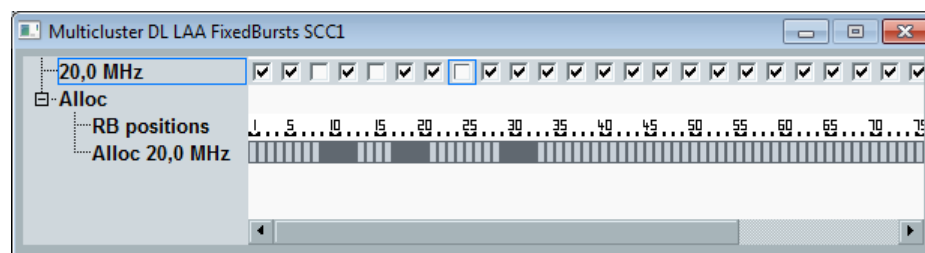
Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:PBTR
```

Multicenter DL

You can enable DL multi-cluster allocation only for resource allocation type 0 (resource block groups). Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

To use DL multi-cluster allocation, enable the checkbox. The button "RB Clusters" is displayed. Press the button to open the following configuration window.



In this dialog box, you can enable or disable each individual RBG via the checkboxes.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:MCLuster:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:MCLuster:DL
```

Plus corresponding PCC commands.

#RB / Start RB

Select the number of allocated resource blocks and the number of the first allocated resource block (without DL multi-cluster allocation).

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL
```

Mod. / TBS Idx

Select or display the modulation type and the transport block size index.

Remote command:

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL
```

TBS

Displays the transport block size in bits.

Code Rate

See ["Code Rate"](#) on page 269.

Throughput

See ["Throughput"](#) on page 269.

2.4.14 CQI reporting

Access: "Config" hotkey > "CQI Reporting" node.

The node configures channel quality indication (CQI) reporting. The reporting procedures are specified in 3GPP TS 36.213, section 7.2.

For scenarios without carrier aggregation, option R&S CMW-KS510 is required. For scenarios with carrier aggregation, R&S CMW-KS512 is required.

Which reports are sent by the UE depends on the transmission mode. If you want to see CQI, PMI and RI reports, you can use for example transmission mode 4.

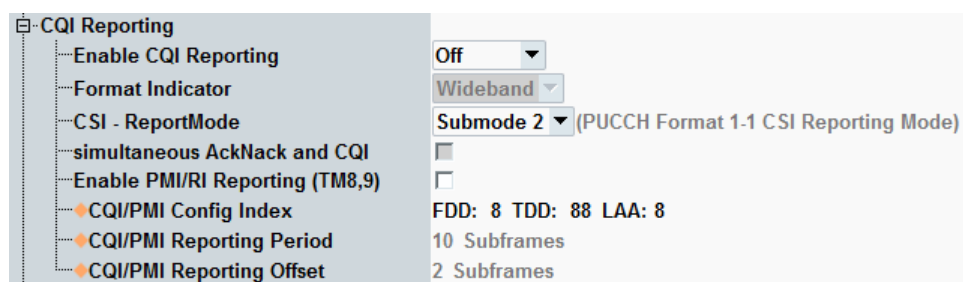


Figure 2-107: CQI reporting settings

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Enable CQI Reporting

Enables periodic CQI reporting ("Periodic") or disables CQI reporting completely ("Off").

For the scheduling types "Follow...", periodic CQI reporting is enabled automatically.

Periodic CQI reporting is described in 3GPP TS 36.213, section 7.2.2.

Remote command:

`CONFigure:LTE:SIGN<i>:CQIReporting:ENABLE`

Format Indicator

The current software version supports only wideband CQI reporting. Subband CQI reporting is not supported.

Remote command:
n/a

CSI - Report Mode

Configures the parameter "PUCCH_format1-1_CSI_reporting_mode", signaled to the UE.

The parameter is only relevant for transmission mode 9 with 8 CSI-RS antenna ports and wideband CQI reporting. See also 3GPP TS 36.213, section 7.2.2.

Remote command:

`CONFigure:LTE:SIGN<i>:CQIReporting:CSIRmode`

Simultaneous AckNack and CQI

Configures the parameter "simultaneousAckNackAndCQI", signaled to the UE.

The parameter indicates whether the simultaneous transmission of ACK/NACK and CQI is allowed.

For carrier aggregation scenarios, the checkbox is always disabled.

Remote command:

`CONFigure:LTE:SIGN<i>:CQIReporting:SANCqi`

Enable PMI/RI Reporting (TM 8, 9)

In transmission mode 8 and 9, PMI and RI reports must be explicitly requested from the UE.

For the scheduling types "Follow...", PMI/RI reporting is enabled automatically.

"OFF"	The UE does not report PMI and RI values. For TM 9, reported CQI values are based on the cell-specific reference signal.
"ON"	The UE reports also PMI and RI values (if reporting is enabled at all). For TM 9, reported values are based on the CSI reference signal. For PMI and RI reports, you must enable at least two CSI-RS antenna ports. For CQI reports, at least one antenna port is required. See "CSI-RS Antenna Ports" on page 219.

Remote command:

`CONFigure:LTE:SIGN<i>:CQIReporting:PRIReporting:ENABLE`

CQI/PMI Config Index

Specifies the "cqi-pmi-ConfigIndex" ($I_{CQI/PMI}$) for FDD, TDD and LAA.

If you want to evaluate reports for several component carriers, you must configure compatible values, so that the carriers use different subframes for reporting. See ["Configuration Hints"](#) on page 280.

Remote command:

`CONFigure:LTE:SIGN<i>:CQIReporting[:PCC]:CINDEX[:FDD]`
`CONFigure:LTE:SIGN<i>:CQIReporting[:PCC]:CINDEX:TDD`
`CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINDEX[:FDD]`
`CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINDEX:TDD`
`CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINDEX:LAA`

CQI/PMI Reporting Period / Offset

The displayed reporting period N_p and the reporting offset $N_{\text{OFFSET,CQI}}$ result from the configured "cqi-pmi-ConfigIndex" and the duplex mode. They are derived using the mapping tables 7.2.2-1A and 7.2.2-1C in 3GPP TS 36.213.

For carrier aggregation scenarios, the information is available per component carrier.

Remote command:

```
SENSe:LTE:SIGN<i>:CQIReporting[:PCC]:RPERiod?
SENSe:LTE:SIGN<i>:CQIReporting[:PCC]:ROFFset?
SENSe:LTE:SIGN<i>:CQIReporting:SCC<c>:RPERiod?
SENSe:LTE:SIGN<i>:CQIReporting:SCC<c>:ROFFset?
```

Configuration Hints

The eNodeB determines which subframes the UE can use for CQI, PMI and RI reporting, as specified in 3GPP TS 36.213, section 7.2.2.

The subframes for CQI and PMI reporting are determined by the parameter "cqi-pmi-ConfigIndex". This parameter is configurable. Via a mapping table, the "cqi-pmi-ConfigIndex" determines the reporting period N_p and the reporting offset $N_{\text{OFFSET,CQI}}$.

The subframes for RI reporting are determined by the parameters " M_{RI} " and " $N_{\text{OFFSET,RI}}$ ". The LTE signaling application uses " $M_{\text{RI}} = 1$ " and " $N_{\text{OFFSET,RI}} = -1$ ".

Thus the subframes for RI reporting are located immediately before the subframes for CQI/PMI reporting.

If you want to evaluate reports for several component carriers, the carriers must use different subframes for reporting. The carrier configurations must not overlap. This rule is also important for scheduling types with a DL configuration that reacts to the reports, for example "Follow WB CQI".

Example, dual-carrier FDD, wrong configuration:

PCC: "cqi-pmi-ConfigIndex" = 3 means offset = 1, period = 5

SCC: "cqi-pmi-ConfigIndex" = 9 means offset = 2, period = 10

The following table shows the resulting configuration for two frames. There is a collision in subframe number 1, with CQI/PMI reporting for the PCC and RI reporting for the SCC.

Subframe no.	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9
PCC	R	C				R	C				R	C				R	C			
SCC		R	C									R	C							
C = CQI/PMI reporting, R = RI reporting																				

Example, dual-carrier FDD, correct configuration:

PCC: "cqi-pmi-ConfigIndex" = 3 means offset = 1, period = 5

SCC: "cqi-pmi-ConfigIndex" = 5 means offset = 3, period = 5

The following table shows the resulting configuration. There is no collision.

Subframe no.	0	1	2	3	4	5	6	7	8	9	0	1	2	3	4	5	6	7	8	9
PCC	R	C				R	C				R	C				R	C			
SCC			R	C				R	C				R	C				R	C	

C = CQI/PMI reporting, R = RI reporting

2.4.15 UE measurement report settings

Access: "Config" hotkey > "UE Measurement Report" node.

The node configures the UE measurement report. The report is shown in the main signaling view, see [Section 2.4.1.3, "UE measurement report"](#), on page 121.

For enabling of neighbor cell measurements, see [Section 2.4.10.1, "Neighbor cell settings"](#), on page 179.

Figure 2-108: UE measurement report settings

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L RMTC Period, RMTC Subframe Offset, Measurement Duration.....	284
L Channel Occupancy Threshold.....	284

Report

Enables or disables the UE measurement report.

Configure the other settings before enabling this parameter.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:ENABLe`

Reporting Interval

Sets the interval between two consecutive measurement reports. Reduce the interval to check whether the UE can cope with a high repetition rate.

The value is signaled to the UE as "reportInterval".

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:RINTerval`

Measurement Gap Enable

Enables or disables transmission gaps that can be used by the UE for neighbor cell measurements.

Measurement gaps occur only if at least one neighbor cell measurement is active.

Option R&S CMW-KS510 is required for neighbor cell measurements.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:MGENable`

Measurement Gap Period

Specifies the periodicity of transmission gaps. A measurement gap of 6 ms can occur every 40 ms or every 80 ms.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:MGPeriod`

Filter Coefficient RSRP/RSRQ

Selects the filter coefficients used by the UE to measure the reference signal received power (RSRP) and the reference signal received quality (RSRQ).

The values are signaled to the UE via the information elements "filterCoefficientRSRP" and "filterCoefficientRSRQ". Do not confuse these elements with the "filterCoefficient" for uplink power control, see "[UE Meas. Filter Coefficient](#)" on page 206.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:FCoefficient:RSRP`

`CONFigure:LTE:SIGN<i>:UEReport:FCoefficient:RSRQ`

WCDMA Measurement Quantity

Selects whether the UE must determine the RSCP or the Ec/No during WCDMA neighbor cell measurements. The setting is signaled to the UE.

Option R&S CMW-KS510 is required for neighbor cell measurements.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:WMQuantity`

Measurement Cycle SCell

Specifies how often the UE must measure the SCC in state "RRC Added". The setting is signaled to the UE as "measCycleSCell". The measurement period is calculated from the configured value. For details, see 3GPP TS 36.133, section 8.3.3.2.

"Off" means that parameter "measCycleSCell" is not signaled to the UE.

The setting is only visible for carrier aggregation scenarios.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:MCSCell`

Allow Interruptions

Specifies whether the UE is allowed to cause PCC interruptions when measuring an SCC. The setting is signaled to the UE as "allowInterruptions".

The setting is only visible for carrier aggregation scenarios.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:AINInterrupt`

DMTC Configuration

The discovery signals measurement timing configuration (DMTC) defines the periodicity and subframe offset of the discovery reference signal (DRS) transmitted by the signaling application for frame structure type 3 (LAA).

The information is signaled to the UE in the IE "MeasDS-Config".

The "DMTC Period Offset" is at least 5 ms smaller than the "DMTC Period".

The settings are configurable per SCC with frame structure type 3. Option R&S CMW-KS514 is required.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:DMTC:PERiod`

`CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:DMTC:POFFset`

RSSI Measurement

This node configures received signal strength indicator (RSSI) measurements and channel occupancy measurements of the UE.

Option R&S CMW-KS514 is required for all settings. All settings except the "Channel Occupancy Threshold" are configurable per SCC with frame structure type 3.

Enable ← RSSI Measurement

Enables the signaling of the IE "measRSSI-ReportConfig" to the UE. If the UE does not get this information, it does not perform RSSI and channel occupancy measurements.

Remote command:

`CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:ENABLE`

RMTC Period, RMTC Subframe Offset, Measurement Duration ← RSSI Measurement

The RSSI measurement timing configuration (RMTC) defines the periodicity, subframe offset and duration of the UE measurements. The settings apply to RSSI and channel occupancy measurements.

The information is signaled to the UE in the IE "RMTC-Config".

The "RMTC Subframe Offset" is at least 5 ms smaller than the "RMTC Period".

Remote command:

```
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:RMTC:PERiod
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:RMTC:SOFFset
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:MDURation
```

Channel Occupancy Threshold ← RSSI Measurement

In the channel occupancy measurement, the UE determines the percentage of samples where the measured RSSI value is greater than the channel occupancy threshold.

The setting is signaled to the UE as "channelOccupancyThreshold". The same value applies to all SCCs with frame structure type 3.

Remote command:

```
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:COThreshhold
```

2.4.16 Messaging (SMS) parameters

Access: "Config" hotkey > "Messaging (SMS)" node.

The node configures parameters of the short message service (SMS). Sending a short message to the UE is triggered via hotkey, see ["Connection control hotkeys"](#) on page 131.

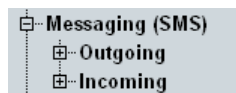


Figure 2-109: SMS parameters

2.4.16.1 Outgoing SMS

Access: "Config" hotkey > "Messaging (SMS)" node > "Outgoing" node.

The node configures outgoing mobile-terminating short messages.

Figure 2-110: Outgoing SMS parameters

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L Message Encoding, Message Text, Message Length.....	288
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Large SMS Handling

Selects the handling of messages exceeding 160 characters.

- "Truncate" The SMS is truncated to 160 characters, the rest is discarded.
- "Multiple SMS" Up to five concatenated messages are sent, consisting in sum of up to 800 characters.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:LHANDling`

Outgoing Message Handling

Selects whether the message text is defined via the GUI or read from a file.

"Use Internal" Use the message text configured via the GUI ([Outgoing SMS](#) or [Outgoing SMS binary](#)).

"From File" Read the text from the file selected via ["From File"](#) on page 288.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:MESHAndling`

Outgoing SMS

Defines the message contents for outgoing 7-bit ASCII messages. Up to 800 characters are allowed.

This text is used for "Outgoing Message Handling" = "Use Internal" plus "Data Coding / Character Set" = "7bit".

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:INTernal`

Outgoing SMS binary

Defines the message contents for outgoing 8-bit binary messages. The contents are entered as hexadecimal numbers. Up to 1400 digits are allowed.

This text is used for "Outgoing Message Handling" = "Use Internal" plus "Data Coding / Character Set" = "8bit".

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:BINary`

Protocol Identifier

Specifies the TP protocol identifier (TP-PID) value to be sent. The meaning of the values is defined in 3GPP TS 23.040 section 9.2.3.9.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:PIDentifier`

Data Coding / Character Set

Selects between the 7-bit ASCII message (["Outgoing SMS"](#) on page 286) and the 8-bit binary message (["Outgoing SMS binary"](#) on page 286).

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:DCODing`

User Data Header

To add a header to the TP user data field, enable the checkbox and configure the header via the hex value. The structure of the user data header is defined in 3GPP TS 23.040, section 9.2.3.24.

The hex value configures the following parts of the header:

- Information element identifier (IEI, one octet)
Currently supported values: #H05, #H70 to #H7F

- Length of information element (IEIDL, one octet)
- Information element data (IED, 0 to n octets)

The signaling application sets the TP-UDHI automatically, according to the checkbox. With enabled checkbox, it sends the hex value to the UE and uses the correct number of fill bits in the header. It evaluates the IEIa and applies the correct message concatenation and header repetition.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:UDHeader`

Coding Group

Selects the coding group to be indicated to the message recipient in the TP-Data-Coding-Scheme field. See also 3GPP TS 23.038.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:CGROUP`

Message Class

Selects the message class to be indicated to the message recipient in the TP-Data-Coding-Scheme field. See also 3GPP TS 23.038.

"None" means that no message class is sent.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:MClass`

Originator SMSC Address

Short message service center address, to be sent to the recipient.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:OSAddress`

Originating Address

Address of the originator of the SMS, to be sent to the recipient.

Remote command:

`CONFigure:LTE:SIGN<i>:SMS:OUTGoing:OAddress`

Service Center Time Stamp



The screenshot shows a configuration window titled "Service Center Time Stamp". Inside, there is a "Time Source" dropdown menu currently set to "CMW Time". Below it, a "Date / Time" field displays the value "11 / 11 / 2011 11 : 11 : 00".

Service center time stamp, to be sent to the recipient.

You can configure the time stamp in two ways:

- Select the time source "CMW Time".
The time stamp is set according to the current date and time of the operation system.
- Select the time source "Date / Time" and configure the time stamp value via the parameter "Date / Time".

Option R&S CMW-KS510 is required.

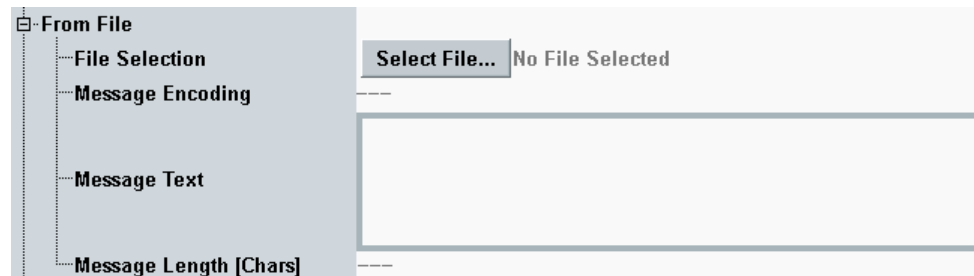
Remote command:

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TSource

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:DATE

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TIME

From File



This node selects a file containing the message to be transmitted.

The file is used for [Outgoing Message Handling](#) = "From File".

Select File ← From File

Selects a file from the system drive folder

D:\Rohde-Schwarz\CMW\Data\SMS\LTE\Send or one of its subfolders. Store your message files there, using the file extension . sms.

The supported encodings are 7-bit ASCII, 8-bit binary and 16-bit Unicode.

Remote command:

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE

Message Encoding, Message Text, Message Length ← From File

Displays information about the message in the selected file.

Remote command:

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE:INFO?

Last Message Sent

Indicates whether the last outgoing short message transfer was "Successful" or "Failed".

The displayed value is reset to the default value "---" whenever the cell is switched on and when a new transfer is initiated. A new session always starts with the default value.

Remote command:

SENSe:LTE:SIGN<i>:SMS:OUTGoing:INFO:LMSent?

2.4.16.2 Incoming SMS

Access: "Config" hotkey > "Messaging (SMS)" node > "Incoming" node.

The node displays information about incoming mobile-originating short messages.

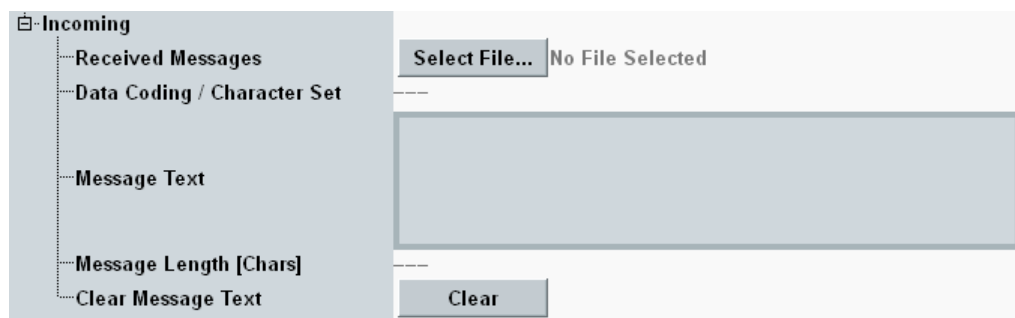


Figure 2-111: Incoming SMS parameters

Select File

Selects a file from the system drive folder

D:\Rohde-Schwarz\CMW\Data\SMS\LTE\Received. Incoming messages are stored there with the file extension .sms.

Only messages received in the current session are available. The files are deleted during startup.

Remote command:

```
CONFigure:LTE:SIGN<i>:SMS:INComing:FILE
```

```
CONFigure:LTE:SIGN<i>:SMS:INComing:FILE:INFO?
```

Data Coding / Character Set

Indicates whether the last received message is a 7-bit ASCII message, an 8-bit binary message or a 16-bit Unicode message.

Remote command:

```
SENSe:LTE:SIGN<i>:SMS:INComing:INFO:DCODing?
```

Message Text / Message Length

Show the text and length of the last received SMS message.

Remote command:

```
SENSe:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT?
```

```
SENSe:LTE:SIGN<i>:SMS:INComing:INFO:MLENgt?
```

Clear Message Text

The button resets all parameters related to a received message.

The message information is deleted. The "message read" flag is set to true.

Remote command:

```
SENSe:LTE:SIGN<i>:SMS:INFO:LRMessage:RFLag?
```

```
CLEAn:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT
```

2.4.17 Messaging (CBS) parameters

Access: "Config" hotkey > "Messaging (CBS)" node.

The node configures the cell broadcast service (CBS) settings.

The supported CBS messages can be divided into two categories:

- Commercial mobile alert system (CMAS) messages:
Message types: presidential alert, extreme alert, severe alert, amber alert and geo fencing
CMAS messages are transmitted via SIB 12.
The term CMAS is used in 3GPP specifications. Other resources also use the newer name wireless emergency alerts (WEA).
- Earthquake and tsunami warning system (ETWS) messages:
Message types: earthquake, tsunami, earthquake + tsunami
ETWS messages are transmitted via SIB 10 and SIB 11.

Option R&S CMW-KS170 is required.

Figure 2-112: CBS settings

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Enable

Enables the transmission of cell broadcast messages via system information messages.

Cell broadcast messages are transmitted if this checkbox is switched on and the cell is switched on.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:ENABLe`

ID

Select a message type. The resulting message ID is displayed for information as decimal and hexadecimal value.

The mapping between message type and message ID is defined in 3GPP TS 23.041, section 9.4.1.2.2.

For "User Defined...", you can enter a message ID.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:ID`

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:IDTYpe`

Serial Number

Sets the serial number, see 3GPP TS 23.041, section 9.4.1.2.1.

The serial number identifies a specific cell broadcast message and consists of three parts:

- **Geographical scope:** message code validity area
- **Message code:** 10-bit number
- **Update number:** version of the message
Enable "Auto Increment" to increase the number automatically when you change the message.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:SERial`

Data Coding Scheme

Specifies the coding scheme octet. Bit 7 to 4 define the coding group. The meaning of bit 3 to 0 depends on the coding group value. For details, see 3GPP TS 23.038, section 5.

The behavior of the settings depends on the configured data source:

- "Use Internal": "Coding Group" bits = 0000, "Language" configurable
You can select a language or enter its hexadecimal or binary code.
- "From File": fixed octet 00010001 (UCS2, message preceded by language indication)
- "User Coded": entire octet configurable as hexadecimal value

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:CGRoup?`

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:LANGuage`

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:DCSCHEME:UCODEd`

Data Source

Selects the source of the message text.

- "Use Internal" Configure the message via the parameter "Data" on page 292.
- "From File" Configure the message via the section "From File" on page 293.
- "User Coded" Configure the message via the parameter "User Coded" on page 292.
Only allowed for CMAS messages.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:SOURce`

Data

Defines the contents of the message as plain text, for the data source "Use Internal".

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:DATA`

User Coded

Defines the contents of the message for the data source "User Coded". You can enter hexadecimal or binary data.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:UCODEd`

Warning Area

Defines a warning area in which the message is relevant.

Warning Area Enable ← Warning Area

Enables the transmission of the warning area coordinates to the UE.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:WAENable`

Warning Area Coordinates ← Warning Area

The data define a warning area, either as a circle or as a polygon. You can enter hexadecimal or binary data.

The UE uses the information and its own position to decide whether a received message must be displayed or not.

The encoding of the information is defined in ATIS 0700041, "WEA 3.0: Device-Based Geo-Fencing".

The basic structure according to WEA 3.0 is as follows:

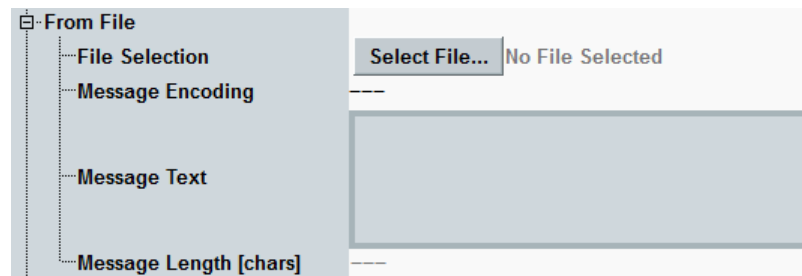
- Circle bits:
 - 0011 0000 0010 1000
 - 22 bits, circle center latitude
 - 22 bits, circle center longitude
 - 20 bits, circle radius
- Polygon bits:
 - 0010
 - 10 bits, length of the warning area coordinates field, in octets
 - 00

- 22 bits, latitude of a point of the polygon
- 22 bits, longitude of a point of the polygon
- Other points of the polygon, 44 bits per point

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:WACoordinate`

From File



Selects a file containing the message to be transmitted. The file is used if the configured data source equals "From File".

Select File... ← From File

Selects a file from the system drive folder

D: \Rohde-Schwarz\CMW\Data\cbs\LTE\ or one of its subfolders. Store your message files there, using the file extension .cbs.

The current software version supports only 16-bit Unicode (UTF16).

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE`

Message Encoding, Message Text, Message Length ← From File

Displays information about the message in the selected file.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE:INFO?`

ETWS

Configures additional settings for the ETWS message types.

- "Activate Alert": The UE must activate an emergency user alert.
- "Activate Popup on Display": The UE must display a warning popup.

The related messages are defined in 3GPP TS 23.041, section 9.3.24.

Remote command:

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:ETWS:ALERT`

`CONFigure:LTE:SIGN<i>:CBS:MESSAge:ETWS:POPup`

2.4.18 Shortcut configuration

Access: "Config" hotkey > "Shortcut Softkey" node.

The node configures three shortcut softkeys that provide a fast way to switch to selectable measurements.

See also [Section 2.4.3, "Using the shortcut softkeys"](#), on page 137.

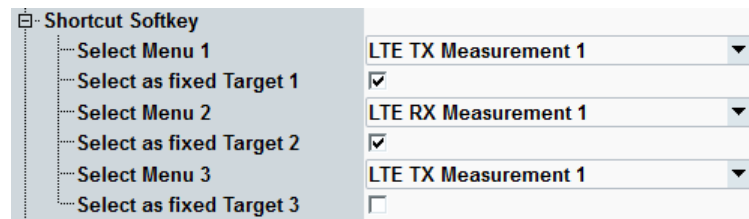


Figure 2-113: Shortcut configuration

Select Menu

Selects a measurement. The corresponding shortcut softkey opens a dialog presenting this measurement as default target or uses the measurement as fixed target.

Select as fixed Target

Configures and renames the corresponding shortcut softkey.

- **Enabled:** The softkey directly opens the measurement selected via [Select Menu](#).
- **Disabled:** The softkey opens a dialog box for selection of the target measurement.

2.4.19 Message monitoring settings

Access: "Config" hotkey > "Message Monitoring" node.

Messages exchanged between the LTE signaling application and the UE can be monitored. For this purpose, the messages are sent to an external PC.

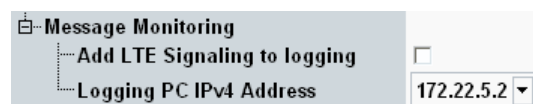


Figure 2-114: Message monitoring settings

Add LTE Signaling to logging

Enables or disables message monitoring for the LTE signaling application.

Remote command:

`CONFigure:LTE:SIGN<i>:MMONitor:ENABLe`

Logging PC IPv4 Address

Selects the IP address to which the messages are sent for monitoring.

The address pool is configured globally, see "Setup" dialog, section "Logging".

Remote command:

`CONFigure:LTE:SIGN<i>:MMONitor:IPADdress`

2.4.20 Debug configuration

Access: "Config" hotkey > "Debug" node.

The node configures the log file generation.



Do not enable the debug mode during regular operation. It impacts the test performance, timing, system drive usage, and so on.

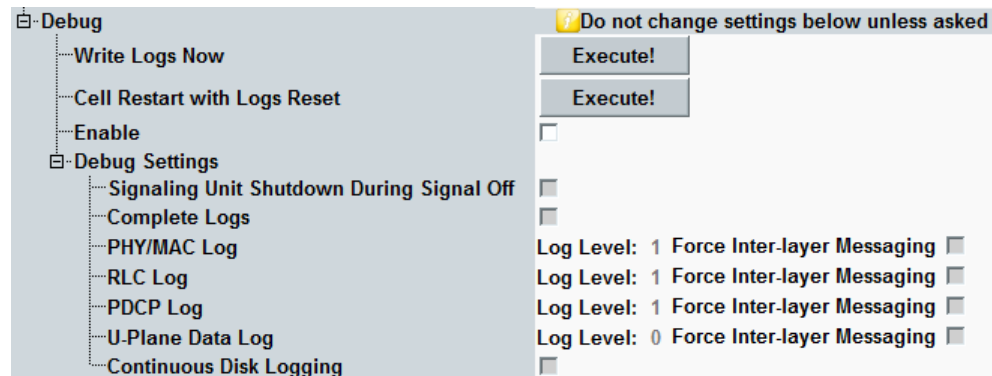


Figure 2-115: Debug configuration

The settings are not affected by a reset or preset. They are not stored in save files. A reboot restores the default settings.

Write Logs Now.....	295
Cell Restart with Logs Reset.....	295
Enable.....	295
Signaling Unit Shutdown During Signal Off.....	295
Complete Logs.....	295
PHY/MAC Log to U-Plane Data Log.....	296
Continuous Disk Logging.....	296

Write Logs Now

Writes the LTE log files to the system drive.

Cell Restart with Logs Reset

Switches the cell signal off and on again with empty logs.

Enable

Enables the configuration of the "Debug Settings".

Signaling Unit Shutdown During Signal Off

The complete log files are written if the signaling unit is shut down. The signaling unit is shut down if the instrument is shut down or if the CMW software is terminated.

With enabled checkbox, the signaling unit is also shut down if the cell signal is switched off. So switching off the cell delivers log files.

Complete Logs

With disabled checkbox, only a reduced set of log files is written if the LTE signaling instance is replaced by another signaling instance due to a resource conflict. This behavior saves time and system drive space.

With enabled checkbox, a complete set of log files is written if the LTE signaling instance is replaced by another signaling instance.

PHY/MAC Log to U-Plane Data Log

These settings configure the level of detail of the log files.

Continuous Disk Logging

With enabled checkbox, the log files are written continuously to the system drive. The only limitation is the size of the system drive.

With disabled checkbox, a ring buffer is used. The monitored time interval is limited and the oldest data is overwritten when the buffer is full.

2.4.21 Annex: UE capabilities

This section describes the UE capability information displayed in the main view, if the UE capabilities area is maximized. See also [Section 2.4.1.4, "UE capabilities"](#), on page 124.

All capability information is transmitted by the UE in the information element *UE-EUTRA-Capability* described in 3GPP TS 36.331.

For some parameters described in the following sections, three remote control commands are listed:

- First command:
Queries the parameter in the upper part of the capability report. Features indicated as "supported" are supported for FDD and TDD.
- Command with additional :FAUeeutra: mnemonic:
Queries the parameter in section "fdd Add UE-EUTRA Capabilities". Features indicated as "supported" are supported only for FDD, not for TDD.
- Command with additional :TAUeeutra: mnemonic:
Queries the parameter in section "tdd Add UE-EUTRA Capabilities". Features indicated as "supported" are supported only for TDD, not for FDD.

2.4.21.1 General UE capability information

At the highest level, the report lists the following information.

Access Stratum Release	Release 11
UE Category	11
UE Category DL	11
UE Category UL	5
Feature Group Indicators	0111 1111 1100 1111 1111 1111 1111 1110 bin
Feature Group Indicators Rel9Add	1100 0000 0000 0000 0000 0000 0000 0000 bin
Feature Group Indicators Rel10	0100 0000 0010 0100 0000 0000 0000 0000 bin
Device Type	----
Crash Report	☒

Figure 2-116: General UE capability information

Access Stratum Release.....	297
UE Category.....	297
Feature Group Indicators.....	297
Feature Group Indicators Rel9 Add.....	297
Feature Group Indicators Rel 10.....	298
Device Type.....	298
rach Report.....	298

Access Stratum Release

Supported release of the E-UTRA layer one, two and three specifications.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:ASRelease?
```

UE Category

UE categories of the UE, defining several transport channel and physical channel parameters, for UL and DL.

For details, see 3GPP TS 36.306:

- "UE Category": see section 4.1 "ue-Category"
- "UE Category DL": see section 4.1A "ue-CategoryDL and UE Category UL"
- "UE Category UL": see section 4.1A "ue-CategoryDL and UE Category UL"

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:UECategory?
```

```
SENSe:LTE:SIGN<i>:UECapability:UECategory:DL:ENHanced?
```

```
SENSe:LTE:SIGN<i>:UECapability:UECategory:UL:ENHanced?
```

Feature Group Indicators

32-bit value, containing one bit per feature group ("1" = supported, "0" = not supported).

The features assigned to the individual bits are listed in the annex of 3GPP TS 36.331, table B.1-1.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:FGIndicators?
```

```
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators?
```

```
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators?
```

Feature Group Indicators Rel9 Add

32-bit value, containing one bit per feature group ("1" = supported, "0" = not supported).

The features assigned to the individual bits are listed in the annex of 3GPP TS 36.331, table B.1-1a.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:FGIndicators:RNADd?
```

```
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators:RNADd?
```

```
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators:RNADd?
```

Feature Group Indicators Rel 10

32-bit value, containing one bit per feature group ("1" = supported, "0" = not supported).

The features assigned to the individual bits are listed in the annex of 3GPP TS 36.331, table C.1-1.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:FGIndicators:RTEN?`

`SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators:RTEN?`

`SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators:RTEN?`

Device Type

Indicates whether the UE benefits from NW-based battery consumption optimization:

- "noBenFromBatConsumpOpt" means that the UE does not benefit from the optimization.
- An empty field means that the UE does benefit from the optimization.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:DTYPE?`

RACH Report

Support of RACH report delivery.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:RREPort?`

2.4.21.2 PDCP UE capabilities

The following parameters indicate in which way the UE supports the packet data convergence protocol (PDCP) described in 3GPP TS 36.323.

PDCP	
ROHC RTP	☒
ROHC UDP	☐
ROHC ESP	☒
ROHC IP	☐
ROHC TCP	☐
ROHCv2 RTP	☒
ROHCv2 UDP	☒
ROHCv2 ESP	☒
ROHCv2 IP	☐
Max Num. ROHC Context Sessions	cs32
PDCP SN Extension	☒
Support ROHC Context Continue	☒

Figure 2-117: PDCP UE capabilities

ROHC RTP to ROHCv2 IP	299
Max Num ROHC Context Sessions	299
PDCP SN Extension	299
Support ROHC Context Continue	299

ROHC RTP to ROHCv2 IP

Support of the robust header compression profiles with the profile identifiers 0x0001 to 0x0004, 0x0006, 0x0101 to 0x0104.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PDCP:SRPProfiles?`

Max Num ROHC Context Sessions

Maximum number of header compression context sessions supported by the UE.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PDCP:MRCSessions?`

PDCP SN Extension

Support of 15-bit PDCP sequence numbers.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PDCP:SNExtension?`

Support ROHC Context Continue

Support of ROHC context continuation during a handover.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PDCP:SRCCContinue?`

2.4.21.3 Physical layer UE capabilities, part 1

The following parameters indicate physical layer capabilities of the UE.

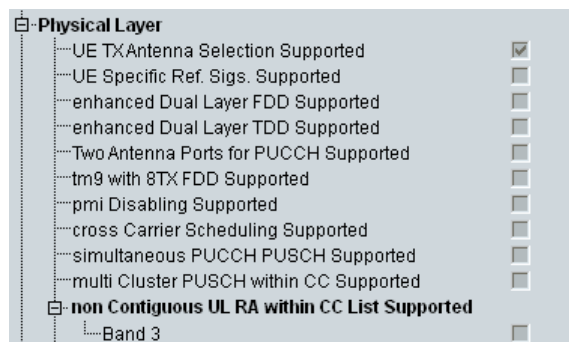


Figure 2-118: Physical layer UE capabilities

UE TX Antenna Selection Supported	300
UE-Specific Ref. Sigs. Supported	300
enhanced Dual layer FDD/TDD Supported	300
Two Antenna Ports for PUCCH Supported	300
tm9 with 8TX FDD Supported	300
pmi Disabling Supported	300
cross Carrier Scheduling Supported	300
simultaneous PUCCH PUSCH Supported	301
multi Cluster PUSCH within CC Supported	301
non Contiguous UL RA within CC List Supported	301

UE TX Antenna Selection Supported

Support of transmit antenna selection, see 3GPP TS 36.213.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:UTASupported?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:UTASupported?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:UTASupported?
```

UE-Specific Ref. Sigs. Supported

Support of PDSCH TM 7 for FDD.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:USRSsupport?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:USRSsupport?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:USRSsupport?
```

enhanced Dual layer FDD/TDD Supported

Support of enhanced dual layer (PDSCH TM 8) for FDD/TDD.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:EDLfsupport?  
SENSe:LTE:SIGN<i>:UECapability:PLAYer:EDLTsupport?
```

Two Antenna Ports for PUCCH Supported

Support of transmit diversity for PUCCH formats 1/1a/1b/2/2a/2b and support of PUCCH format 3 / transmit diversity for PUCCH format 3.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:TAPPsupport?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:TAPPsupport?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:TAPPsupport?
```

tm9 with 8TX FDD Supported

Support of PDSCH TM 9 with 8 CSI-RS ports for FDD.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:TWEFsupport?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:TWEFsupport?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:TWEFsupport?
```

pmi Disabling Supported

Support of PMI disabling.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:PDSsupport?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:PDSsupport?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:PDSsupport?
```

cross Carrier Scheduling Supported

Support of cross-carrier scheduling for CA.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:CCSSupport?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:CCSSupport?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:CCSSupport?
```

simultaneous PUCCH PUSCH Supported

UE baseband supports simultaneous transmission of PUCCH and PUSCH and is band agnostic.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:SPPSupport?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:SPPSupport?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:SPPSupport?
```

multi Cluster PUSCH within CC Supported

UE baseband supports multi-cluster PUSCH transmission within a component carrier (CC), and is band agnostic.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:MCPCsupport?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:MCPCsupport?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:MCPCsupport?
```

non Contiguous UL RA within CC List Supported

For each supported E-UTRA band: UE RF supports non-contiguous UL resource allocations within a CC.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PLAYer:NURClst?  
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:NURClst?  
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:NURClst?
```

2.4.21.4 Physical layer UE capabilities, part 2

The following parameters indicate physical layer capabilities of the UE.

CRS InterfHandl	☐
ePDCCH	☐
multiACK CSI Reporting	☐
SS CCH InterfHandl	☐
TDD SpecialSubframe	☐
txDiv PUCCH1b ChSelect	☐
UL CoMP	☐
interBand TDD CAWith Different Config	00 bin
e HARQ Pattern FDD	☐
enhanced 4 Tx Codebook	☐
tdd FDD CA PCell Duplex	00 bin
phy TDD ReConfig TDD PCell	☐
phy TDD ReConfig FDD PCell	☐
pusch Feedback Mode	☐
pusch SRS PowerControl SubframeSet	☐
CSI Subframe Set	☐
no Resource Restriction For TTI Bundling	☐
discovery Signals In Deact SCell	☐

Figure 2-119: Physical layer UE capabilities

CRS InterfHandl.....	302
ePDCCH.....	302
multiACK CSI Reporting.....	302
SS CCH InterfHandl.....	303
TDD SpecialSubframe.....	303
txDiv PUCCH1b ChSelect.....	303
UL CoMP.....	303
interBand TDD CA With Different Config.....	303
e HARQ Pattern FDD.....	303
enhanced 4 Tx Codebook.....	303
tdd FDD CA PCell Duplex.....	303
phy TDD ReConfig TDD PCell.....	304
phy TDD ReConfig FDD PCell.....	304
pusch Feedback Mode.....	304
pusch SRS PowerControl SubframeSet.....	304
CSI Subframe Set.....	304
no Resource Restriction For TTI Bundling.....	304
discovery Signals In Deact SCell.....	304

CRS InterfHandl

Support of CRS interference handling.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:CIHandl?`

ePDCCH

Support of DCI reception via UE-specific search space on enhanced PDCCH.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:EPDCch?`

multiACK CSI Reporting

Support of multi-cell HARQ ACK, periodic CSI reporting and SR on PUCCH format 3.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:MACReporting?`

SS CCH InterfHandl

Support of synchronization signal and common channel interference handling.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:SCIHandl?`

TDD SpecialSubframe

Support of TDD special subframe as defined in 3GPP TS 36.211.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:TSSubframe?`

txDiv PUCCH1b ChSelect

Support of transmit diversity for PUCCH format 1b with channel selection.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:TDPChselect?`

UL CoMP

Support of UL coordinated multi-point operation.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:ULComp?`

interBand TDD CA With Different Config

Support of inter-band TDD CA with different UL/DL configuration combinations.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:ITCWithdiff?`

e HARQ Pattern FDD

Support of enhanced HARQ pattern for TTI bundling operation for FDD.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:EHPFdd?`

enhanced 4 Tx Codebook

Support of enhanced 4 TX codebook.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:EFTCodebook?`

tdd FDD CA PCell Duplex

Support of PCell in any supported band combination including at least one FDD band and at least one TDD band.

First bit: support of TDD PCell. Second bit: support of FDD PCell.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:TFCPcelldplx?`

phy TDD ReConfig TDD PCell

Support of TDD UL/DL reconfiguration for TDD serving cell via monitoring PDCCH on a TDD PCell.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:TRCTddpcell?`

phy TDD ReConfig FDD PCell

Support of TDD UL/DL reconfiguration for TDD serving cell via monitoring PDCCH on an FDD PCell.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:TRCFddpcell?`

pusch Feedback Mode

Support of PUSCH feedback mode 3-2.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:PFMode?`

pusch SRS PowerControl SubframeSet

Support of subframe set dependent UL power control for PUSCH and SRS.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:PSPSfset?`

CSI Subframe Set

Support of R12 DL CSI subframe set configuration.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:CSFSet?`

no Resource Restriction For TTI Bundling

Support of TTI bundling without resource allocation restriction.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:NRRT?`

discovery Signals In Deact SCell

Support of discovery signal detection for deactivated SCells.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:PLAYer:DSDCell?`

2.4.21.5 RF UE capabilities

The following parameters indicate the supported E-UTRA operating bands and other RF capabilities.

RF Parameters				
Supported Bands				
	Half Duplex	dl-256QAM	ul-64QAM	
Band 38	☒	☐	☐	
Band 39	☒	☐	☐	
Band 40	☒	☐	☐	
Band 41	☒	☐	☐	
Band 1	☐	☐	☐	
Band 3	☐	☐	☐	
Band 20	☐	☐	☐	
Band 26	☐	☐	☐	

Figure 2-120: RF UE capabilities, part 1

Supported Band Combination					
Show	5 Band Combinations from index 0				Total: 63
Band combinations vs Parameter	No.1	No.2	No.3	No.4	BW Comb. Set
0	Band 3	---	---	---	---
1	Band 8	---	---	---	---
2	Band 1	---	---	---	---
3	Band 28	---	---	---	---
4	Band 7	---	---	---	---
No. 1					
Band Combination ...	0	1	2	3	4
No.1	Band 3	Band 8	Band 1	Band 28	Band 7
Bandwidth Class UL 1	a	a	a	a	a
MIMO Capability UL 1	---	---	---	---	---
Bandwidth Class DL 1	a	a	a	a	a
MIMO Capability DL 1	4	2	2	2	4
Supported CSI-Proc	---	---	---	---	---
No. 2					
Band Combination ...	0	1	2	3	4
No.2	---	---	---	---	---
Bandwidth Class UL 2	---	---	---	---	---
MIMO Capability UL 2	---	---	---	---	---
Bandwidth Class DL 2	---	---	---	---	---
MIMO Capability DL 2	---	---	---	---	---
Supported CSI-Proc	---	---	---	---	---
No. 3					
No. 4					

Figure 2-121: RF UE capabilities, part 2

[-] Multiple Timing Advance		0 1 2 3 4				
Band Combination		☐ ☐ ☐ ☐ ☐				
Supported		☐ ☐ ☐ ☐ ☐				
[-] Simultaneous Rx-Tx		0 1 2 3 4				
Band Combination		☐ ☐ ☐ ☐ ☐				
Supported		☐ ☐ ☐ ☐ ☐				
[-] DC-Support		Asynchronous				
0		☐				
1		☐				
2		☐				
3		☐				
4		☐				
[-] supported NAICS-2CRS-AP		Supp Cell Grouping				
0		---				
1		---				
2		---				
3		---				
4		---				
Freq Band Retrieval		☒				
[-] Requested Frequency Bands		☒				
Freq Band Priority Adjustment		☒				
modifiedMPR-Behavior						

Figure 2-122: RF UE capabilities, part 3

Supported Bands.....	306
Supported Band Combination.....	307
L Band Combinations vs Parameter.....	307
L No. n nodes.....	307
L Bandwidth Class.....	307
L MIMO Capability.....	307
L Supported CSI-Proc.....	307
L Multiple Timing Advance.....	308
L Simultaneous Rx-Tx.....	308
L DC-Support.....	308
L Supported NAICS-2CRS-AP.....	308
Freq Band Retrieval.....	308
Requested Frequency Bands.....	308
Freq Band Priority Adjustment.....	308
modifiedMPR-Behavior.....	309

Supported Bands

The UE supports all listed operating bands.

The columns indicate whether the UE supports certain features in each supported band:

- "Half Duplex": The UE supports only half duplex operation in the band.
- "dl-256QAM": The UE supports DL 256-QAM in the band.
- "ul-64-QAM": The UE supports UL 64-QAM in the band.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:SUPPorted?
SENSe:LTE:SIGN<i>:UECapability:RF:HDUPlex?
SENSe:LTE:SIGN<i>:UECapability:RF:DL<qam>?
SENSe:LTE:SIGN<i>:UECapability:RF:UL<qam>?
```

Supported Band Combination

This node lists the band combinations supported by the UE for carrier aggregation. The band combinations are numbered from 0 to n.

The row "Show" selects which of the supported combinations are displayed. You can select the total number of shown combinations and the first shown combination. In the figure, five combinations are shown, starting with combination number 0. In total, the UE has reported 63 supported combinations.

Band Combinations vs Parameter ← Supported Band Combination

The table "Band combinations vs Parameter" contains one row per shown band combination. The columns indicate the bands. Column "BW Comb. Set" indicates which bandwidth combination sets are supported for a band combination. The leftmost bit corresponds to set 0, the next bit to set 1, and so on. "0" means that the set is not supported. "1" means that the set is supported. The sets are specified in 3GPP TS 36.101, section 5.6A.1.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>?
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:BCSet?
```

No. n nodes ← Supported Band Combination

The "No. n" nodes contain one column per band combination. There is one node ("No. 1", "No. 2", ...) per band of the band combination.

For each band, the information listed in the following is given.

Bandwidth Class... ← No. n nodes ← Supported Band Combination

Supported bandwidth classes for UL and DL.

The bandwidth classes are defined in 3GPP TS 36.101, section 5.6A.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:BCLass:UL?
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:BCLass:DL?
```

MIMO Capability... ← No. n nodes ← Supported Band Combination

MIMO capability for each supported bandwidth class, UL and DL. It indicates the number of supported layers for spatial multiplexing.

Example: bandwidth class = "abc", MIMO capability = "244" means 2 layers for class a, 4 layers for class b and c.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:MCAPability:UL?
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:MCAPability:DL?
```

Supported CSI-Proc ← No. n nodes ← Supported Band Combination

Maximum number of supported CSI processes.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:SCPRoc?
```

Multiple Timing Advance ← Supported Band Combination

Support of multiple timing advances, indicated per shown band combination.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:MTADvance?
```

Simultaneous Rx-Tx ← Supported Band Combination

Support of simultaneous reception and transmission on different bands, indicated per shown band combination. The information element is specified by 3GPP only for TDD band combinations.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:SRTX?
```

DC-Support ← Supported Band Combination

There is one row per shown band combination. The columns provide the following information:

- "Asynchronous":
Support of asynchronous DC and power control mode 2.
- "Supp Cell Grouping":
Indicates for which mapping of serving cells to the first and second cell groups the UE supports asynchronous DC.
The length and meaning of the bitstring depends on the number of bands in the band combination. For details, see 3GPP TS 36.331, "supportedCellGrouping".

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:DCSupport:ASYNchronous?
SENSe:LTE:SIGN<i>:UECapability:RF:DCSupport:SCGRouping?
```

Supported NAICS-2CRS-AP ← Supported Band Combination

Bitstring from the element "supportedNAICS-2CRS-AP", per band combination.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:SNCap?
```

Freq Band Retrieval

Indicates whether the UE supports the reception of "requestedFrequencyBands".

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:FBRetrieval?
```

Requested Frequency Bands

Lists all frequency bands requested by E-UTRAN.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:RF:RBANDs?
```

Freq Band Priority Adjustment

Support of prioritization of frequency bands as requested by "freqBandIndicatorPriority-r12".

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:RF:FBPadjust?`

modifiedMPR-Behavior

Support of modified MPR/A-MPR behaviors. The leftmost bit refers to behavior 0, the next bit to behavior 1, and so on. 1 means supported. 0 means not supported.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:RF:MMPRbehavior?`

2.4.21.6 Measurement UE capabilities

The following parameters indicate the UE capabilities for channel measurements, for example on other operating bands or on other radio access technologies (RAT).

Meas Parameters

- Inter Freq Need for Gaps**

Band ...	3	8	1	28	7
Band 3	☒	☒	☒	☒	☒
Band 8	☒	☒	☒	☒	☒
Band 1	☒	☒	☒	☒	☒
Band 28	☒	☒	☒	☒	☒
Band 7	☒	☒	☒	☒	☒
- Inter RAT Need for Gaps**

EUTRA Band ...	3	8	1	28	7
UTRA FDD :					
Band 1	☒	☒	☒	☒	☒
Band 2	☒	☒	☒	☒	☒
Band 5	☒	☒	☒	☒	☒
Band 8	☒	☒	☒	☒	☒
UTRA TDD128 :					
GERAN :					
CDMA2000 HRPD :					
CDMA2000 1XRTT :					

Figure 2-123: Measurement UE capabilities, part 1

Inter Freq Need for Gaps v1020

Band Combination ...	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Band 3	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
Band 8	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
Band 1	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
Band 28	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒
Band 7	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒

Inter RAT Need for Gaps v1020

EUTRA Band Combination ...	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	
UTRA FDD :																	
Band 1	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	
Band 2	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	
Band 5	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	
Band 8	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	☒	
UTRA TDD128 :																	
GERAN :																	
CDMA2000 HRPD :																	
CDMA2000 1XRTT :																	
RSRQ Meas Wideband	☐																
Benefits from Interruption	---																
rsrssi-And Channel Occupancy Reporting	☐																

Figure 2-124: Measurement UE capabilities, part 3

Inter-Freq Need for Gaps.....	310
Inter-RAT Need for Gaps.....	310
Inter-Freq Need for Gaps v1020.....	310
Inter-RAT Need for Gaps v1020.....	310
RSRQ Meas Wideband.....	311
Benefits from Interruption.....	311
rsqi-And Chanell Occupancy Reporting.....	311

Inter-Freq Need for Gaps

Indicates the need for downlink measurement gaps when using a certain E-UTRA band and measuring a certain E-UTRA band. Each column corresponds to a specific used E-UTRA band, each row to a specific measured E-UTRA band.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:MEAS:IFNGaps?
```

Inter-RAT Need for Gaps

Indicates the need for downlink measurement gaps when using a certain E-UTRA band and measuring on a supported band of another RAT. Each column corresponds to a specific used E-UTRA band, each row to a specific measured band of another RAT.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:UFDD?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:UTDD<n>?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:GERan?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:CHRPd?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:CXRTt?
```

Inter-Freq Need for Gaps v1020

Indicates the need for downlink measurement gaps when using a certain E-UTRA band combination for carrier aggregation and measuring a certain E-UTRA band. Each column corresponds to a specific used band combination, each row to a specific measured E-UTRA band.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:MEAS:IFNGaps:V<number>?
```

Inter-RAT Need for Gaps v1020

Indicates the need for downlink measurement gaps when using a certain E-UTRA band combination for carrier aggregation and measuring on a supported band of another RAT. Each column corresponds to a specific used band combination, each row to a specific measured band of another RAT.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:UFDD?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:UTDD<n>?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:GERan?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:CHRPd?
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:CXRTt?
```

RSRQ Meas Wideband

Support of RSRQ measurements with wider bandwidth.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:MEAS:RMWideband?
```

Benefits from Interruption

Indicates whether the UE power consumption can be reduced by allowing the UE to cause interruptions to serving cells. The information applies to measurements of deactivated SCell carriers for measurement cycles of less than 640 ms, as specified in 3GPP TS 36.133.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:MEAS:BFInterrupt?
```

rssi-And Chanel Occupancy Reporting

Indicates whether the UE supports RSSI and channel occupancy measurements for LAA.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:MEAS:RCOReporting?
```

2.4.21.7 Inter-RAT UE capabilities

The following parameters indicate the inter-RAT handover capabilities of the UE.

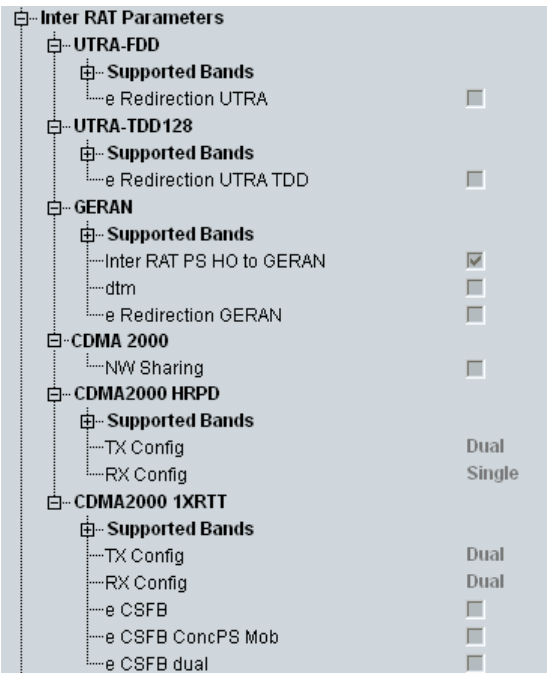


Figure 2-125: Inter-RAT handover UE capabilities

UTRA-FDD.....	312
L Supported Bands.....	312
L e Redirection UTRA.....	312

UTRA-TDD128.....	312
L Supported Bands.....	312
L e Redirection UTRA TDD.....	313
GERAN.....	313
L Supported Bands.....	313
L Inter-RAT PS HO to GERAN.....	313
L dtm.....	313
L e Redirection GERAN.....	313
CDMA2000 NW Sharing.....	313
CDMA2000 HRPD.....	313
L Supported Bands.....	313
L TX Config.....	314
L RX Config.....	314
CDMA2000 1xRTT.....	314
L Supported Bands.....	314
L TX Config.....	314
L RX Config.....	314
L e CSFB.....	314
L e CSFB ConcPS Mob.....	314
L e CSFB dual.....	315

UTRA-FDD

This node contains capability information for handover to UTRA FDD.

Supported Bands ← UTRA-FDD

Lists the supported UTRA FDD operating bands. A handover to these bands is supported by the UE.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:UFDD:SUPPorted?`

e Redirection UTRA ← UTRA-FDD

Support of enhanced redirection to UTRA FDD, using system information provided upon redirection.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:UFDD:EREDirection:UTRA?`

`SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:EREDirection:UTRA?`

`SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:EREDirection:UTRA?`

UTRA-TDD128

This node contains capability information for handover to UTRA TDD with a chip rate of 1.28 Mcps.

Supported Bands ← UTRA-TDD128

Lists the supported UTRA TDD operating bands. A handover to these bands is supported by the UE.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:UTDD128:SUPPorted?`

e Redirection UTRA TDD ← UTRA-TDD128

Support of enhanced redirection to UTRA TDD, using system information provided upon redirection.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:UTDD128:EREDirection:UTDD?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:EREDirection:UTDD?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:EREDirection:UTDD?
```

GERAN

This node contains capability information for handover to GERAN.

Supported Bands ← GERAN

Lists the supported GERAN operating bands.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:SUPPorted?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:GERan:SUPPorted?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:SUPPorted?
```

Inter-RAT PS HO to GERAN ← GERAN

Support of a handover to GERAN.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:PHGeran?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:GERan:PHGeran?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:PHGeran?
```

dtm ← GERAN

Support of the dual transfer mode (DTM) in GERAN.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:DTM?
```

e Redirection GERAN ← GERAN

Support of an enhanced redirection to GERAN, using system information provided upon redirection.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:EREDirection?
```

CDMA2000 NW Sharing

Support of network sharing for CDMA2000.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:CDMA2000:NWSHaring?
```

CDMA2000 HRPD

This node contains capability information for handover to CDMA2000 HRPD.

Supported Bands ← CDMA2000 HRPD

Lists the supported HRPD band classes.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:SUPPorted?`

TX Config ← CDMA2000 HRPD

Support of single/dual transmitter. Dual transmitter allows the UE to transmit simultaneously on E-UTRAN and HRPD.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:TCONfig?`

RX Config ← CDMA2000 HRPD

Support of single/dual receiver. Dual receiver allows the UE to receive simultaneously on E-UTRAN and HRPD.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:RCONfig?`

CDMA2000 1xRTT

This node contains capability information for handover to CDMA2000 1xRTT.

Supported Bands ← CDMA2000 1xRTT

Lists the supported 1xRTT band classes.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRtT:SUPPorted?`

TX Config ← CDMA2000 1xRTT

Support of single/dual transmitter. Dual transmitter allows the UE to transmit simultaneously on E-UTRAN and 1xRTT.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRtT:TCONfig?`

RX Config ← CDMA2000 1xRTT

Support of single/dual receiver. Dual receiver allows the UE to receive simultaneously on E-UTRAN and 1xRTT.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRtT:RCONfig?`

e CSFB ← CDMA2000 1xRTT

Support of enhanced CS fallback to CDMA2000 1xRTT.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRtT:ECSFb?`

`SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:CXRtT:ECSFb?`

`SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:CXRtT:ECSFb?`

e CSFB ConcPS Mob ← CDMA2000 1xRTT

Support of concurrent enhanced CS fallback to CDMA2000 1xRTT and handover/redirection to CDMA2000 HRPD.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:XRtT:ECCMob?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:XRtT:ECCMob?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:XRtT:ECCMob?
```

e CSFB dual ← CDMA2000 1xRTT

Support of enhanced CS fallback to CDMA2000 1xRTT for dual Rx/Tx configuration.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IRAT:XRtT:ECDual?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:XRtT:ECDual?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:XRtT:ECDual?
```

2.4.21.8 Other UE capabilities

The following parameters indicate the UE capabilities categorized by 3GPP as "other parameters".

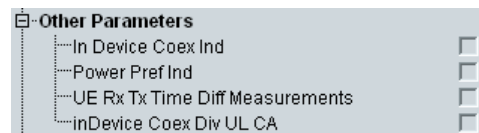


Figure 2-126: Other capabilities

In Device Coex Ind

Support of in-device coexistence indication and autonomous denial functionality.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:IDCindex?
```

Power Pref Ind

Support of power preference indication.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:PPIndex?
```

UE Rx Tx Time Diff Measurements

Support of RX-TX time difference measurements.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:URTTimediff?
```

In Device Coex Div UL CA

Support of in-device coexistence indication for UL CA.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:DCIulca?
```

2.4.21.9 MBMS UE capabilities

The following parameters indicate the UE capabilities for MBMS reception.

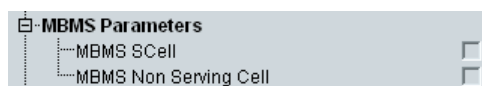


Figure 2-127: MBMS capabilities

MBMS SCell

Support of MBMS reception via an SCell.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:MBMS:SCEl1?`

MBMS Non Serving Cell

Support of MBMS reception via a serving cell to be added.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:MBMS:NSCell?`

2.4.21.10 CSG proximity indication UE capabilities

The following parameters indicate the support of proximity indications by the UE.

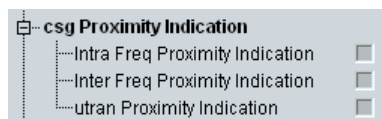


Figure 2-128: Proximity indication capabilities

Intra Freq Proximity Indication

Support of proximity indications for intra-frequency E-UTRAN CSG member cells.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:CPINdication:FREQuency:INTRa?`

Inter Freq Proximity Indication

Support of proximity indications for inter-frequency E-UTRAN CSG member cells.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:CPINdication:FREQuency:INTER?`

utran Proximity Indication

Support of proximity indications for UTRAN CSG member cells.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:CPINdication:UTRan?`

2.4.21.11 Neighbor cell SI-acquisition UE capabilities

The following parameters indicate whether the UE supports system information requests for handover. When the UE receives such a request, it must use autonomous gaps to read and report the system information of neighbor cells.

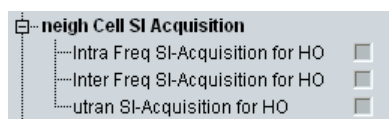


Figure 2-129: SI-Acquisition capabilities

Intra Freq SI-Acquisition for HO

Support of system information acquisition for intra-frequency neighbor cells.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:NCSacq:FREQUENCY:INTRa?
```

```
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:FREQUENCY:INTRa?
```

```
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQUENCY:INTRa?
```

Inter Freq SI-Acquisition for HO

Support of system information acquisition for inter-frequency neighbor cells.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:NCSacq:FREQUENCY:INTER?
```

```
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:FREQUENCY:INTER?
```

```
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQUENCY:INTER?
```

utran SI-Acquisition for HO

Support of system information acquisition for UMTS neighbor cells.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:NCSacq:UTRan?
```

```
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:UTRan?
```

```
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:UTRan?
```

2.4.21.12 UE-based network performance measurement UE capabilities

The following parameters indicate UE capabilities for UE-based network performance measurements.

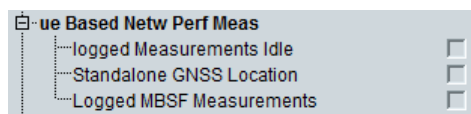


Figure 2-130: Network performance measurement capabilities

logged Measurements Idle

Support of logged measurements in idle mode.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:UBNPmeas:LMIDle?
```

Standalone GNSS Location

Indicates whether the UE is equipped with a GNSS receiver or not.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:UBNPmeas:SGLocation?
```

Logged MBSFN Measurements

Support of logged MBSFN measurements in RRC idle and connected mode.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:LMMeas?`

2.4.21.13 RLC UE capabilities

The following parameters indicate UE capabilities for RLC.



Figure 2-131: RLC capabilities

extended RLC LI Fields

Support of 15-bit RLC length indicator.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:ERLField?`

2.4.21.14 WLAN IW UE capabilities

The following parameters indicate UE capabilities for WLAN interworking.



Figure 2-132: WLAN interworking capabilities

WLAN IW RAN Rules

Support of RAN-assisted WLAN interworking based on access network selection and traffic steering rules specified in 3GPP TS 36.304.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:WIW:WIRRules?`

WLAN IW ANDSF Policies

Support of RAN-assisted WLAN interworking based on ANDSF policies specified in 3GPP TS 24.312.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:WIW:WIAPolicies?`

2.4.21.15 CE UE capabilities

The following parameters indicate UE capabilities for the eMTC CE modes.



Figure 2-133: CE capabilities

ce-ModeA

Support of operation in CE mode A.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:CEParameters:MODE:A?`

ce-ModeB

Support of operation in CE mode B.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:CEParameters:MODE:B?`

2.4.21.16 LAA UE capabilities

The following parameters indicate UE capabilities for licensed-assisted access.

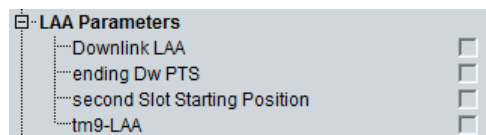


Figure 2-134: LAA capabilities

Downlink LAA

Support of DL LAA operation.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:LAA:DL?`

ending Dw PTS

Support of partial allocation in the ending subframe of an LAA burst.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:LAA:EDPTs?`

second Slot Starting Position

Support of partial allocation in the initial subframe of an LAA burst.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:LAA:SSSPosition?`

tm9-LAA

Support of transmission mode 9 for LAA downlinks.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:LAA:TM<TMnr>?`

2.4.21.17 DC UE capabilities

The following parameters indicate UE capabilities for dual connectivity.



Figure 2-135: Dual connectivity capabilities

DRB TypeSplit

Support of DRB type of split bearer.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:DCParameters:DTSplit?`

DRB TypeSCG

Support of DRB type of SCG bearer.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:DCParameters:DTSCg?`

2.4.21.18 MAC UE capabilities

The following parameters indicate UE capabilities for the MAC layer.

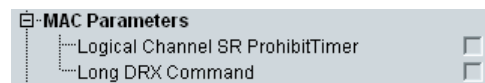


Figure 2-136: MAC capabilities

Logical Channel SR Prohibit Timer

UE supports the logical channel SR prohibit timer as specified in 3GPP TS 36.321.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:MAC:LCSptimer?`

Long DRX Command

UE supports the long DRX command MAC control element as specified in 3GPP TS 36.321.

Remote command:

`SENSe:LTE:SIGN<i>:UECapability:MAC:LDRXcommand?`

2.4.21.19 FDD/TDD additional UE E-UTRA capabilities

The capability report sections "fdd-Add-UE-EUTRA-Capabilities" and "tdd-Add-UE-EUTRA-Capabilities" are only relevant for UEs that support both FDD and TDD.

In that case, the previous sections contain information that applies to FDD and TDD. If a feature is listed as supported, it is supported in both modes.

The additional section "fdd-Add-UE-EUTRA-Capabilities" contains a subset of the parameters described in the previous sections. If a feature is listed as supported, it is only supported for FDD but not for TDD.

The additional section "tdd-Add-UE-EUTRA-Capabilities" contains the same subset of parameters. If a feature is listed as supported, it is only supported for TDD but not for FDD.

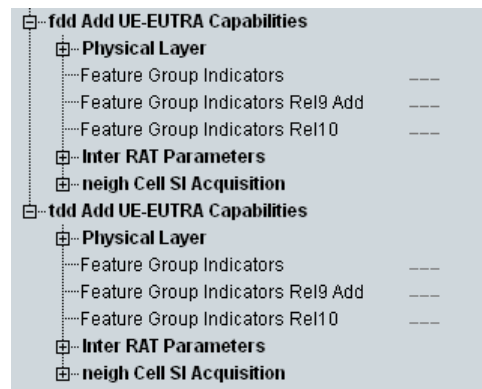


Figure 2-137: Additional FDD/TDD-only capabilities

For the description of the individual parameters, refer to the previous sections. The remote control commands for the additional sections are also referenced there.

2.4.21.20 SL UE capabilities

The following parameters indicate UE capabilities for sidelink communication.

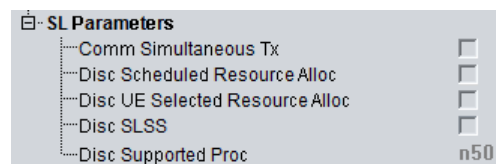


Figure 2-138: Sidelink capabilities

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Comm Simultaneous Tx

Support of simultaneous transmission of EUTRA and sidelink communication on different carriers.

Remote command:

SENSe:LTE:SIGN<i>:UECapability:SL:CSTX?

Disc Scheduled Resource Alloc

Support of transmission of discovery announcements based on network scheduled resource allocation.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:SL:DSRalloc?
```

Disc UE Selected Resource Alloc

Support of transmission of discovery announcements based on UE autonomous resource selection.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:SL:DUSRalloc?
```

Disc SLSS

Support of sidelink synchronization signal (SLSS) transmission and reception for sidelink discovery.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:SL:DSLSSs?
```

Disc Supported Proc

Number of processes supported for sidelink discovery.

Remote command:

```
SENSe:LTE:SIGN<i>:UECapability:SL:DSPProc?
```

2.5 Programming

The following sections provide programming examples for the LTE signaling application.

The LTE signaling application is programmed as follows:

- The application is controlled by SCPI commands with the following syntax: `...:LTE:SIGN:...`
- After a `*RST`, the DL signal is switched off.
To activate the DL signal, use `SOURCE:LTE:SIGN:CELL:STATE ON`.
Query the cell state using `SOURCE:LTE:SIGN:CELL:STATE:ALL?`. The result `ON,ADJ` indicates that the DL signal is available.
- To initiate a connection setup, use `CALL:LTE:SIGN:PSWitched:ACTION CONNECT`.
To query the connection state, use `FETCH:LTE:SIGN:PSWitched:STATE?`.

The examples contain SCPI commands supported by the R&S CMW and the following symbolic scripting commands:

- `// <comment>:`
A `<comment>` ignored by the used programming tool
- `WHILE <query> <> <value>:`
Waits until the `<query>` returns a certain `<value>`, e.g. a specific state is reached.

• WAITKEY <message>:	
Displays a dialog box with a <message> and waits until you close the box.	
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2.5.1 Initialization

```
// *****
// Initial system-reset
// *****
```

```
*RST; *OPC?
*CLS; *OPC?
```

2.5.2 Selecting a scenario

Each scenario is activated via a different `ROUTE` command. Some examples are given in this section.

Each of the following command blocks activates a scenario, selects the related signal paths and configures the related external attenuations. Execute only one command block.

```
// *****
// SISO, no CA, no fading
// *****
ROUTE:LTE:SIGN:SCENario:SCEl1:FLEXible SUA1,RF2C,RX1,RF2C,TX1
CONFIGure:LTE:SIGN:RFSettings:EATTenuation:OUTPut 2
CONFIGure:LTE:SIGN:RFSettings:EATTenuation:INPut 2

// *****
// MIMO nx2, DL CA two carriers, internal fading
// *****
ROUTE:LTE:SIGN:SCENario:CAFF:FLEXible:INTernal
      SUA1,RF1C,RX1,RF1C,TX1,RF2C,TX3,SUA2,RF3C,TX2,RF4C,TX4
CONFIGure:LTE:SIGN:RFSettings:PCC:EATTenuation:OUTPut1 2
CONFIGure:LTE:SIGN:RFSettings:PCC:EATTenuation:OUTPut2 2
CONFIGure:LTE:SIGN:RFSettings:SCC:EATTenuation:OUTPut1 2
CONFIGure:LTE:SIGN:RFSettings:SCC:EATTenuation:OUTPut2 2
CONFIGure:LTE:SIGN:RFSettings:PCC:EATTenuation:INPut 2

// *****
// DL CA three carriers, SCC1 MIMO nx2, PCC/SCC2 SISO, no fading
// *****
ROUTE:LTE:SIGN:SCENario:CCMS1:FLEXible
      SUA1,RF2C,RX3,RF2C,TX3,SUA2,RF1C,TX1,RF3C,TX2,SUA3,RF4C,TX4
CONFIGure:LTE:SIGN:RFSettings:PCC:EATTenuation:OUTPut1 2
CONFIGure:LTE:SIGN:RFSettings:SCC1:EATTenuation:OUTPut1 2
CONFIGure:LTE:SIGN:RFSettings:SCC1:EATTenuation:OUTPut2 2
CONFIGure:LTE:SIGN:RFSettings:SCC2:EATTenuation:OUTPut1 2
CONFIGure:LTE:SIGN:RFSettings:PCC:EATTenuation:INPut 2

// *****
// MIMO nx2, DL CA four carriers, internal fading, UL CA
// *****
ROUTE:LTE:SIGN:SCENario:DHF:INTernal SUA11,R11C,RX11,R11C,TX11,R12C,TX13,
      SUA12,R13C,TX12,R14C,TX14,SUA21,R21C,TX21,R22C,TX23,SUA22,R23C,TX22,R24C,TX24
CONFIGure:LTE:SIGN:RFSettings:PCC:EATTenuation:OUTPut1 2
CONFIGure:LTE:SIGN:RFSettings:PCC:EATTenuation:OUTPut2 2
CONFIGure:LTE:SIGN:RFSettings:SCC1:EATTenuation:OUTPut1 2
CONFIGure:LTE:SIGN:RFSettings:SCC1:EATTenuation:OUTPut2 2
```



```

CONFigure:LTE:SIGN:RFSettings:SCC2:EATTenuation:OUTPut1 2
CONFigure:LTE:SIGN:RFSettings:SCC2:EATTenuation:OUTPut2 2
CONFigure:LTE:SIGN:RFSettings:SCC3:EATTenuation:OUTPut1 2
CONFigure:LTE:SIGN:RFSettings:SCC3:EATTenuation:OUTPut2 2
CONFigure:LTE:SIGN:RFSettings:PCC:EATTenuation:INPut 2
CONFigure:LTE:SIGN:RFSettings:SCC1:EATTenuation:INPut 2

```

2.5.3 Specifying general settings

```

// *****
// Select duplex mode FDD for all carriers and automatic SCC activation mode.
// Activate the uplink for SCC1.
// *****
CONFigure:LTE:SIGN:DMode:UCSPecific OFF
CONFigure:LTE:SIGN:DMode FDD
CONFigure:LTE:SIGN:SCC:AMode AUTO
CONFigure:LTE:SIGN:SCC:UUL ON

// *****
// Define the time delay to be compensated in output and input paths.
// *****
CONFigure:LTE:SIGN:RFSettings:EDC:OUTPut 5E-9
CONFigure:LTE:SIGN:RFSettings:EDC:INPut 5E-9

// *****
// Select manual expected nominal power mode and specify the expected power,
// the user margin and the mixer level offset for the PCC.
// Configure the SCC1.
// *****
CONFigure:LTE:SIGN:RFSettings:PCC:ENPMode MANual
CONFigure:LTE:SIGN:RFSettings:PCC:ENPower -25
CONFigure:LTE:SIGN:RFSettings:PCC:UMARgin 3
CONFigure:LTE:SIGN:RFSettings:PCC:MLOffset 1

CONFigure:LTE:SIGN:RFSettings:SCC:ENPMode MANual
CONFigure:LTE:SIGN:RFSettings:SCC:ENPower -25
CONFigure:LTE:SIGN:RFSettings:SCC:UMARgin 3
CONFigure:LTE:SIGN:RFSettings:SCC:MLOffset 1

```

2.5.4 Configuring operating bands and channels

```

// *****
// Specify PCC operating band plus DL channel number and query the
// automatically calculated UL channel number and the UL frequency in Hz.
// Configure PCC frequency offsets for DL and UL.
// Configure the SCC1.
// *****
CONFigure:LTE:SIGN:PCC:BAND OB5

```

```

CONFigure:LTE:SIGN:RFSettings:PCC:CHANnel:DL 2400 ;UL?
CONFigure:LTE:SIGN:RFSettings:PCC:CHANnel:UL? Hz
CONFigure:LTE:SIGN:RFSettings:PCC:FOFFset:DL 100
CONFigure:LTE:SIGN:RFSettings:PCC:FOFFset:UL -200
CONFigure:LTE:SIGN:SCC:BAND OB5
CONFigure:LTE:SIGN:RFSettings:SCC:CHANnel:DL 2600 ;UL?
CONFigure:LTE:SIGN:RFSettings:SCC:CHANnel:UL? Hz
CONFigure:LTE:SIGN:RFSettings:PCC:FOFFset:DL:UCSPecific ON
CONFigure:LTE:SIGN:RFSettings:SCC:FOFFset:DL 200
CONFigure:LTE:SIGN:RFSettings:PCC:FOFFset:UL:UCSPecific ON
CONFigure:LTE:SIGN:RFSettings:SCC:FOFFset:UL -100

// *****
// Specify additional bands supported by the cell.
// *****
CONFigure:LTE:SIGN:RFSettings:AFBands:ALL ON, OB18, ON, OB19, ON, OB26, OFF,
    OB1, OFF, OB1, OFF, OB1, OFF, OB1, OFF, OB1

// *****
// Configure the PCC user-defined band: UL/DL separation, band indicator,
// minimum DL frequency and channel numbers. Query resulting calculated settings.
// Configure the SCC1 user-defined band.
// *****
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:UDSeparation 200E+6
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:BINdicator 3
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:FREQuency:DL:MINimum 1E+9
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:CHANnel:DL:MINimum 100
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:CHANnel:DL:MAXimum 999
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:CHANnel:UL:MINimum 18000
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:CHANnel:UL:MAXimum?
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:FREQuency:DL:MAXimum?
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:FREQuency:UL:MINimum?
CONFigure:LTE:SIGN:RFSettings:PCC:UDEfined:FREQuency:UL:MAXimum?

CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:UDSeparation 200E+6
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:BINdicator 4
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:FREQuency:DL:MINimum 3E+9
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:CHANnel:DL:MINimum 100
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:CHANnel:DL:MAXimum 999
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:CHANnel:UL:MINimum 18000
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:CHANnel:UL:MAXimum?
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:FREQuency:DL:MAXimum?
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:FREQuency:UL:MINimum?
CONFigure:LTE:SIGN:RFSettings:SCC:UDEfined:FREQuency:UL:MAXimum?

```

2.5.5 Configuring internal fading

The following commands configure the PCC. To configure the SCC number <n>, substitute :PCC: by :SCC<n>: (:SSC1:, :SCC2:, ...).

```
// *****
// Configure fading for the PCC downlink:
// Enable it, select a fading profile, start fading automatically,
// set start seed, calculate insertion loss automatically,
// specify max Doppler frequency.
// *****
Configure:LTE:SIGN:FADing:PCC:FSIMulator:ENABle ON
Configure:LTE:SIGN:FADing:PCC:FSIMulator:PROFile EP5Low
Configure:LTE:SIGN:FADing:PCC:FSIMulator:REStart:MODE AUTO
Configure:LTE:SIGN:FADing:PCC:FSIMulator:GLOBal:SEED 0
Configure:LTE:SIGN:FADing:PCC:FSIMulator:ILOss:MODE NORMAl
Configure:LTE:SIGN:FADing:PCC:FSIMulator:DSHift:MODE USER
Configure:LTE:SIGN:FADing:PCC:FSIMulator:DSHift 6

// *****
// Configure AWGN insertion for the PCC downlink:
// Enable AWGN, no frequency offset, set min noise/system BW ratio,
// set signal to noise ratio and query calculated noise power.
// *****
Configure:LTE:SIGN:FADing:PCC:AWGN:ENABle ON
Configure:LTE:SIGN:FADing:PCC:AWGN:FOFFset 0
Configure:LTE:SIGN:FADing:PCC:AWGN:BWIDth:RATio 1
Configure:LTE:SIGN:FADing:PCC:AWGN:SNRatio 1
Configure:LTE:SIGN:FADing:PCC:POWer:NOISe?
```

2.5.6 Configuring DL power levels

The following commands configure the PCC. To configure the SCC number <n>, substitute :PCC: by :SCC<n>: (:SSC1:, :SCC2:, ...).

```
// *****
// Define the RS EPRE level and the power of the DL channels and signals
// relative to the RS EPRE level. Query full cell BW power.
// *****
Configure:LTE:SIGN:DL:PCC:RSEPre:LEVel -80
Configure:LTE:SIGN:DL:PCC:PSS:POFFset -3
Configure:LTE:SIGN:DL:PCC:SSS:POFFset -3
Configure:LTE:SIGN:DL:PCC:PBCH:POFFset 0
Configure:LTE:SIGN:DL:PCC:PCFich:POFFset 0
Configure:LTE:SIGN:DL:PCC:PHIch:POFFset 0
Configure:LTE:SIGN:DL:PCC:PDCCh:POFFset -3
SENSe:LTE:SIGN:DL:PCC:FCPower?

// *****
// Configure the used CSI-RS power offset manually,
// independent from the signaled power offset.
// *****
Configure:LTE:SIGN:DL:PCC:CSIRs:MODE MANual
Configure:LTE:SIGN:DL:PCC:CSIRs:POFFset -10
```

```
// *****
// Enable the OFDMA channel noise generator.
// *****
CONFIGure:LTE:SIGN:DL:PCC:OCNG ON

// *****
// Define power offset, power ratio index and antenna port power
// for calculation of the PDSCH power level.
// *****
CONFIGure:LTE:SIGN:DL:PCC:PDSch:PA ZERO
CONFIGure:LTE:SIGN:DL:PCC:POWER:PORTs 0
CONFIGure:LTE:SIGN:DL:PCC:PDSch:RINdEx 1

// *****
// Define the AWGN power level (also activates the AWGN interferer).
// *****
CONFIGure:LTE:SIGN:DL:PCC:AWGN -100
```

2.5.7 Configuring UL power control for call setup

```
// *****
// Enable joint UL power control.
// *****
CONFIGure:LTE:SIGN:UL:JUPower ON

// *****
// Define the open loop nominal power directly (basic UL power configuration).
// Query the resulting parameter values signaled to the UE.
// *****
CONFIGure:LTE:SIGN:UL:PCC:PUSCH:OLNPower -30
SENSe:LTE:SIGN:UL:PCC:APPower:RSPower:BASic?
SENSe:LTE:SIGN:UL:PCC:APPower:PNPusch:BASic?
SENSe:LTE:SIGN:UL:PCC:APPower:PCALpha:BASic?
SENSe:LTE:SIGN:UL:PCC:APPower:TPRRcsetup:BASic?

// *****
// Alternatively, enable advanced UL power configuration,
// define the parameter values to be signaled to the UE and
// query values calculated from the parameters.
// *****
CONFIGure:LTE:SIGN:UL:PCC:APPower:EASettings ON
CONFIGure:LTE:SIGN:UL:PCC:APPower:RSPower:ADVanced 15
CONFIGure:LTE:SIGN:UL:PCC:APPower:PIRPower:ADVanced -100
CONFIGure:LTE:SIGN:UL:PCC:APPower:PNPusch:ADVanced -81
CONFIGure:LTE:SIGN:UL:PCC:APPower:PCALpha:ADVanced DOT7
CONFIGure:LTE:SIGN:UL:PCC:APPower:TPRRcsetup:ADVanced OFF
SENSe:LTE:SIGN:UL:PCC:APPower:PATHloss?
```

```

SENSe:LTE:SIGN:UL:PCC:APPower:EPPPower?
SENSe:LTE:SIGN:UL:PCC:APPower:EOPower?

// *****
// Define maximum allowed UE power.
// *****
CONFigure:LTE:SIGN:UL:PCC:PMAX 3

```

2.5.8 Configuring physical cell setup

```

// *****
// Define cell bandwidth and physical cell ID for PCC and SCC1.
// Select cyclic prefix. Allow 64-QAM and 256-QAM in the uplink.
// *****
CONFigure:LTE:SIGN:CELL:BANDwidth:PCC:DL B100
CONFigure:LTE:SIGN:CELL:BANDwidth:SCC:DL B100
CONFigure:LTE:SIGN:CELL:PCC:PCID 10
CONFigure:LTE:SIGN:CELL:SCC:PCID 15
CONFigure:LTE:SIGN:CELL:CPRefix NORM
CONFigure:LTE:SIGN:CELL:ULSupport:QAM64:ENABLE ON

// *****
// Configure normal uplink signal (not SRS).
// Configure PCC SRS settings for later use.
// *****
CONFigure:LTE:SIGN:CELL:PCC:SRS:ENABLE OFF
CONFigure:LTE:SIGN:CELL:PCC:SRS:MCENable ON
CONFigure:LTE:SIGN:CELL:PCC:SRS:BWConfig 7
CONFigure:LTE:SIGN:CELL:PCC:SRS:SFCConfig 3
CONFigure:LTE:SIGN:CELL:PCC:SRS:DCCONfig ON
CONFigure:LTE:SIGN:CELL:PCC:SRS:DBANDwidth 3
CONFigure:LTE:SIGN:CELL:PCC:SRS:HBANDwidth 3
CONFigure:LTE:SIGN:CELL:PCC:SRS:SCINdex:TDD 0
CONFigure:LTE:SIGN:CELL:PCC:SRS:SCINdex:FDD 7
CONFigure:LTE:SIGN:CELL:PCC:SRS:POFFset 4

// *****
// Select UL/DL config and special subframe config for PCC and SCC1.
// *****
CONFigure:LTE:SIGN:CELL:TDD:SPECific ON
CONFigure:LTE:SIGN:CELL:ULDL 1
CONFigure:LTE:SIGN:CELL:SCC1:ULDL 3
CONFigure:LTE:SIGN:CELL:SSUBframe 5
CONFigure:LTE:SIGN:CELL:SCC1:SSUBframe 7

// *****
// PRACH settings: answer preambles, set power step size,
// PRACH configuration index, frequency offset,
// logical root sequence index and zero correlation zone config.

```

```
// Define MCS for RAR messages.
// *****
CONFIGure:LTE:SIGN:CELL:PRACH:NRPreambles OFF
CONFIGure:LTE:SIGN:CELL:PRACH:PRStep P4DB
CONFIGure:LTE:SIGN:CELL:PRACH:PCIndex:FDD 15
CONFIGure:LTE:SIGN:CELL:PRACH:PCIndex:TDD 15
CONFIGure:LTE:SIGN:CELL:PRACH:PFOffset 10
CONFIGure:LTE:SIGN:CELL:PRACH:LRSindex 120
CONFIGure:LTE:SIGN:CELL:PRACH:ZCZConfig 5
CONFIGure:LTE:SIGN:CELL:RAR:CMCS:ENABLE ON
CONFIGure:LTE:SIGN:CELL:RAR:CMCS 13

// *****
// CSAT settings: enable CSAT for SCCL1, LDS every 80 ms,
// ON/OFF 20 ms, periodic MAC activation.
// *****
CONFIGure:LTE:SIGN:CELL:SCCL1:CSAT:ENABLE ON
CONFIGure:LTE:SIGN:CELL:SCCL1:CSAT:DMTCperiod M80
CONFIGure:LTE:SIGN:CELL:SCCL1:SCMuting:ONSDuration 20
CONFIGure:LTE:SIGN:CELL:SCCL1:SCMuting:OFFSduration 20
CONFIGure:LTE:SIGN:CELL:SCCL1:SCMuting:PMAC ON
```

2.5.9 Configuring neighbor cells and reselection

```
// *****
// Specify 2 neighbor cell entries for LTE, GSM, WCDMA, CDMA2000,
// 1xEV-DO and TD-SCDMA.
// *****
CONFIGure:LTE:SIGN:NCELL:LTE:CELL1 ON, OB1, 10, 1, ZERO, ON
CONFIGure:LTE:SIGN:NCELL:LTE:CELL2 ON, OB2, 700, 3, P1, OFF
CONFIGure:LTE:SIGN:NCELL:GSM:CELL1 ON, G09, 0, ON
CONFIGure:LTE:SIGN:NCELL:GSM:CELL2 ON, G09, 124, OFF
CONFIGure:LTE:SIGN:NCELL:WCDMA:CELL1 ON, OB1, 10562, #H10A, ON
CONFIGure:LTE:SIGN:NCELL:WCDMA:CELL2 ON, OB2, 412, #H10B, OFF
CONFIGure:LTE:SIGN:NCELL:CDMA:CELL1 ON, USC, 1, 5, ON
CONFIGure:LTE:SIGN:NCELL:CDMA:CELL2 ON, USC, 799, 6, OFF
CONFIGure:LTE:SIGN:NCELL:EVDO:CELL1 ON, USC, 1, 5, ON
CONFIGure:LTE:SIGN:NCELL:EVDO:CELL2 ON, USC, 799, 6, OFF
CONFIGure:LTE:SIGN:NCELL:TDSCdma:CELL1 ON, OB1, 9500, #H1, ON
CONFIGure:LTE:SIGN:NCELL:TDSCdma:CELL2 ON, OB2, 10100, #H2, ON

// *****
// Specify neighbor cell reselection thresholds per technology.
// *****
CONFIGure:LTE:SIGN:NCELL:LTE:THResholds:LOW 5
CONFIGure:LTE:SIGN:NCELL:GSM:THResholds:LOW 5
CONFIGure:LTE:SIGN:NCELL:WCDMA:THResholds:LOW 5
CONFIGure:LTE:SIGN:NCELL:CDMA:THResholds:LOW 10
CONFIGure:LTE:SIGN:NCELL:EVDO:THResholds:LOW 10
```

```

CONFigure:LTE:SIGN:NCELL:TDSCdma:THResholds:LOW 5

// *****
// Specify general reselection parameters for the cell.
// *****
CONFigure:LTE:SIGN:CELL:RESelection:SEARCh:INTRasearch OFF
CONFigure:LTE:SIGN:CELL:RESelection:SEARCh:NINTRasearch OFF
CONFigure:LTE:SIGN:CELL:RESelection:TSLow 14
CONFigure:LTE:SIGN:CELL:RESelection:QUALity:RXLevmin -130

```

2.5.10 Configuring other network settings

```

// *****
// Specify MCC, 2-digit MNC and tracking area code.
// Configure E-UTRAN cell ID for PCC and SCC1.
// *****
CONFigure:LTE:SIGN:CELL:MCC 262
CONFigure:LTE:SIGN:CELL:MNC:DIGits TWO
CONFigure:LTE:SIGN:CELL:MNC 30
CONFigure:LTE:SIGN:CELL:TAC 1384
CONFigure:LTE:SIGN:CELL:PCC:CID:EUTRan #B10000010001000
CONFigure:LTE:SIGN:CELL:SCC:CID:EUTRan #B10000010001001

// *****
// Enable authentication, NAS security, AS security and milenage.
// Define algorithms, OPc, secret key and RAND mode.
// *****
CONFigure:LTE:SIGN:CELL:SECurity:AUTHenticat ON
CONFigure:LTE:SIGN:CELL:SECurity:NAS ON
CONFigure:LTE:SIGN:CELL:SECurity:AS ON
CONFigure:LTE:SIGN:CELL:SECurity:MILenage ON
CONFigure:LTE:SIGN:CELL:SECurity:IALgorithm S3G
CONFigure:LTE:SIGN:CELL:SECurity:NCALgorithm NULL
CONFigure:LTE:SIGN:CELL:SECurity:OPC #H000010B00091006000F00000A0005000
CONFigure:LTE:SIGN:CELL:SECurity:SKEY #H000102030405060708090A0B0C0D0E0F
CONFigure:LTE:SIGN:CELL:SECurity:RVALue EVEN

// *****
// Configure timers.
// *****
CONFigure:LTE:SIGN:CELL:TOUT:OSYNch 40
CONFigure:LTE:SIGN:CELL:TOUT:T3412 180
CONFigure:LTE:SIGN:CELL:TOUT:TEXT3412 18600

// *****
// Configure NAS signaling settings:
// Disable sending of a DNS IP address to the UE, configure reject causes and
// the ATTACH ACCEPT message, configure IE "EPS Network Feature Support".
// *****

```

```

CONFigure:LTE:SIGN:CONNection:SDNSpco OFF
CONFigure:LTE:SIGN:CELL:RCAuse:ATTach CONG22
CONFigure:LTE:SIGN:CELL:ACAuse:ATTach ON
CONFigure:LTE:SIGN:CELL:EAResult EONLY
CONFigure:LTE:SIGN:CELL:AUResult NAINform
CONFigure:LTE:SIGN:CELL:RCAuse:TAU TANAl2
CONFigure:LTE:SIGN:CELL:NAS:EPSNetwork ON
CONFigure:LTE:SIGN:CELL:NAS:IMSVops NSUP
CONFigure:LTE:SIGN:CELL:NAS:EMCBs SUPP
CONFigure:LTE:SIGN:CELL:NAS:EPCLcs NSUP
CONFigure:LTE:SIGN:CELL:NAS:CSLcs NINF

// *****
// Synchronize the signaling application and the PCC to zone 1.
// Apply an offset of 30 µs to the SCC1.
// *****
CONFigure:LTE:SIGN:CELL:SYNC:ZONE Z1
CONFigure:LTE:SIGN:CELL:PCC:SYNC:OFFSet 0
CONFigure:LTE:SIGN:CELL:SCC:SYNC:OFFSet 30E-6

```

2.5.11 Configuring eMTC settings

```

// *****
// Enable eMTC. Put MPDCCH in UE-specific search space. Allow up to 5 MHz BW.
// *****
CONFigure:LTE:SIGN:EMTC:ENABle ON
CONFigure:LTE:SIGN:EMTC:MPDCch:SSPace UESP
CONFigure:LTE:SIGN:EMTC:MB5 ON

// *****
// Configure repetition settings.
// *****
CONFigure:LTE:SIGN:EMTC:MPDCch:MRPaging MR2
CONFigure:LTE:SIGN:EMTC:MPDCch:MREPetitions MR2
CONFigure:LTE:SIGN:EMTC:MPDCch:RLEVel RL2

CONFigure:LTE:SIGN:EMTC:PDSch:A:MRCE MR16
CONFigure:LTE:SIGN:EMTC:PDSch:A:CERepetition R4
CONFigure:LTE:SIGN:EMTC:PDSch:B:MRCE MR192
CONFigure:LTE:SIGN:EMTC:PDSch:B:CERepetition R4

CONFigure:LTE:SIGN:EMTC:PUSch:A:MRCE MR16
CONFigure:LTE:SIGN:EMTC:PUSch:A:CERepetition R4
CONFigure:LTE:SIGN:EMTC:PUSch:B:MRCE MR192
CONFigure:LTE:SIGN:EMTC:PUSch:B:CERepetition R4

CONFigure:LTE:SIGN:EMTC:PUCCh:A:CERepetition R2
CONFigure:LTE:SIGN:EMTC:PUCCh:B:CERepetition R8

```



```
// *****
// Configure coverage enhancement settings:
// Global settings, CE level 1 and 2, level 3 disabled.
// *****
Configure:LTE:SIGN:EMTC:CE:MODE A

Configure:LTE:SIGN:EMTC:CE:LEVel:ENABLE 3, OFF
Configure:LTE:SIGN:EMTC:CE:LEVel:QRXLevmin 1, -50
Configure:LTE:SIGN:EMTC:CE:LEVel:QRXLevmin 2, -70
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:CINdEx 1, 4
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:CINdEx 2, 6
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:FOFFset 1, 2
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:FOFFset 2, 2
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:MPATtempts 1, A4
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:MPATtempts 2, A8
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:RPATtempt 1, R2
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:RPATtempt 2, R4
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:MMRRepetitio 1, R4
Configure:LTE:SIGN:EMTC:CE:LEVel:PRACH:MMRRepetitio 2, R8

// *****
// Configure narrowband hopping.
// *****
Configure:LTE:SIGN:EMTC:HOPPing:DL:ENABLE ON
Configure:LTE:SIGN:EMTC:HOPPing:DL:HOFFset 2
Configure:LTE:SIGN:EMTC:HOPPing:DL:A:INTerval I2
Configure:LTE:SIGN:EMTC:HOPPing:DL:B:INTerval I4
Configure:LTE:SIGN:EMTC:HOPPing:UL:ENABLE ON
Configure:LTE:SIGN:EMTC:HOPPing:UL:HOFFset 2
Configure:LTE:SIGN:EMTC:HOPPing:UL:A:INTerval I2
Configure:LTE:SIGN:EMTC:HOPPing:UL:B:INTerval I4

// *****
// Configure eMTC auto mode.
// *****
Configure:LTE:SIGN:CONNection:STYPe EMAMode
Configure:LTE:SIGN:CONNection:EMAMode:A:DL:ANB1 ON
Configure:LTE:SIGN:CONNection:EMAMode:A:DL:ALL N4, 1, 2, Q16, 11
Configure:LTE:SIGN:CONNection:EMAMode:A:UL:ALL N4, 1, 2, Q16, 11
Configure:LTE:SIGN:CONNection:EMAMode:B:DL:ALL N4, 0, 2, QPSK, 9
Configure:LTE:SIGN:CONNection:EMAMode:B:UL:ALL N1, 2, 2, QPSK, 9

// *****
// Configure eMTC compact scheduling.
// *****
Configure:LTE:SIGN:CONNection:STYPe EMCS
Configure:LTE:SIGN:CONNection:CSCHeduling:SFPattern HAB10
Configure:LTE:SIGN:CONNection:CSCHeduling:A:DL:ANB1 ON
Configure:LTE:SIGN:CONNection:CSCHeduling:A:DL:ALL 4, 0, Q16, 14
Configure:LTE:SIGN:CONNection:CSCHeduling:A:UL:ALL 6, 0, Q16, 14
```

```
// *****
// Configure eMTC user-defined channels.
// *****
CONFIGure:LTE:SIGN:CONNection:STYPe UDCH
CONFIGure:LTE:SIGN:CONNection:UDCHannels:EMTC:SFPattern HAB10
CONFIGure:LTE:SIGN:CONNection:UDCHannels:EMTC:A:DL:ANB1 ON
CONFIGure:LTE:SIGN:CONNection:UDCHannels:EMTC:A:DL 4, 0, QPSK, 5
CONFIGure:LTE:SIGN:CONNection:UDCHannels:EMTC:NBPosition:DL LOW
CONFIGure:LTE:SIGN:CONNection:UDCHannels:EMTC:A:UL 6, 0, QPSK, 6
CONFIGure:LTE:SIGN:CONNection:UDCHannels:EMTC:NBPosition:UL LOW

// *****
// Configure eMTC RMCs.
// *****
CONFIGure:LTE:SIGN:CONNection:STYPe RMC
CONFIGure:LTE:SIGN:CONNection:RMC:EMTC:NBPosition:UL LOW
CONFIGure:LTE:SIGN:CONNection:RMC:EMTC:NBPosition:DL GPP3
CONFIGure:LTE:SIGN:CONNection:RMC:EMTC:SFPattern P1
CONFIGure:LTE:SIGN:CONNection:PCC:RMC:DL N4,QPSK,KEEP
CONFIGure:LTE:SIGN:CONNection:PCC:RMC:RBPosition:DL LOW
CONFIGure:LTE:SIGN:CONNection:PCC:RMC:UL N6,QPSK,KEEP
CONFIGure:LTE:SIGN:CONNection:PCC:RMC:RBPosition:UL LOW
```

2.5.12 Configuring LAA settings

```
// *****
// To use LAA on SCC2 together with FDD on the PCC:
// Enable carrier-specific duplex mode configuration.
// Configure the duplex modes and the LAA band.
// Query the frame structure for SCC2 - must be T3.
// *****
CONFIGure:LTE:SIGN:DMode:UCSPeCific ON
CONFIGure:LTE:SIGN:PCC:DMode FDD
CONFIGure:LTE:SIGN:SCC2:DMode TDD
CONFIGure:LTE:SIGN:SCC2:BAND OB46
CONFIGure:LTE:SIGN:SCC2:FSTRucture?

// *****
// Configure user-defined channels with fixed bursts for SCC2.
// Prepare contiguous allocation and multi-cluster allocation.
// Use multi-cluster allocation and query resulting code rates.
// *****
CONF:LTE:SIGN:CONN:SCC2:STYPe UDCH
CONF:LTE:SIGN:CONN:SCC2:LAA:TBURsts FBURst
CONF:LTE:SIGN:CONN:SCC2:LAA:FBURst:BLength 5
CONF:LTE:SIGN:CONN:SCC2:LAA:FBURst:PBTR 10
CONF:LTE:SIGN:CONN:SCC2:LAA:FBURst:SPFSubframe OFDM7
CONF:LTE:SIGN:CONN:SCC2:LAA:FBURst:OSLSubframe SYM9
```

```

CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:FSUBframes:DL 48, 8, Q16, 11
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:PIPSubframes:DL 48, 8, QPSK, 4
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:PEPSubframes:DL 48, 8, QPSK, 6
CONF:LTE:SIGN:CONN:SCC2:MCLuster:DL ON
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:FSUBframes:MCLuster:DL
    #B10101000000000000000000011111, Q16, 11
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:PIPSubframes:MCLuster:DL
    #B10101000000000000000000011111, QPSK, 4
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:PEPSubframes:MCLuster:DL
    #B10101000000000000000000011111, QPSK, 6
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:DL:FSUBframes:CRATe?
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:DL:PIPSubframes:CRATe?
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:FBURst:DL:PEPSubframes:CRATe?

// *****
// Configure user-defined channels with random bursts for SCC2.
// Prepare contiguous allocation and multi-cluster allocation.
// Use multi-cluster allocation and query resulting code rates.
// *****
CONF:LTE:SIGN:CONN:SCC2:STYPe UDCH
CONF:LTE:SIGN:CONN:SCC2:LAA:TBURsts RBURst
CONF:LTE:SIGN:CONN:SCC2:LAA:RBURst:BLENGth ON,ON,ON,ON,ON,OFF,OFF,OFF,OFF,OFF
CONF:LTE:SIGN:CONN:SCC2:LAA:RBURst:LSConfig ON,OFF,ON,OFF,ON,OFF
CONF:LTE:SIGN:CONN:SCC2:LAA:RBURst:IPSubframe ON
CONF:LTE:SIGN:CONN:SCC2:LAA:RBURst:TProbability 60
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:FSUBframes:DL 48, 8, Q16, 11
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PIPSubframes:DL 48, 8, QPSK, 4
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:DL S6, 48, 8, Q16, 10
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:DL S9, 48, 8, Q16, 11
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:DL S10, 48, 8, Q16, 12
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:DL S11, 48, 8, Q16, 13
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:DL S12, 48, 8, Q16, 14
CONF:LTE:SIGN:CONN:SCC2:MCLuster:DL ON
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:FSUBframes:MCLuster:DL
    #B10101000000000000000000011111, Q16, 11
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PIPSubframes:MCLuster:DL
    #B10101000000000000000000011111, QPSK, 4
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:MCLuster:DL
    S6, #B10101000000000000000000011111, Q16, 10
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:MCLuster:DL
    S9, #B10101000000000000000000011111, Q16, 11
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:MCLuster:DL
    S10, #B10101000000000000000000011111, Q16, 12
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:MCLuster:DL
    S11, #B10101000000000000000000011111, Q16, 13
CONF:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:PEPSubframes:MCLuster:DL
    S12, #B10101000000000000000000011111, Q16, 14
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:DL:FSUBframes:CRATe?
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:DL:PIPSubframes:CRATe?
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:DL:PEPSubframes:CRATe? S6

```

```

SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:DL:PEPSubframes:CRATe? S9
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:DL:PEPSubframes:CRATe? S10
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:DL:PEPSubframes:CRATe? S11
SENSe:LTE:SIGN:CONN:SCC2:UDCH:LAA:RBURst:DL:PEPSubframes:CRATe? S12

// *****
// Configure RMC with random bursts for SCC2.
// *****
Configure:LTE:SIGN:CONNection:SCC2:STYPe RMC
Configure:LTE:SIGN:CONNection:SCC2:LAA:TBURsts RBURst
Configure:LTE:SIGN:CONNection:SCC2:LAA:RBURst:PSFConfig BOTH
Configure:LTE:SIGN:CONNection:SCC2:RMC:DL N100,Q64,KEEP
Configure:LTE:SIGN:CONNection:SCC2:RMC:RBPosition:DL HIGH

// *****
// Configure follow WB scheduling for SCC2.
// Use the same commands as without LAA, plus the following commands.
// *****
Configure:LTE:SIGN:CONNection:SCC2:LAA:FBURst:BLENgth 5
Configure:LTE:SIGN:CONNection:SCC2:LAA:FBURst:PBTR 10

// *****
// Configure measurement report settings for SCC2.
// *****
Configure:LTE:SIGN:UEReport:SCC2:DMTC:PERiod MS80
Configure:LTE:SIGN:UEReport:SCC2:DMTC:POFFset 40
Configure:LTE:SIGN:UEReport:SCC2:RSSI:ENABLe ON
Configure:LTE:SIGN:UEReport:SCC2:RSSI:RMTC:PERiod MS80
Configure:LTE:SIGN:UEReport:SCC2:RSSI:RMTC:SOFFset 40
Configure:LTE:SIGN:UEReport:SCC2:RSSI:MDURation S14
Configure:LTE:SIGN:UEReport:SCC2:RSSI:COThreshhold 53

```

2.5.13 Configuring general connection settings part 1

```

// *****
// Enable the easy mode and group hopping. Configure the UE category manually.
// Allow UE category 0. Signal support of IMS emergency call and eCall over IMS.
// *****
Configure:LTE:SIGN:CONNection:EASY:BFBW ON
Configure:LTE:SIGN:CONNection:GHOPping ON
Configure:LTE:SIGN:CONNection:UECategory:MANual:ENHanced 5
Configure:LTE:SIGN:CONNection:UECategory:REPorted:ENHanced OFF
Configure:LTE:SIGN:CONNection:UECategory:CZAllowed ON
Configure:LTE:SIGN:CONNection:IEmergency ON
Configure:LTE:SIGN:CONNection:EOISupport ON

// *****
// Allow power saving mode. Configure nB. Select default paging cycle,
// additional spectrum emission requirements and filter coefficient.

```

```
// *****
CONFIGure:LTE:SIGN:CONNection:PSMallowed ON
CONFIGure:LTE:SIGN:CONNection:PCNB NBT
CONFIGure:LTE:SIGN:CONNection:DPCycle P128
CONFIGure:LTE:SIGN:CONNection:ASEMission NS02
CONFIGure:LTE:SIGN:CONNection:SCC:ASEMission:CAGGregation NS03
CONFIGure:LTE:SIGN:CONNection:FCOefficient FC4

// *****
// Set connection type and request test mode.
// Query default bearer RLC mode, allowed IP versions, APN and QCI.
// Configure SIB reconfiguration. Disable keeping the RRC connection with a
// timeout of 20 s.
// *****
CONFIGure:LTE:SIGN:CONNection:CTYPE TEST
CONFIGure:LTE:SIGN:CONNection:TMode ON
CONFIGure:LTE:SIGN:CONNection:RLCMode?
CONFIGure:LTE:SIGN:CONNection:IPVersion?
CONFIGure:LTE:SIGN:CONNection:APN?
CONFIGure:LTE:SIGN:CONNection:QCI?
CONFIGure:LTE:SIGN:CONNection:TDBearer:RLCMode UM
CONFIGure:LTE:SIGN:CONNection:SIBReconfig RRCReconfig
CONFIGure:LTE:SIGN:CONNection:KRRC OFF
CONFIGure:LTE:SIGN:CONNection:RTimer 20

// *****
// Enable usage of external DAU and specify network segment and network ID
// of instrument with external DAU.
// *****
CONFIGure:LTE:SIGN:CONNection:EDAU:ENABLE ON
CONFIGure:LTE:SIGN:CONNection:EDAU:NSEgment A
CONFIGure:LTE:SIGN:CONNection:EDAU:NID 5

// *****
// Activate DL padding and insert 10% transport block errors.
// *****
CONFIGure:LTE:SIGN:CONNection:DLPadding ON
CONFIGure:LTE:SIGN:CONNection:DLEinsertion 10
```

2.5.14 Configuring general connection settings part 2

```
// *****
// Disable UE TX antenna selection, TTI bundling and alternative TBS index,
// use redirection for intra-LTE handover,
// do not accept multiple default bearer requests,
// disable timing advance control, disable skipping UL transmissions.
// *****
CONFIGure:LTE:SIGN:CONNection:UETSelection OFF
CONFIGure:LTE:SIGN:CONNection:TTIBundling OFF
```

```

CONFigure:LTE:SIGN:CONNection:PCC:TIA2 OFF
CONFigure:LTE:SIGN:CONNection:SCC1:TIA2 OFF
CONFigure:LTE:SIGN:CONNection:OBCHange REDirection
CONFigure:LTE:SIGN:CONNection:FCHange REDirection
CONFigure:LTE:SIGN:CONNection:AMDBearer OFF
CONFigure:LTE:SIGN:CONNection:TAControl OFF
CONFigure:LTE:SIGN:CONNection:SUITx OFF

// *****
// Enable PUSCH hopping and query whether PUSCH hopping is active.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:HPUSch:ENABle ON
CONFigure:LTE:SIGN:CONNection:SCC1:HPUSch:ENABle ON
SENSe:LTE:SIGN:CONNection:PCC:HPUSch:ACTive?
SENSe:LTE:SIGN:CONNection:SCC1:HPUSch:ACTive?

// *****
// Enable bidirectional header compression with all profiles.
// Disable but prepare UL-only header compression.
// *****
CONFigure:LTE:SIGN:CONNection:ROHC:ENABle ON
CONFigure:LTE:SIGN:CONNection:ROHC:EFOR VVB
CONFigure:LTE:SIGN:CONNection:ROHC:PROFiles ON, ON, ON, ON
CONFigure:LTE:SIGN:CONNection:ROHC:ULONLY:ENABle OFF
CONFigure:LTE:SIGN:CONNection:ROHC:ULONLY:PROFiles ON

// *****
// Set and query number of PDCCH symbols.
// Set PDCCH C-RNTI aggregation levels. Query all used aggregation levels.
// For PCC and SCC1.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:PDCCh:SYMBol P3
SENSe:LTE:SIGN:CONNection:PCC:PDCCh:PSYMBols?
CONFigure:LTE:SIGN:CONNection:PCC:PDCCh:ALEVel AUTO
SENSe:LTE:SIGN:CONNection:PCC:PDCCh:ALEVel?

CONFigure:LTE:SIGN:CONNection:SCC:PDCCh:SYMBol P3
SENSe:LTE:SIGN:CONNection:SCC:PDCCh:PSYMBols?
CONFigure:LTE:SIGN:CONNection:SCC:PDCCh:ALEVel AUTO
SENSe:LTE:SIGN:CONNection:SCC:PDCCh:ALEVel?

// *****
// Set the PUCCH format for carrier aggregation.
// *****
CONFigure:LTE:SIGN:CONNection:PUCCh:FFCA F3

// *****
// Configure GSM, WCDMA and TD-SCDMA target for MO CSFB.
// Activate the GSM target.
// *****

```

```

CONFigure:LTE:SIGN:CONNection:CSFB:GSM G09, 1000, G18
CONFigure:LTE:SIGN:CONNection:CSFB:WCDma OB1, 10600
CONFigure:LTE:SIGN:CONNection:CSFB:TDSCdma OB1, 9500
CONFigure:LTE:SIGN:CONNection:CSFB:DESTination GSM

// *****
// Select user-defined channels as scheduling type for PCC and SCC1.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:STYPe UDCH
CONFigure:LTE:SIGN:CONNection:SCC:STYPe UDCH

```

2.5.15 Configuring HARQ

```

// *****
// Configure HARQ for the uplink: enable HARQ, allow 4 transmissions,
// use DCI 0 and PHICH.
// *****
CONFigure:LTE:SIGN:CONNection:HARQ:UL:ENABle ON
CONFigure:LTE:SIGN:CONNection:HARQ:UL:NHT 4
CONFigure:LTE:SIGN:CONNection:HARQ:UL:DPHich D0PHich

// *****
// Configure HARQ for the downlink: enable HARQ, allow 4 transmissions and
// specify user-defined redundancy version sequence.
// *****
CONFigure:LTE:SIGN:CONNection:HARQ:DL:ENABle ON
CONFigure:LTE:SIGN:CONNection:HARQ:DL:NHT 4
CONFigure:LTE:SIGN:CONNection:HARQ:DL:RVCSequence UDEF
CONFigure:LTE:SIGN:CONNection:HARQ:DL:UDSequence:LENGth 4
CONFigure:LTE:SIGN:CONNection:HARQ:DL:UDSequence 0,0,2,3

```

2.5.16 Configuring connected DRX

```

// *****
// Enable connected DRX with user-defined settings.
// *****
CONFigure:LTE:SIGN:CONNection:CDRX:ENABle UDEF
CONFigure:LTE:SIGN:CONNection:CDRX:ODTimer PSF3
CONFigure:LTE:SIGN:CONNection:CDRX:ITIMer PSF80
CONFigure:LTE:SIGN:CONNection:CDRX:RTIMer PSF8
CONFigure:LTE:SIGN:CONNection:CDRX:LDCYcle SF80
CONFigure:LTE:SIGN:CONNection:CDRX:SOFfset 10
CONFigure:LTE:SIGN:CONNection:CDRX:SCENable ON
CONFigure:LTE:SIGN:CONNection:CDRX:SDCYcle SF5
CONFigure:LTE:SIGN:CONNection:CDRX:SCTimer 5

// *****
// Configure UL dynamic scheduling.

```

```
// *****
CONFIGure:LTE:SIGN:CONNection:UDScheduling ON
CONFIGure:LTE:SIGN:CONNection:IUGMcsidx 10
CONFIGure:LTE:SIGN:CONNection:IUGNrnb 20

// *****
// Configure PUCCH resources for scheduling requests.
// *****
CONFIGure:LTE:SIGN:CONNection:SRPRindex 50
CONFIGure:LTE:SIGN:CONNection:SRCindex 35
```

2.5.17 Configuring extended DRX

```
// *****
// Enable extended DRX and configure the eDRX settings for the idle mode.
// *****
CONFIGure:LTE:SIGN:CONNection:CDRX:IMODE:ENABle ON
CONFIGure:LTE:SIGN:CONNection:CDRX:IMODE:PTWindow W2560
CONFIGure:LTE:SIGN:CONNection:CDRX:IMODE:CLENgth L8192
```

2.5.18 Configuring capability requests

```
// *****
// Configure field "ue-CapabilityRequest" of "ueCapabilityEnquiry" message:
// Add "utra", "geran-cs" and "geran-ps".
// *****
CONFIGure:LTE:SIGN:UECapability:RUTRa ON
CONFIGure:LTE:SIGN:UECapability:RGCS ON
CONFIGure:LTE:SIGN:UECapability:RGPS ON

// *****
// Configure list of requested frequency bands for UE capability report.
// *****
CONFIGure:LTE:SIGN:UECapability:RFBands:ALL ON,OB5,ON,OB6,OFF,OB1,OFF,OB1,
OFF,OB1,ON,OB13,ON,OB14,OFF,OB1,OFF,OB1,OFF,OB1,OFF,OB1,OFF,OB1,OFF,
OB1,OFF,OB1,OFF,OB1,OFF,OB1

// *****
// Add "requestReducedFormat-r13" and "requestSkipFallbackComb-r13"
// to the "ueCapabilityEnquiry" message.
// *****
CONFIGure:LTE:SIGN:UECapability:RRFormat ON
CONFIGure:LTE:SIGN:UECapability:SFC ON
```


2.5.19 Configuring MIMO settings

The following commands configure the PCC. To configure the SCC number <n>, substitute :PCC: by :SCC<n>:.

```
// *****
// Select mode 4 (closed loop spatial multiplexing), DCI format 2 and
// two transmit antennas (MIMO 2x2). Query resulting transmission scheme.
// Select precoding matrix. Select number of layers.
// *****
CONFIGure:LTE:SIGN:CONNection:PCC:TRANsmission TM4
CONFIGure:LTE:SIGN:CONNection:PCC:DCIFormat D2
CONFIGure:LTE:SIGN:CONNection:PCC:NENBantennas TWO
SENSe:LTE:SIGN:CONNection:PCC:TSCHeme?
CONFIGure:LTE:SIGN:CONNection:PCC:PMATrix PMI1
CONFIGure:LTE:SIGN:CONNection:PCC:NOLayers L2

// *****
// Enable and specify channel matrix for MIMO 2x2.
// Specify channel matrix for MIMO 4x2.
// Enable and specify channel matrix for MIMO 4x4.
// *****
CONF:LTE:SIGN:CONN:PCC:SCHModel:ENABle ON
CONF:LTE:SIGN:CONN:PCC:SCHModel 0.9,0,45,0.1,45,0
CONF:LTE:SIGN:CONN:PCC:SCHModel:MIMO42 0.1,0,0.2,0,0.3,0,0.4,270,0.4,270,0.3,90,
    0.2,90,0.1,180
CONF:LTE:SIGN:CONN:PCC:SCHModel:ENABle:MIMO44 ON
CONF:LTE:SIGN:CONN:PCC:SCHModel:MIMO44 0.4,0,0.3,0,0.2,0,0.1,270,0.1,270,0.2,90,
    0.3,90,0.4,180,0.4,45,0.3,45,0.2,270,0.1,270,0.1,45,0.2,45,0.3,90,0.4,90

// *****
// Configure port 0 mapping for TM 7.
// *****
CONFIGure:LTE:SIGN:CONNection:PCC:PZERo:MAPPing R1R2

// *****
// Enable beamforming, configure dual-layer beamforming for TM 8. Configure
// the beamforming matrix and the channel matrix for TM 8.
// *****
CONFIGure:LTE:SIGN:CONNection:PCC:BEAMforming:MODE ON
CONFIGure:LTE:SIGN:CONNection:PCC:BEAMforming:NOLayers L2
CONFIGure:LTE:SIGN:CONNection:PCC:BEAMforming:MATRIx 0,0,0.4,0.6,270,15
CONFIGure:LTE:SIGN:CONNection:PCC:TM8:CHMatrix 0.9,45,0.1,45,0.9,0,0.1,0

// *****
// Configure parameters for TM 9:
// Select number of antennas, number of code words, second precoding matrix.
// Set signaled CSI-RS power, antenna ports, configuration and subframe
// configuration. Set zero power CSI-RS bitmap and subframe configuration.
// *****
```

```

CONFigure:LTE:SIGN:CONNection:PCC:TM9:NTXantennas FOUR
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CODewords TWO
CONFigure:LTE:SIGN:CONNection:PCC:TM9:PMATrix PMI3

CONFigure:LTE:SIGN:CONNection:PCC:TM9:CSIRs:POWer 0
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CSIRs:APORTs P1516
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CSIRs:RESource 11
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CSIRs:SUBFrame 8
CONFigure:LTE:SIGN:CONNection:PCC:TM9:ZP:BITS #B1010000000000000
CONFigure:LTE:SIGN:CONNection:PCC:TM9:ZP:CSIRs:SUBFrame 8

// *****
// Configure channel matrix for TM 9: 2x2, 4x2, 8x2 and 4x4.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:TWO1 0.9,0,45
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:TWO2 0.1,45,0

CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:FOUR1 0.1,0,0.2,0,0.3,0,270
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:FOUR2 0.4,270,0.3,90,0.2,90,180

CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:EIGHT1 0.05,15,0.05,30,0.1,15,0.1,
30,0.15,15,0.15,30,0.2,15,30
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:EIGHT2 0.2,30,0.2,15,0.15,30,0.15,
15,0.1,30,0.1,15,0.05,30,15

CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:MIMO44:LINE1 0.1,0,0.2,0,0.3,0,270
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:MIMO44:LINE2 0.2,0,0.3,0,0.4,270,0
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:MIMO44:LINE3 0.3,0,0.4,270,0.1,0,0
CONFigure:LTE:SIGN:CONNection:PCC:TM9:CMATrix:MIMO44:LINE4 0.4,270,0.1,0,0.2,0,0

```

2.5.20 Configuring RMCs

The following commands configure the PCC. To configure the SCC number <n>, substitute :PCC: by :SCC<n>:.

```

// *****
// Define 3GPP-compliant RMCs with contiguous allocation:
// Select scheduling type, configure DL RMC with 50 RBs and 16-QAM modulation,
// UL RMC with 12 RBs and QPSK modulation. The transport block size index
// is selected automatically.
// Configure the upper end of the cell bandwidth as RB position (not really
// relevant for DL - the 50 RBs use the entire bandwidth).
// Select version 1 of ambiguous RMCs.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:STYPe RMC
CONFigure:LTE:SIGN:CONNection:PCC:MCLuster:UL OFF
CONFigure:LTE:SIGN:CONNection:PCC:RMC:DL N50,Q16,KEEP
CONFigure:LTE:SIGN:CONNection:PCC:RMC:RBPosition:DL HIGH
CONFigure:LTE:SIGN:CONNection:PCC:RMC:UL N12,QPSK,KEEP

```

```

CONFigure:LTE:SIGN:CONNection:PCC:RMC:RBPosition:UL HIGH
CONFigure:LTE:SIGN:CONNection:PCC:RMC:VERSion:DL 1

// *****
// Define the same RMC for the second MIMO downlink stream.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:RMC:DL2 N50,Q16,KEEP
CONFigure:LTE:SIGN:CONNection:PCC:RMC:RBPosition:DL2 HIGH
CONFigure:LTE:SIGN:CONNection:PCC:RMC:VERSion:DL2 1

// *****
// Instead of configuring the same settings for stream 1 and stream 2, you can
// apply the stream 1 settings to all streams and skip the "DL2" commands.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:DLEQual ON

// *****
// Configure an UL RMC with multi-cluster allocation instead of
// contiguous allocation.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:MCLuster:UL ON
CONFigure:LTE:SIGN:CONNection:PCC:RMC:MCLuster:UL N3,P0,N42,P6,Q16,T11

```

2.5.21 Configuring user-defined channels

```

// *****
// Specify user-defined channels: Select scheduling type,
// configure DL PCC channel with 15 RBs starting with RB number 12,
// 64-QAM modulation, TBS index 26. UL channel with 12 RBs starting with
// RB number 21, QPSK modulation, TBS index 19.
// Query the resulting maximum expected throughput and the code rate.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:SType UDChannels
CONFigure:LTE:SIGN:CONNection:PCC:UDChannels:DL1 15,12,Q64,26
CONFigure:LTE:SIGN:CONNection:PCC:UDChannels:UL 12,21,QPSK,9
SENSe:LTE:SIGN:CONNection:ETHroughput:UL:PCC?
SENSe:LTE:SIGN:CONNection:ETHroughput:DL:PCC:STream1?
SENSe:LTE:SIGN:CONNection:PCC:UDChannels:UL:CRATe:ALL?
SENSe:LTE:SIGN:CONNection:PCC:UDChannels:DL1:CRATe:ALL?

// *****
// Define the same user-defined channel for the second MIMO downlink stream.
// Query the resulting maximum expected throughput for that stream and for
// both PCC downlink streams together. Query the code rate.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:UDChannels:DL2 15,12,Q64,26
SENSe:LTE:SIGN:CONNection:ETHroughput:DL:PCC:STream2?
SENSe:LTE:SIGN:CONNection:ETHroughput:DL:PCC?
SENSe:LTE:SIGN:CONNection:PCC:UDChannels:DL2:CRATe:ALL?

```

```
// *****
// Configure the same user-defined channel for the SCC1.
// Query the code rates for the streams.
// Query the maximum expected throughput for each SCC1 stream, for the sum of
// the SCC1 DL streams and for the sum of all PCC and SCC DL streams.
// *****
CONFIGure:LTE:SIGN:CONNection:SCC:STYPe UDCHannels
CONFIGure:LTE:SIGN:CONNection:SCC:UDCHannels:DL1 15,12,Q64,26
CONFIGure:LTE:SIGN:CONNection:SCC:UDCHannels:DL2 15,12,Q64,26
CONFIGure:LTE:SIGN:CONNection:SCC:UDCHannels:UL 12,21,QPSK,9
SENSe:LTE:SIGN:CONNection:SCC:UDCHannels:DL1:CRATe:ALL?
SENSe:LTE:SIGN:CONNection:SCC:UDCHannels:DL2:CRATe:ALL?
SENSe:LTE:SIGN:CONNection:SCC:UDCHannels:UL:CRATe:ALL?
SENSe:LTE:SIGN:CONNection:ETHROUGHput:UL:SCC?
SENSe:LTE:SIGN:CONNection:ETHROUGHput:DL:SCC:STReam1?
SENSe:LTE:SIGN:CONNection:ETHROUGHput:DL:SCC:STReam2?
SENSe:LTE:SIGN:CONNection:ETHROUGHput:DL:SCC?
SENSe:LTE:SIGN:CONNection:ETHROUGHput:DL:ALL?

// *****
// Instead of configuring the same settings for stream 1 and stream 2, you can
// apply the stream 1 settings to all streams and skip the "DL2" commands.
// *****
CONFIGure:LTE:SIGN:CONNection:PCC:DLEQual ON
CONFIGure:LTE:SIGN:CONNection:SCC:DLEQual ON

// *****
// Configure a PCC user-defined channel with multi-cluster allocation instead of
// contiguous allocation.
// *****
CONF:LTE:SIGN:CONNection:PCC:MCLuster:UL ON
CONF:LTE:SIGN:CONNection:PCC:UDCHannels:MCLuster:UL 6,21,9,30,QPSK,9
CONF:LTE:SIGN:CONNection:PCC:MCLuster:DL ON
CONF:LTE:SIGN:CONNection:PCC:UDCHannels:MCLuster:DL #B11100011100011111,Q64,26
```

2.5.22 Configuring TTI-based user-defined channels

The following commands configure the PCC. To configure the SCC number <n>, substitute :PCC: by :SCC<n>:.

```
// *****
// TTI-based user-defined channels: Select scheduling type,
// configure subframe number 3 and 4, UL and both PCC DL streams.
// *****
CONFIGure:LTE:SIGN:CONNection:PCC:STYPe UDTTibased
CONFIGure:LTE:SIGN:CONNection:PCC:UDTTibased:UL 3,12,21,QPSK,9
CONFIGure:LTE:SIGN:CONNection:PCC:UDTTibased:UL 4,12,21,QPSK,9
CONFIGure:LTE:SIGN:CONNection:PCC:UDTTibased:DL1 3,15,12,Q64,26
```

```

CONFigure:LTE:SIGN:CONNection:PCC:UDTTibased:DL1 4,15,12,Q64,26
CONFigure:LTE:SIGN:CONNection:PCC:UDTTibased:DL2 3,15,12,Q64,26
CONFigure:LTE:SIGN:CONNection:PCC:UDTTibased:DL2 4,15,12,Q64,26

// *****
// TTI-based user-defined channels: configure all subframes for PCC DL stream 1
// decreasing number of RBs (50 to 3 for subframe 0 to 9),
// position approximately in middle of bandwidth (start RB = 0 to 24),
// all subframes use 64-QAM and TBS index 26.
// Configure the same settings for DL stream 2.
// Configure also the UL.
// Query the resulting code rates.
// *****
CONF:LTE:SIGN:CONN:PCC:UDTT:DL1:ALL 50,48,45,39,33,27,21,15,9,3,0,0,3,6,9,12,15,
18,21,24,Q64,Q64,Q64,Q64,Q64,Q64,Q64,Q64,Q64,Q64,26,26,26,26,26,26,26,26,26
CONF:LTE:SIGN:CONN:PCC:UDTT:DL2:ALL 50,48,45,39,33,27,21,15,9,3,0,0,3,6,9,12,15,
18,21,24,Q64,Q64,Q64,Q64,Q64,Q64,Q64,Q64,Q64,Q64,26,26,26,26,26,26,26,26,26
CONF:LTE:SIGN:CONN:PCC:UDTT:UL:ALL 50,45,40,36,30,24,18,12,8,3,0,2,5,7,10,13,16,
19,21,24,Q16,Q16,Q16,Q16,Q16,Q16,Q16,Q16,Q16,Q16,19,19,19,19,19,19,19,19,19
SENSe:LTE:SIGN:CONNection:PCC:UDTTibased:DL1:CRATe:ALL?
SENSe:LTE:SIGN:CONNection:PCC:UDTTibased:DL2:CRATe:ALL?
SENSe:LTE:SIGN:CONNection:PCC:UDTTibased:UL:CRATe:ALL?

// *****
// Instead of configuring the same settings for stream 1 and stream 2, you can
// apply the stream 1 settings to all streams and skip the "DL2" commands.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:DLEQual ON

```

2.5.23 Configuring CQI DL channels

The following commands configure the PCC. To configure the SCC number <n>, substitute :PCC: by :SCC<n>:.

```

// *****
// Use tables with 256-QAM.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:QAM256:DL ON

// *****
// Fixed CQI downlink channels: Select scheduling type.
// Configure subframe number 3 and 4, both PCC DL streams.
// For uplink, see TTI-based user-defined channels.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:STYPe CQI, TTIBased
CONFigure:LTE:SIGN:CONNection:PCC:FCTTibased:DL1 3,15,12,10
CONFigure:LTE:SIGN:CONNection:PCC:FCTTibased:DL1 4,15,12,10
CONFigure:LTE:SIGN:CONNection:PCC:FCTTibased:DL2 3,15,12,10
CONFigure:LTE:SIGN:CONNection:PCC:FCTTibased:DL2 4,15,12,10

```

```

// *****
// Fixed CQI downlink channels - configure all subframes for PCC DL stream 1:
// decreasing number of RBs (50 to 3 for subframe 0 to 9),
// position approximately in middle of bandwidth (start RB = 0 to 24),
// CQI index from 15 to 6.
// Configure the same settings for DL stream 2.
// *****
CONF:LTE:SIGN:CONN:PCC:FCTT:DL1:ALL 50,48,45,39,33,27,21,15,9,3,0,0,3,6,9,12,15,
    18,21,24,15,14,13,12,11,10,9,8,7,6
CONF:LTE:SIGN:CONN:PCC:FCTT:DL2:ALL 50,48,45,39,33,27,21,15,9,3,0,0,3,6,9,12,15,
    18,21,24,15,14,13,12,11,10,9,8,7,6

// *****
// Instead of configuring the same settings for stream 1 and stream 2, you can
// apply the stream 1 settings to all streams and skip the "DL2" commands.
// *****
CONF:LTE:SIGN:CONN:PCC:DLEQual ON

// *****
// Follow WB CQI PCC downlink channels: Select scheduling type.
// Configure the global settings, including user-defined mapping tables.
// Contiguous RB allocation and alternatively multi-cluster DL allocation.
// *****
CONF:LTE:SIGN:CONN:PCC:SType CQI,FWB
CONF:LTE:SIGN:CONN:PCC:FWBCqi:DL:STTI ON,ON,ON,ON,ON,OFF,ON,ON,ON,ON
CONF:LTE:SIGN:CONN:PCC:FWBCqi:DL 50,0,UDEF //or multi-cluster:
CONF:LTE:SIGN:CONN:PCC:FWBCqi:MCLuster:DL #B1010110101101011,UDEF
CONF:LTE:SIGN:CONN:PCC:FWBC:DL:MCST:UDEF 0,1,2,3,6,7,9,13,14,16,19,22,24,27,27
CONF:LTE:SIGN:CONN:PCC:FWBCqi:DL:MCSTable:CSIRs:UDEFined
    0,1,3,5,7,10,12,14,17,19,21,22,24,25,25
CONF:LTE:SIGN:CONN:PCC:FWBCqi:DL:MCSTable:SSUBframe:UDEFined
    0,1,3,5,7,10,12,14,17,19,21,22,24,25,25

// *****
// Follow WB PMI PCC downlink channels: Select scheduling type.
// Configure the scheduling.
// Contiguous RB allocation and alternatively multi-cluster DL allocation.
// *****
CONF:LTE:SIGN:CONN:PCC:SType CQI,FPMI ON,ON,ON,ON,ON,ON,ON,ON,ON,ON
CONF:LTE:SIGN:CONN:PCC:FPMI:DL:STTI
CONF:LTE:SIGN:CONN:PCC:FPMI:DL 50,0,Q16,11 //or multi-cluster:
CONF:LTE:SIGN:CONN:PCC:FPMI:MCLuster:DL #B1010110101101011,Q16,11

// *****
// Follow WB PMI-RI PCC downlink channels: Select scheduling type.
// Configure the RB allocation.
// Contiguous RB allocation and alternatively multi-cluster DL allocation.
// *****
CONF:LTE:SIGN:CONN:PCC:SType CQI,FPRI

```

```

CONFigure:LTE:SIGN:CONNection:PCC:FPRI:DL:STTI ON,ON,ON,ON,ON,ON,ON,ON,ON,ON,ON
CONFigure:LTE:SIGN:CONNection:PCC:FPRI:DL 50,0,Q16,11 //or multi-cluster:
CONFigure:LTE:SIGN:CONNection:PCC:FPRI:MCLuster:DL #B10101101011010111,Q16,11

// *****
// Follow WB CQI-RI PCC downlink channels: Select scheduling type.
// Configure the global settings, including user-defined mapping tables.
// Contiguous RB allocation and alternatively multi-cluster DL allocation.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:STYPe CQI,FCRI
CONFigure:LTE:SIGN:CONNection:PCC:FCRI:DL:STTI ON,ON,ON,ON,ON,ON,OFF,ON,ON,ON,ON
CONFigure:LTE:SIGN:CONNection:PCC:FCRI:DL 50,0,UDEF //or multi-cluster:
CONFigure:LTE:SIGN:CONNection:PCC:FCRI:MCLuster:DL #B10101101011010111,UDEF
CONF:LTE:SIGN:CONN:PCC:FCRI:DL:MCST:UDEF 0,1,2,3,6,7,9,13,14,16,19,22,24,27,28
CONF:LTE:SIGN:CONN:PCC:FCRI:DL:MCSTable:SSUBframe:UDEFined
    0,1,2,3,6,7,9,13,14,16,19,22,24,27,28

// *****
// Follow WB CQI-PMI-RI PCC downlink channels: Select scheduling type.
// Configure the global settings, including user-defined mapping tables.
// Contiguous RB allocation and alternatively multi-cluster DL allocation.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:STYPe CQI,FCPri
CONFigure:LTE:SIGN:CONNection:PCC:FCPri:DL:STTI ON,ON,ON,ON,ON,ON,OFF,ON,ON,ON,ON
CONFigure:LTE:SIGN:CONNection:PCC:FCPri:DL 50,0,UDEF //or multi-cluster:
CONFigure:LTE:SIGN:CONNection:PCC:FCPri:MCLuster:DL #B10101101011010111,UDEF
CONF:LTE:SIGN:CONN:PCC:FCPri:DL:MCST:UDEF 0,1,2,3,6,7,9,13,14,16,19,22,24,27,28
CONF:LTE:SIGN:CONN:PCC:FCPri:DL:MCST:CSIRs:UDEFined
    0,1,3,5,7,10,12,14,17,19,21,22,24,25,25
CONF:LTE:SIGN:CONN:PCC:FCPri:DL:MCST:SSUBframe:UDEFined
    0,1,2,3,6,7,9,13,14,16,19,22,24,27,28

```

2.5.24 Configuring SPS

```

// *****
// Select the scheduling type and configure twoIntervalsConfig for TDD.
// Configure the periodicity and RB allocation.
// Query the resulting code rates and expected throughput.
// *****
CONFigure:LTE:SIGN:CONNection:PCC:STYPe SPS
CONFigure:LTE:SIGN:CONNection:PCC:SPS:TIConfig ON
CONFigure:LTE:SIGN:CONNection:PCC:SPS:SINterval:DL S20
CONFigure:LTE:SIGN:CONNection:PCC:SPS:SINterval:UL SADL
CONFigure:LTE:SIGN:CONNection:PCC:SPS:DL 10,20,QPSK,9
CONFigure:LTE:SIGN:CONNection:PCC:SPS:UL 10,20,QPSK,9
SENSe:LTE:SIGN:CONNection:PCC:SPS:DL:CRATe:ALL?
SENSe:LTE:SIGN:CONNection:PCC:SPS:UL:CRATe:ALL?

```

```
SENSe:LTE:SIGN:CONNection:ETHroughput:DL:ALL?
SENSe:LTE:SIGN:CONNection:ETHroughput:UL?
```

2.5.25 Configuring CQI reporting

```
// *****
// Enable periodic CQI reporting, set the CSI reporting mode,
// allow simultaneous ACK/NACK and CQI transmission,
// enable PMI/RI reporting and set the cqi-pmi-ConfigIndex for PCC
// and SCC1. Query the reporting period and reporting offset
// resulting for the active duplex mode.
// *****
CONFIgure:LTE:SIGN:CQIReporting:ENABle PER
CONFIgure:LTE:SIGN:CQIReporting:CSIRmode S2
CONFIgure:LTE:SIGN:CQIReporting:SANCqi ON
CONFIgure:LTE:SIGN:CQIReporting:PRIReporting:ENABle ON
CONFIgure:LTE:SIGN:CQIReporting:PCC:CINDeX:FDD 17
CONFIgure:LTE:SIGN:CQIReporting:SCC:CINDeX:FDD 13
CONFIgure:LTE:SIGN:CQIReporting:PCC:CINDeX:TDD 17
CONFIgure:LTE:SIGN:CQIReporting:SCC:CINDeX:TDD 13
CONFIgure:LTE:SIGN:CQIReporting:SCC:CINDeX:LAA 13
SENSe:LTE:SIGN:CQIReporting:PCC:RPERiod?
SENSe:LTE:SIGN:CQIReporting:PCC:ROFFset?
SENSe:LTE:SIGN:CQIReporting:SCC:RPERiod?
SENSe:LTE:SIGN:CQIReporting:SCC:ROFFset?
```

2.5.26 Configuring measurement reports

```
// *****
// Set reporting interval, enable measurement gaps, set gap period,
// filter coefficient RSRP, filter coefficient RSRQ, WCDMA measurement quantity
// and SCell measurement cycle. Do not allow interruptions.
// Enable measurement reports.
// *****
CONFIgure:LTE:SIGN:UEReport:RINteRval I640
CONFIgure:LTE:SIGN:UEReport:MGENABle ON
CONFIgure:LTE:SIGN:UEReport:MGPeRiod G080
CONFIgure:LTE:SIGN:UEReport:FCOefficient:RSRP FC4
CONFIgure:LTE:SIGN:UEReport:FCOefficient:RSRQ FC4
CONFIgure:LTE:SIGN:UEReport:WMQuantity ECNO
CONFIgure:LTE:SIGN:UEReport:MCSCell SF640
CONFIgure:LTE:SIGN:UEReport:AINteRrupt OFF
CONFIgure:LTE:SIGN:UEReport:ENABle ON
```


2.5.27 Configuring message monitoring

```
// *****
// Enable message monitoring for LTE, select address number 2 from the global
// logging PC address pool and query the corresponding IP address string.
// *****
CONFIGure:LTE:SIGN:MMONitor:ENABLE ON
CONFIGure:LTE:SIGN:MMONitor:IPAdress IP2
CONFIGure:LTE:SIGN:MMONitor:IPAdress?
```

2.5.28 Configuring the cell broadcast service

```
// *****
// Set the message type. Query the resulting message ID and configure the
// serial number.
// *****
CONFIGure:LTE:SIGN:CBS:MESSAge:IDType AAMB
CONFIGure:LTE:SIGN:CBS:MESSAge:ID?
CONFIGure:LTE:SIGN:CBS:MESSAge:SERial PLMN, 1, OFF, 0

// *****
// Select internal data source, enter the message text, query the coding group and
// set the language.
// *****
CONFIGure:LTE:SIGN:CBS:MESSAge:SOURce INTERNAL
CONFIGure:LTE:SIGN:CBS:MESSAge:DATA "Warning! This is a test!"
CONFIGure:LTE:SIGN:CBS:MESSAge:CGRoup?
CONFIGure:LTE:SIGN:CBS:MESSAge:LANGUage 1,"English"

// *****
// Select user-coded data source, define coding scheme, enter message contents.
// *****
CONFIGure:LTE:SIGN:CBS:MESSAge:SOURce UCODed
CONFIGure:LTE:SIGN:CBS:MESSAge:DCSCHEME:UCODed #H24
CONFIGure:LTE:SIGN:CBS:MESSAge:UCODed #H30280000234000000040

// *****
// Enable transfer of the warning area coordinates and define them.
// *****
CONFIGure:LTE:SIGN:CBS:MESSAge:WAENable ON
CONFIGure:LTE:SIGN:CBS:MESSAge:WACoordinate #H3028000000

// *****
// Configure ETWS-specific settings.
// *****
CONFIGure:LTE:SIGN:CBS:MESSAge:ETWS:ALERT ON
CONFIGure:LTE:SIGN:CBS:MESSAge:ETWS:POPup ON

// *****
```

```
// Enable cell broadcast messages.
// *****
CONFIGure:LTE:SIGN:CB:MESSagE:ENABle ON
```

2.5.29 Attaching the UE and activating SCCs

```
// *****
// Connect the UE (switched off). Switch on the DL signal. Query the cell
// state until it equals ON,ADJ (DL signal available at RF connector).
// *****
WAITKEY >Ensure that the UE is connected to the instrument and switched off<
SOURCE:LTE:SIGN:CELL:STATe ON
WHILE SOURCE:LTE:SIGN:CELL:STATe:ALL? <> "ON,ADJ"

// *****
// Switch on the UE and wait until it is attached (connection state = ATT).
// *****
WAITKEY >Switch on the UE<
WHILE FETCh:LTE:SIGN:PSWitched:STATe? <> "ATT"

// *****
// Query the RRC connection state.
// *****
SENSE:LTE:SIGN:RRCState?

// *****
// Query the SCC1 state. The SCC connection is set up automatically after attach
// if: a carrier aggregation scenario is active, the SCC activation mode is
// "AUTO" and keep RRC connection is enabled.
// *****
FETCh:LTE:SIGN:SCC1:STATe?
```

2.5.30 Setting up a test mode connection

This section is only relevant for test mode connections, not for data application tests.

```
// *****
// Set up a mobile-terminated test mode connection.
// Query the connection state until it equals CEST (connection established).
// Query the IPv4 addresses assigned to the UE.
// *****
CALL:LTE:SIGN:PSWitched:ACTion CONNect
WHILE FETCh:LTE:SIGN:PSWitched:STATe? <> "CEST"
SENSE:LTE:SIGN:UESinfo:UEAddress:IPV4?
```

After the connection has been established, you can e.g.:

- Perform an LTE multi-evaluation measurement (option R&S CMW-KM500/-KM550)
- Perform a BLER measurement, see [Section 3.3, "Programming"](#), on page 939

- Modify parameters, see [Section 2.5.40, "Modifying parameters for an established connection"](#), on page 360

2.5.31 Connecting/releasing dedicated bearers

This section is only relevant for data application tests, not for test mode connections.

```
// *****
// Query a list of established default bearers.
// Configure a dedicated bearer.
// Establish the dedicated bearer.
// *****
CATalog:LTE:SIGN:CONNection:DEFBearer?
PREPare:LTE:SIGN:CONN:DEDBearer "5 (cmw500.rohde-schwarz.com)", DRAM, 10, 20
CALL:LTE:SIGN:PSWitched:ACTion CONNect

// *****
// Query a list of established dedicated bearers.
// Select a dedicated bearer.
// Release the selected dedicated bearer.
// *****
CATalog:LTE:SIGN:CONNection:DEDBearer?
CONFIgure:LTE:SIGN:CONNection:DEDBearer "6 (->5, DRAM)"
CALL:LTE:SIGN:PSWitched:ACTion DISConnect
```

2.5.32 Querying UE measurement report contents

```
// *****
// Query UE measurement report values for the serving LTE cell and
// for the first neighbor cell of each technology, PCC and SCC1.
// *****
SENSe:LTE:SIGN:UEReport:PCC:SCEl1?
SENSe:LTE:SIGN:UEReport:SCC:SCEl1?
SENSe:LTE:SIGN:UEReport:PCC:SCEl1:RANGe?
SENSe:LTE:SIGN:UEReport:SCC:SCEl1:RANGe?
SENSe:LTE:SIGN:UEReport:NCEl1:LTE:CEl1?
SENSe:LTE:SIGN:UEReport:NCEl1:LTE:CEl1:RANGe?
SENSe:LTE:SIGN:UEReport:NCEl1:GSM:CEl1?
SENSe:LTE:SIGN:UEReport:NCEl1:GSM:CEl1:RANGe?
SENSe:LTE:SIGN:UEReport:NCEl1:WCDMa:CEl1?
SENSe:LTE:SIGN:UEReport:NCEl1:WCDMa:CEl1:RANGe?
SENSe:LTE:SIGN:UEReport:NCEl1:CDMA:CEl1?
SENSe:LTE:SIGN:UEReport:NCEl1:EVDO:CEl1?
SENSe:LTE:SIGN:UEReport:NCEl1:TDSCdma:CEl1?
SENSe:LTE:SIGN:UEReport:NCEl1:TDSCdma:CEl1:RANGe?
SENSe:LTE:SIGN:UEReport:SCC:RRESult?
SENSe:LTE:SIGN:UEReport:SCC:COCC?
```

2.5.33 Querying UE information

```
// *****
// Query all UE information results.
// *****
SENSe:LTE:SIGN:UESinfo:IMEI?
SENSe:LTE:SIGN:UESinfo:IMSI?
SENSe:LTE:SIGN:UESinfo:VDPReference?
SENSe:LTE:SIGN:UESinfo:UEUSage?
SENSe:LTE:SIGN:UESinfo:UEAddress:IPV4?
SENSe:LTE:SIGN:UESinfo:UEAddress:IPV6?
SENSe:LTE:SIGN:UESinfo:UEAddress:DEDBearer:SEParate?
```

2.5.34 Querying UE capability report contents

```
// *****
// Query general UE capability information.
// *****
SENSe:LTE:SIGN:UECapability:ASRelease?
SENSe:LTE:SIGN:UECapability:UECategory?
SENSe:LTE:SIGN:UECapability:UECategory:UL:ENHanced?
SENSe:LTE:SIGN:UECapability:UECategory:DL:ENHanced?
SENSe:LTE:SIGN:UECapability:FGINdicators?
SENSe:LTE:SIGN:UECapability:FAUeeutra:FGINdicators?
SENSe:LTE:SIGN:UECapability:TAUeeutra:FGINdicators?
SENSe:LTE:SIGN:UECapability:FGINdicators:RNADd?
SENSe:LTE:SIGN:UECapability:FAUeeutra:FGINdicators:RNADd?
SENSe:LTE:SIGN:UECapability:TAUeeutra:FGINdicators:RNADd?
SENSe:LTE:SIGN:UECapability:FGINdicators:RTEN?
SENSe:LTE:SIGN:UECapability:FAUeeutra:FGINdicators:RTEN?
SENSe:LTE:SIGN:UECapability:TAUeeutra:FGINdicators:RTEN?
SENSe:LTE:SIGN:UECapability:DTYPe?
SENSe:LTE:SIGN:UECapability:RREPort?

// *****
// Query PDCP UE capabilities.
// *****
SENSe:LTE:SIGN:UECapability:PDCP:SRPProfiles?
SENSe:LTE:SIGN:UECapability:PDCP:MRCSessions?
SENSe:LTE:SIGN:UECapability:PDCP:SNEXtension?
SENSe:LTE:SIGN:UECapability:PDCP:SRCContinue?

// *****
// Query physical layer UE capabilities.
// *****
SENSe:LTE:SIGN:UECapability:PLAYer:UTASupported?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAYer:UTASupported?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAYer:UTASupported?
SENSe:LTE:SIGN:UECapability:PLAYer:USRSsupport?
```

```

SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:USRSsupport?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:USRSsupport?
SENSe:LTE:SIGN:UECapability:PLAyER:EDLFSsupport?
SENSe:LTE:SIGN:UECapability:PLAyER:EDLTsupport?
SENSe:LTE:SIGN:UECapability:PLAyER:TAPPSupport?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:TAPPSupport?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:TAPPSupport?
SENSe:LTE:SIGN:UECapability:PLAyER:TWEFSsupport?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:TWEFSsupport?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:TWEFSsupport?
SENSe:LTE:SIGN:UECapability:PLAyER:PDsupport?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:PDsupport?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:PDsupport?
SENSe:LTE:SIGN:UECapability:PLAyER:CCSSupport?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:CCSSupport?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:CCSSupport?
SENSe:LTE:SIGN:UECapability:PLAyER:SPPSupport?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:SPPSupport?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:SPPSupport?
SENSe:LTE:SIGN:UECapability:PLAyER:MCPCsupport?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:MCPCsupport?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:MCPCsupport?
SENSe:LTE:SIGN:UECapability:PLAyER:NURClst?
SENSe:LTE:SIGN:UECapability:FAUeeutra:PLAyER:NURClst?
SENSe:LTE:SIGN:UECapability:TAUeeutra:PLAyER:NURClst?
SENSe:LTE:SIGN:UECapability:PLAyER:CIHandl?
SENSe:LTE:SIGN:UECapability:PLAyER:EPDCch?
SENSe:LTE:SIGN:UECapability:PLAyER:MACReporting?
SENSe:LTE:SIGN:UECapability:PLAyER:SCIHandl?
SENSe:LTE:SIGN:UECapability:PLAyER:TSSubframe?
SENSe:LTE:SIGN:UECapability:PLAyER:TDPChselect?
SENSe:LTE:SIGN:UECapability:PLAyER:ULComp?
SENSe:LTE:SIGN:UECapability:PLAyER:ITCWithdiff?
SENSe:LTE:SIGN:UECapability:PLAyER:EHPFdd?
SENSe:LTE:SIGN:UECapability:PLAyER:EFTCodebook?
SENSe:LTE:SIGN:UECapability:PLAyER:TFCPCelldplx?
SENSe:LTE:SIGN:UECapability:PLAyER:TRCTddpcell?
SENSe:LTE:SIGN:UECapability:PLAyER:TRCFddpcell?
SENSe:LTE:SIGN:UECapability:PLAyER:PFMode?
SENSe:LTE:SIGN:UECapability:PLAyER:PSPSfset?
SENSe:LTE:SIGN:UECapability:PLAyER:CSFSet?
SENSe:LTE:SIGN:UECapability:PLAyER:NRRT?
SENSe:LTE:SIGN:UECapability:PLAyER:DSDCell?

// *****
// Query RF UE capabilities.
// *****
SENSe:LTE:SIGN:UECapability:RF:SUPPorted?
SENSe:LTE:SIGN:UECapability:RF:HDUPlex?
SENSe:LTE:SIGN:UECapability:RF:DL256?

```

```

SENSe:LTE:SIGN:UECapability:RF:UL64?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:BCSet?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa1?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa2?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa3?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa4?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa1:BClass:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa2:BClass:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa3:BClass:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa4:BClass:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa1:BClass:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa2:BClass:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa3:BClass:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa4:BClass:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa1:MCAPability:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa2:MCAPability:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa3:MCAPability:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa4:MCAPability:UL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa1:MCAPability:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa2:MCAPability:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa3:MCAPability:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa4:MCAPability:DL?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa1:SCPRoc?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa2:SCPRoc?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa3:SCPRoc?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1020:EUTRa4:SCPRoc?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1090:EUTRa1?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1090:EUTRa2?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1090:EUTRa3?
SENSe:LTE:SIGN:UECapability:RF:BCOMbination:V1090:EUTRa4?
SENSe:LTE:SIGN:UECapability:RF:MTADvance?
SENSe:LTE:SIGN:UECapability:RF:SRTX?
SENSe:LTE:SIGN:UECapability:RF:DCSupport:ASYNchronous?
SENSe:LTE:SIGN:UECapability:RF:DCSupport:SCGRouping?
SENSe:LTE:SIGN:UECapability:RF:SNCap?
SENSe:LTE:SIGN:UECapability:RF:FBRetrieval?
SENSe:LTE:SIGN:UECapability:RF:RBANds?
SENSe:LTE:SIGN:UECapability:RF:FBPadjust?
SENSe:LTE:SIGN:UECapability:RF:MMPRbehavior?

// *****
// Query measurement UE capabilities.
// *****

SENSe:LTE:SIGN:UECapability:MEAS:IFNGaps?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:UFDD?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:UTDD128?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:GERan?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:CHRPd?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:CXRTt?
SENSe:LTE:SIGN:UECapability:MEAS:IFNGaps:V1020?

```

```

SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:V1020:UFDD?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:V1020:UTDD128?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:V1020:GERan?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:V1020:CHRPd?
SENSe:LTE:SIGN:UECapability:MEAS:IRNGaps:V1020:CXRTt?
SENSe:LTE:SIGN:UECapability:MEAS:RMWideband?
SENSe:LTE:SIGN:UECapability:MEAS:BFInterrupt?
SENSe:LTE:SIGN:UECapability:MEAS:RCOREporting?

// *****
// Query inter-RAT UE capabilities.
// *****

SENSe:LTE:SIGN:UECapability:IRAT:UFDD:SUPPorted?
SENSe:LTE:SIGN:UECapability:IRAT:UFDD:EREDirection:UTRA?
SENSe:LTE:SIGN:UECapability:FAUeeutra:IRAT:EREDirection:UTRA?
SENSe:LTE:SIGN:UECapability:TAUeeutra:IRAT:EREDirection:UTRA?
SENSe:LTE:SIGN:UECapability:IRAT:UTDD128:SUPPorted?
SENSe:LTE:SIGN:UECapability:IRAT:UTDD128:EREDirection:UTDD?
SENSe:LTE:SIGN:UECapability:FAUeeutra:IRAT:EREDirection:UTDD?
SENSe:LTE:SIGN:UECapability:TAUeeutra:IRAT:EREDirection:UTDD?
SENSe:LTE:SIGN:UECapability:IRAT:GERan:SUPPorted?
SENSe:LTE:SIGN:UECapability:FAUeeutra:IRAT:GERan:SUPPorted?
SENSe:LTE:SIGN:UECapability:TAUeeutra:IRAT:GERan:SUPPorted?
SENSe:LTE:SIGN:UECapability:IRAT:GERan:PHGeran?
SENSe:LTE:SIGN:UECapability:FAUeeutra:IRAT:GERan:PHGeran?
SENSe:LTE:SIGN:UECapability:TAUeeutra:IRAT:GERan:PHGeran?
SENSe:LTE:SIGN:UECapability:IRAT:GERan:DTM?
SENSe:LTE:SIGN:UECapability:IRAT:GERan:EREDirection?
SENSe:LTE:SIGN:UECapability:IRAT:CDMA2000:NWSHaring?
SENSe:LTE:SIGN:UECapability:IRAT:CHRPd:SUPPorted?
SENSe:LTE:SIGN:UECapability:IRAT:CHRPd:TCONfig?
SENSe:LTE:SIGN:UECapability:IRAT:CHRPd:RCONfig?
SENSe:LTE:SIGN:UECapability:IRAT:CXRTt:SUPPorted?
SENSe:LTE:SIGN:UECapability:IRAT:CXRTt:TCONfig?
SENSe:LTE:SIGN:UECapability:IRAT:CXRTt:RCONfig?
SENSe:LTE:SIGN:UECapability:IRAT:CXRTt:ECSFb?
SENSe:LTE:SIGN:UECapability:FAUeeutra:IRAT:CXRTt:ECSFb?
SENSe:LTE:SIGN:UECapability:TAUeeutra:IRAT:CXRTt:ECSFb?
SENSe:LTE:SIGN:UECapability:IRAT:CXRTt:ECCMob?
SENSe:LTE:SIGN:UECapability:FAUeeutra:IRAT:CXRTt:ECCMob?
SENSe:LTE:SIGN:UECapability:TAUeeutra:IRAT:CXRTt:ECCMob?
SENSe:LTE:SIGN:UECapability:IRAT:CXRTt:ECDual?
SENSe:LTE:SIGN:UECapability:FAUeeutra:IRAT:CXRTt:ECDual?
SENSe:LTE:SIGN:UECapability:TAUeeutra:IRAT:CXRTt:ECDual?

// *****
// Query other UE capabilities.
// *****

SENSe:LTE:SIGN:UECapability:IDCindex?
SENSe:LTE:SIGN:UECapability:PPINdex?

```

```

SENSe:LTE:SIGN:UECapability:URTTimediff?
SENSe:LTE:SIGN:UECapability:DCIulca?

// *****
// Query MBMS UE capabilities.
// *****
SENSe:LTE:SIGN:UECapability:MBMS:SCElL?
SENSe:LTE:SIGN:UECapability:MBMS:NSCell?

// *****
// Query CSG proximity indication UE capabilities.
// *****
SENSe:LTE:SIGN:UECapability:CPINdication:FREQuency:INTRa?
SENSe:LTE:SIGN:UECapability:CPINdication:FREQuency:INTER?
SENSe:LTE:SIGN:UECapability:CPINdication:UTRan?

// *****
// Query neighbor cell SI-acquisition UE capabilities.
// *****
SENSe:LTE:SIGN:UECapability:NCSacq:FREQuency:INTRa?
SENSe:LTE:SIGN:UECapability:FAUeeutra:NCSacq:FREQuency:INTRa?
SENSe:LTE:SIGN:UECapability:TAUeeutra:NCSacq:FREQuency:INTRa?
SENSe:LTE:SIGN:UECapability:NCSacq:FREQuency:INTER?
SENSe:LTE:SIGN:UECapability:FAUeeutra:NCSacq:FREQuency:INTER?
SENSe:LTE:SIGN:UECapability:TAUeeutra:NCSacq:FREQuency:INTER?
SENSe:LTE:SIGN:UECapability:NCSacq:UTRan?
SENSe:LTE:SIGN:UECapability:FAUeeutra:NCSacq:UTRan?
SENSe:LTE:SIGN:UECapability:TAUeeutra:NCSacq:UTRan?

// *****
// Query UE-based network performance measurement capabilities.
// *****
SENSe:LTE:SIGN:UECapability:UBNPmeas:LMIDle?
SENSe:LTE:SIGN:UECapability:UBNPmeas:SGLocation?
SENSe:LTE:SIGN:UECapability:LMMeas?

// *****
// Query UE capabilities for RLC.
// *****
SENSe:LTE:SIGN:UECapability:ERLField?

// *****
// Query UE capabilities for WLAN interworking.
// *****
SENSe:LTE:SIGN:UECapability:WIW:WIRRules?
SENSe:LTE:SIGN:UECapability:WIW:WIAPolicies?

// *****
// Query UE capabilities for eMTC CE modes.
// *****

```



```

SENSe:LTE:SIGN:UECapability:CEParameters:MODE:A?
SENSe:LTE:SIGN:UECapability:CEParameters:MODE:B?

// *****
// Query UE capabilities for LAA.
// *****
SENSe:LTE:SIGN:UECapability:LAA:DL?
SENSe:LTE:SIGN:UECapability:LAA:EDPTs?
SENSe:LTE:SIGN:UECapability:LAA:SSSPosition?
SENSe:LTE:SIGN:UECapability:LAA:TM9?

// *****
// Query UE capabilities for dual connectivity.
// *****
SENSe:LTE:SIGN:UECapability:DCParameters:DTSPplit?
SENSe:LTE:SIGN:UECapability:DCParameters:DTSCg?

// *****
// Query UE capabilities for MAC layer.
// *****
SENSe:LTE:SIGN:UECapability:MAC:LCSPtimer?
SENSe:LTE:SIGN:UECapability:MAC:LDRXcommand?

// *****
// Query UE capabilities for sidelink communication.
// *****
SENSe:LTE:SIGN:UECapability:SL:CSTX?
SENSe:LTE:SIGN:UECapability:SL:DSRalloc?
SENSe:LTE:SIGN:UECapability:SL:DUSRalloc?
SENSe:LTE:SIGN:UECapability:SL:DSLSSs?
SENSe:LTE:SIGN:UECapability:SL:DSPRoc?

```

2.5.35 Performing an intra-RAT handover

```

// *****
// An intra-RAT handover is a handover within the LTE signaling application.
//
// Select the LTE signaling application as handover destination.
// Define the destination parameters: band 2, DL channel no. 910,
// 10 MHz cell bandwidth, additional requirement NS_03.
// Initiate the handover.
// *****
PREPare:LTE:SIGN:HANdOver:DESTination "LTE Sig1"
PREPare:LTE:SIGN:HANdOver OB2, 910, B100, NS03
CALL:LTE:SIGN:PSWitched:ACTion HANdOver

```

2.5.36 Performing an inter-RAT handover

```
// *****
// An inter-RAT handover is a handover to another signaling application.
//
// Query a list of possible handover destinations (signaling applications).
// Select a handover destination from the list.
// Select the handover mechanism.
// Wait until the destination is ready to receive a handover.
// Initiate the handover.
//
// Destination parameters like operating band or channel can be changed using
// commands provided by the destination signaling application. Adjust these
// parameters before executing the following commands.
// *****
PREPare:LTE:SIGN:HANdOver:CATalog:DESTination?
PREPare:LTE:SIGN:HANdOver:DESTination "GSM Sig1"
PREPare:LTE:SIGN:HANdOver:MMODE REDirection
WHILE SOURce:GSM:SIGN:CELL:STATe? <> "RFH"
CALL:LTE:SIGN:PSWitched:ACTion HANdOver
```

2.5.37 Performing a handover to another instrument

```
// *****
// Select handover to other instrument ("No Connection").
// Select target RAT (WCDMA) and configure the other destination settings.
// Initiate the handover.
// Prepare also the settings for other target RATs.
// *****
PREPare:LTE:SIGN:HANdOver:DESTination "No Connection"
PREPare:LTE:SIGN:HANdOver:EXTernal:DESTination WCDMa
PREPare:LTE:SIGN:HANdOver:EXTernal:WCDMa OB1, 10565
CALL:LTE:SIGN:PSWitched:ACTion HANdOver

PREPare:LTE:SIGN:HANdOver:EXTernal:CDMA USC, 500
PREPare:LTE:SIGN:HANdOver:EXTernal:EVDO USC, 500
PREPare:LTE:SIGN:HANdOver:EXTernal:GSM G09, 55, G18
PREPare:LTE:SIGN:HANdOver:EXTernal:LTE OB1, 300
PREPare:LTE:SIGN:HANdOver:EXTernal:TDSCdma OB1, 10565
```

2.5.38 Sending/receiving a short message

```
// *****
// Configure support of concatenated SMS.
// *****
CONFigure:LTE:SIGN:SMS:OUTGoing:LHANDling MSMS

// *****
```

```

// Configure outgoing messages directly:
// ASCII message text, binary message contents, TP-PID, data coding,
// user data header, coding group, message class,
// SC address, originating address, service center time stamp.
// Send the message and query whether the transmission was successful.
// *****
CONFIGure:LTE:SIGN:SMS:OUTGoing:MESHandling INTERNAL
CONFIGure:LTE:SIGN:SMS:OUTGoing:INTERNAL "Testing SMS 012!.#\*%+-( )<>?=@$, "
CONFIGure:LTE:SIGN:SMS:OUTGoing:BINary #H0125498fa3bc8d348
CONFIGure:LTE:SIGN:SMS:OUTGoing:PIDentifier #H0
CONFIGure:LTE:SIGN:SMS:OUTGoing:DCODing BIT7
CONFIGure:LTE:SIGN:SMS:OUTGoing:UDHeader #H050415820000
CONFIGure:LTE:SIGN:SMS:OUTGoing:CGRoup GDCoding
CONFIGure:LTE:SIGN:SMS:OUTGoing:MCLass CL2
CONFIGure:LTE:SIGN:SMS:OUTGoing:OSAddress '543221'
CONFIGure:LTE:SIGN:SMS:OUTGoing:OAddress '3526735'
CONFIGure:LTE:SIGN:SMS:OUTGoing:SCTStamp:TSource DATE
CONFIGure:LTE:SIGN:SMS:OUTGoing:SCTStamp:DATE 15,5,2014
CONFIGure:LTE:SIGN:SMS:OUTGoing:SCTStamp:TIME 14,40,50
CALL:LTE:SIGN:PSWitched:ACTion SMS; *OPC?
SENSe:LTE:SIGN:SMS:OUTGoing:INFO:LMSent?

// *****
// Reset parameters related to an already received short message.
// Wait until a message from the UE has been received.
// Query information about the received message: encoding, message content,
// message length, message segments.
// *****
CLEan:LTE:SIGN:SMS:INComing:INFO:MTEXT
WAITKEY >Send short message from UE<
WHILE SENSe:LTE:SIGN:SMS:INFO:LRMessage:RFLag? <> "OFF"
SENSe:LTE:SIGN:SMS:INComing:INFO:DCODing?
SENSe:LTE:SIGN:SMS:INComing:INFO:MTEXT?
SENSe:LTE:SIGN:SMS:INComing:INFO:MLENGTH?
SENSe:LTE:SIGN:SMS:INComing:INFO:SEGMENT?

// *****
// Configure outgoing messages via a file:
// Select message file and query file information.
// *****
CONFIGure:LTE:SIGN:SMS:OUTGoing:MESHandling FILE
CONFIGure:LTE:SIGN:SMS:OUTGoing:FILE "@USERDATA\sms\LTE\Send\myfile.sms"
CONFIGure:LTE:SIGN:SMS:OUTGoing:FILE:INFO?

// *****
// Get information about an old received short message:
// Select a message file and query information about it.
// *****

```

```
CONFigure:LTE:SIGN:SMS:INComing:FILE "@USERDATA\sms\LTE\Received\rx_001.sms"
CONFigure:LTE:SIGN:SMS:INComing:FILE:INFO?
```

2.5.39 Sending date and time information to the UE

```
// *****
// Select a time source and configure date, time, DST and time zone offset.
// Enable sending of the information during attach.
// Disable sending the network name.
// Send the information to the UE now.
// *****
CONFigure:LTE:SIGN:CELL:TIME:TSource DATE
CONFigure:LTE:SIGN:CELL:TIME:DATE 24,10,2012
CONFigure:LTE:SIGN:CELL:TIME:TIME 12,40,30
CONFigure:LTE:SIGN:CELL:TIME:DSTime P1H
CONFigure:LTE:SIGN:CELL:TIME:LTzoffset 1
CONFigure:LTE:SIGN:CELL:TIME:SATTach ON
CONFigure:LTE:SIGN:CELL:TIME:SNNName OFF
CONFigure:LTE:SIGN:CELL:TIME:SNOW
```

2.5.40 Modifying parameters for an established connection

```
// *****
// Modify inserted transport block errors and additional spectrum emission
// requirements. Set PCC scheduling type to 3GPP-compliant RMC.
// *****
CONFigure:LTE:SIGN:CONNECTION:DLEinsertion 20
CONFigure:LTE:SIGN:CONNECTION:ASEmission NS03
CONFigure:LTE:SIGN:CONNECTION:PCC:STYPe RMC

// *****
// Redefine the PCC RMCs: DL RMC with 50 RBs, 64-QAM modulation and
// transport block size index 5. UL RMC with 50 RBs, QPSK modulation and
// block size index 6.
// Position the resource blocks at the lower end of the cell bandwidth.
// *****
CONFigure:LTE:SIGN:CONNECTION:PCC:RMC:DL N50,QPSK,T5
CONFigure:LTE:SIGN:CONNECTION:PCC:RMC:UL N50,QPSK,T6
CONFigure:LTE:SIGN:CONNECTION:PCC:RMC:RBPosition:DL LOW
CONFigure:LTE:SIGN:CONNECTION:PCC:RMC:RBPosition:UL LOW

// *****
// Modify the RS EPRE level for the PCC.
// *****
CONFigure:LTE:SIGN:DL:PCC:RSEPre:LEVel -83

// *****
// Command the UE to change the UL power of the PCC by +3 dB:
```

```
// Select single pattern as active TPC setup, define a single pattern of
// 3 steps UP, execute the pattern.
// *****
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:SET SINGLE
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:SINGLE 3, UP
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:PEXecute

// *****
// Apply a continuous TPC command pattern of 5x -1 dB and 4x +1 dB:
// Define the pattern and select the active TPC setup.
// *****
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:UDPattern 9,-1,-1,-1,-1,-1,1,1,1,1
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:SET UDContinuous

// *****
// Command the UE to a target power of -10 dBm.
// For the SCC, use a 3 dB higher power.
// *****
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:CLTPower -10
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:SET CLOop
CONFIGure:LTE:SIGN:UL:SCC:PUSCh:TPC:CLTPower:OFFSet 3

// *****
// Execute ramping up pattern B for a 3GPP relative power control test.
// *****
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:SET RPControl
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:RPControl RUB
CONFIGure:LTE:SIGN:UL:PCC:PUSCh:TPC:PEXecute
```

2.6 Command reference

The following sections provide detailed reference information on the remote control commands of the LTE signaling application.

• Conventions and general information	362
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• UE measurement report contents	388
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• Uplink power control	662
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• eMTC settings	722

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2.6.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

2.6.1.1 SIGN<i>

SIGN<i> is used as abbreviation of SIGNaling<instance>. For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same firmware application. It can be omitted if the instrument supports only one instance, or to address the first instance.

2.6.1.2 STReam<s> and DL<s>

Commands that contain a STReam<s> or a DL<s> mnemonic are related to a single downlink stream.

The suffix <s> selects the downlink stream and is only relevant for scenarios with MIMO. For SISO scenarios, you can omit the suffix.

2.6.1.3 PCC and SCC commands

For carrier aggregation scenarios, many settings are configurable per carrier. The following command variants are relevant in this context:

- A [:PCC] command configures the PCC. The command can also be relevant for scenarios without carrier aggregation.
- An :SCC<c> command configures the SCC number <c>. For the SCC number 1, you can omit the suffix 1.
- An :SCC command configures the SCC number 1. Other SCCs cannot be configured via such a command.
- A :SETA or :SETB command configures all carriers of set A or set B (inband contiguous UL carrier allocation).

2.6.1.4 Values for signal path selection

Which connectors and modules can be specified in a command, depends on the installed hardware and the active subinstrument or instance <i>.

This section lists all values available for path selection. Depending on your configuration, only a subset is relevant for you. Virtual connector names are only relevant for setting commands. Queries return the physical connector names.

Additional information is available in the base software documentation. It describes typical instrument configurations with the allowed RF connector - TX/RX module combinations and the mapping of virtual connector names to physical connectors.

Single-CMW setup

Selection values:

- SUA as baseband unit:
SUA1 | SUA2 | SUA3 | SUA4 | SUA5 | SUA6
SUA012 | SUA034 | SUA056
SUA partition 1, 2, 3, 4, 5, 6, 1&2, 3&4, 5&6
- SUA as coprocessor:
SUA012 | SUA034 | SUA056
SUA partition 1&2, 3&4, 5&6
 - Scenario with optional coprocessing: To disable coprocessing, skip the optional parameter. If coprocessing is not used, a query does not return the parameter.
 - Scenario with mandatory coprocessing: If you skip the optional parameter, the default selection is used. A query always returns the used SUA.
- RX module:
RX1 | RX2 | RX3 | RX4
- TX module:
TX1 | TX2 | TX3 | TX4
- RX connector:
RF1C | RF2C | RF3C | RF4C | RFAC | RFBC
RF 1 COM to RF 4 COM plus virtual connector names
- TX connector:
RF1C | RF2C | RF3C | RF4C | RFAC | RFBC
RF1O | RF3O | RFAO
RF 1 COM to RF 4 COM plus virtual connector names
RF 1 OUT and RF 3 OUT plus virtual connector name
- SUA as fader:
FAD1 | FAD2 | FAD3 | FAD4 | FAD5 | FAD6
FAD012 | FAD034 | FAD056
SUA fader 1, 2, 3, 4, 5, 6, 1&2, 3&4, 5&6
The SUA fader numbering is reversed. The highest SUA fader uses the lowest SUA partition.

CMWflexx setup with R&S CMWC and several R&S CMW500

Selection values:

- **SUA as baseband unit:**

SUA11 | SUA12 | SUA13 | SUA14 | SUA15 | SUA16
 SUA21 | SUA22 | SUA23 | SUA24 | SUA25 | SUA26
 SUA31 | SUA32 | SUA33 | SUA34 | SUA35 | SUA36
 SUA41 | SUA42 | SUA43 | SUA44 | SUA45 | SUA46
 SUA112 | SUA134 | SUA156 | SUA212 | SUA234 | SUA256
 SUA312 | SUA334 | SUA356 | SUA412 | SUA434 | SUA456

**SUA<a>: CMW <a>, SUA partition **

Example SUA21: CMW 2, SUA partition 1

SUA<a><c>: CMW <a>, SUA partitions &<c>

Example SUA312: CMW 3, SUA partitions 1&2

- **SUA as coprocessor:**

SUA112 | SUA134 | SUA156 | SUA212 | SUA234 | SUA256
 SUA312 | SUA334 | SUA356 | SUA412 | SUA434 | SUA456

SUA<a><c>: CMW <a>, SUA partitions &<c>

- Scenario with optional coprocessing: To disable coprocessing, skip the optional parameter. If coprocessing is not used, a query does not return the parameter.

- Scenario with mandatory coprocessing: If you skip the optional parameter, the default selection is used. A query always returns the used SUA.

- **RX module:**

RX11 | RX12 | RX13 | RX14 | RX21 | RX22 | RX23 | RX24
 RX31 | RX32 | RX33 | RX34 | RX41 | RX42 | RX43 | RX44

**RX<a>: CMW <a>, RX **

Example RX24: CMW 2, RX module 4

- **TX module:**

TX11 | TX12 | TX13 | TX14 | TX21 | TX22 | TX23 | TX24
 TX31 | TX32 | TX33 | TX34 | TX41 | TX42 | TX43 | TX44

**TX<a>: CMW <a>, TX **

Example TX13: CMW 1, TX module 3

- **RX connector:**

R11C | R12C | R13C | R14C | R21C | R22C | R23C | R24C
 R31C | R32C | R33C | R34C | R41C | R42C | R43C | R44C

R<a>C: CMW <a>, connector RF COM

Example R12C: CMW 1, RF 2 COM

- **TX connector:**

R11C | R12C | R13C | R14C | R21C | R22C | R23C | R24C
 R31C | R32C | R33C | R34C | R41C | R42C | R43C | R44C
 R11O | R13O | R21O | R23O | R31O | R33O | R41O | R43O

R<a>C: CMW <a>, connector RF COM

R<a>O: CMW <a>, connector RF OUT

Example R21O: CMW 2, RF 1 OUT

- **SUA as fader:**

FAD11 | FAD12 | FAD13 | FAD14 | FAD15 | FAD16
 FAD21 | FAD22 | FAD23 | FAD24 | FAD25 | FAD26
 FAD31 | FAD32 | FAD33 | FAD34 | FAD35 | FAD36

FAD41 | FAD42 | FAD43 | FAD44 | FAD45 | FAD46
 FAD112 | FAD134 | FAD156 | FAD212 | FAD234 | FAD256
 FAD312 | FAD334 | FAD356 | FAD412 | FAD434 | FAD456

FAD<a>: CMW <a>, SUA fader

Example FAD21: CMW 2, SUA fader 1

FAD<a><c>: CMW <a>, SUA fader &<c>

Example FAD312: CMW 3, SUA fader 1&2

The SUA fader numbering is reversed. The highest SUA fader uses the lowest SUA partition.

For CMW 1, you can alternatively use the single-CMW selection values.

2.6.1.5 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**

If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:

- "Query only": The command can only be used to query parameters. No setting.
- "Event": The command initiates an event. No query.

- **Parameter usage**

The parameter usage is indicated by the keyword preceding the parameters:

- "Parameters" are sent with a setting or query command and are returned as the result of a query.
- "Setting parameters" are sent with a setting command.
- "Query parameters" are sent with a query command, to refine the query.
- "Return values" are returned as the result of a query.

- **Parameter type**

The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:

- "string": characters enclosed in quotation marks (" or ')
- "numeric", "decimal": decimal number
- "integer": integer number
- "float": floating point number
- "hex": hexadecimal number starting with #H
- "binary": binary number starting with #B
- "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).

- **Firmware/Software:**

Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

2.6.2 General settings

The following commands configure general settings at the top of the configuration dialog.

For scenarios, see [Section 2.6.8.1, "Scenario selection and signal routing"](#), on page 451.

CONFigure:LTE:SIGN<i>[:PCC]:DMODE	366
CONFigure:LTE:SIGN<i>:SCC<c>:DMODE	366
CONFigure:LTE:SIGN<i>[:PCC]:DMODE:UCSPecific	366
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:ENABLE	367
CONFigure:LTE:SIGN<i>:SCC<c>:FSTRucture	367
CONFigure:LTE:SIGN<i>:SCC:AMODE	367
CONFigure:LTE:SIGN<i>:SCC<c>:UUL	368
CONFigure:LTE:SIGN<i>:CAGGregation:SET	368
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SEXecute	370
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:CEXecute	370

CONFigure:LTE:SIGN<i>[:PCC]:DMODE <Mode>

CONFigure:LTE:SIGN<i>:SCC<c>:DMODE <Mode>

Selects the duplex mode of the LTE signal: FDD or TDD.

See also [CONFigure:LTE:SIGN<i>\[:PCC\]:DMODE:UCSPecific](#) on page 366.

Suffix:

<c> 1..7

Parameters:

<Mode> FDD | TDD

Example: See [Specifying general settings](#)

Firmware/software: V2.1.20, SCC command V3.5.10

Options: R&S CMW-KS500/-KS550 for FDD/TDD

Manual operation: See ["Duplex Mode"](#) on page 140

CONFigure:LTE:SIGN<i>[:PCC]:DMODE:UCSPecific <Enable>

Enables the carrier-specific duplex mode configuration.

- Enabled: The duplex mode is configured per carrier via:
[CONFigure:LTE:SIGN<i>\[:PCC\]:DMODE](#)
[CONFigure:LTE:SIGN<i>:SCC<c>:DMODE](#)
- Disabled: All carriers have the same duplex mode, configured via:
[CONFigure:LTE:SIGN<i>\[:PCC\]:DMODE](#)

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Specifying general settings](#)

Firmware/software: V3.5.10

Manual operation: See ["Duplex Mode"](#) on page 140

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:ENABLE <Enable>

Enables or disables eMTC.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["eMTC"](#) on page 142

CONFigure:LTE:SIGN<i>:SCC<c>:FSTRucture <Structure>

The frame structure is only configurable for the TDD user-defined band. In any other case, only the query is relevant.

Suffix:

<c> 1..7

Parameters:

<Structure> T1 | T2 | T3
T1: Type 1 - FDD
T2: Type 2 - TDD normal operation
T3: Type 3 - LAA operation mode

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS500 for T1, R&S CMW-KS550 for T2
R&S CMW-KS514 for T3

Manual operation: See ["Frame Structure"](#) on page 142

CONFigure:LTE:SIGN<i>:SCC:AMODE <Mode>

Selects the SCC activation mode. For manual triggering of a state transition, see [CALL:LTE:SIGN<i>:SCC<c>:ACTion](#).

Parameters:

<Mode> AUTO | MANual | SEMiauto

AUTO

All SCCs are activated automatically at RRC connection establishment, so that the state "MAC Activated" is reached.

MANual

Each state transition step must be initiated separately for each SCC. So several actions are required to reach the state "MAC Activated".

SEMiauto

The activation must be initiated manually for each SCC. As a result, all state transitions required to reach the state "MAC Activated" are performed.

*RST: AUTO

Example: See [Specifying general settings](#)

Firmware/software: V3.2.50

Options: R&S CMW-KS512 for MANual and SEMiauto

Manual operation: See ["SCC Activation Mode"](#) on page 142

CONFigure:LTE:SIGN<i>:SCC<c>:UUL <UseUplink>[, <SCCRXConnector>, <SCCRXConverter>]

Activates the uplink for the SCC number <c> and optionally selects the signal path.

For possible connector and converter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<c> 1..7

Parameters:

<UseUplink> OFF | ON

*RST: OFF

<SCCRXConnector> RF connector for the SCC input path

<SCCRXConverter> RX module for the SCC input path

Example: See [Specifying general settings](#)

Firmware/software: V3.5.20

Options: R&S CMW-KS512, for more than two UL also R&S CMW-KS514

Manual operation: See ["Use UL"](#) on page 143

CONFigure:LTE:SIGN<i>:CAGGregation:SET <SetApos1>, <SetApos2>, <SetApos3>, <SetApos4>, <SetBpos1>, <SetBpos2>, <SetBpos3>, <SetBpos4>[, <SetCpos1>, <SetCpos2>, <SetCpos3>, <SetCpos4>]

Configures the alignment of UL component carriers for intraband contiguous UL CA.

The command configures set A, set B and set C. It aligns all component carriers of a set for contiguous UL CA.

Rules for valid parameter combinations:

- Enable the UL of a component carrier before adding it to a set.
- Use set C only for scenarios with at least six carriers.
Use set B only for scenarios with at least four carriers.
- To disable a set, select INV for all four parameters of the set.
Omitting the <SetC...> parameters, sets them to INV.
- To use a set, select the primary carrier for <...pos1> and a second carrier for <...pos2>.
To align only two carriers, set <...pos3> and <...pos4> to INV.
To align only three carriers, select <...pos3> and set <...pos4> to INV.
To align four carriers, select <...pos3> and <...pos4>.
- All carriers of a set must fit into the band of the primary carrier, without changing the frequency of the primary carrier.

Parameters:

<SetApos1>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Primary carrier of set A.
<SetApos2>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Second carrier of set A.
<SetApos3>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Third carrier of set A.
<SetApos4>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Fourth carrier of set A.
<SetBpos1>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Primary carrier of set B.
<SetBpos2>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Second carrier of set B.
<SetBpos3>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Third carrier of set B.
<SetBpos4>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Fourth carrier of set B.
<SetCpos1>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Primary carrier of set C.

<SetCpos2>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Second carrier of set C.
<SetCpos3>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Third carrier of set C.
<SetCpos4>	INV PCC SCC1 SCC2 SCC3 SCC4 SCC5 SCC6 SCC7 Fourth carrier of set C.

Firmware/software: V3.7.30, V3.7.70 added set C.

Options: R&S CMW-KS512

Manual operation: See ["Intraband Contiguous UL"](#) on page 143

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SEXecute

Initiates a swap of settings between the PCC and the SCC number <c>.

Suffix:

<c> 1..7

Usage: Event

Firmware/software: V3.2.60

Manual operation: See ["Swap button"](#) on page 128

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:CEXecute

Copies PCC DL settings to the SCC number <c>.

Suffix:

<c> 1..7

Usage: Event

Firmware/software: V3.2.82

Manual operation: See ["Copy button"](#) on page 129

2.6.3 Connection control and states

The commands in the following sections control the connection to the UE and retrieve connection-related information.

- [General commands](#).....371
- [Connect/disconnect preparation](#)..... 375
- [Handover preparation](#)..... 379

2.6.3.1 General commands

The following commands control the connection to the UE and retrieve connection-related information.

Preparation commands for `CALL:LTE:SIGN<i>:PSWitched:ACTion` are described in the subsequent sections.

<code>SOURce:LTE:SIGN<i>:CELL:STATe</code>	371
<code>SOURce:LTE:SIGN<i>:CELL:STATe:ALL?</code>	371
<code>SENSe:LTE:SIGN<i>:RRcState?</code>	372
<code>CALL:LTE:SIGN<i>:PSWitched:ACTion</code>	372
<code>CALL:LTE:SIGN<i>:SCC<c>:ACTion</code>	373
<code>CONFigure:LTE:SIGN<i>:A:SCC<c>:ENABLE</code>	373
<code>CONFigure:LTE:SIGN<i>:B:SCC<c>:ENABLE</code>	373
<code>CALL:LTE:SIGN<i>:A:ACTion</code>	374
<code>CALL:LTE:SIGN<i>:B:ACTion</code>	374
<code>FETCh:LTE:SIGN<i>:PSWitched:STATe?</code>	374
<code>FETCh:LTE:SIGN<i>:SCC<c>:STATe?</code>	375
<code>FETCh:LTE:SIGN<i>:A:STATe?</code>	375
<code>FETCh:LTE:SIGN<i>:B:STATe?</code>	375

SOURce:LTE:SIGN<i>:CELL:STATe <Control>

Turns the generator (the cell) on or off.

Setting parameters:

<Control> OFF | ON
 Switch generator **ON** or **OFF**
 *RST: OFF

Return values:

<GeneratorState> OFF | PENDING | ON
 OFF: generator switched off
 PEND: generator switched on but no signal available yet
 ON: generator switched on, signal available
 *RST: OFF

Example: See [Attaching the UE and activating SCCs](#)

Firmware/software: V1.0.15.20

Manual operation: See ["Cell"](#) on page 120

SOURce:LTE:SIGN<i>:CELL:STATe:ALL?

Returns detailed information about the signaling generator state.

Return values:

<MainState>	OFF ON RFHandover OFF: generator switched off ON: generator switched on RFHandover: ready to receive a handover from another signaling application
<SyncState>	PENDING ADJUSTED PENDING: generator turned on (off) but signal not yet (still) available ADJUSTED: physical output signal corresponds to main generator state

Example: See [Attaching the UE and activating SCCs](#)

Usage: Query only

Firmware/software: V1.0.15.20
V3.0.10: RFHandover added

Manual operation: See "[ON | OFF] key / LTE Signaling softkey" on page 130

SENSe:LTE:SIGN<i>:RRCState?

Queries whether an RRC connection is established (connected) or not (idle).

Return values:

<State> IDLE | CONNECTED

Example: See [Attaching the UE and activating SCCs](#)

Usage: Query only

Firmware/software: V3.0.20

Manual operation: See "[RRC State](#)" on page 120

CALL:LTE:SIGN<i>:PSWitched:ACTion <PSAction>

Controls the PS connection state. As a prerequisite, the DL signal must be switched on, see [SOURCE:LTE:SIGN<i>:CELL:STATE](#).

Setting parameters:

<PSAction>	CONNECT DISCONNECT SMS DETACH HANDOVER CONNECT: Initiate a mobile-terminated connection setup DISCONNECT: Release the connection SMS: Send an SMS DETACH: Detach the UE HANDOVER: Initiate a handover (within the LTE signaling application or to another signaling application)
------------	--

Example: See [Setting up a test mode connection](#)

Usage: Event

Firmware/software: V2.1.20
V3.0.10: added DETach and HANDover

Manual operation: See ["Connection control hotkeys"](#) on page 131

CALL:LTE:SIGN<i>:SCC<c>:ACTion <SCCAction>

Controls the state of the secondary component carrier (SCC) number <c>.

Suffix:

<c> 1..7

Setting parameters:

<SCCAction> OFF | ON | RRCadd | MACactivate | MACDeactivat | RRCDelete

OFF: Switch off SCC

ON: Switch on SCC

RRCadd: Add SCC RRC connection

MACactivate: Activate MAC for the SCC

MACDeactivat: Deactivate MAC for the SCC

RRCDelete: Delete SCC RRC connection

Usage: Event

Firmware/software: V3.2.50

Manual operation: See ["Connection control hotkeys"](#) on page 131

CONFigure:LTE:SIGN<i>:A:SCC<c>:ENABLE <Enable>

CONFigure:LTE:SIGN<i>:B:SCC<c>:ENABLE <Enable>

Configures whether the SCC<c> belongs to the SCC synchronization set A/B or not.

An SCC can only belong to one of the sets. Adding it to one set, removes it from the other set (if applicable).

Adding an SCC to a set is only possible, if the set and the SCC have the same state (for example "RRC added").

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON

OFF: The SCC does not belong to the set.

ON: The SCC belongs to the set.

*RST: OFF

Firmware/software: V3.7.10

Options: R&S CMW-KS512

Manual operation: See ["Multiple SCC Actions hotkey"](#) on page 132

CALL:LTE:SIGN<i>:A:ACTion <SCCAction>

CALL:LTE:SIGN<i>:B:ACTion <SCCAction>

Control the state of all SCCs assigned to the synchronization set A or B.

Setting parameters:

<SCCAction> OFF | ON | RRCadd | MACactivate | MACDeactivat | RRCDelete

OFF: Switch off SCC

ON: Switch on SCC

RRCadd: Add SCC RRC connection

MACactivate: Activate MAC for the SCC

MACDeactivat: Deactivate MAC for the SCC

RRCDelete: Delete SCC RRC connection

Usage: Event

Firmware/software: V3.7.10

Manual operation: See ["Connection control hotkeys"](#) on page 131

FETCh:LTE:SIGN<i>:PSWitched:STATE?

Queries the PS domain state, see also [Section 2.2.9.1, "Packet-switched states"](#), on page 38.

Return values:

<PS State> OFF | ON | ATTached | CESTablished | DISConnect |
CONNECTing | SIGNaling | SMESsage | RMESsage |
IHANdover | OHANdover

OFF: signal off

ON: signal on

ATTached: UE attached

CEStablished: connection established

DISConnect: disconnect in progress

CONNECTing: connection setup in progress

SIGNaling: signaling in progress

SMESsage: sending message

RMESsage: receiving message

IHANdover: incoming handover in progress

OHANdover: outgoing handover in progress

*RST: OFF

Example: See [Setting up a test mode connection](#)

Usage: Query only

Firmware/software: V2.1.20
V3.0.10: added IHANdover, OHANdover

Manual operation: See ["Packet Switched"](#) on page 120

FETCH:LTE:SIGN<i>:SCC<c>:STATE?

Queries the state of the SCC number <c>, see also [Section 2.2.9.2, "SCC states"](#), on page 40.

Suffix:

<c> 1..7

Return values:

<SCC State> OFF | ON | RRCadded | MACactivated

OFF: SCC off

ON: SCC on

RRCadded: RRC added

MACactivated: MAC activated

Example:

See [Attaching the UE and activating SCCs](#)

Usage:

Query only

Firmware/software: V3.2.50

Manual operation: See ["SCC<n> State"](#) on page 120

FETCH:LTE:SIGN<i>:A:STATE?**FETCH:LTE:SIGN<i>:B:STATE?**

Query the state of the SCC synchronization set A or B, see also [Section 2.2.9.2, "SCC states"](#), on page 40.

Return values:

<SyncSetState> OFF | ON | RRCadded | MACactivated

OFF: SCC off

ON: SCC on

RRCadded: RRC added

MACactivated: MAC activated

Usage:

Query only

Firmware/software: V3.7.10

Manual operation: See ["Connection control hotkeys"](#) on page 131

2.6.3.2 Connect/disconnect preparation

The following commands configure the dedicated bearer settings as a preparation for a dedicated bearer connect or disconnect, initiated via [CALL:LTE:SIGN<i>:PSwitched:ACTion](#).

The commands are only relevant for the connection type "Data Application". For the connection type "Testmode", no preparation is required.

CATalog:LTE:SIGN<i>:CONNection:DEFBearer?	376
CATalog:LTE:SIGN<i>:CONNection:DEDBearer?	376
PREPare:LTE:SIGN<i>:CONNection:DEDBearer	376
PREPare:LTE:SIGN<i>:CONNection:DEDBearer:SEParate	377
CONFigure:LTE:SIGN<i>:CONNection:DEDBearer	378

CATalog:LTE:SIGN<i>:CONNection:DEFBearer?

Queries a list of all established default bearers.

Return values:

<ID> string
 Comma-separated list of bearer IDs as strings
 String example: "5 (cmw500.rohde-schwarz.com)"

Example: See [Connecting/releasing dedicated bearers](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Connect button"](#) on page 133

CATalog:LTE:SIGN<i>:CONNection:DEDBearer?

Queries a list of all established dedicated bearers.

Return values:

<ID> string
 Comma-separated list of bearer IDs as strings
 String example: "6 (->5, Voice)"

Example: See [Connecting/releasing dedicated bearers](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Disconnect button"](#) on page 134

PREPare:LTE:SIGN<i>:CONNection:DEDBearer <DefBearerID>, <Profile>, <TFTportLow>, <TFTportHigh>

Configures dedicated bearer settings as a preparation for a bearer setup via
 CALL:LTE:SIGN:PSWitched:ACTion CONNect.

The same port range is used for the uplink and for the downlink.

Parameters:

<DefBearerID>	string Bearer ID, selecting the default bearer, to which the dedicated bearer is mapped. Example: "5 (cmw500.rohde-schwarz.com)" To query a list of IDs for all established default bearers, see CATalog:LTE:SIGN<i>:CONNectioN:DEFBearer? .
<Profile>	VOICE VIDEO DRAM DRUM Selects a dedicated bearer profile VOICE : for voice connections VIDEO : for video connections DRAM : for data connections with RLC acknowledged mode DRUM : for data connections with RLC unacknowledged mode *RST: DRUM
<TFTportLow>	numeric Selects the lower end of the port range, for which traffic is routed to the dedicated bearer Range: 1 to 65535 *RST: 1
<TFTportHigh>	numeric Selects the upper end of the port range Range: 1 to 65535 *RST: 65535
Example:	See Connecting/releasing dedicated bearers
Firmware/software:	V3.2.80
Options:	R&S CMW-KS510
Manual operation:	See " Connect button " on page 133

PREPare:LTE:SIGN<i>:CONNectioN:DEDBearer:SEParate <DefBearerID>, <Profile>, <TFTportLowDL>, <TFTportHighDL>, <TFTportLowUL>, <TFTportHighUL>

Configures dedicated bearer settings as a preparation for a bearer setup via `CALL:LTE:SIGN:PSWitched:ACTioN CONNect`.

Different port ranges can be set for the uplink and for the downlink.

Parameters:

<DefBearerID>	string Bearer ID, selecting the default bearer, to which the dedicated bearer is mapped. Example: "5 (cmw500.rohde-schwarz.com)" To query a list of IDs for all established default bearers, see CATalog:LTE:SIGN<i>:CONNection:DEFBearer? .
<Profile>	VOICE VIDEO DRAM DRUM Selects a dedicated bearer profile VOICE : for voice connections VIDEO : for video connections DRAM : for data connections with RLC acknowledged mode DRUM : for data connections with RLC unacknowledged mode Range: DRUM
<TFTportLowDL>	numeric Selects the lower end of the port range for downlink traffic Range: 1 to 65535 *RST: 1
<TFTportHighDL>	numeric Selects the upper end of the port range for downlink traffic Range: 1 to 65535 *RST: 65535
<TFTportLowUL>	numeric Selects the lower end of the port range for uplink traffic Range: 1 to 65535 *RST: 1
<TFTportHighUL>	numeric Selects the upper end of the port range for uplink traffic Range: 1 to 65535 *RST: 65535

Firmware/software: V3.5.30**Options:** R&S CMW-KS510**Manual operation:** See "[Connect button](#)" on page 133

CONFigure:LTE:SIGN<i>:CONNection:DEDBearer <ID>

Selects a dedicated bearer as a preparation for a bearer release via
 CALL:LTE:SIGN:PSWitched:ACTion DISConnect.

Parameters:

<ID> string
 Dedicated bearer ID
 Example: "6 (->5, Voice)"
 To query a list of IDs for all established dedicated bearers, see
[CATalog:LTE:SIGN<i>:CONNection:DEDBearer?](#).

Example: See [Connecting/releasing dedicated bearers](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["Disconnect button"](#) on page 134

2.6.3.3 Handover preparation

The following commands configure the handover settings as a preparation for a hand-over action, initiated via [CALL:LTE:SIGN<i>:PSWitched:ACTion](#).

PREPare:LTE:SIGN<i>:HANDover:DESTination	379
PREPare:LTE:SIGN<i>:HANDover:CATalog:DESTination?	379
PREPare:LTE:SIGN<i>:HANDover:MMODE	380
PREPare:LTE:SIGN<i>:HANDover:CTYPE	380
PREPare:LTE:SIGN<i>:HANDover	380
PREPare:LTE:SIGN<i>:HANDover:ENHanced	381
PREPare:LTE:SIGN<i>:HANDover:EXternal:DESTination	382
PREPare:LTE:SIGN<i>:HANDover:EXternal:CDMA	383
PREPare:LTE:SIGN<i>:HANDover:EXternal:EVDO	383
PREPare:LTE:SIGN<i>:HANDover:EXternal:GSM	384
PREPare:LTE:SIGN<i>:HANDover:EXternal:LTE	385
PREPare:LTE:SIGN<i>:HANDover:EXternal:TDSCdma	385
PREPare:LTE:SIGN<i>:HANDover:EXternal:WCDma	386

PREPare:LTE:SIGN<i>:HANDover:DESTination <Destination>

Selects the handover destination. A complete list of all supported values can be displayed using [PREPare:LTE:SIGN<i>:HANDover:CATalog:DESTination?](#) on page 379.

Parameters:

<Destination> string

Example: See [Performing an inter-RAT handover](#)

Firmware/software: V3.0.10

Manual operation: See ["Target"](#) on page 135

PREPare:LTE:SIGN<i>:HANDover:CATalog:DESTination?

Lists all handover destinations that can be selected using [PREPare:LTE:SIGN<i>:HANDover:DESTination](#).

Return values:

<Destination> string
 Comma-separated list of all supported destinations. Each destination is represented as a string.
 "No Connection" means handover to another instrument.
 The "...Sig..." strings refer to signaling applications at the same instrument.

Example: See [Performing an inter-RAT handover](#)

Usage: Query only

Firmware/software: V3.0.10

Manual operation: See ["Target"](#) on page 135

PREPare:LTE:SIGN<i>:HANDover:MMODE <Mode>

Selects the mechanism to be used for handover to another signaling application.

Parameters:

<Mode> REDirection | MTCSfallback | HANDover
***RST:** RED

Example: See [Performing an inter-RAT handover](#)

Firmware/software: V3.2.50, V3.5.20 added HANDover

Options: R&S CMW-KS510 for MTCSfallback

Manual operation: See ["Mobility Mode"](#) on page 136

PREPare:LTE:SIGN<i>:HANDover:CTYPE <Type>

Selects the call type to be set up at the destination, for handover of VoLTE calls.

Parameters:

<Type> PSData | PSVolte
PSData: E2E packet data connection
PSVolte: Voice call, use handover with SRVCC
***RST:** PSD

Firmware/software: V3.5.40

Options: R&S CMW-KS510

Manual operation: See ["Connection Type"](#) on page 136

PREPare:LTE:SIGN<i>:HANDover <Band>, <DLChannel>, <DLBandwidth>, <AddSpecEmission>

Configures the destination parameters for an intra-RAT handover within the LTE signaling application.

The duplex mode of the destination is the same as the duplex mode of the source. For a handover with duplex mode change, see [PREPare:LTE:SIGN<i>:HANDOver:ENHanced](#) on page 381.

Parameters:

<Band>	FDD: UDEFined OB1 ... OB28 OB30 OB31 OB65 OB66 OB68 OB70 ... OB74 OB85 OB87 OB88 OB106 TDD: UDEFined OB33 ... OB45 OB48 OB50 ... OB53 OB250 Operating band of the handover destination
<DLChannel>	decimal DL channel number valid for the selected operating band. The related UL channel number is calculated and set automatically. For channel numbers depending on operating bands, see Section 2.2.23, "Operating bands" , on page 91. Range: depends on operating band
<DLBandwidth>	B014 B030 B050 B100 B150 B200 DL cell bandwidth (also used for UL) 1.4 MHz, 3 MHz, 5 MHz, 10 MHz, 15 MHz, 20 MHz *RST: B100
<AddSpecEmission>	NS01 ... NS288 Value signaled to the UE as additional ACLR and spectrum emission requirement *RST: NS01

Example: See [Performing an intra-RAT handover](#)

Firmware/software: V3.0.10, some bands added in later versions
V3.2.70: NS16 to NS18, NS20
V3.2.82: NS19, NS21 to NS24
V3.7.20: NS25 to NS32
V3.7.60: NS33 to NS288

Options: R&S CMW-KS525 for UDEFined, OB250

Manual operation: See ["Destination Parameters"](#) on page 136

PREPare:LTE:SIGN<i>:HANDOver:ENHanced <DuplexMode>, <Band>, <DLChannel>, <DLBandwidth>, <AddSpecEmission>

Configures the destination parameters for an intra-RAT handover within the LTE signaling application. The duplex mode of the destination is configurable.

Parameters:

<DuplexMode>	FDD TDD Duplex mode of the handover destination
--------------	--

<Band>	FDD: UDEFined OB1 ... OB28 OB30 OB31 OB65 OB66 OB68 OB70 ... OB74 OB85 OB87 OB88 OB106 TDD: UDEFined OB33 ... OB45 OB48 OB50 ... OB53 OB250 Operating band of the handover destination
<DLChannel>	decimal DL channel number valid for the selected operating band. The related UL channel number is calculated and set automatically. For channel numbers depending on operating bands, see Section 2.2.23, "Operating bands", on page 91. Range: depends on operating band
<DLBandwidth>	B014 B030 B050 B100 B150 B200 DL cell bandwidth (also used for UL) 1.4 MHz, 3 MHz, 5 MHz, 10 MHz, 15 MHz, 20 MHz *RST: B100
<AddSpecEmission>	NS01 ... NS288 Value signaled to the UE as additional ACLR and spectrum emission requirement *RST: NS01
Firmware/software:	V3.5.10, some bands added in later versions V3.7.20: NS25 to NS32, V3.7.60: NS33 to NS288
Options:	R&S CMW-KS525 for UDEFined, OB250
Manual operation:	See "Destination Parameters" on page 136

PREPare:LTE:SIGN<i>:HANDover:EXternal:DESTination <Destination>

Selects the target radio access technology for handover to another instrument.

Parameters:

<Destination>	LTE EVDO CDMA GSM WCDMA TDSCdma
*RST:	LTE

Example: See [Performing a handover to another instrument](#)

Firmware/software: V3.0.20

Manual operation: See ["Target"](#) on page 135

PREPare:LTE:SIGN<i>:HANDover:EXternal:CDMA <BandClass>, <DLChannel>
PREPare:LTE:SIGN<i>:HANDover:EXternal:EVDO <BandClass>, <DLChannel>

Configures the destination parameters for handover to a CDMA2000 or 1xEV-DO destination at another instrument.

Parameters:

<BandClass> USC | KCEL | NAPC | TACS | JTAC | KPCS | N45T | IM2K |
 NA7C | B18M | NA8S | PA4M | PA8M | IEXT | USPC | AWS |
 U25B | U25F | NA9C | PS7C | LO7C
USC: BC 0, US cellular
KCEL: BC 0, Korean cellular
NAPC: BC 1, North American PCS
TACS: BC 2, TACS band
JTAC: BC 3, JTACS band
KPCS: BC 4, Korean PCS
N45T: BC 5, NMT-450
IM2K: BC 6, IMT-2000
NA7C: BC 7, upper 700 MHz
B18M: BC 8, 1800-MHz band
NA9C: BC 9, North American 900 MHz
NA8S: BC 10, secondary 800 MHz
PA4M: BC 11, European 400-MHz PAMR
PA8M: BC 12, 800-MHz PAMR
IEXT: BC 13, IMT-2000 2.5-GHz extension
USPC: BC 14, US PCS 1900 MHz
AWS: BC 15, AWS band
U25B: BC 16, US 2.5-GHz band
U25F: BC 17, US 2.5 GHz forward
PS7C: BC 18, public safety band 700 MHz
LO7C: BC 19, lower 700 MHz
 *RST: USC

<DLChannel> decimal
 Channel number
 Range: depends on the band class, see table below
 *RST: 283

Example: See [Performing a handover to another instrument](#)

Firmware/software: V3.0.20

Manual operation: See "[Destination Parameters](#)" on page 136

Table 2-62: Channel number range depending on band class

Band class	Channel number
USC, KCEL	1 to 799, 991 to 1323
NAPC, IM2K	0 to 1199
TACS	0 to 1000, 1329 to 2108
JTAC	1 to 799, 801 to 1039, 1041 to 1199, 1201 to 1600

Band class	Channel number
KPCS	0 to 599
N45T	1 to 400, 472 to 871, 1039 to 1473, 1536 to 1715, 1792 to 2016
NA7C, PS7C	0 to 240
B18M	0 to 1499
NA9C	0 to 699
NA8S	0 to 919
PA4M	1 to 400, 472 to 871, 1536 to 1715
PA8M	0 to 239
IEXT	0 to 1399
USPC	0 to 1299
AWS	0 to 899
U25B, U25F	140 to 1459
LO7C	0 to 360

PREPare:LTE:SIGN<i>:HANDover:EXternal:GSM <Band>, <DLChannel>, <BandIndicator>

Configures the destination parameters for handover to a GSM destination at another instrument.

Parameters:

<Band>	G085 G09 G18 G19 GSM 850, GSM 900, GSM 1800, GSM 1900 *RST: G09
<DLChannel>	decimal Channel number used for the BCCH Range: depends on GSM band, see table below *RST: 20
<BandIndicator>	G18 G19 Band indicator for distinction of GSM 1800 and GSM 1900 bands. The two bands partially use the same channel numbers for different frequencies. *RST: G18

Example: See [Performing a handover to another instrument](#)

Firmware/software: V3.0.20

Manual operation: See ["Destination Parameters"](#) on page 136

Table 2-63: Channel number range depending on GSM band

Band	Channel number
G085	128 to 251
G09	0 to 124, 940 to 1023
G18	512 to 885
G19	512 to 810

PREPare:LTE:SIGN<i>:HANDover:EXternal:LTE <Band>, <DLChannel>

Configures the destination parameters for handover to an LTE destination at another instrument.

For channel number ranges depending on operating bands, see [Section 2.2.23, "Operating bands"](#), on page 91.

Parameters:

<Band>	UDEFined OB1 ... OB46 OB48 ... OB53 OB65 ... OB76 OB85 OB87 OB88 OB106 OB250 OB252 OB255 Operating band *RST: OB1
<DLChannel>	decimal Downlink channel number Range: depends on operating band *RST: 300

Example: See [Performing a handover to another instrument](#)

Firmware/software: V3.0.20, some bands added in later versions

Manual operation: See ["Destination Parameters"](#) on page 136

PREPare:LTE:SIGN<i>:HANDover:EXternal:TDSCdma <Band>, <DLChannel>

Configures the destination parameters for handover to a TD-SCDMA destination at another instrument.

Parameters:

<Band>	OB1 OB2 OB3 OB1: Band 1 (F), 1880 MHz to 1920 MHz OB2: Band 2 (A), 2010 MHz to 2025 MHz OB3: Band 3 (E), 2300 MHz to 2400 MHz *RST: OB1
<DLChannel>	decimal Downlink channel number The allowed range depends on the frequency band: OB1: 9400 to 9600 OB2: 10050 to 10125

OB3: 11500 to 12000

*RST: 10563

Example: See [Performing a handover to another instrument](#)

Firmware/software: V3.0.50

Manual operation: See ["Destination Parameters"](#) on page 136

PREPare:LTE:SIGN<i>:HANDover:EXternal:WCDMa <Band>, <DLChannel>

Configures the destination parameters for handover to a WCDMA destination at another instrument.

Parameters:

<Band> OB1 | OB2 | OB3 | OB4 | OB5 | OB6 | OB7 | OB8 | OB9 | OB10 |
OB11 | OB12 | OB13 | OB14 | OB19 | OB20 | OB21 | OB22 |
OB25 | OBS1 | OBS2 | OBS3 | OBL1 | OB26

OB1, ..., OB14: band I to XIV

OB19, ..., OB22: band XIX to XXII

OB25, OB26: band XXV, XXVI

OBS1: band S

OBS2: band S 170 MHz

OBS3: band S 190 MHz

OBL1: band L

*RST: OB1

<DLChannel> decimal
Downlink channel number
Range: Depends on operating band, see table below

Example: See [Performing a handover to another instrument](#)

Firmware/software: V3.0.20, V3.5.40: OB22/OB25, V3.7.10: OB26

Manual operation: See ["Destination Parameters"](#) on page 136

Table 2-64: Channel number range depending on operating band

Operating band	Channel number
OB1	10562 to 10838
OB2	412 to 687 (step 25), 9662 to 9938
OB3	1162 to 1513
OB4	1537 to 1738, 1887 to 2087 (step 25)
OB5	1007, 1012, 1032, 1037, 1062, 1087, 4357 to 4458
OB6	1037, 1062, 4387 to 4413
OB7	2237 to 2563, 2587 to 2912 (step 25)
OB8	2937 to 3088
OB9	9237 to 9387

Operating band	Channel number
OB10	3112 to 3388, 3412 to 3687 (step 25)
OB11	3712 to 3787
OB12	3842 to 3903, 3932, 3957, 3962, 3987, 3992
OB13	4017 to 4043, 4067, 4092
OB14	4117 to 4143, 4167, 4192
OB19	712 to 763, 787, 812, 837
OB20	4512 to 4638
OB21	862 to 912
OB22	4662 to 5038
OB25	5112 to 5413, 6292 to 6592 (step 25)
OB26	5762 to 5913, 5937, 5962, 5987, 5992, 6012, 6017, 6037, 6042, 6062, 6067, 6087
OBS1	10912 to 10988
OBS2	10900 to 10950
OBS3	10950 to 11000
OBL1	7637 to 7783, 7788 to 7933

2.6.4 Event log

The following commands retrieve event log entries and clear the log.

SENSe:LTE:SIGN<i>:ELOG:LAST?	387
SENSe:LTE:SIGN<i>:ELOG:ALL?	388
CLEAn:LTE:SIGN<i>:ELOG	388

SENSe:LTE:SIGN<i>:ELOG:LAST? [<HRES>]

Queries the latest entry of the event log.

Query parameters:

<HRES>	HRES
	If you omit this parameter, the timestamp resolution is 1 s (format "hh:mm:ss").
	If you send the value HRES, the timestamp resolution is 1 ms (format "hh:mm:ss.sss").

Return values:

<Timestamp>	string
	Timestamp of the entry
<Category>	INFO WARNing ERRor CONTinue
	Category of the entry, as indicated in the main view by an icon

<Event>	string Describes the event, e.g. "RRC Connection Established"
Usage:	Query only
Firmware/software:	V2.1.30, V3.7.10 <HRES>
Manual operation:	See "Event log entries" on page 121

SENSe:LTE:SIGN<i>:ELOG:ALL? [<HRES>]

Queries all entries of the event log.

For each entry, three parameters are returned, from oldest to latest entry: {<Timestamp>, <Category>, <Event>}_{entry 1}, {<Timestamp>, <Category>, <Event>}_{entry 2}, ...

Query parameters:

<HRES>	HRES If you omit this parameter, the timestamp resolution is 1 s (format "hh:mm:ss"). If you send the value HRES, the timestamp resolution is 1 ms (format "hh:mm:ss.sss").
--------	---

Return values:

<Timestamp>	string Timestamp of the entry
<Category>	INFO WARNING ERROR CONTINUE Category of the entry, as indicated in the main view by an icon
<Event>	string Describes the event, e.g. "RRC Connection Established"
Usage:	Query only
Firmware/software:	V2.1.30, V3.7.10 <HRES>
Manual operation:	See "Event log entries" on page 121

CLEAn:LTE:SIGN<i>:ELOG

Clears the event log.

Usage:	Event
Firmware/software:	V3.5.30
Manual operation:	See "Event log entries" on page 121

2.6.5 UE measurement report contents

The following commands retrieve UE measurement report contents for the serving cell and for measured neighbor cells.

SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRP?	389
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRP?	389
SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRP:RANGe?	389
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRP:RANGe?	389
SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRQ?	390
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRQ?	390
SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRQ:RANGe?	390
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRQ:RANGe?	390
SENSe:LTE:SIGN<i>:UEReport[:PCC]:SCeLI?	391
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCeLI?	391
SENSe:LTE:SIGN<i>:UEReport[:PCC]:SCeLI:RANGe?	391
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCeLI:RANGe?	391
SENSe:LTE:SIGN<i>:UEReport:NCELI:LTE:CELL<no>?	392
SENSe:LTE:SIGN<i>:UEReport:NCELI:LTE:CELL<no>:RANGe?	392
SENSe:LTE:SIGN<i>:UEReport:NCELI:GSM:CELL<no>?	393
SENSe:LTE:SIGN<i>:UEReport:NCELI:GSM:CELL<no>:RANGe?	393
SENSe:LTE:SIGN<i>:UEReport:NCELI:WCDMA:CELL<no>?	394
SENSe:LTE:SIGN<i>:UEReport:NCELI:WCDMA:CELL<no>:RANGe?	394
SENSe:LTE:SIGN<i>:UEReport:NCELI:CDMA:CELL<no>?	395
SENSe:LTE:SIGN<i>:UEReport:NCELI:EVDO:CELL<no>?	396
SENSe:LTE:SIGN<i>:UEReport:NCELI:TDSCdma:CELL<no>?	396
SENSe:LTE:SIGN<i>:UEReport:NCELI:TDSCdma:CELL<no>:RANGe?	397
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RRESult?	397
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:COCC?	397

SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRP?**SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRP?**

Returns the RSRP reported by the UE as dimensionless index.

Suffix:

<c> 1..7

Return values:

<RSRP> decimal
Range: 0 to 97

Usage: Query only

Firmware/software: V2.0.10, SCC command V3.2.70

Manual operation: See "[RSRP for serving LTE cell](#)" on page 122

SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRP:RANGe?**SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRP:RANGe?**

Returns the RSRP value range, corresponding to the RSRP index reported by the UE.

Suffix:

<c> 1..7

Return values:

<Lower> decimal
 Range: -140 dBm to -44 dBm
 Default unit: dBm

<Upper> decimal
 Range: -140 dBm to -44 dBm
 Default unit: dBm

Usage: Query only

Firmware/software: V2.0.10, SCC command V3.2.70

Manual operation: See ["RSRP for serving LTE cell"](#) on page 122

SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRQ?**SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRQ?**

Returns the RSRQ reported by the UE as dimensionless index.

Suffix:

<c> 1..7

Return values:

<RSRQ> decimal
 Range: -30 to 46

Usage: Query only

Firmware/software: V2.0.10, SCC command V3.2.70

Manual operation: See ["RSRQ for serving LTE cell"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRQ:RANGe?**SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RSRQ:RANGe?**

Returns the RSRQ value range, corresponding to the RSRQ index reported by the UE.

Suffix:

<c> 1..7

Return values:

<Lower> float
 Range: -34 dB to 2.5 dB
 Default unit: dB

<Upper> float
 Range: -34 dB to 2.5 dB
 Default unit: dB

Usage: Query only

Firmware/software: V2.0.10, SCC command V3.2.70

Manual operation: See ["RSRQ for serving LTE cell"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport[:PCC]:SCELI?
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCELI?

Returns measurement report values for the serving LTE cell.

Suffix:

<c> 1..7

Return values:

<RSRP> decimal
 RSRP as dimensionless index
 Range: 0 to 97

<RSRQ> decimal
 RSRQ as dimensionless index
 Range: -30 to 46

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50, SCC command V3.2.70

Manual operation: See ["RSRP for serving LTE cell"](#) on page 122

SENSe:LTE:SIGN<i>:UEReport[:PCC]:SCELI:RANGe?
SENSe:LTE:SIGN<i>:UEReport:SCC<c>:SCELI:RANGe?

Returns the value ranges corresponding to the dimensionless index values reported for the serving LTE cell.

Suffix:

<c> 1..7

Return values:

<RSRPlower> decimal
 RSRP minimum value
 Range: -140 dBm to -44 dBm
 Default unit: dBm

<RSRPupper> decimal
 RSRP maximum value
 Range: -140 dBm to -44 dBm
 Default unit: dBm

<RSRQlower> float
 RSRQ minimum value
 Range: -34 dB to 2.5 dB
 Default unit: dB

<RSRQupper>	float RSRQ maximum value Range: -34 dB to 2.5 dB Default unit: dB
Example:	See Querying UE measurement report contents
Usage:	Query only
Firmware/software:	V3.0.50, SCC command V3.2.70
Manual operation:	See "RSRP for serving LTE cell" on page 122

SENSe:LTE:SIGN<i>:UEReport:NCELL:LTE:CELL<no>?

Returns measurement report values for the LTE neighbor cell number <no>.

Suffix:	
<no>	1..16
Return values:	
<RSRP>	decimal RSRP as dimensionless index Range: 0 to 97
<RSRQ>	decimal RSRQ as dimensionless index Range: 0 to 34
Example:	See Querying UE measurement report contents
Usage:	Query only
Firmware/software:	V3.0.50
Options:	R&S CMW-KS510
Manual operation:	See "RSRP and RSRQ for LTE neighbor cells" on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELL:LTE:CELL<no>:RANGe?

Returns the value ranges corresponding to the dimensionless index values reported for the LTE neighbor cell number <no>.

Suffix:	
<no>	1..16
Return values:	
<RSRPlower>	decimal RSRP minimum value Range: -140 dBm to -44 dBm Default unit: dBm

<RSRPupper> decimal
 RSRP maximum value
 Range: -140 dBm to -44 dBm
 Default unit: dBm

<RSRQlower> float
 RSRQ minimum value
 Range: -19.5 dB to -3 dB
 Default unit: dB

<RSRQupper> float
 RSRQ maximum value
 Range: -19.5 dB to -3 dB
 Default unit: dB

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["RSRP and RSRQ for LTE neighbor cells"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELL:GSM:CELL<no>?

Returns the RSSI value reported as dimensionless index for the GSM neighbor cell number <no>.

Suffix:
 <no> 1..4

Return values:
 <RSSI> decimal
 Range: 0 to 63

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["RSSI for GSM"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELL:GSM:CELL<no>:RANGe?

Returns the value range corresponding to the dimensionless RSSI index value reported for the GSM neighbor cell number <no>.

Suffix:
 <no> 1..4

Return values:

<RSSIlower>	decimal RSSI minimum value Range: -110 dBm to -48 dBm Default unit: dBm
<RSSIupper>	decimal RSSI maximum value Range: -110 dBm to -48 dBm Default unit: dBm

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["RSSI for GSM"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELL:WCDMA:CELL<no>?

Returns measurement report values for the WCDMA neighbor cell number <no>.

Suffix:

<no> 1..4

Return values:

<RSCP>	decimal RSCP as dimensionless index Range: -5 to 91
<EcNO>	decimal Ec/No as dimensionless index Range: 0 to 49

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["RSCP and EcNO for WCDMA"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELL:WCDMA:CELL<no>:RANGe?

Returns the value ranges corresponding to the dimensionless index values reported for the WCDMA neighbor cell number <no>.

Suffix:

<no> 1..4

Return values:

<RSCPlower>	decimal RSCP minimum value Range: -120 dBm to -25 dBm Default unit: dBm
<RSCPupper>	decimal RSCP maximum value Range: -120 dBm to -25 dBm Default unit: dBm
<EcNolower>	float Ec/No minimum value Range: -24 dB to 0 dB Default unit: dB
<EcNOupper>	float Ec/No maximum value Range: -24 dB to 0 dB Default unit: dB

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["RSCP and EcNO for WCDMA"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELI:CDMA:CELL<no>?

Returns measurement report values for the CDMA2000 neighbor cell number <no>.

Suffix:

<no> 1..4

Return values:

<pilotPnPhase>	decimal Reported pilot PN phase value Range: 0 PN chips to 32767 PN chips Default unit: PN chips
<pilotStrength>	decimal Reported pilot strength value Range: 0 to 63

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["pilot Pn Phase and pilot Strength for CDMA2000 and 1xEV-DO"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELI:EVDO:CELL<no>?

Returns measurement report values for the 1xEV-DO neighbor cell number <no>.

Suffix:

<no> 1..4

Return values:

<pilotPnPhase> decimal
Reported pilot PN phase value
Range: 0 PN chips to 32767 PN chips
Default unit: PN chips

<pilotStrength> decimal
Reported pilot strength value
Range: 0 to 63

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["pilot Pn Phase and pilot Strength for CDMA2000 and 1xEV-DO"](#) on page 123

SENSe:LTE:SIGN<i>:UEReport:NCELI:TDSCdma:CELL<no>?

Returns measurement report values for the TD-SCDMA neighbor cell number <no>.

Suffix:

<no> 1..4

Return values:

<RSCP> decimal
RSCP as dimensionless index
Range: -5 to 91

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.2.20

Options: R&S CMW-KS510

Manual operation: See ["RSCP for TD-SCDMA"](#) on page 124

SENSe:LTE:SIGN<i>:UEReport:NCELI:TDSCdma:CELL<no>:RANGe?

Returns the value ranges corresponding to the dimensionless index values reported for the TD-SCDMA neighbor cell number <no>.

Suffix:

<no> 1..4

Return values:

<RSCPlower> decimal
RSCP minimum value
Range: -120 dBm to -25 dBm
Default unit: dBm

<RSCPupper> decimal
RSCP maximum value
Range: -120 dBm to -25 dBm
Default unit: dBm

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.2.20

Options: R&S CMW-KS510

Manual operation: See ["RSCP for TD-SCDMA"](#) on page 124

SENSe:LTE:SIGN<i>:UEReport:SCC<c>:RRESult?

Returns the RSSI value reported as dimensionless index for an SCC DL with LAA.

Suffix:

<c> 1..7

Return values:

<Result> decimal
Range: 0 to 76

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["RSSI Measurements for LAA"](#) on page 124

SENSe:LTE:SIGN<i>:UEReport:SCC<c>:COCC?

Returns the channel occupancy value reported for an SCC DL with LAA.

Suffix:

<c> 1..7

Return values:

<Occupancy> decimal
 Range: 0 % to 100 %
 Default unit: %

Example: See [Querying UE measurement report contents](#)

Usage: Query only

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Channel Occupancy for LAA"](#) on page 124

2.6.6 UE capabilities

The commands in this section retrieve information about the connected mobile as shown in the "UE Capabilities" area of the main view.

For some parameters, three remote control commands are listed:

- First command:
Queries the parameter in the upper part of the capability report. Features indicated as "supported" are supported for FDD and TDD.
- Command with additional :FAUeeutra: mnemonic:
Queries the parameter in section "fdd Add UE-EUTRA Capabilities". Features indicated as "supported" are supported only for FDD, not for TDD.
- Command with additional :TAUeeutra: mnemonic:
Queries the parameter in section "tdd Add UE-EUTRA Capabilities". Features indicated as "supported" are supported only for TDD, not for FDD.

2.6.6.1 General UE capability information

The following commands query the information at the highest level of the "UE Capabilities" section (or the highest level of the additional FDD section and additional TDD section).

SENSe:LTE:SIGN<i>:UECapability:ASRelease?	399
SENSe:LTE:SIGN<i>:UECapability:UECategory?	399
SENSe:LTE:SIGN<i>:UECapability:UECategory:DL:ENHanced?	399
SENSe:LTE:SIGN<i>:UECapability:UECategory:UL:ENHanced?	399
SENSe:LTE:SIGN<i>:UECapability:FGIndicators?	400
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators?	400
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators?	400
SENSe:LTE:SIGN<i>:UECapability:FGIndicators:RNADd?	400
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators:RNADd?	400
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators:RNADd?	400
SENSe:LTE:SIGN<i>:UECapability:FGIndicators:RTEN?	400
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators:RTEN?	400

SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGINdicator:RTEN?	400
SENSe:LTE:SIGN<i>:UECapability:DTYPE?	401
SENSe:LTE:SIGN<i>:UECapability:RREPort?	401

SENSe:LTE:SIGN<i>:UECapability:ASRelease?

Returns the "Access Stratum Release" according to the UE capability information.

Return values:

<AccStratRelease> REL8 | REL9 | REL10 | REL11 | REL12 | REL13 | REL14 |
REL15 | REL16

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.10, V3.7.31 added REL13 and REL 14
V4.0.110 added REL15 and REL16

Manual operation: See ["Access Stratum Release"](#) on page 297

SENSe:LTE:SIGN<i>:UECapability:UECategory?

Returns the UE category according to the UE capability information.

Return values:

<UEcategory> decimal

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.70

Manual operation: See ["UE Category"](#) on page 297

SENSe:LTE:SIGN<i>:UECapability:UECategory:DL:ENHanced?

Returns the DL UE category according to the UE capability information.

Return values:

<UEcategory> string

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.60

Manual operation: See ["UE Category"](#) on page 297

SENSe:LTE:SIGN<i>:UECapability:UECategory:UL:ENHanced?

Returns the UL UE category according to the UE capability information.

Return values:

<UEcategory> string

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.7.60**Manual operation:** See ["UE Category"](#) on page 297**SENSe:LTE:SIGN<i>:UECapability:FGIndicators?****SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators?**

Returns the "featureGroupIndicators" contained in the UE capability information.

The 32-bit value contains one bit per feature group (1 = supported, 0 = not supported).

Return values:

<FeatureGroupInd> binary

Range: #B0 to #B11111111111111111111111111111111

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V2.1.30, FAUeeutra / TAUeeutra V3.2.80**Manual operation:** See ["Feature Group Indicators"](#) on page 297**SENSe:LTE:SIGN<i>:UECapability:FGIndicators:RNADd?****SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators:RNADd?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators:RNADd?**

Returns the "featureGroupIndRel9Add-r9" contained in the UE capability information.

The 32-bit value contains one bit per feature group (1 = supported, 0 = not supported).

Return values:

<FeatureGroupInd> binary

Range: #B0 to #B11111111111111111111111111111111

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.2.80**Manual operation:** See ["Feature Group Indicators Rel9 Add"](#) on page 297**SENSe:LTE:SIGN<i>:UECapability:FGIndicators:RTEN?****SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:FGIndicators:RTEN?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:FGIndicators:RTEN?**

Returns the "featureGroupIndRel10-r10" contained in the UE capability information.

The 32-bit value contains one bit per feature group (1 = supported, 0 = not supported).

Return values:

<FeatureGroupInd> binary
Range: #B0 to #B11111111111111111111111111111111

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Feature Group Indicators Rel 10"](#) on page 298

SENSe:LTE:SIGN<i>:UECapability:DTYPE?

Returns whether the UE benefits from NW-based battery consumption optimization or not.

Return values:

<DeviceType> NFBcopt | NAV
NFBcopt: UE does not benefit
NAV: UE does benefit

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Device Type"](#) on page 298

SENSe:LTE:SIGN<i>:UECapability:RREPort?

Returns whether the UE supports the delivery of RACH reports or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["rach Report"](#) on page 298

2.6.6.2 PDCP UE capabilities

The following commands query in which way the UE supports the PDCP protocol.

SENSe:LTE:SIGN<i>:UECapability:PDCP:SRPProfiles?	402
SENSe:LTE:SIGN<i>:UECapability:PDCP:MRCSessions?	402
SENSe:LTE:SIGN<i>:UECapability:PDCP:SNExtension?	403
SENSe:LTE:SIGN<i>:UECapability:PDCP:SRCCContinue?	403

SENSe:LTE:SIGN<i>:UECapability:PDCCP:SRPRofiles?

Returns UE capability information indicating the support of the individual robust header compression (ROHC) profiles.

Return values:

<ROHC_RTP>	OFF ON Support of profile 0x0001, ROHC RTP
<ROHC_UDP>	OFF ON Support of profile 0x0002, ROHC UDP
<ROHC_ESP>	OFF ON Support of profile 0x0003, ROHC ESP
<ROHC_IP>	OFF ON Support of profile 0x0004, ROHC IP
<ROHC_TCP>	OFF ON Support of profile 0x0006, ROHC TCP
<ROHCv2_RTP>	OFF ON Support of profile 0x0101, ROHCv2 RTP
<ROHCv2_UDP>	OFF ON Support of profile 0x0102, ROHCv2 UDP
<ROHCv2_ESP>	OFF ON Support of profile 0x0103, ROHCv2 ESP
<ROHCv2_IP>	OFF ON Support of profile 0x0104, ROHCv2 IP

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["ROHC RTP to ROHCv2 IP"](#) on page 299

SENSe:LTE:SIGN<i>:UECapability:PDCCP:MRCSessions?

Returns the maximum number of ROHC context sessions supported by the UE.

Return values:

<MaxSessions>	CS2 CS4 CS8 CS12 CS16 CS24 CS32 CS48 CS64 CS128 CS256 CS512 CS1024 CS16384
---------------	--

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["Max Num ROHC Context Sessions"](#) on page 299

SENSe:LTE:SIGN<i>:UECapability:PDCCP:SNExtension?

Returns whether the UE supports PDCCP SN extension.

Return values:

<Extension> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["PDCCP SN Extension"](#) on page 299

SENSe:LTE:SIGN<i>:UECapability:PDCCP:SRCCContinue?

Returns whether the UE supports ROHC context continuation during handover.

Return values:

<Supportrrcc> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Support ROHC Context Continue"](#) on page 299

2.6.6.3 Physical layer UE capabilities

The following commands query the physical layer capabilities of the UE.

SENSe:LTE:SIGN<i>:UECapability:PLAYEr:UTASupported?	404
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:UTASupported?	404
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:UTASupported?	404
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:USRSsupport?	404
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:USRSsupport?	404
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:USRSsupport?	404
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:EDLFSupport?	405
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:EDLTsupport?	405
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:TAPPSupport?	405
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:TAPPSupport?	405
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:TAPPSupport?	405
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:TWEFSupport?	406
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:TWEFSupport?	406
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:TWEFSupport?	406
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:PDSsupport?	406
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:PDSsupport?	406
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:PDSsupport?	406
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:CCSSupport?	406
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:CCSSupport?	406
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:CCSSupport?	406

SENSe:LTE:SIGN<i>:UECapability:PLAYEr:SPPSupport?	407
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:SPPSupport?	407
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:SPPSupport?	407
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:MCPCsupport?	407
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:MCPCsupport?	407
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:MCPCsupport?	407
SENSe:LTE:SIGN<i>:UECapability:PLAYEr:NURClst?	407
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:NURClst?	407
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Returns whether the UE supports transmit antenna selection or not.

Return values:

<UE_TXantSelSupp> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30, FAUeeutra / TAUeeutra V3.2.80

Manual operation: See ["UE TX Antenna Selection Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYEr:USRSsupport?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYEr:USRSsupport?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYEr:USRSsupport?**

Returns whether the UE supports PDSCH transmission mode 7 for FDD or not.

Return values:

<UEspRefSigsSupp> OFF | ON

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V2.1.30, FAUeeutra / TAUeeutra V3.2.80
Manual operation: See ["UE-Specific Ref. Sigs. Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYer:EDLFSupport?

Returns whether the UE supports enhanced dual layer (PDSCH TM 8) for FDD or not.

Return values:

<EnDualLayFDDsup> OFF | ON

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V3.2.80
Manual operation: See ["enhanced Dual layer FDD/TDD Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYer:EDLTSupport?

Returns whether the UE supports enhanced dual layer (PDSCH TM 8) for TDD or not.

Return values:

<EnDualLayTDDsup> OFF | ON

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V3.2.80
Manual operation: See ["enhanced Dual layer FDD/TDD Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYer:TAPPSupport?

SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:TAPPSupport?

SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:TAPPSupport?

Returns whether the UE supports transmit diversity for specific PUCCH formats.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V3.2.80
Manual operation: See ["Two Antenna Ports for PUCCH Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYer:TWEFsupport?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:TWEFsupport?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:TWEFsupport?**

Returns whether the UE supports PDSCH TM 9 with 8 CSI reference signal ports for FDD.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.2.80**Manual operation:** See ["tm9 with 8TX FDD Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYer:PDSsupport?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:PDSsupport?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:PDSsupport?**

Returns whether the UE supports PMI disabling.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.2.80**Manual operation:** See ["pmi Disabling Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYer:CCSSupport?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:CCSSupport?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:CCSSupport?**

Returns whether the UE supports cross-carrier scheduling for CA.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.2.80**Manual operation:** See ["cross Carrier Scheduling Supported"](#) on page 300

SENSe:LTE:SIGN<i>:UECapability:PLAYer:SPPSupport?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:SPPSupport?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:SPPSupport?

Returns whether the UE supports the simultaneous transmission of PUCCH and PUSCH.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See "[simultaneous PUCCH PUSCH Supported](#)" on page 301

SENSe:LTE:SIGN<i>:UECapability:PLAYer:MCPCsupport?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:MCPCsupport?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:MCPCsupport?

Returns whether the UE supports multi-cluster PUSCH transmission within a CC.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See "[multi Cluster PUSCH within CC Supported](#)" on page 301

SENSe:LTE:SIGN<i>:UECapability:PLAYer:NURClst?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:PLAYer:NURClst?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:NURClst?

Returns a list of values, indicating whether the UE supports non-contiguous UL resource allocations within a CC for the individual E-UTRA operating bands.

Return values:

<SupportedBand> OFF | ON

256 values: user-defined band, band 1 to band 255

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

V3.5.40: result array restructured

Manual operation: See "[non Contiguous UL RA within CC List Supported](#)" on page 301

SENSe:LTE:SIGN<i>:UECapability:PLAYer:CIHandl?

Returns whether the UE supports CRS interference handling.

Return values:

<InterfHandl> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[CRS InterfHandl](#)" on page 302

SENSe:LTE:SIGN<i>:UECapability:PLAYer:EPDCch?

Returns whether the UE supports DCI reception via UE-specific search space on enhanced PDCCH.

Return values:

<Epdccch> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[ePDCCH](#)" on page 302

SENSe:LTE:SIGN<i>:UECapability:PLAYer:MACReporting?

Returns whether the UE supports multi-cell HARQ ACK, periodic CSI reporting and SR on PUCCH format 3.

Return values:

<Reporting> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[multiACK CSI Reporting](#)" on page 302

SENSe:LTE:SIGN<i>:UECapability:PLAYer:SCIHandl?

Returns whether the UE supports synchronization signal and common channel interference handling.

Return values:

<Handl> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["SS CCH InterfHandl"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:TSSubframe?

Returns whether the UE supports TDD special subframe as defined in 3GPP TS 36.211.

Return values:
<TddSpecialSf> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["TDD SpecialSubframe"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:TDPChselect?

Returns whether the UE supports transmit diversity for PUCCH format 1b with channel selection.

Return values:
<Tdpch> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["txDiv PUCCH1b ChSelect"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:ULComp?

Returns whether the UE supports UL coordinated multi-point operation.

Return values:
<Comp> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["UL CoMP"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:ITCWithdiff?

Returns whether the UE supports inter-band TDD CA with different UL/DL configuration combinations.

Return values:

<InterBand> string
Two bits, for example "b00" or "b01"

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["interBand TDD CA With Different Config"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:EHPFdd?

Returns whether the UE supports enhanced HARQ pattern for TTI bundling operation for FDD.

Return values:

<EHPattern> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["e HARQ Pattern FDD"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:EFTCodebook?

Returns whether the UE supports enhanced 4 TX codebook.

Return values:

<Codebook> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["enhanced 4 Tx Codebook"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:TFCPCellDplx?

Returns whether the UE supports PCell in any supported band combination including at least one FDD band and at least one TDD band.

Return values:

<CellDplx> string
Two bits, for example "b00" or "b10"

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V3.5.50
Manual operation: See ["tdd FDD CA PCell Duplex"](#) on page 303

SENSe:LTE:SIGN<i>:UECapability:PLAYer:TRCTddpcell?

Returns whether the UE supports TDD UL/DL reconfiguration for TDD serving cell via monitoring PDCCH on a TDD PCell.

Return values:

<Pcell> OFF | ON

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V3.5.50
Manual operation: See ["phy TDD ReConfig TDD PCell"](#) on page 304

SENSe:LTE:SIGN<i>:UECapability:PLAYer:TRCFddpcell?

Returns whether the UE supports TDD UL/DL reconfiguration for TDD serving cell via monitoring PDCCH on an FDD PCell.

Return values:

<Pcell> OFF | ON

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V3.5.50
Manual operation: See ["phy TDD ReConfig FDD PCell"](#) on page 304

SENSe:LTE:SIGN<i>:UECapability:PLAYer:PFMode?

Returns whether the UE supports PUSCH feedback mode 3-2.

Return values:

<Mode> OFF | ON

Example: See [Querying UE capability report contents](#)
Usage: Query only
Firmware/software: V3.5.50
Manual operation: See ["pusch Feedback Mode"](#) on page 304

SENSe:LTE:SIGN<i>:UECapability:PLAYer:PSPSfset?

Returns whether the UE supports subframe set dependent UL power control for PUSCH and SRS.

Return values:

<SfSet> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[pusch SRS PowerControl SubframeSet](#)" on page 304

SENSe:LTE:SIGN<i>:UECapability:PLAYer:CSFSet?

Returns whether the UE supports R12 DL CSI subframe set configuration.

Return values:

<Subframeset> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[CSI Subframe Set](#)" on page 304

SENSe:LTE:SIGN<i>:UECapability:PLAYer:NRRT?

Returns whether the UE supports TTI bundling without resource allocation restriction.

Return values:

<Norr> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[no Resource Restriction For TTI Bundling](#)" on page 304

SENSe:LTE:SIGN<i>:UECapability:PLAYer:DSDCell?

Returns whether the UE supports discovery signal detection for deactivated SCells.

Return values:

<Discovery> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["discovery Signals In Deact SCell"](#) on page 304

2.6.6.4 RF UE capabilities

The following commands query the supported E-UTRA operating bands, band combinations and other RF capabilities.

SENSe:LTE:SIGN<i>:UECapability:RF:SUPPorted?	413
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SENSe:LTE:SIGN<i>:UECapability:RF:DL<qam>?	414
SENSe:LTE:SIGN<i>:UECapability:RF:UL<qam>?	414
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:BCSet?	415
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:EUTRa<BandNr>?	415
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:EUTRa<BandNr>: BClass:UL?	416
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:EUTRa<BandNr>: BClass:DL?	416
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:EUTRa<BandNr>: MCAPability:UL?	416
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:EUTRa<BandNr>: MCAPability:DL?	416
SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:EUTRa<BandNr>: SCPRoc?	417
SENSe:LTE:SIGN<i>:UECapability:RF:MTADvance?	417
SENSe:LTE:SIGN<i>:UECapability:RF:SRTX?	417
SENSe:LTE:SIGN<i>:UECapability:RF:DCSupport:ASYNchronous?	418
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SENSe:LTE:SIGN<i>:UECapability:RF:MMPRbehavior?	420

SENSe:LTE:SIGN<i>:UECapability:RF:SUPPorted?

Returns a list of values indicating the support of the individual E-UTRA operating bands by the UE.

Return values:

<SupportedBand> OFF | ON
256 values: user-defined band, band 1 to band 255

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.0.30
V3.5.40: result array restructured

Manual operation: See ["Supported Bands"](#) on page 306

SENSe:LTE:SIGN<i>:UECapability:RF:HDUPlex?

Returns a list of values indicating whether the UE supports only half duplex operation for the individual E-UTRA operating bands.

Return values:

<HalfDuplex> OFF | ON
256 values: user-defined band, band 1 to band 255

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.0.30
V3.5.40: result array restructured

Manual operation: See ["Supported Bands"](#) on page 306

SENSe:LTE:SIGN<i>:UECapability:RF:DL<qam>?

Returns a list of values indicating whether the UE supports DL 256-QAM in the individual E-UTRA operating bands.

Suffix:

<qam> 256

Return values:

<Capabilities> 0 | 1
256 values: user-defined band, band 1 to band 255

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Supported Bands"](#) on page 306

SENSe:LTE:SIGN<i>:UECapability:RF:UL<qam>?

Returns a list of values indicating whether the UE supports UL 64-QAM in the individual E-UTRA operating bands.

Suffix:

<qam> 64

Return values:

<Capabilities> 0 | 1
256 values: user-defined band, band 1 to band 255

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Supported Bands"](#) on page 306

SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:BCSet?

Returns a list of binary numbers, indicating which bandwidth combination sets the UE supports for the individual carrier aggregation band combinations.

Suffix:

<Number> 1020

Return values:

<Band> binary

Comma-separated list of binary numbers, one binary number per band combination (combination 0 to n)

Each binary number indicates which bandwidth combination sets are supported for the band combination. The leftmost bit corresponds to set 0, the next bit to set 1, and so on. "0" means not supported. "1" means supported.

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Band Combinations vs Parameter"](#) on page 307

SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:EUTRa<BandNr>?

Returns the operating band combinations supported for carrier aggregation.

Suffix:

<Number> 1020, 1090
Selects the UE capability report element to be evaluated:
RF-Parameters-v1020 or RF-Parameters-v1090

<BandNr> 1..4
Selects which band of the band combinations is returned

Return values:

<Band> UDEFined | OB1 | ... | OB46 | OB48 | ... | OB53 | OB65 | ... | OB76 | OB85 | OB87 | OB88 | OB106 | OB250 | OB252 | OB255
Comma-separated list of bands, one band per band combination (combination 0 to n)

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80, some bands added in later versions
V3.5.20: band no 3+4

Manual operation: See ["Band Combinations vs Parameter"](#) on page 307

**SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:BCLass:UL?**

**SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:BCLass:DL?**

Returns the bandwidth classes supported by the UE in the uplink or downlink. The information is returned for a selected band of all supported carrier aggregation band combinations.

Suffix:

<Number> 1020

<BandNr> 1..4

Selects for which band of the band combinations the information is returned

Return values:

<BandwidthClass> string

Comma-separated list of strings, one string per band combination (combination 0 to n)

Each string indicates the bandwidth classes supported for the selected band (<BandNr>) of the combination, for example "abc".

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80, V3.5.20 band 3+4

Manual operation: See "[Bandwidth Class...](#)" on page 307

**SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:MCAPability:UL?**

**SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>:
EUTRa<BandNr>:MCAPability:DL?**

Returns the number of layers supported by the UE for spatial multiplexing in the uplink or downlink. The information is returned for a selected band of all supported carrier aggregation band combinations.

Suffix:

<Number> 1020

<BandNr> 1..4

Selects for which band of the band combinations the information is returned

Return values:

<MIMOCapability> decimal

Comma-separated list of numbers, 26 numbers per band combination (combination 0 to n)

The 26 numbers indicate the supported number of layers for bandwidth class "a" to "z".

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80, V3.5.20 band 3+4

Manual operation: See ["MIMO Capability..."](#) on page 307

SENSe:LTE:SIGN<i>:UECapability:RF:BCOMbination:V<Number>: EUTRa<BandNr>:SCPRoc?

Returns the maximum number of CSI processes supported by the UE. The information is returned for a selected band of all supported carrier aggregation band combinations.

Suffix:

<Number> 1020

<BandNr> 1..4

Selects for which band of the band combinations the information is returned

Return values:

<Proc> N1 | N3 | N4

Comma-separated list of values, one value per band combination (combination 0 to n)

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Supported CSI-Proc"](#) on page 307

SENSe:LTE:SIGN<i>:UECapability:RF:MTADvance?

Returns whether the UE supports multiple timing advances.

Return values:

<Timing> decimal

Comma-separated list of values, one value per band combination (combination 0 to n)

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Multiple Timing Advance"](#) on page 308

SENSe:LTE:SIGN<i>:UECapability:RF:SRTX?

Returns whether the UE supports the simultaneous reception and transmission on different bands.

Return values:

<Simultaneous> 0 | 1

Comma-separated list of values, one value per band combination (combination 0 to n)

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.7.30**Manual operation:** See ["Simultaneous Rx-Tx"](#) on page 308**SENSe:LTE:SIGN<i>:UECapability:RF:DCSupport:ASYNchronous?**

Returns whether the UE supports asynchronous DC and power control mode 2.

Return values:

<Support> 0 | 1

Comma-separated list of values, one value per band combination (combination 0 to n)

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.7.30**Manual operation:** See ["DC-Support"](#) on page 308**SENSe:LTE:SIGN<i>:UECapability:RF:DCSupport:SCGRouping?**

Returns the value received as "supportedCellGrouping". It indicates for which mapping of serving cells to the first and second cell groups the UE supports asynchronous DC.

Return values:

<Support> decimal

Comma-separated list of values, one value per band combination (combination 0 to n)

Example: See [Querying UE capability report contents](#)**Usage:** Query only**Firmware/software:** V3.7.30**Manual operation:** See ["DC-Support"](#) on page 308**SENSe:LTE:SIGN<i>:UECapability:RF:SNCap?**

Returns the bitstring from the element "supportedNAICS-2CRS-AP".

Return values:

<Naics> string
Comma-separated list of strings, one string per band combination (combination 0 to n)

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Supported NAICS-2CRS-AP"](#) on page 308

SENSe:LTE:SIGN<i>:UECapability:RF:FBRetrieval?

Returns whether the UE supports the reception of "requestedFrequencyBands".

Return values:

<Retrieval> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Freq Band Retrieval"](#) on page 308

SENSe:LTE:SIGN<i>:UECapability:RF:RBANDs?

Returns all frequency bands requested by E-UTRAN.

Return values:

<RequestedBands> UDEFined | OB1 | ... | OB46 | OB48 | ... | OB53 | OB65 | ... | OB76 | OB85 | OB87 | OB88 | OB106 | OB250 | OB252 | OB255
Comma-separated list of 64 values
Typically, fewer than 64 bands are requested and the remaining values are filled with NAV.

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50, some bands added in later versions

Manual operation: See ["Requested Frequency Bands"](#) on page 308

SENSe:LTE:SIGN<i>:UECapability:RF:FBPadjust?

Returns whether the UE supports the prioritization of frequency bands as requested by "freqBandIndicatorPriority-r12".

Return values:

<Adjustment> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Freq Band Priority Adjustment"](#) on page 308

SENSe:LTE:SIGN<i>:UECapability:RF:MMPRbehavior?

Returns which MPR/A-MPR behaviors the UE supports.

Return values:

<Behavior> string
Sequence of bits
The leftmost bit refers to behavior 0, the next bit to behavior 1, and so on.
1 means supported. 0 means not supported.

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.20

Manual operation: See ["modifiedMPR-Behavior"](#) on page 309

2.6.6.5 Measurement UE capabilities

The following commands query the UE capabilities for channel measurements.

SENSe:LTE:SIGN<i>:UECapability:MEAS:IFNGaps?	420
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:UFDD?	421
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:UTDD<n>?	422
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:GERan?	423
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:CHRPd?	423
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:CXRTt?	424
SENSe:LTE:SIGN<i>:UECapability:MEAS:IFNGaps:V<number>?	425
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:UFDD?	426
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:UTDD<n>?	426
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:GERan?	427
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:CHRPd?	428
SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:CXRTt?	429
SENSe:LTE:SIGN<i>:UECapability:MEAS:RMWideband?	429
SENSe:LTE:SIGN<i>:UECapability:MEAS:BFInterrupt?	430
SENSe:LTE:SIGN<i>:UECapability:MEAS:RCORreporting?	430

SENSe:LTE:SIGN<i>:UECapability:MEAS:IFNGaps? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band and measuring on (another) specific E-UTRA band.

The full list contains 256 times 256 values. The 256 values/repetitions correspond to the LTE bands. The list is ordered as follows:

{measured band: user-defined, 1, 2, ..., 255}_{used band: user-defined}

{measured band: user-defined, 1, 2, ..., 255}_{used band: 1, ...,}

{measured band: user-defined, 1, 2, ..., 255}_{used band: 255}

Via the optional parameter <Index>, you can alternatively query the list for one measured band:

{used band: user-defined, 1, 2, ..., 255}_{measured band <Index>}

Query parameters:

<Index> UDEFined | OB1 | ... | OB46 | OB48 | ... | OB53 | OB65 | ... |
OB76 | OB85 | OB87 | OB88 | OB106 | OB250 | OB252 | OB255
Selects the measured E-UTRA band, for which the list is returned.

Return values:

<Value> OFF | ON
Without <Index>: 256 x 256 = 65536 values
With <Index>: 256 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.0.30
V3.2.80: added <Index>, some bands added in later versions
V3.5.40: result array restructured

Manual operation: See ["Inter-Freq Need for Gaps"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:UFDD? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band and measuring on a specific UTRA FDD band.

The full list contains 32 times 256 values. Each block of 32 values corresponds to the UTRA FDD bands. The 256 repetitions correspond to the E-UTRA bands:

{measured band: 1, 2, ..., 32}_{used band: user-defined}

{measured band: 1, 2, ..., 32}_{used band: 1, ...,}

{measured band: 1, 2, ..., 32}_{used band: 255}

Via the optional parameter <Index>, you can alternatively query the list for a single UTRA FDD band:

{used band: user-defined, 1, 2, ..., 255}_{measured band <Index>}

Query parameters:

<Index> OB1 | OB2 | ... | OB32
Selects the measured UTRA FDD band, for which the list is returned.

Return values:

<Value> OFF | ON
 Without <Index>: 32 x 256 = 8192 values
 With <Index>: 256 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.0.30
 V3.2.80: added <Index>
 V3.5.40: result array restructured

Manual operation: See ["Inter-RAT Need for Gaps"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:UTDD<n>? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band and measuring on a specific UTRA TDD band.

The full list contains 32 times 256 values. Each block of 32 values corresponds to the UTRA TDD bands. The 256 repetitions correspond to the E-UTRA bands:

{measured band: 1, 2, ..., 32}_{used band: user-defined}

{measured band: 1, 2, ..., 32}_{used band: 1, ...,}

{measured band: 1, 2, ..., 32}_{used band: 255}

Via the optional parameter <Index>, you can alternatively query the list for a single UTRA TDD band:

{used band: user-defined, 1, 2, ..., 255}_{measured band <Index>}

Suffix:

<n> 128

Query parameters:

<Index> OB1 | OB2 | ... | OB32
 Selects the measured UTRA TDD band, for which the list is returned.

Return values:

<Value> OFF | ON
 Without <Index>: 32 x 256 = 8192 values
 With <Index>: 256 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80
 V3.5.40: result array restructured

Manual operation: See ["Inter-RAT Need for Gaps"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:GERan? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band and measuring on a specific GERAN band.

The full list contains 11 times 256 values. Each block of 11 values corresponds to the following GERAN bands: GSM 450, GSM 480, GSM 710, GSM 750, GSM 810, GSM 850, P-GSM 900, E-GSM 900, R-GSM 900, GSM 1800, GSM 1900. The 256 repetitions correspond to the E-UTRA bands:

{measured band: GSM 450, GSM 480, ..., GSM 1900}_{used band: user-defined,}

{measured band: GSM 450, GSM 480, ..., GSM 1900}_{used band: 1, ...,}

{measured band: GSM 450, GSM 480, ..., GSM 1900}_{used band: 256}

Via the optional parameter <Index>, you can alternatively query the list for a single GERAN band:

{used band: user-defined, 1, 2, ..., 255}_{measured band <Index>}

Query parameters:

<Index> G045 | G048 | G071 | G075 | G081 | G085 | G09P | G09E |
G09R | G18 | G19

Selects the measured GERAN band, for which the list is returned.

Return values:

<Value> OFF | ON
Without <Index>: 11 x 256 = 2816 values
With <Index>: 256 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.0.30
V3.2.80: added <Index>
V3.5.40: result array restructured

Manual operation: See ["Inter-RAT Need for Gaps"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:CHRPd? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band and measuring on a specific CDMA2000 HRPD band.

The full list contains 18 times 256 values. Each block of 18 values corresponds to the CDMA2000 band classes. The 256 repetitions correspond to the E-UTRA bands:

{measured band: 0, 1, ..., 17}_{used band: user-defined,}

{measured band: 0, 1, ..., 17}_{used band: 1, ...,}

{measured band: 0, 1, ..., 17}_{used band: 256}

Via the optional parameter <Index>, you can alternatively query the list for a single CDMA2000 band class:

{used band: user-defined, 1, 2, ..., 255}_{measured band <Index>}

Query parameters:

<Index> BC0 | BC1 | ... | BC17
Selects the measured CDMA2000 band class, for which the list is returned.

Return values:

<Value> OFF | ON
Without <Index>: 18 x 256 = 4608 values
With <Index>: 256 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.0.30
V3.2.80: added <Index>
V3.5.40: result array restructured

Manual operation: See ["Inter-RAT Need for Gaps"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:CXRTt? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band and measuring on a specific CDMA2000 1xRTT band class.

The full list contains 18 times 256 values. Each block of 18 values corresponds to the CDMA2000 band classes. The 256 repetitions correspond to the E-UTRA bands:

{measured band: 0, 1, ..., 17}_{used band: user-defined}

{measured band: 0, 1, ..., 17}_{used band: 1, ...,}

{measured band: 0, 1, ..., 17}_{used band: 256}

Via the optional parameter <Index>, you can alternatively query the list for a single CDMA2000 band class:

{used band: user-defined, 1, 2, ..., 255}_{measured band <Index>}

Query parameters:

<Index> BC0 | BC1 | ... | BC17
Selects the measured CDMA2000 band class, for which the list is returned.

Return values:

<Value> OFF | ON
Without <Index>: 18 x 256 = 4608 values
With <Index>: 256 values

Example: See [Querying UE capability report contents](#)

Usage:	Query only
Firmware/software:	V3.0.30 V3.2.80: added <Index> V3.5.40: result array restructured
Manual operation:	See "Inter-RAT Need for Gaps" on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IFNGaps:V<number>? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band combination and measuring on a specific E-UTRA band.

The full list contains 256 times n+1 values. Each block of 256 values corresponds to the measured E-UTRA bands. Each repetition corresponds to a supported band combination.

The list is ordered as follows:

{measured band: user-defined, 1, 2, ..., 255}_{used band combination 0},

{measured band: user-defined, 1, 2, ..., 255}_{used band combination 1, ...,}

{measured band: user-defined, 1, 2, ..., 255}_{used band combination n}

Via the optional parameter <Index>, you can alternatively query the list for a single measured E-UTRA band:

{used combination: 0, 1, ..., n}_{measured band <Index>}

Suffix:
<number> 1020

Query parameters:
<Index> UDEFined | OB1 | ... | OB46 | OB48 | ... | OB53 | OB65 | ... | OB76 | OB85 | OB87 | OB88 | OB106 | OB250 | OB252 | OB255
Selects the measured E-UTRA band, for which the list is returned.

Return values:
<Value> OFF | ON
Without <Index>: 256 x (n+1) values
With <Index>: n+1 values

Example: See [Querying UE capability report contents](#)

Usage:	Query only
Firmware/software:	V3.2.80, some bands added in later versions V3.5.40: result array restructured
Manual operation:	See "Inter-Freq Need for Gaps v1020" on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:UFDD? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band combination and measuring on a specific UTRA FDD band.

The full list contains 32 times n+1 values. Each block of 32 values corresponds to the UTRA FDD bands. Each repetition corresponds to a supported band combination:

{measured band: 1, 2, ..., 32}_{used band combination 0},

{measured band: 1, 2, ..., 32}_{used band combination 1}, ...,

{measured band: 1, 2, ..., 32}_{used band combination n}

Via the optional parameter <Index>, you can alternatively query the list for a single UTRA FDD band:

{used combination: 0, 1, ..., n}_{measured band <Index>}

Suffix:

<number> 1020

Query parameters:

<Index> OB1 | OB2 | ... | OB32

Selects the measured UTRA FDD band, for which the list is returned.

Return values:

<Value> OFF | ON

Without <Index>: 32 x (n+1) values

With <Index>: n+1 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Inter-RAT Need for Gaps v1020"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:UTDD<n>? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band combination and measuring on a specific UTRA TDD band.

The full list contains 32 times n+1 values. Each block of 32 values corresponds to the UTRA TDD bands. Each repetition corresponds to a supported band combination:

{measured band: 1, 2, ..., 32}_{used band combination 0},

{measured band: 1, 2, ..., 32}_{used band combination 1}, ...,

{measured band: 1, 2, ..., 32}_{used band combination n}

Via the optional parameter <Index>, you can alternatively query the list for a single UTRA TDD band:

{used combination: 0, 1, ..., n}_{measured band <Index>}

Suffix:

<n> 128

<number> 1020

Query parameters:

<Index> OB1 | OB2 | ... | OB32

Selects the measured UTRA TDD band, for which the list is returned.

Return values:

<Value> OFF | ON

Without <Index>: 32 x (n+1) values

With <Index>: n+1 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Inter-RAT Need for Gaps v1020"](#) on page 310

SENSE:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:GERan? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band combination and measuring on a specific GERAN band.

The full list contains 11 times n+1 values. Each block of 11 values corresponds to the following GERAN bands: GSM 450, GSM 480, GSM 710, GSM 750, GSM 810, GSM 850, P-GSM 900, E-GSM 900, R-GSM 900, GSM 1800, GSM 1900. Each repetition corresponds to a supported band combination:

{measured band: GSM 450, GSM 480, ..., GSM 1900}_{used band combination 0},

{measured band: GSM 450, GSM 480, ..., GSM 1900}_{used band combination 1, ...,}

{measured band: GSM 450, GSM 480, ..., GSM 1900}_{used band combination n}

Via the optional parameter <Index>, you can alternatively query the list for a single GERAN band:

{used combination: 0, 1, ..., n}_{measured band <Index>}

Suffix:

<number> 1020

Query parameters:

<Index> G045 | G048 | G071 | G075 | G081 | G085 | G09P | G09E | G09R | G18 | G19

Selects the measured GERAN band, for which the list is returned.

Return values:

<Value> OFF | ON
 Without <Index>: 11 x (n+1) values
 With <Index>: n+1 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Inter-RAT Need for Gaps v1020"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:CHRPd?
 [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band combination and measuring on a specific CDMA2000 HRPD band class.

The full list contains 18 times n+1 values. Each block of 18 values corresponds to the CDMA2000 band classes. Each repetition corresponds to a supported band combination:

{measured band: 0, 1, ..., 17}_{used band combination 0},

{measured band: 0, 1, ..., 17}_{used band combination 1, ...,}

{measured band: 0, 1, ..., 17}_{used band combination n}

Via the optional parameter <Index>, you can alternatively query the list for a single CDMA2000 band class:

{used combination: 0, 1, ..., n}_{measured band <Index>}

Suffix:

<number> 1020

Query parameters:

<Index> BC0 | BC1 | ... | BC17

Selects the measured CDMA2000 band class, for which the list is returned.

Return values:

<Value> OFF | ON
 Without <Index>: 18 x (n+1) values
 With <Index>: n+1 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Inter-RAT Need for Gaps v1020"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:IRNGaps:V<number>:CXRTt? [<Index>]

Returns a list of values indicating the need for downlink measurement gaps when operating on a specific E-UTRA band combination and measuring on a specific CDMA2000 1xRTT band class.

The full list contains 18 times n+1 values. Each block of 18 values corresponds to the CDMA2000 band classes. Each repetition corresponds to a supported band combination:

{measured band: 0, 1, ..., 17}_{used band combination 0},
 {measured band: 0, 1, ..., 17}_{used band combination 1}, ...,
 {measured band: 0, 1, ..., 17}_{used band combination n}

Via the optional parameter <Index>, you can alternatively query the list for a single CDMA2000 band class:

{used combination: 0, 1, ..., n}_{measured band <Index>}

Suffix:
 <number> 1020

Query parameters:
 <Index> BC0 | BC1 | ... | BC17
 Selects the measured CDMA2000 band class, for which the list is returned.

Return values:
 <Value> OFF | ON
 Without <Index>: 18 x (n+1) values
 With <Index>: n+1 values

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Inter-RAT Need for Gaps v1020"](#) on page 310

SENSe:LTE:SIGN<i>:UECapability:MEAS:RMWideband?

Returns whether the UE supports RSRQ measurements with wider bandwidth.

Return values:
 <Wideband> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[RSRQ Meas Wideband](#)" on page 311

SENSe:LTE:SIGN<i>:UECapability:MEAS:BFInterrupt?

Returns whether the UE power consumption can be reduced by allowing the UE to cause interruptions to serving cells during measurements.

Return values:
 <Benefits> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[Benefits from Interruption](#)" on page 311

SENSe:LTE:SIGN<i>:UECapability:MEAS:RCOReporting?

Returns whether the UE supports RSSI and channel occupancy measurements for LAA.

Return values:
 <Reporting> 0 | 1

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See "[rssi-And Chanel Occupancy Reporting](#)" on page 311

2.6.6.6 Inter-RAT UE capabilities

The following commands query the inter-RAT handover capabilities of the UE.

SENSe:LTE:SIGN<i>:UECapability:IRAT:UFDD:SUPPorted?	431
SENSe:LTE:SIGN<i>:UECapability:IRAT:UFDD:EREDirection:UTRA?	431
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:EREDirection:UTRA?	431
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:EREDirection:UTRA?	431
SENSe:LTE:SIGN<i>:UECapability:IRAT:UTDD128:SUPPorted?	432
SENSe:LTE:SIGN<i>:UECapability:IRAT:UTDD128:EREDirection:UTDD?	432
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:EREDirection:UTDD?	432
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:EREDirection:UTDD?	432
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:SUPPorted?	432
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:GERan:SUPPorted?	432
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:SUPPorted?	432
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:PHGeran?	433
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:GERan:PHGeran?	433

SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:PHGeran?	433
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:DTM?	433
SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:EREDirection?	433
SENSe:LTE:SIGN<i>:UECapability:IRAT:CDMA2000:NWSHaring?	433
SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:SUPPorted?	434
SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:TCONfig?	434
SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:RCONfig?	434
SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:SUPPorted?	435
SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:TCONfig?	435
SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:RCONfig?	435
SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:ECSFb?	435
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:CXRTt:ECSFb?	435
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:CXRTt:ECSFb?	435
SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:ECCMob?	436
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:CXRTt:ECCMob?	436
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:CXRTt:ECCMob?	436
SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:ECDual?	436
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:CXRTt:ECDual?	436
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:CXRTt:ECDual?	436

SENSe:LTE:SIGN<i>:UECapability:IRAT:UFDD:SUPPorted?

Returns a list of values indicating the support of the individual UTRA FDD operating bands by the UE.

Return values:

<SupportedBand> OFF | ON
 32 values: band 1, ..., band 32

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["Supported Bands"](#) on page 312

SENSe:LTE:SIGN<i>:UECapability:IRAT:UFDD:EREDirection:UTRA?

SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:EREDirection:UTRA?

SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:EREDirection:UTRA?

Returns whether the UE supports an enhanced redirection to UTRA FDD or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["e Redirection UTRA"](#) on page 312

SENSe:LTE:SIGN<i>:UECapability:IRAT:UTDD128:SUPPorted?

Returns a list of values indicating the support of the individual UTRA TDD operating bands by the UE, according to the UE capability information.

Return values:

<SupportedBand> OFF | ON
26 values: band a to band z

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Supported Bands"](#) on page 312

SENSe:LTE:SIGN<i>:UECapability:IRAT:UTDD128:EREDirection:UTDD?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:EREDirection:UTDD?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:EREDirection:UTDD?**

Returns whether the UE supports an enhanced redirection to UTRA TDD or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["e Redirection UTRA TDD"](#) on page 313

SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:SUPPorted?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:GERan:SUPPorted?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:SUPPorted?**

Returns a list of values indicating the support of the individual GERAN operating bands by the UE.

Return values:

<SupportedBand> OFF | ON
11 values: GSM 450, GSM 480, GSM 710, GSM 750, GSM 810, GSM 850, P-GSM 900, E-GSM 900, R-GSM 900, GSM 1800, GSM 1900

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30, FAUeeutra / TAUeeutra V3.2.80

Manual operation: See ["Supported Bands"](#) on page 313

SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:PHGeran?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:GERan:PHGeran?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:PHGeran?**

Returns whether the UE supports handover to GERAN or not.

Return values:

<PS_HO_GERAN> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30, FAUeeutra / TAUeeutra V3.2.80

Manual operation: See ["Inter-RAT PS HO to GERAN"](#) on page 313

SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:DTM?

Returns whether the UE supports DTM in GERAN or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["dtm"](#) on page 313

SENSe:LTE:SIGN<i>:UECapability:IRAT:GERan:EREDirection?

Returns whether the UE supports an enhanced redirection to GERAN or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["e Redirection GERAN"](#) on page 313

SENSe:LTE:SIGN<i>:UECapability:IRAT:CDMA2000:NWSHaring?

Returns whether the UE supports network sharing for CDMA2000.

Return values:

<Sharing> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["CDMA2000 NW Sharing"](#) on page 313

SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:SUPPorted?

Returns a list of values indicating the support of the individual CDMA2000 HRPD band classes by the UE.

Return values:

<SupportedBand> OFF | ON
18 values: band class 0 to 17

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["Supported Bands"](#) on page 313

SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:TCONfig?

Returns whether the UE supports dual transmitter for HRPD/E-UTRAN or only single transmitter.

Return values:

<TXconfiguration> SINGLE | DUAL

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["TX Config"](#) on page 314

SENSe:LTE:SIGN<i>:UECapability:IRAT:CHRPd:RCONfig?

Returns whether the UE supports dual receiver for HRPD/E-UTRAN or only single receiver.

Return values:

<RXconfiguration> SINGLE | DUAL

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["RX Config"](#) on page 314

SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:SUPPorted?

Returns a list of values indicating the support of the individual CDMA2000 1xRTT band classes by the UE.

Return values:

<SupportedBand> OFF | ON
 18 values: band class 0 to 17

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["Supported Bands"](#) on page 314

SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:TCONfig?

Returns whether the UE supports dual transmitter for 1xRTT/E-UTRAN or only single transmitter.

Return values:

<TXconfiguration> SINGle | DUAL

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["TX Config"](#) on page 314

SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:RCONfig?

Returns whether the UE supports dual receiver for 1xRTT/E-UTRAN or only single receiver.

Return values:

<RXconfiguration> SINGle | DUAL

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["RX Config"](#) on page 314

SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:ECSFb?**SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:CXRTt:ECSFb?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:CXRTt:ECSFb?**

Returns whether the UE supports enhanced CS fallback to CDMA2000 1xRTT or not.

Return values:

<Supported> OFF | ON

Example:See [Querying UE capability report contents](#)**Usage:**

Query only

Firmware/software: V3.2.80**Manual operation:** See ["e CSFB"](#) on page 314**SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:ECCMob?****SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:CXRTt:ECCMob?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:CXRTt:ECCMob?**

Returns whether the UE supports concurrent enhanced CS fallback to CDMA2000 1xRTT and handover/redirection to CDMA2000 HRPD or not.

Return values:

<Supported> OFF | ON

Example:See [Querying UE capability report contents](#)**Usage:**

Query only

Firmware/software: V3.2.80**Manual operation:** See ["e CSFB ConcPS Mob"](#) on page 314**SENSe:LTE:SIGN<i>:UECapability:IRAT:CXRTt:ECDual?****SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:IRAT:CXRTt:ECDual?****SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:CXRTt:ECDual?**

Returns whether the UE supports enhanced CS fallback to CDMA2000 1xRTT for dual Rx/Tx configuration or not.

Return values:

<Supported> OFF | ON

Example:See [Querying UE capability report contents](#)**Usage:**

Query only

Firmware/software: V3.2.80**Manual operation:** See ["e CSFB dual"](#) on page 315

2.6.6.7 Other UE capabilities

The following commands query the UE capabilities categorized by 3GPP as "other parameters".

SENSe:LTE:SIGN<i>:UECapability:IDCIndex?	437
SENSe:LTE:SIGN<i>:UECapability:PPIndex?	437
SENSe:LTE:SIGN<i>:UECapability:URTTimediff?	437
SENSe:LTE:SIGN<i>:UECapability:DCIulca?	437

SENSe:LTE:SIGN<i>:UECapability:IDCindex?

Returns whether the UE supports in-device coexistence indication and autonomous denial functionality.

Return values:

<Index> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["In Device Coex Ind"](#) on page 315

SENSe:LTE:SIGN<i>:UECapability:PPIndex?

Returns whether the UE supports power preference indication.

Return values:

<Index> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Power Pref Ind"](#) on page 315

SENSe:LTE:SIGN<i>:UECapability:URTTimediff?

Returns whether the UE supports RX-TX time difference measurements.

Return values:

<Timediff> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["UE Rx Tx Time Diff Measurements"](#) on page 315

SENSe:LTE:SIGN<i>:UECapability:DClulca?

Returns whether the UE supports in-device coexistence indication for UL CA.

Return values:

<Index> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["In Device Coex Div UL CA"](#) on page 315

2.6.6.8 MBMS UE capabilities

The following commands query the UE capabilities for MBMS reception.

SENSe:LTE:SIGN<i>:UECapability:MBMS:SCELI?	438
SENSe:LTE:SIGN<i>:UECapability:MBMS:NSCell?	438

SENSe:LTE:SIGN<i>:UECapability:MBMS:SCELI?

Returns whether the UE supports MBMS reception via an SCell.

Return values:

<Scell> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["MBMS SCell"](#) on page 316

SENSe:LTE:SIGN<i>:UECapability:MBMS:NSCell?

Returns whether the UE supports MBMS reception via a serving cell to be added.

Return values:

<Cell> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["MBMS Non Serving Cell"](#) on page 316

2.6.6.9 CSG proximity indication capabilities

The following commands query whether the UE supports proximity indications or not.

SENSe:LTE:SIGN<i>:UECapability:CPINdication:FREQUENCY:INTRa?	438
SENSe:LTE:SIGN<i>:UECapability:CPINdication:FREQUENCY:INTer?	439
SENSe:LTE:SIGN<i>:UECapability:CPINdication:UTRan?	439

SENSe:LTE:SIGN<i>:UECapability:CPINdication:FREQUENCY:INTRa?

Returns whether the UE supports proximity indications for intra-frequency E-UTRAN CSG member cells or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Intra Freq Proximity Indication"](#) on page 316

SENSe:LTE:SIGN<i>:UECapability:CPINdication:FREQuency:INTer?

Returns whether the UE supports proximity indications for inter-frequency E-UTRAN CSG member cells or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Inter Freq Proximity Indication"](#) on page 316

SENSe:LTE:SIGN<i>:UECapability:CPINdication:UTRan?

Returns whether the UE supports proximity indications for UTRAN CSG member cells or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["utran Proximity Indication"](#) on page 316

2.6.6.10 Neighbor cell SI-acquisition capabilities

The following commands query whether the UE supports system information requests for handover or not.

SENSe:LTE:SIGN<i>:UECapability:NCSacq:FREQuency:INTRa?	440
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:FREQuency:INTRa?	440
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQuency:INTRa?	440
SENSe:LTE:SIGN<i>:UECapability:NCSacq:FREQuency:INTer?	440
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:FREQuency:INTer?	440
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQuency:INTer?	440
SENSe:LTE:SIGN<i>:UECapability:NCSacq:UTRan?	440
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:UTRan?	440
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:UTRan?	440

SENSe:LTE:SIGN<i>:UECapability:NCSacq:FREQUENCY:INTRa?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:FREQUENCY:INTRa?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQUENCY:INTRa?

Returns whether the UE supports system information acquisition for intra-frequency neighbor cells or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Intra Freq SI-Acquisition for HO"](#) on page 317

SENSe:LTE:SIGN<i>:UECapability:NCSacq:FREQUENCY:INTER?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:FREQUENCY:INTER?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQUENCY:INTER?

Returns whether the UE supports system information acquisition for inter-frequency neighbor cells or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Inter Freq SI-Acquisition for HO"](#) on page 317

SENSe:LTE:SIGN<i>:UECapability:NCSacq:UTRan?
SENSe:LTE:SIGN<i>:UECapability:FAUeeutra:NCSacq:UTRan?
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:UTRan?

Returns whether the UE supports system information acquisition for UMTS neighbor cells or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["utran SI-Acquisition for HO"](#) on page 317

2.6.6.11 UE-based network performance measurement capabilities

The following commands query the UE capabilities for UE-based network performance measurements.

SENSe:LTE:SIGN<i>:UECapability:UBNPmeas:LMIDle?	441
SENSe:LTE:SIGN<i>:UECapability:UBNPmeas:SGLocation?	441
SENSe:LTE:SIGN<i>:UECapability:LMMeas?	441

SENSe:LTE:SIGN<i>:UECapability:UBNPmeas:LMIDle?

Returns whether the UE supports logged measurements in idle mode or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["logged Measurements Idle"](#) on page 317

SENSe:LTE:SIGN<i>:UECapability:UBNPmeas:SGLocation?

Returns whether the UE is equipped with a GNSS receiver or not.

Return values:

<Supported> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Standalone GNSS Location"](#) on page 317

SENSe:LTE:SIGN<i>:UECapability:LMMeas?

Returns whether the UE supports logged MBSFN measurements in RRC idle and connected mode.

Return values:

<Lmbsfn> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Logged MBSFN Measurements"](#) on page 318

2.6.6.12 RLC UE capabilities

The following commands query the UE capabilities for RLC.

SENSe:LTE:SIGN<i>:UECapability:ERLField?

Returns whether the UE supports 15-bit RLC length indicators.

Return values:

<Field> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["extended RLC LI Fields"](#) on page 318

2.6.6.13 WLAN interworking UE capabilities

The following commands query the UE capabilities for WLAN interworking.

[SENSe:LTE:SIGN<i>:UECapability:WIW:WIRRules?](#)..... 442

[SENSe:LTE:SIGN<i>:UECapability:WIW:WIAPolicies?](#)..... 442

SENSe:LTE:SIGN<i>:UECapability:WIW:WIRRules?

Returns whether the UE supports RAN-assisted WLAN interworking based on access network selection and traffic steering rules specified in 3GPP TS 36.304.

Return values:

<rules> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["WLAN IW RAN Rules"](#) on page 318

SENSe:LTE:SIGN<i>:UECapability:WIW:WIAPolicies?

Returns whether the UE supports RAN-assisted WLAN interworking based on ANDSF policies specified in 3GPP TS 24.312.

Return values:

<Policies> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["WLAN IW ANDSF Policies"](#) on page 318

2.6.6.14 CE UE capabilities

The following commands query the UE capabilities for the eMTC CE modes.

SENSe:LTE:SIGN<i>:UECapability:CEParameters:MODE:A?	443
SENSe:LTE:SIGN<i>:UECapability:CEParameters:MODE:B?	443

SENSe:LTE:SIGN<i>:UECapability:CEParameters:MODE:A?

Returns whether the UE supports operation in CE mode A.

Return values:

<Parameter> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["ce-ModeA"](#) on page 319

SENSe:LTE:SIGN<i>:UECapability:CEParameters:MODE:B?

Returns whether the UE supports operation in CE mode B.

Return values:

<Parameter> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["ce-ModeB"](#) on page 319

2.6.6.15 LAA UE capabilities

The following commands query the UE capabilities for licensed-assisted access (LAA).

SENSe:LTE:SIGN<i>:UECapability:LAA:DL?	443
SENSe:LTE:SIGN<i>:UECapability:LAA:EDPTs?	444
SENSe:LTE:SIGN<i>:UECapability:LAA:SSSPosition?	444
SENSe:LTE:SIGN<i>:UECapability:LAA:TM<TMnr>?	444

SENSe:LTE:SIGN<i>:UECapability:LAA:DL?

Returns whether the UE supports DL LAA operation.

Return values:

<Downlink> 0 | 1

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Downlink LAA"](#) on page 319

SENSe:LTE:SIGN<i>:UECapability:LAA:EDPTs?

Returns whether the UE supports partial allocation in the ending subframe of an LAA burst.

Return values:

<Pts> 0 | 1

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["ending Dw PTS"](#) on page 319

SENSe:LTE:SIGN<i>:UECapability:LAA:SSSPosition?

Returns whether the UE supports partial allocation in the initial subframe of an LAA burst.

Return values:

<Position> 0 | 1

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["second Slot Starting Position"](#) on page 319

SENSe:LTE:SIGN<i>:UECapability:LAA:TM<TMnr>?

Returns whether the UE supports transmission mode 9 for LAA downlinks.

Suffix:

<TMnr> 9

Return values:

<Tm> 0 | 1

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["tm9-LAA"](#) on page 319

2.6.6.16 Dual connectivity UE capabilities

The following commands query the UE capabilities for dual connectivity (DC).

[SENSe:LTE:SIGN<i>:UECapability:DCParameters:DTSPplit?](#)..... 445

[SENSe:LTE:SIGN<i>:UECapability:DCParameters:DTSCg?](#)..... 445

SENSe:LTE:SIGN<i>:UECapability:DCParameters:DTSPplit?

Returns whether the UE supports the DRB type of split bearer.

Return values:

<Typesplit> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[DRB TypeSplit](#)" on page 320

SENSe:LTE:SIGN<i>:UECapability:DCParameters:DTSCg?

Returns whether the UE supports the DRB type of SCG bearer.

Return values:

<Typescgv> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See "[DRB TypeSCG](#)" on page 320

2.6.6.17 MAC UE capabilities

The following commands query the UE capabilities for the MAC layer.

[SENSe:LTE:SIGN<i>:UECapability:MAC:LCSPtimer?](#)..... 445

[SENSe:LTE:SIGN<i>:UECapability:MAC:LDRXcommand?](#)..... 446

SENSe:LTE:SIGN<i>:UECapability:MAC:LCSPtimer?

Returns whether the UE supports the logical channel SR prohibit timer as specified in 3GPP TS 36.321.

Return values:

<Channel> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Logical Channel SR Prohibit Timer"](#) on page 320

SENSe:LTE:SIGN<i>:UECapability:MAC:LDRXcommand?

Returns whether the UE supports the long DRX command MAC control element as specified in 3GPP TS 36.321.

Return values:

<Command> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Long DRX Command"](#) on page 320

2.6.6.18 SL UE capabilities

The following commands query the UE capabilities for sidelink communication.

SENSe:LTE:SIGN<i>:UECapability:SL:CSTX?	446
SENSe:LTE:SIGN<i>:UECapability:SL:DSRalloc?	446
SENSe:LTE:SIGN<i>:UECapability:SL:DUSRalloc?	447
SENSe:LTE:SIGN<i>:UECapability:SL:DSLSS?	447
SENSe:LTE:SIGN<i>:UECapability:SL:DSPRoc?	447

SENSe:LTE:SIGN<i>:UECapability:SL:CSTX?

Returns whether the UE supports simultaneous transmission of EUTRA and sidelink communication on different carriers.

Return values:

<Simultaneous> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Comm Simultaneous Tx"](#) on page 321

SENSe:LTE:SIGN<i>:UECapability:SL:DSRalloc?

Returns whether the UE supports transmission of discovery announcements based on network scheduled resource allocation.

Return values:

<Alloc> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Disc Scheduled Resource Alloc"](#) on page 322

SENSe:LTE:SIGN<i>:UECapability:SL:DUSRalloc?

Returns whether the UE supports transmission of discovery announcements based on UE autonomous resource selection.

Return values:

<Alloc> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Disc UE Selected Resource Alloc"](#) on page 322

SENSe:LTE:SIGN<i>:UECapability:SL:DSLSSs?

Returns whether the UE supports SLSS transmission and reception.

Return values:

<Slss> OFF | ON

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Disc SLSS"](#) on page 322

SENSe:LTE:SIGN<i>:UECapability:SL:DSPRoc?

Returns the number of processes supported by the UE for sidelink discovery.

Return values:

<Proc> N50 | N400

Example: See [Querying UE capability report contents](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Disc Supported Proc"](#) on page 322

2.6.7 UE info

The following queries retrieve information about the connected mobile as shown in the "UE Info" area of the main view.

SENSe:LTE:SIGN<i>:UESInfo:IMEI?	448
SENSe:LTE:SIGN<i>:UESInfo:IMSI?	448
SENSe:LTE:SIGN<i>:UESInfo:VDPReference?	448
SENSe:LTE:SIGN<i>:UESInfo:UEUSage?	449
SENSe:LTE:SIGN<i>:UESInfo:UEADdress:IPV<n>?	449
SENSe:LTE:SIGN<i>:UESInfo:UEADdress:DEDBearer?	449
SENSe:LTE:SIGN<i>:UESInfo:UEADdress:DEDBearer:SEParate?	450

SENSe:LTE:SIGN<i>:UESInfo:IMEI?

Queries the IMEI of the UE.

Return values:

<IMEI> string
Up to 18 digits

Example: See [Querying UE information](#)

Usage: Query only

Firmware/software: V2.0.20

Manual operation: See ["IMEI"](#) on page 126

SENSe:LTE:SIGN<i>:UESInfo:IMSI?

Queries the IMSI of the UE.

Return values:

<IMSI> string
Up to 16 digits

Example: See [Querying UE information](#)

Usage: Query only

Firmware/software: V2.0.20

Manual operation: See ["IMSI"](#) on page 126

SENSe:LTE:SIGN<i>:UESInfo:VDPReference?

Queries the voice domain preference of the UE.

Return values:

<Value> CVONLY | IPVonly | CVPReferred | IPVPreferred
CVONLY: CS voice only
IPVonly: IMS PS voice only
CVPReferred: CS voice preferred, IMS PS voice as secondary
IPVPreferred: IMS PS voice preferred, CS voice as secondary

Example: See [Querying UE information](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Voice Domain Preference"](#) on page 126

SENSe:LTE:SIGN<i>:UESinfo:UEUSage?

Queries the usage setting of the UE.

Return values:

<Usage> VCENtric | DCENtric
VCENtric: Voice centric
DCENtric: Data centric

Example: See [Querying UE information](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["UE's Usage Setting"](#) on page 126

SENSe:LTE:SIGN<i>:UESinfo:UEAddress:IPV<n>?

Returns the IPv4 addresses (<n> = 4) or the IPv6 prefixes (<n> = 6) assigned to the UE by the R&S CMW.

Suffix:

<n> 4, 6

Return values:

<IPAddresses> string
 Comma-separated list of IP address/prefix strings

Example: See [Setting up a test mode connection](#)

Usage: Query only

Firmware/software: V3.0.20

Manual operation: See ["Default Bearer"](#) on page 126

SENSe:LTE:SIGN<i>:UESinfo:UEAddress:DEDBearer?

Returns information about all established dedicated bearers. Three values are returned per bearer:

{<ID>, <TFTPPortLow>, <TFTPPortHigh>}_{Bearer 1}, ..., {...}_{Bearer n}

Use this command if you have configured a single port range per bearer, applicable to the uplink and the downlink.

Return values:

<ID>	string Dedicated bearer ID Example: "6 (->5, Voice)" means dedicated bearer 6, mapped to default bearer 5, using dedicated bearer profile "Voice"
<TFTPortLow>	decimal Lower end of TFT port range assigned to the dedicated bearer Range: 1 to 65535
<TFTPortHigh>	decimal Upper end of TFT port range assigned to the dedicated bearer Range: 1 to 65535

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["Dedicated Bearer"](#) on page 127

SENSe:LTE:SIGN<i>:UESinfo:UEAddress:DEDBearer:SEparate?

Returns information about all established dedicated bearers. Five values are returned per bearer:

```
{<ID>, <TFTPortLowDL>, <TFTPortHighDL>, <TFTPortLowUL>,
<TFTPortHighUL>}Bearer 1, ..., {...}Bearer n
```

Use this command if you have configured separate port ranges for the uplink and the downlink.

Return values:

<ID>	string Dedicated bearer ID Example: "6 (->5, Voice)" means dedicated bearer 6, mapped to default bearer 5, using dedicated bearer profile "Voice"
<TFTPortLowDL>	decimal Lower end of TFT port range assigned to the downlink Range: 1 to 65535
<TFTPortHighDL>	decimal Upper end of TFT port range assigned to the downlink Range: 1 to 65535
<TFTPortLowUL>	decimal Lower end of TFT port range assigned to the uplink Range: 1 to 65535
<TFTPortHighUL>	decimal Upper end of TFT port range assigned to the uplink Range: 1 to 65535

Example: See [Querying UE information](#)

Usage: Query only

Firmware/software: V3.5.30

Manual operation: See ["Dedicated Bearer"](#) on page 127

2.6.8 Routing settings

The following commands configure the signal input and output paths.

- [Scenario selection and signal routing](#)..... 451
- [RF path properties](#)..... 631
- [Signal settings](#)..... 633
- [User-defined band](#)..... 641

2.6.8.1 Scenario selection and signal routing

The commands in this section configure the scenario and select the paths for the generated downlink signal (output) and the analyzed signal (input).

LTE-specific configuration rules:

- For scenarios with carrier aggregation, select a different signaling unit for each carrier.
- For scenarios with several output paths due to MIMO, select a different TX module and connector for each output path of a carrier.
- For scenarios with several output paths due to carrier aggregation, select a different TX module for each output path.

With co-location, you can use the same TX module for two carriers. For details, see [Section 2.2.5.1, "RF paths and co-location"](#), on page 33.

The following table lists all scenarios and the related commands, so that you can easily find the relevant command.

For configuration of the SCC uplink routing, see [CONFigure:LTE:SIGN<i>:SCC<c>:UUL](#) on page 368.

Scenario name in the GUI	Command
"1CC - 1x1"	ROUTE:LTE:SIGN<i>:SCENario:SCell:FLEXible
"1CC - nx2"	ROUTE:LTE:SIGN<i>:SCENario:TRO:FLEXible
"1CC - nx4"	ROUTE:LTE:SIGN<i>:SCENario:AD[:FLEXible]
"1CC - Fading - 1x1"	ROUTE:LTE:SIGN<i>:SCENario:SCFading:FLEXible:INTernal
"1CC - Fading - nx2"	ROUTE:LTE:SIGN<i>:SCENario:TROFading:FLEXible:INTernal
"1CC - Fading - nx4"	ROUTE:LTE:SIGN<i>:SCENario:ADF[:FLEXible]:INTernal
"2CC - 1x1 1x1"	ROUTE:LTE:SIGN<i>:SCENario:CATRfout:FLEXible
"2CC - nx2 nx2"	ROUTE:LTE:SIGN<i>:SCENario:CAFRfout:FLEXible

Scenario name in the GUI	Command
"2CC - nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:BF[:FLEXible]</code>
"2CC - nx2 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:BFSM<MIMO4x4>[:FLEXible]</code>
"2CC - nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:BH[:FLEXible]</code>
"2CC - Fading - 1x1 1x1"	<code>ROUTe:LTE:SIGN<i>:SCENario:CATF:FLEXible:INTernal</code>
"2CC - Fading - nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CAFF:FLEXible:INTernal</code>
"2CC - Fading - nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:BFF[:FLEXible]:INTernal</code>
"2CC - Fading - nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:BHF[:FLEXible]:INTernal</code>
"3CC - 1x1 1x1 1x1"	<code>ROUTe:LTE:SIGN<i>:SCENario:CC:FLEXible</code>
"3CC - nx2 1x1 1x1"	<code>ROUTe:LTE:SIGN<i>:SCENario:CCMP:FLEXible</code>
"3CC - 1x1 nx2 1x1"	<code>ROUTe:LTE:SIGN<i>:SCENario:CCMS<c>:FLEXible</code>
"3CC - nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CF[:FLEXible]</code>
"3CC - nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CH[:FLEXible]</code>
"3CC - nx2 nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CHSM<MIMO44>[:FLEXible]</code>
"3CC - nx4 nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CJ[:FLEXible]</code>
"3CC - nx2 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:CJSM<MIMO44>[:FLEXible]</code>
"3CC - nx4 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:CL[:FLEXible]</code>
"3CC - Fading - nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CFF[:FLEXible]:INTernal</code>
"3CC - Fading - nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CHF[:FLEXible]:INTernal</code>
"3CC - Fading - nx4 nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:CJF[:FLEXible]:INTernal</code>
"3CC - Fading - nx2 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:CJFS<MIMO44>[:FLEXible]:INTernal</code>
"4CC - 1x1 1x1 1x1 1x1"	<code>ROUTe:LTE:SIGN<i>:SCENario:DD:FLEXible</code>
"4CC - nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:DH[:FLEXible]</code>
"4CC - nx4 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:DJ[:FLEXible]</code>
"4CC - nx2 nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:DJSM<MIMO4x4>[:FLEXible]</code>
"4CC - nx4 nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:DL[:FLEXible]</code>
"4CC - nx2 nx4 nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:DLSM<MIMO4x4>[:FLEXible]</code>
"4CC - nx4 nx4 nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:DN[:FLEXible]</code>
"4CC - nx2 nx4 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:DNSM<MIMO4x4>[:FLEXible]</code>
"4CC - nx4 nx4 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:DP[:FLEXible]</code>
"4CC - Fading - nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:DHF[:FLEXible]:INTernal</code>
"4CC - Fading - nx4 nx4 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:DPF[:FLEXible]:INTernal</code>
"5CC - 1x1 1x1 1x1 1x1 1x1"	<code>ROUTe:LTE:SIGN<i>:SCENario:EE[:FLEXible]</code>

Scenario name in the GUI	Command
"5CC - nx2 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EJ[:FLEXible]
"5CC - nx4 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EL[:FLEXible]
"5CC - nx2 nx4 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:ELSM<MIMO4x4>[:FLEXible]
"5CC - nx4 nx4 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EN[:FLEXible]
"5CC - nx2 nx4 nx4 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:ENSM<MIMO4x4>[:FLEXible]
"5CC - nx4 nx4 nx4 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EP[:FLEXible]
"5CC - nx2 nx4 nx4 nx4 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EPSM<MIMO4x4>[:FLEXible]
"5CC - nx4 nx4 nx4 nx4 nx2"	ROUTe:LTE:SIGN<i>:SCENario:ER[:FLEXible]
"5CC - nx2 nx4 nx4 nx4 nx4"	ROUTe:LTE:SIGN<i>:SCENario:ERSM<MIMO4x4>[:FLEXible]
"5CC - nx4 nx4 nx4 nx4 nx4"	ROUTe:LTE:SIGN<i>:SCENario:ET[:FLEXible]
"5CC - Fading - nx2 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EJF[:FLEXible]:INTernal
"5CC - Fading - nx4 nx4 nx4 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EPF[:FLEXible]:INTernal
"5CC - Fading - nx2 nx4 nx4 nx4 nx2"	ROUTe:LTE:SIGN<i>:SCENario:EPFS<MIMO4x4>[:FLEXible]:INTernal
"6CC - 1x1 1x1 1x1 1x1 1x1 1x1"	ROUTe:LTE:SIGN<i>:SCENario:FF[:FLEXible]
"6CC - nx2 nx2 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FL[:FLEXible]
"6CC - nx4 nx2 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FN[:FLEXible]
"6CC - nx2 nx4 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FNSM<MIMO4x4>[:FLEXible]
"6CC - nx4 nx4 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FP[:FLEXible]
"6CC - nx2 nx4 nx4 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FPSM<MIMO4x4>[:FLEXible]
"6CC - nx4 nx4 nx4 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FR[:FLEXible]
"6CC - nx2 nx4 nx4 nx4 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FRSM<MIMO4x4>[:FLEXible]
"6CC - nx4 nx4 nx4 nx4 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FT[:FLEXible]
"6CC - nx2 nx4 nx4 nx4 nx4 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FTSM<MIMO4x4>[:FLEXible]
"6CC - nx4 nx4 nx4 nx4 nx4 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FV[:FLEXible]
"6CC - nx2 nx4 nx4 nx4 nx4 nx4"	ROUTe:LTE:SIGN<i>:SCENario:FVSM<MIMO4x4>[:FLEXible]
"6CC - nx4 nx4 nx4 nx4 nx4 nx4"	ROUTe:LTE:SIGN<i>:SCENario:FX[:FLEXible]
"6CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FLF[:FLEXible]:INTernal
"6CC - Fading - nx4 nx4 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FPF[:FLEXible]:INTernal
"6CC - Fading - nx2 nx4 nx4 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:FPFS<MIMO4x4>[:FLEXible]:INTernal
"7CC - 1x1 1x1 1x1 1x1 1x1 1x1 1x1"	ROUTe:LTE:SIGN<i>:SCENario:GG[:FLEXible]
"7CC - nx2 nx2 nx2 nx2 nx2 nx2 nx2"	ROUTe:LTE:SIGN<i>:SCENario:GN[:FLEXible]

Scenario name in the GUI	Command
"7CC - nx4 nx2 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GP[:FLEXible]</code>
"7CC - nx2 nx4 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GPSM<MIMO4x4>[:FLEXible]</code>
"7CC - nx4 nx4 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GR[:FLEXible]</code>
"7CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GRSM<MIMO4>[:FLEXible]</code>
"7CC - nx4 nx4 nx4 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GT[:FLEXible]</code>
"7CC - nx2 nx4 nx4 nx4 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GTSM<Mimo4x4>[:FLEXible]</code>
"7CC - nx4 nx4 nx4 nx4 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GV[:FLEXible]</code>
"7CC - nx2 nx4 nx4 nx4 nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GVSM<MIMO4x4>[:FLEXible]</code>
"7CC - nx4 nx4 nx4 nx4 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GX[:FLEXible]</code>
"7CC - nx2 nx4 nx4 nx4 nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GXSM<MIMO4x4>[:FLEXible]</code>
"7CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GYA[:FLEXible]</code>
"7CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:GYAS<MIMO4x4>[:FLEXible]</code>
"7CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:GYC[:FLEXible]</code>
"7CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GNF[:FLEXible]:INTernal</code>
"7CC - Fading - nx4 nx2 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GPF[:FLEXible]:INTernal</code>
"7CC - Fading - nx2 nx4 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:GPFS<MIMO4x4>[:FLEXible]:INTernal</code>
"8CC - 1x1 1x1 1x1 1x1 1x1 1x1 1x1"	<code>ROUTe:LTE:SIGN<i>:SCENario:HH[:FLEXible]</code>
"8CC - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HP[:FLEXible]</code>
"8CC - nx4 nx2 nx2 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HR[:FLEXible]</code>
"8CC - nx2 nx4 nx2 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HRSM<MIMO4x4>[:FLEXible]</code>
"8CC - nx4 nx4 nx2 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HT[:FLEXible]</code>
"8CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HTSM<MIMO4x4>[:FLEXible]</code>
"8CC - nx4 nx4 nx4 nx2 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HV[:FLEXible]</code>
"8CC - nx2 nx4 nx4 nx4 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HVSM<Mimo4x4>[:FLEXible]</code>
"8CC - nx4 nx4 nx4 nx4 nx2 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HX[:FLEXible]</code>
"8CC - nx2 nx4 nx4 nx4 nx4 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HXSM<MIMO4x4>[:FLEXible]</code>
"8CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HYA[:FLEXible]</code>
"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HYAS<MIMO4x4>[:FLEXible]</code>
"8CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HYC[:FLEXible]</code>
"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HYCS<MIMO4x4>[:FLEXible]</code>
"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4 nx2"	<code>ROUTe:LTE:SIGN<i>:SCENario:HYE[:FLEXible]</code>
"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4 nx4"	<code>ROUTe:LTE:SIGN<i>:SCENario:HYES<MIMO4x4>[:FLEXible]</code>

Scenario name in the GUI	Command
"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4 nx4"	ROUTE:LTE:SIGN<i>:SCENario:HYG[:FLEXible]
"8CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2"	ROUTE:LTE:SIGN<i>:SCENario:HPF[:FLEXible]:INTernal

CATalog:LTE:SIGN<i>:SCENario?

Queries a list of all supported scenarios, depending on the available hardware and licenses.

Return values:

<Scenarios>

NAV | SCEL | TRO | AD | SCF | TROF | ADF | CATR | CAFR |
 BF | BFSM4 | BH | CATF | CAFF | BFF | BHF | CC | CCMP |
 CCMS1 | CF | CH | CHSM4 | CJ | CJSM4 | CL | CFF | CHF |
 CJF | CJFS4 | DD | DH | DJ | DJSM4 | DL | DLSM4 | DN |
 DNSM4 | DP | DHF | DPF | EE | EJ | EL | ELSM4 | EN |
 ENSM4 | EP | EPSM4 | ER | ERS4 | ET | EJF | EPF | EPFS4 |
 FF | FL | FN | FNSM4 | FP | FPSM4 | FR | FRSM4 | FT |
 FTSM4 | FV | FVSM4 | FX | FLF | FPF | FPFS4 | GG | GN | GP |
 GPSM4 | GR | GRSM4 | GT | GTSM4 | GV | GVSM4 | GX |
 GXSM4 | GYA | GYAS4 | GYC | GNF | GPF | GPFS4 | HH | HP |
 HR | HRSM4 | HT | HTSM4 | HV | HVSM4 | HX | HXSM4 | HYA |
 HYAS4 | HYC | HYCS4 | HYE | HYES4 | HYG | HPF

Comma-separated list of all supported scenarios

For mapping of the values to scenario names, see [ROUTE:LTE:SIGN<i>:SCENario?](#).

Usage:

Query only

Firmware/software: V3.5.50, some scenarios added in later versions

ROUTE:LTE:SIGN<i>:SCENario?

Returns the active scenario.

Return values:

<Scenario>

NAV | SCEL | TRO | AD | SCF | TROF | ADF | CATR | CAFR |
 BF | BFSM4 | BH | CATF | CAFF | BFF | BHF | CC | CCMP |
 CCMS1 | CF | CH | CHSM4 | CJ | CJSM4 | CL | CFF | CHF |
 CJF | CJFS4 | DD | DH | DJ | DJSM4 | DL | DLSM4 | DN |
 DNSM4 | DP | DHF | DPF | EE | EJ | EL | ELSM4 | EN |
 ENSM4 | EP | EPSM4 | ER | ERS4 | ET | EJF | EPF | EPFS4 |
 FF | FL | FN | FNSM4 | FP | FPSM4 | FR | FRSM4 | FT |
 FTSM4 | FV | FVSM4 | FX | FLF | FPF | FPFS4 | GG | GN | GP |
 GPSM4 | GR | GRSM4 | GT | GTSM4 | GV | GVSM4 | GX |
 GXSM4 | GYA | GYAS4 | GYC | GNF | GPF | GPFS4 | HH | HP |
 HR | HRSM4 | HT | HTSM4 | HV | HVSM4 | HX | HXSM4 | HYA |
 HYAS4 | HYC | HYCS4 | HYE | HYES4 | HYG | HPF

For mapping of the values to scenario names, see [Table 2-65](#).

<Fader>	EXternal INternal Only returned for fading scenarios Indicates whether internal or external fading is active.
Usage:	Query only
Firmware/software:	V2.0.10 V3.0.20: added <Fader> V3.5.20: added scenarios CF, CFF, DD, DH, DHF V3.7.10: AD, ADF, BF, BFF, BFSM4, BH, BHF, CH, CHSM4, CJ, CJF, CJSM4, DJ, EE, EJ, EJF V3.7.30: CHF, CJFS4, CL, DJSM4, DL, DLSM4, DN, DNSM4, DP, EL, ELSM4, ET V3.7.50: DPF, EP, EPF, EPFS4, EPSM4, FF, FL, FP, FPSM4, GG, GN, GP, GPSM4, HH, HP, HPF V3.7.60: EN, ENSM4, ER, ERS4, FN, FNSM4, FPF, FPFS4, FR, FRSM4, FT, FTSM4, GPF, GPFS4, GR, GRSM4, GT, GTSM4 V3.7.70: FLF, FV, FVSM4, FX, GNF, GV, GVSM4, GX, GXSM4 V3.8.10: GYA, GYAS4, GYC, HR, HRSM4, HT, HTSM4, HV, HVSM4, HX, HXSM4, HYA, HYAS4, HYC, HYCS4, HYE, HYES4, HYG

Table 2-65: Mapping of <Scenario> to scenario names

	<Scenario>	Scenario name
A	AD	"1CC - nx4"
	ADF	"1CC - Fading - nx4"
B	BF	"2CC - nx4 nx2"
	BFF	"2CC - Fading - nx4 nx2"
	BFSM4	"2CC - nx2 nx4"
	BH	"2CC - nx4 nx4"
	BHF	"2CC - Fading - nx4 nx4"
C	CAFF	"2CC - Fading - nx2 nx2"
	CAFR	"2CC - nx2 nx2"
	CATF	"2CC - Fading - 1x1 1x1"
	CATR	"2CC - 1x1 1x1"
	CC	"3CC - 1x1 1x1 1x1"
	CCMP	"3CC - nx2 1x1 1x1"
	CCMS1	"3CC - 1x1 nx2 1x1"
	CF	"3CC - nx2 nx2 nx2"
	CFF	"3CC - Fading - nx2 nx2 nx2"
	CH	"3CC - nx4 nx2 nx2"
	CHF	"3CC - Fading - nx4 nx2 nx2"

	<Scenario>	Scenario name
	CHSM4	"3CC - nx2 nx4 nx2"
	CJ	"3CC - nx4 nx4 nx2"
	CJF	"3CC - Fading - nx4 nx4 nx2"
	CJFS4	"3CC - Fading - nx2 nx4 nx4"
	CJSM4	"3CC - nx2 nx4 nx4"
	CL	"3CC - nx4 nx4 nx4"
D	DD	"4CC - 1x1 1x1 1x1 1x1"
	DH	"4CC - nx2 nx2 nx2 nx2"
	DHF	"4CC - Fading - nx2 nx2 nx2 nx2"
	DJ	"4CC - nx4 nx2 nx2 nx2"
	DJSM4	"4CC - nx2 nx4 nx2 nx2"
	DL	"4CC - nx4 nx4 nx2 nx2"
	DLSM4	"4CC - nx2 nx4 nx4 nx2"
	DN	"4CC - nx4 nx4 nx4 nx2"
	DNSM4	"4CC - nx2 nx4 nx4 nx4"
	DP	"4CC - nx4 nx4 nx4 nx4"
	DPF	"4CC - Fading - nx4 nx4 nx4 nx4"
E	EE	"5CC - 1x1 1x1 1x1 1x1 1x1"
	EJ	"5CC - nx2 nx2 nx2 nx2 nx2"
	EJF	"5CC - Fading - nx2 nx2 nx2 nx2 nx2"
	EL	"5CC - nx4 nx2 nx2 nx2 nx2"
	ELSM4	"5CC - nx2 nx4 nx2 nx2 nx2"
	EN	"5CC - nx4 nx4 nx2 nx2 nx2"
	ENSM4	"5CC - nx2 nx4 nx4 nx2 nx2"
	EP	"5CC - nx4 nx4 nx4 nx2 nx2"
	EPF	"5CC - Fading - nx4 nx4 nx4 nx2 nx2"
	EPFS4	"5CC - Fading - nx2 nx4 nx4 nx4 nx2"
	EPSM4	"5CC - nx2 nx4 nx4 nx4 nx2"
	ER	"5CC - nx4 nx4 nx4 nx4 nx2"
	ERSM4	"5CC - nx2 nx4 nx4 nx4 nx4"
	ET	"5CC - nx4 nx4 nx4 nx4 nx4"
F	FF	"6CC - 1x1 1x1 1x1 1x1 1x1 1x1"
	FL	"6CC - nx2 nx2 nx2 nx2 nx2 nx2"

	<Scenario>	Scenario name
	FLF	"6CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2"
	FN	"6CC - nx4 nx2 nx2 nx2 nx2 nx2"
	FNSM4	"6CC - nx2 nx4 nx2 nx2 nx2 nx2"
	FP	"6CC - nx4 nx4 nx2 nx2 nx2 nx2"
	FPF	"6CC - Fading - nx4 nx4 nx2 nx2 nx2 nx2"
	FPFS4	"6CC - Fading - nx2 nx4 nx4 nx2 nx2 nx2"
	FPSM4	"6CC - nx2 nx4 nx4 nx2 nx2 nx2"
	FR	"6CC - nx4 nx4 nx4 nx2 nx2 nx2"
	FRSM4	"6CC - nx2 nx4 nx4 nx4 nx2 nx2"
	FT	"6CC - nx4 nx4 nx4 nx4 nx2 nx2"
	FTSM4	"6CC - nx2 nx4 nx4 nx4 nx4 nx2"
	FV	"6CC - nx4 nx4 nx4 nx4 nx4 nx2"
	FVSM4	"6CC - nx2 nx4 nx4 nx4 nx4 nx4"
	FX	"6CC - nx4 nx4 nx4 nx4 nx4 nx4"
G	GG	"7CC - 1x1 1x1 1x1 1x1 1x1 1x1"
	GN	"7CC - nx2 nx2 nx2 nx2 nx2 nx2"
	GNF	"7CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2"
	GP	"7CC - nx4 nx2 nx2 nx2 nx2 nx2"
	GPF	"7CC - Fading - nx4 nx2 nx2 nx2 nx2 nx2"
	GPFS4	"7CC - Fading - nx2 nx4 nx2 nx2 nx2 nx2"
	GPSM4	"7CC - nx2 nx4 nx2 nx2 nx2 nx2"
	GR	"7CC - nx4 nx4 nx2 nx2 nx2 nx2"
	GRSM4	"7CC - nx2 nx4 nx4 nx2 nx2 nx2"
	GT	"7CC - nx4 nx4 nx4 nx2 nx2 nx2"
	GTSM4	"7CC - nx2 nx4 nx4 nx4 nx2 nx2"
	GV	"7CC - nx4 nx4 nx4 nx4 nx2 nx2"
	GVSM4	"7CC - nx2 nx4 nx4 nx4 nx4 nx2"
	GX	"7CC - nx4 nx4 nx4 nx4 nx4 nx2"
	GXSM4	"7CC - nx2 nx4 nx4 nx4 nx4 nx2"
	GYA	"7CC - nx4 nx4 nx4 nx4 nx4 nx2"
	GYAS4	"7CC - nx2 nx4 nx4 nx4 nx4 nx4"
	GYC	"7CC - nx4 nx4 nx4 nx4 nx4 nx4"
H	HH	"8CC - 1x1 1x1 1x1 1x1 1x1 1x1 1x1 1x1"

	<Scenario>	Scenario name
	HP	"8CC - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2"
	HPF	"8CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2"
	HR	"8CC - nx4 nx2 nx2 nx2 nx2 nx2 nx2 nx2"
	HRSM4	"8CC - nx2 nx4 nx2 nx2 nx2 nx2 nx2 nx2"
	HT	"8CC - nx4 nx4 nx2 nx2 nx2 nx2 nx2 nx2"
	HTSM4	"8CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2 nx2"
	HV	"8CC - nx4 nx4 nx4 nx2 nx2 nx2 nx2 nx2"
	HVSM4	"8CC - nx2 nx4 nx4 nx4 nx2 nx2 nx2 nx2"
	HX	"8CC - nx4 nx4 nx4 nx4 nx2 nx2 nx2 nx2"
	HXSM4	"8CC - nx2 nx4 nx4 nx4 nx4 nx2 nx2 nx2"
	HYA	"8CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2 nx2"
	HYAS4	"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx2 nx2"
	HYC	"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx2 nx2"
	HYCS4	"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4 nx2"
	HYE	"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4 nx2"
	HYES4	"8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4 nx4"
	HYG	"8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4 nx4"
S	SCEL	"1CC - 1x1"
	SCF	"1CC - Fading - 1x1"
T	TRO	"1CC - nx2"
	TROF	"1CC - Fading - nx2"

ROUTE:LTE:SIGN<i>?</i>

Returns the configured routing settings. The parameters <Scenario> and <Controller> are always returned. From the other parameters, only the subset relevant for the active scenario is returned.

For possible connector and converter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Return values:

<Scenario>	NAV SCEL TRO AD SCF TROF ADF CATR CAFR BF BFSM4 BH CATF CAFF BFF BHF CC CCMP CCMS1 CF CH CHSM4 CJ CJSM4 CL CFF CHF CJF CJFS4 DD DH DJ DJSM4 DL DLMS4 DN DNSM4 DP DHF DPF EE EJ EL ELSM4 EN ENSM4 EP EPSM4 ER ERS4 ET EJF EPF EPFS4 FF FL FN FNSM4 FP FPSM4 FR FRSM4 FT FTSM4 FV FVSM4 FX FLF FPF FPFS4 GG GN GP GPSM4 GR GRSM4 GT GTSM4 GV GVSM4 GX GXSM4 GYA GYAS4 GYC GNF GPF GPFS4 HH HP HR HRSM4 HT HTSM4 HV HVSM4 HX HXSM4 HYA HYAS4 HYC HYCS4 HYE HYES4 HYG HPF Active scenario For mapping of the values to scenario names, see ROUTE:LTE:SIGN<i>:SCENario? .
<Controller>	string For future use - returned value not relevant
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<TXConnector1>	RF connector for output path 1
<TXConverter1>	TX module for output path 1
<TXConnector2>	RF connector for output path 2
<TXConverter2>	TX module for output path 2
<TXConnector3>	RF connector for output path 3
<TXConverter3>	TX module for output path 3
<TXConnector4>	RF connector for output path 4
<TXConverter4>	TX module for output path 4
<TXConnector5>	RF connector for output path 5
<TXConverter5>	TX module for output path 5
<TXConnector6>	RF connector for output path 6
<TXConverter6>	TX module for output path 6
<TXConnector7>	RF connector for output path 7
<TXConverter7>	TX module for output path 7
<TXConnector8>	RF connector for output path 8
<TXConverter8>	TX module for output path 8
<TXConnector9>	RF connector for output path 9
<TXConverter9>	TX module for output path 9

<TXConnector10>	RF connector for output path 10
<TXConverter10>	TX module for output path 10
<TXConnector11>	RF connector for output path 11
<TXConverter11>	TX module for output path 11
<TXConnector12>	RF connector for output path 12
<TXConverter12>	TX module for output path 12
<TXConnector13>	RF connector for output path 13
<TXConverter13>	TX module for output path 13
<TXConnector14>	RF connector for output path 14
<TXConverter14>	TX module for output path 14
<TXConnector15>	RF connector for output path 15
<TXConverter15>	TX module for output path 15
<TXConnector16>	RF connector for output path 16
<TXConverter16>	TX module for output path 16
<IQConnector1>	No longer relevant
<IQConnector2>	No longer relevant
<IQConnector3>	No longer relevant
<IQConnector4>	No longer relevant
<IQConnector5>	No longer relevant
<IQConnector6>	No longer relevant
<IQConnector7>	No longer relevant
<IQConnector8>	No longer relevant

Usage: Query only

Firmware/software: V2.0.10, some return values and scenarios added in later versions

ROUTE:LTE:SIGN<i>:SCENario:AD[:FLEXible] <PCCBBBoard>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <TX2Connector>, <TX2Converter>, <TX3Connector>, <TX3Converter>, <TX4Connector>, <TX4Converter>

Activates the scenario "1CC - nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for all paths

<RXConnector>	RF connector for the input path
<RXConverter>	RX module for the input path
<TX1Connector>	RF connector for the first output path
<TX1Converter>	TX module for the first output path
<TX2Connector>	RF connector for the second output path
<TX2Converter>	TX module for the second output path
<TX3Connector>	RF connector for the third output path
<TX3Converter>	TX module for the third output path
<TX4Connector>	RF connector for the fourth output path
<TX4Converter>	TX module for the fourth output path

Firmware/software: V3.7.10

Options: R&S CMW-KS521 or R&S CMW-KS540

ROUTE:LTE:SIGN<i>:SCENario:ADF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <TX2Connector>, <TX2Converter>, <TX3Connector>, <TX3Converter>, <TX4Connector>, <TX4Converter>, <PCCFadingBoard>

Activates the scenario "1CC - Fading - nx4" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit
<RXConnector>	RF connector for the input path
<RXConverter>	RX module for the input path
<TX1Connector>	RF connector for the first output path
<TX1Converter>	TX module for the first output path
<TX2Connector>	RF connector for the second output path
<TX2Converter>	TX module for the second output path
<TX3Connector>	RF connector for the third output path
<TX3Converter>	TX module for the third output path
<TX4Connector>	RF connector for the fourth output path
<TX4Converter>	TX module for the fourth output path
<PCCFadingBoard>	Internal fader

Firmware/software: V3.7.10

Options: R&S CMW-KS510, -KS540, -KE504

ROUTE:LTE:SIGN<i>:SCENario:BF[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>

Activates the scenario "2CC - nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<PCCTX3Connector> RF connector for the third PCC output path

<PCCTX3Converter> TX module for the third PCC output path

<PCCTX4Connector> RF connector for the fourth PCC output path

<PCCTX4Converter> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC

<SCC1TX1Conn> RF connector for the first SCC output path

<SCC1TX1Conv> TX module for the first SCC output path

<SCC1TX2Conn> RF connector for the second SCC output path

<SCC1TX2Conv> TX module for the second SCC output path

Firmware/software: V3.7.10

Options: R&S CMW-KS502 for FDD / R&S CMW-KS552 for TDD
 R&S CMW-KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:BFF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <PCCFadingBoard>, <SCC1FadingBoard>

Activates the scenario "2CC - Fading - nx4 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC
<SCC1TX1Conn>	RF connector for the first SCC output path
<SCC1TX1Conv>	TX module for the first SCC output path
<SCC1TX2Conn>	RF connector for the second SCC output path
<SCC1TX2Conv>	TX module for the second SCC output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC

Firmware/software: V3.7.10

Options: R&S CMW-KS512, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i></i>:SCENario:BFSM<MIMO4x4>[:FLEXible] <PCCBBBoard>,<PCCRConnector>,<PCCRConverter>,<PCCTX1Connector>,<PCCTX1Converter>,<PCCTX2Connector>,<PCCTX2Converter>,<SCCBBBoard>,<SCCTX1Connector>,<SCCTX1Converter>,<SCCTX2Connector>,<SCCTX2Converter>,<SCCTX3Connector>,<SCCTX3Converter>,<SCCTX4Connector>,<SCCTX4Converter>

Activates the scenario "2CC - nx2 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCCBBBoard> Signaling unit for the SCC

<SCCTX1Connector> RF connector for the first SCC output path

<SCCTX1Converter> TX module for the first SCC output path

<SCCTX2Connector> RF connector for the second SCC output path

<SCCTX2Converter> TX module for the second SCC output path

<SCCTX3Connector> RF connector for the third SCC output path

<SCCTX3Converter> TX module for the third SCC output path

<SCCTX4Connector> RF connector for the fourth SCC output path

<SCCTX4Converter> TX module for the fourth SCC output path

Firmware/software: V3.7.10

Options: R&S CMW-KS502 for FDD / R&S CMW-KS552 for TDD
R&S CMW-KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:BH[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>

Activates the scenario "2CC - nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<PCCTX3Connector> RF connector for the third PCC output path

<PCCTX3Converter> TX module for the third PCC output path

<PCCTX4Connector> RF connector for the fourth PCC output path

<PCCTX4Converter> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC

<SCC1TX1Conn> RF connector for the first SCC output path

<SCC1TX1Conv> TX module for the first SCC output path

<SCC1TX2Conn> RF connector for the second SCC output path

<SCC1TX2Conv> TX module for the second SCC output path

<SCC1TX3Conn> RF connector for the third SCC output path

<SCC1TX3Conv> TX module for the third SCC output path

<SCC1TX4Conn> RF connector for the fourth SCC output path

<SCC1TX4Conv> TX module for the fourth SCC output path

Firmware/software: V3.7.10

Options: R&S CMW-KS502 for FDD / R&S CMW-KS552 for TDD
 R&S CMW-KS540

ROUTE:LTE:SIGN<i>:SCENario:BHF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <PCCFadingBoard>, <SCC1FadingBoard>

Activates the scenario "2CC - Fading - nx4 nx4" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC
<SCC1TX1Conn>	RF connector for the first SCC output path
<SCC1TX1Conv>	TX module for the first SCC output path
<SCC1TX2Conn>	RF connector for the second SCC output path
<SCC1TX2Conv>	TX module for the second SCC output path
<SCC1TX3Conn>	RF connector for the third SCC output path
<SCC1TX3Conv>	TX module for the third SCC output path
<SCC1TX4Conn>	RF connector for the fourth SCC output path
<SCC1TX4Conv>	TX module for the fourth SCC output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC

Firmware/software: V3.7.10

Options: R&S CMW-KS512, -KS540, -KE504

ROUTE:LTE:SIGN<i>:SCENario:CAFF:FLEXible:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>]

Activates the scenario "2CC - Fading - nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC
<SCC1TX1Conn>	RF connector for the first SCC output path
<SCC1TX1Conv>	TX module for the first SCC output path
<SCC1TX2Conn>	RF connector for the second SCC output path
<SCC1TX2Conv>	TX module for the second SCC output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC

Example: See [Selecting a scenario](#)

Firmware/software: V3.5.40, V3.7.10 added fader selection

Options: R&S CMW-KS512, -KS520, -KE500

ROUTE:LTE:SIGN<i>:SCENario:CAFRfout:FLEXible <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCCBBBoard>, <SCCTX1Connector>, <SCCTX1Converter>, <SCCTX2Connector>, <SCCTX2Converter>[, <SCCRConnector>, <SCCRConverter>]

Activates the scenario "2CC - nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCCBBBoard>	Signaling unit for the SCC
<SCCTX1Connector>	RF connector for the first SCC output path
<SCCTX1Converter>	TX module for the first SCC output path
<SCCTX2Connector>	RF connector for the second SCC output path
<SCCTX2Converter>	TX module for the second SCC output path
<SCCRConnector>	RF connector for the SCC input path, for UL CA only
<SCCRConverter>	RX module for the SCC input path, for UL CA only

Firmware/software: V3.5.40

Options: R&S CMW-KS502 for FDD / R&S CMW-KS552 for TDD
R&S CMW-KS520

ROUTE:LTE:SIGN<i>:SCENario:CATF:FLEXible:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTXConnector>, <PCCTXConverter>, <SCC1BBBoard>, <SCCTXConnector>, <SCCTXConverter>[, <PCCFadingBoard>, <SCC1FadingBoard>]

Activates the scenario "2CC - Fading - 1x1 1x1" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTXConnector>	RF connector for the PCC output path
<PCCTXConverter>	TX module for the PCC output path
<SCC1BBBoard>	Signaling unit for the SCC
<SCCTXConnector>	RF connector for the SCC output path
<SCCTXConverter>	TX module for the SCC output path

<PCCFadingBoard> Internal fader for the PCC

<SCC1FadingBoard> Internal fader for the SCC

Firmware/software: V3.5.40, V3.7.10 added fader selection

Options: R&S CMW-KS512, R&S CMW-KE500

ROUTE:LTE:SIGN<i>:SCENario:CATRfout:FLEXible <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTXConnector>, <PCCTXConverter>, <SCCBBBoard>, <SCCTXConnector>, <SCCTXConverter>[, <SCCRXConnector>, <SCCRXConverter>]

Activates the scenario "2CC - 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<RXConnector> RF connector for the PCC input path

<RXConverter> RX module for the PCC input path

<PCCTXConnector> RF connector for the PCC output path

<PCCTXConverter> TX module for the PCC output path

<SCCBBBoard> Signaling unit for the SCC

<SCCTXConnector> RF connector for the SCC output path

<SCCTXConverter> TX module for the SCC output path

<SCCRXConnector> RF connector for the SCC input path, for UL CA only

<SCCRXConverter> RX module for the SCC input path, for UL CA only

Firmware/software: V3.5.40

Options: R&S CMW-KS502 for FDD / R&S CMW-KS552 for TDD

ROUTE:LTE:SIGN<i>:SCENario:CC:FLEXible <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTXConnector>, <PCCTXConverter>, <SCC1BBBoard>, <SCC1TXConnector>, <SCC1TXConverter>, <SCC2BBBoard>, <SCC2TXConnector>, <SCC2TXConverter>

Activates the scenario "3CC - 1x1 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<RXConnector> RF connector for the PCC input path

<RXConverter> RX module for the PCC input path

<PCCTXConnector> RF connector for the PCC output path
 <PCCTXConverter> TX module for the PCC output path
 <SCC1BBBoard> Signaling unit for the SCC1
 <SCC1TXConnector> RF connector for the SCC1 output path
 <SCC1TXConverter> TX module for the SCC1 output path
 <SCC2BBBoard> Signaling unit for the SCC2
 <SCC2TXConnector> RF connector for the SCC2 output path
 <SCC2TXConverter> TX module for the SCC2 output path

Firmware/software: V3.5.40

Options: R&S CMW-KS512

ROUTE:LTE:SIGN<i>:SCENario:CCMP:FLEXible <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TXConnector>, <SCC1TXConverter>, <SCC2BBBoard>, <SCC2TXConnector>, <SCC2TXConverter>

Activates the scenario "3CC - nx2 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC
 <RXConnector> RF connector for the PCC input path
 <RXConverter> RX module for the PCC input path
 <PCCTX1Connector> RF connector for the first PCC output path
 <PCCTX1Converter> TX module for the first PCC output path
 <PCCTX2Connector> RF connector for the second PCC output path
 <PCCTX2Converter> TX module for the second PCC output path
 <SCC1BBBoard> Signaling unit for the SCC1
 <SCC1TXConnector> RF connector for the SCC1 output path
 <SCC1TXConverter> TX module for the SCC1 output path
 <SCC2BBBoard> Signaling unit for the SCC2
 <SCC2TXConnector> RF connector for the SCC2 output path
 <SCC2TXConverter> TX module for the SCC2 output path

Firmware/software: V3.5.40

Options: R&S CMW-KS512, -KS520

ROUTE:LTE:SIGN<i>:SCENario:CCMS<c>:FLEXible <PCCBBBoard>,
 <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>,
 <SCC1BBBoard>, <SCC1TX1Connect>, <SCC1TX1Convert>,
 <SCC1TX2Connect>, <SCC1TX2Convert>, <SCC2BBBoard>,
 <SCC2TXConnector>, <SCC2TXConverter>

Activates the scenario "3CC - 1x1 nx2 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<c> 1
 MIMO SCC number

Parameters:

<PCCBBBoard> Signaling unit for the PCC
 <RXConnector> RF connector for the PCC input path
 <RXConverter> RX module for the PCC input path
 <PCCTX1Connector> RF connector for the PCC output path
 <PCCTX1Converter> TX module for the PCC output path
 <SCC1BBBoard> Signaling unit for the SCC1
 <SCC1TX1Connect> RF connector for the first SCC1 output path
 <SCC1TX1Convert> TX module for the first SCC1 output path
 <SCC1TX2Connect> RF connector for the second SCC1 output path
 <SCC1TX2Convert> TX module for the second SCC1 output path
 <SCC2BBBoard> Signaling unit for the SCC2
 <SCC2TXConnector> RF connector for the SCC2 output path
 <SCC2TXConverter> TX module for the SCC2 output path

Example: See [Selecting a scenario](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS512, -KS520

ROUTE:LTE:SIGN<i>:SCENario:CF[:FLEXible] <PCCBBBoard>, <PCCRConnect>,
 <PCCRConvert>, <PCCTX1Connect>, <PCCTX1Convert>,
 <PCCTX2Connect>, <PCCTX2Convert>, <SCC1BBBoard>,
 <SCC1TX1Connect>, <SCC1TX1Convert>, <SCC1TX2Connect>,
 <SCC1TX2Convert>, <SCC2BBBoard>, <SCC2TX1Connect>,
 <SCC2TX1Convert>, <SCC2TX2Connect>, <SCC2TX2Convert>[,
 <Coprocesor>]

Activates the scenario "3CC - nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnect>	RF connector for the PCC input path
<PCCRConvert>	RX module for the PCC input path
<PCCTX1Connect>	RF connector for the first PCC output path
<PCCTX1Convert>	TX module for the first PCC output path
<PCCTX2Connect>	RF connector for the second PCC output path
<PCCTX2Convert>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Connect>	RF connector for the first SCC1 output path
<SCC1TX1Convert>	TX module for the first SCC1 output path
<SCC1TX2Connect>	RF connector for the second SCC1 output path
<SCC1TX2Convert>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Connect>	RF connector for the first SCC2 output path
<SCC2TX1Convert>	TX module for the first SCC2 output path
<SCC2TX2Connect>	RF connector for the second SCC2 output path
<SCC2TX2Convert>	TX module for the second SCC2 output path
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.5.30, V3.7.30 added <Coproprocessor>
Options:	R&S CMW-KS512, -KS520

ROUTE:LTE:SIGN<i>:SCENario:CFF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnect>, <RXConvert>, <PCCTX1Connect>, <PCCTX1Convert>, <PCCTX2Connect>, <PCCTX2Convert>, <SCC1BBBoard>, <SCC1TX1Connect>, <SCC1TX1Convert>, <SCC1TX2Connect>, <SCC1TX2Convert>, <SCC2BBBoard>, <SCC2TX1Connect>, <SCC2TX1Convert>, <SCC2TX2Connect>, <SCC2TX2Convert>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>[, <Coproprocessor>]]

Activates the scenario "3CC - Fading - nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnect>	RF connector for the PCC input path
<RXConvert>	RX module for the PCC input path
<PCCTX1Connect>	RF connector for the first PCC output path
<PCCTX1Convert>	TX module for the first PCC output path
<PCCTX2Connect>	RF connector for the second PCC output path
<PCCTX2Convert>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Connect>	RF connector for the first SCC1 output path
<SCC1TX1Convert>	TX module for the first SCC1 output path
<SCC1TX2Connect>	RF connector for the second SCC1 output path
<SCC1TX2Convert>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Connect>	RF connector for the first SCC2 output path
<SCC2TX1Convert>	TX module for the first SCC2 output path
<SCC2TX2Connect>	RF connector for the second SCC2 output path
<SCC2TX2Convert>	TX module for the second SCC2 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.5.30, V3.7.10 added fader selection V3.7.30 added <Coproprocessor>
Options:	R&S CMW-KS512, -KS520, -KE500

ROUTE:LTE:SIGN<i>:SCENario:CH[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>[, <Coproprocessor>]

Activates the scenario "3CC - nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:CHF[:FLEXible]:Internal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>[, <Coproprocessor>]

Activates the scenario "3CC - Fading - nx4 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.30

Options: R&S CMW-KS512, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:SCENario:CHSM<MIMO44>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>[, <Coproprocessor>]

Activates the scenario "3CC - nx2 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO44> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv> TX module for the first SCC2 output path

<SCC2TX2Conn> RF connector for the second SCC2 output path

<SCC2TX2Conv> TX module for the second SCC2 output path

<Coproprocessor> SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:CJ[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>[, <Coprocessor>]

Activates the scenario "3CC - nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<PCCTX3Connector> RF connector for the third PCC output path

<PCCTX3Converter> TX module for the third PCC output path

<PCCTX4Connector> RF connector for the fourth PCC output path

<PCCTX4Converter> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.7.10, V3.7.30 added <Coproprocessor>
Options:	R&S CMW-KS512, -KS520, -KS540

ROUTe:LTE:SIGN<i>:SCENario:CJF[:FLEXible]:INteRnal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>[, <Coproprocessor>]

Activates the scenario "3CC - Fading - nx4 nx4 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path

<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.7.10, V3.7.30 added <Coproprocessor>
Options:	R&S CMW-KS512, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:SCENario:CJFS<MIMO44>[:FLEXible]:INTernal

<PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>[, <Coproprocessor>]

Activates the scenario "3CC - Fading - nx2 nx4 nx4" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO44> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path

<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.30

Options: R&S CMW-KS512, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:SCENario:CJSM<MIMO44>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>[, <Coprocessor>]

Activates the scenario "3CC - nx2 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO44> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRXCConnector> RF connector for the PCC input path

<PCCRXCConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv> TX module for the first SCC2 output path

<SCC2TX2Conn> RF connector for the second SCC2 output path

<SCC2TX2Conv> TX module for the second SCC2 output path

<SCC2TX3Conn> RF connector for the third SCC2 output path

<SCC2TX3Conv> TX module for the third SCC2 output path

<SCC2TX4Conn> RF connector for the fourth SCC2 output path

<SCC2TX4Conv> TX module for the fourth SCC2 output path

<Coproprocessor> SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:CL[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>,
 <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>[, <Coprocesor>]

Activates the scenario "3CC - nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path

<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS540

ROUTE:LTE:SIGN<i>:SCENario:DD:FLEXible <PCCBBBoard>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <SCC1BBBoard>, <TX2Connector>, <TX2Converter>, <SCCBBBoard2>, <TX3Connector>, <TX3Converter>, <SCCBBBoard3>, <TX4Connector>, <TX4Converter>[, <Coproprocessor>]

Activates the scenario "4CC - 1x1 1x1 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<TX1Connector>	RF connector for the PCC output path
<TX1Converter>	TX module for the PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<TX2Connector>	RF connector for the SCC1 output path
<TX2Converter>	TX module for the SCC1 output path
<SCCBBBoard2>	Signaling unit for the SCC2
<TX3Connector>	RF connector for the SCC2 output path
<TX3Converter>	TX module for the SCC2 output path
<SCCBBBoard3>	Signaling unit for the SCC3
<TX4Connector>	RF connector for the SCC3 output path
<TX4Converter>	TX module for the SCC3 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.5.40, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512

ROUTE:LTE:SIGN<i>:SCENario:DH[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>,
 <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>,
 <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>,
 <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>[, <Coproprocessor>]

Activates the scenario "4CC - nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.5.30, V3.7.30 added <Coprocesor>

Options: R&S CMW-KS512, -KS520

ROUTE:LTE:SIGN<i>:SCENario:DHf[:FLEXible]:INTernal <PCCBBBoard>, <RXConnect>, <RXConvert>, <PCCTX1Connect>, <PCCTX1Convert>, <PCCTX2Connect>, <PCCTX2Convert>, <SCC1BBBoard>, <SCC1TX1Connect>, <SCC1TX1Convert>, <SCC1TX2Connect>, <SCC1TX2Convert>, <SCC2BBBoard>, <SCC2TX1Connect>, <SCC2TX1Convert>, <SCC2TX2Connect>, <SCC2TX2Convert>, <SCC3BBBoard>, <SCC3TX1Connect>, <SCC3TX1Convert>, <SCC3TX2Connect>, <SCC3TX2Convert>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>[, <Coprocesor>]]

Activates the scenario "4CC - Fading - nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnect>	RF connector for the PCC input path
<RXConvert>	RX module for the PCC input path
<PCCTX1Connect>	RF connector for the first PCC output path
<PCCTX1Convert>	TX module for the first PCC output path
<PCCTX2Connect>	RF connector for the second PCC output path
<PCCTX2Convert>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Connect>	RF connector for the first SCC1 output path
<SCC1TX1Convert>	TX module for the first SCC1 output path
<SCC1TX2Connect>	RF connector for the second SCC1 output path
<SCC1TX2Convert>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Connect>	RF connector for the first SCC2 output path
<SCC2TX1Convert>	TX module for the first SCC2 output path
<SCC2TX2Connect>	RF connector for the second SCC2 output path
<SCC2TX2Convert>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Connect>	RF connector for the first SCC3 output path

<SCC3TX1Convert> TX module for the first SCC3 output path
 <SCC3TX2Connect> RF connector for the second SCC3 output path
 <SCC3TX2Convert> TX module for the second SCC3 output path
 <PCCFadingBoard> Internal fader for the PCC
 <SCC1FadingBoard> Internal fader for the SCC1
 <SCC2FadingBoard> Internal fader for the SCC2
 <SCC3FadingBoard> Internal fader for the SCC3
 <Coprocessor> SUA for coprocessing

Example: See [Selecting a scenario](#)

Firmware/software: V3.5.30, V3.7.10 added fader selection
 V3.7.30 added <Coprocessor>

Options: R&S CMW-KS512, -KS520, -KE500

ROUTE:LTE:SIGN<i>:SCENario:DJ[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>[,
 <Coprocessor>]

Activates the scenario "4CC - nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC
 <PCCRConnector> RF connector for the PCC input path
 <PCCRConverter> RX module for the PCC input path
 <PCCTX1Connector> RF connector for the first PCC output path
 <PCCTX1Converter> TX module for the first PCC output path
 <PCCTX2Connector> RF connector for the second PCC output path
 <PCCTX2Converter> TX module for the second PCC output path
 <PCCTX3Connector> RF connector for the third PCC output path
 <PCCTX3Converter> TX module for the third PCC output path
 <PCCTX4Connector> RF connector for the fourth PCC output path
 <PCCTX4Converter> TX module for the fourth PCC output path

<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.7.10, V3.7.30 added <Coproprocessor>
Options:	R&S CMW-KS512, -KS520, -KS540

ROUTe:LTE:SIGN<i>:SCENario:DJSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>[, <Coproprocessor>]

Activates the scenario "4CC - nx2 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path

<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.30

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:DL[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>[,
 <Coprocesor>]

Activates the scenario "4CC - nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path

<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:DLSM<MIMO4x4>[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>,
 <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>,
 <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>,
 <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>,
 <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>,
 <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>[, <Coproprocessor>]

Activates the scenario "4CC - nx2 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path

<SCC1TX2Conn>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.30

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:DN[:FLEXible] <PCCBBBoard>, <PCCRXConnector>, <PCCRXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>[, <Coprocessor>]

Activates the scenario "4CC - nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path

<SCC3TX2Conv> TX module for the second SCC3 output path

<Coproprocessor> SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:DNSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>[, <Coproprocessor>]

Activates the scenario "4CC - nx2 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.30

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:DP[:FLExiBle] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>[, <Coprocessor>]

Activates the scenario "4CC - nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path

<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<Coprocesor>	SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coprocesor>

Options: R&S CMW-KS512, -KS540

ROUTE:LTE:SIGN<i>:SCENario:DPF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>[, <Coprocesor>]]

Activates the scenario "4CC - Fading - nx4 nx4 nx4 nx4" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path

<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS512, -KS540, -KE504

ROUTE:LTE:SIGN<i>:SCENario:EE[:FLEXible] <PCCBBBoard>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <SCC1BBBoard>, <TX2Connector>, <TX2Converter>, <SCCBBBoard2>, <TX3Connector>, <TX3Converter>, <SCCBBBoard3>, <TX4Connector>, <TX4Converter>, <SCCBBBoard4>, <TX5Connector>, <TX5Converter>[, <Coproprocessor>]

Activates the scenario "5CC - 1x1 1x1 1x1 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<TX1Connector>	RF connector for the PCC output path
<TX1Converter>	TX module for the PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<TX2Connector>	RF connector for the SCC1 output path
<TX2Converter>	TX module for the SCC1 output path
<SCCBBBoard2>	Signaling unit for the SCC2
<TX3Connector>	RF connector for the SCC2 output path
<TX3Converter>	TX module for the SCC2 output path
<SCCBBBoard3>	Signaling unit for the SCC3
<TX4Connector>	RF connector for the SCC3 output path
<TX4Converter>	TX module for the SCC3 output path
<SCCBBBoard4>	Signaling unit for the SCC4
<TX5Connector>	RF connector for the SCC4 output path
<TX5Converter>	TX module for the SCC4 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512

ROUTE:LTE:SIGN<i>:SCENario:EJ[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>,
 <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>,
 <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>,
 <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>,
 <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[,
 <Coprocesor>]

Activates the scenario "5CC - nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4

<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coprocessor>

Options: R&S CMW-KS512, -KS520

ROUTE:LTE:SIGN<i>:SCENario:EJF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnect>, <RXConvert>, <PCCTX1Connect>, <PCCTX1Convert>, <PCCTX2Connect>, <PCCTX2Convert>, <SCC1BBBoard>, <SCC1TX1Connect>, <SCC1TX1Convert>, <SCC1TX2Connect>, <SCC1TX2Convert>, <SCC2BBBoard>, <SCC2TX1Connect>, <SCC2TX1Convert>, <SCC2TX2Connect>, <SCC2TX2Convert>, <SCC3BBBoard>, <SCC3TX1Connect>, <SCC3TX1Convert>, <SCC3TX2Connect>, <SCC3TX2Convert>, <SCC4BBBoard>, <SCC4TX1Connect>, <SCC4TX1Convert>, <SCC4TX2Connect>, <SCC4TX2Convert>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>[, <Coprocessor>]]

Activates the scenario "5CC - Fading - nx2 nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnect>	RF connector for the PCC input path
<RXConvert>	RX module for the PCC input path
<PCCTX1Connect>	RF connector for the first PCC output path
<PCCTX1Convert>	TX module for the first PCC output path
<PCCTX2Connect>	RF connector for the second PCC output path
<PCCTX2Convert>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Connect>	RF connector for the first SCC1 output path
<SCC1TX1Convert>	TX module for the first SCC1 output path
<SCC1TX2Connect>	RF connector for the second SCC1 output path
<SCC1TX2Convert>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2

<SCC2TX1Connect> RF connector for the first SCC2 output path
 <SCC2TX1Convert> TX module for the first SCC2 output path
 <SCC2TX2Connect> RF connector for the second SCC2 output path
 <SCC2TX2Convert> TX module for the second SCC2 output path
 <SCC3BBBoard> Signaling unit for the SCC3
 <SCC3TX1Connect> RF connector for the first SCC3 output path
 <SCC3TX1Convert> TX module for the first SCC3 output path
 <SCC3TX2Connect> RF connector for the second SCC3 output path
 <SCC3TX2Convert> TX module for the second SCC3 output path
 <SCC4BBBoard> Signaling unit for the SCC4
 <SCC4TX1Connect> RF connector for the first SCC4 output path
 <SCC4TX1Convert> TX module for the first SCC4 output path
 <SCC4TX2Connect> RF connector for the second SCC4 output path
 <SCC4TX2Convert> TX module for the second SCC4 output path
 <PCCFadingBoard> Internal fader for the PCC
 <SCC1FadingBoard> Internal fader for the SCC1
 <SCC2FadingBoard> Internal fader for the SCC2
 <SCC3FadingBoard> Internal fader for the SCC3
 <SCC4FadingBoard> Internal fader for the SCC4
 <Coproprocessor> SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS520, -KE500

ROUTE:LTE:SIGN<i>:SCENario:EL[:FLEXible] <PCCBBBoard>,
 <PCCRXConnector>, <PCCRXConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>,
 <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>,
 <SCC4TX2Conv>[, <Coproprocessor>]

Activates the scenario "5CC - nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.30

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:ELSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[, <Coprocessor>]

Activates the scenario "5CC - nx2 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv> TX module for the first SCC2 output path

<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coprocesor>	SUA for coprocessing

Firmware/software: V3.7.30

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:EN[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>,
 <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>,
 <SCC4TX2Conv>[, <Coprocesor>]

Activates the scenario "5CC - nx4 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path

<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS512, -KS520, -KS540

ROUTe:LTE:SIGN<i>:SCENario:ENSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[, <Coprocesor>]

Activates the scenario "5CC - nx2 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv> TX module for the first SCC2 output path

<SCC2TX2Conn> RF connector for the second SCC2 output path

<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coprocesor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:EP[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[, <Coprocesor>]

Activates the scenario "5CC - nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path

<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path

<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS512, -KS520, -KS540

ROUTe:LTE:SIGN<i>:SCENario:EPF[:FLEXible]:INTErnal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>[, <Coproprocessor>]]

Activates the scenario "5CC - Fading - nx4 nx4 nx4 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path

<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS512, -KS520, -KS540, -KE500, -KE504

ROUTe:LTE:SIGN<i>:SCENario:EPFS<MIMO4x4>[:FLEXible]:INTernal
 <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>,
 <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>,
 <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>,
 <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>,
 <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>,
 <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>,
 <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>,
 <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[,
 <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>,
 <SCC3FadingBoard>, <SCC4FadingBoard>[, <Coprocessor>]]

Activates the scenario "5CC - Fading - nx2 nx4 nx4 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC
 <RXConnector> RF connector for the PCC input path
 <RXConverter> RX module for the PCC input path
 <PCCTX1Connector> RF connector for the first PCC output path
 <PCCTX1Converter> TX module for the first PCC output path
 <PCCTX2Connector> RF connector for the second PCC output path
 <PCCTX2Converter> TX module for the second PCC output path
 <SCC1BBBoard> Signaling unit for the SCC1
 <SCC1TX1Conn> RF connector for the first SCC1 output path
 <SCC1TX1Conv> TX module for the first SCC1 output path
 <SCC1TX2Conn> RF connector for the second SCC1 output path
 <SCC1TX2Conv> TX module for the second SCC1 output path
 <SCC1TX3Conn> RF connector for the third SCC1 output path
 <SCC1TX3Conv> TX module for the third SCC1 output path
 <SCC1TX4Conn> RF connector for the fourth SCC1 output path
 <SCC1TX4Conv> TX module for the fourth SCC1 output path
 <SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.7.50
Options:	R&S CMW-KS512, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:SCENario:EPSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[, <Coprocessor>]

Activates the scenario "5CC - nx2 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv> TX module for the first SCC2 output path

<SCC2TX2Conn> RF connector for the second SCC2 output path

<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:ER[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>[, <Coproprocessor>]

Activates the scenario "5CC - nx4 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path

<SCC3TX1Conn>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<Coprocesor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:ERSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>[, <Coprocesor>]

Activates the scenario "5CC - nx2 nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path

<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path

<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS512, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:ET[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>[, <Coprocessor>]

Activates the scenario "5CC - nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path

<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path

<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.10, V3.7.30 added <Coproprocessor>

Options: R&S CMW-KS512, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FF[:FLEXible] <BBBoard1>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <BBBoard2>, <TX2Connector>, <TX2Converter>, <BBBoard3>, <TX3Connector>, <TX3Converter>, <BBBoard4>, <TX4Connector>, <TX4Converter>, <BBBoard5>, <TX5Connector>, <TX5Converter>, <BBBoard6>, <TX6Connector>, <TX6Converter>[, <Coproprocessor>]

Activates the scenario "6CC - 1x1 1x1 1x1 1x1 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<BBBoard1>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<TX1Connector>	RF connector for the PCC output path
<TX1Converter>	TX module for the PCC output path
<BBBoard2>	Signaling unit for the SCC1
<TX2Connector>	RF connector for the SCC1 output path
<TX2Converter>	TX module for the SCC1 output path
<BBBoard3>	Signaling unit for the SCC2
<TX3Connector>	RF connector for the SCC2 output path
<TX3Converter>	TX module for the SCC2 output path
<BBBoard4>	Signaling unit for the SCC3
<TX4Connector>	RF connector for the SCC3 output path
<TX4Converter>	TX module for the SCC3 output path
<BBBoard5>	Signaling unit for the SCC4
<TX5Connector>	RF connector for the SCC4 output path
<TX5Converter>	TX module for the SCC4 output path

<BBBoard6>	Signaling unit for the SCC5
<TX6Connector>	RF connector for the SCC5 output path
<TX6Converter>	TX module for the SCC5 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514

ROUTe:LTE:SIGN<i>:SCENario:FL[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coproprocessor>]

Activates the scenario "6CC - nx2 nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path

<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520

ROUTE:LTE:SIGN<i>:SCENario:FN[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coprocessor>]

Activates the scenario "6CC - nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path

<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path

<SCC5TX2Conv> TX module for the second SCC5 output path

<Coproprocessor> SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FNSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coproprocessor>]

Activates the scenario "6CC - nx2 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coprocesor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTe:LTE:SIGN<i>:SCENario:FP[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coprocesor>]

Activates the scenario "6CC - nx4 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4

<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FLF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>, <SCC5FadingBoard>[, <Coprocessor>]]

Activates the scenario "6CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1

<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<SCC5FadingBoard>	Internal fader for the SCC5
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS514, -KS520, -KE500

ROUTE:LTE:SIGN<i></i>:SCENario:FPF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>, <SCC5FadingBoard>[, <Coprocessor>]]

Activates the scenario "6CC - Fading - nx4 nx4 nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path

<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<SCC5FadingBoard>	Internal fader for the SCC5
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:SCENario:FPFS<MIMO4x4>[:FLEXible]:INTernal

<PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>, <SCC5FadingBoard>[, <Coprocessor>]]

Activates the scenario "6CC - Fading - nx2 nx4 nx4 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC
 <RXConnector> RF connector for the PCC input path
 <RXConverter> RX module for the PCC input path
 <PCCTX1Connector> RF connector for the first PCC output path
 <PCCTX1Converter> TX module for the first PCC output path
 <PCCTX2Connector> RF connector for the second PCC output path
 <PCCTX2Converter> TX module for the second PCC output path
 <SCC1BBBoard> Signaling unit for the SCC1
 <SCC1TX1Conn> RF connector for the first SCC1 output path
 <SCC1TX1Conv> TX module for the first SCC1 output path
 <SCC1TX2Conn> RF connector for the second SCC1 output path
 <SCC1TX2Conv> TX module for the second SCC1 output path
 <SCC1TX3Conn> RF connector for the third SCC1 output path
 <SCC1TX3Conv> TX module for the third SCC1 output path
 <SCC1TX4Conn> RF connector for the fourth SCC1 output path
 <SCC1TX4Conv> TX module for the fourth SCC1 output path
 <SCC2BBBoard> Signaling unit for the SCC2
 <SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<SCC5FadingBoard>	Internal fader for the SCC5
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.7.60
Options:	R&S CMW-KS514, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:SCENario:FPSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coprocessor>]

Activates the scenario "6CC - nx2 nx4 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv> TX module for the first SCC2 output path

<SCC2TX2Conn> RF connector for the second SCC2 output path

<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FR[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>,
 <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>,
 <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>,
 <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>,
 <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coprocesor>]

Activates the scenario "6CC - nx4 nx4 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2

<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FRSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coprocessor>]

Activates the scenario "6CC - nx2 nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv> TX module for the fourth SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn> RF connector for the first SCC2 output path

<SCC2TX1Conv> TX module for the first SCC2 output path

<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coprocesor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FT[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConn>, <PCCTX1Conn>, <PCCTX1Conn>, <PCCTX2Conn>, <PCCTX2Conn>, <PCCTX3Conn>, <PCCTX3Conn>, <PCCTX4Conn>, <PCCTX4Conn>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conn>, <SCC1TX2Conn>, <SCC1TX2Conn>, <SCC1TX3Conn>, <SCC1TX3Conn>, <SCC1TX4Conn>, <SCC1TX4Conn>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conn>, <SCC2TX2Conn>, <SCC2TX2Conn>, <SCC2TX3Conn>, <SCC2TX3Conn>, <SCC2TX4Conn>, <SCC2TX4Conn>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conn>, <SCC3TX2Conn>, <SCC3TX2Conn>, <SCC3TX3Conn>, <SCC3TX3Conn>, <SCC3TX4Conn>, <SCC3TX4Conn>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conn>, <SCC4TX2Conn>, <SCC4TX2Conn>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conn>, <SCC5TX2Conn>, <SCC5TX2Conn>[, <Coprocesor>]

Activates the scenario "6CC - nx4 nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConn>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conn>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conn>	TX module for the second PCC output path
<PCCTX3Conn>	RF connector for the third PCC output path
<PCCTX3Conn>	TX module for the third PCC output path
<PCCTX4Conn>	RF connector for the fourth PCC output path
<PCCTX4Conn>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conn>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conn>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conn>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conn>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2

<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FTSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coprocessor>]

Activates the scenario "6CC - nx2 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path

<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FV[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>,
 <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>,
 <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>,
 <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>,
 <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>,
 <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>,
 <SCC5TX2Conn>, <SCC5TX2Conv>[, <Coprocesor>]

Activates the scenario "6CC - nx4 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<PCCTX3Connector> RF connector for the third PCC output path

<PCCTX3Converter> TX module for the third PCC output path

<PCCTX4Connector> RF connector for the fourth PCC output path

<PCCTX4Converter> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path

<Coprocesor> SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FVSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>[, <Coprocesor>]

Activates the scenario "6CC - nx2 nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRXCConn> RF connector for the PCC input path

<PCCRXCConv> RX module for the PCC input path

<PCCTX1Conn> RF connector for the first PCC output path

<PCCTX1Conv> TX module for the first PCC output path

<PCCTX2Conn> RF connector for the second PCC output path

<PCCTX2Conv> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC1TX3Conn> RF connector for the third SCC1 output path

<SCC1TX3Conv> TX module for the third SCC1 output path

<SCC1TX4Conn> RF connector for the fourth SCC1 output path

<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path

<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:FX[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>[, <Coprocessor>]

Activates the scenario "6CC - nx4 nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path

<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path

<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS515, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GG[:FLEXible] <BBBoard1>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <BBBoard2>, <TX2Connector>, <TX2Converter>, <BBBoard3>, <TX3Connector>, <TX3Converter>, <BBBoard4>, <TX4Connector>, <TX4Converter>, <BBBoard5>, <TX5Connector>, <TX5Converter>, <BBBoard6>, <TX6Connector>, <TX6Converter>, <BBBoard7>, <TX7Connector>, <TX7Converter>[, <Coproprocessor>]

Activates the scenario "7CC - 1x1 1x1 1x1 1x1 1x1 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<BBBoard1>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<TX1Connector>	RF connector for the PCC output path
<TX1Converter>	TX module for the PCC output path
<BBBoard2>	Signaling unit for the SCC1
<TX2Connector>	RF connector for the SCC1 output path
<TX2Converter>	TX module for the SCC1 output path

<BBBoard3>	Signaling unit for the SCC2
<TX3Connector>	RF connector for the SCC2 output path
<TX3Converter>	TX module for the SCC2 output path
<BBBoard4>	Signaling unit for the SCC3
<TX4Connector>	RF connector for the SCC3 output path
<TX4Converter>	TX module for the SCC3 output path
<BBBoard5>	Signaling unit for the SCC4
<TX5Connector>	RF connector for the SCC4 output path
<TX5Converter>	TX module for the SCC4 output path
<BBBoard6>	Signaling unit for the SCC5
<TX6Connector>	RF connector for the SCC5 output path
<TX6Converter>	TX module for the SCC5 output path
<BBBoard7>	Signaling unit for the SCC6
<TX7Connector>	RF connector for the SCC6 output path
<TX7Converter>	TX module for the SCC6 output path
<Coprocesor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514

ROUTE:LTE:SIGN<i>:SCENario:GN[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>,
 <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>,
 <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>,
 <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>,
 <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>,
 <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>,
 <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>,
 <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocesor>]

Activates the scenario "7CC - nx2 nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path

<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path

<SCC6TX2Conv> TX module for the second SCC6 output path

<Coprocessor> SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520

ROUTE:LTE:SIGN<i>:SCENario:GP[:FLEXible] <PCCBBBoard>, <PCCRConn>, <PCCRConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <PCCTX3Conn>, <PCCTX3Conv>, <PCCTX4Conn>, <PCCTX4Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx4 nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConn> RF connector for the PCC input path

<PCCRConv> RX module for the PCC input path

<PCCTX1Conn> RF connector for the first PCC output path

<PCCTX1Conv> TX module for the first PCC output path

<PCCTX2Conn> RF connector for the second PCC output path

<PCCTX2Conv> TX module for the second PCC output path

<PCCTX3Conn> RF connector for the third PCC output path

<PCCTX3Conv> TX module for the third PCC output path

<PCCTX4Conn> RF connector for the fourth PCC output path

<PCCTX4Conv> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv> TX module for the second SCC1 output path

<SCC2BBBoard> Signaling unit for the SCC2

<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GNF[:FLEXible]:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>, <SCC5FadingBoard>, <SCC6FadingBoard>[, <Coprocesor>]]

Activates the scenario "7CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path

<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<SCC5FadingBoard>	Internal fader for the SCC5
<SCC6FadingBoard>	Internal fader for the SCC6
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS514, -KS520, -KE500

ROUTE:LTE:SIGN<i>:SCENario:GPF[:FLEXible]:INteRnal <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <PCCTX3Conn>, <PCCTX3Conv>, <PCCTX4Conn>, <PCCTX4Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>, <SCC5FadingBoard>, <SCC6FadingBoard>[, <Coprocessor>]]

Activates the scenario "7CC - Fading - nx4 nx2 nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<PCCTX3Conn>	RF connector for the third PCC output path
<PCCTX3Conv>	TX module for the third PCC output path
<PCCTX4Conn>	RF connector for the fourth PCC output path
<PCCTX4Conv>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path

<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<SCC5FadingBoard>	Internal fader for the SCC5
<SCC6FadingBoard>	Internal fader for the SCC6
<Coproprocessor>	SUA for coprocessing
Firmware/software:	V3.7.60
Options:	R&S CMW-KS514, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:</i>SCENario:GPFS<MIMO4x4>[:FLEXible]:INTernal

<PCCBBBoard>, <PCCRConn>, <PCCRConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <PCCFadingBoard>, <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>, <SCC4FadingBoard>, <SCC5FadingBoard>, <SCC6FadingBoard>[, <Coprocesor>]]

Activates the scenario "7CC - Fading - nx2 nx4 nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConn>	RF connector for the PCC input path
<PCCRConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path

<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<SCC5FadingBoard>	Internal fader for the SCC5
<SCC6FadingBoard>	Internal fader for the SCC6
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540, -KE500, -KE504

ROUTE:LTE:SIGN<i>:SCENario:GPSM<MIMO4x4>[:**FLEXible**] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocesor>]

Activates the scenario "7CC - nx2 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path

<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GR[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConn>, <PCCTX1Conn>, <PCCTX1Conn>, <PCCTX2Conn>, <PCCTX2Conn>, <PCCTX3Conn>, <PCCTX3Conn>, <PCCTX4Conn>, <PCCTX4Conn>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conn>, <SCC1TX2Conn>, <SCC1TX2Conn>, <SCC1TX3Conn>, <SCC1TX3Conn>, <SCC1TX4Conn>, <SCC1TX4Conn>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conn>, <SCC2TX2Conn>, <SCC2TX2Conn>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conn>, <SCC3TX2Conn>, <SCC3TX2Conn>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conn>, <SCC4TX2Conn>, <SCC4TX2Conn>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conn>, <SCC5TX2Conn>, <SCC5TX2Conn>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conn>, <SCC6TX2Conn>, <SCC6TX2Conn>[, <Coprocessor>]

Activates the scenario "7CC - nx4 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConn>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conn>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conn>	TX module for the second PCC output path
<PCCTX3Conn>	RF connector for the third PCC output path
<PCCTX3Conn>	TX module for the third PCC output path
<PCCTX4Conn>	RF connector for the fourth PCC output path
<PCCTX4Conn>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conn>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conn>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conn>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conn>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2

<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GRSM<MIMO4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path

<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GT[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <PCCTX3Conn>, <PCCTX3Conv>, <PCCTX4Conn>, <PCCTX4Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx4 nx4 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<PCCTX3Conn>	RF connector for the third PCC output path
<PCCTX3Conv>	TX module for the third PCC output path
<PCCTX4Conn>	RF connector for the fourth PCC output path
<PCCTX4Conv>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GTSM<Mimo4x4>[:FLEXible] <PCCBBBoard>, <PCCRConn>, <PCCRConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx2 nx4 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<Mimo4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConn>	RF connector for the PCC input path
<PCCRConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path

<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.60

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GV[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConn>, <PCCTX1Conn>, <PCCTX1Conn>, <PCCTX2Conn>, <PCCTX2Conn>, <PCCTX3Conn>, <PCCTX3Conn>, <PCCTX4Conn>, <PCCTX4Conn>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conn>, <SCC1TX2Conn>, <SCC1TX2Conn>, <SCC1TX3Conn>, <SCC1TX3Conn>, <SCC1TX4Conn>, <SCC1TX4Conn>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conn>, <SCC2TX2Conn>, <SCC2TX2Conn>, <SCC2TX3Conn>, <SCC2TX3Conn>, <SCC2TX4Conn>, <SCC2TX4Conn>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conn>, <SCC3TX2Conn>, <SCC3TX2Conn>, <SCC3TX3Conn>, <SCC3TX3Conn>, <SCC3TX4Conn>, <SCC3TX4Conn>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conn>, <SCC4TX2Conn>, <SCC4TX2Conn>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conn>, <SCC5TX2Conn>, <SCC5TX2Conn>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conn>, <SCC6TX2Conn>, <SCC6TX2Conn>[, <Coprocessor>]

Activates the scenario "7CC - nx4 nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConn>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conn>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conn>	TX module for the second PCC output path
<PCCTX3Conn>	RF connector for the third PCC output path
<PCCTX3Conn>	TX module for the third PCC output path
<PCCTX4Conn>	RF connector for the fourth PCC output path
<PCCTX4Conn>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conn>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conn>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conn>	TX module for the third SCC1 output path

<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path

<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GVSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx2 nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path

<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5

<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GX[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx4 nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path

<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path

<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GXSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRConn>, <PCCRConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx2 nx4 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConn>	RF connector for the PCC input path
<PCCRConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path

<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.70

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GYA[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>[, <Coproprocessor>]

Activates the scenario "7CC - nx4 nx4 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path

<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4

<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GYAS<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC6TX3Conn>, <SCC6TX3Conv>, <SCC6TX4Conn>, <SCC6TX4Conv>[, <Coprocessor>]

Activates the scenario "7CC - nx2 nx4 nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path

<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC6TX3Conn>	RF connector for the third SCC6 output path
<SCC6TX3Conv>	TX module for the third SCC6 output path
<SCC6TX4Conn>	RF connector for the fourth SCC6 output path
<SCC6TX4Conv>	TX module for the fourth SCC6 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:GYC[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC6TX3Conn>, <SCC6TX3Conv>, <SCC6TX4Conn>, <SCC6TX4Conv>[, <Coproprocessor>]

Activates the scenario "7CC - nx4 nx4 nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path

<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC6TX3Conn>	RF connector for the third SCC6 output path
<SCC6TX3Conv>	TX module for the third SCC6 output path
<SCC6TX4Conn>	RF connector for the fourth SCC6 output path
<SCC6TX4Conv>	TX module for the fourth SCC6 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HH[:FLEXible] <BBBoard1>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <BBBoard2>, <TX2Connector>, <TX2Converter>, <BBBoard3>, <TX3Connector>, <TX3Converter>, <BBBoard4>, <TX4Connector>, <TX4Converter>, <BBBoard5>, <TX5Connector>, <TX5Converter>, <BBBoard6>, <TX6Connector>, <TX6Converter>, <BBBoard7>, <TX7Connector>, <TX7Converter>, <BBBoard8>, <TX8Connector>, <TX8Converter>[, <Coprocesor>]

Activates the scenario "8CC - 1x1 1x1 1x1 1x1 1x1 1x1 1x1 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<BBBoard1>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<TX1Connector>	RF connector for the PCC output path
<TX1Converter>	TX module for the PCC output path
<BBBoard2>	Signaling unit for the SCC1
<TX2Connector>	RF connector for the SCC1 output path
<TX2Converter>	TX module for the SCC1 output path
<BBBoard3>	Signaling unit for the SCC2
<TX3Connector>	RF connector for the SCC2 output path
<TX3Converter>	TX module for the SCC2 output path
<BBBoard4>	Signaling unit for the SCC3
<TX4Connector>	RF connector for the SCC3 output path
<TX4Converter>	TX module for the SCC3 output path
<BBBoard5>	Signaling unit for the SCC4
<TX5Connector>	RF connector for the SCC4 output path
<TX5Converter>	TX module for the SCC4 output path
<BBBoard6>	Signaling unit for the SCC5
<TX6Connector>	RF connector for the SCC5 output path
<TX6Converter>	TX module for the SCC5 output path
<BBBoard7>	Signaling unit for the SCC6
<TX7Connector>	RF connector for the SCC6 output path

<TX7Converter>	TX module for the SCC6 output path
<BBBoard8>	Signaling unit for the SCC7
<TX8Connector>	RF connector for the SCC7 output path
<TX8Converter>	TX module for the SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514

ROUTE:LTE:SIGN<i>:SCENario:HP[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2

<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520

ROUTE:LTE:SIGN<i>:SCENario:HR[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <PCCTX3Conn>, <PCCTX3Conv>, <PCCTX4Conn>, <PCCTX4Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx4 nx2 nx2 nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<PCCTX3Conn>	RF connector for the third PCC output path
<PCCTX3Conv>	TX module for the third PCC output path
<PCCTX4Conn>	RF connector for the fourth PCC output path
<PCCTX4Conv>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path

<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HRSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx2 nx4 nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path

<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HT[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <PCCTX3Conn>, <PCCTX3Conv>, <PCCTX4Conn>, <PCCTX4Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx4 nx4 nx2 nx2 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<PCCTX3Conn>	RF connector for the third PCC output path
<PCCTX3Conv>	TX module for the third PCC output path
<PCCTX4Conn>	RF connector for the fourth PCC output path
<PCCTX4Conv>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path

<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HTSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXTConn>, <PCCRXTConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocesor>]

Activates the scenario "8CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXTConn>	RF connector for the PCC input path
<PCCRXTConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path

<SCC7TX2Conv> TX module for the second SCC7 output path

<Coproprocessor> SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS514, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HV[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <PCCTX3Conn>, <PCCTX3Conv>, <PCCTX4Conn>, <PCCTX4Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coproprocessor>]

Activates the scenario "8CC - nx4 nx4 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRXCConn> RF connector for the PCC input path

<PCCRXCConv> RX module for the PCC input path

<PCCTX1Conn> RF connector for the first PCC output path

<PCCTX1Conv> TX module for the first PCC output path

<PCCTX2Conn> RF connector for the second PCC output path

<PCCTX2Conv> TX module for the second PCC output path

<PCCTX3Conn> RF connector for the third PCC output path

<PCCTX3Conv> TX module for the third PCC output path

<PCCTX4Conn> RF connector for the fourth PCC output path

<PCCTX4Conv> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path

<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HVSM<Mimo4x4>[:FLEXible] <PCCBBBoard>, <PCCRConn>, <PCCRConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx2 nx4 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<Mimo4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConn>	RF connector for the PCC input path
<PCCRConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path

<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path

<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HX[:FLEXible] <PCCBBBoard>, <PCCRXTConn>, <PCCRXTConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <PCCTX3Conn>, <PCCTX3Conv>, <PCCTX4Conn>, <PCCTX4Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx4 nx4 nx4 nx4 nx2 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCnn>	RF connector for the PCC input path
<PCCRXCnnv>	RX module for the PCC input path
<PCCTX1Cnn>	RF connector for the first PCC output path
<PCCTX1Cnnv>	TX module for the first PCC output path
<PCCTX2Cnn>	RF connector for the second PCC output path
<PCCTX2Cnnv>	TX module for the second PCC output path
<PCCTX3Cnn>	RF connector for the third PCC output path
<PCCTX3Cnnv>	TX module for the third PCC output path
<PCCTX4Cnn>	RF connector for the fourth PCC output path
<PCCTX4Cnnv>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Cnn>	RF connector for the first SCC1 output path
<SCC1TX1Cnnv>	TX module for the first SCC1 output path
<SCC1TX2Cnn>	RF connector for the second SCC1 output path
<SCC1TX2Cnnv>	TX module for the second SCC1 output path
<SCC1TX3Cnn>	RF connector for the third SCC1 output path
<SCC1TX3Cnnv>	TX module for the third SCC1 output path
<SCC1TX4Cnn>	RF connector for the fourth SCC1 output path
<SCC1TX4Cnnv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Cnn>	RF connector for the first SCC2 output path
<SCC2TX1Cnnv>	TX module for the first SCC2 output path
<SCC2TX2Cnn>	RF connector for the second SCC2 output path
<SCC2TX2Cnnv>	TX module for the second SCC2 output path
<SCC2TX3Cnn>	RF connector for the third SCC2 output path
<SCC2TX3Cnnv>	TX module for the third SCC2 output path
<SCC2TX4Cnn>	RF connector for the fourth SCC2 output path
<SCC2TX4Cnnv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Cnn>	RF connector for the first SCC3 output path
<SCC3TX1Cnnv>	TX module for the first SCC3 output path
<SCC3TX2Cnn>	RF connector for the second SCC3 output path

<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HXSM<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx2 nx4 nx4 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6

<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HYA[:FLEXible] <PCCBBBoard>, <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path

<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path

<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HYAS<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocesor>]

Activates the scenario "8CC - nx2 nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path

<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path

<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTe:LTE:SIGN<i>:SCENario:HYC[:FLEXible] <PCCBBBoard>, <PCCRXConnector>, <PCCRXConverter>, <PCCTX1Connector>, <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>, <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>, <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC6TX3Conn>, <SCC6TX3Conv>, <SCC6TX4Conn>, <SCC6TX4Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx4 nx4 nx4 nx4 nx4 nx2 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRConnector>	RF connector for the PCC input path
<PCCRConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<PCCTX3Connector>	RF connector for the third PCC output path
<PCCTX3Converter>	TX module for the third PCC output path
<PCCTX4Connector>	RF connector for the fourth PCC output path
<PCCTX4Converter>	TX module for the fourth PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path

<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path

<Coprocesor> SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HYCS<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC6TX3Conn>, <SCC6TX3Conv>, <SCC6TX4Conn>, <SCC6TX4Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[, <Coprocesor>]

Activates the scenario "8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRXCConn> RF connector for the PCC input path

<PCCRXCConv> RX module for the PCC input path

<PCCTX1Conn> RF connector for the first PCC output path

<PCCTX1Conv> TX module for the first PCC output path

<PCCTX2Conn> RF connector for the second PCC output path

<PCCTX2Conv> TX module for the second PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5

<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC6TX3Conn>	RF connector for the third SCC6 output path
<SCC6TX3Conv>	TX module for the third SCC6 output path
<SCC6TX4Conn>	RF connector for the fourth SCC6 output path
<SCC6TX4Conv>	TX module for the fourth SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coproprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i></i>:SCENario:HYE[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>,
 <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>,
 <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>,
 <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>,
 <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>,
 <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>,
 <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>,
 <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>,
 <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC6TX3Conn>,
 <SCC6TX3Conv>, <SCC6TX4Conn>, <SCC6TX4Conv>, <SCC7BBBoard>,
 <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>[,
 <Coprocesor>]

Activates the scenario "8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<PCCTX3Connector> RF connector for the third PCC output path

<PCCTX3Converter> TX module for the third PCC output path

<PCCTX4Connector> RF connector for the fourth PCC output path

<PCCTX4Converter> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conn>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5

<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC6TX3Conn>	RF connector for the third SCC6 output path
<SCC6TX3Conv>	TX module for the third SCC6 output path
<SCC6TX4Conn>	RF connector for the fourth SCC6 output path
<SCC6TX4Conv>	TX module for the fourth SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HYES<MIMO4x4>[:FLEXible] <PCCBBBoard>, <PCCRXCConn>, <PCCRXCConv>, <PCCTX1Conn>, <PCCTX1Conv>, <PCCTX2Conn>, <PCCTX2Conv>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>, <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>, <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>, <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>, <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>, <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC6TX3Conn>, <SCC6TX3Conv>, <SCC6TX4Conn>, <SCC6TX4Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>, <SCC7TX3Conn>, <SCC7TX3Conv>, <SCC7TX4Conn>, <SCC7TX4Conv>[, <Coprocessor>]

Activates the scenario "8CC - nx2 nx4 nx4 nx4 nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Suffix:

<MIMO4x4> 4

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<PCCRXCConn>	RF connector for the PCC input path
<PCCRXCConv>	RX module for the PCC input path
<PCCTX1Conn>	RF connector for the first PCC output path
<PCCTX1Conv>	TX module for the first PCC output path
<PCCTX2Conn>	RF connector for the second PCC output path
<PCCTX2Conv>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path

<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path

<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC6TX3Conn>	RF connector for the third SCC6 output path
<SCC6TX3Conv>	TX module for the third SCC6 output path
<SCC6TX4Conn>	RF connector for the fourth SCC6 output path
<SCC6TX4Conv>	TX module for the fourth SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<SCC7TX3Conn>	RF connector for the third SCC7 output path
<SCC7TX3Conv>	TX module for the third SCC7 output path
<SCC7TX4Conn>	RF connector for the fourth SCC7 output path
<SCC7TX4Conv>	TX module for the fourth SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS520, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HYG[:FLEXible] <PCCBBBoard>,
 <PCCRConnector>, <PCCRConverter>, <PCCTX1Connector>,
 <PCCTX1Converter>, <PCCTX2Connector>, <PCCTX2Converter>,
 <PCCTX3Connector>, <PCCTX3Converter>, <PCCTX4Connector>,
 <PCCTX4Converter>, <SCC1BBBoard>, <SCC1TX1Conn>, <SCC1TX1Conv>,
 <SCC1TX2Conn>, <SCC1TX2Conv>, <SCC1TX3Conn>, <SCC1TX3Conv>,
 <SCC1TX4Conn>, <SCC1TX4Conv>, <SCC2BBBoard>, <SCC2TX1Conn>,
 <SCC2TX1Conv>, <SCC2TX2Conn>, <SCC2TX2Conv>, <SCC2TX3Conn>,
 <SCC2TX3Conv>, <SCC2TX4Conn>, <SCC2TX4Conv>, <SCC3BBBoard>,
 <SCC3TX1Conn>, <SCC3TX1Conv>, <SCC3TX2Conn>, <SCC3TX2Conv>,
 <SCC3TX3Conn>, <SCC3TX3Conv>, <SCC3TX4Conn>, <SCC3TX4Conv>,
 <SCC4BBBoard>, <SCC4TX1Conn>, <SCC4TX1Conv>, <SCC4TX2Conn>,
 <SCC4TX2Conv>, <SCC4TX3Conn>, <SCC4TX3Conv>, <SCC4TX4Conn>,
 <SCC4TX4Conv>, <SCC5BBBoard>, <SCC5TX1Conn>, <SCC5TX1Conv>,
 <SCC5TX2Conn>, <SCC5TX2Conv>, <SCC5TX3Conn>, <SCC5TX3Conv>,
 <SCC5TX4Conn>, <SCC5TX4Conv>, <SCC6BBBoard>, <SCC6TX1Conn>,
 <SCC6TX1Conv>, <SCC6TX2Conn>, <SCC6TX2Conv>, <SCC6TX3Conn>,
 <SCC6TX3Conv>, <SCC6TX4Conn>, <SCC6TX4Conv>, <SCC7BBBoard>,
 <SCC7TX1Conn>, <SCC7TX1Conv>, <SCC7TX2Conn>, <SCC7TX2Conv>,
 <SCC7TX3Conn>, <SCC7TX3Conv>, <SCC7TX4Conn>, <SCC7TX4Conv>[,
 <Coprocesor>]

Activates the scenario "8CC - nx4 nx4 nx4 nx4 nx4 nx4 nx4" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit for the PCC

<PCCRConnector> RF connector for the PCC input path

<PCCRConverter> RX module for the PCC input path

<PCCTX1Connector> RF connector for the first PCC output path

<PCCTX1Converter> TX module for the first PCC output path

<PCCTX2Connector> RF connector for the second PCC output path

<PCCTX2Converter> TX module for the second PCC output path

<PCCTX3Connector> RF connector for the third PCC output path

<PCCTX3Converter> TX module for the third PCC output path

<PCCTX4Connector> RF connector for the fourth PCC output path

<PCCTX4Converter> TX module for the fourth PCC output path

<SCC1BBBoard> Signaling unit for the SCC1

<SCC1TX1Conn> RF connector for the first SCC1 output path

<SCC1TX1Conv> TX module for the first SCC1 output path

<SCC1TX2Conn> RF connector for the second SCC1 output path

<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC1TX3Conn>	RF connector for the third SCC1 output path
<SCC1TX3Conv>	TX module for the third SCC1 output path
<SCC1TX4Conn>	RF connector for the fourth SCC1 output path
<SCC1TX4Conv>	TX module for the fourth SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC2TX3Conn>	RF connector for the third SCC2 output path
<SCC2TX3Conv>	TX module for the third SCC2 output path
<SCC2TX4Conn>	RF connector for the fourth SCC2 output path
<SCC2TX4Conv>	TX module for the fourth SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path
<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC3TX3Conn>	RF connector for the third SCC3 output path
<SCC3TX3Conv>	TX module for the third SCC3 output path
<SCC3TX4Conn>	RF connector for the fourth SCC3 output path
<SCC3TX4Conv>	TX module for the fourth SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC4TX3Conn>	RF connector for the third SCC4 output path
<SCC4TX3Conv>	TX module for the third SCC4 output path
<SCC4TX4Conn>	RF connector for the fourth SCC4 output path
<SCC4TX4Conv>	TX module for the fourth SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5

<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC5TX3Conn>	RF connector for the third SCC5 output path
<SCC5TX3Conv>	TX module for the third SCC5 output path
<SCC5TX4Conn>	RF connector for the fourth SCC5 output path
<SCC5TX4Conv>	TX module for the fourth SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC6TX3Conn>	RF connector for the third SCC6 output path
<SCC6TX3Conv>	TX module for the third SCC6 output path
<SCC6TX4Conn>	RF connector for the fourth SCC6 output path
<SCC6TX4Conv>	TX module for the fourth SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<SCC7TX3Conn>	RF connector for the third SCC7 output path
<SCC7TX3Conv>	TX module for the third SCC7 output path
<SCC7TX4Conn>	RF connector for the fourth SCC7 output path
<SCC7TX4Conv>	TX module for the fourth SCC7 output path
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.8.10

Options: R&S CMW-KS515, -KS540

ROUTE:LTE:SIGN<i>:SCENario:HPF[:FLEXible]:INTernal <PCCBBBoard>,
 <RXConnector>, <RXConverter>, <PCCTX1Connector>, <PCCTX1Converter>,
 <PCCTX2Connector>, <PCCTX2Converter>, <SCC1BBBoard>,
 <SCC1TX1Conn>, <SCC1TX1Conv>, <SCC1TX2Conn>, <SCC1TX2Conv>,
 <SCC2BBBoard>, <SCC2TX1Conn>, <SCC2TX1Conv>, <SCC2TX2Conn>,
 <SCC2TX2Conv>, <SCC3BBBoard>, <SCC3TX1Conn>, <SCC3TX1Conv>,
 <SCC3TX2Conn>, <SCC3TX2Conv>, <SCC4BBBoard>, <SCC4TX1Conn>,
 <SCC4TX1Conv>, <SCC4TX2Conn>, <SCC4TX2Conv>, <SCC5BBBoard>,
 <SCC5TX1Conn>, <SCC5TX1Conv>, <SCC5TX2Conn>, <SCC5TX2Conv>,
 <SCC6BBBoard>, <SCC6TX1Conn>, <SCC6TX1Conv>, <SCC6TX2Conn>,
 <SCC6TX2Conv>, <SCC7BBBoard>, <SCC7TX1Conn>, <SCC7TX1Conv>,
 <SCC7TX2Conn>, <SCC7TX2Conv>[, <PCCFadingBoard>,
 <SCC1FadingBoard>, <SCC2FadingBoard>, <SCC3FadingBoard>,
 <SCC4FadingBoard>, <SCC5FadingBoard>, <SCC6FadingBoard>,
 <SCC7FadingBoard>[, <Coprocessor>]]

Activates the scenario "8CC - Fading - nx2 nx2 nx2 nx2 nx2 nx2 nx2 nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit for the PCC
<RXConnector>	RF connector for the PCC input path
<RXConverter>	RX module for the PCC input path
<PCCTX1Connector>	RF connector for the first PCC output path
<PCCTX1Converter>	TX module for the first PCC output path
<PCCTX2Connector>	RF connector for the second PCC output path
<PCCTX2Converter>	TX module for the second PCC output path
<SCC1BBBoard>	Signaling unit for the SCC1
<SCC1TX1Conn>	RF connector for the first SCC1 output path
<SCC1TX1Conv>	TX module for the first SCC1 output path
<SCC1TX2Conn>	RF connector for the second SCC1 output path
<SCC1TX2Conv>	TX module for the second SCC1 output path
<SCC2BBBoard>	Signaling unit for the SCC2
<SCC2TX1Conn>	RF connector for the first SCC2 output path
<SCC2TX1Conv>	TX module for the first SCC2 output path
<SCC2TX2Conn>	RF connector for the second SCC2 output path
<SCC2TX2Conv>	TX module for the second SCC2 output path
<SCC3BBBoard>	Signaling unit for the SCC3
<SCC3TX1Conn>	RF connector for the first SCC3 output path

<SCC3TX1Conv>	TX module for the first SCC3 output path
<SCC3TX2Conn>	RF connector for the second SCC3 output path
<SCC3TX2Conv>	TX module for the second SCC3 output path
<SCC4BBBoard>	Signaling unit for the SCC4
<SCC4TX1Conn>	RF connector for the first SCC4 output path
<SCC4TX1Conv>	TX module for the first SCC4 output path
<SCC4TX2Conn>	RF connector for the second SCC4 output path
<SCC4TX2Conv>	TX module for the second SCC4 output path
<SCC5BBBoard>	Signaling unit for the SCC5
<SCC5TX1Conn>	RF connector for the first SCC5 output path
<SCC5TX1Conv>	TX module for the first SCC5 output path
<SCC5TX2Conn>	RF connector for the second SCC5 output path
<SCC5TX2Conv>	TX module for the second SCC5 output path
<SCC6BBBoard>	Signaling unit for the SCC6
<SCC6TX1Conn>	RF connector for the first SCC6 output path
<SCC6TX1Conv>	TX module for the first SCC6 output path
<SCC6TX2Conn>	RF connector for the second SCC6 output path
<SCC6TX2Conv>	TX module for the second SCC6 output path
<SCC7BBBoard>	Signaling unit for the SCC7
<SCC7TX1Conn>	RF connector for the first SCC7 output path
<SCC7TX1Conv>	TX module for the first SCC7 output path
<SCC7TX2Conn>	RF connector for the second SCC7 output path
<SCC7TX2Conv>	TX module for the second SCC7 output path
<PCCFadingBoard>	Internal fader for the PCC
<SCC1FadingBoard>	Internal fader for the SCC1
<SCC2FadingBoard>	Internal fader for the SCC2
<SCC3FadingBoard>	Internal fader for the SCC3
<SCC4FadingBoard>	Internal fader for the SCC4
<SCC5FadingBoard>	Internal fader for the SCC5
<SCC6FadingBoard>	Internal fader for the SCC6
<SCC7FadingBoard>	Internal fader for the SCC7
<Coprocessor>	SUA for coprocessing

Firmware/software: V3.7.50

Options: R&S CMW-KS514, -KS520, -KE500

ROUTE:LTE:SIGN<i>:SCENario:SCELI:FLEXible <PCCBBBoard>, <RXConnector>, <RXConverter>, <TXConnector>, <TXConverter>

Activates the scenario "1CC - 1x1" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit
 <RXConnector> RF connector for the input path
 <RXConverter> RX module for the input path
 <TXConnector> RF connector for the output path
 <TXConverter> TX module for the output path

Example: See [Selecting a scenario](#)

Firmware/software: V3.5.40

ROUTE:LTE:SIGN<i>:SCENario:SCFading:FLEXible:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <TXConnector>, <TXConverter>, <PCCFadingBoard>

Activates the scenario "1CC - Fading - 1x1" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard> Signaling unit
 <RXConnector> RF connector for the input path
 <RXConverter> RX module for the input path
 <TXConnector> RF connector for the output path
 <TXConverter> TX module for the output path
 <PCCFadingBoard> Internal fader

Firmware/software: V3.5.40

Options: R&S CMW-KS510, -KE500

ROUTE:LTE:SIGN<i>:SCENario:TRO:FLEXible <PCCBBBoard>, <RXConnector>, <RXConverter>, <TXConnector>, <TXConverter>, <TX2Connector>, <TX2Converter>

Activates the scenario "1CC - nx2" and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit
<RXConnector>	RF connector for the input path
<RXConverter>	RX module for the input path
<TXConnector>	RF connector for the first output path
<TXConverter>	TX module for the first output path
<TX2Connector>	RF connector for the second output path
<TX2Converter>	TX module for the second output path

Firmware/software: V3.5.40

Options: R&S CMW-KS520

ROUTE:LTE:SIGN<i>:SCENario:TROFading:FLEXible:INTernal <PCCBBBoard>, <RXConnector>, <RXConverter>, <TX1Connector>, <TX1Converter>, <TX2Connector>, <TX2Converter>, <PCCFadingBoard>

Activates the scenario "1CC - Fading - nx2" with internal fading and selects the signal paths.

For possible parameter values, see [Section 2.6.1.4, "Values for signal path selection"](#), on page 363.

Parameters:

<PCCBBBoard>	Signaling unit
<RXConnector>	RF connector for the input path
<RXConverter>	RX module for the input path
<TX1Connector>	RF connector for the first output path
<TX1Converter>	TX module for the first output path
<TX2Connector>	RF connector for the second output path
<TX2Converter>	TX module for the second output path
<PCCFadingBoard>	Internal fader

Firmware/software: V3.5.40

Options: R&S CMW-KS510, -KS520, -KE500

2.6.8.2 RF path properties

The following commands define path properties to be compensated (external attenuation and time delay).

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:OUTPut<n>.....	632
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:OUTPut<n>.....	632
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:INPut.....	632
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:INPut.....	632
CONFigure:LTE:SIGN<i>:RFSettings:EDC:OUTPut.....	633
CONFigure:LTE:SIGN<i>:RFSettings:EDC:INPut.....	633

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:OUTPut<n>

<ExtRFOutAtt>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:OUTPut<n>

<ExtRFOutAtt>

Defines an external attenuation (or gain, if the value is negative), to be applied to the RF output path number <n>. Depending on the transmission scheme, several output paths are used for each carrier and the attenuation can be configured per output path.

Suffix:

<n> 1..4

<c> 1..7

Parameters:

<ExtRFOutAtt> numeric
 Range: -50 dB to 90 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Selecting a scenario](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["External Attenuation"](#) on page 145

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:INPut <ExtRFInAtt>
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:INPut <ExtRFInAtt>

Defines an external attenuation (or gain, if the value is negative), to be applied to the input connector.

Suffix:

<c> 1..7

Parameters:

<ExtRFInAtt> numeric
 Range: -50 dB to 90 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Selecting a scenario](#)

Firmware/software: V1.0.15.20, SCC command V3.5.20

Manual operation: See ["External Attenuation"](#) on page 146

CONFigure:LTE:SIGN<i>:RFSettings:EDC:OUTPut <Time>

CONFigure:LTE:SIGN<i>:RFSettings:EDC:INPut <Time>

Defines the value of an external time delay in the output path and in the input path, so that it can be compensated.

Parameters:

<Time> numeric
 Range: 0 s to 20E-6 s
 *RST: 0 s
 Default unit: s

Example: See [Specifying general settings](#)

Firmware/software: V3.2.50

Manual operation: See ["External Delay Compensation"](#) on page 146

2.6.8.3 Signal settings

The following commands provide settings for the downlink and uplink signals.

CONFigure:LTE:SIGN<i>:RFSettings:ALL:BWCHannel.....	634
CONFigure:LTE:SIGN<i>[:PCC]:BAND.....	636
CONFigure:LTE:SIGN<i>:SCC<c>:BAND.....	636
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:AFBands:ALL.....	636
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDSeparation.....	636
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDSeparation.....	636
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:DL.....	637
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:DL.....	637
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:UL.....	637
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:UL.....	637
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:DL.....	638
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:DL.....	638
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:DL:UCSPecific.....	638
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL:UCSPecific.....	638
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL.....	639
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:UL.....	639
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPMode.....	639
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPMode.....	639
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPower.....	640
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPower.....	640
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UMARgin.....	640
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UMARgin.....	640
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:MLOffset.....	641
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:MLOffset.....	641

CONFigure:LTE:SIGN<i>:RFSettings:ALL:BWChannel <BandPCC>,
 <DLChannelPCC>, <BandwithPCC>[, <BandSCC1>, <DLChannelSCC1>,
 <BandwithSCC1>[, <BandSCC2>, <DLChannelSCC2>, <BandwithSCC2>[,
 <BandSCC3>, <DLChannelSCC3>, <BandwithSCC3>[, <BandSCC4>,
 <DLChannelSCC4>, <BandwithSCC4>]]]]

Selects the operating band, the downlink channel number and the cell bandwidth for the PCC and optionally for the SCCs.

A query returns only the component carriers that are supported by the current scenario.

Parameters:

<BandPCC>	FDD: UDEFined OB1 ... OB28 OB30 OB31 OB65 OB66 OB68 OB70 ... OB74 OB85 OB87 OB88 OB106 TDD: UDEFined OB33 ... OB46 OB48 ... OB53 OB250 Selects the PCC operating band *RST: OB1 (OB33 for TDD)
<DLChannelPCC>	decimal PCC DL channel number Range: depends on operating band
<BandwithPCC>	B014 B030 B050 B100 B150 B200 PCC cell bandwidth B014: 1.4 MHz B030: 3 MHz B050: 5 MHz B100: 10 MHz B150: 15 MHz B200: 20 MHz *RST: B100
<BandSCC1>	FDD: UDEFined OB1 ... OB32 OB65 ... OB76 OB85 OB87 OB88 OB106 OB252 OB255 TDD: UDEFined OB33 ... OB46 OB48 ... OB53 OB250 SCC1 operating band *RST: OB1 (OB33 for TDD)
<DLChannelSCC1>	decimal SCC1 DL channel number Range: depends on operating band
<BandwithSCC1>	B014 B030 B050 B100 B150 B200 SCC1 cell bandwidth *RST: B100

<BandSCC2>	FDD: UDEFined OB1 ... OB32 OB65 ... OB76 OB85 OB87 OB88 OB106 OB252 OB255 TDD: UDEFined OB33 ... OB46 OB48 ... OB53 OB250 SCC2 operating band *RST: OB1 (OB33 for TDD)
<DLChannelSCC2>	decimal SCC2 DL channel number Range: depends on operating band
<BandwidthSCC2>	B014 B030 B050 B100 B150 B200 SCC2 cell bandwidth *RST: B100
<BandSCC3>	FDD: UDEFined OB1 ... OB32 OB65 ... OB76 OB85 OB87 OB88 OB106 OB252 OB255 TDD: UDEFined OB33 ... OB46 OB48 ... OB53 OB250 SCC3 operating band *RST: OB1 (OB33 for TDD)
<DLChannelSCC3>	decimal SCC3 DL channel number Range: depends on operating band
<BandwidthSCC3>	B014 B030 B050 B100 B150 B200 SCC3 cell bandwidth *RST: B100
<BandSCC4>	FDD: UDEFined OB1 ... OB32 OB65 ... OB76 OB85 OB87 OB88 OB106 OB252 OB255 TDD: UDEFined OB33 ... OB46 OB48 ... OB53 OB250 SCC4 operating band *RST: OB1 (OB33 for TDD)
<DLChannelSCC4>	decimal SCC4 DL channel number Range: depends on operating band
<BandwidthSCC4>	B014 B030 B050 B100 B150 B200 SCC4 cell bandwidth *RST: B100
Firmware/software:	V3.2.80, some bands added in later versions V3.5.10: added SCC2/3, V3.7.10: added SCC4
Options:	R&S CMW-KS525 for UDEFined, OB250, OB252, OB255 R&S CMW-KS514 for OB46, OB49
Manual operation:	See "Operating Band, Channel, Frequency" on page 148

CONFigure:LTE:SIGN<i>[:PCC]:BAND <Band>
CONFigure:LTE:SIGN<i>:SCC<c>:BAND <Band>

Selects the operating band (OB). The allowed input range depends on the duplex mode (FDD or TDD).

Suffix:

<c> 1..7

Parameters:

<Band> FDD: UDEFined | OB1 | ... | OB32 | OB65 | ... | OB76 | OB85 | OB87 | OB88 | OB106 | OB252 | OB255
(OB29/32/67/69/75/76/252/255 only for SCC DL)
TDD: UDEFined | OB33 | ... | OB46 | OB48 | ... | OB53 | OB250
(OB46/49 only for SCC DL)

Example: See [Configuring operating bands and channels](#)

Firmware/software: V3.0.10, V3.2.50 SCC command, some bands added in later versions.

Options: R&S CMW-KS525 for UDEFined, OB250, OB252, OB255
R&S CMW-KS514 for OB46, OB49

Manual operation: See ["Operating Band, Channel, Frequency"](#) on page 148

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:AFBands:ALL <Enable1>, <Band1>, ..., <Enable8>, <Band8>

Configures additional frequency bands supported by the cell ("multiBandInfoList").

There are eight entries. You can enable/disable each entry and assign a band to each entry.

Parameters:

<Enable> OFF | ON
Enables/disables the entry.
*RST: OFF

<Bands> OB1 | ... | OB46 | OB48 | ... | OB53 | OB65 | ... | OB76 | OB85 | OB87 | OB88 | OB106 | OB250 | OB252 | OB255
*RST: OB1

Example: See [Configuring operating bands and channels](#)

Firmware/software: V3.7.31, some bands added in later versions

Manual operation: See ["Additional Frequency Bands"](#) on page 148

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDSeparation <Frequency>
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDSeparation <Frequency>

Configures the UL/DL separation. For most operating bands, this setting is fixed.

Suffix:

<c> 1..7

Parameters:

<Frequency> numeric
 UL/DL separation
 Range: see table
 *RST: see table
 Default unit: Hz

Firmware/software: V3.7.10**Manual operation:** See ["UL/DL Separation"](#) on page 149*Table 2-66: Range and *RST*

Band	Allowed values	*RST
70	295E+6 Hz, 300E+6 Hz	300E+6 Hz

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:DL <Channel>**CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:DL** <Channel>

Selects the DL channel number. It must be valid for the current operating band. The related UL channel number is calculated and set automatically.

By appending a Hz unit (e.g. Hz, kHz, MHz) to a setting command, you can set the channel via its center frequency (only integer numbers accepted). By appending a Hz unit to a query command, you can query the center frequency instead of the channel number.

For channel numbers and frequencies depending on operating bands, see [Section 2.2.23, "Operating bands"](#), on page 91.

Suffix:

<c> 1..7

Parameters:

<Channel> decimal
 Range: depends on operating band

Example: See [Configuring operating bands and channels](#)**Firmware/software:** V1.0.15.20, SCC command V3.2.50**Manual operation:** See ["Operating Band, Channel, Frequency"](#) on page 148**CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:UL** <Channel>**CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:UL** <Channel>

Selects the UL channel number. It must be valid for the current operating band. The related DL channel number is calculated and set automatically.

By appending a Hz unit (e.g. Hz, kHz, MHz) to a setting command, you can set the channel via its center frequency (only integer numbers accepted). By appending a Hz unit to a query command, you can query the center frequency instead of the channel number.

For channel numbers and frequencies depending on operating bands, see [Section 2.2.23, "Operating bands"](#), on page 91.

Suffix:

<c> 1..7

Parameters:

<Channel> decimal
Range: depends on operating band

Example: See [Configuring operating bands and channels](#)

Firmware/software: V1.0.15.20, SCC command V3.5.20

Manual operation: See ["Operating Band, Channel, Frequency"](#) on page 148

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:DL <Offset>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:DL <Offset>

Specifies a positive or negative frequency offset to be added to the center frequency of the configured downlink channel.

You can use the PCC command to configure the same offset for the PCC and all SCCs. Or you can use the PCC and SCC command to configure different values. See also [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:FOFFset:DL:UCSPecific](#) on page 638.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
Range: -100E+3 Hz to 100E+3 Hz
*RST: 0 Hz
Default unit: Hz

Example: See [Configuring operating bands and channels](#)

Firmware/software: V3.2.10, SCC command V3.2.81

Manual operation: See ["Frequency Offset"](#) on page 149

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:DL:UCSPecific <Enable>

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL:UCSPecific <Enable>

Enables or disables the usage of different frequency offset values for the individual downlink or uplink component carriers.

Parameters:

<Enable> OFF | ON

OFF: The configured PCC offset is also used for the SCCs. The configured SCC offsets have no effect.

ON: You can configure the frequency offset per carrier.

*RST: OFF

Example: See [Configuring operating bands and channels](#)

Firmware/software: V3.2.80, UL command V3.5.20

Manual operation: See ["Frequency Offset"](#) on page 149

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL <Offset>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:UL <Offset>

Specifies a positive or negative frequency offset to be added to the center frequency of the configured uplink channel.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric

Range: -100E+3 Hz to 100E+3 Hz

*RST: 0 Hz

Default unit: Hz

Example: See [Configuring operating bands and channels](#)

Firmware/software: V3.2.10, SCC command V3.5.20

Manual operation: See ["Frequency Offset"](#) on page 149

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPMode <Mode>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPMode <Mode>

Selects the expected nominal power mode. The expected nominal power of the UL signal can be defined manually or calculated automatically, according to the UL power control settings.

For manual configuration, see:

- [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:ENPower](#) on page 640
- [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:UMARgin](#) on page 640

For UL power control settings, see [Section 2.6.11, "Uplink power control"](#), on page 662.

Suffix:

<c> 1..7

Parameters:

<Mode> MANual | ULPC

MANual

The expected nominal power and margin are specified manually.

ULPC

The expected nominal power is calculated according to the UL power control settings. The margin is set to 12 dB.

*RST: ULPC

Example: See [Specifying general settings](#)

Firmware/software: V1.0.15.21, SCC command V3.5.20

Manual operation: See ["Exp. Nominal Power..., Margin"](#) on page 151

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPower <ExpectedPower>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPower <ExpectedPower>

Sets the expected nominal power of the UL signal in manual mode.

If the expected nominal power is calculated automatically according to the UL power control settings, you can only query the result.

To configure the expected nominal power mode, see [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:ENPMode](#) on page 639.

Suffix:

<c> 1..7

Parameters:

<ExpectedPower> numeric

In manual mode, the range of the expected nominal power can be calculated as follows:

Range (expected nominal power) = range (input power) + external attenuation - margin

The input power range is stated in the specifications document.

*RST: -20 dBm

Default unit: dBm

Example: See [Specifying general settings](#)

Firmware/software: V1.0.15.21, SCC command V3.5.20

Manual operation: See ["Exp. Nominal Power..., Margin"](#) on page 151

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UMARgin <UserMargin>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UMARgin <UserMargin>

Sets the margin that the R&S CMW adds to the expected nominal power to determine the reference level in manual mode. If the expected nominal power is calculated automatically according to the UL power control settings, a fix margin of 12 dB is used instead.

The reference level minus the external input attenuation must be within the power range of the selected input connector; refer to the specifications document.

Refer also to the following commands:

- [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:ENPMode](#) on page 639
- [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:ENPower](#) on page 640
- [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:EATTenuation:INPut](#) on page 632

Suffix:

<c> 1..7

Parameters:

<UserMargin> numeric
 Range: 0 dB to (42 dB + external attenuation - expected nominal power)
 *RST: 12 dB
 Default unit: dB

Example: See [Specifying general settings](#)

Firmware/software: V1.0.15.21, SCC command V3.5.20

Manual operation: See ["Exp. Nominal Power..., Margin"](#) on page 151

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:MLOffset <MixLevOffset>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:MLOffset <MixLevOffset>

Varies the input level of the mixer in the analyzer path.

Suffix:

<c> 1..7

Parameters:

<MixLevOffset> numeric
 Range: -10 dB to 10 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Specifying general settings](#)

Firmware/software: V3.2.10, SCC command V3.5.20

Manual operation: See ["Mixer Level Offset"](#) on page 152

2.6.8.4 User-defined band

The following commands configure a user-defined band.

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFineD:UDSeparation	642
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFineD:UDSeparation	642
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFineD:BINDicator	642
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFineD:BINDicator	642
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFineD:CHANnel:DL:MINimum	643
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFineD:CHANnel:DL:MAXimum	643
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFineD:CHANnel:DL:MINimum	643

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:DL:MAXimum.....	643
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:DL:MINimum.....	643
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:DL:MINimum.....	643
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:DL:MAXimum?.....	644
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:DL:MAXimum?.....	644
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:CHANnel:UL:MINimum.....	645
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:UL:MINimum.....	645
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:CHANnel:UL:MAXimum?.....	645
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:UL:MAXimum?.....	645
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:UL:MINimum?.....	645
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:UL:MAXimum?.....	645
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:UL:MINimum?.....	646
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:UL:MAXimum?.....	646

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:UDSeparation

<Frequency>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:UDSeparation

<Frequency>

Configures the UL/DL separation $F_{DL} - F_{UL}$ for the user-defined band.

The allowed range depends on the remaining user-defined band settings: The resulting uplink carrier center frequencies must be within the allowed frequency range. For calculations, see [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:UDEFined:FREQuency:DL:MINimum](#).

Suffix:

<c> 1..7

Parameters:

<Frequency>

numeric

Depending on the other settings, only a part of the following range is allowed.

Range: -5930E+6 Hz to 5930E+6 Hz

*RST: 190E+6 Hz

Default unit: Hz

Example:

See [Configuring operating bands and channels](#)

Firmware/software: V2.1.30, SCC command V3.5.20

Options: R&S CMW-KS525

Manual operation: See ["UL/DL Separation"](#) on page 149

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:BINDicator

<BandIndicator>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:BINDicator

<BandIndicator>

Configures the frequency band indicator, identifying the user-defined band in signaling messages.

Suffix:

<c> 1..7

Parameters:

<BandIndicator> numeric
 Range: 1 to 256
 *RST: 1

Example: See [Configuring operating bands and channels](#)**Firmware/software:** V2.1.30, SCC command V3.2.50
V3.5.10: changed maximum from 64 to 256**Options:** R&S CMW-KS525**Manual operation:** See ["Band Indicator"](#) on page 150

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:CHANnel:DL:MINimum
 <Channel>

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:CHANnel:DL:MAXimum
 <Channel>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:DL:MINimum
 <Channel>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:DL:MAXimum
 <Channel>

Configures channel numbers for the user-defined band: the minimum downlink channel number and the maximum downlink channel number.

Combinations that result in frequencies outside of the allowed range are corrected automatically.

Suffix:

<c> 1..7

Parameters:

<Channel> decimal
 Range: 0 to 262143

Example: See [Configuring operating bands and channels](#)**Firmware/software:** V2.1.30, SCC commands V3.2.50**Options:** R&S CMW-KS525**Manual operation:** See ["DL Channel, Frequency"](#) on page 150

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:DL:MINimum
 <Frequency>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:DL:MINimum <Frequency>

Configures the carrier center frequency corresponding to the minimum downlink channel number for the user-defined band.

The other frequencies are calculated from the settings as follows:

$$FREQ:DL:MAX = FREQ:DL:MIN + (CHAN:DL:MAX - CHAN:DL:MIN) * 100 \text{ kHz}$$

$$FREQ:UL:MIN = FREQ:DL:MIN - UDSeparation$$

$$FREQ:UL:MAX = FREQ:DL:MIN - UDSeparation + (CHAN:DL:MAX - CHAN:DL:MIN) * 100 \text{ kHz}$$

Suffix:

<c> 1..7

Parameters:

<Frequency> numeric

The allowed range depends on the remaining user-defined band settings. All frequencies resulting from the calculations stated above must be located within the following frequency range.

Range: 70E+6 Hz to 6E+9 Hz

*RST: 2110E+6 Hz

Default unit: Hz

Example: See [Configuring operating bands and channels](#)

Firmware/software: V2.1.30, SCC command V3.2.50

Options: R&S CMW-KS525

Manual operation: See ["DL Channel, Frequency"](#) on page 150

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:DL:MAXimum?

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:DL:MAXimum?

Queries the maximum downlink carrier center frequency resulting from the user-defined band settings. For calculation, see [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:UDEFined:FREQuency:DL:MINimum](#).

Suffix:

<c> 1..7

Return values:

<Frequency> decimal

Range: 70E+6 Hz to 6E+9 Hz

Default unit: Hz

Example: See [Configuring operating bands and channels](#)

Usage: Query only

Firmware/software: V2.1.30, SCC command V3.2.50

Options: R&S CMW-KS525

Manual operation: See ["DL Channel, Frequency"](#) on page 150

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:CHANnel:UL:MINimum
 <Channel>

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:UL:MINimum
 <Channel>

Configures the minimum uplink channel number for the user-defined band.

Combinations that result in frequencies outside of the allowed range are corrected automatically.

Suffix:

<c> 1..7

Parameters:

<Channel> decimal

Range: 0 to 262143

Example: See [Configuring operating bands and channels](#)

Firmware/software: V2.1.30, SCC command V3.5.20

Options: R&S CMW-KS525

Manual operation: See "[UL Channel, Frequency](#)" on page 150

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:CHANnel:UL:MAXimum?

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:CHANnel:UL:
MAXimum?

Queries the maximum uplink channel number for the user-defined band, resulting from the other channel number settings.

Suffix:

<c> 1..7

Return values:

<Channel> decimal

Maximum uplink channel number

$CHAN:UL:MAX = CHAN:UL:MIN + CHAN:DL:MAX - CHAN:DL:MIN$

Example: See [Configuring operating bands and channels](#)

Usage: Query only

Firmware/software: V2.1.30, SCC command V3.5.20

Options: R&S CMW-KS525

Manual operation: See "[UL Channel, Frequency](#)" on page 150

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:UL:
MINimum?

CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UDEFined:FREQuency:UL:
MAXimum?

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:UL:MINimum?

CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UDEFined:FREQuency:UL:MAXimum?

Query the minimum and maximum uplink carrier center frequencies resulting from the user-defined band settings. For calculations, see [CONFigure:LTE:SIGN<i>:RFSettings\[:PCC\]:UDEFined:FREQuency:DL:MINimum](#).

Suffix:

<c> 1..7

Return values:

<Frequency> decimal
Range: 70E+6 Hz to 6E+9 Hz
Default unit: Hz

Example: See [Configuring operating bands and channels](#)

Usage: Query only

Firmware/software: V2.1.30, SCC commands V3.5.20

Options: R&S CMW-KS525

Manual operation: See ["UL Channel, Frequency"](#) on page 150

2.6.9 Internal fading

The following commands configure the internal fader of the R&S CMW.

2.6.9.1 Fading simulator

The following commands configure the fading simulator of the internal fader for fading of the PCC and SCC downlink.

CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ENABLE	647
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ENABLE	647
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:PROFile	647
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:PROFile	647
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:REStart:MODE	648
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:REStart:MODE	648
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:REStart	648
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:REStart	648
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:GLOBal:SEED	649
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:GLOBal:SEED	649
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ILOSs:MODE	649
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:MODE	649
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ILOSs:LOSS	650
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:LOSS	650
SENSe:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ILOSs:CSAMples<n>?	650
SENSe:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:CSAMples<n>?	650
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:DSHift:MODE	650

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:DSHift:MODE.....	650
CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:DSHift.....	651
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:DSHift.....	651

CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ENABLE <Enable>
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ENABLE <Enable>

Enables/disables the fading simulator.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring internal fading](#)

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Enable"](#) on page 153

CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:PROFile <profile>
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:PROFile <Profile>

Selects a propagation condition profile for fading.

Suffix:

<c> 1..7

Parameters:

<Profile> EP5Low | EP5Medium | EP5High | EV5Low | EV5Medium |
 EV5High | EV7Low | EV7Medium | EV7High | ET7Low |
 ET7Medium | ET7High | ET3Low | ET3Medium | ET3High |
 HSTRain | HST | CTEST | ETL30 | ETM30 | ETH30 | EVL200 |
 EVM200 | EVH200 | UMI3 | UMI30 | UMA3 | UMA30

EP5Low | EP5Medium | EP5High

EPA, 5-Hz Doppler, low/medium/high correlation

ETL30 | ETM30 | ETH30

ETU, 30-Hz Doppler, low/medium/high correlation

ET7Low | ET7Medium | ET7High

ETU, 70-Hz Doppler, low/medium/high correlation

ET3Low | ET3Medium | ET3High

ETU, 300-Hz Doppler, low/medium/high correlation

EV5Low | EV5Medium | EV5High

EVA, 5-Hz Doppler, low/medium/high correlation

EV7Low | EV7Medium | EV7High

EVA, 70-Hz Doppler, low/medium/high correlation

EVL200 | EVM200 | EVH200

EVA, 200-Hz Doppler, low/medium/high correlation

HSTRain | HST

High-speed train scenario (both values have the same effect)

CTEST

Multi-path profile for CQI tests

UMI3 | UMI30

SCME UMi, 3 km/h or 30 km/h

UMA3 | UMA30

SCME UMa, 3 km/h or 30 km/h

*RST: EP5Low

Example: See [Configuring internal fading](#)

Firmware/software: V3.5.10
V3.5.30: EVL200, EVM200, EVH200
V3.7.50: UMI3, UMI30, UMA3, UMA30

Options: See relevant fading scenario
R&S CMW-KE504 for UMI3, UMI30, UMA3, UMA30

Manual operation: See ["Profile"](#) on page 153

CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:REStart:MODE
<RestartMode>

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:REStart:MODE
<RestartMode>

Sets the restart mode of the fading simulator.

Suffix:

<c> 1..7

Parameters:

<RestartMode> AUTO | MANual

AUTO: fading automatically starts with the DL signal

MANual: fading is started and restarted manually (see
CONFigure:...:FSIMulator:REStart)

*RST: AUTO

Example: See [Configuring internal fading](#)

Firmware/software: V3.0.20, V3.2.70 SCC command

Options: See relevant fading scenario

Manual operation: See ["Restart Event"](#) on page 154

CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:REStart

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:REStart

Restarts the fading process in MANual mode (see also
CONFigure:...:FSIMulator:REStart:MODE).

Suffix:
 <c> 1..7

Usage: Event

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Restart Event"](#) on page 154

CONFIGure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:GLOBal:SEED <Seed>
CONFIGure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:GLOBal:SEED <Seed>

Sets the start seed for the pseudo-random fading algorithm.

Suffix:
 <c> 1..7

Parameters:
 <Seed> numeric
 Range: 0 to 9
 *RST: 0

Example: See [Configuring internal fading](#)

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Start Seed"](#) on page 154

CONFIGure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ILOSs:MODE
 <InsertLossMode>
CONFIGure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:MODE
 <InsertLossMode>

Sets the insertion loss mode.

Suffix:
 <c> 1..7

Parameters:
 <InsertLossMode> NORMal | USER
NORMal: The insertion loss is determined by the fading profile.
USER: The insertion loss is configurable.
 *RST: NORM

Example: See [Configuring internal fading](#)

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Insertion Loss"](#) on page 154

CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ILOSs:LOSS <InsertionLoss>
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:LOSS
 <InsertionLoss>

Sets the insertion loss for the fading simulator.

A setting is only allowed in USER mode (see
 CONFigure:...:FSIMulator:ILOSs:MODE).

Suffix:

<c> 1..7

Parameters:

<InsertionLoss> numeric
 Range: 0 dB to 30 dB
 *RST: 0 dB
 Default unit: dB

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Insertion Loss"](#) on page 154

SENSe:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:ILOSs:CSAMples<n>?
SENSe:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:ILOSs:CSAMples<n>?

Returns the percentage of clipped samples for the output path number <n>.

Suffix:

<n> 1..4

<c> 1..7

Return values:

<ClippedSamples> float
 Range: 0 % to 100 %
 Default unit: %

Usage: Query only

Firmware/software: V3.2.10, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Insertion Loss"](#) on page 154

CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:DSHift:MODE <Mode>
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:DSHift:MODE <Mode>

Sets the Doppler shift mode.

Suffix:

<c> 1..7

Parameters:

<Mode> NORMal | USER

NORMal: The maximum Doppler frequency is determined by the fading profile.**USER:** The maximum Doppler frequency is configurable.

*RST: NORM

Example:See [Configuring internal fading](#)**Firmware/software:** V3.2.20, SCC command V3.2.70**Options:**

See relevant fading scenario

Manual operation:See ["Doppler Frequency Mode, Doppler Frequency"](#) on page 154**CONFigure:LTE:SIGN<i>:FADing[:PCC]:FSIMulator:DSHift** <Frequency>**CONFigure:LTE:SIGN<i>:FADing:SCC<c>:FSIMulator:DSHift** <Frequency>

Sets the maximum Doppler frequency for the fading simulator.

A setting is only allowed in USER mode (see [CONFigure:LTE:SIGN<i>:FADing\[:PCC\]:FSIMulator:DSHift:MODE](#)).**Suffix:**

<c> 1..7

Parameters:

<Frequency> numeric

Range: 1 Hz to 2000 Hz

*RST: 5 Hz

Default unit: Hz

Example:See [Configuring internal fading](#)**Firmware/software:** V3.2.20, SCC command V3.2.70**Options:**

See relevant fading scenario

Manual operation:See ["Doppler Frequency Mode, Doppler Frequency"](#) on page 154**2.6.9.2 DL settings**

The following commands query noise power information for the PCC and SCC down-link.

[CONFigure:LTE:SIGN<i>:FADing\[:PCC\]:POWer:NOISe?](#)..... 652[CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:NOISe?](#)..... 652[CONFigure:LTE:SIGN<i>:FADing\[:PCC\]:POWer:NOISe:TOTal?](#)..... 652[CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:NOISe:TOTal?](#)..... 652[CONFigure:LTE:SIGN<i>:FADing\[:PCC\]:POWer:SUM?](#)..... 652[CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:SUM?](#)..... 652

CONFigure:LTE:SIGN<i>:FADing[:PCC]:POWer:NOISe?**CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:NOISe?**

Queries the calculated noise power on the DL channel, i.e. within the cell bandwidth.

Suffix:

<c> 1..7

Return values:

<NoisePower> float
Default unit: dBm

Example: See [Configuring internal fading](#)

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Noise \(System BW\) Power"](#) on page 155

CONFigure:LTE:SIGN<i>:FADing[:PCC]:POWer:NOISe:TOTal?**CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:NOISe:TOTal?**

Queries the total noise power for one carrier.

Suffix:

<c> 1..7

Return values:

<NoisePower> float
Default unit: dBm

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Noise \(Total BW\) Power"](#) on page 155

CONFigure:LTE:SIGN<i>:FADing[:PCC]:POWer:SUM?**CONFigure:LTE:SIGN<i>:FADing:SCC<c>:POWer:SUM?**

Queries the calculated total power (signal + noise) on the DL channel, i.e. within the cell bandwidth.

Suffix:

<c> 1..7

Return values:

<Power> float
Default unit: dBm

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Signal + Noise \(System BW\) Power"](#) on page 155

2.6.9.3 Fading module AWGN

The following commands configure the AWGN generator of the internal fader for the PCC and SCC downlink.

CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:ENABLE.....	653
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:ENABLE.....	653
CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:FOFFset.....	653
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:FOFFset.....	653
CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:BWIDth:RATio.....	654
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:BWIDth:RATio.....	654
CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:BWIDth:NOISe?.....	654
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:BWIDth:NOISe?.....	654
CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:SNRatio.....	655
CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:SNRatio.....	655

CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:ENABLE <Enable>

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:ENABLE <Enable>

Enables or disables AWGN insertion via the fading module.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON

*RST: OFF

Example: See [Configuring internal fading](#)

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Enable"](#) on page 156

CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:FOFFset <Offset>

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:FOFFset <Offset>

Shifts the center frequency of the noise bandwidth relative to the carrier center frequency.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 Range: -40E+6 Hz to 40E+6 Hz
 *RST: 0 Hz
 Default unit: Hz

Example: See [Configuring internal fading](#)

Firmware/software: V3.5.40

Options: See relevant fading scenario

Manual operation: See ["Frequency Offset"](#) on page 156

CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:BWIDth:RATio <Ratio>

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:BWIDth:RATio <Ratio>

Specifies the minimum ratio between the noise bandwidth and the cell bandwidth.

Suffix:

<c> 1..7

Parameters:

<Ratio> numeric
 Range: 1 to 1000
 *RST: 1

Example: See [Configuring internal fading](#)

Firmware/software: V3.0.20, SCC command V3.2.70
 V3.5.40 range enhanced

Options: See relevant fading scenario

Manual operation: See ["Min. Noise/System BW Ratio"](#) on page 156

CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:BWIDth:NOISe?

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:BWIDth:NOISe?

Queries the noise bandwidth.

Suffix:

<c> 1..7

Return values:

<NoiseBandwidth> float
 Range: 0 Hz to 80E+6 Hz
 Default unit: Hz

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Noise Bandwidth"](#) on page 156

CONFigure:LTE:SIGN<i>:FADing[:PCC]:AWGN:SNRatio <Ratio>

CONFigure:LTE:SIGN<i>:FADing:SCC<c>:AWGN:SNRatio <Ratio>

Specifies the signal to noise ratio for the AWGN inserted on the internal fading module.

Suffix:

<c> 1..7

Parameters:

<Ratio> numeric
 Range: -50 dB to 40 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring internal fading](#)

Firmware/software: V3.0.20, SCC command V3.2.70

Options: See relevant fading scenario

Manual operation: See ["Signal/Noise Ratio"](#) on page 157

2.6.10 Downlink power levels

The following commands define power levels of physical downlink channels and signals.

CONFigure:LTE:SIGN<i>:DL[:PCC]:RSEPre:LEVel.....	656
CONFigure:LTE:SIGN<i>:DL:SCC<c>:RSEPre:LEVel.....	656
SENSe:LTE:SIGN<i>:DL[:PCC]:FCPower?.....	656
SENSe:LTE:SIGN<i>:DL:SCC<c>:FCPower?.....	656
CONFigure:LTE:SIGN<i>:DL[:PCC]:PSS:POFFset.....	657
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PSS:POFFset.....	657
CONFigure:LTE:SIGN<i>:DL[:PCC]:SSS:POFFset.....	657
CONFigure:LTE:SIGN<i>:DL:SCC<c>:SSS:POFFset.....	657
CONFigure:LTE:SIGN<i>:DL[:PCC]:PBCH:POFFset.....	658
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PBCH:POFFset.....	658
CONFigure:LTE:SIGN<i>:DL[:PCC]:PCFich:POFFset.....	658
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PCFich:POFFset.....	658
CONFigure:LTE:SIGN<i>:DL[:PCC]:PHICH:POFFset.....	658
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PHICH:POFFset.....	658
CONFigure:LTE:SIGN<i>:DL[:PCC]:PDCCh:POFFset.....	659
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDCCh:POFFset.....	659
CONFigure:LTE:SIGN<i>:DL[:PCC]:OCNG.....	659
CONFigure:LTE:SIGN<i>:DL:SCC<c>:OCNG.....	659
CONFigure:LTE:SIGN<i>:DL[:PCC]:PDSCh:PA.....	660
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDSCh:PA.....	660
CONFigure:LTE:SIGN<i>:DL[:PCC]:PDSCh:RINdex.....	660
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDSCh:RINdex.....	660
CONFigure:LTE:SIGN<i>:DL[:PCC]:POWER:PORTs.....	660
CONFigure:LTE:SIGN<i>:DL:SCC<c>:POWER:PORTs.....	660
CONFigure:LTE:SIGN<i>:DL[:PCC]:CSIRs:MODE.....	661

CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:MODE.....	661
CONFigure:LTE:SIGN<i>:DL[:PCC]:CSIRs:POFFset.....	661
CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:POFFset.....	661
CONFigure:LTE:SIGN<i>:DL[:PCC]:AWGN.....	662
CONFigure:LTE:SIGN<i>:DL:SCC<c>:AWGN.....	662

CONFigure:LTE:SIGN<i>:DL[:PCC]:RSEPre:LEVel <Level>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:RSEPre:LEVel <Level>

Defines the energy per resource element (EPRE) of the cell-specific reference signal (C-RS). The power levels of resource elements used for other channels/signals are defined relative to this power level.

The allowed value range depends basically on the used connector, the number of allocated resource blocks (specified via the cell bandwidth) and the external attenuation in the output path.

$$level_{RS\ EPRE, min} = level_{connector, min} - 10 \cdot \log_{10}(12 \cdot N_{RB}) - ext\ att_{out}$$

$$level_{RS\ EPRE, max} = level_{connector, max} - 10 \cdot \log_{10}(12 \cdot N_{RB}) - ext\ att_{out} - 15\ dB$$

With $level_{connector, min} = -130\ dBm$ ($-120\ dBm$), $level_{connector, max} = -5\ dBm$ ($8\ dBm$) for [RF COM] ([RF OUT]). Notice also the ranges quoted in the data sheet.

The range is also affected by active AWGN ("Downlink Power Levels" parameter), internal fading (insertion loss value), CSI-RS power, number of MIMO transmit antennas.

Suffix:

<c> 1..7

Parameters:

<Level> numeric
 Range: see above
 *RST: -85 dBm/15kHz
 Default unit: dBm/15kHz

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["RS EPRE"](#) on page 158

SENSe:LTE:SIGN<i>:DL[:PCC]:FCPower?

SENSe:LTE:SIGN<i>:DL:SCC<c>:FCPower?

Queries the "Full Cell BW Power". The power results from the configured RS EPRE and the cell bandwidth.

Suffix:

<c> 1..7

Return values:

<Level> float
 Range: -220 dBm to 48 dBm
 Default unit: dBm

Example: See [Configuring DL power levels](#)

Usage: Query only

Firmware/software: V3.0.10, SCC command V3.2.50

Manual operation: See ["RS EPRE"](#) on page 158

CONFigure:LTE:SIGN<i>:DL[:PCC]:PSS:POFFset <Offset>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:PSS:POFFset <Offset>

Defines the power level of a primary synchronization signal (PSS) resource element.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 PSS power relative to RS EPRE
 Range: -30 dB to 0 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["PSS Power Offset"](#) on page 158

CONFigure:LTE:SIGN<i>:DL[:PCC]:SSS:POFFset <Offset>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:SSS:POFFset <Offset>

Defines the power level of a secondary synchronization signal (SSS) resource element.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 SSS power relative to RS EPRE
 Range: -30 dB to 0 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["SSS Power Offset"](#) on page 158

CONFigure:LTE:SIGN<i>:DL[:PCC]:PBCH:POFFset <Offset>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:PBCH:POFFset <Offset>

Defines the power level of a physical broadcast channel (PBCH) resource element.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 PBCH power relative to RS EPRE
 Range: -30 dB to 0 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["PBCH Power Offset"](#) on page 158

CONFigure:LTE:SIGN<i>:DL[:PCC]:PCFich:POFFset <Offset>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:PCFich:POFFset <Offset>

Defines the power level of a physical control format indicator channel (PCFICH) resource element.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 PCFICH power relative to RS EPRE
 Range: -30 dB to 0 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["PCFICH Power Offset"](#) on page 159

CONFigure:LTE:SIGN<i>:DL[:PCC]:PHICH:POFFset <Offset>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:PHICH:POFFset <Offset>

Defines the power level of a physical hybrid ARQ indicator channel (PHICH) resource element.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 PHICH power relative to RS EPRE
 Range: -30 dB to 0 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V3.2.10, SCC command V3.2.50

Manual operation: See ["PHICH Power Offset"](#) on page 159

CONFigure:LTE:SIGN<i>:DL[:PCC]:PDCCh:POFFset <Offset>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDCCh:POFFset <Offset>

Defines the power level of a physical downlink control channel (PDCCH) resource element.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 PDCCH power relative to RS EPRE
 Range: -30 dB to 0 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["PDCCH Power Offset"](#) on page 159

CONFigure:LTE:SIGN<i>:DL[:PCC]:OCNG <Enable>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:OCNG <Enable>

Enables or disables the OFDMA channel noise generator (OCNG).

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["OCNG"](#) on page 159

CONFigure:LTE:SIGN<i>:DL[:PCC]:PDSCh:PA <PA>
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDSCh:PA <PA>

Defines the power offset P_A .

Suffix:

<c> 1..7

Parameters:

<PA> ZERO | N3DB | N6DB
 Power offset of 0 dB | -3 dB | -6 dB
 *RST: ZERO

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.21, SCC command V3.2.50

Manual operation: See "PDSCH" on page 159

CONFigure:LTE:SIGN<i>:DL[:PCC]:PDSCh:RINdex <RatioIndex>
CONFigure:LTE:SIGN<i>:DL:SCC<c>:PDSCh:RINdex <RatioIndex>

Defines the power ratio index P_B . The index is required for calculation of the power level of a PDSCH resource element.

Suffix:

<c> 1..7

Parameters:

<RatioIndex> numeric
 Range: 0 to 3
 *RST: 0

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See "PDSCH" on page 159

CONFigure:LTE:SIGN<i>:DL[:PCC]:POWER:PORTs <power>
CONFigure:LTE:SIGN<i>:DL:SCC<c>:POWER:PORTs <power>

Defines the power offset for the antenna ports 7 to 10.

Suffix:

<c> 1..7

Parameters:

<power> numeric
 Range: -12 dB to 0 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS520 or -KS540

Manual operation: See "[PDSCH](#)" on page 159

CONFigure:LTE:SIGN<i>:DL[:PCC]:CSIRs:MODE <Mode>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:MODE <Mode>

Selects a configuration mode for the used CSI-RS power offset.

Suffix:

<c> 1..7

Parameters:

<Mode> ACSirs | MANual

ACSirs

The used power offset matches the signaled value.

For configuration of the signaled value, see [CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:POWer](#) on page 768.

MANual

The used power offset is independent from the signaled value.

For configuration of the used power offset, see [CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:POFFset](#) on page 661.

*RST: ACS

Example: See [Configuring DL power levels](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS520 or -KS540

Manual operation: See "[CSI-RS](#)" on page 160

CONFigure:LTE:SIGN<i>:DL[:PCC]:CSIRs:POFFset <Offset>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:POFFset <Offset>

Sets the EPRE of the PDSCH relative to the EPRE of the CSI reference signal.

The value is only used for [CONFigure:LTE:SIGN<i>:DL:SCC<c>:CSIRs:MODE](#) = ACSirs.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric

Range: -30 dB to 8 dB

*RST: 0 dB

Default unit: dB

Example: See [Configuring DL power levels](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS520 or -KS540

Manual operation: See ["CSI-RS"](#) on page 160

CONFigure:LTE:SIGN<i>:DL[:PCC]:AWGN <AWGN>

CONFigure:LTE:SIGN<i>:DL:SCC<c>:AWGN <AWGN>

Specifies the total level of the additive white Gaussian noise (AWGN) interferer. The unit dBm/15 kHz indicates the spectral density integrated across one subcarrier.

The range depends on several parameters. It either equals the range of the RS EPRE or is a part of this range.

Suffix:

<c> 1..7

Parameters:

<AWGN> numeric | ON | OFF

Range: depends on many parameters

*RST: OFF

Default unit: dBm/15kHz

ON | OFF enables or disables the AWGN interferer.

Example: See [Configuring DL power levels](#)

Firmware/software: V1.0.15.21, SCC command V3.2.50

Options: R&S CMW-KS510

Manual operation: See ["AWGN"](#) on page 160

2.6.11 Uplink power control

The following commands define parameters related to UE uplink power control by the instrument.

2.6.11.1 General power control parameters

The following commands configure the power control mode for UL CA and specify the cell-specific maximum allowed power.

CONFigure:LTE:SIGN<i>:UL[:PCC]:JUPower.....	663
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PMCC?.....	663
CONFigure:LTE:SIGN<i>:UL:SETA:PMAX.....	663
CONFigure:LTE:SIGN<i>:UL:SETB:PMAX.....	663
CONFigure:LTE:SIGN<i>:UL:SETC:PMAX.....	663
CONFigure:LTE:SIGN<i>:UL[:PCC]:PMAX.....	663
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PMAX.....	663

CONFigure:LTE:SIGN<i>:UL[:PCC]:JUPower <Enable>

Enables or disables joint uplink power control for uplink carrier aggregation.

Parameters:

<Enable> OFF | ON
*RST: ON

Example: See [Configuring UL power control for call setup](#)

Firmware/software: V3.5.30

Manual operation: See ["Joint UL Power, Power Master CC"](#) on page 161

CONFigure:LTE:SIGN<i>:UL:SCC<c>:PMCC?

Queries the UL power primary carrier for the SCC<c>.

Suffix:

<c> 1..7

Return values:

<Primary> PCC | SCC1 | SCC2 | SCC3 | SCC4 | SCC5 | SCC6 | SCC7

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Joint UL Power, Power Master CC"](#) on page 161

CONFigure:LTE:SIGN<i>:UL:SETA:PMAX <Power>**CONFigure:LTE:SIGN<i>:UL:SETB:PMAX <Power>****CONFigure:LTE:SIGN<i>:UL:SETC:PMAX <Power>****CONFigure:LTE:SIGN<i>:UL[:PCC]:PMAX <Power>****CONFigure:LTE:SIGN<i>:UL:SCC<c>:PMAX <Power>**

Specifies the maximum allowed UE power.

Suffix:

<c> 1..7

Parameters:

<Power> numeric | ON | OFF
Range: -30 dBm to 33 dBm
*RST: 24 dBm
Default unit: dBm
ON | OFF enables or disables signaling of the value to the UE.

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.0.10, SCC V3.5.20, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Max. allowed Power P-Max"](#) on page 161

2.6.11.2 PRACH and initial PUSCH power

The following commands define parameters related to PRACH and initial PUSCH power configuration.

CONFigure:LTE:SIGN<i>:UL:SETA:APower:EASettings	665
CONFigure:LTE:SIGN<i>:UL:SETB:APower:EASettings	665
CONFigure:LTE:SIGN<i>:UL:SETC:APower:EASettings	665
CONFigure:LTE:SIGN<i>:UL[:PCC]:APower:EASettings	665
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APower:EASettings	665
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:OLNPower	666
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:OLNPower	666
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:OLNPower	666
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:OLNPower	666
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:OLNPower	666
SENSe:LTE:SIGN<i>:UL:SETA:APower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SETB:APower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SETC:APower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL[:PCC]:APower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SCC<c>:APower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SETA:APower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETB:APower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETC:APower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL[:PCC]:APower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SCC<c>:APower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETA:APower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETB:APower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETC:APower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL[:PCC]:APower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SCC<c>:APower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETA:APower:PCALpha:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETB:APower:PCALpha:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETC:APower:PCALpha:BASic?	667
SENSe:LTE:SIGN<i>:UL[:PCC]:APower:PCALpha:BASic?	668
SENSe:LTE:SIGN<i>:UL:SCC<c>:APower:PCALpha:BASic?	668
SENSe:LTE:SIGN<i>:UL:SETA:APower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL:SETB:APower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL:SETC:APower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL[:PCC]:APower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL:SCC<c>:APower:TPRRcsetup:BASic?	668
CONFigure:LTE:SIGN<i>:UL:SETA:APower:RSPower:ADVanced	668
CONFigure:LTE:SIGN<i>:UL:SETB:APower:RSPower:ADVanced	668
CONFigure:LTE:SIGN<i>:UL:SETC:APower:RSPower:ADVanced	668
CONFigure:LTE:SIGN<i>:UL[:PCC]:APower:RSPower:ADVanced	669
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APower:RSPower:ADVanced	669
CONFigure:LTE:SIGN<i>:UL:SETA:APower:PIRPower:ADVanced	669
CONFigure:LTE:SIGN<i>:UL:SETB:APower:PIRPower:ADVanced	669
CONFigure:LTE:SIGN<i>:UL:SETC:APower:PIRPower:ADVanced	669
CONFigure:LTE:SIGN<i>:UL[:PCC]:APower:PIRPower:ADVanced	669
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APower:PIRPower:ADVanced	669

CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PNPusch:ADVanced.....	669
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PNPusch:ADVanced.....	669
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PNPusch:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PNPusch:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PNPusch:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PCALpha:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PCALpha:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PCALpha:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PCALpha:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PCALpha:ADVanced.....	670
CONFigure:LTE:SIGN<i>:UL:SETA:APPower:TPRRcsetup:ADVanced.....	671
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:TPRRcsetup:ADVanced.....	671
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:TPRRcsetup:ADVanced.....	671
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:TPRRcsetup:ADVanced.....	671
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:TPRRcsetup:ADVanced.....	671
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PATHloss?.....	671
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PATHloss?.....	671
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PATHloss?.....	671
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PATHloss?.....	671
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PATHloss?.....	671
SENSe:LTE:SIGN<i>:UL:SETA:APPower:EPPower?.....	671
SENSe:LTE:SIGN<i>:UL:SETB:APPower:EPPower?.....	671
SENSe:LTE:SIGN<i>:UL:SETC:APPower:EPPower?.....	671
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:EPPower?.....	671
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:EPPower?.....	671
SENSe:LTE:SIGN<i>:UL:SETA:APPower:EOPower?.....	672
SENSe:LTE:SIGN<i>:UL:SETB:APPower:EOPower?.....	672
SENSe:LTE:SIGN<i>:UL:SETC:APPower:EOPower?.....	672
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:EOPower?.....	672
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:EOPower?.....	672

CONFigure:LTE:SIGN<i>:UL:SETA:APPower:EASettings <Enable>
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:EASettings <Enable>
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:EASettings <Enable>
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:EASettings <Enable>
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:EASettings <Enable>

Enables or disables advanced configuration of the PRACH and open loop power settings via the other CONFigure:LTE:SIGN:UL:PCC/SCC<c>:APPower:... commands.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Enable Advanced Settings"](#) on page 163

CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:OLNPower <Power>
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:OLNPower <Power>
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:OLNPower <Power>
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:OLNPower <Power>
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:OLNPower <Power>

Defines a cell-specific nominal power value for full resource block allocation in the UL (entire cell bandwidth used). From this value, the cell-specific nominal power value $P_{O_NOMINAL_PUSCH}$ related to one resource block is determined and sent to all UEs via broadcast.

This command is only relevant for basic configuration and rejected if advanced configuration is active.

Suffix:

<c> 1..7

Parameters:

<Power> numeric
 Range: -50 dBm to 23 dBm
 *RST: -20 dBm
 Default unit: dBm

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.5.10, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Open Loop Nominal Power"](#) on page 162

SENSe:LTE:SIGN<i>:UL:SETA:APPower:RSPower:BASic?
SENSe:LTE:SIGN<i>:UL:SETB:APPower:RSPower:BASic?
SENSe:LTE:SIGN<i>:UL:SETC:APPower:RSPower:BASic?
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:RSPower:BASic?
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:RSPower:BASic?

Queries the "referenceSignalPower" value, signaled to the UE if basic UL power configuration applies.

Suffix:

<c> 1..7

Return values:

<RefSignalPower> float
 Range: -60 dBm to 50 dBm
 Default unit: dBm

Example: See [Configuring UL power control for call setup](#)

Usage: Query only

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Reference Signal Power"](#) on page 163

SENSe:LTE:SIGN<i>:UL:SETA:APPower:PIRPower:BASic?
 SENSe:LTE:SIGN<i>:UL:SETB:APPower:PIRPower:BASic?
 SENSe:LTE:SIGN<i>:UL:SETC:APPower:PIRPower:BASic?
 SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PIRPower:BASic?
 SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PIRPower:BASic?

Queries the "preambleInitialReceivedTargetPower" value, signaled to the UE if basic UL power configuration applies.

Suffix:

<c> 1..7

Return values:

<TargetPower> float
 Range: -120 dBm to -90 dBm
 Default unit: dBm

Example: See [Configuring UL power control for call setup](#)

Usage: Query only

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Preamble Initial Received Target Power"](#) on page 163

SENSe:LTE:SIGN<i>:UL:SETA:APPower:PNPusch:BASic?
 SENSe:LTE:SIGN<i>:UL:SETB:APPower:PNPusch:BASic?
 SENSe:LTE:SIGN<i>:UL:SETC:APPower:PNPusch:BASic?
 SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PNPusch:BASic?
 SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PNPusch:BASic?

Queries the "p0-NominalPUSCH" value, signaled to the UE if basic UL power configuration applies.

Suffix:

<c> 1..7

Return values:

<P0NominalPUSCH> float
 Range: -126 dBm to 24 dBm
 Default unit: dBm

Example: See [Configuring UL power control for call setup](#)

Usage: Query only

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["P0 Nominal PUSCH"](#) on page 164

SENSe:LTE:SIGN<i>:UL:SETA:APPower:PCALpha:BASic?
 SENSe:LTE:SIGN<i>:UL:SETB:APPower:PCALpha:BASic?
 SENSe:LTE:SIGN<i>:UL:SETC:APPower:PCALpha:BASic?

SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PCALpha:BASic?

SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PCALpha:BASic?

Queries the value of parameter "alpha", signaled to the UE if basic UL power configuration applies.

Suffix:

<c> 1..7

Return values:

<PathCompAlpha> ZERO | DOT4 | DOT5 | DOT6 | DOT7 | DOT8 | DOT9 | ONE

ZERO: 0

DOT4 ... DOT9: 0.4 ... 0.9

ONE: 1.0

Example: See [Configuring UL power control for call setup](#)

Usage: Query only

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Pathloss Compensation Alpha"](#) on page 164

SENSe:LTE:SIGN<i>:UL:SETA:APPower:TPRRcsetup:BASic?

SENSe:LTE:SIGN<i>:UL:SETB:APPower:TPRRcsetup:BASic?

SENSe:LTE:SIGN<i>:UL:SETC:APPower:TPRRcsetup:BASic?

SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:TPRRcsetup:BASic?

SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:TPRRcsetup:BASic?

Queries the state of P0-UE-PUSCH toggling, determining the P0-UE-PUSCH values signaled to the UE during RRC connection setup if basic UL power configuration applies.

Suffix:

<c> 1..7

Return values:

<Enable> OFF | ON

*RST: OFF

Example: See [Configuring UL power control for call setup](#)

Usage: Query only

Firmware/software: PCC V3.2.20, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Toggle P0-UE-PUSCH at RRC Setup"](#) on page 165

CONFigure:LTE:SIGN<i>:UL:SETA:APPower:RSPower:ADVanced

<RefSignalPower>

CONFigure:LTE:SIGN<i>:UL:SETB:APPower:RSPower:ADVanced

<RefSignalPower>

CONFigure:LTE:SIGN<i>:UL:SETC:APPower:RSPower:ADVanced

<RefSignalPower>

CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:RSPower:ADVanced
 <RefSignalPower>

CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:RSPower:ADVanced
 <RefSignalPower>

Specifies the "referenceSignalPower" value, signaled to the UE if advanced UL power configuration applies.

Suffix:

<c> 1..7

Parameters:

<RefSignalPower> numeric
 Range: -60 dBm to 50 dBm
 *RST: 18 dBm
 Default unit: dBm

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Reference Signal Power"](#) on page 163

CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PIRPower:ADVanced <TargetPower>

CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PIRPower:ADVanced <TargetPower>

CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PIRPower:ADVanced <TargetPower>

CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PIRPower:ADVanced <TargetPower>

CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PIRPower:ADVanced
 <TargetPower>

Specifies the "preambleInitialReceivedTargetPower" value, signaled to the UE if advanced UL power configuration applies.

Suffix:

<c> 1..7

Parameters:

<TargetPower> numeric
 Range: -120 dBm to -90 dBm
 Increment: 2 dB
 *RST: -104 dBm
 Default unit: dBm

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Preamble Initial Received Target Power"](#) on page 163

CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PNPusch:ADVanced
 <P0NominalPUSCH>

CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PNPusch:ADVanced
 <P0NominalPUSCH>

CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PNPusch:ADVanced
 <P0NominalPUSCH>

CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PNPusch:ADVanced
 <P0NominalPUSCH>

CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PNPusch:ADVanced
 <P0NominalPUSCH>

Specifies the "p0-NominalPUSCH" value, signaled to the UE if advanced UL power configuration applies.

Suffix:

<c> 1..7

Parameters:

<P0NominalPUSCH> numeric

Range: -126 dBm to 24 dBm

*RST: -85 dBm

Default unit: dBm

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["P0 Nominal PUSCH"](#) on page 164

CONFigure:LTE:SIGN<i>:UL:SETA:APPower:PCALpha:ADVanced
 <PathCompAlpha>

CONFigure:LTE:SIGN<i>:UL:SETB:APPower:PCALpha:ADVanced
 <PathCompAlpha>

CONFigure:LTE:SIGN<i>:UL:SETC:APPower:PCALpha:ADVanced
 <PathCompAlpha>

CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:PCALpha:ADVanced
 <PathCompAlpha>

CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:PCALpha:ADVanced
 <PathCompAlpha>

Specifies the value of parameter "alpha", signaled to the UE if advanced UL power configuration applies.

Suffix:

<c> 1..7

Parameters:

<PathCompAlpha> ZERO | DOT4 | DOT5 | DOT6 | DOT7 | DOT8 | DOT9 | ONE

ZERO: 0

DOT4 ... DOT9: 0.4 ... 0.9

ONE: 1.0

*RST: DOT8

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.0.50, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Pathloss Compensation Alpha"](#) on page 164

CONFigure:LTE:SIGN<i>:UL:SETA:APPower:TPRRcsetup:ADVanced <Enable>
CONFigure:LTE:SIGN<i>:UL:SETB:APPower:TPRRcsetup:ADVanced <Enable>
CONFigure:LTE:SIGN<i>:UL:SETC:APPower:TPRRcsetup:ADVanced <Enable>
CONFigure:LTE:SIGN<i>:UL[:PCC]:APPower:TPRRcsetup:ADVanced <Enable>
CONFigure:LTE:SIGN<i>:UL:SCC<c>:APPower:TPRRcsetup:ADVanced <Enable>

Enables or disables P0-UE-PUSCH toggling and thus determines the P0-UE-PUSCH values signaled to the UE during RRC connection setup if advanced UL power configuration applies.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring UL power control for call setup](#)

Firmware/software: PCC V3.2.20, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Toggle P0-UE-PUSCH at RRC Setup"](#) on page 165

SENSe:LTE:SIGN<i>:UL:SETA:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PATHloss?
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PATHloss?

Queries the pathloss resulting from the advanced UL power settings.

Suffix:

<c> 1..7

Return values:

<Pathloss> float
 Default unit: dB

Example: See [Configuring UL power control for call setup](#)

Usage: Query only

Firmware/software: PCC V3.0.50, SCC V3.5.20, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Pathloss"](#) on page 165

SENSe:LTE:SIGN<i>:UL:SETA:APPower:EPPPower?
SENSe:LTE:SIGN<i>:UL:SETB:APPower:EPPPower?
SENSe:LTE:SIGN<i>:UL:SETC:APPower:EPPPower?
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:EPPPower?
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:EPPPower?

Queries the expected power of the first preamble, resulting from the advanced UL power settings.

Suffix:

<c> 1..7

Return values:<Power> float
Default unit: dBm**Example:** See [Configuring UL power control for call setup](#)**Usage:** Query only**Firmware/software:** PCC V3.0.50, SCC V3.5.20, SETA/B V3.7.30, SETC V3.7.70**Manual operation:** See ["Expected PRACH Preamble Power"](#) on page 165**SENSE:LTE:SIGN<i>:UL:SETA:APower:EOPower?****SENSE:LTE:SIGN<i>:UL:SETB:APower:EOPower?****SENSE:LTE:SIGN<i>:UL:SETC:APower:EOPower?****SENSE:LTE:SIGN<i>:UL[:PCC]:APower:EOPower?****SENSE:LTE:SIGN<i>:UL:SCC<c>:APower:EOPower?**

Queries the expected initial PUSCH power, resulting from the advanced UL power settings.

Suffix:

<c> 1..7

Return values:<ExpectedOLpower> float
Default unit: dBm**Example:** See [Configuring UL power control for call setup](#)**Usage:** Query only**Firmware/software:** PCC V3.0.50, SCC V3.5.20, SETA/B V3.7.30, SETC V3.7.70**Manual operation:** See ["Expected OL Power"](#) on page 166**2.6.11.3 TX power control (TPC)**

The following commands define parameters related to UE uplink power control via TPC commands.

CONFfigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:SET	673
CONFfigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:SET	673
CONFfigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:SET	673
CONFfigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:SET	673
CONFfigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:SET	673
CONFfigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:PEXecute	674
CONFfigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:PEXecute	674
CONFfigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:PEXecute	674
CONFfigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:PEXecute	674
CONFfigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:PEXecute	674
CONFfigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:RPControl	674

CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:RPCControl.....	674
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:RPCControl.....	674
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:RPCControl.....	674
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:RPCControl.....	674
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:TPOWer.....	675
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:TPOWer.....	675
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:TPOWer.....	675
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:TPOWer.....	675
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:TPOWer.....	675
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:CLTPower.....	675
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:CLTPower.....	675
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:CLTPower.....	675
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:CLTPower.....	675
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:CLTPower.....	675
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:CLTPower:OFFSet.....	675
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:CLTPower:OFFSet.....	675
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:CLTPower:OFFSet.....	676
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:CLTPower:OFFSet.....	676
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:SINGLE.....	676
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:SINGLE.....	676
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:SINGLE.....	676
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:SINGLE.....	676
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:SINGLE.....	676
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:UDPattern.....	677
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:UDPattern.....	677
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:UDPattern.....	677
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:UDPattern.....	677
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:UDPattern.....	677

CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:SET <SetType>

CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:SET <SetType>

CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:SET <SetType>

CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:SET <SetType>

CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:SET <SetType>

Selects the active TPC setup to be executed for power control of the PUSCH.

For some TPC setups, the execution must be explicitly triggered via [CONFigure:LTE:SIGN<i>:UL\[:PCC\]:PUSCh:TPC:PEXecute](#).

Suffix:

<c> 1..7

Parameters:

<SetType> MINPower | MAXPower | CONSTant | SINGLE | UDSingle | UDContinuous | ALT0 | CLOop | RPControl | FULPower

MINPower: command the UE to minimum power

MAXPower: command the UE to maximum power

CONSTant: command the UE to keep the power constant

SINGLE: send a pattern once (only one type of TPC command)

UDSingle: send a pattern once (mixed TPC commands allowed)

UDContinuous: send a pattern continuously
ALT0: send an alternating pattern continuously
CLOop: command the UE to a configurable target power
RPControl: patterns for 3GPP relative power control test
FULPower: flexible uplink power
***RST:** CLO

Example: See [Modifying parameters for an established connection](#)

Firmware/software: PCC V3.2.50, SCC and FULPower V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Active TPC Setup"](#) on page 167

CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:PEXecute
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:PEXecute

Execute the active TPC setup for power control of the PUSCH. This command is only relevant for setups which are not executed automatically (SINGLE, UDSingle, RPControl, FULPower).

Suffix:
 <c> 1..7

Example: See [Modifying parameters for an established connection](#)

Usage: Event

Firmware/software: PCC V2.0.20, SCC V3.5.20, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Active TPC Setup"](#) on page 167

CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:RPControl <Pattern>
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:RPControl <Pattern>
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:RPControl <Pattern>
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:RPControl <Pattern>
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:RPControl <Pattern>

Selects a TPC pattern for 3GPP relative power control tests with the TPC setup RPControl.

Suffix:
 <c> 1..7

Parameters:
 <Pattern> RUA | RDA | RUB | RDB | RUC | RDC
RUA | RUB | RUC: ramping up A | B | C
RDA | RDB | RDC: ramping down A | B | C
***RST:** RUA

Example: See [Modifying parameters for an established connection](#)

Firmware/software: PCC V3.2.50, SCC V3.5.20, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["3GPP Rel. Pow. Ctrl. Pattern"](#) on page 168

```
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:TPOWer <Power>
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:TPOWer <Power>
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:TPOWer <Power>
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:TPOWer <Power>
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:TPOWer <Power>
```

Defines the target powers for power control with the TPC setup `FULPower`.

Suffix:

<c> 1..7

Parameters:

<Power> numeric
 Range: -50 dBm to 33 dBm
 *RST: -20 dBm
 Default unit: dBm

Firmware/software: V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Target Power PCC / SCC"](#) on page 168

```
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:CLTPower <Power>
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:CLTPower <Power>
CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:CLTPower <Power>
CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:CLTPower <Power>
CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:CLTPower <Power>
```

Defines the target power for power control with the TPC setup `CLoop`.

Suffix:

<c> 1..7

Parameters:

<Power> numeric
 Range: -50 dBm to 33 dBm
 *RST: -20 dBm
 Default unit: dBm

Example: See [Modifying parameters for an established connection](#)

Firmware/software: PCC V3.2.60, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Closed Loop Target Power"](#) on page 169

```
CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:CLTPower:OFFSet <Offset>
CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:CLTPower:OFFSet <Offset>
```

CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:CLTPower:OFFSet <Offset>

CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:CLTPower:OFFSet <Offset>

Defines a target power offset relative to the power primary carrier, for power control with the TPC setup `CLoOp`.

The setting is irrelevant for carriers with independent UL power control.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 Target power = primary carrier target power + <Offset>
 Range: -7 dB to 7 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Modifying parameters for an established connection](#)

Firmware/software: V3.5.50, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Closed Loop Target Power"](#) on page 169

CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:SINGLE <NoOfSteps>,
 <StepDirection>

CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:SINGLE <NoOfSteps>,
 <StepDirection>

CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:SINGLE <NoOfSteps>,
 <StepDirection>

CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:SINGLE <NoOfSteps>,
 <StepDirection>

CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:SINGLE <NoOfSteps>,
 <StepDirection>

Defines a pattern for power control of the PUSCH with the TPC setup `SINGLE`. The pattern consists of 1 to 35 up (+1 dB) or down (-1 dB) commands, followed by "constant power" commands (0 dB).

Suffix:

<c> 1..7

Parameters:

<NoOfSteps> numeric
 Range: 1 to 35
 *RST: 1

<StepDirection> UP | DOWN
 *RST: UP

Example: See [Modifying parameters for an established connection](#)

Firmware/software: PCC V3.0.10, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["Single Pattern"](#) on page 169

CONFigure:LTE:SIGN<i>:UL:SETA:PUSCh:TPC:UDPattern <PatternLength>, <Value1>[, <Value2>, <Value3>, <Value4>, <Value5>, <Value6>, <Value7>, <Value8>, <Value9>, <Value10>, <Value11>, <Value12>, <Value13>, <Value14>, <Value15>, <Value16>, <Value17>, <Value18>, <Value19>, <Value20>]

CONFigure:LTE:SIGN<i>:UL:SETB:PUSCh:TPC:UDPattern <PatternLength>, <Value1>[, <Value2>, <Value3>, <Value4>, <Value5>, <Value6>, <Value7>, <Value8>, <Value9>, <Value10>, <Value11>, <Value12>, <Value13>, <Value14>, <Value15>, <Value16>, <Value17>, <Value18>, <Value19>, <Value20>]

CONFigure:LTE:SIGN<i>:UL:SETC:PUSCh:TPC:UDPattern <PatternLength>, <Value1>[, <Value2>, <Value3>, <Value4>, <Value5>, <Value6>, <Value7>, <Value8>, <Value9>, <Value10>, <Value11>, <Value12>, <Value13>, <Value14>, <Value15>, <Value16>, <Value17>, <Value18>, <Value19>, <Value20>]

CONFigure:LTE:SIGN<i>:UL[:PCC]:PUSCh:TPC:UDPattern <PatternLength>, <Value1>[, <Value2>, <Value3>, <Value4>, <Value5>, <Value6>, <Value7>, <Value8>, <Value9>, <Value10>, <Value11>, <Value12>, <Value13>, <Value14>, <Value15>, <Value16>, <Value17>, <Value18>, <Value19>, <Value20>]

CONFigure:LTE:SIGN<i>:UL:SCC<c>:PUSCh:TPC:UDPattern <PatternLength>, <Value1>[, <Value2>, <Value3>, <Value4>, <Value5>, <Value6>, <Value7>, <Value8>, <Value9>, <Value10>, <Value11>, <Value12>, <Value13>, <Value14>, <Value15>, <Value16>, <Value17>, <Value18>, <Value19>, <Value20>]

Defines a pattern for power control of the PUSCH with the TPC setup `UDSingle` or `UDContinuous`.

The pattern consists of 1 to 20 TPC commands. To configure the pattern, specify the pattern length and a corresponding number of TPC commands.

If you specify fewer TPC commands than required according to the pattern length, the previously defined values are used for the remaining commands. If you specify more TPC commands than required according to the pattern length, all values are set, but only the values corresponding to the pattern length are used.

Suffix:

<c> 1..7

Parameters:

<PatternLength>	numeric
	Number of values to be considered for the pattern
Range:	1 to 20
*RST:	10
<Value1>	numeric
Range:	-1 dB to 3 dB
*RST:	1 dB
	Default unit: dB

<Value2>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value3>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value4>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value5>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value6>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value7>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value8>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value9>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value10>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value11>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB

<Value12>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value13>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value14>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value15>	numeric Range: -1 dB to 3 dB *RST: 1 dB Default unit: dB
<Value16>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value17>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value18>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value19>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB
<Value20>	numeric Range: -1 dB to 3 dB *RST: -1 dB Default unit: dB

Example: See [Modifying parameters for an established connection](#)

Firmware/software: PCC V2.0.20, SCC V3.5.30, SETA/B V3.7.30, SETC V3.7.70

Manual operation: See ["User-Defined Pattern"](#) on page 169

2.6.12 Physical cell setup

The commands in this section configure physical layer attributes of the simulated cell.

• General settings	680
• SRS settings	682
• TDD settings	687
• PRACH settings	688
• RAR settings	692
• CSAT/SCell muting settings	692

2.6.12.1 General settings

This section describes the following cell setup commands.

CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL	680
CONFigure:LTE:SIGN<i>:CELL:BANDwidth:SCC<c>:DL	680
CONFigure:LTE:SIGN<i>:CELL[:PCC]:PCID	681
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:PCID	681
CONFigure:LTE:SIGN<i>:CELL:CPRfix	681
CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULSupport:QAM<m>:ENABle	681
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULSupport:QAM<m>:ENABle	681

CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL <Bandwidth>

CONFigure:LTE:SIGN<i>:CELL:BANDwidth:SCC<c>:DL <Bandwidth>

Defines the DL cell bandwidth. The PCC DL bandwidth is also used for the UL.

Suffix:

<c> 1..7

Parameters:

<Bandwidth> B014 | B030 | B050 | B100 | B150 | B200

B014: 1.4 MHz

B030: 3 MHz

B050: 5 MHz

B100: 10 MHz

B150: 15 MHz

B200: 20 MHz

*RST: B100

Example: See [Configuring physical cell setup](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See "[DL / UL Cell Bandwidth](#)" on page 170

CONFigure:LTE:SIGN<i>:CELL[:PCC]:PCID <ID>
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:PCID <ID>

Defines the physical cell ID used for generation of the DL physical synchronization signals. If you use carrier aggregation, configure different values for the component carriers.

Suffix:

<c> 1..7

Parameters:

<ID> decimal

Range: 0 to 503

*RST: 0 for PCC / 1 for SCC1 / 2 for SCC2

Example: See [Configuring physical cell setup](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["Physical Cell ID"](#) on page 171

CONFigure:LTE:SIGN<i>:CELL:CPRefix <CyclicPrefix>

Defines whether a normal or extended cyclic prefix (CP) is used.

Parameters:

<CyclicPrefix> NORMAl | EXTended

*RST: NORM

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.0.50

Options: R&S CMW-KS510 for EXTended

Manual operation: See ["Cyclic Prefix"](#) on page 171

CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULSupport:QAM<m>:ENABle <Enable>
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULSupport:QAM<m>:ENABle <Enable>

Selects whether 64-QAM and 256-QAM are allowed in the uplink or not.

Suffix:

<m> 64, 256
Both values apply to 64-QAM and 256-QAM. They have the same effect.

<c> 1..7

Parameters:

<Enable> OFF | ON

ON: 64-QAM and 256-QAM allowed

OFF: 64-QAM and 256-QAM not allowed

*RST: OFF

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.7.10, V3.7.60 added <m>=256

Options: R&S CMW-KS504/-KS554 for FDD/TDD

Manual operation: See ["Enable 64/256-QAM UL Support"](#) on page 171

2.6.12.2 SRS settings

The following commands are related to sounding RS (SRS).

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:ENABLE	682
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:ENABLE	682
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:MCENable	683
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:MCENable	683
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:BWConfig	683
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:BWConfig	683
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SFCConfig	683
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SFCConfig	683
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:DCCONfig	684
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:DCCONfig	684
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:DBANdwidth	684
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:DBANdwidth	684
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:HBANdwidth	685
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:HBANdwidth	685
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SCINdex:FDD	685
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SCINdex:FDD	685
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SCINdex:TDD	686
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SCINdex:TDD	686
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:POFFset	686
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:POFFset	686

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:ENABLE <Enable>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:ENABLE <Enable>

Enables support of SRS.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring physical cell setup](#)

Firmware/software: V2.0.20, SCC command V3.7.10

Manual operation: See ["Sounding RS \(SRS\)"](#) on page 172

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:MCENable <Enable>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:MCENable <Enable>

Enables or disables the manual configuration of signaled values for SRS configuration.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.2.60, SCC command V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Manual Configuration"](#) on page 172

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:BWConfig <BWConfiguration>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:BWConfig <BWConfiguration>

Specifies the "srs-BandwidthConfig" value.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

Suffix:

<c> 1..7

Parameters:

<BWConfiguration> numeric
Range: 0 to 7
*RST: 7

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.40, SCC command V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Bandwidth Config \(Common\)"](#) on page 173

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SFCConfig <Subframe>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SFCConfig <Subframe>

Specifies the "srs-SubframeConfig" value.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

Suffix:

<c> 1..7

Parameters:

<Subframe> decimal
 Range: 0 to 15
 *RST: 3

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.2.60, SCC command V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Subframe Configuration"](#) on page 173

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:DConfig <DConfiguration>
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:DConfig <DConfiguration>

Selects whether the UE-specific SRS parameters are signaled to the UE or not.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

Suffix:

<c> 1..7

Parameters:

<DConfiguration> OFF | ON
OFF: send only cell-specific SRS parameters
ON: send also UE-specific SRS parameters
 *RST: ON

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Dedicated Configuration"](#) on page 173

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:DBANdwidth <DedicatedBW>
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:DBANdwidth <DedicatedBW>

Specifies the "srs-Bandwidth" value.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

Suffix:

<c> 1..7

Parameters:

<DedicatedBW> numeric
 Range: 0 to 3
 *RST: 3

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.40, SCC command V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Bandwidth \(Dedicated\)"](#) on page 173

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:HBANDwidth <HoppingBW>
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:HBANDwidth <HoppingBW>

Specifies the "srs-HoppingBandwidth" value.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

Suffix:

<c> 1..7

Parameters:

<HoppingBW> numeric
 Range: 0 to 3
 *RST: 3

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.40, SCC command V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Hopping Bandwidth"](#) on page 173

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SCIndex:FDD <Index>
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SCIndex:FDD <Index>

Specifies the "srs-ConfigIndex" value for FDD.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

Suffix:

<c> 1..7

Parameters:

<Index> numeric
 Range: 0 to 636
 *RST: 7

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.2.70, SCC command V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Configuration Index"](#) on page 174

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:SCIndex:TDD <Index>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:SCIndex:TDD <Index>

Specifies the "srs-ConfigIndex" value for TDD.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

Suffix:

<c> 1..7

Parameters:

<Index> numeric
 Range: 0 to 644
 *RST: 0

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.2.70, SCC command V3.7.10

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["Configuration Index"](#) on page 174

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:POFFset <Offset>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SRS:POFFset <Offset>

Specifies the "pSRS-Offset" value.

The setting is only used if manual configuration is enabled, see [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:MCENable](#).

A query returns <Offset>, <Value>.

Suffix:

<c> 1..7

Parameters:

<Offset> decimal
 "pSRS-Offset" value
 Range: 0 to 15
 *RST: 3

Return values:

<Value> float
 Offset in dB, corresponding to the configured "pSRS-Offset" value
 Range: -10.5 dB to 12 dB
 *RST: -6 dB
 Default unit: dB

Example: See [Configuring physical cell setup](#)

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See ["pSRS-Offset"](#) on page 174

2.6.12.3 TDD settings

The following commands configure TDD-specific settings, not relevant for FDD.

CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific	687
CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULDL	687
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULDL	687
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SSUBframe	688
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SSUBframe	688

CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific <UseSpecific>

Enables or disables the carrier-specific configuration of the UL/DL configuration and of the special subframe configuration.

- Enabled: Configuration per carrier via
[CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:ULDL](#)
[CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULDL](#)
[CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SSUBframe](#)
[CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SSUBframe](#)
- Disabled: Global configuration via
[CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:ULDL](#)
[CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SSUBframe](#)

Parameters:

<UseSpecific> OFF | ON
 *RST: OFF

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.20

Options: R&S CMW-KS550 and R&S CMW-KS512

Manual operation: See "[Use Carrier Specific](#)" on page 174

CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULDL <UplinkDownlink>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULDL <UplinkDownlink>

Selects a UL-DL configuration, defining the combination of UL, DL and special subframes within a radio frame. This command is only relevant for duplex mode TDD.

See also [CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific](#) on page 687.

Suffix:

<c> 1..7

Parameters:

<UplinkDownlink> integer
 Range: 0 to 6
 *RST: 1

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.0.10, V3.0.50 value 0, 2, 3, 4, 6
V3.5.20 SCC command

Options: R&S CMW-KS550 and R&S CMW-KS510
R&S CMW-KS512 for carrier-specific configuration

Manual operation: See ["Uplink Downlink Configuration"](#) on page 175

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SSUBframe <SpecialSubframe>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SSUBframe <SpecialSubframe>

Selects a special subframe configuration, defining the inner structure of special subframes. This parameter is only relevant for TDD signals.

The special subframe configurations are defined in 3GPP TS 36.211, section 4, "Frame Structure".

See also [CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific](#) on page 687.

Suffix:

<c> 1..7

Parameters:

<SpecialSubframe> integer

Value 8 and 9 can only be used with the normal cyclic prefix.

Range: 0 to 9

*RST: 7

Example: See [Configuring physical cell setup](#)

Firmware/software: V2.1.20, V3.5.10 value 9, V3.5.20 SCC command

Options: R&S CMW-KS550
R&S CMW-KS512 for value 7 plus extended cyclic prefix / for value 9 / for carrier-specific configuration.

Manual operation: See ["Special Subframe"](#) on page 175

2.6.12.4 PRACH settings

The following commands configure settings related to the random access procedure.

CONFigure:LTE:SIGN<i>:CELL:PRACH:NRPreambles	688
CONFigure:LTE:SIGN<i>:CELL:PRACH:NIPRach	689
CONFigure:LTE:SIGN<i>:CELL:PRACH:PRSTep	689
CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:FDD	689
CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:TDD	690
CONFigure:LTE:SIGN<i>:CELL:PRACH:PFOffset	691
CONFigure:LTE:SIGN<i>:CELL:PRACH:LRIndex	691
CONFigure:LTE:SIGN<i>:CELL:PRACH:ZCZConfig	691

CONFigure:LTE:SIGN<i>:CELL:PRACH:NRPreambles <Enable>

Selects whether the application ignores received preambles or not.

Parameters:

<Enable> OFF | ON | NIPReambles
OFF: respond to received preambles
ON: ignore received preambles
NIPReambles: ignore a configured number of preambles, then respond to subsequent preambles - for configuration see [CONFigure:LTE:SIGN<i>:CELL:PRACH:NIPRach](#), only allowed for power ramping step size 0 dB
 *RST: OFF

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.2.20

Manual operation: See ["No Response to Preambles, # Ignored Preambles"](#) on page 176

CONFigure:LTE:SIGN<i>:CELL:PRACH:NIPRach <Count>

Configures the number of preambles to be ignored if the mode NIPReambles is active, see [CONFigure:LTE:SIGN<i>:CELL:PRACH:NRPPreambles](#).

Parameters:

<Count> numeric
 Range: 1 to 250
 *RST: 1

Firmware/software: V3.2.20

Manual operation: See ["No Response to Preambles, # Ignored Preambles"](#) on page 176

CONFigure:LTE:SIGN<i>:CELL:PRACH:PRSTep <Step>

Specifies the transmit power difference between two consecutive preambles.

Parameters:

<Step> ZERO | P2DB | P4DB | P6DB
 0 dB, 2 dB, 4 dB, 6 dB
 *RST: P2DB

Example: See [Configuring physical cell setup](#)

Firmware/software: V2.0.20

Manual operation: See ["Power Ramping Step"](#) on page 176

CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:FDD <PRACHconfIndex>

Selects the PRACH configuration index for FDD.

Parameters:

<PRACHconfIndex> numeric
 Range: 0 to 63
 *RST: 12

Example: See [Configuring physical cell setup](#)

Firmware/software: V2.1.20

Options: R&S CMW-KS500

Manual operation: See ["Configuration Index"](#) on page 176

CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:TDD <PRACHconfIndex>

Selects the PRACH configuration index for TDD.

Parameters:

<PRACHconfIndex> numeric
 Range: depends on UL-DL configuration, see tables below
 *RST: 12

Example: See [Configuring physical cell setup](#)

Firmware/software: V2.1.20

Options: R&S CMW-KS550

Manual operation: See ["Configuration Index"](#) on page 176

The general range for the TDD PRACH configuration index equals 0 to 57. Within this range, 3GPP defines forbidden values depending on the UL-DL configuration, see 3GPP TS 36.211, table 5.7.1-4. The following tables list the allowed values.

Table 2-67: Allowed values for cell bandwidth > 1.4 MHz

UL-DL configuration	Allowed PRACH configuration indices
0	0-10, 12-18, 20-57
1	0-7, 9-12, 15-39, 48-57
2	0-4, 6, 9, 10, 12, 15, 16, 18, 48-57
3	0-9, 12-18, 20, 21, 23, 25-29, 30, 31, 33, 35-39, 40, 41, 43, 45-49, 51, 53-57
4	0-4, 6, 9, 10, 12, 15, 16, 18, 20, 21, 23, 25-29, 30, 31, 33, 35-39, 48, 49, 51, 53-57
5	0, 1, 3, 6, 9, 12, 15, 18, 48, 49, 51, 53-57
6	0-15, 18-41, 43, 45-57

Table 2-68: Allowed values for cell bandwidth = 1.4 MHz

UL-DL configuration	Allowed PRACH configuration indices
0	0-10, 12-18, 20-26, 30-36, 40-45, 48-53
1	0-7, 9-12, 20-25, 30-35, 48-53
2	0-4, 6, 48-53

UL-DL configuration	Allowed PRACH configuration indices
3	0-9, 20, 21, 23, 30, 31, 33, 40, 41, 43, 48, 49, 51
4	0-4, 6, 20, 21, 23, 30, 31, 33, 48, 49, 51
5	0, 1, 3, 48, 49, 51
6	0-15, 20-25, 30-35, 40, 41, 43, 48-53

CONFigure:LTE:SIGN<i>:CELL:PRACH:PFOffset <PRACHfreqOffset>

Specifies the PRACH frequency offset.

Parameters:

<PRACHfreqOffset> numeric
 Range: 0 to <total RB - 6> depending on cell bandwidth,
 see table below
 *RST: 0

Example: See [Configuring physical cell setup](#)

Firmware/software: V2.0.10

Manual operation: See "[Frequency Offset](#)" on page 176

Table 2-69: Maximum input value depending on cell bandwidth

Cell bandwidth / MHz	1.4	3	5	10	15	20
<total RB - 6>	0	9	19	44	69	94

CONFigure:LTE:SIGN<i>:CELL:PRACH:LRSIndex <LogRootSeqIndex>

Specifies the logical root sequence index to be used by the UE for generation of the preamble sequence.

Parameters:

<LogRootSeqIndex> numeric
 Range: 0 to 837
 *RST: 123

Example: See [Configuring physical cell setup](#)

Firmware/software: V2.0.10

Manual operation: See "[Logical Root Sequ.Idx](#)" on page 177

CONFigure:LTE:SIGN<i>:CELL:PRACH:ZCZConfig <ZeroCorrZoneCon>

Specifies the zero correlation zone config.

Parameters:

<ZeroCorrZoneCon> numeric
 Range: 0 to 15
 *RST: 9

Example: See [Configuring physical cell setup](#)
Firmware/software: V2.0.10
Manual operation: See ["Zero Corr. Zone Conf."](#) on page 177

2.6.12.5 RAR settings

The following commands configure settings related to the random access response message.

[CONFigure:LTE:SIGN<i>:CELL:RAR:CMCS:ENABLE](#).....692
[CONFigure:LTE:SIGN<i>:CELL:RAR:CMCS](#).....692

CONFigure:LTE:SIGN<i>:CELL:RAR:CMCS:ENABLE <Enable>

Enables the custom MCS index definition for RAR messages.

Parameters:

<Enable> OFF | ON
OFF: MCS index selected automatically
ON: MCS index set to configured custom value
 *RST: OFF

Example: See [Configuring physical cell setup](#)
Firmware/software: V3.7.70
Options: R&S CMW-KS510
Manual operation: See ["Custom MCS"](#) on page 177

CONFigure:LTE:SIGN<i>:CELL:RAR:CMCS <CustomMCS>

Configures a custom MCS index for RAR messages.

Parameters:

<CustomMCS> numeric
 Range: 0 to 13
 *RST: 9

Example: See [Configuring physical cell setup](#)
Firmware/software: V3.7.70
Options: R&S CMW-KS510
Manual operation: See ["Custom MCS"](#) on page 177

2.6.12.6 CSAT/SCell muting settings

The following commands configure settings for CSAT and SCell muting.

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CSAT:ENABLe.....	693
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CSAT:DMTCperiod.....	693
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:ONSDuration.....	693
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:OFFSduration.....	694
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:PMAC.....	694

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CSAT:ENABLe <Enable>

Enables CSAT, including LDS transmission and SCell muting.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS525

Manual operation: See ["Enable"](#) on page 178

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CSAT:DMTCperiod <Period>

Configures the LDS periodicity.

Suffix:

<c> 1..7

Parameters:

<Period> M40 | M80 | M160
40 ms, 80 ms, 160 ms
*RST: M40

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS525

Manual operation: See ["DMTC Period"](#) on page 178

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:ONSDuration <Duration>

Configures the ON state duration for SCell muting (SCC DL muting).

Suffix:

<c> 1..7

Parameters:

<Duration> numeric
 Range: 1 ms to 1000 ms
 *RST: 10 ms
 Default unit: ms

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS525

Manual operation: See ["ON State Duration / OFF State Duration"](#) on page 178

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:OFFSduration <Duration>

Configures the OFF state duration for SCell muting (SCC DL muting).

Suffix:

<c> 1..7

Parameters:

<Duration> numeric
 Range: 1 ms to 1000 ms
 *RST: 10 ms
 Default unit: ms

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS525

Manual operation: See ["ON State Duration / OFF State Duration"](#) on page 178

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SCMuting:PMAC <Activation>

Enables periodic MAC activation.

Suffix:

<c> 1..7

Parameters:

<Activation> OFF | ON
 *RST: OFF

Example: See [Configuring physical cell setup](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS525

Manual operation: See ["Periodic MAC Activation"](#) on page 179

2.6.13 Network settings

The commands in this section configure parameters of the simulated radio network.

• Neighbor cell settings	695
• Cell reselection settings	704
• Identity settings	706
• Security settings	707
• UE identity	710
• Timer and constants	710
• Time	712
• NAS signaling settings	716
• Synchronization settings	721

2.6.13.1 Neighbor cell settings

The following commands define neighbor cell information to be broadcasted to the UE.

CONFigure:LTE:SIGN<i>:NCELL:LTE:THResholds:LOW	695
CONFigure:LTE:SIGN<i>:NCELL:GSM:THResholds:LOW	695
CONFigure:LTE:SIGN<i>:NCELL:WCDma:THResholds:LOW	696
CONFigure:LTE:SIGN<i>:NCELL:CDMA:THResholds:LOW	696
CONFigure:LTE:SIGN<i>:NCELL:EVDO:THResholds:LOW	696
CONFigure:LTE:SIGN<i>:NCELL:TDSCdma:THResholds:LOW	697
CONFigure:LTE:SIGN<i>:NCELL:ALL:THResholds:LOW	697
CONFigure:LTE:SIGN<i>:NCELL:LTE:CELL<n>	697
CONFigure:LTE:SIGN<i>:NCELL:GSM:CELL<n>	698
CONFigure:LTE:SIGN<i>:NCELL:WCDma:CELL<n>	699
CONFigure:LTE:SIGN<i>:NCELL:CDMA:CELL<n>	701
CONFigure:LTE:SIGN<i>:NCELL:EVDO:CELL<n>	701
CONFigure:LTE:SIGN<i>:NCELL:TDSCdma:CELL<n>	703

CONFigure:LTE:SIGN<i>:NCELL:LTE:THResholds:LOW <Low>

Configures the reselection threshold value "threshX-Low" for LTE neighbor cells.

Parameters:

<Low>	numeric
Range:	0 to 31
*RST:	5

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V2.1.30

Manual operation: See ["Threshold"](#) on page 181

CONFigure:LTE:SIGN<i>:NCELL:GSM:THResholds:LOW <Low>

Configures the reselection threshold value "threshX-Low" for GSM neighbor cells.

Parameters:

<Low> numeric
Range: 0 to 31
*RST: 0

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V2.1.30

Manual operation: See ["Threshold"](#) on page 181

CONFigure:LTE:SIGN<i>:NCELL:WCDMA:THResholds:LOW <Low>

Configures the reselection threshold value "threshX-Low" for WCDMA neighbor cells.

Parameters:

<Low> numeric
Range: 0 to 31
*RST: 5

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V2.1.30

Manual operation: See ["Threshold"](#) on page 181

CONFigure:LTE:SIGN<i>:NCELL:CDMA:THResholds:LOW <Low>

Configures the reselection threshold value "threshX-Low" for CDMA2000 neighbor cells.

Parameters:

<Low> numeric
Range: 0 to 63
*RST: 0

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V2.1.30

Manual operation: See ["Threshold"](#) on page 181

CONFigure:LTE:SIGN<i>:NCELL:EVDO:THResholds:LOW <Low>

Configures the reselection threshold value "threshX-Low" for 1xEV-DO neighbor cells.

Parameters:

<Low> numeric
Range: 0 to 63
*RST: 0

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V2.1.30

Manual operation: See ["Threshold"](#) on page 181

CONFigure:LTE:SIGN<i>:NCELL:TDSCdma:THResholds:LOW <Low>

Configures the reselection threshold value "threshX-Low" for TD-SCDMA neighbor cells.

Parameters:

<Low> numeric
 Range: 0 to 31
 *RST: 5

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.2.20

Manual operation: See ["Threshold"](#) on page 181

CONFigure:LTE:SIGN<i>:NCELL:ALL:THResholds:LOW <Valid>, <Low>

Configures a common reselection threshold value "threshX-Low" applicable to all technologies.

Alternatively to a common threshold you can also use individual thresholds. They are defined per technology via the commands

CONFigure:LTE:SIGN<i>:NCELL:<Technology>:THResholds:LOW. The parameter <Valid> selects whether common or individual thresholds are used.

Parameters:

<Valid> OFF | ON
 OFF: use individual thresholds defined by separate commands
 ON: use common threshold defined by this command
 *RST: OFF

 <Low> numeric
 Range: 0 to 31
 *RST: 5

Firmware/software: V2.1.30

Manual operation: See ["Threshold"](#) on page 181

CONFigure:LTE:SIGN<i>:NCELL:LTE:CELL<n> <Enable>, <Band>, <Channel>, <CellID>, <QOffset>[, <Measurement>]

Configures the entry number <n> of the neighbor cell list for LTE.

For channel number ranges depending on operating bands see [Section 2.2.23, "Operating bands"](#), on page 91.

Note that only 5 entries with different channel numbers can be active at a time. Entries with the same channel number must have different cell IDs.

Suffix:	
<n>	1..16
Parameters:	
<Enable>	OFF ON Enables or disables the entry *RST: OFF
<Band>	OB1 ... OB46 OB48 ... OB53 OB65 ... OB76 OB85 OB87 OB88 OB106 OB250 OB252 OB255 *RST: OB1
<Channel>	integer Downlink channel number Range: depends on operating band *RST: 300
<CellID>	integer Physical layer cell ID Range: 0 to 503 *RST: 0
<QOffset>	N24 N22 N20 N18 N16 N14 N12 N10 N8 N6 N5 N4 N3 N2 N1 ZERO P1 P2 P3 P4 P5 P6 P8 P10 P12 P14 P16 P18 P20 P22 P24 Corresponds to value "q-OffsetCell" in 3GPP TS 36.331 N24 to N1: -24 dB to -1 dB ZERO: 0 dB P1 to P24: 1 dB to 24 dB *RST: ZERO
<Measurement>	OFF ON Disables / enables neighbor cell measurements for the entry ON is only allowed if also <Enable> = ON *RST: OFF
Example:	See Configuring neighbor cells and reselection
Firmware/software:	V3.0.10, some bands added in later versions V3.0.50: added <Measurement>
Options:	R&S CMW-KS510 for neighbor cell measurements
Manual operation:	See " LTE " on page 181

CONFigure: LTE:SIGN<i>:NCELL:GSM:CELL<n> <Enable>, <Band>, <Channel>[, <Measurement>]

Configures the entry number <n> of the neighbor cell list for GSM.

Suffix:
<n> 1..4

Parameters:

<Enable>	OFF ON Enables or disables the entry *RST: OFF
<Band>	G085 G09 G18 G19 GSM 850, GSM 900, GSM 1800, GSM 1900 *RST: G09
<Channel>	integer Channel number used for the broadcast control channel (BCCH) Range: 0 to 1023, depending on GSM band, see table below *RST: 20
<Measurement>	OFF ON Disables / enables neighbor cell measurements for the entry ON is only allowed if also <Enable> = ON *RST: OFF

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.0.50

Options: R&S CMW-KS510 for neighbor cell measurements

Manual operation: See "GSM" on page 181

Table 2-70: Channel number range depending on GSM band

Band	Channel number
G085	128 to 251
G09	0 to 124, 940 to 1023
G18	512 to 885
G19	512 to 810

CONFigure: LTE:SIGN<i>:NCELL:WCDMa:CELL<n> <Enable>, <Band>, <Channel>, <ScramblingCode>[, <Measurement>]

Configures the entry number <n> of the neighbor cell list for WCDMA.

Suffix:

<n> 1..4

Parameters:

<Enable>	OFF ON Enables or disables the entry *RST: OFF
<Band>	OB1 OB2 OB3 OB4 OB5 OB6 OB7 OB8 OB9 OB10 OB11 OB12 OB13 OB14 OB19 OB20 OB21 OB22 OB25 OBS1 OBS2 OBS3 OBL1 OB26 OB1, ..., OB14: band I to XIV OB19, ..., OB22: band XIX to XXII OB25, OB26: band XXV, XXVI OBS1: band S OBS2: band S 170 MHz OBS3: band S 190 MHz OBL1: band L *RST: OB1
<Channel>	integer Downlink channel number Range: 412 to 11000, depending on operating band, see table below *RST: 10563
<ScramblingCode>	hex Primary scrambling code Range: #H0 to #H1FF *RST: #H0
<Measurement>	OFF ON Disables / enables neighbor cell measurements for the entry ON is only allowed if also <Enable> = ON *RST: OFF

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.0.50, V3.5.40: OB22/OB25, V3.7.10: OB26

Options: R&S CMW-KS510 for neighbor cell measurements

Manual operation: See ["WCDMA FDD"](#) on page 181

Table 2-71: Channel number range depending on operating band

Operating band	Channel number
OB1	10562 to 10838
OB2	412 to 687 (step 25), 9662 to 9938
OB3	1162 to 1513
OB4	1537 to 1738, 1887 to 2087 (step 25)
OB5	1007, 1012, 1032, 1037, 1062, 1087, 4357 to 4458
OB6	1037, 1062, 4387 to 4413

Operating band	Channel number
OB7	2237 to 2563, 2587 to 2912 (step 25)
OB8	2937 to 3088
OB9	9237 to 9387
OB10	3112 to 3388, 3412 to 3687 (step 25)
OB11	3712 to 3787
OB12	3842 to 3903, 3932, 3957, 3962, 3987, 3992
OB13	4017 to 4043, 4067, 4092
OB14	4117 to 4143, 4167, 4192
OB19	712 to 763, 787, 812, 837
OB20	4512 to 4638
OB21	862 to 912
OB22	4662 to 5038
OB25	5112 to 5413, 6292 to 6592 (step 25)
OB26	5762 to 5913, 5937, 5962, 5987, 5992, 6012, 6017, 6037, 6042, 6062, 6067, 6087
OBS1	10912 to 10988
OBS2	10900 to 10950
OBS3	10950 to 11000
OBL1	7637 to 7783, 7788 to 7933

CONFIGure:LTE:SIGN<i>:NCELL:CDMA:CELL<n> <Enable>, <BandClass>, <Channel>, <CellID>[, <Measurement>]

CONFIGure:LTE:SIGN<i>:NCELL:EVDO:CELL<n> <Enable>, <BandClass>, <Channel>, <CellID>[, <Measurement>]

Configures the entry number <n> of the neighbor cell list for CDMA2000 (1xRTT) or 1xEV-DO (HRPD).

Suffix:

<n> 1..4

Parameters:

<Enable> OFF | ON
Enables or disables the entry

*RST: OFF

<BandClass> USC | KCEL | NAPC | TACS | JTAC | KPCS | N45T | IM2K | NA7C | B18M | NA8S | PA4M | PA8M | IEXT | USPC | AWS | U25B | U25F | NA9C | PS7C | LO7C

USC: BC 0, US cellular

KCEL: BC 0, Korean cellular

NAPC: BC 1, North American PCS

TACS: BC 2, TACS band
JTAC: BC 3, JTACS band
KPCS: BC 4, Korean PCS
N45T: BC 5, NMT-450
IM2K: BC 6, IMT-2000
NA7C: BC 7, upper 700 MHz
B18M: BC 8, 1800-MHz band
NA9C: BC 9, North American 900 MHz
NA8S: BC 10, secondary 800 MHz
PA4M: BC 11, European 400-MHz PAMR
PA8M: BC 12, 800-MHz PAMR
IEXT: BC 13, IMT-2000 2.5-GHz extension
USPC: BC 14, US PCS 1900 MHz
AWS: BC 15, AWS band
U25B: BC 16, US 2.5-GHz band
U25F: BC 17, US 2.5 GHz forward
PS7C: BC 18, public safety band 700 MHz
LO7C: BC 19, lower 700 MHz
***RST:** USC

<Channel> integer
 Channel number
 Range: 0 to 2108, depending on band class, see table below
***RST:** 283

<CellID> integer
 Physical cell ID
 Range: 0 to 511
***RST:** 0

<Measurement> OFF | ON
 Disables / enables neighbor cell measurements for the entry
 ON is only allowed if also <Enable> = ON
***RST:** OFF

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.0.50

Options: R&S CMW-KS510 for neighbor cell measurements

Manual operation: See "[CDMA2000, 1xEV-DO](#)" on page 182

Table 2-72: Channel number range depending on band class

Band class	Channel number
USC, KCEL	1 to 799, 991 to 1323
NAPC, IM2K	0 to 1199
TACS	0 to 1000, 1329 to 2108
JTAC	1 to 799, 801 to 1039, 1041 to 1199, 1201 to 1600

Band class	Channel number
KPCS	0 to 599
N45T	1 to 400, 472 to 871, 1039 to 1473, 1536 to 1715, 1792 to 2016
NA7C, PS7C	0 to 240
B18M	0 to 1499
NA9C	0 to 699
NA8S	0 to 919
PA4M	1 to 400, 472 to 871, 1536 to 1715
PA8M	0 to 239
IEXT	0 to 1399
USPC	0 to 1299
AWS	0 to 899
U25B, U25F	140 to 1459
LO7C	0 to 360

CONFigure:LTE:SIGN<i>:NCELL:TDSCdma:CELL<n> <Enable>, <Band>, <Channel>, <ScramblingCode>[, <Measurement>]

Configures the entry number <n> of the neighbor cell list for TD-SCDMA.

Suffix:

<n> 1..4

Parameters:

<Enable> OFF | ON
Enables or disables the entry
*RST: OFF

<Band> OB1 | OB2 | OB3
OB1: Band 1 (F), channel 9400 to 9600
OB2: Band 2 (A), channel 10050 to 10125
OB3: Band 3 (E), channel 11500 to 12000
*RST: OB1

<Channel> integer
Channel number
Range: 9400 to 12000, depending on operating band
*RST: 9600

<ScramblingCode> hex
Cell parameter ID
Range: #H0 to #H7F
*RST: #H0

<Measurement>	OFF ON Disables / enables neighbor cell measurements for the entry ON is only allowed if also <Enable> = ON *RST: OFF
Example:	See Configuring neighbor cells and reselection
Firmware/software:	V3.2.20
Options:	R&S CMW-KS510 for neighbor cell measurements
Manual operation:	See " TD-SCDMA " on page 182

2.6.13.2 Cell reselection settings

The following commands define cell reselection information to be broadcasted to the UE.

CONFigure:LTE:SIGN<i>:CELL:RESelection:SEARch:INTRasearch	704
CONFigure:LTE:SIGN<i>:CELL:RESelection:SEARch:NINTRasearch	704
CONFigure:LTE:SIGN<i>:CELL:RESelection:TSLow	705
CONFigure:LTE:SIGN<i>:CELL:RESelection:QUALity:RXLevmin	705

CONFigure:LTE:SIGN<i>:CELL:RESelection:SEARch:INTRasearch <Sintrasearch>

Defines the threshold $S_{\text{IntraSearch}}$. The value divided by 2 is broadcasted to the UE in SIB3.

Parameters:

<Sintrasearch>	numeric ON OFF Range: 0 dB to 62 dB Increment: 2 dB *RST: 32 dB Default unit: dB ON OFF enables or disables transmission of the information element.
----------------	---

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.2.70

Options: R&S CMW-KS510

Manual operation: See "[S IntraSearch](#)" on page 182

CONFigure:LTE:SIGN<i>:CELL:RESelection:SEARch:NINTRasearch <Snonintrasearch>

Defines the threshold $S_{\text{nonIntraSearch}}$. The value divided by 2 is broadcasted to the UE in SIB3.

Parameters:

<Snonintrasearch> numeric | ON | OFF
 Range: 0 dB to 62 dB
 Increment: 2 dB
 *RST: 32 dB
 Default unit: dB
 ON | OFF enables or disables transmission of the information element.

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.2.70

Options: R&S CMW-KS510

Manual operation: See "[S NonIntraSearch](#)" on page 183

CONFigure:LTE:SIGN<i>:CELL:RESelection:TSLow <Value>

Defines $\text{Thresh}_{\text{Serving,Low}}$. The value divided by 2 is broadcasted to the UE in SIB3.

Parameters:

<Value> numeric
 Range: 0 dB to 62 dB
 Increment: 2 dB
 *RST: 16 dB
 Default unit: dB

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.2.70

Options: R&S CMW-KS510

Manual operation: See "[ThreshServingLow](#)" on page 183

CONFigure:LTE:SIGN<i>:CELL:RESelection:QUALity:RXLevmin <Qrxlevmin>

Defines the level Q_{rxlevmin} . The value divided by 2 is broadcasted to the UE in SIB1.

Parameters:

<Qrxlevmin> numeric
 Range: -140 dBm to -44 dBm
 Increment: 2 dB
 *RST: -132 dBm
 Default unit: dBm

Example: See [Configuring neighbor cells and reselection](#)

Firmware/software: V3.0.10

Options: R&S CMW-KS510

Manual operation: See "[Q rxlevmin](#)" on page 183

2.6.13.3 Identity settings

The following commands configure identities of the simulated radio network.

CONFigure:LTE:SIGN<i>:CELL:MCC	706
CONFigure:LTE:SIGN<i>:CELL:MNC	706
CONFigure:LTE:SIGN<i>:CELL:MNC:DIGits	706
CONFigure:LTE:SIGN<i>:CELL:TAC	707
CONFigure:LTE:SIGN<i>:CELL[:PCC]:CID:EUTRan	707
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CID:EUTRan	707

CONFigure:LTE:SIGN<i>:CELL:MCC <MCC>

Specifies the three-digit mobile country code (MCC). You can omit leading zeros.

Parameters:

<MCC>	decimal
Range:	0 to 999
*RST:	1

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["MCC"](#) on page 184

CONFigure:LTE:SIGN<i>:CELL:MNC <MNC>

Specifies the mobile network code (MNC). You can omit leading zeros.

A two or three-digit MNC can be set, see [CONFigure:LTE:SIGN<i>:CELL:MNC:DIGits](#).

Parameters:

<MNC>	decimal
Range:	0 to 99 or 999
*RST:	1

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["MNC"](#) on page 184

CONFigure:LTE:SIGN<i>:CELL:MNC:DIGits <NoDigits>

Specifies the number of digits of the mobile network code (MNC).

For setting the MNC, see [CONFigure:LTE:SIGN<i>:CELL:MNC](#).

Parameters:

<NoDigits>	TWO THRee
*RST:	TWO

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["MNC"](#) on page 184

CONFigure:LTE:SIGN<i>:CELL:TAC <TAC>

Specifies the tracking area code.

Parameters:

<TAC>	decimal
Range:	0 to 65535
*RST:	1

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["TAC"](#) on page 184

CONFigure:LTE:SIGN<i>:CELL[:PCC]:CID:EUTRan <CID>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:CID:EUTRan <CID>

Specifies the E-UTRAN cell identifier (28-digit binary number). If you use carrier aggregation, configure different values for the component carriers.

Suffix:

<c>	1..7
-----	------

Parameters:

<CID>	binary
Range:	#B0 to #B11111111111111111111111111111111

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20, SCC command V3.2.50

Manual operation: See ["E-UTRAN Cell Identifier"](#) on page 184

2.6.13.4 Security settings

The following commands configure parameters related to the authentication procedure and other security procedures.

CONFigure:LTE:SIGN<i>:CELL:SECurity:AUTHenticat	708
CONFigure:LTE:SIGN<i>:CELL:SECurity:NAS	708
CONFigure:LTE:SIGN<i>:CELL:SECurity:AS	708
CONFigure:LTE:SIGN<i>:CELL:SECurity:IALgorithm	708
CONFigure:LTE:SIGN<i>:CELL:SECurity:NCALgorithm	709
CONFigure:LTE:SIGN<i>:CELL:SECurity:MILenage	709
CONFigure:LTE:SIGN<i>:CELL:SECurity:OPC	709
CONFigure:LTE:SIGN<i>:CELL:SECurity:SKEY	709
CONFigure:LTE:SIGN<i>:CELL:SECurity:RVALue	710

CONFigure:LTE:SIGN<i>:CELL:SECurity:AUTHenticat <Enable>

Enables or disables authentication, to be performed during the attach procedure.

Parameters:

<Enable> OFF | ON
*RST: ON

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["Authentication"](#) on page 185

CONFigure:LTE:SIGN<i>:CELL:SECurity:NAS <Enable>

Enables or disables the NAS security mode.

Parameters:

<Enable> OFF | ON
*RST: ON

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["NAS Security"](#) on page 185

CONFigure:LTE:SIGN<i>:CELL:SECurity:AS <Enable>

Enables or disables the AS security mode.

Parameters:

<Enable> OFF | ON
*RST: ON

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["AS Security"](#) on page 185

CONFigure:LTE:SIGN<i>:CELL:SECurity:IALGorithm <Algorithm>

Selects an algorithm for integrity protection.

Parameters:

<Algorithm> NULL | S3G
NULL: no integrity protection
S3G: SNOW3G (EIA1) algorithm
*RST: S3G

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.50

Manual operation: See "Integrity Algorithm" on page 185

CONFigure:LTE:SIGN<i></i>:CELL:SECurity:NCALgorithm <Algorithm>

Selects an algorithm for ciphering of NAS signaling.

Parameters:

<Algorithm> NULL | S3G
NULL: no ciphering
S3G: SNOW3G (EIA1) algorithm
***RST**: NULL

Example: See [Configuring other network settings](#)

Firmware/software: V4.0.20

Manual operation: See ["NAS Cipherng Algorithm"](#) on page 186

CONFigure:LTE:SIGN<i>:CELL:SECurity:MLEnage <Enable>

Enables or disables using the MILENAGE algorithm set instead of the standard algorithms.

Parameters:

<Enable>	OFF ON
*RST:	OFF

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See "Milénage" on page 186

CONFigure:LTE:SIGN*<i>*:CELL:SECurity:OPC <OPC>

Specifies the key OP_c as 32-digit hexadecimal number.

Parameters:

```
<OPC>          hex
                Range:      #H0 to
                                #HFFFFFFFFFFFFFFFFFFFFFFFFFFFFFFF
                *RST:      #H00000000000000000000000000000000
```

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See "OPc" on page 186

CONFigure:LTE:SIGN<i></i>:CELL:SECurity:SKEY <SecretKey>

Defines the secret key K as 32-digit hexadecimal number. You can omit leading zeros.

K is used for the authentication procedure including a possible integrity check.

Parameters:

<SecretKey> hex
 Range: #H0 to #FFFFFFFFFFFFFFFFFFFFFFFFFFFFF
 *RST: #H000102030405060708090A0B0C0D0E0F

Example: See [Configuring other network settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["Secret Key"](#) on page 186

CONFigure:LTE:SIGN<i>:CELL:SECurity:RVALue <Mode>

Selects whether an even or odd RAND value is used.

Parameters:

<Mode> EVEN | ODD

Example: See [Configuring other network settings](#)

Firmware/software: V2.1.20

Manual operation: See ["RAND Value"](#) on page 186

2.6.13.5 UE identity

The following command configures the default IMSI.

CONFigure:LTE:SIGN<i>:CELL:UEIDentity:IMSI <Value>

Specifies the default IMSI.

Parameters:

<Value> string
 14 to 16 digits
 *RST: '001010123456063'

Firmware/software: V1.0.15.20

Manual operation: See ["Default IMSI"](#) on page 186

2.6.13.6 Timer and constants

The commands in this section configure timers.

[CONFigure:LTE:SIGN<i>:CELL:TOUT:OSYNch](#)..... 711
[CONFigure:LTE:SIGN<i>:CELL:TOUT:T<no>](#)..... 711
[CONFigure:LTE:SIGN<i>:CELL:TOUT:TEXT<no>](#)..... 711

CONFigure:LTE:SIGN<i>:CELL:TOUT:OSYNch <Value>

Specifies the time after which the instrument, having waited for a signal from the connected UE, releases the connection.

Parameters:

<Value> numeric
 Range: 1 s to 50 s
 *RST: 5 s
 Default unit: s

Example: See [Configuring other network settings](#)

Firmware/software: V2.1.30

Manual operation: See ["Out of Sync"](#) on page 187

CONFigure:LTE:SIGN<i>:CELL:TOUT:T<no> <Value>

Configures one of the following timers:

- T3402, attach/TAU reattempts
- T3412, periodic tracking area updates

The information elements support the values 1 to 31 combined with the units 2 seconds, 1 minute and 6 minutes.

This command configures the timer value in seconds. So there are three subranges with different increments.

Suffix:

<no> 3402, 3412

Parameters:

<Value> numeric | ON | OFF
 Range: 2 s to 11160 s
 Increment: 2 s (2 s to 62 s), 60 s (120 s to 1860 s), 360 s (2160 s to 11160 s)
 *RST: 720 s for T3402, 2 s for T3412
 ON | OFF enables or disables the timer.

Example: See [Configuring other network settings](#)

Firmware/software: V3.0.10, V3.5.40: added T3402

Manual operation: See ["T3402"](#) on page 187

CONFigure:LTE:SIGN<i>:CELL:TOUT:TEXT<no> <Value>

Configures an extended value for timer T3412.

The information element supports the values 1 to 31 combined with the units 2 s, 30 s, 1 min, 10 min, 1 h, 10 h and 320 h.

This command configures the timer value in seconds. So there are subranges with different increments.

Suffix:

<no> 3412

Parameters:

<Value> numeric | ON | OFF
 Range: 2 s to 35712000 s
 Increment: see table
 *RST: 2 s
 ON | OFF enables or disables the timer.

Example: See [Configuring other network settings](#)

Firmware/software: V3.7.10

Manual operation: See "[T3412 Extended](#)" on page 187

Range	Increment	Remark
2 s to 62 s	2 s	
90 s to 930 s	30 s	900 s = 15 min
960 s to 1860 s	60 s	1800 s = 30 min
2400 s to 18600 s	600 s	18000 s = 5 h
21600 s to 111600 s	3600 s	111600 s = 1 d, 7 h
144000 s to 1116000 s	36000 s	1116000 s = 12 d, 22 h
1152000 s to 35712000 s	1152000 s	35712000 s = 413 d, 8 h

2.6.13.7 Time

The commands in this section configure and send date and time information to the UE.

CONFigure:LTE:SIGN<i>:CELL:TIME:TSource	712
CONFigure:LTE:SIGN<i>:CELL:TIME:DATE	713
CONFigure:LTE:SIGN<i>:CELL:TIME:TIME	713
CONFigure:LTE:SIGN<i>:CELL:TIME:DSTime	714
CONFigure:LTE:SIGN<i>:CELL:TIME:LTZoffset	714
CONFigure:LTE:SIGN<i>:CELL:TIME:SNOW	715
CONFigure:LTE:SIGN<i>:CELL:TIME:SATTach	715
CONFigure:LTE:SIGN<i>:CELL:TIME:SNName	715

CONFigure:LTE:SIGN<i>:CELL:TIME:TSource <SourceTime>

Selects the date and time source.

The time source DATE is configured via the following commands:

- [CONFigure:LTE:SIGN<i>:CELL:TIME:DATE](#)
- [CONFigure:LTE:SIGN<i>:CELL:TIME:TIME](#)

- [CONFigure:LTE:SIGN<i>:CELL:TIME:DSTime](#)
- [CONFigure:LTE:SIGN<i>:CELL:TIME:LTZoffset](#)

Parameters:

<SourceTime> CMWTime | DATE
CMWTime: Windows date and time
DATE: Date and time specified via remote commands
 *RST: CMWT

Example: See [Sending date and time information to the UE](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510

Manual operation: See ["Time Source"](#) on page 188

CONFigure:LTE:SIGN<i>:CELL:TIME:DATE <Day>, <Month>, <Year>

Specifies the UTC date for the time source DATE (see [CONFigure:LTE:SIGN<i>:CELL:TIME:TSource](#) on page 712).

Parameters:

<Day> integer
 Range: 1 to 31
 *RST: 11

<Month> integer
 Range: 1 to 12
 *RST: 11

<Year> integer
 Range: 2011 to 9999
 *RST: 2011

Example: See [Sending date and time information to the UE](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510

Manual operation: See ["Date / Time \(UTC\)"](#) on page 188

CONFigure:LTE:SIGN<i>:CELL:TIME:TIME <Hour>, <Minute>, <Second>

Specifies the UTC time for the time source DATE (see [CONFigure:LTE:SIGN<i>:CELL:TIME:TSource](#) on page 712).

Parameters:

<Hour> integer
 Range: 0 to 23
 *RST: 11

<Minute> integer
 Range: 0 to 59
 *RST: 11

<Second> integer
 Range: 0 to 59
 *RST: 0

Example: See [Sending date and time information to the UE](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510

Manual operation: See ["Date / Time \(UTC\)"](#) on page 188

CONFigure:LTE:SIGN<i>:CELL:TIME:DSTime <Enable>

Specifies a daylight saving time (DST) offset for the time source `DATE` (see [CONFigure:LTE:SIGN<i>:CELL:TIME:TSource](#) on page 712).

Parameters:

<Enable> P1H | P2H
P1H: +1h offset if DST is ON
P2H: +2h offset if DST is ON
 *RST: OFF (P1H)
 Additional parameters: OFF | ON (disables | enables DST)

Example: See [Sending date and time information to the UE](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510

Manual operation: See ["Daylight Saving Time"](#) on page 188

CONFigure:LTE:SIGN<i>:CELL:TIME:LTZoffset <TimeZoneOffset>

Specifies the time zone offset for the time source `DATE` (see [CONFigure:LTE:SIGN<i>:CELL:TIME:TSource](#) on page 712).

Parameters:

<TimeZoneOffset> numeric
 Range: -19.75 h to 19.75 h
 Increment: 0.25 h
 *RST: 0 h
 Default unit: h

Example: See [Sending date and time information to the UE](#)

Firmware/software: V3.5.30

Options: R&S CMW-KS510

Manual operation: See ["Local Time Zone Offset"](#) on page 189

CONFigure:LTE:SIGN<i>:CELL:TIME:SNOW

Triggers the transfer of the date and time information to the UE.

Example: See [Sending date and time information to the UE](#)

Usage: Event

Firmware/software: V3.0.30

Options: R&S CMW-KS510

Manual operation: See ["Send Time"](#) on page 189

CONFigure:LTE:SIGN<i>:CELL:TIME:SATTach <Enable>

Specifies whether the date and time information is sent to the UE during the attach procedure or not.

Parameters:

<Enable> OFF | ON
ON: send date and time at attach
OFF: do not send date and time at attach
***RST:** OFF

Example: See [Sending date and time information to the UE](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510

Manual operation: See ["Send Time"](#) on page 189

CONFigure:LTE:SIGN<i>:CELL:TIME:SNName <Enable>

Selects whether the network name is sent together with the date and time information.

Parameters:

<Enable> OFF | ON
OFF: Do not send name
ON: Send full and short network name
***RST:** ON

Example: See [Sending date and time information to the UE](#)

Firmware/software: V3.7.60

Options: R&S CMW-KS510

Manual operation: See ["Send Network Name"](#) on page 189

2.6.13.8 NAS signaling settings

The commands in this section configure settings related to NAS signaling messages to be sent to the UE.

CONFigure:LTE:SIGN<i>:CONNection:SDNSpco.....	716
CONFigure:LTE:SIGN<i>:CELL:RCAuse:ATTach.....	716
CONFigure:LTE:SIGN<i>:CELL:RCAuse:TAU.....	716
CONFigure:LTE:SIGN<i>:CELL:ACAuse:ATTach.....	718
CONFigure:LTE:SIGN<i>:CELL:EAResult.....	718
CONFigure:LTE:SIGN<i>:CELL:AUResult.....	719
CONFigure:LTE:SIGN<i>:CELL:NAS:EPSNetwork.....	719
CONFigure:LTE:SIGN<i>:CELL:NAS:IMSVops.....	719
CONFigure:LTE:SIGN<i>:CELL:NAS:EMCBs.....	720
CONFigure:LTE:SIGN<i>:CELL:NAS:EPCLcs.....	720
CONFigure:LTE:SIGN<i>:CELL:NAS:CSLCs.....	720

CONFigure:LTE:SIGN<i>:CONNection:SDNSpco <Enable>

Enables or disables sending of a DNS IP address to the UE.

Parameters:

<Enable> OFF | ON
 *RST: ON

Example: See [Configuring other network settings](#)

Firmware/software: V3.0.10

Manual operation: See "[Send DNS PCO](#)" on page 190

CONFigure:LTE:SIGN<i>:CELL:RCAuse:ATTach <Cause>

CONFigure:LTE:SIGN<i>:CELL:RCAuse:TAU <Cause>

Enables or disables the rejection of attach requests and tracking area update requests and selects the rejection cause to be transmitted.

Parameters:

<Cause> IUE3 | EPS7 | PLMN11 | TANA12 | C13 | C17 | CONG22 | C2 |
 C5 | C6 | C8 | C9 | C10 | C14 | C15 | C16 | C18 | C19 | C20 |
 C21 | C23 | C24 | C25 | C26 | C35 | C39 | C40 | C42 | C95 |
 C96 | C97 | C98 | C99 | C100 | C101 | C111 | ON | OFF
 See table for explanation of values
 *RST: OFF

Example: See [Configuring other network settings](#)

Firmware/software: V3.0.20

V3.2.70: added C13, V3.5.40: added C17

V3.5.50: added C2, C5, C6, C8 to C10, C14 to C16, C18 to C21,
 C23 to C26, C35, C39, C40, C42, C95 to C101, C111

Options: R&S CMW-KS510

Manual operation: See ["TAU Reject Cause"](#) on page 191

<Cause>	Number	Meaning
C2	2	IMSI unknown in HSS
IUE3	3	Illegal UE
C5	5	IMEI not accepted
C6	6	Illegal mobile equipment
EPS7	7	EPS services not allowed
C8	8	EPS and non-EPS services not allowed
C9	9	UE identity cannot be derived by the network
C10	10	Implicitly detached
PLMN11	11	PLMN not allowed
TANA12	12	Tracking area not allowed
C13	13	Roaming not allowed in this tracking area
C14	14	EPS services not allowed in this PLMN
C15	15	No suitable cells in tracking area
C16	16	MSC temporarily not reachable
C17	17	Network failure
C18	18	CS domain not available
C19	19	ESM failure
C20	20	MAC failure
C21	21	Synch failure
CONG22	22	Congestion
C23	23	UE security capabilities mismatch
C24	24	Security mode rejected, unspecified
C25	25	Not authorized for this CSG
C26	26	Non-EPS authentication unacceptable
C35	35	Requested service option not authorized in this PLMN
C39	39	CS service temporarily not available
C40	40	No EPS bearer context activated
C42	42	Severe network failure
C95	95	Semantically incorrect message
C96	96	Invalid mandatory information
C97	97	Message type non-existent or not implemented
C98	98	Message type not compatible with protocol state

<Cause>	Number	Meaning
C99	99	Information element non-existent or not implemented
C100	100	Conditional information element error
C101	101	Message not compatible with protocol state
C111	111	Protocol error, unspecified
ON	-	Enables the rejection with previously selected cause
OFF	-	Disables the rejection

CONFigure:LTE:SIGN<i>:CELL:ACause:ATTach <Cause>

Selects whether a cause is added to ATTACH ACCEPT messages or not.

Parameters:

<Cause> C18 | ON | OFF
OFF: disables sending of a cause
ON / C18: enables sending of attach accept cause 18 (CS domain not available)
 *RST: OFF

Example: See [Configuring other network settings](#)

Firmware/software: V3.7.20

Manual operation: See ["Attach Accept Cause"](#) on page 190

CONFigure:LTE:SIGN<i>:CELL:EAResult <Result>

Enables or disables the configuration of the EPS attach result IE and selects the value to be transmitted.

Parameters:

<Result> EONLY | CEIattach | ON | OFF
EONLY: value "EPS only"
CEIattach: value "combined EPS/IMSI attach"
ON: Enables sending the previously selected value.
OFF: Disables the configuration.
 *RST: CEI

Example: See [Configuring other network settings](#)

Firmware/software: V4.0.250

Manual operation: See ["EPS Attach Result"](#) on page 191

CONFigure:LTE:SIGN<i>:CELL:AUResult <Result>

Enables or disables sending the IE "additional update result" and selects the value to be transmitted.

Parameters:

<Result> NAINform | CFNPreferred | SONLy | ON | OFF
NAINform: value "no additional information"
CFNPreferred: value "CS fallback not preferred"
SONLy: value "SMS only"
ON: Enables sending the previously selected value.
OFF: Disables sending the IE.
 *RST: NAIN

Example: See [Configuring other network settings](#)

Firmware/software: V4.0.250

Manual operation: See ["Additional Update Result"](#) on page 191

CONFigure:LTE:SIGN<i>:CELL:NAS:EPSNetwork <Enable>

Enables or disables sending of the information element "EPS Network Feature Support" to the UE in the ATTACH ACCEPT message.

For configuration of the information element contents, see other
 CONFigure:LTE:SIGN<i>:CELL:NAS:... commands.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.20

Manual operation: See ["EPS Network Feature Support"](#) on page 191

CONFigure:LTE:SIGN<i>:CELL:NAS:IMSVops <Support>

Configures the field "IMS voice over PS session indicator" of the information element "EPS Network Feature Support".

Parameters:

<Support> NSUPported | SUPPorted
NSUPported: not supported
SUPPorted: supported
 *RST: NSUP

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.20

Manual operation: See ["IMS Voice Over PS Session Indicator"](#) on page 191

CONFigure:LTE:SIGN<i>:CELL:NAS:EMCBs <Support>

Configures the field "Emergency bearer services indicator" of the information element "EPS Network Feature Support".

Parameters:

<Support> NSUPported | SUPPorted
 NSUPported: not supported
 SUPPorted: supported
 *RST: NSUP

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.20

Options: R&S CMW-KS510

Manual operation: See ["Emergency Bearer Services Indicator"](#) on page 191

CONFigure:LTE:SIGN<i>:CELL:NAS:EPCLcs <Support>

Configures the field "Location services indicator in EPC" of the information element "EPS Network Feature Support".

Parameters:

<Support> NSUPported | SUPPorted
 NSUPported: not supported
 SUPPorted: supported
 *RST: NSUP

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.20

Options: R&S CMW-KS510

Manual operation: See ["Location Service Indicator in EPC"](#) on page 192

CONFigure:LTE:SIGN<i>:CELL:NAS:CSLCs <Support>

Configures the field "Location services indicator in CS" of the information element "EPS Network Feature Support".

Parameters:

<Support> NSUPported | SUPPorted | NINFormation
 NSUPported: not supported
 SUPPorted: supported
 NINFormation: no information
 *RST: NSUP

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.20

Options: R&S CMW-KS510

Manual operation: See ["Location Service Indicator in CS"](#) on page 192

2.6.13.9 Synchronization settings

The commands in this section configure the synchronization to other signaling applications and the synchronization between the component carriers.

[CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SYNC:ZONE](#).....721

[CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SYNC:OFFSet](#)..... 721

[CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SYNC:OFFSet](#).....721

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SYNC:ZONE <Zone>

Selects the synchronization zone for the signaling application.

Parameters:

<Zone> NONE | Z1
NONE: no synchronization
Z1: synchronization to zone 1
***RST:** Z1

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.50

Manual operation: See ["Synchronization Zone"](#) on page 192

CONFigure:LTE:SIGN<i>:CELL[:PCC]:SYNC:OFFSet <Offset>

CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SYNC:OFFSet <Offset>

Configures the timing offset relative to the time zone.

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 Range: 0 s to 1E-3 s
***RST:** 0 s
 Default unit: s

Example: See [Configuring other network settings](#)

Firmware/software: V3.2.50

Options: R&S CMW-KS510 for PCC, R&S CMW-KS512 for SCC

Manual operation: See ["Synchronization Offset"](#) on page 192

2.6.14 eMTC settings

The commands in this section configure settings that are only relevant for enhanced machine type communication (eMTC).

• General settings	722
• Repetition settings	723
• CE settings	726
• Hopping settings	730

2.6.14.1 General settings

The commands in this section configure general eMTC settings.

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MB<number>	722
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:SSPace	722

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MB<number> <Enable>

Selects the maximum bandwidth for an eMTC connection.

Suffix:

<number> 5

Parameters:

<Enable> OFF | ON
OFF: Max bandwidth 1.4 MHz
ON: Max bandwidth 5 MHz
 *RST: OFF

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS590

Manual operation: See ["Max Bandwidth 5 MHz"](#) on page 194

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:SSPace <SearchSpace>

Selects where the signaling application puts the MPDCCH.

Parameters:

<SearchSpace> COMM | UESP
COMM: common search space
UESP: UE-specific search space
 *RST: UESP

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Search Space"](#) on page 194

2.6.14.2 Repetition settings

The commands in this section configure repetition settings for eMTC, for the individual channel types.

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:MRPaging.....	723
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:MREPetitions.....	723
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:RLEVel.....	724
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDSch:A:MRCE.....	724
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:A:MRCE.....	724
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDSch:A:CERepetition.....	724
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:A:CERepetition.....	724
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDSch:B:MRCE.....	725
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:B:MRCE.....	725
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDSch:B:CERepetition.....	725
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:B:CERepetition.....	725
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUCCh:A:CERepetition.....	725
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUCCh:B:CERepetition.....	726

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:MRPaging <MaxRepetitions>

Configures the maximum number of MPDCCH repetitions for paging.

Parameters:

<MaxRepetitions> MR1 | MR2 | MR4 | MR8 | MR16 | MR32 | MR64 | MR128 |
MR256
1, 2, 4, ..., 128, 256 repetitions
*RST: MR1

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Max Repetition Paging"](#) on page 195

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:MREPetitions <MaxRepetitions>

Configures the maximum number of MPDCCH repetitions (no paging).

Parameters:

<MaxRepetitions> MR1 | MR2 | MR4 | MR8 | MR16 | MR32 | MR64 | MR128 |
MR256
1, 2, 4, ..., 128, 256 repetitions
*RST: MR1

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Max Repetition"](#) on page 195

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MPDCch:RLEVel <RepLevel>

Configures the repetition level for MPDCCH repetitions.

Parameters:

<RepLevel> RL1 | RL2 | RL3 | RL4
*RST: RL1

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Repetition Level"](#) on page 195

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDSch:A:MRCE <MaxRepetition>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:A:MRCE <MaxRepetitions>

Configures the maximum number of PDSCH or PUSCH repetitions for CE mode A.

Parameters:

<MaxRepetitions> NCON | MR16 | MR32
Not configured (omit field), 16, 32
*RST: NCON

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Max Repetitions CE Mode A"](#) on page 197

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDSch:A:CERepetition <Repetitions>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:A:CERepetition <Repetitions>

Configures the number of PDSCH or PUSCH repetitions for CE mode A.

Parameters:

<Repetitions> R1 | R2 | R4 | R8 | R16 | R32
*RST: R1

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Repetition CE Mode A"](#) on page 197

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDsch:B:MRCE <MaxRepetitions>
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:B:MRCE <MaxRepetitions>

Configures the maximum number of PDSCH or PUSCH repetitions for CE mode B.

Parameters:

<MaxRepetitions> NCON | MR192 | MR256 | MR384 | MR512 | MR768 | MR1024 | MR1536 | MR2048

Not configured (omit field), 192, 256, ..., 2048

*RST: NCON

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See "[Max Repetitions CE Mode B](#)" on page 198

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PDsch:B:CERepetition <Repetitions>
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUSch:B:CERepetition <Repetitions>

Configures the number of PDSCH or PUSCH repetitions for CE mode B.

Parameters:

<Repetitions> R1 | R4 | R8 | R16 | R32 | R64 | R128 | R192 | R256 | R384 | R512 | R768 | R1024 | R1536 | R2048

*RST: R4

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20, V3.8.10 added R512 | R768 | R1024 | R1536 | R2048

Options: R&S CMW-KS590

Manual operation: See "[Repetition CE Mode B](#)" on page 198

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUCCh:A:CERepetition <Repetitions>

Configures the number of PUCCH repetitions for CE mode A.

Parameters:

<Repetitions> R1 | R2 | R4 | R8

*RST: R1

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See "[Repetitions CE Mode A](#)" on page 197

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:PUCCh:B:CERepetition <Repetitions>

Configures the number of PUCCH repetitions for CE mode B.

Parameters:

<Repetitions> R4 | R8 | R16 | R32 | R64 | R128
*RST: R4

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20, V3.7.70 added R64 and R128

Options: R&S CMW-KS590

Manual operation: See ["Repetitions CE Mode B"](#) on page 197

2.6.14.3 CE settings

The commands in this section configure the coverage enhancement (CE) levels and modes for eMTC.

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:MODE.....	726
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:ENABLE.....	726
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:QRXLevmin.....	727
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:CINdex.....	727
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:FOFFset.....	728
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MPATtempts.....	728
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:RPATtempt.....	729
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MMRRepetitio.....	729

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:MODE <Mode>

Selects the coverage enhancement mode.

Parameters:

<Mode> A | B
*RST: A

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20, V3.7.70 added mode B

Options: R&S CMW-KS590

Manual operation: See ["CE Mode"](#) on page 199

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:ENABLE <Level>, <Enable>**CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:ENABLE? <Level>**

Selects whether the eNodeB supports a certain CE level.

If you disable a CE level, the higher CE levels are disabled automatically. You can enable a CE level only if all lower CE levels are enabled.

Parameters:

<Enable> OFF | ON
 Disables or enables the selected CE <Level>
 *RST: ON

Parameters for setting and query:

<Level> integer
 Selects a CE level
 Range: 1 to 3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Enable"](#) on page 199

**CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:QRXLevmin <level>, <Qrxlevmin>
 CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:QRXLevmin? <level>**

Defines the $Q_{rxlevmin}$ for CE level selection for PRACH. The value divided by two is signaled to the UE.

The value is defined per CE level. With increasing CE level, the $Q_{rxlevmin}$ must decrease.

Parameters:

<Qrxlevmin> numeric
 $Q_{rxlevmin}$ for the selected CE <level>
 Range: -140 dBm to -44 dBm
 Increment: 2 dB
 *RST: <level> 1 to 3: -136 dBm, -138 dBm, -140 dBm

Parameters for setting and query:

<level> integer
 Selects a CE level
 Range: 1 to 3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Q rxlevmin"](#) on page 200

**CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:CINdex <Level>, <Index>
 CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:CINdex? <Level>**

Sets the PRACH configuration index for a certain CE level.

Parameters:

<Index> numeric
 PRACH configuration index for the selected CE <Level>
 Range: 0 to 63
 *RST: 3

Parameters for setting and query:

<Level> integer
 Selects a CE level
 Range: 0 to 3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Configuration Index"](#) on page 200

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:FOFFset <Level>, <Offset>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:FOFFset? <Level>

Sets the frequency offset for the preamble RBs, for a certain CE level.

Parameters:

<Offset> numeric
 Frequency offset for the selected CE <Level>
 Range: 0 to 94
 *RST: 0

Parameters for setting and query:

<Level> integer
 Selects a CE level
 Range: 0 to 3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Frequency Offset"](#) on page 200

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MPATtempts <Level>, <Attempts>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MPATtempts? <Level>

Specifies the maximum number of preamble transmission attempts for a certain CE level.

Parameters:

<Attempts> A3 | A4 | A5 | A6 | A7 | A8 | A10
 Maximum attempts for the selected CE <Level>
 *RST: A3

Parameters for setting and query:

<Level> integer
 Selects a CE level
 Range: 0 to 3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Max Preamble Attempts"](#) on page 200

CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:RPATtempt <Level>, <Repetitions>

CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:RPATtempt? <Level>

Specifies the number of repetitions per preamble transmission attempt, for a certain CE level.

Parameters:

<Repetitions> R1 | R2 | R4 | R8 | R16 | R32 | R64 | R128
 Maximum repetitions for the selected CE <Level>
 *RST: R1

Parameters for setting and query:

<Level> integer
 Selects a CE level
 Range: 0 to 3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Repetitions per Preamble Attempt"](#) on page 201

CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MMRRepetitio <Level>, <MaxRepetitions>

CONFIGure:LTE:SIGN<i>[:PCC]:EMTC:CE:LEVel:PRACH:MMRRepetitio? <Level>

Specifies the maximum number of MPDCCH repetitions for the random access response, for a certain CE level.

Parameters:

<MaxRepetitions> R1 | R2 | R4 | R8 | R16 | R32 | R64 | R128 | R256
 Maximum repetitions for the selected CE <Level>
 *RST: R1

Parameters for setting and query:

<Level> integer
 Selects a CE level
 Range: 0 to 3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Max MPDCCH Repetitions RA"](#) on page 201

2.6.14.4 Hopping settings

The commands in this section configure frequency hopping between narrowbands.

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:DL:ENABle.....	730
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:UL:ENABle.....	730
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:DL:HOFFset.....	730
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:UL:HOFFset.....	730
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:DL:A:INterval.....	731
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:UL:A:INterval.....	731
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:DL:B:INterval.....	731
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:UL:B:INterval.....	731

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:DL:ENABle <Enable>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:UL:ENABle <Enable>

Enables or disables frequency hopping for eMTC, DL or UL.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["Enable"](#) on page 201

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:DL:HOFFset <Offset>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPping:UL:HOFFset <Offset>

Specifies the size of one frequency hop, DL or UL.

For the number of narrowbands per cell bandwidth, see [Table 2-8](#).

Parameters:

<Offset> numeric
 Hop size in narrowbands
 Range: 1 to 16 (depends on cell BW)
 *RST: 1

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See "[Hopping Offset](#)" on page 202

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:DL:A:INTERval <Interval>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:UL:A:INTERval <Interval>

Specifies the time interval between two hops for CE mode A, DL or UL.

Parameters:

<Interval> I1 | I2 | I4 | I8
 Time interval in subframes
 *RST: I1

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See "[Interval Mode A](#)" on page 202

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:DL:B:INTERval <Interval>

CONFigure:LTE:SIGN<i>[:PCC]:EMTC:HOPPing:UL:B:INTERval <Interval>

Specifies the time interval between two hops for CE mode B, DL or UL.

Parameters:

<Interval> I2 | I4 | I8 | I16
 Time interval in subframes
 *RST: I2

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See "[Interval Mode B](#)" on page 202

2.6.15 Connection configuration

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2.6.15.1 General connection settings

The following commands define general connection parameters.

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CONFigure:LTE:SIGN<i>:CONNection:EOISupport.....	736
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CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:DL.....	744
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:NOLayers.....	745
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CONFigure:LTE:SIGN<i>:CONNection:TAControl.....	751
CONFigure:LTE:SIGN<i>:CONNection:NTA:ENABle.....	751
CONFigure:LTE:SIGN<i>:CONNection:IDCHsindic.....	751

CONFigure:LTE:SIGN<i>:CONNection:EASY:BFBW <Enable>

Specifies whether the easy mode is used if the band or the frequency or the cell bandwidth is changed.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.5.30

Manual operation: See ["Band/Frequency/BW Change"](#) on page 204

CONFigure:LTE:SIGN<i>:CONNection:GHOPping <Enable>

Enables or disables group hopping.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.0.30

Manual operation: See ["Group Hopping"](#) on page 204

CONFigure:LTE:SIGN<i>:CONNection:UECategory:MANual:ENHanced <UECatManual>

Configures the UE category to be used by the R&S CMW if no reported value is available or usage of the reported value is disabled, see [CONFigure:LTE:SIGN<i>:CONNection:UECategory:REPorted:ENHanced](#).

Parameters:

<UECatManual> M1 | M2 | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12
*RST: 3

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.8.10

Manual operation: See ["UE Category"](#) on page 205

CONFigure:LTE:SIGN<i>:CONNection:UECategory:REPorted:ENHanced <UseReported>

Enables or disables the usage of the UE category value reported by the UE.

When disabled, the UE category must be set manually, see [CONFigure:LTE:SIGN<i>:CONNection:UECategory:MANual:ENHanced](#). The manually set value is also used if no reported value is available.

A query returns two parameters. The second parameter depends on <UseReported> as follows:

- If <UseReported> = ON: Query returns <UseReported>, <UECatReported>.
- If <UseReported> = OFF: Query returns <UseReported>, <UECatManual>.

Parameters:

<UseReported> OFF | ON
*RST: ON

Return values:

<UECatReported>	decimal UE category reported by the UE (NAV indicates that none has been reported) Range: 1 to 12
<UECatManual>	M1 M2 0 1 2 3 4 5 6 7 8 9 10 11 12 UE category configured manually.

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.8.10

Manual operation: See ["UE Category"](#) on page 205

CONFigure:LTE:SIGN<i>:CONNection:UECategory:CZAllowed <Allowed>

Specifies whether category 0 UEs are allowed to access the cell. This information is sent to the UE via broadcast in system information block 1.

Parameters:

<Allowed>	OFF ON
*RST:	OFF

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.5.30

Options: R&S CMW-KS590

Manual operation: See ["UE Category 0 Allowed"](#) on page 205

CONFigure:LTE:SIGN<i>:CONNection:PSMAllowed <Allowed>

Specifies whether a UE request for power-saving mode is accepted or rejected.

Parameters:

<Allowed>	OFF ON
*RST:	OFF

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS590

Manual operation: See ["PSM Allowed"](#) on page 205

CONFigure:LTE:SIGN<i>:CONNection:IEmergency <Enable>

Enables the optional field "ims-EmergencySupport-r9" in system information block 1.

Parameters:

<Enable> OFF | ON
OFF: field omitted
ON: field included
 *RST: ON

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.7.10

Manual operation: See ["IMS Emergency Support"](#) on page 205

CONFigure:LTE:SIGN<i>:CONNection:EOISupport <Support>

Enables the optional field "eCallOverIMS-Support-r14" in system information block 1.

Parameters:

<Support> OFF | ON
OFF: field omitted
ON: field included
 *RST: OFF

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.7.60

Manual operation: See ["eCall Over IMS Support"](#) on page 205

CONFigure:LTE:SIGN<i>:CONNection:DPCycle <Cycle>

Selects the cell-specific default paging cycle.

Parameters:

<Cycle> P032 | P064 | P128 | P256
 32, 64, 128 or 256 radio frames
 *RST: P064

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V2.1.30

Manual operation: See ["Default Paging Cycle"](#) on page 206

CONFigure:LTE:SIGN<i>:CONNection:PCNB <Value>

Configures the field "nB" in the "PCCH-Config" in system information block 2.

Parameters:

<Value> NB4T | NB2T | NBT | NBT2 | NBT4 | NBT8 | NBT16 | NBT32 | NBT64 | NBT128 | NBT256
 4T, 2T, T, T/2, T/4, T/8, T/16, T/32, T/64, T/128, T/256
 The values NBT64, NBT128 and NBT256 are only allowed for eMTC.
 And they are only allowed, if the default paging cycle has the same or a greater value. Example: NBT64 needs cycle $\geq$ P064.
 *RST: NBT2

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.7.10, V3.7.20 added NBT64 | NBT128 | NBT256

Manual operation: See ["PCCH-Config nB"](#) on page 206

CONFigure:LTE:SIGN<i>:CONNection:ASEMission <Value>

Selects a value signaled to the UE as an additional ACLR and spectrum emission requirement.

Parameters:

<Value> NS01 | ... | NS288
 Value NS_01 to NS_288
 *RST: NS01

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.0.10
 V3.2.70: added NS16, NS17, NS18, NS20
 V3.2.82: added NS19, NS21, NS22, NS23, NS24
 V3.7.20: added NS25 to NS32
 V3.7.60: added NS33 to NS288

Manual operation: See ["Additional Spectrum Emission"](#) on page 206

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:ASEMission:CAGGregation <Value>

Selects a value signaled to the UE as an additional ACLR and spectrum emission requirement for the SCC<c>.

The setting is only relevant if the SCC uplink is enabled.

Suffix:

<c> 1..7

Parameters:

<Value> NS01 | ... | NS32
 Value CA_NS_01 to CA_NS_32
 *RST: NS01

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.5.30

Options: R&S CMW-KS512

Manual operation: See ["Add. Spectrum Emission Scell"](#) on page 206

CONFigure:LTE:SIGN<i>:CONNection:FCoefficient <Filter>

Selects the value to be sent to the UE as "filterCoefficient" in RRC messages containing this information element.

Parameters:

<Filter> FC4 | FC8
*RST: FC4

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V1.0.15.21

Manual operation: See ["UE Meas. Filter Coefficient"](#) on page 206

CONFigure:LTE:SIGN<i>:CONNection:CTYPE <Type>

Selects the connection type to be applied.

Parameters:

<Type> TESTmode | DAPPLication
TESTmode: for signaling tests not involving the DAU
DAPPLication: for data application measurements using the DAU
*RST: TEST

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V2.0.20

Options: For DAPPLication: installed DAU or external DAU

Manual operation: See ["Connection Type"](#) on page 207

CONFigure:LTE:SIGN<i>:CONNection:EDAU:ENABLE <Enable>

Enables usage of an external DAU.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.2.80

Options: R&S CMW-KA120

Manual operation: See ["Use external DAU"](#) on page 207

CONFigure:LTE:SIGN<i>:CONNection:EDAU:NSEgment <NetworkSegment>

Specifies the network segment of the instrument where the external DAU is installed.

Parameters:

<NetworkSegment> A | B | C
 *RST: A

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.2.80

Options: R&S CMW-KA120

Manual operation: See ["Network Segment, Subnet Node ID"](#) on page 207

CONFigure:LTE:SIGN<i>:CONNection:EDAU:NID <ID>

Specifies the subnet node ID of the instrument where the external DAU is installed.

Parameters:

<ID> integer
 Range: 1 to 254
 *RST: 1

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.2.80

Options: R&S CMW-KA120

Manual operation: See ["Network Segment, Subnet Node ID"](#) on page 207

CONFigure:LTE:SIGN<i>:CONNection:TMODE <Enable>

Specifies whether the UE is forced into a test mode. If enabled, the message "ACTIVATE TEST MODE" is sent to the UE.

Parameters:

<Enable> OFF | ON
 *RST: ON

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V2.1.20

Manual operation: See ["Test Mode > Use "Activate Testmode" Message"](#) on page 207

CONFigure:LTE:SIGN<i>:CONNection:RLCMode <Mode>

Selects the RLC mode for downlink transmissions (default bearer).

Parameters:

<Mode> UM | AM
UM: unacknowledged mode
AM: acknowledged mode
 *RST: UM

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V2.1.20

Options: R&S CMW-KS510

Manual operation: See ["RLC Mode"](#) on page 208

CONFigure:LTE:SIGN<i>:CONNection:IPVersion <IPversion>

Configures the allowed IP versions for default bearers and data application tests. In test mode, the setting is fixed and can only be queried.

Parameters:

<IPversion> IPV4 | IPV6 | IPV46
IPV4: IPV4 only
IPV6: IPV6 only
IPV46: IPV4 and IPV6
 *RST: IPV46

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["IP Version"](#) on page 208

CONFigure:LTE:SIGN<i>:CONNection:APN <APN>

Configures the default APN for default bearers and data application tests. In test mode, the setting is fixed and can only be queried.

Parameters:

<APN> string
 APN default value
 *RST: "cmw500.rohde-schwarz.com"

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.2.80

Manual operation: See ["APN"](#) on page 208

CONFigure:LTE:SIGN<i>:CONNection:QCI <QCI>

Configures the QCI value for default bearers and data application tests. In test mode, the setting is fixed and can only be queried.

Parameters:

<QCI> numeric
 Quality-of-service class identifier
 Range: 5 to 9
 *RST: 5

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["QCI"](#) on page 208

CONFigure:LTE:SIGN<i>:CONNection:TDBearer:RLCMode <Mode>

Selects the RLC mode for downlink transmissions (dedicated bearer in test mode).

Parameters:

<Mode> UM | AM
 UM: unacknowledged mode
 AM: acknowledged mode
 *RST: UM

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.7.70

Manual operation: See ["RLC Mode"](#) on page 209

CONFigure:LTE:SIGN<i>:CONNection:SIBReconfig <Type>

Selects a method for information of the UE about changes in the system information, resulting from modified parameters: SIB paging or RRC reconfiguration.

Parameters:

<Type> SIBPaging | RRCReconfig
 *RST: SIBP

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.0.20

Manual operation: See ["SIB Reconfiguration"](#) on page 209

CONFigure:LTE:SIGN<i>:CONNection:KRRC <Enable>

Selects whether the RRC connection is kept or released after attach.

Parameters:

<Enable> OFF | ON
OFF: The RRC connection is released.
ON: The RRC connection is kept.
 *RST: ON

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V2.1.20

Manual operation: See ["Keep RRC Connection"](#) on page 209

CONFigure:LTE:SIGN<i>:CONNection:RITimer <Time>

Configures the inactivity timeout for disabled "Keep RRC Connection" (CONFigure:LTE:SIGN:CONNection:KRRC OFF).

Parameters:

<Time> integer
 Range: 1 s to 255 s
 *RST: 5 s
 Default unit: s

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V3.2.50

Options: R&S CMW-KS510

Manual operation: See ["Keep RRC Connection"](#) on page 209

CONFigure:LTE:SIGN<i>:CONNection:DLPadding <Value>

Activates or deactivates downlink padding at the MAC layer (filling an allocated RMC with padding bits when no data is available from higher layers).

Parameters:

<Value> OFF | ON
 *RST: ON

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V1.0.15.20

Manual operation: See ["Downlink MAC Padding"](#) on page 210

CONFigure:LTE:SIGN<i>:CONNection:DLEinsertion <Value>

Configures the rate of transport block errors to be inserted into the downlink data.

Parameters:

<Value> numeric
 Range: 0 % to 100 %
 Increment: 1 %
 *RST: 0 %
 Default unit: %

Example: See [Configuring general connection settings part 1](#)

Firmware/software: V1.0.15.20, V3.8.10 changed increment

Manual operation: See ["Downlink MAC Error Insertion"](#) on page 210

CONFigure:LTE:SIGN<i>:CONNection:UETSelection <Selection>

Configures the parameter "ue-TransmitAntennaSelection" signaled to the UE.

Parameters:

<Selection> OFF | OLOop
OFF
 UE transmit antenna selection not allowed
OLOop
 Open-loop UE transmit antenna selection
 *RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.5.40

Manual operation: See ["UE Transmit Antenna Selection"](#) on page 223

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:STYPe <Type>[, <CQIMode>]**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:STYPe <Type>[, <CQIMode>]**

Selects the scheduling type.

Suffix:

<c> 1..7

Parameters:

<Type> RMC | UDCHannels | UDTTibased | CQI | SPS | EMAMode | EMCSched
RMC: 3GPP-compliant reference measurement channel
UDCHannels: user-defined channel
UDTTibased: user-defined channel configurable per TTI
CQI: CQI channel, as specified by next parameter
SPS: semi-persistent scheduling (only PCC, not SCC)
EMAMode: eMTC auto mode
EMCSched: eMTC compact scheduling
 *RST: RMC
 <CQIMode> TTIBased | FWB | FPMI | FCPRi | FCRI | FPRi

Only relevant for <Type> = CQI

TTIBased: fixed CQI

FWB: follow wideband CQI

FPMI: follow wideband PMI

FCPRi: follow wideband CQI-PMI-RI

FCRI: follow wideband CQI-RI

FPRI: follow wideband PMI-RI

*RST: TTIB

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.0.10, V3.0.50 <CQIMode>, V3.2.50 SCC command
V3.2.80 SPS, FPMI, FCPRi and FCRI
V3.5.10 FPRI
V3.7.20 EMAMode, EMCSched

Options: R&S CMW-KS510 for UDTT, SPS, CQI-TTIB
R&S CMW-KS510 for UDCH, except for eMTC
R&S CMW-KS510/-KS512 (without CA/with CA) for CQI other than TTIB
R&S CMW-KS590 for EMAMode, EMCSched

Manual operation: See ["Scheduling Type"](#) on page 223

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:MCLuster:UL <Multicluster>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:UL <Multicluster>

Enables/disables multi-cluster allocation for the UL.

Suffix:

<c> 1..7

Parameters:

<Multicluster> OFF | ON

OFF: contiguous allocation, resource allocation type 0

ON: multi-cluster allocation, resource allocation type 1

Example: See [Configuring RMCs](#)

Firmware/software: V3.5.20

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Multicluster UL"](#) on page 224

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:MCLuster:DL <Multicluster>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:DL <Multicluster>

Enables/disables multi-cluster allocation for the DL.

Suffix:

<c> 1..7

Parameters:

<Multicluster> OFF | ON
OFF: contiguous allocation
ON: multi-cluster allocation
 *RST: OFF

Example: See [Configuring user-defined channels](#)

Firmware/software: V3.5.50

Manual operation: See ["Multicluster"](#) on page 247

Option R&S CMW-KS510/-KS512 is required (without CA/with CA).

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:NOLayers <Number>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:NOLayers <Number>

Selects the number of layers for MIMO 4x4 with spatial multiplexing (TM 3 and 4).

Suffix:

<c> 1..7

Parameters:

<Number> L2 | L4
 Two layers or four layers
 *RST: Depends on available options

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS540 for L4, R&S CMW-KS521 for L2

Manual operation: See ["Number of Layers"](#) on page 225

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:QAM<m>:DL <Enable>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:QAM<m>:DL <Enable>

Selects which 3GPP tables are used for CQI scheduling: tables up to 64-QAM, tables up to 256-QAM or tables up to 1024-QAM.

Suffix:

<m> 256, 1024

<c> 1..7

Parameters:

<Enable> OFF | ON
ON, QAM256: use tables with 256-QAM
OFF, QAM256: use tables without 256-QAM
ON, QAM1024: use tables with 1024-QAM
OFF, QAM1024: use tables without 1024-QAM
 *RST: OFF

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.40, V3.7.70 QAM1024

Options: QAM256: R&S CMW-KS504/-KS554
QAM1024: R&S CMW-KS505/-KS555

Manual operation: See ["Use 256-QAM, Use 1024-QAM"](#) on page 268

CONFigure:LTE:SIGN<i>:CONNection:SUITx <SkipUITx>

Enables or disables the fields "skipUplinkTxSPS" and "skipUplinkTxDynamic".

Parameters:

<SkipUITx> OFF | ON
*RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.8.10

Manual operation: See ["SkipUITx"](#) on page 225

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:HPUSch:ENABLE <Hopping>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:HPUSch:ENABLE <Hopping>

Enables inter-subframe PUSCH frequency hopping, type 2.

Suffix:

<c> 1..7

Parameters:

<Hopping> OFF | ON
*RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.7.50

Manual operation: See ["PUSCH Hopping"](#) on page 225

SENSe:LTE:SIGN<i>:CONNection[:PCC]:HPUSch:ACTive?

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:HPUSch:ACTive?

Queries whether PUSCH frequency hopping is active.

Suffix:

<c> 1..7

Return values:

<Active> OFF | ON

Example: See [Configuring general connection settings part 2](#)

Usage: Query only

Firmware/software: V3.7.50

Manual operation: See ["PUSCH Hopping"](#) on page 225

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TTIBundling <Enable>

Enables or disables TTI bundling for the uplink.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["TTI Bundling"](#) on page 225

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TIA<no> <Enable>**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TIA<no> <Enable>**

Enables or disables sending the "tbsIndexAlt<no>..." parameter to the UE.

Suffix:

<no> 2, 3
<c> 1..7

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.7.70

Options: R&S CMW-KS514 for <no>=2
 R&S CMW-KS505/-KS555 for <no>=3

Manual operation: See ["tbsIndexAlt2"](#) on page 226

CONFigure:LTE:SIGN<i>:CONNection:OBCHange <Mode>

Selects the mechanism to be used for inter-band handover.

Parameters:

<Mode> BHANdover | REDirection
 Blind handover or redirection
*RST: RED

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V2.1.30

Manual operation: See ["Operating Band Change, Frequency Change"](#) on page 136

CONFigure:LTE:SIGN<i>:CONNection:FCChange <Mode>

Selects the mechanism to be used for inter-frequency handover (operating band not changed).

Parameters:

<Mode> BHANdover | REDirection
Blind handover or redirection
*RST: BHAN

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V2.1.20

Manual operation: See ["Operating Band Change, Frequency Change"](#) on page 136

CONFigure:LTE:SIGN<i>:CONNection:CSFB:DESTination <Destination>

Selects the target radio access technology for MO CSFB.

Parameters:

<Destination> GSM | WCDMa | TDSCdma | NONE
*RST: GSM

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.20
V3.5.20: added TDSCdma, V3.5.40: added NONE

Manual operation: See ["CS Fallback \(MO\)"](#) on page 229

CONFigure:LTE:SIGN<i>:CONNection:CSFB:GSM <Band>, <DLChannel>, <BandIndicator>

Configures the GSM target for MO CSFB.

Parameters:

<Band> G085 | G09 | G18 | G19
GSM 850, GSM 900, GSM 1800, GSM 1900
*RST: G09

<DLChannel> decimal
Channel number used for the broadcast control channel (BCCH)
Range: 0 to 1023, depending on GSM band, see table below
*RST: 20

<BandIndicator> G18 | G19
Band indicator for distinction of GSM 1800 and GSM 1900 bands. The two bands partially use the same channel numbers for different frequencies.
*RST: G18

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.20

Manual operation: See "[CS Fallback \(MO\)](#)" on page 229

Table 2-73: Channel number range depending on GSM band

Band	Channel number
G085	128 to 251
G09	0 to 124, 940 to 1023
G18	512 to 885
G19	512 to 810

CONFigure:LTE:SIGN<i>:CONNection:CSFB:WCDMa <Band>, <DLChannel>

Configures the WCDMA target for MO CSFB.

Parameters:

<Band> OB1 | OB2 | OB3 | OB4 | OB5 | OB6 | OB7 | OB8 | OB9 | OB10 |
OB11 | OB12 | OB13 | OB14 | OB19 | OB20 | OB21 | OB22 |
OB25 | OBS1 | OBS2 | OBS3 | OBL1 | OB26

OB1, ..., OB14: band I to XIV

OB19, ..., OB22: band XIX to XXII

OB25, OB26: band XXV, XXVI

OBS1: band S

OBS2: band S 170 MHz

OBS3: band S 190 MHz

OBL1: band L

*RST: OB1

<DLChannel> decimal

Downlink channel number

Range: 412 to 11000, depending on operating band, see
table below

*RST: 10563

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.20, V3.5.40: OB22/OB25, V3.7.10: OB26

Manual operation: See "[CS Fallback \(MO\)](#)" on page 229

Table 2-74: Channel number range depending on operating band

Operating band	Channel number
OB1	10562 to 10838
OB2	412 to 687 (step 25), 9662 to 9938
OB3	1162 to 1513
OB4	1537 to 1738, 1887 to 2087 (step 25)

Operating band	Channel number
OB5	1007, 1012, 1032, 1037, 1062, 1087, 4357 to 4458
OB6	1037, 1062, 4387 to 4413
OB7	2237 to 2563, 2587 to 2912 (step 25)
OB8	2937 to 3088
OB9	9237 to 9387
OB10	3112 to 3388, 3412 to 3687 (step 25)
OB11	3712 to 3812
OB12	3837 to 3903, 3927, 3932, 3957, 3962, 3987, 3992
OB13	4017 to 4043, 4067, 4092
OB14	4117 to 4143, 4167, 4192
OB19	712 to 763, 787, 812, 837
OB20	4512 to 4638
OB21	862 to 912
OB22	4662 to 5038
OB25	5112 to 5413, 6292 to 6592 (step 25)
OB26	5762 to 5913, 5937, 5962, 5987, 5992, 6012, 6017, 6037, 6042, 6062, 6067, 6087
OBS1	5912 to 5987 (step 25), 10912 to 10988
OBS2	10900 to 10950
OBS3	5962, 5987, 10950 to 11000
OBL1	7637 to 7783, 7788 to 7933

CONFigure:LTE:SIGN<i>:CONNection:CSFB:TDSCdma <Band>, <DLChannel>

Configures the TD-SCDMA target for MO CSFB.

Parameters:

<Band>

OB1 | OB2 | OB3

OB1: Band 1 (F), 1880 MHz to 1920 MHz**OB2:** Band 2 (A), 2010 MHz to 2025 MHz**OB3:** Band 3 (E), 2300 MHz to 2400 MHz***RST:** OB1

<DLChannel>

decimal

Downlink channel number

The allowed range depends on the frequency band:

OB1: 9400 to 9600

OB2: 10050 to 10125

OB3: 11500 to 12000

*RST: 9400

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.5.20

Manual operation: See ["CS Fallback \(MO\)"](#) on page 229

CONFigure:LTE:SIGN<i>:CONNection:AMDBearer <Enable>

Enables/disables accepting multiple default bearer requests.

Parameters:

<Enable> OFF | ON

*RST: ON

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.50

Manual operation: See ["Accept Multiple Default Bearer"](#) on page 229

CONFigure:LTE:SIGN<i>:CONNection:TAControl <Enable>

Enables the correction of a changing UL frame timing via timing advance commands.

Parameters:

<Enable> OFF | ON

*RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.5.10

Manual operation: See ["Timing Advance Control"](#) on page 232

CONFigure:LTE:SIGN<i>:CONNection:NTA:ENABle <Enable>

Enables setting the N_{TA} value at the UE, for SCCs in TDD mode if the PCC is in FDD mode. N_{TA} is defined in 3GPP TS 36.213 and 36.211.

Parameters:

<Enable> OFF | ON

ON: The N_{TA} value at the UE is set to at least 624.

OFF: The N_{TA} is not configured.

*RST: OFF

Firmware/software: V4.0.200

CONFigure:LTE:SIGN<i>:CONNection:IDCHsindic <Indication>

Enables sending the information element "idc-HardwareSharingIndication" to the UE.

Parameters:

<Indication> OFF | ON
 *RST: OFF

Firmware/software: V3.7.50

Options: R&S CMW-KS514

Manual operation: See ["IDC - Hardware Sharing Indication"](#) on page 233

2.6.15.2 General MIMO settings

The following commands configure MIMO connection settings applicable to most transmission modes.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TRANsmission	752
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TRANsmission	752
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:DCIFormat	752
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:DCIFormat	752
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:NENBantennas	753
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:NENBantennas	753
SENSe:LTE:SIGN<i>:CONNection[:PCC]:TSCHeMe?	753
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:TSCHeMe?	753
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PMATrix	754
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PMATrix	754

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TRANsmission <Mode>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TRANsmission <Mode>

Selects the LTE transmission mode. The value must be compatible to the active scenario, see [Table 2-32](#).

Suffix:

<c> 1..7

Parameters:

<Mode> TM1 | TM2 | TM3 | TM4 | TM6 | TM7 | TM8 | TM9
 Transmission mode 1, 2, 3, 4, 6, 7, 8, 9
 *RST: TM1

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.2.70, V3.5.10: TM9 added

Options: R&S CMW-KS520 or -KS540 for TM 2, 3, 4, 6, 7, 9
 R&S CMW-KS520 for TM 8

Manual operation: See ["Transmission Mode"](#) on page 211

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:DCIFormat <DCI>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:DCIFormat <DCI>

Selects the DCI format. The value must be compatible to the transmission mode, see [Table 2-32](#).

Suffix:

<c> 1..7

Parameters:

<DCI> D1 | D1A | D1B | D2 | D2A | D2B | D2C | D61
 Format 1, 1A, 1B, 2, 2A, 2B, 2C, 6-1A/B
 *RST: D1A

Example: See [Configuring MIMO settings](#)**Firmware/software:** V3.2.70, V3.5.10 added D2C, V3.7.10 added D61**Manual operation:** See ["DCI Format"](#) on page 211**CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:NENBantennas** <Antennas>**CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:NENBantennas** <Antennas>

Selects the number of downlink TX antennas for transmission mode 1 to 6. The value must be compatible to the active scenario and transmission mode, see [Table 2-32](#).

Suffix:

<c> 1..7

Parameters:

<Antennas> ONE | TWO | FOUR
 *RST: ONE

Example: See [Configuring MIMO settings](#)**Firmware/software:** V3.0.50, SCC command V3.2.50

Options: TWO (2x2): R&S CMW-KS520
 TWO (2x4): R&S CMW-KS540
 FOUR (4x2): R&S CMW-KS521
 FOUR (4x4): R&S CMW-KS540

Manual operation: See ["Antenna Configuration"](#) on page 211**SENSe:LTE:SIGN<i>:CONNection[:PCC]:TSCHeMeme?****SENSe:LTE:SIGN<i>:CONNection:SCC<c>:TSCHeMeme?**

Queries the transmission scheme.

Suffix:

<c> 1..7

Return values:

<Scheme> SISO | SIMO | TXDiversity | OLSMultiplex | CLSMultiplex |
 CLSingle | SBF5 | SBF8 | DBF78 | FBF710
SISO: single input single output
SIMO: single input multiple outputs (receive diversity)
TXDiversity: transmit diversity
OLSMultiplex: open loop spatial multiplexing
CLSMultiplex: closed loop spatial multiplexing

CLSingle: closed loop spatial multiplexing, single layer

SBF5: single-layer beamforming (port 5)

SBF8: single-layer beamforming (port 8)

DBF78: dual-layer beamforming (ports 7, 8)

FBF710: four-layer beamforming (ports 7 to 10)

Example: See [Configuring MIMO settings](#)

Usage: Query only

Firmware/software: V3.2.70, V3.7.10 added FBF710

Manual operation: See ["Transmission Scheme"](#) on page 212

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PMATrix <Mode>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PMATrix <Mode>

Selects the precoding matrix. The value must be compatible to the active scenario and transmission mode, see [Table 2-32](#).

For TM 8 and TM 9, the matrix is used as beamforming matrix, not for precoding.

Suffix:

<c> 1..7

Parameters:

<Mode> PMI0 | PMI1 | PMI2 | PMI3 | PMI4 | PMI5 | PMI6 | PMI7 | PMI8 |
PMI9 | PMI10 | PMI11 | PMI12 | PMI13 | PMI14 | PMI15 |
RANDom_pmi

Matrix according to PMI 0, PMI 1, ... PMI15.

RANDom_pmi: The PMI value is selected randomly as defined in 3GPP TS 36.521, annex B.4.1 and B.4.2.

*RST: PMI0

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.0.10
V3.0.50: PMI2 to PMI15
V3.2.50: SCC command
V3.5.10: RANDom_pmi added

Options: R&S CMW-KS520 or -KS540

Manual operation: See ["Precoding Matrix"](#) on page 211

2.6.15.3 MIMO channel matrix TM 2 to TM 6

The following commands configure the channel matrix for transmission mode 2 to 6.

[CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:SCHModel:ENABLE.....755](#)

[CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:ENABLE.....755](#)

[CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:SCHModel:ENABLE:MIMO<Mimo>.....755](#)

[CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:ENABLE:MIMO<Mimo>.....755](#)

[CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:SCHModel:MSELection:MIMO<Mimo>.....756](#)

[CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:MSELection:MIMO<Mimo>.....756](#)

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel.....	756
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel.....	756
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:MIMO<Mimo>.....	757
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:MIMO<Mimo>.....	757

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:ENABLE <Enable>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:ENABLE <Enable>

Enables or disables the MIMO 2x2 static channel matrix. Disabling the channel matrix results in an ideal radio channel without any coupling between the downlink signals.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON

*RST: OFF

Example: See [Configuring MIMO settings](#)

Firmware/software: V2.0.10, SCC command V3.2.50

Options: R&S CMW-KS520

Manual operation: See ["2x2 channel matrix"](#) on page 213

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:ENABLE:MIMO<Mimo>
<Enable>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:ENABLE:MIMO<Mimo>
<Enable>

Enables or disables the MIMO 4x4 static channel matrix. Disabling the channel matrix results in an ideal radio channel without any coupling between the downlink signals.

Suffix:

<c> 1..7

<Mimo> 44

Parameters:

<Enable> OFF | ON

*RST: OFF

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS540

Manual operation: See ["Enable"](#) on page 213

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:MSElection:
MIMO<Mimo> <Selection>**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:MSElection:****MIMO<Mimo>** <Selection>

Selects a predefined channel matrix or the user-defined channel matrix for MIMO 4x4.

Suffix:

<c> 1..7

<Mimo> 44

Parameters:

<Selection> UDEFined | CM3Gpp | HADamard | IDENtity
 User-defined matrix, 3GPP channel matrix, Hadamard matrix, identity matrix
 *RST: UDEF

Firmware/software: V3.7.10**Options:** R&S CMW-KS540**Manual operation:** See ["Matrix Selection"](#) on page 213

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel <h11abs>, <h11phi>, <h12phi>, <h21abs>, <h21phi>, <h22phi>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel <h11abs>, <h11phi>, <h12phi>, <h21abs>, <h21phi>, <h22phi>

Configures the channel coefficients, characterizing the radio channel for MIMO 2x2.

Suffix:

<c> 1..7

Parameters:

<h11abs> numeric
 Square of magnitude of h_{11}
 Range: 0 to 1
 *RST: 1

<h11phi> numeric
 Phase of h_{11}
 Range: 0 deg to 345 deg
 Increment: 15 deg
 *RST: 0 deg
 Default unit: deg

<h12phi> numeric
 Phase of h_{12}
 Range: 0 deg to 345 deg
 Increment: 15 deg
 *RST: 0 deg
 Default unit: deg

<h21abs>	numeric Square of magnitude of h_{21} Range: 0 to 1 *RST: 0
<h21phi>	numeric Phase of h_{21} Range: 0 deg to 345 deg Increment: 15 deg *RST: 0 deg Default unit: deg
<h22phi>	numeric Phase of h_{22} Range: 0 deg to 345 deg Increment: 15 deg *RST: 0 deg Default unit: deg

Example: See [Configuring MIMO settings](#)

Firmware/software: V2.0.10, SCC command V3.2.50

Options: R&S CMW-KS520

Manual operation: See ["2x2 channel matrix"](#) on page 213

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SCHModel:MIMO<Mimo> <h11abs>, <h11phi>, <h12abs>, <h12phi>, <h13abs>, <h13phi>, <h14abs>, <h14phi>, <h21abs>, <h21phi>, <h22abs>, <h22phi>, <h23abs>, <h23phi>, <h24abs>, <h24phi>[, <h31abs>, <h31phi>, <h32abs>, <h32phi>, <h33abs>, <h33phi>, <h34abs>, <h34phi>, <h41abs>, <h41phi>, <h42abs>, <h42phi>, <h43abs>, <h43phi>, <h44abs>, <h44phi>]

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:SCHModel:MIMO<Mimo> <h11abs>, <h11phi>, <h12abs>, <h12phi>, <h13abs>, <h13phi>, <h14abs>, <h14phi>, <h21abs>, <h21phi>, <h22abs>, <h22phi>, <h23abs>, <h23phi>, <h24abs>, <h24phi>[, <h31abs>, <h31phi>, <h32abs>, <h32phi>, <h33abs>, <h33phi>, <h34abs>, <h34phi>, <h41abs>, <h41phi>, <h42abs>, <h42phi>, <h43abs>, <h43phi>, <h44abs>, <h44phi>]

Configures the coefficients of the user-defined channel matrix, characterizing the radio channel for MIMO 4x2 or MIMO 4x4.

There are two types of parameters:

- <hnmabs> defines the square of the magnitude of the channel coefficient nm:

$$\langle h_{nm}abs \rangle = (h_{nm})^2$$
The sum of all <h1mabs> must equal 1:

$$\langle h_{11}abs \rangle + \langle h_{12}abs \rangle + \langle h_{13}abs \rangle + \langle h_{14}abs \rangle = 1$$
The same applies to <h2mabs>, <h3mabs> and <h4mabs>.
- <hnmphi> defines the phase of the channel coefficient nm:

$$\langle h_{nm}phi \rangle = \varphi(h_{nm})$$

The phase can be entered in steps of 15 degrees. The setting is rounded, if necessary.

The *RST values depend on <Mimo> and are listed as *RST 4x2 / *RST 4x4.

Suffix:

<Mimo> 42, 44
MIMO 4x2 or 4x4
The optional parameters in [...] are only relevant for 4x4.

<c> 1..7

Parameters:

<h11abs>	numeric Range: 0 to 1 *RST: 0.25 / 1
<h11phi>	numeric Range: 0 deg to 345 deg *RST: 0 deg / 0 deg Default unit: deg
<h12abs>	numeric Range: 0 to 1 *RST: 0.25 / 0
<h12phi>	numeric Range: 0 deg to 345 deg *RST: 0 deg / 0 deg Default unit: deg
<h13abs>	numeric Range: 0 to 1 *RST: 0.25 / 0
<h13phi>	numeric Range: 0 deg to 345 deg *RST: 0 deg / 0 deg Default unit: deg
<h14abs>	numeric Range: 0 to 1 *RST: 0.25 / 0
<h14phi>	numeric Range: 0 deg to 345 deg *RST: 270 deg / 0 deg Default unit: deg
<h21abs>	numeric Range: 0 to 1 *RST: 0.25 / 0

<h21phi>	numeric Range: 0 deg to 345 deg *RST: 270 deg / 0 deg Default unit: deg
<h22abs>	numeric Range: 0 to 1 *RST: 0.25 / 1
<h22phi>	numeric Range: 0 deg to 345 deg *RST: 90 deg / 0 deg Default unit: deg
<h23abs>	numeric Range: 0 to 1 *RST: 0.25 / 0
<h23phi>	numeric Range: 0 deg to 345 deg *RST: 90 deg / 0 deg Default unit: deg
<h24abs>	numeric Range: 0 to 1 *RST: 0.25 / 0
<h24phi>	numeric Range: 0 deg to 345 deg *RST: 180 deg / 0 deg Default unit: deg
<h31abs>	numeric Range: 0 to 1 *RST: - / 0
<h31phi>	numeric Range: 0 deg to 345 deg *RST: - / 0 deg Default unit: deg
<h32abs>	numeric Range: 0 to 1 *RST: - / 0
<h32phi>	numeric Range: 0 deg to 345 deg *RST: - / 0 deg Default unit: deg

<h33abs>	numeric Range: 0 to 1 *RST: - / 1
<h33phi>	numeric Range: 0 deg to 345 deg *RST: - / 0 deg Default unit: deg
<h34abs>	numeric Range: 0 to 1 *RST: - / 0
<h34phi>	numeric Range: 0 deg to 345 deg *RST: - / 0 deg Default unit: deg
<h41abs>	numeric Range: 0 to 1 *RST: - / 0
<h41phi>	numeric Range: 0 deg to 345 deg *RST: - / 0 deg Default unit: deg
<h42abs>	numeric Range: 0 to 1 *RST: - / 0
<h42phi>	numeric Range: 0 deg to 345 deg *RST: - / 0 deg Default unit: deg
<h43abs>	numeric Range: 0 to 1 *RST: - / 0
<h43phi>	numeric Range: 0 deg to 345 deg *RST: - / 0 deg Default unit: deg
<h44abs>	numeric Range: 0 to 1 *RST: - / 1

<h44phi>	numeric
	Range: 0 deg to 345 deg
	*RST: - / 0 deg
	Default unit: deg
Example:	See Configuring MIMO settings
Firmware/software:	V3.0.50, SCC command V3.2.50 V3.7.10 MIMO 4x4
Options:	R&S CMW-KS521 for MIMO 4x2 R&S CMW-KS540 for MIMO 4x4
Manual operation:	See "4x2 channel matrix" on page 213

2.6.15.4 MIMO beamforming settings TM 7/8

The following commands configure the beamforming settings for transmission mode 7 and 8.

CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:PZERo:MAPPing.....	761
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:PZERo:MAPPing.....	761
CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MODE.....	762
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MODE.....	762
CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:NOLayers.....	762
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:NOLayers.....	762
CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MATRix.....	763
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MATRix.....	763
CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:TM<8>:CHMatrix.....	764
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<8>:CHMatrix.....	764

CONFfigure:LTE:SIGN<i>:CONNection[:PCC]:PZERo:MAPPing <Port>
CONFfigure:LTE:SIGN<i>:CONNection:SCC<c>:PZERo:MAPPing <Port>

Selects the mapping of antenna port 0 to the RF output paths.

Only for TM 7 in scenarios with two RF output paths, without fading.

Suffix:

<c> 1..7

Parameters:

<Port> R1 | R1R2

R1: Map port 0 to the first RF output path.

R1R2: Map port 0 to both RF output paths.

*RST: R1

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS520

Manual operation: See ["Port 0 Mapping"](#) on page 214

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MODE <Mode>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MODE <Mode>

Selects the beamforming mode for TM 7 and 8.

Depending on other settings, only a subset of the values is allowed, see:

- TM 7: ["Beamforming Mode"](#) on page 215
- TM 8: ["Beamforming Mode"](#) on page 216

Suffix:

<c> 1..7

Parameters:

<Mode>

OFF | ON | TSBF | PMAT

OFF: Beamforming is disabled

ON: Beamforming is enabled. The configured beamforming matrix is used.

TSBF: Beamforming is enabled. The beamforming matrix is selected randomly as defined in 3GPP TS 36.521, annex B.4.1 and B.4.2.

PMAT: Beamforming is enabled. A precoding matrix is used as beamforming matrix, see [CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:PMATrix](#).

*RST: OFF

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.2.70, V3.5.40 PMAT added

Options: R&S CMW-KS520

Manual operation: See ["Beamforming Mode"](#) on page 215

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:NOLayers <Number>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:NOLayers <Number>

Selects the number of layers for transmission mode 8.

Suffix:

<c> 1..7

Parameters:

<Number>

L1 | L2

L1: single-layer beamforming

L2: dual-layer beamforming

*RST: L2

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.2.70

Options: R&S CMW-KS520

Manual operation: See ["Number of Layers"](#) on page 217

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:BEAMforming:MATRix <b11phi>[, <b12phi>[, <b11abs>, <b12abs>, <b21phi>, <b22phi>[, <b13phi>, <b14phi>[, <b13abs>, <b14abs>, <b23phi>, <b24phi>]]]]

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:BEAMforming:MATRix <b11phi>[, <b12phi>[, <b11abs>, <b12abs>, <b21phi>, <b22phi>[, <b13phi>, <b14phi>[, <b13abs>, <b14abs>, <b23phi>, <b24phi>]]]]

Configures the beamforming matrix coefficients for TM 7 and TM 8.

There are two types of parameters:

- <bnmabs> defines the square of the magnitude of the coefficient nm:
 $\text{<bnmabs>} = (b_{nm})^2$
- <bnmphi> defines the phase of the coefficient nm:
 $\text{<bnmphi>} = \varphi(b_{nm})$
 The phase can be entered in steps of 15 degrees. The setting is rounded, if necessary.

Depending on the size of your matrix, use the following parameters:

- 1x1: <b11phi>
- 1x2: <b11phi>, <b12phi>
- 2x2: <b11phi>, <b12phi>, <b11abs>, <b12abs>, <b21phi>, <b22phi>

The last six parameters are for future use and can always be omitted.

Suffix:

<c> 1..7

Parameters:

<b11phi>	numeric Range: 0 deg to 345 deg Default unit: deg
<b12phi>	numeric Range: 0 deg to 345 deg Default unit: deg
<b11abs>	numeric Range: 0 to 1
<b12abs>	numeric Range: 0 to 1
<b21phi>	numeric Range: 0 deg to 345 deg Default unit: deg
<b22phi>	numeric Range: 0 deg to 345 deg Default unit: deg

<b13phi>	numeric Range: 0 deg to 345 deg Default unit: deg
<b14phi>	numeric Range: 0 deg to 345 deg Default unit: deg
<b13abs>	numeric Range: 0 to 1
<b14abs>	numeric Range: 0 to 1
<b23phi>	numeric Range: 0 deg to 345 deg Default unit: deg
<b24phi>	numeric Range: 0 deg to 345 deg Default unit: deg

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.2.70

Options: R&S CMW-KS520

Manual operation: See ["Beamforming Matrix"](#) on page 215

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<8>:CHMatrix <abs11>, <phase11>, <abs12>, <phase12>, <abs21>, <phase21>, <abs22>, <phase22>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<8>:CHMatrix <abs11>, <phase11>, <abs12>, <phase12>, <abs21>, <phase21>, <abs22>, <phase22>

Configures the channel coefficients, characterizing the radio channel for TM 8.

Suffix:

<c> 1..7

<8> 8

Parameters:

<abs11> numeric
Square of magnitude of h_{11}
<abs11> + <abs12> must equal 1
Range: 0 to 1
*RST: 1

<phase11>	numeric Phase of h_{11} Range: 0 deg to 345 deg Increment: 15 deg *RST: 0 deg Default unit: deg
<abs12>	numeric Square of magnitude of h_{12} Range: 0 to 1 *RST: 1
<phase12>	numeric Phase of h_{12} Range: 0 deg to 345 deg Increment: 15 deg *RST: 0 deg Default unit: deg
<abs21>	numeric Square of magnitude of h_{21} <abs21> + <abs22> must equal 1 Range: 0 to 1 *RST: 1
<phase21>	numeric Phase of h_{21} Range: 0 deg to 345 deg Increment: 15 deg *RST: 0 deg Default unit: deg
<abs22>	numeric Square of magnitude of h_{22} Range: 0 to 1 *RST: 1
<phase22>	numeric Phase of h_{22} Range: 0 deg to 345 deg Increment: 15 deg *RST: 0 deg Default unit: deg

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS520

Manual operation: See ["Channel Matrix"](#) on page 217

2.6.15.5 MIMO TM 9 settings

The following commands configure settings that are specific for transmission mode 9.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:NTXantennas	766
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:NTXantennas	766
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CODewords	767
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CODewords	767
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:PMATrix	767
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:PMATrix	767
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:POWer	768
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:POWer	768
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:APORts	768
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:APORts	768
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:RESource	769
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:RESource	769
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:SUBFrame	769
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:SUBFrame	769
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:ZP:BITS	769
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:ZP:BITS	769
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:ZP:CSIRs:SUBFrame	770
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:ZP:CSIRs:SUBFrame	770
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:TWO<line>	770
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:TWO<line>	770
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:FOUR<line>	771
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:FOUR<line>	771
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:EIGHt<line>	773
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:EIGHt<line>	773
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:MIMO<Mimo>: MSELection	775
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:MIMO<Mimo>: MSELection	775
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:MIMO<Mimo>:LINE<line>...	775
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:MIMO<Mimo>: LINE<line>	775

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:NTXantennas <Antennas>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:NTXantennas

<Antennas>

Selects the number of downlink TX antennas for TM 9.

Suffix:

<no> 9

<c> 1..7

Parameters:

<Antennas> TWO | FOUR | EIGHt

*RST: TWO

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: TWO (2x2): R&S CMW-KS520
 TWO (2x4): R&S CMW-KS540
 FOUR (4x2): R&S CMW-KS521
 FOUR (4x4): R&S CMW-KS540
 EIGHT (8x2): R&S CMW-KS522
 EIGHT (8x4): R&S CMW-KS541

Manual operation: See ["Number Tx Antennas TM 9"](#) on page 218

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CODewords <Codewords>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CODewords
 <Codewords>

Selects the number of code words for TM 9.

Suffix:

<no> 9
 <c> 1..7

Parameters:

<Codewords> ONE | TWO | FOUR
 *RST: TWO

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10, V3.7.10 FOUR

Options: R&S CMW-KS520 or -KS540

Manual operation: See ["Codewords / Layers"](#) on page 219

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:PMATrix <Mode>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:PMATrix <Mode>

Selects the second precoding matrix for TM 9.

Suffix:

<c> 1..7
 <no> 9

Parameters:

<Mode> PMI0 | PMI1 | PMI2 | PMI3 | PMI4 | PMI5 | PMI6 | PMI7 | PMI8 |
 PMI9 | PMI10 | PMI11 | PMI12 | PMI13 | PMI14 | PMI15
 Matrix according to PMI 0, PMI 1, ... PMI 15.
 *RST: PMI0

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS522 or -KS541

Manual operation: See ["Precoding Matrix 2"](#) on page 219

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:POWer <Power>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:POWer <Power>

Sets the value P_c to be signaled to the UE. P_c is the assumed ratio of the RS EPRE to the CSI-RS EPRE.

Suffix:

<no> 9

<c> 1..7

Parameters:

<Power> integer
 Range: -8 dB to 15 dB
 *RST: 0 dB
 Default unit: dB

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS520 or -KS540

Manual operation: See "[CSI-RS Power](#)" on page 219

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:APORts <Ports>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:APORts <Ports>

Selects the antenna ports used for the CSI-RS for TM 9.

Suffix:

<no> 9

<c> 1..7

Parameters:

<Ports> NONE | P15 | P1516 | P1518 | P1522
NONE: no CSI-RS
P15: port 15
P1516: port 15 and 16
P1518: port 15 to 18
P1522: port 15 to 22
 *RST: P1516

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: P1516 (nx2): R&S CMW-KS520
 P1516 (nx4): R&S CMW-KS540
 P1518 (nx2): R&S CMW-KS521
 P1518 (nx4): R&S CMW-KS540
 P1522 (nx2): R&S CMW-KS522
 P1522 (nx4): R&S CMW-KS541

Manual operation: See "[CSI-RS Antenna Ports](#)" on page 219

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:RESource
 <Resource>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:RESource
 <Resource>

Selects the CSI reference signal configuration.

Suffix:

<no> 9

<c> 1..7

Parameters:

<Resource> numeric
 Range: 0 to 31
 *RST: 0

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS520 or -KS540

Manual operation: See "[CSI-RS Configuration](#)" on page 220

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CSIRs:SUBFrame <Config>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CSIRs:SUBFrame
 <Config>

Selects the CSI-RS subframe configuration.

Suffix:

<no> 9

<c> 1..7

Parameters:

<Config> numeric
 Range: 0 to 154
 *RST: 2

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS520 or -KS540

Manual operation: See "[CSI-RS Subframe Cfg](#)" on page 220

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:ZP:BITS <Bits>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:ZP:BITS <Bits>

Specifies the bitmap "ZeroPowerCSI-RS".

Suffix:

<no> 9

<c> 1..7

Parameters:

<Bits> binary
16-bit value
Range: #B00000000000000000 to #B1111111111111111
*RST: #B00000000000000000

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS520 or -KS540

Manual operation: See ["Zero Power CSI-RS"](#) on page 220

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:ZP:CSIRs:SUBFrame
<Config>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:ZP:CSIRs:SUBFrame
<Config>

Selects the zero power CSI-RS subframe configuration.

Suffix:

<no> 9
<c> 1..7

Parameters:

<Config> numeric
Range: 0 to 154
*RST: 2

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS520 or -KS540

Manual operation: See ["ZP CSI-RS Subframe Cfg"](#) on page 220

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:TWO<line>
<h1xabs>, <h1xphi>, <h2xphi>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:TWO<line>
<h1xabs>, <h1xphi>, <h2xphi>

Configures the 2x2 channel coefficients for TM 9.

The value <h2xabs> is calculated automatically from <h1xabs>, so that the sum of the values equals 1.

A query returns <h1xabs>, <h1xphi>, <h2xabs>, <h2xphi>.

Suffix:

<no> 9

<line> 1, 2
Selects the matrix line (value "x" in the other parameters)

<c> 1..7

Parameters:

<h1xabs> numeric
Square of magnitude of h_{1x}
Range: 0 to 1

<h1xphi> numeric
Phase of h_{1x}
Range: 0 deg to 345 deg
Increment: 15 deg
Default unit: deg

<h2xphi> numeric
Phase of h_{2x}
Range: 0 deg to 345 deg
Increment: 15 deg
Default unit: deg

Return values:

<h2xabs> float
Square of magnitude of h_{2x}
Range: 0 to 1

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS520

Manual operation: See ["Channel Matrix"](#) on page 220

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:FOUR<line>
<h1xabs>, <h1xphi>, <h2xabs>, <h2xphi>, <h3xabs>, <h3xphi>, <h4xphi>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:FOUR<line>
<h1xabs>, <h1xphi>, <h2xabs>, <h2xphi>, <h3xabs>, <h3xphi>, <h4xphi>

Configures the 4x2 channel coefficients for TM 9.

There are two types of parameters:

- <hnmabs> defines the square of the magnitude of the channel coefficient nm:
 $\text{<hnmabs>} = (h_{nm})^2$
The sum of all values in one matrix line must not be greater than 1. <h4xabs> is calculated automatically, so that the sum equals 1.
- <hnmphi> defines the phase of the channel coefficient nm:
 $\text{<hnmphi>} = \varphi(h_{nm})$

A query returns <h1xabs>, <h1xphi>, <h2xabs>, ..., <h4xabs>, <h4xphi>.

Suffix:

<no>	9
<line>	1, 2 Selects the matrix line (value "x" in the other parameters)
<c>	1..7

Parameters:

<h1xabs>	numeric Range: 0 to 1
<h1xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h2xabs>	numeric Range: 0 to 1
<h2xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h3xabs>	numeric Range: 0 to 1
<h3xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h4xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg

Return values:

<h4xabs>	float Range: 0 to 1
----------	------------------------

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS521

Manual operation: See ["Channel Matrix"](#) on page 220

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:EIGHt<line>

<h1xabs>, <h1xphi>, <h2xabs>, <h2xphi>, <h3xabs>, <h3xphi>, <h4xabs>, <h4xphi>, <h5xabs>, <h5xphi>, <h6xabs>, <h6xphi>, <h7xabs>, <h7xphi>, <h8xphi>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:EIGHt<line>

<h1xabs>, <h1xphi>, <h2xabs>, <h2xphi>, <h3xabs>, <h3xphi>, <h4xabs>, <h4xphi>, <h5xabs>, <h5xphi>, <h6xabs>, <h6xphi>, <h7xabs>, <h7xphi>, <h8xphi>

Configures the 8x2 channel coefficients for TM 9.

There are two types of parameters:

- <hnmabs> defines the square of the magnitude of the channel coefficient nm:
 $\text{<hnmabs>} = (h_{nm})^2$
 The sum of all values in one matrix line must not be greater than 1. <h8xabs> is calculated automatically, so that the sum equals 1.
- <hnmphi> defines the phase of the channel coefficient nm:
 $\text{<hnmphi>} = \varphi(h_{nm})$

A query returns <h1xabs>, <h1xphi>, <h2xabs>, ..., <h8xabs>, <h8xphi>.

Suffix:

<no>	9
<line>	1, 2 Selects the matrix line (value "x" in the other parameters)
<c>	1..7

Parameters:

<h1xabs>	numeric Range: 0 to 1
<h1xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h2xabs>	numeric Range: 0 to 1
<h2xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h3xabs>	numeric Range: 0 to 1
<h3xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg

<h4xabs>	numeric Range: 0 to 1
<h4xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h5xabs>	numeric Range: 0 to 1
<h5xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h6xabs>	numeric Range: 0 to 1
<h6xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h7xabs>	numeric Range: 0 to 1
<h7xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h8xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg

Return values:

<h8xabs>	float Range: 0 to 1
----------	------------------------

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS522

Manual operation: See ["Channel Matrix"](#) on page 220

**CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:MIMO<Mimo>:
MSElection <Selection>**

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:MIMO<Mimo>:
MSElection <Selection>**

Selects a predefined channel matrix or the user-defined channel matrix for MIMO 4x4 plus TM 9.

Suffix:

<c> 1..7

<Mimo> 44

<no> 9

Parameters:

<Selection> UDEFined | CM3Gpp | HADamard | IDENtity
User-defined matrix, 3GPP channel matrix, Hadamard matrix,
identity matrix
*RST: UDEF

Firmware/software: V3.7.10

Options: R&S CMW-KS540

Manual operation: See "[Matrix Selection](#)" on page 221

**CONFigure:LTE:SIGN<i>:CONNection[:PCC]:TM<no>:CMATrix:MIMO<Mimo>:
LINE<line> <h1xabs>, <h1xphi>, <h2xabs>, <h2xphi>, <h3xabs>, <h3xphi>,
<h4xabs>, <h4xphi>**

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:TM<no>:CMATrix:MIMO<Mimo>:
LINE<line> <h1xabs>, <h1xphi>, <h2xabs>, <h2xphi>, <h3xabs>, <h3xphi>,
<h4xabs>, <h4xphi>**

Configures the coefficients of the user-defined 4x4 channel matrix for TM 9.

There are two types of parameters:

- <hnmabs> defines the square of the magnitude of the channel coefficient nm:
 $\text{<hnmabs>} = (h_{nm})^2$
The sum of all values in one matrix line must not be greater than 1. <h4xabs> is calculated automatically, so that the sum equals 1.
- <hnmphi> defines the phase of the channel coefficient nm:
 $\text{<hnmphi>} = \varphi(h_{nm})$

A query returns <h1xabs>, <h1xphi>, <h2xabs>, ..., <h4xabs>, <h4xphi>.

Suffix:

<no> 9

<line> 1..4
Selects the matrix line (value "x" in the other parameters)

<c> 1..7

<Mimo> 44

Parameters:

<h1xabs>	numeric Range: 0 to 1
<h1xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h2xabs>	numeric Range: 0 to 1
<h2xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h3xabs>	numeric Range: 0 to 1
<h3xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg
<h4xphi>	numeric Range: 0 deg to 345 deg Increment: 15 deg Default unit: deg

Return values:

<h4xabs>	float Range: 0 to 1
----------	------------------------

Example: See [Configuring MIMO settings](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS540

Manual operation: See "[Channel Matrix](#)" on page 220

2.6.15.6 ROHC settings

The following commands configure robust header compression.

CONFigure:LTE:SIGN<i>:CONNection:ROHC:ENABLE	777
CONFigure:LTE:SIGN<i>:CONNection:ROHC:EFOR	777
CONFigure:LTE:SIGN<i>:CONNection:ROHC:PROFiles	777
CONFigure:LTE:SIGN<i>:CONNection:ROHC:ULONLY:ENABle	778
CONFigure:LTE:SIGN<i>:CONNection:ROHC:ULONLY:PROFiles	778

CONFigure:LTE:SIGN<i>:CONNection:ROHC:ENABLE <Enable>

Enables or disables bidirectional header compression.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["Enable Header Compression"](#) on page 233

CONFigure:LTE:SIGN<i>:CONNection:ROHC:EFOR <For>

Selects for which types of dedicated bearers header compression is enabled.

Parameters:

<For> VVB | ADB
VVB: voice and video bearers
ADB: all dedicated bearers
*RST: VVB

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS510

Manual operation: See ["Enable for"](#) on page 234

CONFigure:LTE:SIGN<i>:CONNection:ROHC:PROFiles <Profile0x0001>, <Profile0x0002>, <Profile0x0004>[, <Profile0x0006>]

Enables header compression profiles for bidirectional header compression.

Parameters:

<Profile0x0001> OFF | ON
Profile 1, for IP/UDP/RTP
*RST: OFF

<Profile0x0002> OFF | ON
Profile 2, for IP/UDP/...
*RST: OFF

<Profile0x0004> OFF | ON
Profile 4, for IP/...
*RST: OFF

<Profile0x0006> OFF | ON
 Profile 6, for IP/TCP/...
 *RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.80, V3.7.60 added <Profile0x0006>

Options: R&S CMW-KS510

Manual operation: See ["Profile ..."](#) on page 234

CONFigure:LTE:SIGN<i>:CONNection:ROHC:ULONly:ENABLE <Enable>

Enables or disables uplink-only header compression.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.7.60

Options: R&S CMW-KS510

Manual operation: See ["Enable Uplink Only Header Compression"](#) on page 234

CONFigure:LTE:SIGN<i>:CONNection:ROHC:ULONly:PROFiles <profile0x0006>

Enables header compression profiles for uplink-only header compression.

Parameters:

<profile0x0006> OFF | ON
 Profile 6, for IP/TCP/...
 *RST: OFF

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.7.60

Options: R&S CMW-KS510

Manual operation: See ["Profile ..."](#) on page 234

2.6.15.7 PDCCH settings

The following commands configure the used PDCCH configuration.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:SYMBol	779
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:SYMBol	779
SENSe:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:PSYMBOLs?	779
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:PSYMBOLs?	779
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:ALEVel	780
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:ALEVel	780

SENSe:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:ALEVel?	780
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:ALEVel?	780
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:RPDCch	781
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:RPDCch	781

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:SYMBol <PDCCH>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:SYMBol <PDCCH>

Configures the number of PDCCH symbols per normal subframe.

Suffix:

<c> 1..7

Parameters:

<PDCCH> AUTO | P1 | P2 | P3 | P4

AUTO: automatic configuration depending on scheduling type

P1 to P4: 1, 2, 3, 4 symbols

***RST:** AUTO

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.82

Manual operation: See ["PDCCH Symbol Config, #PDCCH Symbols"](#) on page 227

Table 2-75: Allowed values, depending on cell bandwidth

	1.4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
AUTO				X	X	X
P1				X	X	X
P2		X	X	X	X	X
P3		X	X	X	X	X
P4	X					

SENSe:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:PSYMBOLs?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:PSYMBOLs?

Queries the number of PDCCH symbols per normal subframe.

Suffix:

<c> 1..7

Return values:

<PDCCHsymbols> decimal

Range: 1 to 4

Example: See [Configuring general connection settings part 2](#)

Usage: Query only

Firmware/software: V3.0.50, SCC command V3.2.50

Manual operation: See ["PDCCH Symbol Config, #PDCCH Symbols"](#) on page 227

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:ALEVel <Aggregationlevel>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:ALEVel <Aggregationlevel>

Configures the aggregation levels for DCI messages with C-RNTI.

The individual values have prerequisites, see manual operation.

Suffix:

<c> 1..7

Parameters:

<Aggregationlevel> AUTO | D8U4 | D4U4 | D4U2 | D1U1 | D8U8

AUTO: automatic configuration

D<a>U: <a> CCE for DCI messages for the DL, CCE for messages for the UL

*RST: AUTO

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.2.82, V3.7.10 added D8U8

Manual operation: See ["Aggr. Level DL/UL Config, Aggreg. Level ..."](#) on page 227

SENSe:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:ALEVel?

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:ALEVel?

Queries the used PDCCH aggregation levels.

Suffix:

<c> 1..7

Return values:

<DLDCI_CRNTI> decimal
DCI for DL with C-RNTI
Range: 1 to 8

<ULDCI_CRNTI> decimal
DCI for UL with C-RNTI
Range: 1 to 8

<DLDCI_SIRNTI> decimal
DCI for DL with SI-RNTI
Range: 1 to 8

Example: See [Configuring general connection settings part 2](#)

Usage: Query only

Firmware/software: V3.0.50, SCC command V3.2.50

Manual operation: See ["Aggr. Level DL/UL Config, Aggreg. Level ..."](#) on page 227

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PDCCh:RPDCch <ReducedPDCCH>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:PDCCh:RPDCch
 <ReducedPDCCH>

Enables / disables the reduction of PDCCH resources.

Suffix:

<c> 1..7

Parameters:

<ReducedPDCCH> OFF | ON
 *RST: OFF

Firmware/software: V3.0.50, SCC command V3.2.50

Manual operation: See ["Reduced PDCCH"](#) on page 228

2.6.15.8 PUCCH settings

The following commands configure the used PUCCH configuration.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:PUCCh:FFCA <Format>

Selects the PUCCH format used for carrier aggregation scenarios.

Parameters:

<Format> F1BCs | F3 | F4 | F5
F1BCs: PUCCH format 1b with channel selection (if allowed)
F3: PUCCH format 3
F4: PUCCH format 4
F5: PUCCH format 5
 *RST: F1BC

Example: See [Configuring general connection settings part 2](#)

Firmware/software: V3.7.10, V3.7.50 added F4 and F5

Manual operation: See ["PUCCH"](#) on page 228

SENSe:LTE:SIGN<i>:CONNection[:PCC]:PUCCh:FFCA?

Queries the PUCCH format used for carrier aggregation scenarios.

Return values:

<Format> F1BCs | F3 | F4 | F5
F1BCs: PUCCH format 1b with channel selection
F3: PUCCH format 3
F4: PUCCH format 4
F5: PUCCH format 5

Usage: Query only

Firmware/software: V3.7.10, V3.7.50 added F4 and F5

Manual operation: See ["PUCCH"](#) on page 228

2.6.15.9 HARQ connection settings

The following commands configure HARQ connection settings.

CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:ENABLE.....	782
CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:NHT.....	782
CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:DPHich.....	783
CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:MAXTx.....	783
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:ENABLE.....	783
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:NHT.....	784
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:RVSequence.....	784
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence:LENGth.....	784
CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence.....	785

CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:ENABLE <Enable>

Enables or disables HARQ for uplink transmissions.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring HARQ](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS510

Manual operation: See ["UL HARQ"](#) on page 230

CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:NHT <Number>

Specifies the maximum number of uplink transmissions, including initial transmissions and retransmissions.

Parameters:

<Number> numeric
 Range: 1 to 5
 *RST: 4

Example: See [Configuring HARQ](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS510

Manual operation: See ["Number of HARQ Transmissions"](#) on page 230

CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:DPHich <Mode>

Selects how the UE is informed about required UL retransmissions / successful UL transmissions, for UL HARQ.

Parameters:

<Mode> D0ONLY | PHICHonly | D0PHich | PNACK | PND0

D0ONLY

DCI-0 only, normal operation

PHICHonly

PHICH only, normal operation

D0PHich

DCI-0 & PHICH, normal operation

PNACK

PHICH NACK, always retransmission

PND0

PHICH NACK & DCI-0, always retransmission

*RST: D0ONLY

Example: See [Configuring HARQ](#)

Firmware/software: V3.5.50, V3.7.50 PNACK and PND0

Options: R&S CMW-KS510

Manual operation: See "[DCI-0 / PHICH](#)" on page 230

CONFigure:LTE:SIGN<i>:CONNection:HARQ:UL:MAXTx <Number>

Specifies the signaling parameter "maxHARQ-Tx". The setting is only relevant for the scheduling type SPS.

Parameters:

<Number> numeric

Range: 1 to 4

*RST: 1

Firmware/software: V3.5.30

Options: R&S CMW-KS510

Manual operation: See "[maxHARQ-Tx](#)" on page 230

CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:ENABLE <Enable>

Enables or disables HARQ for downlink transmissions.

Parameters:

<Enable> OFF | ON

*RST: OFF

Example: See [Configuring HARQ](#)

Firmware/software: V3.0.50

Options: R&S CMW-KS510 for scenarios without carrier aggregation
R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["DL HARQ"](#) on page 231

CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:NHT <Number>

Specifies the maximum number of downlink transmissions, including initial transmissions and retransmissions.

Parameters:

<Number> numeric
Range: 2 to 4
*RST: 2

Example: See [Configuring HARQ](#)

Firmware/software: V3.0.50

Manual operation: See ["Number of HARQ Transmissions"](#) on page 231

CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:RVSequence <Sequence>

Selects the redundancy version sequence for DL HARQ.

Parameters:

<Sequence> TS1 | TS4 | UDEFined
TS1: according to 3GPP TS 36.101
TS4: according to 3GPP TS 36.104
UDEFined: user-defined sequence, see [CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence](#)
*RST: TS1

Example: See [Configuring HARQ](#)

Firmware/software: V3.0.50

Manual operation: See ["Redundancy Version Coding Sequence, User Defined Sequence"](#) on page 231

CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence:LENGth <Length>

Specifies the length of the user-defined redundancy version sequence.

Parameters:

<Length> numeric
Range: 1 to 4
*RST: 4

Example: See [Configuring HARQ](#)

Firmware/software: V3.0.50

Manual operation: See ["Redundancy Version Coding Sequence, User Defined Sequence"](#) on page 231

CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence <Value1>[, <Value2>, <Value3>, <Value4>]

Specifies the user-defined redundancy version sequence. Only the first n values are used, according to the specified length, see [CONFigure:LTE:SIGN<i>:CONNection:HARQ:DL:UDSequence:LENGTh](#).

You can either set the first value only (relevant for initial transmissions) or all four values.

Parameters:

<Value1>	numeric
	In this software version fixed set to 0
	Range: 0
	*RST: 0
<Value2>	numeric
	Range: 0 to 3
	*RST: 0
<Value3>	numeric
	Range: 0 to 3
	*RST: 0
<Value4>	numeric
	Range: 0 to 3
	*RST: 0

Example: See [Configuring HARQ](#)

Firmware/software: V3.0.50

Manual operation: See ["Redundancy Version Coding Sequence, User Defined Sequence"](#) on page 231

2.6.15.10 Connected DRX settings

The following commands configure discontinuous reception (DRX) of the UE during an established connection.

CONFigure:LTE:SIGN<i>:CONNection:CDRX:ENABLE	786
CONFigure:LTE:SIGN<i>:CONNection:CDRX:ODTimer	786
CONFigure:LTE:SIGN<i>:CONNection:CDRX:ITIMER	786
CONFigure:LTE:SIGN<i>:CONNection:CDRX:RTIMER	787
CONFigure:LTE:SIGN<i>:CONNection:CDRX:LDCYcle	787
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SOFFset	788
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SCENable	788
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SDCYcle	788
CONFigure:LTE:SIGN<i>:CONNection:CDRX:SCTimer	788

CONFigure:LTE:SIGN<i>:CONNection:UDScheduling.....	789
CONFigure:LTE:SIGN<i>:CONNection:IUGNrB.....	789
CONFigure:LTE:SIGN<i>:CONNection:IUGMcsidx.....	789
CONFigure:LTE:SIGN<i>:CONNection:SRPRIndex.....	790
CONFigure:LTE:SIGN<i>:CONNection:SRCIndex.....	790

CONFigure:LTE:SIGN<i>:CONNection:CDRX:ENABLE <Enable>

Enables or disables DRX and selects a set of DRX settings.

Parameters:

<Enable> DRXS | DRXL | UDEFined | ON | OFF
DRXS: DRX_S, 3GPP TS 36.521-3, table H.3.6-1
DRXL: DRX_L, 3GPP TS 36.521-3, table H.3.6-2
UDEFined: user-defined DRX settings
ON: enables DRX with previously selected set
OFF: disables DRX
 *RST: OFF

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20

Manual operation: See ["Connected DRX"](#) on page 236

CONFigure:LTE:SIGN<i>:CONNection:CDRX:ODTimer <Timer>

Configures the onDurationTimer. The value must be smaller than or equal to the long DRX cycle duration.

Parameters:

<Timer> PSF1 | PSF2 | PSF3 | PSF4 | PSF5 | PSF6 | PSF8 | PSF10 |
 PSF20 | PSF30 | PSF40 | PSF50 | PSF60 | PSF80 | PSF100 |
 PSF200 | PSF300 | PSF400 | PSF500 | PSF600 | PSF800 |
 PSF1000 | PSF1200 | PSF1600
 PSFn means n PDCCH subframes
 *RST: PSF2

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20, V3.7.10: PSF300 to PSF1600

Manual operation: See ["On Duration Timer"](#) on page 236

CONFigure:LTE:SIGN<i>:CONNection:CDRX:ITIMer <Timer>

Configures the "drx-InactivityTimer".

Parameters:

<Timer> PSF1 | PSF2 | PSF3 | PSF4 | PSF5 | PSF6 | PSF8 | PSF10 | PSF20 | PSF30 | PSF40 | PSF50 | PSF60 | PSF80 | PSF100 | PSF200 | PSF300 | PSF500 | PSF750 | PSF1280 | PSF1920 | PSF2560
 PSFn means n PDCCH subframes
 *RST: PSF100

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20

Manual operation: See ["Inactivity Timer"](#) on page 236

CONFigure:LTE:SIGN<i>:CONNection:CDRX:RTIMEr <Timer>

Configures the "drx-RetransmissionTimer".

Parameters:

<Timer> PSF0 | PSF1 | PSF2 | PSF4 | PSF6 | PSF8 | PSF16 | PSF24 | PSF33 | PSF40 | PSF64 | PSF80 | PSF96 | PSF112 | PSF128 | PSF160 | PSF320
 PSFn means n PDCCH subframes
 *RST: PSF16

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20, V3.7.10: PSF0 and PSF40 to PSF320

Manual operation: See ["Retransmission Timer"](#) on page 236

CONFigure:LTE:SIGN<i>:CONNection:CDRX:LDCYcle <Cycle>

Configures the duration of one long DRX cycle. If short DRX cycles are enabled, the long DRX cycle duration must be divisible by the short DRX cycle duration.

Parameters:

<Cycle> SF10 | SF20 | SF32 | SF40 | SF60 | SF64 | SF70 | SF80 | SF128 | SF160 | SF256 | SF320 | SF512 | SF640 | SF1024 | SF1280 | SF2048 | SF2560 | SF5120 | SF10240
 SFn means n subframes
 *RST: SF40

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20
 V3.7.10: SF60, SF70, SF5120, SF10240

Options: R&S CMW-KS590 for SF5120, SF10240

Manual operation: See ["Long DRX Cycle"](#) on page 236

CONFigure:LTE:SIGN<i>:CONNection:CDRX:SOFFset <Offset>

Configures the "drxStartOffset", shifting all DRX cycles.

Parameters:

<Offset> numeric
 Range: 0 to length of long DRX cycle - 1
 *RST: 0

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20

Manual operation: See ["Start Offset"](#) on page 237

CONFigure:LTE:SIGN<i>:CONNection:CDRX:SCENable <Enable>

Enables or disables short DRX cycles.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20

Manual operation: See ["Enable"](#) on page 237

CONFigure:LTE:SIGN<i>:CONNection:CDRX:SDCYcle <Cycle>

Configures the duration of one short DRX cycle. The long DRX cycle duration must be divisible by the short DRX cycle duration.

Parameters:

<Cycle> SF2 | SF4 | SF5 | SF8 | SF10 | SF16 | SF20 | SF32 | SF40 |
 SF64 | SF80 | SF128 | SF160 | SF256 | SF320 | SF512 | SF640
 SFn means n subframes
 If a query returns NAV, short cycles are disabled.
 *RST: SF2

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20, V3.7.10 SF4

Manual operation: See ["Short DRX Cycle"](#) on page 237

CONFigure:LTE:SIGN<i>:CONNection:CDRX:SCTimer <Timer>

Configures the short cycle timer.

Parameters:

<Timer> numeric
 Number of short DRX cycles
 Range: 1 to 16
 *RST: 1

Example: See [Configuring connected DRX](#)

Firmware/software: V3.2.20

Manual operation: See ["Short Cycle Timer"](#) on page 237

CONFigure:LTE:SIGN<i>:CONNection:UDScheduling <Enable>

Enables or disables uplink dynamic scheduling.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring connected DRX](#)

Firmware/software: V3.7.60

Manual operation: See ["UL Dynamic Scheduling"](#) on page 237

CONFigure:LTE:SIGN<i>:CONNection:IUGNrb <nrb>

Configures the number of resource blocks for the first UL grant after an UL grant request of the UE (uplink dynamic scheduling).

Parameters:

<nrb> decimal
 Range: 0 to 100 (depends on cell BW)
 *RST: 6

Example: See [Configuring connected DRX](#)

Firmware/software: V3.7.60

Manual operation: See ["Initial UL Grant #RBs / MCS Index"](#) on page 237

CONFigure:LTE:SIGN<i>:CONNection:IUGMcsidx <McsIndex>

Configures the MCS index value for the first UL grant after an UL grant request of the UE (uplink dynamic scheduling).

Parameters:

<McsIndex> numeric
 Range: 0 to 28
 *RST: 6

Example: See [Configuring connected DRX](#)

Firmware/software: V3.7.60

Manual operation: See ["Initial UL Grant #RBs / MCS Index"](#) on page 237

CONFigure:LTE:SIGN<i>:CONNection:SRPRIndex <Index>

Specifies the "sr-PUCCH ResourceIndex".

Parameters:

<Index>	integer
Range:	0 to 2047
*RST:	41

Example: See [Configuring connected DRX](#)

Firmware/software: V3.5.10

Manual operation: See ["sr-PUCCH ResourceIndex"](#) on page 238

CONFigure:LTE:SIGN<i>:CONNection:SRCIndex <Index>

Specifies the "sr-ConfigIndex".

Parameters:

<Index>	integer
Range:	0 to 157
*RST:	27

Example: See [Configuring connected DRX](#)

Firmware/software: V3.5.10

Manual operation: See ["sr-ConfigIndex"](#) on page 238

2.6.15.11 eDRX settings

The following commands configure extended discontinuous reception (eDRX) of the UE.

CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:ENABLE	790
CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:PTWindow	791
CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:CLENGTH	791

CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:ENABLE <Enable>

Enables or disables eDRX.

Parameters:

<Enable>	OFF ON
*RST:	OFF

Example: See [Configuring extended DRX](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS590

Manual operation: See ["Enabled"](#) on page 232

CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:PTWindow <Window>

Configures the duration of one paging time window in idle mode.

Parameters:

<Window> W1280 | W2560 | W3840 | W5120 | W6400 | W7680 | W8960 |
W10240 | W11520 | W12800 | W14080 | W15360 | W16640 |
W17920 | W19200 | W20480
Wn means n subframes
*RST: W128

Example: See [Configuring extended DRX](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS590

Manual operation: See ["Paging Time Window"](#) on page 232

CONFigure:LTE:SIGN<i>:CONNection:CDRX:IMODE:CLENGTH <Length>

Configures the duration of one eDRX cycle in idle mode.

Parameters:

<Length> L512 | L1024 | L2048 | L4096 | L6144 | L8192 | L10240 |
L12288 | L14336 | L16384 | L32768 | L65536 | L131072 |
L262144
Ln means n radio frames
*RST: L512

Example: See [Configuring extended DRX](#)

Firmware/software: V3.7.10

Options: R&S CMW-KS590

Manual operation: See ["eDRX Cycle Length"](#) on page 232

2.6.15.12 Capability request settings

The following commands define which capability information is requested from the UE.

CONFigure:LTE:SIGN<i>:UECapability:RUTra	792
CONFigure:LTE:SIGN<i>:UECapability:RGCS	792
CONFigure:LTE:SIGN<i>:UECapability:RGPS	792
CONFigure:LTE:SIGN<i>:UECapability:RFBands:ALL	792
CONFigure:LTE:SIGN<i>:UECapability:RRFormat	793
CONFigure:LTE:SIGN<i>:UECapability:SFC	793

CONFigure:LTE:SIGN<i>:UECapability:RUTRa <Enable>

Selects whether UTRA capabilities are requested from the UE (entry "utra" in field "ue-CapabilityRequest" of "ueCapabilityEnquiry" message).

Parameters:

<Enable> OFF | ON
 *RST: ON

Example: See [Configuring capability requests](#)

Firmware/software: V3.7.60

Manual operation: See ["Request UTRA Capabilities"](#) on page 238

CONFigure:LTE:SIGN<i>:UECapability:RGCS <Enable>

Selects whether GERAN CS capabilities are requested from the UE (entry "geran-cs" in field "ue-CapabilityRequest" of "ueCapabilityEnquiry" message).

Parameters:

<Enable> OFF | ON
 *RST: ON

Example: See [Configuring capability requests](#)

Firmware/software: V3.7.60

Manual operation: See ["Request GERAN CS Capabilities"](#) on page 239

CONFigure:LTE:SIGN<i>:UECapability:RGPS <Enable>

Selects whether GERAN PS capabilities are requested from the UE (entry "geran-ps" in field "ue-CapabilityRequest" of "ueCapabilityEnquiry" message).

Parameters:

<Enable> OFF | ON
 *RST: ON

Example: See [Configuring capability requests](#)

Firmware/software: V3.7.60

Manual operation: See ["Request GERAN PS Capabilities"](#) on page 239

CONFigure:LTE:SIGN<i>:UECapability:RFBands:ALL {<Enable>, <Band>}...

Configures the list of operating bands for the information element "requestedFrequencyBands" of the "ueCapabilityEnquiry" message.

The command has 32 parameters, for 16 entries with two parameters each:

{<Enable>, <Band>}_{entry 1}, {<Enable>, <Band>}_{entry 2}, ..., {<Enable>, <Band>}_{entry 16}

Parameters:

<Enable>	OFF ON Disables or enables the entry *RST: OFF
<Band>	UDEFined OB1 ... OB46 OB48 ... OB53 OB65 ... OB76 OB85 OB87 OB88 OB106 OB250 OB252 OB255 Assigns a band to the entry

Example: See [Configuring capability requests](#)

Firmware/software: V3.5.40, some bands added in later versions

Manual operation: See ["Requested Frequency Bands, LTE"](#) on page 239

CONFigure:LTE:SIGN<i>:UECapability:RRFormat <Enable>

Enables the optional field "requestReducedFormat-r13" in the "ueCapabilityEnquiry" message.

Parameters:

<Enable>	OFF ON OFF: field omitted ON: field included *RST: ON
----------	--

Example: See [Configuring capability requests](#)

Firmware/software: V3.7.60

Manual operation: See ["Request Reduced Format"](#) on page 239

CONFigure:LTE:SIGN<i>:UECapability:SFC <Enable>

Enables the optional field "requestSkipFallbackComb-r13" in the "ueCapabilityEnquiry" message.

Parameters:

<Enable>	OFF ON OFF: field omitted ON: field included *RST: OFF
----------	---

Example: See [Configuring capability requests](#)

Firmware/software: V3.7.60

Manual operation: See ["Skip Fallback Combinations"](#) on page 239

2.6.15.13 General scheduling commands

The following commands are related to the scheduling settings and are independent of the scheduling type.

Scheduling-type specific commands are described in the subsequent sections.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:DLEQual.....	794
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:DLEQual.....	794
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:HdUPlex.....	794
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]:STReam<s>?.....	795
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>:STReam<s>?.....	795
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]?.....	795
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>?.....	795
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:ALL?.....	795
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?.....	796
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:SCC<c>?.....	796
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:ALL?.....	796
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:TBURsts.....	796
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:BLENgth.....	797
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:PBTR.....	797
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:BLENgth.....	798

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:DLEQual <Enable>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:DLEQual <Enable>

Enables or disables the coupling of all MIMO downlink streams.

When you switch on the coupling, the settings for DL stream 1 are applied to all DL streams.

With enabled coupling, commands of the format `CONFigure: . . . :DL<s> . . .` configure all DL streams at once, independent of the specified `<s>`.

With disabled coupling, such commands configure a single selected DL stream `<s>`. However, some settings are never configurable per stream and are always coupled.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
*RST: ON

Example: See [Configuring RMCs](#)

Firmware/software: V3.2.60

Manual operation: See ["Use Stream 1 Settings"](#) on page 241

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:HdUPlex <HalfDuplex>

Selects between half-duplex operation and full-duplex operation.

Parameters:

<HalfDuplex> OFF | ON
OFF: full-duplex operation
ON: half-duplex operation
*RST: ON

Firmware/software: V3.7.10

Manual operation: See ["Half Duplex"](#) on page 244

SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]:STReam<s>?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>:STReam<s>?

Returns the expected maximum throughput (averaged over one frame) for one DL stream of one component carrier. The throughput is calculated for the currently selected scheduling type.

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Throughput> float

Default unit: Mbit/s

Example: See [Configuring user-defined channels](#)

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.2.50

Manual operation: See ["Throughput"](#) on page 243

SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL[:PCC]?
SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:SCC<c>?

Returns the expected maximum throughput (averaged over one frame) for the sum of all DL streams of one component carrier. The throughput is calculated for the currently selected scheduling type.

Suffix:

<c> 1..7

Return values:

<Throughput> float

Default unit: Mbit/s

Example: See [Configuring user-defined channels](#)

Usage: Query only

Firmware/software: V3.2.50

Manual operation: See ["Max Throughput"](#) on page 226

SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:DL:ALL?

Returns the expected maximum throughput (averaged over one frame) for the sum of all DL streams of all component carriers. The throughput is calculated for the currently selected scheduling type.

Return values:

<Throughput> float
Default unit: Mbit/s

Example: See [Configuring user-defined channels](#)

Usage: Query only

Firmware/software: V3.0.20

SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL[:PCC]?**SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:SCC<c>?**

Returns the expected maximum throughput (averaged over one frame) for the uplink of one component carrier. The throughput is calculated for the currently selected scheduling type.

Suffix:

<c> 1..7

Return values:

<Throughput> float
Default unit: Mbit/s

Example: See [Configuring user-defined channels](#)

Usage: Query only

Firmware/software: V3.0.20

Manual operation: See ["Max Throughput"](#) on page 226

SENSe:LTE:SIGN<i>:CONNection:ETHRoughput:UL:ALL?

Returns the expected maximum uplink throughput (averaged over one frame) for the sum of all component carriers. The throughput is calculated for the currently selected scheduling type.

Return values:

<Throughput> float
Default unit: Mbit/s

Example: See [Configuring user-defined channels](#)

Usage: Query only

Firmware/software: V3.7.50

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:TBURsts <Bursts>

Selects between fixed transmission bursts and random transmission bursts for LAA.

Suffix:

<c> 1..7

Parameters:

<Bursts> FBURst | RBURst
FBURst: fixed transmission bursts
RBURst: random transmission bursts
 *RST: FBUR

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Burst Type"](#) on page 241

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:BLENgth
 <BurstLength>

Specifies the number of subframes per burst, for LAA with fixed bursts.

Suffix:

<c> 1..7

Parameters:

<BurstLength> numeric
 Range: 1 to 10
 *RST: 10

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Length of Burst"](#) on page 253

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:PBTR <Periodicity>

Specifies the burst transmission periodicity, for LAA with fixed bursts.

The minimum allowed value equals the configured burst length, see [CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:BLENgth](#).

Suffix:

<c> 1..7

Parameters:

<Periodicity> numeric
 Range: 1 to 100
 *RST: 10

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Periodicity of Burst Transmissions"](#) on page 253

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:BLENgth
 <BurstLength>(10)

Specifies the possible burst lengths for LAA with random bursts. At least one value must be allowed (ON).

Suffix:

<c> 1..7

Parameters:

<BurstLength> OFF | ON
 Comma-separated list of 10 values
 Allowing lengths of (1, 2, 3, 4, 5, 6, 7, 8, 9, 10) subframes
 *RST: ON,OFF,ON,OFF,ON,OFF,OFF,ON,OFF,OFF

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Possible Burst Lengths"](#) on page 258

2.6.15.14 RMC settings

The following commands define parameters for the scheduling type "RMC".

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:PSFConfig	798
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:SFPattern	799
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:DL<s>	799
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:DL<s>	799
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:UL	800
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:UL	800
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:RBPosition:DL<s>	801
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:RBPosition:DL<s>	801
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:RBPosition:UL	802
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:RBPosition:UL	802
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:NBPosition:DL	802
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:NBPosition:UL	803
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:VERSion:DL<s>	803
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:VERSion:DL<s>	803
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:MCLuster:UL	803
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:MCLuster:UL	803

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:PSFConfig
 <Configuration>

Configures in which subframes partial allocation is allowed, for LAA with random bursts.

Suffix:

<c> 1..7

Parameters:

<Configuration> NO | INIT | END | BOTH

NO: only full allocation**INIT:** partial allocation allowed for initial subframes**END:** partial allocation allowed for ending subframes**BOTH:** partial allocation allowed for initial and ending subframes

*RST: NO

Example:See [Configuring LAA settings](#)**Firmware/software:** V3.7.20**Options:** R&S CMW-KS514**Manual operation:** See ["Partial Subframe Configuration"](#) on page 241**CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:SFPattern <Pattern>**

Determines the subframe pattern for eMTC RMCs.

Parameters:

<Pattern> P1 | P2 | P3 | P4 | P5

P1: standard**P2:** section 6.3.4EA**P3:** section 6.3.5EA.3**P4:** section 6.5.2.1EA.2-A**P5:** section 6.5.2.1EA.2-B

*RST: P1

Example:See [Configuring eMTC settings](#)**Firmware/software:** V3.7.20**Options:** R&S CMW-KS590**Manual operation:** See ["SF Pattern"](#) on page 244**CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:DL<s> <NumberRB>, <Modulation>, <TransBlockSizeIdx>****CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:DL<s> <NumberRB>, <Modulation>, <TransBlockSizeIdx>**

Configures a downlink reference measurement channel (RMC).

Depending on other settings, only a subset of the listed values is allowed, see:

- [Section 2.2.19, "Scheduling type RMC"](#), on page 69
- [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45
- [Section 2.2.16.2, "Scheduling type RMC for LAA"](#), on page 58

Suffix:

<s> 1..2

<c> 1..7

Parameters:

<NumberRB> ZERO | N1 | N2 | N3 | N4 | N5 | N6 | N7 | N8 | N9 | N10 | N12 | N15 | N16 | N17 | N18 | N20 | N21 | N24 | N25 | N27 | N30 | N32 | N36 | N40 | N42 | N45 | N48 | N50 | N54 | N60 | N64 | N72 | N75 | N80 | N81 | N83 | N90 | N92 | N96 | N100

Number of allocated resource blocks. The same value must be configured for all streams of the carrier.

*RST: N50

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024

QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM

*RST: QPSK

<TransBlockSizeldx> ZERO | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 | T9 | T10 | T11 | T12 | T13 | T14 | T15 | T16 | T17 | T18 | T19 | T20 | T21 | T22 | T23 | T24 | T25 | T26 | T27 | T28 | T29 | T30 | T31 | T32 | T33 | T34 | T35 | T36 | T37

Transport block size index. Use KEEP to select a compatible value.

*RST: T5

Example: See [Configuring RMCs](#)

Firmware/software: V3.0.20, some Nx and Tx added in later versions
V3.2.50: SCC command, V3.5.30: 256-QAM
V3.7.70: 1024-QAM

Options: 256-QAM: R&S CMW-KS504/-KS554 for FDD/TDD
1024-QAM: R&S CMW-KS505/-KS555 for FDD/TDD

Manual operation: See ["# Resource Blocks"](#) on page 241

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:UL <NumberRB>, <Modulation>, <TransBlockSizeldx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:UL <NumberRB>, <Modulation>, <TransBlockSizeldx>

Configures an uplink reference measurement channel (RMC) with contiguous allocation.

Depending on other settings, only a subset of the listed values is allowed, see:

- [Section 2.2.19, "Scheduling type RMC"](#), on page 69
- [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45
- [Section 2.2.16.2, "Scheduling type RMC for LAA"](#), on page 58

Suffix:

<c> 1..7

Parameters:

<NumberRB>	ZERO N1 N2 N3 N4 N5 N6 N7 N8 N9 N10 N12 N15 N16 N17 N18 N20 N21 N24 N25 N27 N30 N32 N36 N40 N42 N45 N48 N50 N54 N60 N64 N72 N75 N80 N81 N83 N90 N92 N96 N100 Number of allocated resource blocks *RST: N50
<Modulation>	QPSK Q16 Q64 Q256 Modulation type QPSK 16-QAM 64-QAM 256-QAM *RST: QPSK
<TransBlockSizIdx>	ZERO T1 T2 T3 T4 T5 T6 T7 T8 T9 T10 T11 T12 T13 T14 T15 T16 T17 T18 T19 T20 T21 T22 T23 T24 T25 T26 T27 T28 T29 T30 T31 T32 T33 T34 T35 T36 T37 Transport block size index. Use KEEP to select a compatible value. *RST: T6

Example: See [Configuring RMCs](#)

Firmware/software: V3.0.20, some Nx and Tx added in later versions
V3.5.10: SCC command, V3.7.10: 64-QAM, V3.7.60: 256-QAM

Options: 64-QAM/256-QAM: R&S CMW-KS504/-KS554 for FDD/TDD

Manual operation: See "[# Resource Blocks](#)" on page 241

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:RBPosition:DL<s> <Position>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:RBPosition:DL<s> <Position>

Selects the position of the allocated downlink resource blocks. Set the same value for both streams of a carrier.

Depending on other settings, only a subset of the listed values is allowed, see:

- [Section 2.2.19, "Scheduling type RMC"](#), on page 69
- [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45
- [Section 2.2.16.2, "Scheduling type RMC for LAA"](#), on page 58

Suffix:

<s>	1..2
<c>	1..7

Parameters:

<Position>	LOW HIGH P5 P10 P23 P35 P48 *RST: LOW
------------	--

Example: See [Configuring RMCs](#)

Firmware/software: V3.2.50

Manual operation: See ["RB Position/Start RB"](#) on page 242

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:RBPosition:UL <Position>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:RBPosition:UL <Position>

Selects the position of the allocated uplink resource blocks, for contiguous allocation.

Depending on other settings, only a subset of the listed values is allowed, see:

- [Section 2.2.19, "Scheduling type RMC"](#), on page 69
- [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45
- [Section 2.2.16.2, "Scheduling type RMC for LAA"](#), on page 58

Suffix:

<c> 1..7

Parameters:

<Position> LOW | HIGH | MID | P0 | P1 | P2 | P3 | P4 | P6 | P7 | P8 | P9 |
P10 | P11 | P12 | P13 | P14 | P15 | P16 | P19 | P20 | P21 | P22 |
P24 | P25 | P28 | P30 | P31 | P33 | P36 | P37 | P39 | P40 | P43 |
P44 | P45 | P48 | P49 | P50 | P51 | P52 | P54 | P56 | P57 | P58 |
P62 | P63 | P66 | P68 | P70 | P74 | P75 | P83 | P96 | P99
*RST: LOW

Example: See [Configuring RMCs](#)

Firmware/software: V3.0.20, SCC command V3.5.10
V3.2.70: values P1 - P4, P7, P9, P10, P14, P15, P36, P39, P44,
P49, P74, P75, P99
V3.5.10: values P0, P6, P8, P12, P22, P24, P45, P68, P96
V3.5.30: values P20, P21, P28, P30, P31, P33, P40, P50-P52,
P54, P57, P58, P62, P63, P66, P70, P83

Manual operation: See ["RB Position/Start RB"](#) on page 242

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:NBPosition:DL
<Position>

Selects the lowest used narrowband for an eMTC DL RMC.

Depending on other settings, only a subset of the listed values is allowed, see [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45.

Parameters:

<Position> LOW | MID | HIGH | GPP3
*RST: GPP3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.30

Options: R&S CMW-KS590

Manual operation: See ["NB Position"](#) on page 245

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:EMTC:NBPosition:UL
 <Position>

Selects the lowest used narrowband for an eMTC UL RMC.

Depending on other settings, only a subset of the listed values is allowed, see [Section 2.2.12, "Scheduling type RMC for eMTC"](#), on page 45.

Parameters:

<Position> LOW | HIGH | NB1 | NB2 | NB3 | NB4 | NB5 | NB6 | NB7 | NB8 | NB9 | NB10 | NB11 | NB12 | NB13 | NB14

LOW: NB0

HIGH: highest NB within the cell bandwidth

*RST: LOW

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20, V3.7.30 added NB1 to NB14

Options: R&S CMW-KS590

Manual operation: See ["NB Position"](#) on page 245

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:VERSion:DL<s> <Version>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:VERSion:DL<s> <Version>

Selects the version to distinguish ambiguous RMCs. This command is only relevant for certain downlink RMCs for TDD multiple antenna configurations, see [Section 2.2.19.4, "DL RMCs, multiple TX antennas \(TM 2 to 6\)"](#), on page 77.

Suffix:

<s> 1..2

<c> 1..7

Parameters:

<Version> integer

Range: 0 to 1

*RST: 0

Example: See [Configuring RMCs](#)

Firmware/software: V3.2.70

Manual operation: See ["Version"](#) on page 242

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:MCLuster:UL <NumberRB1>,
 <PositionRB1>, <NumberRB2>, <PositionRB2>, <Modulation>,
 <TransBlockSizeIdx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:MCLuster:UL <NumberRB1>,
 <PositionRB1>, <NumberRB2>, <PositionRB2>, <Modulation>,
 <TransBlockSizeIdx>

Configures an uplink reference measurement channel (RMC) with multi-cluster allocation.

Only certain value combinations are accepted, see [Section 2.2.19, "Scheduling type RMC"](#), on page 69.

Suffix:

<c> 1..7

Parameters:

<NumberRB1> ZERO | N1 | N2 | N3 | N4 | N5 | N6 | N7 | N8 | N9 | N10 | N12 | N15 | N16 | N17 | N18 | N20 | N21 | N24 | N25 | N27 | N30 | N32 | N36 | N40 | N42 | N45 | N48 | N50 | N54 | N60 | N64 | N72 | N75 | N80 | N81 | N83 | N90 | N92 | N96 | N100
Number of allocated resource blocks, cluster 1
*RST: N42

<PositionRB1> FULL | LOW | HIGH | MID | P0 | P1 | P2 | P3 | P4 | P6 | P7 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 | P19 | P20 | P21 | P22 | P24 | P25 | P28 | P30 | P31 | P33 | P36 | P37 | P39 | P40 | P43 | P44 | P45 | P48 | P49 | P50 | P51 | P52 | P54 | P56 | P57 | P58 | P62 | P63 | P66 | P68 | P70 | P74 | P75 | P83 | P96 | P99
Position of first RB, cluster 1
*RST: P0

<NumberRB2> ZERO | N1 | N2 | N3 | N4 | N5 | N6 | N7 | N8 | N9 | N10 | N12 | N15 | N16 | N17 | N18 | N20 | N21 | N24 | N25 | N27 | N30 | N32 | N36 | N40 | N42 | N45 | N48 | N50 | N54 | N60 | N64 | N72 | N75 | N80 | N81 | N83 | N90 | N92 | N96 | N100
Number of allocated resource blocks, cluster 2
*RST: N3

<PositionRB2> FULL | LOW | HIGH | MID | P0 | P1 | P2 | P3 | P4 | P6 | P7 | P8 | P9 | P10 | P11 | P12 | P13 | P14 | P15 | P16 | P19 | P20 | P21 | P22 | P24 | P25 | P28 | P30 | P31 | P33 | P36 | P37 | P39 | P40 | P43 | P44 | P45 | P48 | P49 | P50 | P51 | P52 | P54 | P56 | P57 | P58 | P62 | P63 | P66 | P68 | P70 | P74 | P75 | P83 | P96 | P99
Position of first RB, cluster 2
*RST: P45

<Modulation> Q16 | Q64
Modulation scheme 16QAM | 64QAM
*RST: Q16

<TransBlockSizeIdx> ZERO | T1 | T2 | T3 | T4 | T5 | T6 | T7 | T8 | T9 | T10 | T11 | T12 | T13 | T14 | T15 | T16 | T17 | T18 | T19 | T20 | T21 | T22 | T23 | T24 | T25 | T26 | T27 | T28 | T29 | T30 | T31 | T32 | T33 | T34 | T35 | T36 | T37
Transport block size index. Use KEEP to select a compatible value.
*RST: T11

Example: See [Configuring RMCs](#)

Firmware/software: V3.5.20, some Nx and Tx added in later versions
V3.7.60: 64QAM

Options: 64QAM: R&S CMW-KS504/-KS554 for FDD/TDD

Manual operation: See "[# Resource Blocks](#)" on page 241

2.6.15.15 User-defined channels (no LAA)

The following commands define parameters for the scheduling type "User-Defined Channels" in normal operation mode (no LAA).

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:DL<s>	805
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:DL<s>	805
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:UL	806
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:UL	806
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:SFPattern	807
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:DL	807
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:UL	808
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:NBPosition:DL	809
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:NBPosition:UL	809
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:DL:ANB<Number>	809
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:MCLuster:DL<s>	810
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:MCLuster:DL<s>	810
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:MCLuster:UL	811
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:MCLuster:UL	811
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:DL<s>:CRATe:ALL?	812
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:DL<s>:CRATe:ALL?	812
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:UL:CRATe:ALL?	812
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:UL:CRATe:ALL?	812

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:DL<s> <NumberRB>,
<StartRB>, <Modulation>, <TransBlockSizeldx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:DL<s> <NumberRB>,
<StartRB>, <Modulation>, <TransBlockSizeldx>

Configures a user-defined downlink channel with contiguous allocation (no LAA, no eMTC).

The <NumberRB> and <StartRB> settings apply to all DL streams. The other settings apply to DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

Suffix:

<s>	1..2
<c>	1..7

Parameters:

<NumberRB>	numeric
	Number of allocated resource blocks
	*RST: 50
<StartRB>	numeric
	Position of first resource block
	*RST: 0
<Modulation>	QPSK Q16 Q64 Q256 Q1024
	QPSK 16-QAM 64-QAM 256-QAM 1024-QAM
	*RST: QPSK
<TransBlockSizeIdx>	numeric
	Transport block size index
	*RST: 5

Example: See [Configuring user-defined channels](#)

Firmware/software: V2.0.10, SCC V3.2.50, V3.5.30 256-QAM, V3.7.70 1024-QAM

Options: R&S CMW-KS510
 256-QAM: R&S CMW-KS504/-KS554
 1024-QAM: R&S CMW-KS505/-KS555

Manual operation: See ["# Resource Blocks ... Transport Block Size"](#) on page 248

CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:UL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>

CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:UL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>

Configures a user-defined uplink channel with contiguous allocation (no LAA, no eMTC). The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

Suffix:

<c> 1..7

Parameters:

<NumberRB>	numeric
	Number of allocated resource blocks
	*RST: 50
<StartRB>	numeric
	Position of first resource block
	*RST: 0
<Modulation>	QPSK Q16 Q64 Q256
	Modulation type QPSK 16-QAM 64-QAM 256-QAM
	*RST: QPSK

<TransBlockSizeldx> numeric

Transport block size index

*RST: 6

Example: See [Configuring user-defined channels](#)

Firmware/software: V2.0.20, SCC command V3.5.20
V3.7.10: 64-QAM, V3.7.60: 256-QAM

Options: R&S CMW-KS510
64-QAM/256-QAM: R&S CMW-KS504/-KS554 for FDD/TDD

Manual operation: See "[# Resource Blocks ... Transport Block Size](#)" on page 248

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:SFPattern <Pattern>

Selects the subframe pattern for user-defined channels, half-duplex. There are patterns with and without PDSCH HARQ-ACK bundling.

Parameters:

<Pattern> STANdard | HAB8 | HAB10

STANdard: no bundling

HAB8: bundling, 8 HARQ processes

HAB10: bundling, 10 HARQ processes

*RST: STAN

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS590

Manual operation: See "[SF Pattern](#)" on page 250

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:DL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeldx>

Configures a user-defined downlink channel for eMTC, CE mode A. The ranges have dependencies described in the background information, see [Section 2.2.13, "User-defined channels for eMTC"](#), on page 49.

Parameters:

<NumberRB> numeric

Number of allocated resource blocks

Range: 0 to 4

*RST: 4

<StartRB> numeric

Range: 0 to 4

*RST: 0

<Modulation> QPSK | Q16
 Modulation type QPSK | 16-QAM
 *RST: QPSK

<TransBlockSizeldx> numeric
 Range: 0 to 14
 *RST: 5

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See "[DL, # Resource Blocks ... Transport Block Size](#)" on page 250

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:UL
 <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeldx>

Configures a user-defined uplink channel for eMTC, CE mode A. The ranges have dependencies described in the background information, see [Section 2.2.13, "User-defined channels for eMTC"](#), on page 49.

Parameters:

<NumberRB> numeric
 Number of allocated resource blocks
 Range: 0 to 24
 *RST: 6

<StartRB> integer
 Range: 0 to 6
 *RST: 0

<Modulation> QPSK | Q16
 Modulation type QPSK | 16-QAM
 *RST: QPSK

<TransBlockSizeldx> numeric
 Transport block size index
 Range: 0 to 14, 16 to 21
 *RST: 6

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See "[UL, # Resource Blocks ... Transport Block Size](#)" on page 250

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:NBPosition:DL
 <Position>

Selects the narrowband position for a user-defined downlink channel for eMTC.

The allowed values have dependencies described in the background information, see [Section 2.2.13, "User-defined channels for eMTC"](#), on page 49.

Parameters:

<Position> LOW | MID | HIGH | GPP3
 *RST: GPP3

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.30

Options: R&S CMW-KS590

Manual operation: See ["DL, # Resource Blocks ... Transport Block Size"](#)
 on page 250

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:NBPosition:UL
 <Position>

Selects the lowest narrowband for a user-defined uplink channel for eMTC.

The allowed values have dependencies described in the background information, see [Section 2.2.13, "User-defined channels for eMTC"](#), on page 49.

Parameters:

<Position> LOW | HIGH | NB1 | NB2 | NB3 | NB4 | NB5 | NB6 | NB7 | NB8 |
 NB9 | NB10 | NB11 | NB12 | NB13 | NB14
 *RST: LOW

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.30

Options: R&S CMW-KS590

Manual operation: See ["UL, # Resource Blocks ... Transport Block Size"](#)
 on page 250

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:EMTC:A:DL:
ANB<Number> <AdditionalNB>

Enables or disables additional narrowbands for a user-defined downlink channel, for a maximum eMTC bandwidth of 5 MHz.

Suffix:

<Number> 1..3
 Number of the additional narrowband (1st, 2nd or 3rd narrow-
 band above the lowest used narrowband).

Parameters:

<AdditionalNB> OFF | ON
ON: Use the additional narrowband.
OFF: Do not use the additional narrowband.
 *RST: OFF

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS590

Manual operation: See "[DL, # Resource Blocks ... Transport Block Size](#)" on page 250

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:MCLuster:DL<s>
 <Cluster>, <Modulation>, <TransBlockSizeIdx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:MCLuster:DL<s>
 <Cluster>, <Modulation>, <TransBlockSizeIdx>

Configures a user-defined downlink channel with multi-cluster allocation (no LAA, no eMTC).

The <Cluster> setting applies to all DL streams. The other settings apply to DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83 and especially [Table 2-46](#).

Suffix:

<s> 1..2
 <c> 1..7

Parameters:

<Cluster> binary
 Bitmap, enabling or disabling the individual RBGs
 1 means RBG is allocated, 0 means RBG is not allocated
 The number of bits depends on the cell bandwidth and equals the total number of RBGs.
 The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.
 Example for BW 1.4 MHz:
 #B101010 means that the RBGs 0, 2 and 4 are allocated

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024
 QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM
 *RST: QPSK

<TransBlockSizeIdx> numeric
 Transport block size index
 *RST: 5

Example: See [Configuring user-defined channels](#)

Firmware/software: V3.5.50, V3.7.70 1024-QAM

Options: R&S CMW-KS510
256-QAM: R&S CMW-KS504/-KS554
1024-QAM: R&S CMW-KS505/-KS555

Manual operation: See ["Multicluster"](#) on page 247

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:MCLuster:UL
<NumberRB1>, <StartRB1>, <NumberRB2>, <StartRB2>, <Modulation>,
<TransBlockSizeIdx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:MCLuster:UL
<NumberRB1>, <StartRB1>, <NumberRB2>, <StartRB2>, <Modulation>,
<TransBlockSizeIdx>

Configures a user-defined uplink channel with multi-cluster allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

Suffix:

<c> 1..7

Parameters:

<NumberRB1>	numeric Number of allocated resource blocks, cluster 1 *RST: 45
<StartRB1>	numeric Position of first RB, cluster 1 *RST: 0
<NumberRB2>	numeric Number of allocated resource blocks, cluster 2 *RST: 2
<StartRB2>	numeric Position of first RB, cluster 2 *RST: 48
<Modulation>	QPSK Q16 Q64 Q256 Modulation scheme QPSK 16QAM 64QAM 256QAM *RST: QPSK
<TransBlockSizeIdx>	numeric Transport block size index *RST: 6

Example: See [Configuring user-defined channels](#)

Firmware/software: V3.5.20, V3.7.60 64QAM and 256QAM

Options: R&S CMW-KS510/-KS512 (without CA/with CA)
64QAM/256QAM: R&S CMW-KS504/-KS554 for FDD/TDD

Manual operation: See "[# Resource Blocks ... Transport Block Size](#)" on page 248

SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:DL<s>:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:DL<s>:CRATe:ALL?

Queries the code rate for all downlink subframes for the scheduling type "User-defined Channels".

Suffix:

<s> 1..2

<c> 1..7

Return values:

<CodeRate> float
Comma-separated list of 10 values (subframe 0 to subframe 9)
Range: 0 to 50

Example: See [Configuring user-defined channels](#)

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.2.50

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See "[Code Rate](#)" on page 248

SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDCHannels:UL:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:UL:CRATe:ALL?

Queries the code rate for all uplink subframes for the scheduling type "User-defined Channels".

Suffix:

<c> 1..7

Return values:

<CodeRate> float
Comma-separated list of 10 values (subframe 0 to subframe 9)
Range: 0 to 10

Example: See [Configuring user-defined channels](#)

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.5.20

Options: R&S CMW-KS510 (except for eMTC)

Manual operation: See "[Code Rate](#)" on page 248

2.6.15.16 User-defined channels for LAA

The following commands define parameters for the scheduling type "User-Defined Channels" for LAA.

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:SPFSubframe.....	814
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:OSLSubframe.....	814
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:LSConfig.....	814
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CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst: FSUBframes:DL<s>.....	816
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst: PIPSubframes:DL<s>.....	817
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst: PEPSubframes:DL<s>.....	817
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst: FSUBframes:MCLuster:DL<s>.....	818
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst: PIPSubframes:MCLuster:DL<s>.....	819
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst: PEPSubframes:MCLuster:DL<s>.....	820
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst:DL<s>: FSUBframes:CRATe?.....	821
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:FBURst:DL<s>: PIPSubframes:CRATe?.....	821
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CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:RBURst: FSUBframes:DL<s>.....	822
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CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:RBURst: FSUBframes:MCLuster:DL<s>.....	825
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:RBURst: PIPSubframes:MCLuster:DL<s>.....	826
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:RBURst: PEPSubframes:MCLuster:DL<s>.....	827
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:RBURst:DL<s>: FSUBframes:CRATe?.....	828
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:RBURst:DL<s>: PIPSubframes:CRATe?.....	828
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:LAA:RBURst:DL<s>: PEPSubframes:CRATe?.....	828

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:SPFSubframe
 <StartingPos>

Selects the first allocated OFDM symbol for initial subframes, for LAA with fixed bursts.

Suffix:

<c> 1..7

Parameters:

<StartingPos> OFDM0 | OFDM7

OFDM0: symbol 0 (full allocation)

OFDM7: symbol 7 (partial allocation, needs burst length > 1)

*RST: OFDM0

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Start Position of First Subframe of Burst"](#) on page 253

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:FBURst:OSLSubframe
 <occOFDMsymbols>

Specifies the number of allocated OFDM symbols for ending subframes, for LAA with fixed bursts.

At least one subframe of each burst must have full allocation. This rule restricts the allowed values for the burst lengths 1 and 2.

Suffix:

<c> 1..7

Parameters:

<occOFDMsymbols> SYM6 | SYM9 | SYM10 | SYM11 | SYM12 | SYM14

6 to 12 OFDM symbols (partial allocation)

14 OFDM symbols (full allocation)

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Occupied OFDM Symbols in Last Subframe of Burst"](#) on page 253

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:LSConfig
 <Config>(6)

Specifies the possible number of allocated OFDM symbols in ending subframes for LAA with random bursts. At least one value must be allowed (ON).

Suffix:

<c> 1..7

Parameters:

<Config> OFF | ON
 Comma-separated list of 6 values
 Allowing (6, 9, 10, 11, 12, 14) symbols
 *RST: OFF,OFF,OFF,OFF,OFF,ON

Example: See [Configuring LAA settings](#)**Firmware/software:** V3.7.20**Options:** R&S CMW-KS514**Manual operation:** See ["Possible Last Subframe of Burst"](#) on page 258
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:IPSubframe
 <Enable>

Allows or forbids partial allocation for initial subframes, for LAA with random bursts.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
ON: initial partial subframes allowed
OFF: only full allocation in initial subframes
 *RST: OFF

Example: See [Configuring LAA settings](#)**Firmware/software:** V3.7.20**Options:** R&S CMW-KS514**Manual operation:** See ["Initial Partial Subframes"](#) on page 258
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:LAA:RBURst:TPRobability
 <Probability>

Specifies the burst transmission probability, for LAA with random bursts.

Suffix:

<c> 1..7

Parameters:

<Probability> numeric
 Range: 0 % to 100 %
 *RST: 50 %
 Default unit: %

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Burst Transmission Probability"](#) on page 258

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:
FSUBframes:DL<s>** <NumberRB>, <StartRB>, <Modulation>,
<TransBlockSizeIdx>

Configures a user-defined downlink channel with contiguous allocation, for LAA, fixed bursts.

The <NumberRB> and <StartRB> settings apply to all subframes of the burst and to all DL streams. The other settings apply to subframes with full allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56.

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<NumberRB> numeric
Number of allocated resource blocks
*RST: 100

<StartRB> numeric
Position of first resource block
*RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024
QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM
*RST: QPSK

<TransBlockSizeIdx> numeric
Transport block size index
*RST: 5

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20, V3.7.70 1024-QAM

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554
1024-QAM: R&S CMW-KS555

Manual operation: See ["# Resource Blocks ... TBS \(Ending Partial SF\)"](#) on page 253

**CONFigure:LTE:SIGN<i>:CONNectio:n:SCC<c>:UDCHannels:LAA:FBURst:
PIPSubframes:DL<s>** <NumberRB>, <StartRB>, <Modulation>,
<TransBlockSizeIdx>

Configures a user-defined downlink channel with contiguous allocation, for LAA, fixed bursts.

The <NumberRB> and <StartRB> settings apply to all subframes of the burst and to all DL streams. The other settings apply to initial subframes with partial allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56.

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<NumberRB> numeric
Number of allocated resource blocks
*RST: 100

<StartRB> numeric
Position of first resource block
*RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024
QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM
*RST: QPSK

<TransBlockSizeIdx> numeric
Transport block size index
*RST: 4

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20, V3.7.70 1024-QAM

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554
1024-QAM: R&S CMW-KS555

Manual operation: See ["# Resource Blocks ... TBS \(Ending Partial SF\)"](#)
on page 253

**CONFigure:LTE:SIGN<i>:CONNectio:n:SCC<c>:UDCHannels:LAA:FBURst:
PEPSubframes:DL<s>** <NumberRB>, <StartRB>, <Modulation>,
<TransBlockSizeIdx>

Configures a user-defined downlink channel with contiguous allocation, for LAA, fixed bursts.

The <NumberRB> and <StartRB> settings apply to all subframes of the burst and to all DL streams. The other settings apply to ending subframes with partial allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56.

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<NumberRB> numeric
Number of allocated resource blocks
*RST: 100

<StartRB> numeric
Position of first resource block
*RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024
QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM
*RST: QPSK

<TransBlockSizeIdx> numeric
Transport block size index
*RST: 5

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20, V3.7.70 1024-QAM

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554
1024-QAM: R&S CMW-KS555

Manual operation: See ["# Resource Blocks ... TBS \(Ending Partial SF\)"](#) on page 253

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:
FSUBframes:MCLuster:DL<s>** <Cluster>, <Modulation>, <TransBlockSizeIdx>

Configures a user-defined downlink channel with multi-cluster allocation, for LAA, fixed bursts.

The <Cluster> setting applies to all subframes of the burst and to all DL streams. The other settings apply to subframes with full allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56 and especially [Table 2-46](#).

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<Cluster> binary

Bitmap, enabling or disabling the individual RBGs

1 means RBG is allocated, 0 means RBG is not allocated

The number of bits depends on the cell bandwidth and equals the total number of RBGs.

The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.

Example for BW 20 MHz:

#B101010000000000000000011 allocates RBG 0, 2, 4, 23, 24

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024

QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM

*RST: QPSK

<TransBlockSizeIdx> numeric

Transport block size index

*RST: 5

Example: See [Configuring LAA settings](#)**Firmware/software:** V3.7.20, V3.7.70 1024-QAM**Options:** R&S CMW-KS510 and R&S CMW-KS514

256-QAM: R&S CMW-KS554

1024-QAM: R&S CMW-KS555

Manual operation: See ["Multicluster"](#) on page 252

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:
PIPSubframes:MCLuster:DL<s> <Cluster>, <Modulation>,
<TransBlockSizeIdx>**

Configures a user-defined downlink channel with multi-cluster allocation, for LAA, fixed bursts.

The <Cluster> setting applies to all subframes of the burst and to all DL streams. The other settings apply to initial subframes with partial allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56 and especially [Table 2-46](#).

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<Cluster> binary

	Bitmap, enabling or disabling the individual RBGs 1 means RBG is allocated, 0 means RBG is not allocated The number of bits depends on the cell bandwidth and equals the total number of RBGs. The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency. Example for BW 20 MHz: #B1010100000000000000000011 allocates RBG 0, 2, 4, 23, 24
<Modulation>	QPSK Q16 Q64 Q256 Q1024 QPSK 16-QAM 64-QAM 256-QAM 1024-QAM *RST: QPSK
<TransBlockSizeIdx>	numeric Transport block size index *RST: 4
Example:	See Configuring LAA settings
Firmware/software:	V3.7.20, V3.7.70 1024-QAM
Options:	R&S CMW-KS510 and R&S CMW-KS514 256-QAM: R&S CMW-KS554 1024-QAM: R&S CMW-KS555
Manual operation:	See "Multicluster" on page 252

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:PEPSubframes:MCLuster:DL<s> <Cluster>, <Modulation>, <TransBlockSizeIdx>

Configures a user-defined downlink channel with multi-cluster allocation, for LAA, fixed bursts.

The <Cluster> setting applies to all subframes of the burst and to all DL streams. The other settings apply to ending subframes with partial allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56 and especially [Table 2-46](#).

Suffix:

<c>	1..7
<s>	1..2

Parameters:

<Cluster>	binary Bitmap, enabling or disabling the individual RBGs 1 means RBG is allocated, 0 means RBG is not allocated The number of bits depends on the cell bandwidth and equals the total number of RBGs. The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.
-----------	---

Example for BW 20 MHz:
 #B101010000000000000000011 allocates RBG 0, 2, 4, 23, 24

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024
 QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM
 *RST: QPSK

<TransBlockSizeldx> numeric
 Transport block size index
 *RST: 5

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20, V3.7.70 1024-QAM

Options: R&S CMW-KS510 and R&S CMW-KS514
 256-QAM: R&S CMW-KS554
 1024-QAM: R&S CMW-KS555

Manual operation: See ["Multicluster"](#) on page 252

**SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:DL<s>:
 FSUBframes:CRATe?**

Queries the code rate for subframes with full allocation, for LAA, fixed bursts, scheduling type "User-defined Channels".

Suffix:

<c> 1..7

<s> 1..2

Return values:

<CodeRate> float
 Range: 0 to 50

Example: See [Configuring LAA settings](#)

Usage: Query only

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514

Manual operation: See ["DL Code Rate \(...\)"](#) on page 254

**SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:DL<s>:
 PIPSubframes:CRATe?**

Queries the code rate for initial subframes with partial allocation, for LAA, fixed bursts, scheduling type "User-defined Channels".

Suffix:

<c> 1..7

<s> 1..2

Return values:

<CodeRate> float
Range: 0 to 50

Example: See [Configuring LAA settings](#)

Usage: Query only

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514

Manual operation: See "[DL Code Rate \(...\)](#)" on page 254

**SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:FBURst:DL<s>:
PEPSubframes:CRATe?**

Queries the code rate for ending subframes with partial allocation, for LAA, fixed bursts, scheduling type "User-defined Channels".

Suffix:

<c> 1..7

<s> 1..2

Return values:

<CodeRate> float
Range: 0 to 50

Example: See [Configuring LAA settings](#)

Usage: Query only

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514

Manual operation: See "[DL Code Rate \(...\)](#)" on page 254

**CONFIgure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
FSUBframes:DL<s> <NumberRB>, <StartRB>, <Modulation>,
<TransBlockSizeldx>**

Configures a user-defined downlink channel with contiguous allocation, for LAA, random bursts.

The <NumberRB> and <StartRB> settings apply to all subframes of the burst and to all DL streams. The other settings apply to subframes with full allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56.

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<NumberRB>	numeric Number of allocated resource blocks *RST: 100
<StartRB>	numeric Position of first resource block *RST: 0
<Modulation>	QPSK Q16 Q64 Q256 QPSK 16-QAM 64-QAM 256-QAM *RST: QPSK
<TransBlockSizeIdx>	numeric Transport block size index *RST: 5

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554

Manual operation: See ["#RB ... TBS \(PEPS 12 Symbols\)"](#) on page 257

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
PIPSubframes:DL<s> <NumberRB>, <StartRB>, <Modulation>,
<TransBlockSizeIdx>**

Configures a user-defined downlink channel with contiguous allocation, for LAA, random bursts.

The <NumberRB> and <StartRB> settings apply to all subframes of the burst and to all DL streams. The other settings apply to initial subframes with partial allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56.

Suffix:

<c>	1..7
<s>	1..2

Parameters:

<NumberRB>	numeric Number of allocated resource blocks *RST: 100
<StartRB>	numeric Position of first resource block *RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256
 QPSK | 16-QAM | 64-QAM | 256-QAM
 *RST: QPSK

<TransBlockSizeIdx> numeric
 Transport block size index
 *RST: 4

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514
 256-QAM: R&S CMW-KS554

Manual operation: See ["#RB ... TBS \(PEPS 12 Symbols\)"](#) on page 257

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
 PEPSubframes:DL<s>** <Symbols>, <NumberRB>, <StartRB>, <Modulation>,
 <TransBlockSizeIdx>

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
 PEPSubframes:DL<s>?** <Symbols>

Configures a user-defined downlink channel with contiguous allocation, for LAA, random bursts.

The <NumberRB> and <StartRB> settings apply to all subframes of the burst and to all DL streams.

The <Modulation> and <TransBlockSizeIdx> settings apply to partial ending subframes with the specified number of allocated OFDM <Symbols> and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56.

Suffix:

<c> 1..7
 <s> 1..2

Parameters:

<NumberRB> numeric
 Number of allocated resource blocks
 *RST: 100

<StartRB> numeric
 Position of first resource block
 *RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256
 QPSK | 16-QAM | 64-QAM | 256-QAM
 *RST: QPSK

<TransBlockSizeldx> numeric

Transport block size index

*RST: 5

Parameters for setting and query:

<Symbols> S6 | S9 | S10 | S11 | S12

Number of OFDM symbols allocated in the ending subframe

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554

Manual operation: See ["#RB ... TBS \(PEPS 12 Symbols\)"](#) on page 257

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:
FSUBframes:MCLuster:DL<s>** <Cluster>, <Modulation>, <TransBlockSizeldx>

Configures a user-defined downlink channel with multi-cluster allocation, for LAA, random bursts.

The <Cluster> setting applies to all subframes of the burst and to all DL streams. The other settings apply to subframes with full allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56 and especially [Table 2-46](#).

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<Cluster> binary

Bitmap, enabling or disabling the individual RBGs

1 means RBG is allocated, 0 means RBG is not allocated

The number of bits depends on the cell bandwidth and equals the total number of RBGs.

The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.

Example for BW 20 MHz:

#B101010000000000000000011 allocates RBG 0, 2, 4, 23, 24

<Modulation> QPSK | Q16 | Q64 | Q256

QPSK | 16-QAM | 64-QAM | 256-QAM

*RST: QPSK

<TransBlockSizeldx> numeric

Transport block size index

*RST: 5

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554

Manual operation: See ["Multicluster"](#) on page 256

**CONFigure:LTE:SIGN<i>:CONNectio:n:SCC<c>:UDCHannels:LAA:RBURst:
PIPSubframes:MCLuster:DL<s> <Cluster>, <Modulation>,
<TransBlockSizeldx>**

Configures a user-defined downlink channel with multi-cluster allocation, for LAA, random bursts.

The <Cluster> setting applies to all subframes of the burst and to all DL streams. The other settings apply to initial subframes with partial allocation and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56 and especially [Table 2-46](#).

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<Cluster> binary

Bitmap, enabling or disabling the individual RBGs

1 means RBG is allocated, 0 means RBG is not allocated

The number of bits depends on the cell bandwidth and equals the total number of RBGs.

The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.

Example for BW 20 MHz:

#B1010100000000000000000011 allocates RBG 0, 2, 4, 23, 24

<Modulation> QPSK | Q16 | Q64 | Q256

QPSK | 16-QAM | 64-QAM | 256-QAM

*RST: QPSK

<TransBlockSizeldx> numeric

Transport block size index

*RST: 4

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554

Manual operation: See ["Multicluster"](#) on page 256

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:PEPSubframes:MCLuster:DL<s> <Symbols>, <Cluster>, <Modulation>, <TransBlockSizeldx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:PEPSubframes:MCLuster:DL<s>? <Symbols>

Configures a user-defined downlink channel with multi-cluster allocation, for LAA, random bursts.

The <Cluster> setting applies to all subframes of the burst and to all DL streams.

The <Modulation> and <TransBlockSizeldx> settings apply to partial ending subframes with the specified number of allocated OFDM <Symbols> and DL stream <s>.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.16.1, "User-defined channels for LAA"](#), on page 56.

Suffix:

<c> 1..7

<s> 1..2

Parameters:

<Cluster> binary

Bitmap, enabling or disabling the individual RBGs

1 means RBG is allocated, 0 means RBG is not allocated

The number of bits depends on the cell bandwidth and equals the total number of RBGs.

The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.

Example for BW 20 MHz:

#B101010000000000000000011 allocates RBG 0, 2, 4, 23, 24

<Modulation> QPSK | Q16 | Q64 | Q256

QPSK | 16-QAM | 64-QAM | 256-QAM

*RST: QPSK

<TransBlockSizeldx> numeric

Transport block size index

*RST: 5

Parameters for setting and query:

<Symbols> S6 | S9 | S10 | S11 | S12

Number of OFDM symbols allocated in the ending subframe

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514
256-QAM: R&S CMW-KS554

Manual operation: See ["Multicluster"](#) on page 256

**SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:DL<s>:
FSUBframes:CRATe?**

Queries the code rate for subframes with full allocation, for LAA, random bursts, scheduling type "User-defined Channels".

Suffix:

<c> 1..7

<s> 1..2

Return values:

<CodeRate> float
Range: 0 to 10

Example: See [Configuring LAA settings](#)

Usage: Query only

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514

Manual operation: See "[Code Rate \(...\)](#)" on page 257

**SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:DL<s>:
PIPSubframes:CRATe?**

Queries the code rate for initial subframes with partial allocation, for LAA, random bursts, scheduling type "User-defined Channels".

Suffix:

<c> 1..7

<s> 1..2

Return values:

<CodeRate> float
Range: 0 to 10

Example: See [Configuring LAA settings](#)

Usage: Query only

Firmware/software: V3.7.20

Options: R&S CMW-KS510 and R&S CMW-KS514

Manual operation: See "[Code Rate \(...\)](#)" on page 257

**SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDCHannels:LAA:RBURst:DL<s>:
PEPSubframes:CRATe? <Symbols>**

Queries the code rate for ending subframes with a certain partial allocation, for LAA, random bursts, scheduling type "User-defined Channels".

Suffix:

<c> 1..7

<s> 1..2

Query parameters:

<Symbols> S6 | S9 | S10 | S11 | S12
 Number of OFDM symbols allocated in the ending subframe

Return values:

<CodeRate> float
 Range: 0 to 10

Example: See [Configuring LAA settings](#)**Usage:** Query only**Firmware/software:** V3.7.20**Options:** R&S CMW-KS510 and R&S CMW-KS514**Manual operation:** See ["Code Rate \(...\)"](#) on page 257**2.6.15.17 User-defined TTI-based channels**

The following commands define parameters for the scheduling type "User-Defined TTI-Based Channels".

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>	829
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>	829
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>:ALL	830
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>:ALL	830
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL	832
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL	832
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL:ALL	833
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL:ALL	833
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>:CRATe:ALL?	834
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>:CRATe:ALL?	834
SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL:CRATe:ALL?	834
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL:CRATe:ALL?	834

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s> <TTI>,
 <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>? <TTI>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s> <TTI>,
 <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>? <TTI>

Configures a selected downlink subframe for the scheduling type "User-defined TTI-Based".

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

A query for TDD can also return `OFF, OFF, OFF, OFF`, indicating that the queried subframe is no DL subframe.

Suffix:

<s> 1..2

<c> 1..7

Parameters:

<NumberRB> numeric | OFF

Number of allocated resource blocks. The same value must be configured for all streams.

*RST: 50

<StartRB> numeric | OFF

Position of first resource block. The same value must be configured for all streams of the carrier.

*RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024 | OFF

QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM | no DL subframe

*RST: QPSK

<TransBlockSizeIdx> numeric | OFF

Transport block size index

*RST: 5

Parameters for setting and query:

<TTI> numeric

Number of the subframe to be configured/queried.

Range: 0 to 9

*RST: 0

Example: See [Configuring TTI-based user-defined channels](#)

Firmware/software: V3.2.50, V3.5.10 `OFF`, V3.5.40 256-QAM, V3.7.70 1024-QAM

Options: R&S CMW-KS510

256-QAM: R&S CMW-KS504/-KS554

1024-QAM: R&S CMW-KS505/-KS555

Manual operation: See ["User-Defined TTI-Based"](#) on page 269

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>:ALL

<NumberRB>(10), <StartRB>(10), <Modulation>(10), <TransBlockSizeIdx>(10)

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>:ALL

<NumberRB>(10), <StartRB>(10), <Modulation>(10), <TransBlockSizeIdx>(10)

Configures all downlink subframes for the scheduling type "User-defined TTI-Based".

The parameters are entered 10 times, so that all subframes are configured by a single command (index = subframe number 0 to 9):

<NumberRB>₀, ..., <NumberRB>₉, <StartRB>₀, ..., <StartRB>₉, <Modulation>₀, ..., <Modulation>₉, <TransBlockSizeldx>₀, ..., <TransBlockSizeldx>₉

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

For TDD UL and special subframes, you can set `OFF` or specify a number from the allowed input range. The effect is the same. A query returns `OFF` for non-DL subframes.

Suffix:

<s> 1..2

<c> 1..7

Parameters:

<NumberRB> numeric | `OFF`
 Number of allocated resource blocks. The same value must be configured for all streams of the carrier.
 *RST: 50

<StartRB> numeric | `OFF`
 Position of first resource block. The same value must be configured for all streams of the carrier.
 *RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024 | `OFF`
 QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM | no DL sub-frame
 *RST: QPSK

<TransBlockSizeldx> numeric | `OFF`
 Transport block size index
 *RST: 5

Example: See [Configuring TTI-based user-defined channels](#)

Firmware/software: V2.1.30, SCC V3.2.50, V3.5.10 `OFF`, V3.5.40 256-QAM
 V3.7.70 1024-QAM

Options: R&S CMW-KS510
 256-QAM: R&S CMW-KS504/-KS554
 1024-QAM: R&S CMW-KS505/-KS555

Manual operation: See ["User-Defined TTI-Based"](#) on page 269

```

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL <TTI>,
    <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL? <TTI>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL <TTI>,
    <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL? <TTI>

```

Configures a selected uplink subframe for all scheduling types with a TTI-based UL definition.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

A query for TDD can also return `OFF`, `OFF`, `OFF`, `OFF`, indicating that the queried subframe is no UL subframe.

For UL-DL configuration 0, use the command `CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL:ALL` on page 833.

Suffix:

<c> 1..7

Parameters:

<NumberRB> numeric | OFF
Number of allocated resource blocks
*RST: 50

<StartRB> numeric | OFF
Position of first resource block
*RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256 | OFF
QPSK | 16-QAM | 64-QAM | 256-QAM | no UL subframe
*RST: QPSK

<TransBlockSizeIdx> numeric | OFF
Transport block size index
*RST: 6

Parameters for setting and query:

<TTI> numeric
Number of the subframe to be configured/queried.
Range: 0 to 9

Example: See [Configuring TTI-based user-defined channels](#)

Firmware/software: V2.0.20, V3.5.10: value `OFF`, V3.5.20: SCC command
V3.7.10: 64-QAM, V3.7.60: 256-QAM

Options: R&S CMW-KS510
64-QAM/256-QAM: R&S CMW-KS504/-KS554 for FDD/TDD

Manual operation: See ["UL configuration commands"](#) on page 268

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL:ALL

<NumberRB>(10), <StartRB>(10), <Modulation>(10), <TransBlockSizeldx>(10)

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL:ALL

<NumberRB>(10), <StartRB>(10), <Modulation>(10), <TransBlockSizeldx>(10)

Configures the uplink channel for all scheduling types with a TTI-based UL definition.

The parameters are entered 10 times, so that all subframes are configured by a single command (index = subframe number 0 to 9):

<NumberRB>₀, ..., <NumberRB>₉, <StartRB>₀, ..., <StartRB>₉, <Modulation>₀, ..., <Modulation>₉, <TransBlockSizeldx>₀, ..., <TransBlockSizeldx>₉

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.20, "User-defined channels"](#), on page 83.

For TDD DL and special subframes, you can set `OFF` or specify a number from the allowed input range. The effect is the same. A query returns `OFF` for non-UL subframes.

For UL-DL configuration 0, the settings specified for subframe number 2 are automatically applied to all UL subframes.

Suffix:

<c> 1..7

Parameters:

<NumberRB>	numeric OFF
	Number of allocated resource blocks
	*RST: 50
<StartRB>	numeric OFF
	Position of first resource block
	*RST: 0
<Modulation>	QPSK Q16 Q64 Q256 OFF
	QPSK 16-QAM 64-QAM 256-QAM no UL subframe
	*RST: QPSK
<TransBlockSizeldx>	numeric OFF
	Transport block size index
	*RST: 6

Example: See [Configuring TTI-based user-defined channels](#)

Firmware/software: V2.1.30

V3.5.10: value `OFF`, V3.5.20: SCC command

V3.7.10: 64-QAM, V3.7.60: 256-QAM

Options: R&S CMW-KS510

64-QAM/256-QAM: R&S CMW-KS504/-KS554 for FDD/TDD

Manual operation: See ["UL configuration commands"](#) on page 268

SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:DL<s>:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:DL<s>:CRATe:ALL?

Queries the code rate for all downlink subframes for the scheduling type "User-defined TTI-Based".

Suffix:

<s> 1..2

<c> 1..7

Return values:

<CodeRate> float

Comma-separated list of 10 values (subframe 0 to subframe 9)

Range: 0 to 50

Example: See [Configuring TTI-based user-defined channels](#)

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.2.50

Options: R&S CMW-KS510

Manual operation: See ["Code Rate"](#) on page 269

SENSe:LTE:SIGN<i>:CONNection[:PCC]:UDTTibased:UL:CRATe:ALL?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:UDTTibased:UL:CRATe:ALL?

Queries the code rate for all uplink subframes, applicable to all scheduling types with a TTI-based UL definition.

Suffix:

<c> 1..7

Return values:

<CodeRate> float

Comma-separated list of 10 values (subframe 0 to subframe 9)

Range: 0 to 10

Example: See [Configuring TTI-based user-defined channels](#)

Usage: Query only

Firmware/software: V3.0.20, SCC command V3.5.20

Options: R&S CMW-KS510

Manual operation: See ["Code Rate"](#) on page 269

2.6.15.18 Fixed CQI settings

The following commands define DL settings for the scheduling type "Fixed CQI".

For the query of code rates and all UL settings, use the "User-Defined TTI-Based" commands, see [Section 2.6.15.17, "User-defined TTI-based channels"](#), on page 829.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCTTibased:DL<s>.....	835
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCTTibased:DL<s>.....	835
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCTTibased:DL<s>:ALL.....	836
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCTTibased:DL<s>:ALL.....	836

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCTTibased:DL<s> <TTI>,
<NumberRB>, <StartRB>, <CQIIdx>

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCTTibased:DL<s>? <TTI>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCTTibased:DL<s> <TTI>,
<NumberRB>, <StartRB>, <CQIIdx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCTTibased:DL<s>? <TTI>

Configures a selected downlink subframe for the scheduling type "Fixed CQI".

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86.

A query for TDD can also return *OFF*, *OFF*, *OFF*, *OFF*, indicating that the queried subframe is no DL subframe.

Suffix:

<s> 1..2

<c> 1..7

Parameters:

<NumberRB> numeric | OFF

Number of allocated resource blocks. The same value must be configured for all streams of the carrier.

*RST: 50

<StartRB> numeric | OFF

Position of first resource block. The same value must be configured for all streams of the carrier.

*RST: 0

<CQIIdx> numeric | OFF

CQI index

Range: 1 to 15

*RST: 1

Parameters for setting and query:

<TTI> numeric

Number of the subframe to be configured/queried

Range: 0 to 9

*RST: 0

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.0.10, SCC command V3.2.50

V3.5.10: Value *OFF* added

Options: R&S CMW-KS510

Manual operation: See ["Fixed CQI"](#) on page 269

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCTTibased:DL<s>:ALL

<NumberRB>(10), <StartRB>(10), <CQIIdx>(10)

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCTTibased:DL<s>:ALL

<NumberRB>(10), <StartRB>(10), <CQIIdx>(10)

Configures the downlink channel for the scheduling type "Fixed CQI".

The parameters are entered 10 times, so that all subframes are configured by a single command (index = subframe number 0 to 9):

<NumberRB>₀, ..., <NumberRB>₉, <StartRB>₀, ..., <StartRB>₉, <CQIIdx>₀, ..., <CQIIdx>₉

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86.

For TDD UL and special subframes, you can set `OFF` or specify a number from the allowed input range. The effect is the same. A query returns `OFF` for non-DL subframes.

Suffix:

<s> 1..2

<c> 1..7

Parameters:

<NumberRB>	numeric OFF Number of allocated resource blocks. The same value must be configured for all streams of the carrier. *RST: 50
<StartRB>	numeric OFF Position of first resource block. The same value must be configured for all streams of the carrier. *RST: 0
<CQIIdx>	numeric OFF CQI index Range: 1 to 15 *RST: 1

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.0.10, SCC command V3.2.50
V3.5.10: Value `OFF` added

Options: R&S CMW-KS510

Manual operation: See ["Fixed CQI"](#) on page 269

2.6.15.19 Follow WB CQI settings

The following commands define DL settings for the scheduling type "Follow WB CQI".

For the UL settings, use the "User-Defined TTI-Based" commands, see [Section 2.6.15.17, "User-defined TTI-based channels"](#), on page 829.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:STTI.....	837
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:STTI.....	837
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL.....	838
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL.....	838
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:MCLuster:DL.....	838
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:MCLuster:DL.....	838
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:UDEFined.....	839
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:UDEFined.....	839
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:CSIRs:UDEFined.....	840
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:CSIRs:UDEFined....	840
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:SSUBframe: UDEFined.....	840
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:SSUBframe: UDEFined.....	840
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCS:ATABle:LIST?.....	841
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCS:ATABle:LIST?.....	841
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:DETermined?.....	841
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:DETermined?.....	841
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:CSIRs:DETermined?.....	842
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:CSIRs:DETermined?.....	842
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:SSUBframe: DETermined?.....	843
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:SSUBframe: DETermined?.....	843

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:STTI <Scheduled>...

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:STTI <Scheduled>...

Configures which subframes are scheduled for the DL of the scheduling type "Follow WB CQI".

For most subframes, the setting is fixed, depending on the duplex mode and the UL-DL configuration. For these subframes, your setting is ignored.

Suffix:

<c> 1..7

Parameters:

<Scheduled> OFF | ON
Comma-separated list of 10 values, for subframe 0 to 9

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Scheduled CQI"](#) on page 271

CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL <NumberRB>, <StartRB>, <Table>

CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL <NumberRB>, <StartRB>, <Table>

Configures the downlink for the scheduling type "Follow WB CQI", with contiguous RB allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86.

Suffix:

<c> 1..7

Parameters:

<NumberRB> numeric
Number of allocated resource blocks
*RST: 50

<StartRB> numeric
Position of first resource block
*RST: 0

<Table> DETermined | UDEFined
DETermined: Automatic CQI-MCS mapping table
UDEFined: User-defined mapping table
*RST: DET

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.0.50, SCC command V3.2.70

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["#RB / Start RB"](#) on page 272

CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:MCLuster:DL <Cluster>, <Table>

CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:MCLuster:DL <Cluster>, <Table>

Configures the downlink for the scheduling type "Follow WB CQI", with multi-cluster allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86 and especially [Table 2-46](#).

Suffix:

<c> 1..7

Parameters:

<Cluster> binary

Bitmap, enabling or disabling the individual RBGs
 1 means RBG is allocated, 0 means RBG is not allocated
 The number of bits depends on the cell bandwidth and equals the total number of RBGs.
 The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.
 Example for BW 1.4 MHz:
 #B101010 means that the RBGs 0, 2 and 4 are allocated

<Table>

DETermined | UDEFined

DETermined: Automatic CQI-MCS mapping table**UDEFined:** User-defined mapping table

*RST: DET

Example:See [Configuring CQI DL channels](#)**Firmware/software:** V3.5.50**Options:**

R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation:See ["Multicluster DL"](#) on page 271

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:UDEFined
 <MCS>(15)

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:UDEFined
 <MCS>(15)

Configures a user-defined mapping table that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI" if the table mode is set to UDEFined.

Suffix:

<c> 1..7

Parameters:

<MCS> numeric

Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15

The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.

Range: 0 to 28

*RST: 0, 1, 2, 3, 5, 7, 9, 12, 14, 16, 19, 22, 24, 27, 27

Example:See [Configuring CQI DL channels](#)**Firmware/software:** V3.0.50, SCC command V3.2.70**Options:**

R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation:See ["Commands for Follow WB CQI"](#) on page 274

**CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:CSIRs:
UDEFined <MCS>(15)**

**CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:CSIRs:
UDEFined <MCS>(15)**

Configures a user-defined mapping table for subframes with CSI-RS that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI" if the table mode is set to UDEFined.

Suffix:

<c> 1..7

Parameters:

<MCS> numeric
Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.
Range: 0 to 28
*RST: 0, 1, 3, 5, 7, 10, 12, 14, 17, 19, 21, 22, 24, 25, 25

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI"](#) on page 274

**CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:SSUBframe:
UDEFined <MCS>(15)**

**CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:
SSUBframe:UDEFined <MCS>(15)**

Configures a user-defined mapping table for special subframes that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI" if the table mode is set to UDEFined.

Suffix:

<c> 1..7

Parameters:

<MCS> numeric
Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.
Range: 0 to 28
*RST: 0, 1, 3, 5, 7, 10, 12, 14, 17, 19, 21, 22, 24, 25, 25

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI"](#) on page 274

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCS:ATABle:LIST?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCS:ATABle:LIST?

Returns a list of all mapping tables available for the scheduling type "Follow WB CQI" in the table mode **DETermined**.

Suffix:

<c> 1..7

Return values:

<Tables> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Comma-separated list of table identifiers that can be used in the mapping table commands
 SENSe:LTE:SIGN<i>:CONNection:...:FWBCqi:DL:MCSTable[:...]:DETermined.

Usage: Query only

Firmware/software: V3.7.70

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI"](#) on page 274

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:DETermined?
 [<Tablename>]
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:DETermined?
 [<Tablename>]

Queries an automatically determined mapping table for normal subframes. For the scheduling type "Follow WB CQI" in the table mode **DETermined**.

Suffix:

<c> 1..7

Query parameters:

<Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Selects which mapping table is queried. To check which tables are available, use [SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCS:ATABle:LIST?](#).
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total

TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
If the <Tablename> is omitted, ANY is used.

Return values:

<MCS> decimal
Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
Range: 0 to 31

Usage: Query only

Firmware/software: V3.0.50, SCC command V3.2.70, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI"](#) on page 274

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:CSIRs:

DETermined? [<Tablename>]

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:CSIRs:

DETermined? [<Tablename>]

Queries an automatically determined mapping table for subframes with CSI-RS. For the scheduling type "Follow WB CQI" in the table mode **DETermined**.

Suffix:

<c> 1..7

Query parameters:

<Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
Selects which mapping table is queried. To check which tables are available, use [SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCS:ATable:LIST?](#).
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total
TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
If the <Tablename> is omitted, ANY is used.

Return values:

<MCS> decimal
Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
Range: 0 to 31

Usage: Query only

Firmware/software: V3.5.40, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI"](#) on page 274

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FWBCqi:DL:MCSTable:SSUBframe:DETermined? [<Tablename>]

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCSTable:SSUBframe:DETermined? [<Tablename>]

Queries an automatically determined mapping table for special subframes. For the scheduling type "Follow WB CQI" in the table mode **DETermined**.

Suffix:

<c> 1..7

Query parameters:

<Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Selects which mapping table is queried. To check which tables are available, use [SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FWBCqi:DL:MCS:ATable:LIST?](#).
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total
TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
 If the <Tablename> is omitted, ANY is used.

Return values:

<MCS> decimal
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 Range: 0 to 31

Usage: Query only

Firmware/software: V3.5.50, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI"](#) on page 274

2.6.15.20 Follow WB PMI settings

The following commands define DL settings for the scheduling type "Follow WB PMI".

For the UL settings, use the "User-Defined TTI-Based" commands, see [Section 2.6.15.17, "User-defined TTI-based channels"](#), on page 829.

[CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:FPMI:DL:STTI](#).....844
[CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPMI:DL:STTI](#).....844
[CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:FPMI:DL](#).....844
[CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPMI:DL](#).....844

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPMI:MCLuster:DL.....	845
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPMI:MCLuster:DL.....	845

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPMI:DL:STTI <Scheduled>...

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPMI:DL:STTI <Scheduled>...

Configures which subframes are scheduled for the DL of the scheduling type "Follow WB PMI".

For most subframes, the setting is fixed, depending on the duplex mode and the UL-DL configuration. For these subframes, your setting is ignored.

Suffix:

<c> 1..7

Parameters:

<Scheduled> OFF | ON
Comma-separated list of 10 values, for subframe 0 to 9

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Scheduled CQI"](#) on page 276

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPMI:DL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeldx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPMI:DL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeldx>

Configures the downlink for the scheduling type "Follow WB PMI", with contiguous allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86.

Suffix:

<c> 1..7

Parameters:

<NumberRB> numeric
Number of allocated resource blocks
*RST: 50

<StartRB> numeric
Position of first resource block
*RST: 0

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024
QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM
*RST: QPSK

<TransBlockSizeldx> numeric

Transport block size index

*RST: 5

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.2.80, V3.5.40 256-QAM, V3.7.70 1024-QAM

Options: R&S CMW-KS510/-KS512 (without CA/with CA)
256-QAM: R&S CMW-KS504/-KS554
1024-QAM: R&S CMW-KS505/-KS555

Manual operation: See "[#RB / Start RB](#)" on page 277

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPMI:MCLuster:DL <Cluster>,
<Modulation>, <TransBlockSizeldx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPMI:MCLuster:DL <Cluster>,
<Modulation>, <TransBlockSizeldx>

Configures the downlink for the scheduling type "Follow WB PMI", with multi-cluster allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86 and especially [Table 2-46](#).

Suffix:

<c> 1..7

Parameters:

<Cluster> binary

Bitmap, enabling or disabling the individual RBGs

1 means RBG is allocated, 0 means RBG is not allocated

The number of bits depends on the cell bandwidth and equals the total number of RBGs.

The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.

Example for BW 1.4 MHz:

#B101010 means that the RBGs 0, 2 and 4 are allocated

<Modulation> QPSK | Q16 | Q64 | Q256 | Q1024

QPSK | 16-QAM | 64-QAM | 256-QAM | 1024-QAM

*RST: QPSK

<TransBlockSizeldx> numeric

Transport block size index

*RST: 5

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50, V3.7.70 1024-QAM

Options: R&S CMW-KS510/-KS512 (without CA/with CA)
256-QAM: R&S CMW-KS504/-KS554
1024-QAM: R&S CMW-KS505/-KS555

Manual operation: See ["Multicluster DL"](#) on page 277

2.6.15.21 Follow WB PMI-RI settings

The following commands define DL settings for the scheduling type "Follow WB PMI-RI".

For the UL settings, use the "User-Defined TTI-Based" commands, see [Section 2.6.15.17, "User-defined TTI-based channels"](#), on page 829.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL:STTI.....	846
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL:STTI.....	846
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL.....	846
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL.....	846
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:MCLuster:DL.....	847
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:MCLuster:DL.....	847

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL:STTI <Scheduled>...

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL:STTI <Scheduled>...

Configures which subframes are scheduled for the DL of the scheduling type "Follow WB PMI-RI".

For most subframes, the setting is fixed, depending on the duplex mode and the UL-DL configuration. For these subframes, your setting is ignored.

Suffix:

<c> 1..7

Parameters:

<Scheduled> OFF | ON
Comma-separated list of 10 values, for subframe 0 to 9

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Scheduled CQI"](#) on page 276

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:DL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeldx>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:DL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeldx>

Configures the downlink for the scheduling type "Follow WB PMI-RI", with contiguous allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86.

Suffix:

<c> 1..7

Parameters:

<NumberRB>	numeric
	Number of allocated resource blocks
	*RST: 50
<StartRB>	numeric
	Position of first resource block
	*RST: 0
<Modulation>	QPSK Q16 Q64 Q256 Q1024
	QPSK 16-QAM 64-QAM 256-QAM 1024-QAM
	*RST: QPSK
<TransBlockSizeIdx>	numeric
	Transport block size index
	*RST: 5

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.10, V3.5.40 256-QAM, V3.7.70 1024-QAM

Options: R&S CMW-KS510/-KS512 (without CA/with CA)
 256-QAM: R&S CMW-KS504/-KS554
 1024-QAM: R&S CMW-KS505/-KS555

Manual operation: See ["#RB / Start RB"](#) on page 277

CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FPRI:MCLuster:DL <Cluster>,
 <Modulation>, <TransBlockSizeIdx>

CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FPRI:MCLuster:DL <Cluster>,
 <Modulation>, <TransBlockSizeIdx>

Configures the downlink for the scheduling type "Follow WB PMI-RI", with multi-cluster allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86 and especially [Table 2-46](#).

Suffix:

<c> 1..7

Parameters:

<Cluster>	binary
	Bitmap, enabling or disabling the individual RBGs
	1 means RBG is allocated, 0 means RBG is not allocated
	The number of bits depends on the cell bandwidth and equals the total number of RBGs.
	The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.
	Example for BW 1.4 MHz: #B101010 means that the RBGs 0, 2 and 4 are allocated

<Modulation>	QPSK Q16 Q64 Q256 Q1024 QPSK 16-QAM 64-QAM 256-QAM 1024-QAM *RST: QPSK
<TransBlockSizeIdx>	numeric Transport block size index *RST: 5
Example:	See Configuring CQI DL channels
Firmware/software:	V3.5.50, V3.7.70 1024-QAM
Options:	R&S CMW-KS510/-KS512 (without CA/with CA) 256-QAM: R&S CMW-KS504/-KS554 1024-QAM: R&S CMW-KS505/-KS555
Manual operation:	See "Multicluseter DL" on page 277

2.6.15.22 Follow WB CQI-RI settings

The following commands define DL settings for the scheduling type "Follow WB CQI-RI".

For the UL settings, use the "User-Defined TTI-Based" commands, see [Section 2.6.15.17, "User-defined TTI-based channels"](#), on page 829.

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:STTI.....	848
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:STTI.....	848
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL.....	849
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL.....	849
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:MCLuster:DL.....	850
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:MCLuster:DL.....	850
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:UDEFineD.....	850
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:UDEFineD.....	850
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:SSUBframe:UDEFineD....	851
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:SSUBframe:UDEFineD.....	851
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCS:ATABle:LIST?.....	851
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCS:ATABle:LIST?.....	851
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:DETermined?.....	852
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:DETermined?.....	852
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:SSUBframe:DETermined?.....	853
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:SSUBframe:DETermined?..	853

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:STTI <Scheduled>...

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:STTI <Scheduled>...

Configures which subframes are scheduled for the DL of the scheduling type "Follow WB CQI-RI".

For most subframes, the setting is fixed, depending on the duplex mode and the UL-DL configuration. For these subframes, your setting is ignored.

Suffix:	
<c>	1..7
Parameters:	
<Scheduled>	OFF ON Comma-separated list of 10 values, for subframe 0 to 9
Example:	See Configuring CQI DL channels
Firmware/software:	V3.5.50
Options:	R&S CMW-KS510/-KS512 (without CA/with CA)
Manual operation:	See "Scheduled CQI" on page 271

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL <NumberRB>, <StartRB>, <Table>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL <NumberRB>, <StartRB>, <Table>

Configures the downlink for the scheduling type "Follow WB CQI-RI", with contiguous allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86.

Suffix:	
<c>	1..7
Parameters:	
<NumberRB>	numeric Number of allocated resource blocks *RST: 50
<StartRB>	numeric Position of first resource block *RST: 0
<Table>	DETermined UDEFined DETermined: Automatic CQI-MCS mapping table UDEFined: User-defined mapping table *RST: DET
Example:	See Configuring CQI DL channels
Firmware/software:	V3.2.80
Options:	R&S CMW-KS510/-KS512 (without CA/with CA)
Manual operation:	See "#RB / Start RB" on page 272

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:MCLuster:DL <Cluster>,
<Table>

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:MCLuster:DL <Cluster>,
<Table>

Configures the downlink for the scheduling type "Follow WB CQI-RI", with multi-cluster allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86 and especially [Table 2-46](#).

Suffix:

<c> 1..7

Parameters:

<Cluster> binary
 Bitmap, enabling or disabling the individual RBGs
 1 means RBG is allocated, 0 means RBG is not allocated
 The number of bits depends on the cell bandwidth and equals the total number of RBGs.
 The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency.
 Example for BW 1.4 MHz:
 #B101010 means that the RBGs 0, 2 and 4 are allocated

<Table> DETermined | UDEFined
DETermined: Automatic CQI-MCS mapping table
UDEFined: User-defined mapping table
 *RST: DET

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Multicluster DL"](#) on page 271

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:UDEFined
<MCS>(15)

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:UDEFined
<MCS>(15)

Configures a user-defined mapping table that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI-RI" if the table mode is set to UDEFined.

Suffix:

<c> 1..7

Parameters:

<MCS> numeric
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.
 Range: 0 to 28
 *RST: 0, 1, 2, 3, 5, 7, 9, 12, 14, 16, 19, 22, 24, 27, 28

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI-RI"](#) on page 275

CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:SSUBframe:UDEFined <MCS>(15)

CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:SSUBframe:UDEFined <MCS>(15)

Configures a user-defined mapping table for special subframes that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI-RI" if the table mode is set to UDEFined.

Suffix:

<c> 1..7

Parameters:

<MCS> numeric
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.
 Range: 0 to 28
 *RST: 0, 1, 3, 5, 7, 10, 12, 14, 17, 19, 21, 22, 24, 25, 25

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI-RI"](#) on page 275

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCS:ATABle:LIST?

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCS:ATABle:LIST?

Returns a list of all mapping tables available for the scheduling type "Follow WB CQI-RI" in the table mode DETermined.

Suffix:	
<c>	1..7
Return values:	
<Tables>	ANY CW1 CW2 OTLC1 OTLC2 TFLC1 TFLC2 Comma-separated list of table identifiers that can be used in the mapping table commands SENSE:LTE:SIGN<i>:CONNection:...:FCRI:DL: MCSTable[:...]:DETermined.
Usage:	Query only
Firmware/software:	V3.7.70
Options:	R&S CMW-KS510/-KS512 (without CA/with CA)
Manual operation:	See "Commands for Follow WB CQI-RI" on page 275

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:DETermined?
 [<Tablename>]

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:DETermined?
 [<Tablename>]

Queries an automatically determined mapping table for normal subframes. For the scheduling type "Follow WB CQI-RI" in the table mode DETermined.

Suffix:
 <c> 1..7

Query parameters:
 <Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Selects which mapping table is queried. To check which tables are available, use [SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCS:ATABLE:LIST?](#).
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total
TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
 If the <Tablename> is omitted, CW1 is used.

Return values:
 <MCS> decimal
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 Range: 0 to 31

Usage: Query only

Firmware/software: V3.2.80, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI-RI"](#) on page 275

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCRI:DL:MCSTable:SSUBframe:DETermined? [<Tablename>]

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCSTable:SSUBframe:DETermined? [<Tablename>]

Queries an automatically determined mapping table for special subframes. For the scheduling type "Follow WB CQI-RI" in the table mode **DETermined**.

Suffix:

<c> 1..7

Query parameters:

<Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Selects which mapping table is queried. To check which tables are available, use [SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCRI:DL:MCS:ATABLE:LIST?](#).
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total
TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
 If the <Tablename> is omitted, CW1 is used.

Return values:

<MCS> decimal
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 Range: 0 to 31

Usage: Query only

Firmware/software: V3.5.50, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI-RI"](#) on page 275

2.6.15.23 Follow WB CQI-PMI-RI settings

The following commands define DL settings for the scheduling type "Follow WB CQI-PMI-RI".

For the UL settings, use the "User-Defined TTI-Based" commands, see [Section 2.6.15.17, "User-defined TTI-based channels"](#), on page 829.

[CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:FCPRi:DL:STTI](#)..... 854
[CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:STTI](#)..... 854
[CONFigure:LTE:SIGN<i>:CONNection\[:PCC\]:FCPRi:DL](#)..... 854
[CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL](#)..... 854

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:MCLister:DL.....	855
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:MCLister:DL.....	855
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:UDEFined.....	856
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:UDEFined.....	856
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:CSIRs:UDEFined.....	856
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:CSIRs:UDEFined.....	856
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:SSUBframe:UDEFined..	857
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:SSUBframe: UDEFined.....	857
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCS:ATable:LIST?.....	858
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCS:ATable:LIST?.....	858
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:DETermined?.....	858
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:DETermined?.....	858
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:CSIRs:DETermined?.....	859
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:CSIRs:DETermined?.....	859
SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:SSUBframe:DETermined?...	860
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:SSUBframe: DETermined?.....	860

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:STTI <Scheduled>...

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:STTI <Scheduled>...

Configures which subframes are scheduled for the DL of the scheduling type "Follow WB CQI-PMI-RI".

For most subframes, the setting is fixed, depending on the duplex mode and the UL-DL configuration. For these subframes, your setting is ignored.

Suffix:

<c> 1..7

Parameters:

<Scheduled> OFF | ON
Comma-separated list of 10 values, for subframe 0 to 9

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Scheduled CQI"](#) on page 271

**CONFigure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL <NumberRB>, <StartRB>,
<Table>**

**CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL <NumberRB>,
<StartRB>, <Table>**

Configures the downlink for the scheduling type "Follow WB CQI-PMI-RI", with contiguous allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86.

Suffix:	
<c>	1..7
Parameters:	
<NumberRB>	numeric Number of allocated resource blocks *RST: 50
<StartRB>	numeric Position of first resource block *RST: 0
<Table>	DETermined UDEFined DETermined: Automatic CQI-MCS mapping table UDEFined: User-defined mapping table *RST: DET
Example:	See Configuring CQI DL channels
Firmware/software:	V3.2.80
Options:	R&S CMW-KS510/-KS512 (without CA/with CA)
Manual operation:	See " #RB / Start RB " on page 272

CONFigure:LTE:SIGN<i>:CONNectiOn[:PCC]:FCPRi:MCLister:DL <Cluster>,
<Table>

CONFigure:LTE:SIGN<i>:CONNectiOn:SCC<c>:FCPRi:MCLister:DL <Cluster>,
<Table>

Configures the downlink for the scheduling type "Follow WB CQI-PMI-RI", with multi-cluster allocation.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.21, "CQI channels"](#), on page 86 and especially [Table 2-46](#).

Suffix:	
<c>	1..7
Parameters:	
<Cluster>	binary Bitmap, enabling or disabling the individual RBGs 1 means RBG is allocated, 0 means RBG is not allocated The number of bits depends on the cell bandwidth and equals the total number of RBGs. The bitmap starts with RBG 0 (most significant bit) and continues with increasing RBG index / frequency. Example for BW 1.4 MHz: #B101010 means that the RBGs 0, 2 and 4 are allocated

<Table> DETermined | UDEFined
DETermined: Automatic CQI-MCS mapping table
UDEFined: User-defined mapping table
 *RST: DET

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.5.50

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Multicluster DL"](#) on page 271

CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:UDEFined
 <MCS>(15)

CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:UDEFined
 <MCS>(15)

Configures a user-defined mapping table that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI-PMI-RI" if the table mode is set to UDEFined.

Suffix:
 <c> 1..7

Parameters:
 <MCS> numeric
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.
 Range: 0 to 28
 *RST: 0, 1, 2, 3, 5, 7, 9, 12, 14, 16, 19, 22, 24, 27, 27

Example: See [Configuring CQI DL channels](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI-PMI-RI"](#) on page 275

CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:CSIRs:
 UDEFined <MCS>(15)

CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:CSIRs:
 UDEFined <MCS>(15)

Configures a user-defined mapping table for subframes with CSI-RS that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI-PMI-RI" if the table mode is set to UDEFined.

Suffix:

<c> 1..7

Parameters:

<MCS> numeric

Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15

The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.

Range: 0 to 28

*RST: 0, 1, 3, 5, 7, 10, 12, 14, 17, 19, 21, 22, 24, 25, 25

Example:See [Configuring CQI DL channels](#)**Firmware/software:** V3.5.40**Options:** R&S CMW-KS510/-KS512 (without CA/with CA)**Manual operation:** See ["Commands for Follow WB CQI-PMI-RI"](#) on page 275**CONFIGure:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:SSUBframe: UDEFined <MCS>(15)****CONFIGure:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:SSUBframe: UDEFined <MCS>(15)**

Configures a user-defined mapping table for special subframes that assigns an MCS index value to each possible reported wideband CQI index value.

The table is used for the scheduling type "Follow WB CQI-PMI-RI" if the table mode is set to UDEFined.

Suffix:

<c> 1..7

Parameters:

<MCS> numeric

Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15

The range is restricted by the highest allowed modulation scheme: max 27 with 256-QAM, max 26 with 1024-QAM.

Range: 0 to 28

*RST: 0, 1, 3, 5, 7, 10, 12, 14, 17, 19, 21, 22, 24, 25, 25

Example:See [Configuring CQI DL channels](#)**Firmware/software:** V3.5.50**Options:** R&S CMW-KS510/-KS512 (without CA/with CA)**Manual operation:** See ["Commands for Follow WB CQI-PMI-RI"](#) on page 275

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCS:ATABle:LIST?
SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCS:ATABle:LIST?

Returns a list of all mapping tables available for the scheduling type "Follow WB CQI-PMI-RI" in the table mode `DETermined`.

Suffix:

<c> 1..7

Return values:

<Tables> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Comma-separated list of table identifiers that can be used in the mapping table commands
 SENSE:LTE:SIGN<i>:CONNection:...:FCPRi:DL:MCSTable[:...]:DETermined.

Usage: Query only

Firmware/software: V3.7.70

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Commands for Follow WB CQI-PMI-RI"](#) on page 275

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:DETermined?
 [<Tablename>]

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:DETermined?
 [<Tablename>]

Queries an automatically determined mapping table for normal subframes. For the scheduling type "Follow WB CQI-PMI-RI" in the table mode `DETermined`.

Suffix:

<c> 1..7

Query parameters:

<Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Selects which mapping table is queried. To check which tables are available, use [SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCS:ATABle:LIST?](#).
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total
TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
 If the <Tablename> is omitted, OTLC1 is used.

Return values:

<MCS> decimal
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 Range: 0 to 31

Usage: Query only

Firmware/software: V3.2.80, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See "[Commands for Follow WB CQI-PMI-RI](#)" on page 275

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:CSIRs:DETermined? [<Tablename>]

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:CSIRs:DETermined? [<Tablename>]

Queries an automatically determined mapping table for subframes with CSI-RS. For the scheduling type "Follow WB CQI-PMI-RI" in the table mode DETermined.

Suffix:

<c> 1..7

Query parameters:

<Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Selects which mapping table is queried. To check which tables are available, use [SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCS:ATABLE:LIST?](#).
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total
TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
 If the <Tablename> is omitted, OTLC1 is used.

Return values:

<MCS> decimal
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 Range: 0 to 31

Usage: Query only

Firmware/software: V3.5.40, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See "[Commands for Follow WB CQI-PMI-RI](#)" on page 275

SENSe:LTE:SIGN<i>:CONNection[:PCC]:FCPRi:DL:MCSTable:SSUBframe:DETermined? [<Tablename>]

SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCSTable:SSUBframe:DETermined? [<Tablename>]

Queries an automatically determined mapping table for special subframes. For the scheduling type "Follow WB CQI-PMI-RI" in the table mode **DETermined**.

Suffix:

<c> 1..7

Query parameters:

<Tablename> ANY | CW1 | CW2 | OTLC1 | OTLC2 | TFLC1 | TFLC2
 Selects which mapping table is queried. To check which tables are available, use **SENSe:LTE:SIGN<i>:CONNection:SCC<c>:FCPRi:DL:MCS:ATABLE:LIST?**.
ANY: table used for all code word / layer constellations
CW1: codeword mapped to 1 layer
CW2: codeword mapped to 2 layers
OTLC1: codeword mapped to 1 layer / 1 or 2 layers in total
OTLC2: codeword mapped to 2 layers / 2 layers in total
TFLC1: codeword mapped to 1 layer / 3 or 4 layers in total
TFLC2: codeword mapped to 2 layers / 3 or 4 layers in total
 If the <Tablename> is omitted, OTLC1 is used.

Return values:

<MCS> decimal
 Comma-separated list of 15 MCS values, for reported CQI index value 1 to 15
 Range: 0 to 31

Usage: Query only

Firmware/software: V3.5.50, V3.7.70 added <Tablename>

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See "Commands for Follow WB CQI-PMI-RI" on page 275

2.6.15.24 SPS settings

The following commands define parameters for the scheduling type "SPS".

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:TIconfig..... 861
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:SINterval[:DL]..... 861
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:SINterval:UL..... 861
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:DL<s>..... 862
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:UL..... 862
SENSe:LTE:SIGN<i>:CONNection[:PCC]:SPS:DL<s>:CRATe:ALL?..... 863
SENSe:LTE:SIGN<i>:CONNection[:PCC]:SPS:UL:CRATe:ALL?..... 863

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:TIConfig <Enable>

Configures the parameter "twoIntervalsConfig", signaled to the UE for the scheduling type SPS in TDD mode.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring SPS](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See "[SPS TwoIntervalsConfig TDD](#)" on page 225

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:SINteRval[:DL] <Interval>

Configures the DL subframe periodicity n for the scheduling type SPS. The UE is granted the configured RB allocation in every nth subframe.

For TDD, the selected value is internally rounded down to a multiple of 10. Example: S128 means every 120th subframe.

Parameters:

<Interval> S10 | S20 | S32 | S40 | S64 | S80 | S128 | S160 | S320 | S640
Every 10th subframe to every 640th subframe
*RST: S20

Example: See [Configuring SPS](#)

Firmware/software: V3.2.80
V3.2.82: changed *RST value

Options: R&S CMW-KS510

Manual operation: See "[Interval DL / UL](#)" on page 261

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:SINteRval:UL <Interval>

Configures the UL subframe periodicity n for the scheduling type SPS. The UE is granted the configured RB allocation in every nth subframe.

For TDD, the selected value is internally rounded down to a multiple of 10. Example: S128 means every 120th subframe.

Parameters:

<Interval> SADL | S1 | S2 | S3 | S4 | S5 | S10 | S20 | S32 | S40 | S64 | S80 | S128 | S160 | S320 | S640
SADL: same as downlink
S1, S2 to S640: Every subframe, every 2nd subframe to every 640th subframe
*RST: SADL

Example: See [Configuring SPS](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS510

Manual operation: See ["Interval DL / UL"](#) on page 261

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:DL<s> <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>

Configures the downlink RB allocation for the scheduling type SPS.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.22, "Semi-persistent scheduling \(SPS\)"](#), on page 90.

Suffix:

<s> 1..2

Parameters:

<NumberRB> numeric
Number of allocated resource blocks
*RST: 50

<StartRB> numeric
Position of first resource block
*RST: 0

<Modulation> QPSK | Q16
Modulation type QPSK | 16-QAM
*RST: QPSK

<TransBlockSizeIdx> numeric
Transport block size index
*RST: 5

Example: See [Configuring SPS](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["# RB ... TBSI /TBS"](#) on page 261

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:SPS:UL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>

Configures the uplink RB allocation for the scheduling type SPS.

The allowed input ranges have dependencies and are described in the background information, see [Section 2.2.22, "Semi-persistent scheduling \(SPS\)"](#), on page 90.

Parameters:

<NumberRB>	numeric
	Number of allocated resource blocks
	*RST: 50
<StartRB>	numeric
	Position of first resource block
	*RST: 0
<Modulation>	QPSK Q16
	Modulation type QPSK 16-QAM
	*RST: QPSK
<TransBlockSizeIdx>	numeric
	Transport block size index
	*RST: 6

Example: See [Configuring SPS](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["# RB ... TBSI /TBS"](#) on page 261

SENSe:LTE:SIGN<i>:CONNection[:PCC]:SPS:DL<s>:CRATe:ALL?

Queries the code rate for all downlink subframes for the scheduling type SPS.

Suffix:

<s>	1..2
	Selects the downlink stream

Return values:

<CodeRate>	float
	Comma-separated list of 10 values (subframe 0 to subframe 9)
	Range: 0 to 50

Example: See [Configuring SPS](#)

Usage: Query only

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["Code Rate"](#) on page 261

SENSe:LTE:SIGN<i>:CONNection[:PCC]:SPS:UL:CRATe:ALL?

Queries the code rate for all uplink subframes for the scheduling type SPS.

Return values:

<CodeRate> float
 Comma-separated list of 10 values (subframe 0 to subframe 9)
 Range: 0 to 10

Example: See [Configuring SPS](#)

Usage: Query only

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["Code Rate"](#) on page 261

2.6.15.25 eMTC auto mode settings

The following commands define parameters for the scheduling type "eMTC Auto Mode".

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:DL:ANB<Number>	864
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:DL:ALL	865
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:UL:ALL	865
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:B:DL:ALL	866
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:B:UL:ALL	867

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:DL:ANB<Number>
 <AdditionalNB>

Enables or disables additional DL narrowbands for the eMTC auto mode, for a maximum eMTC bandwidth of 5 MHz.

Suffix:

<Number> 1..3
 Number of the additional narrowband, relative to the lowest used narrowband (1st, 2nd or 3rd higher narrowband).

Parameters:

<AdditionalNB> OFF | ON
ON: Use the additional narrowband.
OFF: Do not use the additional narrowband.
 *RST: OFF

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS590

Manual operation: See ["DL, # Resource Blocks ... Transport Block Size"](#) on page 263

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:DL:ALL <NumberRB>, <StartRB>, <NarrowBand>, <Modulation>, <TranspBlockSizeldx>

Configures the eMTC auto mode, downlink, for CE mode A.

The indicated input ranges list all possible values. The ranges have dependencies described in the background information, see [Section 2.2.15, "eMTC auto mode"](#), on page 53.

Parameters:

<NumberRB>	ZERO N1 N2 N3 N4 N5 N6 Number of allocated resource blocks *RST: N6
<StartRB>	numeric Position of first resource block Range: 0 to 6 *RST: 0
<NarrowBand>	numeric Narrowband for the first transmission Range: 0 to 15 *RST: 0
<Modulation>	QPSK Q16 Modulation type QPSK 16-QAM *RST: QPSK
<TranspBlockSizeldx>	numeric Transport block size index Range: 0 to 14 *RST: 6

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["DL, # Resource Blocks ... Transport Block Size"](#) on page 263

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:A:UL:ALL <NumberRB>, <StartRB>, <NarrowBand>, <Modulation>, <TranspBlockSizeldx>

Configures the eMTC auto mode, uplink, for CE mode A.

The indicated input ranges list all possible values. The ranges have dependencies described in the background information, see [Section 2.2.15, "eMTC auto mode"](#), on page 53.

Parameters:

<NumberRB>	ZERO N1 N2 N3 N4 N5 N6 N9 N12 N15 N18 N21 N24 Number of allocated resource blocks *RST: N6
<StartRB>	numeric Position of first resource block Range: 0 to 6 *RST: 0
<NarrowBand>	numeric Narrowband for the first transmission Range: 0 to 15 *RST: 0
<Modulation>	QPSK Q16 Modulation type QPSK 16-QAM *RST: QPSK
<TranspBlockSizeIdx>	numeric Transport block size index Range: 0 to 14, 16 to 21 *RST: 6

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20, V3.8.10 N9 to N24

Options: R&S CMW-KS590

Manual operation: See "[UL, # Resource Blocks ... Transport Block Size](#)" on page 263

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:B:DL:ALL <NumberRB>, <StartRB>, <NarrowBand>, <Modulation>, <TBSIndex>

Configures the eMTC auto mode, downlink, for CE mode B.

The indicated input ranges list all possible values. The ranges have dependencies described in the background information, see [Section 2.2.15, "eMTC auto mode"](#), on page 53.

Parameters:

<NumberRB>	numeric ZERO N4 N6 Number of allocated resource blocks *RST: N6
------------	---

<StartRB>	numeric Position of first resource block Range: 0 to 0 *RST: 0
<NarrowBand>	numeric Narrowband for the first transmission Range: 0 to 15 *RST: 0
<Modulation>	QPSK Modulation type QPSK *RST: QPSK
<TBSindex>	numeric Transport block size index Range: 0 to 9 *RST: 6

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.70

Options: R&S CMW-KS590

Manual operation: See ["DL, # Resource Blocks ... Transport Block Size"](#) on page 263

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:EMAMode:B:UL:ALL <NumberRB>, <StartRB>, <NarrowBand>, <Modulation>, <TransBlockSizeIdx>

Configures the eMTC auto mode, uplink, for CE mode B.

The indicated input ranges list all possible values. The ranges have dependencies described in the background information, see [Section 2.2.15, "eMTC auto mode"](#), on page 53.

Parameters:

<NumberRB>	ZERO N1 N2 Number of allocated resource blocks *RST: N2
<StartRB>	numeric Position of first resource block Range: 0 to 5 *RST: 0
<NarrowBand>	numeric Narrowband for the first transmission Range: 0 to 15 *RST: 0

<Modulation>	QPSK Modulation type QPSK *RST: QPSK
<TransBlockSizeIdx>	numeric Transport block size index Range: 0 to 10 *RST: 5
Example:	See Configuring eMTC settings
Firmware/software:	V3.7.70
Options:	R&S CMW-KS590
Manual operation:	See "UL, # Resource Blocks ... Transport Block Size" on page 263

2.6.15.26 eMTC compact scheduling settings

The following commands define parameters for the scheduling type "eMTC Compact Scheduling".

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:SFPattern	868
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:A:DL:ALL	868
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:A:UL:ALL	869
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:A:DL:ANB<Number>	870

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:SFPattern <Pattern>

Selects the subframe pattern for eMTC compact scheduling, half-duplex. There are patterns with and without PDSCH HARQ-ACK bundling.

Parameters:

<Pattern>	STANdard HAB8 HAB10 STANdard: no bundling HAB8: bundling, 8 HARQ processes HAB10: bundling, 10 HARQ processes *RST: STAN
-----------	---

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS590

Manual operation: See "[SF Pattern](#)" on page 264

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:A:DL:ALL <NumberRB>, <StartRB>, <Modulation>, <TransBlockSizeIdx>

Configures eMTC compact scheduling, downlink, for CE mode A.

The indicated input ranges list all possible values. The ranges have dependencies described in the background information, see [Section 2.2.14, "eMTC compact scheduling"](#), on page 51.

Parameters:

<NumberRB>	numeric
	Number of allocated resource blocks
	Range: 0 to 4
	*RST: 4
<StartRB>	numeric
	Position of first resource block
	Range: 0 to 4
	*RST: 0
<Modulation>	QPSK Q16
	Modulation type QPSK 16-QAM
	*RST: QPSK
<TransBlockSizIdx>	numeric
	Transport block size index
	Range: 0 to 14
	*RST: 5

Example: See [Configuring eMTC settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS590

Manual operation: See ["DL, # Resource Blocks ... Transport Block Size"](#) on page 265

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:A:UL:ALL

<NumberRB>, <StartRB>, <Modulation>, <TransBlockSizIdx>

Configures eMTC compact scheduling, uplink, for CE mode A.

The indicated input ranges list all possible values. The ranges have dependencies described in the background information, see [Section 2.2.14, "eMTC compact scheduling"](#), on page 51.

Parameters:

<NumberRB>	numeric
	Number of allocated resource blocks
	Range: 0 to 24
	*RST: 6
<StartRB>	numeric
	Position of first resource block
	Range: 0 to 6
	*RST: 0

<Modulation>	QPSK Q16 Modulation type QPSK 16-QAM *RST: QPSK
<TransBlockSizeIdx>	numeric Range: 0 to 14, 16 to 21 *RST: 6
Example:	See Configuring eMTC settings
Firmware/software:	V3.7.20
Options:	R&S CMW-KS590
Manual operation:	See " UL, # Resource Blocks ... Transport Block Size " on page 265

CONFigure:LTE:SIGN<i>:CONNection[:PCC]:CSCHeduling:A:DL:ANB<Number> <AdditionalNB>

Enables or disables additional narrowbands for compact scheduling downlink, for a maximum eMTC bandwidth of 5 MHz.

Suffix:	
<Number>	1..3 Number of the additional narrowband (1st, 2nd or 3rd narrowband above the lowest used narrowband).
Parameters:	
<AdditionalNB>	OFF ON ON: Use the additional narrowband. OFF: Do not use the additional narrowband. *RST: OFF
Example:	See Configuring eMTC settings
Firmware/software:	V3.8.10
Options:	R&S CMW-KS590
Manual operation:	See " DL, # Resource Blocks ... Transport Block Size " on page 265

2.6.16 CQI reporting settings

The following commands configure CQI reporting.

CONFigure:LTE:SIGN<i>:CQIReporting:ENABLE	871
CONFigure:LTE:SIGN<i>:CQIReporting:CSIRmode	871
CONFigure:LTE:SIGN<i>:CQIReporting:SANCqi	871
CONFigure:LTE:SIGN<i>:CQIReporting:PRIReporting:ENABLE	872
CONFigure:LTE:SIGN<i>:CQIReporting[:PCC]:CINdex[:FDD]	872
CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINdex[:FDD]	872

CONFigure:LTE:SIGN<i>:CQIReporting[:PCC]:CINDeX:TDD.....	872
CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINDeX:TDD.....	872
CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINDeX:LAA.....	873
SENSe:LTE:SIGN<i>:CQIReporting[:PCC]:RPERiod?.....	873
SENSe:LTE:SIGN<i>:CQIReporting:SCC<c>:RPERiod?.....	873
SENSe:LTE:SIGN<i>:CQIReporting[:PCC]:ROFFset?.....	874
SENSe:LTE:SIGN<i>:CQIReporting:SCC<c>:ROFFset?.....	874

CONFigure:LTE:SIGN<i>:CQIReporting:ENABLE <Enable>

Enables/disables periodic CQI reporting.

Parameters:

<Enable> OFF | PERiodic
OFF: no CQI reporting
PERiodic: periodic CQI reporting
 *RST: OFF

Example: See [Configuring CQI reporting](#)

Firmware/software: V3.0.10

Options: R&S CMW-KS510 for scenarios without carrier aggregation
 R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["Enable CQI Reporting"](#) on page 278

CONFigure:LTE:SIGN<i>:CQIReporting:CSIRmode <Mode>

Configures the CSI reporting mode.

Parameters:

<Mode> S1 | S2
S1: submode 1
S2: submode 2
 *RST: S2

Example: See [Configuring CQI reporting](#)

Firmware/software: V3.5.10

Options: R&S CMW-KS510 for scenarios without carrier aggregation
 R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["CSI - Report Mode"](#) on page 279

CONFigure:LTE:SIGN<i>:CQIReporting:SANCqi <Enable>

Configures whether the simultaneous transmission of ACK/NACK and CQI is allowed.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring CQI reporting](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS510

Manual operation: See ["Simultaneous AckNack and CQI"](#) on page 279

CONFigure:LTE:SIGN<i>:CQIReporting:PRIReporting:ENABLE <Enable>

Enables/disables PMI/RI reporting for transmission mode 8 and 9. As a prerequisite for PMI and RI reporting, CQI reporting must also be enabled.

Parameters:

<Enable> OFF | ON
OFF: only CQI reporting
ON: CQI, PMI and RI reporting
 *RST: OFF

Example: See [Configuring CQI reporting](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510 for scenarios without carrier aggregation
 R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["Enable PMI/RI Reporting \(TM 8, 9\)"](#) on page 279

CONFigure:LTE:SIGN<i>:CQIReporting[:PCC]:CINDeX[:FDD] <Index>

CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINDeX[:FDD] <Index>

Specifies the FDD "cqi-pmi-ConfigIndex" ($I_{CQI/PMI}$).

Suffix:

<c> 1..7

Parameters:

<Index> integer
 Range: 0 to 316, 318 to 541

Example: See [Configuring CQI reporting](#)

Firmware/software: V3.0.10, SCC command V3.2.70

Options: R&S CMW-KS500
 R&S CMW-KS510 for scenarios without carrier aggregation
 R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["CQI/PMI Config Index"](#) on page 279

CONFigure:LTE:SIGN<i>:CQIReporting[:PCC]:CINDeX:TDD <Index>

CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINDeX:TDD <Index>

Specifies the TDD "cqi-pmi-ConfigIndex" ($I_{CQI/PMI}$).

Suffix:
 <c> 1..7

Parameters:
 <Index> integer
 Range: 1 to 315

Example: See [Configuring CQI reporting](#)

Firmware/software: V3.0.50, SCC command V3.2.70

Options: R&S CMW-KS550
 R&S CMW-KS510 for scenarios without carrier aggregation
 R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["CQI/PMI Config Index"](#) on page 279

CONFigure:LTE:SIGN<i>:CQIReporting:SCC<c>:CINdex:LAA <ConflIndex>

Specifies the LAA "cqi-pmi-ConfigIndex" ($I_{\text{CQI/PMI}}$).

Suffix:
 <c> 1..7

Parameters:
 <ConflIndex> numeric
 Range: 0 to 316, 318 to 541

Example: See [Configuring CQI reporting](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS512 and R&S CMW-KS514

Manual operation: See ["CQI/PMI Config Index"](#) on page 279

SENSe:LTE:SIGN<i>:CQIReporting[:PCC]:RPERiod? **SENSe:LTE:SIGN<i>:CQIReporting:SCC<c>:RPERiod?**

Queries the reporting period N_p in subframes, resulting from the configured "cqi-pmi-ConfigIndex".

Suffix:
 <c> 1..7

Return values:
 <Period> decimal
 Range: 1 to 160

Example: See [Configuring CQI reporting](#)

Usage: Query only

Firmware/software: V3.0.10, SCC command V3.2.70

Options: R&S CMW-KS510 for scenarios without carrier aggregation
 R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["CQI/PMI Reporting Period / Offset"](#) on page 280

SENSe:LTE:SIGN<i>:CQIReporting[:PCC]:ROFFset?
SENSe:LTE:SIGN<i>:CQIReporting:SCC<c>:ROFFset?

Queries the reporting offset $N_{\text{OFFSET,CQI}}$ in subframes, resulting from the configured "cqi-pmi-ConfigIndex".

Suffix:

<c> 1..7

Return values:

<Offset> decimal
Range: 0 to 159

Example: See [Configuring CQI reporting](#)

Usage: Query only

Firmware/software: V3.0.10, SCC command V3.2.70

Options: R&S CMW-KS510 for scenarios without carrier aggregation
R&S CMW-KS512 for scenarios with carrier aggregation

Manual operation: See ["CQI/PMI Reporting Period / Offset"](#) on page 280

2.6.17 UE measurement report settings

The following commands configure UE measurement reports.

CONFigure:LTE:SIGN<i>:UEReport:ENABLE.....	874
CONFigure:LTE:SIGN<i>:UEReport:RINTerval.....	875
CONFigure:LTE:SIGN<i>:UEReport:MGENable.....	875
CONFigure:LTE:SIGN<i>:UEReport:MGPeriod.....	875
CONFigure:LTE:SIGN<i>:UEReport:FCoefficient:RSRP.....	876
CONFigure:LTE:SIGN<i>:UEReport:FCoefficient:RSRQ.....	876
CONFigure:LTE:SIGN<i>:UEReport:WMQuantity.....	876
CONFigure:LTE:SIGN<i>:UEReport:MCSCell.....	876
CONFigure:LTE:SIGN<i>:UEReport:AINTErrupt.....	877
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:DMTC:PERiod.....	877
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:DMTC:POFFset.....	877
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:ENABLE.....	878
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:RMTC:PERiod.....	878
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:RMTC:SOFFset.....	879
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:MDURATION.....	879
CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:COTHreshold.....	879

CONFigure:LTE:SIGN<i>:UEReport:ENABLE <Enable>

Enables or disables UE measurement reports.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring measurement reports](#)

Firmware/software: V2.0.10

Manual operation: See ["Report"](#) on page 282

CONFigure:LTE:SIGN<i>:UEReport:RINterval <Interval>

Sets the interval between two consecutive measurement reports.

Parameters:

<Interval> I120 | I240 | I480 | I640 | I1024 | I2048 | I5120 | I10240
 Interval in ms, e.g. I240 = 240 ms
 *RST: I1024

Example: See [Configuring measurement reports](#)

Firmware/software: V2.1.20

Manual operation: See ["Reporting Interval"](#) on page 282

CONFigure:LTE:SIGN<i>:UEReport:MGENable <Enable>

Enables or disables transmission gaps for neighbor cell measurements.

Parameters:

<Enable> OFF | ON
 *RST: ON

Example: See [Configuring measurement reports](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS510

Manual operation: See ["Measurement Gap Enable"](#) on page 282

CONFigure:LTE:SIGN<i>:UEReport:MGPeriod <Gap>

Specifies the periodicity of transmission gaps for neighbor cell measurements.

Parameters:

<Gap> G040 | G080
G040: one gap per 40 ms
G080: one gap per 80 ms
 *RST: G040

Example: See [Configuring measurement reports](#)

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["Measurement Gap Period"](#) on page 282

CONFigure:LTE:SIGN<i>:UEReport:FCoefficient:RSRP <Filter>

Selects the value to be sent to the UE as "filterCoefficientRSRP". It is used by the UE to measure the reference signal received power (RSRP).

Parameters:

<Filter> FC0 | FC4
*RST: FC4

Example: See [Configuring measurement reports](#)

Firmware/software: V3.2.81

Manual operation: See ["Filter Coefficient RSRP/RSRQ"](#) on page 282

CONFigure:LTE:SIGN<i>:UEReport:FCoefficient:RSRQ <Filter>

Selects the value to be sent to the UE as "filterCoefficientRSRQ". It is used by the UE to measure the reference signal received quality (RSRQ).

Parameters:

<Filter> FC0 | FC4
*RST: FC4

Example: See [Configuring measurement reports](#)

Firmware/software: V3.2.81

Manual operation: See ["Filter Coefficient RSRP/RSRQ"](#) on page 282

CONFigure:LTE:SIGN<i>:UEReport:WMQuantity <Quantity>

Selects whether the UE must determine the RSCP or the Ec/No during WCDMA neighbor cell measurements.

Parameters:

<Quantity> RSCP | ECNO
*RST: RSCP

Example: See [Configuring measurement reports](#)

Firmware/software: V3.0.50

Options: R&S CMW-KS510

Manual operation: See ["WCDMA Measurement Quantity"](#) on page 282

CONFigure:LTE:SIGN<i>:UEReport:MCSCell <Cycle>

Specifies the signaling parameter "measCycleSCell".

Parameters:

<Cycle> OFF | SF160 | SF256 | SF320 | SF512 | SF640 | SF1024 | SF1280
OFF: Do not signal "measCycleSCell"
SFn: n subframes
 *RST: OFF

Example: See [Configuring measurement reports](#)

Firmware/software: V3.2.70

Manual operation: See ["Measurement Cycle SCell"](#) on page 283

CONFigure:LTE:SIGN<i>:UEReport:AINterrupt <Enable>

Specifies the signaling parameter "allowInterruptions".

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring measurement reports](#)

Firmware/software: V3.7.31

Manual operation: See ["Allow Interruptions"](#) on page 283

CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:DMTC:PERiod <Period>

Specifies the periodicity of the DRS for LAA.

Suffix:

<c> 1..7

Parameters:

<Period> MS40 | MS80 | MS160
 Periodicity in ms
 *RST: MS40

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["DMTC Configuration"](#) on page 283

CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:DMTC:POFFset <Offset>

Specifies the offset of the DRS for LAA.

The offset must be at least 5 ms smaller than the configured periodicity, see [CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:DMTC:PERiod](#).

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 Range: 0 ms to 155 ms
 Increment: 5 ms
 *RST: 0 ms
 Default unit: ms

Example: See [Configuring LAA settings](#)**Firmware/software:** V3.7.20**Options:** R&S CMW-KS514**Manual operation:** See ["DMTC Configuration"](#) on page 283**CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:ENABLE <Enable>**

Enables or disables the signaling of the IE "measRSSI-ReportConfig" to the UE.

Suffix:

<c> 1..7

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring LAA settings](#)**Firmware/software:** V3.7.20**Options:** R&S CMW-KS514**Manual operation:** See ["Enable"](#) on page 283**CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:RMTC:PERiod <Period>**

Specifies the periodicity of UE measurements for LAA.

Suffix:

<c> 1..7

Parameters:

<Period> MS40 | MS80 | MS160 | MS320 | MS640
 Periodicity in ms
 *RST: MS40

Example: See [Configuring LAA settings](#)**Firmware/software:** V3.7.20**Options:** R&S CMW-KS514

Manual operation: See ["RMTC Period, RMTC Subframe Offset, Measurement Duration"](#) on page 284

CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:RMTC:SOFFset <Offset>

Specifies the offset of UE measurements for LAA.

The offset must be at least 5 ms smaller than the configured periodicity, see [CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:RMTC:PERiod](#).

Suffix:

<c> 1..7

Parameters:

<Offset> numeric
 Range: 0 ms to 635 ms
 Increment: 5 ms
 *RST: 20 ms
 Default unit: ms

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["RMTC Period, RMTC Subframe Offset, Measurement Duration"](#) on page 284

CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:MDURATION <Duration>

Specifies the duration of UE measurements for LAA.

Suffix:

<c> 1..7

Parameters:

<Duration> S1 | S14 | S28 | S42 | S70
 Duration in OFDM symbols
 *RST: S14

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["RMTC Period, RMTC Subframe Offset, Measurement Duration"](#) on page 284

CONFigure:LTE:SIGN<i>:UEReport:SCC<c>:RSSI:COTHreshold <Threshold>

Specifies a threshold for channel occupancy measurements for LAA. The setting is signaled to the UE as "channelOccupancyThreshold".

The same value applies to all SCCs with frame structure type 3.

Suffix:

<c> 1

Parameters:

<Threshold> numeric
Range: 0 to 76
*RST: 63

Example: See [Configuring LAA settings](#)

Firmware/software: V3.7.20

Options: R&S CMW-KS514

Manual operation: See ["Channel Occupancy Threshold"](#) on page 284

2.6.18 Messaging (SMS)

The following commands configure parameters of the short message service (SMS) and return information about received short messages.

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:LHAndling.....	881
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:MEShAndling.....	881
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:INTErnal.....	881
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:BINary.....	882
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:PIDentifier.....	882
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:DCODing.....	882
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:UDHeader.....	882
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:CGRoup.....	883
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:MCLass.....	883
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:OSADdress.....	883
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:OADdress.....	884
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TSource.....	884
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:DATE.....	884
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TIME.....	885
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE.....	885
CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE:INFO?.....	886
SENSe:LTE:SIGN<i>:SMS:OUTGoing:INFO:LMSent?.....	886
CONFigure:LTE:SIGN<i>:SMS:INComing:FILE.....	887
CONFigure:LTE:SIGN<i>:SMS:INComing:FILE:INFO?.....	887
SENSe:LTE:SIGN<i>:SMS:INComing:INFO:DCODing?.....	887
SENSe:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT?.....	888
SENSe:LTE:SIGN<i>:SMS:INComing:INFO:MLENgtH?.....	888
CLEAn:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT.....	888
SENSe:LTE:SIGN<i>:SMS:INFO:LRMessage:RFLag?.....	888

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:LHANDling <LSMSHandling>

Selects the handling of messages exceeding 160 characters.

Parameters:

<LSMSHandling> TRUNcate | MSMS

TRUNcate

The SMS is truncated to 160 characters, the rest is discarded.

MSMS

Up to five concatenated messages are sent, consisting in sum of up to 800 characters.

*RST: TRUN

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.5.30

Manual operation: See ["Large SMS Handling"](#) on page 285

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:MESHandling <MessageHandling>

Selects whether an outgoing message is defined directly via the GUI/commands or read from a file.

For file selection, see [CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE](#) on page 885.

Parameters:

<MessageHandling> INTernal | FILE

INTernal: message defined directly

FILE: message specified via a file

*RST: INT

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.5.50

Manual operation: See ["Outgoing Message Handling"](#) on page 286

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:INTernal <SMSInternal>

Defines the message text for outgoing 7-bit ASCII messages.

Parameters:

<SMSInternal> string

Message contents, up to 800 characters

RST: "R&S Short Message Service Text.The quick brown fox jumps over the lazy dog.THE QUICK BROWN FOX JUMPS OVER THE LAZY DOG.0123456789 !"#%+-(/)<>?=:,@\$,,"

Example: See [Sending/receiving a short message](#)

Firmware/software: V2.1.20

Manual operation: See ["Outgoing SMS"](#) on page 286

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:BINary <SMSbinary>

Defines the message contents for outgoing 8-bit binary messages.

Parameters:

<SMSbinary> hex
 Message contents, up to 1400 digits

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.2.70

Manual operation: See ["Outgoing SMS binary"](#) on page 286

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:PIDentifier <ID>

Specifies the TP protocol identifier (TP-PID) value to be sent.

Parameters:

<ID> numeric
 Range: #H0 to #HFF
 *RST: #H0

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.5.20

Manual operation: See ["Protocol Identifier"](#) on page 286

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:DCODing <DataCoding>

Selects the data coding for outgoing messages.

Parameters:

<DataCoding> BIT7 | BIT8
 BIT7: 7-bit encoded ASCII message
 BIT8: 8-bit encoded binary message
 *RST: BIT7

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.2.70

Manual operation: See ["Data Coding / Character Set"](#) on page 286

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:UDHeader <Header>

Configures the TP user data header.

Parameters:

<Header> hex | ON | OFF
 Up to 16 hexadecimal digits
 Range: #H0 to #FFFFFFFFFFFFFFFF
 *RST: OFF
 ON | OFF enables or disables sending the header.

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.5.40

Manual operation: See ["User Data Header"](#) on page 286

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:CGRoup <CodingGroup>

Selects the coding group to be indicated to the message recipient in the TP-Data-Coding-Scheme field.

Parameters:

<CodingGroup> GDCoding | DCMClass
GDCoding: general data coding
DCMClass: data coding / message class
 *RST: GDC

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.2.70

Manual operation: See ["Coding Group"](#) on page 287

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:MCLass <MessageClass>

Selects the message class to be indicated to the message recipient in the TP-Data-Coding-Scheme field.

Parameters:

<MessageClass> CL0 | CL1 | CL2 | CL3 | NONE
CL0, CL1, CL2, CL3: Class 0 to 3
NONE: Do not send message class
 *RST: NONE

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.2.70

Manual operation: See ["Message Class"](#) on page 287

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:OSADdress <OrigSMSCAddress>

Specifies the originator short message service center address to be sent to the recipient.

Parameters:

<OrigSMSCAddress> string

*RST: '764332637249279'

Example: See [Sending/receiving a short message](#)**Firmware/software:** V3.2.70**Manual operation:** See ["Originator SMSC Address"](#) on page 287**CONFigure:LTE:SIGN<i>:SMS:OUTGoing:OAddress <OrigAddress>**

Specifies the originating address to be sent to the message recipient.

Parameters:

<OrigAddress> string

*RST: '764332637249279'

Example: See [Sending/receiving a short message](#)**Firmware/software:** V3.2.70**Manual operation:** See ["Originating Address"](#) on page 287**CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TSource <SourceTime>**

Selects the source for the service center time stamp.

The date and time for the source DATE is configured via the following commands:

- [CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:DATE](#)
- [CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TIME](#)

Parameters:

<SourceTime> CMWTime | DATE

CMWTime: Current date and time of the operation system**DATE:** Date and time specified via remote commands

*RST: CMWT

Example: See [Sending/receiving a short message](#)**Firmware/software:** V3.2.80**Options:** R&S CMW-KS510**Manual operation:** See ["Service Center Time Stamp"](#) on page 287**CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:DATE <Day>, <Month>, <Year>**Specifies the date of the service center time stamp for the time source DATE (see [CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TSource](#)).

Parameters:

<Day>	integer
Range:	1 to 31
*RST:	11
<Month>	integer
Range:	1 to 12
*RST:	11
<Year>	integer
Range:	2011 to 9999
*RST:	2011

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["Service Center Time Stamp"](#) on page 287

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TIME <Hour>, <Minute>, <Second>

Specifies the time of the service center time stamp for the time source DATE (see [CONFigure:LTE:SIGN<i>:SMS:OUTGoing:SCTStamp:TSource](#)).

Parameters:

<Hour>	integer
Range:	0 to 23
*RST:	11
<Minute>	integer
Range:	0 to 59
*RST:	11
<Second>	integer
Range:	0 to 59
*RST:	0

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.2.80

Options: R&S CMW-KS510

Manual operation: See ["Service Center Time Stamp"](#) on page 287

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE <SMSFile>

Selects a file containing the message to be transmitted.

Parameters:

<SMSFile> string
 Path of the file, for example:
 "@USERDATA\sms\LTE\Send\example_ascii.sms"
 *RST: No File Selected

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.5.50

Manual operation: See ["Select File"](#) on page 288

CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE:INFO?

Displays information about the file selected via [CONFigure:LTE:SIGN<i>:SMS:OUTGoing:FILE](#).

Return values:

<MessageEncoding> string
 Encoding of the message (7-bit ASCII, 8-bit binary, 16-bit Unicode)
 <MessageText> string
 <MessageLength> decimal
 Number of characters in the message
 Range: 0 to 10E+3

Example: See [Sending/receiving a short message](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Message Encoding, Message Text, Message Length"](#) on page 288

SENSe:LTE:SIGN<i>:SMS:OUTGoing:INFO:LMSent?

Queries whether the last outgoing short message transfer was successful or not.

Return values:

<LastMessageSent> SUCcEssful | FAILED | NAV
 NAV is returned during an outgoing short message transfer and if there has been no transfer since the cell was switched on / the session has been started.

Example: See [Sending/receiving a short message](#)

Usage: Query only

Firmware/software: V3.5.20

Manual operation: See ["Last Message Sent"](#) on page 288

CONFigure:LTE:SIGN<i>:SMS:INComing:FILE <SMSFile>

Selects the file of a received message. You can display information about the selected file via the command `CONFigure:LTE:SIGN<i>:SMS:INComing:FILE:INFO?`.

Parameters:

<SMSFile> string
 Path of the file, for example:
 "@USERDATA\sms\LTE\Received\rx_001.sms"
 *RST: No File Selected

Example: See [Sending/receiving a short message](#)

Firmware/software: V3.5.50

Manual operation: See ["Select File"](#) on page 289

CONFigure:LTE:SIGN<i>:SMS:INComing:FILE:INFO?

Displays information about the file selected via `CONFigure:LTE:SIGN<i>:SMS:INComing:FILE`.

Return values:

<MessageEncoding> string
 Encoding of the message (7-bit "ascii", 8-bit "binary", 16-bit "Unicode")
 <MessageText> string
 <MessageLength> decimal
 Number of characters in the message
 Range: 0 to 10E+3
 <MessageSegments> decimal
 Number of segments
 Range: 0 to 1000

Example: See [Sending/receiving a short message](#)

Usage: Query only

Firmware/software: V3.5.50

Manual operation: See ["Select File"](#) on page 289

SENSe:LTE:SIGN<i>:SMS:INComing:INFO:DCODing?

Returns the data coding of the last message received from the UE.

Return values:

<MessageEncoding> string
 Encoding ("7bit" ASCII, "8bit" binary, "16bit" Unicode)

Example: See [Sending/receiving a short message](#)

Usage: Query only

Firmware/software: V3.2.70

Manual operation: See ["Data Coding / Character Set"](#) on page 289

SENSe:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT?

Returns the text of the last SMS message received from the UE.

Return values:

<MessageText> string

Example: See [Sending/receiving a short message](#)

Usage: Query only

Firmware/software: V2.1.20

Manual operation: See ["Message Text / Message Length"](#) on page 289

SENSe:LTE:SIGN<i>:SMS:INComing:INFO:MLENGTH?

Returns the length of the last SMS message received from the UE.

Return values:

<MessageLength> decimal
Number of characters of the message

Example: See [Sending/receiving a short message](#)

Usage: Query only

Firmware/software: V2.1.20

Manual operation: See ["Message Text / Message Length"](#) on page 289

CLEAn:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT

Resets all parameters related to a received SMS message. The "message read" flag is set to true.

Example: See [Sending/receiving a short message](#)

Usage: Event

Firmware/software: V2.1.20

Manual operation: See ["Clear Message Text"](#) on page 289

SENSe:LTE:SIGN<i>:SMS:INFO:LRMessage:RFLag?

Queries the "message read" flag for the last received message.

The flag is true (ON) in the following cases:

- No SMS message has been received.

- The last received SMS message has been read, see [SENSe:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT?](#) on page 888.
- The last received SMS message has been deleted, see [CLEan:LTE:SIGN<i>:SMS:INComing:INFO:MTEXT](#) on page 888.

Return values:

<LastRecMessRead> OFF | ON

OFF: unread message available

ON: no unread message available

*RST: ON

Example: See [Sending/receiving a short message](#)

Usage: Query only

Firmware/software: V2.1.20

Manual operation: See ["Clear Message Text"](#) on page 289

2.6.19 Messaging (CBS)

The following commands configure the cell broadcast service (CBS).

CONFigure:LTE:SIGN<i>:CBS:MESSAge:ENABLE	889
CONFigure:LTE:SIGN<i>:CBS:MESSAge:ID	890
CONFigure:LTE:SIGN<i>:CBS:MESSAge:IDTYpe	890
CONFigure:LTE:SIGN<i>:CBS:MESSAge:SERIAL	891
CONFigure:LTE:SIGN<i>:CBS:MESSAge:CGRoup?	891
CONFigure:LTE:SIGN<i>:CBS:MESSAge:LANGUage	892
CONFigure:LTE:SIGN<i>:CBS:MESSAge:DCSCHEME:UCODEd	893
CONFigure:LTE:SIGN<i>:CBS:MESSAge:SOURce	893
CONFigure:LTE:SIGN<i>:CBS:MESSAge:DATA	893
CONFigure:LTE:SIGN<i>:CBS:MESSAge:UCODEd	894
CONFigure:LTE:SIGN<i>:CBS:MESSAge:WAENable	894
CONFigure:LTE:SIGN<i>:CBS:MESSAge:WACoordinate	894
CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE	895
CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE:INFO?	895
CONFigure:LTE:SIGN<i>:CBS:MESSAge:ETWS:ALERT	895
CONFigure:LTE:SIGN<i>:CBS:MESSAge:ETWS:POPup	896

CONFigure:LTE:SIGN<i>:CBS:MESSAge:ENABLE <Enable>

Enables the transmission of cell broadcast messages.

Parameters:

<Enable> OFF | ON

*RST: OFF

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["Enable"](#) on page 291

CONFigure:LTE:SIGN<i>:CBS:MESSage:ID <ID>

Specifies the message ID as decimal value. The related message type is set automatically.

Parameters:

<ID> numeric
 Range: 0 to 65535
 *RST: 4370

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40, V3.5.50 enhanced range for user-defined values

Options: R&S CMW-KS170

Manual operation: See ["ID"](#) on page 291

CONFigure:LTE:SIGN<i>:CBS:MESSage:IDType <Type>

Selects the message type. The related message ID is set automatically.

For user-defined CMAS/ETWS, specify the message ID via [CONFigure:LTE:SIGN<i>:CBS:MESSage:ID](#).

Parameters:

<Type> APResidentia | AEXTreme | ASEVere | AAMBer | EARTHquake |
 TSUNami | ETWarning | ETWTest | UDCMas | UDETwS |
 GFENcing
APResidentia: presidential alert
AEXTreme: extreme alert
ASEVere: severe alert
AAMBer: amber alert
EARTHquake: earthquake
TSUNami: tsunami
ETWarning: earthquake + tsunami
ETWTest: ETWS test
UDCMas: user-defined CMAS
UDETwS: user-defined ETWS
GFENcing: geo fencing
 *RST: APR

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40, V3.5.50 added UDEFined, UDETwS
 V3.8.10 added GFENcing

Options: R&S CMW-KS170

Manual operation: See ["ID"](#) on page 291

CONFigure:LTE:SIGN<i>:CBS:MESSAge:SERial <GeoScope>, <MessageCode>, <AutoIncr>[, <UpdateNumber>]

Specifies the serial number, consisting of the geographical scope, the message code and the update number.

Parameters:

<GeoScope>	CIMMediate PLMN LOcation CNORmal Geographical scope CIMMediate : cell immediate PLMN : PLMN normal LOcation : tracking area normal CNORmal : cell normal *RST: CIMM
<MessageCode>	numeric Range: 0 to 1023 *RST: 0
<AutoIncr>	OFF ON OFF : <UpdateNumber> is not changed automatically ON : <UpdateNumber> is increased if message is changed *RST: OFF
<UpdateNumber>	numeric Range: 0 to 15 *RST: 0

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["Serial Number"](#) on page 291

CONFigure:LTE:SIGN<i>:CBS:MESSAge:CGRoup?

Queries the coding group of the message, for the data sources `INTernal` and `FILE`.

Return values:

<CodingGroup>	0 1 0 : coding group bits 0000, used for internal data source 1 : coding group bits 0001, used for file data source *RST: 0
---------------	--

Example: See [Configuring the cell broadcast service](#)

Usage: Query only

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["Data Coding Scheme"](#) on page 291

CONFigure:LTE:SIGN<i>:CBS:MESSAge:LANGUage <Language>, <LngIndication>

Specifies the language of the message for the data source `INTernal`.

The mapping of language codes to language indication strings is listed in the table below. If you specify a value pair that does not match, the specified code is used and the correct string is set automatically.

Parameters:

<Language> numeric
 Range: 0 to 15
 *RST: 1

<LngIndication> string
 Language indication
 *RST: English

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["Data Coding Scheme"](#) on page 291

<Language>	<LngIndication>
0	"German"
1	"English"
2	"Italian"
3	"French"
4	"Spanish"
5	"Dutch"
6	"Swedish"
7	"Danish"
8	"Portuguese"
9	"Finnish"
10	"Norwegian"
11	"Greek"
12	"Turkish"
13	"Hungarian"
14	"Polish"
15	"Language unspecified"

CONFigure:LTE:SIGN<i>:CBS:MESSAge:DCSCHEME:UCODEd <DCodingScheme>

Defines the data coding scheme octet for the data source UCODEd.

Parameters:

<DCodingScheme> hex
 Range: #H0 to #HFF
 *RST: #H44

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS170

Manual operation: See ["Data Coding Scheme"](#) on page 291

CONFigure:LTE:SIGN<i>:CBS:MESSAge:SOURCe <MessageHandling>

Selects the source of the message text.

Parameters:

<MessageHandling> INTernal | FILE | UCODEd

INTernal

The message text is defined via [CONFigure:LTE:SIGN<i>:CBS:MESSAge:DATA](#).

FILE

The message text is read from a file, selected via [CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE](#).

UCODEd

The message contents are defined via [CONFigure:LTE:SIGN<i>:CBS:MESSAge:UCODEd](#)

*RST: INT

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40, V3.8.10 added UCODEd

Options: R&S CMW-KS170

Manual operation: See ["Data Source"](#) on page 291

CONFigure:LTE:SIGN<i>:CBS:MESSAge:DATA <Data>

Defines the message text for the data source INTernal.

Parameters:

<Data> string
 Up to 1395 characters
 *RST: Coffee is running out!!!

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["Data"](#) on page 292

CONFigure:LTE:SIGN<i>:CBS:MESSage:UCODed <UserCoded>

Defines the message contents for the data source UCODed.

Parameters:

<UserCoded> numeric
 0 to 56 binary octets, as hexadecimal or binary number
 Only complete octets are allowed (binary n*8 bits or hexadecimal n*2 digits).
 *RST: #H30280000000000000040

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS170

Manual operation: See ["User Coded"](#) on page 292

CONFigure:LTE:SIGN<i>:CBS:MESSage:WAENable <Enable>

Enables or disables the transmission of the warning area coordinates to the UE.

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS170

Manual operation: See ["Warning Area Enable"](#) on page 292

CONFigure:LTE:SIGN<i>:CBS:MESSage:WACoordinate <WACoordinates>

Defines the contents of the warning area coordinates field.

Parameters:

<WACoordinates> numeric
 0 to 56 binary octets, as hexadecimal or binary number
 Only complete octets are allowed (binary n*8 bits or hexadecimal n*2 digits).
 *RST: #H3028000000000000000040

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.8.10

Options: R&S CMW-KS170

Manual operation: See ["Warning Area Coordinates"](#) on page 292

CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE <File>

Selects a message file for the data source `FILE`.

Store your message files in the directory

D:\Rohde-Schwarz\CMW\Data\cbs\LTE\.

Parameters:

<File> string
Path and filename
*RST: No File Selected

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["Select File..."](#) on page 293

CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE:INFO?

Queries information about the message in the selected file (see [CONFigure:LTE:SIGN<i>:CBS:MESSAge:FILE](#)).

Return values:

<MessageEncoding> string
<MessageText> string
<MessageLength> decimal
Number of characters in the message
Range: 0 to 600

Usage: Query only

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["Message Encoding, Message Text, Message Length"](#) on page 293

CONFigure:LTE:SIGN<i>:CBS:MESSAge:ETWS:ALERT <Enable>

Deactivates / activates ETWS emergency user alerting.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["ETWS"](#) on page 293

CONFigure:LTE:SIGN<i>:CBS:MESSAge:ETWS:POPup <Enable>

Deactivates / activates ETWS warning popups.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring the cell broadcast service](#)

Firmware/software: V3.5.40

Options: R&S CMW-KS170

Manual operation: See ["ETWS"](#) on page 293

2.6.20 Message monitoring settings

The following commands configure message monitoring for LTE.

CONFigure:LTE:SIGN<i>:MMONitor:ENABLE	896
CONFigure:LTE:SIGN<i>:MMONitor:IPADdress	896

CONFigure:LTE:SIGN<i>:MMONitor:ENABLE <Enable>

Enables or disables message monitoring for the LTE signaling application.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Configuring message monitoring](#)

Firmware/software: V2.1.30

Manual operation: See ["Add LTE Signaling to logging"](#) on page 294

CONFigure:LTE:SIGN<i>:MMONitor:IPADdress <Index>

Selects the IP address to which signaling messages are sent for message monitoring. The address pool is configured globally via

CONFigure:BASE:MMONitor:IPADdress<n>.

A query returns both the current index and the resulting IP address.

Parameters:

<Index> IP1 | IP2 | IP3
Address pool index

Return values:

<IPAddress> string
 Used IP address

Example: See [Configuring message monitoring](#)

Firmware/software: V3.0.10

Manual operation: See "[Logging PC IPv4 Address](#)" on page 294

2.7 List of commands

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SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:PHGeran?	433
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:IRAT:GERan:SUPPorted?	432
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQuency:INTer?	440
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:FREQuency:INTRa?	440
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:NCSacq:UTRan?	440
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:CCSSupport?	406
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:MCPCsupport?	407
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:NURClst?	407
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:PDSupport?	406
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:SPPSupport?	407
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SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:USRSsupport?	404
SENSe:LTE:SIGN<i>:UECapability:TAUeeutra:PLAYer:UTASupported?	404
SENSe:LTE:SIGN<i>:UECapability:UBNPmeas:LMIDle?	441
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SENSe:LTE:SIGN<i>:UECapability:UECategory:DL:ENHanced?	399
SENSe:LTE:SIGN<i>:UECapability:UECategory:UL:ENHanced?	399
SENSe:LTE:SIGN<i>:UECapability:UECategory?	399
SENSe:LTE:SIGN<i>:UECapability:URTTimediff?	437
SENSe:LTE:SIGN<i>:UECapability:WIW:WIAPolicies?	442
SENSe:LTE:SIGN<i>:UECapability:WIW:WIRRules?	442
SENSe:LTE:SIGN<i>:UERReport:NCELI:CDMA:CELL<no>?	395
SENSe:LTE:SIGN<i>:UERReport:NCELI:EVDO:CELL<no>?	396
SENSe:LTE:SIGN<i>:UERReport:NCELI:GSM:CELL<no>:RANGe?	393
SENSe:LTE:SIGN<i>:UERReport:NCELI:GSM:CELL<no>?	393
SENSe:LTE:SIGN<i>:UERReport:NCELI:LTE:CELL<no>:RANGe?	392
SENSe:LTE:SIGN<i>:UERReport:NCELI:LTE:CELL<no>?	392
SENSe:LTE:SIGN<i>:UERReport:NCELI:TDSCDma:CELL<no>:RANGe?	397
SENSe:LTE:SIGN<i>:UERReport:NCELI:TDSCDma:CELL<no>?	396
SENSe:LTE:SIGN<i>:UERReport:NCELI:WCDMa:CELL<no>:RANGe?	394
SENSe:LTE:SIGN<i>:UERReport:NCELI:WCDMa:CELL<no>?	394
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:COCC?	397
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:RRESult?	397
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:RSRP:RANGe?	389
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:RSRP?	389
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:RSRQ:RANGe?	390
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:RSRQ?	390
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:SCELI:RANGe?	391
SENSe:LTE:SIGN<i>:UERReport:SCC<c>:SCELI?	391
SENSe:LTE:SIGN<i>:UERReport[:PCC]:RSRP:RANGe?	389

SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRP?	389
SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRQ:RANge?	390
SENSe:LTE:SIGN<i>:UEReport[:PCC]:RSRQ?	390
SENSe:LTE:SIGN<i>:UEReport[:PCC]:SCEL:RANge?	391
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SENSe:LTE:SIGN<i>:UESInfo:UEADdress:DEDBearer?	449
SENSe:LTE:SIGN<i>:UESInfo:UEADdress:IPV<n>?	449
SENSe:LTE:SIGN<i>:UESInfo:UEUSage?	449
SENSe:LTE:SIGN<i>:UESInfo:VDPreference?	448
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:EOPower?	672
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:EPPPower?	671
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PATHloss?	671
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PCALpha:BASic?	668
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SCC<c>:APPower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL:SETA:APPower:EOPower?	672
SENSe:LTE:SIGN<i>:UL:SETA:APPower:EPPPower?	671
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PATHloss?	671
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PCALpha:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETA:APPower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETA:APPower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SETA:APPower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL:SETB:APPower:EOPower?	672
SENSe:LTE:SIGN<i>:UL:SETB:APPower:EPPPower?	671
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PATHloss?	671
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PCALpha:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETB:APPower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETB:APPower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SETB:APPower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL:SETC:APPower:EOPower?	672
SENSe:LTE:SIGN<i>:UL:SETC:APPower:EPPPower?	671
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PATHloss?	671
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PCALpha:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETC:APPower:PNPusch:BASic?	667
SENSe:LTE:SIGN<i>:UL:SETC:APPower:RSPower:BASic?	666
SENSe:LTE:SIGN<i>:UL:SETC:APPower:TPRRcsetup:BASic?	668
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:EOPower?	672
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:EPPPower?	671
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PATHloss?	671
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PCALpha:BASic?	668
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PIRPower:BASic?	667
SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:PNPusch:BASic?	667

SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:RSPower:BASic?.....666

SENSe:LTE:SIGN<i>:UL[:PCC]:APPower:TPRRcsetup:BASic?.....668

SOURce:LTE:SIGN<i>:CELL:STATe.....371

SOURce:LTE:SIGN<i>:CELL:STATe:ALL?.....371

3 LTE BLER measurement

The extended BLER measurement is installed together with the LTE signaling application. To access the measurement, press the softkey "LTE RX Meas" in the LTE signaling main view. Then select the tab "Extended BLER".

The measurement sends data to the UE via PDSCH subframes. It evaluates the positive acknowledgments (ACK) and negative acknowledgments (NACK) returned by the UE to determine the block error ratio (BLER) and the throughput. The CQI indices reported by the UE are also evaluated.

With enabled HARQ / CQI reporting, the HARQ retransmissions, reported PMI values and rank indicators are also evaluated. For the uplink, the results of a cyclic redundancy check (CRC) are provided.

The measurement is especially suitable to assess the characteristics and the performance of the UE receiver at low RF power levels.

The LTE standard does not request the implementation of test loops at the UE. Bit error rate (BER) measurements are therefore no issue for LTE.

3.1 Performing a BLER measurement

To measure the downlink BLER, you must set up a connection and transfer data via the downlink. In test mode, activate downlink padding to transfer data (see ["Downlink MAC Padding"](#) on page 210). In data application mode, transfer data, for example via FTP.

If you want to measure the throughput, activate downlink padding also in data application mode.

If you want to set up a downlink signal with a specific CQI index value, use a "CQI" scheduling type.

When you start the measurement, the R&S CMW sends data to the UE via PDSCH subframes. The UE confirms each received subframe with an ACK or NACK via the PUSCH. The R&S CMW calculates the DL BLER from the received ACKs and NACKs. It determines the CQI, PMI and RI results from the corresponding reported values. For the uplink, the R&S CMW performs a CRC check and calculates the UL BLER from the results of the check.

For transmission schemes using several downlink streams, the ACKs, NACKs and CQI indices reported for the streams are evaluated separately.

Note: As the transmit time interval for LTE equals one subframe, a subframe corresponds to one transport block.

It is possible to insert block errors into the downlink data (send wrong CRC value). For configuration, see ["Downlink MAC Error Insertion"](#) on page 210.

Active neighbor cell measurements cause an increased number of NACKs. So it is recommended to disable neighbor cell measurements for BLER evaluation, see [Neighbor cell settings](#).

Reduced PDCCH resources can also result in an increased number of NACKs. So it is recommended to disable "Reduced PDCCH", see [PDCCH](#).

For configuration of CQI, PMI and RI reporting, see [CQI reporting](#).

For configuration of DL HARQ settings, see [DL HARQ](#).

Confidence BLER measurements

In a normal BLER measurement, typically a fixed number of subframes is transmitted, leading to a fixed test time. The idea behind confidence BLER measurements is to apply probability theory and predict a range for the BLER at an early stage of the measurement. The measurement is stopped if the probability of the UE to pass or fail the receiver quality test is large enough (early pass or early fail decision).

Confidence BLER measurements can reduce test times considerably, especially if the BLER of a receiver is extremely low or high. They are specified in annex G of 3GPP TS 36.521 as "Statistical Testing". Option R&S CMW-KS510/-KS512 (without CA/with CA) is required for confidence BLER measurements.

To perform a confidence BLER measurement, configure especially the following measurement settings (see also [Section 3.2.4, "Settings"](#), on page 936):

- Set "Stop Condition" to "Confidence Level"
- Select the "Error Ratio Calculation" according to the test case
- Configure the "Minimum Test Time" according to the test case
- Select a "Limit Error Rate"
Depending on this selection, the pass/fail decision is made according to the rules specified in 3GPP TS 36.521 annex G.2 or annex G.4.
- For tests with carrier aggregation, select the "Over All Stop Decision"

3.2 GUI reference

The BLER measurement is part of the RX measurements that are installed with the LTE signaling application.

Access: LTE signaling main view > "LTE RX Meas" softkey > "Extended BLER" tab.

The views and configuration dialogs of the tab are described in this section.

- [Tab overview](#)..... 923
- [Measurement control](#)..... 923
- [Measurement results](#)..... 924
- [Settings](#)..... 936

3.2.1 Tab overview

The "Extended BLER" tab provides several views with measurement results. The connection status information displayed at the bottom is the same as in the LTE signaling main view.

The most important settings of the LTE signaling application can be accessed via the "Signaling Parameters" softkey and the related hotkeys.

To switch to the signaling application, press the "LTE Signaling" softkey two times.

The "Config" hotkey opens either the configuration dialog of the measurement or the configuration dialog of the signaling application, depending on which softkey is active.

The following figure shows the result overview, providing access to the individual detailed result views.

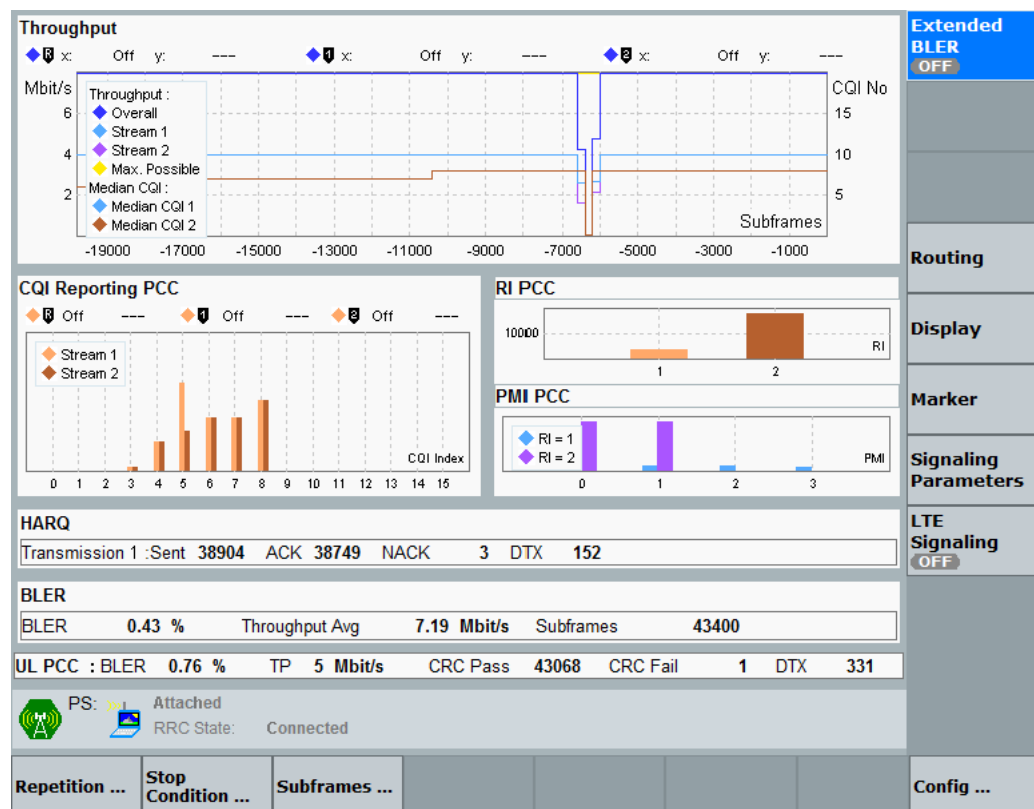
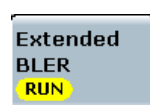


Figure 3-1: Extended BLER tab, result overview

3.2.2 Measurement control

To turn the measurement on or off, select the control softkey and press [ON | OFF] or [RESTART | STOP]. Alternatively, right-click the control softkey.



Extended BLER softkey

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

```
INITiate:LTE:SIGN<i>:EBLer
STOP:LTE:SIGN<i>:EBLer
ABORt:LTE:SIGN<i>:EBLer
FETCh:LTE:SIGN<i>:EBLer:STATe?
FETCh:LTE:SIGN<i>:EBLer:STATe:ALL?
```

3.2.3 Measurement results

The results of the LTE BLER measurement are displayed in several different views, described in the following sections.

• Overview	924
• View BLER	926
• View Throughput	929
• View CQI Reporting	930
• View PMI - RI	931
• View HARQ	933
• Common view elements	935
• Selecting and modifying views	936
• Using markers	936

3.2.3.1 Overview

Access: "Extended BLER" tab > "Display" softkey > "Select View" hotkey > "Overview".



Figure 3-2: Result overview

You can enlarge one of the diagrams in the overview and show a detailed view with additional measurement results, see [Section 3.2.3.8, "Selecting and modifying views"](#), on page 936. The individual results are described in the subsequent sections.

Uplink results

The "UL PCC" / "UL SCCn" BLER results are only available in the overview. Option R&S CMW-KS510 is required for these results. They are displayed per uplink component carrier.

- **"BLER":**

Block error ratio, percentage of received uplink subframes with failed CRC check.

The following formula is used:

$$BLER = CRC\ fail / (CRC\ pass + CRC\ fail)$$

- **"TP":**

Throughput calculated from the "CRC Pass / Fail" results and the theoretical maximum uplink throughput. The following formula is used:

$$Throughput = CRC\ pass / (CRC\ pass + CRC\ fail) * theoretical\ max\ throughput$$

The uplink throughput indicates the average uplink throughput since the start of the measurement.

The theoretical maximum throughput is calculated from the UL settings. It is displayed, for example, in the signaling application main view.

- **"CRC Pass / Fail":**

Number of uplink subframes with passed / failed CRC check, received since the start of the measurement.

- **"DTX"/"Skipped":**

Number of scheduled uplink subframes skipped by the UE.

The label depends on the setting "[SkipUITx](#)" on page 225. If skipping UL transmissions is allowed, the label is "Skipped". If it is not allowed, the label is "DTX".

The result is calculated from the number of expected uplink subframes and the number of received uplink subframes (CRC pass / fail results). For some configurations, the number of expected subframes is unknown and NCAP is displayed as result (for example for CDRX or scheduling type SPS).

Remote command:

```
FETCh:LTE:SIGN<i>:EBLer[:PCC]:UPLink?
```

```
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:UPLink?
```

3.2.3.2 View BLER

The "BLER" view provides a table of downlink BLER and throughput results. The table contains always an overall result section at the top (results over all streams of one carrier). For configurations with several downlink streams, it provides the results also per downlink stream.

For carrier aggregation scenarios, the results for the individual component carriers are presented on different tabs. The overall results for the sum of all component carriers are provided in the "Throughput" view.

The BLER view shows also common settings of the LTE signaling application, see [Section 2.4.1.6, "Settings"](#), on page 127.

The following figure shows the results of two measurements. On the left, you see the results of a measurement without stop condition and two downlink streams. On the right, you see the results of a confidence BLER measurement with one downlink stream.

PCC	SCC1	SCC2	SCC3
	Relative	Absolute	
ACK	99.95 %	99946	
NACK	0.00 %	0	
DTX	0.05 %	54	
BLER	0.05 %		
Throughput	Relative	Mbit/s	
Average	99.94 %	23.17	
Stream 1	Relative	Absolute	
ACK	99.95 %	49973	
NACK	0.00 %	0	
DTX	0.05 %	27	
BLER	0.05 %		
Throughput	Relative	Mbit/s	
Average	99.94 %	7.71	
Stream 2	Relative	Absolute	
ACK	99.95 %	49973	
NACK	0.00 %	0	
DTX	0.05 %	27	
BLER	0.05 %		
Throughput	Relative	Mbit/s	
Average	99.94 %	15.45	
Subframes	50 000 / 50000		
Scheduled:	50000		

PCC	SCC1	SCC2	SCC3
	Relative	Absolute	
ACK	96.25 %	385	
NACK	3.75 %	15	
DTX	0.00 %	0	
BLER	3.75 %		
Throughput	Relative	Mbit/s	
Average	96.25 %	11.02	
Minimum		10.99	
Maximum		11.05	
Subframes	400		
Scheduled:	400		
PCC:	Early Pass		
Over All:	Early Pass		
Specified BLER Level =	5 %		

Figure 3-3: BLER view results (two streams normal measurement / one stream confidence BLER)

Result tables

- "ACK / NACK / DTX":**
 Number of acknowledgments and negative acknowledgments received via the PUSCH since the start of the measurement. No answer at all (no ACK, no NACK) is counted as DTX.
 The results are presented as absolute number and as percentage relative to the number of sent scheduled subframes.
- "BLER":**
 Block error ratio, percentage of sent scheduled subframes for which no acknowledgment has been received. The formula used to calculate the BLER is configurable, see ["Error Ratio Calculation"](#) on page 938. By default the following formula is used:

$$BLER = (NACK + DTX) / (ACK + NACK + DTX)$$
- "Throughput":**
 Throughput calculated from the number of acknowledged transport blocks (number of received ACK multiplied with bits per transport block, divided by the time).
 The throughput is calculated for each 200 processed subframes. From the resulting throughput values, the instrument determines the average value and the minimum and maximum values.

The "Relative" average throughput indicates the average throughput as percentage of the theoretical maximum throughput, i.e. of the throughput that would be reached with ACK = 100%.

It is recommended to always activate downlink padding for throughput measurements, even in data application mode. Otherwise, the throughput results can be hard to interpret.

Remote command:

`FETCh:LTE:SIGN<i>:EBLer:SCC<c>:ABSolute? etc.`

`FETCh:LTE:SIGN<i>:EBLer:SCC<c>:RELative? etc.`

`FETCh:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:ABSolute? etc.`

`FETCh:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:RELative? etc.`

`FETCh:LTE:SIGN<i>:EBLer:ALL:ABSolute? etc.`

`FETCh:LTE:SIGN<i>:EBLer:ALL:RELative? etc.`

Confidence BLER results

- **"PCC or SCC Pass/Fail":**

When a pass/fail decision has been made for the carrier, one of the following result values is displayed:

- "Early Pass" / "Early Fail":
An early pass or early fail limit was exceeded and an early decision has been made.
- "Pass" / "Fail":
The configured minimum test time is larger than the early decision table. The pass/fail decision has been made using the test limit.

The pass/fail decision is always based on the overall BLER results of the carrier, independent of the number of downlink streams.

When a pass/fail result is displayed for a carrier, the measurement can continue to derive the pass/fail result for another carrier and the overall pass/fail result.

- **"Over All Pass/Fail":**

For measurements with carrier aggregation, an overall pass/fail result is derived from the PCC/SCC pass/fail results.

The meaning of the overall result depends on the configured "Over All Stop Decision". For the stop decision "PCC only", the overall result equals the PCC result.

For the stop decision "SCC<n> only", the overall result equals the SCC<n> result.

For the "All Carrier, ..." stop decisions, the overall result has the following meaning:

- "Early Fail": at least one carrier "Early Fail"
- "Fail": no carrier "Early Fail", at least one carrier "Fail"
- "Early Pass": all carriers "Early Pass"
- "Pass": at least one carrier "Pass", no "Fail", no "Early Fail"

- **"Specified BLER Level":**

Indicates the currently configured "Limit Error Rate".

Remote command:

`FETCh:LTE:SIGN<i>:EBLer[:PCC]:CONFidence?`

`FETCh:LTE:SIGN<i>:EBLer:SCC<c>:CONFidence?`

`FETCh:LTE:SIGN<i>:EBLer:ALL:CONFidence?`

3.2.3.3 View Throughput

The "Throughput" view shows a history of the downlink throughput and the median CQI for the previous measurement cycle.

For carrier aggregation scenarios, the view presents the results on several tabs. The "PCC" and "SCC<n>" tabs provide the results for the PCC downlink and for the SCC downlinks. The results for the sum of all PCC plus SCC downlink streams are provided on the "Overall" tab.

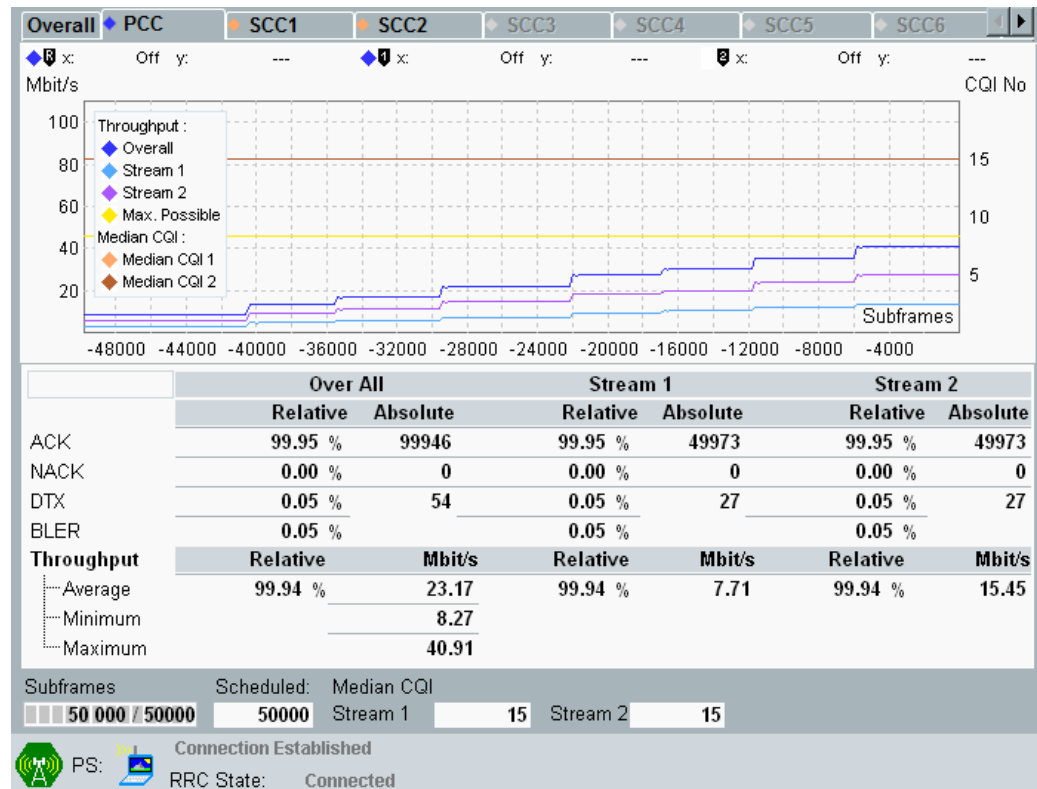


Figure 3-4: Throughput view (two streams)

Result diagram

Graphical presentation of downlink throughput and median CQI results for the previous measurement cycle. The X-axis indicates the processed subframes, with the last processed subframe labeled 0, the previously processed subframe labeled -1, and so on. The diagram displays one result value per 200 processed subframes.

Remote command:

`FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>:STReam<s>? etc.`

`FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>? etc.`

`FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:ALL?`

`FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>:MCQI:STReam<s>?`

etc.

Result tables

The BLER, throughput and median CQI results below the diagram are related to the entire measurement duration. They are also provided in the "BLER" or "CQI Reporting" views. For details, refer to the descriptions of these views.

3.2.3.4 View CQI Reporting

The main focus of the "CQI Reporting" view is on evaluation of the CQI index values reported by the UE. This view is only available if CQI reporting is enabled in the signaling application (see ["Enable CQI Reporting"](#) on page 278).

If the UE reports CQI values per stream (for example for TM 4 or TM 9), all results are displayed per stream as in the following figure. Otherwise, the results are displayed as "Stream 1" results.

For carrier aggregation scenarios, the results for the individual component carriers are presented on different tabs. If you want to evaluate CQI reports for several carriers, ensure that the configured "CQI/PMI Config Index" of the carriers is compatible, see ["CQI/PMI Config Index"](#) on page 279.

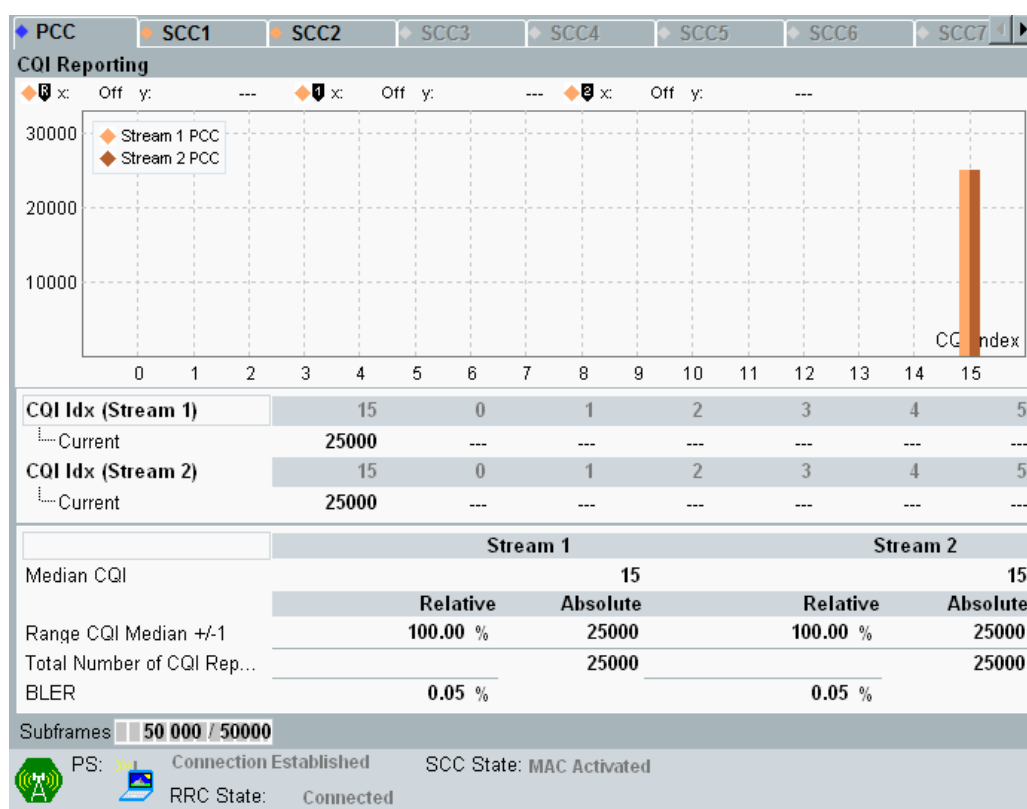


Figure 3-5: CQI Reporting view (two streams)

Bar graph and upper result table

The bar graph shows how often the CQI indices 0 to 15 have been reported by the UE since the measurement has been started.

The table directly below the bar graph provides the index value and the height of the seven highest bars in the diagram.

Remote command:

```
FETCh:LTE:SIGN<i>:EBLer:TRACe:CQIReporting[:PCC]:STReam<s>?
FETCh:LTE:SIGN<i>:EBLer:TRACe:CQIReporting:SCC<c>:STReam<s>?
```

Lower result table

The table at the bottom provides the following information:

- **"Median CQI"**: median of the CQI values reported by the UE
Add the bars of one stream from left to right. The median CQI value is the index value where the sum reaches/crosses 50% of the total sum. This definition of the median value is specified in 3GPP TS 36.521, section 9.
- **"Range CQI Median +/-1"**: evaluation of three adjacent bars, from median CQI - 1 to median CQI + 1
The absolute value indicates the sum of the three bars. The relative value indicates the percentage of the absolute value relative to the sum of all bars of the stream.
- **"Total Number of CQI Reports"**: total number of received CQI index values (sum of all bars per stream)
- **"BLER"**: relative BLER result per stream, as shown in the "BLER" view

Remote command:

```
FETCh:LTE:SIGN<i>:EBLer[:PCC]:CQIReporting:STReam<s>?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:CQIReporting:STReam<s>?
```

3.2.3.5 View PMI - RI

The "PMI - RI" view provides a statistical evaluation of the precoding matrix indicator (PMI) values and rank indicator (RI) values reported by the UE.

This view is only available if CQI reporting is enabled in the signaling application, see ["Enable CQI Reporting"](#) on page 278.

Whether PMI and/or RI reports are sent by the UE depends on the transmission mode. For TM 8 and 9, you must explicitly request reports from the UE, see ["Enable PMI/RI Reporting \(TM 8, 9\)"](#) on page 279.

For carrier aggregation scenarios, the results for the individual component carriers are presented on different tabs. If you want to evaluate PMI reports for several carriers, ensure that the configured "CQI/PMI Config Index" of the carriers is compatible, see ["CQI/PMI Config Index"](#) on page 279.

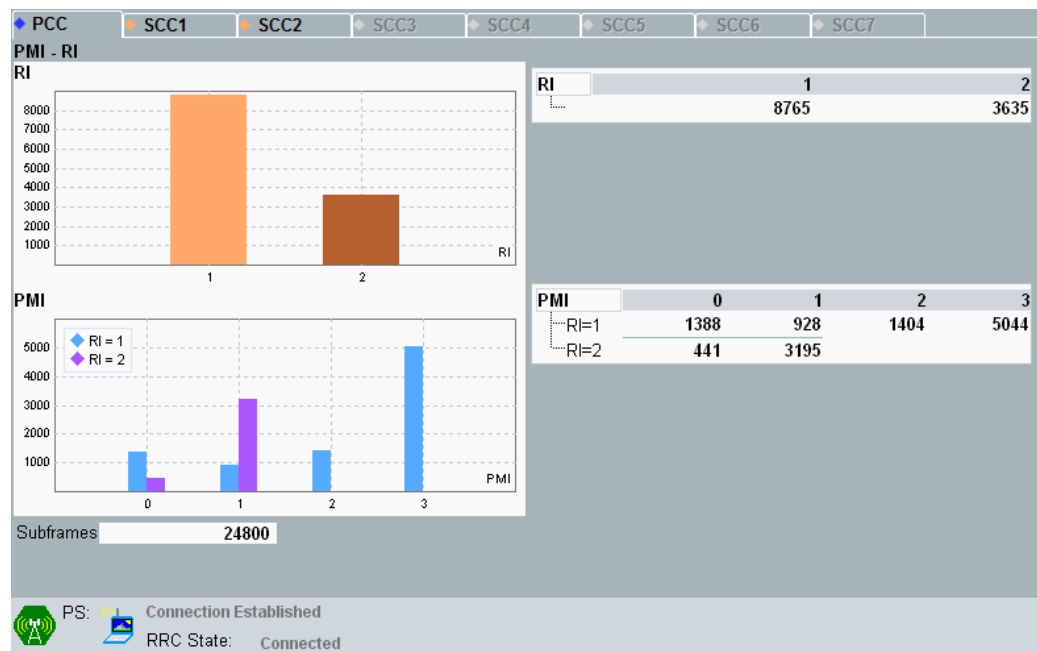


Figure 3-6: PMI - RI view (TM 4, 2 antennas)

RI bar graph / table

The "RI" bar graph indicates how often the individual RI values have been reported by the UE since the measurement has been started.

There is one bar / one table column per RI value.

Remote command:

```
FETCh:LTE:SIGN<i>:EBLer[:PCC]:RI?
```

```
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:RI?
```

PMI bar graph / table

The "PMI" bar graph indicates how often specific PMI - RI combinations have been reported by the UE since the measurement has been started.

There is one bar / one table cell per PMI - RI combination.

The possible PMI values depend on the number of DL TX antennas, for TM 9 on the number of CSI-RS antenna ports:

- Two antennas / two CSI-RS ports: PMI = 0 to 3 for RI = 1, PMI = 0 to 1 for RI = 2
 - Four antennas / four CSI-RS ports: PMI = 0 to 15
 - Eight CSI-RS ports: two PMI values are reported, PMI = 0 to 15
- The view shows results for the most frequently reported PMI value combinations (up to 10 combinations).

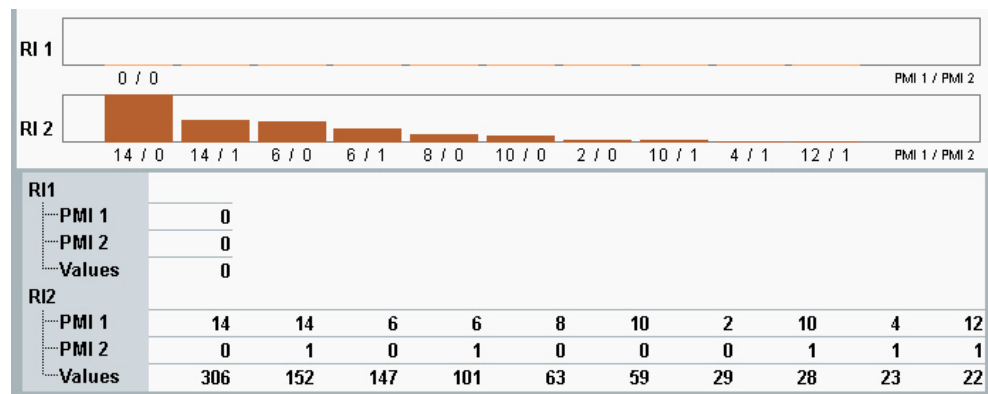


Figure 3-7: Lower part for eight CSI-RS ports

To display result tables for all possible RI - PMI 1 - PMI 2 combinations, press the softkey - hotkey combination "Display > PMI 1 x PMI 2 Results".

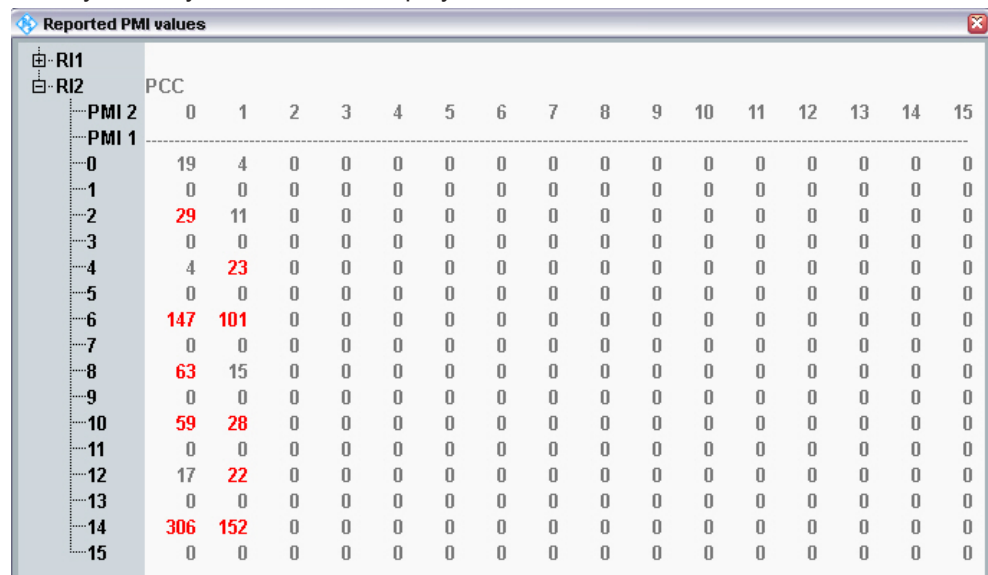


Figure 3-8: Complete tables, RI - PMI 1 - PMI 2 combinations

The highlighted values in the complete tables are also displayed in the bar graphs.

Remote command:

FETCH:LTE:SIGN<i>:EBLer[:PCC]:PMI:RI<no>?

FETCH:LTE:SIGN<i>:EBLer:SCC<c>:PMI:RI<no>?

3.2.3.6 View HARQ

The "HARQ" view provides an overview of all downlink subframe transmissions performed since the measurement was started.

If you cannot access the view, check whether HARQ is enabled, see [Section 2.4.12.7, "Miscellaneous connection settings part 2"](#), on page 221.

For carrier aggregation scenarios, the results for the individual component carriers are presented on different tabs.

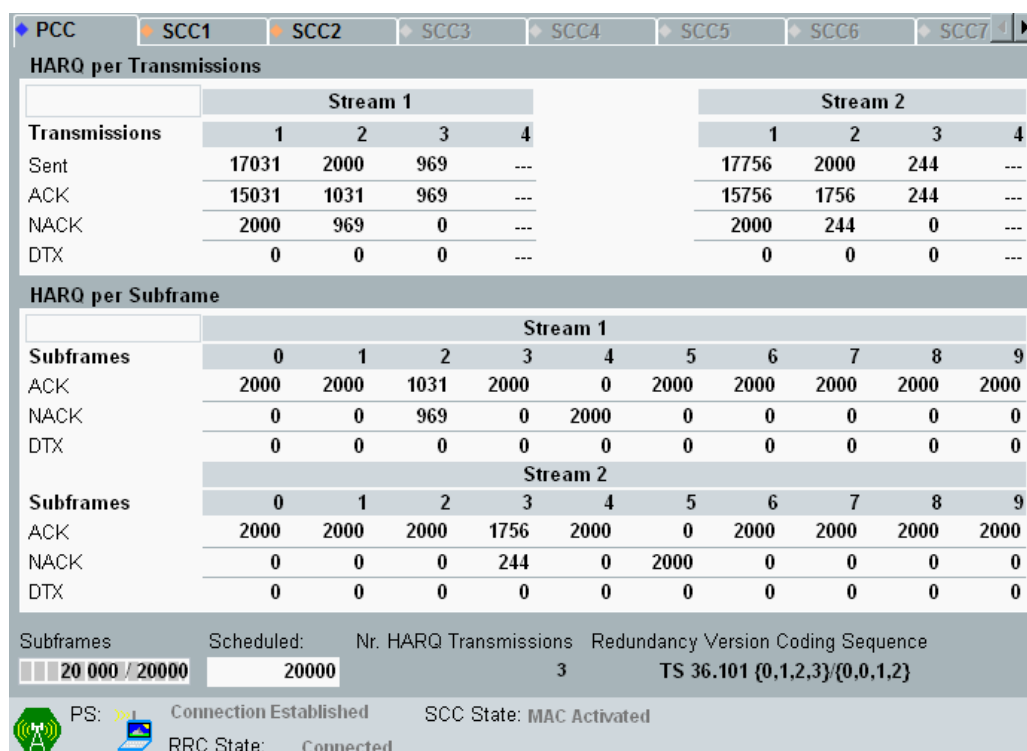


Figure 3-9: HARQ view (two streams, absolute results)

The view displays two tables for each downlink stream.

HARQ per Transmissions table	934
HARQ per Subframe table	935
Relative and absolute results	935
Bottom of the view	935

HARQ per Transmissions table

The table lists the initial transmissions and all retransmissions. Column 1 indicates the initial transmissions (first redundancy version), column 2 the first retransmission (second redundancy version), column 3 the second retransmission (third redundancy version), and so on.

For each transmission type, the number of sent subframes, received ACK, received NACK and DTX is displayed.

Remote command:

```
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:TRANsmiSSion:
ABSolute? etc.
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:TRANsmiSSion:
RELative? etc.
```

HARQ per Subframe table

The "HARQ per Subframe" table lists the ACK, NACK and DTX received in the individual subframes of the radio frame. So column 0 indicates the ACK/NACK/DTX received in the first subframe of a radio frame, column 9 the ACK/NACK/DTX received in the last subframe of a radio frame.

Remote command:

```
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:SUBFrame:ABSolute?
```

etc.

```
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:SUBFrame:RELative?
```

etc.

Relative and absolute results

You can display all results in the tables as absolute numbers or as percentages. To toggle the presentation, use the softkey > hotkey combination "Display" > "Absolute / Relative".

A relative "Sent" value indicates the percentage of the absolute "Sent" value relative to the total number of sent subframes (sum of the table row). A relative ACK, NACK or DTX value indicates the percentage of the absolute value, relative to the sum of the absolute ACK, NACK and DTX values in the table column.

Bottom of the view

At the bottom of the view, some HARQ settings are displayed for information, in addition to the view-independent GUI elements. For a description, see ["DL HARQ"](#) on page 231.

3.2.3.7 Common view elements

Each view displays at least one of the following elements to indicate the progress of the measurement:

- **"Subframes":**

This value indicates the number of already processed subframes. For single-shot measurements with fixed length, also the total number of subframes to be processed is displayed.

For FDD, all downlink subframes are counted, scheduled subframes and subframes without allocated resource blocks. For TDD, special subframes and uplink subframes are also counted.

A scheduled downlink subframe that is sent via several downlink streams in parallel is counted as one subframe.

- **"Scheduled":**

Displays the number of already measured subframes (scheduled downlink subframes). This number equals the sum of the absolute values for ACK, NACK and DTX for one downlink stream.

Examples for a single-shot measurement with fixed length:

- FDD: You set the no. of subframes to 10000, schedule every fifth subframe and use two downlink streams. Thus 2000 subframes are scheduled. The UE is expected to send 2000 responses per stream, 4000 in total. At the end of the measurement, the parameter "Subframes" displays 10000/10000 and the parameter "Scheduled" displays 2000.

- TDD: You set the no. of subframes to 10000. You use UL/DL configuration 1 and transfer downlink data via subframe 0, 1, 4, 6 and 9 (subframe 5 not scheduled, subframe 2, 3, 7, 8 = uplink). Thus 5 of 10 subframes are used for downlink data transfer and 5000 of 10000 subframes are scheduled. At the end of the measurement, the parameter "Subframes" displays 10000/10000 and the parameter "Scheduled" displays 5000.

3.2.3.8 Selecting and modifying views

Press the "Display" softkey to show hotkeys for view configuration. They change the appearance and contents of the active view. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Select View ..."	Switch to a certain detailed view or to the overview. Alternatively, select a diagram in the overview and press ENTER or the rotary knob.
"Select Trace ..."	Select the trace types to be displayed in the view.
"X Scale... / Y Scale..."	Modify the ranges of the X-axis and the Y-axis.
"Absolute Relative"	Toggles between display of absolute and relative results in the HARQ detailed view.

3.2.3.9 Using markers

Press the "Marker" softkey to show hotkeys for marker configuration. They activate markers and modify their position. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Ref. Marker"	Enable or disable the reference marker and select the marker position. If several traces can be displayed, a trace can also be selected.
"Marker 1/2"	Enable or disable marker 1 or 2 and define the marker position (absolute or relative to the reference marker). Depending on the trace mode, a trace can also be selected.
"Select Trace Mode"	Define whether markers 1 and 2 are set to the same trace as the reference marker (collective) or to selectable individual traces.

3.2.4 Settings

Access: "Extended BLER" tab > "Config" hotkey.

If you want to insert block errors into the downlink signal, see ["Downlink MAC Error Insertion"](#) on page 210.

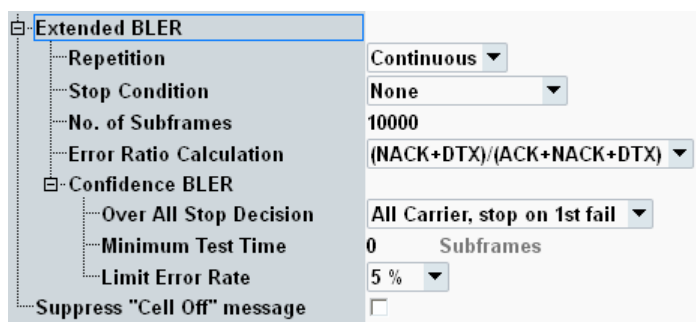


Figure 3-10: Extended BLER settings

Repetition.....	937
Stop Condition.....	937
No. of Subframes.....	938
Error Ratio Calculation.....	938
Confidence BLER.....	938
L Over All Stop Decision.....	938
L Minimum Test Time.....	939
L Limit Error Rate.....	939
Suppress "Cell Off" Message.....	939

Repetition

Defines how often the measurement is repeated.

"Continuous"	The measurement is continued until it is explicitly terminated. The results are periodically updated.
"Single-Shot"	The measurement stops automatically, depending on the configured "Stop Condition" either when the configured "No. of Subframes" has been processed or when a confidence BLER measurement result is available.

Remote command:

CONFigure:LTE:SIGN<i>:EBLer:REPetition

Stop Condition

Selects whether a BLER measurement without stop condition or a confidence BLER measurement with early decision concept is performed.

"None"	The measurement is performed according to its "Repetition" mode and the specified "No. of Subframes". No confidence BLER result is determined.
"Confidence Level"	A confidence BLER measurement is performed. The measurement stops when the configured "Minimum Test Time" has passed and a pass/fail decision has been made, so that a confidence BLER result (e.g. "Early Pass") is available. The "Repetition" is automatically set to "Single-Shot". Option R&S CMW-KS510/-KS512 (without CA/with CA) is required.

Remote command:

CONFigure:LTE:SIGN<i>:EBLer:SCONdition

No. of Subframes

For measurements without stop condition, this parameter defines the number of subframes to be processed per measurement cycle (a single-shot measurement covers one measurement cycle).

For confidence BLER measurements, this parameter specifies only the length of the throughput result trace. It does not influence the duration of the measurement.

For FDD, all scheduled and unscheduled downlink subframes are considered. For TDD, all downlink, uplink and special subframes are considered.

A scheduled downlink subframe that is sent via several downlink streams in parallel is counted as one subframe.

For examples, see [Section 3.2.3.7, "Common view elements"](#), on page 935.

Remote command:

`CONFigure:LTE:SIGN<i>:EBLer:SFRames`

Error Ratio Calculation

Selects the formula to be used for calculation of the BLER from the number of ACK, NACK and DTX. PDSCH decoding errors result in NACK while PDCCH decoding errors result in DTX.

3GPP TS 36.521 specifies which formula must be used for a certain performance test or CQI test.

The following formulae are available:

- $BLER = (NACK + DTX) / (ACK + NACK + DTX)$
- $BLER = DTX / (ACK + NACK + DTX)$
- $BLER = NACK / (ACK + NACK + DTX)$
- $BLER = NACK / (ACK + NACK)$

Remote command:

`CONFigure:LTE:SIGN<i>:EBLer:ERCalc`

Confidence BLER

The following parameters configure parameters only relevant for confidence BLER measurements.

Over All Stop Decision ← Confidence BLER

This setting configures the stop decision and the overall confidence BLER result calculation for measurements with carrier aggregation.

Depending on the selection, the measurement stops when the following condition is fulfilled:

- "PCC only": A result is available for the PCC.
- "SCC<n> only": A result is available for SCC number <n>.
- "All Carrier, stop on 1st fail": Results are available for all component carriers or at least one result equals "Fail" or "Early Fail".
- "All Carrier, wait for all CCs": Results are available for all component carriers.

Depending on the selection, the overall confidence BLER result is calculated as follows:

- "PCC only": Overall result = PCC result
- "SCC<n> only": Overall result = SCC<n> result
- "All Carrier, ...": Overall result = ...

- If at least one result = "Early Fail": Overall result = "Early Fail"
- Else, if at least one result = "Fail": Overall result = "Fail"
- Else, if all results = "Early Pass": Overall result = "Early Pass"
- Else: Overall result = "Pass"

Remote command:

`CONFigure:LTE:SIGN<i>:EBLer:CONFidence:OASCondition`

Minimum Test Time ← Confidence BLER

Specifies the minimum test time of a confidence BLER measurement as number of processed subframes. During this time, no pass/fail decision is allowed.

Minimum test times are, for example, specified in 3GPP TS 36.521 annex G.4. They are necessary if the test conditions introduce fluctuations disturbing the statistical independence of the single bit error events (e.g. multipath fading). A minimum test time ensures that the fluctuations are averaged out.

For FDD, all scheduled and unscheduled downlink subframes are considered. For TDD, all downlink, uplink and special subframes are considered.

Remote command:

`CONFigure:LTE:SIGN<i>:EBLer:CONFidence:MTTime`

Limit Error Rate ← Confidence BLER

Selects the limit error ratio for a confidence BLER measurement. The selection determines, for example, the used pass/fail decision rules.

The minimum test time and the error ratio calculation formula are independent of this selection, so configure also these parameters as desired.

"0.1%, 1%" Pass/fail decision according to 3GPP TS 36.521 annex G.4

"5%" Pass/fail decision according to 3GPP TS 36.521 annex G.2

Remote command:

`CONFigure:LTE:SIGN<i>:EBLer:CONFidence:LERate`

Suppress "Cell Off" Message

If you press [ON | OFF] while the "LTE Signaling" softkey is selected and the cell signal is on, a warning can be displayed. It asks you whether you really want to switch off the cell signal.

The checkbox enables/disables the warning.

Remote command:

n/a

3.3 Programming

The BLER measurement is programmed as follows:

- The application is controlled by SCPI commands with the following syntax: `...:LTE:SIGN:EBLer:...`
- After a `*RST`, the measurement is switched off.

You can start the measurement using `INIT:LTE:SIGN:EBLer` and retrieve the results using `FETCH` commands.

The examples in this section focus on commands directly related to the BLER measurement. For general configuration of the signaling application and setting up the connection, refer to [Section 2.5, "Programming"](#), on page 322.

- [Configuring a BLER measurement](#)..... 940
- [Performing a confidence BLER measurement](#).....940
- [Performing a continuous BLER measurement](#).....941
- [Performing a single-shot BLER measurement](#).....942

3.3.1 Configuring a BLER measurement

```
// *****
// Activate DL padding.
// *****
Configure:LTE:SIGN:CONNECTION:DLPadding ON

// *****
// Configure a continuous BLER measurement without stop condition and
// 1000 subframes per measurement cycle. NACK and DTX contribute to the BLER.
// *****
Configure:LTE:SIGN:EBLer:SCONdition NONE
Configure:LTE:SIGN:EBLer:REPetition CONT
Configure:LTE:SIGN:EBLer:SFRames 1000
Configure:LTE:SIGN:EBLer:ERCalc ERC1

// *****
// Configure a confidence BLER measurement.
// The stop condition "Confidence Level" sets also the repetition.
// NACK and DTX contribute to the BLER, wait for all carriers,
// no minimum test time, ER limit 5%.
// *****
Configure:LTE:SIGN:EBLer:SCONdition CLEV
Configure:LTE:SIGN:EBLer:ERCalc ERC1
Configure:LTE:SIGN:EBLer:CONFidence:OASCondition ACWait
Configure:LTE:SIGN:EBLer:CONFidence:MTTime 0
Configure:LTE:SIGN:EBLer:CONFidence:LERate P050
```

3.3.2 Performing a confidence BLER measurement

```
// *****
// Start a confidence BLER measurement and return the pass/fail results
// for PCC and SCC1.
// *****
Configure:LTE:SIGN:EBLer:SCONdition CLEV
INIT:LTE:SIGN:EBLer
FETCH:LTE:SIGN:EBLer:PCC:CONFidence?
```



```

FETCh:LTE:SIGN:EBLer:SCC:CONFidence?
FETCh:LTE:SIGN:EBLer:ALL:CONFidence?

```

3.3.3 Performing a continuous BLER measurement

```

// *****
// Start a continuous BLER measurement.
// Wait until the measurement is really running.
// *****
CONFIGure:LTE:SIGN:EBLer:REPetition CONT
CONFIGure:LTE:SIGN:EBLer:SCONdition NONE
INIT:LTE:SIGN:EBLer
WHILE FETCh:LTE:SIGN:EBLer:STATE:ALL? <> "RUN","ADJ","ACT"

// *****
// Monitor intermediate BLER results for a SISO single carrier configuration,
// by sending the following commands periodically.
// *****
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:RELative?

// *****
// Monitor intermediate BLER results for a MIMO configuration with carrier
// aggregation and two carriers, by sending the following commands periodically.
// *****
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:RELative?
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:STReam1:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:STReam2:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:STReam1:RELative?
FETCh:INTERmediate:LTE:SIGN:EBLer:PCC:STReam2:RELative?

FETCh:INTERmediate:LTE:SIGN:EBLer:SCC:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:SCC:RELative?
FETCh:INTERmediate:LTE:SIGN:EBLer:SCC:STReam1:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:SCC:STReam2:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:SCC:STReam1:RELative?
FETCh:INTERmediate:LTE:SIGN:EBLer:SCC:STReam2:RELative?

// *****
// Alternatively monitor the sum of all PCC and SCC streams.
// *****
FETCh:INTERmediate:LTE:SIGN:EBLer:ALL:ABSolute?
FETCh:INTERmediate:LTE:SIGN:EBLer:ALL:RELative?

```

3.3.4 Performing a single-shot BLER measurement

```
// *****
// Start a single-shot BLER measurement (one measurement cycle).
// *****
CONFIGure:LTE:SIGN:EBLer:REPetition SING
CONFIGure:LTE:SIGN:EBLer:SCONdition NONE
INIT:LTE:SIGN:EBLer

// *****
// Return BLER and throughput single results for a SISO single carrier
// configuration.
// *****
FETCh:LTE:SIGN:EBLer:ABSolute?
FETCh:LTE:SIGN:EBLer:RELative?

// *****
// Return BLER and throughput single results for a MIMO single carrier
// configuration:
// Overall absolute and relative results, absolute results for stream 1 and
// stream 2, relative results for stream 1 and stream 2.
// *****
FETCh:LTE:SIGN:EBLer:PCC:ABSolute?
FETCh:LTE:SIGN:EBLer:PCC:RELative?
FETCh:LTE:SIGN:EBLer:PCC:STReam1:ABSolute?
FETCh:LTE:SIGN:EBLer:PCC:STReam2:ABSolute?
FETCh:LTE:SIGN:EBLer:PCC:STReam1:RELative?
FETCh:LTE:SIGN:EBLer:PCC:STReam2:RELative?

// *****
// Return throughput traces for a MIMO configuration with CA and two carriers:
// Sum of all PCC plus SCC streams, sum of all PCC streams,
// sum of all SCC streams, throughput per stream.
// *****
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:ALL?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:PCC?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:SCC?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:PCC:STReam1?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:PCC:STReam2?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:SCC:STReam1?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:SCC:STReam2?

// *****
// Return CQI single results for a MIMO configuration with PCC and SCC1.
// *****
FETCh:LTE:SIGN:EBLer:PCC:CQIReporting:STReam1?
FETCh:LTE:SIGN:EBLer:PCC:CQIReporting:STReam2?
FETCh:LTE:SIGN:EBLer:SCC:CQIReporting:STReam1?
FETCh:LTE:SIGN:EBLer:SCC:CQIReporting:STReam2?
```

```

// *****
// Return CQI bar graphs and median CQI traces for a MIMO configuration with
// carrier aggregation (PCC and SCC1).
// *****
FETCh:LTE:SIGN:EBLer:TRACe:CQIReporting:PCC:STReam1?
FETCh:LTE:SIGN:EBLer:TRACe:CQIReporting:PCC:STReam2?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:PCC:MCQI:STReam1?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:PCC:MCQI:STReam2?

FETCh:LTE:SIGN:EBLer:TRACe:CQIReporting:SCC:STReam1?
FETCh:LTE:SIGN:EBLer:TRACe:CQIReporting:SCC:STReam2?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:SCC:MCQI:STReam1?
FETCh:LTE:SIGN:EBLer:TRACe:THROUGHput:SCC:MCQI:STReam2?

// *****
// Return RI and PMI bar graphs for PCC and SCC1.
// *****
FETCh:LTE:SIGN:EBLer:PCC:RI?
FETCh:LTE:SIGN:EBLer:PCC:PMI:RI1?
FETCh:LTE:SIGN:EBLer:PCC:PMI:RI2?
FETCh:LTE:SIGN:EBLer:PCC:PMI:RI3?
FETCh:LTE:SIGN:EBLer:PCC:PMI:RI4?

FETCh:LTE:SIGN:EBLer:SCC:RI?
FETCh:LTE:SIGN:EBLer:SCC:PMI:RI1?
FETCh:LTE:SIGN:EBLer:SCC:PMI:RI2?
FETCh:LTE:SIGN:EBLer:SCC:PMI:RI3?
FETCh:LTE:SIGN:EBLer:SCC:PMI:RI4?

// *****
// Return uplink BLER results for PCC and SCC1.
// *****
FETCh:LTE:SIGN:EBLer:PCC:UPLink?
FETCh:LTE:SIGN:EBLer:SCC:UPLink?

// *****
// Return entire HARQ tables for a MIMO configuration with PCC and SCC1.
// *****
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam1:TRANsmission:ABSolute?
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam1:TRANsmission:RELative?
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam2:TRANsmission:ABSolute?
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam2:TRANsmission:RELative?
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam1:SUBFrame:ABSolute?
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam1:SUBFrame:RELative?
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam2:SUBFrame:ABSolute?
FETCh:LTE:SIGN:EBLer:PCC:HARQ:STReam2:SUBFrame:RELative?

FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam1:TRANsmission:ABSolute?
FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam1:TRANsmission:RELative?
FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam2:TRANsmission:ABSolute?

```

```

FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam2:TRANsmission:RELative?
FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam1:SUBFrame:ABSolute?
FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam1:SUBFrame:RELative?
FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam2:SUBFrame:ABSolute?
FETCh:LTE:SIGN:EBLer:SCC:HARQ:STReam2:SUBFrame:RELative?

```

3.4 Command reference

The following sections describe the remote control commands of the "Extended BLER" measurement.

- [Conventions and general information](#)..... 944
- [Measurement control and states](#)..... 950
- [Measurement settings](#)..... 952
- [Measurement results](#)..... 955

3.4.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

3.4.1.1 SIGN<i>

SIGN<i> is used as abbreviation of SIGNaling<instance>. For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same firmware application. It can be omitted if the instrument supports only one instance, or to address the first instance.

3.4.1.2 STReam<s> and DL<s>

Commands that contain a STReam<s> mnemonic are related to a single downlink stream.

The suffix <s> selects the downlink stream and is only relevant for scenarios with MIMO. For SISO scenarios, you can omit the suffix.

3.4.1.3 PCC and SCC commands

For carrier aggregation scenarios, many settings are configurable per carrier. The following command variants are relevant in this context:

- A [:PCC] command configures the PCC. The command can also be relevant for scenarios without carrier aggregation.

- An `:SCC<c>` command configures the SCC number `<c>`. For the SCC number 1, you can omit the suffix 1.
- An `:SCC` command configures the SCC number 1. Other SCCs cannot be configured via such a command.
- A `:SETA` or `:SETB` command configures all carriers of set A or set B (inband contiguous UL carrier allocation).

3.4.1.4 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**

If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:

- "Query only": The command can only be used to query parameters. No setting.
- "Event": The command initiates an event. No query.

- **Parameter usage**

The parameter usage is indicated by the keyword preceding the parameters:

- "Parameters" are sent with a setting or query command and are returned as the result of a query.
- "Setting parameters" are sent with a setting command.
- "Query parameters" are sent with a query command, to refine the query.
- "Return values" are returned as the result of a query.

- **Parameter type**

The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:

- "string": characters enclosed in quotation marks (" or ')
- "numeric", "decimal": decimal number
- "integer": integer number
- "float": floating point number
- "hex": hexadecimal number starting with #H
- "binary": binary number starting with #B
- "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).

- **Firmware/Software:**

Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

3.4.1.5 Reliability indicator

The first value in the sequences returned by measurement result queries indicates the most severe error that has occurred during the measurement.

Example for an output array: 0, 10.22, 10.15, 10.01, 10.29, 100 (reliability = 0, followed by 5 numeric measurement values).

The reliability indicator has one of the following values:

- **0 - "no error"**
No error detected.
- **1 - "measurement timeout"**
The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INValid` results or has not completed the full statistic count.
- **2 - "capture buffer overflow"**
The measurement configuration results in a capture length that exceeds the available memory.
- **3 - "input overdriven" / 4 - "input underdriven"**
The accuracy of measurement results can be impaired because the input signal level was too high / too low.
- **6 - "trigger timeout"**
The measurement could not be started or continued because no trigger event was detected.
For 1xEV-DO: A test application report was not received from the mobile within three update time periods.
- **7 - "acquisition error"**
The R&S CMW could not properly decode the RF input signal.
- **8 - "sync error"**
The R&S CMW could not synchronize to the RF input signal.
- **9 - "uncalibrated"**
Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the technical specifications.
- **15 - "reference frequency error"**
The instrument has been configured to use an external reference signal. But the reference oscillator could not be locked to the external signal (for example signal level too low, frequency out of range or reference signal not available at all).
- **16 - "RF not available"**
The measurement could not be started because the configured RF input path was not active.
This problem can occur if a measurement is started in combined signal path mode and the controlling application has not yet activated the input path.
- **17 - "RF level not settled" / 18 - "RF frequency not settled"**
The measurement could not be started because the R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 - "call not established"**
For measurements: The measurement could not be started because no signaling connection to the DUT was established.
For IMS service: Establishing a voice over IMS call failed.
- **20 - "call type not usable"**

For measurements: The measurement could not be started because the established signaling connection had the wrong properties.

For IMS service: The voice over IMS settings could not be applied.

- **21 - "call lost"**

For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.

For IMS service: The VoIMS call was lost.

- **23 - "missing option"**

An action cannot be executed due to a missing option.

For GPRF generator: The ARB file cannot be played due to a missing option.

- **24 - "invalid RF setting"**

The desired RF TX level or RF RX reference level could not be applied.

- **25 - "level overrange"**

The RF TX level is in overrange. The signal quality can be degraded.

- **26 - "resource conflict"**

The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application.

Stop the application that has allocated the conflicting resources and try again.

- **27 - "no sensor connected"**

The GPRF external power sensor measurement could not be started due to a missing power sensor.

- **28 - "unexpected parameter change"**

One or more measurement configuration parameters were changed while the measurement was running. The results were not obtained with these new parameter values. Repeat the measurement. This situation can only occur in remote single-shot mode.

- **29 - "invalid RF frequency setting"**

The desired RF TX frequency or RF RX frequency could not be applied.

- **30 - "file not found"**

The specified file could not be found.

- **31 - "no DTM reply"**

The EUT did not reply to the direct test mode (DTM) command.

- **32 - "ACL disconnected"**

The ACL connection has been disconnected or lost.

- **40 - "ARB file CRC error"**

The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.

- **42 - "ARB header tag invalid"**

The ARB file selected in the GPRF generator contains an invalid header tag.

- **43 - "ARB segment overflow"**

The number of segments in the multi-segment ARB file is higher than the allowed maximum.

- **44 - "ARB file not found"**

The selected ARB file could not be found.

- **45 - "ARB memory overflow"**

The ARB file length is greater than the available memory.

- **46 - "ARB sample rate out of range"**
The clock rate of the ARB file is either too high or too low.
- **47 - "ARB cycles out of range"**
The repetition mode equals "Single Shot" and the playback length is too long.
Reduce the playback length or set the repetition mode to "Continuous".
$$\text{<Length>} = (\text{<Cycles>} * \text{<Samples>} + \text{<Additional Samples>}) / \text{<Clock Rate>}$$
- **48 - "ARB pool invalid"**
At least one of the ARB files in the ARB file pool is invalid.
- **49 - "sequencer list invalid"**
The sequencer list configuration has at least one invalid entry.
- **50 - "startup error"**
The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 - "no reply"**
The DAU has received no response, for example for a ping request.
- **52 - "connection error"**
A connection setup failed or a connection was lost.
For DAU: The DAU could not establish a connection to internal components.
Restart the instrument.
For Bluetooth: The EUT has been disconnected from the R&S CMW.
- **53 - "configuration error"**
The current DAU configuration is incomplete or wrong and could not be applied.
Check especially the IP address configuration.
- **54 - "file system error"**
The system drive of the DAU is full or corrupt. Execute a DAU self-test.
- **64 - "missing pathloss correction data"**
The pathloss measurement cannot be executed due to missing connector correction data. Order a calibration of the instrument by Rohde & Schwarz.
- **70 - "wrong standard"**
The standard of the measured signal does not match the configured standard.
- **71 - "wrong bandwidth"**
The bandwidth of the measured signal does not match the configured bandwidth.
- **72 - "wrong burst type"**
The burst type of the measured signal does not match the configured burst type.
- **73 - "MIMO signal detected"**
The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.
- **74 - "more streams than antennas"**
The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.
- **75 - "matrix inversion failed"**
The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.
- **76 - "SIG CRC failed"**
The cyclic redundancy check of a SIGNAL field failed.

- **77 - "parity check failed"**
The parity check of a SIGNAL field failed.
- **78 - "bursts not identical"**
In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.
- **79 - "wrong modulation"**
The modulation scheme of the measured signal does not match the configured modulation scheme.
- **80 - "SISO signal detected"**
Received signal contains only one stream.
- **81 - "fewer streams than antennas"**
Received signal has fewer streams than configured antennas. Reconfigure the number of antennas.
- **82 - "signal fields of different antennas are not identical"**
The signal fields of all captured antennas are compared. In a valid MIMO transmission, all signal fields are identical. If not, some basic parameters such as MCS or data length are not the same and a MIMO measurement is not possible.
- **83 - "bit errors in some of the received symbols"**
Adjust input signal power and repeat training.
- **84 - "first and last symbols need to be skipped"**
Due to a sequence counter in the MAC header, the first symbol and the last symbol (due to the FCS) change from one burst to another. You can configure the first and the last symbol to be skipped.
- **85 - "received and saved bursts are not identical"**
The signal field of the received burst is not identical to that of the saved burst. Some basic parameters such as MCS or data length are not the same and a composite MIMO measurement is not possible.
- **86 - "no symbols for measurement"**
Too many head and tail symbols have been skipped.
- **93 - "OCXO oven temperature too low"**
The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **96 - "frequency/level change too fast"**
There was a too fast frequency and/or level change. If you use GPRF with list mode, increase the duration of the list entry.
- **101 - "firmware error"**
This value indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.
If the error occurs again, consider the following hints:
 - Firmware errors can often be repaired by restoring the factory default settings.
 - If a software package has not been properly installed, this failure is often indicated in the "Setup" dialog > section "SW/HW-Equipment > Installed Software".

- Check for software updates correcting the error. Updates are provided in the customer web on GLORIS (registration required): <https://gloris.rohde-schwarz.com>.

If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.

- **102 - "unidentified error"**

This value indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 firmware error".

- **103 - "parameter error"**

This value indicates that the measurement could not be performed due to internal conflicting parameter settings.

For 1xEV-DO:

- The 1xEV-DO signaling test application is not enabled.
- Forward link measurement data were requested, but the test application direction is not set to "Forward" or "Forward & Reverse".
- Reverse link measurement data were requested, but the test application direction is not set to "reverse" or "forward & reverse".
- Measurement data were requested, but the corresponding measurement is not assigned.

For CDMA2000: The "1st Service Option" is not set to a loopback service option.

A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.

If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see <https://www.service.rohde-schwarz.com>).

- **104 - "not functional"**

The application could not be started with the configured parameter set.

3.4.2 Measurement control and states

The following commands control the measurement and return the current measurement state.

<code>INITiate:LTE:SIGN<i>:EBLer</code>	950
<code>STOP:LTE:SIGN<i>:EBLer</code>	950
<code>ABORT:LTE:SIGN<i>:EBLer</code>	950
<code>FETCH:LTE:SIGN<i>:EBLer:STATE?</code>	951
<code>FETCH:LTE:SIGN<i>:EBLer:STATE:ALL?</code>	951

INITiate:LTE:SIGN<i>:EBLer

STOP:LTE:SIGN<i>:EBLer

ABORT:LTE:SIGN<i>:EBLer

Starts, stops or aborts the measurement:

- The `INITiate:...` command starts or restarts the measurement. The measurement enters the "RUN" state.

- The `STOP:...` command halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- The `ABORT:...` command halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

To query the current measurement state, use `FETCH:..:STATE?`.

Example: See [Performing a single-shot BLER measurement](#)

Usage: Event

Firmware/software: V1.0.15.20

Manual operation: See ["Extended BLER softkey"](#) on page 924

FETCH:LTE:SIGN<i>:EBLer:STATE?

Queries the main measurement state. Use `FETCH:..:STATE:ALL?` to query the measurement state including the substates. The substates tell you, for example, whether a state transition is ongoing.

Return values:

<State> OFF | RUN | RDY
 Current state or target state of ongoing state transition
OFF: measurement off
RUN: measurement running
RDY: measurement completed
 *RST: OFF

Usage: Query only

Firmware/software: V1.0.15.20

Manual operation: See ["Extended BLER softkey"](#) on page 924

FETCH:LTE:SIGN<i>:EBLer:STATE:ALL?

Queries the main measurement state and the measurement substates. A really running measurement returns RUN, ADJ, ACT.

Return values:

<MainState> OFF | RUN | RDY
 Current state or target state of ongoing state transition
OFF: measurement off
RUN: measurement running
RDY: measurement completed
 *RST: OFF

<SyncState> PEND | ADJ
PEND: transition to <MainState> ongoing
ADJ: <MainState> reached

<ResourceState>	QUE ACT INV QUE: waiting for resource allocation ACT: resources allocated INV: no resources allocated
Example:	See Performing a continuous BLER measurement
Usage:	Query only
Firmware/software:	V1.0.15.20
Manual operation:	See "Extended BLER softkey" on page 924

3.4.3 Measurement settings

The following commands configure the BLER measurement.

CONFigure:LTE:SIGN<i>:EBLer:TOUT	952
CONFigure:LTE:SIGN<i>:EBLer:REPetition	953
CONFigure:LTE:SIGN<i>:EBLer:SCONdition	953
CONFigure:LTE:SIGN<i>:EBLer:SFRames	953
CONFigure:LTE:SIGN<i>:EBLer:ERCalc	954
CONFigure:LTE:SIGN<i>:EBLer:CONFidence:OASCondition	954
CONFigure:LTE:SIGN<i>:EBLer:CONFidence:MTTime	954
CONFigure:LTE:SIGN<i>:EBLer:CONFidence:LERate	955

CONFigure:LTE:SIGN<i>:EBLer:TOUT <Timeout>

Defines a timeout for the measurement. The timer is started when the measurement is initiated via a `READ` or `INIT` command. It is not started if the measurement is initiated manually.

When the measurement has completed the first measurement cycle (first single shot), the statistical depth is reached and the timer is reset.

If the first measurement cycle has not been completed when the timer expires, the measurement is stopped. The measurement state changes to `RDY`. The reliability indicator is set to 1, indicating that a measurement timeout occurred. Still running `READ`, `FETCH` or `CALCulate` commands are completed, returning the available results. At least for some results, there are no values at all or the statistical depth has not been reached.

A timeout of 0 s corresponds to an infinite measurement timeout.

Parameters:

<Timeout>	numeric
	Default unit: s

Firmware/software: V2.0.10

CONFigure:LTE:SIGN<i>:EBLer:REPetition <Repetition>

Specifies whether the measurement is stopped after a single shot or repeated continuously.

Parameters:

<Repetition> SINGleshot | CONTInuous
SINGleshot: Single-shot measurement
CONTInuous: Continuous measurement
 *RST: SING

Example: See [Configuring a BLER measurement](#)

Firmware/software: V3.0.30

Manual operation: See ["Repetition"](#) on page 937

CONFigure:LTE:SIGN<i>:EBLer:SCONdition <StopCondition>

Selects whether a BLER measurement without stop condition or a confidence BLER measurement with early decision concept is performed.

Parameters:

<StopCondition> NONE | CLEVel
NONE: no stop condition, no early termination of measurement
CLEVel: confidence BLER measurement
 *RST: NONE

Example: See [Configuring a BLER measurement](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510/-KS512 for CLEVel (without CA/with CA)

Manual operation: See ["Stop Condition"](#) on page 937

CONFigure:LTE:SIGN<i>:EBLer:SFRames <Subframes>

Defines the number of subframes (= number of transport blocks) to be processed per measurement cycle.

For confidence BLER measurements, this parameter specifies only the length of the throughput result trace but does not influence the duration of the measurement.

Parameters:

<Subframes> integer
 Range: 100 to 400E+3
 *RST: 10E+3

Example: See [Configuring a BLER measurement](#)

Firmware/software: V3.0.30

Manual operation: See ["No. of Subframes"](#) on page 938

CONFigure:LTE:SIGN<i>:EBLer:ERCalc <Algorithm>

Selects the formula to be used for calculation of the BLER from the number of ACK, NACK and DTX.

Parameters:

<Algorithm> ERC1 | ERC2 | ERC3 | ERC4
ERC1: $BLER = (NACK + DTX) / (ACK + NACK + DTX)$
ERC2: $BLER = DTX / (ACK + NACK + DTX)$
ERC3: $BLER = NACK / (ACK + NACK + DTX)$
ERC4: $BLER = NACK / (ACK + NACK)$
 *RST: ERC1

Example: See [Configuring a BLER measurement](#)

Firmware/software: V3.0.30

Manual operation: See ["Error Ratio Calculation"](#) on page 938

CONFigure:LTE:SIGN<i>:EBLer:CONFidence:OASCondition <Condition>

Configures the stop decision and the overall result calculation for confidence BLER measurements with carrier aggregation.

Parameters:

<Condition> PCC | SCC1 | SCC2 | AC1St | ACWait
PCC: PCC only
SCC1: SCC1 only
SCC2: SCC2 only
AC1St: all carriers, stop on 1st fail
ACWait: all carriers, wait for all CCs
 *RST: AC1S

Example: See [Configuring a BLER measurement](#)

Firmware/software: V3.2.80
 V3.5.10: added SCC2

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Over All Stop Decision"](#) on page 938

CONFigure:LTE:SIGN<i>:EBLer:CONFidence:MTTime <Time>

Specifies a minimum test time for confidence BLER measurements.

Parameters:

<Time> numeric
 Minimum number of processed subframes
 Range: 0 to 500E+3
 Increment: 200
 *RST: 0

Example: See [Configuring a BLER measurement](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Minimum Test Time"](#) on page 939

CONFigure:LTE:SIGN<i>:EBLer:CONFidence:LERate <Rate>

Selects the limit error ratio for a confidence BLER measurement.

Parameters:

<Rate> P001 | P010 | P050

P001: 0.1 %, 3GPP TS 36.521 annex G.4

P010: 1 %, 3GPP TS 36.521 annex G.4

P050: 5 %, 3GPP TS 36.521 annex G.2

*RST: P050

Example: See [Configuring a BLER measurement](#)

Firmware/software: V3.0.30

Options: R&S CMW-KS510/-KS512 (without CA/with CA)

Manual operation: See ["Limit Error Rate"](#) on page 939

3.4.4 Measurement results

All results of the BLER measurement are retrieved via `FETCH` commands.

There are two types of `FETCH` commands:

- `FETCH:LTE:...` waits until the current measurement cycle is complete and returns the final results of the measurement cycle.
This type of command is available for all results.
- `FETCH:INTERmediate:LTE:...` returns the already available measurement results immediately, even if the first measurement cycle is not yet complete.
This type of command is only available for selected results.

The following commands retrieve the results of the BLER measurement.

<code>FETCH:LTE:SIGN<i>:EBLer[:PCC]:CONFidence?</code>	956
<code>FETCH:LTE:SIGN<i>:EBLer:SCC<c>:CONFidence?</code>	956
<code>FETCH:LTE:SIGN<i>:EBLer:ALL:CONFidence?</code>	957
<code>FETCH:INTERmediate:LTE:SIGN<i>:EBLer[:PCC]:ABSolute?</code>	957
<code>FETCH:INTERmediate:LTE:SIGN<i>:EBLer:SCC<c>:ABSolute?</code>	957
<code>FETCH:LTE:SIGN<i>:EBLer[:PCC]:ABSolute?</code>	957
<code>FETCH:LTE:SIGN<i>:EBLer:SCC<c>:ABSolute?</code>	957
<code>FETCH:INTERmediate:LTE:SIGN<i>:EBLer:ALL:ABSolute?</code>	958
<code>FETCH:LTE:SIGN<i>:EBLer:ALL:ABSolute?</code>	958
<code>FETCH:INTERmediate:LTE:SIGN<i>:EBLer[:PCC]:RELative?</code>	959
<code>FETCH:INTERmediate:LTE:SIGN<i>:EBLer:SCC<c>:RELative?</code>	959
<code>FETCH:LTE:SIGN<i>:EBLer[:PCC]:RELative?</code>	959

FETCh:LTE:SIGN<i>:EBLer:SCC<c>:RELative?	959
FETCh:INTErmediate:LTE:SIGN<i>:EBLer:ALL:RELative?	960
FETCh:LTE:SIGN<i>:EBLer:ALL:RELative?	960
FETCh:INTErmediate:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:ABSolute?	961
FETCh:INTErmediate:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:ABSolute?	961
FETCh:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:ABSolute?	961
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:ABSolute?	961
FETCh:INTErmediate:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:RELative?	962
FETCh:INTErmediate:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:RELative?	962
FETCh:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:RELative?	962
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:RELative?	962
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput[:PCC]:STReam<s>?	963
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>:STReam<s>?	963
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput[:PCC]?	963
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>?	963
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:ALL?	964
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput[:PCC]:MCQI:STReam<s>?	965
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>:MCQI:STReam<s>?	965
FETCh:LTE:SIGN<i>:EBLer:TRACe:CQIReporting[:PCC]:STReam<s>?	965
FETCh:LTE:SIGN<i>:EBLer:TRACe:CQIReporting:SCC<c>:STReam<s>?	965
FETCh:LTE:SIGN<i>:EBLer[:PCC]:CQIReporting:STReam<s>?	966
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:CQIReporting:STReam<s>?	966
FETCh:LTE:SIGN<i>:EBLer[:PCC]:RI?	967
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:RI?	967
FETCh:LTE:SIGN<i>:EBLer[:PCC]:PMI:RI<no>?	967
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:PMI:RI<no>?	967
FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:TRANsmiSSion:ABSolute?	968
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:TRANsmiSSion:ABSolute?	968
FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:TRANsmiSSion:RELative?	969
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:TRANsmiSSion:RELative?	969
FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:SUBFrame:ABSolute?	969
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:SUBFrame:ABSolute?	969
FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:SUBFrame:RELative?	970
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:SUBFrame:RELative?	970
FETCh:LTE:SIGN<i>:EBLer[:PCC]:UPLink?	971
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:UPLink?	971

FETCh:LTE:SIGN<i>:EBLer[:PCC]:CONFidence?**FETCh:LTE:SIGN<i>:EBLer:SCC<c>:CONFidence?**

Returns the pass/fail result of a confidence BLER measurement, for one carrier.

Suffix:

<c> 1..7

Return values:

<Reliability> decimal
See [Reliability indicator](#)

<Confidence>	EPASSs EFAil PASS FAIL UNDecided EPASSs, EFAil: early pass, early fail PASS, FAIL: pass, fail UNDecided: undecided
Example:	See Performing a confidence BLER measurement
Usage:	Query only
Firmware/software:	V3.0.30 V3.2.80: added UNDecided and SCC command
Manual operation:	See "Confidence BLER results" on page 928

FETCh:LTE:SIGN<i>:EBLer:ALL:CONFidence?

Returns the overall pass/fail result of a confidence BLER measurement.

Return values:

<Reliability>	decimal See Reliability indicator
<Confidence>	EPASSs EFAil PASS FAIL UNDecided EPASSs, EFAil: early pass, early fail PASS, FAIL: pass, fail UNDecided: undecided
Example:	See Performing a confidence BLER measurement
Usage:	Query only
Firmware/software:	V3.2.80
Manual operation:	See "Confidence BLER results" on page 928

FETCh:INTermediate:LTE:SIGN<i>:EBLer[:PCC]:ABSolute? FETCh:INTermediate:LTE:SIGN<i>:EBLer:SCC<c>:ABSolute? FETCh:LTE:SIGN<i>:EBLer[:PCC]:ABSolute? FETCh:LTE:SIGN<i>:EBLer:SCC<c>:ABSolute?

Returns the absolute overall results of the BLER measurement for the sum of all DL streams of one carrier.

Suffix:

<c>	1..7
-----	------

Return values:

<1_Reliability>	decimal See Reliability indicator
<2_ACK>	decimal Number of received acknowledgments (sum of all downlink streams)

<3_NACK>	decimal Number of received negative acknowledgments (sum of all downlink streams)
<4_Subframes>	decimal Number of already processed subframes (per downlink stream)
<5_ThroughputAver>	float
<6_ThroughputMin>	Average, minimum and maximum throughput (sum of all downlink streams) Default unit: kbit/s
<7_ThroughputMax>	
<8_DTX>	decimal Number of sent scheduled subframes for which no ACK and no NACK has been received (sum of all downlink streams)
<9_Scheduled>	decimal Number of already sent scheduled subframes (per downlink stream)
<10_MedianCQI>	decimal Median value of received CQI indices
Example:	See Performing a continuous BLER measurement
Usage:	Query only
Firmware/software:	V3.0.10 V3.0.30: INtermediate command, V3.2.50: SCC command
Manual operation:	See "Result tables" on page 927

FETCh:INtermediate:LTE:SIGN<i>:EBLer:ALL:ABSolute?**FETCh:LTE:SIGN<i>:EBLer:ALL:ABSolute?**

Returns the absolute overall results of the BLER measurement for the sum of all downlink streams of all carriers.

Return values:

<1_Reliability>	decimal See Reliability indicator
<2_ACK>	decimal Number of received acknowledgments (sum of all downlink streams)
<3_NACK>	decimal Number of received negative acknowledgments (sum of all downlink streams)
<4_Subframes>	decimal Number of already processed subframes (per downlink stream)

<5_ThroughputAver>	float
<6_ThroughputMin>	Average, minimum and maximum throughput (sum of all downlink streams)
<7_ThroughputMax>	Default unit: kbit/s
<8_DTX>	decimal Number of sent scheduled subframes for which no ACK and no NACK has been received (sum of all downlink streams)
<9_Scheduled>	decimal Number of already sent scheduled subframes (per downlink stream)
<10_MedianCQI>	decimal Median value of received CQI indices
Example:	See Performing a continuous BLER measurement
Usage:	Query only
Firmware/software:	V3.2.50
Manual operation:	See "Result tables" on page 927

FETCh:INTermediate:LTE:SIGN<i>:EBLer[:PCC]:RELative?
FETCh:INTermediate:LTE:SIGN<i>:EBLer:SCC<c>:RELative?
FETCh:LTE:SIGN<i>:EBLer[:PCC]:RELative?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:RELative?

Returns the relative overall results of the BLER measurement for the sum of all DL streams of one carrier.

Suffix:

<c> 1..7

Return values:

<1_Reliability>	decimal See Reliability indicator
<2_ACK>	float Received acknowledgments (percentage of sent scheduled subframes) Default unit: %
<3_NACK>	float Received negative acknowledgments (percentage of sent scheduled subframes) Default unit: %
<4_BLER>	float Block error ratio (percentage of sent scheduled subframes for which no ACK has been received) Default unit: %

<5_ThroughputAver>	float
	Average DL throughput (as percentage of maximum reachable throughput)
	Default unit: %
<6_DTX>	float
	Percentage of sent scheduled subframes for which no ACK and no NACK has been received
	Default unit: %
Example:	See Performing a continuous BLER measurement
Usage:	Query only
Firmware/software:	V3.0.30, SCC command V3.2.50
Manual operation:	See "Result tables" on page 927

FETCh:INtermediate:LTE:SIGN<i>:EBLer:ALL:RELative?**FETCh:LTE:SIGN<i>:EBLer:ALL:RELative?**

Returns the relative overall results of the BLER measurement for the sum of all down-link streams of all carriers.

Return values:

<1_Reliability>	decimal
	See Reliability indicator
<2_ACK>	float
	Received acknowledgments (percentage of sent scheduled subframes)
	Default unit: %
<3_NACK>	float
	Received negative acknowledgments (percentage of sent scheduled subframes)
	Default unit: %
<4_BLER>	float
	Block error ratio (percentage of sent scheduled subframes for which no ACK has been received)
	Default unit: %
<5_ThroughputAver>	float
	Average DL throughput (as percentage of maximum reachable throughput)
	Default unit: %

<6_DTX> float
 Percentage of sent scheduled subframes for which no ACK and no NACK has been received
 Default unit: %

Example: See [Performing a continuous BLER measurement](#)

Usage: Query only

Firmware/software: V3.2.50

Manual operation: See ["Result tables"](#) on page 927

FETCH:INTERmediate:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:ABSolute?
FETCH:INTERmediate:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:ABSolute?
FETCH:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:ABSolute?
FETCH:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:ABSolute?

Returns the absolute results of the BLER measurement for one downlink stream of one carrier.

Suffix:

<s> 1..2

<c> 1..7

Return values:

<1_Reliability> decimal
 See [Reliability indicator](#)

<2_ACK> decimal
 Number of received acknowledgments

<3_NACK> decimal
 Number of received negative acknowledgments

<4_Subframes> decimal
 Number of already processed subframes

<5_Throughput> float
 Average DL throughput
 Default unit: kbit/s

<6_DTX> decimal
 Number of sent scheduled subframes for which no ACK and no NACK has been received

<7_Scheduled> decimal
 Number of already sent scheduled subframes

<8_MedianCQI> decimal
 Median value of received CQI indices

Example: See [Performing a continuous BLER measurement](#)

Usage:	Query only
Firmware/software:	V3.0.10 V3.0.30: INTermediate command, V3.2.50: SCC command
Options:	R&S CMW-KS520 or -KS540
Manual operation:	See "Result tables" on page 927

FETCh:INTermediate:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:RELative?
FETCh:INTermediate:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:RELative?
FETCh:LTE:SIGN<i>:EBLer[:PCC]:STReam<s>:RELative?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:STReam<s>:RELative?

Returns the relative results of the BLER measurement for one downlink stream of one carrier.

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<ACK> float

Received acknowledgments (percentage of sent scheduled subframes)

Default unit: %

<NACK> float

Received negative acknowledgments (percentage of sent scheduled subframes)

Default unit: %

<BLER> float

Block error ratio (percentage of sent scheduled subframes for which no ACK has been received)

Default unit: %

<Throughput> float

Average DL throughput (percentage of maximum reachable throughput)

Default unit: %

<DTX> float

Percentage of sent scheduled subframes for which no ACK and no NACK has been received

Default unit: %

Example: See [Performing a continuous BLER measurement](#)

Usage: Query only

Firmware/software: V3.0.30, SCC command V3.2.50

Options: R&S CMW-KS520

Manual operation: See ["Result tables"](#) on page 927

FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput[:PCC]:STReam<s>?
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>:STReam<s>?

Returns the throughput trace for one downlink stream of one carrier.

Each value is returned as a pair of X-value and Y-value. The number of result pairs n equals the number of subframes to be processed per measurement cycle, divided by 200.

Returned results: <Reliability>, <XValue>₁, <YValue>₁, ..., <XValue>_n, <YValue>_n

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<XValue> float

Subframe label, 0 = last processed subframe, -1 = previously processed subframe, and so on

<YValue> float

Throughput value calculated from the BLER result of 200 processed subframes (the labeled subframe and the previous 199 subframes)

Default unit: kbit/s

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V2.0.10, SCC command V3.2.50

Options: R&S CMW-KS520 or -KS540

Manual operation: See ["Result diagram"](#) on page 929

FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput[:PCC]?
FETCh:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>?

Returns the throughput trace for the sum of all DL streams of one carrier.

Each value is returned as a pair of X-value and Y-value. The number of result pairs n equals the number of subframes to be processed per measurement cycle, divided by 200.

Returned results: <Reliability>, <XValue>₁, <YValue>₁, ..., <XValue>_n, <YValue>_n

Suffix:

<c> 1..7

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<XValue> float

Subframe label, 0 = last processed subframe, -1 = previously processed subframe, and so on

<YValue> float

Throughput value calculated from the BLER result of 200 processed subframes (the labeled subframe and the previous 199 subframes)

Default unit: kbit/s

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V2.0.10. SCC command V3.2.50

Manual operation: See ["Result diagram"](#) on page 929

FETCh:LTE:SIGN<i>:EBLer:TRACe:THROughput:ALL?

Returns the throughput trace for the sum of all downlink streams of all carriers.

Each value is returned as a pair of X-value and Y-value. The number of result pairs n equals the number of subframes to be processed per measurement cycle, divided by 200.

Returned results: <Reliability>, <XValue>₁, <YValue>₁, ..., <XValue>_n, <YValue>_n

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<XValue> float

Subframe label, 0 = last processed subframe, -1 = previously processed subframe, and so on

<YValue> float

Throughput value calculated from the BLER result of 200 processed subframes (the labeled subframe and the previous 199 subframes)

Default unit: kbit/s

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.2.50

Manual operation: See ["Result diagram"](#) on page 929

FETCH:LTE:SIGN<i>:EBLer:TRACe:THRoughput[:PCC]:MCQI:STReam<s>?
FETCH:LTE:SIGN<i>:EBLer:TRACe:THRoughput:SCC<c>:MCQI:STReam<s>?

Returns the median CQI trace for one downlink stream.

Each value is returned as a pair of X-value and Y-value. The number of result pairs n equals the number of subframes to be processed per measurement cycle, divided by 200.

Returned results: <Reliability>, <XValue>₁, <YValue>₁, ..., <XValue>_n, <YValue>_n

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal
 See [Reliability indicator](#)

<XValue> float
 Subframe label, 0 = last processed subframe, -1 = previously processed subframe, and so on

<YValue> float
 Median CQI value calculated from the CQI indices reported within 200 processed subframes (the labeled subframe and the previous 199 subframes)

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.0.10, SCC command V3.2.70

Manual operation: See ["Result diagram"](#) on page 929

FETCH:LTE:SIGN<i>:EBLer:TRACe:CQIReporting[:PCC]:STReam<s>?
FETCH:LTE:SIGN<i>:EBLer:TRACe:CQIReporting:SCC<c>:STReam<s>?

Returns the Y-values of the CQI index bar graph for one downlink stream.

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal
 See [Reliability indicator](#)

<YValue> float
Comma-separated list of 16 Y-values, for CQI index 0 to 15

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.0.10, SCC command V3.2.70

Manual operation: See ["Bar graph and upper result table"](#) on page 930

FETCH:LTE:SIGN<i>:EBLer[:PCC]:CQIReporting:STReam<s>?

FETCH:LTE:SIGN<i>:EBLer:SCC<c>:CQIReporting:STReam<s>?

Returns the single results of the CQI reporting view for one downlink stream of one carrier.

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal
See [Reliability indicator](#)

<CQImedian> decimal
Median reported CQI value

<RangeAbsolute> decimal
Number of reports received for the range from median CQI - 1 to median CQI + 1

<RangeRelative> float
<RangeAbsolute> as percentage of total number of received reports
Default unit: %

<BLER> float
Block error ratio (percentage of sent scheduled subframes for which no ACK has been received)
Default unit: %

<TotalNumber> decimal
Total number of received CQI reports

<ExpiredSubframes> decimal
Number of already sent scheduled subframes

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.0.30, SCC command V3.2.70

Manual operation: See ["Lower result table"](#) on page 931

FETCh:LTE:SIGN<i>:EBLer[:PCC]:RI?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:RI?

Returns the rank indicator (RI) results.

Suffix:

<c> 1..7

Return values:

<Reliability> decimal
 See [Reliability indicator](#)

<RI> decimal
 Comma-separated list of four values:
 Number of received "RI = 1",
 Number of received "RI = 2",
 Number of received "RI = 3",
 Number of received "RI = 4"

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.2.20, SCC command V3.2.70
 V3.7.10 RI = 3, 4

Manual operation: See ["RI bar graph / table"](#) on page 932

FETCh:LTE:SIGN<i>:EBLer[:PCC]:PMI:RI<no>?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:PMI:RI<no>?

Returns the PMI results for the RI value <no>.

Suffix:

<no> 1..4

<c> 1..7

Return values:

<Reliability> decimal
 See [Reliability indicator](#)

<PMI> decimal
 Comma-separated list of values, indicating the number of received PMI values, see table

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.2.20, SCC command V3.2.70
 V3.7.10 <no> 3, 4

Manual operation: See ["PMI bar graph / table"](#) on page 932

Table 3-1: <PMI> results depending on TX antennas / CSI-RS ports, RX antennas and <no>

Antennas	RI<no>	Number of <PMI> results
2x2	1	4 values for PMI = 0, 1, 2, 3
2x4	2	2 values for PMI = 0, 1
4x2	1, 2	16 values for PMI = 0, 1, ..., 15
4x4	1, 2, 3, 4	
8x2	1, 2	256 values for {PMI2 = 0, ..., 15} _{PMI1 = 0, ..., {...}PMI1 = 15}
8x4	1, 2	
	3	
	4	32 values for {PMI2 = 0, ..., 7} _{PMI1 = 0, ..., {...}PMI1 = 3}

FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:TRANsmiSSion:ABSolute?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:TRANsmiSSion:
ABSolute?

Returns absolute HARQ results for one downlink stream. All columns of the "HARQ per Transmissions" result table are returned:

<Reliability>, {<Sent>, <ACK>, <NACK>, <DTX>}_{column 1}, {...}_{col. 2}, {...}_{col. 3}, {...}_{col. 4}

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<Sent> decimal

Number of sent subframes

<ACK> decimal

Number of received acknowledgments

<NACK> decimal

Number of received negative acknowledgments

<DTX> decimal

Number of sent subframes for which no ACK and no NACK has been received

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.2.10, SCC command V3.2.70

Manual operation: See "[HARQ per Transmissions table](#)" on page 934

FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:TRANsmiSSion:RELative?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:TRANsmiSSion:RELative?

Returns relative HARQ results for one downlink stream. All columns of the "HARQ per Transmissions" result table are returned:

<Reliability>, {<Sent>, <ACK>, <NACK>, <DTX>}_{column 1}, {...}_{col. 2}, {...}_{col. 3}, {...}_{col. 4}

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<Sent> float

Sent subframes (percentage of sum of sent subframes over all transmissions)

Default unit: %

<ACK> float

Received acknowledgments (percentage of ACK+NACK+DTX in the column)

Default unit: %

<NACK> float

Received negative acknowledgments (percentage of ACK+NACK+DTX in the column)

Default unit: %

<DTX> float

Sent subframes for which no ACK and no NACK has been received (percentage of ACK+NACK+DTX in the column)

Default unit: %

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.2.10, SCC command V3.2.70

Manual operation: See ["HARQ per Transmissions table"](#) on page 934

FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:SUBFrame:ABSolute?
FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:SUBFrame:ABSolute?

Returns absolute HARQ results for one downlink stream. All columns of the "HARQ per Subframe" result table are returned:

<Reliability>, {<Sent>, <ACK>, <NACK>, <DTX>}_{column 0}, {...}_{column 1}, ..., {...}_{column 9}

Suffix:

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<Sent> NAV returned, for future use

<ACK> decimal

Number of received acknowledgments

<NACK> decimal

Number of received negative acknowledgments

<DTX> decimal

Number of sent subframes for which no ACK and no NACK has been received

Example: See [Performing a single-shot BLER measurement](#)**Usage:** Query only**Firmware/software:** V3.2.10, SCC command V3.2.70**Manual operation:** See ["HARQ per Subframe table"](#) on page 935**FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:SUBFrame:RELative?****FETCh:LTE:SIGN<i>:EBLer:SCC<c>:HARQ:STReam<s>:SUBFrame:RELative?**

Returns relative HARQ results for one downlink stream. All columns of the "HARQ per Subframe" result table are returned:

<Reliability>, {<Sent>, <ACK>, <NACK>, <DTX>}_{column 0}, {...}_{column 1}, ..., {...}_{column 9}**Suffix:**

<s> 1..2

<c> 1..7

Return values:

<Reliability> decimal

See [Reliability indicator](#)

<Sent> NAV returned, for future use

<ACK> float

Received acknowledgments (percentage of ACK+NACK+DTX in the column)

Default unit: %

<NACK> float
Received negative acknowledgments (percentage of ACK+NACK+DTX in the column)
Default unit: %

<DTX> float
Sent subframes for which no ACK and no NACK has been received (percentage of ACK+NACK+DTX in the column)
Default unit: %

Example: See [Performing a single-shot BLER measurement](#)

Usage: Query only

Firmware/software: V3.2.10, SCC command V3.2.70

Manual operation: See ["HARQ per Subframe table"](#) on page 935

FETCh:LTE:SIGN<i>:EBLer[:PCC]:UPLink?

FETCh:LTE:SIGN<i>:EBLer:SCC<c>:UPLink?

Returns the uplink results of the BLER measurement.

Suffix:

<c> 1..7

Return values:

<Reliability> decimal
See [Reliability indicator](#)

<BLER> decimal
Block error ratio (percentage of received uplink subframes with failed CRC check)
Default unit: %

<Throughput> decimal
Average uplink throughput
Default unit: kbit/s

<CRCPass> decimal
Number of received subframes with passed CRC check

<CRCFail> decimal
Number of received subframes with failed CRC check

<DTX> decimal
Number of scheduled UL subframes not sent by the UE
Only evaluated if skipping UL transmissions is not allowed.

<Skipped> decimal
Number of scheduled UL subframes not sent by the UE
Only evaluated if skipping UL transmissions is allowed.

Example:	See Performing a single-shot BLER measurement
Usage:	Query only
Firmware/software:	V3.0.50, SCC command V3.5.20 V3.5.50 added <DTX>, V3.8.10 added <Skipped>
Options:	R&S CMW-KS510
Manual operation:	See "Uplink results" on page 925

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FETCh:LTE:SIGN<i>:EBLer[:PCC]:HARQ:STReam<s>:TRANsmision:ABSolute?	968
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STOP:LTE:SIGN<i>:EBLer	950

4 LTE RLC throughput measurement

The RLC throughput measurement is installed with the LTE signaling application. To access the measurement, press the softkey "LTE RX Meas" in the LTE signaling main view. Then select the tab "RLC Throughput".

The RLC throughput measurement evaluates the total data throughput in the downlink and in the uplink.

4.1 Performing RLC throughput measurements

To measure the RLC throughput, you must set up a connection in data application mode and generate IP traffic.

1. Set up a connection as described in [Section 2.3.2, "LTE IP-based data tests"](#), on page 103.
2. Use the data application unit (DAU) to generate IP traffic in the measured direction (uplink and/or downlink). You can, for example, start an iperf measurement. Or you can transfer data via FTP.
For details, refer to the DAU documentation.
3. Start the RLC throughput measurement.
4. Evaluate the results.

4.2 GUI reference

The RLC throughput measurement is part of the RX measurements that are installed with the LTE signaling application.

Access: LTE signaling main view > "LTE RX Meas" softkey > "RLC Throughput" tab.

The tab is described in this section.

• Tab overview	974
• Measurement control	975
• Measurement results	976
• Settings	977

4.2.1 Tab overview

The "RLC Throughput" tab shows the measurement results in a diagram and a table. The connection status information displayed at the bottom is the same as in the LTE signaling main view.

The most important settings of the LTE signaling application can be accessed via the "Signaling Parameters" softkey and the related hotkeys.

To switch to the signaling application, press the "LTE Signaling" softkey two times.

The "Config" hotkey opens either the configuration dialog of the measurement or the configuration dialog of the signaling application, depending on which softkey is active.

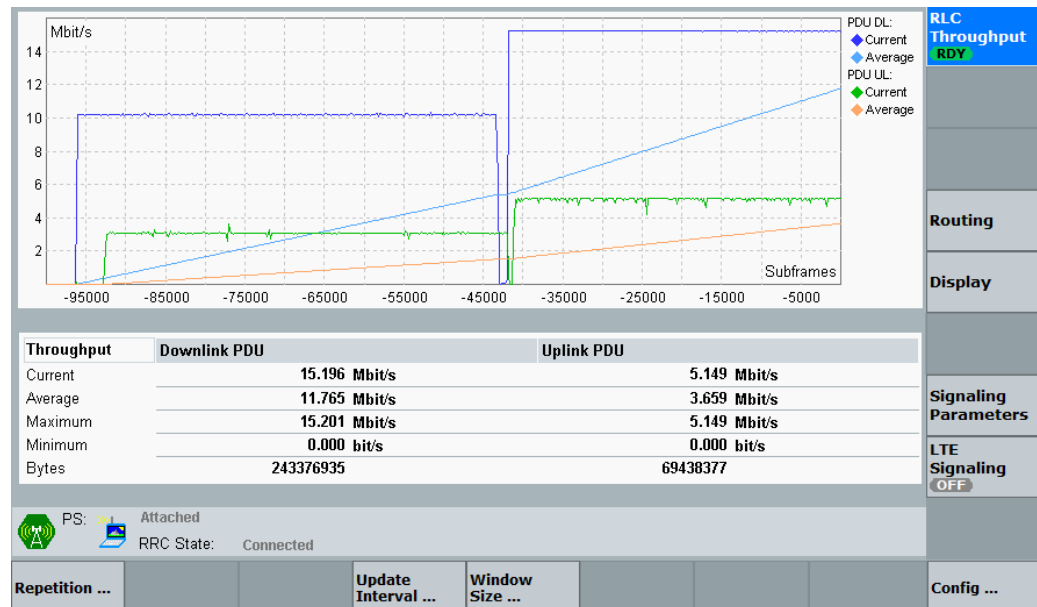


Figure 4-1: RLC throughput tab

4.2.2 Measurement control

To turn the measurement on or off, select the control softkey and press [ON | OFF] or [RESTART | STOP]. Alternatively, right-click the control softkey.



RLC Throughput softkey

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

INITiate:LTE:SIGN<i>:THROUGHput

STOP:LTE:SIGN<i>:THROUGHput

ABORT:LTE:SIGN<i>:THROUGHput

FETCH:LTE:SIGN<i>:THROUGHput:STATE?

FETCH:LTE:SIGN<i>:THROUGHput:STATE:ALL?

4.2.3 Measurement results

All results of the RLC throughput measurement are displayed on the "RLC Throughput" tab. The results are described below.

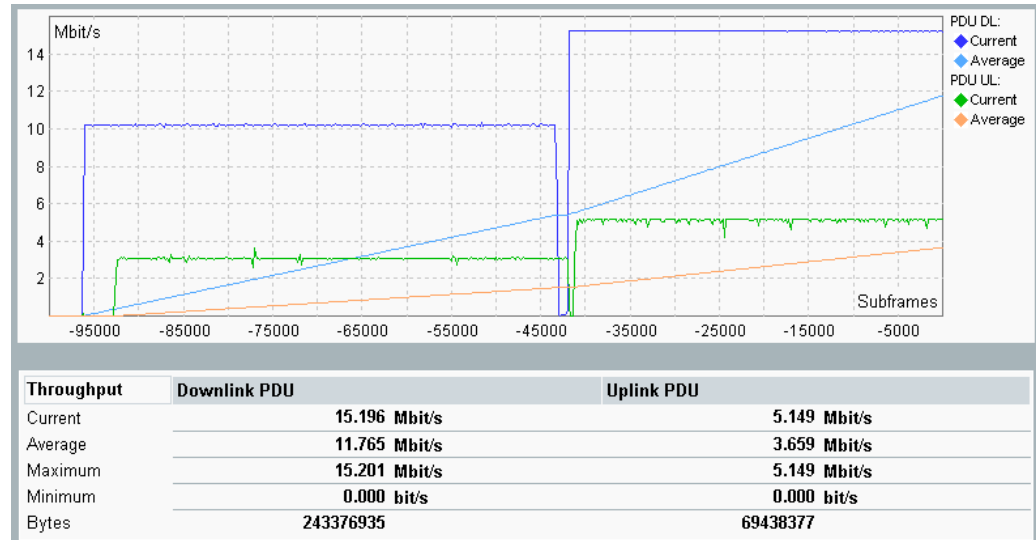


Figure 4-2: RLC throughput results

Diagram

The diagram provides a graphical presentation of the data throughput results for the previous measurement cycle. The X-axis indicates the processed subframes, with the last processed subframe labeled 0, the previously processed subframe labeled -1, and so on.

The diagram displays one result value per "Update Interval". The number of subframes per interval is configurable. The number of subframes per measurement cycle is also configurable ("Window Size", number of subframes on the X-axis).

You can enable/disable the display of the individual traces via the softkey - hotkey combination "Display > Select Trace".

Remote command:

`FETCh:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:CURRent?` etc.

`FETCh:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:CURRent?` etc.

Table

The throughput table indicates statistical throughput results for downlink and uplink.

The "Bytes" value indicates the number of bytes that the R&S CMW has transmitted (DL) or received (UL) since the measurement was started.

Remote command:

`FETCh:LTE:SIGN<i>:THRoughput?` etc.

Statistical Results

The statistical values are calculated as follows:

- **"Current"**: value obtained in the last update interval

- **"Average"**: average of all "Current" values within the last measurement cycle (window size)
- **"Minimum", "Maximum"**: largest or smallest "Current" value since the start of the measurement

4.2.4 Settings

Access: "RLC Throughput" tab > "Config" hotkey.

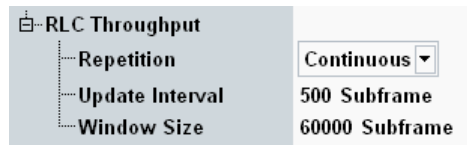


Figure 4-3: RLC throughput settings

Repetition.....	977
Update Interval.....	977
Window Size.....	977

Repetition

Defines how often the measurement is repeated.

- | | |
|---------------|---|
| "Continuous" | The measurement is continued until it is explicitly terminated. The results are periodically updated. |
| "Single-Shot" | The measurement stops automatically when the configured "Window Size" has been processed. |

Remote command:

CONFigure:LTE:SIGN<i>:THRoughput:REPetition

Update Interval

Number of subframes used to derive a single throughput result (multiple of 100 subframes).

Remote command:

CONFigure:LTE:SIGN<i>:THRoughput:UPDate

Window Size

Number of subframes on the X-axis of the throughput diagram.

The number of results in the diagram equals the window size divided by the update interval rounded down to the next integer value, plus one:

$$\text{Number of results} = \text{integer} (<\text{Window Size}> / <\text{Update Interval}>) + 1$$

The window size cannot be smaller than the update interval.

Remote command:

CONFigure:LTE:SIGN<i>:THRoughput:WINDow

4.3 Programming

The RLC throughput measurement is programmed as follows:

- The application is controlled by SCPI commands with the following syntax: `...:LTE:SIGN:THROUGHput:...`
- After a `*RST`, the measurement is switched off.
You can start the measurement using `INIT:LTE:SIGN:THROUGHput` and retrieve the results using `FETCh:LTE:SIGN:THROUGHput:...`.

The examples in this section focus on commands directly related to the RLC throughput measurement. For general configuration of the signaling application, refer to [Section 2.5, "Programming"](#), on page 322.

- [Configuring an RLC throughput measurement](#)..... 978
- [Setting up a data connection](#).....978
- [Performing an RLC throughput measurement](#)..... 979

4.3.1 Configuring an RLC throughput measurement

```
// *****
// System-Reset
// *****
*RST; *OPC?
*CLS; *OPC?

// *****
// Configure repetition mode, update interval and window size.
// *****
CONFIGure:LTE:SIGN:THROUGHput:REPetition SINGleshot
CONFIGure:LTE:SIGN:THROUGHput:UPDate 200
CONFIGure:LTE:SIGN:THROUGHput:WINDow 20000

// *****
// Select the data application mode
// *****
CONFIGure:LTE:SIGN:CONNECTION:CTYPE DAPplication
```

4.3.2 Setting up a data connection

Proceed as follows:

1. Configure the other settings of the signaling application compatible to your UE.
2. Configure the data application unit (see DAU documentation).
3. Switch on the cell signal and attach the UE.
4. Set up a data connection.

5. Generate IP traffic, for example using the iperf measurement provided by the DAU.

4.3.3 Performing an RLC throughput measurement

```
// *****
// Start the measurement and return the contents of the result table.
// Query the measurement state (should be "RDY").
// *****
INIT:LTE:SIGN:THROUGHput
FETCh:LTE:SIGN:THROUGHput?
FETCh:LTE:SIGN:THROUGHput:STATe?

// *****
// Query the result traces.
// *****
FETCh:LTE:SIGN:THROUGHput:TRACe:DL:PDU:CURRent?
FETCh:LTE:SIGN:THROUGHput:TRACe:DL:PDU:AVERage?
FETCh:LTE:SIGN:THROUGHput:TRACe:UL:PDU:CURRent?
FETCh:LTE:SIGN:THROUGHput:TRACe:UL:PDU:AVERage?
```

4.4 Command reference

The following sections describe the remote control commands of the "RLC Throughput" measurement.

- [Conventions and general information](#)..... 979
- [Measurement control and states](#)..... 985
- [Measurement settings](#)..... 987
- [Measurement results](#)..... 988

4.4.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

4.4.1.1 SIGN<i>

SIGN<i> is used as abbreviation of SIGNaling<instance>. For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same firmware application. It can be omitted if the instrument supports only one instance, or to address the first instance.

4.4.1.2 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**

If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:

- "Query only": The command can only be used to query parameters. No setting.
- "Event": The command initiates an event. No query.

- **Parameter usage**

The parameter usage is indicated by the keyword preceding the parameters:

- "Parameters" are sent with a setting or query command and are returned as the result of a query.
- "Setting parameters" are sent with a setting command.
- "Query parameters" are sent with a query command, to refine the query.
- "Return values" are returned as the result of a query.

- **Parameter type**

The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:

- "string": characters enclosed in quotation marks (" or ')
- "numeric", "decimal": decimal number
- "integer": integer number
- "float": floating point number
- "hex": hexadecimal number starting with #H
- "binary": binary number starting with #B
- "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).

- **Firmware/Software:**

Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

4.4.1.3 Reliability indicator

The first value in the sequences returned by measurement result queries indicates the most severe error that has occurred during the measurement.

Example for an output array: 0, 10.22, 10.15, 10.01, 10.29, 100 (reliability = 0, followed by 5 numeric measurement values).

The reliability indicator has one of the following values:

- **0 - "no error"**
No error detected.
- **1 - "measurement timeout"**

The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INValid` results or has not completed the full statistic count.

- **2 - "capture buffer overflow"**
The measurement configuration results in a capture length that exceeds the available memory.
- **3 - "input overdriven" / 4 - "input underdriven"**
The accuracy of measurement results can be impaired because the input signal level was too high / too low.
- **6 - "trigger timeout"**
The measurement could not be started or continued because no trigger event was detected.
For 1xEV-DO: A test application report was not received from the mobile within three update time periods.
- **7 - "acquisition error"**
The R&S CMW could not properly decode the RF input signal.
- **8 - "sync error"**
The R&S CMW could not synchronize to the RF input signal.
- **9 - "uncalibrated"**
Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the technical specifications.
- **15 - "reference frequency error"**
The instrument has been configured to use an external reference signal. But the reference oscillator could not be locked to the external signal (for example signal level too low, frequency out of range or reference signal not available at all).
- **16 - "RF not available"**
The measurement could not be started because the configured RF input path was not active.
This problem can occur if a measurement is started in combined signal path mode and the controlling application has not yet activated the input path.
- **17 - "RF level not settled" / 18 - "RF frequency not settled"**
The measurement could not be started because the R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 - "call not established"**
For measurements: The measurement could not be started because no signaling connection to the DUT was established.
For IMS service: Establishing a voice over IMS call failed.
- **20 - "call type not usable"**
For measurements: The measurement could not be started because the established signaling connection had the wrong properties.
For IMS service: The voice over IMS settings could not be applied.
- **21 - "call lost"**
For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.
For IMS service: The VoIMS call was lost.
- **23 - "missing option"**

An action cannot be executed due to a missing option.

For GPRF generator: The ARB file cannot be played due to a missing option.

- **24 - "invalid RF setting"**
The desired RF TX level or RF RX reference level could not be applied.
- **25 - "level overrange"**
The RF TX level is in overrange. The signal quality can be degraded.
- **26 - "resource conflict"**
The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application.
Stop the application that has allocated the conflicting resources and try again.
- **27 - "no sensor connected"**
The GPRF external power sensor measurement could not be started due to a missing power sensor.
- **28 - "unexpected parameter change"**
One or more measurement configuration parameters were changed while the measurement was running. The results were not obtained with these new parameter values. Repeat the measurement. This situation can only occur in remote single-shot mode.
- **29 - "invalid RF frequency setting"**
The desired RF TX frequency or RF RX frequency could not be applied.
- **30 - "file not found"**
The specified file could not be found.
- **31 - "no DTM reply"**
The EUT did not reply to the direct test mode (DTM) command.
- **32 - "ACL disconnected"**
The ACL connection has been disconnected or lost.
- **40 - "ARB file CRC error"**
The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.
- **42 - "ARB header tag invalid"**
The ARB file selected in the GPRF generator contains an invalid header tag.
- **43 - "ARB segment overflow"**
The number of segments in the multi-segment ARB file is higher than the allowed maximum.
- **44 - "ARB file not found"**
The selected ARB file could not be found.
- **45 - "ARB memory overflow"**
The ARB file length is greater than the available memory.
- **46 - "ARB sample rate out of range"**
The clock rate of the ARB file is either too high or too low.
- **47 - "ARB cycles out of range"**
The repetition mode equals "Single Shot" and the playback length is too long.
Reduce the playback length or set the repetition mode to "Continuous".
$$\langle \text{Length} \rangle = (\langle \text{Cycles} \rangle * \langle \text{Samples} \rangle + \langle \text{Additional Samples} \rangle) / \langle \text{Clock Rate} \rangle$$
- **48 - "ARB pool invalid"**
At least one of the ARB files in the ARB file pool is invalid.

- **49 - "sequencer list invalid"**
The sequencer list configuration has at least one invalid entry.
- **50 - "startup error"**
The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 - "no reply"**
The DAU has received no response, for example for a ping request.
- **52 - "connection error"**
A connection setup failed or a connection was lost.
For DAU: The DAU could not establish a connection to internal components. Restart the instrument.
For Bluetooth: The EUT has been disconnected from the R&S CMW.
- **53 - "configuration error"**
The current DAU configuration is incomplete or wrong and could not be applied. Check especially the IP address configuration.
- **54 - "file system error"**
The system drive of the DAU is full or corrupt. Execute a DAU self-test.
- **64 - "missing pathloss correction data"**
The pathloss measurement cannot be executed due to missing connector correction data. Order a calibration of the instrument by Rohde & Schwarz.
- **70 - "wrong standard"**
The standard of the measured signal does not match the configured standard.
- **71 - "wrong bandwidth"**
The bandwidth of the measured signal does not match the configured bandwidth.
- **72 - "wrong burst type"**
The burst type of the measured signal does not match the configured burst type.
- **73 - "MIMO signal detected"**
The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.
- **74 - "more streams than antennas"**
The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.
- **75 - "matrix inversion failed"**
The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.
- **76 - "SIG CRC failed"**
The cyclic redundancy check of a SIGNAL field failed.
- **77 - "parity check failed"**
The parity check of a SIGNAL field failed.
- **78 - "bursts not identical"**
In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.
- **79 - "wrong modulation"**

The modulation scheme of the measured signal does not match the configured modulation scheme.

- **80 - "SISO signal detected"**
Received signal contains only one stream.
- **81 - "fewer streams than antennas"**
Received signal has fewer streams than configured antennas. Reconfigure the number of antennas.
- **82 - "signal fields of different antennas are not identical"**
The signal fields of all captured antennas are compared. In a valid MIMO transmission, all signal fields are identical. If not, some basic parameters such as MCS or data length are not the same and a MIMO measurement is not possible.
- **83 - "bit errors in some of the received symbols"**
Adjust input signal power and repeat training.
- **84 - "first and last symbols need to be skipped"**
Due to a sequence counter in the MAC header, the first symbol and the last symbol (due to the FCS) change from one burst to another. You can configure the first and the last symbol to be skipped.
- **85 - "received and saved bursts are not identical"**
The signal field of the received burst is not identical to that of the saved burst. Some basic parameters such as MCS or data length are not the same and a composite MIMO measurement is not possible.
- **86 - "no symbols for measurement"**
Too many head and tail symbols have been skipped.
- **93 - "OCXO oven temperature too low"**
The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **96 - "frequency/level change too fast"**
There was a too fast frequency and/or level change. If you use GPRF with list mode, increase the duration of the list entry.
- **101 - "firmware error"**
This value indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.
If the error occurs again, consider the following hints:
 - Firmware errors can often be repaired by restoring the factory default settings.
 - If a software package has not been properly installed, this failure is often indicated in the "Setup" dialog > section "SW/HW-Equipment > Installed Software".
 - Check for software updates correcting the error. Updates are provided in the customer web on GLORIS (registration required): <https://gloris.rohde-schwarz.com>.

If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.
- **102 - "unidentified error"**
This value indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 firmware error".
- **103 - "parameter error"**

This value indicates that the measurement could not be performed due to internal conflicting parameter settings.

For 1xEV-DO:

- The 1xEV-DO signaling test application is not enabled.
- Forward link measurement data were requested, but the test application direction is not set to "Forward" or "Forward & Reverse".
- Reverse link measurement data were requested, but the test application direction is not set to "reverse" or "forward & reverse".
- Measurement data were requested, but the corresponding measurement is not assigned.

For CDMA2000: The "1st Service Option" is not set to a loopback service option.

A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.

If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see <https://www.service.rohde-schwarz.com>).

- **104 - "not functional"**

The application could not be started with the configured parameter set.

4.4.2 Measurement control and states

The following commands control the measurement and return the current measurement state.

<code>INITiate:LTE:SIGN<i>:THRoughput</code>	985
<code>STOP:LTE:SIGN<i>:THRoughput</code>	985
<code>ABORT:LTE:SIGN<i>:THRoughput</code>	985
<code>FETCh:LTE:SIGN<i>:THRoughput:STATe?</code>	986
<code>FETCh:LTE:SIGN<i>:THRoughput:STATe:ALL?</code>	986

INITiate:LTE:SIGN<i>:THRoughput

STOP:LTE:SIGN<i>:THRoughput

ABORT:LTE:SIGN<i>:THRoughput

Starts, stops or aborts the measurement:

- The `INITiate:...` command starts or restarts the measurement. The measurement enters the "RUN" state.
- The `STOP:...` command halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- The `ABORT:...` command halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

To query the current measurement state, use `FETCh...STATe?`.

Example: See [Performing an RLC throughput measurement](#)

Usage: Event

Firmware/software: V3.2.80

Manual operation: See ["RLC Throughput softkey"](#) on page 975

FETCh:LTE:SIGN<i>:THRoughput:STATe?

Queries the main measurement state. Use FETCh: . . . :STATe:ALL? to query the measurement state including the substates. The substates tell you, for example, whether a state transition is ongoing.

Return values:

<State> OFF | RUN | RDY
 Current state or target state of ongoing state transition
OFF: measurement off
RUN: measurement running
RDY: measurement completed
 *RST: OFF

Example: See [Performing an RLC throughput measurement](#)

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["RLC Throughput softkey"](#) on page 975

FETCh:LTE:SIGN<i>:THRoughput:STATe:ALL?

Queries the main measurement state and the measurement substates. A really running measurement returns RUN, ADJ, ACT.

Return values:

<MainState> OFF | RUN | RDY
 Current state or target state of ongoing state transition
OFF: measurement off
RUN: measurement running
RDY: measurement completed
 *RST: OFF

<SyncState> PEND | ADJ
PEND: transition to <MainState> ongoing
ADJ: <MainState> reached

<ResourceState> QUE | ACT | INV
QUE: waiting for resource allocation
ACT: resources allocated
INV: no resources allocated

Usage: Query only

Firmware/software: V3.2.80

Manual operation: See ["RLC Throughput softkey"](#) on page 975

4.4.3 Measurement settings

The following commands configure the "RLC Throughput" measurement.

CONFigure:LTE:SIGN<i>:THRoughput:TOUT	987
CONFigure:LTE:SIGN<i>:THRoughput:REPetition	987
CONFigure:LTE:SIGN<i>:THRoughput:UPDate	988
CONFigure:LTE:SIGN<i>:THRoughput:WINDow	988

CONFigure:LTE:SIGN<i>:THRoughput:TOUT <Timeout>

Defines a timeout for the measurement. The timer is started when the measurement is initiated via a `READ` or `INIT` command. It is not started if the measurement is initiated manually.

When the measurement has completed the first measurement cycle (first single shot), the statistical depth is reached and the timer is reset.

If the first measurement cycle has not been completed when the timer expires, the measurement is stopped. The measurement state changes to `RDY`. The reliability indicator is set to 1, indicating that a measurement timeout occurred. Still running `READ`, `FETCh` or `CALCulate` commands are completed, returning the available results. At least for some results, there are no values at all or the statistical depth has not been reached.

A timeout of 0 s corresponds to an infinite measurement timeout.

Parameters:

<Timeout> numeric
 Default unit: s

Firmware/software: V3.2.80

CONFigure:LTE:SIGN<i>:THRoughput:REPetition <Repetition>

Specifies whether the measurement is stopped after a single shot (window size) or repeated continuously.

Parameters:

<Repetition> SINGleshot | CONTInuous
 SINGleshot: Single-shot measurement
 CONTInuous: Continuous measurement
 *RST: SING

Example: See [Configuring an RLC throughput measurement](#)

Firmware/software: V3.2.80

Manual operation: See ["Repetition"](#) on page 977

CONFigure:LTE:SIGN<i>:THRoughput:UPDate <Interval>

Configures the number of subframes used to derive a single throughput result.

Parameters:

<Interval>	numeric
Range:	200 to 10000
Increment:	100
*RST:	500

Example: See [Configuring an RLC throughput measurement](#)

Firmware/software: V3.2.80

Manual operation: See ["Update Interval"](#) on page 977

CONFigure:LTE:SIGN<i>:THRoughput:WINDow <Size>

Configures the number of subframes on the X-axis of the throughput diagram (duration of a single-shot measurement).

The size cannot be smaller than the update interval.

Parameters:

<Size>	numeric
Range:	200 to 120000
*RST:	60000

Example: See [Configuring an RLC throughput measurement](#)

Firmware/software: V3.2.80

Manual operation: See ["Window Size"](#) on page 977

4.4.4 Measurement results

The following commands retrieve the results of the "RLC Throughput" measurement.

FETCh:LTE:SIGN<i>:THRoughput?	988
READ:LTE:SIGN<i>:THRoughput?	988
FETCh:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:CURRent?	989
FETCh:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:AVERAge?	989
READ:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:CURRent?	990
READ:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:AVERAge?	990
FETCh:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:CURRent?	990
FETCh:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:AVERAge?	990
READ:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:CURRent?	990
READ:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:AVERAge?	990

FETCh:LTE:SIGN<i>:THRoughput?**READ:LTE:SIGN<i>:THRoughput?**

Returns the contents of the RLC throughput result table.

Return values:

<Reliability>	decimal See Reliability indicator
<CurrDIPDU>	float Current downlink throughput Default unit: bit/s
<AvgDIPDU>	float Average downlink throughput Default unit: bit/s
<MaxDIPDU>	float Maximum downlink throughput Default unit: bit/s
<MinDIPDU>	float Minimum downlink throughput Default unit: bit/s
<BytesDIPDU>	decimal Number of bytes transmitted in the downlink
<CurrUIPDU>	float Current uplink throughput Default unit: bit/s
<AvgUIPDU>	float Average uplink throughput Default unit: bit/s
<MaxUIPDU>	float Maximum uplink throughput Default unit: bit/s
<MinUIPDU>	float Minimum uplink throughput Default unit: bit/s
<BytesUIPDU>	float Number of bytes received in the uplink

Example: See [Performing an RLC throughput measurement](#)

Usage: Query only

Firmware/software: V3.2.80

FETCh:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:CURRent?
FETCh:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:AVERAge?

READ:LTE:SIGN<i>:THROUGHput:TRACe:UL:PDU:CURRent?

READ:LTE:SIGN<i>:THROUGHput:TRACe:UL:PDU:AVERAge?

Returns the values of the uplink throughput traces. The results of the current and average traces can be retrieved.

The number of trace values n depends on the configured update interval and window size: $n = \text{integer} (<\text{window size}> / <\text{update interval}>) + 1$

Return values:

<Reliability> decimal
See [Reliability indicator](#)

<UplinkPDU> float
Comma-separated list of n throughput values
Default unit: bit/s

Example: See [Performing an RLC throughput measurement](#)

Usage: Query only

Firmware/software: V3.2.80

FETCh:LTE:SIGN<i>:THROUGHput:TRACe:DL:PDU:CURRent?

FETCh:LTE:SIGN<i>:THROUGHput:TRACe:DL:PDU:AVERAge?

READ:LTE:SIGN<i>:THROUGHput:TRACe:DL:PDU:CURRent?

READ:LTE:SIGN<i>:THROUGHput:TRACe:DL:PDU:AVERAge?

Returns the values of the downlink throughput traces. The results of the current and average traces can be retrieved.

The number of trace values n depends on the configured update interval and window size: $n = \text{integer} (<\text{window size}> / <\text{update interval}>) + 1$

Return values:

<Reliability> decimal
See [Reliability indicator](#)

<DownlinkPDU> float
Comma-separated list of n throughput values
Default unit: bit/s

Example: See [Performing an RLC throughput measurement](#)

Usage: Query only

Firmware/software: V3.2.80

4.5 List of commands

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FETCh:LTE:SIGN<i>:THRoughput:STATe?.....	986
FETCh:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:AVERage?.....	990
FETCh:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:CURRent?.....	990
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READ:LTE:SIGN<i>:THRoughput:TRACe:DL:PDU:CURRent?.....	990
READ:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:AVERage?.....	990
READ:LTE:SIGN<i>:THRoughput:TRACe:UL:PDU:CURRent?.....	990
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5 LTE multi-evaluation measurement

The LTE multi-evaluation measurement provides quick and flexible TX tests on LTE uplink and sidelink signals.

A downlink RX test (block error ratio test) is also included.

The TX tests cover the following UE transmitter properties:

- UE output power per resource block (in-band emissions)
- UE output power per subframe (power monitor)
- ON/OFF power (power dynamics)
- UE power variation across the allocated subcarriers (equalizer spectrum flatness)
- Modulation accuracy per SC-FDMA symbol (EVM, magnitude error, phase error)
- Modulation accuracy per subcarrier (EVM)
- I/Q constellation diagram
- Overview of allocated resource blocks (RB allocation table)
- Out-of-band emissions (spectrum emission mask and ACLR)

Many of the tests and conformance requirements are specified in 3GPP TS 36.101 or 3GPP TS 36.521.

The multi-evaluation measurement requires option R&S CMW-KM500 for R8 FDD uplink signals, R&S CMW-KM550 for R8 TDD uplink signals and R&S CMW-KM570 for R14 sidelink signals.

5.1 What's new

This documentation describes version 4.0.250 and higher of the LTE multi-evaluation measurement. Compared to version 4.0.230, it provides the following new features:

- Editorial changes only.

5.2 General description

The LTE multi-evaluation measurement captures an LTE signal and provides the TX measurement results for a configurable subframe / range of subframes.

The basic functionality provided by R&S CMW-KM500/550 can be enhanced via the following options:

- R&S CMW-KM502 for R10 FDD uplink signals with carrier aggregation.
- R&S CMW-KM552 for R10 TDD uplink signals with carrier aggregation.
- R&S CMW-KM504 for R12 FDD uplink signals with 3 or 4 carriers.
R&S CMW-KM502 is a prerequisite.
- R&S CMW-KM554 for R12 TDD uplink signals with 3 or 4 carriers.
R&S CMW-KM552 is a prerequisite.

For downlink block error ratio (BLER) measurements (RX tests), the GPRF generator and an LTE waveform file are used to generate a downlink signal.

The following sections describe how to perform and configure the measurement.

• Test setups	993
• V2X sidelink measurements	993
• How to measure a sidelink LTE signal	994
• How to measure an uplink LTE signal	994
• How to perform an RX test	995
• Defining the scope of a TX measurement	996
• Uplink carrier aggregation	998
• Parallel signaling and measurement	999
• Calculation of modulation results	1001
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• Trigger signals	1015

5.2.1 Test setups

Test setup for TX tests

Connect the external RF signal source (mobile station, signal generator etc.) to one of the bidirectional RF connectors on the front panel of the R&S CMW.

Test setup for RX tests

The downlink RF generator signal of the R&S CMW is fed to the input of the DUT. The R&S CMW measures the uplink signal.

5.2.2 V2X sidelink measurements

The option R&S CMW-KM570 enables the LTE multi-evaluation measurement to measure sidelink signals.

Sidelink signals are part of the vehicle to everything (V2X) communication, defined by 3GPP. The signals use the PC5 interface between a vehicle and another vehicle or UE.

The LTE measurement can measure the physical sidelink control channel (PSCCH) and the physical sidelink shared channel (PSSCH).

Many uplink measurement settings are not relevant for sidelink measurements. A common setting of the LTE multi-evaluation measurement selects between uplink and sidelink. If you select the signal type "Sidelink", most not relevant settings are either hidden or the only allowed value is displayed and grayed out.

Compared to uplink signals, the most important sidelink signal characteristics are:

- Direct communication between two units - no cell, no network, no uplink or downlink.

- The duplex mode is always TDD. There are no special subframes and no UL-DL configuration.
- Only band 47 is supported.
- The channel types are PSCCH and PSSCH instead of PUCCH and PUSCH. For RB allocations, see [Section 5.2.11.3, "Sidelink resource block allocation"](#), on page 1012.
- No eMTC, no frequency hopping, no SRS signals.

Additional differences to uplink measurements:

- There is no signaling application for sidelink.
- The current software version does not support carrier aggregation (CA).
- No RX measurements (BLER), only TX measurements.
- No view filter, no EVM exclusion periods, no high dynamic mode.

5.2.3 How to measure a sidelink LTE signal

After connecting your DUT, you have to adjust at least the following analyzer settings to the properties of the analyzed sidelink LTE signal:

- Signal type = sidelink
- Analyzer frequency
- Expected nominal power, external attenuation and user margin
Recommended values: expected nominal power = peak power of the signal during the measurement. User margin = 0 dB. The smallest possible value of the expected nominal power plus the user margin ensures a maximum dynamic range.

For synchronization to the received signal and proper decoding, some settings must be in accordance with the measured signal. In particular, ensure that the following parameters match up:

- Channel bandwidth
- Measured subframe

With matching "Measurement Control" settings, the measurement is able to decode the signal and determine its frame timing.

Non-matching "Measurement Control" settings generally result in large EVM results.

5.2.4 How to measure an uplink LTE signal

After connecting your LTE UE, you have to adjust at least the following analyzer settings to the properties of the analyzed UL LTE signal:

- Duplex mode
- Analyzer frequency
- Expected nominal power, external attenuation and user margin
Recommended values: expected nominal power = peak power of the UE signal during the measurement. User margin = 0 dB. The smallest possible value of the expected nominal power plus the user margin ensures a maximum dynamic range.

For synchronization to the received signal and proper decoding, some settings must be in accordance with the measured signal. In particular, ensure that the following parameters match up:

- For TDD signals: UL-DL configuration, special subframe configuration and measured subframe
Ensure that you select an UL subframe for the measured subframe. When a frame trigger signal is used, the subframe 0 selected by default is a DL subframe.
- For eMTC signals: enable eMTC, max. eMTC bandwidth, LO location
- Length of the cyclic prefix
- Channel bandwidth
- For signals with PUCCH: PUCCH format
- Physical cell ID
- Delta sequence shift PUSCH
- Group hopping

The measurement can auto-detect the modulation scheme of the received signal.

With matching "Measurement Control" settings, the measurement is able to decode the signal and to determine its frame timing. Non-matching "Measurement Control" settings generally result in large EVM results.

5.2.5 How to perform an RX test

The multi-evaluation measurement supports RX tests in parallel to uplink TX tests. You need the GPRF generator and an LTE waveform file. Option R&S CMW-KW500 is required to process LTE waveform files.

In signaling mode (combined signal path), use the BLER measurement provided by the LTE signaling application and ignore this section.

Measurement principle

The RX test uses the GPRF generator to send data to the receiver of the UE under test, transmitting a configurable number of PDSCH subframes. The UE is requested to acknowledge the correct reception of each subframe. It must return the positive acknowledgment (ACK) or negative acknowledgment (NACK) messages over the PUCCH using format F1a.

The measurement calculates the BLER from the received ACKs, NACKs and missing answers (DTX). As the transmit time interval for LTE equals one subframe, a subframe corresponds to one transport block.

To synchronize the measurement to the generator signal, an appropriate marker in the ARB file is recommended as a trigger source.

Measurement procedure

To perform an LTE BLER test, proceed as follows:

1. Establish the basic test setup for RX tests, connecting the UE under test to an RF input connector.
2. Generator:
 - a) Access the GPRF generator application.
 - b) Load an LTE ARB file.
 - c) Turn on the RF generator.
3. Access the LTE multi-evaluation measurement application.
4. Enable the BLER measurement ("Multi Evaluation > Assign Views > BLER: ON").
5. Select "Measurement Control" > "PUCCH Format: F1a".
This setting also ensures that a TX measurement which runs in parallel to the BLER test is configured in accordance with the expected UL signal.
6. Select "Measurement Control" > "Measurement Subframe" > "No. of Subframes" = 320.
This action configures the maximum value resulting in the maximum measurement speed. And it is a multiple of 10, ensuring that the number of scheduled subframes per radio frame can be evaluated correctly.
7. Click "Measurement Control" > "BLER" > "No. of Subframes".
Configure the number of subframes that you want to evaluate in total.
8. Click "Measurement Control" > "BLER" > "Scheduled Subframes/Frame".
Specify the number of scheduled subframes per radio frame according to the generated downlink signal.
9. Click "Trigger" > "Trigger Source".
Select one of the marker signals in the ARB file to trigger the BLER measurement.
10. Select the appropriate view ("Display > Select View: BLER"). And start the measurement.

5.2.6 Defining the scope of a TX measurement

The LTE multi-evaluation measurement can capture up to 320 consecutive subframes, starting at a configurable offset from the trigger event.

The results "Power Monitor" and "RB Allocation Table" are available for the entire range of captured subframes. All other TX measurement results are derived from a selected subframe within the range of captured subframes (setting "Measure Subframe"). You can select which slot of the selected subframe is evaluated: the first one only, the second one only or both (setting "Measure Slot").

For TDD signals, you must ensure that the selected subframe contains a signal. Which subframe is labeled 0, depends on the type of trigger signal and the subframe offset.

For subframe offset 0, the subframe labeled 0 is suitable for power trigger signals (first detected subframe with signal).

For TDD frame trigger signals in a combined signal path scenario, the subframe labeled 0 is a downlink subframe (the first subframe of the frame).

For signals with carrier aggregation, some results are available per component carrier. Other results are measured on only one component carrier, the selected "Measurement Carrier". For details, see [Section 5.2.7, "Uplink carrier aggregation"](#), on page 998.

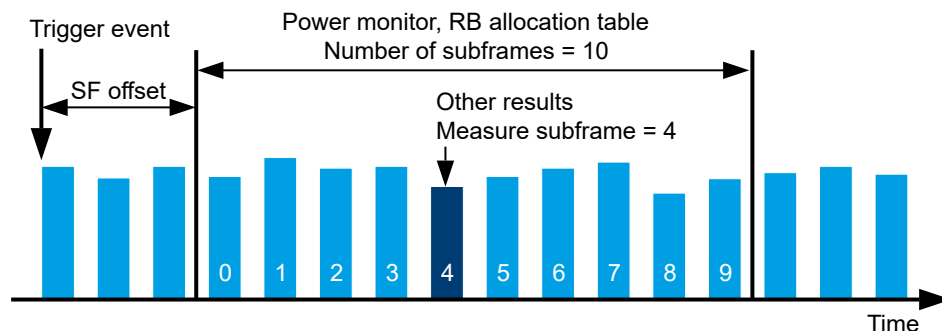
Depending on the measurement, the statistic count is defined in subframes or in slots.

Example

Assume the following settings (see [Section 5.3.5, "Measurement control settings"](#), on page 1050):

- "Subframe Offset" = 3
- "Number of Subframes" = 10
- "Measure Subframe" = 4
- "Measure Slot" = 0 / 1 / all
- All statistic counts = 5

The following figure shows the resulting measured subframes.



The measurement scope is explained in the following:

- The power monitor and the RB allocation table show 10 subframes ("Number of Subframes"), labeled 0 to 9. The subframe labeled 0 is the fourth subframe (first + "Subframe Offset" 3) captured after the trigger event. For the power monitor, the 10 subframes are captured five times (statistic count) to complete one measurement cycle.
- For the power dynamics measurement, the subframe labeled 4 ("Measure Subframe") is captured five times (statistic count) to complete one measurement cycle.
- For all other results, five slots (statistic count) have to be measured per cycle. Only slots within the subframe labeled 4 ("Measure Subframe") contribute to the results. Each subframe contains two slots. To measure five slots with "Measure Slot" = all, the entire subframe range has to be captured three times. The first and second time, both slots of subframe 4 are measured. The third time, only the first slot of

subframe 4 is measured. Now the measurement cycle is complete and a single-shot measurement stops.

With "Measure Slot" = 0, only the first slot is evaluated. With "Measure Slot" = 1, only the second slot is evaluated. In both cases, the entire subframe range has to be captured five times to measure five slots.

5.2.7 Uplink carrier aggregation

Most tests specified for signals with uplink carrier aggregation (CA) measure one carrier at a time. The tests per carrier are the same as for signals without carrier aggregation.

Only for signals with intraband contiguous uplink CA, some tests measure all aggregated component carriers at the same time.

CA bandwidth classes

The BW classes are defined in 3GPP TS 36.101, section 5.6A.

Table 5-1: BW classes supported by the measurement

Class	Carriers	Allocated RB	CA mode	CA options (R&S CMW-...)
A	1	0 to 100	"Off"	none
B	2	26 to 100	"Intraband Cont. (BW Class B & C)"	FDD: KM502
C	2	101 to 200		TDD: KM552
D	3	201 to 300	"Intraband Cont. (BW Class D)"	FDD: KM504 TDD: KM554
E	4	301 to 400	"Intraband Cont. (BW Class E)"	FDD: KM504 TDD: KM554

Measurement results for intraband contiguous uplink CA

The following results are measured for one component carrier at a time. For synchronization, this carrier must contain allocated RBs.

- EVM, magnitude error, phase error
- I/Q diagram
- Modulation result tables
- Equalizer spectrum flatness
- Power dynamics

The spectrum results are measured for the combination of all carriers, the aggregated bandwidth.

- Spectrum ACLR
- Spectrum emission mask

The following results are measured for all carriers in parallel. There are separate results per component carrier.

- In-band emission diagram

- Power monitor
- RB allocation table

How to measure an LTE signal with intraband contiguous uplink CA

The basic measurement procedure is the same as for signals without carrier aggregation. Even the test setup is the same.

In addition, consider the following points:

- For a standalone measurement:
 - Select the "Carrier Aggregation Mode" to enable carrier aggregation.
 - Configure the carrier center frequencies according to the 3GPP rules for intraband contiguous CA. An automatism facilitates the configuration, so that you need to configure only the frequency of one carrier and can adjust the other carriers via a button.
- For a combined signal path measurement:
 - In the signaling application, configure an intraband contiguous uplink CA set. Refer to the signaling application documentation for details.
 - In the measurement, select the carrier set via "Controlled by".



The measurement automatically sets the "Carrier Aggregation Mode" accordingly.

- Note the carrier displayed at the right end of the line "Controlled by". As a trigger source, use the frame trigger signal of that carrier, provided by the signaling application.
- Select a "Measurement Carrier". This carrier is used for synchronization, so it must have allocated RBs. The selected carrier is evaluated by single carrier measurements, for example for measurement of modulation or power dynamics results.
- Select the local oscillator location of the UE transmitter.

See also:

- [Section 5.3.4, "Carrier aggregation settings"](#), on page 1046
- ["LO Location for CA"](#) on page 1064

5.2.8 Parallel signaling and measurement

You can use the multi-evaluation measurement in parallel to the LTE signaling application (option R&S CMW-KS500/-KS550). Set up a connection to the UE with the signaling application. Then measure the resulting uplink signal with the multi-evaluation measurement.

To use both applications in parallel, the combined signal path scenario must be activated (see ["Scenario = Combined Signal Path"](#) on page 1042). Most signal routing and analyzer settings and some measurement control settings are then configured by the signaling application. The multi-evaluation measurement displays the corresponding

signaling settings instead of its own settings. These signaling settings can be configured both in the measurement GUI and in the GUI of the signaling application.

To configure these settings via remote commands, use the commands of the signaling application. For a command-mapping table, see [Section 5.5.4, "Combined signal path commands"](#), on page 1287.

The most important signaling parameters not relevant for standalone measurements can nevertheless be configured both in the measurement GUI and in the GUI of the signaling application. In the measurement GUI, they can be accessed via hotkeys.

Whenever the combined signal path scenario is activated or the controlling application is changed, the trigger source for the measurement is set automatically. The frame trigger signal provided by the controlling signaling application is used.

Carrier names

In a standalone measurement, the measurement does not know which carrier it measures - whether it is a PCC or a certain SCC. You can measure any uplink carrier.

Even with contiguous uplink carrier aggregation, the measurement does not care which carriers you aggregate. The carriers are numbered in the GUI as CC1, CC2, and so on.

In a combined signal path measurement, the signaling application informs the measurement about the carrier configuration. The measurement displays the mapping between CC<n> and PCC/SCC<m>.

Measurement instances in multi-CMW setups

Each measurement instance is running on a specific measurement board and thus in a specific CMW. The measurement instance can only measure carriers that are available in the same CMW.

In standalone measurements, you can only select RF input connectors that are compatible to the measurement instance.

But in combined signal path measurements, you can select carriers that use RF input connectors that are incompatible to the measurement instance. In that case, the measurement fails, showing the reliability indicator value 104 ("not functional").

To identify the measurement instance compatible to a certain carrier, you can proceed as follows.

1. In the standalone scenario, check for each measurement instance which RF connectors are available in the GUI.
In each measurement instance, you see the connectors of one CMW. Write a mapping list between measurement instances and CMW numbers.
The mapping remains valid as long as you do not change your multi-CMW configuration (that means the cabling between CMWs and R&S CMWC and the hardware equipped in each CMW).
2. Check at which RF input connector the carrier is available.

You can check it in the signaling application. Or you can check it in the measurement, in the combined signal path scenario, after selecting the carrier via the setting "Controlled by".

5.2.9 Calculation of modulation results

Modulation results are based on a comparison of an ideal reference waveform with the measured signal. The measured signal is corrected by the average frequency and timing offset for each measured subframe.

3GPP TS 36.101, annex F, specifies the following additional conditions:

- Before calculating the EVM and the in-band emissions, the I/Q origin offset is removed from the evaluated signal.
- The EVM calculation is based on two different FFT processing windows in the time domain, separated by the "EVM window length". The minimum test requirement applies to the larger of the two obtained EVM values.

The two FFT processing windows start in the cyclic prefix (CP) at the beginning of the SC-FDMA symbol. The cyclic prefix is a copy of the end of the SC-FDMA symbol.

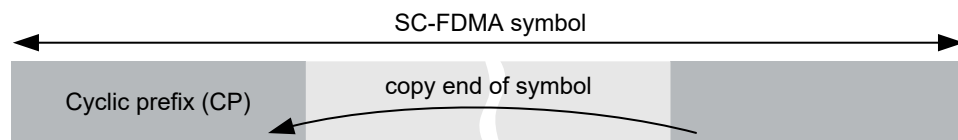


Figure 5-1: Cyclic extension of SC-FDMA symbol

The EVM window between the start of the two FFT processing windows is centered on the cyclic prefix. The EVM window length is specified in the standard, depending on the channel bandwidth and the CP type (see 3GPP TS 36.101, tables F.5.3-1 and F.5.4-1).

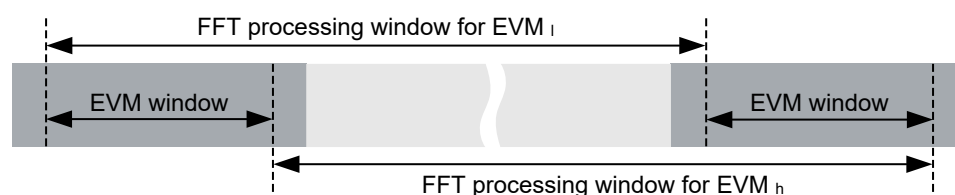


Figure 5-2: EVM window definition

The EVM for a signal with good modulation accuracy is expected to be largely independent of the EVM window size. Differences between EVM_h and EVM_l are caused, for example, by the effects of time domain windowing of FIR pulse shaping.



EVM window length settings

Use the [EVM Window Length](#) settings to adjust the length of the EVM window. The minimum value of one FFT symbol means that EVM_h equals EVM_l.

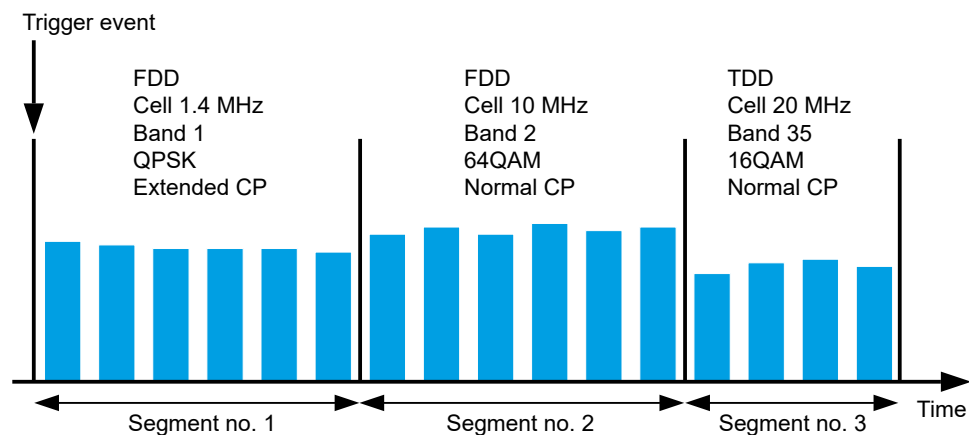
5.2.10 List mode

The LTE multi-evaluation list mode requires option R&S CMW-KM012. In list mode, the TX measurement interval is subdivided into segments, according to basic signal properties like expected power, frequency, channel bandwidth and type of cyclic prefix.

The list mode supports uplink and sidelink measurements. Most examples in the following sections are for uplink measurements.

5.2.10.1 List mode configuration

Each segment contains an integer number of subframes and is measured at constant analyzer settings. The figure below shows a series of three segments with different lengths, expected powers and the listed signal properties. The bars depict subframes.



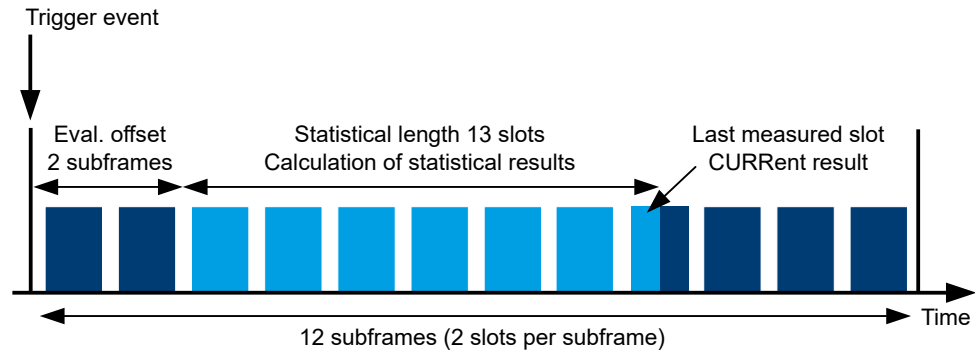
You can configure up to 2000 segments in total as preparation for list mode measurements. One list mode measurement can cover up to 1000 segments with in total up to 4000 measured subframes. For details, see [Maximum number of measured subframes](#). So for each measurement, you select a range of up to 1000 segments to be measured from the total range of up to 2000 configured segments.

In list mode, the multi-evaluation measurement can measure modulation results (including in-band emission and equalizer spectrum flatness results), ACLR, spectrum emission and power monitor results. The measured quantities can be enabled or disabled individually for each segment.

A segment without any enabled measurements is called an inactive segment. Inactive segments are useful for time-consuming UE reconfiguration. For that purpose, you define alternating active and inactive segments. In the active segments, you measure the signal. In the inactive segments, you reconfigure the UE for the next active segment.

It is possible to measure all subframes of a segment or to exclude subframes or slots at the beginning and/or the end of the segment. The evaluation offset specifies how many subframes are excluded at the beginning of the segment. The statistical length defines the number of slots to be measured. The "current" result of a segment refers to the last measured slot of the statistical length. Additional statistical values (average,

minimum, maximum and standard deviation) are calculated for the entire statistical length. Statistical length and statistical values are not relevant for power monitor measurements. The following figure shows a summary.



Two consecutive segments are often measured at different analyzer settings. The measurement changes the analyzer settings in the last subframes of the first segment. This change impairs the accuracy of the measurement results for up to three subframes at the end of the segment. The affected slots are automatically excluded from the measurement.

Results of other slots that cannot be measured accurately can also be discarded automatically (for example, overflow, low signal or synchronization error). For automatic discarding, set parameter [Measure on Exception](#) = off.

The reached statistical length, a reliability indicator for the measurement and a reliability indicator for the segment are included in most measurement results. TDD downlink subframes and special subframes are automatically ignored and do not contribute to the (reached) statistical length.

Maximum number of measured subframes

The limit of 4000 measured subframes comprises measured subframes included in a statistical length and subframes measured with the power monitor. The not measured subframes of an active segment do not contribute. Other contributions that "cost" subframes are inactive segments and triggering of a segment.

All segments selected for the measurement are considered. In the worst case, the sum of the following components must not exceed 4000:

- One subframe per inactive segment
- Two times the number of triggered segments
- For each active segment without power monitor:
Subframes in the statistic count of the segment plus three subframes
- For each active segment with power monitor:
All subframes in the segment

Trigger modes

A list mode measurement can either be triggered only once, or it can be retriggered at the beginning of specified segments.

In "Once" mode, a trigger event is only required to start the measurement. As a result, the entire range of segments (up to 1000) is measured without additional trigger event. The trigger is rearmed after the measurement has been finished. The global trigger source is applied and the retrigger flags of the segments are ignored.

We recommend the "Once" mode for UL signals with accurate timing over the entire range of segments.

In "Segment" mode, the retrigger flag of each segment is evaluated. It defines whether the measurement waits for a trigger event before measuring the segment and which trigger source is used. We recommend retriggering the measurement if the timing of the first subframe of a segment is inaccurate, for example because of signal reconfiguration at the UE. Furthermore, retriggering from time to time can compensate for a possible time drift of the UE.

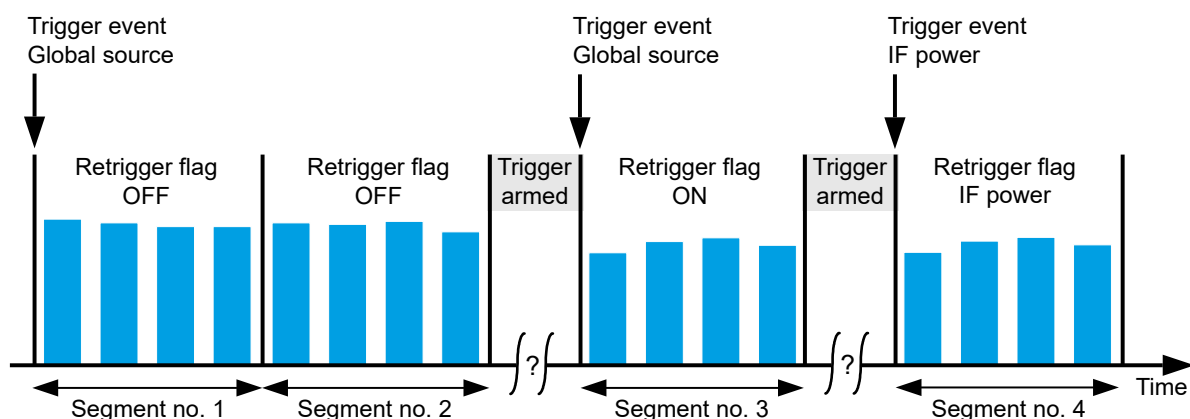
The retrigger flag allows you to select from three trigger sources:

- Global trigger source configured for measurements without list mode.
- Wideband IF power trigger, with a bandwidth of 160 MHz.
- Narrowband IF power trigger, with a configurable bandwidth.
Useful for over-the-air measurements, to prevent an additional signal (for example, a BT signal) from triggering the measurement.

For segments using this trigger source, consider the following:

- The segments must have a length of at least two subframes.
- Active segments must have an evaluation offset of at least two subframes.

In the following example, the "Segment" mode is enabled. The measurement stops when the second segment has been captured and waits for a trigger event from the globally configured trigger source, before capturing the third segment. After the third segment, it waits for a trigger event from the wideband IF power trigger source, before capturing the fourth segment.



Spectrum measurements

For carrier aggregation class D/E, segments with active spectrum measurements are divided into three parts.

The first part of a divided segment is measured using the nominal carrier frequency and is evaluated for all enabled measurements. The other parts are only evaluated for spectrum measurements.

The maximum statistical length reachable in a divided segment is smaller than the statistical length reachable with disabled spectrum measurements. If the segment is, for example, divided into three parts, the maximum statistical length is reduced to 1/3.

Assign more subframes to reach a higher statistical length. The minimum length of a divided segment is three subframes per part. If you configure a shorter segment length, the effect is the same as if you disable the spectrum measurements for the segment.

Configuration of segments and measurement

Segment configuration and measurement are independent from each other. To perform a sequence of measurements at maximum speed, proceed as follows:

1. Configure all segments ever needed.

The measurement supports a range of up to 2000 configured segments.

2. Select up to 1000 consecutive segments within the configured segment range (consider the maximum number of measured subframes).
3. Measure the selected segments.
4. Repeat steps 2 and 3 as often as needed.

The list mode is essentially a single-shot remote control application. When a measurement is initiated in list mode, all defined segments are measured once. Afterwards, the results can be retrieved using `FETCH` commands.

The following remote control commands are used for list mode settings and result retrieval.

Table 5-2: List mode commands

Parameters	SCPI commands
Activate / deactivate list mode	<code>CONFigure:LTE:MEAS<i>:MEValuation:LIST</code>
Range of measured segments	<code>CONFigure:LTE:MEAS<i>:MEValuation:LIST:LRANge</code>

Parameters	SCPI commands
Segment configuration (duplex mode, channel bandwidth, number of subframes, ...)	<pre> CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: SETup CONFigure:LTE:MEAS<i>:MEValuation:LIST:PLCMode CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: PLCid CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: CC<c> CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: CAGGregation:ACSPacing CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: CAGGregation:MCARrier:ENHanced CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TDD CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: EMTC:NBAND CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: RBALlocation CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: RBALlocation:SIDelink </pre>
Statistical length and result activation	<pre> CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>: MODulation etc. </pre>
Trigger mode	<pre> TRIGger:LTE:MEAS<i>:MEValuation:LIST:MODE TRIGger:LTE:MEAS<i>:MEValuation:LIST:NBANDwidth </pre>
Retrieve results	<pre> FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:... FETCh:LTE:MEAS<i>:MEValuation:LIST:... </pre> See: • Section 5.5.3.34, "List mode results (one segment)", on page 1242 • Section 5.5.3.35, "List mode results (all segments, one result)", on page 1263 The segment number <no> for configure commands is an absolute number (1 to 2000). The segment number <no> for result retrieval is a relative number within the range of measured segments (1 to 1000). Example: Segment 1 to 100 configured. Segment 50 to 59 measured. For result retrieval, <no> = 1 refers to segment 50, <no> = 10 to segment 59.

You can deactivate the list mode via a command and also via the GUI:

1. Go to local using the corresponding hotkey.
The active list mode is indicated in the upper right corner of the current view by the words "List Mode!".
2. Open the configuration dialog box.
3. In the section "Measurement Control", select another measurement mode.



Global and list mode parameters

Some settings are available as special list mode settings and as multi-evaluation settings (e.g. duplex mode, channel bandwidth and modulation scheme). In list mode, the measurement ignores these multi-evaluation parameters. All other settings not available as special list mode settings are taken from the multi-evaluation measurement, e.g.:

- "Signal Type"
- "External Attenuation"
- "Measure on Exception"
- "Carrier Aggregation Mode"
- "Subframe Offset", "Number of Subframes", "Measure Subframe": Always use the *RST values (0, 1, 0) for list mode measurements.
- Limit settings

5.2.10.2 Offline mode and offline segment

In list mode, it is possible to select the results of a single segment for display in the measurement diagram (see also "Offline Segment No." on page 1069). This "offline mode" offers several advantages:

- Check of the measurement results (e.g. while developing test scripts).
- Calculation and display of additional measurement results (traces). Use the `FETCH:LTE:MEAS:MEvaluation:TRACE:...` commands to retrieve these results.

After a list mode measurement is completed, the measurement diagrams show the last segment measured. Calculation of the offline results in an arbitrary segment requires a two-stage measurement:

1. Start a single-shot list mode measurement
(`INITiate:LTE:MEAS:MEvaluation`) to collect all measurement data.
2. Select the offline segment
(`CONFigure:LTE:MEAS:MEvaluation:LIST:OSIndex <Segment>`).
3. Initiate a second measurement (repeat `INITiate:LTE:MEAS:MEvaluation`).
4. View the result diagrams.

The second measurement stage implies a calculation of all measurement results in the offline segment from the existing raw data. No new measurement data is acquired. By repeating this second stage for different segments, you can obtain a complete set of measurement results over the entire measurement length.



Reconfiguration of the measurement, example

To obtain consistent results in the second measurement stage, the raw data must still correspond to the measurement settings. If you wish to reuse your data in offline mode, avoid any reconfiguration that requires a new measurement.

5.2.11 LTE signal properties

This section describes the following selected topics related to LTE uplink and sidelink signal properties.

- [Resources in time and frequency domain](#).....1008
- [Uplink resource block allocation](#).....1009
- [Sidelink resource block allocation](#).....1012
- [Frequency bands](#).....1012

5.2.11.1 Resources in time and frequency domain

The radio resources are divided into time-frequency units called resource elements. In the time domain, a resource element corresponds to one SC-FDMA symbol. In the frequency domain, it corresponds to one subcarrier.

For the mapping of physical channels to resources, the resource elements are grouped into resource blocks (RB). Each RB consists of 12 consecutive subcarriers (180 kHz) and 6 or 7 consecutive SC-FDMA symbols (0.5 ms).

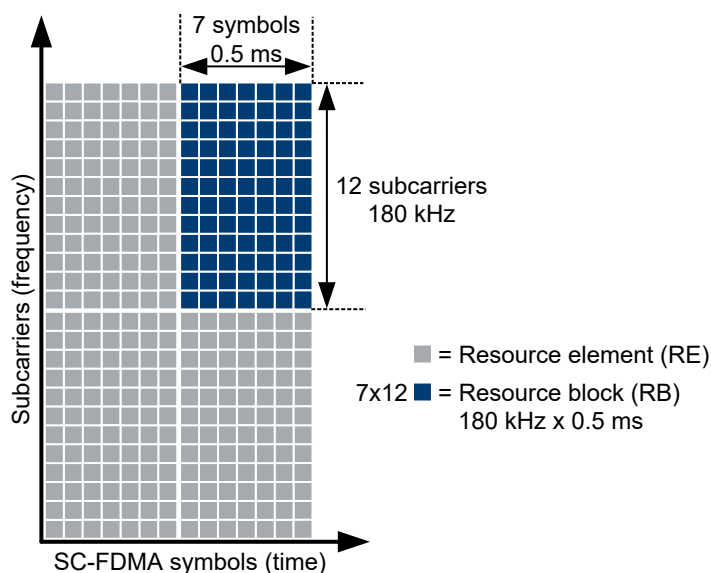


Figure 5-3: Resource element grid (seven symbols per RB)

In the time domain, the additional units radio frame, subframe and slot (containing the SC-FDMA symbols) are defined, see figures below. Each SC-FDMA symbol contains a guard time called cyclic prefix (CP). Depending on the duration of the guard time, it is

either called normal CP or extended CP. A slot contains either seven SC-FDMA symbols with normal CP or six SC-FDMA symbols with extended CP.

The basic time unit in LTE is the sample interval T_s . T_s can be calculated from the sampling frequency 30.72 MHz: $1 T_s = 1/30.72 \text{ MHz} \approx 32.55 \text{ ns}$.

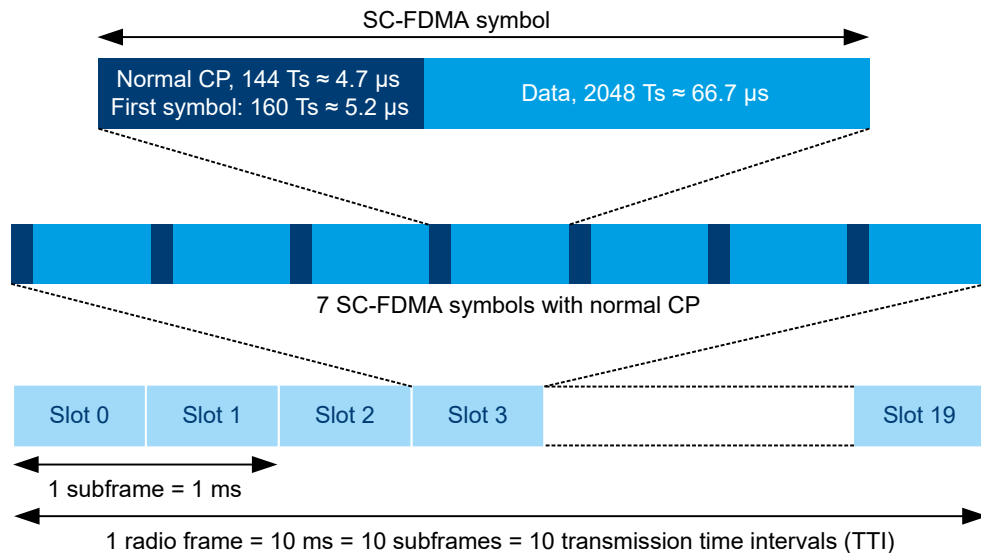


Figure 5-4: Frame structure for FDD with normal CP

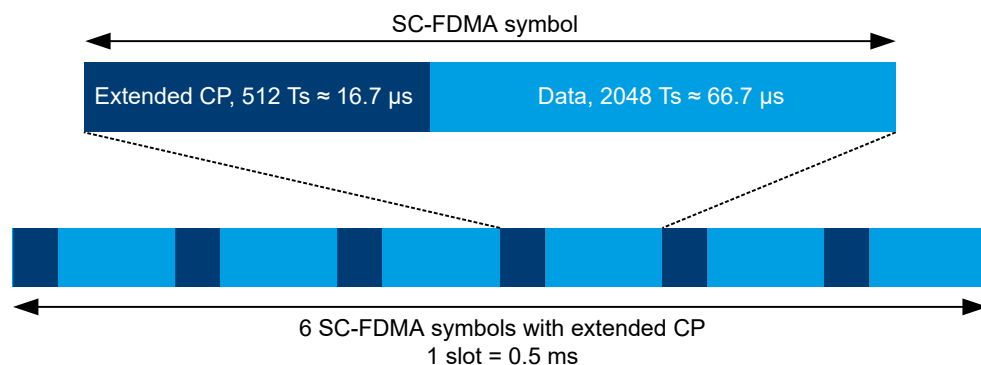


Figure 5-5: Slot structure for FDD with extended CP

The figures above are for FDD, but the shown structure of SC-FDMA symbols and slots applies also to TDD.

5.2.11.2 Uplink resource block allocation

The total number of resource blocks available per channel depends on the channel bandwidth as shown in the following table.

Channel bandwidth / MHz	1.4	3	5	10	15	20
Resource blocks (RB)	6	15	25	50	75	100

The total number of allocated resource blocks is restricted by the following formula, specified in 3GPP TS 36.211. For multi-cluster RB allocation, it applies to the sum of both clusters.

$$N_{RB} = 2^i \times 3^j \times 5^k, \text{ with } i, j, k \text{ being non-negative integer values}$$

Example, channel bandwidth 3 MHz:

- Allowed values: 0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15 RB.
- Not allowed: 7, 11, 13, 14 RB.

The other rules depend on the usage of multi-cluster allocation, as described in the following.

Contiguous RB allocation, single cluster, allocation type 0

This resource allocation type is defined in 3GPP TS 36.213 section 8.1.1.

The position of the first allocated RB is freely selectable within the channel bandwidth. The number of allocated RBs is only restricted by the channel bandwidth and the stated formula.

Multi-cluster RB allocation, two clusters, allocation type 1

This resource allocation type is defined in 3GPP TS 36.213 section 8.1.2.

The available resource blocks are divided into resource block groups (RBG) as specified in 3GPP TS 36.213, section 7.1.6.1. The RBG size depends on the channel bandwidth. The last RBG can be smaller than the other RBGs. Only entire RBGs can be allocated.

The allocated RBGs are divided into two clusters. Within each cluster, the RBG allocation is contiguous. There must be at least one unused RBG between the two clusters.

The following table provides an overview of the resource block groups.

Table 5-3: RBG parameters

Channel bandwidth	Maximum no. of RBs	RBG size	Size of last RBG	Allowed start RB
1.4 MHz	6	1	1	0, 1, 2, 3, 4, 5
3 MHz	15	2	1	0, 2, 4, 6, 8, 10, 12, 14
5 MHz	25	2	1	0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24
10 MHz	50	3	2	0, 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, 39, 42, 45, 48
15 MHz	75	4	3	0, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72
20 MHz	100	4	4	0, 4, 8, 12, 16, 20, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 64, 68, 72, 76, 80, 84, 88, 92, 96

RB allocation, narrowbands and widebands for eMTC

eMTC uses narrowband operation. A narrowband covers 1.4 MHz and contains six resource blocks. An LTE channel can contain several eMTC narrowbands. A 10-MHz LTE channel, for example, contains eight eMTC narrowbands.

The number and location of the narrowbands within the channel bandwidth is specified by 3GPP. There are as many narrowbands as fit into the bandwidth. Depending on the bandwidth, the center RB is not used. If there are more unused RBs, they are at the edge of the bandwidth. The narrowbands are numbered 0 to n, with increasing frequency. The following table shows the assignment of resource blocks to narrowbands.

Table 5-4: Narrowbands and resource blocks per channel BW

Channel BW / #RB	#NB	Narrowband positions	Unused RB
1.4 MHz / 6	1	NB 0 (RB 0 to 5)	-
3 MHz / 15	2	NB 0 (RB 1 to 6), NB 1 (RB 8 to 13)	0, 7, 14
5 MHz / 25	4	NB 0 (RB 0 to 5), NB 1 (RB 6 to 11), NB 2 (RB 13 to 18), NB 3 (RB 19 to 24)	12
10 MHz / 50	8	NB 0 (RB 1 to 6), ..., NB 3 (RB 19 to 24) NB 4 (RB 25 to 30), ..., NB 7 (RB 43 to 48)	0, 49
15 MHz / 75	12	NB 0 (RB 1 to 6), ..., NB 5 (RB 31 to 36), NB 6 (RB 38 to 43), ..., NB 11 (RB 68 to 73)	0, 37, 74
20 MHz / 100	16	NB 0 (RB 2 to 7), ..., NB 7 (RB 44 to 49), NB 8 (RB 50 to 55), ..., NB 15 (RB 92 to 97)	0, 1, 98, 99

There are two allowed maximum eMTC bandwidths: 1.4 MHz and 5 MHz. With a maximum eMTC BW of 1.4 MHz, a single narrowband is used. With a maximum eMTC BW of 5 MHz, up to four narrowbands are used (called wideband). In both cases, the RBs are allocated contiguously, in a single narrowband or over narrowband-borders.

In the RB allocation settings, you can select the lowest used narrowband the position of the lowest RB in that narrowband and the number of RBs.

Table 5-5: RB allocation settings depending on maximum eMTC bandwidth

	Maximum eMTC BW	
	1.4 MHz or 5 MHz	5 MHz
Narrowband	0 to (n-1) ¹⁾	0 to (n – ceiling(#RB+offset RB) / 6)
Allocated RBs (#RB)	1 to 6	9, 12, 15, 18, 21, 24 ²⁾
Offset RB	0 to (6 - #RB)	0, 3
¹⁾ n = number of narrowbands fitting into the channel BW, see Table 5-4 . ²⁾ for 1.4-MHz channel BW max 6 RB, for 3-MHz channel BW max 12 RB.		

5.2.11.3 Sidelink resource block allocation

The total number of resource blocks available per channel depends on the channel bandwidth as shown in the following table.

Table 5-6: Total number of RBs

Channel bandwidth / MHz	1.4	3	5	10	15	20
Resource blocks (RB)	6	15	25	50	75	100

The PSCCH occupies always two RBs. The remaining RBs are available for the PSSCH, as shown in the following table.

Table 5-7: RBs available for PSSCH

Channel bandwidth / MHz	1.4	3	5	10	15	20
Resource blocks (RB)	4	13	23	48	73	98

The number of allocated PSSCH RBs is restricted by the following formula, specified in 3GPP TS 36.211.

$$N_{RB} = 2^i \times 3^j \times 5^k, \text{ with } i, j, k \text{ being non-negative integer values}$$

Example, channel bandwidth 3 MHz:

- Allowed values: 1, 2, 3, 4, 5, 6, 8, 9, 10, 12 RB.
- Not allowed: 7, 11, 13 RB.

So the number of allocated RBs for the PSCCH is two. The number of allocated RBs for the PSSCH is 1 to n, restricted by the channel bandwidth and the stated formula.

Position of the RBs

The position of the first allocated PSCCH RB and the first allocated PSSCH RB is selectable within the channel bandwidth. The PSCCH is located below the PSSCH (lower RB numbers and frequencies). There can be unused RBs between the PSCCH and the PSSCH.

Example, channel bandwidth 20 MHz (100 RBs):

- No of PSCCH RBs = 2 (as always)
No of PSSCH RBs = 10 (for example)
- The resulting range for the PSCCH RB offset is:
Minimum = 0 (always)
Maximum = <available RBs> – <allocated RBs> = 100 - 10 - 2 = 88
PSCCH RB offset = 40 (for example)
- The resulting range for the PSSCH RB offset is:
Minimum = <PSCCH offset> + 2 = 42
Maximum = <available RBs> – <allocated PSSCH RBs> = 100 - 10 = 90

5.2.11.4 Frequency bands

The carrier frequencies for LTE signals are defined in 3GPP TS 36.101. Each operating band contains carrier frequencies identified by channel numbers (EARFCN, E-

UTRA absolute radio frequency channel number). The assignment between channel numbers N and carrier center frequencies F is defined as:

$$N = 10 \times (F - F_{\text{Offset}}) / \text{MHz} + N_{\text{Offset}}$$

The tables below provide an overview of all supported bands. For each band, they list the channel numbers N and the carrier center frequencies F. The offsets F_{Offset} and N_{Offset} equal the lower boundaries of the listed ranges (example: channel 18000 to 18599 means $N_{\text{Offset}} = 18000$).

For the supported frequency range, see [Section 5.5.1.5, "Frequency ranges"](#), on page 1105.

Table 5-8: FDD bands, uplink measurement

Band	Channel no N	F / MHz
1	18000 to 18599	1920 to 1979.9
2	18600 to 19199	1850 to 1909.9
3	19200 to 19949	1710 to 1784.9
4	19950 to 20399	1710 to 1754.9
5	20400 to 20649	824 to 848.9
6	20650 to 20749	830 to 839.9
7	20750 to 21449	2500 to 2569.9
8	21450 to 21799	880 to 914.9
9	21800 to 22149	1749.9 to 1784.8
10	22150 to 22749	1710 to 1769.9
11	22750 to 22949	1427.9 to 1447.8
12	23010 to 23179	699 to 715.9
13	23180 to 23279	777 to 786.9
14	23280 to 23379	788 to 797.9
15	23380 to 23579	1900 to 1919.9
16	23580 to 23729	2010 to 2024.9
17	23730 to 23849	704 to 715.9
18	23850 to 23999	815 to 829.9
19	24000 to 24149	830 to 844.9
20	24150 to 24449	832 to 861.9
21	24450 to 24599	1447.9 to 1462.8
22	24600 to 25399	3410 to 3489.9
23	25500 to 25699	2000 to 2019.9
24	25700 to 26039	1626.5 to 1660.4
25	26040 to 26689	1850 to 1914.9

Band	Channel no N	F / MHz
26	26690 to 27039	814 to 848.9
27	27040 to 27209	807 to 823.9
28	27210 to 27659	703 to 747.9
30	27660 to 27759	2305 to 2314.9
31	27760 to 27809	452.5 to 457.4
65	131072 to 131971	1920 to 2009.9
66	131972 to 132671	1710 to 1779.9
68	132672 to 132971	698 to 727.9
70	132972 to 133121	1695 to 1709.9
71	133122 to 133471	663 to 697.9
72	133472 to 133521	451 to 455.9
73	133522 to 133571	450 to 454.9
74	133572 to 134001	1427 to 1469.9
85	134002 to 134181	698 to 715.9
87	134182 to 134231	410 to 414.9
88	134231 to 134280	412 to 416.9
106	134292 to 134341	896 to 900.9

Table 5-9: TDD bands, uplink measurement

Band	Channel no N	F / MHz
33	36000 to 36199	1900 to 1919.9
34	36200 to 36349	2010 to 2024.9
35	36350 to 36949	1850 to 1909.9
36	36950 to 37549	1930 to 1989.9
37	37550 to 37749	1910 to 1929.9
38	37750 to 38249	2570 to 2619.9
39	38250 to 38649	1880 to 1919.9
40	38650 to 39649	2300 to 2399.9
41	39650 to 41589	2496 to 2689.9
42	41590 to 43589	3400 to 3599.9
43	43590 to 45589	3600 to 3799.9
44	45590 to 46589	703 to 802.9
45	46590 to 46789	1447 to 1466.9
48	55240 to 56739	3550 to 3699.9

Band	Channel no N	F / MHz
50	58240 to 59089	1432 to 1516.9
51	59090 to 59139	1427 to 1431.9
52	59140 to 60139	3300 to 3399.9
53	60140 to 60254	2483.5 to 2494.9
250	253644 to 255143	3550 to 3699.9

Table 5-10: TDD bands, sidelink measurement

Band	Channel no N	F / MHz
47	54540 to 55239	5855 to 5924.9

5.2.12 Trigger signals

The LTE multi-evaluation measurement provides a trigger signal that can be used by other R&S CMW applications to synchronize to the measurement.

To address the trigger signal in remote commands, use the following string, with <i> replaced by the instance number of the measurement application:

- "LTE Meas<i>: Multi Evaluation"

The trigger signal is synchronized with the trigger event starting the multi-evaluation measurement.

5.3 GUI reference

The following sections provide detailed reference information on the graphical user interface (GUI) of the LTE multi-evaluation measurement.

To configure measurement settings, select the "Multi Evaluation" tab and press the hotkey "Config". It opens the main configuration dialog box. Some settings are also accessible via softkeys and hotkeys.

The measurement results are presented on the "Multi Evaluation" tab.

- [Measurement control](#)..... 1016
- [Measurement results](#).....1016
- [Signal routing and analyzer settings](#)..... 1040
- [Carrier aggregation settings](#)..... 1046
- [Measurement control settings](#)..... 1050
- [Trigger settings](#)..... 1069
- [Limit settings](#)..... 1072
- [Shortcut configuration](#)..... 1086
- [Additional softkeys and hotkeys](#)..... 1086

5.3.1 Measurement control

To turn the measurement on or off, select the control softkey and press [ON | OFF] or [RESTART | STOP]. Alternatively, right-click the control softkey.



Multi Evaluation (Softkey)

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

```
INITiate:LTE:MEAS<i>:MEvaluation
ABORt:LTE:MEAS<i>:MEvaluation
STOP:LTE:MEAS<i>:MEvaluation
FETCh:LTE:MEAS<i>:MEvaluation:STATe?
FETCh:LTE:MEAS<i>:MEvaluation:STATe:ALL?
```

5.3.2 Measurement results

The results of the LTE multi-evaluation measurement are displayed in several different views. The results are described in the following sections.

• Overview.....	1016
• View EVM.....	1017
• View EVM vs Subcarrier.....	1019
• View Magnitude Error, Phase Error.....	1020
• View Inband Emissions.....	1021
• View Equalizer Spectrum Flatness.....	1022
• View Spectrum Emission Mask.....	1024
• View Spectrum ACLR.....	1026
• View I/Q Constellation.....	1027
• View RB Allocation Table.....	1029
• View Power Monitor.....	1031
• View Power Dynamics.....	1032
• View TX Measurement and BLER.....	1035
• Statistical results.....	1038
• Detected configuration and general statistical information.....	1038
• Selecting and modifying views.....	1039
• Using markers.....	1040

5.3.2.1 Overview

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "Overview".

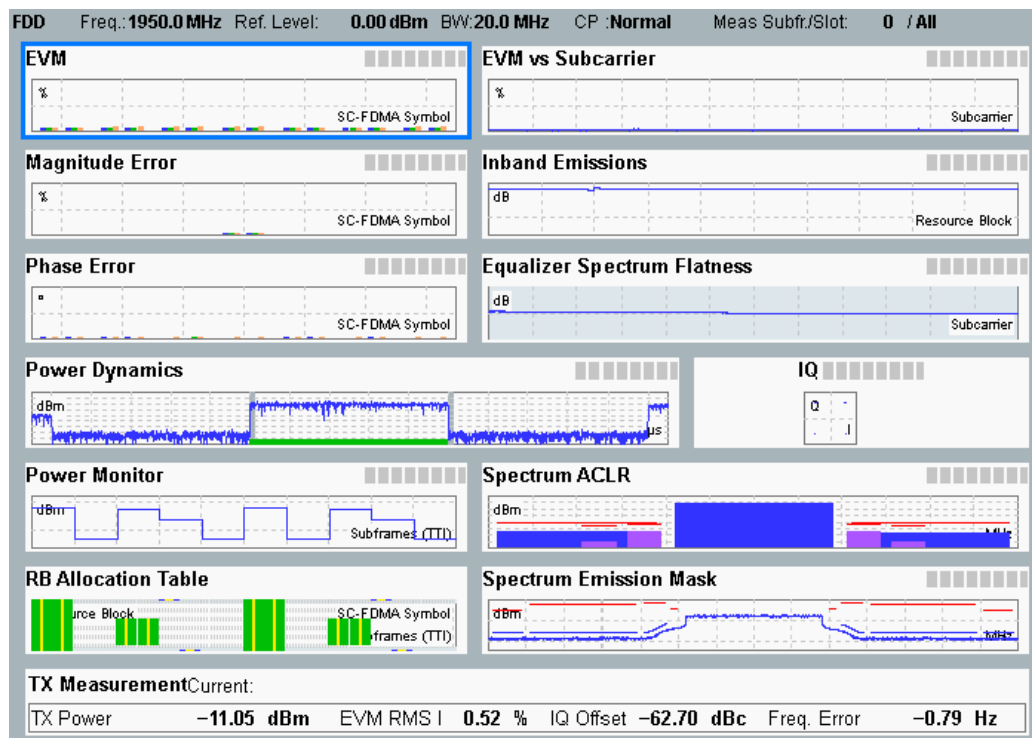


Figure 5-6: Result overview

You can limit the results to be measured and displayed via the hotkey "Assign Views", see [Section 5.3.9, "Additional softkeys and hotkeys"](#), on page 1086.

You can enlarge one of the diagrams in the overview and show a detailed view with additional measurement results, see [Section 5.3.2.16, "Selecting and modifying views"](#), on page 1039. The traces and diagrams are described in the following sections.

5.3.2.2 View EVM

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "EVM".

The view shows one or two diagrams and a statistical overview of results per slot.

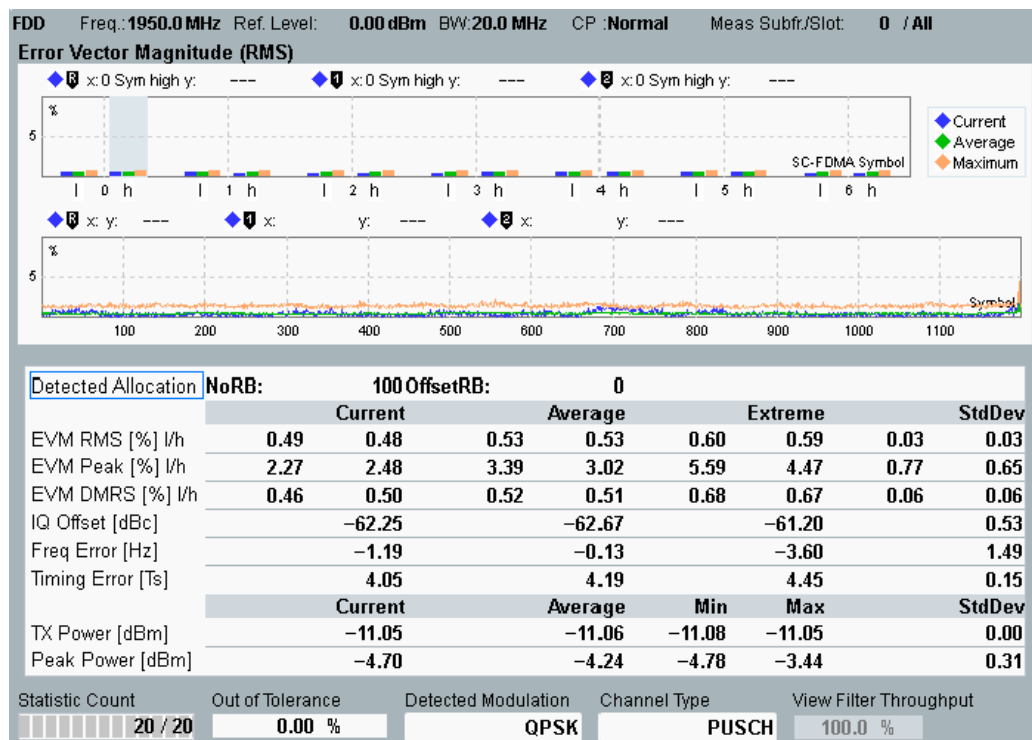


Figure 5-7: EVM view

All results are derived from the measured slot located in the configured "Measure Subframe". Only a single carrier is analyzed.

For signals with carrier aggregation, you can select which carrier is measured (setting "Measurement Carrier").

For sidelink signals, you can select which channel type is measured (setting "Sidelink Channel Type").

- **Upper diagram**

The diagram shows the EVM for each SC-FDMA symbol in the measured slot. The symbols are numbered from 0 to 6 (5) for normal (extended) cyclic prefix duration, with the reference symbol labeled 3 (2).

The measurement calculates two values (EVM_I and EVM_Q) for each SC-FDMA symbol and displays both values. See also [Section 5.2.9, "Calculation of modulation results"](#), on page 1001.

Each value is an RMS value or a peak value over all modulation symbols in the SC-FDMA symbol. The title above the diagram indicates whether RMS or peak values are currently displayed. To switch between RMS and peak values, use the soft-key/hotkey combination "Display" > "Select Trace".

If SRS signals are enabled, the diagram shows no results for the last SC-FDMA symbol of each subframe (second slot, last symbol).

- **Lower diagram**

The lower diagram shows the EVM for all modulation symbols in a selected SC-FDMA symbol.

The results are either EVM_l or EVM_h values. The selected SC-FDMA symbol and the selected low/high position are indicated in the upper diagram by a shadowed rectangle.

The lower diagram is only shown if the measurement of "EVM vs symbol" results is enabled. For enabling the diagram and configuring the SC-FDMA symbol selection, see ["Multi Evaluation softkey > EVM vs Symbol hotkey"](#) on page 1087.

The lower diagram shows results even for SRS signals. To use the diagram for SRS, restrict the measurement to the second slot of the measured subframe ("Measure Slot" = 1) and select the last SC-FDMA symbol for display.

For a description of the statistical overview, see [Section 5.3.2.13, "View TX Measurement and BLER"](#), on page 1035.

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the diagram contents via remote control, see [Section 5.5.3.21, "EVM results \(diagrams\)"](#), on page 1211.

5.3.2.3 View EVM vs Subcarrier

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "EVMvsC".

The view shows a diagram and a statistical overview of results per slot.

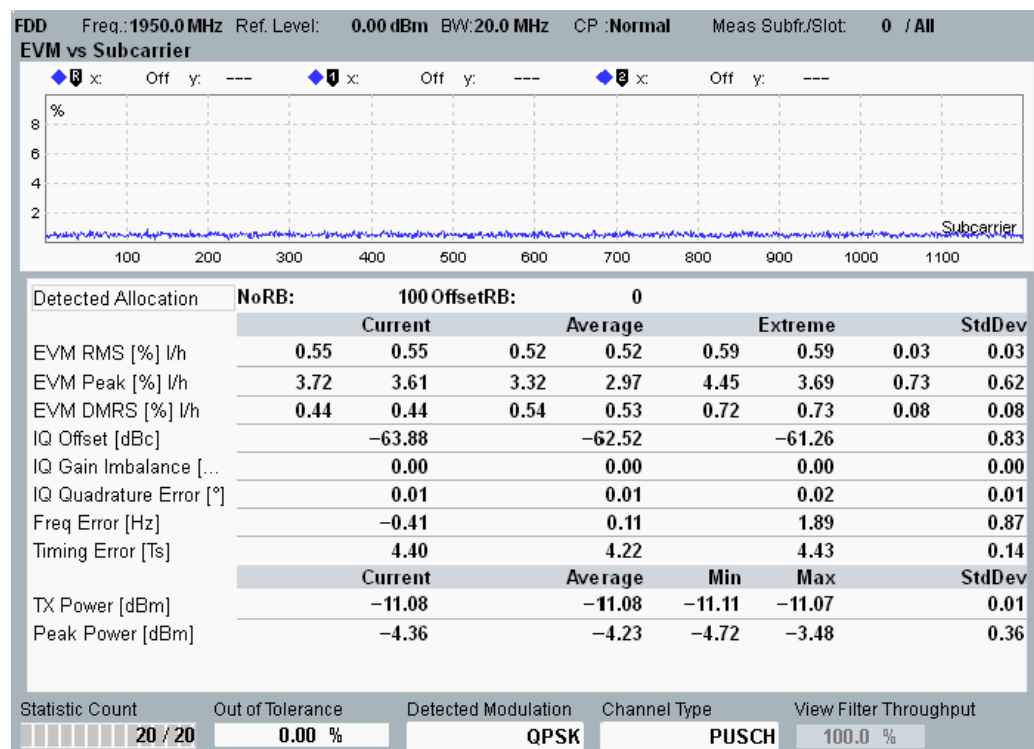


Figure 5-8: EVM vs Subcarrier view

The diagram shows the EVM in all allocated subcarriers of the measured slot. Only a single carrier is analyzed.

For signals with carrier aggregation, you can select which carrier is measured (setting "Measurement Carrier").

For sidelink signals, you can select which channel type is measured (setting "Sidelink Channel Type").

The values are RMS averaged over the SC-FDMA data symbols in the subcarrier.

For a description of the statistical overview, see [Section 5.3.2.13, "View TX Measurement and BLER"](#), on page 1035.

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the diagram contents via remote control, see [Section 5.5.3.21, "EVM results \(diagrams\)"](#), on page 1211.

5.3.2.4 View Magnitude Error, Phase Error

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "MErr" / "PhErr".

Both views show a diagram and a statistical overview of results per slot.

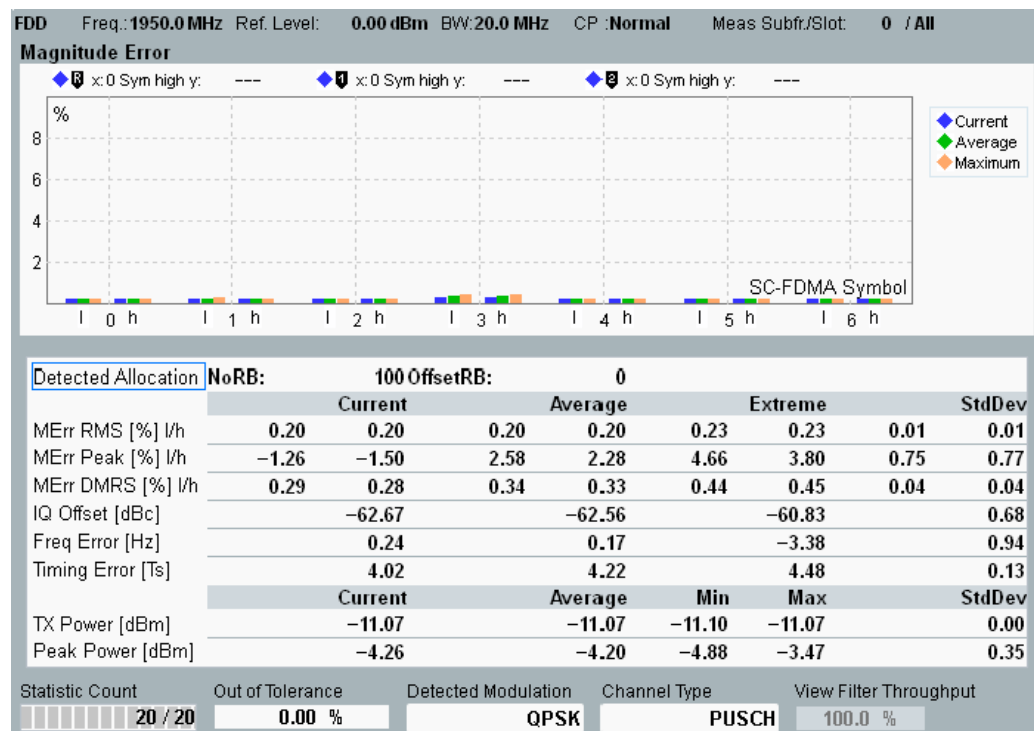


Figure 5-9: Magnitude Error view

The diagrams show the average magnitude error and phase error for each SC-FDMA symbol in the measured slot. The symbols are numbered from 0 to 6 (5) for normal (extended) cyclic prefix duration, with the reference symbol labeled 3 (2).

The measured slot is located in the configured "Measure Subframe". Only a single carrier is analyzed.

For signals with carrier aggregation, you can select which carrier is measured (setting "Measurement Carrier").

For sidelink signals, you can select which channel type is measured (setting "Sidelink Channel Type").

The diagrams show RMS and therefore positive values. The average runs over all modulation symbols in the SC-FDMA symbol. The measurement calculates two values for each SC-FDMA symbol and displays both values, see also [Section 5.2.9, "Calculation of modulation results"](#), on page 1001.

For a description of the statistical overview, see [Section 5.3.2.13, "View TX Measurement and BLER"](#), on page 1035.

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the diagram contents via remote control, see:

- [Section 5.5.3.22, "Magnitude error results \(diagrams\)"](#), on page 1214
- [Section 5.5.3.23, "Phase error results \(diagrams\)"](#), on page 1215

5.3.2.5 View Inband Emissions

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "InBandEmissions".

The view shows one diagram per component carrier and a statistical overview of results per slot.

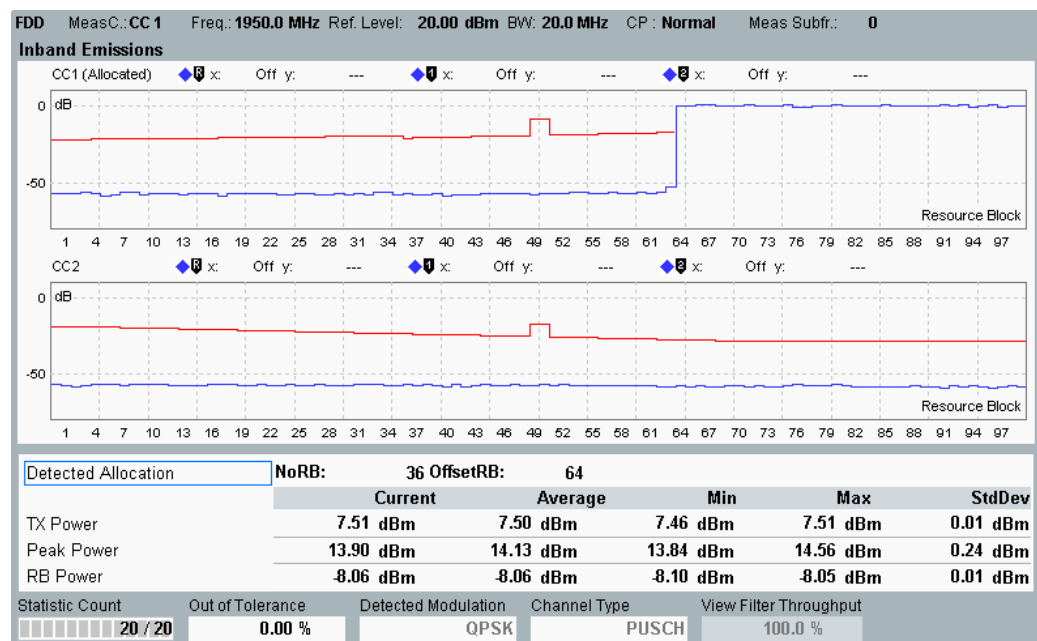


Figure 5-10: Inband Emissions view

For measurements without carrier aggregation, a single diagram is displayed. For measurements with carrier aggregation, all component carriers are measured and the results are displayed in one diagram per carrier.

Each diagram shows the average power in all resource blocks (RBs) of the measured slot. So it shows a power vs frequency representation where every RB comprises 12 subcarriers. The RB powers are normalized to the current RB power.

The figure shows a measurement with carrier aggregation. In the CC1 diagram, 36 out of 100 resource blocks are allocated. The power of the remaining, non-allocated blocks must be below the red limit line. For the CC2, no resource blocks are allocated. So the entire trace must be below the red limit line.

The carrier currently selected as "Measurement Carrier" is marked by the string "(Allocated)". It must contain allocated resource blocks. Otherwise, the measurement fails ("Underdriven" or "Sync Error").

The in-band emissions are calculated after I/Q origin offset correction, see [Section 5.2.9, "Calculation of modulation results"](#), on page 1001.

The result table refers to the configured "Measurement Carrier". For details about the table results, see [Section 5.3.2.13, "View TX Measurement and BLER"](#), on page 1035.

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the diagram contents via remote control, see [Section 5.5.3.24, "In-band emission results"](#), on page 1216.

5.3.2.6 View Equalizer Spectrum Flatness

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "Equalizer-SpectrumFlatness".

The equalizer spectrum flatness is determined for the post-FFT equalization step, as specified in the standard. It reflects the spectrum flatness of the equalizer.

The results are displayed in a diagram and as a table of statistical results.

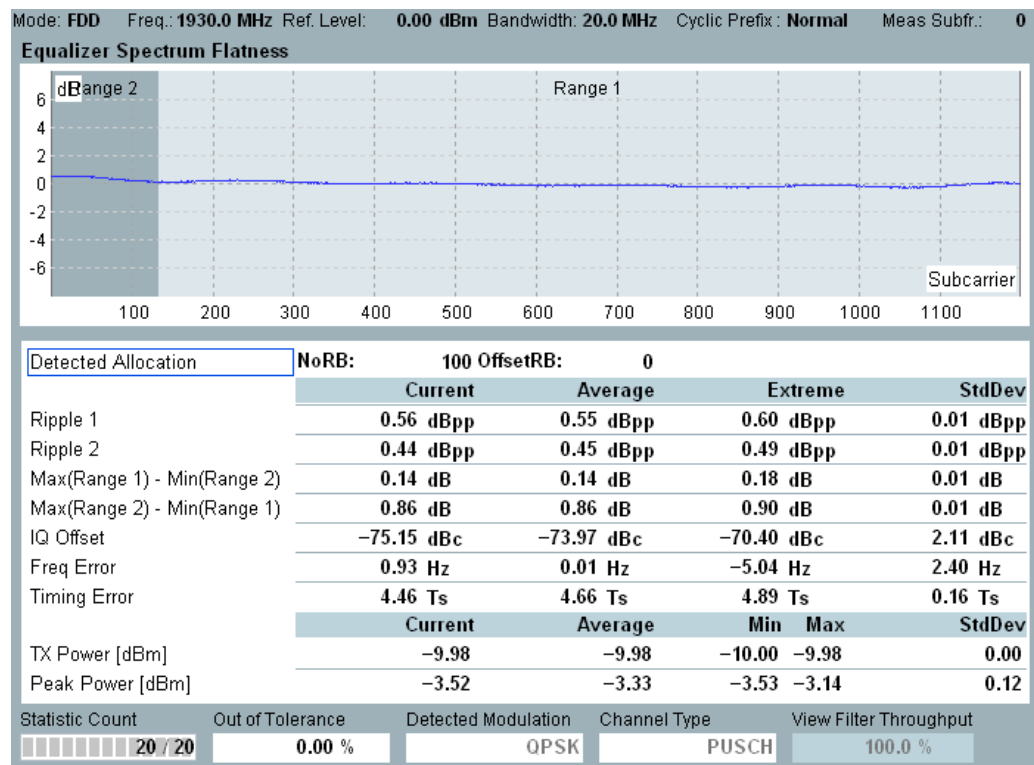


Figure 5-11: Equalizer Spectrum Flatness view

The diagram shows the power of the equalizer coefficients for all allocated subcarriers of the measured slot. The measured slot is located in the configured "Measure Sub-frame". Only a single carrier is analyzed.

For signals with carrier aggregation, you can select which carrier is measured (setting "Measurement Carrier").

For sidelink signals, you can select which channel type is measured (setting "Sidelink Channel Type").

The results are normalized to the arithmetic mean value of all results. Hence the measured curve is centered on the 0-dB line.

The table below the diagram presents statistical values. The first four values are related to the spectrum flatness limits. They indicate the difference between maximum and minimum values within range 1 ("Ripple 1"), within range 2 ("Ripple 2") and between the ranges. For details, see [Section 5.3.7.6, "Equalizer spectrum flatness limits"](#), on page 1077.

For the other results in the table, see [Section 5.3.2.13, "View TX Measurement and BLER"](#), on page 1035.

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the spectrum flatness results, see [Section 5.5.3.25, "Equalizer spectrum flatness results"](#), on page 1218.

5.3.2.7 View Spectrum Emission Mask

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "SpectrumEmissionMask".

The spectrum emission mask results are displayed in a diagram and as a table of statistical results. Additional margin results can be shown/hidden via hotkey.

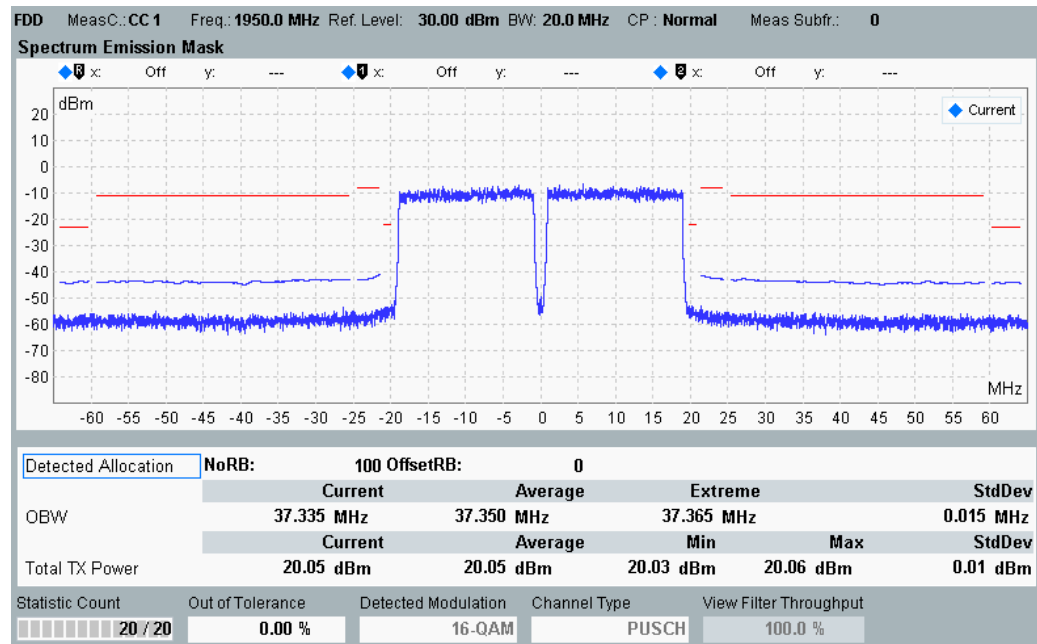


Figure 5-12: Spectrum Emission Mask view

Without carrier aggregation, the measurement covers a symmetric frequency range around the carrier center frequency. With carrier aggregation, all component carriers are measured. So the measurement covers a symmetric frequency range around the center frequency of the aggregated bandwidth.

Up to three traces are measured in parallel, with different resolution bandwidths (RBW). A continuous trace with an RBW of 30 kHz or 50 kHz is always measured. The other two traces are only displayed for frequency ranges with active limits. Depending on the limits, the traces are hidden completely, displayed partially or displayed for the entire out-of-band range. The resolution bandwidths for the three traces are listed in the following table.

Table 5-11: RBW depending on "network signaled value" and channel BW

Trace	NS_x value	Channel BW / MHz	RBW
1	NS_27	1.4, 3	30 kHz
		5, 10, 15, 20	50 kHz
	other NS_x	any	30 kHz
2	NS_27	1.4, 3, 5, 10	100 kHz
		15	150 kHz

Trace	NS_x value	Channel BW / MHz	RBW
		20	200 kHz
	other NS_x	any	100 kHz
3	any	any	1 MHz

The RBW increases from trace 1 to trace 3, so the measured power increases too. Trace 1 is always at the bottom, trace 3 at the top and trace 2 in-between (if trace 2 and trace 3 are visible at all).

The shape of the resolution filter is configurable for trace 2 and trace 3 (Gaussian or bandpass filter). For trace 1, a Gaussian filter is used.

According to 3GPP TS 36.521, the measurement must be carried out at the maximum output power of the UE.

See also [Section 5.3.7.9, "Spectrum emission mask"](#), on page 1080.

Margin results

To show/hide the margin results, use the softkey/hotkey combination "Display" > "Margin On | Off".

	Area 1	Area 2	Area 3	Area 4	Area 5	Area 6	Area 7	Area 8	Area 9	Area 10	Area 11	Area 12
Margin												
Curr neg												
X [MHz]	-5.01	-6.50	-10.80	-19.08	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP
Y [dB]	36.51	43.62	55.38	43.27	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP
Curr pos												
X [MHz]	5.01	6.50	11.88	15.57	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP
Y [dB]	39.97	45.22	55.48	43.68	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP	NCAP

Figure 5-13: Emission mask, margin results

A margin indicates the vertical distance between the spectrum emission mask and a trace. For each emission mask area, the margin represents the "worst" value, i.e. the minimum determined for the frequencies of the area:

$$\text{Margin} = \text{minimum} (P(f)_{\text{mask}} - P(f)_{\text{trace}})$$

A negative margin indicates that the trace is located above the limit line and the limit is exceeded.

For each active area, there is an "X [MHz]" and a "Y [dB]" value for negative and positive offset frequencies. The y-value indicates the margin (worst value within area). The x-value indicates the frequency offset at which the margin value was found. These frequency offsets are indicated relative to the center frequency.

The screenshot above shows the margin results for the current trace. Corresponding margin results are displayed for all active traces (current, average, maximum trace). In this context, "Average" means "margins for average trace" and not "average values of margins for current trace". "Minimum" means "margins for maximum trace" (resulting in minimum margins).

Result table

The "Total TX Power" in the result table indicates the sum of the TX power of all component carriers. For other table results, see [Section 5.3.2.13, "View TX Measurement and BLER"](#), on page 1035.

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the spectrum emission mask results via remote control, see [Section 5.5.3.26, "Spectrum emission results"](#), on page 1221.

5.3.2.8 View Spectrum ACLR

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "Spectrum ACLR".

The adjacent channel leakage ratio (ACLR) results are displayed in a diagram and as a table of statistical results.

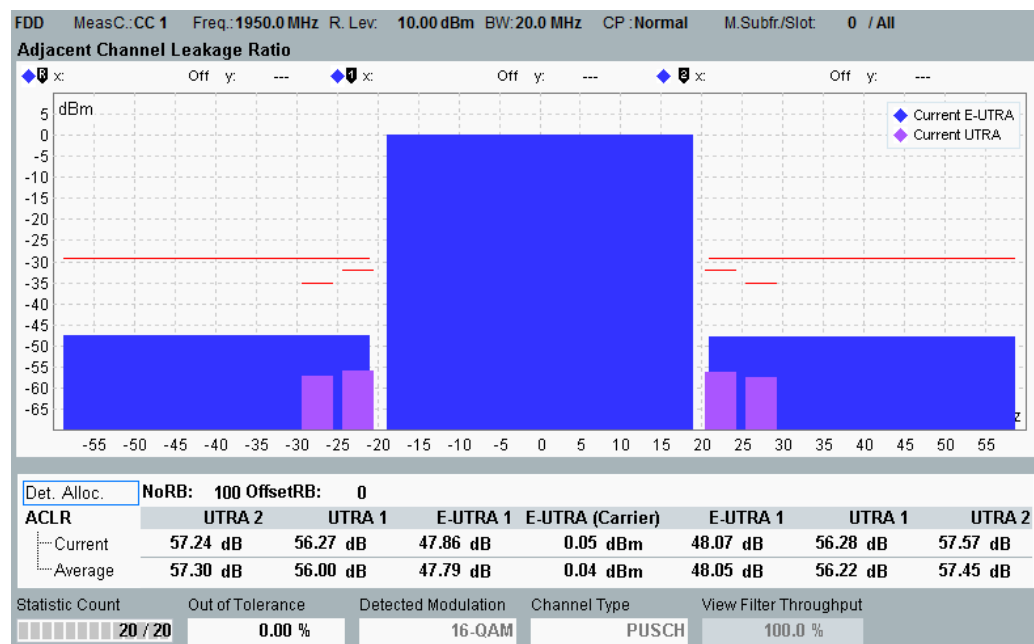


Figure 5-14: ACLR view

The central bar in the diagram indicates the mean power in the assigned E-UTRA channel. The other bars indicate the mean powers in the first adjacent E-UTRA channels and in the first and second adjacent UTRA channels. All values are absolute power levels in dBm.

With carrier aggregation, an E-UTRA channel comprises the aggregated bandwidth minus a guard band. The figure shows a measurement with carrier aggregation and a channel bandwidth of 20 MHz for two carriers. The width of an E-UTRA bar is 38 MHz.

The displayed absolute limit lines are calculated from the configured relative ACLR limits. They are only displayed for adjacent channels with a power level above the configured absolute limit.

The ACLR values resulting from the measured powers are displayed in the table below the diagram. According to 3GPP TS 36.521, the ACLR is defined as the mean power in the assigned E-UTRA channel divided by the mean power in an adjacent channel. The mean power in the assigned E-UTRA channel is also displayed.

The filters used for the ACLR measurement are compliant with 3GPP TS 36.521. According to this specification, ACLR tests must be carried out at the maximum output power of the UE.

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the ACLR results via remote control, see [Section 5.5.3.27, "ACLR spectrum results"](#), on page 1226.

5.3.2.9 View I/Q Constellation

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "IQConst".

The constellation diagram shows the modulation symbols in the measured slot of the measured component carrier as points in the I/Q plane.

For sidelink signals, you can select which channel type is measured (setting "Sidelink Channel Type").

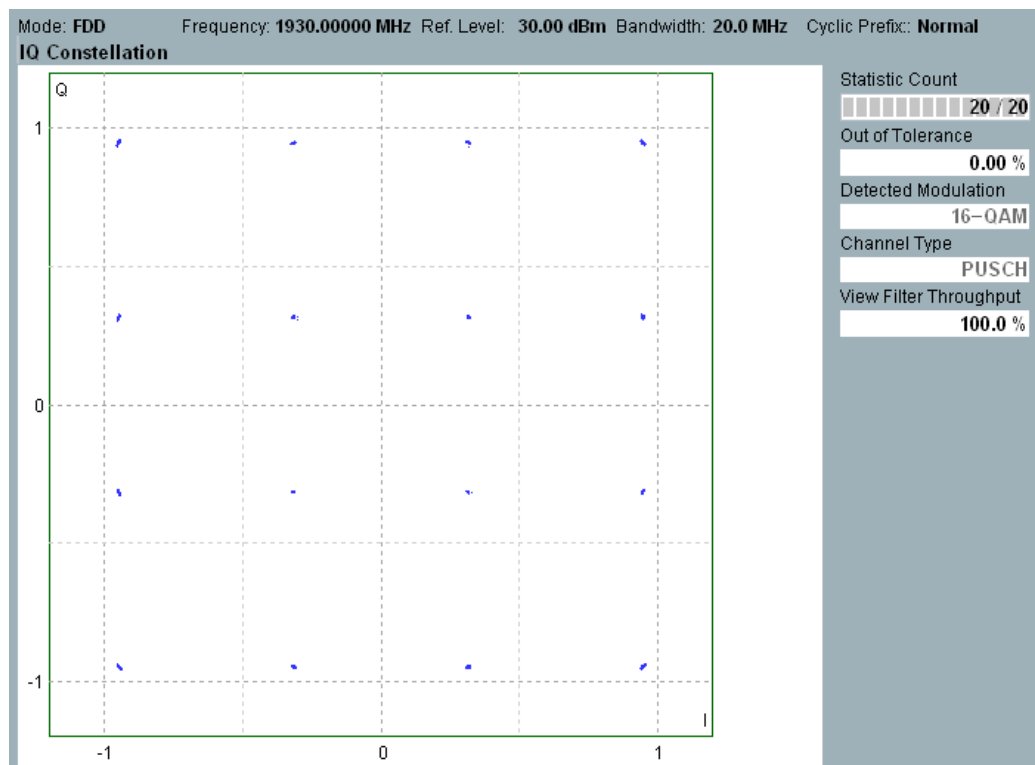


Figure 5-15: I/Q constellation diagram

The constellation diagrams depend on the modulation scheme. For an ideal signal, the QPSK constellation diagram consists of four points, on a circle around the origin, with relative phase angles of 90 deg. The QAM modulation schemes produce rectangular patterns with 16, 64 or 256 points; see example above.

If channel type PUCCH is detected, the constellation diagram shows the SC-FDMA symbols in the frequency domain.

Constellation diagrams give a graphical representation of the signal quality and can help to reveal typical modulation errors causing signal distortions.

The LTE constellation diagram shows the following peculiarities:

- The I/Q amplitudes correspond to the values that the measurement uses for the EVM calculation: A possible I/Q origin offset is already subtracted out.
- Due to the properties of the UL LTE signal, a pure I/Q imbalance causes circular constellation points.

For the parameters on the right, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the diagram contents, see [Section 5.5.3.28, "I/Q constellation results"](#), on page 1229.

5.3.2.10 View RB Allocation Table

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "RBAllocationTable".

The resource block (RB) allocation table provides an overview of the detected RB allocation, including the detected channel type as color-coded information.

With uplink carrier aggregation, there is a separate table for each component carrier.

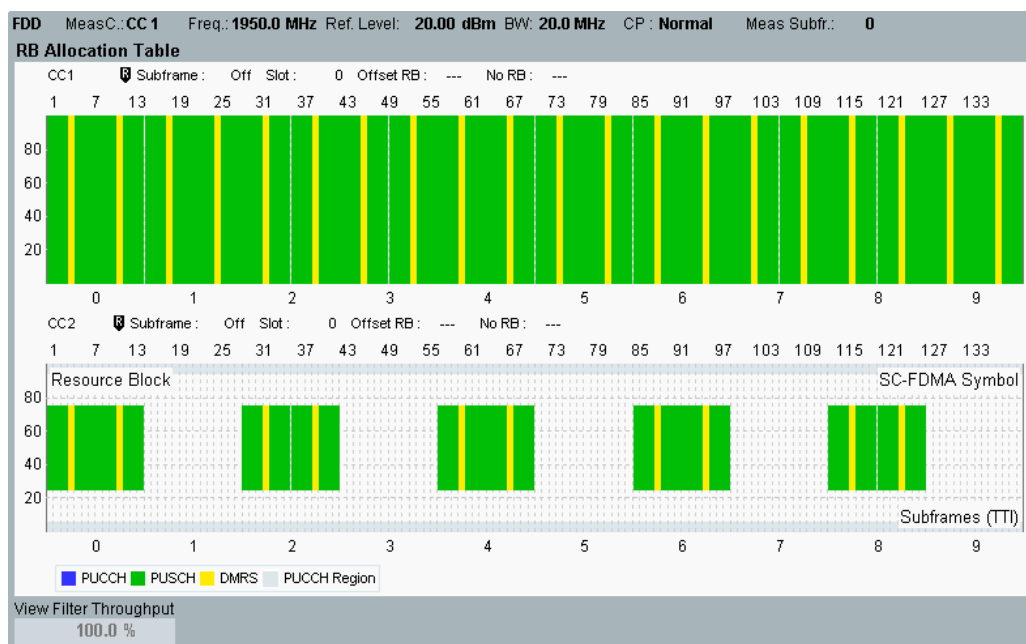


Figure 5-16: RB allocation table, UL carrier aggregation

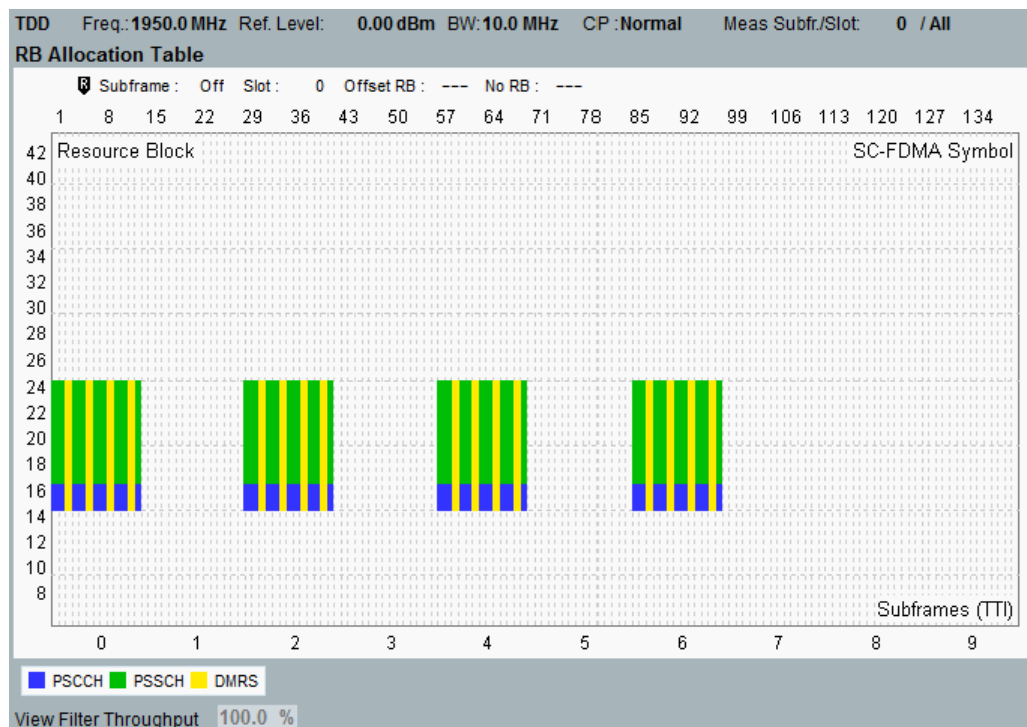


Figure 5-17: RB allocation table, sidelink

Each table presents the time domain on the x-axis (subframes and SC-FDMA symbols) and the frequency domain on the y-axis (resource blocks). The allocated resource blocks are marked by colored bars. The colors indicate the detected channel type as listed in the legend.

"DMRS" refers to the demodulation reference signal within a control channel or shared channel.

For uplink measurements, the PUCCH region indicates the resource block regions at the edges of the channel where a PUCCH can be located. Within this region, automatic channel detection distinguishes between PUCCH and PUSCH. Outside of this region, automatic channel detection always assumes PUSCH.

Related settings and dependencies:

- The number of subframes to be captured and shown by default on the x-axis is configurable. The offset of subframe 0 relative to the trigger event is also configurable. One specific subframe can be selected for slot measurements. For settings, see [Section 5.3.5.6, "Measurement subframe settings"](#), on page 1060.
- The number of resource blocks displayed by default on the y-axis depends on the channel bandwidth.
- The number of SC-FDMA symbols per slot depends on the cyclic prefix type.
- Uplink measurements: The automatic detection of the channel type can be disabled for measurements that evaluate the "Measure Subframe". The detected channel type is then set manually as PUCCH or PUSCH irrespective of the PUCCH region and displayed accordingly in the table. For configuration, see ["Uplink Channel Type"](#) on page 1055.

The range of allocated resource blocks in the measured subframe can also be defined manually. For configuration, see [Section 5.3.5.3, "Uplink RB allocation settings"](#), on page 1057 or [Section 5.3.5.4, "Sidelink RB allocation settings"](#), on page 1059.

- The location of downlink and special subframes for uplink TDD signals is always defined manually and never detected automatically. For configuration, see ["Uplink Downlink"](#) on page 1054.

For background information, see [Section 5.2.11.1, "Resources in time and frequency domain"](#), on page 1008.

For "View Filter Throughput", see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the allocation table contents, see [Section 5.5.3.29, "RB allocation table results"](#), on page 1229.

5.3.2.11 View Power Monitor

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "PowerMonitor".

The view displays the power in the captured subframes. With carrier aggregation, the component carriers are measured separately. One trace per carrier is displayed.

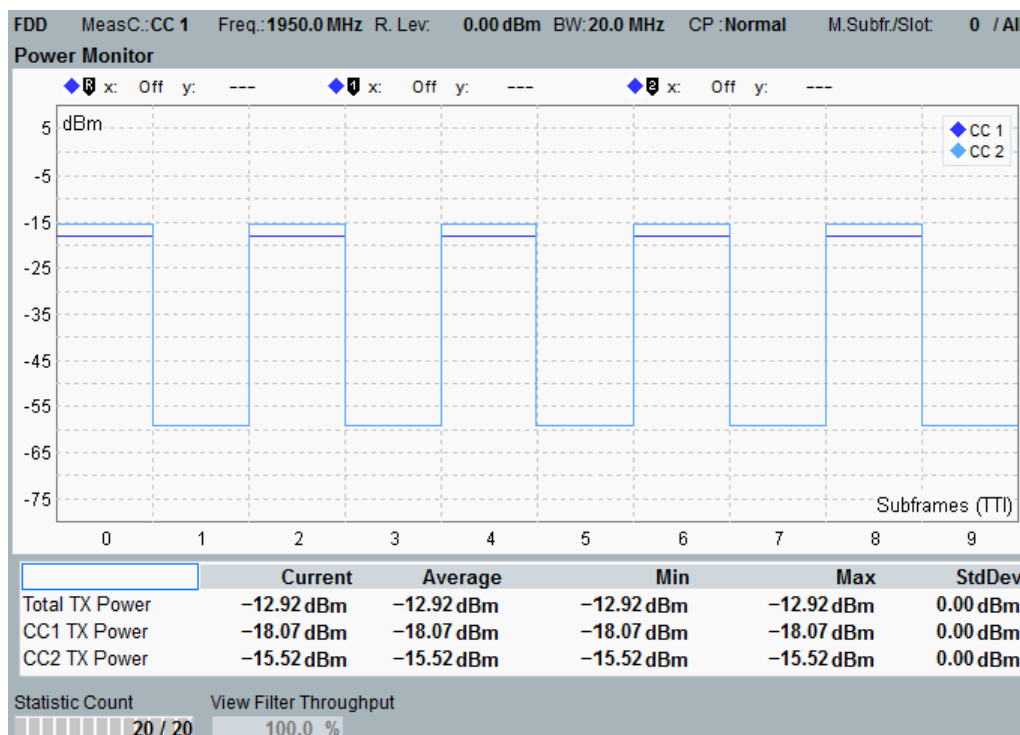


Figure 5-18: Power monitor

Each subframe power value is RMS averaged over the entire subframe, excluding 20 μ s at each subframe border.

The number of subframes to be captured and shown on the x-axis is configurable. The offset of subframe 0 relative to the trigger event is also configurable. For settings, see [Section 5.3.5.6, "Measurement subframe settings"](#), on page 1060.

The table provides a statistical evaluation of the power in the configured "Measure Subframe". It lists the TX power per component carrier and the sum of all component carriers ("Total TX Power").

The power monitor can be used for power control tests, especially for aggregate power control tests (see 3GPP TS 36.521, 6.3.5 "Power Control").

For the parameters at the bottom, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the power monitor results via remote control, see [Section 5.5.3.30, "Power monitor results"](#), on page 1230.

5.3.2.12 View Power Dynamics

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "Power Dynamics".

The view shows a diagram and a statistical overview of related power results. For signals with carrier aggregation, only the configured "Measurement Carrier" is measured.

The following figure shows measurement results for the general time mask.



Figure 5-19: Power Dynamics view, general time mask

The following statements also apply to power dynamics measurements with other time masks.

The diagram shows the UE output power vs time, sampled with $48 T_s$ ($1.5625 \mu s$). The measured trace covers the range from $-1100 \mu s$ to $+2098.4375 \mu s$ relative to the start of the "Measure Subframe" (see [Section 5.3.5.6, "Measurement subframe settings"](#), on page 1060). The diagram shows a subsection of the measured trace, depending on the configured time mask.

The individual ON and OFF power areas are labeled in the diagram. ON power periods are marked by a green horizontal bar. The measured subframe and an eventual SRS period at the end of the "Measure Subframe" are indicated. Exclusion periods are marked by gray vertical bars.

The table below the diagram shows a statistical evaluation of the ON power and OFF power values. The table columns are not necessarily ordered in the same way as the related diagram sections.

Depending on the selected time mask, the following results are available:

- "ON Power, RMS/Peak":
Mean value and peak value of the UE output power over the "Measure Subframe" (minus exclusion period and SRS period)
- "OFF Power (After)", "ON Power (After)":
Mean power in one subframe after the "Measure Subframe" (minus exclusion period)
- "OFF Power (Before)":
Mean power in one subframe before the "Measure Subframe"
- "SRS ON/OFF":
Mean power in the SRS period at the end of the "Measure Subframe"

For the parameters at the bottom of the view, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

For query of the diagram and table contents via remote control, see [Section 5.5.3.31, "Power dynamics results"](#), on page 1233.



Signal configuration for general time mask

The "Measure Subframe" must be on. One subframe before and after the "Measure Subframe" must be off.

However, we recommend configuring two subframes off before and after the "Measure Subframe" (OFF, OFF, ON, OFF, OFF). This configuration guarantees that the "OFF Power (before)" result is not falsified by power contributions of a preceding ON subframe ramped down too slowly. And that the "OFF Power (after)" is not falsified by a subsequent ON subframe ramped up too early.

The trace sections below $-1000 \mu s$ and above $2000 \mu s$ belong to the second subframes before and after the "Measure Subframe". In the figure above, they are on.



Signal configuration for SRS time masks

The "Measure Subframe" and one subframe after the "Measure Subframe" must be on. At the end of the "Measure Subframe", there must be an SRS. Depending on the time mask, it must be a UE-specific SRS (SRS ON, UE transmitting) or a cell-specific SRS (SRS OFF, another UE assumed to transmit).

For FDD, you can achieve this SRS configuration with the combined signal path scenario. In the signaling application, enable SRS and use the default SRS subframe configuration and SRS configuration index. In the measurement, set the "Measure Subframe" to subframe 0 for SRS ON and to subframe 5 for SRS OFF.

The following figures show measurement results for the PUCCH / PUSCH / SRS time mask and the SRS blanking time mask.

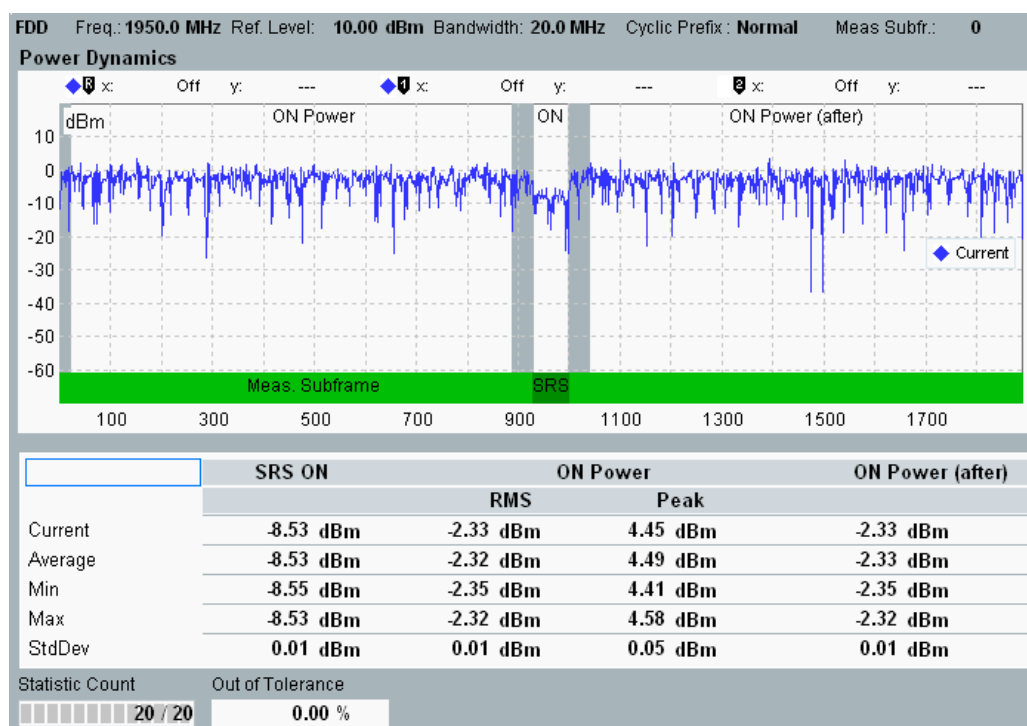


Figure 5-20: Power Dynamics view, PUCCH / PUSCH / SRS time mask

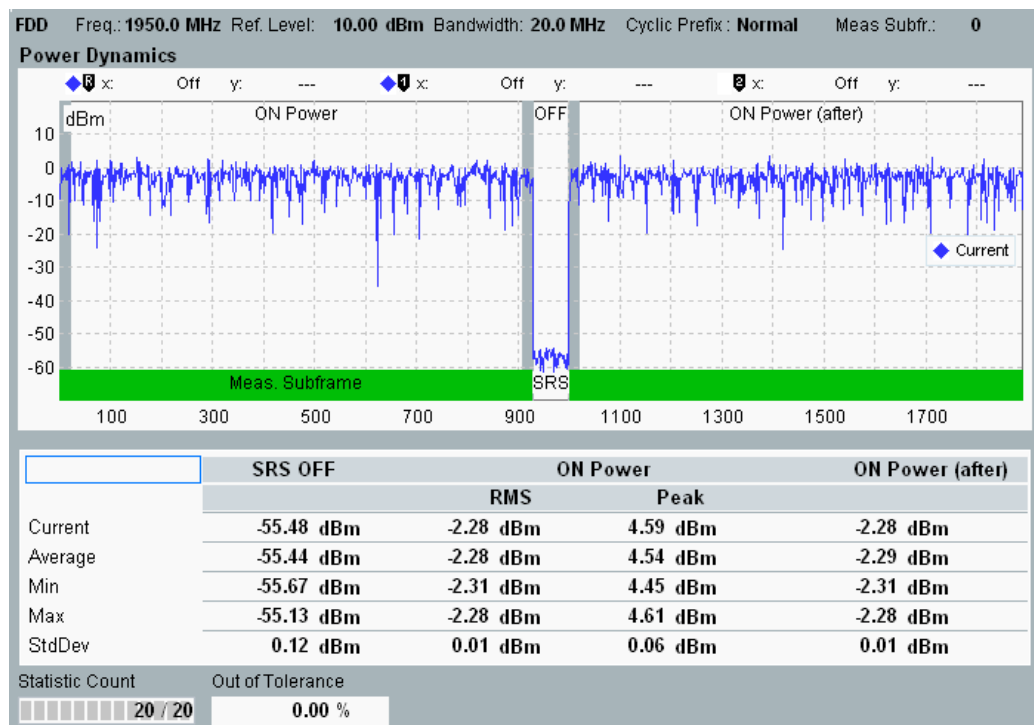


Figure 5-21: Power Dynamics view, SRS blanking time mask

5.3.2.13 View TX Measurement and BLER

Access: "Multi Evaluation" tab > "Display" softkey > "Select View" hotkey > "TX Measurement" or "BLER".

This view contains tables of statistical results for the TX and RX measurements.

FDD Freq.:1950.0 MHz Ref. Level: 0.00 dBm BW:20.0 MHz CP :Normal Meas Subfr./Slot: 0 / All								
TX Measurement								
Detected Allocation	NoRB:	100 OffsetRB:		0				
		Current		Average		Extreme		StdDev
EVM RMS [%] I/h	0.47	0.48	0.51	0.52	0.64	0.66	0.03	0.03
EVM Peak [%] I/h	2.73	2.48	3.42	2.99	5.50	4.28	0.91	0.73
EVM DMRS [%] I/h	0.47	0.49	0.53	0.52	0.82	0.86	0.07	0.08
EVM SRS [%]	---	---	---	---	---	---	---	---
MErr RMS [%] I/h	0.19	0.19	0.20	0.20	0.23	0.23	0.01	0.01
MErr Peak [%] I/h	2.16	1.68	2.51	2.35	4.71	3.94	0.76	0.83
MErr DMRS [%] I/h	0.32	0.36	0.34	0.34	0.43	0.45	0.04	0.05
PhErr RMS [°] I/h	0.25	0.25	0.27	0.27	0.34	0.36	0.02	0.02
PhErr Peak [°] I/h	-1.55	1.39	1.80	1.40	-3.12	2.39	0.57	0.32
PhErr DMRS [°] I/h	0.19	0.19	0.23	0.23	0.42	0.44	0.04	0.04
IQ Offset [dBc]	-62.53		-62.43		-60.98		0.70	
IQ Gain Imbalance [dB]	0.00		0.00		0.00		0.00	
IQ Quadrature Error [°]	0.00		0.01		0.02		0.01	
Freq Error [Hz]	0.99		-0.03		-3.50		1.22	
Timing Error [Ts]	4.08		4.24		4.48		0.15	
OBW [MHz]	17.78		17.79		17.82		0.02	
	Current		Average		Min	Max	StdDev	
TX Power [dBm]	-11.07		-11.07		-11.10	-11.07	0.00	
Peak Power [dBm]	-4.37		-4.24		-4.86	-3.46	0.38	
Statistic Count Out of Tolerance Detected Modulation Detected Channel Type View Filter Throughput								
20 / 20		0.00 %		QPSK		PUSCH		100.0 %

Figure 5-22: TX measurement

TX measurement

The table provides an overview of statistical values obtained in the measured slot. Except for the OBW result, only a single carrier is analyzed. For signals with carrier aggregation, you can select which carrier is measured (setting "Measurement Carrier").

Other detailed views provide a subset of these values:

- Detected allocation:
 "NoRB" is the number of allocated resource blocks.
 "OffsetRB" is the offset of the first allocated resource block from the edge of the allocated UL transmission bandwidth.
- EVM, magnitude error ("MErr"), phase error ("PhErr"):
 The measurement calculates and displays two values (I/h) for low and high EVM window position, see [Section 5.2.9, "Calculation of modulation results"](#), on page 1001.
 For the SC-FDMA data symbols (all symbols except reference symbol), RMS and peak values are provided. The DMRS value refers to the reference symbol (demodulation reference signal).
 An "EVM SRS" line is only shown if SRS is enabled in the modulation settings. And result values are only shown if an SRS signal is detected in the last SC-FDMA symbol of a subframe.
- I/Q offset: The I/Q offset is estimated from the distribution of the constellation points.
- Gain imbalance: absolute value of $(1 + \text{I/Q imbalance})$
 Quadrature error: angular component of $(1 + \text{I/Q imbalance})$

These results are only available if the view "EVM vs subcarrier" is enabled and the measured carrier has a full RB allocation (for example 50 RB for BW 10 MHz). Otherwise, these results are not calculated.

- Carrier frequency error: offset between the measured carrier frequency and the nominal RF frequency of the measured radio channel
- Timing error: difference between the actual timing and the expected timing
The timing error measurement requires a trigger signal from another application or instrument, to derive the expected timing.
A suitable trigger signal is, for example, the frame trigger signal provided by the signaling application.
The unit T_s stands for the basic LTE time unit, see [Section 5.2.11.1, "Resources in time and frequency domain"](#), on page 1008.
- Occupied bandwidth (OBW): width of the frequency range that contains 99 % of the total integrated power of the transmitted spectrum
With carrier aggregation, the transmitted spectrum comprises all carriers. The center of the frequency range is the carrier center frequency / the aggregated bandwidth center frequency.
- TX power: RMS averaged power over all samples in the slot, equivalent to the sum of the powers of all (allocated and non-allocated) RBs.
- Peak power: peak power of all samples in the slot
- RB power: arithmetic mean value of the average powers in all allocated resource blocks

For query of the results via remote control, see [Section 5.5.3.32, "Modulation results \(single values\)"](#), on page 1235.

For the parameters below the "TX Measurement" table, see [Section 5.3.2.15, "Detected configuration and general statistical information"](#), on page 1038.

RX measurement

The RX measurement results are displayed at the bottom of the view (only for uplink measurements, not for sidelink measurements).

RX Measurement										
Subframes				2 800 / 10000		Scheduled		2800		
BLER	8.00 %		ACK	2576		92.00 %		NACK	224	
								8.00 %	DTX	0
										0.00 %

The table provides the following results:

- Subframes: number of already processed subframes and total number of subframes to be processed
All downlink subframes are counted, scheduled subframes and subframes without allocated resource blocks.
- Scheduled: number of already measured subframes (scheduled downlink subframes)
If the measurement is configured correctly, this number equals the sum of the absolute values for ACK, NACK and DTX.
- BLER: block error ratio, percentage of sent scheduled subframes for which no acknowledgment has been received
The result considers received NACK and PDCCH decoding errors (DTX):

$$BLER = (\#NACK + \#DTX) / (\#ACK + \#NACK + \#DTX)$$

- **ACK / NACK / DTX:** number of acknowledgments and negative acknowledgments received over the PUCCH
No answer at all (no ACK and no NACK) is counted as DTX. If a PUSCH is detected instead of a PUCCH, a NACK is counted.
The results are presented as absolute number and as percentage relative to the number of sent scheduled subframes.

For query of the results via remote control, see [Section 5.5.3.33, "BLER results"](#), on page 1242.

For information about this measurement, see [Section 5.2.5, "How to perform an RX test"](#), on page 995.

5.3.2.14 Statistical results

Most result diagrams and result tables provide statistical results.

The statistical values are calculated as follows:

- **"Current":** Result obtained in the last measurement interval. For some modulation results, an RMS value and a peak value are available. They are calculated as the average and peak of all samples in the measured slot.
- **"Average":** Weighted average of all "Current" values since the start of the measurement.
- **"Extreme", "Min", "Max":** Largest or smallest "Current" value since the start of the measurement. For EVM, magnitude error and phase error, the extreme value of all modulation symbols in the slot. No average over data symbols is involved.
- **"Std Dev":** Standard deviation of all "Current" values since the start of the measurement.

5.3.2.15 Detected configuration and general statistical information

The following information is available in most result views.

Statistic Count	1038
Modulation	1039
Channel Type (uplink)	1039
Channel Type (sidelink)	1039
Out of Tolerance	1039
View Filter Throughput	1039

Statistic Count

Progress bar for the measurement. During the first single shot after the start of the measurement, the bar shows the number of elapsed measured slots or subframes relative to the configured "Statistic Count". A filled progress bar indicates that the first shot is complete and the statistical depth has been reached.

Note that two slots can be measured when the defined number of subframes is captured once. For an example, see [Section 5.2.6, "Defining the scope of a TX measurement"](#), on page 996.

Modulation

Modulation scheme in the measured slot. The modulation scheme is either detected automatically or selected manually (see ["Modulation Scheme"](#) on page 1062). If channel type PUCCH is detected, QPSK is displayed for the modulation scheme, because the QPSK limits are applied in that case.

Remote command:

```
FETCh:LTE:MEAS<i>:MEValuation:MODulation:DMODulation?
```

Channel Type (uplink)

Channel type (PUCCH or PUSCH) for the measured slot. The channel type is either detected automatically or selected manually, see ["Uplink Channel Type"](#) on page 1055.

Remote command:

```
FETCh:LTE:MEAS<i>:MEValuation:MODulation:DCHType?
```

```
FETCh:LTE:MEAS<i>:MEValuation:SEMask:DCHType?
```

```
FETCh:LTE:MEAS<i>:MEValuation:ACLR:DCHType?
```

Channel Type (sidelink)

Channel type (PSCCH or PSSCH) evaluated for modulation results. The channel type is selected manually, see ["Sidelink Channel Type"](#) on page 1055.

Remote command:

```
FETCh:LTE:MEAS<i>:MEValuation:MODulation:SCHType?
```

Out of Tolerance

Percentage of measurement intervals that were failed because at least one result exceeded the limits.

All results that use the same statistic count setting contribute to the same "Out of Tolerance" result.

View Filter Throughput

Percentage of measurement intervals in which the detected signal configuration matched the "View Filter" settings. Only slots which contribute to the "View Filter Throughput" are displayed, counted, and used for the statistical results. Other slots are rejected. For filter settings, see [Section 5.3.5, "Measurement control settings"](#), on page 1050.

The "View Filter Throughput" is evaluated in a moving window of 1000 frames. If a non-matching signal configuration is changed according to the "View Filter" settings, the throughput increases linearly from 0 % to 100 %.

Remote command:

```
FETCh:LTE:MEAS<i>:MEValuation:VFTHroughput?
```

5.3.2.16 Selecting and modifying views

Press the "Display" softkey to show hotkeys for view configuration. They change the appearance and contents of the active view. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Select View ..."	Switch to a certain detailed view or overview. Alternatively, select a diagram in the overview and press [ENTER] or the rotary knob.
"Select Trace ..."	Select the trace types to be displayed in the view.
"X Scale... / Y Scale..."	Modify the ranges of the x-axis and the y-axis.
"EVM Window Position ..."	Select which of the results calculated at the low and high extremities of the EVM window is displayed, see Section 5.2.9, "Calculation of modulation results" , on page 1001.
"Margin On Off"	For the spectrum emission mask measurement: Show or hide margin results.

Additional options are available in the "Measurement Control" section of the configuration dialog.

5.3.2.17 Using markers

Press the "Marker" softkey to show hotkeys for marker configuration. They activate markers and modify their position. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Ref. Marker..."	Enable or disable the reference marker and select the marker position. If several traces can be displayed, a trace can also be selected.
"Marker 1/2..."	Enable or disable marker 1 or 2 and define the marker position (absolute or relative to the reference marker). Depending on the trace mode, a trace can also be selected.
"Select Trace Mode"	Define whether markers 1 and 2 are set to the same trace as the reference marker (collective) or to selectable individual traces.

5.3.3 Signal routing and analyzer settings

Access: "Multi Evaluation" tab > "Config" hotkey.

The parameters at the top of the configuration tree configure the RF input path. Most parameters are common measurement settings. They have the same value in all measurements (for example PRACH measurement and multi-evaluation measurement).

Signal Type	Uplink
Duplex Mode	FDD
Scenario	StandAlone (Non Signaling)
eMTC	☒
MaxBandwidth 5 MHz	☐
RF Routing	Connector: 1.RF1COM Converter: 1.RFRX1
External Attenuation (Input)	0.00 dB
Carrier Aggregation Mode	Off
Band - Channel	Band 1 18300 Ch
Frequency	1950.000000 MHz
Frequency Offset	0 Hz
Expected Nominal Power	0.00 dBm Ref.Level: 0.00 dBm
User Margin	0.00 dB
Mixer Level Offset	0 dB

Figure 5-23: Uplink signal routing and analyzer settings

For sidelink, some settings are fixed.

Signal Type	Sidelink
Duplex Mode	TDD
Scenario	StandAlone (Non Signaling)
RF Routing	Connector: RF1COM Converter: RFRX1
External Attenuation (Input)	0.00 dB
Carrier Aggregation Mode	Off
Band - Channel	Band 47 ---
Frequency	1950.000000 MHz
Frequency Offset	0 Hz
Expected Nominal Power	0.00 dBm Ref.Level: 0.00 dBm
User Margin	0.00 dB
Mixer Level Offset	8 dB

Figure 5-24: Sidelink signal routing and analyzer settings

Signal Type.....	1041
Duplex Mode.....	1042
Scenario = StandAlone.....	1042
Scenario = Combined Signal Path.....	1042
Scenario = Measure@ProtocolTest.....	1043
eMTC.....	1043
Max Bandwidth 5 MHz.....	1043
RF Routing.....	1044
Device Mode.....	1044
External Attenuation (Input).....	1044
Carrier Aggregation Mode.....	1044
Band, Channel, Frequency.....	1045
Frequency Offset.....	1045
Expected Nominal Power.....	1045
User Margin.....	1046
Mixer Level Offset.....	1046

Signal Type

Selects the type of signal to be measured.

"Uplink"	LTE uplink signal with PUSCH or PUCCH. Option R&S CMW-KM500 or -KM550 is required.
"Sidelink"	V2X sidelink signal with PSSCH and PSCCH. Option R&S CMW-KM570 is required.

Remote command:

CONFigure:LTE:MEAS<i>:STYPe

Duplex Mode

Selects the duplex mode of the LTE signal: FDD or TDD.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

CONFigure:LTE:MEAS<i>:DMODE (SA)

CONFigure:LTE:SIGN<i>[:PCC]:DMODE (CSP)

CONFigure:LTE:SIGN<i>:SCC<c>:DMODE (CSP)

CONFigure:LTE:SIGN<i>[:PCC]:DMODE:UCSPecific (CSP)

Scenario = StandAlone

The measurement is used as a standalone application.

Remote command:

ROUTE:LTE:MEAS<i>:SCENario:SALone

ROUTE:LTE:MEAS<i>:SCENario?

ROUTE:LTE:MEAS<i>?

Scenario = Combined Signal Path

Scenario	Combined Signal Path		
Controlled by	LTE Sig1	Set A	SCC2
CC - SCC	CC1 - SCC1	CC2 - SCC2	

You can use an LTE signaling application (option R&S CMW-KS500/-KS550) in parallel to the LTE measurement.

The additional parameter "Controlled by" selects the signaling application and the uplink carrier to be measured.

For contiguous uplink CA, you select a set of carriers (for example "Set A") instead of a single carrier. An additional field to the right indicates the component carrier used by the measurement to select the RF path. A single input path is used to measure all aggregated carriers. The carriers of the set (PCC / SCC<n>) are listed in an additional line. The name "CCn" is used within the measurement to identify the carriers.

In a multi-CMW setup, ensure that the selected carrier or set and the measurement instance are in the same CMW. Otherwise, the measurement fails. See also "[Measurement instances in multi-CMW setups](#)" on page 1000.

Most parameters described in this section and some measurement control settings display values determined by the signaling application. The corresponding measurement settings are remembered in the background and displayed again when switching back to the standalone scenario.

Connection status information of the signaling application is displayed at the bottom of the measurement views. Softkeys and hotkeys provide access to the settings of the signaling application and allow you to switch the downlink signal on or off, see [Section 5.3.9, "Additional softkeys and hotkeys"](#), on page 1086.

For additional information, see [Section 5.2.8, "Parallel signaling and measurement"](#), on page 999.

Remote command:

```
ROUTE:LTE:MEAS<i>:SCENario:CSPath
ROUTE:LTE:MEAS<i>:SCENario?
ROUTE:LTE:MEAS<i>?
CONFigure:LTE:MEAS<i>:CAGGregation:MAPing?
CONFigure:LTE:MEAS<i>:CAGGregation:MAPing:PCC?
CONFigure:LTE:MEAS<i>:CAGGregation:MAPing:SCC<c>?
SENSe:LTE:MEAS<i>:CAGGregation:FSHWare?
```

Scenario = Measure@ProtocolTest

You can use an LTE protocol test application in parallel to the LTE measurement. The protocol test application is selected by the additional parameter "Controlled by".

The signal routing and analyzer settings described in this section are ignored by the measurement application. Configure the corresponding settings within the protocol test application.

Remote command:

```
ROUTE:LTE:MEAS<i>:SCENario:MAPRotocol
ROUTE:LTE:MEAS<i>:SCENario?
```

eMTC

Enables enhanced machine type communication (eMTC).

Effects:

- The measurement expects an eMTC narrowband signal. To select the narrowband, see ["Narrowband"](#) on page 1057.
- The in-band emission limits are related to the narrowband, not to the channel, see [Section 5.3.7.5, "In-band emissions limits"](#), on page 1075.
- SC-FDMA symbols in guard periods for narrowband retuning are detected and excluded from the modulation results.
- Some features are not relevant for eMTC and cannot be enabled (carrier aggregation, multicluster, SRS, automatic channel type detection).

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:EMTC:ENABle (SA)
CONFigure:LTE:SIGN<i>[:PCC]:EMTC:ENABle (CSP)
```

Max Bandwidth 5 MHz

Selects whether the maximum bandwidth for eMTC is enhanced from 1.4 MHz to 5 MHz.

With enabled checkbox, the RB allocation settings allow up to 24 RBs instead of max 6 RBs, see ["No. of RBs, Offset RB - no multicluster"](#) on page 1058.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFigure:LTE:MEAS<i>:EMTC:MB<number> (SA)`

`CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MB<number> (CSP)`

RF Routing

Selects the input path for the measured RF signal, i.e. the input connector and the RX module to be used.

In the standalone (SA) scenario, these parameters are controlled by the measurement. In the combined signal path (CSP) scenario, they are controlled by the signaling application.

Remote command:

`ROUTE:LTE:MEAS<i>:SCENario:SALone (SA)`

`ROUTE:LTE:SIGN<i>:SCENario:... (CSP)`

Device Mode

The device mode is relevant if you have an R&S CMX500 connected to the R&S CMW. Some measurement applications are then accessible via the R&S CMsquares GUI described in the R&S CMX500 documentation and via the GUI described in this documentation. Depending on the GUI used, the software must adapt its behavior. The behavior is controlled via the "Device Mode" setting.

Only the value "CMW" is compatible to the GUI and SCPI commands described in this document. So never use the value "CMX" in this context. If you do not see the setting in the GUI, it is set to "CMW".

Configure the setting before starting the measurement.

External Attenuation (Input)

Defines the value of an external attenuation (or gain, if the value is negative) in the input path. The power readings are corrected by the external attenuation value.

The external attenuation value is also used in the calculation of the maximum configurable input power value.

If a correction table for frequency-dependent attenuation is active for the chosen connector, then the table name and a button are displayed. Press the button to display the table entries.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFigure:LTE:MEAS<i>:RFSettings:EATTenuation (SA)`

`CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:INPut (CSP)`

`CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:INPut (CSP)`

Carrier Aggregation Mode

Selects whether the measured signal uses carrier aggregation or not, see [Section 5.3.4, "Carrier aggregation settings"](#), on page 1046.

Band, Channel, Frequency

Center frequency of the RF analyzer for measurements without carrier aggregation. Set this frequency to the frequency of the measured RF signal to obtain a meaningful measurement result. The relation between operating band, frequency and channel number is defined by 3GPP (see [Section 5.2.11.4, "Frequency bands"](#), on page 1012).

You can specify the RF frequency in two ways:

- Enter the frequency directly. The band and channel settings can be ignored or used for validation of the entered frequency. For validation, select the designated band. The channel number resulting from the selected band and frequency is displayed. For an invalid combination, no channel number is displayed.
- Select a band and enter a channel number valid for this band. The measurement calculates the resulting frequency.

In the standalone (SA) scenario, these parameters are controlled by the measurement. In the combined signal path (CSP) scenario, they are controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:BAND (SA)
CONFigure:LTE:MEAS<i>:RFSettings:CC<no>:FREQuency (SA)
CONFigure:LTE:SIGN<i>[:PCC]:BAND (CSP)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:UL (CSP)
CONFigure:LTE:SIGN<i>:SCC<c>:BAND (CSP)
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:UL (CSP)
```

Frequency Offset

Positive or negative frequency offset to be added to the specified center frequency of the RF analyzer.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:RFSettings:FOFFset (SA)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL (CSP)
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:UL (CSP)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL:UCSPecific (CSP)
```

Expected Nominal Power

Defines the nominal power of the RF signal to be measured. An appropriate value for LTE signals is the peak output power at the DUT during the measurement.

The reference level is calculated as follows:

Reference level = Expected Nominal Power + User Margin

Note: The actual input power at the connectors must be within the level range of the selected RF input connector; refer to the specifications document. If all power settings are configured correctly, the actual power equals the reference level minus the external attenuation setting.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:RFSettings:ENPower (SA)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPower (CSP)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPower (CSP)
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPower (CSP)
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPower (CSP)
```

User Margin

Margin that the measurement adds to the expected nominal power to determine its reference power (reference level). The margin is typically used to account for the known variations of the RF input signal power, for example, the variations due to a specific channel configuration.

The variations (crest factor) depend on the LTE signal parameters, in particular the modulation scheme. If the expected nominal power is set to the peak power during the measurement, a 0-dB user margin is sufficient.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:RFSettings:UMARgin (SA)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UMARgin (CSP)
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UMARgin (CSP)
```

Mixer Level Offset

Varies the input level of the mixer in the analyzer path. A negative offset reduces the mixer input level. A positive offset increases the mixer input level. Optimize the mixer input level according to the properties of the measured signal.

Mixer level offset	Advantages	Possible shortcomings
< 0 dB	Suppression of distortion (e.g. of the inter-modulation products generated in the mixer).	Lower dynamic range (due to smaller signal-to-noise ratio).
> 0 dB	High signal-to-noise ratio, higher dynamic range.	Risk of intermodulation, smaller overdrive reserve.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:RFSettings:MLOffset (SA)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:MLOffset (CSP)
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:MLOffset (CSP)
```

5.3.4 Carrier aggregation settings

Access: "Multi Evaluation" tab > "Config" hotkey.

Carrier aggregation (CA) is only supported for UL signals, not for sidelink signals.

If the CA mode is set to contiguous uplink carrier aggregation, additional settings are displayed for configuration of the individual component carriers (CC1, CC2, ...). The band, channel and frequency settings are moved into this node, as well as the channel bandwidth setting and the physical cell ID setting from the "Measurement Control" section.

These settings are described in the following. Other parameters that depend on the carrier aggregation mode are the spectrum limits and the "network signaled value". Some measurement result views are also modified depending on the carrier aggregation mode.

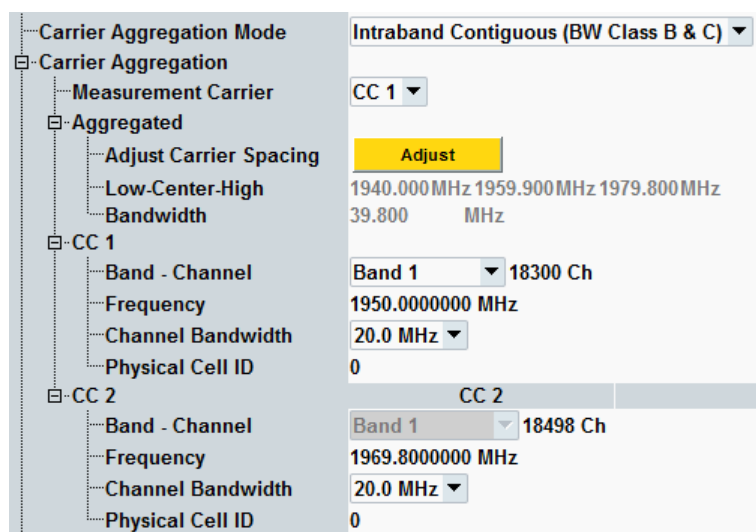


Figure 5-25: Carrier aggregation settings

Configuring the component carrier frequencies

To ensure that the component carriers are aggregated contiguously in a nonsignaling measurement, proceed as follows:

1. Select the channel bandwidths of all carriers (CC1, CC2, ...).
2. Specify the CC1 center frequency.
3. Select whether the CC2 center frequency is lower or higher than the CC1 center frequency:
Specify any CC2 frequency above or below the CC1 frequency.
4. Press the "Adjust" button.

The CC2 center frequency is adjusted automatically, so that the spacing between CC1 and CC2 is correct. The channel spacing is defined in 3GPP TS 36.101, section 5.7.1A.

If you aggregate three carriers, CC3 is placed next to CC2. In the spectrum, you get the sequence CC3-CC2-CC1 or CC1-CC2-CC3.

If you aggregate four carriers, CC4 is placed next to CC3. In the spectrum, you get the sequence CC4-CC3-CC2-CC1 or CC1-CC2-CC3-CC4.

For background information, see [Section 5.2.7, "Uplink carrier aggregation"](#), on page 998.

The settings in [Figure 5-25](#) are described in the following.

Carrier Aggregation Mode	1048
Measurement Carrier	1048
Adjust Carrier Spacing	1049
Low, Center, High, Bandwidth	1049
Band, Channel, Frequency	1049
Channel Bandwidth	1050
Physical Cell ID	1050

Carrier Aggregation Mode

Selects how many component carriers with intraband contiguous aggregation are measured.

The following modes are supported:

- **"Off"**:
Only one carrier is measured. The other CA settings are hidden.
- **"Intraband Contiguous (BW Class B & C)"**:
Two carriers with intraband contiguous aggregation are measured.
Option R&S CMW-KM502/552 is required for FDD/TDD.
- **"Intraband Contiguous (BW Class D)"**:
Three carriers with intraband contiguous aggregation are measured.
Option R&S CMW-KM504/554 is required for FDD/TDD.
- **"Intraband Contiguous (BW Class E)"**:
Four carriers with intraband contiguous aggregation are measured.
Option R&S CMW-KM504/554 is required for FDD/TDD.

Carrier aggregation is not allowed for eMTC.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is set indirectly via the "Controlled by" setting, see ["Scenario = Combined Signal Path"](#) on page 1042.

Remote command:

`CONFigure:LTE:MEAS<i>:CAGGregation:MODE (SA)`

`ROUTe:LTE:MEAS<i>:SCENario:CSPath (CSP)`

`CONFigure:LTE:MEAS<i>:CAGGregation:MODE:CSPath? (CSP)`

Measurement Carrier

Selects a component carrier for single carrier measurements. If the selected carrier is not available, a "Sync Error" or an "Underdriven" error is displayed.

Single carrier measurement views, evaluating the "Measurement Carrier":

- EVM vs symbol / vs subcarrier, magnitude error, phase error
- Equalizer spectrum flatness
- Power dynamics
- I/Q diagram
- TX measurement

Multicarrier measurements, evaluating all carriers:

- In-band emission diagrams
- Spectrum ACLR

- Spectrum emission mask
- Power monitor
- RB allocation table

Remote command:

```
CONFigure:LTE:MEAS<i>:CAGGregation:MCARrier:ENHanced
```

Adjust Carrier Spacing

Adjusts the carrier frequencies for intraband contiguous aggregation, see ["Configuring the component carrier frequencies"](#) on page 1047.

The button is only relevant and visible for the standalone (SA) scenario. In the combined signal path (CSP) scenario with contiguous UL CA, the signaling application adjusts the frequencies.

Remote command:

```
CONFigure:LTE:MEAS<i>:CAGGregation:ACSPacing (SA)
```

```
CONFigure:LTE:SIGN<i>:CAGGregation:SET (CSP)
```

Low, Center, High, Bandwidth

Information about the aggregated bandwidth, resulting from the configured frequencies and the channel bandwidths of the carriers.

"Low", "Center", "High" indicates the lower edge of the aggregated bandwidth, the center frequency and the higher edge. "Bandwidth" indicates the width of the aggregated bandwidth (higher edge minus lower edge).

The displayed bandwidth can be slightly less than the sum of the carrier channel bandwidths. For details, see 3GPP TS 36.101, sections 5.6A and 5.7.1A.

Remote command:

```
CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:LOW?
```

```
CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:CENTer?
```

```
CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:HIGH?
```

```
CONFigure:LTE:MEAS<i>:CAGGregation:CBANdwidth:AGGRegated?
```

Band, Channel, Frequency

The "Frequency" defines the carrier center frequency.

With carrier aggregation, the carriers must be aggregated contiguously. The correct channel spacing depends on the bandwidths. For standalone measurements, see also ["Configuring the component carrier frequencies"](#) on page 1047.

The "Band" setting is identical for all carriers. The relation between operating band, frequency and channel number is defined by 3GPP (see [Section 5.2.11.4, "Frequency bands"](#), on page 1012).

You can specify the RF frequency in two ways:

- Enter the frequency directly. The band and channel settings can be ignored or used for validation of the entered frequency. For validation, select the designated band. The channel number resulting from the selected band and frequency is displayed. For an invalid combination, no channel number is displayed.
- Select a band and enter a channel number valid for this band. The measurement calculates the resulting frequency.

In the standalone (SA) scenario, these parameters are controlled by the measurement. In the combined signal path (CSP) scenario, they are controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:BAND (SA)
CONFigure:LTE:MEAS<i>:RFSettings:CC<no>:FREQuency (SA)
CONFigure:LTE:SIGN<i>[:PCC]:BAND (CSP)
CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:UL (CSP)
CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:UL (CSP)
```

Channel Bandwidth

Set the bandwidths in accordance with the measured LTE signal.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:CC<no>:CBANDwidth (SA)
CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL (CSP)
CONFigure:LTE:SIGN<i>:CELL:BANDwidth:SCC<c>:DL (CSP)
```

Physical Cell ID

Set the physical cell IDs in accordance with the measured LTE signal.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:CC<no>:PLCid (SA)
CONFigure:LTE:SIGN<i>:CELL[:PCC]:PCID (CSP)
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:PCID (CSP)
```

5.3.5 Measurement control settings

The "Measurement Control" parameters configure the scope of the LTE multi-evaluation measurement.

• Miscellaneous, part 1.....	1051
• View filter settings.....	1056
• Uplink RB allocation settings.....	1057
• Sidelink RB allocation settings.....	1059
• Miscellaneous, part 2.....	1059
• Measurement subframe settings.....	1060
• Modulation settings.....	1061
• Spectrum settings.....	1065
• Power settings.....	1066
• BLER settings.....	1068
• List mode settings.....	1069

5.3.5.1 Miscellaneous, part 1

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node.

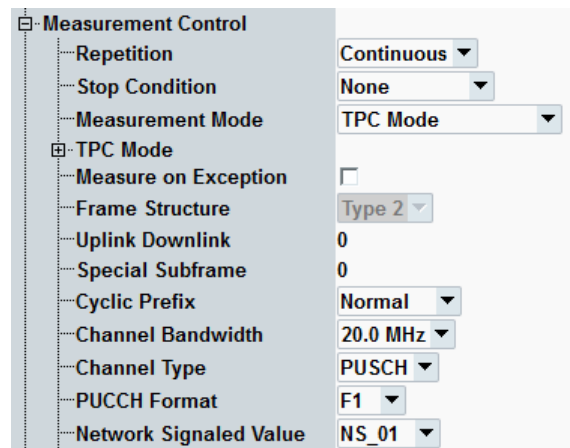


Figure 5-26: Uplink measurement control settings - part 1

For sidelink, some settings are not relevant and hidden.

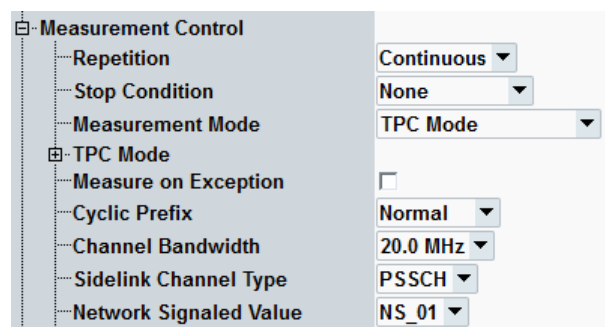


Figure 5-27: Sidelink measurement control settings - part 1

Repetition.....	1052
Stop Condition.....	1052
Measurement Mode.....	1052
TPC Mode.....	1053
Measure on Exception.....	1053
Frame Structure.....	1054
Uplink Downlink.....	1054
Special Subframe.....	1054
Cyclic Prefix.....	1054
Channel Bandwidth.....	1055
Uplink Channel Type.....	1055
Sidelink Channel Type.....	1055
PUCCH Format.....	1055
Network Signaled Value.....	1056

Repetition

Defines how often the measurement is repeated if it is not stopped explicitly or by a failed limit check.

- **"Continuous"**: The measurement is continued until it is explicitly terminated. The results are periodically updated.
- **"Single Shot"**: The measurement is stopped after one measurement cycle.

"Single Shot" is preferable if only a single measurement result is required under fixed conditions, which is typical for remote-controlled measurements. The continuous mode is suitable to monitor the evolution of the measurement results and to observe how they depend on the measurement configuration, which is typically done in manual control.

Remote command:

`CONFigure:LTE:MEAS<i>:MEValuation:REPetition`

Stop Condition

Specifies the conditions for an early termination of the measurement:

- **"None"**: The measurement is performed according to its "Repetition" mode and "Statistic Count", irrespective of the limit check results.
- **"On Limit Failure"**: The measurement is stopped when one of the limits is exceeded, irrespective of the repetition mode set. If no limit failure occurs, it is performed according to its "Repetition" mode and "Statistic Count". Use this setting for measurements that are intended for checking limits, e.g. production tests.

Remote command:

`CONFigure:LTE:MEAS<i>:MEValuation:SCONdition`

Measurement Mode

Selects/displays the measurement mode.

- **"Normal"**:
In normal mode, the measurement uses a constant reference level for the entire measurement duration.
Use the normal mode for nonsignaling tests. For signaling tests, the normal mode is only suitable if the controlling application does not modify the reference level during the measurement.
- **"TPC Mode"**:
In TPC mode, the measurement configures the reference level once per subframe. In the combined signal path scenario, the reference level is controlled by the signaling application. In the standalone scenario, the reference levels must be configured, see parameter [TPC Mode](#).
The TPC mode has been designed for TPC measurements where the UE power is ramped up or down over a wide range, exceeding the dynamic range of the measurement. A typical example is a conformance test case according to 3GPP TS 36.521, section 6.3.5.2 "Power Control Relative power tolerance".
Note that for spectrum measurements, we recommend the normal mode.
- **"Multi Eval List Mode"**:
The list mode can only be enabled via a remote control command. You can disable it by selecting one of the other modes.
For an introduction to the list mode, see [Section 5.2.10, "List mode"](#), on page 1002. Option R&S CMW-KM012 is required.

While the combined signal path scenario is active, this parameter is not configurable.

Remote command:

```
CONFfigure:LTE:MEAS<i>:MEvaluation:MMODE
```

```
CONFfigure:LTE:MEAS<i>:MEvaluation:LIST
```

TPC Mode

The node configures a sequence of expected nominal power values to be applied during a nonsignaling measurement in TPC mode. The node is only visible if the measurement mode "TPC Mode" is selected.

TPC Mode	Nr of Subframes	Exp.Nom. Power	Reference Level
[0]	320	0.00 dBm	0.00 dBm
[1]	320	0.00 dBm	0.00 dBm
[2]	320	0.00 dBm	0.00 dBm

For each list entry, you can configure the number of subframes and the expected nominal power applicable to these subframes. The resulting reference level is calculated and displayed for information. The first entry is coupled to the standard expected nominal power setting, see [Duplex Mode](#).

The list defines the expected nominal powers to be applied within one sequence of measured subframes, configured via parameter "[No. of Subframes](#)" on page 1061.

Example:

- Configured number of subframes = 30, to be measured two times (defined via statistic count)
- List entry 0 = 10 subframes, 0 dBm
List entry 1 = 15 subframes, 10 dBm
List entry 2 = 10 subframes, 20 dBm
- Resulting used expected nominal power:
Measured subframe 1 to 10: entry 0, 0 dBm
Measured subframe 11 to 25: entry 1, 10 dBm
Measured subframe 26 to 30: entry 2, 20 dBm
Second measurement of 30 subframes: again entries 0 to 2 are applied

If the configured number of subframes to be measured is longer than the sequence defined via the list entries, the last list entry is used for the remaining subframes.

Note: Between two list entries, the measurement reconfigures the reference level according to the next entry. This action can impair the accuracy of measurement results for the preceding and the subsequent subframe. Keep this effect in mind when configuring the measurement and evaluating the results. Do not reconfigure the reference level directly before/after the "Measure Subframe".

Remote command:

```
CONFfigure:LTE:MEAS<i>:MEvaluation:TMODE:SCount
```

```
CONFfigure:LTE:MEAS<i>:MEvaluation:TMODE:ENPower
```

```
CONFfigure:LTE:MEAS<i>:MEvaluation:TMODE:RLEVEL?
```

Measure on Exception

Specifies whether results that the measurement identifies as faulty or inaccurate are rejected. A faulty result occurs, for example, when an overload is detected. In remote control, the cause of the error is indicated by the "reliability indicator".

- **Off:** Faulty results are rejected. The measurement is continued. The statistical counters are not reset. Using this mode ensures that a single faulty result does not affect the entire measurement.
- **On:** Results are never rejected. Use this mode for development purposes, if you want to analyze the reasons for occasional wrong transmissions.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:MOEXception
```

Frame Structure

Displays the frame structure of the uplink signal as defined in 3GPP TS 36.211. The value is set implicitly via the **Duplex Mode** ("Type 1" = FDD, "Type 2" = TDD).

For a TDD signal with signal type uplink, the additional parameters "Uplink Downlink" and "Special Subframe" have to be set to specify the frame structure.

Remote command:

```
CONFigure:LTE:MEAS<i>:FSTRucture?
```

Uplink Downlink

UL-DL configuration of a TDD signal, as defined in 3GPP TS 36.211, section 4, "Frame Structure". Each configuration defines a combination of uplink subframes, downlink subframes and special subframes within a radio frame.

The selected configuration is visualized in the "RB Allocation Table".

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:ULDL (SA)
CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULDL (CSP)
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULDL (CSP)
CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific (CSP)
```

Special Subframe

Configuration of the special subframes of a TDD signal, as defined in 3GPP TS 36.211, section 4, "Frame Structure". Each configuration defines the inner structure of a special subframe, i.e. the lengths of DwPTS, GP and UpPTS.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:SSUBframe (SA)
CONFigure:LTE:SIGN<i>:CELL[:PCC]:SSUBframe (CSP)
CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SSUBframe (CSP)
CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific (CSP)
```

Cyclic Prefix

Normal or extended cyclic prefix. Set this parameter in accordance with the cyclic prefix of the measured LTE signal. The number of measured data symbols is 7 (6) for normal (extended) cyclic prefix duration.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:CPRefix (SA)
CONFigure:LTE:SIGN<i>:CELL:CPRefix (CSP)
```

Channel Bandwidth

Channel bandwidth in the range 1.4 MHz to 20 MHz. Set the bandwidth in accordance with the measured LTE signal.

To configure the channel bandwidth for carrier aggregation, see [Section 5.3.4, "Carrier aggregation settings"](#), on page 1046.

The parameter is a common measurement setting, i.e. it has the same value in all measurements (e.g. PRACH measurement and multi-evaluation measurement).

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:CC<no>:CBANDwidth (SA)
CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL (CSP)
CONFigure:LTE:SIGN<i>:CELL:BANDwidth:SCC<c>:DL (CSP)
```

Uplink Channel Type

Configures the channel type detection for the measured subframe of the measured carrier. With carrier aggregation, the measured carrier is the configured "Measurement Carrier".

The channel type can usually be detected automatically. If automatic detection fails, for example, because the signal contains a PUCCH outside of the PUCCH region, select the channel type manually.

- **AUTO**: The channel type is detected automatically. Within the PUCCH region, it can be PUCCH or PUSCH. Outside of the PUCCH region, PUSCH is always assumed.
Not supported for eMTC.
- **PUSCH**: Measure a PUSCH.
- **PUCCH**: Measure a PUCCH.

To verify that the automatic detection works, you can use the RB allocation table, see [Section 5.3.2.10, "View RB Allocation Table"](#), on page 1029.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:CTYPE
```

Sidelink Channel Type

Selects the channel type to be evaluated for modulation results.

Modulation results in this context are: EVM, magnitude error, phase error, equalizer spectrum flatness, I/Q constellation and modulation result tables.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:SCType
```

PUCCH Format

Specifies the format of the PUCCH. With carrier aggregation, only the PUCCH of the "Measurement Carrier" is relevant.

If your signal contains a PUCCH, adjust this value to ensure that the measurement can synchronize to the signal and perform channel estimation.

All PUCCH contained in the measured carrier must have the same format. The PUCCH formats are defined in 3GPP TS 36.211.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:PFORmat
```

Network Signaled Value

Adjust this value to the "network signaled value" used by the UE. The selected value impacts the spectrum emission mask limit check, see [Section 5.3.7.9, "Spectrum emission mask"](#), on page 1080.

There are different values with and without carrier aggregation ("CA_NS_xx" values or "NS_xx" values).

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:NSValue (SA)
```

```
CONFigure:LTE:MEAS<i>:MEvaluation:NSValue:CAGgregation (SA)
```

```
CONFigure:LTE:SIGN<i>:CONNction:ASEmission (CSP)
```

```
CONFigure:LTE:SIGN<i>:CONNction:SCC<c>:ASEmission:CAGgregation (CSP)
```

5.3.5.2 View filter settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "View Filter" node.

The view filter restricts an uplink measurement to signal periods with certain properties. Periods that do not match the filter settings, do not contribute to the measurement results.

With carrier aggregation, the view filter is applied to the "Measurement Carrier".

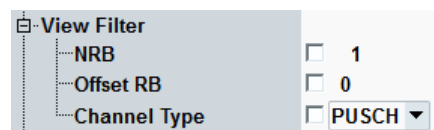


Figure 5-28: View filter settings

NRB

Specifies the number of allocated resource blocks (RB) in the measured slot. Slots with a different number of allocated blocks are rejected.

The allowed values for this parameter are restricted by 3GPP TS 36.211. For details, see [Section 5.2.11.1, "Resources in time and frequency domain"](#), on page 1008.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:NVFilter
```

Offset RB
 Specifies the offset of the first allocated RB in the measured slot. Slots with a different RB offset are rejected.
 Remote command:
 CONFigure:LTE:MEAS<i>:MEvaluation:ORVFilter

Channel Type
 You can measure only slots for which the detected channel type equals PUSCH or only slots with channel type PUCCH. The detected channel type is influenced by the parameters [Uplink Channel Type](#) and [PUCCH Format](#).
 Remote command:
 CONFigure:LTE:MEAS<i>:MEvaluation:CTVFilter

5.3.5.3 Uplink RB allocation settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "RB Allocation" node.
 Configure the RB allocation settings according to the resource block allocation of the measured uplink signal.
 With carrier aggregation, the settings apply to the "Measurement Carrier".
 For background information, see [Section 5.2.11.2, "Uplink resource block allocation"](#), on page 1009.



Figure 5-29: Uplink RB allocation settings

Auto	1057
Narrowband	1057
Multiclust	1058
No. of RBs, Offset RB - no multiclust	1058
No. of RBs, Offset RB - with multiclust	1058

Auto
 If "Auto" is enabled, the RB configuration is detected automatically. The settings "Narrowband", "No. of RBs" and "Offset RB" are ignored.
 Automatic detection is not possible for multiclust allocation.
 Remote command:
 CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:AUTO

Narrowband
 Select the lowest narrowband used for eMTC.

For eMTC signals with narrowband hopping, ignore this setting and use the "Auto" mode.

Remote command:

```
CONFigure:LTE:MEAS<i>:EMTC:NBAND
```

Multicluster

Specify whether the UL signal uses multicluster allocation or not. Multicluster allocation is not allowed for eMTC.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

"OFF"	All allocated RBs are grouped in a single contiguous cluster. Allocation type 0 is used (no resource block groups).
"ON"	The allocated RBs are divided into two clusters. Within each cluster, the allocation is contiguous. Allocation type 1 is used (resource block groups). The view "RB Allocation Table" is not available. For in-band emission results, the limit check is disabled.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:MCLuster (SA)
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:MCLuster:UL (CSP)
```

```
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:UL (CSP)
```

No. of RBs, Offset RB - no multicluster

Number of allocated RBs in the measured slot and offset of the first allocated RB from the edge of the allocated transmission bandwidth.

For eMTC, the "Offset RB" defines the offset of the first allocated RB within the configured narrowband. The allowed "No. of RBs" depends on the maximum eMTC bandwidth, see ["Max Bandwidth 5 MHz"](#) on page 1043 and ["RB allocation, narrowbands and widebands for eMTC"](#) on page 1011.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:NRB
```

```
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:ORB
```

No. of RBs, Offset RB - with multicluster

The settings are configurable per cluster.

In the standalone (SA) scenario, these parameters are controlled by the measurement. In the combined signal path (CSP) scenario, they are controlled by the signaling application.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:MCLuster:NRB<Number> (SA)
```

```
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:MCLuster:ORB<Number> (SA)
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:MCLuster:UL (CSP)
```

```
CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDChannels:MCLuster:UL (CSP)
```

```

CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:MCLuster:UL (CSP)
CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:MCLuster:UL
(CSP)
    
```

5.3.5.4 Sidelink RB allocation settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "RB Allocation" node.

Configure the RB allocation settings according to the resource block allocation of the measured sidelink signal.

For background information, see [Section 5.2.11.3, "Sidelink resource block allocation"](#), on page 1012.

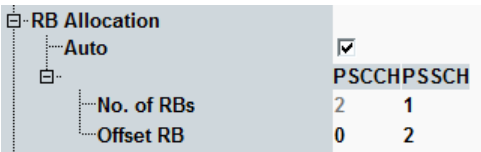


Figure 5-30: Sidelink RB allocation settings

Auto.....	1059
No. of RBs, Offset RB.....	1059

Auto

If "Auto" is enabled, the RB configuration is detected automatically. The settings "No. of RBs" and "Offset RB" are ignored.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:AUTO
```

No. of RBs, Offset RB

Number of allocated RBs in the measured slot and offset of the first allocated RB from the edge of the allocated transmission bandwidth.

The settings are configured separately for PSCCH and PSSCH. The measured slot contains both channel types.

Remote command:

```

CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:NRB:PSCCh
CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:NRB:PSSCh
CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:ORB:PSCCh
CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:ORB:PSSCh
    
```

5.3.5.5 Miscellaneous, part 2

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node.

For sidelink, the settings are not relevant and hidden.

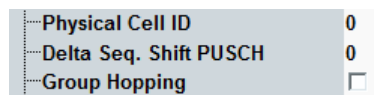


Figure 5-31: Uplink measurement control settings - part 2

Physical Cell ID.....	1060
Delta Sequence Shift PUSCH.....	1060
Group Hopping.....	1060

Physical Cell ID

Adjust this parameter to the physical cell ID used by the measured uplink signal. This setting is required to ensure that the measurement can synchronize to the signal and perform channel estimation.

To configure the physical cell ID for carrier aggregation, see [Section 5.3.4, "Carrier aggregation settings"](#), on page 1046.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFfigure:LTE:MEAS<i>:MEvaluation:CC<no>:PLCid` (SA)

`CONFfigure:LTE:SIGN<i>:CELL[:PCC]:PCID` (CSP)

`CONFfigure:LTE:SIGN<i>:CELL:SCC<c>:PCID` (CSP)

Delta Sequence Shift PUSCH

Delta sequence shift value (Δ_{ss}) used to calculate the sequence shift pattern for PUSCH from the sequence shift pattern for PUCCH (refer to "group hopping" in 3GPP TS 36.211). Adjust this value to the measured LTE signal to ensure that the measurement can synchronize to the signal and perform channel estimation.

Remote command:

`CONFfigure:LTE:MEAS<i>:MEvaluation:DSSPusch`

Group Hopping

Specifies whether group hopping is used (refer to "group hopping" in 3GPP TS 36.211). Set this parameter according to the measured uplink signal to ensure that the measurement can synchronize to the signal and perform channel estimation.

Measurements with group hopping require a frame trigger signal.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFfigure:LTE:MEAS<i>:MEvaluation:GHOPping` (SA)

`CONFfigure:LTE:SIGN<i>:CONNection:GHOPping` (CSP)

5.3.5.6 Measurement subframe settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "Measurement Subframe" node.

The settings configure the scope of the measurement, i.e. which subframes are measured.

See also [Section 5.2.6, "Defining the scope of a TX measurement"](#), on page 996.

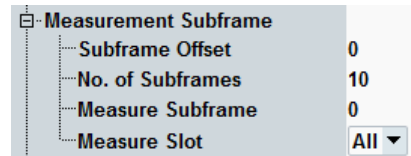


Figure 5-32: Measurement subframe settings

Subframe Offset.....	1061
No. of Subframes.....	1061
Measure Subframe.....	1061
Measure Slot.....	1061

Subframe Offset

Specifies the start of the measured subframe range relative to the trigger event.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:MSUBframes`

No. of Subframes

Specifies the length of the measured subframe range, i.e.:

- Number of subframes to be measured and displayed in the views "Power Monitor" and "RB Allocation Table".
- Number of subframes to be evaluated for BLER measurements. Select a high value being a multiple of 10 to speed up BLER measurements and ensure that the number of scheduled subframes per radio frame can be evaluated correctly.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:MSUBframes`

Measure Subframe

Selects one subframe of the measured subframe range containing the measured slots for all other results, e.g. modulation and spectrum results. For "Power Dynamics" measurements, the ON power is measured in this subframe.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:MSUBframes`

Measure Slot

Selects which slots of the "Measure Subframe" are measured: both slots, the first slot only (slot #0) or the second slot only (slot #1).

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:MSLot`

5.3.5.7 Modulation settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "Modulation" node.

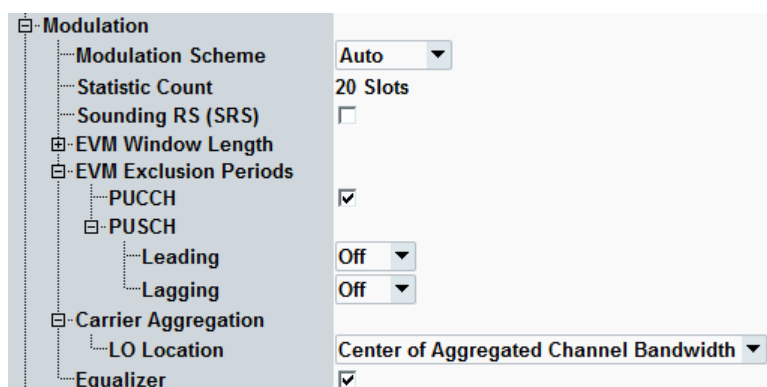


Figure 5-33: Uplink modulation measurement settings

For sidelink, some settings are not relevant and hidden.

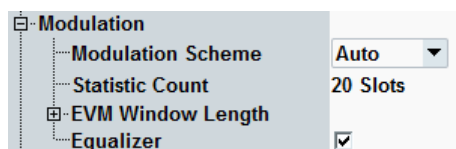


Figure 5-34: Sidelink modulation measurement settings

Modulation Scheme.....	1062
Statistic Count.....	1063
Sounding RS (SRS).....	1063
EVM Window Length.....	1063
EVM Exclusion Periods.....	1064
L PUCCH.....	1064
L PUSCH Leading/Lagging.....	1064
LO Location for CA.....	1064
LO Location for eMTC.....	1065
Equalizer.....	1065

Modulation Scheme

Selects the modulation scheme (QPSK, 16QAM, 64QAM, 256QAM) of the measured signal. The measurement can also auto-detect the modulation scheme (setting: "AUTO").

Auto-detection can fail for signals with a poor modulation accuracy. And auto-detection of 256QAM signals is not supported.

In the combined signal path scenario, you can nevertheless use "AUTO" even for 256QAM signals. The signaling application then tells the measurement that the measured subframe uses 256QAM.

For remote operation, we recommend selecting the modulation scheme manually to ensure correct detection and speed up the measurement. The reset value differs from the preset value.

Option R&S CMW-KM504/554 is required for 256QAM.

Remote command:

CONFigure:LTE:MEAS<i>:MEvaluation:MODulation:MScheme

Statistic Count

Defines the number of measurement intervals per measurement cycle for modulation measurements. The following views use this statistic count: EVM, magnitude error, phase error, in-band emissions, equalizer spectrum flatness, I/Q diagram and TX measurement.

One measurement interval is completed when the measurement has measured a full slot of the configured "Measure Subframe". The measurement always extends over the whole subcarrier range.

The measurement provides independent statistic counts for different results. In single-shot mode, a measurement with a completed statistic count is stopped, even if another measurement with an incomplete statistic count is still running.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:SCount:MODulation`

Sounding RS (SRS)

Specifies whether the uplink signal can contain a sounding reference signal (SRS) or not.

- **Disabled:** The uplink signal must not contain an SRS. Otherwise the modulation results are distorted.
- **Enabled:** The uplink signal can contain an SRS in the last SC-FDMA symbol of a subframe.
This SC-FDMA symbol is analyzed to derive an EVM result for SRS (EVM SRS).
For all other modulation results, the SC-FDMA symbol is not evaluated.

SRS is not allowed for eMTC.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, this parameter is set by the signaling application. A subsequent modification in the signaling application is also applied to the measurement. But changing the setting in the measurement does not affect the signaling application.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:SRS:ENABLE` (SA)

`CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:ENABLE` (CSP)

EVM Window Length

Length of the EVM window in samples. The window length defines the two sets of "I / h" modulation quantities in the result tables. See [Section 5.2.9, "Calculation of modulation results"](#), on page 1001. The minimum value of one FFT symbol means that EVM_h equals EVM_l .

The standard (3GPP TS 36.101, annex F) specifies the window length as a function of the bandwidth and of the cyclic prefix.

EVM Window Length						
Current	136					
Bandwidth	1.4	3	5	10	15	20
@ CP Normal	5	12	32	66	102	136
@ CP Extended	28	58	124	250	374	504

The measurement uses the cyclic prefix and bandwidth settings to select the relevant table cell. The "Current" setting is linked to this table cell.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EWLength
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EWLength:
CBANdwidth<Band>
```

EVM Exclusion Periods

Exclusion periods to be considered for calculation of EVM, magnitude error and phase error results (quantities with "low" and "high" values).

PUCCH ← EVM Exclusion Periods

Enables exclusion periods for slots with detected channel type "PUCCH".

The exclusion periods affect low and high single-value results (tables below diagrams). The first and the last SC-FDMA symbol of each slot is excluded from the calculation of these results. If the last symbol of a slot is already excluded because SRS signals are allowed, the second but last symbol is also excluded.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUCCh
```

PUSCH Leading/Lagging ← EVM Exclusion Periods

Define exclusion periods for slots with detected channel type "PUSCH".

The exclusion periods affect all low and high results (single values and diagrams). The selected time intervals are excluded from the calculation of these results.

The "Leading" time interval is excluded at the beginning of each subframe.

The "Lagging" time interval is excluded at the end of each subframe; if SRS signals are allowed, at the end of each shortened subframe.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUSCh:
LEADing
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUSCh:
LAGGing
```

LO Location for CA

Selects the local oscillator (LO) location for carrier aggregation. The setting is hidden if carrier aggregation is disabled.

Configure the parameter according to the UE transmitter architecture:

- **"Center of Aggregated Channel Bandwidth":**
The transmitter uses a single local oscillator (LO). The LO frequency is the center frequency of the aggregated channel bandwidth.
- **"Center of each Component Carrier":**
The transmitter uses a separate LO for each component carrier. The LO frequencies are the center frequencies of the component carriers.

The setting influences the measured I/Q offset and the in-band emission mask. The mask depends on the location of the RF carriers ("IQ Offset" component) and on the location of the image frequencies ("IQ Image" component), see also [Section 5.3.7.5, "In-band emissions limits"](#), on page 1075.

Remote command:
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:CAGGregation:
LLOCation

LO Location for eMTC
Selects the local oscillator (LO) location for eMTC. The setting is hidden if eMTC is disabled.

- Configure it according to the UE transmitter architecture:
- **"Center of Narrowband / Wideband":**
The LO frequency of the transmitter is in the center of the narrowband (with max bandwidth 1.4 MHz) or in the center of the wideband (with max bandwidth 5 MHz).
 - **"Center of Channel Bandwidth":**
The LO frequency of the transmitter is in the center of the channel bandwidth.

The setting influences the measured I/Q offset and the in-band emission mask. The mask depends on the location of the RF carrier ("IQ Offset" component) and on the location of the image frequencies ("IQ Image" component), see also [Section 5.3.7.5, "In-band emissions limits"](#), on page 1075.

Remote command:
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:LLOCation

Equalizer
Enables or disables the post-FFT equalization step. The equalization step is specified in the standard for the measurement of modulation results like the EVM. If you disable the checkbox, spectrum ripples contribute to the modulation results and increase the EVM.

Remote command:
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EQualizer

5.3.5.8 Spectrum settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "Spectrum" node.

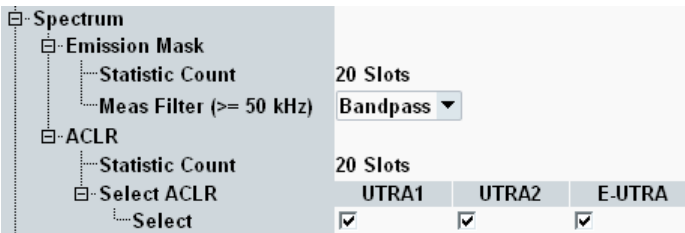


Figure 5-35: Spectrum measurement settings

Statistic Count.....	1066
Measurement Filter (>= 50 kHz).....	1066
UTRA1, UTRA2, E-UTRA.....	1066

Statistic Count

Defines the number of measurement intervals per measurement cycle. There are separate settings for emission mask and ACLR measurements.

One measurement interval is completed when the measurement has measured a full slot of the configured "Measure Subframe".

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:SCount:SPECTrum:SEMask
```

```
CONFigure:LTE:MEAS<i>:MEvaluation:SCount:SPECTrum:ACLR
```

Measurement Filter (>= 50 kHz)

Selects the resolution filter type for emission mask measurements, for filter bandwidths of 50 kHz and greater (Gaussian or bandpass filter). For filters with smaller bandwidth, the type is fixed (Gaussian). The filter bandwidth can be configured per emission mask area as part of the limits.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:SPECTrum:SEMask:MFIltter
```

UTRA1, UTRA2, E-UTRA

Specifies which adjacent channels are evaluated for ACLR measurements: first adjacent UTRA channels, second adjacent UTRA channels or first adjacent E-UTRA channels. Any combinations are supported, including the activation of all three options in parallel.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:SPECTrum:ACLR:ENABLE
```

5.3.5.9 Power settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "Power" node.

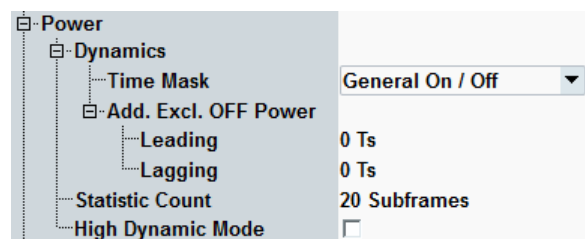


Figure 5-36: Uplink power measurement settings

For sidelink, some settings are not relevant and hidden.

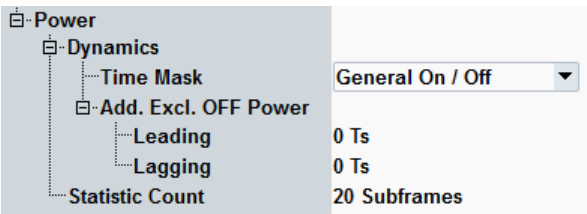


Figure 5-37: Sidelink power measurement settings

Time Mask.....	1067
Add Exclusive OFF Power.....	1067
Statistic Count.....	1068
High Dynamic Mode.....	1068

Time Mask

Selects the time mask for power dynamics measurements. The time masks are defined by 3GPP. Each mask is designed for a specific sequence of powers.

"ON" and "OFF" indicate the power of the control channel or shared channel. "SRS ON" and "SRS OFF" indicate the SRS power.

- **General on/off:** OFF - ON - OFF
See 3GPP TS 36.521, section 6.3.4.1 / 6.3.4A.1.
- **PUCCH / PUSCH / SRS:** ON - SRS ON - ON
See 3GPP TS 36.101, figure 6.3.4.4-2.
- **SRS blanking:** ON - SRS OFF - ON
See 3GPP TS 36.101, figure 6.3.4.4-4.

See also [Section 5.3.7.11, "Power dynamics limits"](#), on page 1084.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:PDYNamics:TMASK
```

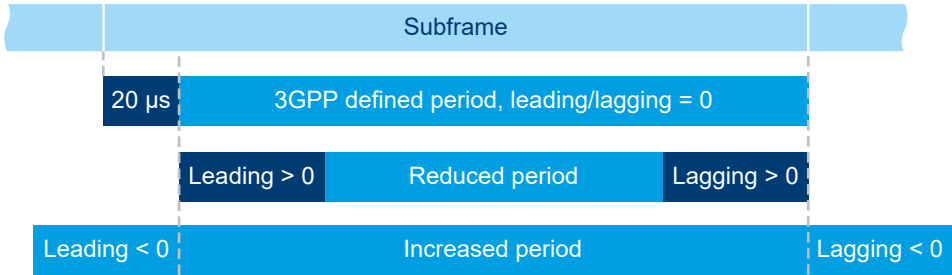
Add Exclusive OFF Power

Shifts the OFF power evaluation periods in power dynamics measurements.

With the default value 0 Ts, the OFF power is measured over the evaluation period defined by 3GPP. The leading value moves the beginning of the evaluation period, the lagging value moves the end. Positive values decrease the evaluation period. Negative values increase it.

This setting affects the measurement of PUCCH, PUSCH and SRS OFF powers.

Example with a default period of one subframe and 20-μs exclusion at the beginning:



Remote command:
CONFigure:LTE:MEAS<i>:MEvaluation:PDYNamics:AEOPower:LEADing
CONFigure:LTE:MEAS<i>:MEvaluation:PDYNamics:AEOPower:LAGGing

Statistic Count

Defines the number of measurement intervals per measurement cycle for power measurements. The following views use this statistic count: power dynamics and power monitor.

For power dynamics results, one measurement interval is completed when the measurement has measured one complete diagram.

For power monitor results, one measurement interval is completed when the number of subframes specified via parameter [No. of Subframes](#) has been captured.

Remote command:
CONFigure:LTE:MEAS<i>:MEvaluation:SCount:POWer

High Dynamic Mode

Enables or disables the high dynamic mode for uplink power measurements.

The high dynamic mode is suitable for power dynamics measurements involving high ON powers. In that case, the dynamic range of the measurement is eventually not sufficient to measure both the high ON powers and the low OFF powers accurately.

In high dynamic mode, the dynamic range is increased by measuring the results in two shots. One shot uses the configured settings to measure the ON power. The other shot uses a lower "Expected Nominal Power" value to measure the OFF power results.

Disable the high dynamic mode if you do not need it, to optimize the measurement speed. Disable it especially if you are not interested in power dynamics results or if you measure low ON powers using a low "Expected Nominal Power" setting.

While the combined signal path scenario is active, this parameter is only configurable for the measurement mode "Normal", see ["Measurement Mode"](#) on page 1052.

Remote command:
CONFigure:LTE:MEAS<i>:MEvaluation:POWer:HDMode

5.3.5.10 BLER settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "BLER" node.

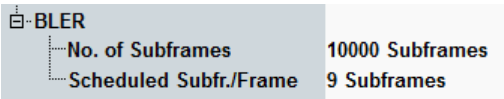


Figure 5-38: BLER measurement control settings

[No. of Subframes](#)..... 1069

[Scheduled Subfr./Frame](#)..... 1069

No. of Subframes

Defines the number of subframes to be evaluated per measurement cycle (statistics cycle). This value includes all downlink subframes (scheduled and unscheduled). The BLER measurement is a single-shot measurement.

All subframes of the configured subframe range contribute to the measurement, see [Section 5.3.5.6, "Measurement subframe settings"](#), on page 1060.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:BLER:SFRames`

Scheduled Subfr./Frame

Number of scheduled subframes per radio frame. Configure this parameter according to the properties of the generated downlink signal. A wrong value can result in wrong measurement results (BLER, DTX, relative ACK, relative NACK, number of processed subframes).

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:BLER:SFRames`

5.3.5.11 List mode settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Measurement Control" node > "List Mode" node.



Figure 5-39: Measurement control settings - list mode

[Offline Segment No.](#)..... 1069

Offline Segment No.

Selects the list mode segment to be displayed in the result diagrams. To select a segment number, the list mode must be enabled.

Option R&S CMW-KM012 is required.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:LIST:OSIndex`

5.3.6 Trigger settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Trigger" node.

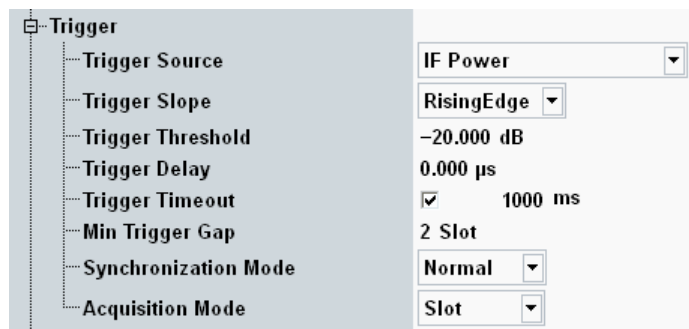


Figure 5-40: Trigger settings

Trigger Source.....	1070
Trigger Slope.....	1071
Trigger Threshold.....	1071
Trigger Delay.....	1071
Trigger Timeout.....	1071
Min Trigger Gap.....	1071
Synchronization Mode.....	1072
Acquisition Mode.....	1072

Trigger Source

Selects the source of the trigger event.

Only the measurement-specific trigger sources are listed here. Additional trigger sources are provided by the base software and by other applications like the GPRF generator. For a description of these trigger sources, refer to the documentation of the corresponding software part.

"Free Run (Fast Sync)"	The measurement starts immediately after it is initiated. The measurement decodes the signal to derive its slot timing and frame timing so that the "Measurement Length" can start at a slot or subframe boundary of the LTE signal. This procedure is repeated after each measurement cycle. This value is not suitable for list mode measurements.
"Free Run (No Sync)"	The measurement starts immediately after it is initiated, without any synchronization. This trigger mode is faster than the "Free Run (Fast Sync)" mode. It can be used for power monitor, ACLR and emission mask measurements, if the resource block allocation and power are continuous. Ignore all other results (e.g. modulation results, RB allocation and power dynamics).
"IF Power"	The measurement is triggered by the power of the received signal, converted into an IF signal. The trigger event coincides with the rising or falling edge of the detected LTE burst.

Remote command:

```
TRIGger:LTE:MEAS<i>:MEValuation:SOURce
```

```
TRIGger:LTE:MEAS<i>:MEValuation:CATalog:SOURce?
```

Trigger Slope

Selects whether the trigger event is generated at the rising edge or at the falling edge of the trigger pulse. This setting is relevant for power trigger sources.

Remote command:

```
TRIGger:LTE:MEAS<i>:MEvaluation:SLOPe
```

Trigger Threshold

Defines the input signal power where a trigger event is generated for power trigger sources.

The threshold is a dB value, relative to the reference level minus the external attenuation. If the reference level equals the maximum output power of the DUT and the external attenuation settings are correct, the trigger threshold is relative to the maximum input power.

A low threshold ensures that the input signal is always detected. A higher threshold can prevent unintended trigger events.

Remote command:

```
TRIGger:LTE:MEAS<i>:MEvaluation:THReshold
```

Trigger Delay

Delays the start of the measurement relative to the trigger event. This setting has no influence on "Free Run" measurements.

Remote command:

```
TRIGger:LTE:MEAS<i>:MEvaluation:DELay
```

Trigger Timeout

Sets a time after which a started measurement must have received a trigger event. If no trigger event is received, a trigger timeout is indicated in manual operation mode. In remote control mode, the measurement is stopped. The parameter can be disabled so that no timeout occurs.

This setting has no influence on "Free Run" measurements.

Remote command:

```
TRIGger:LTE:MEAS<i>:MEvaluation:TOUT
```

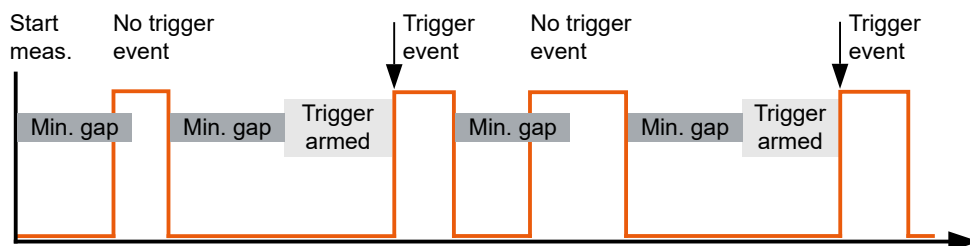
Min Trigger Gap

Defines a minimum duration of the power-down periods (gaps) between two triggered power pulses. This setting is valid for "(IF) Power" trigger sources.

The trigger system is controlled by a timer which is reset to zero in the following instances:

- At the IF power-down ramp of each triggered or untriggered pulse, even if the previous counter has not yet elapsed. A power-down ramp is detected when the signal power falls below the trigger threshold.
- At the beginning of each measurement. The minimum gap defines the minimum time between the start of the measurement and the first trigger event.

The trigger system is rearmed when the timer has reached the specified minimum gap.



This parameter can be used to prevent unwanted trigger events due to fast power variations.

Remote command:

`TRIGger:LTE:MEAS<i>:MEvaluation:MGAP`

Synchronization Mode

Selects the size of the search window for synchronization - normal or enhanced.

If the trigger signal is timed accurately, the normal synchronization mode is sufficient and recommended for performance reasons. With increasing inaccuracy of the trigger signal timing, the search window in normal mode can become too small, resulting in a "Synchronization Problem". In that case, select the enhanced mode to increase the search window size.

For "Free Run (Fast Sync)", an enhanced search window is used irrespective of this parameter.

Remote command:

`TRIGger:LTE:MEAS<i>:MEvaluation:SMODE`

Acquisition Mode

Selects whether the measurement synchronizes to a slot boundary or to a subframe boundary. The parameter is relevant for "Free Run (Fast Sync)" and for list mode measurements with "Synchronization Mode" = "Enhanced".

Remote command:

`TRIGger:LTE:MEAS<i>:MEvaluation:AMODE`

5.3.7 Limit settings

The following sections give an overview of the LTE multi-evaluation limit settings and of the related conformance test requirements.

Conformance requirements for LTE UE transmitter tests are specified in 3GPP TS 36.521, section 6, "Transmitter Characteristics".

- [Modulation limits](#)..... 1073
- [EVM, magnitude error and phase error limits](#)..... 1073
- [Frequency error limits](#)..... 1074
- [I/Q origin offset limits](#)..... 1074
- [In-band emissions limits](#)..... 1075
- [Equalizer spectrum flatness limits](#)..... 1077
- [Spectrum limit concept](#)..... 1078

• Occupied bandwidth limits.....	1079
• Spectrum emission mask.....	1080
• ACLR limits.....	1082
• Power dynamics limits.....	1084

5.3.7.1 Modulation limits

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Modulation" node.

All modulation limits can be configured per modulation scheme.

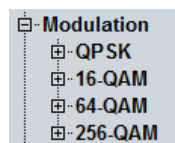


Figure 5-41: One node per modulation scheme

The individual modulation limits are described in the following sections.

5.3.7.2 EVM, magnitude error and phase error limits

A poor modulation accuracy of the UE transmitter increases the transmission errors and decreases the system capacity. The error vector magnitude (EVM) is the critical quantity to assess the modulation accuracy of an LTE UE.

According to 3GPP, the EVM measured at UE output powers ≥ -40 dBm must not exceed a certain percentage, depending on the modulation scheme (see table).

Characteristics	Refer to 3GPP TS 36.521 V14.5.0, section...	Specified limit
EVM (RMS)	6.5.2.1 Error Vector Magnitude (EVM)	≤ 17.5 % (QPSK)
	6.5.2.1_1 Error Vector Magnitude (EVM) for UL 64QAM	≤ 12.5 % (16QAM)
	6.5.2.1_2 Error Vector Magnitude (EVM) for UL 256QAM	≤ 8 % (64QAM)
	6.5.2A.1 Error Vector Magnitude (EVM) for CA	≤ 3.5 % (256QAM)

EVM, magnitude error, phase error limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Modulation" node > all subnodes.

The EVM limits can be set depending on the modulation scheme and along with limits for the other measured quantities.

EVM RMS	☒	17.50 %
EVM Peak	☐	35.00 %
MErr RMS	☐	17.50 %
MErr Peak	☐	35.00 %
PhErr RMS	☐	17.50 °
PhErr Peak	☐	35.00 °

Figure 5-42: Limit settings for QPSK

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:EVMagnitude
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:MERRor
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:PERRor
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:EVMagnitude
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:MERRor
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:PERRor
```

5.3.7.3 Frequency error limits

According to 3GPP, the UE modulated carrier frequency must be accurate within ± 0.1 ppm, compared to the nominal carrier frequency.

Characteristics	Refer to 3GPP TS 36.521 V14.5.0, section...	Specified limit
Frequency error	6.5.1 Frequency Error 6.5.1A Frequency Error for CA	< 0.1 ppm

Frequency error limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Modulation" node > all subnodes.

The frequency error limit can be set depending on the modulation scheme. The same default value is used for all schemes.

Frequency Error [ppm] ☒ 0.100

Figure 5-43: Frequency error limit setting

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:FERRor
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:FERRor
```

5.3.7.4 I/Q origin offset limits

An I/Q origin offset is due to an additive sinusoid waveform at the frequency of the reference signal. The standard specifies the I/Q origin offset power limit as a function of the output power of the UE transmitter (see "TX power" in the table).

Characteristics	Refer to 3GPP TS 36.521 V14.5.0, section...	TX power	Specified limit
I/Q origin offset	6.5.2.2 Carrier Leakage 6.5.2A.2 Carrier Leakage for CA	> 0 dBm 0 dBm to -30 dBm < -30 dBm to -40 dBm	-24.2 dBc -19.2 dBc -9.2 dBc

I/Q origin offset limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Modulation" node > all subnodes > "IQ Offset" node.

I/Q origin offset limits for three different TX power ranges can be set depending on the modulation scheme. The same default values are used for all schemes.



Figure 5-44: I/Q origin offset limit settings

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:IQOffset
```

```
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IQOffset
```

5.3.7.5 In-band emissions limits

The in-band emission is the relative UE output power in non-allocated resource blocks (RBs). In-band emissions are interferers in the subcarrier range that is potentially used by other connected UEs. 3GPP defines a complex combined in-band emission limit described below.

The combined limit can be determined via the following rules:

- The minimum of the combined limit is -29.2 dB. In the measurement, this general minimum is configurable.
- A general component is considered for all non-allocated RBs.
- For non-allocated RBs at image frequencies of allocated RBs, an I/Q image component is added to the general component.
- For RBs at the carrier frequency or next to the carrier frequency, an I/Q offset component is added to the general component.

So the most complex limit is determined for a non-allocated RB located both at an image frequency and next to the carrier frequency. In that case, the general component, the I/Q image component and the I/Q offset component are added. Then the result is compared to the general minimum and the bigger value of both applies.

The following table provides an overview of the three components.

Component	Value	Applicable RBs
General	See formula below table.	Non-allocated RBs
I/Q image	-24.2 dB or -27.2 dB, depending on carrier frequency and TX power	Image frequencies
I/Q offset	-27.2 dBc to -9.2 dBc, depending on carrier frequency and TX power See Section 5.3.7.4, "I/Q origin offset limits" , on page 1074	Carrier frequency
For contiguous carrier aggregation and for eMTC, the carrier frequency and thus the image frequencies depend on the transmitter architecture. The measurement is informed about the architecture via settings, see: • CA: "LO Location for CA" on page 1064 • eMTC: "LO Location for eMTC" on page 1065		

The general limit is derived using the following formula:

$$\text{Limit}_{\text{General}} = \max \left[\begin{array}{l} -25 - 10 * \log_{10} \left(\frac{\langle \text{All RB} \rangle}{\langle \text{No RB} \rangle} \right), \\ 20 * \log_{10} \langle \text{EVM} \rangle - 3 - \frac{5 * (\langle \text{Offset} \rangle - 1)}{\langle \text{No RB} \rangle}, \\ \langle \text{RB Power} \rangle - \langle \text{RB Power Meas} \rangle \end{array} \right], \text{ dB} + 0.8 \text{ dB}$$

The variables are defined as follows:

- $\langle \text{All RB} \rangle$ = total number of RBs within the channel bandwidth of the allocated component carrier (as defined by 3GPP, e.g. 75 RBs for BW 15 MHz).
- $\langle \text{No RB} \rangle$ = number of allocated RBs in the slot.
- $\langle \text{EVM} \rangle$ = maximum allowed EVM in percent (configurable), see [Section 5.3.7.2, "EVM, magnitude error and phase error limits"](#), on page 1073.
- $\langle \text{Offset} \rangle$ = distance of the RB from the closest allocated RB.
- $\langle \text{RB Power} \rangle$ = -57 (configurable).
- $\langle \text{RB Power Meas} \rangle$ = arithmetic mean value of the average powers in all allocated resource blocks (dBm/RB).

Characteristics	Refer to 3GPP TS 36.521 V14.5.0, section...	Specified limit
IBE min	6.5.2.3 In-band emissions for non-allocated RB 6.5.2A.3 In-band emissions for non-allocated RB for CA 6.5.2.3G In-band emissions for non-allocated RB for V2X communication	See table above

In-band emission limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Modulation" node > all subnodes > "IBE" node.

The general minimum, the variables $\langle \text{EVM} \rangle$ and $\langle \text{RB Power} \rangle$, the I/Q image component and the I/Q offset component can be set depending on the modulation scheme.

IBE	Min	EVM	RB Power
Enable	☒		
General	-29.20 dB	17.50 %	-57.00 dBm
IQ Image	-24.20 dB		
IQ Offset			
Output Power			
> 0 dBm	-24.20 dBc		
> -30 dBm	-19.20 dBc		
> -40 dBm	-9.20 dBc		

Figure 5-45: In-band emissions limit settings for QPSK

With carrier aggregation, RBs are allocated for one CC only. The limits apply to all component carriers. The measurement displays in-band emission result diagrams for all carriers.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:IBE
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:IBE:IQOffset
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IBE
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IBE:IQOffset
```

5.3.7.6 Equalizer spectrum flatness limits

For EVM measurements, an equalization step has to be performed after the FFT. The spectrum flatness of this equalizer has to be verified to validate the EVM results.

For equalizer spectrum flatness measurements, 3GPP divides each frequency band into two ranges. For normal conditions, range 1 contains the subcarriers that are at least 3 MHz away from the band edges. And range 2 contains all subcarriers within 3 MHz from the band edges. For extreme conditions, 5 MHz instead of 3 MHz apply.

The conformance requirements define limits for the maximum power variations within each range and between the ranges. These limits are listed in the table below. The following figure illustrates the limits for normal conditions.

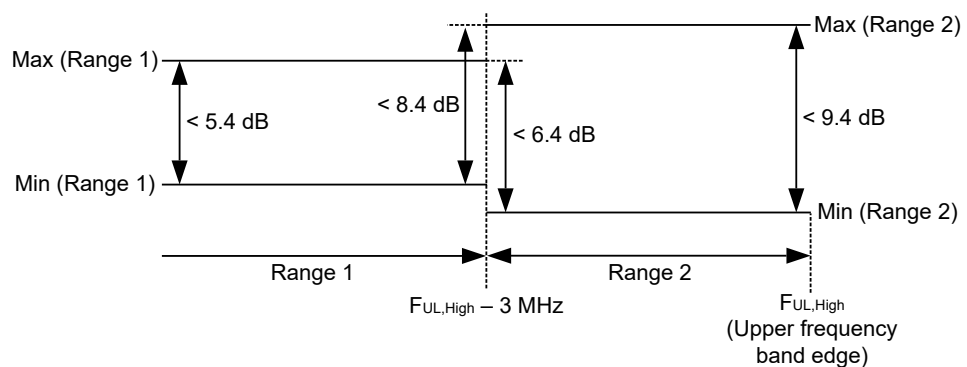


Figure 5-46: Limits for normal conditions (vertical arrows)

Characteristics	Refer to 3GPP TS 36.521 V14.5.0, section...	Result	Upper limits normal / extreme
Spectrum flatness	6.5.2.4 EVM equalizer spectrum flatness	max (range 1) - min (range 1)	5.4 dB / 5.4 dB
		max (range 2) - min (range 2)	9.4 dB / 13.4 dB
	6.5.2.4G EVM equalizer spectrum flatness for V2X transmission	max (range 1) - min (range 2)	6.4 dB / 7.4 dB
		max (range 2) - min (range 1)	8.4 dB / 11.4 dB

Spectrum flatness limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Modulation" node > all subnodes > "Spectrum Flatness" node.

The limits and the band edge distance can be set depending on the modulation scheme. The same default values are used for all schemes.

Spectrum Flatness	
Enable	☒
Range 1	5.40 dBpp
Range 2	9.40 dBpp
Max(Range 1) - Min(Range 2)	6.40 dB
Max(Range 2) - Min(Range 1)	8.40 dB
Band Edge Distance	3.00 MHz

Figure 5-47: Spectrum flatness limit settings

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:QPSK:ESFlatness
```

```
CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:QAM<m>:ESFlatness
```

5.3.7.7 Spectrum limit concept

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Spectrum" node.

Without carrier aggregation, all spectrum limits are configured per channel bandwidth.

Spectrum	
☐ 1.4 MHz	
☐ 3.0 MHz	
☐ 5.0 MHz	
☐ 10.0 MHz	
☐ 15.0 MHz	
☐ 20.0 MHz	
☐ Add. Test Tolerance SEM	

Figure 5-48: Limits per bandwidth, no CA

With carrier aggregation, the spectrum limits are configured per channel bandwidth combination. There are limit sets for specific channel bandwidth combinations required for 3GPP tests. An additional limit set applies to all other bandwidth combinations.

Example: "CA 20 MHz + 10 MHz" applies, if the CC1 has a bandwidth of 20 MHz and the CC2 a bandwidth of 10 MHz, or vice versa. "Other Combination" is used, for example, for 15 MHz + 10 MHz or 1.4 MHz + 5 MHz.

Spectrum	
☐ CA 10MHz + 5MHz	
☐ CA 10MHz + 10MHz	
☐ CA 20MHz + 5MHz	
☐ CA 20MHz + 10MHz	
☐ CA 15MHz + 15MHz	
☐ CA 20MHz + 15MHz	
☐ CA 20MHz + 20MHz	
☐ Other Combination	
☐ Add. Test Tolerance SEM	

Figure 5-49: Limits per BW combination, two carriers

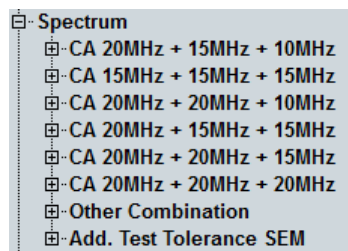


Figure 5-50: Limits per BW combination, three carriers

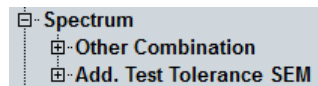


Figure 5-51: Only other limits, four carriers

The individual spectrum limits are described in the following sections.

5.3.7.8 Occupied bandwidth limits

The occupied bandwidth is the bandwidth that contains 99 % of the total integrated power of the transmitted spectrum. According to 3GPP, the occupied bandwidth must be less than the theoretical (aggregated) channel bandwidth.

Characteristics	Refer to 3GPP TS 36.521 V14.5.0, section...	Specified limit
Occupied bandwidth	6.6.1 Occupied bandwidth 6.6.1G Occupied bandwidth for V2X communication	< channel bandwidth
	6.6.1A Occupied bandwidth for CA	< aggregated channel bandwidth

OBW limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Spectrum" node > all subnodes.

The limit is set per channel bandwidth / per channel bandwidth combination. See also [Section 5.3.7.7, "Spectrum limit concept"](#), on page 1078.



Figure 5-52: OBW limit for BW 1.4 MHz

Remote command:

```

CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:SEMask:OBWLimit:
CBANdwidth<Band>
CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:SEMask:OBWLimit:
CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>
  
```

```

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:
CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>:
CBANdwidth<Band3>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:
CAGGregation:OCOMbination

```

5.3.7.9 Spectrum emission mask

The energy that spills outside the designated radio channel, increases the interference with adjacent channels and decreases the system capacity. The amount of unwanted off-carrier energy is assessed by the out-of-band emissions (excluding spurious emissions). They are specified in terms of the spectrum emission mask and of the adjacent channel leakage power ratio (ACLR).

3GPP specifies a general spectrum emission mask and several additional spectrum emission masks. Which of the emission masks is applicable depends on the parameter [Network Signaled Value](#). The masks differ for signals without carrier aggregation and for signals with different numbers of aggregated carriers.

For carrier frequencies between 3 GHz and 4.2 GHz, 3GPP relaxes the limits by an additional tolerance of 0.3 dB, compared to the limits for carrier frequencies below 3 GHz. For carrier frequencies above 5 GHz, the additional tolerance is 0.5 dB.

Characteristics	Refer to 3GPP TS 36.521 V14.5.0, section...
General emission mask	6.6.2.1 Spectrum Emission Mask 6.6.2.1A Spectrum Emission Mask for CA 6.6.2.1G Spectrum Emission Mask for V2X Communication
Additional emission masks	6.6.2.2 Additional Spectrum Emission Mask 6.6.2.2A Additional Spectrum Emission Mask for CA 6.6.2.2G Additional Spectrum Emission Mask for V2X Communication

The emission mask settings are described in the following.

General and additional limit settings	1080
Emission mask configuration for uplink measurements	1081
Emission mask configuration for sidelink measurements	1082
Additional test tolerance settings	1082

General and additional limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Spectrum" node > all subnodes > "Emission Mask" node.

Both the general requirements and the additional requirements can be defined depending on the channel bandwidth / the channel bandwidth combination. See also [Section 5.3.7.7, "Spectrum limit concept"](#), on page 1078.

The information displayed above the general limits indicates whether the general limits are applicable, or which set of additional limits applies instead.

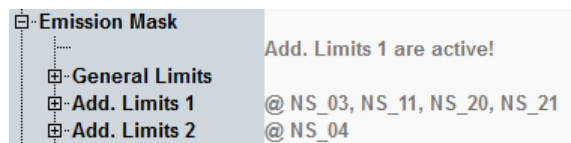


Figure 5-53: General and additional limit sets

Emission mask configuration for uplink measurements

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Spectrum" node > all subnodes > "Emission Mask" node.

For uplink measurements, all emission masks consist of horizontal and vertical lines. There are no slopes.

For each emission mask area, you define the frequencies of the borders (vertical lines), set an upper power limit (horizontal line) and select the resolution bandwidth to be used.

General Limits		Start	Stop	Power	RBW
Area 1	☒	0.015 MHz	0.985 MHz	-8.5 dBm	30kHz
Area 2	☒	1.500 MHz	2.000 MHz	-8.5 dBm	1MHz
Area 3	☒	3.000 MHz	3.000 MHz	-23.5 dBm	1MHz
Area 4	☐	5.000 MHz	5.000 MHz	-25.0 dBm	1MHz

Figure 5-54: Example for UL emission mask areas

The start and stop frequencies of each emission mask area are defined relative to the edge of the assigned channel bandwidth. For carrier aggregation, they are defined relative to the edge of the assigned aggregated channel bandwidth.

Example: Assume 0 MHz as start frequency and 1 MHz as stop frequency for a channel bandwidth of 1.4 MHz. The resulting area ranges from +0.7 MHz to +1.7 MHz relative to the carrier frequency. As all ranges are symmetrical, it ranges also from -0.7 MHz to -1.7 MHz relative to the carrier frequency.

Remote command:

```

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
CBANdwidth<Band>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
ADDITIONal<Table>:CBANdwidth<Band>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
ADDITIONal<Table>:CAGGregation:CBANdwidth<Band1>:
CBANdwidth<Band2>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>:
CBANdwidth<Band3>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
CAGGregation:OCOMbination
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
ADDITIONal<Table>:CAGGregation:OCOMbination

```

Emission mask configuration for sidelink measurements

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Spectrum" node > all subnodes > "Emission Mask" node.

The general limits for sidelink measurements are the same as for uplink measurements.

The additional emission masks for sidelink are different. The individual mask areas have slopes instead of horizontal lines. The following figure shows the additional limit settings for BW 10 MHz and NS_33. The "Power" setting applies to the "Start" frequency. With increasing distance from the carrier, the power limit decreases as defined by the "Slope".

Emission Mask							
Add. Limits 1 are active!							
General Limits							
Add. Limits 1							
Area 1	☒	0.000 MHz	0.500 MHz	-11.5 dBm	-12.00 dB/MHz	100kHz	▼
Area 2	☒	0.500 MHz	5.000 MHz	-17.5 dBm	-1.78 dB/MHz	100kHz	▼
Area 3	☒	5.000 MHz	10.000 MHz	-25.5 dBm	-2.00 dB/MHz	100kHz	▼
Area 4	☐	10.000 MHz	10.000 MHz	-35.5 dBm	-2.00 dB/MHz	100kHz	▼

Figure 5-55: Additional limits 1 for sidelink 10 MHz, NS_33

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:SEMask:LIMit<no>:
CBANdwidth<Band>
CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:SEMask:LIMit<no>:
ADDitional<Table>:CBANdwidth<Band>:SIDelink
```

Additional test tolerance settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Spectrum" node > "Add. Test Tolerance SEM" node.

The additional test tolerances for high carrier frequencies apply to all emission masks. Below 3 GHz, there is no additional tolerance. Above 3 GHz, there are two configurable tolerances.

Add. Test Tolerance SEM	
3GHz < E-UTRA Bands ≤ 4.2GHz	0.3 dB
E-UTRA Bands ≥ 5GHz	0.5 dB

Figure 5-56: Test tolerance settings

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:SEMask:
ATTolerance<EUTRABand>
```

5.3.7.10 ACLR limits

The adjacent channel leakage power ratio (ACLR) limits complement the spectrum emission mask. The ACLR is defined as the mean power in the assigned E-UTRA channel divided by the mean power in an adjacent channel.

According to 3GPP, the ACLR must be evaluated for the first adjacent UTRA channel (UTRA1) and the first adjacent E-UTRA channel (E-UTRA). For channel bandwidths > 3 MHz, also the ACLR for the second adjacent UTRA channel (UTRA2) must be evaluated.

The relative ACLR limit is evaluated only if the measured adjacent channel power is greater than -50 dBm (absolute limit). In that case, the ACLR must be greater than the limits listed in the following table.

Characteristics	Refer to 3GPP TS 36.521 V10.3.0, section...	Relative limit
ACLR	6.6.2.3 Adjacent Channel Leakage power Ratio 6.6.2.3A Adjacent Channel Leakage power Ratio for CA	UTRA1: 32.2 dB UTRA2: 35.2 dB E-UTRA: 29.2 dB

ACLR limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Spectrum" node > all subnodes > "ACLR" node.

The limits can be set depending on the channel bandwidth / the channel bandwidth combination. See also [Section 5.3.7.7, "Spectrum limit concept"](#), on page 1078.

ACLR	Rel	Abs
UTRA1	☒ 32.20 dB	☒ -50.00 dBm
UTRA2	☐ 35.20 dB	☐ -50.00 dBm
E-UTRA	☒ 29.20 dB	☒ -50.00 dBm

Figure 5-57: ACLR limit settings for BW 1.4 MHz

The default values are identical for all bandwidths and for signals with and without carrier aggregation.

For carrier aggregation, the term "E-UTRA" refers to the aggregated bandwidth. Adjacent E-UTRA channels have the same bandwidth as the assigned E-UTRA channel.

The default settings are suitable to check the 3GPP requirements. The relative limits are only evaluated if the corresponding absolute limit is exceeded. If you disable an absolute limit, the corresponding enabled relative limit is always evaluated.

Remote command:

```

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:
CBANdwidth<Band>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:
CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:
CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>:
CBANdwidth<Band3>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:
CAGGregation:OCOMbination
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:
CBANdwidth<Band>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation:
CBANdwidth<Band1>:CBANdwidth<Band2>

```

```

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation:
CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation:
OCOMbination

```

5.3.7.11 Power dynamics limits

Transmission at excessive power increases interference to other channels. A too low power increases transmission errors. 3GPP defines a general time mask. The mask contains limits for the power in subframes not used for transmission (transmit OFF power), subframes used for transmission (transmit ON power) and the power ramping between them.

The ON power is specified as the mean UE output power within a subframe used for transmission, excluding a transient period of 20 µs at the beginning of the subframe. According to 3GPP, subframe number 3 has to be used. For the measurement, the used subframe is configurable, see ["Measure Subframe"](#) on page 1061.

For the OFF power, the mean power has to be measured both in the preceding subframe and in the subsequent subframe. For the subsequent subframe, a transient period of 20 µs at the beginning is excluded.

The following figure provides a summary of the time periods relevant for ON and OFF power limits.

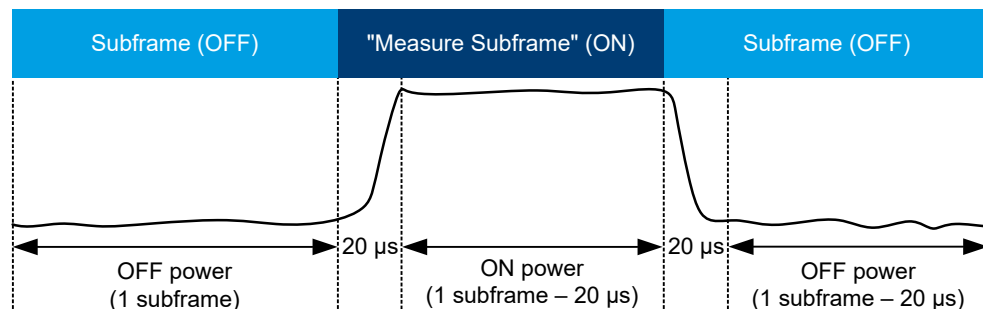


Figure 5-58: Measurement periods ON/OFF power

The OFF power must not exceed -48.5 dBm. The ON power limit must be within a range depending on the channel bandwidth.

Characteristics	Refer to 3GPP TS 36.521 V10.3.0, section...	BW	Specified limit
ON power	6.3.4.1 General ON/OFF time mask 6.3.4A.1 General ON/OFF time mask for CA 6.3.4G ON/OFF time mask for V2X communication	1.4 MHz	≥ -22.3 dBm, ≤ -7.3 dBm
		3 MHz	≥ -18.3 dBm, ≤ -3.3 dBm
		5 MHz	≥ -16.1 dBm, ≤ -1.1 dBm
		10 MHz	≥ -13.1 dBm, ≤ 1.9 dBm
		15 MHz	≥ -11.4 dBm, ≤ 3.6 dBm
		20 MHz	≥ -10.1 dBm, ≤ 4.9 dBm
OFF power		any	≤ -48.5 dBm

The power dynamics measurement supports additional time masks, defined in 3GPP TS 36.101. The following list provides an overview of all supported time masks:

- "General ON/OFF" time mask described in detail in the preceding section and defined in 3GPP TS 36.521, section 6.3.4.1 / 6.3.4A.1 / 6.3.4G.
Three subframes with the power sequence OFF - ON - OFF are measured. The exclusion periods are 20 μ s. The limits are evaluated for all ON and OFF powers.

Subframe	"Measure Subframe"		Subframe
OFF power	20 μ s	ON power	20 μ s OFF power

- "PUCCH / PUSCH / SRS" time mask with transmission before and after the SRS, see 3GPP TS 36.101, figure 6.3.4.4-2.
Two subframes with ON power are measured. The UE sends an SRS at the end of the first subframe. So the power sequence is ON - SRS ON - ON.
Exclusion periods of 40 μ s are used around the SRS. The limits are evaluated for the "ON power", but not for the "SRS ON" power.

"Measure Subframe"			Subframe	
20 μ s	ON power	40 μ s	SRS ON	40 μ s ON power

- SRS time mask with "SRS blanking", see 3GPP TS 36.101, figure 6.3.4.4-4.
Two subframes with ON power are measured. The UE assumes that another UE sends an SRS at the end of the first subframe. It does not transmit at all during this SRS period. So the power sequence is ON - SRS OFF - ON.
Exclusion periods of 20 μ s are used around the SRS. The limits are evaluated for all ON and OFF powers, including SRS OFF.

"Measure Subframe"			Subframe	
20 μ s	ON power	20 μ s	SRS OFF	20 μ s ON power

Note that you can shift the borders of all OFF power evaluation periods, see ["Add Exclusive OFF Power"](#) on page 1067.

Power dynamics limit settings

Access: "Multi Evaluation" tab > "Config" hotkey > "Limits" node > "Power" node > all subnodes > "Dynamics" node.

Dynamics	Upper	Lower
Enable	☒	
ON Power	-7.3 dBm	-22.3 dBm
OFF Power	-48.5 dBm	

Figure 5-59: Power dynamics limit settings (BW 1.4 MHz)

The "ON Power" limits apply to the ON power periods, but not to the SRS ON power period.

The "OFF Power" limit applies to the OFF and SRS OFF power periods.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:PDYNamics:
CBANdwidth<Band>
```

5.3.8 Shortcut configuration

Access: "Config" hotkey > bottom of the tree.

This section configures a shortcut softkey that provides a fast way to access the GPRF generator from the measurement.

The setting is a common measurement setting. It has the same value in all measurements (for example, PRACH measurement and multi-evaluation measurement).



Figure 5-60: Shortcut configuration

Generator Shortcut

Selects an instance of the GPRF generator. Softkeys for the selected instance are added to the softkey panel.

5.3.9 Additional softkeys and hotkeys

The measurement provides some softkey/hotkey combinations which have no equivalent in the configuration dialog. Most of these hotkeys provide display configurations (like diagram scaling). They are self-explanatory and do not have any remote control commands assigned.

The remaining softkeys > hotkeys are described below. Most of them are available for multi-evaluation, PRACH and SRS measurements.

Multi Evaluation, PRACH softkeys > Assign Views hotkey.....	1086
Multi Evaluation softkey > EVM vs Symbol hotkey.....	1087
Signaling softkeys.....	1087
L Signaling Parameter softkey.....	1087
L LTE Signaling softkey.....	1087
Generator softkeys.....	1087
L ARB / List Mode softkey.....	1088
L GPRF Generator softkey.....	1088
Suppress "Cell Off" Message.....	1088

Multi Evaluation, PRACH softkeys > Assign Views hotkey

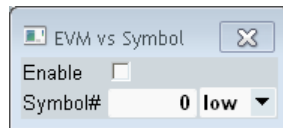
Selects the view types to be displayed in the overview. The R&S CMW does not evaluate the results for disabled views. Therefore, limiting the number of assigned views can speed up the measurement.

Remote command:

```
CONFigure:LTE:MEAS<i>:MEvaluation:RESult[:ALL]
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:PDYNamics etc.
CONFigure:LTE:MEAS<i>:PRACH:RESult[:ALL]
CONFigure:LTE:MEAS<i>:PRACH:RESult:PVPreamble etc.
```

Multi Evaluation softkey > EVM vs Symbol hotkey

The hotkey is only available in the view "Error Vector Magnitude". It opens the following dialog box.



The checkbox enables the measurement of EVM vs modulation symbol results. They are shown in the lower diagram in the "Error Vector Magnitude" view.

"Symbol#" selects the SC-FDMA symbol to be evaluated. "low" and "high" select between EVM_l and EVM_h results.

For SRS signals, the low/high setting is irrelevant. The FFT processing window begins at the center of the cyclic prefix, not at the low or high position.

Remote command:

`CONFigure:LTE:MEAS<i>:MEvaluation:RESult:EVMagnitude:EVMSymbol`

Signaling softkeys

The softkeys "Signaling Parameter" and "LTE Signaling" are displayed if the combined signal path scenario is active. The softkeys are provided by the "LTE Signaling" application selected as the controlling application. See also "[Scenario = Combined Signal Path](#)" on page 1042.

While one of the signaling softkeys is selected, the "Config" hotkey opens the configuration dialog of the signaling application, not the configuration dialog of the measurement.

Signaling Parameter softkey ← Signaling softkeys

Provides hotkeys for access to the most essential settings of the "LTE Signaling" application.

Remote command:

Use the commands of the signaling application.

LTE Signaling softkey ← Signaling softkeys

Select this softkey and press [ON | OFF] to turn the downlink signal transmission on or off. Alternatively, right-click the softkey.

Press the softkey two times (select it and press it again) to switch to the signaling application.

Remote command:

Use the commands of the signaling application.

Generator softkeys

To display the softkeys "ARB / List Mode" and "GPRF Generator" enable the generator shortcut, see [Section 5.3.8, "Shortcut configuration"](#), on page 1086.

If the combined signal path scenario is active, the generator softkeys are hidden.

While one of the generator softkeys is selected, the "Config" hotkey opens the configuration dialog of the generator, not the configuration dialog of the measurement.

ARB / List Mode softkey ← Generator softkeys

Provides hotkeys for access to the most important ARB and list mode settings of the GPRF generator.

Remote command:

Use the commands of the GPRF generator.

GPRF Generator softkey ← Generator softkeys

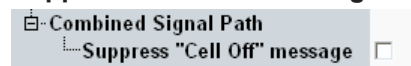
Select this softkey and press [ON | OFF] to turn the GPRF generator on or off. Alternatively, right-click the softkey.

Press the softkey two times (select it and press it again) to switch to the generator application.

The hotkeys assigned to this softkey provide access to the most important GPRF generator settings.

Remote command:

Use the commands of the GPRF generator.

Suppress "Cell Off" Message

This setting is located in the configuration tree. It is only visible if the combined signal path scenario is active.

The checkbox suppresses a warning that is displayed if the "... Signaling" softkey is selected and the cell signal is on and you press [ON | OFF]. The warning asks you whether you really want to switch off the cell signal.

Remote command:

n.a.

5.4 Programming

The following sections provide programming examples for the LTE multi-evaluation measurement.

- [General examples](#)..... 1088
- [Using the list mode](#).....1097

5.4.1 General examples

The LTE multi-evaluation measurement is programmed as follows:

- The measurement is controlled by SCPI commands with the following syntax: `...LTE:MEAS:MEvaluation...`
- After a `*RST`, the measurement is switched off. Use `READ:LTE:MEAS:MEvaluation...?` to initiate a measurement and to retrieve the results. You can also start the measurement using

INIT:LTE:MEAS:MEValuation and retrieve the results using
FETCh:LTE:MEAS:MEValuation...?.

- For synchronization and proper decoding, some UE signal settings must be in accordance with the measured signal; see [Section 5.4.1.2, "Specifying required settings"](#), on page 1090.

5.4.1.1 Specifying general and common measurement settings

```
// *****
// Initial system-reset
// *****
*RST; *OPC?
*CLS; *OPC?

// *****
// Set signal type UL, duplex mode TDD and
// query the resulting frame structure type (T2).
// *****
CONFigure:LTE:MEAS:STYPe UL
CONFigure:LTE:MEAS:DMode TDD
CONFigure:LTE:MEAS:FSTRucture?

// *****
// Define the RF input path.
// *****
ROUTe:LTE:MEAS:SCENario:SALone RF1C, RX1

// *****
// Configure RF and analyzer settings:
// External attenuation 2 dB, CC1 center frequency 1850 MHz,
// peak power 7 dBm, 5 dB user margin, CC1 channel bandwidth 1.4 MHz.
// *****
CONFigure:LTE:MEAS:RFSettings:EATTenuation 2
CONFigure:LTE:MEAS:RFSettings:CC1:FREQuency 1850E+6
CONFigure:LTE:MEAS:RFSettings:ENPower 7
CONFigure:LTE:MEAS:RFSettings:UMARgin 5
CONFigure:LTE:MEAS:CC1:CBANDwidth B014

// *****
// Disable eMTC.
// *****
CONFigure:LTE:MEAS:EMTC:ENABle OFF

// *****
// Enable carrier aggregation class B/C.
// Define the CC2 channel bandwidth. Adjust and query the CC2 frequency.
// Query information about the resulting aggregated bandwidth.
// *****
CONFigure:LTE:MEAS:CAGGregation:MODE INTRaband
CONFigure:LTE:MEAS:CC2:CBANDwidth B014
```

```

CONFigure:LTE:MEAS:CAGGregation:ACSPacing
CONFigure:LTE:MEAS:RFSettings:CC2:FREquency?

CONFigure:LTE:MEAS:CAGGregation:FREquency:AGGRegated:LOW?
CONFigure:LTE:MEAS:CAGGregation:FREquency:AGGRegated:CENTer?
CONFigure:LTE:MEAS:CAGGregation:FREquency:AGGRegated:HIGh?
CONFigure:LTE:MEAS:CAGGregation:CBANdwidth:AGGRegated?

// *****
// Enable carrier aggregation class D.
// Define the CC2/3 channel bandwidths. Adjust and query the CC2/3 frequencies.
// *****
CONFigure:LTE:MEAS:CAGGregation:MODE ICD
CONFigure:LTE:MEAS:CC2:CBANdwidth B014
CONFigure:LTE:MEAS:CC3:CBANdwidth B030
CONFigure:LTE:MEAS:CAGGregation:ACSPacing
CONFigure:LTE:MEAS:RFSettings:CC2:FREquency?
CONFigure:LTE:MEAS:RFSettings:CC3:FREquency?

// *****
// Select the CC1 for single-carrier measurements.
// *****
CONFigure:LTE:MEAS:CAGGregation:MCARrier:ENHanced CC1

```

5.4.1.2 Specifying required settings

```

// *****
// Specify required UE signal settings: UL-DL configuration 1,
// special subframe configuration 1, extended cyclic prefix,
// contains PUCCH with format 1a, physical layer cell IDs 5 and 10,
// delta sequence shift value 10, no group hopping.
// *****
CONFigure:LTE:MEAS:MEvaluation:ULDL 1
CONFigure:LTE:MEAS:MEvaluation:SSUBframe 1
CONFigure:LTE:MEAS:MEvaluation:CPRefix EXTended
CONFigure:LTE:MEAS:MEvaluation:PFormat F1A
CONFigure:LTE:MEAS:MEvaluation:CC1:PLCid 5
CONFigure:LTE:MEAS:MEvaluation:CC2:PLCid 10
CONFigure:LTE:MEAS:MEvaluation:CC3:PLCid 11
CONFigure:LTE:MEAS:MEvaluation:DSSPusch 10
CONFigure:LTE:MEAS:MEvaluation:GHOPping OFF

```

5.4.1.3 Specifying measurement-specific settings

```

// *****
// Define stop condition (stop on limit failure), measurement mode and error
// handling, measured subframe contains only PUCCH (no automatic detection),
// network signaled value NS_10 (without CA) and CA_NS_11 (with CA),
// set no. of resource blocks filter to 10, set RB offset filter,

```

```

// set a channel type filter.
// *****
CONFIGure:LTE:MEAS:MEvaluation:SCONdition SLFail
CONFIGure:LTE:MEAS:MEvaluation:MMODE NORMAl
CONFIGure:LTE:MEAS:MEvaluation:MOEXception ON
CONFIGure:LTE:MEAS:MEvaluation:TOUT 3600
CONFIGure:LTE:MEAS:MEvaluation:CTYPE PUCCH
CONFIGure:LTE:MEAS:MEvaluation:NSValue NS10
CONFIGure:LTE:MEAS:MEvaluation:NSValue:CAGGregation NS11
CONFIGure:LTE:MEAS:MEvaluation:NVFilter 10
CONFIGure:LTE:MEAS:MEvaluation:ORVFilter 3
CONFIGure:LTE:MEAS:MEvaluation:CTVFilter PUSCh

// *****
// Define the "TPC Mode" list of expected nominal powers.
// *****
CONFIGure:LTE:MEAS:MEvaluation:TMODE:SCount 5,8,4,4,1,1,1,1,1,1,1,1,1,1,1,1
CONFIGure:LTE:MEAS:MEvaluation:TMODE:ENPower 3,9,12,15,0,0,0,0,0,0,0,0,0,0,0,0
CONFIGure:LTE:MEAS:MEvaluation:TMODE:RLEVel?

// *****
// Disable automatic detection of RB configuration. Select eMTC narrowband.
// Set number of resource blocks and RB offset manually.
// No multi-cluster allocation.
// *****
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:AUTO OFF
CONFIGure:LTE:MEAS:EMTC:NBAND 2
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:NRB 5
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:ORB 2

// *****
// Enable multi-cluster allocation.
// Configure the number of resource blocks and the RB offset for both clusters.
// *****
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:MCLuster ON
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:MCLuster:NRB1 2
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:MCLuster:ORB1 0
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:MCLuster:NRB2 3
CONFIGure:LTE:MEAS:MEvaluation:RBAllocation:MCLuster:ORB2 3

// *****
// Define the scope of the measurement:
// Skip 2 subframes after the trigger event, then capture 20 subframes,
// use subframe number 9 for slot measurements (both slots).
// *****
CONFIGure:LTE:MEAS:MEvaluation:MSUBframes 2, 20, 9
CONFIGure:LTE:MEAS:MEvaluation:MSlot ALL

// *****
// Specify modulation measurement settings:

```

```

// measurement over 20 statistics cycles, modulation scheme QPSK, no SRS,
// EVM window lengths depending on cyclic prefix length and bandwidth.
// Correct EVM window length for 1.4 MHz BW.
// Enable/specify exclusion periods for PUCCH and PUSCH.
// Local oscillator at center of aggregated bandwidth.
// *****
CONFIGure:LTE:MEAS:MEvaluation:SCount:MODulation 20
CONFIGure:LTE:MEAS:MEvaluation:MODulation:MSCHEME QPSK
CONFIGure:LTE:MEAS:MEvaluation:SRS:ENABLE OFF
CONF:LTE:MEAS:MEV:MODulation:EWLength 5,12,35,66,102,136,28,58,124,250,374,504
CONFIGure:LTE:MEAS:MEvaluation:MODulation:EWLength:CBANDwidth14 6,13
CONFIGure:LTE:MEAS:MEvaluation:MODulation:EEPeriods:PUCCh ON
CONFIGure:LTE:MEAS:MEvaluation:MODulation:EEPeriods:PUSCh:LEADing MS25
CONFIGure:LTE:MEAS:MEvaluation:MODulation:EEPeriods:PUSCh:LAGging MS25
CONFIGure:LTE:MEAS:MEvaluation:MODulation:CAGGregation:LLOCation CACB

// *****
// Specify spectrum measurement settings:
// select a measurement length of 30 slots, use Gaussian resolution filter,
// for ACLR measure all adjacent channels.
// *****
CONFIGure:LTE:MEAS:MEvaluation:SCount:SPECTrum:ACLR 30
CONFIGure:LTE:MEAS:MEvaluation:SCount:SPECTrum:SEMask 30
CONFIGure:LTE:MEAS:MEvaluation:SPECTrum:SEMask:MFILter GAUSS
CONFIGure:LTE:MEAS:MEvaluation:SPECTrum:ACLR:ENABLE ON,ON,ON

// *****
// Specify power measurement settings:
// time mask, OFF power exclusion, high dynamic mode, statistic count
// *****
CONFIGure:LTE:MEAS:MEvaluation:PDYnamicS:TMASK SBLanking
CONFIGure:LTE:MEAS:MEvaluation:PDYnamicS:AEOPower:LAGging -5
CONFIGure:LTE:MEAS:MEvaluation:PDYnamicS:AEOPower:LEADing -5
CONFIGure:LTE:MEAS:MEvaluation:POWER:HDMODE ON
CONFIGure:LTE:MEAS:MEvaluation:SCount:POWER 30

// *****
// Specify BLER measurement settings: number of subframes to be measured
// *****
CONFIGure:LTE:MEAS:MEvaluation:BLER:SFRAMES 5000, 10

```

5.4.1.4 Configuring the trigger system

```

// *****
// Set trigger source, timeout, trigger level, slope, delay,
// minimum trigger gap, synchronization mode and acquisition mode.
// *****
TRIGger:LTE:MEAS:MEvaluation:SOURce 'IF Power'
TRIGger:LTE:MEAS:MEvaluation:TOUT 1
TRIGger:LTE:MEAS:MEvaluation:THReshold -30

```



```

TRIGger:LTE:MEAS:MEvaluation:SLOPe FEDGe
TRIGger:LTE:MEAS:MEvaluation:DElay 0.0001
TRIGger:LTE:MEAS:MEvaluation:MGAP 3
TRIGger:LTE:MEAS:MEvaluation:SMODE ENHanced
TRIGger:LTE:MEAS:MEvaluation:AMODE SLOT

```

5.4.1.5 Specifying limits

```

// *****
// Define QPSK modulation limits
// Commands for 16QAM, 64QAM and 256QAM are analogous with
// mnemonic QAM16, QAM64 or QAM256 instead of QPSK.
// *****
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:EVMagnitude 20, 40
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:MERRor 20, OFF
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:PERRor 20, OFF
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:FERRor 0.15
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:IQOffset ON, -26, -21, -11
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:IBE ON, -20, 20, -60, -27
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:IBE:IQOffset -26, -21, -11
CONFIGure:LTE:MEAS:MEvaluation:LIMit:QPSK:ESFlatness ON, 5, 9, 6, 8, 3MHz

// *****
// Define ACLR limits for measurements without carrier aggregation.
// *****
CONFIGure:LTE:MEAS:MEvaluation:LIMit:ACLR:UTRA1:CBANDwidth14 35, -50
CONFIGure:LTE:MEAS:MEvaluation:LIMit:ACLR:EUTRa:CBANDwidth14 32, -50

// *****
// Define ACLR limits for measurements with two carriers.
// *****
CONF:LTE:MEAS:MEV:LIMit:ACLR:UTRA1:CAGG:CBANDwidth200:CBANDwidth100 35, -50
CONF:LTE:MEAS:MEV:LIMit:ACLR:UTRA1:CAGG:OCOMbination 35, -50
CONF:LTE:MEAS:MEV:LIMit:ACLR:EUTRa:CAGG:CBANDwidth200:CBANDwidth100 32, -50
CONF:LTE:MEAS:MEV:LIMit:ACLR:EUTRa:CAGG:OCOMbination 32, -50

// *****
// Define ACLR limits for measurements with three carriers.
// *****
CONF:LTE:MEAS:MEV:LIMit:ACLR:UTRA1:CAGG:CBAN200:CBAN150:CBAN100 35, -50
CONF:LTE:MEAS:MEV:LIMit:ACLR:UTRA1:CAGG:OCOMbination 35, -50
CONF:LTE:MEAS:MEV:LIMit:ACLR:EUTRa:CAGG:CBAN200:CBAN150:CBAN100 32, -50
CONF:LTE:MEAS:MEV:LIMit:ACLR:EUTRa:CAGG:OCOMbination 32, -50

// *****
// Define spectrum emission limits for measurements without carrier aggregation.
// *****
CONF:LTE:MEAS:MEvaluation:LIMit:SEMask:OBWLimit:CBANDwidth14 1.2E+6
CONF:LTE:MEAS:MEV:LIMit:SEM:LIMit1:CBANDwidth14 ON, 2E+4, 1E+6, -15, K030

```

```

CONF:LTE:MEAS:MEV:LIMit:SEM:LIMit1:ADD3:CBANdwidth14 ON,2E+4,1E+6,-15,K030

// *****
// Define spectrum emission limits for measurements with two carriers.
// *****
CONF:LTE:MEAS:MEV:LIMit:SEMask:OBWLimit:CAGG:CBAN200:CBAN100 29.8E+6
CONF:LTE:MEAS:MEV:LIMit:SEMask:OBWLimit:CAGG:OCOMbination 2.3E+6
CONF:LTE:MEAS:MEV:LIMit:SEMask:LIMit1:CAGG:CBAN200:CBAN100 ON,2E+4,1E+6,-20,K030
CONF:LTE:MEAS:MEV:LIMit:SEMask:LIMit1:CAGG:OCOMbination ON,2E+4,1E+6,-20,K030
CONF:LTE:MEAS:MEV:LIMit:SEMask:LIM1:ADD1:CAGG:CBAN200:CBAN100 ON,2E+4,1E+6,
-20,K030
CONF:LTE:MEAS:MEV:LIMit:SEMask:LIM1:ADD1:CAGG:OCOMbination ON,2E+4,1E+6,-20,K030

// *****
// Define spectrum emission limits for measurements with three carriers.
// *****
CONF:LTE:MEAS:MEV:LIMit:SEMask:OBWLimit:CAGG:CBAN200:CBAN150:CBAN100 29.8E+6
CONF:LTE:MEAS:MEV:LIMit:SEMask:OBWLimit:CAGG:OCOMbination 2.3E+6
CONF:LTE:MEAS:MEV:LIMit:SEMask:LIMit1:CAGG:CBAN200:CBAN150:CBAN100 ON,2E+4,1E+6,
-20,K030
CONF:LTE:MEAS:MEV:LIMit:SEMask:LIMit1:CAGG:OCOMbination ON,2E+4,1E+6,-20,K030

// *****
// Define additional tolerances for spectrum emission limits.
// *****
CONF:Figure:LTE:MEAS:MEValuation:LIMit:SEMask:ATTolerance30 0.3
CONF:Figure:LTE:MEAS:MEValuation:LIMit:SEMask:ATTolerance50 0.5

// *****
// Define power dynamics limits
// *****
CONF:Figure:LTE:MEAS:MEValuation:LIMit:PDYNamics:CBANdwidth14 ON,-7.6,-22.6,-48.8

```

5.4.1.6 Specifying V2X sidelink settings

```

// *****
// Set signal type SL.
// *****
CONF:Figure:LTE:MEAS:STYPe SL

// *****
// For modulation results, measure the PSSCH.
// *****
CONF:Figure:LTE:MEAS:MEValuation:SCTYPe PSSCh

// *****
// Set channel BW to 10 MHz.
// Disable automatic detection of RB configuration.
// Set number of resource blocks and RB offset manually.

```

```
// *****
CONFiGure:LTE:MEAS:CC1:CBANdwidth B100
CONFiGure:LTE:MEAS:MEValuation:RBAllocation:AUTO OFF
CONFiGure:LTE:MEAS:MEValuation:RBAllocation:NRB:PSSCh 10
CONFiGure:LTE:MEAS:MEValuation:RBAllocation:ORB:PSSCh 5
CONFiGure:LTE:MEAS:MEValuation:RBAllocation:ORB:PSSCh 20

// *****
// Configure sidelink-specific limits.
// *****
CONF:LTE:MEAS:MEV:LIM:SEM:LIMit1:ADD1:CBAN100:SID ON,0,5E+5,-11.5,-12,K100
CONF:LTE:MEAS:MEV:LIM:SEM:LIMit2:ADD1:CBAN100:SID ON,5E+5,5E+6,-17.5,-1.78,K100
CONF:LTE:MEAS:MEV:LIM:SEM:LIMit3:ADD1:CBAN100:SID ON,5E+6,1E+7,-25.5,-2,K100
```

5.4.1.7 Running a measurement

```
// *****
// Enable all measurements.
// *****
CONF:LTE:MEAS:MEvaluation:RESult:ALL ON,ON,ON,ON,ON,ON,ON,ON,ON,ON,ON,ON,ON,ON,ON,ON
CONFigure:LTE:MEAS:MEvaluation:RESult:EVMagnitude:EVMSymbol ON, 3, LOW

// *****
// Start the measurement.
// *****
INIT:LTE:MEAS:MEvaluation

// *****
// Query detected allocation, modulation scheme and channel type for the
// measured slot of modulation measurements.
// Query the View Filter Throughput.
// *****
FETCh:LTE:MEAS:MEvaluation:MODulation:DALlocation?
FETCh:LTE:MEAS:MEvaluation:MODulation:DMODulation?
FETCh:LTE:MEAS:MEvaluation:MODulation:DCHType?
FETCh:LTE:MEAS:MEvaluation:VFThroughput?

// *****
// Query modulation diagram contents and statistical values.
// *****
FETCh:LTE:MEAS:MEvaluation:EVMagnitude:MAXimum?
FETCh:LTE:MEAS:MEvaluation:EVMagnitude:PEAK:MAXimum?
FETCh:LTE:MEAS:MEvaluation:TRACe:EVMSymbol:MAXimum?
FETCh:LTE:MEAS:MEvaluation:TRACe:EVMC?
FETCh:LTE:MEAS:MEvaluation:MERRor:AVERage?
FETCh:LTE:MEAS:MEvaluation:MERRor:MAXimum?
FETCh:LTE:MEAS:MEvaluation:PERRor:AVERage?
FETCh:LTE:MEAS:MEvaluation:PERRor:MAXimum?
FETCh:LTE:MEAS:MEvaluation:TRACe:IQ:LOW?
```

```

FETCh:LTE:MEAS:MEvaluation:TRACe:IQ:HIGH?
FETCh:LTE:MEAS:MEvaluation:MODulation:CURRent?
FETCh:LTE:MEAS:MEvaluation:MODulation:EXTReMe?
FETCh:LTE:MEAS:MEvaluation:EVMC:PEAK:MAXimum?

// *****
// Query in-band emission results for CC1, CC2 and CC3.
// *****
FETCh:LTE:MEAS:MEvaluation:TRACe:IEMissions:CC1?
FETCh:LTE:MEAS:MEvaluation:TRACe:IEMissions:CC2?
FETCh:LTE:MEAS:MEvaluation:TRACe:IEMissions:CC3?
FETCh:LTE:MEAS:MEvaluation:IEMission:CC1:MARGIn:AVERage?
FETCh:LTE:MEAS:MEvaluation:IEMission:CC2:MARGIn:AVERage?
FETCh:LTE:MEAS:MEvaluation:IEMission:CC3:MARGIn:AVERage?
FETCh:LTE:MEAS:MEvaluation:IEMission:CC1:MARGIn:EXTReMe:RBINdex?
FETCh:LTE:MEAS:MEvaluation:IEMission:CC2:MARGIn:EXTReMe:RBINdex?
FETCh:LTE:MEAS:MEvaluation:IEMission:CC3:MARGIn:EXTReMe:RBINdex?

// *****
// Query equalizer spectrum flatness results.
// *****
FETCh:LTE:MEAS:MEvaluation:TRACe:ESFLatness?
FETCh:LTE:MEAS:MEvaluation:ESFLatness:AVERage?
FETCh:LTE:MEAS:MEvaluation:ESFLatness:CURRent:SCINdex?

// *****
// Query spectrum emission results.
// *****
SENSE:LTE:MEAS:MEvaluation:SPECTrum:SEMask:RBW:USED?
FETCh:LTE:MEAS:MEvaluation:TRACe:SEMask:RBW30:CURRent?
FETCh:LTE:MEAS:MEvaluation:TRACe:SEMask:RBW100:AVERage?
FETCh:LTE:MEAS:MEvaluation:TRACe:SEMask:RBW1000:MAXimum?
FETCh:LTE:MEAS:MEvaluation:SEMask:CURRent?
FETCh:LTE:MEAS:MEvaluation:SEMask:EXTReMe?
FETCh:LTE:MEAS:MEvaluation:SEMask:MARGIn:ALL?
FETCh:LTE:MEAS:MEvaluation:SEMask:MARGIn:MINimum:POSitiv?
FETCh:LTE:MEAS:MEvaluation:SEMask:MARGIn:MINimum:NEGativ?

// *****
// Query ACLR results.
// *****
FETCh:LTE:MEAS:MEvaluation:TRACe:ACLR:CURRent?
FETCh:LTE:MEAS:MEvaluation:TRACe:ACLR:AVERage?
FETCh:LTE:MEAS:MEvaluation:ACLR:CURRent?
FETCh:LTE:MEAS:MEvaluation:ACLR:AVERage?

// *****
// Query RB allocation table and power monitor results for CC1 and CC2.
// *****
FETCh:LTE:MEAS:MEvaluation:TRACe:RBATable:CC1?

```

```

FETCh:LTE:MEAS:MEvaluation:TRACe:RBATable:CC2?
FETCh:LTE:MEAS:MEvaluation:TRACe:PMONitor:CC1?
FETCh:LTE:MEAS:MEvaluation:TRACe:PMONitor:CC2?
FETCh:LTE:MEAS:MEvaluation:PMONitor:AVERage?
FETCh:LTE:MEAS:MEvaluation:PMONitor:CC1:AVERage?
FETCh:LTE:MEAS:MEvaluation:PMONitor:CC2:AVERage?

// *****
// Query power dynamics results.
// *****
FETCh:LTE:MEAS:MEvaluation:TRACe:PDYNamics:AVERage?
FETCh:LTE:MEAS:MEvaluation:TRACe:PDYNamics:MAXimum?
FETCh:LTE:MEAS:MEvaluation:PDYNamics:AVERage?
FETCh:LTE:MEAS:MEvaluation:PDYNamics:MINimum?
FETCh:LTE:MEAS:MEvaluation:PDYNamics:MAXimum?

// *****
// Query BLER results.
// *****
FETCh:LTE:MEAS:MEvaluation:BLER?

// *****
// Query limit check results.
// *****
CALCulate:LTE:MEAS:MEvaluation:ACLR:CURRent?
CALCulate:LTE:MEAS:MEvaluation:SEMask:CURRent?
CALCulate:LTE:MEAS:MEvaluation:ESFlatness:CURRent?
CALCulate:LTE:MEAS:MEvaluation:MODulation:CURRent?
CALCulate:LTE:MEAS:MEvaluation:PDYNamics:AVERage?
CALCulate:LTE:MEAS:MEvaluation:PDYNamics:MINimum?
CALCulate:LTE:MEAS:MEvaluation:PDYNamics:MAXimum?

```

5.4.2 Using the list mode

The LTE multi-evaluation list mode is programmed as follows:

- The measurement is controlled by SCPI commands with the following syntax: `...LTE:MEAS:MEvaluation:LIST...`
- After a `*RST`, the measurement is switched off and the list mode is disabled. Use `CONFigure:LTE:MEAS:MEvaluation:LIST ON` to enable the list mode and `INIT:LTE:MEAS:MEvaluation` to initiate a single-shot measurement.
- Use `FETCh:LTE:MEAS:MEvaluation:LIST:...?` commands to retrieve the results.

Speeding up list mode measurements

List mode tests are often used in production lines. The same tests are repeated for different UEs. Optimizing the test duration is important.

Consider the following rules to minimize the test duration:

- Reduce the number and size of the segments and the statistical length as far as possible for your use case.
- Enable only the results that you need / disable the other result calculations.
- There are two groups of `FETCH` commands. One for result retrieval per segment and one for result retrieval over all segments. Which of the two groups is more effective, depends on the enabled results and on the number of segments. Minimize the number of `FETCH` commands needed to query results.
- To reset the system repeatedly, use partial reset commands instead of the global reset command `*RST`.

Example: Use `SYSTEM:RESet 'LTE Meas1'` to reset the LTE measurement instance 1.

- To configure the same list mode settings repeatedly, use partial save/recall commands:
 - Configure all list mode settings.
 - Save the list mode settings to a file, for example to the file `ListModeLTE1.dfl` located in the directory assigned to the `@SAVE` alias:


```
MMEMory:STORe:ITEM 'LTE Meas:MEV:LIST',
 '@SAVE\ListModeLTE1.dfl'
```
 - If you want to configure the same settings again, recall the stored settings:


```
MMEMory:LOAD:ITEM 'LTE Meas:MEV:LIST',
 '@SAVE\ListModeLTE1.dfl'
```

5.4.2.1 Specifying global measurement settings

```
// *****
// Reset
// *****
SYSTEM:RESet 'LTE Meas1'; *OPC?
*CLS; *OPC?

// *****
// Define the RF input path.
// *****
ROUTE:LTE:MEAS:SCENario:SALone RF1C, RX1

// *****
// Define external attenuation and carrier aggregation.
// *****
CONFIGure:LTE:MEAS:RFSettings:EATTenuation 2
CONFIGure:LTE:MEAS:CAGGregation:MODE INTRaband

// *****
// Specify required UE signal settings not configurable per segment:
// signal contains PUCCH with format 1a, delta sequence shift value 10.
// *****
CONFIGure:LTE:MEAS:MEvaluation:PFormat F1A
CONFIGure:LTE:MEAS:MEvaluation:DSSPusch 10
```

5.4.2.2 Specifying list mode settings

```
// *****
// For uplink measurements:
// Define 2 segments with a length of 10 subframes each (1 radio frame),
// different expected nominal power and identical remaining settings.
// Select eMTC narrowband. Define automatic RB allocation detection.
// *****
CONF:LTE:MEAS:MEV:LIST:SEGM1:SET 10,1,FDD,OB1,2E+9,B100,NORM,AUTO,OFF,0,NS04
CONF:LTE:MEAS:MEV:LIST:SEGM2:SET 10,-19,FDD,OB1,2E+9,B100,NORM,AUTO,OFF,0,NS04
CONF:LTE:MEAS:MEV:LIST:SEGMENT1:EMTC:NBAND 2
CONF:LTE:MEAS:MEV:LIST:SEGMENT2:EMTC:NBAND 2
CONF:LTE:MEAS:MEV:LIST:SEGMENT1:RBALLOCATION ON, KEEP, KEEP
CONF:LTE:MEAS:MEV:LIST:SEGMENT2:RBALLOCATION ON, KEEP, KEEP

// *****
// For sidelink measurements:
// Define 2 segments with a length of 10 subframes each (1 radio frame),
// different expected nominal power and identical remaining settings.
// Define automatic RB allocation detection.
// *****
CONF:LTE:MEAS:MEV:LIST:SEGM1:SET 10,1,TDD,OB47,59E+8,B100,NORM,PSSch,OFF,0,NS33
CONF:LTE:MEAS:MEV:LIST:SEGM2:SET 10,-9,TDD,OB47,59E+8,B100,NORM,PSSch,OFF,0,NS33
CONF:LTE:MEAS:MEV:LIST:SEGMENT1:RBALLOCATION:SIDELINK OFF, KEEP, KEEP, KEEP
CONF:LTE:MEAS:MEV:LIST:SEGMENT2:RBALLOCATION:SIDELINK OFF, KEEP, KEEP, KEEP

// *****
// Define a second UL carrier with lower bandwidth, located above the first
// UL carrier and adjust the frequency of the second carrier automatically.
// *****
CONF:LTE:MEAS:MEV:LIST:SEGMENT1:CC2 3E+9,B050
CONF:LTE:MEAS:MEV:LIST:SEGMENT2:CC2 3E+9,B050
CONF:LTE:MEAS:MEV:LIST:SEGMENT1:CAGGREGATION:ACSPACING
CONF:LTE:MEAS:MEV:LIST:SEGMENT2:CAGGREGATION:ACSPACING

// *****
// For single-carrier UL measurements, measure the CC1.
// *****
CONF:LTE:MEAS:MEV:LIST:SEGMENT1:CAGGREGATION:MCCARRIER:ENHANCED CC1
CONF:LTE:MEAS:MEV:LIST:SEGMENT2:CAGGREGATION:MCCARRIER:ENHANCED CC1

// *****
// Configure the physical layer cell ID per segment.
// *****
CONF:LTE:MEAS:MEV:LIST:PLCMode LIST
CONF:LTE:MEAS:MEV:LIST:SEGMENT1:PLCID 5
CONF:LTE:MEAS:MEV:LIST:SEGMENT2:PLCID 6

// *****
// Enable modulation results except phase error for both segments. Use
// a statistical length of 10 slots and select automatic detection of the
```

```
// modulation scheme.
// *****
CONF:LTE:MEAS:MEvaluation:LIST:SEGment1:MODulation 10,ON,ON,ON,OFF,ON,ON,AUTO
CONF:LTE:MEAS:MEvaluation:LIST:SEGment2:MODulation 10,ON,ON,ON,OFF,ON,ON,AUTO

// *****
// Enable all spectrum emission results for segment 2.
// Enable the E-UTRA ACLR results for segment 2 (no UTRA results).
// Use a statistical length of 10 slots.
// *****
CONFigure:LTE:MEAS:MEvaluation:LIST:SEGment2:SEMask 10,ON,ON,ON
CONFigure:LTE:MEAS:MEvaluation:LIST:SEGment2:ACLR 10,ON,OFF,OFF,ON

// *****
// Enable power results for segment 1 and 2.
// *****
CONFigure:LTE:MEAS:MEvaluation:LIST:SEGment1:PMONitor ON
CONFigure:LTE:MEAS:MEvaluation:LIST:SEGment2:PMONitor ON
CONFigure:LTE:MEAS:MEvaluation:LIST:SEGment1:POWer 5,ON
CONFigure:LTE:MEAS:MEvaluation:LIST:SEGment2:POWer 5,ON

// *****
// Select both segments for measurement and select trigger mode "ONCE".
// *****
CONFigure:LTE:MEAS:MEvaluation:LIST:LRANge 1,2
TRIGger:LTE:MEAS:MEvaluation:LIST:MODE ONCE

// *****
// Enable the list mode.
// *****
CONFigure:LTE:MEAS:MEvaluation:LIST ON
```

5.4.2.3 Running a list mode measurement

```
// *****
// Start the measurement.
// *****
INIT:LTE:MEAS:MEvaluation

// *****
// Return average and extreme modulation results.
// Return extreme in-band emission and spectrum flatness results.
// Return limit check results for the average results.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:MODulation:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:MODulation:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:MODulation:EXTReme?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:MODulation:EXTReme?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:IEmission:CC1:MARGin:EXTReme?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:IEmission:CC1:MARGin:EXTReme?
```



```

FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:IEmission:CC1:MARGin:EXTreme:RBINdex?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:IEmission:CC1:MARGin:EXTreme:RBINdex?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:IEmission:CC2:MARGin:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:IEmission:CC2:MARGin:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:IEmission:CC2:MARGin:EXTreme:RBINdex?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:IEmission:CC2:MARGin:EXTreme:RBINdex?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:ESFlatness:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:ESFlatness:EXTreme?
CALCulate:LTE:MEAS:MEvaluation:LIST:SEGment1:MODulation:AVERage?
CALCulate:LTE:MEAS:MEvaluation:LIST:SEGment2:MODulation:AVERage?
CALCulate:LTE:MEAS:MEvaluation:LIST:SEGment1:ESFlatness:AVERage?
CALCulate:LTE:MEAS:MEvaluation:LIST:SEGment2:ESFlatness:AVERage?

// *****
// Return average and extreme spectrum emission results.
// Return spectrum emission margins: x- and y-values of minimum margins and
// y-values of all margins.
// Return limit check results for the average spectrum emission results.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:SEMask:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:SEMask:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:SEMask:MARGin:MINimum:POSitiv?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:SEMask:MARGin:MINimum:NEGativ?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:SEMask:MARGin:ALL?
CALCulate:LTE:MEAS:MEvaluation:LIST:SEGment2:SEMask:AVERage?

// *****
// Return current and average ACLR results.
// Return limit check results for the average ACLR results.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:ACLR:CURRent?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:ACLR:AVERage?
CALCulate:LTE:MEAS:MEvaluation:LIST:SEGment2:ACLR:AVERage?

// *****
// Return RMS power monitor results for segment 1 and 2.
// Return current and average total TX power results for segment 2.
// Return current and average TX power results for segment 1, carrier 2.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:PMONitor:RMS?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:PMONitor:RMS?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:POWer:CURRent?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment2:POWer:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:POWer:CC2:CURRent?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGment1:POWer:CC2:AVERage?

// *****
// Return detected allocation, modulation scheme and channel type of the
// last slot within the statistical length of the modulation measurement.
// *****

```

```

FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT1:MODulation:DALLocation?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT1:MODulation:DMODulation?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT1:MODulation:DCHType?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT2:MODulation:DALLocation?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT2:MODulation:DMODulation?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT2:MODulation:DCHType?

// *****
// Select segment no. 2 as offline segment (and implicitly enable the
// offline mode). Restart the measurement to calculate all results in
// segment 2. Go to local to view the results.
// *****
CONFigure:LTE:MEAS:MEvaluation:LIST:OSIndex 2
INIT:LTE:MEAS:MEvaluation
&GTL

```

5.4.2.4 Retrieving single results for all segments

```

// *****
// Return selected modulation results.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:EVM:RMS:LOW:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:EVM:PEAK:LOW:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:EVM:DMRS:LOW:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:MERRor:RMS:LOW:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:MERRor:PEAK:LOW:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:MERRor:DMRS:LOW:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:PERRor:RMS:LOW:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:PERRor:PEAK:HIGh:CURRent?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:PERRor:DMRS:HIGh:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:IQOffset:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:FERRor:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:TERRor:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:TPower:MAXimum?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:PPower:MINimum?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:PSD:MINimum?

// *****
// Return selected in-band emission and spectrum flatness results.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:IEmission:MARGin:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:IEmission:MARGin:RBIndex:EXTreme?
FETCh:LTE:MEAS:MEvaluation:LIST:ESFLatness:RIPple1:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:ESFLatness:DIFFerence2:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:ESFLatness:MINR1:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:ESFLatness:MAXR2:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:ESFLatness:SCIndex:MINimum1:CURRent?

// *****
// Return selected spectrum emission and ACLR results.

```

```
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:SEMask:OBW:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:SEMask:TXPower:MAXimum?
FETCh:LTE:MEAS:MEvaluation:LIST:SEMask:MARGin:AREA5:NEGativ:MINimum?
FETCh:LTE:MEAS:MEvaluation:LIST:ACLR:EUTRa:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:ACLR:EUTRa:NEGativ:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:ACLR:UTRA2:POSitiv:AVERage?

// *****
// Query the RMS power monitor results.
// Query the offset of the first result for segment 2 within the
// returned list of power values.
// Query the number of power results related to segment 2.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:PMONitor:RMS?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT2:PMONitor:ARRay:START?
FETCh:LTE:MEAS:MEvaluation:LIST:SEGMENT2:PMONitor:ARRay:LENGth?

// *****
// Return selected total TX power results (all carriers).
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:POWer:TXPower:AVERage?
FETCh:LTE:MEAS:MEvaluation:LIST:POWer:TXPower:MAXimum?

// *****
// Return detected allocation, modulation scheme and channel type of the
// last slot within the statistical length of the modulation measurement.
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:DALLocation?
FETCh:LTE:MEAS:MEvaluation:LIST:MODulation:DMODulation?
FETCh:LTE:MEAS:MEvaluation:LIST:SEMask:DCHType?

// *****
// Return the individual segment reliability indicators
// *****
FETCh:LTE:MEAS:MEvaluation:LIST:SRELiability?
```

5.5 Command reference

The following sections describe the remote control commands of the LTE multi-evaluation measurement. The common commands applicable to all LTE measurements are also described here.

- [Conventions and general information](#)..... 1104
- [Common measurement settings](#)..... 1111
- [Multi-evaluation measurement commands](#)..... 1124
- [Combined signal path commands](#)..... 1287

5.5.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

5.5.1.1 MEAS<i>

MEAS<i> is used as an abbreviation of "MEASurement<instance>". For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same measurement application. It can be omitted if the instrument supports only one instance, or to address the first instance.

5.5.1.2 FETCh, READ and CALCulate commands

The commands are used to retrieve measurement results:

- A **FETCh...** command returns the results of the measurement cycle after the cycle is complete. Before sending a **FETCh...**, start the measurement via an **INITiate...**
- A **READ...** command starts a new single-shot measurement and returns the results.
- A **CALCulate...** command returns one limit check result per **FETCh** result:
 - **OK**: The **FETCh** result is located within the limits or no limit has been defined/enabled for this result.
 - **ULEU** ("User limit exceeded upper"): An upper limit is violated. The **FETCh** result is located above the limit.
 - **ULEL** ("User limit exceeded lower"): A lower limit is violated. The **FETCh** result is located below the limit.

5.5.1.3 Current and statistical results

The measurement is repeated according to the selected statistic count and repetition mode. Consecutive measurement values are stored and used to calculate statistical results.

Commands for result query often contain a mnemonic that indicates which results are returned: **CURRent**, **AVERage**, **EXTReme**, **MINimum**, **MAXimum** or **SDEViation**.

5.5.1.4 Values for RF path selection

To select an RF input path, you must specify an RF connector and an RX module (converter).

Which connectors and modules can be specified in a command, depends on the installed hardware, the test setup and the active subinstrument or instance <i>.

This section lists all values available for path selection. Depending on your configuration, only a subset is relevant for you. Virtual connector names are only relevant for setting commands. Queries return the physical connector names.

Additional information is available in the base unit documentation. It describes typical instrument configurations with the allowed RF connector - TX/RX module combinations and the mapping of virtual connector names to physical connectors.

Single R&S CMW500 / 270 / 290

RF path selection values:

- RX module:
RX1 | RX2 | RX3 | RX4
- RX connector:
RF1C | RF2C | RF3C | RF4C | RFAC | RFBC
RF 1 COM to RF 4 COM plus virtual connector names

CMWflexx setup with R&S CMWC and several R&S CMW500

RF path selection values:

- RX module:
RX11 | RX12 | RX13 | RX14 | RX21 | RX22 | RX23 | RX24
RX31 | RX32 | RX33 | RX34 | RX41 | RX42 | RX43 | RX44
RX<a>: CMW <a>, RX
Example RX34: RX module 4 of CMW 3
- RX connector:
R11C | R12C | R13C | R14C | R21C | R22C | R23C | R24C
R31C | R32C | R33C | R34C | R41C | R42C | R43C | R44C
R<a>C: CMW <a>, connector RF COM
Example R34C: RF 4 COM of CMW 3

For CMW 1, you can alternatively use the single-CMW RF path selection values.

5.5.1.5 Frequency ranges

The following frequency ranges are supported:

- Basic frequency range RX: 150 MHz to 3.3 GHz
- Basic frequency range TX: 70 MHz to 3.3 GHz
- Frequency range extension up to 6 GHz: R&S CMW-KB036

5.5.1.6 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**
If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:
 - "Query only": The command can only be used to query parameters. No setting.
 - "Event": The command initiates an event. No query.

- **Parameter usage**

The parameter usage is indicated by the keyword preceding the parameters:

- "Parameters" are sent with a setting or query command and are returned as the result of a query.
- "Setting parameters" are sent with a setting command.
- "Query parameters" are sent with a query command, to refine the query.
- "Return values" are returned as the result of a query.

- **Parameter type**

The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:

- "string": characters enclosed in quotation marks (" or ')
- "numeric", "decimal": decimal number
- "integer": integer number
- "float": floating point number
- "hex": hexadecimal number starting with #H
- "binary": binary number starting with #B
- "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).

- **Firmware/Software:**

Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

5.5.1.7 Reliability indicator

The first value in the sequences returned by measurement result queries indicates the most severe error that has occurred during the measurement.

Example for an output array: 0, 10.22, 10.15, 10.01, 10.29, 100 (reliability = 0, followed by 5 numeric measurement values).

In list mode, a zero reliability indicator indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments.

The reliability indicator has one of the following values:

- **0 - "no error"**
No error detected.
- **1 - "measurement timeout"**
The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INValid` results or has not completed the full statistic count.
- **2 - "capture buffer overflow"**
The measurement configuration results in a capture length that exceeds the available memory.
- **3 - "input overdriven" / 4 - "input underdriven"**

The accuracy of measurement results can be impaired because the input signal level was too high / too low.

- **6 - "trigger timeout"**
The measurement could not be started or continued because no trigger event was detected.
For 1xEV-DO: A test application report was not received from the mobile within three update time periods.
- **7 - "acquisition error"**
The R&S CMW could not properly decode the RF input signal.
- **8 - "sync error"**
The R&S CMW could not synchronize to the RF input signal.
- **9 - "uncalibrated"**
Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the technical specifications.
- **15 - "reference frequency error"**
The instrument has been configured to use an external reference signal. But the reference oscillator could not be locked to the external signal. For example, signal level too low, frequency out of range or reference signal not available at all.
- **16 - "RF not available"**
The measurement could not be started because the configured RF input path was not active.
This problem can occur if a measurement is started in combined signal path mode and the controlling application has not yet activated the input path.
- **17 - "RF level not settled" / 18 - "RF frequency not settled"**
The measurement could not be started. The R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 - "call not established"**
For measurements: The measurement could not be started because no signaling connection to the DUT was established.
For IMS service: Establishing a voice over IMS call failed.
- **20 - "call type not usable"**
For measurements: The measurement could not be started because the established signaling connection had the wrong properties.
For IMS service: The voice over IMS settings could not be applied.
- **21 - "call lost"**
For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.
For IMS service: The VoIMS call was lost.
- **23 - "missing option"**
An action cannot be executed due to a missing option.
For GPRF generator: The ARB file cannot be played due to a missing option.
- **24 - "invalid RF setting"**
The desired RF TX level or RF RX reference level could not be applied.
- **25 - "level overrange"**
The RF TX level is in overrange. The signal quality can be degraded.

- **26 - "resource conflict"**
The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application. Stop the application that has allocated the conflicting resources and try again.
- **27 - "no sensor connected"**
The GPRF external power sensor measurement could not be started due to a missing power sensor.
- **28 - "unexpected parameter change"**
One or more measurement configuration parameters were changed while the measurement was running. The results were not obtained with these new parameter values. Repeat the measurement. This situation can only occur in remote single-shot mode.
- **29 - "invalid RF frequency setting"**
The desired RF TX frequency or RF RX frequency could not be applied.
- **30 - "file not found"**
The specified file could not be found.
- **31 - "no DTM reply"**
The EUT did not reply to the direct test mode (DTM) command.
- **32 - "ACL disconnected"**
The ACL connection has been disconnected or lost.
- **40 - "ARB file CRC error"**
The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.
- **42 - "ARB header tag invalid"**
The ARB file selected in the GPRF generator contains an invalid header tag.
- **43 - "ARB segment overflow"**
The number of segments in the multi-segment ARB file is higher than the allowed maximum.
- **44 - "ARB file not found"**
The selected ARB file could not be found.
- **45 - "ARB memory overflow"**
The ARB file length is greater than the available memory.
- **46 - "ARB sample rate out of range"**
The clock rate of the ARB file is either too high or too low.
- **47 - "ARB cycles out of range"**
The repetition mode equals "Single Shot" and the playback length is too long. Reduce the playback length or set the repetition mode to "Continuous".
$$\text{<Length>} = (\text{<Cycles>} * \text{<Samples>} + \text{<Additional Samples>}) / \text{<Clock Rate>}$$
- **48 - "ARB pool invalid"**
At least one of the ARB files in the ARB file pool is invalid.
- **49 - "sequencer list invalid"**
The sequencer list configuration has at least one invalid entry.
- **50 - "startup error"**
The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 - "no reply"**

The DAU has received no response, for example for a ping request.

- **52 - "connection error"**

A connection setup failed or a connection was lost.

For DAU: The DAU could not establish a connection to internal components.

Restart the instrument.

For Bluetooth: The EUT has been disconnected from the R&S CMW.

- **53 - "configuration error"**

The current DAU configuration is incomplete or wrong and could not be applied.

Check especially the IP address configuration.

- **54 - "file system error"**

The system drive of the DAU is full or corrupt. Execute a DAU self-test.

- **64 - "missing pathloss correction data"**

The pathloss measurement cannot be executed due to missing connector correction data. Order a calibration of the instrument by Rohde & Schwarz.

- **70 - "wrong standard"**

The standard of the measured signal does not match the configured standard.

- **71 - "wrong bandwidth"**

The bandwidth of the measured signal does not match the configured bandwidth.

- **72 - "wrong burst type"**

The burst type of the measured signal does not match the configured burst type.

- **73 - "MIMO signal detected"**

The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.

- **74 - "more streams than antennas"**

The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.

- **75 - "matrix inversion failed"**

The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.

- **76 - "SIG CRC failed"**

The cyclic redundancy check of a SIGNAL field failed.

- **77 - "parity check failed"**

The parity check of a SIGNAL field failed.

- **78 - "bursts not identical"**

In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.

- **79 - "wrong modulation"**

The modulation scheme of the measured signal does not match the configured modulation scheme.

- **80 - "SISO signal detected"**

Received signal contains only one stream.

- **81 - "fewer streams than antennas"**

Received signal has fewer streams than configured antennas. Reconfigure the number of antennas.

- **82 - "signal fields of different antennas are not identical"**
The signal fields of all captured antennas are compared. In a valid MIMO transmission, all signal fields are identical. If not, some basic parameters such as MCS or data length are not the same and a MIMO measurement is not possible.
- **83 - "bit errors in some of the received symbols"**
Adjust input signal power and repeat training.
- **84 - "first and last symbols need to be skipped"**
Due to a sequence counter in the MAC header, the first symbol and the last symbol change from one burst to another. You can configure the first and the last symbol to be skipped.
- **85 - "received and saved bursts are not identical"**
The signal field of the received burst is not identical to that of the saved burst. Some basic parameters such as MCS or data length are not the same and a composite MIMO measurement is not possible.
- **86 - "no symbols for measurement"**
Too many head and tail symbols have been skipped.
- **93 - "OCXO oven temperature too low"**
The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **96 - "frequency/level change too fast"**
There was a too fast frequency and/or level change. If you use GPRF with list mode, increase the duration of the list entry.
- **101 - "firmware error"**
This value indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.
If the error occurs again, consider the following hints:
 - Firmware errors can often be repaired by restoring the factory default settings.
 - If a software package has not been properly installed, this failure is often indicated in the "Setup" dialog > section "SW/HW-Equipment > Installed Software".
 - Check for software updates correcting the error. Updates are provided in the customer web on GLORIS (registration required): <https://gloris.rohde-schwarz.com>.

If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.
- **102 - "unidentified error"**
This value indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 firmware error".
- **103 - "parameter error"**
This value indicates that the measurement could not be performed due to internal conflicting parameter settings.
For 1xEV-DO:
 - The 1xEV-DO signaling test application is not enabled.
 - Forward link measurement data were requested, but the test application direction is not set to "Forward" or "Forward & Reverse".

- Reverse link measurement data were requested, but the test application direction is not set to "reverse" or "forward & reverse".
- Measurement data were requested, but the corresponding measurement is not assigned.

For CDMA2000: The "1st Service Option" is not set to a loopback service option. A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.

If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see <https://www.service.rohde-schwarz.com>).

- **104 - "not functional"**

The application could not be started with the configured parameter set.

5.5.2 Common measurement settings

The commands valid for all LTE measurements are divided into the groups listed below.

- [Basic decisions](#)..... 1111
- [Signal routing](#)..... 1112
- [Analyzer settings](#)..... 1117
- [Common measurement control settings](#)..... 1123

5.5.2.1 Basic decisions

The following commands select the signal type and the duplex mode and query the resulting frame structure.

- [CONFigure:LTE:MEAS<i>:STYPe](#)..... 1111
- [CONFigure:LTE:MEAS<i>:DMODE](#)..... 1112
- [CONFigure:LTE:MEAS<i>:FSTRucture?](#)..... 1112

CONFigure:LTE:MEAS<i>:STYPe <SignalType>

Selects the type of the measured signal.

Parameters:

<SignalType> UL | SL
 UL: LTE uplink signal
 SL: V2X sidelink signal

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V3.7.55

Options: UL: R&S CMW-KM500 or -KM550
 SL: R&S CMW-KM570

Manual operation: See ["Signal Type"](#) on page 1041

CONFigure:LTE:MEAS<i>:DMODE <Mode>

Selects the duplex mode of the LTE signal: FDD or TDD.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>[:PCC]:DMODE`
- `CONFigure:LTE:SIGN<i>:SCC<c>:DMODE`
- `CONFigure:LTE:SIGN<i>[:PCC]:DMODE:UCSPecific`

Parameters:

<Mode> FDD | TDD

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V2.0.10

Options: FDD requires R&S CMW-KM500.
TDD requires R&S CMW-KM550 or R&S CMW-KM570.

Manual operation: See ["Duplex Mode"](#) on page 1042

CONFigure:LTE:MEAS<i>:FSTRucture?

Queries the frame structure type of the LTE signal. The value depends on the duplex mode (`CONFigure:LTE:MEAS<i>:DMODE`).

Return values:

<FrameStructure> T1 | T2
T1: Type 1, FDD signal
T2: Type 2, TDD signal

Example: See [Specifying general and common measurement settings](#)

Usage: Query only

Firmware/software: V2.0.20

Manual operation: See ["Frame Structure"](#) on page 1054

5.5.2.2 Signal routing

The following commands configure the scenario and select the input path for the measured signal.

<code>ROUTe:LTE:MEAS<i>:SCENario:SALone</code>	1113
<code>ROUTe:LTE:MEAS<i>:SCENario:CSPath</code>	1113
<code>CONFigure:LTE:MEAS<i>:CAGGregation:MAPing?</code>	1114
<code>CONFigure:LTE:MEAS<i>:CAGGregation:MAPing:PCC?</code>	1114
<code>CONFigure:LTE:MEAS<i>:CAGGregation:MAPing:SCC<c>?</code>	1115
<code>SENSe:LTE:MEAS<i>:CAGGregation:FSHWare?</code>	1115
<code>ROUTe:LTE:MEAS<i>:SCENario:MAPRotocol</code>	1116
<code>ROUTe:LTE:MEAS<i>:SCENario?</code>	1116
<code>ROUTe:LTE:MEAS<i>?</code>	1116

ROUTe:LTE:MEAS<i>:SCENario:SALone <RXConnector>, <RFConverter>

Activates the standalone scenario and selects the RF input path for the measured RF signal.

For possible connector and converter values, see [Section 5.5.1.4, "Values for RF path selection"](#), on page 1104.

Parameters:

<RXConnector> RF connector for the input path

<RFConverter> RX module for the input path

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V1.0.15.0

Manual operation: See ["Scenario = StandAlone"](#) on page 1042

ROUTe:LTE:MEAS<i>:SCENario:CSPath <Controller>[, <Carrier>]

Activates the combined signal path scenario and selects the controlling application and primary carrier. The selected application controls most signal routing settings, analyzer settings and some measurement control settings while the combined signal path scenario is active.

The command usage depends on the carrier aggregation mode of the measured signal: no UL carrier aggregation, non-contiguous UL carrier aggregation or intraband contiguous UL carrier aggregation. The following table provides an overview.

CA type	Setting command	Query returns
No UL CA	ROUT:LTE:MEAS:SCEN:CSP <Controller> <Carrier> can be skipped and equals "PCC".	<Controller>, "PCC"
Non-contiguous UL CA	ROUT:LTE:MEAS:SCEN:CSP <Controller>, <Carrier> <Carrier>: measured carrier ("PCC", "SCC2", ...)	<Controller>, <Carrier>
Intraband contiguous UL CA	ROUT:LTE:MEAS:SCEN:CSP <Controller>, <Carrier> <Carrier>: set of carriers ("Set A", "Set B", ...)	<Controller>, <Carrier>, <Set> <Carrier>: carrier selecting RF path ("PCC", "SCC2", ...) <Set>: measured set of carriers ("Set A", "Set B", ...)

Parameters:

<Controller> string
Controlling application
Example: 'LTE Sig1' or 'LTE Sig2'

<Carrier> string
Uplink carrier or set of uplink carriers configured in the controlling application
Examples: 'PCC', 'SCC2', 'Set A', 'Set B'
If a set is selected, a query returns the carrier of the set that is used by the measurement to select the RF path.

Return values:

<Set> string
 Measured set of uplink carriers
 Example: 'Set A', 'Set B'
 Only returned for intraband contiguous UL CA.

Firmware/software: V3.5.10, changed CA functionality in V3.7.30

Manual operation: See "[Scenario = Combined Signal Path](#)" on page 1042

CONFigure:LTE:MEAS<i>:CAGGregation:MAPing?

This command is only relevant for combined signal path measurements with contiguous uplink CA.

It queries which carriers are mapped to CC1 to CC4.

The measurement identifies the aggregated carriers as CC1 to CC4. The signaling application uses PCC and SCC<n>.

A returned INV means that no carrier is mapped to the CC.

Return values:

<CC1> INV | PCC | SCC1 | SCC2 | SCC3 | SCC4 | SCC5 | SCC6 | SCC7
 Carrier mapped to CC1

<CC2> INV | PCC | SCC1 | SCC2 | SCC3 | SCC4 | SCC5 | SCC6 | SCC7
 Carrier mapped to CC2

<CC3> INV | PCC | SCC1 | SCC2 | SCC3 | SCC4 | SCC5 | SCC6 | SCC7
 Carrier mapped to CC3

<CC4> INV | PCC | SCC1 | SCC2 | SCC3 | SCC4 | SCC5 | SCC6 | SCC7
 Carrier mapped to CC4

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See "[Scenario = Combined Signal Path](#)" on page 1042

CONFigure:LTE:MEAS<i>:CAGGregation:MAPing:PCC?

This command is only relevant for combined signal path measurements with contiguous uplink CA.

It queries to which CC the PCC is mapped.

The measurement identifies the aggregated carriers as CC1 to CC4. The signaling application uses PCC and SCC<n>.

Return values:

<CC> string
 Examples: "CC1", "CC2", "INV"
 "INV" means that the PCC is not contained in the measured set of aggregated uplink carriers.

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Scenario = Combined Signal Path"](#) on page 1042

CONFigure:LTE:MEAS<i>:CAGGregation:MAPing:SCC<c>?

This command is only relevant for combined signal path measurements with contiguous uplink CA.

It queries to which CC the SCC<c> is mapped.

The measurement identifies the aggregated carriers as CC1 to CC4. The signaling application uses PCC and SCC<c>.

Suffix:

<c> 1..7

Return values:

<CC> string
 Examples: "CC1", "CC2", "INV"
 "INV" means that the SCC<c> is not contained in the measured set of aggregated uplink carriers.

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See ["Scenario = Combined Signal Path"](#) on page 1042

SENSe:LTE:MEAS<i>:CAGGregation:FSHWare?

This command is only relevant for combined signal path measurements in multi-CMW setups.

It queries whether the measurement instance and the carrier to be measured are in the same CMW. If they are in different CMWs, the measurement fails.

To correct the problem, use another measurement instance or select another carrier, so that both are in the same CMW.

Return values:

<Value> OFF | ON
OFF: Different CMWs - error
ON: Same CMW - ok

Usage: Query only

Firmware/software: V3.7.30

Manual operation: See "[Scenario = Combined Signal Path](#)" on page 1042

ROUTe:LTE:MEAS<i>:SCENario:MAPRotocol [<Controler>]

Activates the "Measure@ProtocolTest" scenario and optionally selects the controlling protocol test application.

The signal routing and analyzer settings are ignored by the measurement application. Configure the corresponding settings within the protocol test application used in parallel.

Setting parameters:

<Controler> string
 Protocol test application
 Example: 'Protocol Test1'

Usage: Event

Firmware/software: V1.0.15.20
 V2.1.25: added <Controler>

Manual operation: See "[Scenario = Measure@ProtocolTest](#)" on page 1043

ROUTe:LTE:MEAS<i>:SCENario?

Returns the active scenario.

Return values:

<Scenario> SALone | CSPath | MAPRotocol
 SALone: Standalone (nonsignaling)
 CSPath: Combined signal path
 MAPRotocol: Measure at protocol test

Usage: Query only

Firmware/software: V2.0.10

Manual operation: See "[Scenario = StandAlone](#)" on page 1042

ROUTe:LTE:MEAS<i>?

Returns the configured routing settings.

For possible connector and converter values, see [Section 5.5.1.4, "Values for RF path selection"](#), on page 1104.

Return values:

<Scenario> SALone | CSPath | MAPRotocol
 SALone: Standalone (nonsignaling)
 CSPath: Combined signal path
 MAPRotocol: Measure at protocol test

<Controller>	string
	Controlling application for scenario CSPath or MAPRotocol
<RXConnector>	RF connector for the input path
<RFConverter>	RX module for the input path
Usage:	Query only
Firmware/software:	V2.0.10
Manual operation:	See " Scenario = StandAlone " on page 1042

5.5.2.3 Analyzer settings

The following commands configure the RF input path.

CONFigure:LTE:MEAS<i>:RFSettings:EATTenuation.....	1117
CONFigure:LTE:MEAS<i>:RFSettings:ENPower.....	1118
CONFigure:LTE:MEAS<i>:RFSettings:UMARgin.....	1118
CONFigure:LTE:MEAS<i>:EMTC:ENABLE.....	1119
CONFigure:LTE:MEAS<i>:EMTC:MB<number>.....	1119
CONFigure:LTE:MEAS<i>:CAGGregation:MODE.....	1119
CONFigure:LTE:MEAS<i>:CAGGregation:MODE:CSPath?.....	1120
CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:LOW?.....	1120
CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:CENTer?.....	1120
CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:HIGh?.....	1121
CONFigure:LTE:MEAS<i>:CAGGregation:CBANdwidth:AGGRegated?.....	1121
CONFigure:LTE:MEAS<i>:BAND.....	1121
CONFigure:LTE:MEAS<i>:RFSettings:CC<no>:FREQuency.....	1122
CONFigure:LTE:MEAS<i>:CAGGregation:ACSPacing.....	1122
CONFigure:LTE:MEAS<i>:RFSettings:FOFFset.....	1122
CONFigure:LTE:MEAS<i>:RFSettings:MLOFFset.....	1123

CONFigure:LTE:MEAS<i>:RFSettings:EATTenuation <RFinputExtAtt>

Defines an external attenuation (or gain, if the value is negative), to be applied to the input connector.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:EATTenuation:INPut`
- `CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:EATTenuation:INPut`

Parameters:

<RFinputExtAtt>	numeric
Range:	-50 dB to 90 dB
*RST:	0 dB
Default unit:	dB

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V1.0.10.1

Manual operation: See "[External Attenuation \(Input\)](#)" on page 1044

CONFigure:LTE:MEAS<i>:RFSettings:ENPower <ExpNomPow>

Sets the expected nominal power of the measured RF signal.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPMODE`
- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPower`
- `CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPMODE`
- `CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPower`

Parameters:

<ExpNomPow> numeric

The range of the expected nominal power can be calculated as follows:

$$\text{Range (Expected Nominal Power)} = \text{Range (Input Power)} + \text{External Attenuation} - \text{User Margin}$$

The input power range is stated in the specifications document.

*RST: 0 dBm

Default unit: dBm

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V1.0.10.1
V3.0.10: enhanced range

Manual operation: See ["Expected Nominal Power"](#) on page 1045

CONFigure:LTE:MEAS<i>:RFSettings:UMArgin <UserMargin>

Sets the margin that the measurement adds to the expected nominal power to determine the reference power. The reference power minus the external input attenuation must be within the power range of the selected input connector. Refer to the specifications document.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UMArgin`
- `CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UMArgin`

Parameters:

<UserMargin> numeric

Range: 0 dB to (55 dB + external attenuation - expected nominal power)

*RST: 0 dB

Default unit: dB

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V1.0.10.1
V3.0.10: enhanced range

Manual operation: See ["User Margin"](#) on page 1046

CONFigure:LTE:MEAS<i>:EMTC:ENABLE <Enable>

Enables or disables eMTC.

For the combined signal path scenario, use **CONFigure:LTE:SIGN<i>[:PCC]:EMTC:ENABLE**.

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V3.7.10

Manual operation: See "eMTC" on page 1043

CONFigure:LTE:MEAS<i>:EMTC:MB<number> <Enable>

Selects the maximum eMTC bandwidth.

For the combined signal path scenario, use **CONFigure:LTE:SIGN<i>[:PCC]:EMTC:MB<number>**.

Suffix:

<number> 5

Parameters:

<Enable> OFF | ON
OFF: Max bandwidth 1.4 MHz
ON: Max bandwidth 5 MHz
*RST: OFF

Firmware/software: V3.8.10

Manual operation: See "Max Bandwidth 5 MHz" on page 1043

CONFigure:LTE:MEAS<i>:CAGGregation:MODE <CAmode>

Selects how many component carriers with intraband contiguous aggregation are measured.

For the combined signal path scenario, use **ROUTE:LTE:MEAS<i>:SCENario:CSPath**.

Parameters:

<CAmode> OFF | INTRaband | ICD | ICE
OFF: Only one carrier is measured.
INTRaband: two carriers (BW class B & C)
ICD: three carriers (BW class D)
ICE: four carriers (BW class E)
*RST: OFF

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V3.2.70, V3.7.50: ICD, V3.7.55: ICE
Options: R&S CMW-KM502/-KM552 for INTRaband
 R&S CMW-KM504/-KM554 for ICD, ICE
Manual operation: See ["Carrier Aggregation Mode"](#) on page 1048

CONFigure:LTE:MEAS<i>:CAGGregation:MODE:CSPath?

Queries the carrier aggregation mode in the CSP scenario. The mode is configured indirectly via [ROUTE:LTE:MEAS<i>:SCENario:CSPath](#) on page 1113.

Return values:

<CAmode> OFF | INTRaband | ICD | ICE
OFF: no carrier aggregation
INTRaband: Intraband contiguous CA (BW class B & C).
ICD: Intraband contiguous CA (BW class D).
ICE: Intraband contiguous CA (BW class E).

Usage: Query only
Firmware/software: V3.7.30, V3.7.50: ICD, V3.7.55: ICE
Manual operation: See ["Carrier Aggregation Mode"](#) on page 1048

CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:LOW?

Queries the lower edge of the aggregated bandwidth.

Return values:

<FrequencyLow> float
 Default unit: Hz
Example: See [Specifying general and common measurement settings](#)
Usage: Query only
Firmware/software: V3.2.70
Manual operation: See ["Low, Center, High, Bandwidth"](#) on page 1049

CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:CENTER?

Queries the center frequency of the aggregated bandwidth.

Return values:

<FrequencyCenter> float
 Default unit: Hz
Example: See [Specifying general and common measurement settings](#)
Usage: Query only
Firmware/software: V3.2.70
Manual operation: See ["Low, Center, High, Bandwidth"](#) on page 1049

CONFigure:LTE:MEAS<i>:CAGGregation:FREQuency:AGGRegated:HIGH?

Queries the upper edge of the aggregated bandwidth.

Return values:

<FrequencyHigh> float
Default unit: Hz

Example: See [Specifying general and common measurement settings](#)

Usage: Query only

Firmware/software: V3.2.70

Manual operation: See ["Low, Center, High, Bandwidth"](#) on page 1049

CONFigure:LTE:MEAS<i>:CAGGregation:CBANDwidth:AGGRegated?

Queries the width of the aggregated channel bandwidth.

Return values:

<ChBandwidth> float
Default unit: Hz

Example: See [Specifying general and common measurement settings](#)

Usage: Query only

Firmware/software: V3.2.70

Manual operation: See ["Low, Center, High, Bandwidth"](#) on page 1049

CONFigure:LTE:MEAS<i>:BAND <Band>

Selects the operating band (OB).

The allowed input range has dependencies:

- FDD UL: OB1 | ... | OB28 | OB30 | OB31 | OB65 | OB66 | OB68 | OB70 | ... | OB74 | OB85 | OB87 | OB88
- TDD UL: OB33 | ... | OB45 | OB48 | OB50 | ... | OB53 | OB250
- Sidelink: OB47

For the combined signal path scenario, use:

- [CONFigure:LTE:SIGN<i>\[:PCC\]:BAND](#)
- [CONFigure:LTE:SIGN<i>:SCC<c>:BAND](#)

Parameters:

<Band> OB1 to OB250, see list above

Firmware/software: V1.0.10.1, some bands added in later versions.

Manual operation: See ["Band, Channel, Frequency"](#) on page 1045

CONFigure:LTE:MEAS<i>:RFSettings:CC<no>:FREQUENCY <AnalyzerFreq>

Selects the center frequency of component carrier CC<no>. Without carrier aggregation, you can omit <no>.

Using the unit CH, the frequency can be set via the channel number. The allowed channel number range depends on the operating band, see [Section 5.2.11.4, "Frequency bands"](#), on page 1012.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:UL`
- `CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:UL`

For the supported frequency range, see [Section 5.5.1.5, "Frequency ranges"](#), on page 1105.

Suffix:

<no> 1..4

Parameters:

<AnalyzerFreq> numeric
 *RST: Depends on <no>
 Default unit: Hz

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V3.7.30

Manual operation: See ["Band, Channel, Frequency"](#) on page 1045

CONFigure:LTE:MEAS<i>:CAGGregation:ACSPacing

Adjusts the component carrier frequencies, so that the carriers are aggregated contiguously.

For the combined signal path scenario, use `CONFigure:LTE:SIGN<i>:CAGGregation:SET`.

Example: See [Specifying general and common measurement settings](#)

Usage: Event

Firmware/software: V3.7.30

Manual operation: See ["Adjust Carrier Spacing"](#) on page 1049

CONFigure:LTE:MEAS<i>:RFSettings:FOFFset <Offset>

Specifies a positive or negative frequency offset to be added to the carrier center frequency (`CONFigure:LTE:MEAS<i>:RFSettings:CC<no>:FREQUENCY`).

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL`
- `CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:UL`

- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOffset:UL:UCSPecific`

Parameters:

<Offset> numeric
 Range: -100E+3 Hz to 100E+3 Hz
 *RST: 0 Hz
 Default unit: Hz

Firmware/software: V3.2.10

Manual operation: See ["Frequency Offset"](#) on page 1045

CONFigure:LTE:MEAS<i>:RFSettings:MLOffset <MixLevOffset>

Varies the input level of the mixer in the analyzer path.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:MLOffset`
- `CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:MLOffset`

Parameters:

<MixLevOffset> numeric
 Range: -10 dB to 10 dB
 *RST: 0 dB
 Default unit: dB

Firmware/software: V1.0.15.20

Manual operation: See ["Mixer Level Offset"](#) on page 1046

5.5.2.4 Common measurement control settings

The following commands configure measurement control settings, valid for all LTE measurements.

<code>CONFigure:LTE:MEAS<i>:CAGGregation:MCARrier:ENHanced</code>	1123
<code>CONFigure:LTE:MEAS<i>:CC<no>:CBANdwidth</code>	1124

CONFigure:LTE:MEAS<i>:CAGGregation:MCARrier:ENHanced <MeasCarrier>

Selects a component carrier for single carrier measurements.

Parameters:

<MeasCarrier> CC1 | CC2 | CC3 | CC4
 *RST: CC1

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V3.7.30

Manual operation: See ["Measurement Carrier"](#) on page 1048

CONFigure:LTE:MEAS<i>:CC<no>:CBANdwidth <ChannelBW>

Selects the channel bandwidth of component carrier CC<no>. Without carrier aggregation, you can omit <no>.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL`
- `CONFigure:LTE:SIGN<i>:CELL:BANDwidth:SCC<c>:DL`

Suffix:

<no> 1..4

Parameters:

<ChannelBW> B014 | B030 | B050 | B100 | B150 | B200

B014: 1.4 MHz

B030: 3 MHz

B050: 5 MHz

B100: 10 MHz

B150: 15 MHz

B200: 20 MHz

*RST: B200

Example: See [Specifying general and common measurement settings](#)

Firmware/software: V3.7.30

Manual operation: See ["Channel Bandwidth"](#) on page 1050

5.5.3 Multi-evaluation measurement commands

The commands for the LTE multi-evaluation measurement are divided into the groups listed below. The common measurement settings also affect the measurement, see [Section 5.5.2, "Common measurement settings"](#), on page 1111.

• Measurement control and states	1125
• Enabling results	1127
• Measurement parameters - general part	1131
• RB allocation settings	1140
• Modulation measurement settings	1145
• Spectrum measurement settings	1150
• Power measurement settings	1152
• BLER measurement settings	1153
• List mode settings	1154
• Trigger settings	1167
• Limits (modulation, QPSK)	1170
• Limits (modulation, 16QAM / 64QAM / 256QAM)	1175
• Limits (OBW and spectrum emission, no carrier aggregation)	1181
• Limits (ACLR, no carrier aggregation)	1187
• Limits (OBW, with carrier aggregation)	1189
• Limits (spectrum emission, with carrier aggregation)	1191
• Limits (ACLR, with carrier aggregation)	1197

• Limits (power).....	1202
• Result diagram markers.....	1203
• Detected signal configuration.....	1209
• EVM results (diagrams).....	1211
• Magnitude error results (diagrams).....	1214
• Phase error results (diagrams).....	1215
• In-band emission results.....	1216
• Equalizer spectrum flatness results.....	1218
• Spectrum emission results.....	1221
• ACLR spectrum results.....	1226
• I/Q constellation results.....	1229
• RB allocation table results.....	1229
• Power monitor results.....	1230
• Power dynamics results.....	1233
• Modulation results (single values).....	1235
• BLER results.....	1242
• List mode results (one segment).....	1242
• List mode results (all segments, one result).....	1263

5.5.3.1 Measurement control and states

The following commands control the measurement and return the current measurement state.

INITiate:LTE:MEAS<i>:MEValuation.....	1125
STOP:LTE:MEAS<i>:MEValuation.....	1125
ABORT:LTE:MEAS<i>:MEValuation.....	1125
FETCh:LTE:MEAS<i>:MEValuation:STATe?.....	1126
FETCh:LTE:MEAS<i>:MEValuation:STATe:ALL?.....	1126

INITiate:LTE:MEAS<i>:MEValuation

STOP:LTE:MEAS<i>:MEValuation

ABORT:LTE:MEAS<i>:MEValuation

Starts, stops or aborts the measurement:

- The **INITiate:...** command starts or restarts the measurement. The measurement enters the "RUN" state.
- The **STOP:...** command halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- The **ABORT:...** command halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

To query the current measurement state, use **FETCh...STATe?**.

Example: See [Running a measurement](#)

Usage: Event

Firmware/software: V1.0.10.1

Manual operation: See ["Multi Evaluation \(Softkey\)"](#) on page 1016

FETCh:LTE:MEAS<i>:MEValuation:STATe? [<Timeout>[, <TargetMainState>[, <TargetSyncState>]]]

Queries the main measurement state.

Without query parameters, the state is returned immediately.

With query parameters, the state is returned when the <TargetMainState> and the <TargetSyncState> are reached or when the <Timeout> expires.

Query parameters:

<Timeout>	numeric Default unit: ms
<TargetMainState>	OFF RUN RDY Target <MainState> for the query Default is RUN.
<TargetSyncState>	PENDING ADJusted Target <SyncState> for the query Default is ADJ.

Return values:

<MainState>	OFF RUN RDY Current state or target state of ongoing state transition OFF: measurement off RUN: measurement running RDY: measurement completed *RST: OFF
-------------	---

Usage: Query only

Firmware/software: V1.0.10.1
V4.0.140 added optional parameters.

Manual operation: See ["Multi Evaluation \(Softkey\)"](#) on page 1016

FETCh:LTE:MEAS<i>:MEValuation:STATe:ALL? [<Timeout>[, <TargetMainState>[, <TargetSyncState>]]]

Queries the main measurement state and the measurement substates.

Without query parameters, the states are returned immediately.

With query parameters, the states are returned when the <TargetMainState> and the <TargetSyncState> are reached or when the <Timeout> expires.

Query parameters:

<Timeout>	numeric Default unit: ms
-----------	-----------------------------

<TargetMainState>	OFF RUN RDY Target <MainState> for the query Default is RUN.
<TargetSyncState>	PENDING ADJUSTED Target <SyncState> for the query Default is ADJ.
Return values:	
<MainState>	OFF RUN RDY Current state or target state of ongoing state transition OFF: measurement off RUN: measurement running RDY: measurement completed *RST: OFF
<SyncState>	PEND ADJ PEND: transition to <MainState> ongoing ADJ: <MainState> reached
<ResourceState>	QUE ACT INV QUE: waiting for resource allocation ACT: resources allocated INV: no resources allocated
Usage:	Query only
Firmware/software:	V1.0.10.1 V4.0.140 added optional parameters.
Manual operation:	See " Multi Evaluation (Softkey) " on page 1016

5.5.3.2 Enabling results

The following commands select which results are evaluated if the measurement is started via the command `INITiate:LTE:MEAS<i>:MEValuation` or via the GUI.

The GUI shows only result views for enabled results.

<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult[:ALL]</code>	1128
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:EVMagnitude</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:EVMC</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:MERRor</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:PERRRor</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:IEMissions</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:IQ</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:ESFLatness</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:TXM</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:SEMask</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:ACLR</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:RBATable</code>	1129
<code>CONFigure:LTE:MEAS<i>:MEValuation:RESult:PMONitor</code>	1129

CONFigure:LTE:MEAS<i>:MEvaluation:RESult:BLER.....	1129
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:PDYNamics.....	1129
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:EVMagnitude:EVMsymbol.....	1130

CONFigure:LTE:MEAS<i>:MEvaluation:RESult[:ALL] <EVM>, <MagnitudeError>, <PhaseError>, <InbandEmissions>, <EVMversusC>, <IQ>, <EquSpecFlatness>, <TXMeasurement>, <SpecEmMask>, <ACLR>[, <RBAllocTable>, <PowerMonitor>, <BLER>, <PowerDynamics>]

Enables or disables the evaluation of results in the multi-evaluation measurement. This command combines most other

CONFigure:LTE:MEAS<i>:MEvaluation:RESult... commands.

Parameters:

<EVM>	OFF ON Error vector magnitude OFF: Do not evaluate the results. ON: Evaluate the results. *RST: ON
<MagnitudeError>	OFF ON *RST: OFF
<PhaseError>	OFF ON *RST: OFF
<InbandEmissions>	OFF ON *RST: ON
<EVMversusC>	OFF ON EVM vs subcarrier *RST: OFF
<IQ>	OFF ON I/Q constellation diagram *RST: OFF
<EquSpecFlatness>	OFF ON Equalizer spectrum flatness *RST: ON
<TXMeasurement>	OFF ON TX measurement statistical overview *RST: ON
<SpecEmMask>	OFF ON Spectrum emission mask *RST: ON

<ACLR>	OFF ON Adjacent channel leakage power ratio *RST: ON
<RBAllocTable>	OFF ON Resource block allocation table *RST: OFF
<PowerMonitor>	OFF ON *RST: OFF
<BLER>	OFF ON Block error ratio *RST: OFF
<PowerDynamics>	OFF ON *RST: OFF

Example: See [Running a measurement](#)

Firmware/software: V1.0.10.1
V1.0.15.20: added <RBAllocTable>, <PowerMonitor>
V1.0.15.21: added <BLER>
V2.0.10: added <PowerDynamics>

Manual operation: See ["Multi Evaluation, PRACH softkeys > Assign Views hotkey"](#) on page 1086

CONFigure:LTE:MEAS<i>:MEvaluation:RESult:EVMagnitude <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:EVMC <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:MERRor <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:PERRor <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:IEmissions <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:IQ <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:ESFlatness <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:TXM <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:SEMask <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:ACLR <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:RBATable <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:PMONitor <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:BLER <Enable>
CONFigure:LTE:MEAS<i>:MEvaluation:RESult:PDYNamics <Enable>

Enables or disables the evaluation of results in the multi-evaluation measurement.

Mnemonic	Description	Mnemonic	Description
EVMagnitude	Error vector magnitude	EVMC	EVM vs subcarrier
MERRor	Magnitude error	PERRor	Phase error
IEmissions	In-band emissions	IQ	I/Q constellation diagram

Mnemonic	Description	Mnemonic	Description
ESFlatness	Equalizer spectrum flatness	TXM	TX meas. statistical overview
SEMask	Spectrum emission mask	ACLR	Adj. channel leakage power ratio
RBATable	Resource block allocation table	PMONitor	Power monitor
BLER	Block error ratio	PDYNamics	Power dynamics

For reset values, see [CONFigure:LTE:MEAS<i>:MEvaluation:RESult\[:ALL\]](#).

Parameters:

<Enable> OFF | ON
OFF: Do not evaluate the results.
ON: Evaluate the results.
 *RST: Depends on measurement

Firmware/software: V2.0.10

Manual operation: See ["Multi Evaluation, PRACH softkeys > Assign Views hotkey"](#) on page 1086

CONFigure:LTE:MEAS<i>:MEvaluation:RESult:EVMagnitude:EVMsymbol
 <Enable>, <Symbol>, <LowHigh>

Enables or disables the measurement of EVM vs modulation symbol results and configures the scope of the measurement.

Parameters:

<Enable> OFF | ON
OFF: Do not measure the results.
ON: Measure the results.
 *RST: OFF

<Symbol> decimal
 SC-FDMA symbol to be evaluated.
 Range: 0 to 6
 *RST: 0

<LowHigh> LOW | HIGH
 Low or high EVM window position.
 *RST: LOW

Example: See [Running a measurement](#)

Firmware/software: V3.5.40

Manual operation: See ["Multi Evaluation softkey > EVM vs Symbol hotkey"](#) on page 1087

5.5.3.3 Measurement parameters - general part

The following commands define general measurement control parameters for the multi-evaluation measurement.

CONFigure:LTE:MEAS<i>:MEValuation:TOUT	1131
CONFigure:LTE:MEAS<i>:MEValuation:REPetition	1132
CONFigure:LTE:MEAS<i>:MEValuation:SCONdition	1132
CONFigure:LTE:MEAS<i>:MEValuation:MMODE	1132
CONFigure:LTE:MEAS<i>:MEValuation:TMODE:SCOnt	1133
CONFigure:LTE:MEAS<i>:MEValuation:TMODE:ENPower	1133
CONFigure:LTE:MEAS<i>:MEValuation:TMODE:RLEVel?	1134
CONFigure:LTE:MEAS<i>:MEValuation:MOEXception	1134
CONFigure:LTE:MEAS<i>:MEValuation:ULDL	1134
CONFigure:LTE:MEAS<i>:MEValuation:SSUBframe	1135
CONFigure:LTE:MEAS<i>:MEValuation:CPRefix	1135
CONFigure:LTE:MEAS<i>:MEValuation:CTYPe	1136
CONFigure:LTE:MEAS<i>:MEValuation:SCTYPe	1136
CONFigure:LTE:MEAS<i>:MEValuation:PFORmat	1136
CONFigure:LTE:MEAS<i>:MEValuation:NSValue	1136
CONFigure:LTE:MEAS<i>:MEValuation:NSValue:CAGGregation	1137
CONFigure:LTE:MEAS<i>:MEValuation:NVFilter	1137
CONFigure:LTE:MEAS<i>:MEValuation:ORVFilter	1138
CONFigure:LTE:MEAS<i>:MEValuation:CTVFilter	1138
CONFigure:LTE:MEAS<i>:MEValuation:CC<no>:PLCid	1138
CONFigure:LTE:MEAS<i>:MEValuation:DSSPusch	1139
CONFigure:LTE:MEAS<i>:MEValuation:GHOPping	1139
CONFigure:LTE:MEAS<i>:MEValuation:MSUBframes	1139
CONFigure:LTE:MEAS<i>:MEValuation:MSLot	1140

CONFigure:LTE:MEAS<i>:MEValuation:TOUT <Timeout>

Defines a timeout for the measurement. The timer is started when the measurement is initiated via a **READ** or **INIT** command. It is not started if the measurement is initiated manually.

When the measurement has completed the first measurement cycle (first single shot), the statistical depth is reached and the timer is reset.

If the first measurement cycle has not been completed when the timer expires, the measurement is stopped. The measurement state changes to **RDY**. The reliability indicator is set to 1, indicating that a measurement timeout occurred. Still running **READ**, **FETCH** or **CALCulate** commands are completed, returning the available results. At least for some results, there are no values at all or the statistical depth has not been reached.

A timeout of 0 s corresponds to an infinite measurement timeout.

Parameters:

<Timeout> numeric
 Default unit: s

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

CONFigure:LTE:MEAS<i>:MEValuation:REPetition <Repetition>

Specifies the repetition mode of the measurement. The repetition mode specifies whether the measurement is stopped after a single shot or repeated continuously. Use the **CONFigure:...:MEAS<i>:...:SCount** commands to specify the number of measurement intervals per single shot.

Parameters:

<Repetition> SINGleshot | CONTInuous
SINGleshot: Single-shot measurement
CONTInuous: Continuous measurement

Firmware/software: V1.0.10.1

Manual operation: See ["Repetition"](#) on page 1052

CONFigure:LTE:MEAS<i>:MEValuation:SCONdition <StopCondition>

Qualifies whether the measurement is stopped after a failed limit check or continued. With **SLFail**, the measurement is stopped and reaches the **RDY** state when one of the results exceeds the limits.

Parameters:

<StopCondition> NONE | SLFail
NONE: Continue measurement irrespective of the limit check.
SLFail: Stop measurement on limit failure.
***RST:** NONE

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Stop Condition"](#) on page 1052

CONFigure:LTE:MEAS<i>:MEValuation:MMODE <MeasurementMode>

Selects the measurement mode.

Parameters:

<MeasurementMode> NORMAl | TMODE | MELMode
NORMAl: normal mode
TMODE: TPC mode
MELMode: multi-evaluation list mode
***RST:** NORM

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.1.20

Options: R&S CMW-KM012 for MELMode

Manual operation: See ["Measurement Mode"](#) on page 1052

CONFigure:LTE:MEAS<i>:MEvaluation:TMODe:SCount <SubframeCount>...

Defines the subframe counts for all entries of the "TPC Mode" list.

For the definition of the corresponding expected nominal power values, see [CONFigure:LTE:MEAS<i>:MEvaluation:TMODe:ENPower](#).

Parameters:

<SubframeCount> decimal
 Comma-separated list of 16 values, for list entry number 0 to 15
 Range: 1 to 320
 *RST: 320

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.1.30

Manual operation: See ["TPC Mode"](#) on page 1053

CONFigure:LTE:MEAS<i>:MEvaluation:TMODe:ENPower <ExpNomPow>...

Defines the expected nominal power values for all entries of the "TPC Mode" list.

For the definition of the corresponding subframe count values, see [CONFigure:LTE:MEAS<i>:MEvaluation:TMODe:SCount](#).

Parameters:

<ExpNomPow> numeric
 Comma-separated list of 16 values, for list entry number 0 to 15
 The range of the expected nominal power can be calculated as follows:
 Range (Expected Nominal Power) = Range (Input Power) + External Attenuation - User Margin
 The input power range is stated in the specifications document.
 *RST: 0 dBm
 Default unit: dBm

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.1.30

Manual operation: See ["TPC Mode"](#) on page 1053

CONFigure:LTE:MEAS<i>:MEValuation:TMODe:RLEVel?

Queries the reference level for all entries of the "TPC Mode" list. The reference level is calculated from the expected nominal power of each entry and the user margin.

Return values:

<ReferenceLevel> float

Comma-separated list of 16 values, for list entry number 0 to 15

The range of the reference levels can be calculated as follows:

Range (Reference Level) = Range (Input Power) + External Attenuation

The input power range is stated in the specifications document.

Default unit: dBm

Example: See [Specifying measurement-specific settings](#)

Usage: Query only

Firmware/software: V2.1.30

Manual operation: See ["TPC Mode"](#) on page 1053

CONFigure:LTE:MEAS<i>:MEValuation:MOEXception <MeasOnException>

Specifies whether measurement results identified as faulty or inaccurate are rejected.

Parameters:

<MeasOnException> OFF | ON

OFF: Faulty results are rejected.

ON: Results are never rejected.

*RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Measure on Exception"](#) on page 1053

CONFigure:LTE:MEAS<i>:MEValuation:ULDL <UplinkDownlink>

Selects an UL-DL configuration, defining the combination of uplink, downlink and special subframes within a radio frame. This parameter is only relevant for frame structure "Type 2" ([CONFigure:LTE:MEAS<i>:FSTRucture?](#)).

The UL-DL configurations are defined in 3GPP TS 36.211, section 4, "Frame Structure".

For the combined signal path scenario, use:

- [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:ULDL](#)
- [CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULDL](#)
- [CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific](#)

Parameters:

<UplinkDownlink> integer
 Range: 0 to 6
 *RST: 0

Example: See [Specifying required settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Uplink Downlink"](#) on page 1054

CONFigure:LTE:MEAS<i>:MEvaluation:SSUBframe <SpecialSubframe>

Selects a special subframe configuration, defining the inner structure of special subframes. This parameter is only relevant for frame structure "Type 2" ([CONFigure:LTE:MEAS<i>:FSTRucture?](#)).

The special subframe configurations are defined in 3GPP TS 36.211, section 4, "Frame Structure".

For the combined signal path scenario, use:

- [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SSUBframe](#)
- [CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SSUBframe](#)
- [CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific](#)

Parameters:

<SpecialSubframe> integer
 Range: 0 to 8
 *RST: 0

Example: See [Specifying required settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Special Subframe"](#) on page 1054

CONFigure:LTE:MEAS<i>:MEvaluation:CPRefix <CyclicPrefix>

Selects the type of cyclic prefix of the LTE signal.

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CELL:CPRefix](#).

Parameters:

<CyclicPrefix> NORMal | EXTended
 *RST: NORM

Example: See [Specifying required settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Cyclic Prefix"](#) on page 1054

CONFigure:LTE:MEAS<i>:MEvaluation:CTYPe <ChannelType>

Configures the channel type detection for uplink measurements.

Parameters:

<ChannelType> AUTO | PUSCh | PUCCh
Automatic detection of channel type or manual selection
*RST: PUSC

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["Uplink Channel Type"](#) on page 1055

CONFigure:LTE:MEAS<i>:MEvaluation:SCTYPe <ChannelType>

Configures the channel type for modulation results of sidelink measurements.

Parameters:

<ChannelType> PSSCh | PSCCh
*RST: PSSC

Example: See [Specifying V2X sidelink settings](#)

Firmware/software: V3.7.55

Manual operation: See ["Sidelink Channel Type"](#) on page 1055

CONFigure:LTE:MEAS<i>:MEvaluation:PFORmat <PUCCHFormat>

Specifies the PUCCH format (only relevant for signals containing a PUCCH). The formats are defined in 3GPP TS 36.211.

Parameters:

<PUCCHFormat> F1 | F1A | F1B | F2 | F2A | F2B | F3
*RST: F1

Example: See [Specifying required settings](#)

Firmware/software: V1.0.15.0
V3.5.40: added F3

Manual operation: See ["PUCCH Format"](#) on page 1055

CONFigure:LTE:MEAS<i>:MEvaluation:NSValue <Value>

Selects the "network signaled value" for measurements without carrier aggregation.

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CONNection:ASEMission](#).

Parameters:

<Value> NS01 | ... | NS288
 Value NS_01 to NS_288
 *RST: NS01

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Options: R&S CMW-KM570 for NS33

Manual operation: See ["Network Signaled Value"](#) on page 1056

CONFigure:LTE:MEAS<i>:MEvaluation:NSValue:CAGGregation <Value>

Selects the "network signaled value" for measurements with carrier aggregation.

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CONNecTion:SCC<c>:ASEmission:CAGGregation](#).

Parameters:

<Value> NS01 | NS02 | NS03 | NS04 | NS05 | NS06 | NS07 | NS08 |
 NS09 | NS10 | NS11 | NS12 | NS13 | NS14 | NS15 | NS16 |
 NS17 | NS18 | NS19 | NS20 | NS21 | NS22 | NS23 | NS24 |
 NS25 | NS26 | NS27 | NS28 | NS29 | NS30 | NS31 | NS32
 Value CA_NS_01 to CA_NS_32
 *RST: NS01

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.2.70

Manual operation: See ["Network Signaled Value"](#) on page 1056

CONFigure:LTE:MEAS<i>:MEvaluation:NVFilter <NRBViewFilter>

Specifies, enables or disables the number of resource blocks (NRB) view filter. If the filter is active, only slots with a matching number of allocated resource blocks are measured.

Within the indicated input range, only specific numbers are allowed as defined in 3GPP TS 36.211. For details, see [Section 5.2.11.1, "Resources in time and frequency domain"](#), on page 1008.

Parameters:

<NRBViewFilter> numeric | ON | OFF
 Number of allocated resource blocks
 Range: 1 to 100
 *RST: OFF
 ON | OFF enables or disables the filter.

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["NRB"](#) on page 1056

CONFigure:LTE:MEAS<i>:MEvaluation:ORVFilter <OffsetRB>

Specifies, enables or disables the RB offset view filter. If the filter is active, only slots with a matching number of RB offset are measured.

The indicated input range applies to a 20-MHz channel bandwidth. The maximum value depends on the bandwidth (maximum number of RBs minus one).

Parameters:

<OffsetRB> numeric | ON | OFF
 Offset of the first allocated RB
 Range: 0 to 99
 *RST: OFF
 ON | OFF enables or disables the filter.

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.7.50

Manual operation: See ["Offset RB"](#) on page 1057

CONFigure:LTE:MEAS<i>:MEvaluation:CTVFilter <ChannelType>

Specifies, enables or disables the channel type view filter. If the filter is active, only slots with detected channel type PUSCH or PUCCH are measured.

Parameters:

<ChannelType> PUSCh | PUCCh | ON | OFF
 PUSCh: measure only PUSCH
 PUCCh: measure only PUCCH
 ON: enable the filter
 OFF: disable the filter
 *RST: OFF (PUSC)

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.15.0

Manual operation: See ["Channel Type"](#) on page 1057

CONFigure:LTE:MEAS<i>:MEvaluation:CC<no>:PLCid <PhsLayerCellID>

Specifies the physical layer cell ID of component carrier CC<no>. Without carrier aggregation, you can omit <no>.

For the combined signal path scenario, use:

- [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:PCID](#)
- [CONFigure:LTE:SIGN<i>:CELL:SCC<c>:PCID](#)

Suffix:

<no> 1..4

Parameters:

<PhsLayerCellID> integer
 Range: 0 to 503
 *RST: 0

Example: See [Specifying required settings](#)**Firmware/software:** V3.7.30**Manual operation:** See ["Physical Cell ID"](#) on page 1050**CONFigure:LTE:MEAS<i>:MEValuation:DSSPusch** <DeltaSeqShPUSCH>

Specifies the delta sequence shift value (Δ_{ss}) used to calculate the sequence shift pattern for PUSCH.

Parameters:

<DeltaSeqShPUSCH>integer
 Range: 0 to 29
 *RST: 0

Example: See [Specifying required settings](#)**Firmware/software:** V1.0.10.1**Manual operation:** See ["Delta Sequence Shift PUSCH"](#) on page 1060**CONFigure:LTE:MEAS<i>:MEValuation:GHOPping** <Value>

Specifies whether group hopping is used.

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CONNection:GHOPping](#).

Parameters:

<Value> OFF | ON
 *RST: OFF

Example: See [Specifying required settings](#)**Firmware/software:** V2.0.20**Manual operation:** See ["Group Hopping"](#) on page 1060**CONFigure:LTE:MEAS<i>:MEValuation:MSUBframes** <SubframeOffset>, <SubframeCount>, <MeasSubframe>

Configures the scope of the measurement, i.e. which subframes are measured.

Parameters:

<SubframeOffset>	decimal
	Start of the measured subframe range relative to the trigger event.
Range:	0 to 9
*RST:	0
<SubframeCount>	decimal
	Length of the measured subframe range.
Range:	1 to 320
*RST:	1
<MeasSubframe>	decimal
	Subframe containing the measured slots for modulation and spectrum results.
Range:	0 to <SubframeCount>-1
*RST:	0

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.15.20
V2.1.10: <SubframeCount> maximum increased to 320.

Manual operation: See ["Subframe Offset"](#) on page 1061

CONFigure:LTE:MEAS<i>:MEValuation:MSLot <MeasureSlot>

Selects which slots of the "Measure Subframe" are measured.

Parameters:

<MeasureSlot>	MS0 MS1 ALL
	MS0: slot number 0 only
	MS1: slot number 1 only
	ALL: both slots
*RST:	ALL

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.5.40

Manual operation: See ["Measure Slot"](#) on page 1061

5.5.3.4 RB allocation settings

The following commands specify the RB allocation of the measured signal.

CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:AUTO.....	1141
CONFigure:LTE:MEAS<i>:EMTC:NBANd.....	1141
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:MCLuster.....	1141
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:NRB.....	1142
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:NRB:PSCCh.....	1142
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:NRB:PSSCh.....	1142

CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:MCLuster:NRB<Number>.....	1143
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:ORB.....	1143
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:ORB:PSCCh.....	1144
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:ORB:PSSCh.....	1144
CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:MCLuster:ORB<Number>.....	1144

CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:AUTO <Auto>

Enables or disables the automatic detection of the RB configuration.

Parameters:

<Auto> OFF | ON
 OFF: manual definition
 ON: automatic detection
 *RST: ON

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Auto"](#) on page 1057

CONFigure:LTE:MEAS<i>:EMTC:NBAND <Number>

Selects the narrowband used for eMTC.

Parameters:

<Number> numeric
 The maximum depends on the channel BW, see ["RB allocation, narrowbands and widebands for eMTC"](#) on page 1011.
 Range: 0 to 15
 *RST: 0

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.7.10

Manual operation: See ["Narrowband"](#) on page 1057

CONFigure:LTE:MEAS<i>:MEValuation:RBALlocation:MCLuster <Enable>

Specifies whether the UL signal uses multi-cluster allocation or not.

For the combined signal path scenario, use:

- [CONFigure:LTE:SIGN<i>:CONNECTION\[:PCC\]:MCLuster:UL](#)
- [CONFigure:LTE:SIGN<i>:CONNECTION:SCC<c>:MCLuster:UL](#)

Parameters:

<Enable> OFF | ON
 OFF: contiguous allocation, resource allocation type 0
 ON: multi-cluster allocation, resource allocation type 1
 *RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.5.50

Manual operation: See ["Multicluster"](#) on page 1058

CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:NRB <NoRB>

Specifies the number of allocated RBs in the measured slot. For manual RB allocation definition, for uplink signals without multi-cluster allocation.

Parameters:

<NoRB> numeric
 For the allowed input range, see [Section 5.2.11.2, "Uplink resource block allocation"](#), on page 1009.
 *RST: 100

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["No. of RBs, Offset RB - no multicluster"](#) on page 1058

CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:NRB:PSCCh <NoRB>

Specifies the number of allocated RBs for the PSCCH in the measured slot. For manual RB allocation definition, for sidelink signals.

Parameters:

<NoRB> numeric
 The value is fixed.
 Range: 2 to 2
 *RST: 2

Firmware/software: V3.7.55

Manual operation: See ["No. of RBs, Offset RB"](#) on page 1059

CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:NRB:PSSCh <NoRB>

Specifies the number of allocated RBs for the PSSCH in the measured slot. For manual RB allocation definition, for sidelink signals.

Parameters:

<NoRB> numeric
 For the allowed input range, see [Section 5.2.11.3, "Sidelink resource block allocation"](#), on page 1012.
 *RST: 1

Example: See [Specifying V2X sidelink settings](#)

Firmware/software: V3.7.55

Manual operation: See ["No. of RBs, Offset RB"](#) on page 1059

**CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:MCLuster:NRB<Number>
<NoRB>**

Specifies the number of allocated RBs in the measured slot, for multi-cluster allocation.

For the combined signal path scenario, use:

- `CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:MCLuster:UL`
- `CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDChannels:MCLuster:UL`
- `CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:MCLuster:UL`
- `CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:MCLuster:UL`

Suffix:

<Number> 1, 2
Selects the cluster to be configured.

Parameters:

<NoRB> numeric
For the allowed input ranges, see [Section 5.2.11.2, "Uplink resource block allocation"](#), on page 1009.

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.5.50

Manual operation: See ["No. of RBs, Offset RB - with multicluster"](#) on page 1058

CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:ORB <OffsetRB>

Specifies the offset of the first allocated resource block for manual RB allocation definition, for uplink signals without multi-cluster allocation.

Parameters:

<OffsetRB> numeric
For the maximum number of RBs depending on the channel BW, see [Section 5.2.11.2, "Uplink resource block allocation"](#), on page 1009.
Range: 0 to maximum number of RBs minus 1
*RST: 0

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["No. of RBs, Offset RB - no multicluster"](#) on page 1058

CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:ORB:PSCCh <OffsetRB>

Specifies the offset of the first allocated PSCCH resource block for manual RB allocation definition, for sidelink signals.

Parameters:

<OffsetRB> numeric

For the maximum number of RBs depending on the channel BW, see [Section 5.2.11.2, "Uplink resource block allocation"](#), on page 1009.

Range: 0 to max no of RBs minus total allocated RBs

*RST: 0

Example: See [Specifying V2X sidelink settings](#)

Firmware/software: V3.7.55

Manual operation: See ["No. of RBs, Offset RB"](#) on page 1059

CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:ORB:PSSCh <OffsetRB>

Specifies the offset of the first allocated PSSCH resource block for manual RB allocation definition, for sidelink signals.

Parameters:

<OffsetRB> numeric

The range depends on the <OffsetRB> for the PSCCH, the channel BW and the number of allocated PSSCH RBs, see [Section 5.2.11.3, "Sidelink resource block allocation"](#), on page 1012.

*RST: 2

Example: See [Specifying V2X sidelink settings](#)

Firmware/software: V3.7.55

Manual operation: See ["No. of RBs, Offset RB"](#) on page 1059

**CONFigure:LTE:MEAS<i>:MEvaluation:RBAllocation:MCLuster:ORB<Number>
<OffsetRB>**

Specifies the offset of the first allocated resource block, for multi-cluster allocation.

For the combined signal path scenario, use:

- [CONFigure:LTE:SIGN<i>:CONNECTION\[:PCC\]:RMC:MCLuster:UL](#)
- [CONFigure:LTE:SIGN<i>:CONNECTION\[:PCC\]:UDChannels:MCLuster:UL](#)
- [CONFigure:LTE:SIGN<i>:CONNECTION:SCC<c>:RMC:MCLuster:UL](#)
- [CONFigure:LTE:SIGN<i>:CONNECTION:SCC<c>:UDChannels:MCLuster:UL](#)

Suffix:

<Number> 1, 2

Selects the cluster to be configured.

Parameters:

<OffsetRB> numeric
 For the allowed input ranges, see [Section 5.2.11.2, "Uplink resource block allocation"](#), on page 1009.

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.5.50

Manual operation: See ["No. of RBs, Offset RB - with multicluster"](#) on page 1058

5.5.3.5 Modulation measurement settings

The following commands specify settings for the modulation measurements.

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:MSCHeme.....	1145
CONFigure:LTE:MEAS<i>:MEValuation:SCount:MODulation.....	1145
CONFigure:LTE:MEAS<i>:MEValuation:SRS:ENABLE.....	1146
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EWLength.....	1146
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EWLength:CBANDwidth<Band>.....	1148
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUCCh.....	1148
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUSCh:LEADing.....	1149
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUSCh:LAGGing.....	1149
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:CAGGregation:LLOCation.....	1149
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:LLOCation.....	1150
CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EQualizer.....	1150

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:MSCHeme <ModScheme>

Selects the modulation scheme used by the measured signal.

Parameters:

<ModScheme> AUTO | QPSK | Q16 | Q64 | Q256
 Auto-detection, QPSK, 16QAM, 64QAM, 256QAM
 *RST: QPSK

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1, V3.7.50: 256QAM

Options: R&S CMW-KM504/-KM554 for 256QAM

Manual operation: See ["Modulation Scheme"](#) on page 1062

CONFigure:LTE:MEAS<i>:MEValuation:SCount:MODulation <StatisticCount>

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

Parameters:

<StatisticCount> numeric
 Range: 1 slot to 1000 slots
 *RST: 20 slots

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Statistic Count"](#) on page 1063

CONFigure:LTE:MEAS<i>:MEvaluation:SRS:ENABLE <Enable>

Specifies whether a sounding reference signal is allowed (ON) or not (OFF).

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CELL\[:PCC\]:SRS:ENABLE](#).

Parameters:

<Enable> OFF | ON
*RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.0.10

Manual operation: See ["Sounding RS \(SRS\)"](#) on page 1063

CONFigure:LTE:MEAS<i>:MEvaluation:MODulation:EWLength <CPnormB014>, <CPnormB030>, <CPnormB050>, <CPnormB100>, <CPnormB150>, <CPnormB200>, <CPextB014>, <CPextB030>, <CPextB050>, <CPextB100>, <CPextB150>, <CPextB200>

Specifies the EVM window length in samples for all channel bandwidths, depending on the cyclic prefix (CP) type.

Parameters:

<CPnormB014>	integer Length for normal CP, BW = 1.4 MHz Range: 1 to 9 *RST: 5
<CPnormB030>	integer Length for normal CP, BW = 3 MHz Range: 1 to 18 *RST: 12
<CPnormB050>	integer Length for normal CP, BW = 5 MHz Range: 1 to 36 *RST: 32
<CPnormB100>	integer Length for normal CP, BW = 10 MHz Range: 1 to 72 *RST: 66

<CPnormB150>	integer Length for normal CP, BW = 15 MHz Range: 1 to 108 *RST: 102
<CPnormB200>	integer Length for normal CP, BW = 20 MHz Range: 1 to 144 *RST: 136
<CPextB014>	integer Length for extended CP, BW = 1.4 MHz Range: 1 to 32 *RST: 28
<CPextB030>	integer Length for extended CP, BW = 3 MHz Range: 1 to 64 *RST: 58
<CPextB050>	integer Length for extended CP, BW = 5 MHz Range: 1 to 128 *RST: 124
<CPextB100>	integer Length for extended CP, BW = 10 MHz Range: 1 to 256 *RST: 250
<CPextB150>	integer Length for extended CP, BW = 15 MHz Range: 1 to 384 *RST: 374
<CPextB200>	integer Length for extended CP, BW = 20 MHz Range: 1 to 512 *RST: 504
Example:	See Specifying measurement-specific settings
Firmware/software:	V1.0.10.1
Manual operation:	See "EVM Window Length" on page 1063

**CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EWLength:
CBANdwidth<Band> <CycPrefixNormal>, <CycPrefixExtend>**

Specifies the EVM window length in samples for a selected channel bandwidth, depending on the cyclic prefix (CP) type.

Suffix:

<Band> 14, 30, 50, 100, 150, 200
Channel bandwidth in 0.1 MHz

Parameters:

<CycPrefixNormal> integer
Samples for normal CP
Range: see below
*RST: see below

<CycPrefixExtend> integer
Samples for extended CP
Range: see below
*RST: see below

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["EVM Window Length"](#) on page 1063

<Band>	Range <CycPrefixNormal>	*RST <CycPrefixNormal>	Range <CycPrefixExtend>	*RST <CycPrefixExtend>
14	1 to 9	5	1 to 32	28
30	1 to 18	12	1 to 64	58
50	1 to 36	32	1 to 128	124
100	1 to 72	66	1 to 256	250
150	1 to 108	102	1 to 384	374
200	1 to 144	136	1 to 512	504

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUCCh <PUCCH>

Enables or disables EVM exclusion periods for slots with detected channel type "PUCCH". If enabled, the first and the last SC-FDMA symbol of each slot is excluded from the calculation of EVM, magnitude error and phase error single value results. If the last symbol of a slot is already excluded because SRS signals are allowed, the second but last symbol is also excluded.

Parameters:

<PUCCH> OFF | ON
*RST: ON

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Manual operation: See ["PUCCH"](#) on page 1064

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUSCh:LEADing
<Leading>

Specifies an EVM exclusion period at the beginning of a subframe (detected channel type "PUSCH"). The specified period is excluded from the calculation of EVM, magnitude error and phase error results.

Parameters:

<Leading> OFF | MS25
OFF: no exclusion
MS25: 25 µs excluded
 *RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Manual operation: See ["PUSCH Leading/Lagging"](#) on page 1064

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EEPeriods:PUSCh:LAGGing
<Lagging>

Specifies an EVM exclusion period at the end of each subframe (detected channel type "PUSCH"); if SRS signals are allowed, at the end of each shortened subframe. The specified period is excluded from the calculation of EVM, magnitude error and phase error results.

Parameters:

<Lagging> OFF | MS05 | MS25
OFF: no exclusion
MS05: 5 µs excluded
MS25: 25 µs excluded
 *RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Manual operation: See ["PUSCH Leading/Lagging"](#) on page 1064

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:CAGGregation:LLOCation
<Value>

Specifies the UE transmitter architecture (local oscillator location) used for contiguous carrier aggregation.

Parameters:

<Value> CACB | CECC
CACB: Center of aggregated channel bandwidth
CECC: Center of each component carrier
 *RST: CACB

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.5.10

Manual operation: See ["LO Location for CA"](#) on page 1064

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:LLOCation <Value>

Specifies the UE transmitter architecture (local oscillator location) used for eMTC.

Parameters:

<Value> CN | CCB
CN: Center of narrowband/wideband
CCB: Center of channel bandwidth
 *RST: CN

Firmware/software: V3.7.10

Manual operation: See ["LO Location for eMTC"](#) on page 1065

CONFigure:LTE:MEAS<i>:MEValuation:MODulation:EQualizer <Enable>

Enables or disables the post-FFT equalization step for the measurement of modulation results.

Parameters:

<Enable> OFF | ON
 *RST: ON

Firmware/software: V3.7.50

Manual operation: See ["Equalizer"](#) on page 1065

5.5.3.6 Spectrum measurement settings

The following commands specify settings for the spectrum measurements.

CONFigure:LTE:MEAS<i>:MEValuation:SCount:SPECtrum:ACLR	1151
CONFigure:LTE:MEAS<i>:MEValuation:SCount:SPECtrum:SEMask	1151
CONFigure:LTE:MEAS<i>:MEValuation:SPECtrum:SEMask:MFILter	1151
CONFigure:LTE:MEAS<i>:MEValuation:SPECtrum:ACLR:ENABle	1151

CONFigure:LTE:MEAS<i>:MEValuation:SCount:SPECtrum:ACLR <StatisticCount>
CONFigure:LTE:MEAS<i>:MEValuation:SCount:SPECtrum:SEMask
 <StatisticCount>

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

Separate statistic counts for ACLR and spectrum emission mask measurements are supported.

Parameters:

<StatisticCount> numeric
 Range: 1 slot to 1000 slots
 *RST: 20 slots

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Statistic Count"](#) on page 1066

CONFigure:LTE:MEAS<i>:MEValuation:SPECtrum:SEMask:MFILter <MeasFilter>

Selects the resolution filter type for filter bandwidths of 50 kHz and greater.

Parameters:

<MeasFilter> BANDpass | GAUSs
 *RST: BAND

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.15.20

Manual operation: See ["Measurement Filter \(>= 50 kHz\)"](#) on page 1066

**CONFigure:LTE:MEAS<i>:MEValuation:SPECtrum:ACLR:ENABLE <UTRA1>,
 <UTRA2>, <EUTRA>**

Enables or disables the evaluation of the first adjacent UTRA channels, second adjacent UTRA channels and first adjacent E-UTRA channels.

Parameters:

<UTRA1> OFF | ON
 *RST: ON

 <UTRA2> OFF | ON
 *RST: ON

 <EUTRA> OFF | ON
 *RST: ON

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.15.0

Manual operation: See ["UTRA1, UTRA2, E-UTRA"](#) on page 1066

5.5.3.7 Power measurement settings

The following commands specify settings for the power measurements.

CONFigure:LTE:MEAS<i>:MEValuation:PDYNamics:TMASK	1152
CONFigure:LTE:MEAS<i>:MEValuation:PDYNamics:AEOPower:LEADing	1152
CONFigure:LTE:MEAS<i>:MEValuation:PDYNamics:AEOPower:LAGGing	1152
CONFigure:LTE:MEAS<i>:MEValuation:SCOUNt:POWer	1153
CONFigure:LTE:MEAS<i>:MEValuation:POWer:HDMODE	1153

CONFigure:LTE:MEAS<i>:MEValuation:PDYNamics:TMASK <TimeMask>

Selects the time mask for power dynamics measurements.

Parameters:

<TimeMask> GOO | PPSRs | SBLanking
GOO: General time mask
PPSRs: PUCCH/PUSCH transmission before and after an SRS
SBLanking: SRS blanking time mask
 *RST: GOO

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.2.70

Manual operation: See ["Time Mask"](#) on page 1067

CONFigure:LTE:MEAS<i>:MEValuation:PDYNamics:AEOPower:LEADing <Leading>

Shifts the beginning of the evaluation period for OFF power measurements.

Parameters:

<Leading> integer
 Positive values reduce the evaluation period (starts later).
 Negative values increase the evaluation period (starts earlier).
 Range: -1000 Ts to 1000 Ts
 *RST: 0 Ts
 Default unit: Ts

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.2.70

Manual operation: See ["Add Exclusive OFF Power"](#) on page 1067

CONFigure:LTE:MEAS<i>:MEValuation:PDYNamics:AEOPower:LAGGing <Lagging>

Shifts the end of the evaluation period for OFF power measurements.

Parameters:

<Lagging> integer
 Positive values reduce the evaluation period (ends earlier).
 Negative values increase the evaluation period (ends later).
 Range: -1000 Ts to 1000 Ts
 *RST: 0 Ts
 Default unit: Ts

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.2.70

Manual operation: See ["Add Exclusive OFF Power"](#) on page 1067

CONFigure:LTE:MEAS<i>:MEvaluation:SCount:POWer <StatisticCount>

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

Parameters:

<StatisticCount> numeric
 Range: 1 subframe to 1000 subframes
 *RST: 20 subframes

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Manual operation: See ["Statistic Count"](#) on page 1068

CONFigure:LTE:MEAS<i>:MEvaluation:POWer:HDMode <HighDynamicMode>

Enables or disables the high dynamic mode for power dynamics measurements.

Parameters:

<HighDynamicMode> OFF | ON
 *RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.1.25

Manual operation: See ["High Dynamic Mode"](#) on page 1068

5.5.3.8 BLER measurement settings

The following commands specify settings for the block error ratio measurement.

CONFigure:LTE:MEAS<i>:MEValuation:BLER:SFRames <Subframes>[,
<SchedSubfrPerFr>]

Specifies the statistic count (number of measured subframes) and the number of scheduled subframes per radio frame for the BLER measurement. BLER is a single shot measurement.

Parameters:

<Subframes>	integer
	Number of subframes to be measured.
Range:	1 subframe to 200E+3 subframes
*RST:	10E+3 subframes
<SchedSubfrPerFr>	integer
	Number of scheduled subframes per radio frame in the generated downlink signal.
Range:	1 subframe to 10 subframes
*RST:	9 subframes

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V1.0.15.21
V3.0.10: added <SchedSubfrPerFr>

Manual operation: See ["No. of Subframes"](#) on page 1069

5.5.3.9 List mode settings

The following commands configure the list mode. For retrieving list mode results, see [Section 5.5.3.34, "List mode results \(one segment\)"](#), on page 1242 and [Section 5.5.3.35, "List mode results \(all segments, one result\)"](#), on page 1263.

For a description of the list mode, see [Section 5.2.10, "List mode"](#), on page 1002.

The segment number <no> in the following commands refers to the complete range of configured segments (1..2000).

CONFigure:LTE:MEAS<i>:MEValuation:LIST	1155
CONFigure:LTE:MEAS<i>:MEValuation:LIST:LRANge	1155
CONFigure:LTE:MEAS<i>:MEValuation:LIST:OSINdex	1155
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SETup	1156
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:TDD	1158
CONFigure:LTE:MEAS<i>:MEValuation:LIST:PLCMode	1159
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:PLCid	1159
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:CC<c>	1159
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:CAGGregation:ACSPacing	1160
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:CAGGregation:MCARrier: ENHanced	1160
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:EMTC:NBAND	1161
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:RBALlocation	1161
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:RBALlocation:SIDelink	1162
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:MODulation	1163

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:SEMAsk.....	1164
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:ACLR.....	1165
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:PMONitor.....	1165
CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:POWer.....	1166
TRIGger:LTE:MEAS<i>:MEValuation:LIST:MODE.....	1166
TRIGger:LTE:MEAS<i>:MEValuation:LIST:NBANdwidth.....	1167

CONFigure:LTE:MEAS<i>:MEValuation:LIST <Enable>

Enables or disables the list mode.

Parameters:

<Enable> OFF | ON
 OFF: Disable list mode.
 ON: Enable list mode.
 *RST: OFF

Example: See [Specifying list mode settings](#)

Firmware/software: V2.0.10

Options: R&S CMW-KM012

Manual operation: See "[Measurement Mode](#)" on page 1052

CONFigure:LTE:MEAS<i>:MEValuation:LIST:LRANge <StartIndex>, <NrSegments>

Selects a range of measured segments. Configure the segments via [CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:SETup](#).

Parameters:

<StartIndex> numeric
 First measured segment in the range of configured segments
 Range: 1 to 2000
 *RST: 1

<NrSegments> numeric
 Number of measured segments
 Range: 1 to 1000
 *RST: 10

Example: See [Specifying list mode settings](#)

Firmware/software: V2.0.10

CONFigure:LTE:MEAS<i>:MEValuation:LIST:OSINdex <OfflineSegIndex>

Selects the number of the segment to be displayed in offline mode. The index refers to the range of measured segments, see [CONFigure:LTE:MEAS<i>:MEValuation:LIST:LRANge](#).

Setting a value also enables the offline mode.

Parameters:

<OfflineSegIndex> numeric | OFF
 Range: 1 to <number of measured segments>
 *RST: OFF
 OFF disables the offline mode.

Example: See [Running a list mode measurement](#)

Firmware/software: V2.0.10

Manual operation: See "Offline Segment No." on page 1069

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:SETup

<SegmentLength>, <Level>, <DuplexMode>, <Band>, <Frequency>,
 <ChBandwidth>, <CyclicPrefix>, <ChannelType>, <RetriggerFlag>,
 <EvaluatOffset>[, <NetworkSigValue>]

Defines the length and analyzer settings of segment <no>. Send this command for all segments to be measured ([CONFigure:LTE:MEAS<i>:MEValuation:LIST:LRANge](#)).

For uplink signals with TDD mode, see also [CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:TDD](#).

For carrier-specific settings for carrier aggregation, see [CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:CC<c>](#).

Suffix:

<no> 1..2000
 Segment number

Parameters:

<SegmentLength> integer
 Number of subframes in the segment
 Range: 1 to 2000
 *RST: 1

<Level> numeric
 Expected nominal power in the segment. The range can be calculated as follows:
 $\text{Range (Expected Nominal Power)} = \text{Range (Input Power)} + \text{External Attenuation} - \text{User Margin}$
 The input power range is stated in the specifications document.
 *RST: 0 dBm
 Default unit: dBm

<DuplexMode> FDD | TDD
 Duplex mode used in the segment

<Band>	FDD UL: OB1 ... OB28 OB30 OB31 OB65 OB66 OB68 OB70 ... OB74 OB85 OB87 OB88 TDD UL: OB33 ... OB45 OB48 OB50 ... OB53 OB250 Sidelink: OB47 Operating band used in the segment
<Frequency>	numeric Center frequency of CC1 used in the segment For the supported range, see Section 5.5.1.5, "Frequency ranges", on page 1105. Default unit: Hz
<ChBandwidth>	B014 B030 B050 B100 B150 B200 Channel bandwidth of CC1 used in the segment. B014: 1.4 MHz B030: 3 MHz B050: 5 MHz B100: 10 MHz B150: 15 MHz B200: 20 MHz *RST: B200
<CyclicPrefix>	NORMal EXTended Type of cyclic prefix used in the segment *RST: NORM
<ChannelType>	AUTO PUSCh PUCCh PSSCh PSCCh Channel type to be measured in the segment (AUTO for automatic detection). Uplink: AUTO, PUSCh, PUCCh Sidelink: PSSCh, PSCCh
<RetriggerFlag>	OFF ON IFPower IFPNarrow Specifies whether the measurement waits for a trigger event before measuring the segment, or not. The retrigger flag is ignored for trigger mode ONCE and evaluated for trigger mode SEGMENT, see TRIGGER:LTE:MEAS<i><:MEvaluation:LIST:MODE on page 1166. OFF Measure the segment without retrigger. For the first segment, the value OFF is interpreted as ON. ON Wait for a trigger event from the trigger source configured via TRIGGER:LTE:MEAS<i><:MEvaluation:SOURce on page 1168. IFPower Wait for a trigger event from the trigger source "IF Power". The trigger evaluation bandwidth is 160 MHz.

IFPNarrowband

Wait for a trigger event from the trigger source "IF Power".

The trigger evaluation bandwidth is configured via [TRIGGER:LTE:MEAS<i>:MEvaluation:LIST:NBANDwidth](#)

on page 1167.

*RST: ON

<EvaluatOffset> integer

Number of subframes at the beginning of the segment that are not evaluated.

Range: 0 to 1000

*RST: 0

<NetworkSigValue> NS01 | ... | NS288

Network signaled value to be used for the segment.

*RST: NS01

Example: See [Specifying list mode settings](#)

Firmware/software: V3.0.10, some NS values and bands have been added in later versions.

V3.2.70 added <RetriggerFlag> value IFPower.

V3.7.55 added <ChannelType> values PSSCh, PSCCh.

V4.0.20 added <RetriggerFlag> = IFPNarrowband.

CONFigure:LTE:MEAS<i>:MEvaluation:LIST:SEGment<no>:TDD

<UplinkDownlink>, <SpecialSubframe>

Defines segment settings only relevant for uplink measurements with the duplex mode TDD.

For general segment configuration, see [CONFigure:LTE:MEAS<i>:MEvaluation:LIST:SEGment<no>:SETup](#).

Suffix:

<no> 1..2000

Segment number

Parameters:

<UplinkDownlink> integer

UL-DL configuration, defining the combination of uplink, downlink and special subframes within a radio frame

Range: 0 to 6

*RST: 0

<SpecialSubframe> integer

Special subframe configuration, defining the inner structure of special subframes

Range: 0 to 8

*RST: 0

Firmware/software: V2.0.10

CONFigure:LTE:MEAS<i>:MEValuation:LIST:PLCMode <PLCIDmode>

Selects which physical cell ID setting is used for list mode measurements.

Parameters:

<PLCIDmode> GLOBal | LIST

GLOBal

The global setting is used for all segments, see [CONFigure:LTE:MEAS<i>:MEValuation:CC<no>:PLCid](#) on page 1138.

LIST

The cell ID is configured per segment, see [CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:PLCid](#) on page 1159.

*RST: GLOB

Example: See [Specifying list mode settings](#)

Firmware/software: V4.0.20

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:PLCid

<PhysLayerCellID>

Specifies the physical cell ID for segment <no>.

See also [CONFigure:LTE:MEAS<i>:MEValuation:LIST:PLCMode](#) on page 1159.

Suffix:

<no> 1..2000

Parameters:

<PhysLayerCellID> integer

Range: 0 to 503

*RST: 0

Example: See [Specifying list mode settings](#)

Firmware/software: V4.0.20

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:CC<c> <Frequency>, <ChBandwidth>

Defines carrier-specific analyzer settings for component carrier CC<c>, in segment <no>. This command is only relevant for carrier aggregation.

Suffix:

<no> 1..2000

<c> 1..4

Parameters:

<Frequency>	numeric Center frequency of the component carrier For the supported range, see Section 5.5.1.5, "Frequency ranges" , on page 1105. *RST: Depends on <c> Default unit: Hz
<ChBandwidth>	B014 B030 B050 B100 B150 B200 Channel bandwidth of the component carrier B014: 1.4 MHz B030: 3 MHz B050: 5 MHz B100: 10 MHz B150: 15 MHz B200: 20 MHz *RST: B200

Example: See [Specifying list mode settings](#)

Firmware/software: V3.7.55

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:CAGGregation:ACSPacing

Adjusts the component carrier frequencies in segment <no>, so that the carriers are aggregated contiguously.

Suffix:

<no>	1..2000 Segment number
------	---------------------------

Example: See [Specifying list mode settings](#)

Usage: Event

Firmware/software: V3.5.30

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:CAGGregation:MCARrier:ENHanced <MeasCarrier>

Selects a component carrier for single carrier measurements in segment <no>.

Suffix:

<no>	1..2000 Segment number
------	---------------------------

Parameters:

<MeasCarrier>	CC1 CC2 CC3 CC4 *RST: CC1
---------------	------------------------------------

Example: See [Specifying list mode settings](#)

Firmware/software: V3.7.55

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:EMTC:NBAND
 <Number>

Selects the eMTC narrowband for segment <no>.

Suffix:

<no> 1..2000
 Segment number

Parameters:

<Number> numeric
 The maximum depends on the channel BW, see ["RB allocation, narrowbands and widebands for eMTC"](#) on page 1011.
 Range: 0 to 15
 *RST: 0

Example: See [Specifying list mode settings](#)

Firmware/software: V3.7.20

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:RBALlocation
 <Auto>, <NoRB>, <Offset>

Defines the uplink resource block allocation manually for segment <no>. By default, the RB allocation is detected automatically.

Suffix:

<no> 1..2000
 Segment number

Parameters:

<Auto> OFF | ON
OFF: manual definition via <NoRB> and <Offset>
ON: automatic detection of RB allocation
 *RST: ON

<NoRB> integer
 Number of allocated resource blocks in each measured slot
 Range: see table below
 *RST: 100

<Offset> integer
 Offset of first allocated resource block from edge of allocated UL transmission bandwidth
 Range: 0 to max(<NoRB>) - <NoRB>
 *RST: 0

Example: See [Specifying list mode settings](#)

Firmware/software: V3.0.10

Channel bandwidth	Range <NoRB>
1.4 MHz	1 to 6
3 MHz	1 to 15
5 MHz	1 to 25
10 MHz	1 to 50
15 MHz	1 to 75
20 MHz	1 to 100
For eMTC, the relevant channel bandwidth is always 1.4 MHz, because the RB allocation is defined within the narrowband.	

**CONFigure:LTE:MEAS<i></i>:MEValuation:LIST:SEGment<no>:RBAllocation:
SIDelink <Auto>, <NoRB_PSSCH>, <Offset_PSSCH>, <Offset_PSCCH>**

Defines the sidelink resource block allocation manually for segment <no>. By default, the RB allocation is detected automatically.

Most allowed input ranges depend on other settings, see [Section 5.2.11.3, "Sidelink resource block allocation"](#), on page 1012.

Suffix:

<no> 1..2000
Segment number

Parameters:

<Auto> OFF | ON
OFF: manual definition via the other settings
ON: automatic detection of RB allocation
*RST: ON

<NoRB_PSSCH> integer
Number of allocated RBs for the PSSCH in each measured slot
*RST: 1

<Offset_PSSCH> integer
Offset of the first allocated PSSCH resource block
*RST: 2

<Offset_PSCCH> integer
Offset of the first allocated PSCCH resource block
*RST: 0

Example: See [Specifying list mode settings](#)

Firmware/software: V3.7.55

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:MODulation

<ModStatistics>, <ModEnable>, <EVMenable>, <MagErrorEnable>,
<PhaseErrEnable>, <IBEnable>, <EqSpFlatEnable>, <ModScheme>

Defines settings for modulation measurements in list mode for segment <no>.

Suffix:

<no> 1..2000
Segment number

Parameters:

<ModStatistics> integer
Statistical length in slots.
Range: 1 to 1000
*RST: 20

<ModEnable> OFF | ON
Enable or disable the measurement of modulation results.
ON: Modulation results are measured according to the other enable flags in this command. Modulation results for which there is no explicit enable flag are also measured (e.g. I/Q offset, frequency error and timing error).
OFF: No modulation results at all are measured. The other enable flags in this command are ignored.
*RST: OFF

<EVMenable> OFF | ON
Enable or disable measurement of EVM.
*RST: OFF

<MagErrorEnable> OFF | ON
Enable or disable measurement of magnitude error.
*RST: OFF

<PhaseErrEnable> OFF | ON
Enable or disable measurement of phase error.
*RST: OFF

<IBEnable> OFF | ON
Enable or disable measurement of in-band emissions.
*RST: OFF

<EqSpFlatEnable> OFF | ON
Enable or disable measurement of equalizer spectrum flatness results.
*RST: OFF

<ModScheme> AUTO | QPSK | Q16 | Q64 | Q256
The modulation scheme used by the LTE uplink signal.
AUTO: automatic detection
QPSK: QPSK

Q16: 16QAM**Q64:** 64QAM**Q256:** 256QAM***RST:** QPSK**Example:** See [Specifying list mode settings](#)**Firmware/software:** V2.0.10, V3.7.50: 256QAM**Options:** R&S CMW-KM504/-KM554 for 256QAM**CONFigure:LTE:MEAS<i>:MEvaluation:LIST:SEGment<no>:SEMAsk**

<SEMstatistics>, <SEenable>, <OBWenable>, <SEMenable>

Defines settings for spectrum emission measurements in list mode for segment <no>.

Suffix:<no> 1..2000
Segment number**Parameters:**<SEMstatistics> integer
Statistical length in slots.
Range: 1 to 1000
***RST:** 20<SEenable> OFF | ON
Enable or disable the measurement of spectrum emission results.
ON: Spectrum emission results are measured according to the other <...enable> flags in this command. Results for which there is no explicit enable flag are also measured.
OFF: No spectrum emission results at all are measured. The other enable flags in this command are ignored.
***RST:** OFF<OBWenable> OFF | ON
Enable or disable measurement of occupied bandwidth.
***RST:** OFF<SEMenable> OFF | ON
Enable or disable measurement of spectrum emission trace and margin results.
***RST:** OFF**Example:** See [Specifying list mode settings](#)**Firmware/software:** V2.0.10

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ACLR

<ACLRstatistics>, <ACLRenable>, <UTRA1enable>, <UTRA2enable>,
<EUTRAenable>

Defines settings for ACLR measurements in list mode for segment <no>.

Suffix:

<no> 1..2000
Segment number

Parameters:

<ACLRstatistics> integer
Statistical length in slots
Range: 1 to 1000
*RST: 20

<ACLRenable> OFF | ON
Enable or disable the measurement of ACLR results.
ON: ACLR results are measured according to the other enable flags in this command. ACLR results for which there is no explicit enable flag are also measured (e.g. power in the assigned E-UTRA channel).
OFF: No ACLR results at all are measured. The other enable flags in this command are ignored.
*RST: OFF

<UTRA1enable> OFF | ON
Enable or disable evaluation of first adjacent UTRA channels.
*RST: OFF

<UTRA2enable> OFF | ON
Enable or disable evaluation of second adjacent UTRA channels.
*RST: OFF

<EUTRAenable> OFF | ON
Enable or disable evaluation of first adjacent E-UTRA channels.
*RST: OFF

Example: See [Specifying list mode settings](#)

Firmware/software: V2.0.10

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:PMONitor <Enable>

Enables or disables the measurement of power monitor results (power of one carrier) for segment <no>.

Suffix:

<no> 1..2000
Segment number

Parameters:

<Enable> OFF | ON
 *RST: OFF

Example: See [Specifying list mode settings](#)

Firmware/software: V2.0.20

CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWER
 <PowerStatistics>, <PowerEnable>

Defines settings for the measurement of the total TX power of all carriers for segment <no>.

Suffix:

<no> 1..2000
 Segment number

Parameters:

<PowerStatistics> integer
 Statistical length in subframes
 Range: 1 to 1000
 *RST: 20

<PowerEnable> OFF | ON
 Enables or disables the measurement of the total TX power.
 *RST: OFF

Example: See [Specifying list mode settings](#)

Firmware/software: V3.5.50

TRIGger:LTE:MEAS<i>:MEValuation:LIST:MODE <Mode>

Specifies the trigger mode for list mode measurements.

For configuration of retrigger flags, see [CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SETup](#) on page 1156.

For configuration of the global trigger source, see [TRIGger:LTE:MEAS<i>:MEValuation:SOURce](#) on page 1168.

Parameters:

<Mode> ONCE | SEGMENT

ONCE

A trigger event is only required to start the measurement. The entire range of segments to be measured is captured without additional trigger event. The global trigger source is used.

SEGMENT

The retrigger flag of each segment is evaluated. It defines whether a trigger event is required and which trigger source is used.

*RST: ONCE

Example: See [Specifying list mode settings](#)

Firmware/software: V2.0.10

TRIGger:LTE:MEAS<i>:MEValuation:LIST:NBANdwidth <Bandwidth>

Selects the trigger evaluation bandwidth for the retrigger source IFPNarrowband. Select the retrigger source via [CONFigure:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:SETup](#) on page 1156.

Parameters:

<Bandwidth> M010 | M020 | M040 | M080
Evaluation bandwidth 10 MHz to 80 MHz
*RST: M020

Firmware/software: V4.0.20

5.5.3.10 Trigger settings

The following commands define the trigger parameters.

TRIGger:LTE:MEAS<i>:MEValuation:CATalog:SOURce?	1167
TRIGger:LTE:MEAS<i>:MEValuation:SOURce	1168
TRIGger:LTE:MEAS<i>:MEValuation:SLOPe	1168
TRIGger:LTE:MEAS<i>:MEValuation:THReshold	1168
TRIGger:LTE:MEAS<i>:MEValuation:DELay	1169
TRIGger:LTE:MEAS<i>:MEValuation:TOUT	1169
TRIGger:LTE:MEAS<i>:MEValuation:MGAP	1169
TRIGger:LTE:MEAS<i>:MEValuation:SMODE	1170
TRIGger:LTE:MEAS<i>:MEValuation:AMODE	1170

TRIGger:LTE:MEAS<i>:MEValuation:CATalog:SOURce?

Lists all trigger source values that can be set using [TRIGger:LTE:MEAS<i>:MEValuation:SOURce](#).

Return values:

<Sourcelist> string
Comma-separated list of all supported values. Each value is represented as a string.

Usage: Query only

Firmware/software: V1.0.10.1

Manual operation: See ["Trigger Source"](#) on page 1070

TRIGger:LTE:MEAS<i>:MEValuation:SOURce <Source>

Selects the source of the trigger events. Some values are always available. They are listed below. Depending on the installed options, additional values are available. You can query a list of all supported values via the `TRIGger: . . . :CATalog:SOURce?` command.

Parameters:

<Source> string

'Free Run (Fast Sync)'
Free run with synchronization

'Free Run (No Sync)'
Free run without synchronization

'IF Power'
Power trigger (received RF power)

 *RST: 'IF Power'

Example: See [Configuring the trigger system](#)

Firmware/software: V1.0.10.1
V2.0.10: 'Free Run (No Sync)' added

Manual operation: See ["Trigger Source"](#) on page 1070

TRIGger:LTE:MEAS<i>:MEValuation:SLOPe <Slope>

Qualifies whether the trigger event is generated at the rising or at the falling edge of the trigger pulse (valid for external and power trigger sources).

Parameters:

<Slope> REDGe | FEDGe

REDGe: Rising edge

FEDGe: Falling edge

 *RST: REDG

Example: See [Configuring the trigger system](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Trigger Slope"](#) on page 1071

TRIGger:LTE:MEAS<i>:MEValuation:THReshold <TrigThreshold>

Defines the trigger threshold for power trigger sources.

Parameters:

<TrigThreshold> numeric

 Range: -50 dB to 0 dB

 *RST: -20 dB

 Default unit: dB (full scale, i.e. relative to reference level minus external attenuation)

Example: See [Configuring the trigger system](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Trigger Threshold"](#) on page 1071

TRIGger:LTE:MEAS<i>:MEValuation:DElay <Delay>

Defines a time delaying the start of the measurement relative to the trigger event. This setting has no influence on free run measurements.

Parameters:

<Delay> numeric
 Range: -250E-6 s to 0.02 s
 *RST: 0 s
 Default unit: s

Example: See [Configuring the trigger system](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Trigger Delay"](#) on page 1071

TRIGger:LTE:MEAS<i>:MEValuation:TOUT <TriggerTimeOut>

Selects the maximum time that the measurement waits for a trigger event before it stops in remote control mode or indicates a trigger timeout in manual operation mode. This setting has no influence on "Free Run" measurements.

Parameters:

<TriggerTimeOut> numeric | ON | OFF
 Range: 0.01 s to 167772.15 s
 *RST: 0.1 s
 Default unit: s
 ON | OFF enables or disables the timeout.

Example: See [Configuring the trigger system](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Trigger Timeout"](#) on page 1071

TRIGger:LTE:MEAS<i>:MEValuation:MGAP <MinTrigGap>

Sets a minimum time during which the IF signal must be below the trigger threshold before the trigger is armed so that an IF power trigger event can be generated.

Parameters:

<MinTrigGap> integer
 Range: 0 slots to 20 slots
 *RST: 2 slots
 Default unit: slots

Example: See [Configuring the trigger system](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Min Trigger Gap"](#) on page 1071

TRIGger:LTE:MEAS<i>:MEValuation:SMODE <SyncMode>

Selects the size of the search window for synchronization - normal or enhanced.

Parameters:

<SyncMode> NORMAl | ENHanced
*RST: NORM

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.10

Manual operation: See ["Synchronization Mode"](#) on page 1072

TRIGger:LTE:MEAS<i>:MEValuation:AMODE <AcquisitionMode>

Selects whether the measurement synchronizes to a slot boundary or to a subframe boundary. The parameter is relevant for "Free Run (Fast Sync)" and for list mode measurements with "Synchronization Mode" = "Enhanced".

Parameters:

<AcquisitionMode> SLOT | SUBFrame
*RST: SLOT

Example: See [Configuring the trigger system](#)

Firmware/software: V2.1.25

Manual operation: See ["Acquisition Mode"](#) on page 1072

5.5.3.11 Limits (modulation, QPSK)

The following commands define limits for results which characterize the modulation accuracy of signals with the modulation scheme QPSK.

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:EVMagnitude	1170
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:MERRor	1171
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:PERRor	1171
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:FERRor	1172
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:IQOFfset	1172
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:IBE	1173
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:IBE:IQOFfset	1174
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:ESFLatness	1174

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:EVMagnitude <RMS>, <Peak>

Defines upper limits for the RMS and peak values of the error vector magnitude (EVM) for QPSK.

Parameters:

<RMS> numeric | ON | OFF
 Range: 0 % to 100 %
 *RST: 17.5 %
 Default unit: %
 ON | OFF enables or disables the limit check.

<Peak> numeric | ON | OFF
 Range: 0 % to 100 %
 *RST: OFF
 Default unit: %
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See ["EVM, magnitude error, phase error limit settings"](#) on page 1073

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:MERRor <RMS>, <Peak>

Defines upper limits for the RMS and peak values of the magnitude error for QPSK.

Parameters:

<RMS> numeric | ON | OFF
 Range: 0 % to 100 %
 *RST: OFF
 Default unit: %
 ON | OFF enables or disables the limit check.

<Peak> numeric | ON | OFF
 Range: 0 % to 100 %
 *RST: OFF
 Default unit: %
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See ["EVM, magnitude error, phase error limit settings"](#) on page 1073

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:PERRor <RMS>, <Peak>

Defines symmetric limits for the RMS and peak values of the phase error for QPSK. The limit check fails if the absolute value of the measured phase error exceeds the specified limit.

Parameters:

<RMS> numeric | ON | OFF
 Range: 0 deg to 180 deg
 *RST: OFF
 Default unit: deg
 ON | OFF enables or disables the limit check.

<Peak> numeric | ON | OFF
 Range: 0 deg to 180 deg
 *RST: OFF
 Default unit: deg
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See ["EVM, magnitude error, phase error limit settings"](#) on page 1073

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:FERRor <FrequencyError>

Defines an upper limit for the carrier frequency error (QPSK modulation).

Parameters:

<FrequencyError> numeric | ON | OFF
 Range: 0 ppm to 1 ppm
 *RST: 0.1 ppm
 Default unit: ppm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See ["Frequency error limit settings"](#) on page 1074

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QPSK:IQOffset <Enable>, <Offset1>, <Offset2>, <Offset3>

Defines upper limits for the I/Q origin offset (QPSK modulation). Three different I/Q origin offset limits can be set for three TX power ranges. For details, see [Section 5.3.7.4, "I/Q origin offset limits"](#), on page 1074.

Parameters:

<Enable> OFF | ON
 OFF: disables the limit check
 ON: enables the limit check
 *RST: ON

<Offset1>	numeric I/Q origin offset limit for high TX power range Range: -256 dBc to 256 dBc *RST: -24.2 dBc Default unit: dB
<Offset2>	numeric I/Q origin offset limit for intermediate TX power range Range: -256 dBc to 256 dBc *RST: -19.2 dBc Default unit: dB
<Offset3>	numeric I/Q origin offset limit for low TX power range Range: -256 dBc to 256 dBc *RST: -9.2 dBc Default unit: dB
Example:	See Specifying limits
Firmware/software:	V1.0.10.1
Manual operation:	See "I/Q origin offset limit settings" on page 1074

CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:QPSK:IBE <Enable>, <Minimum>, <EVM>, <RBPower>, <IQImage>

Defines parameters used for calculation of an upper limit for the in-band emission (QPSK modulation), see [Section 5.3.7.5, "In-band emissions limits"](#), on page 1075.

Parameters:

<Enable>	OFF ON OFF: disables the limit check ON: enables the limit check *RST: ON
<Minimum>	numeric Range: -256 dB to 256 dB *RST: -29.2 dB Default unit: dB
<EVM>	numeric Range: 0 % to 100 % *RST: 17.5 % Default unit: %
<RBPower>	numeric Range: -256 dBm to 256 dBm *RST: -57 dBm Default unit: dBm

<IQImage> numeric
 Range: -256 dB to 256 dB
 *RST: -24.2 dB
 Default unit: dB

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See ["In-band emission limit settings"](#) on page 1076

CONFIGure:LTE:MEAS<i>:MEvaluation:LIMit:QPSK:IBE:IQOffset <Offset1>, <Offset2>, <Offset3>

Defines I/Q origin offset values used for calculation of an upper limit for the in-band emission (QPSK modulation). Three different values can be set for three TX power ranges, see [Section 5.3.7.5, "In-band emissions limits"](#), on page 1075.

Parameters:

<Offset1> numeric
 Offset for high TX power range
 Range: -256 dBc to 256 dBc
 *RST: -24.2 dBc
 Default unit: dB

<Offset2> numeric
 Offset for intermediate TX power range
 Range: -256 dBc to 256 dBc
 *RST: -19.2 dBc
 Default unit: dB

<Offset3> numeric
 Offset for low TX power range
 Range: -256 dBc to 256 dBc
 *RST: -9.2 dBc
 Default unit: dB

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See ["In-band emission limit settings"](#) on page 1076

CONFIGure:LTE:MEAS<i>:MEvaluation:LIMit:QPSK:ESFlatness <Enable>, <Range1>, <Range2>, <Max1Min2>, <Max2Min1>, <EdgeFrequency>

Defines limits for the equalizer spectrum flatness (QPSK modulation).

Parameters:

<Enable>	OFF ON OFF: disables the limit check ON: enables the limit check *RST: ON
<Range1>	numeric Upper limit for max(range 1) - min(range 1) Range: -256 dBpp to 256 dBpp *RST: 5.4 dBpp Default unit: dBpp
<Range2>	numeric Upper limit for max(range 2) - min(range 2) Range: -256 dBpp to 256 dBpp *RST: 9.4 dBpp Default unit: dBpp
<Max1Min2>	numeric Upper limit for max(range 1) - min(range 2) Range: -256 dB to 256 dB *RST: 6.4 dB Default unit: dB
<Max2Min1>	numeric Upper limit for max(range 2) - min(range 1) Range: -256 dB to 256 dB *RST: 8.4 dB Default unit: dB
<EdgeFrequency>	numeric Frequency band edge distance of border between range 1 and range 2 Range: 0 MHz to 20 MHz *RST: 3 MHz Default unit: Hz

Example: See [Specifying limits](#)

Firmware/software: V2.0.10

Manual operation: See "[Spectrum flatness limit settings](#)" on page 1077

5.5.3.12 Limits (modulation, 16QAM / 64QAM / 256QAM)

The following commands define limits for results which characterize the modulation accuracy of signals with QAM modulation schemes.

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:EVMagnitude.....	1176
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:MERRor.....	1176
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:PERRor.....	1177

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:FERRor.....	1177
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IQOffset.....	1178
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IBE.....	1179
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IBE:IQOffset.....	1179
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:ESFLatness.....	1180

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:EVMagnitude <RMS>, <Peak>

Defines upper limits for the RMS and peak values of the error vector magnitude (EVM), for QAM modulations.

Suffix:

<m> 16, 64, 256
16QAM, 64QAM, 256QAM

Parameters:

<RMS> numeric | ON | OFF
Range: 0 % to 100 %
*RST: 12.5 % / 8 % / 3.5 %
Default unit: %
ON | OFF enables or disables the limit check.

<Peak> numeric | ON | OFF
Range: 0 % to 100 %
*RST: OFF
Default unit: %
ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1, V3.7.50: added 256QAM and changed *RST values

Manual operation: See ["EVM, magnitude error, phase error limit settings"](#) on page 1073

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:MERRor <RMS>, <Peak>

Defines upper limits for the RMS and peak values of the magnitude error, for QAM modulations.

Suffix:

<m> 16, 64, 256
16QAM, 64QAM, 256QAM

Parameters:

<RMS> numeric | ON | OFF
Range: 0 % to 100 %
*RST: OFF
Default unit: %
ON | OFF enables or disables the limit check.

<Peak> numeric | ON | OFF
 Range: 0 % to 100 %
 *RST: OFF
 Default unit: %
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1, V3.7.50: added 256QAM and changed *RST values

Manual operation: See ["EVM, magnitude error, phase error limit settings"](#) on page 1073

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:PERRor <RMS>, <Peak>

Defines symmetric limits for the RMS and peak values of the phase error, for QAM modulations. The limit check fails if the absolute value of the measured phase error exceeds the specified limit.

Suffix:

<m> 16, 64, 256
 16QAM, 64QAM, 256QAM

Parameters:

<RMS> numeric | ON | OFF
 Range: 0 deg to 180 deg
 *RST: OFF
 Default unit: deg
 ON | OFF enables or disables the limit check.

<Peak> numeric | ON | OFF
 Range: 0 deg to 180 deg
 *RST: OFF
 Default unit: deg
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1, V3.7.50: added 256QAM

Manual operation: See ["EVM, magnitude error, phase error limit settings"](#) on page 1073

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:FERRor <FrequencyError>

Defines an upper limit for the carrier frequency error for QAM modulations.

Suffix:

<m> 16, 64, 256
 16QAM, 64QAM, 256QAM

Parameters:

<FrequencyError> numeric | ON | OFF
 Range: 0 ppm to 1 ppm
 *RST: 0.1 ppm
 Default unit: ppm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1, V3.7.50: 256QAM

Manual operation: See ["Frequency error limit settings"](#) on page 1074

CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:QAM<m>:IQOffset <Enable>, <Offset1>, <Offset2>, <Offset3>

Defines upper limits for the I/Q origin offset for QAM modulations. Three different I/Q origin offset limits can be set for three TX power ranges. For details, see [Section 5.3.7.4, "I/Q origin offset limits"](#), on page 1074.

Suffix:

<m> 16, 64, 256
 16QAM, 64QAM, 256QAM

Parameters:

<Enable> OFF | ON
OFF: disables the limit check
ON: enables the limit check
 *RST: ON

<Offset1> numeric
 I/Q origin offset limit for high TX power range
 Range: -256 dBc to 256 dBc
 *RST: -24.2 dBc
 Default unit: dBc

<Offset2> numeric
 I/Q origin offset limit for intermediate TX power range
 Range: -256 dBc to 256 dBc
 *RST: -19.2 dBc
 Default unit: dBc

<Offset3> numeric
 I/Q origin offset limit for low TX power range
 Range: -256 dBc to 256 dBc
 *RST: -9.2 dBc
 Default unit: dBc

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1, V3.7.50: 256QAM

Manual operation: See ["I/Q origin offset limit settings"](#) on page 1074

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IBE <Enable>, <Minimum>, <EVM>, <RBPow>, <IQImage>

Defines parameters used for calculation of an upper limit for the in-band emission, for QAM modulations, see [Section 5.3.7.5, "In-band emissions limits"](#), on page 1075.

Suffix:

<m> 16, 64, 256
16QAM, 64QAM, 256QAM

Parameters:

<Enable> OFF | ON
OFF: disables the limit check
ON: enables the limit check
 *RST: ON

<Minimum> numeric
 Range: -256 dB to 256 dB
 *RST: -29.2 dB
 Default unit: dB

<EVM> numeric
 Range: 0 % to 100 %
 *RST: 12.5 % / 8 % / 3.5 %
 Default unit: %

<RBPow> numeric
 Range: -256 dBm to 256 dBm
 *RST: -57 dBm
 Default unit: dBm

<IQImage> numeric
 Range: -256 dB to 256 dB
 *RST: -24.2 dB
 Default unit: dB

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1, V3.7.50: added 256QAM and changed *RST values

Manual operation: See ["In-band emission limit settings"](#) on page 1076

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:IBE:IQOffset <Offset1>, <Offset2>, <Offset3>

Defines I/Q origin offset values used for calculation of an upper limit for the in-band emission, for QAM modulations. Three different values can be set for three TX power ranges, see [Section 5.3.7.5, "In-band emissions limits"](#), on page 1075.

Suffix:

<m> 16, 64, 256
16QAM, 64QAM, 256QAM

Parameters:

<Offset1>	numeric Offset for high TX power range Range: -256 dBc to 256 dBc *RST: -24.2 dBc Default unit: dBc
<Offset2>	numeric Offset for intermediate TX power range Range: -256 dBc to 256 dBc *RST: -19.2 dBc Default unit: dBc
<Offset3>	numeric Offset for low TX power range Range: -256 dBc to 256 dBc *RST: -9.2 dBc Default unit: dBc

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1, V3.7.50: 256QAM

Manual operation: See ["In-band emission limit settings"](#) on page 1076

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:QAM<m>:ESFlatness <Enable>, <Range1>, <Range2>, <Max1Min2>, <Max2Min1>, <EdgeFrequency>

Defines limits for the equalizer spectrum flatness, for QAM modulations.

Suffix:

<m>	16, 64, 256 16QAM, 64QAM, 256QAM
-----	-------------------------------------

Parameters:

<Enable>	OFF ON OFF: disables the limit check ON: enables the limit check *RST: ON
<Range1>	numeric Upper limit for max(range 1) - min(range 1) Range: -256 dBpp to 256 dBpp *RST: 5.4 dBpp Default unit: dBpp
<Range2>	numeric Upper limit for max(range 2) - min(range 2) Range: -256 dBpp to 256 dBpp *RST: 9.4 dBpp Default unit: dBpp

<Max1Min2>	numeric Upper limit for max(range 1) - min(range 2) Range: -256 dB to 256 dB *RST: 6.4 dB Default unit: dB
<Max2Min1>	numeric Upper limit for max(range 2) - min(range 1) Range: -256 dB to 256 dB *RST: 8.4 dB Default unit: dB
<EdgeFrequency>	numeric Frequency band edge distance of border between range 1 and range 2 Range: 0 MHz to 20 MHz *RST: 3 MHz Default unit: Hz
Example:	See Specifying limits
Firmware/software:	V2.0.10, V3.7.50: 256QAM
Manual operation:	See " Spectrum flatness limit settings " on page 1077

5.5.3.13 Limits (OBW and spectrum emission, no carrier aggregation)

The following commands define occupied bandwidth (OBW) and spectrum emission limits for measurements without carrier aggregation.

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:CBANdwidth<Band>.....	1181
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:CBANdwidth<Band>.....	1182
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:ADdItional<Table>: CBANdwidth<Band>.....	1183
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:ADdItional<Table>: CBANdwidth<Band>:SIDelink.....	1185
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:ATTolerance<EUTRAband>.....	1187

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:
 CBANdwidth<Band> <OBWlimit>**

Defines an upper limit for the occupied bandwidth, depending on the channel bandwidth.

Suffix:

<Band> 14, 30, 50, 100, 150, 200
Channel bandwidth in 0.1 MHz

Parameters:

<OBWlimit> numeric | ON | OFF
 Range: 0 MHz to 40 MHz
 *RST: Channel bandwidth
 Default unit: Hz
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See ["OBW limit settings"](#) on page 1079

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:

CBANdwidth<Band> <Enable>, <FrequencyStart>, <FrequencyEnd>, <Level>,
 <RBW>

Defines general requirements for the emission mask area <no>. The activation state, the area borders, an upper limit and the resolution bandwidth must be specified.

The emission mask applies to the channel bandwidth <Band>.

Suffix:

<Band> 14, 30, 50, 100, 150, 200
 <no> 1..12
 Number of the emission mask area

Parameters:

<Enable> OFF | ON
OFF: Disables the check of these requirements.
ON: Enables the check of these requirements.
 *RST: depends on channel bandwidth and area number

<FrequencyStart> numeric
 Lower border of the area, relative to the edges of the channel bandwidth.
 Range: see table below
 *RST: depends on channel bandwidth and area number
 Default unit: Hz

<FrequencyEnd> numeric
 Upper border of the area, relative to the edges of the channel bandwidth.
 Range: see table below
 *RST: depends on channel bandwidth and area number
 Default unit: Hz

<Level> numeric
 Upper limit for the area
 Range: -256 dBm to 256 dBm
 *RST: depends on channel bandwidth and area number
 Default unit: dBm

<RBW> K030 | K100 | M1
 Resolution bandwidth to be used for the area.
K030: 30 kHz
K100: 100 kHz
M1: 1 MHz
 *RST: K030 (<no> = 1) / M1 (<no> = 2 to 12)

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1
 V3.2.80: <no> of areas enhanced to 12

Manual operation: See ["Emission mask configuration for uplink measurements"](#) on page 1081

<Band>	Range for <FrequencyStart> and <FrequencyEnd>
14	0 MHz to 5 MHz
30	0 MHz to 6 MHz
50	0 MHz to 10 MHz
100	0 MHz to 15 MHz
150	0 MHz to 20 MHz
200	0 MHz to 25 MHz

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
ADDITIONal<Table>:CBANDwidth<Band> <Enable>, <FrequencyStart>,
 <FrequencyEnd>, <Level>, <RBW>

Defines additional requirements for the emission mask area <no>, for uplink measurements. The activation state, the area borders, an upper limit and the resolution bandwidth must be specified.

The emission mask applies to the channel bandwidth <Band>. Several tables of additional requirements are available.

Suffix:
 <Band> 14, 30, 50, 100, 150, 200
 <no> 1..12
 Number of the emission mask area

<Table>	1..5 Selects the table via its number. Each table is applicable for certain NS_xx values: 1 = NS_03/11/20/21, 2 = NS_04 3 = NS_06/07, 4 = NS_27/43, 5 = NS_35
Parameters:	
<Enable>	OFF ON OFF: Disables the check of these requirements. ON: Enables the check of these requirements. *RST: depends on channel bandwidth, area number and set of requirements
<FrequencyStart>	numeric Lower border of the area, relative to the edges of the channel bandwidth. Range: see table below *RST: depends on channel bandwidth, area number and set of requirements Default unit: Hz
<FrequencyEnd>	numeric Upper border of the area, relative to the edges of the channel bandwidth. Range: see table below *RST: depends on channel bandwidth, area number and set of requirements Default unit: Hz
<Level>	numeric Upper limit for the area Range: -256 dBm to 256 dBm *RST: depends on channel bandwidth, area number and set of requirements Default unit: dBm
<RBW>	K030 K050 K100 K150 K200 M1 Resolution bandwidth to be used for the area. Only a subset of the values is allowed, depending on <Table> and <Band>, see table below. K030: 30 kHz K050: 50 kHz K100: 100 kHz K150: 150 kHz K200: 200 kHz M1: 1 MHz *RST: depends on area number and set of requirements
Example:	See Specifying limits

- Firmware/software:** V2.0.20
 V3.2.10: table 1 valid for NS_11
 V3.2.80: <no> of areas enhanced to 12
 V3.2.82: table 1 valid for NS_20 and NS_21
 V3.7.10: added table 4 and RBW K050, K150, K200
 V3.7.20: added table 5
 V3.7.60: table 4 valid for NS_43
- Manual operation:** See ["Emission mask configuration for uplink measurements"](#) on page 1081

Table 5-12: Frequency ranges

<Band>	Range for <FrequencyStart> and <FrequencyEnd>
14	0 MHz to 5 MHz
30	0 MHz to 6 MHz
50	0 MHz to 10 MHz
100	0 MHz to 15 MHz
150	0 MHz to 20 MHz
200	0 MHz to 25 MHz

Table 5-13: Resolution bandwidths

<Table>	<Band>	Supported <RBW>
1 to 3, 5	14, 30, 50, 100, 150, 200	K030, K100, M1
4	14, 30	K030, K100, M1
	50, 100	K050, K100, M1
	150	K050, K150, M1
	200	K050, K200, M1

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
ADDITIONal<Table>:CBANdwidth<Band>:SIDelink <Enable>,
 <FrequencyStart>, <FrequencyEnd>, <Level>, <Slope>, <RBW>

Defines additional requirements for the emission mask area <no>, for sidelink measurements. The activation state, the area borders, the start value and slope of the upper limit and the resolution bandwidth must be specified.

The emission mask applies to the channel bandwidth <Band>.

Suffix:

- <Band> 100
 Channel bandwidth in 0.1 MHz
- <no> 1..12
 Number of the emission mask area

<Table>	1 Selects the table via its number. Each table is applicable for certain NS_xx values: 1 = NS_33/34
Parameters:	
<Enable>	OFF ON OFF : Disables the check of these requirements. ON : Enables the check of these requirements. *RST: depends on area number
<FrequencyStart>	numeric Lower border of the area, relative to the edges of the channel bandwidth. Range: 0 MHz to 25 MHz *RST: depends on area number Default unit: Hz
<FrequencyEnd>	numeric Upper border of the area, relative to the edges of the channel bandwidth. Range: 0 MHz to 25 MHz *RST: depends on area number Default unit: Hz
<Level>	numeric Upper limit at <FrequencyStart> Range: -256 dBm to 256 dBm *RST: depends on area number Default unit: dBm
<Slope>	numeric Slope for the upper limit within the area Range: -256 dB/MHz to 256 dB/MHz *RST: depends on area number Default unit: dB/MHz
<RBW>	K030 K100 M1 Resolution bandwidth to be used for the area. K030 : 30 kHz K100 : 100 kHz M1 : 1 MHz *RST: K100
Example:	See Specifying V2X sidelink settings
Firmware/software:	V3.7.55
Manual operation:	See "Emission mask configuration for sidelink measurements" on page 1082

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:ATTolerance<EUTRAband>
<AddTestTol>**

Defines additional test tolerances for the emission masks. The tolerance is added to the power values of all general and additional spectrum emission masks. A positive tolerance value relaxes the limits.

For operating bands below 3 GHz, there is no additional test tolerance. You can define different additional test tolerances for bands above 3 GHz and for bands above 5 GHz.

Suffix:

<EUTRAband> 30, 50
 Selects the frequency range and thus the set of operating bands to which the tolerance applies.
 30: 3 GHz < frequency ≤ 4.2 GHz (examples: band 22, 42, 43)
 50: 5 GHz ≤ frequency (example: band 47)

Parameters:

<AddTestTol> numeric
 Additional test tolerance
 Range: -5 dB to 5 dB
 *RST: <EUTRAband> 30: 0.3 dB, 50: 0.5 dB
 Default unit: dB

Example: See [Specifying limits](#)

Firmware/software: V3.7.55

Manual operation: See ["Additional test tolerance settings"](#) on page 1082

5.5.3.14 Limits (ACLR, no carrier aggregation)

The following commands define ACLR limits for measurements without carrier aggregation.

[CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:CBANDwidth<Band>.....](#) 1187
[CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CBANDwidth<Band>.....](#) 1188

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:
CBANDwidth<Band> <RelativeLevel>, <AbsoluteLevel>**

Defines relative and absolute limits for the ACLR measured in the first or second adjacent UTRA channel, depending on <no>. The settings are defined separately for each channel bandwidth.

Suffix:

<Band> 14, 30, 50, 100, 150, 200
 <no> 1..2
 Selects first or second adjacent UTRA channel.

Parameters:

<RelativeLevel> numeric | ON | OFF
 Range: -256 dB to 256 dB
 *RST: 32.2 dB (<no> = 1) / 35.2 dB (<no> = 2, OFF for <Band> = 14 | 30)
 Default unit: dB
 ON | OFF enables or disables the limit check.

<AbsoluteLevel> numeric | ON | OFF
 Range: -256 dBm to 256 dBm
 *RST: -50 dBm (OFF for <no> = 2 and <Band> = 14 | 30)
 Default unit: dBm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See "[ACLR limit settings](#)" on page 1083

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CBANdwidth<Band>
 <RelativeLevel>, <AbsoluteLevel>

Defines relative and absolute limits for the ACLR measured in an adjacent E-UTRA channel. The settings are defined separately for each channel bandwidth.

Suffix:

<Band> 14, 30, 50, 100, 150, 200
 Channel bandwidth in 0.1 MHz

Parameters:

<RelativeLevel> numeric | ON | OFF
 Range: -256 dB to 256 dB
 *RST: 29.2 dB
 Default unit: dB
 ON | OFF enables or disables the limit check.

<AbsoluteLevel> numeric | ON | OFF
 Range: -256 dBm to 256 dBm
 *RST: -50 dBm
 Default unit: dBm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V1.0.10.1

Manual operation: See "[ACLR limit settings](#)" on page 1083

5.5.3.15 Limits (OBW, with carrier aggregation)

The following commands define occupied bandwidth (OBW) limits for measurements with carrier aggregation.

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:CAGGregation: CBANdwidth<Band1>:CBANdwidth<Band2>.....	1189
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:CAGGregation: CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>.....	1189
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:CAGGregation: OCOMbination.....	1190

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:CAGGregation:
 CBANdwidth<Band1>:CBANdwidth<Band2> <OBWlimit>**

Defines an upper limit for the occupied bandwidth.

The setting is defined separately for each channel bandwidth combination, for two aggregated carriers. The following bandwidth combinations are supported:

CA 10MHz + 5MHz
CA 10MHz + 10MHz
CA 20MHz + 5MHz
CA 20MHz + 10MHz
CA 15MHz + 15MHz
CA 20MHz + 15MHz
CA 20MHz + 20MHz

Example: For the first line in the figure, use . . . :CBANdwidth100:CBANdwidth50.

Suffix:

<Band1>	100, 150, 200 First channel bandwidth in 0.1 MHz
<Band2>	50, 100, 150, 200 Second channel bandwidth in 0.1 MHz

Parameters:

<OBWlimit>	numeric ON OFF Range: 0 MHz to 40 MHz *RST: Depends on the channel bandwidths Default unit: Hz ON OFF enables or disables the limit check.
------------	--

Example: See [Specifying limits](#)

Firmware/software: V3.2.70, V3.5.50 <Band2> = 50, V3.8.10 <Band1> = 100

Manual operation: See ["OBW limit settings"](#) on page 1079

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:OBWLimit:CAGGregation:
 CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>
 <OBWlimit>**

Defines an upper limit for the occupied bandwidth.

The settings are defined separately for each channel bandwidth combination, for three aggregated carriers. The following bandwidth combinations are supported:

```

+ CA 20MHz + 15MHz + 10MHz
+ CA 15MHz + 15MHz + 15MHz
+ CA 20MHz + 20MHz + 10MHz
+ CA 20MHz + 15MHz + 15MHz
+ CA 20MHz + 20MHz + 15MHz
+ CA 20MHz + 20MHz + 20MHz

```

Example: For the first line in the figure,
 use . . . :CBANdwidth200:CBANdwidth150:CBANdwidth100.

Suffix:

<Band1> 150, 200
 First channel bandwidth in 0.1 MHz

<Band2> 150, 200
 Second channel bandwidth in 0.1 MHz

<Band3> 100, 150, 200
 Third channel bandwidth in 0.1 MHz

Parameters:

<OBWlimit> numeric | ON | OFF
 Range: 0 MHz to 40 MHz
 *RST: Depends on the channel bandwidths
 Default unit: Hz
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V3.7.50

Manual operation: See ["OBW limit settings"](#) on page 1079

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMAsk:OBWLimit:CAGGregation:
 OCOMbination <OBWlimit>**

Defines an upper limit for the occupied bandwidth.

The setting applies to all "other" channel bandwidth combinations, not covered by other commands in this section.

Parameters:

<OBWlimit> numeric | ON | OFF
 Range: 0 MHz to 40 MHz
 *RST: 0 MHz
 Default unit: Hz
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V3.2.70

Manual operation: See ["OBW limit settings"](#) on page 1079

5.5.3.16 Limits (spectrum emission, with carrier aggregation)

The following commands define spectrum emission limits for measurements with carrier aggregation.

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>.....	1191
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>.....	1192
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:CAGGregation:OCOMbination.....	1194
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:ADDITIONal<Table>:CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2>.....	1195
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:ADDITIONal<Table>:CAGGregation:OCOMbination.....	1196

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:CAGGregation:CBANdwidth<Band1>:CBANdwidth<Band2> <Enable>, <FrequencyStart>, <FrequencyEnd>, <Level>, <RBW>

Defines general requirements for the emission mask area <no>. The activation state, the area borders, an upper limit and the resolution bandwidth must be specified.

The settings are defined separately for each channel bandwidth combination, for two aggregated carriers. The following bandwidth combinations are supported:

CA 10MHz + 5MHz
CA 10MHz + 10MHz
CA 20MHz + 5MHz
CA 20MHz + 10MHz
CA 15MHz + 15MHz
CA 20MHz + 15MHz
CA 20MHz + 20MHz

Example: For the first line in the figure, use . . . :CBANdwidth100:CBANdwidth50.

Suffix:

<no>	1..12 Number of the emission mask area
<Band1>	100, 150, 200 First channel bandwidth in 0.1 MHz
<Band2>	50, 100, 150, 200 Second channel bandwidth in 0.1 MHz

Parameters:

<Enable>	OFF ON OFF: Disables the check of these requirements. ON: Enables the check of these requirements. *RST: ON (<no> = 1 to 4) / OFF (<no> = 5 to 12)
----------	---

<FrequencyStart>	numeric Start frequency of the area, relative to the edges of the aggregated channel bandwidth. Range: 0 MHz to 65 MHz *RST: depends on the suffixes Default unit: Hz
<FrequencyEnd>	numeric Stop frequency of the area, relative to the edges of the aggregated channel bandwidth. Range: 0 MHz to 65 MHz *RST: depends on the suffixes Default unit: Hz
<Level>	numeric Upper limit for the area Range: -256 dBm to 256 dBm *RST: depends on the suffixes Default unit: dBm
<RBW>	K030 K100 M1 Resolution bandwidth to be used for the area. K030 : 30 kHz K100 : 100 kHz M1 : 1 MHz *RST: K030 (<no> = 1) / M1 (<no> = 2 to 12)
Example:	See Specifying limits
Firmware/software:	V3.5.50, V3.7.30 enhanced frequency ranges, V3.8.10 <Band1> = 100
Manual operation:	See "Emission mask configuration for uplink measurements" on page 1081

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:CAGgregation:CBANDwidth<Band1>:CBANDwidth<Band2>:CBANDwidth<Band3><Enable>, <FrequencyStart>, <FrequencyEnd>, <Level>, <RBW>

Defines general requirements for the emission mask area <no>. The activation state, the area borders, an upper limit and the resolution bandwidth must be specified.

The settings are defined separately for each channel bandwidth combination, for three aggregated carriers. The following bandwidth combinations are supported:

- ☐ CA 20MHz + 15MHz + 10MHz
- ☐ CA 15MHz + 15MHz + 15MHz
- ☐ CA 20MHz + 20MHz + 10MHz
- ☐ CA 20MHz + 15MHz + 15MHz
- ☐ CA 20MHz + 20MHz + 15MHz
- ☐ CA 20MHz + 20MHz + 20MHz

Example: For the first line in the figure,

use . . . :CBANdwidth200:CBANdwidth150:CBANdwidth100.

Suffix:

<no>	1..12
	Number of the emission mask area
<Band1>	150, 200
	First channel bandwidth in 0.1 MHz
<Band2>	150, 200
	Second channel bandwidth in 0.1 MHz
<Band3>	100, 150, 200
	Third channel bandwidth in 0.1 MHz

Parameters:

<Enable>	OFF ON
	OFF: Disables the check of these requirements.
	ON: Enables the check of these requirements.
	*RST: ON (<no> = 1 to 4) / OFF (<no> = 5 to 12)
<FrequencyStart>	numeric
	Start frequency of the area, relative to the edges of the aggregated channel bandwidth.
	Range: 0 MHz to 65 MHz
	*RST: depends on the suffixes
	Default unit: Hz
<FrequencyEnd>	numeric
	Stop frequency of the area, relative to the edges of the aggregated channel bandwidth.
	Range: 0 MHz to 65 MHz
	*RST: depends on the suffixes
	Default unit: Hz
<Level>	numeric
	Upper limit for the area
	Range: -256 dBm to 256 dBm
	*RST: depends on the suffixes
	Default unit: dBm
<RBW>	K030 K100 M1
	Resolution bandwidth to be used for the area.
	K030: 30 kHz
	K100: 100 kHz
	M1: 1 MHz
	*RST: K030 (<no> = 1) / M1 (<no> = 2 to 12)

Example: See [Specifying limits](#)

Firmware/software: V3.7.50

Manual operation: See ["Emission mask configuration for uplink measurements"](#) on page 1081

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:CAGGregation:OCOMbination <Enable>, <FrequencyStart>, <FrequencyEnd>, <Level>, <RBW>

Defines general requirements for the emission mask area <no>. The activation state, the area borders, an upper limit and the resolution bandwidth must be specified.

The settings apply to all "other" channel bandwidth combinations, not covered by other commands in this section.

Suffix:

<no> 1..12
Number of the emission mask area

Parameters:

<Enable> OFF | ON
OFF: Disables the check of these requirements.
ON: Enables the check of these requirements.
*RST: OFF

<FrequencyStart> numeric
Start frequency of the area, relative to the edges of the aggregated channel bandwidth.
Range: 0 MHz to 65 MHz
*RST: depends on <no>
Default unit: Hz

<FrequencyEnd> numeric
Stop frequency of the area, relative to the edges of the aggregated channel bandwidth.
Range: 0 MHz to 65 MHz
*RST: depends on <no>
Default unit: Hz

<Level> numeric
Upper limit for the area
Range: -256 dBm to 256 dBm
*RST: depends on <no>
Default unit: dBm

<RBW> K030 | K100 | M1
Resolution bandwidth to be used for the area.
K030: 30 kHz
K100: 100 kHz
M1: 1 MHz
*RST: K030 (<no> = 1) / M1 (<no> = 2 to 12)

Example: See [Specifying limits](#)

Firmware/software: V3.2.80, V3.7.30 enhanced frequency ranges

Manual operation: See ["Emission mask configuration for uplink measurements"](#) on page 1081

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:SEMask:LIMit<no>:
ADDITIONal<Table>:CAGGregation:CBANdwidth<Band1>:
CBANdwidth<Band2> <Enable>, <FrequencyStart>, <FrequencyEnd>,
<Level>, <RBW>**

Defines additional requirements for the emission mask area <no>. The activation state, the area borders, an upper limit and the resolution bandwidth must be specified.

The settings are defined separately for each channel bandwidth combination, for two aggregated carriers. The following bandwidth combinations are supported:

```

+ CA 10MHz + 5MHz
+ CA 10MHz + 10MHz
+ CA 20MHz + 5MHz
+ CA 20MHz + 10MHz
+ CA 15MHz + 15MHz
+ CA 20MHz + 15MHz
+ CA 20MHz + 20MHz

```

Example: For the first line in the figure, use . . . :CBANdwidth100:CBANdwidth50.

Suffix:

<no>	1..12 Number of the emission mask area
<Band1>	100, 150, 200 First channel bandwidth in 0.1 MHz
<Band2>	50, 100, 150, 200 Second channel bandwidth in 0.1 MHz
<Table>	1 A set of additional requirements 1 = CA_NS_04

Parameters:

<Enable>	OFF ON OFF: Disables the check of these requirements. ON: Enables the check of these requirements. *RST: ON (<no> = 1 to 3) / OFF (<no> = 4 to 12)
<FrequencyStart>	numeric Start frequency of the area, relative to the edges of the aggregated channel bandwidth. Range: 0 MHz to 65 MHz *RST: depends on the suffixes Default unit: Hz

<FrequencyEnd>	numeric Stop frequency of the area, relative to the edges of the aggregated channel bandwidth. Range: 0 MHz to 65 MHz *RST: depends on the suffixes Default unit: Hz
<Level>	numeric Upper limit for the area Range: -256 dBm to 256 dBm *RST: depends on the suffixes Default unit: dBm
<RBW>	K030 K100 M1 Resolution bandwidth to be used for the area. K030 : 30 kHz K100 : 100 kHz M1 : 1 MHz *RST: K030 (<no> = 1) / M1 (<no> = 2 to 12)
Example:	See Specifying limits
Firmware/software:	V3.5.50, V3.7.30 enhanced frequency ranges, V3.8.10 <Band1> = 100
Manual operation:	See "Emission mask configuration for uplink measurements" on page 1081

**CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:SEMask:LIMit<no>:
Additional<Table>:CAGgregation:OCOMbination** <Enable>,
<FrequencyStart>, <FrequencyEnd>, <Level>, <RBW>

Defines additional requirements for the emission mask area <no>. The activation state, the area borders, an upper limit and the resolution bandwidth must be specified.

The settings apply to all "other" channel bandwidth combinations, not covered by other commands in this section.

Suffix:

<no> 1..12
Number of the emission mask area

<Table> 1
A set of additional requirements
1 = CA_NS_04

Parameters:

<Enable> OFF | ON
OFF: Disables the check of these requirements.
ON: Enables the check of these requirements.
*RST: OFF

<FrequencyStart>	numeric Start frequency of the area, relative to the edges of the aggregated channel bandwidth. Range: 0 MHz to 65 MHz *RST: depends on the suffixes Default unit: Hz
<FrequencyEnd>	numeric Stop frequency of the area, relative to the edges of the aggregated channel bandwidth. Range: 0 MHz to 65 MHz *RST: depends on the suffixes Default unit: Hz
<Level>	numeric Upper limit for the area. Range: -256 dBm to 256 dBm *RST: depends on the suffixes Default unit: dBm
<RBW>	K030 K100 M1 Resolution bandwidth to be used for the area. K030 : 30 kHz K100 : 100 kHz M1 : 1 MHz *RST: K030 (<no> = 1) / M1 (<no> = 2 to 12)
Example:	See Specifying limits
Firmware/software:	V3.2.80, V3.7.30 enhanced frequency ranges
Manual operation:	See "Emission mask configuration for uplink measurements" on page 1081

5.5.3.17 Limits (ACLR, with carrier aggregation)

The following commands define ACLR limits for measurements with carrier aggregation.

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:CAGGregation: CBANdwidth<Band1>:CBANdwidth<Band2>.....	1198
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:CAGGregation: CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>.....	1198
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:CAGGregation: OCOMbination.....	1199
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation: CBANdwidth<Band1>:CBANdwidth<Band2>.....	1200
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation: CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>.....	1201
CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation:OCOMbination	1202

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:CAGGregation:
CBANdwidth<Band1>:CBANdwidth<Band2> <RelativeLevel>,
<AbsoluteLevel>**

Defines relative and absolute limits for the ACLR measured in the first or second adjacent UTRA channel, depending on <no>.

The settings are defined separately for each channel bandwidth combination, for two aggregated carriers. The following bandwidth combinations are supported:

CA 10MHz + 5MHz
CA 10MHz + 10MHz
CA 20MHz + 5MHz
CA 20MHz + 10MHz
CA 15MHz + 15MHz
CA 20MHz + 15MHz
CA 20MHz + 20MHz

Example: For the first line in the figure, use . . . :CBANdwidth100:CBANdwidth50.

Suffix:

<no>	1..2 Selects first or second adjacent UTRA channel.
<Band1>	100, 150, 200 First channel bandwidth in 0.1 MHz
<Band2>	50, 100, 150, 200 Second channel bandwidth in 0.1 MHz

Parameters:

<RelativeLevel>	numeric ON OFF Range: -256 dB to 256 dB *RST: 32.2 dB (<no> = 1) / 35.2 dB (<no> = 2) Default unit: dB ON OFF enables or disables the limit check.
<AbsoluteLevel>	numeric ON OFF Range: -256 dBm to 256 dBm *RST: -50 dBm Default unit: dBm ON OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V3.2.70, V3.5.50 <Band2> = 50, V3.8.10 <Band1> = 100

Manual operation: See ["ACLR limit settings"](#) on page 1083

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:CAGGregation:
CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>
<RelativeLevel>, <AbsoluteLevel>**

Defines relative and absolute limits for the ACLR measured in the first or second adjacent UTRA channel, depending on <no>.

The settings are defined separately for each channel bandwidth combination, for three aggregated carriers. The following bandwidth combinations are supported:

```

+ CA 20MHz + 15MHz + 10MHz
+ CA 15MHz + 15MHz + 15MHz
+ CA 20MHz + 20MHz + 10MHz
+ CA 20MHz + 15MHz + 15MHz
+ CA 20MHz + 20MHz + 15MHz
+ CA 20MHz + 20MHz + 20MHz

```

Example: For the first line in the figure,
 use . . . :CBANdwidth200:CBANdwidth150:CBANdwidth100.

Suffix:

<no> 1..2
 Selects first or second adjacent UTRA channel.

<Band1> 150, 200
 First channel bandwidth in 0.1 MHz

<Band2> 150, 200
 Second channel bandwidth in 0.1 MHz

<Band3> 100, 150, 200
 Third channel bandwidth in 0.1 MHz

Parameters:

<RelativeLevel> numeric | ON | OFF
 Range: -256 dB to 256 dB
 *RST: 32.2 dB (<no> = 1) / 35.2 dB (<no> = 2)
 Default unit: dB
 ON | OFF enables or disables the limit check.

<AbsoluteLevel> numeric | ON | OFF
 Range: -256 dBm to 256 dBm
 *RST: -50 dBm
 Default unit: dBm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V3.7.50

Manual operation: See ["ACLR limit settings"](#) on page 1083

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:UTRA<no>:CAGGregation:
 OCOMbination <RelativeLevel>, <AbsoluteLevel>**

Defines relative and absolute limits for the ACLR measured in the first or second adjacent UTRA channel, depending on <no>.

The settings apply to all "other" channel bandwidth combinations, not covered by other commands in this section.

Suffix:

<no> 1..2
Selects first or second adjacent UTRA channel.

Parameters:

<RelativeLevel> numeric | ON | OFF
Range: -256 dB to 256 dB
*RST: 32.2 dB (<no> = 1) / 35.2 dB (<no> = 2)
Default unit: dB
ON | OFF enables or disables the limit check.

<AbsoluteLevel> numeric | ON | OFF
Range: -256 dBm to 256 dBm
*RST: -50 dBm
Default unit: dBm
ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V3.2.70

Manual operation: See "[ACLR limit settings](#)" on page 1083

CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation:
CBANdwidth<Band1>:CBANdwidth<Band2> <RelativeLevel> ,
<AbsoluteLevel>

Defines relative and absolute limits for the ACLR measured in an adjacent E-UTRA channel.

The settings are defined separately for each channel bandwidth combination, for two aggregated carriers. The following bandwidth combinations are supported:

CA 10MHz + 5MHz
CA 10MHz + 10MHz
CA 20MHz + 5MHz
CA 20MHz + 10MHz
CA 15MHz + 15MHz
CA 20MHz + 15MHz
CA 20MHz + 20MHz

Example: For the first line in the figure, use . . . :CBANdwidth100:CBANdwidth50.

Suffix:

<Band1> 100, 150, 200
First channel bandwidth in 0.1 MHz

<Band2> 50, 100, 150, 200
Second channel bandwidth in 0.1 MHz

Parameters:

<RelativeLevel> numeric | ON | OFF
 Range: -256 dB to 256 dB
 *RST: 29.2 dB
 Default unit: dB
 ON | OFF enables or disables the limit check.

<AbsoluteLevel> numeric | ON | OFF
 Range: -256 dBm to 256 dBm
 *RST: -50 dBm
 Default unit: dBm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

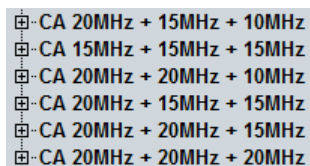
Firmware/software: V3.2.70, V3.5.50 <Band2> = 50, V3.8.10 <Band1> = 100

Manual operation: See ["ACLR limit settings"](#) on page 1083

**CONFigure:LTE:MEAS<i>:MEvaluation:LIMit:ACLR:EUTRa:CAGGregation:
 CBANdwidth<Band1>:CBANdwidth<Band2>:CBANdwidth<Band3>
 <RelativeLevel>, <AbsoluteLevel>**

Defines relative and absolute limits for the ACLR measured in an adjacent E-UTRA channel.

The settings are defined separately for each channel bandwidth combination, for three aggregated carriers. The following bandwidth combinations are supported:



CA 20MHz + 15MHz + 10MHz
 CA 15MHz + 15MHz + 15MHz
 CA 20MHz + 20MHz + 10MHz
 CA 20MHz + 15MHz + 15MHz
 CA 20MHz + 20MHz + 15MHz
 CA 20MHz + 20MHz + 20MHz

Example: For the first line in the figure,
 use . . . :CBANdwidth200:CBANdwidth150:CBANdwidth100.

Suffix:

<Band1> 150, 200
 First channel bandwidth in 0.1 MHz

<Band2> 150, 200
 Second channel bandwidth in 0.1 MHz

<Band3> 100, 150, 200
 Third channel bandwidth in 0.1 MHz

Parameters:

<RelativeLevel> numeric | ON | OFF
 Range: -256 dB to 256 dB
 *RST: 29.2 dB
 Default unit: dB
 ON | OFF enables or disables the limit check.

<AbsoluteLevel> numeric | ON | OFF
 Range: -256 dBm to 256 dBm
 *RST: -50 dBm
 Default unit: dBm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V3.7.50

Manual operation: See "[ACLR limit settings](#)" on page 1083

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:ACLR:EUTRa:CAGGregation:
 OCOMbination <RelativeLevel>, <AbsoluteLevel>**

Defines relative and absolute limits for the ACLR measured in an adjacent E-UTRA channel.

The settings apply to all "other" channel bandwidth combinations, not covered by other commands in this section.

Parameters:

<RelativeLevel> numeric | ON | OFF
 Range: -256 dB to 256 dB
 *RST: 29.2 dB
 Default unit: dB
 ON | OFF enables or disables the limit check.

<AbsoluteLevel> numeric | ON | OFF
 Range: -256 dBm to 256 dBm
 *RST: -50 dBm
 Default unit: dBm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V3.2.70

Manual operation: See "[ACLR limit settings](#)" on page 1083

5.5.3.18 Limits (power)

The following command defines limits for results which characterize the power dynamics of the signal.

**CONFigure:LTE:MEAS<i>:MEValuation:LIMit:PDYNamics:CBANDwidth<Band>
 <Enable>, <OnPowerUpper>, <OnPowerLower>, <OffPowerUpper>**

Defines limits for the ON power and OFF power determined with the power dynamics measurement. Separate limits can be defined for each channel bandwidth.

Suffix:

<Band> 14, 30, 50, 100, 150, 200
 Channel bandwidth in 0.1 MHz

FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:EVMagnitude:PEAK?	1204
FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:MERRor?	1205
FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:PERRor?	1205
FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:EVMagnitude?	1205
FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:EVMagnitude:PEAK?	1205
FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:MERRor?	1205
FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:PERRor?	1205
FETCh:LTE:MEAS<i>:MEValuation:REFMarker:PDYNamics?	1206
FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:PDYNamics?	1206
FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:PDYNamics?	1207
FETCh:LTE:MEAS<i>:MEValuation:REFMarker:PMONitor:CC<no>?	1208
FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:PMONitor:CC<no>?	1208
FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:PMONitor:CC<no>?	1209

FETCh:LTE:MEAS<i>:MEValuation:REFMarker:EVMagnitude? <XValue>, <TraceSelect>

FETCh:LTE:MEAS<i>:MEValuation:REFMarker:EVMagnitude:PEAK? <XValue>, <TraceSelect>

FETCh:LTE:MEAS<i>:MEValuation:REFMarker:MERRor? <XValue>, <TraceSelect>

FETCh:LTE:MEAS<i>:MEValuation:REFMarker:PERRor? <XValue>, <TraceSelect>

Uses the reference marker on the diagrams: EVM RMS, EVM peak, magnitude error and phase error vs SC-FDMA symbol.

Query parameters:

<XValue> integer
 Absolute x-value of the marker position
 There are two x-values per SC-FDMA symbol on the x-axis
 (symbol 0 low, symbol 0 high, ..., symbol 6 low, symbol 6 high).
 Range: 0 to 13

<TraceSelect> CURRent | AVERAge | MAXimum

Return values:

<Reliability> decimal
[Reliability indicator](#)

<YValue> float
 Absolute y-value of the marker position

Usage: Query only

Firmware/software: V4.0.110

FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:EVMagnitude? <XValue>, <TraceSelect>

FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:EVMagnitude:PEAK? <XValue>, <TraceSelect>

FETCH:LTE:MEAS<i>:MEValuation:AMARker<No>:MERRor? <XValue>, <TraceSelect>

FETCH:LTE:MEAS<i>:MEValuation:AMARker<No>:PERRor? <XValue>, <TraceSelect>

Uses the markers 1 and 2 with absolute values on the diagrams: EVM RMS, EVM peak, magnitude error and phase error vs SC-FDMA symbol.

Suffix:

<No> 1..2
Selects the marker number.

Query parameters:

<XValue> integer
Absolute x-value of the marker position
There are two x-values per SC-FDMA symbol on the x-axis (symbol 0 low, symbol 0 high, ..., symbol 6 low, symbol 6 high).
Range: 0 to 13

<TraceSelect> CURRent | AVERage | MAXimum

Return values:

<Reliability> decimal
[Reliability indicator](#)

<YValue> float
Absolute y-value of the marker position

Usage: Query only

Firmware/software: V4.0.110

FETCH:LTE:MEAS<i>:MEValuation:DMARker<No>:EVMagnitude? <XValue>, <TraceSelect>

FETCH:LTE:MEAS<i>:MEValuation:DMARker<No>:EVMagnitude:PEAK? <XValue>, <TraceSelect>

FETCH:LTE:MEAS<i>:MEValuation:DMARker<No>:MERRor? <XValue>, <TraceSelect>

FETCH:LTE:MEAS<i>:MEValuation:DMARker<No>:PERRor? <XValue>, <TraceSelect>

Uses the markers 1 and 2 with relative values on the diagrams: EVM RMS, EVM peak, magnitude error and phase error vs SC-FDMA symbol.

Suffix:

<No> 1..2
Selects the marker number.

Query parameters:

<XValue>	integer X-value of the marker position relative to the x-value of the reference marker There are two x-values per SC-FDMA symbol on the x-axis (symbol 0 low, symbol 0 high, ..., symbol 6 low, symbol 6 high). Range: -13 to 13
<TraceSelect>	CURRent AVERage MAXimum

Return values:

<Reliability>	decimal Reliability indicator
<YValue>	float Y-value of the marker position relative to the y-value of the reference marker

Usage: Query only**Firmware/software:** V4.0.110

FETCh:LTE:MEAS<i>:MEValuation:REFMarker:PDYNamics? <XValue>, <TraceSelect>

Uses the reference marker on the power dynamics trace.

Query parameters:

<XValue>	integer Absolute x-value of the marker position Range: -1100 μ s to 2500 μ s Default unit: μ s
<TraceSelect>	CURRent AVERage MAXimum

Return values:

<Reliability>	decimal Reliability indicator
<YValue>	float Absolute y-value of the marker position Default unit: dBm

Usage: Query only**Firmware/software:** V4.0.110

FETCh:LTE:MEAS<i>:MEValuation:AMARker<No>:PDYNamics? <XValue>, <TraceSelect>

Uses the markers 1 and 2 with absolute values on the power dynamics trace.

Suffix:

<No> 1..2
Selects the marker number.

Query parameters:

<XValue> integer
Absolute x-value of the marker position
Range: -1100 μ s to 2500 μ s
Default unit: μ s

<TraceSelect> CURRent | AVERage | MAXimum

Return values:

<Reliability> decimal
[Reliability indicator](#)

<YValue> float
Absolute y-value of the marker position
Default unit: dBm

Usage: Query only

Firmware/software: V4.0.110

FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:PDYNAmics? <XValue>, <TraceSelect>

Uses the markers 1 and 2 with relative values on the power dynamics trace.

Suffix:

<No> 1..2
Selects the marker number.

Query parameters:

<XValue> integer
X-value of the marker position relative to the x-value of the reference marker
Range: -3600 μ s to 3600 μ s
Default unit: μ s

<TraceSelect> CURRent | AVERage | MAXimum

Return values:

<Reliability> decimal
[Reliability indicator](#)

<YValue> float
Y-value of the marker position relative to the y-value of the reference marker
Default unit: dB

Usage: Query only

Firmware/software: V4.0.110

FETCH:LTE:MEAS<i>:MEValuation:REFMarker:PMONitor:CC<no>? <XValue>

Uses the reference marker on the power monitor trace.

Suffix:

<no> 1..4
Selects the component carrier.

Query parameters:

<XValue> integer
Absolute x-value of the marker position (subframe number)
Range: 0 to 319

Return values:

<Reliability> decimal
[Reliability indicator](#)

<YValue> float
Absolute y-value of the marker position
Default unit: dBm

Usage: Query only

Firmware/software: V4.0.110

FETCH:LTE:MEAS<i>:MEValuation:AMARKer<No>:PMONitor:CC<no>? <XValue>

Uses the markers 1 and 2 with absolute values on the power monitor trace.

Suffix:

<No> 1..2
Selects the marker number.

<no> 1..4
Selects the component carrier.

Query parameters:

<XValue> integer
Absolute x-value of the marker position (subframe number)
Range: 0 to 319

Return values:

<Reliability> decimal
[Reliability indicator](#)

<YValue> float
Absolute y-value of the marker position
Default unit: dBm

Usage: Query only

Firmware/software: V4.0.110

FETCh:LTE:MEAS<i>:MEValuation:DMARker<No>:PMONitor:CC<no>? <XValue>

Uses the markers 1 and 2 with relative values on the power monitor trace.

Suffix:

<No>	1..2 Selects the marker number.
<no>	1..4 Selects the component carrier.

Query parameters:

<XValue>	integer X-value of the marker position relative to the x-value of the reference marker (in subframes) Range: -319 to 319
----------	--

Return values:

<Reliability>	decimal Reliability indicator
<YValue>	float Y-value of the marker position relative to the y-value of the reference marker Default unit: dB

Usage: Query only

Firmware/software: V4.0.110

5.5.3.20 Detected signal configuration

The following commands return the detected allocation, modulation scheme and channel type and the view filter throughput.

FETCh:LTE:MEAS<i>:MEValuation:MODulation:DMODulation?	1209
FETCh:LTE:MEAS<i>:MEValuation:MODulation:DCHType?	1210
FETCh:LTE:MEAS<i>:MEValuation:ACLR:DCHType?	1210
FETCh:LTE:MEAS<i>:MEValuation:SEMask:DCHType?	1210
FETCh:LTE:MEAS<i>:MEValuation:MODulation:SCHType?	1210
FETCh:LTE:MEAS<i>:MEValuation:MODulation:DALLocation?	1211
FETCh:LTE:MEAS<i>:MEValuation:ACLR:DALLocation?	1211
FETCh:LTE:MEAS<i>:MEValuation:SEMask:DALLocation?	1211
FETCh:LTE:MEAS<i>:MEValuation:VFTHroughput?	1211

FETCh:LTE:MEAS<i>:MEValuation:MODulation:DMODulation?

Returns the detected modulation scheme in the measured slot. If channel type PUCCH is detected, QPSK is returned for the modulation scheme because the QPSK limits are applied in that case.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Modulation> QPSK | Q16 | Q64 | Q256
 QPSK, 16QAM, 64QAM, 256QAM

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10, V3.7.50: 256QAM

Manual operation: See "[Modulation](#)" on page 1039

FETCh:LTE:MEAS<i>:MEValuation:MODulation:DCHType?**FETCh:LTE:MEAS<i>:MEValuation:ACLR:DCHType?****FETCh:LTE:MEAS<i>:MEValuation:SEMask:DCHType?**

Returns the uplink channel type for the measured slot.

If the same slot is measured by the individual measurements, all commands yield the same result. If different statistic counts are defined for the modulation, ACLR and spectrum emission mask measurements, different slots can be measured and different results can be returned by the individual commands.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<ChannelType> PUSCh | PUCCh

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

Manual operation: See "[Channel Type \(uplink\)](#)" on page 1039

FETCh:LTE:MEAS<i>:MEValuation:MODulation:SCHType?

Returns the sidelink channel type evaluated for modulation results.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<ChannelType> PSSCh | PSCCh

Usage: Query only

Firmware/software: V3.7.55

Manual operation: See "[Channel Type \(sidelink\)](#)" on page 1039

FETCH:LTE:MEAS<i>:MEValuation:MODulation:DALLocation?**FETCH:LTE:MEAS<i>:MEValuation:ACLR:DALLocation?****FETCH:LTE:MEAS<i>:MEValuation:SEMask:DALLocation?**

Returns the detected allocation for the measured slot.

If the same slot is measured by the individual measurements, all commands yield the same result. If different statistic counts are defined for the modulation, ACLR and spectrum emission mask measurements, different slots can be measured and different results can be returned by the individual commands.

Return values:

<Reliability>	decimal Reliability indicator
<NrResBlocks>	decimal Number of allocated resource blocks
<OffsetResBlocks>	decimal Offset of the first allocated resource block from the edge of the allocated UL transmission bandwidth

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCH:LTE:MEAS<i>:MEValuation:VFThroughput?

Queries the "View Filter Throughput".

Return values:

<Reliability>	decimal Reliability indicator
<VFThroughput>	float Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.0.20

Manual operation: See ["View Filter Throughput"](#) on page 1039

5.5.3.21 EVM results (diagrams)

The following commands return the contents of the EVM result diagrams.

FETCH:LTE:MEAS<i>:MEValuation:EVMagnitude:CURRent?	1212
FETCH:LTE:MEAS<i>:MEValuation:EVMagnitude:AVERage?	1212
FETCH:LTE:MEAS<i>:MEValuation:EVMagnitude:MAXimum?	1212
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:CURRent?	1212

READ:LTE:MEAS<i>:MEValuation:EVMagnitude:AVERage?	1212
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:MAXimum?	1212
FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:CURRent?	1213
FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:AVERage?	1213
FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:MAXimum?	1213
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:CURRent?	1213
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:AVERage?	1213
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:MAXimum?	1213
FETCh:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:CURRent?	1213
FETCh:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:AVERage?	1213
FETCh:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:MAXimum?	1213
READ:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:CURRent?	1213
READ:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:AVERage?	1213
READ:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:MAXimum?	1213
FETCh:LTE:MEAS<i>:MEValuation:TRACe:EVMC?	1214
READ:LTE:MEAS<i>:MEValuation:TRACe:EVMC?	1214

FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:MAXimum?
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:CURRent?
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:AVERage?
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:MAXimum?

Returns the values of the EVM RMS diagrams for the SC-FDMA symbols in the measured slot. The results of the current, average and maximum diagrams can be retrieved.

There is one pair of EVM values per SC-FDMA symbol, returned in the following order:

<Reliability>, {<Low>, <High>}_{symbol 0}, {<Low>, <High>}_{symbol 1}, ...

See also [Section 5.3.2.2, "View EVM"](#), on page 1017.

Return values:

<Reliability>	decimal Reliability indicator
<Low>	float EVM value for low EVM window position Default unit: %
<High>	float EVM value for high EVM window position Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V1.0.10.1

```

FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:MAXimum?
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:CURRent?
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:AVERage?
READ:LTE:MEAS<i>:MEValuation:EVMagnitude:PEAK:MAXimum?

```

Returns the values of the EVM peak diagrams for the SC-FDMA symbols in the measured slot. The results of the current, average and maximum diagrams can be retrieved.

There is one pair of EVM values per SC-FDMA symbol, returned in the following order:

<Reliability>, {<Low>, <High>}_{symbol 0}, {<Low>, <High>}_{symbol 1}, ...

See also [Section 5.3.2.2, "View EVM"](#), on page 1017.

Return values:

<Reliability>	decimal Reliability indicator
<Low>	float EVM value for low EVM window position Default unit: %
<High>	float EVM value for high EVM window position Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.5.40

```

FETCh:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:MAXimum?
READ:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:CURRent?
READ:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:AVERage?
READ:LTE:MEAS<i>:MEValuation:TRACe:EVMSymbol:MAXimum?

```

Returns the values of the EVM vs modulation symbol trace. See also [Section 5.3.2.2, "View EVM"](#), on page 1017.

Return values:

<Reliability>	decimal Reliability indicator
<Ratio>	float Comma-separated list of EVM values, one value per modulation symbol Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.5.40

FETCH:LTE:MEAS<i>:MEValuation:TRACe:EVMC?

READ:LTE:MEAS<i>:MEValuation:TRACe:EVMC?

Returns the values of the EVM vs subcarrier trace. See also [Section 5.3.2.3, "View EVM vs Subcarrier"](#), on page 1019.

Return values:

<Reliability> decimal

[Reliability indicator](#)

<Ratio> float

Comma-separated list of EVM values, one value per subcarrier
For not allocated subcarriers, NCAP is returned.

Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V1.0.10.1

5.5.3.22 Magnitude error results (diagrams)

The following commands return the contents of the magnitude error vs SC-FDMA symbol result diagrams.

FETCH:LTE:MEAS<i>:MEValuation:MERRor:CURRent?

FETCH:LTE:MEAS<i>:MEValuation:MERRor:AVERage?

FETCH:LTE:MEAS<i>:MEValuation:MERRor:MAXimum?

READ:LTE:MEAS<i>:MEValuation:MERRor:CURRent?

READ:LTE:MEAS<i>:MEValuation:MERRor:AVERage?

READ:LTE:MEAS<i>:MEValuation:MERRor:MAXimum?

Returns the values of the magnitude error diagrams for the SC-FDMA symbols in the measured slot. The results of the current, average and maximum diagrams can be retrieved.

There is one pair of magnitude error values per SC-FDMA symbol, returned in the following order:

<Reliability>, {<Low>, <High>}_{symbol 0}, {<Low>, <High>}_{symbol 1}, ...

See also [Section 5.3.2.4, "View Magnitude Error, Phase Error"](#), on page 1020.

Return values:

<Reliability> decimal

[Reliability indicator](#)

<Low>	float Magnitude error value for low EVM window position Default unit: %
<High>	float Magnitude error value for high EVM window position Default unit: %
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V1.0.10.1

5.5.3.23 Phase error results (diagrams)

The following commands return the contents of the phase error vs SC-FDMA symbol result diagrams.

```

FETCh:LTE:MEAS<i>:MEValuation:PERRor:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:PERRor:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:PERRor:MAXimum?
READ:LTE:MEAS<i>:MEValuation:PERRor:CURRent?
READ:LTE:MEAS<i>:MEValuation:PERRor:AVERage?
READ:LTE:MEAS<i>:MEValuation:PERRor:MAXimum?

```

Returns the values of the phase error diagrams for the SC-FDMA symbols in the measured slot. The results of the current, average and maximum diagrams can be retrieved.

There is one pair of phase error values per SC-FDMA symbol, returned in the following order:

<Reliability>, {<Low>, <High>}_{symbol 0}, {<Low>, <High>}_{symbol 1}, ...

See also [Section 5.3.2.4, "View Magnitude Error, Phase Error"](#), on page 1020.

Return values:

<Reliability>	decimal Reliability indicator
<Low>	float Phase error value for low EVM window position Default unit: deg
<High>	float Phase error value for high EVM window position Default unit: deg
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V1.0.10.1

5.5.3.24 In-band emission results

The following commands return the in-band emission results.

FETCh:LTE:MEAS<i>:MEValuation:TRACe:IEmissions:CC<no>?	1216
READ:LTE:MEAS<i>:MEValuation:TRACe:IEmissions:CC<no>?	1216
FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:CURRent?	1216
FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:AVERage?	1216
FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:EXTReme?	1216
FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:SDEViation?	1216
FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:CURRent:RBINdex?	1217
FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:EXTReme:RBINdex?	1217

FETCh:LTE:MEAS<i>:MEValuation:TRACe:IEmissions:CC<no>?

READ:LTE:MEAS<i>:MEValuation:TRACe:IEmissions:CC<no>?

Returns the values of the in-band emissions trace for carrier CC<no>. See also [Section 5.3.2.5, "View Inband Emissions"](#), on page 1021.

Suffix:

<no> 1..4

Return values:

<Reliability> decimal

[Reliability indicator](#)

<Power> float

Comma-separated list of power values, one value per resource block

Default unit: dB

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.7.30

FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:CURRent?

FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:EXTReme?

FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:SDEViation?

Return the limit line margin results for the CC<no> diagram. The CURRent margin indicates the minimum (vertical) distance between the in-band emissions limit line and the current trace. A negative result indicates that the limit is exceeded.

The AVERage, EXTReme and SDEViation values are calculated from the current margins. The margin results cannot be displayed at the GUI.

Suffix:

<no> 1..4

Return values:

<Reliability>	decimal Reliability indicator
<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<Margin>	float Default unit: dB

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.7.30

FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:CURRent:RBINdex?

FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:EXTReme:RBINdex?

Return resource block indices for CC<no> in-band emission margins. At these RB indices, the CURRent and EXTReme margins have been detected (see [FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:CURRent?](#) and [FETCh:LTE:MEAS<i>:MEValuation:IEmission:CC<no>:MARGin:EXTReme?](#)).

Suffix:

<no> 1..4

Return values:

<Reliability>	decimal Reliability indicator
<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<RBIndex>	decimal Resource block index

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.7.30

5.5.3.25 Equalizer spectrum flatness results

The following commands return the equalizer spectrum flatness results.

FETCh:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness?	1218
READ:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness?	1218
FETCh:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness:PHASe?	1218
READ:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness:PHASe?	1218
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:CURREnt?	1219
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:AVERAge?	1219
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:EXTReme?	1219
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:SDEViation?	1219
READ:LTE:MEAS<i>:MEValuation:ESFLatness:CURREnt?	1219
READ:LTE:MEAS<i>:MEValuation:ESFLatness:AVERAge?	1219
READ:LTE:MEAS<i>:MEValuation:ESFLatness:EXTReme?	1219
READ:LTE:MEAS<i>:MEValuation:ESFLatness:SDEViation?	1219
CALCulate:LTE:MEAS<i>:MEValuation:ESFLatness:CURREnt?	1220
CALCulate:LTE:MEAS<i>:MEValuation:ESFLatness:AVERAge?	1220
CALCulate:LTE:MEAS<i>:MEValuation:ESFLatness:EXTReme?	1220
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:CURREnt:SCINdex?	1221

FETCh:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness?

READ:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness?

Returns the values of the equalizer spectrum flatness trace. See also [Section 5.3.2.6](#), "View Equalizer Spectrum Flatness", on page 1022.

Return values:

<Reliability>	decimal Reliability indicator
<Power>	float Comma-separated list of power values, one value per subcarrier For not allocated subcarriers, NCAP is returned. Default unit: dB

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCh:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness:PHASe?

READ:LTE:MEAS<i>:MEValuation:TRACe:ESFLatness:PHASe?

Returns the phase values of the equalizer spectrum flatness trace. The GUI shows only the magnitude values.

Return values:

<Reliability>	decimal Reliability indicator
---------------	--

<Phase> float
 Comma-separated list of phase values, one value per subcarrier
 For not allocated subcarriers, NCAP is returned.
 Default unit: deg

Usage: Query only

Firmware/software: V3.7.50

FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:ESFLatness:SDEViation?
READ:LTE:MEAS<i>:MEValuation:ESFLatness:CURRent?
READ:LTE:MEAS<i>:MEValuation:ESFLatness:AVERage?
READ:LTE:MEAS<i>:MEValuation:ESFLatness:EXTReme?
READ:LTE:MEAS<i>:MEValuation:ESFLatness:SDEViation?

Return current, average, extreme and standard deviation single-value results of the equalizer spectrum flatness measurement. See also [Section 5.3.7.6, "Equalizer spectrum flatness limits"](#), on page 1077.

Return values:

<1_Reliability> decimal
[Reliability indicator](#)

<2_OutOfTol> decimal
 Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits.
 Default unit: %

<3_Ripple1> float
 Max (range 1) - min (range 1)
 Default unit: dB

<4_Ripple2> float
 Max (range 2) - min (range 2)
 Default unit: dB

<5_MaxR1MinR2> float
 Max (range 1) - min (range 2)
 Default unit: dB

<6_MaxR2MinR1> float
 Max (range 2) - min (range 1)
 Default unit: dB

<7_MinR1>	float Min (range 1) Default unit: dB
<8_MaxR1>	float Max (range 1) Default unit: dB
<9_MinR2>	float Min (range 2) Default unit: dB
<10_MaxR2>	float Max (range 2) Default unit: dB

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10, V2.1.25 added <7_MinR1> to <10_MaxR2>.

CALCulate:LTE:MEAS<i>:MEValuation:ESFlatness:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:ESFlatness:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:ESFlatness:EXTReme?

Return current, average and extreme single-value results of the equalizer spectrum flatness measurement. See also [Section 5.3.7.6, "Equalizer spectrum flatness limits"](#), on page 1077.

Return values:

<Reliability>	decimal Reliability indicator
<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<Ripple1>	Limit check result for max (range 1) - min (range 1).
<Ripple2>	Limit check result for max (range 2) - min (range 2).
<MaxR1MinR2>	Limit check result for max (range 1) - min (range 2).
<MaxR2MinR1>	Limit check result for max (range 2) - min (range 1).

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.20

FETCh:LTE:MEAS<i>:MEValuation:ESFlatness:CURRent:SCINdex?

Returns subcarrier indices of the equalizer spectrum flatness measurement. At these SC indices, the current minimum and maximum power of the equalizer coefficients have been detected within range 1 and range 2.

Return values:

<Reliability>	decimal Reliability indicator
<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<Maximum1>	decimal SC index of "Max (Range 1)"
<Minimum1>	decimal SC index of "Min (Range 1)"
<Maximum2>	decimal SC index of "Max (Range 2)"
<Minimum2>	decimal SC index of "Min (Range 2)"

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

5.5.3.26 Spectrum emission results

The following commands return the spectrum emission results.

SENSe:LTE:MEAS<i>:MEValuation:SPECTrum:SEMask:RBW:USED?	1222
FETCh:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:CURRent?	1222
FETCh:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:AVERage?	1222
FETCh:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:MAXimum?	1222
READ:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:CURRent?	1222
READ:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:AVERage?	1222
READ:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:MAXimum?	1222
CALCulate:LTE:MEAS<i>:MEValuation:SEMask:CURRent?	1223
CALCulate:LTE:MEAS<i>:MEValuation:SEMask:AVERage?	1223
FETCh:LTE:MEAS<i>:MEValuation:SEMask:CURRent?	1223
FETCh:LTE:MEAS<i>:MEValuation:SEMask:AVERage?	1223
FETCh:LTE:MEAS<i>:MEValuation:SEMask:SDEViation?	1223
READ:LTE:MEAS<i>:MEValuation:SEMask:CURRent?	1223
READ:LTE:MEAS<i>:MEValuation:SEMask:AVERage?	1223
READ:LTE:MEAS<i>:MEValuation:SEMask:SDEViation?	1223

CALCulate:LTE:MEAS<i>:MEValuation:SEMask:EXTReme?	1224
FETCh:LTE:MEAS<i>:MEValuation:SEMask:EXTReme?	1224
READ:LTE:MEAS<i>:MEValuation:SEMask:EXTReme?	1224
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:ALL?	1225
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:CURREnt:NEGativ?	1226
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:CURREnt:POSitiv?	1226
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:AVERage:NEGativ?	1226
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:AVERage:POSitiv?	1226
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:MINimum:NEGativ?	1226
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:MINimum:POSitiv?	1226

SENSe:LTE:MEAS<i>:MEValuation:SPECtrum:SEMask:RBW:USED?

Queries the resolution bandwidths (RBW) allowed for spectrum emission measurements. The RBWs depend on the channel bandwidth and on the "network signaled value".

Return values:

<Trace1>	decimal RBW for trace 1 (smallest RBW) Default unit: kHz
<Trace2>	decimal RBW for trace 2 (intermediate RBW) Default unit: kHz
<Trace3>	decimal RBW for trace 3 (largest RBW) Default unit: kHz

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.7.70

```

FETCh:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:CURREnt?
FETCh:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:MAXimum?
READ:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:CURREnt?
READ:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:AVERage?
READ:LTE:MEAS<i>:MEValuation:TRACe:SEMask:RBW<kHz>:MAXimum?

```

Returns the values of the spectrum emission traces. Separate traces are available for the individual resolution bandwidths (<kHz>). The results of the current, average and maximum traces can be retrieved.

See also [Section 5.3.2.7, "View Spectrum Emission Mask"](#), on page 1024.

Suffix:

<kHz>	30, 50, 100, 150, 200, 1000 Resolution bandwidth in kHz
-------	--

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Power> float
 Comma-separated list of power results
 The value in the middle of the result array corresponds to the center frequency.
 The test point separation between adjacent results depends on the resolution bandwidth, see table below.
 For RBW100 and greater, results are only available for frequencies with active limits using these RBWs. For other frequencies, INV is returned.
 Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V1.0.10.1, V3.7.10 added RBW 50, 150, 200.

RBW	Test point separation
30, 50	15 kHz
100, 150, 200	45 kHz
1000	90 kHz

CALCulate:LTE:MEAS<i>:MEValuation:SEMask:CURRent?

CALCulate:LTE:MEAS<i>:MEValuation:SEMask:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:SEMask:CURRent?

FETCh:LTE:MEAS<i>:MEValuation:SEMask:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:SEMask:SDEViation?

READ:LTE:MEAS<i>:MEValuation:SEMask:CURRent?

READ:LTE:MEAS<i>:MEValuation:SEMask:AVERage?

READ:LTE:MEAS<i>:MEValuation:SEMask:SDEViation?

Return the current, average and standard deviation single-value results of the spectrum emission measurement.

The values described below are returned by FETCh and READ commands. A CALCulate command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified spectrum emission mask limits. Default unit: %
<OBW>	float Occupied bandwidth Default unit: Hz
<TXpower>	float Total TX power in the slot over all component carriers Default unit: dBm
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V1.0.10.1

CALCulate:LTE:MEAS<i>:MEvaluation:SEMask:EXTreme?**FETCh:LTE:MEAS<i>:MEvaluation:SEMask:EXTreme?****READ:LTE:MEAS<i>:MEvaluation:SEMask:EXTreme?**

Return the extreme single-value results of the spectrum emission measurement.

The values described below are returned by **FETCh** and **READ** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified spectrum emission mask limits. Default unit: %
<OBW>	float Occupied bandwidth Default unit: Hz
<TXpowerMin>	float Minimum total TX power in the slot Default unit: dBm
<TXpowerMax>	float Maximum total TX power in the slot Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:ALL?

Returns spectrum emission mask margin results. A negative margin indicates that the trace is located above the limit line, i.e. the limit is exceeded.

Results are provided for the current, average and maximum traces. For each trace, 24 values related to the negative (Neg) and positive (Pos) offset frequencies of emission mask areas 1 to 12 are provided. For inactive areas, NCAP is returned.

Return values:

<1_Reliability>	decimal Reliability indicator
<2_OutOfTol>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified spectrum emission mask limits. Default unit: %
<3_CurrNeg1> ... <14_CurrNeg12>	float Margin results for current trace, negative offsets Default unit: dB
<15_CurrPos1> ... <26_CurrPos12>	float Margin results for current trace, positive offsets Default unit: dB
<27_AvgNeg1> ... <38_AvgNeg12>	float Margin results for average trace, negative offsets Default unit: dB
<39_AvgPos1> ... <50_AvgPos12>	float Margin results for average trace, positive offsets Default unit: dB
<51_MinNeg1> ... <62_MinNeg12>	float Margin results for maximum trace (resulting in minimum margins), negative offsets Default unit: dB
<63_MinPos1> ... <74_MinPos12>	float Margin results for maximum trace (resulting in minimum margins), positive offsets Default unit: dB

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.2.80

FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:CURRent:NEGativ?
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:CURRent:POSitiv?
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:AVERage:NEGativ?
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:AVERage:POSitiv?
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:MINimum:NEGativ?
FETCh:LTE:MEAS<i>:MEValuation:SEMask:MARGin:MINimum:POSitiv?

Returns spectrum emission mask margin results. A negative margin indicates that the trace is located above the limit line, i.e. the limit is exceeded.

The individual commands provide results for the `CURRent`, `AVERage` and maximum traces (resulting in `MINimum` margins).

For each trace, the x- and y-values of the margins for emission mask areas 1 to 12 are provided for `NEGative` and `POSitive` offset frequencies. For inactive areas, `NCAP` is returned.

Returned sequence: `<Reliability>`, `<OutOfTolerance>`, `{<MarginX>, <MarginY>}`_{area1}, `{...}`_{area2}, ..., `{...}`_{area12}

Return values:

<code><Reliability></code>	decimal Reliability indicator
<code><OutOfTolerance></code>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for spectrum emission measurements exceeding the specified spectrum emission mask limits. Default unit: %
<code><MarginX></code>	float X-position of the margin of an area Default unit: Hz
<code><MarginY></code>	float Y-value of the margin of an area Default unit: dB

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10, V3.2.80 number of areas increased to 12

5.5.3.27 ACLR spectrum results

The following commands return the ACLR spectrum results.

FETCh:LTE:MEAS<i>:MEValuation:TRACe:ACLR:CURRent?	1227
FETCh:LTE:MEAS<i>:MEValuation:TRACe:ACLR:AVERage?	1227
READ:LTE:MEAS<i>:MEValuation:TRACe:ACLR:CURRent?	1227
READ:LTE:MEAS<i>:MEValuation:TRACe:ACLR:AVERage?	1227
CALCulate:LTE:MEAS<i>:MEValuation:ACLR:CURRent?	1228
CALCulate:LTE:MEAS<i>:MEValuation:ACLR:AVERage?	1228
FETCh:LTE:MEAS<i>:MEValuation:ACLR:CURRent?	1228
FETCh:LTE:MEAS<i>:MEValuation:ACLR:AVERage?	1228
READ:LTE:MEAS<i>:MEValuation:ACLR:CURRent?	1228
READ:LTE:MEAS<i>:MEValuation:ACLR:AVERage?	1228

FETCh:LTE:MEAS<i>:MEValuation:TRACe:ACLR:CURRent?

FETCh:LTE:MEAS<i>:MEValuation:TRACe:ACLR:AVERage?

READ:LTE:MEAS<i>:MEValuation:TRACe:ACLR:CURRent?

READ:LTE:MEAS<i>:MEValuation:TRACe:ACLR:AVERage?

Returns the absolute powers as displayed in the ACLR diagram. The current and average values can be retrieved. See also [Section 5.3.2.8, "View Spectrum ACLR"](#), on page 1026.

Return values:

<Reliability>	decimal Reliability indicator
<UTRA2neg>	float Power in the second UTRA channel with lower frequency Default unit: dBm
<UTRA1neg>	float Power in the first UTRA channel with lower frequency Default unit: dBm
<EUTRAneg>	float Power in the first E-UTRA channel with lower frequency Default unit: dBm
<EUTRA>	float Power in the allocated E-UTRA channel Default unit: dBm
<EUTRApos>	float Power in the first E-UTRA channel with higher frequency Default unit: dBm
<UTRA1pos>	float Power in the first UTRA channel with higher frequency Default unit: dBm

<UTRA2pos> float
Power in the second UTRA channel with higher frequency
Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V1.0.10.1

CALCulate:LTE:MEAS<i>:MEValuation:ACLR:CURRent?

CALCulate:LTE:MEAS<i>:MEValuation:ACLR:AVERAge?

FETCh:LTE:MEAS<i>:MEValuation:ACLR:CURRent?

FETCh:LTE:MEAS<i>:MEValuation:ACLR:AVERAge?

READ:LTE:MEAS<i>:MEValuation:ACLR:CURRent?

READ:LTE:MEAS<i>:MEValuation:ACLR:AVERAge?

Returns the relative ACLR values as displayed in the table below the ACLR diagram.
The current and average values can be retrieved.

See also [Section 5.3.2.8, "View Spectrum ACLR"](#), on page 1026.

The values described below are returned by **FETCh** and **READ** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<UTRA2neg> float
ACLR for the second UTRA channel with lower frequency
Default unit: dB

<UTRA1neg> float
ACLR for the first UTRA channel with lower frequency
Default unit: dB

<EUTRAneg> float
ACLR for the first E-UTRA channel with lower frequency
Default unit: dB

<EUTRA> float
Power in the allocated E-UTRA channel
Default unit: dBm

<EUTRApos> float
ACLR for the first E-UTRA channel with higher frequency
Default unit: dB

<UTRA1pos>	float ACLR for the first UTRA channel with higher frequency Default unit: dB
<UTRA2pos>	float ACLR for the second UTRA channel with higher frequency Default unit: dB
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V1.0.10.1

5.5.3.28 I/Q constellation results

The following commands return the contents of the I/Q constellation diagram.

FETCh:LTE:MEAS<i>:MEValuation:TRACe:IQ:LOW?
FETCh:LTE:MEAS<i>:MEValuation:TRACe:IQ:HIGh?

Returns the results in the I/Q constellation diagram for low and high EVM window position.

There is one pair of values per modulation symbol. The results are returned in the following order:

<Reliability>, {<IPhase>, <QPhase>}_{symbol 1}, ..., {<IPhase>, <QPhase>}_{symbol n}

See also [Section 5.3.2.9, "View I/Q Constellation"](#), on page 1027

Return values:

<Reliability>	decimal Reliability indicator
<IPhase>	float Normalized I amplitude.
<QPhase>	float Normalized Q amplitude.
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V1.0.10.1

5.5.3.29 RB allocation table results

The following commands return the contents of the RB allocation table.

FETCH:LTE:MEAS<i>:MEValuation:TRACe:RBATable:CC<no>?

READ:LTE:MEAS<i>:MEValuation:TRACe:RBATable:CC<no>?

Returns the information of the CC<no> RB allocation table. See also [Section 5.3.2.10, "View RB Allocation Table"](#), on page 1029.

For uplink measurements, there are three results per captured slot (n = number of captured subframes):

<Reliability>, {<ChannelType>, <OffsetRB>, <NoRB>}_{slot 1}, ..., {...}_{slot (n*2)}

For sidelink measurements, there are six results per captured subframe (SF), three for the PSCCH and three for the PSSCH:

<Reliability>, {...}_{SF 1 (PSCCH)}, {...}_{SF 1 (PSSCH)}, ..., {...}_{SF n (PSCCH)}, {...}_{SF n (PSSCH)}

Suffix:

<no> 1..4

Return values:

<Reliability> decimal

[Reliability indicator](#)

<ChannelType> PUSCh | PUCCh | NONE | DL | SSUB | PSSCh | PSCCh

PUSCh / PUCCh: for UL slot with RB allocation

PSSCh / PSCCh: for SL subframe with RB allocation

NONE: UL slot or SL subframe contains no allocated RBs.

DL: DL slot (only for TDD UL measurements)

SSUB: part of special SF (only for TDD UL measurements)

<OffsetRB> decimal

Offset of first allocated RB for the given channel type

<NoRB> decimal

Number of allocated RBs for the given channel type

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.7.30

5.5.3.30 Power monitor results

The following commands return the power monitor results.

FETCH:LTE:MEAS<i>:MEValuation:TRACe:PMONitor:CC<no>?	1231
READ:LTE:MEAS<i>:MEValuation:TRACe:PMONitor:CC<no>?	1231
FETCH:LTE:MEAS<i>:MEValuation:PMONitor:CURREnt?	1231
FETCH:LTE:MEAS<i>:MEValuation:PMONitor:AVERAge?	1231
FETCH:LTE:MEAS<i>:MEValuation:PMONitor:MINimum?	1231
FETCH:LTE:MEAS<i>:MEValuation:PMONitor:MAXimum?	1231
FETCH:LTE:MEAS<i>:MEValuation:PMONitor:SDEViation?	1231
READ:LTE:MEAS<i>:MEValuation:PMONitor:CURREnt?	1231
READ:LTE:MEAS<i>:MEValuation:PMONitor:AVERAge?	1231

READ:LTE:MEAS<i>:MEValuation:PMONitor:MINimum?	1231
READ:LTE:MEAS<i>:MEValuation:PMONitor:MAXimum?	1231
READ:LTE:MEAS<i>:MEValuation:PMONitor:SDEviation?	1231
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:CURRent?	1232
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:AVERage?	1232
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MINimum?	1232
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MAXimum?	1232
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:SDEviation?	1232
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:CURRent?	1232
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:AVERage?	1232
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MINimum?	1232
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MAXimum?	1232
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:SDEviation?	1232

FETCh:LTE:MEAS<i>:MEValuation:TRACe:PMONitor:CC<no>?

READ:LTE:MEAS<i>:MEValuation:TRACe:PMONitor:CC<no>?

Returns the power monitor results for all captured CC<no> subframes.

Suffix:

<no> 1..4

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Power> float
 Comma-separated list of power values, one value per subframe
 Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.7.30

FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CURRent?
 FETCh:LTE:MEAS<i>:MEValuation:PMONitor:AVERage?
 FETCh:LTE:MEAS<i>:MEValuation:PMONitor:MINimum?
 FETCh:LTE:MEAS<i>:MEValuation:PMONitor:MAXimum?
 FETCh:LTE:MEAS<i>:MEValuation:PMONitor:SDEviation?
 READ:LTE:MEAS<i>:MEValuation:PMONitor:CURRent?
 READ:LTE:MEAS<i>:MEValuation:PMONitor:AVERage?
 READ:LTE:MEAS<i>:MEValuation:PMONitor:MINimum?
 READ:LTE:MEAS<i>:MEValuation:PMONitor:MAXimum?
 READ:LTE:MEAS<i>:MEValuation:PMONitor:SDEviation?

Returns the total TX power of all carriers.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for power measurements exceeding the specified power limits. Default unit: %
<TXpower>	float Default unit: dBm
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V3.5.40

```

FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:AVERAge?
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:SDEViation?
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:CURRent?
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:AVERAge?
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MINimum?
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:MAXimum?
READ:LTE:MEAS<i>:MEValuation:PMONitor:CC<no>:SDEViation?

```

Returns the TX power of carrier CC<no>.

Suffix:

<no> 1..4

Return values:

<Reliability> decimal
[Reliability indicator](#)

<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for power measurements exceeding the specified power limits. Default unit: %
<TXpower>	float Default unit: dBm
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V3.7.30

5.5.3.31 Power dynamics results

The following commands return the power dynamics measurement results.

FETCH:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:CURRent?	1233
FETCH:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:AVERAge?	1233
FETCH:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:MAXimum?	1233
READ:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:CURRent?	1233
READ:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:AVERAge?	1233
READ:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:MAXimum?	1233
CALCulate:LTE:MEAS< i >:MEValuation:PDYNamics:CURRent?	1234
CALCulate:LTE:MEAS< i >:MEValuation:PDYNamics:AVERAge?	1234
CALCulate:LTE:MEAS< i >:MEValuation:PDYNamics:MINimum?	1234
CALCulate:LTE:MEAS< i >:MEValuation:PDYNamics:MAXimum?	1234
FETCH:LTE:MEAS< i >:MEValuation:PDYNamics:CURRent?	1234
FETCH:LTE:MEAS< i >:MEValuation:PDYNamics:AVERAge?	1234
FETCH:LTE:MEAS< i >:MEValuation:PDYNamics:MINimum?	1234
FETCH:LTE:MEAS< i >:MEValuation:PDYNamics:MAXimum?	1234
FETCH:LTE:MEAS< i >:MEValuation:PDYNamics:SDEViation?	1234
READ:LTE:MEAS< i >:MEValuation:PDYNamics:CURRent?	1234
READ:LTE:MEAS< i >:MEValuation:PDYNamics:AVERAge?	1234
READ:LTE:MEAS< i >:MEValuation:PDYNamics:MINimum?	1234
READ:LTE:MEAS< i >:MEValuation:PDYNamics:MAXimum?	1234
READ:LTE:MEAS< i >:MEValuation:PDYNamics:SDEViation?	1234

FETCH:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:CURRent?
FETCH:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:AVERAge?
FETCH:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:MAXimum?
READ:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:CURRent?
READ:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:AVERAge?
READ:LTE:MEAS< i >:MEValuation:TRACe:PDYNamics:MAXimum?

Return the values of the power dynamics traces. Each value is sampled with $48 T_s$, corresponding to $1.5625 \mu s$. The results of the current, average and maximum traces can be retrieved.

See also [Section 5.3.2.12, "View Power Dynamics"](#), on page 1032.

Return values:

<Reliability>	decimal Reliability indicator
<Power>	float 2048 power values, from $-1100 \mu s$ to $+2098.4375 \mu s$ relative to the start of the measure subframe. The values have a spacing of $1.5625 \mu s$. The 705th value is at the start of the "Measure Sub-frame" ($0 \mu s$). The diagram in the GUI shows only a subsection of this trace. Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

CALCulate:LTE:MEAS<i>:MEValuation:PDYNamics:CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:PDYNamics:AVERAge?
CALCulate:LTE:MEAS<i>:MEValuation:PDYNamics:MINimum?
CALCulate:LTE:MEAS<i>:MEValuation:PDYNamics:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:PDYNamics:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:PDYNamics:AVERAge?
FETCh:LTE:MEAS<i>:MEValuation:PDYNamics:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:PDYNamics:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:PDYNamics:SDEViation?
READ:LTE:MEAS<i>:MEValuation:PDYNamics:CURRent?
READ:LTE:MEAS<i>:MEValuation:PDYNamics:AVERAge?
READ:LTE:MEAS<i>:MEValuation:PDYNamics:MINimum?
READ:LTE:MEAS<i>:MEValuation:PDYNamics:MAXimum?
READ:LTE:MEAS<i>:MEValuation:PDYNamics:SDEViation?

Return the current, average, minimum, maximum and standard deviation single-value results of the power dynamics measurement.

A single result table row is returned, from left to right. The meaning of the values depends on the selected time mask, as follows:

Time mask	<Power1>	<Power2>	<Power3>	<Power4>
General on / off	OFF power (before)	ON power RMS	ON power peak	OFF power (after)
PUCCH / PUSCH / SRS	SRS ON	ON power RMS	ON power peak	ON power (after)
SRS blanking	SRS OFF	ON power RMS	ON power peak	ON power (after)

The values described below are returned by **FETCh** and **READ** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal

[Reliability indicator](#)

<OutOfTolerance> decimal

Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for power measurements exceeding the specified power limits.

Default unit: %

<Power1> float

Default unit: dBm

<Power2> float

Default unit: dBm

<Power3>	float
	Default unit: dBm
<Power4>	float
	Default unit: dBm
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V2.0.10

5.5.3.32 Modulation results (single values)

The following commands return the statistical modulation results.

CALCulate:LTE:MEAS< i >:MEValuation:MODulation:CURRent?	1235
CALCulate:LTE:MEAS< i >:MEValuation:MODulation:AVERage?	1235
FETCh:LTE:MEAS< i >:MEValuation:MODulation:CURRent?	1235
FETCh:LTE:MEAS< i >:MEValuation:MODulation:AVERage?	1235
READ:LTE:MEAS< i >:MEValuation:MODulation:CURRent?	1235
READ:LTE:MEAS< i >:MEValuation:MODulation:AVERage?	1235
READ:LTE:MEAS< i >:MEValuation:MODulation:SDEViation?	1235
CALCulate:LTE:MEAS< i >:MEValuation:MODulation:EXTReme?	1238
FETCh:LTE:MEAS< i >:MEValuation:MODulation:EXTReme?	1238
READ:LTE:MEAS< i >:MEValuation:MODulation:EXTReme?	1238
FETCh:LTE:MEAS< i >:MEValuation:EVMC:PEAK:CURRent?	1241
FETCh:LTE:MEAS< i >:MEValuation:EVMC:PEAK:AVERage?	1241
FETCh:LTE:MEAS< i >:MEValuation:EVMC:PEAK:MAXimum?	1241
FETCh:LTE:MEAS< i >:MEValuation:EVMC:PEAK:SDEViation?	1241
READ:LTE:MEAS< i >:MEValuation:EVMC:PEAK:CURRent?	1241
READ:LTE:MEAS< i >:MEValuation:EVMC:PEAK:AVERage?	1241
READ:LTE:MEAS< i >:MEValuation:EVMC:PEAK:MAXimum?	1241
READ:LTE:MEAS< i >:MEValuation:EVMC:PEAK:SDEViation?	1241

CALCulate:LTE:MEAS< i >:MEValuation:MODulation:CURRent?
CALCulate:LTE:MEAS< i >:MEValuation:MODulation:AVERage?
FETCh:LTE:MEAS< i >:MEValuation:MODulation:CURRent?
FETCh:LTE:MEAS< i >:MEValuation:MODulation:AVERage?
READ:LTE:MEAS< i >:MEValuation:MODulation:CURRent?
READ:LTE:MEAS< i >:MEValuation:MODulation:AVERage?
READ:LTE:MEAS< i >:MEValuation:MODulation:SDEViation?

Return the current, average and standard deviation single-value results.

The values described below are returned by **FETCh** and **READ** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<1_Reliability>	decimal
	Reliability indicator

<2_OutOfTol>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<3_EVM_RMSlow>	float EVM RMS value, low EVM window position Default unit: %
<4_EVM_RMShigh>	float EVM RMS value, high EVM window position Default unit: %
<5_EVMpeakLow>	float EVM peak value, low EVM window position Default unit: %
<6_EVMpeakHigh>	float EVM peak value, high EVM window position Default unit: %
<7_MErr_RMSlow>	float Magnitude error RMS value, low EVM window position Default unit: %
<8_MErr_RMShigh>	float Magnitude error RMS value, high EVM window position Default unit: %
<9_MErrPeakLow>	float Magnitude error peak value, low EVM window position Default unit: %
<10_MErrPeakHigh>	float Magnitude error peak value, high EVM window position Default unit: %
<11_PErr_RMSlow>	float Phase error RMS value, low EVM window position Default unit: deg
<12_PErr_RMSh>	float Phase error RMS value, high EVM window position Default unit: deg
<13_PErrPeakLow>	float Phase error peak value, low EVM window position Default unit: deg

<14_PErrPeakHigh>	float	Phase error peak value, high EVM window position Default unit: deg
<15_IQoffset>	float	I/Q origin offset Default unit: dBc
<16_FreqError>	float	Carrier frequency error Default unit: Hz
<17_TimingError>	float	Time error Default unit: Ts (basic LTE time unit)
<18_TXpower>	float	User equipment power Default unit: dBm
<19_PeakPower>	float	User equipment peak power Default unit: dBm
<20_RBpower>	float	RB power Default unit: dBm
<21_EVM_DMRSI>	float	EVM DMRS value, low EVM window position Default unit: %
<22_EVM_DMRSsh>	float	EVM DMRS value, high EVM window position Default unit: %
<23_MErr_DMRSI>	float	Magnitude error DMRS value, low EVM window position Default unit: %
<24_MErr_DMRSsh>	float	Magnitude error DMRS value, high EVM window position Default unit: %
<25_PErr_DMRS>	float	Phase error DMRS value, low EVM window position Default unit: deg

<26_PErr_DMRS>	float Phase error DMRS value, high EVM window position Default unit: deg
<27_GainImbal>	float Gain imbalance Default unit: dB
<28_QuadError>	float Quadrature error Default unit: deg
<29_EVM_SRS>	float EVM value, first SRS symbol Default unit: %
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V1.0.10.1 V3.2.70: added <27_GainImbal> and <28_QuadError> V3.5.40: added <29_EVM_SRS>

CALCulate:LTE:MEAS<i>:MEvaluation:MODulation:EXTreme?**FETCh:LTE:MEAS<i>:MEvaluation:MODulation:EXTreme?****READ:LTE:MEAS<i>:MEvaluation:MODulation:EXTreme?**

Return the extreme single value results.

The values described below are returned by **FETCh** and **READ** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<1_Reliability>	decimal Reliability indicator
<2_OutOfTol>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<3_EVM_RMSlow>	float EVM RMS value, low EVM window position Default unit: %
<4_EVM_RMShigh>	float EVM RMS value, high EVM window position Default unit: %

<5_EVMpeakLow>	float EVM peak value, low EVM window position Default unit: %
<6_EVMpeakHigh>	float EVM peak value, high EVM window position Default unit: %
<7_MErr_RMSlow>	float Magnitude error RMS value, low EVM window position Default unit: %
<8_MErr_RMShigh>	float Magnitude error RMS value, high EVM window position Default unit: %
<9_MErrPeakLow>	float Magnitude error peak value, low EVM window position Default unit: %
<10_MErrPeakHigh>	float Magnitude error peak value, high EVM window position Default unit: %
<11_PErr_RMSlow>	float Phase error RMS value, low EVM window position Default unit: deg
<12_PErr_RMSh>	float Phase error RMS value, high EVM window position Default unit: deg
<13_PErrPeakLow>	float Phase error peak value, low EVM window position Default unit: deg
<14_PErrPeakHigh>	float Phase error peak value, high EVM window position Default unit: deg
<15_IQoffset>	float I/Q origin offset Default unit: dBc
<16_FreqError>	float Carrier frequency error Default unit: Hz

<17_TimingError>	float Time error Default unit: Ts (basic LTE time unit)
<18_TXpowerMin>	float Minimum user equipment power Default unit: dBm
<19_TXpowerMax>	float Maximum user equipment power Default unit: dBm
<20_PeakPowerMin>	float Minimum user equipment peak power Default unit: dBm
<21_PeakPowerMax>	float Maximum user equipment peak power Default unit: dBm
<22_RBpowerMin>	float Minimum RB power Default unit: dBm
<23_RBpowerMax>	float Maximum RB power Default unit: dBm
<24_EVM_DMRSI>	float EVM DMRS value, low EVM window position Default unit: %
<25_EVM_DMRSIh>	float EVM DMRS value, high EVM window position Default unit: %
<26_MErr_DMRSI>	float Magnitude error DMRS value, low EVM window position Default unit: %
<27_MErr_DMRSIh>	float Magnitude error DMRS value, high EVM window position Default unit: %
<28_PErr_DMRSI>	float Phase error DMRS value, low EVM window position Default unit: deg

<29_PErr_DMRS>	float
	Phase error DMRS value, high EVM window position
	Default unit: deg
<30_GainImbal>	float
	Gain imbalance
	Default unit: dB
<31_QuadError>	float
	Quadrature error
	Default unit: deg
<32_EVM_SRS1>	float
	EVM value, first SRS symbol
	Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V1.0.10.1
V3.2.70: added <30_GainImbal> and <31_QuadError>
V3.5.40: added <32_EVM_SRS>

FETCH:LTE:MEAS<i>:MEValuation:EVMC:PEAK:CURRENT?
FETCH:LTE:MEAS<i>:MEValuation:EVMC:PEAK:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:EVMC:PEAK:MAXimum?
FETCH:LTE:MEAS<i>:MEValuation:EVMC:PEAK:SDEViation?
READ:LTE:MEAS<i>:MEValuation:EVMC:PEAK:CURRENT?
READ:LTE:MEAS<i>:MEValuation:EVMC:PEAK:AVERage?
READ:LTE:MEAS<i>:MEValuation:EVMC:PEAK:MAXimum?
READ:LTE:MEAS<i>:MEValuation:EVMC:PEAK:SDEViation?

The **CURRENT** command returns the maximum value of the EVM vs subcarrier trace.

The **AVERage**, **MAXimum** and **SDEViation** values are calculated from the **CURRENT** values.

The peak results cannot be displayed at the GUI.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<EVMCpeakStdDev> float
Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V3.2.80

5.5.3.33 BLER results

The following commands return the results of the block error ratio measurement.

FETCH:LTE:MEAS<i>:MEvaluation:BLER?

READ:LTE:MEAS<i>:MEvaluation:BLER?

Returns the block error ratio results determined from all captured subframes.

Return values:

<Reliability>	decimal Reliability indicator
<ACK>	float Received acknowledgments (percentage of sent scheduled subframes). Default unit: %
<NACK>	float Received negative acknowledgments (percentage of sent scheduled subframes). Default unit: %
<BLER>	float Block error ratio (percentage of sent scheduled subframes for which no ACK has been received). Default unit: %
<DTX>	float Percentage of sent scheduled subframes for which no ACK and no NACK has been received. Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V1.0.15.21, V3.0.10 added <DTX>.

5.5.3.34 List mode results (one segment)

The following commands return the list mode results for a selected segment.

To configure the list mode, use the commands described in [Section 5.5.3.9, "List mode settings"](#), on page 1154.

For a description of the list mode, see [Section 5.2.10, "List mode"](#), on page 1002.

The segment number <no> in the following commands refers to the range of measured segments (1..1000), see [CONFIGure:LTE:MEAS<i>:MEvaluation:LIST:LRANge](#) on page 1155. It can differ from the absolute segment number used for segment configuration.

The commands return two reliability indicators. The first indicator covers all measured segments. The second indicator considers only the selected segment.

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:CURRent?	1244
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:AVERage?	1244
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:CURRent?	1244
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:AVERage?	1244
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:SDEViation?	1244
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:EXTReme?	1247
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:EXTReme?	1247
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:IEMission:CC<c>:MARGin: CURRent?	1250
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:IEMission:CC<c>:MARGin: AVERage?	1250
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:IEMission:CC<c>:MARGin: EXTReme?	1250
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:IEMission:CC<c>:MARGin: SDEViation?	1250
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:IEMission:CC<c>:MARGin: CURRent:RBINdex?	1251
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:IEMission:CC<c>:MARGin: EXTReme:RBINdex?	1251
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:CURRent?	1251
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:AVERage?	1251
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:EXTReme?	1251
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:SDEViation?	1251
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:CURRent?	1253
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:AVERage?	1253
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:EXTReme?	1253
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ESFLatness:CURRent:SCINdex?	1253
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:CURRent?	1254
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:AVERage?	1254
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:CURRent?	1254
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:AVERage?	1254
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:SDEViation?	1254
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:EXTReme?	1255
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:EXTReme?	1255
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:MARGin:ALL?	1256
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:MARGin:CURRent: NEGativ?	1257
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:MARGin:CURRent: POSitiv?	1257
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:MARGin:AVERage: NEGativ?	1257
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:MARGin:AVERage: POSitiv?	1257
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:MARGin:MINimum: NEGativ?	1257
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:MARGin:MINimum: POSitiv?	1257

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ACLR:CURRent?	1258
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ACLR:AVERage?	1258
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ACLR:CURRent?	1258
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ACLR:AVERage?	1258
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:PMONitor:RMS?	1259
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:PMONitor:PEAK?	1259
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:CURRent?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:AVERage?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:MINimum?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:MAXimum?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:SDEViation?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:CC<no>:CURRent?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:CC<no>:AVERage?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:CC<no>:MINimum?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:CC<no>:MAXimum?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:POWer:CC<no>:SDEViation?	1260
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:DALLocation?	1261
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ACLR:DALLocation?	1261
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:DALLocation?	1261
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:DMODulation?	1262
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:SEMask:DCHType?	1262
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:ACLR:DCHType?	1262
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:DCHType?	1262
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:SCHType?	1263

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:CURRent?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:CURRent?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMent<no>:MODulation:SDEViation?

Returns modulation single-value results for segment <no> in list mode.

The values described below are returned by FETCh commands. The first four values (reliability to out of tolerance result) are also returned by CALCulate commands. The remaining values returned by CALCulate commands are limit check results, one value for each result listed below.

Suffix:

<no> 1..1000

Return values:

<1_Reliability>	decimal Reliability indicator
<2_SegReliability>	decimal Reliability indicator for the segment
<3_StatistExpired>	decimal Reached statistical length in slots

<4_OutOfTol>	decimal Percentage of measured subframes with failed limit check Default unit: %
<5_EVM_RMSlow>	float EVM RMS value, low EVM window position Default unit: %
<6_EVM_RMShigh>	float EVM RMS value, high EVM window position Default unit: %
<7_EVMpeakLow>	float EVM peak value, low EVM window position Default unit: %
<8_EVMpeakHigh>	float EVM peak value, high EVM window position Default unit: %
<9_MErr_RMSlow>	float Magnitude error RMS value, low EVM window position Default unit: %
<10_MErr_RMShigh>	float Magnitude error RMS value, high EVM window position Default unit: %
<11_MErrPeakLow>	float Magnitude error peak value, low EVM window position Default unit: %
<12_MErrPeakHigh>	float Magnitude error peak value, high EVM window position Default unit: %
<13_PErr_RMSlow>	float Phase error RMS value, low EVM window position Default unit: deg
<14_PErr_RMSh>	float Phase error RMS value, high EVM window position Default unit: deg
<15_PErrPeakLow>	float Phase error peak value, low EVM window position Default unit: deg

<16_PErrPeakHigh>	float	Phase error peak value, high EVM window position Default unit: deg
<17_IQoffset>	float	I/Q origin offset Default unit: dBc
<18_FreqError>	float	Carrier frequency error Default unit: Hz
<19_TimingError>	float	Time error Default unit: Ts (basic LTE time unit)
<20_TXpower>	float	User equipment power Default unit: dBm
<21_PeakPower>	float	User equipment peak power Default unit: dBm
<22_RBpower>	float	RB power Default unit: dBm
<23_EVM_DMRSI>	float	EVM DMRS value, low EVM window position Default unit: %
<24_EVM_DMRSsh>	float	EVM DMRS value, high EVM window position Default unit: %
<25_MErr_DMRSI>	float	Magnitude error DMRS value, low EVM window position Default unit: %
<26_MErr_DMRSsh>	float	Magnitude error DMRS value, high EVM window position Default unit: %
<27_PErr_DMRS>	float	Phase error DMRS value, low EVM window position Default unit: deg

<28_PErr_DMRSh> float
 Phase error DMRS value, high EVM window position
 Default unit: deg

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:MODulation:EXTReme?

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:MODulation:EXTReme?

Return modulation single-value results for segment <no> in list mode.

The values described below are returned by **FETCH** commands. The first four values (reliability to out of tolerance result) are also returned by **CALCulate** commands. The remaining values returned by **CALCulate** commands are limit check results, one value for each result listed below.

Suffix:

<no> 1..1000

Return values:

<1_Reliability> decimal
[Reliability indicator](#)

<2_SegReliability> decimal
 Reliability indicator for the segment

<3_StatistExpired> decimal
 Reached statistical length in slots

<4_OutOfTol> decimal
 Percentage of measured subframes with failed limit check
 Default unit: %

<5_EVM_RMSlow> float
 EVM RMS value, low EVM window position
 Default unit: %

<6_EVM_RMShigh> float
 EVM RMS value, high EVM window position
 Default unit: %

<7_EVMpeakLow> float
 EVM peak value, low EVM window position
 Default unit: %

<8_EVMpeakHigh>	float EVM peak value, high EVM window position Default unit: %
<9_MErr_RMSlow>	float Magnitude error RMS value, low EVM window position Default unit: %
<10_MErr_RMShigh>	float Magnitude error RMS value, high EVM window position Default unit: %
<11_MErrPeakLow>	float Magnitude error peak value, low EVM window position Default unit: %
<12_MErrPeakHigh>	float Magnitude error peak value, high EVM window position Default unit: %
<13_PErr_RMSlow>	float Phase error RMS value, low EVM window position Default unit: deg
<14_PErr_RMSh>	float Phase error RMS value, high EVM window position Default unit: deg
<15_PErrPeakLow>	float Phase error peak value, low EVM window position Default unit: deg
<16_PErrPeakHigh>	float Phase error peak value, high EVM window position Default unit: deg
<17_IQoffset>	float I/Q origin offset Default unit: dBc
<18_FreqError>	float Carrier frequency error Default unit: Hz
<19_TimingError>	float Time error Default unit: Ts (basic LTE time unit)

<20_TXpowerMin>	float Minimum user equipment power Default unit: dBm
<21_TXpowerMax>	float Maximum user equipment power Default unit: dBm
<22_PeakPowerMin>	float Minimum user equipment peak power Default unit: dBm
<23_PeakPowerMax>	float Maximum user equipment peak power Default unit: dBm
<24_RBpowerMin>	float Minimum RB power Default unit: dBm
<25_RBpowerMax>	float Maximum RB power Default unit: dBm
<26_EVM_DMRSI>	float EVM DMRS value, low EVM window position Default unit: %
<27_EVM_DMRSb>	float EVM DMRS value, high EVM window position Default unit: %
<28_MErr_DMRSI>	float Magnitude error DMRS value, low EVM window position Default unit: %
<29_MErr_DMRSb>	float Magnitude error DMRS value, high EVM window position Default unit: %
<30_PErr_DMRSI>	float Phase error DMRS value, low EVM window position Default unit: deg
<31_PErr_DMRSb>	float Phase error DMRS value, high EVM window position Default unit: deg

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10

**FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:IEMission:CC<c>:
MARGin:CURRent?**

**FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:IEMission:CC<c>:
MARGin:AVERAge?**

**FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:IEMission:CC<c>:
MARGin:EXTRemE?**

**FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:IEMission:CC<c>:
MARGin:SDEViation?**

Return the in-band emission limit line margin results for component carrier CC<c>, segment <no> in list mode.

The CURRent margins indicate the minimum (vertical) distance between the limit line and the current trace. A negative result indicates that the limit is exceeded.

The AVERAge, EXTRemE and SDEViation values are calculated from the current margins.

Suffix:

<no> 1..1000

<c> 1..4

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SegReliability> decimal
Reliability indicator for the segment

<StatistExpired> decimal
Reached statistical length in slots

<OutOfTolerance> decimal
Percentage of measured subframes with failed limit check
Default unit: %

<Margin> float
Default unit: dB

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V3.7.55

**FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:IEMission:CC<c>:
MARGin:CURRent:RBIndex?**

**FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:IEMission:CC<c>:
MARGin:EXTRemE:RBIndex?**

Return resource block indices of the component carrier CC<c> in-band emission measurement for segment <no> in list mode. At these RB indices, the CURRent and EXTRemE margins have been detected.

Suffix:

<no> 1..1000

<c> 1..4

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SegReliability> decimal
Reliability indicator for the segment

<StatistExpired> decimal
Reached statistical length in slots

<OutOfTolerance> decimal
Percentage of measured subframes with failed limit check
Default unit: %

<RBIndex> decimal
Resource block index of margin

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V3.7.55

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFlatness:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFlatness:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFlatness:EXTRemE?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFlatness:SDEVIation?

Return equalizer spectrum flatness single value results for segment <no> in list mode.

Suffix:

<no> 1..1000

Return values:

<1_Reliability> decimal
[Reliability indicator](#)

<2_SegReliability> decimal
Reliability indicator for the segment

<3_StatistExpired>	decimal Reached statistical length in slots
<4_OutOfTol>	decimal Percentage of measured subframes with failed limit check Default unit: %
<5_Ripple1>	float Max (range 1) - min (range 1) Default unit: dB
<6_Ripple2>	float Max (range 2) - min (range 2) Default unit: dB
<7_MaxR1MinR2>	float Max (range 1) - min (range 2) Default unit: dB
<8_MaxR2MinR1>	float Max (range 2) - min (range 1) Default unit: dB
<9_MinR1>	float Min (range 1) Default unit: dB
<10_MaxR1>	float Max (range 1) Default unit: dB
<11_MinR2>	float Min (range 2) Default unit: dB
<12_MaxR2>	float Max (range 2) Default unit: dB

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFLatness:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFLatness:AVERAGE?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFLatness:EXTREME?

Return equalizer spectrum flatness single value results for segment <no> in list mode.

Suffix:

<no> 1..1000

Return values:

<Reliability> decimal

[Reliability indicator](#)

<SegReliability> decimal

Reliability indicator for the segment

<StatistExpired> decimal

Reached statistical length in slots

<OutOfTolerance> decimal

Percentage of measured subframes with failed limit check

Default unit: %

<Ripple1> Limit check result for max (range 1) - min (range 1).

<Ripple2> Limit check result for max (range 2) - min (range 2).

<MaxR1MinR2> Limit check result for max (range 1) - min (range 2).

Default unit: dB

<MaxR2MinR1> Limit check result for max (range 2) - min (range 1).

Default unit: dB

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ESFLatness:CURRENT:SCINDEX?

Return subcarrier indices of the equalizer spectrum flatness measurement for segment <no> in list mode. At these SC indices, the current minimum and maximum power of the equalizer coefficients have been detected within range 1 and range 2.

Suffix:

<no> 1..1000

Return values:

<Reliability> decimal

[Reliability indicator](#)

<SegReliability>	decimal Reliability indicator for the segment
<StatistExpired>	decimal Reached statistical length in slots
<OutOfTolerance>	decimal Percentage of measured subframes with failed limit check Default unit: %
<Maximum1>	decimal SC index of max (range 1)
<Minimum1>	decimal SC index of min (range 1)
<Maximum2>	decimal SC index of max (range 2)
<Minimum2>	decimal SC index of min (range 2)
Usage:	Query only
Firmware/software:	V2.0.10

CALCulate:LTE:MEAS<i>:MEvaluation:LIST:SEGMent<no>:SEMask:CURRent?
CALCulate:LTE:MEAS<i>:MEvaluation:LIST:SEGMent<no>:SEMask:AVERage?
FETCh:LTE:MEAS<i>:MEvaluation:LIST:SEGMent<no>:SEMask:CURRent?
FETCh:LTE:MEAS<i>:MEvaluation:LIST:SEGMent<no>:SEMask:AVERage?
FETCh:LTE:MEAS<i>:MEvaluation:LIST:SEGMent<no>:SEMask:SDEViation?

Return spectrum emission single value results for segment <no> in list mode.

The values described below are returned by **FETCh** commands. The first four values (reliability to out of tolerance result) are also returned by **CALCulate** commands. The remaining values returned by **CALCulate** commands are limit check results, one value for each result listed below.

Suffix:

<no> 1..1000

Return values:

<Reliability>	decimal Reliability indicator
<SegReliability>	decimal Reliability indicator for the segment
<StatistExpired>	decimal Reached statistical length in slots

<OutOfTolerance>	decimal Percentage of measured subframes with failed limit check Default unit: %
<OBW>	float Occupied bandwidth Default unit: Hz
<TXpower>	float Total TX power in the slot Default unit: dBm

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10

**CALCulate:LTE:MEAS<i>:MEvaluation:LIST:SEGment<no>:SEMAsk:EXTReme?
FETCh:LTE:MEAS<i>:MEvaluation:LIST:SEGment<no>:SEMAsk:EXTReme?**

Return spectrum emission extreme results for segment <no> in list mode.

The values described below are returned by FETCh commands. The first four values (reliability to out of tolerance result) are also returned by CALCulate commands. The remaining values returned by CALCulate commands are limit check results, one value for each result listed below.

Suffix:

<no> 1..1000

Return values:

<Reliability>	decimal Reliability indicator
<SegReliability>	decimal Reliability indicator for the segment
<StatistExpired>	decimal Reached statistical length in slots
<OutOfTolerance>	decimal Percentage of measured subframes with failed limit check Default unit: %
<OBW>	float Occupied bandwidth Default unit: Hz
<TXpowerMin>	float Minimum total TX power in the slot Default unit: dBm

<TXpowerMax>	float Maximum total TX power in the slot Default unit: dBm
Example:	See Running a list mode measurement
Usage:	Query only
Firmware/software:	V2.0.10

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:SEMask:MARGin:ALL?

Return limit line margin values, i.e. vertical distances between the spectrum emission mask and a trace, for segment <no> in list mode.

Suffix:

<no> 1..1000

Return values:

<1_Reliability>	decimal Reliability indicator
<2_SegReliability>	decimal Reliability indicator for the segment
<3_StatistExpired>	decimal Reached statistical length in slots
<4_OutOfTol>	decimal Percentage of measured subframes with failed limit check Default unit: %
<5_CurrNeg1> ... <16_CurrNeg12>	float Margin results for current trace, negative offsets Default unit: dB
<17_CurrPos1> ... <28_CurrPos12>	float Margin results for current trace, positive offsets Default unit: dB
<29_AvgNeg1> ... <40_AvgNeg12>	float Margin results for average trace, negative offsets Default unit: dB
<41_AvgPos1> ... <52_AvgPos12>	float Margin results for average trace, positive offsets Default unit: dB
<53_MinNeg1> ... <64_MinNeg12>	float Margin results for maximum trace (resulting in minimum margins), negative offsets Default unit: dB

<65_MinPos1> ... float
 <76_MinPos12> Margin results for maximum trace (resulting in minimum margins), positive offsets
 Default unit: dB

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V3.2.80

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SEMAsk:MARGIn:CURRENT:NEGativ?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SEMAsk:MARGIn:CURRENT:POSitiv?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SEMAsk:MARGIn:AVERAge:NEGativ?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SEMAsk:MARGIn:AVERAge:POSitiv?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SEMAsk:MARGIn:MINimum:NEGativ?

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SEMAsk:MARGIn:MINimum:POSitiv?

Return spectrum emission mask margin results for segment <no> in list mode.

The individual commands provide results for the **CURRENT**, **AVERAge** and maximum traces (resulting in **MINimum** margins) for **NEGative** and **POSitive** offset frequencies.

Returned sequence: <Reliability>, <SegReliability>, <StatistExpired>, <OutOfTolerance>, {<MarginX>, <MarginY>}_{area1}, {...}_{area2}, ..., {...}_{area12}

Suffix:

<no> 1..1000

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SegReliability> decimal
 Reliability indicator for the segment

<StatistExpired> decimal
 Reached statistical length in slots

<OutOfTolerance> decimal
 Percentage of measured subframes with failed limit check
 Default unit: %

<MarginX> float
 X-position of the margin of an area
 Default unit: Hz

<MarginY>	float Y-value of the margin of an area Default unit: dB
Example:	See Running a list mode measurement
Usage:	Query only
Firmware/software:	V2.0.10

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ACLR:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ACLR:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ACLR:CURRENT?
FETCH:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ACLR:AVERage?

Return ACLR single value results for segment <no> in list mode.

The values described below are returned by **FETCH** commands. The first four values (reliability to out of tolerance result) are also returned by **CALCulate** commands. The remaining values returned by **CALCulate** commands are limit check results, one value for each result listed below.

Suffix:

<no> 1..1000

Return values:

<1_Reliability>	decimal Reliability indicator
<2_SegReliability>	decimal Reliability indicator for the segment
<3_StatistExpired>	decimal Reached statistical length in slots
<4_OutOfTol>	decimal Percentage of measured subframes with failed limit check Default unit: %
<5_UTRA2neg>	float ACLR for the second UTRA channel with lower frequency Default unit: dB
<6_UTRA1neg>	float ACLR for the first UTRA channel with lower frequency Default unit: dB
<7_EUTRAneg>	float ACLR for the first E-UTRA channel below the carrier frequency Default unit: dB

<8_EUTRA>	float Power in the allocated E-UTRA channel Default unit: dBm
<9_EUTRApos>	float ACLR for the first E-UTRA channel above the carrier frequency Default unit: dB
<10_UTRA1pos>	float ACLR for the first UTRA channel with higher frequency Default unit: dB
<11_UTRA2pos>	float ACLR for the second UTRA channel with higher frequency Default unit: dB
Example:	See Running a list mode measurement
Usage:	Query only
Firmware/software:	V2.0.10

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:PMONitor:RMS?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:PMONitor:PEAK?

Return the power monitor results for segment <no> in list mode. The commands return one power result for each subframe of the segment for the measured carrier. The power values are RMS averaged over the subframe or represent the peak value within the subframe.

Suffix:

<no> 1..1000

Return values:

<Reliability>	decimal Reliability indicator
<SegReliability>	decimal Reliability indicator for the segment
<Power>	float Comma-separated list of power values, one value per subframe, from first to last subframe of the segment. For an inactive segment, only one INV is returned, independent of the number of configured subframes. Default unit: dBm

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.20

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:SDEViation?

Return total TX power results for segment <no> in list mode.

Suffix:

<no> 1..1000

Return values:

<Reliability>	decimal Reliability indicator
<SegReliability>	decimal Reliability indicator for the segment
<StatistExpired>	decimal Reached statistical length in subframes
<OutOfTolerance>	decimal Percentage of measured subframes with failed limit check Default unit: %
<TXpower>	float Total TX power of all component carriers Default unit: dBm

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V3.5.50

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:CC<no>:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:CC<no>:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:CC<no>:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:CC<no>:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:POWer:CC<no>:SDEViation?

Return TX power results for component carrier CC<no> and a single segment in list mode.

Suffix:

<no> 1..1000
Segment number

<no>	1..4 Component carrier number
Return values:	
<Reliability>	decimal Reliability indicator
<SegReliability>	decimal Reliability indicator for the segment
<StatistExpired>	decimal Reached statistical length in subframes
<OutOfTolerance>	decimal Percentage of measured subframes with failed limit check Default unit: %
<TXpower>	float TX power of the component carrier Default unit: dBm
Example:	See Running a list mode measurement
Usage:	Query only
Firmware/software:	V4.0.20

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:SEMAsk:DALLocation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:ACLR:DALLocation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:MODulation:
DALLocation?

Return the detected allocation for segment <no> in list mode.

The result is determined from the last measured slot of the statistical length. The individual measurements provide the same result when measuring the same slot. However different statistical lengths can be defined for the measurements so that the measured slots and returned results can differ.

Suffix:	
<no>	1..1000
Return values:	
<Reliability>	decimal Reliability indicator
<SegReliability>	decimal Reliability indicator for the segment
<NrResBlocks>	decimal Number of allocated resource blocks

<OffsetResBlocks> decimal
Offset of the first allocated resource block from the edge of the allocated UL transmission bandwidth

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:MODulation:DMODulation?

Return the detected modulation scheme for segment <no> in list mode.

The result is determined from the last measured slot of the statistical length.

If channel type PUCCH is detected, QPSK is returned for the modulation scheme because the QPSK limits are applied in that case.

Suffix:

<no> 1..1000

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SegReliability> decimal
Reliability indicator for the segment

<Modulation> QPSK | Q16 | Q64 | Q256
QPSK, 16QAM, 64QAM, 256QAM

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10
V2.1.25 max <no> 250, V2.1.30 max <no> 512
V3.0.50 max <no> 1000, V3.7.50 256QAM

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:SEMask:DCHType? FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:ACLR:DCHType? FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:MODulation:DCHType?

Return the uplink channel type for segment <no> in list mode.

The result is determined from the last measured slot of the statistical length. The individual measurements provide the same result when measuring the same slot. However different statistical lengths can be defined for the measurements so that the measured slots and returned results can differ.

Suffix:

<no> 1..1000

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SegReliability> decimal
 Reliability indicator for the segment

<ChannelType> PUSCh | PUCCh

Example: See [Running a list mode measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCh:LTE:MEAS<i>:MEvaluation:LIST:SEGMent<no>:MODulation:SCHType?

Returns the sidelink channel type evaluated for modulation results, for segment <no> in list mode.

Suffix:

<no> 1..1000

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SegReliability> decimal
 Reliability indicator for the segment

<ChannelType> PSSCh | PSSCh

Usage: Query only

Firmware/software: V3.7.55

5.5.3.35 List mode results (all segments, one result)

Each of the following commands returns a selected list mode result for all measured segments. The number of results returned depends on the number of measured segments, see [CONFigure:LTE:MEAS<i>:MEvaluation:LIST:LRANge](#) on page 1155.

To configure the list mode, use the commands described in [Section 5.5.3.9, "List mode settings"](#), on page 1154.

For a description of the list mode, see [Section 5.2.10, "List mode"](#), on page 1002.

FETCh:LTE:MEAS<i>:MEvaluation:LIST:SRELIability?

Returns the segment reliability for all measured list mode segments.

A common reliability indicator of zero indicates that the results in all measured segments are valid. A non-zero value indicates that an error occurred in at least one of the measured segments. If you get a non-zero common reliability indicator, you can use this command to retrieve the individual reliability values of all measured segments for further analysis.

Return values:

<Reliability> decimal

[Reliability indicator](#)

<SegReliability> decimal

Comma-separated list of values, one per measured segment
The meaning of the returned values is the same as for the common reliability indicator, see previous parameter.

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V3.0.10

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:LOW:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:LOW:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:LOW:EXTReme?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:HIGH:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:HIGH:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:HIGH:EXTReme?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:LOW:CURRENT?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:LOW:AVERage?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:LOW:EXTReme?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:LOW:SDEViation?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:HIGH:CURRENT?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:HIGH:AVERage?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:HIGH:EXTReme?

FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:RMS:HIGH:SDEViation?

Return error vector magnitude RMS values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by **FETCH** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<EVM_RMS>	float Comma-separated list of values, one per measured segment Default unit: %

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:LOW:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:LOW:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:LOW:EXTreme?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:HIGH:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:HIGH:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:HIGH:EXTreme?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:LOW:CURRENT?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:LOW:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:LOW:EXTreme?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:LOW:SDEViation?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:HIGH:CURRENT?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:HIGH:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:HIGH:EXTreme?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:PEAK:HIGH:SDEViation?

Return error vector magnitude peak values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<EVMpeak>	float Comma-separated list of values, one per measured segment Default unit: %

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:LOW:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:LOW:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:LOW:EXTReme?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:HIGH:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:HIGH:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:HIGH:EXTReme?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:LOW:CURRENT?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:LOW:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:LOW:EXTReme?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:LOW:SDEViation?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:HIGH:CURRENT?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:HIGH:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:HIGH:EXTReme?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:EVM:DMRS:HIGH:SDEViation?

Return error vector magnitude DMRS values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<EVM_DMRS> float
Comma-separated list of values, one per measured segment
Default unit: %

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:LOW:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:LOW:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:LOW:
  EXTReMe?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:HIGh:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:HIGh:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:HIGh:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:LOW:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:LOW:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:LOW:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:LOW:
  SDEViation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:HIGh:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:HIGh:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:HIGh:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:RMS:HIGh:
  SDEViation?

```

Return magnitude error RMS values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by `FETCh` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<MagErrorRMS>	float Comma-separated list of values, one per measured segment Default unit: %

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:LOW:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:LOW:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:LOW:
  EXTReMe?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:HIGh:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:HIGh:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:HIGh:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:LOW:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:LOW:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:LOW:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:LOW:
  SDEViation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:HIGh:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:HIGh:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:HIGh:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:PEAK:HIGh:
  SDEViation?

```

Return magnitude error peak values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by `FETCh` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<MagErrPeak>	float Comma-separated list of values, one per measured segment Default unit: %

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:LOW:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:LOW:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:LOW:
  EXTReMe?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:HIGH:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:HIGH:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:HIGH:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:LOW:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:LOW:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:LOW:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:LOW:
  SDEViation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:HIGH:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:HIGH:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:HIGH:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:MERRor:DMRS:HIGH:
  SDEViation?

```

Return magnitude error DMRS values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by `FETCh` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<MagErrDMRS>	float Comma-separated list of values, one per measured segment Default unit: %

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:LOW:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:LOW:
  AVERAge?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:LOW:
  EXTReMe?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:HIGH:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:HIGH:
  AVERAge?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:HIGH:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:LOW:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:LOW:
  AVERAge?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:LOW:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:LOW:
  SDEViation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:HIGH:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:HIGH:
  AVERAge?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:HIGH:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:RMS:HIGH:
  SDEViation?

```

Return phase error RMS values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by `FETCh` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<PhErrorRMS>	float Comma-separated list of values, one per measured segment Default unit: deg

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:LOW:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:LOW:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:LOW:
  EXTReMe?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:HIGh:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:HIGh:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:HIGh:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:LOW:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:LOW:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:LOW:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:LOW:
  SDEViation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:HIGh:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:HIGh:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:HIGh:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:PEAK:HIGh:
  SDEViation?

```

Return phase error peak values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by `FETCh` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<PhErrorPeak>	float Comma-separated list of values, one per measured segment Default unit: deg

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:LOW:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:LOW:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:LOW:
  EXTReMe?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:HIGH:
  CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:HIGH:
  AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:HIGH:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:LOW:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:LOW:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:LOW:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:LOW:
  SDEViation?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:HIGH:
  CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:HIGH:
  AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:HIGH:
  EXTReMe?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PERRor:DMRS:HIGH:
  SDEViation?

```

Return phase error DMRS values for low and high EVM window position, for all measured list mode segments.

The values described below are returned by `FETCh` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<PhErrorDMRS>	float Comma-separated list of values, one per measured segment Default unit: deg

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:AVERage?

```

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:CURRENT?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:IQOffset:SDEVIation?

Return I/Q origin offset values for all measured list mode segments.

The values described below are returned by **FETCH** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<IQOffset> float
 Comma-separated list of values, one per measured segment
 Default unit: dBc

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:FERRor:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:FERRor:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:FERRor:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:FERRor:CURRENT?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:FERRor:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:FERRor:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:FERRor:SDEVIation?

Return carrier frequency error values for all measured list mode segments.

The values described below are returned by **FETCH** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<FrequencyError> float
 Comma-separated list of values, one per measured segment
 Default unit: Hz

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:TERRor:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:TERRor:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:TERRor:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TERRor:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TERRor:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TERRor:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TERRor:SDEViation?

Return transmit time error values for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<TimingError>	float Comma-separated list of values, one per measured segment Default unit: Ts (basic LTE time unit)

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:MINimum?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:TPOWER:SDEViation?

Return user equipment power values for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<TXpower>	float Comma-separated list of values, one per measured segment Default unit: dBm

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:MINimum?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PPOWer:SDEViation?

Return user equipment peak power values for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<PeakPower>	float Comma-separated list of values, one per measured segment Default unit: dBm

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:MINimum?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:PSD:SDEViation?

Return RB power values for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<PSD>	float Comma-separated list of values, one per measured segment Default unit: dBm

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

FETCh:LTE:MEAS<i>:MEValuation:LIST:IEMission:MARGin:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:IEMission:MARGin:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:IEMission:MARGin:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:LIST:IEMission:MARGin:SDEViation?

Return the in-band emission limit line margin results for all measured list mode segments.

The **CURRent** margins indicate the minimum (vertical) distance between the limit line and the current trace. A negative result indicates that the limit is exceeded.

The **AVERage**, **EXTReme** and **SDEViation** values are calculated from the current margins.

Return values:

<Reliability> decimal

[Reliability indicator](#)

<Margin> float

Comma-separated list of values, one per measured segment

Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

FETCh:LTE:MEAS<i>:MEValuation:LIST:IEMission:MARGin:RBINdex:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:IEMission:MARGin:RBINdex:EXTReme?

Return resource block indices of the in-band emission measurement for all measured list mode segments. At these RB indices, the **CURRent** and **EXTReme** margins have been detected.

Return values:

<Reliability> decimal

[Reliability indicator](#)

<RBIndex> decimal

Comma-separated list of values, one per measured segment

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:RIPPLE<no>:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:RIPPLE<no>:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:RIPPLE<no>:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:RIPPLE<no>:CURRENT?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:RIPPLE<no>:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:RIPPLE<no>:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:RIPPLE<no>:SDEViation?

Return equalizer spectrum flatness single value results (ripple 1 or ripple 2) for all measured list mode segments.

The values described below are returned by **FETCH** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Suffix:

<no> 1..2
 Ripple 1 = max (range 1) - min (range 1)
 Ripple 2 = max (range 2) - min (range 2)

Return values:

<Reliability> decimal
 [Reliability indicator](#)

 <Ripple> float
 Comma-separated list of values, one per measured segment
 Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:DIFFerence<no>:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:DIFFerence<no>:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:DIFFerence<no>:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:DIFFerence<no>:CURRENT?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:DIFFerence<no>:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:DIFFerence<no>:EXTReme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:DIFFerence<no>:SDEViation?

Return equalizer spectrum flatness single value results (differences between ranges) for all measured list mode segments.

The values described below are returned by **FETCH** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Suffix:

<no> 1..2
 Difference 1 = max (range 1) - min (range 2)
 Difference 2 = max (range 2) - min (range 1)

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Difference> float
 Comma-separated list of values, one per measured segment
 Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MINR<no>:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MINR<no>:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MINR<no>:EXTreme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MINR<no>:CURRENT?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MINR<no>:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MINR<no>:EXTreme?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MINR<no>:SDEVIation?

Return equalizer spectrum flatness single value results (minimum within a range) for all measured list mode segments.

The values described below are returned by **FETCH** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Suffix:

<no> 1..2
 Selects the range for which the minimum is returned.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<MinR> float
 Comma-separated list of values, one per measured segment.
 Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MAXR<no>:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFlatness:MAXR<no>:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ESFLatness:MAXR<no>:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ESFLatness:MAXR<no>:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ESFLatness:MAXR<no>:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ESFLatness:MAXR<no>:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ESFLatness:MAXR<no>:SDEViation?

Return equalizer spectrum flatness single value results (maximum within a range) for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Suffix:

<no> 1..2
 Selects the range for which the maximum is returned.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<MaxR> float
 Comma-separated list of values, one per measured segment.
 Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

FETCh:LTE:MEAS<i>:MEValuation:LIST:ESFLatness:SCINdex:MINimum<no>:CURRent?

FETCh:LTE:MEAS<i>:MEValuation:LIST:ESFLatness:SCINdex:MAXimum<no>:CURRent?

Return subcarrier indices of the equalizer spectrum flatness measurement for all measured list mode segments.

At these SC indices, the current **MINimum** or **MAXimum** power of the equalizer coefficients has been detected within the selected range.

Suffix:

<no> 1..2
 Selects the range.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Index> decimal
 Comma-separated list of values, one per measured segment

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEMask:OBW:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEMask:OBW:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEMask:OBW:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:OBW:CURRENT?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:OBW:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:OBW:EXTReme?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:OBW:SDEVIation?

Return the occupied bandwidth for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<OBW>	float Comma-separated list of values, one per measured segment Default unit: Hz

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:CURRENT?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:AVERage?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:MINimum?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:CURRENT?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:AVERage?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:MINimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:MAXimum?
FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:TXPower:SDEVIation?

Return the total TX power in the slot for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<TXpower>	float Comma-separated list of values, one per measured segment Default unit: dBm

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEMask:MARGIN:AREA<no>:NEGativ:CURRENT?

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEMask:MARGIN:AREA<no>:NEGativ:AVERAge?

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEMask:MARGIN:AREA<no>:NEGativ:MINimum?

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEMask:MARGIN:AREA<no>:POSitiv:AVERAge?

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEMask:MARGIN:AREA<no>:POSitiv:CURRENT?

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEMask:MARGIN:AREA<no>:POSitiv:MINimum?

Return spectrum emission mask margin positions for all measured list mode segments.

The individual commands provide results for the **CURRENT**, **AVERAge** and maximum traces (resulting in **MINimum** margins) for **NEGative** and **POSitive** offset frequencies.

The results are returned as pairs per segment: *<Reliability>, {<MarginPosX>, <MarginPosY>}_{Seg 1}, {<MarginPosX>, <MarginPosY>}_{Seg 2}, ...*

Suffix:

<no> 1..12
Selects the emission mask area.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<MarginPosX> float
X-position of margin for selected area
Default unit: Hz

<MarginPosY> float
Y-value of margin for selected area
Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30
V3.2.80: increased <no> of areas to 12

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:AVERAge?

FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:AVERAge?

Return the power in the allocated E-UTRA channel for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<EUTRA> float
 Comma-separated list of values, one per measured segment
 Default unit: dBm

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:NEGativ:CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:NEGativ:AVERAge?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:POSitiv:CURRent?
CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:POSitiv:AVERAge?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:NEGativ:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:NEGativ:AVERAge?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:POSitiv:CURRent?
FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:EUTRa:POSitiv:AVERAge?

Return the ACLR for the first adjacent E-UTRA channel above (**POSitiv**) or below (**NEGativ**) the carrier frequency for all measured list mode segments.

The values described below are returned by **FETCh** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<EUTRA> float
 Comma-separated list of values, one per measured segment
 Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:NEGativ:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:NEGativ:AVERage?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:POSitiv:CURRENT?

CALCulate:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:POSitiv:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:NEGativ:CURRENT?

FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:NEGativ:AVERage?

FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:POSitiv:CURRENT?

FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:UTRA<no>:POSitiv:AVERage?

Return the ACLR for the first or second adjacent UTRA channel above (POSitiv) or below (NEGativ) the carrier frequency for all measured list mode segments.

The values described below are returned by FETCh commands. A CALCulate command returns limit check results instead, one value for each result listed below.

Suffix:

<no> 1..2
Selects first or second adjacent UTRA channel.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<UTRA> float
Comma-separated list of values, one per measured segment
Default unit: dB

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

FETCh:LTE:MEAS<i>:MEValuation:LIST:PMONitor:RMS?

FETCh:LTE:MEAS<i>:MEValuation:LIST:PMONitor:PEAK?

Return the power monitor results for all measured segments in list mode. The commands return one power result per subframe for the measured carrier. The power values are RMS averaged over the subframe or represent the peak value within the subframe.

Commands for querying the result list structure:

- [FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:PMONitor:ARRAY:START?](#)
- [FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGMENT<no>:PMONitor:ARRAY:LENGTH?](#)

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Power>	float Comma-separated list of power values, one value per subframe, from first subframe of first measured segment to last subframe of last measured segment For an inactive segment only one INV is returned, independent of the number of configured subframes. Default unit: dBm
Example:	See Retrieving single results for all segments
Usage:	Query only
Firmware/software:	V2.0.20

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:PMONitor:ARRay:START?

Returns the offset of the first power monitor result for segment <no> within a result list for all measured segments. Such a result list is, for example, returned by the command [FETCh:LTE:MEAS<i>:MEValuation:LIST:PMONitor:RMS?](#).

A returned <Start> value n indicates that the result for the first subframe of the segment is the (n+1)th result in the power result list over all segments.

Suffix:	
<no>	1..1000
Return values:	
<Reliability>	decimal Reliability indicator
<Start>	decimal Offset of the first power monitor result
Example:	See Retrieving single results for all segments
Usage:	Query only
Firmware/software:	V2.0.20

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEGment<no>:PMONitor:ARRay:LENGTH?

Returns the number of power monitor results for segment <no> contained in a result list for all measured segments. Such a result list is, for example, returned by the command [FETCh:LTE:MEAS<i>:MEValuation:LIST:PMONitor:RMS?](#).

Suffix:	
<no>	1..1000
Return values:	
<Reliability>	decimal Reliability indicator

<Length> decimal
Number of power monitor results

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.0.20

FETCH:LTE:MEAS<i>:MEValuation:LIST:POWer:TXPower:CURRent?
FETCH:LTE:MEAS<i>:MEValuation:LIST:POWer:TXPower:AVERage?
FETCH:LTE:MEAS<i>:MEValuation:LIST:POWer:TXPower:MINimum?
FETCH:LTE:MEAS<i>:MEValuation:LIST:POWer:TXPower:MAXimum?
FETCH:LTE:MEAS<i>:MEValuation:LIST:POWer:TXPower:SDEViation?

Return the total TX power of all component carriers, for all measured list mode segments.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<TXpower> float
Comma-separated list of values, one per measured segment
Default unit: dBm

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V3.5.50

FETCH:LTE:MEAS<i>:MEValuation:LIST:SEMask:DALLocation?
FETCH:LTE:MEAS<i>:MEValuation:LIST:ACLR:DALLocation?
FETCH:LTE:MEAS<i>:MEValuation:LIST:MODulation:DALLocation?

Return the detected allocation for all measured list mode segments.

The result is determined from the last measured slot of the statistical length of a segment. The individual measurements provide the same result when measuring the same slot. However different statistical lengths can be defined for the measurements so that the measured slots and returned results can differ.

The results are returned as pairs per segment: <Reliability>, {<NrResBlocks>, <OffsetResBlocks>}_Seg 1, {<NrResBlocks>, <OffsetResBlocks>}_Seg 2, ...

Return values:

<Reliability> decimal
[Reliability indicator](#)

<NrResBlocks> decimal
Number of allocated resource blocks

<OffsetResBlocks> decimal
Offset of the first allocated resource block from the edge of the allocated UL transmission bandwidth

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:DMODulation?

Return the detected modulation scheme for all measured list mode segments.

The result is determined from the last measured slot of the statistical length of a segment.

If channel type PUCCH is detected, QPSK is returned for the modulation scheme because the QPSK limits are applied in that case.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Modulation> QPSK | Q16 | Q64 | Q256
Comma-separated list of values, one per measured segment
QPSK, 16QAM, 64QAM, 256QAM

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30, V3.7.50: 256QAM

FETCh:LTE:MEAS<i>:MEValuation:LIST:SEMask:DCHType?

FETCh:LTE:MEAS<i>:MEValuation:LIST:ACLR:DCHType?

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:DCHType?

Return the uplink channel type for all measured list mode segments.

The result is determined from the last measured slot of the statistical length of a segment. The individual measurements provide the same result when measuring the same slot. However different statistical lengths can be defined for the measurements so that the measured slots and returned results can differ.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<ChannelType> PUSCh | PUCCh
Comma-separated list of values, one per measured segment

Example: See [Retrieving single results for all segments](#)

Usage: Query only

Firmware/software: V2.1.30

FETCh:LTE:MEAS<i>:MEValuation:LIST:MODulation:SCHType?

Returns the sidelink channel type evaluated for modulation results, for all measured list mode segments.

Return values:

<Reliability> decimal

[Reliability indicator](#)

<ChannelType> PSSCh | PSCCh

Comma-separated list of values, one per measured segment

Usage: Query only

Firmware/software: V3.7.55

5.5.4 Combined signal path commands

For some settings, the relevant command depends on the active scenario. While the combined signal path (CSP) scenario is active, these settings are configured via commands of the signaling application. While the standalone (SA) scenario is active, they are configured via measurement commands.

The following mapping tables provide an overview for general measurement settings and for multi-evaluation measurement commands.

Table 5-14: Mapping for general measurement settings

Setting	Commands for SA scenario	Commands for CSP scenario
Duplex mode	CONFigure:LTE:MEAS<i>:DMODE	CONFigure:LTE:SIGN<i>[:PCC]:DMODE CONFigure:LTE:SIGN<i>:SCC<c>:DMODE CONFigure:LTE:SIGN<i>[:PCC]:DMODE: UCSPecific
eMTC	CONFigure:LTE:MEAS<i>:EMTC:ENABle CONFigure:LTE:MEAS<i>:EMTC:MB<number>	CONFigure:LTE:SIGN<i>[:PCC]:EMTC: ENABle CONFigure:LTE:SIGN<i>[:PCC]:EMTC: MB<number>
Use SCC uplink	No setting	CONFigure:LTE:SIGN<i>:SCC<c>:UUL
Connector and con- verter	ROUTe:LTE:MEAS<i>:SCENario:SALone	ROUTe:LTE:MEAS<i>:SCENario:CSPath ROUTe:LTE:SIGN<i>:SCENario:... See " Scenario " on page 140.
External attenuation	CONFigure:LTE:MEAS<i>:RFSettings: EATTenuation	CONFigure:LTE:SIGN<i>:RFSettings[: PCC]:EATTenuation:INPut CONFigure:LTE:SIGN<i>:RFSettings: SCC<c>:EATTenuation:INPut

Setting	Commands for SA scenario	Commands for CSP scenario
Carrier aggregation mode	<code>CONFigure:LTE:MEAS<i>:CAGGregation:MODE</code>	<code>ROUTe:LTE:MEAS<i>:SCENario:CSPath</code> <code>CONFigure:LTE:MEAS<i>:CAGGregation:MODE:CSPath?</code>
Band	<code>CONFigure:LTE:MEAS<i>:BAND</code>	<code>CONFigure:LTE:SIGN<i>[:PCC]:BAND</code> <code>CONFigure:LTE:SIGN<i>:SCC<c>:BAND</code>
Frequency, channel	<code>CONFigure:LTE:MEAS<i>:RFSettings:CC<no>:FREQuency</code> <code>CONFigure:LTE:MEAS<i>:CAGGregation:ACSPacing</code>	<code>CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:CHANnel:UL</code> <code>CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:CHANnel:UL</code> <code>CONFigure:LTE:SIGN<i>:CAGGregation:SET</code>
Frequency offset	<code>CONFigure:LTE:MEAS<i>:RFSettings:FOffset</code>	<code>CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL</code> <code>CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:FOFFset:UL</code> <code>CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:FOFFset:UL:UCSPecific</code>
Expected nominal power	<code>CONFigure:LTE:MEAS<i>:RFSettings:ENPower</code>	<code>CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPMode</code> <code>CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:ENPower</code> <code>CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPMode</code> <code>CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:ENPower</code>
User margin	<code>CONFigure:LTE:MEAS<i>:RFSettings:UMARgin</code>	<code>CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:UMARgin</code> <code>CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:UMARgin</code>
Mixer level offset	<code>CONFigure:LTE:MEAS<i>:RFSettings:MLOffset</code>	<code>CONFigure:LTE:SIGN<i>:RFSettings[:PCC]:MLOffset</code> <code>CONFigure:LTE:SIGN<i>:RFSettings:SCC<c>:MLOffset</code>
Channel bandwidth	<code>CONFigure:LTE:MEAS<i>:CC<no>:CBANDwidth</code>	<code>CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL</code> <code>CONFigure:LTE:SIGN<i>:CELL:BANDwidth:SCC<c>:DL</code>

Table 5-15: Mapping for multi-evaluation measurement commands

Setting	Commands for SA scenario	Commands for CSP scenario
Measurement mode	<code>CONFigure:LTE:MEAS<i>:MEValuation:MMODE</code>	Fixed value <code>TMODE</code>
TPC mode settings	<code>CONFigure:LTE:MEAS<i>:MEValuation:TMODE:SCount</code> <code>CONFigure:LTE:MEAS<i>:MEValuation:TMODE:ENPower</code> <code>CONFigure:LTE:MEAS<i>:MEValuation:TMODE:RLEVel?</code>	Automatic configuration
UL-DL configuration	<code>CONFigure:LTE:MEAS<i>:MEValuation:ULDL</code>	<code>CONFigure:LTE:SIGN<i>:CELL[:PCC]:ULDL</code> <code>CONFigure:LTE:SIGN<i>:CELL:SCC<c>:ULDL</code> <code>CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific</code>
Special subframe	<code>CONFigure:LTE:MEAS<i>:MEValuation:SSUBframe</code>	<code>CONFigure:LTE:SIGN<i>:CELL[:PCC]:SSUBframe</code> <code>CONFigure:LTE:SIGN<i>:CELL:SCC<c>:SSUBframe</code> <code>CONFigure:LTE:SIGN<i>:CELL:TDD:SPECific</code>
Cyclic prefix	<code>CONFigure:LTE:MEAS<i>:MEValuation:CPRefix</code>	<code>CONFigure:LTE:SIGN<i>:CELL:CPRefix</code>
Network signaled value	<code>CONFigure:LTE:MEAS<i>:MEValuation:NSValue</code> <code>CONFigure:LTE:MEAS<i>:MEValuation:NSValue:CAGGregation</code>	<code>CONFigure:LTE:SIGN<i>:CONNection:ASEMission</code> <code>CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:ASEMission:CAGGregation</code>
Multi-cluster on/off	<code>CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:MCLuster</code>	<code>CONFigure:LTE:SIGN<i>:CONNection[:PCC]:MCLuster:UL</code> <code>CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:MCLuster:UL</code>
RBG multi-cluster allocation	<code>CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:MCLuster:NRB<Number></code> <code>CONFigure:LTE:MEAS<i>:MEValuation:RBAllocation:MCLuster:ORB<Number></code>	<code>CONFigure:LTE:SIGN<i>:CONNection[:PCC]:RMC:MCLuster:UL</code> <code>CONFigure:LTE:SIGN<i>:CONNection[:PCC]:UDChannels:MCLuster:UL</code> <code>CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:RMC:MCLuster:UL</code> <code>CONFigure:LTE:SIGN<i>:CONNection:SCC<c>:UDChannels:MCLuster:UL</code>
Physical cell ID	<code>CONFigure:LTE:MEAS<i>:MEValuation:CC<no>:PLCid</code>	<code>CONFigure:LTE:SIGN<i>:CELL[:PCC]:PCID</code> <code>CONFigure:LTE:SIGN<i>:CELL:SCC<c>:PCID</code>

Setting	Commands for SA scenario	Commands for CSP scenario
Group hopping	CONFigure:LTE:MEAS<i>:MEValuation:GHOPping	CONFigure:LTE:SIGN<i>:CONNection:GHOPping
Sounding RS	CONFigure:LTE:MEAS<i>:MEValuation:SRS:ENABle	CONFigure:LTE:SIGN<i>:CELL[:PCC]:SRS:ENABle modifies signaling and measurement setting CONFigure:LTE:MEAS<i>:MEValuation:SRS:ENABle modifies only measurement setting

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TRIGger:LTE:MEAS<i>:MEValuation:SLOPe.....	1168
TRIGger:LTE:MEAS<i>:MEValuation:SMODE.....	1170
TRIGger:LTE:MEAS<i>:MEValuation:SOURce.....	1168
TRIGger:LTE:MEAS<i>:MEValuation:THReshold.....	1168
TRIGger:LTE:MEAS<i>:MEValuation:TOUT.....	1169

6 LTE PRACH measurement

The LTE PRACH measurement provides quick and flexible tests on LTE FDD and TDD random access preambles. The tests cover the following UE transmitter properties:

- Modulation accuracy for each PRACH subcarrier (EVM, magnitude error, phase error)
- Transmit OFF power, transmit ON power and power ramping between them (power dynamics)
- I/Q constellation diagram

The PRACH measurement requires option R&S CMW-KM500 for FDD signals and R&S CMW-KM550 for TDD signals.

6.1 What's new

This documentation describes version 4.0.250 and higher of the LTE PRACH measurement. Compared to version 4.0.140, it provides the following new features:

- Editorial changes only.

6.2 General description

The LTE PRACH measurement captures an uplink (UL) LTE PRACH signal and provides TX measurement results for the random access preambles. Both FDD signals and TDD signals can be measured.

The following sections describe how to perform and configure the measurement.

- [Test setup](#)..... 1305
- [How to measure an uplink PRACH signal](#)..... 1305
- [Defining the scope of the measurement](#)..... 1306
- [Parallel signaling and measurement](#)..... 1307
- [Calculation of modulation results](#)..... 1308
- [LTE PRACH UL signal properties](#)..... 1309

6.2.1 Test setup

Connect the external RF signal source (mobile station, signal generator etc.) to one of the bidirectional RF connectors on the front panel of the R&S CMW.

6.2.2 How to measure an uplink PRACH signal

The measurement expects an LTE PRACH UL signal. Any other signals do not yield measurement results (e.g. an LTE UL signal for an established connection).

After connecting your UE, you have to adjust at least the following analyzer settings to the properties of the analyzed PRACH signal:

- Duplex mode
- Analyzer frequency
- Expected nominal power, external attenuation and user margin
Recommended values: expected nominal power = peak power of the UE signal during the measurement. User margin = 0 dB. The smallest possible value of the expected nominal power plus the user margin ensures maximum dynamic range.

For synchronization to the received signal and proper decoding, some measurement control settings must be in accordance with the measured signal. In particular, ensure that the following parameters match up:

- Channel bandwidth
- PRACH configuration index
- "High Speed Flag": This software version supports only the value FALSE, i.e. the unrestricted set of N_{CS} must be used.
- Logical root sequence index
- Zero correlation zone configuration

The R&S CMW can auto-detect the location of the used resource blocks in the frequency domain ("PRACH Frequency Offset"). It can also auto-detect which of the 64 preamble sequences defined by the parameters above is used ("Sequence Index").

With matching signal settings, the measurement is able to decode the signal and to determine its frame timing. Non-matching signal settings generally result in large EVM results and/or a sequence correlation result close to 0.

6.2.3 Defining the scope of the measurement

A single-shot LTE PRACH measurement comprises one measurement cycle which comprises a configurable number of measurement intervals (statistic count).

In this context, two types of result views must be distinguished:

- **Single-preamble measurements** evaluate one preamble per measurement interval. The related views display results for the current measurement interval (one preamble). They also provide a statistical evaluation for the already measured preambles.
Examples for single-preamble result views are "Error Vector Magnitude" and "Power Dynamics".
- **Multi-preamble measurements** evaluate up to 32 consecutive preambles per measurement interval. The related views present the results per preamble, for all preambles of the current measurement interval. No statistical evaluation over several measurement intervals is performed.
The preambles captured within one measurement interval are labeled 1 to n, with $n \leq 32$. Single-preamble measurements evaluate the preamble labeled 1.
The time between two captured preambles is configurable (10 ms or 20 ms). If no preamble is found, the measurement continues and there are no demodulation results for that label.

Examples for multi-preamble result views are "EVM vs Preamble" and "Power vs Preamble".

The scope of the measurement is configured in section "Measurement Control" of the configuration tree. The most important parameters are:

- "Repetition" on page 1324
- "Number of Preambles" on page 1325
- "Period of Preambles" on page 1325
- "Statistic Count" on page 1327
- "Statistic Count" on page 1329

Example: Limited sequence of preambles with increasing power

Scenario: A sequence of 10 preambles with a period of 20 ms and increasing power is measured. The main result of interest is the TX power of the individual preambles (verify correct ramping up of preamble power by UE).

Used view: "Power vs Preamble"

Settings: statistic count = 1, number of preambles = 10, period of preambles = 20 ms, repetition = single shot

Example: Statistical evaluation for preambles with constant power

Scenario: The EVM is measured for 100 preambles, so that the average and the maximum value can be evaluated.

Used view: "EVM"

Settings: statistic count = 100, number of preambles = 1, repetition = single shot

Example: Multi-preamble measurement with irregular periodicity

Scenario: The EVM is measured for a sequence of 5 preambles with a periodicity of 20 ms. But the third and fourth preambles are missing (not transmitted by UE).

Used view: "EVM vs Preamble"

Settings: statistic count = 1, number of preambles = 5, period of preambles = 20 ms, repetition = single shot

Results: The result diagram and table contain the labels 1 to 5 as for a measurement without missing preambles. For the labels 3 and 4, there are no demodulation results.

6.2.4 Parallel signaling and measurement

The PRACH measurement can be used in parallel to the LTE signaling application (option R&S CMW-KS500/-KS550). The signaling application emulates an LTE cell signal so that the UE tries to attach and sends random access preambles. These preambles can then be measured using the PRACH measurement. You can configure the signaling application so that it does not answer the received preambles.

To use both applications in parallel, the combined signal path scenario must be activated (see ["Scenario = Combined Signal Path"](#) on page 1042). Most signal routing and analyzer settings and some measurement control settings are then configured by the signaling application. The PRACH measurement displays the corresponding signaling settings instead of its own settings. These signaling settings can be configured both in the measurement GUI and in the GUI of the signaling application.

To configure these settings via remote commands, the commands of the signaling application must be used. For a command-mapping table, see [Section 6.5.4, "Combined signal path commands"](#), on page 1377.

The most important signaling parameters not relevant for standalone measurements can nevertheless be configured both in the measurement GUI and in the GUI of the signaling application. In the measurement GUI, they can be accessed via hotkeys.

Whenever the combined signal path scenario is activated or the controlling application is changed, the trigger source for the measurement is set automatically. The PRACH trigger signal provided by the controlling signaling application is used.

6.2.5 Calculation of modulation results

Modulation results are based on a comparison of an ideal reference waveform with the measured signal. The measured signal is corrected by the average frequency and timing offset for each measured subframe.

3GPP TS 36.101, annex F, specifies the following additional conditions:

- Before calculating the EVM, the I/Q origin offset is removed from the evaluated signal.
- The EVM calculation is based on two different FFT processing windows in the time domain, separated by the "EVM window length". The minimum test requirement applies to the larger of the two obtained EVM values.

The two FFT processing windows start in the cyclic prefix (CP) at the beginning of the preamble. The cyclic prefix is a copy of the end of the preamble sequence.

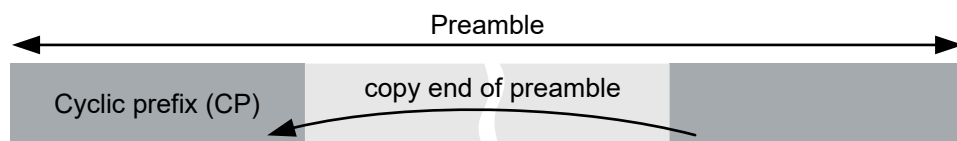


Figure 6-1: Cyclic extension of preamble

The EVM window between the start of the two FFT processing windows is centered on the cyclic prefix. The EVM window length is specified in the standard, depending on the preamble format (see 3GPP TS 36.101, table F.5.5-1).

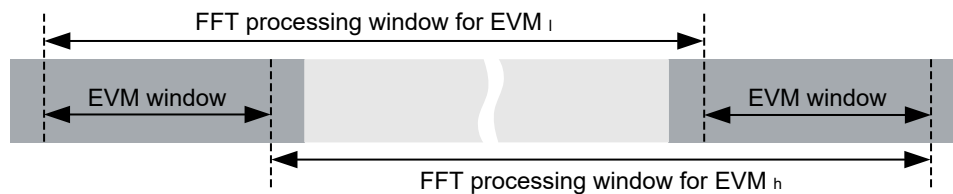


Figure 6-2: EVM window definition

The EVM for a signal with good modulation accuracy is expected to be largely independent of the EVM window size. Differences between EVM_h and EVM_l are caused, for example, by the effects of time domain windowing of FIR pulse shaping.



EVM window settings

Use the [EVM Window Length](#) settings to adjust the length of the EVM window. The minimum value of one FFT symbol means that EVM_h equals EVM_l .

Use the setting [EVM Window Position](#) to select whether the result diagrams (traces) are calculated for low or high EVM window position. Statistical tables provide results for both window positions.

6.2.6 LTE PRACH UL signal properties

This section describes the following selected topics related to LTE UL PRACH signal properties. For a description of the frequency bands, see [Section 5.2.11.4, "Frequency bands"](#), on page 1012.

- [Preambles in the frequency domain](#).....1309
- [Preambles in the time domain](#).....1310
- [Preamble sequences](#).....1311

6.2.6.1 Preambles in the frequency domain

There are five preamble formats defined by 3GPP, see table below. For preamble format 0 to 3, a preamble occupies 839 adjacent subcarriers with a spacing of 1.25 kHz, in total 1.04875 MHz. For preamble format 4, there are 139 subcarriers with a spacing of 7.5 kHz, in total 1.0425 MHz. In both cases, six resource blocks (RB) are reserved in the frequency domain for the preamble.

Table 6-1: Preambles in the frequency domain

Preamble format	Subcarriers (SC)	SC spacing
0 to 3	839	1.25 kHz
4	139	7.5 kHz

The location of the 6 RBs within the channel bandwidth, i.e. the number of the first RB to be used, is determined by higher layers. For FDD, it equals a parameter called PRACH frequency offset, for TDD it is calculated from this parameter and the parameter frequency resource index (determined by the PRACH configuration index). For

details, see 3GPP TS 36.211 section 5.7.1 "Time and frequency structure". Both the PRACH frequency offset and the PRACH configuration index can be configured for the PRACH measurement.

The total number of resource blocks available per channel depends on the channel bandwidth as shown in the following table.

Channel bandwidth / MHz	1.4	3	5	10	15	20
Resource blocks (RB)	6	15	25	50	75	100

6.2.6.2 Preambles in the time domain

In the time domain, a preamble consists of a cyclic prefix (CP) and the preamble sequence.

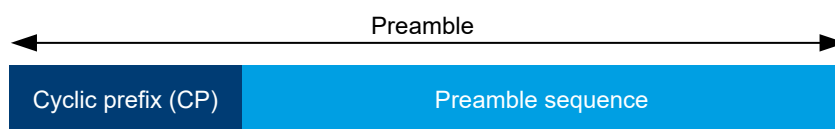


Figure 6-3: Preamble structure

The lengths of CP and preamble sequence depend on the preamble format as shown in the following table. T_s is the basic LTE time unit ($1 T_s = 1/30.72 \text{ MHz} \approx 32.55 \text{ ns}$).

The first multiplier in the listed sequence length indicates the number of OFDM symbols per preamble sequence (1 or 2).

Table 6-2: Preambles in the time domain

Preamble format	CP length / T_s	Sequence length / T_s	Preamble length / T_s
0	3168	1*24576	27744
1	21024	1*24576	45600
2	6240	2*24576	55392
3	21024	2*24576	70176
4	448	1*4096	4544

Thus a preamble with format 0 to 3 requires 2 to 6 UL timeslots for transmission. A preamble with format 4 is only relevant for TDD and is transmitted in uplink pilot timeslots (UpPTS).

The time resources allowed for transmission of preambles are restricted by 3GPP. There are 64 configurations for FDD and 58 configurations for TDD, identified via the PRACH configuration index. The index defines which radio frames, subframes etc. can be used and which preamble format is used. For details, refer to 3GPP TS 36.211 section 5.7.1.

6.2.6.3 Preamble sequences

Random access preambles are generated from Zadoff-Chu sequences with zero correlation zone. For each cell, a set of 64 preamble sequences is generated from one or several root Zadoff-Chu sequences via cyclic shift. The UE decides which of these 64 preamble sequences is used for the PRACH procedure.

A logical root sequence index is broadcasted within the system information as RACH_ROOT_SEQUENCE. The UE uses the logical root sequence index to determine the physical root sequence number u as defined in table 5.7.2-4 and 5.7.2-5 in 3GPP TS 36.211.

From the u^{th} root Zadoff-Chu sequence, random access preambles with zero correlation zones of length $N_{\text{CS}}-1$ are generated by cyclic shift. The parameter N_{CS} depends on the parameters "Zero Correlation Zone Config" and "High Speed Flag", both provided by higher layers. It is also used for the calculation of the cyclic shift.

If the root sequence allows less than 64 cyclic shifts, the next root sequence is used to generate additional preambles via cyclic shift. This process is repeated until the set of 64 preambles is complete.

To allow the R&S CMW to determine the used preamble sequence, the parameters "Logical Root Sequence Index" and "Zero Correlation Zone Config" must be configured correctly. In this software version, the "High Speed Flag" is fixed so that only the unrestricted set of N_{CS} values can be used. Which of the 64 preamble sequences has been selected by the UE can be determined automatically by the R&S CMW or configured manually as "Sequence Index".

For a detailed mathematical description of preamble sequence generation, refer to 3GPP TS 36.211 section 5.7.2 "Preamble sequence generation".

6.3 GUI reference

The following sections provide detailed reference information on the graphical user interface (GUI) of the LTE PRACH measurement.

To configure measurement settings, select the "PRACH" tab and press the hotkey "Config". It opens the main configuration dialog box. Some settings are also accessible via softkeys and hotkeys.

The measurement results are presented on the "PRACH" tab.

• Measurement control	1312
• Measurement results	1312
• Signal routing and analyzer settings	1322
• Measurement control settings	1323
• Trigger settings	1329
• Limit settings	1331
• Shortcut configuration	1333
• Additional softkeys and hotkeys	1334

6.3.1 Measurement control

To turn the measurement on or off, select the control softkey and press [ON | OFF] or [RESTART | STOP]. Alternatively, right-click the control softkey.



PRACH (Softkey)

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

`INITiate:LTE:MEAS<i>:PRACH`

`STOP:LTE:MEAS<i>:PRACH`

`ABORT:LTE:MEAS<i>:PRACH`

`FETCH:LTE:MEAS<i>:PRACH:STATE?`

`FETCH:LTE:MEAS<i>:PRACH:STATE:ALL?`

6.3.2 Measurement results

The results of the LTE PRACH measurement are displayed in several different views. The results are described in the following sections.

• Overview.....	1312
• View EVM, Magnitude Error, Phase Error.....	1313
• View EVM vs Preamble, Power vs Preamble.....	1314
• View EVM vs Symbol.....	1316
• View I/Q Constellation.....	1317
• View Power Dynamics.....	1318
• View TX Measurement.....	1319
• Statistical results.....	1321
• Detected configuration and general statistical information.....	1321
• Selecting and modifying views.....	1322
• Using markers.....	1322

6.3.2.1 Overview

Access: "PRACH" tab > "Display" softkey > "Select View" hotkey > "Overview".

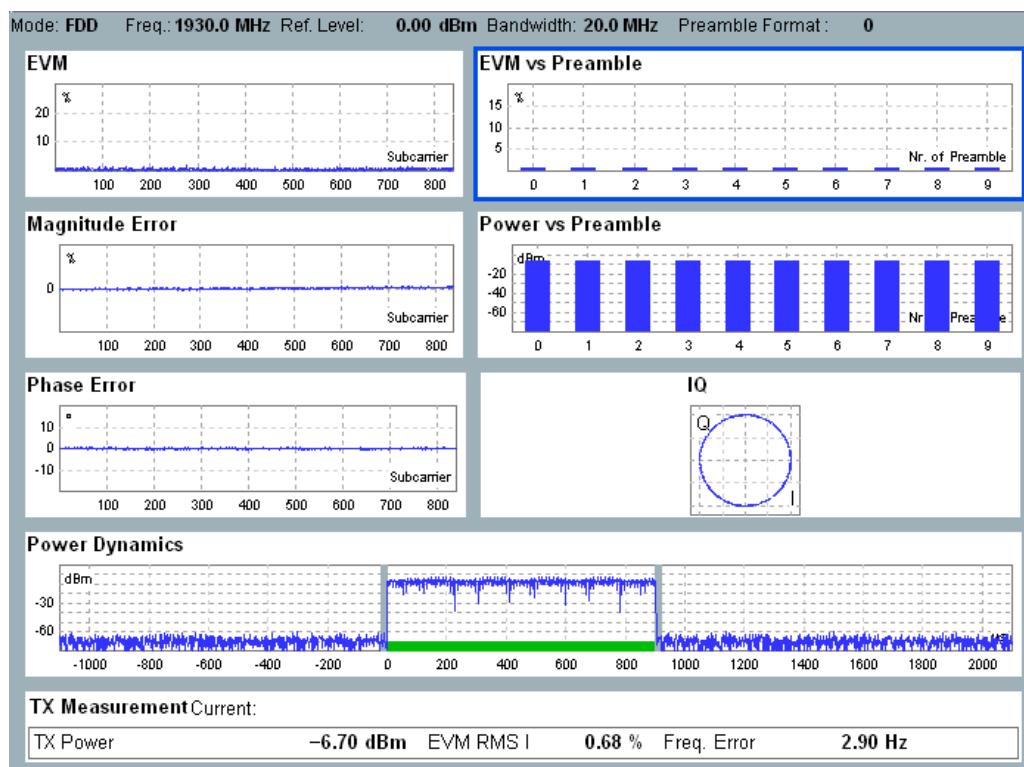


Figure 6-4: PRACH result overview

You can limit the results to be measured and displayed via the hotkey "Assign Views", see [Section 5.3.9, "Additional softkeys and hotkeys"](#), on page 1086.

You can enlarge one of the diagrams in the overview and show a detailed view with additional measurement results, see [Section 6.3.2.10, "Selecting and modifying views"](#), on page 1322. The traces are described in the following sections.

6.3.2.2 View EVM, Magnitude Error, Phase Error

Access: "PRACH" tab > "Display" softkey > "Select View" hotkey > "EVM" / "MErr" / "PhErr".

The views are single-preamble views. They show a diagram and a statistical overview of the results.

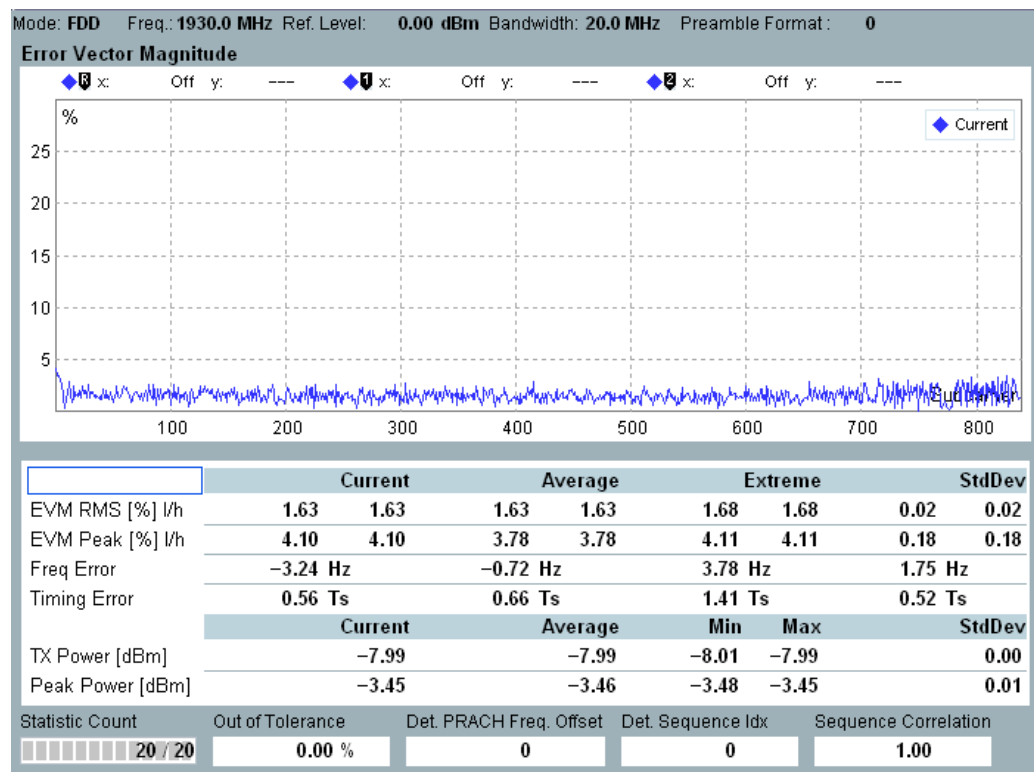


Figure 6-5: Error Vector Magnitude view

The diagram shows the EVM / magnitude error / phase error in all preamble subcarriers. For preamble format 0 to 3, there are 839 subcarriers with a spacing of 1.25 kHz. For preamble format 4, there are 139 subcarriers with a spacing of 7.5 kHz.

You can select which OFDM symbol of the preamble is measured and for which EVM window position results are displayed, see [Section 6.3.4.2, "Modulation settings"](#), on page 1326.

For a description of the statistical overview, see [Section 6.3.2.7, "View TX Measurement"](#), on page 1319.

For the parameters at the bottom, see [Section 6.3.2.9, "Detected configuration and general statistical information"](#), on page 1321.

For query of the diagram contents via remote control, see:

- [Section 6.5.3.10, "EVM vs subcarrier results \(diagrams\)"](#), on page 1364
- [Section 6.5.3.11, "Magnitude error results \(diagrams\)"](#), on page 1365
- [Section 6.5.3.12, "Phase error results \(diagrams\)"](#), on page 1365

6.3.2.3 View EVM vs Preamble, Power vs Preamble

Access: "PRACH" tab > "Display" softkey > "Select View" hotkey > "EVM vs Preamble" / "Power vs Preamble".

The views are multi-preamble views. They show a diagram and a statistical overview of the results.

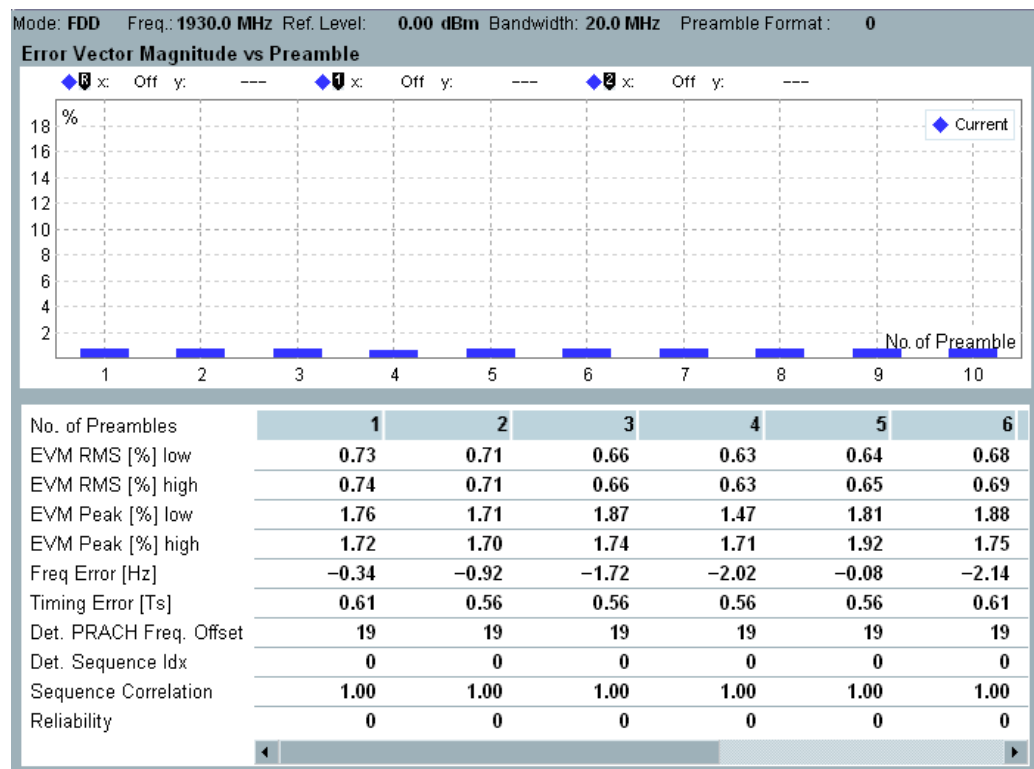


Figure 6-6: EVM vs Preamble view

Result diagram

The diagram shows results for a configurable number of preambles, labeled 1 to n. The preambles are captured within one measurement interval, using a configurable periodicity. A preamble is captured every 10 ms or every 20 ms.

- "Error Vector Magnitude vs Preamble":
The diagram displays EVM RMS values. For each preamble, either the result determined for low EVM window position or the result determined for high EVM window position is shown - whichever is the higher. So for this diagram, the setting [EVM Window Position](#) is ignored.
- "Power vs Preamble":
The diagram displays TX power values (received power, RMS averaged over all samples within the preamble duration).

For query of the diagram contents via remote control, see:

- [Section 6.5.3.13, "EVM vs preamble results \(diagrams\)"](#), on page 1366
- [Section 6.5.3.15, "Power vs preamble results \(diagrams\)"](#), on page 1368

Result table

The table below the diagram shows additional results. Each column corresponds to one preamble with the preamble label indicated in the column title.

The upper rows correspond to the results presented in the "TX Measurement" view. For a description, see [Section 6.3.2.7, "View TX Measurement"](#), on page 1319.

For a description of the lower rows, see [Section 6.3.2.9, "Detected configuration and general statistical information"](#), on page 1321.

The row "Reliability" provides a reliability indicator for each preamble. For a description of the individual values, see [Section 6.5.1.5, "Reliability indicator"](#), on page 1340.

For query of the table contents via remote control, see [Section 6.5.3.19, "EVM/power vs preamble results \(single values\)"](#), on page 1374.

6.3.2.4 View EVM vs Symbol

Access: "PRACH" tab > "Display" softkey > "Select View" hotkey > "EVM vs Symbol".

The view is a single-preamble view. It shows a diagram and a statistical overview of the results.

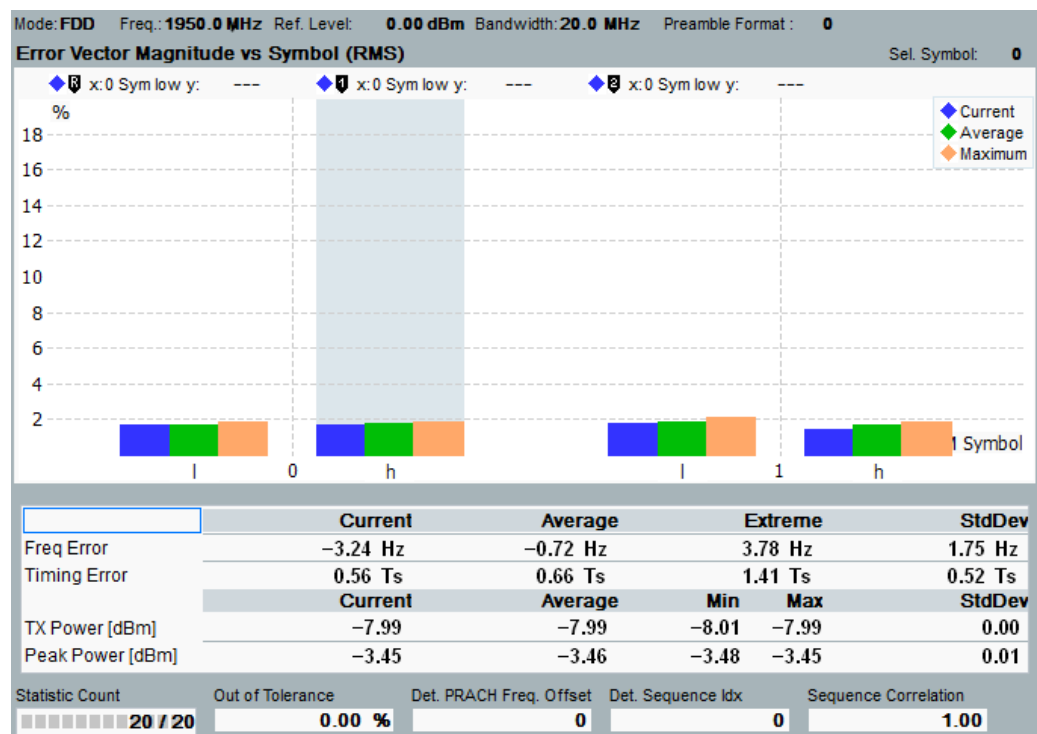


Figure 6-7: EVM vs Symbol view

The diagram shows the EVM RMS value per OFDM symbol of the preamble. Depending on the preamble format, there are one or two OFDM symbols in the preamble, see [Preambles in the time domain](#). The x-axis covers two OFDM symbols and result bars are only displayed for the used OFDM symbols (one or two).

The measurement calculates two values (EVM_l and EVM_h) for each OFDM symbol and displays both values. See also [Calculation of modulation results](#).

The selected OFDM symbol and EVM window position are indicated by a vertical bar. This selection has an impact on other modulation views.

For a description of the statistical overview, see [Section 6.3.2.7, "View TX Measurement"](#), on page 1319.

For the parameters at the bottom, see [Section 6.3.2.9, "Detected configuration and general statistical information"](#), on page 1321.

For query of the diagram contents via remote control, see [Section 6.5.3.14, "EVM vs symbol \(diagrams\)"](#), on page 1366.

6.3.2.5 View I/Q Constellation

Access: "PRACH" tab > "Display" softkey > "Select View" hotkey > "IQConst".

The view is a single-preamble view. It shows the modulation symbols as points in the I/Q plane.

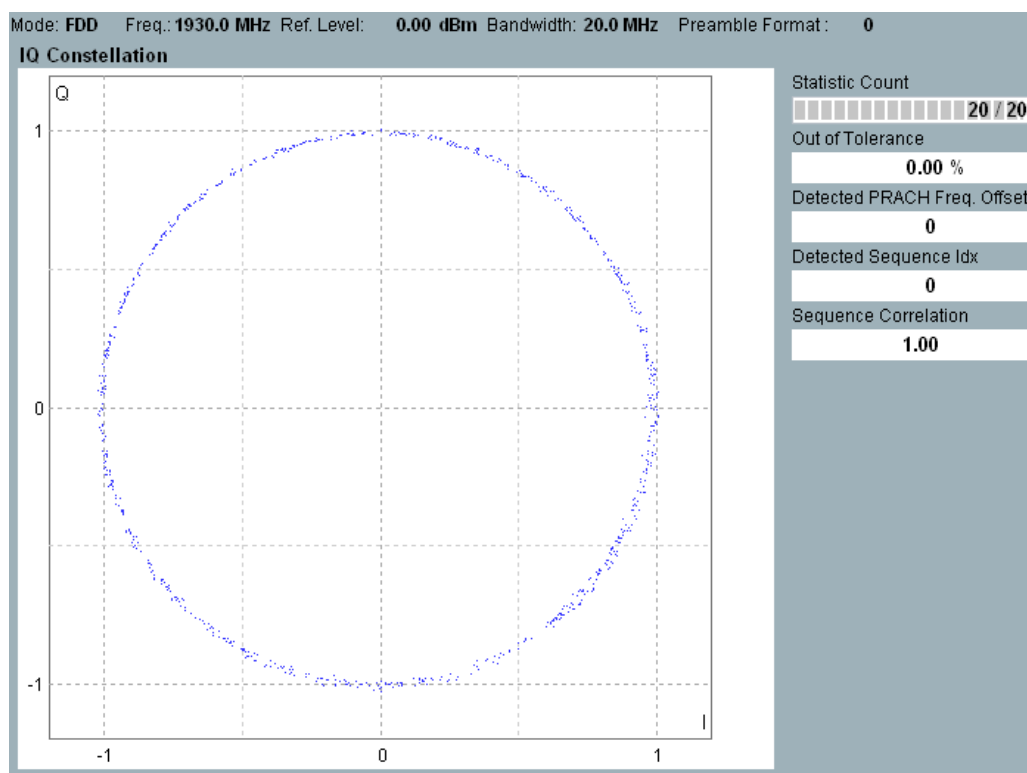


Figure 6-8: I/Q constellation diagram

The typical constellation diagram of a random access preamble shows 839 points (preamble format 4: 139 points) evenly distributed on a circle around the origin. The I/Q amplitudes correspond to the values that the R&S CMW uses for the EVM calculation: A possible I/Q origin offset is already subtracted out.

You can select which OFDM symbol of the preamble is measured and for which EVM window position results are displayed, see [Section 6.3.4.2, "Modulation settings"](#), on page 1326.

For the parameters on the right, see [Section 6.3.2.9, "Detected configuration and general statistical information"](#), on page 1321.

For query of the diagram contents via remote control, see [Section 6.5.3.16, "I/Q constellation results"](#), on page 1369.

6.3.2.6 View Power Dynamics

Access: "PRACH" tab > "Display" softkey > "Select View" hotkey > "Power Dynamics".

The view is a single-preamble view. It shows a diagram and a statistical overview of related power results.



Figure 6-9: Power Dynamics view

The diagram shows the UE output power vs time, sampled with $48 T_s$ ($1.5625 \mu s$). The trace covers the range from $-1100 \mu s$ to $+2098.4375 \mu s$ relative to the start of the preamble.

The table below the diagram shows ON power and OFF power values. According to 3GPP, the ON power is the mean UE output power over the preamble, i.e. over the PRACH measurement period - see [Table 6-3](#). In addition to this value, the table also lists the peak power within the measurement period. The ON power measurement period is marked by a green horizontal bar in the diagram.

The OFF power is the mean power in one subframe (1 ms) before and after the ON power measurement period. A transient period of $20 \mu s$ next to the ON power measurement period is excluded. The transient period is indicated by gray vertical bars in the diagram.

Table 6-3: Measurement period for ON power

Preamble format	Measurement period ON power (µs)
0	903.1
1	1484.4
2	1803.1
3	2284.4
4	147.9

**Signal configuration**

Ensure that the two subframes before the preamble and the two subframes after the preamble are off (OFF, OFF, preamble, OFF, OFF). This configuration guarantees that the "OFF Power (before)" result is not falsified by power contributions of a preceding ON subframe ramped down too slowly. And that the "OFF Power (after)" is not falsified by a subsequent ON subframe ramped up too early.

You can check the first 100 µs of the trace to ensure that the second subframe before the preamble is off. For preamble formats 0 and 4, you can also check the end of the trace to ensure that the second subframe after the preamble is off.

For the parameters at the bottom of the view, see [Section 6.3.2.9, "Detected configuration and general statistical information"](#), on page 1321.

For query of the power dynamics results via remote control, see:

- [Section 6.5.3.17, "Power dynamics results \(diagrams\)"](#), on page 1369
- [Section 6.5.3.20, "Power dynamics results \(single values\)"](#), on page 1376

6.3.2.7 View TX Measurement

Access: "PRACH" tab > "Display" softkey > "Select View" hotkey > "Modulation Scalar".

The view is a single-preamble view. It contains tables of statistical results for the PRACH measurement.

Mode: FDD Freq.: 1930.0 MHz Ref. Level: 0.00 dBm Bandwidth: 20.0 MHz Preamble Format: 0

TX Measurement								
	Current		Average		Extreme		StdDev	
EVM RMS [%] I/h	1.62	1.62	1.63	1.63	1.68	1.68	0.02	0.02
EVM Peak [%] I/h	3.73	3.73	3.84	3.84	4.46	4.46	0.17	0.17
MErr RMS [%] I/h	1.22	1.22	1.22	1.22	1.24	1.24	0.01	0.01
MErr Peak [%] I/h	-3.67	-3.67	3.79	3.79	-4.45	-4.45	0.17	0.17
PhErr RMS [°] I/h	0.61	0.61	0.62	0.62	0.67	0.67	0.01	0.01
PhErr Peak [°] I/h	1.54	1.54	1.50	1.50	1.81	1.81	0.11	0.11
Freq Error	0.00 Hz		-1.03 Hz		-4.65 Hz		1.53 Hz	
Timing Error	0.19 Ts		0.70 Ts		1.41 Ts		0.47 Ts	
	Current		Average		Min		Max	
TX Power [dBm]	-8.00		-8.00		-8.00		-7.99	
Peak Power [dBm]	-3.48		-3.46		-3.48		-3.45	
Statistic Count: 20 / 20 Out of Tolerance: 0.00 % Det. PRACH Freq. Offset: 0 Det. Sequence Idx: 0 Sequence Correlation: 1.00								

Figure 6-10: TX measurement

The table provides an overview of statistical values obtained from the measured preamble. Other detailed views provide a subset of these values.

- EVM, magnitude error ("MErr"), phase error ("PhErr"):
The R&S CMW calculates RMS and peak values for low and high EVM window position (I/h); see [Section 6.2.5, "Calculation of modulation results"](#), on page 1308.
- Carrier frequency error: offset between the measured carrier frequency and the nominal RF frequency of the measured radio channel
- Timing error: difference between the actual timing and the expected timing
The timing error measurement requires a trigger signal from another application or instrument, to derive the expected timing.
A suitable trigger signal is, for example, the PRACH trigger signal provided by the signaling application.
The unit T_s stands for the basic LTE time unit, see [Section 6.2.6.2, "Preambles in the time domain"](#), on page 1310.
- TX power: received power, RMS averaged over all samples within the preamble duration
- Peak power: peak power of all samples within the preamble duration

For additional information, see [Section 6.3.2.9, "Detected configuration and general statistical information"](#), on page 1321.

For query of the table contents via remote control, see [Section 6.5.3.18, "Modulation results \(single values\)"](#), on page 1370.

6.3.2.8 Statistical results

Most result diagrams and result tables provide statistical results.

The statistical values are calculated as follows:

- **"Current"**: Result obtained in the last measurement interval. For some modulation results, an RMS value and a peak value are available. They are calculated as the average and peak of all samples in the measured preamble.
- **"Average"**: Weighted average of all "Current" values since the start of the measurement.
- **"Extreme", "Min", "Max"**: Largest or smallest "Current" value since the start of the measurement.
- **"Std Dev"**: Standard deviation of all "Current" values since the start of the measurement.

6.3.2.9 Detected configuration and general statistical information

The following information is available in most result views.

Statistic Count.....	1321
Out of Tolerance.....	1321
Detected PRACH Frequency Offset.....	1321
Detected Sequence Index.....	1321
Sequence Correlation.....	1322

Statistic Count

Progress bar for the measurement. During the first single shot after the start of the measurement, the bar shows the number of elapsed preambles relative to the configured "Statistic Count". A filled progress bar indicates that the first shot is complete and the statistical depth has been reached.

Out of Tolerance

Percentage of measurement intervals that were failed because at least one result exceeded the limits.

All results that use the same statistic count setting contribute to the same "Out of Tolerance" result.

Detected PRACH Frequency Offset

The PRACH frequency offset determines the position of the six preamble RBs within the channel bandwidth. It is either detected automatically or selected manually.

See also ["PRACH Frequency Offset"](#) on page 1325.

Detected Sequence Index

The sequence index identifies which of the 64 preamble sequences defined for the cell is used by the UE. It is either detected automatically or selected manually.

See also ["Sequence Index"](#) on page 1327.

Sequence Correlation

The sequence correlation indicates the correlation between the ideal preamble sequence determined from the parameter settings and the measured preamble sequence. A value of 1 corresponds to perfect correlation. A value close to 0 indicates that the preamble sequence was not found. In the latter case, check the parameter settings related to the preamble, see also [Section 6.2.2, "How to measure an uplink PRACH signal"](#), on page 1305.

6.3.2.10 Selecting and modifying views

Press the "Display" softkey to show hotkeys for view configuration. They change the appearance and contents of the active view. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Select View ..."	Switch to a certain detailed view or overview. Alternatively, select a diagram in the overview and press [ENTER] or the rotary knob.
"Select Trace ..."	Select the trace types to be displayed in the view.
"X Scale... / Y Scale..."	Modify the ranges of the x-axis and the y-axis.

6.3.2.11 Using markers

Press the "Marker" softkey to show hotkeys for marker configuration. They activate markers and modify their position. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Ref. Marker ..."	Enable or disable the reference marker and select the marker position. If several traces can be displayed, a trace can also be selected.
"Marker 1 / 2 ..."	Enable or disable marker 1 or 2 and define the marker position (absolute or relative to the reference marker). Depending on the trace mode, a trace can also be selected.
"Select Trace Mode"	Define whether markers 1 and 2 are set to the same trace as the reference marker (collective) or to selectable individual traces.

6.3.3 Signal routing and analyzer settings

Access: "PRACH" tab > "Config" hotkey.

The following parameters configure the RF input path. All parameters are common measurement settings. They have the same value in all measurements (for example, PRACH measurement and multi-evaluation measurement).

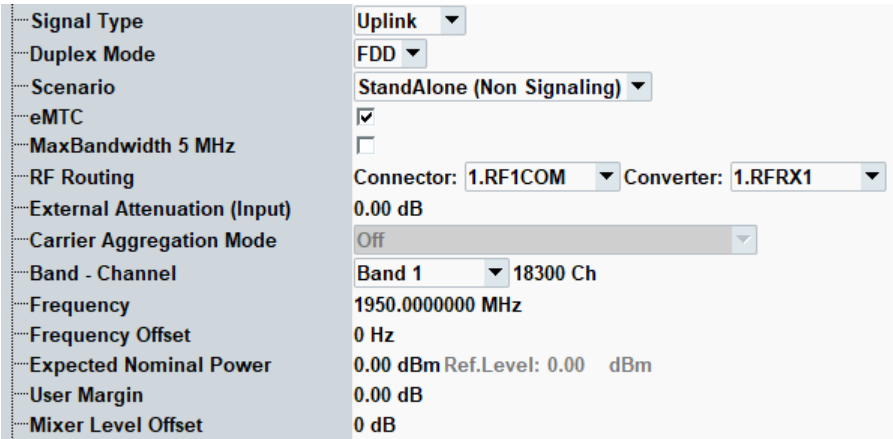


Figure 6-11: Signal routing and analyzer settings

For a description of the settings, see [Section 5.3.3, "Signal routing and analyzer settings"](#), on page 1040.

6.3.4 Measurement control settings

The "Measurement Control" parameters configure the scope of the LTE PRACH measurement and inform the measurement about preamble-related signal properties normally determined by higher layers.

- [General measurement control settings](#)..... 1323
- [Modulation settings](#)..... 1326
- [Power settings](#)..... 1328

6.3.4.1 General measurement control settings

Access: "PRACH" tab > "Config" hotkey > "Measurement Control" node.

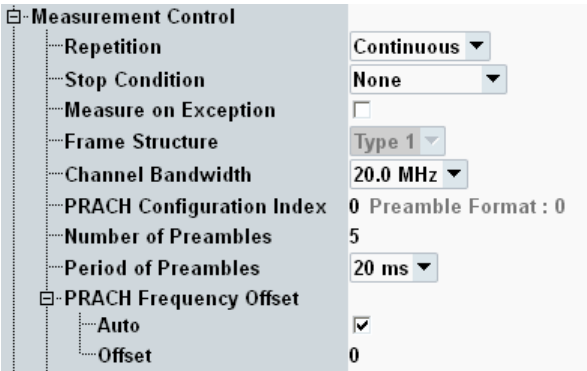


Figure 6-12: General measurement control settings

- [Repetition](#)..... 1324
- [Stop Condition](#)..... 1324
- [Measure on Exception](#)..... 1324

Frame Structure.....	1324
Channel Bandwidth.....	1325
PRACH Configuration Index, Preamble Format.....	1325
Number of Preambles.....	1325
Period of Preambles.....	1325
PRACH Frequency Offset.....	1325

Repetition

Defines how often the measurement is repeated if it is not stopped explicitly or by a failed limit check.

- **"Continuous"**: The measurement is continued until it is explicitly terminated. The results are periodically updated.
- **"Single Shot"**: The measurement is stopped after one measurement cycle.

"Single Shot" is preferable if only a single measurement result is required under fixed conditions, which is typical for remote-controlled measurements. The continuous mode is suitable to monitor the evolution of the measurement results and to observe how they depend on the measurement configuration, which is typically done in manual control.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:REPetition`

Stop Condition

Specifies the conditions for an early termination of the measurement:

- **"None"**: The measurement is performed according to its "Repetition" mode and "Statistic Count", irrespective of the limit check results.
- **"On Limit Failure"**: The measurement is stopped when one of the limits is exceeded, irrespective of the repetition mode set. If no limit failure occurs, it is performed according to its "Repetition" mode and "Statistic Count". Use this setting for measurements that are intended for checking limits, e.g. production tests.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:SCONdition`

Measure on Exception

Specifies whether results that the measurement identifies as faulty or inaccurate are rejected. A faulty result occurs, for example, when an overload is detected. In remote control, the cause of the error is indicated by the "reliability indicator".

- **Off**: Faulty results are rejected. The measurement is continued. The statistical counters are not reset. Using this mode ensures that a single faulty result does not affect the entire measurement.
- **On**: Results are never rejected. Use this mode for development purposes, if you want to analyze the reasons for occasional wrong transmissions.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:MOEXception`

Frame Structure

Displays the frame structure of the uplink signal as defined in 3GPP TS 36.211. The value is set implicitly via the [Duplex Mode](#) ("Type 1" = FDD, "Type 2" = TDD).

Remote command:

`CONFigure:LTE:MEAS<i>:FSTRucture?`

Channel Bandwidth

Channel bandwidth in the range 1.4 MHz to 20 MHz. Set the bandwidth in accordance with the measured LTE signal.

The parameter is a common measurement setting, i.e. it has the same value in all measurements (e.g. PRACH measurement and multi-evaluation measurement).

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFigure:LTE:MEAS<i>:CC<no>:CBANdwidth (SA)`

`CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL (CSP)`

PRACH Configuration Index, Preamble Format

Specifies the PRACH configuration used by the UE. The PRACH configuration defines the preamble format and other signal properties, e.g. which resources in the time domain are allowed for transmission of preambles. The resulting preamble format is displayed for information.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

For background information, see [Section 6.2.6, "LTE PRACH UL signal properties"](#), on page 1309.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:PCINdex (SA)`

`CONFigure:LTE:SIGN<i>:CELL:PRACH:PCINdex:FDD (CSP)`

`CONFigure:LTE:SIGN<i>:CELL:PRACH:PCINdex:TDD (CSP)`

Number of Preambles

Specifies the number of preambles to be captured per measurement interval for multi-preamble result views, e.g. "EVM vs Preamble" and "Power vs Preamble".

Single-preamble views evaluate only one preamble per measurement interval. To achieve maximum measurement speed for these views, set this parameter to 1.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:NOPReambles`

Period of Preambles

Specifies the periodicity of preambles to be captured for multi-preamble result views.

See also [Section 6.2.3, "Defining the scope of the measurement"](#), on page 1306.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:POPReambles`

PRACH Frequency Offset

Enable "Auto" to detect the frequency offset automatically or disable "Auto" and specify the frequency offset used by the UE. A manual configuration of the frequency offset can speed up the measurement.

The frequency offset is required to calculate the location of the six preamble resource blocks (RB) within the channel bandwidth. For FDD, the number of the first RB equals the offset. For TDD, it is calculated from this parameter and the parameter frequency resource index as described in 3GPP TS 36.211 section 5.7.1 "Time and frequency structure".

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

For background information, see [Section 6.2.6.1, "Preambles in the frequency domain"](#), on page 1309.

Remote command:

CONFigure:LTE:MEAS<i>:PRACH:PFOffset:AUTO

CONFigure:LTE:MEAS<i>:PRACH:PFOffset (SA)

CONFigure:LTE:SIGN<i>:CELL:PRACH:PFOffset (CSP)

6.3.4.2 Modulation settings

Access: "PRACH" tab > "Config" hotkey > "Measurement Control" node > "Modulation" node.

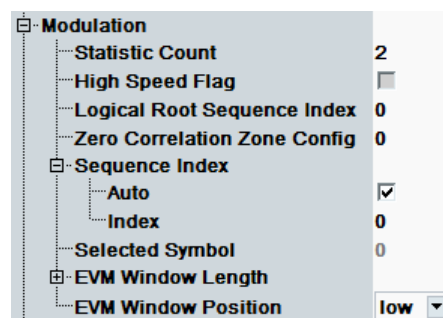


Figure 6-13: Modulation settings

Selected Symbol	1326
Statistic Count	1327
Logical Root Sequence Index	1327
Zero Correlation Zone Config	1327
Sequence Index	1327
EVM Window Length	1328
EVM Window Position	1328

Selected Symbol

Selects the OFDM symbol to be evaluated for modulation result diagrams (EVM, magnitude error, phase error, I/Q diagram).

The setting has no effect on result tables, EVM vs preamble results and EVM vs symbol results. These results consider all OFDM symbols of the preamble.

The setting is only configurable if the preamble contains two symbols (preamble format two and three, see [Preambles in the time domain](#)).

See also settings [PRACH Configuration Index](#), [Preamble Format](#).

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:SSymbol`

Statistic Count

Defines the number of measurement intervals per measurement cycle. A measurement cycle is the duration of a single-shot measurement.

For single-preamble modulation measurements, the measurement interval is completed when the R&S CMW has measured one full preamble (cyclic prefix + preamble sequence). The preamble length depends on the preamble format, see [Section 6.2.6.2, "Preambles in the time domain"](#), on page 1310.

For multi-preamble measurements, the measurement interval is completed when the configured number of preambles has been measured, see ["Number of Preambles"](#) on page 1325.

The measurement provides independent statistic counts for modulation results and power dynamics results. In single-shot mode and with a shorter modulation statistic count, the modulation evaluation is stopped while the R&S CMW still continues providing new power dynamics results.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:SCount:MODulation`

Logical Root Sequence Index

Specifies the logical root sequence index to be used for generation of the preamble sequence. Set this parameter to the value used by the UE.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

For background information, see [Section 6.2.6.3, "Preamble sequences"](#), on page 1311.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:MODulation:LRSindex` (SA)

`CONFigure:LTE:SIGN<i>:CELL:PRACH:LRSindex` (CSP)

Zero Correlation Zone Config

This parameter determines which N_{CS} value of an N_{CS} set is used for generation of the preamble sequence. Set this parameter to the value used by the UE.

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

For background information, see [Section 6.2.6.3, "Preamble sequences"](#), on page 1311.

Remote command:

`CONFigure:LTE:MEAS<i>:PRACH:MODulation:ZCZConfig` (SA)

`CONFigure:LTE:SIGN<i>:CELL:PRACH:ZCZConfig` (CSP)

Sequence Index

Enable "Auto" to detect the sequence index automatically or disable "Auto" and specify the sequence index used by the UE. A manual configuration of the sequence index can speed up the measurement.

The sequence index specifies which of the 64 preamble sequences of the cell is used by the UE.

For background information, see [Section 6.2.6.3, "Preamble sequences"](#), on page 1311.

Remote command:

```
CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex:AUTO
CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex
```

EVM Window Length

Length of the EVM window in samples. The window length defines the two sets of "I / h" modulation quantities in the result tables. See [Section 6.2.5, "Calculation of modulation results"](#), on page 1308. The minimum value of one FFT symbol means that EVM_h equals EVM_l.

The standard (3GPP TS 36.101, annex F) specifies the window length as a function of the preamble format.

EVM Window Length					
Current	3072				
Preamble Format	0	1	2	3	4
EVM Window Length	3072	20928	6144	20928	432

The measurement uses the preamble format to select the relevant table cell. The "Current" setting is linked to this table cell.

Remote command:

```
CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWLength
```

EVM Window Position

Position of the EVM window used for calculation of the trace results. While the result tables show both the results for low and high EVM window position, the diagram results are only determined for the selected window position.

Remote command:

```
CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWPosition
```

6.3.4.3 Power settings

Access: "PRACH" tab > "Config" hotkey > "Measurement Control" node > "Power" node.

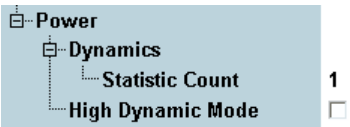


Figure 6-14: Power settings

Statistic Count	1329
High Dynamic Mode	1329

Statistic Count

Defines the number of measurement intervals per measurement cycle. A measurement cycle is the duration of a single-shot measurement.

For power dynamics measurements, the measurement interval is completed when the R&S CMW has measured one preamble plus the preceding and the following power OFF period.

The measurement provides independent statistic counts for modulation results and power dynamics results. In single-shot mode and with a shorter modulation statistic count, the modulation evaluation is stopped while the R&S CMW still continues providing new power dynamics results.

Remote command:
CONFigure:LTE:MEAS<i>:PRACH:SCount:PDYNamics

High Dynamic Mode

Enables or disables the high dynamic mode.

The high dynamic mode is suitable for power dynamics measurements involving high ON powers. In that case, the dynamic range of the R&S CMW is eventually not sufficient to measure both the high ON powers and the low OFF powers accurately.

In high dynamic mode, the dynamic range is increased by measuring the results in two shots. One shot uses the configured settings to measure the ON power. The other shot uses a lower "Expected Nominal Power" value to measure the OFF power results.

Disable the high dynamic mode to optimize the measurement speed. Disable it especially if you are not interested in power dynamics results or if you measure low ON powers using a low "Expected Nominal Power" setting.

While the combined signal path scenario is active, this parameter is not configurable.

Remote command:
CONFigure:LTE:MEAS<i>:PRACH:POWer:HDMode

6.3.5 Trigger settings

Access: "PRACH" tab > "Config" hotkey > "Trigger" node.

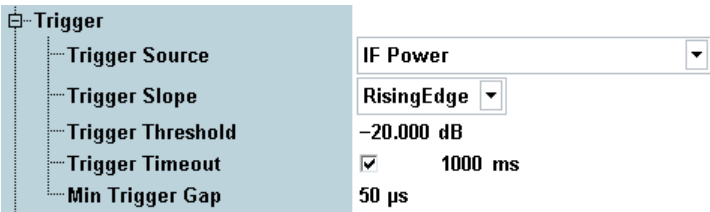


Figure 6-15: Trigger settings

Trigger Source.....	1330
Trigger Slope.....	1330
Trigger Threshold.....	1330
Trigger Timeout.....	1330
Min Trigger Gap.....	1330

Trigger Source

Selects the source of the trigger event.

Only the measurement-specific trigger sources are listed here. Additional trigger sources are provided by the base software and by other applications like the GPRF generator. For a description of these trigger sources, refer to the documentation of the corresponding software part.

"IF Power" The measurement is triggered by the power ramp of the received preambles. The minimum trigger gap must be large enough to ensure that the measurement is not retriggered within a preamble. It must be small enough to ensure that the trigger system is rearmed before the next preamble starts.

Remote command:

`TRIGger:LTE:MEAS<i>:PRACH:SOURce`

`TRIGger:LTE:MEAS<i>:PRACH:CATalog:SOURce?`

Trigger Slope

Qualifies whether the trigger event is generated at the rising or at the falling edge of the trigger pulse. This setting has no influence on the evaluation of trigger pulses provided by other firmware applications.

Remote command:

`TRIGger:LTE:MEAS<i>:PRACH:SLOPe`

Trigger Threshold

Defines the input signal power where a trigger event is generated for power trigger sources.

The threshold is a dB value, relative to the reference level minus the external attenuation. If the reference level equals the maximum output power of the DUT and the external attenuation settings are correct, the trigger threshold is relative to the maximum input power.

A low threshold ensures that the input signal is always detected. A higher threshold can prevent unintended trigger events.

Remote command:

`TRIGger:LTE:MEAS<i>:PRACH:THReshold`

Trigger Timeout

Sets a time after which an initiated measurement must have received a trigger event. If no trigger event is received, a trigger timeout is indicated in manual operation mode. In remote control mode, the measurement is automatically stopped. The parameter can be disabled so that no timeout occurs.

Remote command:

`TRIGger:LTE:MEAS<i>:PRACH:TOUT`

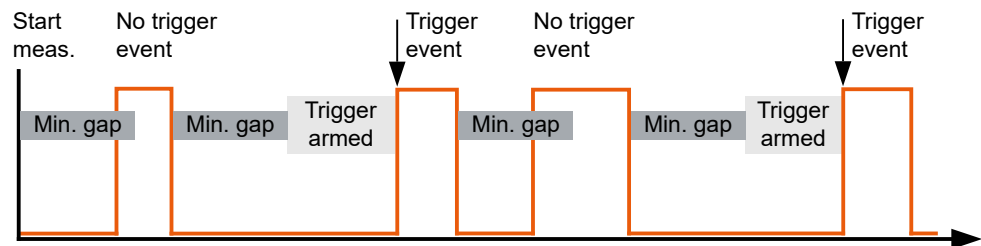
Min Trigger Gap

Defines a minimum duration of the power-down periods (gaps) between two triggered power pulses. This setting is valid for "(IF) Power" trigger sources.

The trigger system is controlled by a timer which is reset to zero in the following instances:

- At the IF power-down ramp of each triggered or untriggered pulse, even if the previous counter has not yet elapsed. A power-down ramp is detected when the signal power falls below the trigger threshold.
- At the beginning of each measurement. The minimum gap defines the minimum time between the start of the measurement and the first trigger event.

The trigger system is rearmed when the timer has reached the specified minimum gap.



This parameter can be used to prevent unwanted trigger events due to fast power variations within a preamble.

The minimum trigger gap must be large enough to ensure that the measurement is not retriggered within a preamble. It must be small enough to ensure that the trigger system is rearmed before the next preamble starts. The default value is often appropriate.

Remote command:

`TRIGger:LTE:MEAS<i>:PRACH:MGAP`

6.3.6 Limit settings

The following sections give an overview of the PRACH measurement limit settings and of the related conformance test requirements.

Conformance requirements for LTE transmitter tests are specified in 3GPP TS 36.521, section 6, "Transmitter Characteristics".

- [Transmit modulation limits](#)..... 1331
- [Power dynamics limits](#).....1332

6.3.6.1 Transmit modulation limits

A poor modulation accuracy of the UE transmitter increases the transmission errors in the uplink channel of the LTE network and decreases the system capacity. The error vector magnitude (EVM) is the critical quantity to assess the modulation accuracy of an LTE UE.

According to 3GPP, the EVM measured at UE output powers ≥ -40 dBm and under normal operating conditions must not exceed 17.5 % for random access preambles.

Characteristics	Refer to 3GPP TS 36.521 V9.3.0, section...	Specified limit
EVM (RMS)	6.5.2.1 Error Vector Magnitude (EVM)	≤ 17.5 %

Modulation limit settings

Access: "PRACH" tab > "Config" hotkey > "Limits" node > "Modulation" node.

The EVM limits can be set in the configuration dialog, along with limits for the other measured quantities.

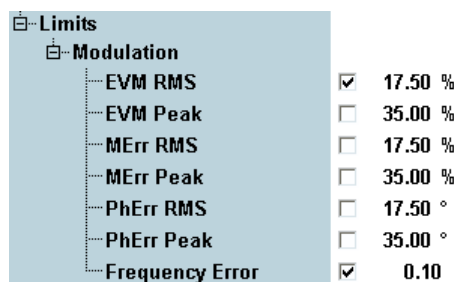


Figure 6-16: Modulation limit settings

Remote command:

```
CONFigure:LTE:MEAS<i>:PRACH:LIMit:EVMagnitude
CONFigure:LTE:MEAS<i>:PRACH:LIMit:MERRor
CONFigure:LTE:MEAS<i>:PRACH:LIMit:PERRor
CONFigure:LTE:MEAS<i>:PRACH:LIMit:FERRor
```

6.3.6.2 Power dynamics limits

Transmission at excessive uplink power increases interference to other channels while a too low uplink power increases transmission errors. For PRACH transmission, 3GPP defines a time mask. The mask contains limits for the UL power during preamble transmission (transmit ON power), the UL power in the adjacent subframes (transmit OFF power) and the power ramping in-between.

The ON power is specified as the mean UE output power over the preamble (more exactly, over the PRACH measurement period, see [Table 6-3](#)). For the OFF power, the mean power has to be measured both in the preceding subframe and in the subsequent subframe. A transient period of 20 μs next to the PRACH measurement period is excluded.

The following figure provides a summary of the time periods relevant for ON and OFF power limits.

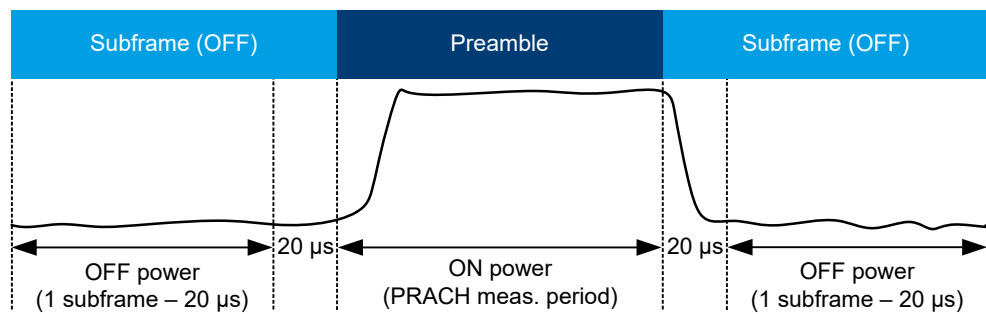


Figure 6-17: Measurement periods ON/OFF power

The OFF power must not exceed -48.5 dBm. The ON power must be in the range -8.5 dBm to 6.5 dBm.

Characteristics	Refer to 3GPP TS 36.521 V9.3.0, section...	Specified limit
ON power	6.3.4.2 PRACH and SRS Time Mask	≥ -8.5 dBm, ≤ 6.5 dBm
OFF power	6.3.4.2 PRACH and SRS Time Mask	≤ -48.5 dBm

Power dynamics limit settings

Access: "PRACH" tab > "Config" hotkey > "Limits" node > "Power" node.

The limits for ON and OFF power can be set in the configuration dialog.

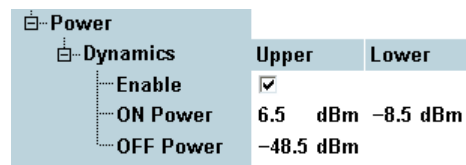


Figure 6-18: Power dynamics limit settings

Remote command:

`CONFfigure:LTE:MEAS<i>:PRACH:LIMit:PDYNamics`

6.3.7 Shortcut configuration

Access: "Config" hotkey > bottom of the tree.

This section configures a shortcut softkey that provides a fast way to access the GPRF generator from the measurement.

The setting is a common measurement setting. It has the same value in all measurements (for example, PRACH measurement and multi-evaluation measurement).



Figure 6-19: Shortcut configuration

Generator Shortcut

Selects an instance of the GPRF generator. Softkeys for the selected instance are added to the softkey panel.

6.3.8 Additional softkeys and hotkeys

See [Section 5.3.9, "Additional softkeys and hotkeys"](#), on page 1086.

6.4 Programming

The following sections provide programming examples for the LTE PRACH measurement.

• Key features	1334
• Specifying general and common measurement settings	1334
• Specifying required PRACH settings	1335
• Specifying measurement-specific settings	1335
• Configuring the trigger system	1336
• Specifying limits	1336
• Running a measurement	1337

6.4.1 Key features

The LTE PRACH measurement is programmed as follows:

- The measurement is controlled by SCPI commands with the following syntax: `...LTE:MEAS:PRACH...`
- Use general commands of the type `...:LTE:MEAS...` (no `:PRACH` mnemonic) to define the signal routing and to perform RF and analyzer settings.
- After a `*RST`, the measurement is switched off. Use `READ:LTE:MEAS:PRACH...?` to initiate a measurement and to retrieve the results. You can also start the measurement using `INIT:LTE:MEAS:PRACH` and retrieve the results using `FETCH:LTE:MEAS:PRACH...?`.
- For synchronization and proper decoding, some UE signal settings must be in accordance with the measured signal; see [Section 6.4.3, "Specifying required PRACH settings"](#), on page 1335.

6.4.2 Specifying general and common measurement settings

```
// *****
// Initial system-reset
// *****
*RST; *OPC?
*CLS; *OPC?
```

```
// *****
// Set duplex mode TDD and query the resulting frame structure type (T2).
// Specify channel bandwidth 1.4 MHz.
// *****
CONFigure:LTE:MEAS:DMODE TDD
CONFigure:LTE:MEAS:FSTRucture?
CONFigure:LTE:MEAS:CC:CBANdwidth B014

// *****
// Configure RF and analyzer settings:
// External attenuation 2 dB, carrier center frequency 1850 MHz,
// peak power 7 dBm, 5 dB user margin.
// *****
CONFigure:LTE:MEAS:RFSettings:EATTenuation 2
CONFigure:LTE:MEAS:RFSettings:CC:FREQuency 1850E+6
CONFigure:LTE:MEAS:RFSettings:ENPower 7
CONFigure:LTE:MEAS:RFSettings:UMARgin 5
```

6.4.3 Specifying required PRACH settings

```
// *****
// Specify required UE signal settings: PRACH configuration index 1,
// logical root sequence index 50, zero correlation zone config 3.
// *****
CONFigure:LTE:MEAS:PRACH:PCINdex 1
CONFigure:LTE:MEAS:PRACH:MODulation:LRSindex 50
CONFigure:LTE:MEAS:PRACH:MODulation:ZCZConfig 3
```

6.4.4 Specifying measurement-specific settings

```
// *****
// Define stop condition (stop on limit failure), statistic counts (20 cycles),
// error handling, number and period of preambles.
// Enable high dynamic mode.
// *****
CONFigure:LTE:MEAS:PRACH:SCONdition SLFail
CONFigure:LTE:MEAS:PRACH:SCount:MODulation 20
CONFigure:LTE:MEAS:PRACH:SCount:PDYNamics 20
CONFigure:LTE:MEAS:PRACH:MOEXception ON
CONFigure:LTE:MEAS:PRACH:TOUT 3600
CONFigure:LTE:MEAS:PRACH:NOPReambles 10
CONFigure:LTE:MEAS:PRACH:POPReambles MS20
CONFigure:LTE:MEAS:PRACH:POWer:HDMODE ON

// *****
// Select OFDM symbol for modulation results.
// *****
CONFigure:LTE:MEAS:PRACH:SSYMBOL 0
```

```
// *****
// Disable automatic detection of the PRACH frequency offset and
// specify it manually.
// *****
CONFIGure:LTE:MEAS:PRACH:PFOffset:AUTO OFF
CONFIGure:LTE:MEAS:PRACH:PFOffset 0

// *****
// Disable automatic detection of the preamble sequence index and
// specify it manually.
// *****
CONFIGure:LTE:MEAS:PRACH:MODulation:SINdex:AUTO OFF
CONFIGure:LTE:MEAS:PRACH:MODulation:SINdex 5

// *****
// Specify EVM window length for all preamble formats.
// Specify the EVM window position for traces.
// *****
CONFIGure:LTE:MEAS:PRACH:MODulation:EWLength 1500,10000,3000,10000,200
CONFIGure:LTE:MEAS:PRACH:MODulation:EWPosition HIGH
```

6.4.5 Configuring the trigger system

```
// *****
// Set trigger source, slope, threshold, timeout and minimum trigger gap.
// *****
TRIGger:LTE:MEAS:PRACH:SOURce 'IF Power'
TRIGger:LTE:MEAS:PRACH:SLOPe FEDGe
TRIGger:LTE:MEAS:PRACH:THReshold -30
TRIGger:LTE:MEAS:PRACH:TOUT 1
TRIGger:LTE:MEAS:PRACH:MGAP 0.00006
```

6.4.6 Specifying limits

```
// *****
// Define modulation limits.
// *****
CONFIGure:LTE:MEAS:PRACH:LIMit:EVMagnitude 20, 40
CONFIGure:LTE:MEAS:PRACH:LIMit:MERRor 20, OFF
CONFIGure:LTE:MEAS:PRACH:LIMit:PERRor 20, OFF
CONFIGure:LTE:MEAS:PRACH:LIMit:FERRor 150

// *****
// Define power dynamics limits.
// *****
CONFIGure:LTE:MEAS:PRACH:LIMit:PDYNamics ON,6.8,-8.8,-48.8
```

6.4.7 Running a measurement

```
// *****
// Enable all measurements.
// *****
CONFIGure:LTE:MEAS:PRACH:RESult:ALL ON,ON,ON,ON,ON,ON,ON,ON,ON

// *****
// Start the measurement.
// *****
INIT:LTE:MEAS:PRACH

// *****
// Query detected PRACH frequency offset, sequence index and
// sequence correlation for single-preamble measurements and for
// preamble number four of multi-preamble measurements.
// *****
FETCH:LTE:MEAS:PRACH:MODulation:DPFOffset?
FETCH:LTE:MEAS:PRACH:MODulation:DSIndex?
FETCH:LTE:MEAS:PRACH:MODulation:SCORrelation?
FETCH:LTE:MEAS:PRACH:MODulation:DPFOffset:PREamble4?
FETCH:LTE:MEAS:PRACH:MODulation:DSIndex:PREamble4?
FETCH:LTE:MEAS:PRACH:MODulation:SCORrelation:PREamble4?

// *****
// Query diagram contents.
// *****
FETCH:LTE:MEAS:PRACH:TRACe:EVM:AVERage?
FETCH:LTE:MEAS:PRACH:TRACe:MERRor:AVERage?
FETCH:LTE:MEAS:PRACH:TRACe:MERRor:MAXimum?
FETCH:LTE:MEAS:PRACH:TRACe:PERRor:AVERage?
FETCH:LTE:MEAS:PRACH:TRACe:PERRor:MAXimum?
FETCH:LTE:MEAS:PRACH:TRACe:EVPreamble?
FETCH:LTE:MEAS:PRACH:TRACe:PVPreamble?

FETCH:LTE:MEAS:PRACH:EVMSymbol:AVERage?
FETCH:LTE:MEAS:PRACH:EVMSymbol:PEAK:MAXimum?

FETCH:LTE:MEAS:PRACH:TRACe:IQ?
FETCH:LTE:MEAS:PRACH:TRACe:PDYNamics:AVERage?
FETCH:LTE:MEAS:PRACH:TRACe:PDYNamics:MAXimum?

// *****
// Query single value results.
// *****
FETCH:LTE:MEAS:PRACH:MODulation:AVERage?
FETCH:LTE:MEAS:PRACH:MODulation:EXTReme?
FETCH:LTE:MEAS:PRACH:MODulation:PREamble4?
FETCH:LTE:MEAS:PRACH:PDYNamics:AVERage?
FETCH:LTE:MEAS:PRACH:PDYNamics:MINimum?
FETCH:LTE:MEAS:PRACH:PDYNamics:MAXimum?
```

```
// *****
// Query limit check results.
// *****
CALCulate:LTE:MEAS:PRACH:MODulation:AVERage?
CALCulate:LTE:MEAS:PRACH:MODulation:EXTreme?
CALCulate:LTE:MEAS:PRACH:PDYNamics:AVERage?
CALCulate:LTE:MEAS:PRACH:PDYNamics:MINimum?
CALCulate:LTE:MEAS:PRACH:PDYNamics:MAXimum?
```

6.5 Command reference

The following sections describe the remote control commands of the LTE PRACH measurement.

- [Conventions and general information](#)..... 1338
- [Common measurement settings](#)..... 1344
- [PRACH measurement commands](#)..... 1344
- [Combined signal path commands](#)..... 1377

6.5.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

6.5.1.1 MEAS<i>

MEAS<i> is used as an abbreviation of "MEASurement<instance>". For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same measurement application. It can be omitted if the instrument supports only one instance, or to address the first instance.

6.5.1.2 FETCh, READ and CALCulate commands

The commands are used to retrieve measurement results:

- A FETCh... command returns the results of the measurement cycle after the cycle is complete. Before sending a FETCh..., start the measurement via an INITiate....
- A READ... command starts a new single-shot measurement and returns the results.
- A CALCulate... command returns one limit check result per FETCh result:
 - **OK:** The FETCh result is located within the limits or no limit has been defined/enabled for this result.

- **ULEU** ("User limit exceeded upper"): An upper limit is violated. The `FETCH` result is located above the limit.
- **ULEL** ("User limit exceeded lower"): A lower limit is violated. The `FETCH` result is located below the limit.

6.5.1.3 Current and statistical results

The measurement is repeated according to the selected statistic count and repetition mode. Consecutive measurement values are stored and used to calculate statistical results.

Commands for result query often contain a mnemonic that indicates which results are returned: `CURRent`, `AVERage`, `EXTReme`, `MINimum`, `MAXimum` or `SDEViation`.

6.5.1.4 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**
If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:
 - "Query only": The command can only be used to query parameters. No setting.
 - "Event": The command initiates an event. No query.
- **Parameter usage**
The parameter usage is indicated by the keyword preceding the parameters:
 - "Parameters" are sent with a setting or query command and are returned as the result of a query.
 - "Setting parameters" are sent with a setting command.
 - "Query parameters" are sent with a query command, to refine the query.
 - "Return values" are returned as the result of a query.
- **Parameter type**
The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:
 - "string": characters enclosed in quotation marks (" or ')
 - "numeric", "decimal": decimal number
 - "integer": integer number
 - "float": floating point number
 - "hex": hexadecimal number starting with #H
 - "binary": binary number starting with #B
 - "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).
- **Firmware/Software:**
Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

6.5.1.5 Reliability indicator

The first value in the sequences returned by measurement result queries indicates the most severe error that has occurred during the measurement.

Example for an output array: 0, 10.22, 10.15, 10.01, 10.29, 100 (reliability = 0, followed by 5 numeric measurement values).

The reliability indicator has one of the following values:

- **0 - "no error"**
No error detected.
- **1 - "measurement timeout"**
The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INVALID` results or has not completed the full statistic count.
- **2 - "capture buffer overflow"**
The measurement configuration results in a capture length that exceeds the available memory.
- **3 - "input overdriven" / 4 - "input underdriven"**
The accuracy of measurement results can be impaired because the input signal level was too high / too low.
- **6 - "trigger timeout"**
The measurement could not be started or continued because no trigger event was detected.
For 1xEV-DO: A test application report was not received from the mobile within three update time periods.
- **7 - "acquisition error"**
The R&S CMW could not properly decode the RF input signal.
- **8 - "sync error"**
The R&S CMW could not synchronize to the RF input signal.
- **9 - "uncalibrated"**
Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the technical specifications.
- **15 - "reference frequency error"**
The instrument has been configured to use an external reference signal. But the reference oscillator could not be locked to the external signal. For example, signal level too low, frequency out of range or reference signal not available at all.
- **16 - "RF not available"**
The measurement could not be started because the configured RF input path was not active.
This problem can occur if a measurement is started in combined signal path mode and the controlling application has not yet activated the input path.
- **17 - "RF level not settled" / 18 - "RF frequency not settled"**
The measurement could not be started. The R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 - "call not established"**

For measurements: The measurement could not be started because no signaling connection to the DUT was established.

For IMS service: Establishing a voice over IMS call failed.

- **20 - "call type not usable"**

For measurements: The measurement could not be started because the established signaling connection had the wrong properties.

For IMS service: The voice over IMS settings could not be applied.

- **21 - "call lost"**

For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.

For IMS service: The VoIMS call was lost.

- **23 - "missing option"**

An action cannot be executed due to a missing option.

For GPRF generator: The ARB file cannot be played due to a missing option.

- **24 - "invalid RF setting"**

The desired RF TX level or RF RX reference level could not be applied.

- **25 - "level overrange"**

The RF TX level is in overrange. The signal quality can be degraded.

- **26 - "resource conflict"**

The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application.

Stop the application that has allocated the conflicting resources and try again.

- **27 - "no sensor connected"**

The GPRF external power sensor measurement could not be started due to a missing power sensor.

- **28 - "unexpected parameter change"**

One or more measurement configuration parameters were changed while the measurement was running. The results were not obtained with these new parameter values. Repeat the measurement. This situation can only occur in remote single-shot mode.

- **29 - "invalid RF frequency setting"**

The desired RF TX frequency or RF RX frequency could not be applied.

- **30 - "file not found"**

The specified file could not be found.

- **31 - "no DTM reply"**

The EUT did not reply to the direct test mode (DTM) command.

- **32 - "ACL disconnected"**

The ACL connection has been disconnected or lost.

- **40 - "ARB file CRC error"**

The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.

- **42 - "ARB header tag invalid"**

The ARB file selected in the GPRF generator contains an invalid header tag.

- **43 - "ARB segment overflow"**

The number of segments in the multi-segment ARB file is higher than the allowed maximum.

- **44 - "ARB file not found"**
The selected ARB file could not be found.
- **45 - "ARB memory overflow"**
The ARB file length is greater than the available memory.
- **46 - "ARB sample rate out of range"**
The clock rate of the ARB file is either too high or too low.
- **47 - "ARB cycles out of range"**
The repetition mode equals "Single Shot" and the playback length is too long.
Reduce the playback length or set the repetition mode to "Continuous".
 $\text{<Length>} = (\text{<Cycles>} * \text{<Samples>} + \text{<Additional Samples>}) / \text{<Clock Rate>}$
- **48 - "ARB pool invalid"**
At least one of the ARB files in the ARB file pool is invalid.
- **49 - "sequencer list invalid"**
The sequencer list configuration has at least one invalid entry.
- **50 - "startup error"**
The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 - "no reply"**
The DAU has received no response, for example for a ping request.
- **52 - "connection error"**
A connection setup failed or a connection was lost.
For DAU: The DAU could not establish a connection to internal components.
Restart the instrument.
For Bluetooth: The EUT has been disconnected from the R&S CMW.
- **53 - "configuration error"**
The current DAU configuration is incomplete or wrong and could not be applied.
Check especially the IP address configuration.
- **54 - "file system error"**
The system drive of the DAU is full or corrupt. Execute a DAU self-test.
- **64 - "missing pathloss correction data"**
The pathloss measurement cannot be executed due to missing connector correction data. Order a calibration of the instrument by Rohde & Schwarz.
- **70 - "wrong standard"**
The standard of the measured signal does not match the configured standard.
- **71 - "wrong bandwidth"**
The bandwidth of the measured signal does not match the configured bandwidth.
- **72 - "wrong burst type"**
The burst type of the measured signal does not match the configured burst type.
- **73 - "MIMO signal detected"**
The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.
- **74 - "more streams than antennas"**
The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.
- **75 - "matrix inversion failed"**

The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.

- **76 - "SIG CRC failed"**
The cyclic redundancy check of a SIGNAL field failed.
- **77 - "parity check failed"**
The parity check of a SIGNAL field failed.
- **78 - "bursts not identical"**
In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.
- **79 - "wrong modulation"**
The modulation scheme of the measured signal does not match the configured modulation scheme.
- **80 - "SISO signal detected"**
Received signal contains only one stream.
- **81 - "fewer streams than antennas"**
Received signal has fewer streams than configured antennas. Reconfigure the number of antennas.
- **82 - "signal fields of different antennas are not identical"**
The signal fields of all captured antennas are compared. In a valid MIMO transmission, all signal fields are identical. If not, some basic parameters such as MCS or data length are not the same and a MIMO measurement is not possible.
- **83 - "bit errors in some of the received symbols"**
Adjust input signal power and repeat training.
- **84 - "first and last symbols need to be skipped"**
Due to a sequence counter in the MAC header, the first symbol and the last symbol change from one burst to another. You can configure the first and the last symbol to be skipped.
- **85 - "received and saved bursts are not identical"**
The signal field of the received burst is not identical to that of the saved burst. Some basic parameters such as MCS or data length are not the same and a composite MIMO measurement is not possible.
- **86 - "no symbols for measurement"**
Too many head and tail symbols have been skipped.
- **93 - "OCXO oven temperature too low"**
The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **96 - "frequency/level change too fast"**
There was a too fast frequency and/or level change. If you use GPRF with list mode, increase the duration of the list entry.
- **101 - "firmware error"**
This value indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.
If the error occurs again, consider the following hints:
 - Firmware errors can often be repaired by restoring the factory default settings.

- If a software package has not been properly installed, this failure is often indicated in the "Setup" dialog > section "SW/HW-Equipment > Installed Software".
- Check for software updates correcting the error. Updates are provided in the customer web on GLORIS (registration required): <https://gloris.rohde-schwarz.com>.

If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.

- **102 - "unidentified error"**

This value indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 firmware error".

- **103 - "parameter error"**

This value indicates that the measurement could not be performed due to internal conflicting parameter settings.

For 1xEV-DO:

- The 1xEV-DO signaling test application is not enabled.
- Forward link measurement data were requested, but the test application direction is not set to "Forward" or "Forward & Reverse".
- Reverse link measurement data were requested, but the test application direction is not set to "reverse" or "forward & reverse".
- Measurement data were requested, but the corresponding measurement is not assigned.

For CDMA2000: The "1st Service Option" is not set to a loopback service option.

A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.

If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see <https://www.service.rohde-schwarz.com>).

- **104 - "not functional"**

The application could not be started with the configured parameter set.

6.5.2 Common measurement settings

The commands valid for all LTE measurements are described here: [Section 5.5.2, "Common measurement settings"](#), on page 1111.

6.5.3 PRACH measurement commands

The commands for the LTE PRACH measurement are divided into the groups listed below.

• Measurement control and states	1345
• Enabling results and views	1347
• Measurement parameters - general part	1349
• Modulation measurement settings	1353
• Power measurement settings	1356
• Trigger settings	1356

• Limits (modulation).....	1359
• Limits (power).....	1360
• Detected signal configuration.....	1361
• EVM vs subcarrier results (diagrams).....	1364
• Magnitude error results (diagrams).....	1365
• Phase error results (diagrams).....	1365
• EVM vs preamble results (diagrams).....	1366
• EVM vs symbol (diagrams).....	1366
• Power vs preamble results (diagrams).....	1368
• I/Q constellation results.....	1369
• Power dynamics results (diagrams).....	1369
• Modulation results (single values).....	1370
• EVM/power vs preamble results (single values).....	1374
• Power dynamics results (single values).....	1376

6.5.3.1 Measurement control and states

The following commands control the measurement and return the current measurement state.

INITiate:LTE:MEAS<i>:PRACH.....	1345
STOP:LTE:MEAS<i>:PRACH.....	1345
ABORT:LTE:MEAS<i>:PRACH.....	1345
FETCh:LTE:MEAS<i>:PRACH:STATE?.....	1346
FETCh:LTE:MEAS<i>:PRACH:STATE:ALL?.....	1346

INITiate:LTE:MEAS<i>:PRACH

STOP:LTE:MEAS<i>:PRACH

ABORT:LTE:MEAS<i>:PRACH

Starts, stops or aborts the measurement:

- The **INITiate:...** command starts or restarts the measurement. The measurement enters the "RUN" state.
- The **STOP:...** command halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- The **ABORT:...** command halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

To query the current measurement state, use **FETCh...STATE?**.

Example: See [Running a measurement](#)

Usage: Event

Firmware/software: V2.0.10

Manual operation: See ["PRACH \(Softkey\)"](#) on page 1312

FETCh:LTE:MEAS<i>:PRACH:STATe? [<Timeout>[, <TargetMainState>[, <TargetSyncState>]]]

Queries the main measurement state.

Without query parameters, the state is returned immediately.

With query parameters, the state is returned when the <TargetMainState> and the <TargetSyncState> are reached or when the <Timeout> expires.

Query parameters:

<Timeout>	numeric Default unit: ms
<TargetMainState>	OFF RUN RDY Target <MainState> for the query Default is RUN.
<TargetSyncState>	PENDING ADJUSTED Target <SyncState> for the query Default is ADJ.

Return values:

<MainState>	OFF RUN RDY Current state or target state of ongoing state transition OFF: measurement off RUN: measurement running RDY: measurement completed *RST: OFF
-------------	--

Usage: Query only

Firmware/software: V2.0.10
V4.0.140 added optional parameters.

Manual operation: See "[PRACH \(Softkey\)](#)" on page 1312

FETCh:LTE:MEAS<i>:PRACH:STATe:ALL? [<Timeout>[, <TargetMainState>[, <TargetSyncState>]]]

Queries the main measurement state and the measurement substates.

Without query parameters, the states are returned immediately.

With query parameters, the states are returned when the <TargetMainState> and the <TargetSyncState> are reached or when the <Timeout> expires.

Query parameters:

<Timeout>	numeric Default unit: ms
<TargetMainState>	OFF RUN RDY Target <MainState> for the query Default is RUN.

<TargetSyncState>	PENDING ADJUSTED Target <SyncState> for the query Default is ADJ.
Return values:	
<MainState>	OFF RUN RDY Current state or target state of ongoing state transition OFF: measurement off RUN: measurement running RDY: measurement completed *RST: OFF
<SyncState>	PEND ADJ PEND: transition to <MainState> ongoing ADJ: <MainState> reached
<ResourceState>	QUE ACT INV QUE: waiting for resource allocation ACT: resources allocated INV: no resources allocated
Usage:	Query only
Firmware/software:	V2.0.10 V4.0.140 added optional parameters.
Manual operation:	See "PRACH (Softkey)" on page 1312

6.5.3.2 Enabling results and views

The following commands select which results are evaluated if the measurement is started via the command `INITiate:LTE:MEAS<i>:PRACH` or via the GUI.

<code>CONFigure:LTE:MEAS<i>:PRACH:RESult[ALL]</code>	1347
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:EVMagnitude</code>	1348
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:MERRor</code>	1348
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:PERRor</code>	1348
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:IQ</code>	1348
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:PDYnamics</code>	1348
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:TXM</code>	1348
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:EVPPreamble</code>	1348
<code>CONFigure:LTE:MEAS<i>:PRACH:RESult:PVPreamble</code>	1348

CONFigure:LTE:MEAS<i>:PRACH:RESult[ALL] <EVM>, <MagnitudeError>, <PhaseError>, <IQ>, <PowerDynamics>, <TXMeasurement>[, <EVMvsPreamble>, <PowervsPreamble>]

Enables or disables the evaluation of results in the PRACH measurement. This command combines all other `CONFigure:LTE:MEAS<i>:PRACH:RESult...` commands.

Parameters:

<EVM>	OFF ON Error vector magnitude OFF: Do not evaluate the results. ON: Evaluate the results. *RST: ON
<MagnitudeError>	OFF ON *RST: OFF
<PhaseError>	OFF ON *RST: OFF
<IQ>	OFF ON I/Q constellation diagram *RST: OFF
<PowerDynamics>	OFF ON *RST: ON
<TXMeasurement>	OFF ON TX measurement statistical overview *RST: ON
<EVMvsPreamble>	OFF ON *RST: OFF
<PowervsPreamble>	OFF ON *RST: OFF

Example: See [Running a measurement](#)

Firmware/software: V2.0.10
V2.1.20: added <EVMvsPreamble> and <PowervsPreamble>

Manual operation: See "[Multi Evaluation, PRACH softkeys > Assign Views hotkey](#)" on page 1086

CONFigure:LTE:MEAS<i>:PRACH:RESult:EVMagnitude <Enable>

CONFigure:LTE:MEAS<i>:PRACH:RESult:MERRor <Enable>

CONFigure:LTE:MEAS<i>:PRACH:RESult:PERRor <Enable>

CONFigure:LTE:MEAS<i>:PRACH:RESult:IQ <Enable>

CONFigure:LTE:MEAS<i>:PRACH:RESult:PDYNamics <Enable>

CONFigure:LTE:MEAS<i>:PRACH:RESult:TXM <Enable>

CONFigure:LTE:MEAS<i>:PRACH:RESult:EVPRreamble <Enable>

CONFigure:LTE:MEAS<i>:PRACH:RESult:PVPreamble <Enable>

Enables or disables the evaluation of results in the PRACH measurement.

Mnemonic	Description	Mnemonic	Description
EVMagnitude	Error vector magnitude	PDYNamics	Power dynamics
MERRor	Magnitude error	TXM	TX meas. statistical overview
PERRor	Phase error	EVPRamble	EVM vs preamble
IQ	I/Q constellation diagram	PVPRamble	Power vs preamble

For reset values, see `CONFigure:LTE:MEAS<i>:PRACH:RESult[:ALL]`.

Parameters:

<Enable> OFF | ON
OFF: Do not evaluate the results.
ON: Evaluate the results.
 *RST: Depends on measurement

Firmware/software: V2.0.10
 V2.1.20: . . . :EVPRamble and . . . :PVPRamble added

Manual operation: See "Multi Evaluation, PRACH softkeys > Assign Views hotkey" on page 1086

6.5.3.3 Measurement parameters - general part

The following commands define general measurement control parameters for the PRACH measurement.

<code>CONFigure:LTE:MEAS<i>:PRACH:TOUT</code>	1349
<code>CONFigure:LTE:MEAS<i>:PRACH:REPetition</code>	1350
<code>CONFigure:LTE:MEAS<i>:PRACH:SCONdition</code>	1350
<code>CONFigure:LTE:MEAS<i>:PRACH:MOEXception</code>	1351
<code>CONFigure:LTE:MEAS<i>:PRACH:PCINdex</code>	1351
<code>CONFigure:LTE:MEAS<i>:PRACH:NOPReambles</code>	1351
<code>CONFigure:LTE:MEAS<i>:PRACH:POPReambles</code>	1352
<code>CONFigure:LTE:MEAS<i>:PRACH:PFOFset:AUTO</code>	1352
<code>CONFigure:LTE:MEAS<i>:PRACH:PFOFset</code>	1352

CONFigure:LTE:MEAS<i>:PRACH:TOUT <Timeout>

Defines a timeout for the measurement. The timer is started when the measurement is initiated via a `READ` or `INIT` command. It is not started if the measurement is initiated manually.

When the measurement has completed the first measurement cycle (first single shot), the statistical depth is reached and the timer is reset.

If the first measurement cycle has not been completed when the timer expires, the measurement is stopped. The measurement state changes to `RDY`. The reliability indicator is set to 1, indicating that a measurement timeout occurred. Still running `READ`, `FETCh` or `CALCulate` commands are completed, returning the available results. At least for some results, there are no values at all or the statistical depth has not been reached.

A timeout of 0 s corresponds to an infinite measurement timeout.

Parameters:

<Timeout> numeric
 Default unit: s

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

CONFigure:LTE:MEAS<i>:PRACH:REPetition <Repetition>

Specifies the repetition mode of the measurement. The repetition mode specifies whether the measurement is stopped after a single shot or repeated continuously. Use the `CONFigure:...:MEAS<i>:...:SCount` commands to specify the number of measurement intervals per single shot.

Parameters:

<Repetition> SINGleshot | CONTInuous
 SINGleshot: Single-shot measurement
 CONTInuous: Continuous measurement

Firmware/software: V2.0.10

Manual operation: See ["Repetition"](#) on page 1324

CONFigure:LTE:MEAS<i>:PRACH:SCONdition <StopCondition>

Qualifies whether the measurement is stopped after a failed limit check or continued. With `SLFail`, the measurement is stopped and reaches the `RDY` state when one of the results exceeds the limits.

Parameters:

<StopCondition> NONE | SLFail
 NONE: Continue measurement irrespective of the limit check.
 SLFail: Stop measurement on limit failure.
 *RST: NONE

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See ["Stop Condition"](#) on page 1324

CONFigure:LTE:MEAS<i>:PRACH:MOEXception <MeasOnException>

Specifies whether measurement results that the R&S CMW identifies as faulty or inaccurate are rejected.

Parameters:

<MeasOnException> OFF | ON

OFF: Faulty results are rejected.

ON: Results are never rejected.

*RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See ["Measure on Exception"](#) on page 1324

CONFigure:LTE:MEAS<i>:PRACH:PCIndex <PRACHconfIndex>

The PRACH configuration index identifies the PRACH configuration used by the UE (preamble format, which resources in the time domain are allowed for transmission of preambles etc.).

For the combined signal path scenario, use:

- [CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:FDD](#)
- [CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:TDD](#)

Parameters:

<PRACHconfIndex> numeric

Range: 0 to 63 for FDD / 57 for TDD

*RST: 0

Example: See [Specifying required PRACH settings](#)

Firmware/software: V2.0.10

Manual operation: See ["PRACH Configuration Index, Preamble Format"](#) on page 1325

CONFigure:LTE:MEAS<i>:PRACH:NOPReambles <NumberPreamble>

Specifies the number of preambles to be captured per measurement interval.

Parameters:

<NumberPreamble> numeric

Range: 1 to 400

*RST: 1

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.1.20, V3.7.10 maximum increased

Manual operation: See ["Number of Preambles"](#) on page 1325

CONFigure:LTE:MEAS<i>:PRACH:POPReambles <PeriodPreamble>

Specifies the periodicity of preambles to be captured for multi-preamble result views.

Parameters:

<PeriodPreamble> MS05 | MS10 | MS20
MS05: 5 ms
MS10: 10 ms
MS20: 20 ms
 *RST: MS20

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V3.5.10, V3.5.20 added MS05.

Manual operation: See ["Period of Preambles"](#) on page 1325

CONFigure:LTE:MEAS<i>:PRACH:PFOffset:AUTO <PRACHfreqAuto>

Enables or disables automatic detection of the PRACH frequency offset. To configure the offset manually for disabled automatic detection, see [CONFigure:LTE:MEAS<i>:PRACH:PFOffset](#) on page 1352.

Parameters:

<PRACHfreqAuto> OFF | ON
 *RST: ON

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See ["PRACH Frequency Offset"](#) on page 1325

CONFigure:LTE:MEAS<i>:PRACH:PFOffset <PRACHfreqOffset>

Specifies the PRACH frequency offset. This setting is only relevant if automatic detection is disabled, see [CONFigure:LTE:MEAS<i>:PRACH:PFOffset:AUTO](#).

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CELL:PRACH:PFOffset](#).

Parameters:

<PRACHfreqOffset> numeric
 Range: 0 to <Total RB - 6> depending on channel bandwidth, see table below
 *RST: 0

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See ["PRACH Frequency Offset"](#) on page 1325

Table 6-4: Maximum input value depending on channel bandwidth

Channel bandwidth / MHz	1.4	3	5	10	15	20
<Total RB - 6>	0	9	19	44	69	94

6.5.3.4 Modulation measurement settings

The following commands specify settings for the modulation measurements.

CONFigure:LTE:MEAS<i>:PRACH:SSyMbol.....	1353
CONFigure:LTE:MEAS<i>:PRACH:SCount:MODulation.....	1353
CONFigure:LTE:MEAS<i>:PRACH:MODulation:LRSindex.....	1354
CONFigure:LTE:MEAS<i>:PRACH:MODulation:ZCZConfig.....	1354
CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex:AUTO.....	1354
CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex.....	1355
CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWLength.....	1355
CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWPosition.....	1356

CONFigure:LTE:MEAS<i>:PRACH:SSyMbol <SelectedSymbol>

Selects the OFDM symbol to be evaluated for single-symbol modulation result diagrams.

The number of OFDM symbols in the preamble (<no of symbols>) depends on the preamble format, see [Table 6-2](#).

Parameters:

<SelectedSymbol> numeric
 Range: 0 to <no of symbols> -1
 *RST: 0

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V4.0.110

Manual operation: See "[Selected Symbol](#)" on page 1326

CONFigure:LTE:MEAS<i>:PRACH:SCount:MODulation <StatisticCount>

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

Parameters:

<StatisticCount> numeric
 Range: 1 to 1000
 *RST: 2

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See "[Statistic Count](#)" on page 1327

CONFigure:LTE:MEAS<i>:PRACH:MODulation:LRSIndex <LogRootSeqIndex>

Specifies the logical root sequence index to be used for generation of the preamble sequence.

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CELL:PRACH:LRSIndex](#).

Parameters:

<LogRootSeqIndex> numeric
 Range: 0 to 837 (for preamble format 4: 0 to 137)
 *RST: 0

Example: See [Specifying required PRACH settings](#)

Firmware/software: V2.0.10

Manual operation: See ["Logical Root Sequence Index"](#) on page 1327

CONFigure:LTE:MEAS<i>:PRACH:MODulation:ZCZConfig <ZeroCorrZoneCon>

Specifies the zero correlation zone config, i.e. which N_{CS} value of an N_{CS} set is used for generation of the preamble sequence.

For the combined signal path scenario, use [CONFigure:LTE:SIGN<i>:CELL:PRACH:ZCZConfig](#).

Parameters:

<ZeroCorrZoneCon> numeric
 Range: 0 to 15 (for preamble format 4: 0 to 6)
 *RST: 0

Example: See [Specifying required PRACH settings](#)

Firmware/software: V2.0.10

Manual operation: See ["Zero Correlation Zone Config"](#) on page 1327

CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex:AUTO <SeqIndexAuto>

Enables or disables automatic detection of the sequence index. To configure the index manually for disabled automatic detection, see [CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex](#) on page 1355.

Parameters:

<SeqIndexAuto> OFF | ON
 *RST: ON

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See ["Sequence Index"](#) on page 1327

CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex <SequenceIndex>

Specifies the sequence index, i.e. which of the 64 preamble sequences of the cell is used by the UE. This setting is only relevant if automatic detection is disabled, see [CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINdex:AUTO](#).

Parameters:

<SequenceIndex> numeric
 Range: 0 to 63
 *RST: 0

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See "[Sequence Index](#)" on page 1327

CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWLength <LengthFormat0>, <LengthFormat1>, <LengthFormat2>, <LengthFormat3>, <LengthFormat4>

Specifies the EVM window length in samples for all preamble formats.

Parameters:

<LengthFormat0> integer
 Length for preamble format 0
 Range: 1 to 3168
 *RST: 3072

<LengthFormat1> integer
 Length for preamble format 1
 Range: 1 to 21024
 *RST: 20928

<LengthFormat2> integer
 Length for preamble format 2
 Range: 1 to 6240
 *RST: 6144

<LengthFormat3> integer
 Length for preamble format 3
 Range: 1 to 21024
 *RST: 20928

<LengthFormat4> integer
 Length for preamble format 4
 Range: 1 to 448
 *RST: 432

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10, V2.0.20 *RST values modified.

Manual operation: See "[EVM Window Length](#)" on page 1328

CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWPosition <EVMwindowPos>

Specifies the position of the EVM window used for calculation of the trace results.

Parameters:

<EVMwindowPos> LOW | HIGH
 *RST: LOW

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See ["EVM Window Position"](#) on page 1328

6.5.3.5 Power measurement settings

The following commands specify settings for power dynamics measurements.

CONFigure:LTE:MEAS<i>:PRACH:SCount:PDYNamics <StatisticCount>

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

Parameters:

<StatisticCount> numeric
 Range: 1 to 1000
 *RST: 1

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.10

Manual operation: See ["Statistic Count"](#) on page 1329

CONFigure:LTE:MEAS<i>:PRACH:POWer:HDMode <HighDynamicMode>

Enables or disables the high dynamic mode for power dynamics measurements.

Parameters:

<HighDynamicMode> OFF | ON
 *RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.1.25

Manual operation: See ["High Dynamic Mode"](#) on page 1329

6.5.3.6 Trigger settings

The following commands define the trigger parameters.

[TRIGger:LTE:MEAS<i>:PRACH:CATalog:SOURce?](#).....1357
[TRIGger:LTE:MEAS<i>:PRACH:SOURce](#).....1357
[TRIGger:LTE:MEAS<i>:PRACH:SLOPe](#).....1357

TRIGger:LTE:MEAS<i>:PRACH:THReshold.....	1358
TRIGger:LTE:MEAS<i>:PRACH:TOUT.....	1358
TRIGger:LTE:MEAS<i>:PRACH:MGAP.....	1358

TRIGger:LTE:MEAS<i>:PRACH:CATalog:SOURce?

Lists all trigger source values that can be set using `TRIGger:LTE:MEAS<i>:PRACH:SOURce`.

Return values:

<Sourcelist> string
Comma-separated list of all supported values. Each value is represented as a string.

Usage: Query only

Firmware/software: V2.0.10

Manual operation: See ["Trigger Source"](#) on page 1330

TRIGger:LTE:MEAS<i>:PRACH:SOURce <Source>

Selects the source of the trigger events. Some values are always available. They are listed below. Depending on the installed options, additional values are available. You can query a list of all supported values via the `TRIGger:...:CATalog:SOURce?` command.

Parameters:

<Source> string
'IF Power': Power trigger (received RF power)
***RST:** 'IF Power'

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.10

Manual operation: See ["Trigger Source"](#) on page 1330

TRIGger:LTE:MEAS<i>:PRACH:SLOPe <Slope>

Qualifies whether the trigger event is generated at the rising or at the falling edge of the trigger pulse (valid for external and power trigger sources).

Parameters:

<Slope> REDGe | FEDGe
REDGe: Rising edge
FEDGe: Falling edge
***RST:** REDG

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.10

Manual operation: See ["Trigger Slope"](#) on page 1330

TRIGger:LTE:MEAS<i>:PRACH:THReshold <TrigThreshold>

Defines the trigger threshold for power trigger sources.

Parameters:

<TrigThreshold> numeric
 Range: -50 dB to 0 dB
 *RST: -20 dB
 Default unit: dB (full scale, i.e. relative to reference level minus external attenuation)

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.10

Manual operation: See ["Trigger Threshold"](#) on page 1330

TRIGger:LTE:MEAS<i>:PRACH:TOUT <TriggerTimeOut>

Selects the maximum time that the measurement waits for a trigger event before it stops in remote control mode or indicates a trigger timeout in manual operation mode. This setting has no influence on "Free Run" measurements.

Parameters:

<TriggerTimeOut> numeric | ON | OFF
 Range: 0.01 s to 167772.15 s
 *RST: 0.1 s
 Default unit: s
 ON | OFF enables or disables the timeout.

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.10

Manual operation: See ["Trigger Timeout"](#) on page 1330

TRIGger:LTE:MEAS<i>:PRACH:MGAP <MinTrigGap>

Sets a minimum time during which the IF signal must be below the trigger threshold before the trigger is armed so that an IF power trigger event can be generated.

Parameters:

<MinTrigGap> float
 Range: 0 s to 1E-3 s
 *RST: 50E-6 s
 Default unit: s

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.10

Manual operation: See ["Min Trigger Gap"](#) on page 1330

6.5.3.7 Limits (modulation)

The following commands define limits for results which characterize the modulation accuracy of the PRACH signal.

CONFigure:LTE:MEAS<i>:PRACH:LIMit:EVMagnitude.....	1359
CONFigure:LTE:MEAS<i>:PRACH:LIMit:MERRor.....	1359
CONFigure:LTE:MEAS<i>:PRACH:LIMit:PERRor.....	1360
CONFigure:LTE:MEAS<i>:PRACH:LIMit:FERRor.....	1360

CONFigure:LTE:MEAS<i>:PRACH:LIMit:EVMagnitude <RMS>, <Peak>

Defines upper limits for the RMS and peak values of the error vector magnitude (EVM).

Parameters:

<RMS>	numeric ON OFF
	Range: 0 % to 100 %
	*RST: 17.5 %
	Default unit: %
	ON OFF enables or disables the limit check.
<Peak>	numeric ON OFF
	Range: 0 % to 100 %
	*RST: OFF
	Default unit: %
	ON OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V2.0.10

Manual operation: See "[Modulation limit settings](#)" on page 1332

CONFigure:LTE:MEAS<i>:PRACH:LIMit:MERRor <RMS>, <Peak>

Defines upper limits for the RMS and peak values of the magnitude error.

Parameters:

<RMS>	numeric ON OFF
	Range: 0 % to 100 %
	*RST: OFF
	Default unit: %
	ON OFF enables or disables the limit check.
<Peak>	numeric ON OFF
	Range: 0 % to 100 %
	*RST: OFF
	Default unit: %
	ON OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V2.0.10

Manual operation: See ["Modulation limit settings"](#) on page 1332

CONFigure:LTE:MEAS<i>:PRACH:LIMit:PERRor <RMS>, <Peak>

Defines symmetric limits for the RMS and peak values of the phase error. The limit check fails if the absolute value of the measured phase error exceeds the specified limit.

Parameters:

<RMS> numeric | ON | OFF
 Range: 0 deg to 180 deg
 *RST: OFF
 Default unit: deg
 ON | OFF enables or disables the limit check.

<Peak> numeric | ON | OFF
 Range: 0 deg to 180 deg
 *RST: OFF
 Default unit: deg
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V2.0.10

Manual operation: See ["Modulation limit settings"](#) on page 1332

CONFigure:LTE:MEAS<i>:PRACH:LIMit:FERRor <FrequencyError>

Defines an upper limit for the carrier frequency error.

Parameters:

<FrequencyError> numeric | ON | OFF
 Range: 0 ppm to 1000 ppm
 *RST: 0.1 ppm
 Default unit: ppm
 ON | OFF enables or disables the limit check.

Example: See [Specifying limits](#)

Firmware/software: V2.0.10

Manual operation: See ["Modulation limit settings"](#) on page 1332

6.5.3.8 Limits (power)

The following command defines limits for results which characterize the power dynamics of the PRACH signal.

CONFigure:LTE:MEAS<i>:PRACH:LIMit:PDYNamics <Enable>, <OnPowerUpper>, <OnPowerLower>, <OffPowerUpper>

Defines limits for the ON power and OFF power determined with the power dynamics measurement.

Parameters:

<Enable>	OFF ON OFF : disables the limit check ON : enables the limit check *RST: ON
<OnPowerUpper>	numeric Upper limit for the ON power Range: -256 dBm to 256 dBm *RST: 6.5 dBm Default unit: dBm
<OnPowerLower>	numeric Lower limit for the ON power Range: -256 dBm to 256 dBm *RST: -8.5 dBm Default unit: dBm
<OffPowerUpper>	numeric Upper limit for the OFF power Range: -256 dBm to 256 dBm *RST: -48.5 dBm Default unit: dBm

Example: See [Specifying limits](#)

Firmware/software: V2.0.10

Manual operation: See ["Power dynamics limit settings"](#) on page 1333

6.5.3.9 Detected signal configuration

The following commands return the detected PRACH frequency offset, sequence index and sequence correlation.

FETCh:LTE:MEAS<i>:PRACH:MODulation:DPFOffset?	1362
FETCh:LTE:MEAS<i>:PRACH:MODulation:DSIndex?	1362
FETCh:LTE:MEAS<i>:PRACH:MODulation:SCORrelation?	1362
FETCh:LTE:MEAS<i>:PRACH:MODulation:DPFOffset:PREamble<Number>?	1363
FETCh:LTE:MEAS<i>:PRACH:MODulation:DSIndex:PREamble<Number>?	1363
FETCh:LTE:MEAS<i>:PRACH:MODulation:SCORrelation:PREamble<Number>?	1363

FETCh:LTE:MEAS<i>:PRACH:MODulation:DPFOffset?

Returns the automatically detected or manually configured PRACH frequency offset for single-preamble measurements.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<PRACHfreqOffset> decimal
PRACH frequency offset

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCh:LTE:MEAS<i>:PRACH:MODulation:DSIndex?

Returns the automatically detected or manually configured sequence index for single-preamble measurements.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SequenceIndex> decimal
Sequence index

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCh:LTE:MEAS<i>:PRACH:MODulation:SCORrelation?

Returns the sequence correlation for single-preamble measurements.

It indicates the correlation between the ideal preamble sequence determined from the parameter settings and the measured preamble sequence. A value of 1 corresponds to perfect correlation. A value close to 0 indicates that the preamble sequence was not found.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SeqCorrelation> float
Sequence correlation

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

FETCh:LTE:MEAS<i>:PRACH:MODulation:DPFoffset:PREamble<Number>?

Returns the automatically detected or manually configured PRACH frequency offset for a selected preamble of multi-preamble measurements.

Suffix:

<Number> 1..400
Number of the preamble for which the results are queried.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<PRACHfreqOffset> decimal
PRACH frequency offset

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.1.20, V3.7.10 number of preambles increased

FETCh:LTE:MEAS<i>:PRACH:MODulation:DSIndex:PREamble<Number>?

Returns the automatically detected or manually configured sequence index for a selected preamble of multi-preamble measurements.

Suffix:

<Number> 1..400
Number of the preamble for which the results are queried.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SequenceIndex> decimal
Sequence index

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.1.20, V3.7.10 number of preambles increased

FETCh:LTE:MEAS<i>:PRACH:MODulation:SCORrelation:PREamble<Number>?

Returns the sequence correlation for a selected preamble of multi-preamble measurements.

It indicates the correlation between the ideal preamble sequence determined from the parameter settings and the measured preamble sequence. A value of 1 corresponds to perfect correlation. A value close to 0 indicates that the preamble sequence was not found.

Suffix:

<Number> 1..400
Number of the preamble for which the results are queried.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<SeqCorrelation> float
Sequence correlation

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.1.20, V3.7.10 number of preambles increased

6.5.3.10 EVM vs subcarrier results (diagrams)

The following commands return the contents of the EVM vs subcarrier result diagram.

```

FETCH:LTE:MEAS<i>:PRACH:TRACe:EVM:CURRent?
FETCH:LTE:MEAS<i>:PRACH:TRACe:EVM:AVERAge?
FETCH:LTE:MEAS<i>:PRACH:TRACe:EVM:MAXimum?
READ:LTE:MEAS<i>:PRACH:TRACe:EVM:CURRent?
READ:LTE:MEAS<i>:PRACH:TRACe:EVM:AVERAge?
READ:LTE:MEAS<i>:PRACH:TRACe:EVM:MAXimum?

```

Return the values of the EVM vs subcarrier traces. Each value is averaged over the samples in one preamble subcarrier. The results of the current, average and maximum traces can be retrieved.

See also [Section 6.3.2.2, "View EVM, Magnitude Error, Phase Error"](#), on page 1313.

Return values:

<Reliability> decimal
[Reliability indicator](#)

<Results> float
The number of results depends on the preamble format.
Format 0 to 3: 839 EVM values, format 4: 139 EVM values
Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

6.5.3.11 Magnitude error results (diagrams)

The following commands return the contents of the magnitude error vs subcarrier result diagram.

```

FETCh:LTE:MEAS<i>:PRACH:TRACe:MERRor:CURRent?
FETCh:LTE:MEAS<i>:PRACH:TRACe:MERRor:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:TRACe:MERRor:MAXimum?
READ:LTE:MEAS<i>:PRACH:TRACe:MERRor:CURRent?
READ:LTE:MEAS<i>:PRACH:TRACe:MERRor:AVERAge?
READ:LTE:MEAS<i>:PRACH:TRACe:MERRor:MAXimum?

```

Return the values of the magnitude error traces. Each value is averaged over the samples in one preamble subcarrier. The results of the current, average and maximum traces can be retrieved.

See also [Section 6.3.2.2, "View EVM, Magnitude Error, Phase Error"](#), on page 1313.

Return values:

<Reliability>	decimal Reliability indicator
<Results>	float The number of results depends on the preamble format. Format 0 to 3: 839 EVM values, format 4: 139 EVM values Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

6.5.3.12 Phase error results (diagrams)

The following commands return the contents of the phase error vs subcarrier result diagram.

```

FETCh:LTE:MEAS<i>:PRACH:TRACe:PERRor:CURRent?
FETCh:LTE:MEAS<i>:PRACH:TRACe:PERRor:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:TRACe:PERRor:MAXimum?
READ:LTE:MEAS<i>:PRACH:TRACe:PERRor:CURRent?
READ:LTE:MEAS<i>:PRACH:TRACe:PERRor:AVERAge?
READ:LTE:MEAS<i>:PRACH:TRACe:PERRor:MAXimum?

```

Return the values of the phase error traces. Each value is averaged over the samples in one preamble subcarrier. The results of the current, average and maximum traces can be retrieved.

See also [Section 6.3.2.2, "View EVM, Magnitude Error, Phase Error"](#), on page 1313.

Return values:

<Reliability>	decimal Reliability indicator
---------------	--

<Results>	float The number of results depends on the preamble format. Format 0 to 3: 839 EVM values, format 4: 139 EVM values Default unit: deg
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V2.0.10

6.5.3.13 EVM vs preamble results (diagrams)

The following commands return the contents of the EVM vs preamble result diagram.

FETCh:LTE:MEAS<i>:PRACH:TRACe:EVPreamble?
READ:LTE:MEAS<i>:PRACH:TRACe:EVPreamble?

Return the values of the EVM vs preamble traces.

See also [Section 6.3.2.3, "View EVM vs Preamble, Power vs Preamble"](#), on page 1314.

Return values:

<Reliability>	decimal Reliability indicator
<Results>	float 32 EVM values, for preamble 1 to 32 (NCAP for not measured preambles) Default unit: %
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V2.1.20, V3.5.20 number of results increased

6.5.3.14 EVM vs symbol (diagrams)

The following commands return the contents of the EVM vs symbol result diagram.

FETCh:LTE:MEAS<i>:PRACH:MODulation:NSYMBOL?	1367
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:CURRent?	1367
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:AVERage?	1367
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:MAXimum?	1367
READ:LTE:MEAS<i>:PRACH:EVMSymbol:CURRent?	1367
READ:LTE:MEAS<i>:PRACH:EVMSymbol:AVERage?	1367
READ:LTE:MEAS<i>:PRACH:EVMSymbol:MAXimum?	1367
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:CURRent?	1368
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:AVERage?	1368
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:MAXimum?	1368

READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:CURRent?	1368
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:AVERAge?	1368
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:MAXimum?	1368

FETCh:LTE:MEAS<i>:PRACH:MODulation:NSYMBOL?

Queries the number of active OFDM symbols (symbols with result bars) in the EVM vs symbol diagram.

Return values:

<Reliability>	decimal Reliability indicator
<NoofSymbols>	decimal

Usage: Query only

Firmware/software: V4.0.110

FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:CURRent?
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:MAXimum?
READ:LTE:MEAS<i>:PRACH:EVMSymbol:CURRent?
READ:LTE:MEAS<i>:PRACH:EVMSymbol:AVERAge?
READ:LTE:MEAS<i>:PRACH:EVMSymbol:MAXimum?

Returns the values of the EVM RMS diagrams for the OFDM symbols in the measured preamble. The results of the current, average and maximum diagrams can be retrieved.

There is one pair of EVM values per OFDM symbol, returned in the following order:

<Reliability>, {<Low>, <High>}_{symbol 0}, {<Low>, <High>}_{symbol 1}

If the preamble contains only one symbol, NCAPs are returned for the remaining symbol.

See also [Section 6.3.2.4, "View EVM vs Symbol"](#), on page 1316.

Return values:

<Reliability>	decimal Reliability indicator
<Low>	float EVM value for low EVM window position. Default unit: %
<High>	float EVM value for high EVM window position. Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V4.0.110

```

FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:CURRent?
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:MAXimum?
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:CURRent?
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:AVERAge?
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:MAXimum?

```

Returns the values of the EVM peak diagrams for the OFDM symbols in the measured preamble. The results of the current, average and maximum diagrams can be retrieved.

There is one pair of EVM values per OFDM symbol, returned in the following order:

<Reliability>, {<Low>, <High>}_{symbol 0}, {<Low>, <High>}_{symbol 1}

If the preamble contains only one symbol, NCAPs are returned for the remaining symbol.

See also [Section 6.3.2.4, "View EVM vs Symbol"](#), on page 1316.

Return values:

<Reliability>	decimal Reliability indicator
<Low>	float EVM value for low EVM window position. Default unit: %
<High>	float EVM value for high EVM window position. Default unit: %

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V4.0.110

6.5.3.15 Power vs preamble results (diagrams)

The following commands return the contents of the power vs preamble result diagram.

```

FETCh:LTE:MEAS<i>:PRACH:TRACe:PVPReamble?
READ:LTE:MEAS<i>:PRACH:TRACe:PVPReamble?

```

Return the values of the power vs preamble traces.

See also [Section 6.3.2.3, "View EVM vs Preamble, Power vs Preamble"](#), on page 1314.

Return values:

<Reliability>	decimal Reliability indicator
<Results>	float 32 power values, for preamble 1 to 32 (NCAP for not measured preambles) Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.1.20, V3.5.20 increased number of preambles

6.5.3.16 I/Q constellation results

The following command returns the contents of the I/Q constellation diagram.

FETCh:LTE:MEAS<i>:PRACH:TRACe:IQ?

Returns the results in the I/Q constellation diagram.

There is one pair of values per modulation symbol. For preamble format 4, there are 139 symbols. For preamble format 0 to 3, there are 839 symbols.

The results are returned in the following order:

<Reliability>, {<IPhase>, <QPhase>}_{symbol 1}, ..., {<IPhase>, <QPhase>}_{symbol n}

See also [Section 6.3.2.5, "View I/Q Constellation"](#), on page 1317.

Return values:

<Reliability>	decimal Reliability indicator
<IPhase>	float Normalized I amplitude.
<QPhase>	float Normalized Q amplitude.

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

6.5.3.17 Power dynamics results (diagrams)

The following commands return the contents of the power dynamics result diagram.

FETCh:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:CURRent?
FETCh:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:MAXimum?

READ:LTE:MEAS<i>:PRACH:TRACE:PDYNamics:CURRent?
READ:LTE:MEAS<i>:PRACH:TRACE:PDYNamics:AVERAge?
READ:LTE:MEAS<i>:PRACH:TRACE:PDYNamics:MAXimum?

Return the values of the power dynamics traces. Each value is sampled with 48 T_s , corresponding to 1.5625 μs . The results of the current, average and maximum traces can be retrieved.

See also [Section 6.3.2.6, "View Power Dynamics"](#), on page 1318.

Return values:

<Reliability>	decimal Reliability indicator
<Power>	float 2048 power values, from -1100 μs to +2098.4375 μs relative to the start of the preamble. The values have a spacing of 1.5625 μs . The 705th value is at the start of the preamble (0 μs). Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

6.5.3.18 Modulation results (single values)

The following commands return the statistical modulation results.

CALCulate:LTE:MEAS<i>:PRACH:MODulation:CURRent?	1370
CALCulate:LTE:MEAS<i>:PRACH:MODulation:AVERAge?	1370
FETCh:LTE:MEAS<i>:PRACH:MODulation:CURRent?	1370
FETCh:LTE:MEAS<i>:PRACH:MODulation:AVERAge?	1370
FETCh:LTE:MEAS<i>:PRACH:MODulation:SDEVIation?	1370
READ:LTE:MEAS<i>:PRACH:MODulation:CURRent?	1370
READ:LTE:MEAS<i>:PRACH:MODulation:AVERAge?	1370
READ:LTE:MEAS<i>:PRACH:MODulation:SDEVIation?	1370
CALCulate:LTE:MEAS<i>:PRACH:MODulation:EXTRemE?	1372
FETCh:LTE:MEAS<i>:PRACH:MODulation:EXTRemE?	1372
READ:LTE:MEAS<i>:PRACH:MODulation:EXTRemE?	1372

CALCulate:LTE:MEAS<i>:PRACH:MODulation:CURRent?
CALCulate:LTE:MEAS<i>:PRACH:MODulation:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:MODulation:CURRent?
FETCh:LTE:MEAS<i>:PRACH:MODulation:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:MODulation:SDEVIation?
READ:LTE:MEAS<i>:PRACH:MODulation:CURRent?
READ:LTE:MEAS<i>:PRACH:MODulation:AVERAge?
READ:LTE:MEAS<i>:PRACH:MODulation:SDEVIation?

Return the current, average and standard deviation single-value results.

The values described below are returned by `FETCH` and `READ` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<1_Reliability>	decimal Reliability indicator
<2_OutOfTol>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<3_EVM_RMSlow>	float EVM RMS value, low EVM window position. Default unit: %
<4_EVM_RMShigh>	float EVM RMS value, high EVM window position. Default unit: %
<5_EVMpeakLow>	float EVM peak value, low EVM window position. Default unit: %
<6_EVMpeakHigh>	float EVM peak value, high EVM window position. Default unit: %
<7_MErr_RMSlow>	float Magnitude error RMS value, low EVM window position. Default unit: %
<8_MErr_RMShigh>	float Magnitude error RMS value, high EVM window position. Default unit: %
<9_MErrPeakLow>	float Magnitude error peak value, low EVM window position. Default unit: %
<10_MErrPeakHigh>	float Magnitude error peak value, high EVM window position. Default unit: %
<11_PErr_RMSlow>	float Phase error RMS value, low EVM window position. Default unit: deg

<12_PErr_RMSh>	float Phase error RMS value, high EVM window position. Default unit: deg
<13_PErrPeakLow>	float Phase error peak value, low EVM window position. Default unit: deg
<14_PErrPeakHigh>	float Phase error peak value, high EVM window position. Default unit: deg
<15_FreqError>	float Carrier frequency error. Default unit: Hz
<16_TimingError>	float Transmit time error. Default unit: Ts (basic LTE time unit)
<17_TXpower>	float UE RMS power. Default unit: dBm
<18_PeakPower>	float UE peak power. Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

CALCulate:LTE:MEAS<i>:PRACH:MODulation:EXTreme?

FETCh:LTE:MEAS<i>:PRACH:MODulation:EXTreme?

READ:LTE:MEAS<i>:PRACH:MODulation:EXTreme?

Returns the extreme single value results.

The values described below are returned by **FETCh** and **READ** commands. A **CALCulate** command returns limit check results instead, one value for each result listed below.

Return values:

<1_Reliability>	decimal Reliability indicator
-----------------	--

<2_OutOfTol>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for modulation measurements exceeding the specified modulation limits. Default unit: %
<3_EVM_RMSlow>	float EVM RMS value, low EVM window position Default unit: %
<4_EVM_RMShigh>	float EVM RMS value, high EVM window position Default unit: %
<5_EVMpeakLow>	float EVM peak value, low EVM window position Default unit: %
<6_EVMpeakHigh>	float EVM peak value, high EVM window position Default unit: %
<7_MErr_RMSlow>	float Magnitude error RMS value, low EVM window position Default unit: %
<8_MErr_RMShigh>	float Magnitude error RMS value, high EVM window position Default unit: %
<9_MErrPeakLow>	float Magnitude error peak value, low EVM window position Default unit: %
<10_MErrPeakHigh>	float Magnitude error peak value, high EVM window position Default unit: %
<11_PErr_RMSlow>	float Phase error RMS value, low EVM window position Default unit: deg
<12_PErr_RMSh>	float Phase error RMS value, high EVM window position Default unit: deg
<13_PErrPeakLow>	float Phase error peak value, low EVM window position Default unit: deg

<14_PErrPeakHigh>	float	Phase error peak value, high EVM window position Default unit: deg
<15_FreqError>	float	Carrier frequency error Default unit: Hz
<16_TimingError>	float	Time error Default unit: Ts (basic LTE time unit)
<17_TXpowerMin>	float	Minimum UE RMS power Default unit: dBm
<18_TXpowerMax>	float	Maximum UE RMS power Default unit: dBm
<19_PeakPowMin>	float	Minimum UE peak power Default unit: dBm
<20_PeakPowMax>	float	Maximum UE peak power Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

6.5.3.19 EVM/power vs preamble results (single values)

The following commands return the single value modulation results as presented in the "EMV vs Preamble" and "Power vs Preamble" views.

For additional commands, see [Section 6.5.3.9, "Detected signal configuration"](#), on page 1361.

FETCH:LTE:MEAS<i>:PRACH:MODulation:PREamble<Number>?

READ:LTE:MEAS<i>:PRACH:MODulation:PREamble<Number>?

Return the single value results of the "EVM vs Preamble" and "Power vs Preamble" views, for a selected preamble.

See also [Section 6.3.2.3, "View EVM vs Preamble, Power vs Preamble"](#), on page 1314.

Suffix:

<Number> 1..400
Number of the preamble for which the results are queried.

Return values:

<1_Reliability> decimal
[Reliability indicator](#)

<2_PreambleRel> decimal
Reliability indicator for the preamble.

<3_EVM_RMSlow> float
EVM RMS value, low EVM window position.
Default unit: %

<4_EVM_RMShigh> float
EVM RMS value, high EVM window position.
Default unit: %

<5_EVMpeakLow> float
EVM peak value, low EVM window position.
Default unit: %

<6_EVMpeakHigh> float
EVM peak value, high EVM window position.
Default unit: %

<7_MErr_RMSlow> float
Magnitude error RMS value, low EVM window position.
Default unit: %

<8_MErr_RMShigh> float
Magnitude error RMS value, high EVM window position.
Default unit: %

<9_MErrPeakLow> float
Magnitude error peak value, low EVM window position.
Default unit: %

<10_MErrPeakHigh> float
Magnitude error peak value, high EVM window position.
Default unit: %

<11_PErr_RMSlow> float
Phase error RMS value, low EVM window position.
Default unit: deg

<12_PErr_RMSh> float
Phase error RMS value, high EVM window position.
Default unit: deg

<13_PErrPeakLow>	float
	Phase error peak value, low EVM window position.
	Default unit: deg
<14_PErrPeakHigh>	float
	Phase error peak value, high EVM window position.
	Default unit: deg
<15_FreqError>	float
	Carrier frequency error.
	Default unit: Hz
<16_TimingError>	float
	Transmit time error.
	Default unit: Ts (basic LTE time unit)
<17_TXpower>	float
	UE RMS power.
	Default unit: dBm
<18_PeakPower>	float
	UE peak power.
	Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.1.20, V3.7.10 number of preambles increased

6.5.3.20 Power dynamics results (single values)

The following commands return the statistical results of the power dynamics measurement.

```

CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:CURRent?
CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:AVERAge?
CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:MINimum?
CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:MAXimum?
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:CURRent?
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:AVERAge?
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:MINimum?
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:MAXimum?
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:SDEViation?
READ:LTE:MEAS<i>:PRACH:PDYNamics:CURRent?
READ:LTE:MEAS<i>:PRACH:PDYNamics:AVERAge?
READ:LTE:MEAS<i>:PRACH:PDYNamics:MINimum?

```

READ:LTE:MEAS<i>:PRACH:PDYNamics:MAXimum?**READ:LTE:MEAS<i>:PRACH:PDYNamics:SDEViation?**

Return the current, average, minimum, maximum and standard deviation single-value results of the power dynamics measurement.

The values described below are returned by `FETCh` and `READ` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for power dynamics measurements exceeding the specified power dynamics limits. Default unit: %
<OffPowerBefore>	float OFF power average value for subframe before preamble without transient period. Default unit: dBm
<OnPowerRMS>	float ON power average value over preamble. Default unit: dBm
<OnPowerPeak>	float ON power peak value within preamble. Default unit: dBm
<OffPowerAfter>	float OFF power average value for subframe after preamble without transient period. Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.10

6.5.4 Combined signal path commands

For some settings, the relevant command depends on the active scenario. While the combined signal path (CSP) scenario is active, these settings are configured via commands of the signaling application. While the standalone (SA) scenario is active, they are configured via measurement commands.

The following table provides the mapping for PRACH measurement commands. For general measurement settings, see [Table 5-14](#).

Table 6-5: Mapping for PRACH measurement commands

Setting	Commands for SA scenario	Commands for CSP scenario
PRACH configuration index	<code>CONFigure:LTE:MEAS<i>:PRACH:PCIndex</code>	<code>CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:FDD</code> <code>CONFigure:LTE:SIGN<i>:CELL:PRACH:PCIndex:TDD</code>
PRACH frequency offset	<code>CONFigure:LTE:MEAS<i>:PRACH:PFOffset</code>	<code>CONFigure:LTE:SIGN<i>:CELL:PRACH:PFOffset</code>
Logical root sequence index	<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:LRSindex</code>	<code>CONFigure:LTE:SIGN<i>:CELL:PRACH:LRSindex</code>
Zero correlation zone config	<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:ZCZConfig</code>	<code>CONFigure:LTE:SIGN<i>:CELL:PRACH:ZCZConfig</code>
High dynamic mode	<code>CONFigure:LTE:MEAS<i>:PRACH:POWer:HDMODE</code>	Fixed value OFF

6.6 List of commands

<code>ABORT:LTE:MEAS<i>:PRACH</code>	1345
<code>CALCulate:LTE:MEAS<i>:PRACH:MODulation:AVERage?</code>	1370
<code>CALCulate:LTE:MEAS<i>:PRACH:MODulation:CURREnt?</code>	1370
<code>CALCulate:LTE:MEAS<i>:PRACH:MODulation:EXTREme?</code>	1372
<code>CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:AVERage?</code>	1376
<code>CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:CURREnt?</code>	1376
<code>CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:MAXimum?</code>	1376
<code>CALCulate:LTE:MEAS<i>:PRACH:PDYNamics:MINimum?</code>	1376
<code>CONFigure:LTE:MEAS<i>:PRACH:LIMit:EVMagnitude</code>	1359
<code>CONFigure:LTE:MEAS<i>:PRACH:LIMit:FERRor</code>	1360
<code>CONFigure:LTE:MEAS<i>:PRACH:LIMit:MERRor</code>	1359
<code>CONFigure:LTE:MEAS<i>:PRACH:LIMit:PDYNamics</code>	1361
<code>CONFigure:LTE:MEAS<i>:PRACH:LIMit:PERRor</code>	1360
<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWLength</code>	1355
<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:EWPosition</code>	1356
<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:LRSindex</code>	1354
<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINDEX</code>	1355
<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:SINDEX:AUTO</code>	1354
<code>CONFigure:LTE:MEAS<i>:PRACH:MODulation:ZCZConfig</code>	1354
<code>CONFigure:LTE:MEAS<i>:PRACH:MOEXception</code>	1351
<code>CONFigure:LTE:MEAS<i>:PRACH:NOPReambles</code>	1351
<code>CONFigure:LTE:MEAS<i>:PRACH:PCIndex</code>	1351
<code>CONFigure:LTE:MEAS<i>:PRACH:PFOffset</code>	1352
<code>CONFigure:LTE:MEAS<i>:PRACH:PFOffset:AUTO</code>	1352
<code>CONFigure:LTE:MEAS<i>:PRACH:POPReambles</code>	1352
<code>CONFigure:LTE:MEAS<i>:PRACH:POWer:HDMODE</code>	1356

CONFigure:LTE:MEAS<i>:PRACH:REPetition.....	1350
CONFigure:LTE:MEAS<i>:PRACH:RESult:EVMagnitude.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult:EVPreable.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult:IQ.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult:MERRor.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult:PDYNamics.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult:PERRor.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult:PVPReable.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult:TXM.....	1348
CONFigure:LTE:MEAS<i>:PRACH:RESult[:ALL].....	1347
CONFigure:LTE:MEAS<i>:PRACH:SCONdition.....	1350
CONFigure:LTE:MEAS<i>:PRACH:SCOUNt:MODulation.....	1353
CONFigure:LTE:MEAS<i>:PRACH:SCOUNt:PDYNamics.....	1356
CONFigure:LTE:MEAS<i>:PRACH:SSYMBOL.....	1353
CONFigure:LTE:MEAS<i>:PRACH:TOUT.....	1349
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:AVERage?.....	1367
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:CURRent?.....	1367
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:MAXimum?.....	1367
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:AVERage?.....	1368
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:CURRent?.....	1368
FETCh:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:MAXimum?.....	1368
FETCh:LTE:MEAS<i>:PRACH:MODulation:AVERage?.....	1370
FETCh:LTE:MEAS<i>:PRACH:MODulation:CURRent?.....	1370
FETCh:LTE:MEAS<i>:PRACH:MODulation:DPFOffset:PREamble<Number>?.....	1363
FETCh:LTE:MEAS<i>:PRACH:MODulation:DPFOffset?.....	1362
FETCh:LTE:MEAS<i>:PRACH:MODulation:DSINDEX:PREamble<Number>?.....	1363
FETCh:LTE:MEAS<i>:PRACH:MODulation:DSINDEX?.....	1362
FETCh:LTE:MEAS<i>:PRACH:MODulation:EXTReme?.....	1372
FETCh:LTE:MEAS<i>:PRACH:MODulation:NSYMBOL?.....	1367
FETCh:LTE:MEAS<i>:PRACH:MODulation:PREamble<Number>?.....	1374
FETCh:LTE:MEAS<i>:PRACH:MODulation:SCORrelation:PREamble<Number>?.....	1363
FETCh:LTE:MEAS<i>:PRACH:MODulation:SCORrelation?.....	1362
FETCh:LTE:MEAS<i>:PRACH:MODulation:SDEViation?.....	1370
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:AVERage?.....	1376
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:CURRent?.....	1376
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:MAXimum?.....	1376
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:MINimum?.....	1376
FETCh:LTE:MEAS<i>:PRACH:PDYNamics:SDEViation?.....	1376
FETCh:LTE:MEAS<i>:PRACH:STATe:ALL?.....	1346
FETCh:LTE:MEAS<i>:PRACH:STATe?.....	1346
FETCh:LTE:MEAS<i>:PRACH:TRACe:EVM:AVERage?.....	1364
FETCh:LTE:MEAS<i>:PRACH:TRACe:EVM:CURRent?.....	1364
FETCh:LTE:MEAS<i>:PRACH:TRACe:EVM:MAXimum?.....	1364
FETCh:LTE:MEAS<i>:PRACH:TRACe:EVPReable?.....	1366
FETCh:LTE:MEAS<i>:PRACH:TRACe:IQ?.....	1369
FETCh:LTE:MEAS<i>:PRACH:TRACe:MERRor:AVERage?.....	1365
FETCh:LTE:MEAS<i>:PRACH:TRACe:MERRor:CURRent?.....	1365
FETCh:LTE:MEAS<i>:PRACH:TRACe:MERRor:MAXimum?.....	1365
FETCh:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:AVERage?.....	1369
FETCh:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:CURRent?.....	1369

FETCh:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:MAXimum?	1369
FETCh:LTE:MEAS<i>:PRACH:TRACe:PERRor:AVERAge?	1365
FETCh:LTE:MEAS<i>:PRACH:TRACe:PERRor:CURRent?	1365
FETCh:LTE:MEAS<i>:PRACH:TRACe:PERRor:MAXimum?	1365
FETCh:LTE:MEAS<i>:PRACH:TRACe:PVPReamble?	1368
INITiate:LTE:MEAS<i>:PRACH	1345
READ:LTE:MEAS<i>:PRACH:EVMSymbol:AVERAge?	1367
READ:LTE:MEAS<i>:PRACH:EVMSymbol:CURRent?	1367
READ:LTE:MEAS<i>:PRACH:EVMSymbol:MAXimum?	1367
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:AVERAge?	1368
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:CURRent?	1368
READ:LTE:MEAS<i>:PRACH:EVMSymbol:PEAK:MAXimum?	1368
READ:LTE:MEAS<i>:PRACH:MODulation:AVERAge?	1370
READ:LTE:MEAS<i>:PRACH:MODulation:CURRent?	1370
READ:LTE:MEAS<i>:PRACH:MODulation:EXTReme?	1372
READ:LTE:MEAS<i>:PRACH:MODulation:PREamble<Number>?	1374
READ:LTE:MEAS<i>:PRACH:MODulation:SDEviation?	1370
READ:LTE:MEAS<i>:PRACH:PDYNamics:AVERAge?	1376
READ:LTE:MEAS<i>:PRACH:PDYNamics:CURRent?	1376
READ:LTE:MEAS<i>:PRACH:PDYNamics:MAXimum?	1377
READ:LTE:MEAS<i>:PRACH:PDYNamics:MINimum?	1376
READ:LTE:MEAS<i>:PRACH:PDYNamics:SDEviation?	1377
READ:LTE:MEAS<i>:PRACH:TRACe:EVM:AVERAge?	1364
READ:LTE:MEAS<i>:PRACH:TRACe:EVM:CURRent?	1364
READ:LTE:MEAS<i>:PRACH:TRACe:EVM:MAXimum?	1364
READ:LTE:MEAS<i>:PRACH:TRACe:EVPreamble?	1366
READ:LTE:MEAS<i>:PRACH:TRACe:MERRor:AVERAge?	1365
READ:LTE:MEAS<i>:PRACH:TRACe:MERRor:CURRent?	1365
READ:LTE:MEAS<i>:PRACH:TRACe:MERRor:MAXimum?	1365
READ:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:AVERAge?	1370
READ:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:CURRent?	1370
READ:LTE:MEAS<i>:PRACH:TRACe:PDYNamics:MAXimum?	1370
READ:LTE:MEAS<i>:PRACH:TRACe:PERRor:AVERAge?	1365
READ:LTE:MEAS<i>:PRACH:TRACe:PERRor:CURRent?	1365
READ:LTE:MEAS<i>:PRACH:TRACe:PERRor:MAXimum?	1365
READ:LTE:MEAS<i>:PRACH:TRACe:PVPReamble?	1368
STOP:LTE:MEAS<i>:PRACH	1345
TRIGger:LTE:MEAS<i>:PRACH:CATalog:SOURce?	1357
TRIGger:LTE:MEAS<i>:PRACH:MGAP	1358
TRIGger:LTE:MEAS<i>:PRACH:SLOPe	1357
TRIGger:LTE:MEAS<i>:PRACH:SOURce	1357
TRIGger:LTE:MEAS<i>:PRACH:THReshold	1358
TRIGger:LTE:MEAS<i>:PRACH:TOUT	1358

7 LTE SRS measurement

The LTE SRS measurement provides quick and flexible tests on LTE FDD and TDD sounding reference signals. The tests cover the following UE transmitter properties:

- Transmit OFF power, transmit ON power and power ramping between them (power dynamics)

The SRS measurement requires option R&S CMW-KM500 for FDD signals and R&S CMW-KM550 for TDD signals.

7.1 What's new

This documentation describes version 4.0.250 and higher of the LTE SRS measurement. Compared to version 4.0.140, it provides the following new features:

- Editorial changes only.

7.2 General description

The LTE SRS measurement captures an uplink sounding reference signal (SRS) and provides power dynamics measurement results. The signal to be measured must not contain any additional components, e.g. a PUSCH or PUCCH.

Both FDD signals and TDD signals can be measured.

The following sections describe how to perform and configure the measurement.

- [Test setup](#)..... 1381
- [How to measure an uplink SRS signal](#)..... 1381
- [Parallel signaling and measurement](#)..... 1382

7.2.1 Test setup

Connect the external RF signal source (mobile station, signal generator etc.) to one of the bidirectional RF connectors on the front panel of the R&S CMW.

7.2.2 How to measure an uplink SRS signal

The measurement expects a pure LTE SRS UL signal. Any other signals, e.g. an LTE SRS UL signal with additional PUSCH does not yield correct measurement results.

After connecting your LTE UE to the R&S CMW, you have to adjust at least the following analyzer settings to the properties of the analyzed SRS signal:

- Duplex mode
- Analyzer "Frequency"

- "Expected Nominal Power"
Optionally "User Margin" and "External Attenuation (Input)"
Recommended values:
 - "Expected Nominal Power" = peak power of the UE signal during the measurement
 - "User Margin" = 0 dB
 The smallest possible value of the "Expected Nominal Power" plus the "User Margin" ensures the maximum dynamic range.

The "Channel Bandwidth" must also be in accordance with the measured signal. For configuration, refer to the "Measurement Control" section of the configuration dialog.

7.2.3 Parallel signaling and measurement

You can use the SRS measurement in parallel to the LTE signaling application (option R&S CMW-KS500/-KS550). Set up a connection to the UE with the signaling application. Then measure the sounding reference signal with the SRS measurement.

To use both applications in parallel, the combined signal path scenario must be activated (see ["Scenario = Combined Signal Path"](#) on page 1042). Most signal routing and analyzer settings and some measurement control settings are then configured by the signaling application. The SRS measurement displays the corresponding signaling settings instead of its own settings. These signaling settings can be configured both in the measurement GUI and in the GUI of the signaling application.

To configure these settings via remote commands, the commands of the signaling application must be used. For a command-mapping table, see [Section 7.5.4, "Combined signal path commands"](#), on page 1411.

The most important signaling parameters not relevant for standalone measurements can nevertheless be configured both in the measurement GUI and in the GUI of the signaling application. In the measurement GUI, they can be accessed via hotkeys.

7.3 GUI reference

The following sections provide detailed reference information on the graphical user interface (GUI) of the LTE SRS measurement.

To configure measurement settings, select the "SRS" tab and press the hotkey "Config". It opens the main configuration dialog box. Some settings are also accessible via softkeys and hotkeys.

The measurement results are presented on the "SRS" tab.

- [Measurement control](#)..... 1383
- [Measurement results](#).....1383
- [Signal routing and analyzer settings](#)..... 1386
- [Measurement control settings](#)..... 1386
- [Trigger settings](#)..... 1389

- [Limit settings](#)..... 1390
- [Shortcut configuration](#)..... 1392
- [Additional softkeys and hotkeys](#)..... 1392

7.3.1 Measurement control

To turn the measurement on or off, select the control softkey and press [ON | OFF] or [RESTART | STOP]. Alternatively, right-click the control softkey.



SRS (Softkey)

The softkey shows the current measurement state. Additional measurement substates can be retrieved via remote control.

Remote command:

```
INITiate:LTE:MEAS<i>:SRS
STOP:LTE:MEAS<i>:SRS
ABORt:LTE:MEAS<i>:SRS
FETCh:LTE:MEAS<i>:SRS:STATe?
FETCh:LTE:MEAS<i>:SRS:STATe:ALL?
```

7.3.2 Measurement results

The LTE SRS measurement presents all measurement results in a single view, described in the following.

7.3.2.1 View power dynamics

Access: "SRS" tab.

The view shows a diagram and a statistical overview of related power results.

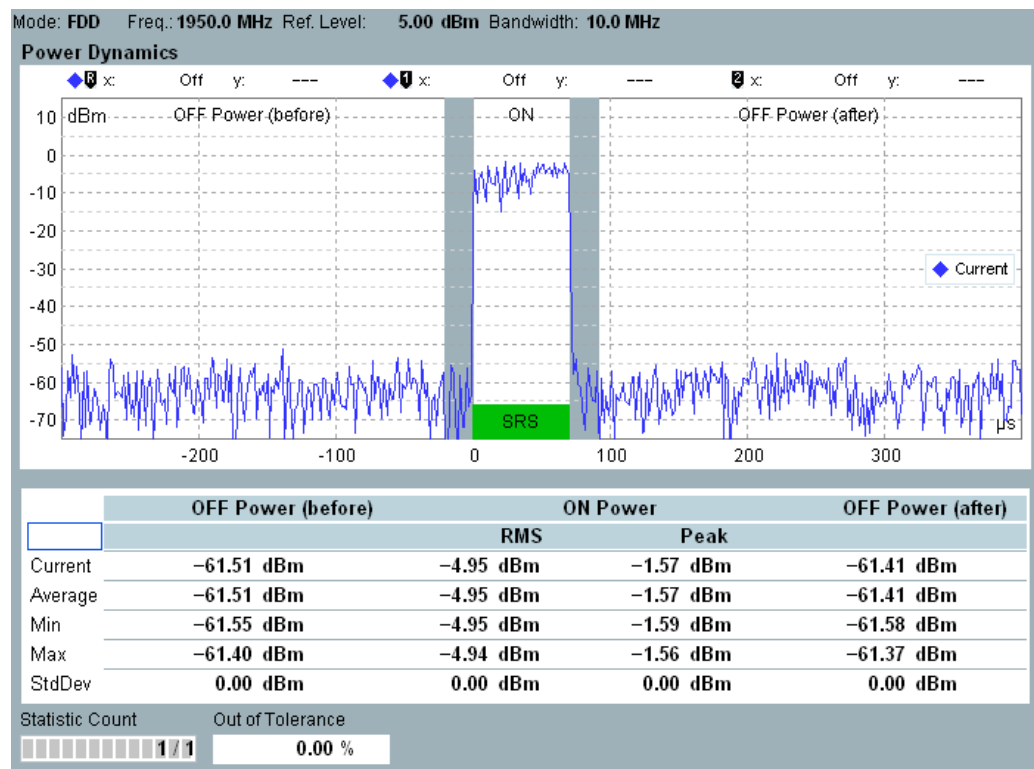


Figure 7-1: LTE SRS power dynamics results, FDD

Result diagram.....	1384
Result table.....	1384
Statistical results.....	1385
Statistic Count.....	1385
Out of Tolerance.....	1385

Result diagram

The diagram shows the UE output power vs time, sampled with $48 T_s$ ($1.5625 \mu s$). The trace covers the range from $-1100 \mu s$ to $+1298.4375 \mu s$ relative to the start of the SRS symbol.

Transient periods are indicated by gray vertical bars in the diagram. They are excluded from the measurement and have a width of $20 \mu s$ each. For TDD signals, there are two exclusion periods between the two SRS symbols.

Remote command:

[Section 7.5.3.5, "Power dynamics results \(diagrams\)"](#), on page 1409

Result table

The table shows ON power and OFF power values:

- **ON power:**

The ON power is the mean UE output power over the SRS symbol, i.e. over the measurement period marked by a green horizontal bar in the diagram. In addition to the mean value, the table also lists the peak power within the measurement period.

For TDD signals, two SRS symbols are measured and separate RMS and peak ON power results are provided for each SRS symbol.

- **OFF power:**

The OFF power represents a mean power value. It is determined before and after the SRS symbols within the following measurement periods:

- "OFF Power (Before)": 13 SC-FDMA symbols for FDD, 8 SC-FDMA symbols for TDD
- "OFF Power (After)": one subframe (FDD and TDD)

Remote command:

[Section 7.5.3.6, "Power dynamics results \(single values\)"](#), on page 1410

Statistical results

The statistical values are calculated as follows:

- **"Current"**: Result obtained in the last measurement interval.
- **"Average"**: Weighted average of all "Current" values since the start of the measurement.
- **"Min", "Max"**: Largest or smallest "Current" value since the start of the measurement.
- **"Std Dev"**: Standard deviation of all "Current" values since the start of the measurement.

Statistic Count

Progress bar for the measurement. During the first single shot after the start of the measurement, the bar shows the number of completed measurement intervals relative to the "Statistic Count". A filled progress bar indicates that the first shot is complete and the statistical depth has been reached.

Out of Tolerance

Percentage of measurement intervals that were failed because at least one result exceeded the limits.

7.3.2.2 Modifying views

Press the "Display" softkey to show hotkeys for view configuration. They change the appearance and contents of the active view. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Select Trace ..."	Select the trace types to be displayed in the view.
"X Scale... / Y Scale..."	Modify the ranges of the x-axis and the y-axis.

7.3.2.3 Using markers

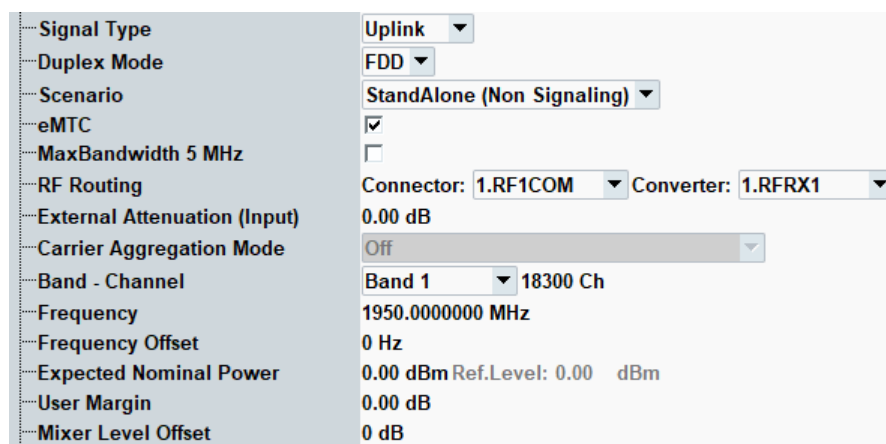
Press the "Marker" softkey to show hotkeys for marker configuration. They activate markers and modify their position. The following hotkeys are available at the bottom of the GUI:

Hotkey	Description
"Ref. Marker ..."	Enable or disable the reference marker and select the marker position. If several traces can be displayed, a trace can also be selected.
"Marker 1/2 ..."	Enable or disable marker 1 or 2 and define the marker position (absolute or relative to the reference marker). Depending on the trace mode, a trace can also be selected.
"Select Trace Mode"	Define whether markers 1 and 2 are set to the same trace as the reference marker (collective) or to selectable individual traces.

7.3.3 Signal routing and analyzer settings

Access: "SRS" tab > "Config" hotkey.

The following parameters configure the RF input path. All parameters are common measurement settings. They have the same value in all measurements (for example SRS measurement and multi-evaluation measurement).



Signal Type	Uplink
Duplex Mode	FDD
Scenario	StandAlone (Non Signaling)
eMTC	☒
MaxBandwidth 5 MHz	☐
RF Routing	Connector: 1.RF1COM Converter: 1.RFRX1
External Attenuation (Input)	0.00 dB
Carrier Aggregation Mode	Off
Band - Channel	Band 1 18300 Ch
Frequency	1950.0000000 MHz
Frequency Offset	0 Hz
Expected Nominal Power	0.00 dBm Ref.Level: 0.00 dBm
User Margin	0.00 dB
Mixer Level Offset	0 dB

Figure 7-2: Signal routing and analyzer settings

For a description of the settings, see [Section 5.3.3, "Signal routing and analyzer settings"](#), on page 1040.

7.3.4 Measurement control settings

Access: "SRS" tab > "Config" hotkey > "Measurement Control" node.

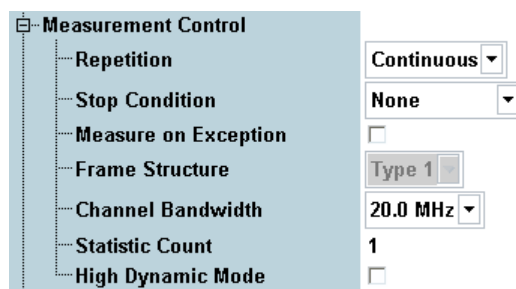


Figure 7-3: Measurement control settings

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Stop Condition.....	1387
Measure on Exception.....	1387
Frame Structure.....	1388
Channel Bandwidth.....	1388
Statistic Count.....	1388
High Dynamic Mode.....	1388

Repetition

Defines how often the measurement is repeated if it is not stopped explicitly or by a failed limit check.

- **"Continuous"**: The measurement is continued until it is explicitly terminated. The results are periodically updated.
- **"Single Shot"**: The measurement is stopped after one measurement cycle.

"Single Shot" is preferable if only a single measurement result is required under fixed conditions, which is typical for remote-controlled measurements. The continuous mode is suitable to monitor the evolution of the measurement results and to observe how they depend on the measurement configuration, which is typically done in manual control.

Remote command:

`CONFigure:LTE:MEAS<i>:SRS:REPetition`

Stop Condition

Specifies the conditions for an early termination of the measurement:

- **"None"**: The measurement is performed according to its "Repetition" mode and "Statistic Count", irrespective of the limit check results.
- **"On Limit Failure"**: The measurement is stopped when one of the limits is exceeded, irrespective of the repetition mode set. If no limit failure occurs, it is performed according to its "Repetition" mode and "Statistic Count". Use this setting for measurements that are intended for checking limits, e.g. production tests.

Remote command:

`CONFigure:LTE:MEAS<i>:SRS:SCONdition`

Measure on Exception

Specifies whether results that the measurement identifies as faulty or inaccurate are rejected. A faulty result occurs, for example, when an overload is detected. In remote control, the cause of the error is indicated by the "reliability indicator".

- **Off:** Faulty results are rejected. The measurement is continued. The statistical counters are not reset. Using this mode ensures that a single faulty result does not affect the entire measurement.
- **On:** Results are never rejected. Use this mode for development purposes, if you want to analyze the reasons for occasional wrong transmissions.

Remote command:

`CONFigure:LTE:MEAS<i>:SRS:MOEXception`

Frame Structure

Displays the frame structure of the uplink signal as defined in 3GPP TS 36.211. The value is set implicitly via the **Duplex Mode** ("Type 1" = FDD, "Type 2" = TDD).

Remote command:

`CONFigure:LTE:MEAS<i>:FSTRucture?`

Channel Bandwidth

Channel bandwidth in the range 1.4 MHz to 20 MHz. Set the bandwidth in accordance with the measured LTE signal.

The parameter is a common measurement setting. It has the same value in all measurements (for example, SRS measurement and multi-evaluation measurement).

In the standalone (SA) scenario, this parameter is controlled by the measurement. In the combined signal path (CSP) scenario, it is controlled by the signaling application.

Remote command:

`CONFigure:LTE:MEAS<i>:CC<no>:CBANDwidth (SA)`

`CONFigure:LTE:SIGN<i>:CELL:BANDwidth[:PCC]:DL (CSP)`

Statistic Count

Defines the number of measurement intervals per measurement cycle. A measurement cycle is the duration of a single-shot measurement.

For SRS measurements, the measurement interval is completed when the R&S CMW has measured a full trace, including one or two SRS symbols and the preceding and following OFF period.

Remote command:

`CONFigure:LTE:MEAS<i>:SRS:SCount:PDYNamics`

High Dynamic Mode

Enables or disables the high dynamic mode.

The high dynamic mode is suitable for power dynamics measurements involving high ON powers. In that case, the dynamic range of the R&S CMW is eventually not sufficient to measure both the high ON powers and the low OFF powers accurately.

In high dynamic mode, the dynamic range is increased by measuring the results in two shots. One shot uses the configured settings to measure the ON power. The other shot uses a lower "Expected Nominal Power" value to measure the OFF power results.

Disable the high dynamic mode to optimize the measurement speed. Disable it especially if you measure low ON powers using a low "Expected Nominal Power" setting, so that the normal dynamic range is sufficient.

While the combined signal path scenario is active, this parameter is not configurable.

Remote command:

CONFigure:LTE:MEAS<i>:SRS:HDMODE

7.3.5 Trigger settings

Access: "SRS" tab > "Config" hotkey > "Trigger" node.

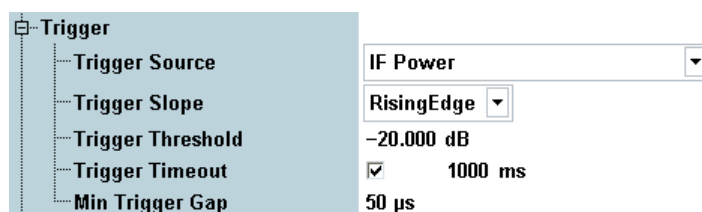


Figure 7-4: Trigger settings

Trigger Source.....	1389
Trigger Slope.....	1389
Trigger Threshold.....	1389
Trigger Timeout.....	1390
Min Trigger Gap.....	1390

Trigger Source

Selects the source of the trigger event.

Only the measurement-specific trigger sources are listed here. Additional trigger sources are provided by the base software and by other applications like the GPRF generator. For a description of these trigger sources, refer to the documentation of the corresponding software part.

"IF Power" The measurement is triggered by the power ramp of the received SRS symbol. The minimum trigger gap must be large enough to ensure that the measurement is not retriggered within an SRS symbol. It must be small enough to ensure that the trigger system is rearmed for the next measurement interval.

Remote command:

TRIGger:LTE:MEAS<i>:SRS:SOURce

TRIGger:LTE:MEAS<i>:SRS:CATalog:SOURce?

Trigger Slope

Qualifies whether the trigger event is generated at the rising or at the falling edge of the trigger pulse. This setting has no influence on the evaluation of trigger pulses provided by other firmware applications.

Remote command:

TRIGger:LTE:MEAS<i>:SRS:SLOPe

Trigger Threshold

Defines the input signal power where a trigger event is generated for power trigger sources.

The threshold is a dB value, relative to the reference level minus the external attenuation. If the reference level equals the maximum output power of the DUT and the external attenuation settings are correct, the trigger threshold is relative to the maximum input power.

A low threshold ensures that the input signal is always detected. A higher threshold can prevent unintended trigger events.

Remote command:

```
TRIGger:LTE:MEAS<i>:SRS:THReshold
```

Trigger Timeout

Sets a time after which an initiated measurement must have received a trigger event. If no trigger event is received, a trigger timeout is indicated in manual operation mode. In remote control mode, the measurement is automatically stopped. The parameter can be disabled so that no timeout occurs.

Remote command:

```
TRIGger:LTE:MEAS<i>:SRS:TOUT
```

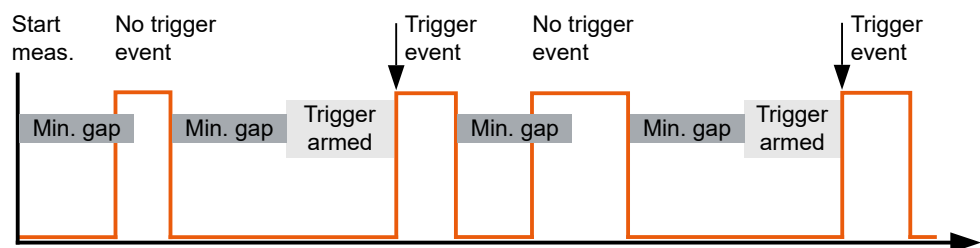
Min Trigger Gap

Defines a minimum duration of the power-down periods (gaps) between two triggered power pulses. This setting is valid for "(IF) Power" trigger sources.

The trigger system is controlled by a timer which is reset to zero in the following instances:

- At the IF power-down ramp of each triggered or untriggered pulse, even if the previous counter has not yet elapsed. A power-down ramp is detected when the signal power falls below the trigger threshold.
- At the beginning of each measurement. The minimum gap defines the minimum time between the start of the measurement and the first trigger event.

The trigger system is rearmed when the timer has reached the specified minimum gap.



You can use this parameter to prevent unwanted trigger events due to fast power variations.

Remote command:

```
TRIGger:LTE:MEAS<i>:SRS:MGAP
```

7.3.6 Limit settings

The following section gives an overview of the SRS measurement limit settings and the related conformance test requirements.

Conformance requirements for LTE transmitter tests are specified in 3GPP TS 36.521, section 6, "Transmitter Characteristics".

7.3.6.1 Power dynamics limits

Transmission at excessive uplink power increases interference to other channels while a too low uplink power increases transmission errors. For SRS transmission, 3GPP defines a time mask. The mask contains limits for the UL power during SRS symbol transmission (ON power), the UL power in the adjacent SC-FDMA symbols (OFF power) and the power ramping in-between.

The ON power is specified as the mean UE output power over the SRS symbols, excluding transient periods of 20 μ s. For the OFF power, the mean power has to be measured both in the preceding symbols and in the subsequent subframe, excluding transient periods of 20 μ s.

The following figures provide a summary of the time periods relevant for ON and OFF power limits for FDD and TDD. The transient periods between the two TDD SRS symbols are only defined for signal configurations with frequency hopping or power change between the two symbols. The measurement assumes that this condition is fulfilled and excludes these transient periods for result calculation.

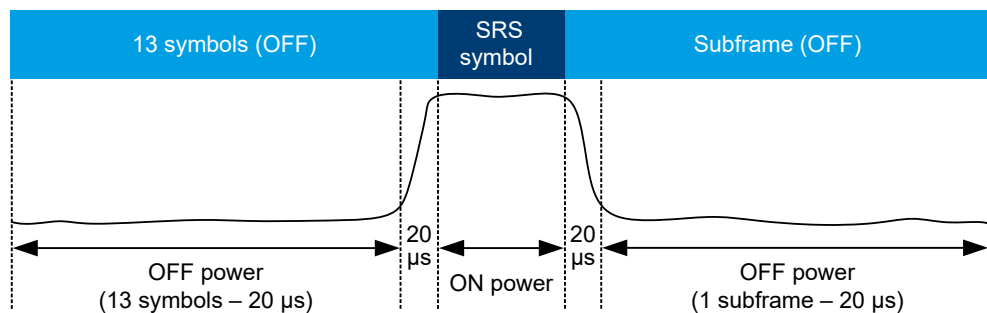


Figure 7-5: Measurement periods ON/OFF power, FDD SRS signal

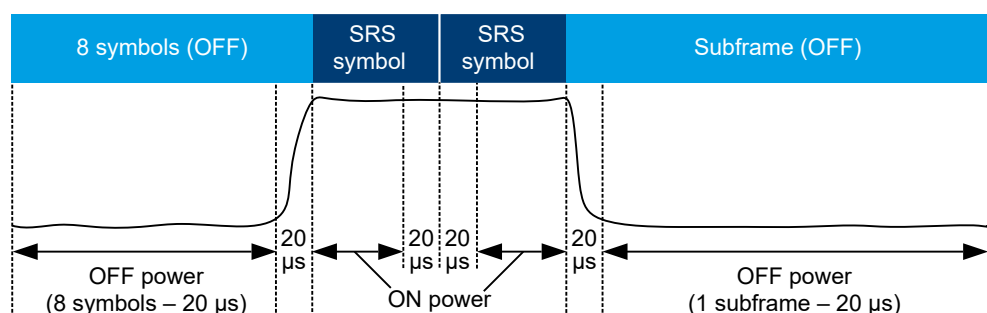


Figure 7-6: Measurement periods ON/OFF power, TDD SRS signal

According to 3GPP, the OFF power must not exceed -48.5 dBm. The ON power must be in the range -10.1 dBm to 4.9 dBm.

Characteristics	Refer to 3GPP TS 36.521 V9.3.0, section...	Specified limit
ON power	6.3.4.2 PRACH and SRS Time Mask	≥ -10.1 dBm, ≤ 4.9 dBm
OFF power	6.3.4.2 PRACH and SRS Time Mask	≤ -48.5 dBm

Power dynamics limit settings

Access: "SRS" tab > "Config" hotkey > "Limits" node.

The limits for ON and OFF power can be set in the configuration dialog.

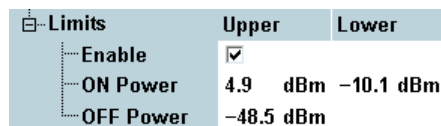


Figure 7-7: Power dynamics limit settings

Remote command:

CONFigure:LTE:MEAS<i>:SRS:LIMit:PDYNamics

7.3.7 Shortcut configuration

Access: "Config" hotkey > bottom of the tree.

This section configures a shortcut softkey that provides a fast way to access the GPRF generator from the measurement.

The setting is a common measurement setting. It has the same value in all measurements (for example, PRACH measurement and multi-evaluation measurement).



Figure 7-8: Shortcut configuration

Generator Shortcut

Selects an instance of the GPRF generator. Softkeys for the selected instance are added to the softkey panel.

7.3.8 Additional softkeys and hotkeys

See [Section 5.3.9, "Additional softkeys and hotkeys"](#), on page 1086.

7.4 Programming

The following sections provide programming examples for the LTE SRS measurement.

- [Key features](#)..... 1393
- [Specifying general and common measurement settings](#)..... 1393
- [Specifying measurement-specific settings](#)..... 1393

• Configuring the trigger system	1394
• Specifying limits	1394
• Running a measurement	1394

7.4.1 Key features

The LTE SRS measurement is programmed as follows:

- The measurement is controlled by SCPI commands with the following syntax: `...LTE:MEAS:SRS...`
- Use general commands of the type `...:LTE:MEAS...` (no `:SRS` mnemonic) to define the signal routing and to perform RF and analyzer settings.
- After a `*RST`, the measurement is switched off. Use `READ:LTE:MEAS:SRS...?` to initiate a measurement and to retrieve the results. You can also start the measurement using `INIT:LTE:MEAS:SRS` and retrieve the results using `FETCh:LTE:MEAS:SRS...?`.

7.4.2 Specifying general and common measurement settings

```
// *****
// Initial system-reset
// *****
*RST; *OPC?
*CLS; *OPC?

// *****
// Set external attenuation 2 dB, carrier center frequency 1850 MHz,
// peak power 7 dBm, 5 dB user margin.
// *****
CONFIGure:LTE:MEAS:RFSettings:EATTenuation 2
CONFIGure:LTE:MEAS:RFSettings:CC:FREQuency 1850E+6
CONFIGure:LTE:MEAS:RFSettings:ENPower 7
CONFIGure:LTE:MEAS:RFSettings:UMARgin 5

// *****
// Set duplex mode TDD and query the resulting frame structure type (T2).
// Specify channel bandwidth 1.4 MHz.
// *****
CONFIGure:LTE:MEAS:DMode TDD
CONFIGure:LTE:MEAS:FSTRucture?
CONFIGure:LTE:MEAS:CC:CBANdwidth B014
```

7.4.3 Specifying measurement-specific settings

```
// *****
// Define stop condition (stop on limit failure), statistic count (20 cycles)
// and error handling. Enable the high dynamic mode.
```

```
// *****
CONFigure:LTE:MEAS:SRS:SCONdition SLFail
CONFigure:LTE:MEAS:SRS:SCount:PDYNamics 20
CONFigure:LTE:MEAS:SRS:MOEXception ON
CONFigure:LTE:MEAS:SRS:TOUT 3600
CONFigure:LTE:MEAS:SRS:HDMODE ON
```

7.4.4 Configuring the trigger system

```
// *****
// Set trigger source, slope, threshold, timeout and minimum trigger gap.
// *****
TRIGger:LTE:MEAS:SRS:SOURce 'IF Power'
TRIGger:LTE:MEAS:SRS:SLOPe FEDGe
TRIGger:LTE:MEAS:SRS:THReshold -30
TRIGger:LTE:MEAS:SRS:TOUT 1
TRIGger:LTE:MEAS:SRS:MGAP 0.00006
```

7.4.5 Specifying limits

```
// *****
// Define power dynamics limits.
// *****
CONFigure:LTE:MEAS:SRS:LIMit:PDYNamics ON,5.1,-10.3,-48.8
```

7.4.6 Running a measurement

```
// *****
// Start the measurement.
// *****
INIT:LTE:MEAS:SRS

// *****
// Query power dynamics traces.
// *****
FETCh:LTE:MEAS:SRS:TRACe:PDYNamics:CURRent?
FETCh:LTE:MEAS:SRS:TRACe:PDYNamics:MAXimum?

// *****
// Query power dynamics statistical results.
// *****
FETCh:LTE:MEAS:SRS:PDYNamics:CURRent?
FETCh:LTE:MEAS:SRS:PDYNamics:AVErAge?
FETCh:LTE:MEAS:SRS:PDYNamics:MINimum?
FETCh:LTE:MEAS:SRS:PDYNamics:MAXimum?
FETCh:LTE:MEAS:SRS:PDYNamics:SDEviation?
```

```
// *****
// Query limit check results.
// *****
CALCulate:LTE:MEAS:SRS:PDYNamics:CURRent?
CALCulate:LTE:MEAS:SRS:PDYNamics:AVERage?
CALCulate:LTE:MEAS:SRS:PDYNamics:MINimum?
CALCulate:LTE:MEAS:SRS:PDYNamics:MAXimum?
```

7.5 Command reference

The following sections describe the remote control commands of the "LTE SRS" measurement.

- [Conventions and general information](#)..... 1395
- [Common measurement settings](#)..... 1401
- [SRS measurement commands](#)..... 1401
- [Combined signal path commands](#)..... 1411

7.5.1 Conventions and general information

The following sections describe the most important conventions and general information concerning the command reference.

7.5.1.1 MEAS<i>

MEAS<i> is used as an abbreviation of "MEASurement<instance>". For better readability, only the abbreviated form (which is also accepted by the instrument) is given in the command reference.

The <instance> is relevant for instruments supporting several instances of the same measurement application. It can be omitted if the instrument supports only one instance, or to address the first instance.

7.5.1.2 FETCh, READ and CALCulate commands

The commands are used to retrieve measurement results:

- A FETCh... command returns the results of the measurement cycle after the cycle is complete. Before sending a FETCh..., start the measurement via an INITiate....
- A READ... command starts a new single-shot measurement and returns the results.
- A CALCulate... command returns one limit check result per FETCh result:
 - **OK:** The FETCh result is located within the limits or no limit has been defined/enabled for this result.

- **ULEU** ("User limit exceeded upper"): An upper limit is violated. The `FETCH` result is located above the limit.
- **ULEL** ("User limit exceeded lower"): A lower limit is violated. The `FETCH` result is located below the limit.

7.5.1.3 Current and statistical results

The measurement is repeated according to the selected statistic count and repetition mode. Consecutive measurement values are stored and used to calculate statistical results.

Commands for result query often contain a mnemonic that indicates which results are returned: `CURRent`, `AVERage`, `EXTReme`, `MINimum`, `MAXimum` or `SDEViation`.

7.5.1.4 Keywords

Selected keywords used in the command description are described in the following.

- **Command usage**

If the usage is not explicitly stated, the command allows you to set parameters and query parameters. Otherwise, the command usage is stated as follows:

- "Query only": The command can only be used to query parameters. No setting.
- "Event": The command initiates an event. No query.

- **Parameter usage**

The parameter usage is indicated by the keyword preceding the parameters:

- "Parameters" are sent with a setting or query command and are returned as the result of a query.
- "Setting parameters" are sent with a setting command.
- "Query parameters" are sent with a query command, to refine the query.
- "Return values" are returned as the result of a query.

- **Parameter type**

The parameter type is indicated by one of the following keywords, at the beginning of the parameter description:

- "string": characters enclosed in quotation marks (" or ')
- "numeric", "decimal": decimal number
- "integer": integer number
- "float": floating point number
- "hex": hexadecimal number starting with #H
- "binary": binary number starting with #B
- "block": binary block data (number of length digits + length digits giving number of data bytes + data bytes + EOS)

Enumerated values are explicitly listed, separated by pipes (example: A | B | C).

- **Firmware/Software:**

Indicates the lowest software version supporting the command. Command enhancements in later software versions are also indicated.

7.5.1.5 Reliability indicator

The first value in the sequences returned by measurement result queries indicates the most severe error that has occurred during the measurement.

Example for an output array: 0, 10.22, 10.15, 10.01, 10.29, 100 (reliability = 0, followed by 5 numeric measurement values).

The reliability indicator has one of the following values:

- **0 - "no error"**
No error detected.
- **1 - "measurement timeout"**
The measurement has been stopped after the configured measurement timeout. Measurement results can be available. However, at least a part of the measurement provides only `INVALID` results or has not completed the full statistic count.
- **2 - "capture buffer overflow"**
The measurement configuration results in a capture length that exceeds the available memory.
- **3 - "input overdriven" / 4 - "input underdriven"**
The accuracy of measurement results can be impaired because the input signal level was too high / too low.
- **6 - "trigger timeout"**
The measurement could not be started or continued because no trigger event was detected.
For 1xEV-DO: A test application report was not received from the mobile within three update time periods.
- **7 - "acquisition error"**
The R&S CMW could not properly decode the RF input signal.
- **8 - "sync error"**
The R&S CMW could not synchronize to the RF input signal.
- **9 - "uncalibrated"**
Due to an inappropriate configuration of resolution bandwidth, video bandwidth or sweep time, the measurement results are not within the technical specifications.
- **15 - "reference frequency error"**
The instrument has been configured to use an external reference signal. But the reference oscillator could not be locked to the external signal. For example, signal level too low, frequency out of range or reference signal not available at all.
- **16 - "RF not available"**
The measurement could not be started because the configured RF input path was not active.
This problem can occur if a measurement is started in combined signal path mode and the controlling application has not yet activated the input path.
- **17 - "RF level not settled" / 18 - "RF frequency not settled"**
The measurement could not be started. The R&S CMW was not yet ready to deliver stable results after a change of the input signal power / the input signal frequency.
- **19 - "call not established"**

For measurements: The measurement could not be started because no signaling connection to the DUT was established.

For IMS service: Establishing a voice over IMS call failed.

- **20 - "call type not usable"**

For measurements: The measurement could not be started because the established signaling connection had the wrong properties.

For IMS service: The voice over IMS settings could not be applied.

- **21 - "call lost"**

For measurements: The measurement was interrupted because the signaling connection to the DUT was lost.

For IMS service: The VoIMS call was lost.

- **23 - "missing option"**

An action cannot be executed due to a missing option.

For GPRF generator: The ARB file cannot be played due to a missing option.

- **24 - "invalid RF setting"**

The desired RF TX level or RF RX reference level could not be applied.

- **25 - "level overrange"**

The RF TX level is in overrange. The signal quality can be degraded.

- **26 - "resource conflict"**

The application could not be started or has been stopped due to a conflicting hardware resource or software option that is allocated by another application.

Stop the application that has allocated the conflicting resources and try again.

- **27 - "no sensor connected"**

The GPRF external power sensor measurement could not be started due to a missing power sensor.

- **28 - "unexpected parameter change"**

One or more measurement configuration parameters were changed while the measurement was running. The results were not obtained with these new parameter values. Repeat the measurement. This situation can only occur in remote single-shot mode.

- **29 - "invalid RF frequency setting"**

The desired RF TX frequency or RF RX frequency could not be applied.

- **30 - "file not found"**

The specified file could not be found.

- **31 - "no DTM reply"**

The EUT did not reply to the direct test mode (DTM) command.

- **32 - "ACL disconnected"**

The ACL connection has been disconnected or lost.

- **40 - "ARB file CRC error"**

The cyclic redundancy check of the ARB file failed. The ARB file is corrupt and not reliable.

- **42 - "ARB header tag invalid"**

The ARB file selected in the GPRF generator contains an invalid header tag.

- **43 - "ARB segment overflow"**

The number of segments in the multi-segment ARB file is higher than the allowed maximum.

- **44 - "ARB file not found"**
The selected ARB file could not be found.
- **45 - "ARB memory overflow"**
The ARB file length is greater than the available memory.
- **46 - "ARB sample rate out of range"**
The clock rate of the ARB file is either too high or too low.
- **47 - "ARB cycles out of range"**
The repetition mode equals "Single Shot" and the playback length is too long.
Reduce the playback length or set the repetition mode to "Continuous".
$$\text{<Length>} = (\text{<Cycles>} * \text{<Samples>} + \text{<Additional Samples>}) / \text{<Clock Rate>}$$
- **48 - "ARB pool invalid"**
At least one of the ARB files in the ARB file pool is invalid.
- **49 - "sequencer list invalid"**
The sequencer list configuration has at least one invalid entry.
- **50 - "startup error"**
The data application unit (DAU), a DAU service or a DAU measurement could not be started. Execute a DAU self-test.
- **51 - "no reply"**
The DAU has received no response, for example for a ping request.
- **52 - "connection error"**
A connection setup failed or a connection was lost.
For DAU: The DAU could not establish a connection to internal components.
Restart the instrument.
For Bluetooth: The EUT has been disconnected from the R&S CMW.
- **53 - "configuration error"**
The current DAU configuration is incomplete or wrong and could not be applied.
Check especially the IP address configuration.
- **54 - "file system error"**
The system drive of the DAU is full or corrupt. Execute a DAU self-test.
- **64 - "missing pathloss correction data"**
The pathloss measurement cannot be executed due to missing connector correction data. Order a calibration of the instrument by Rohde & Schwarz.
- **70 - "wrong standard"**
The standard of the measured signal does not match the configured standard.
- **71 - "wrong bandwidth"**
The bandwidth of the measured signal does not match the configured bandwidth.
- **72 - "wrong burst type"**
The burst type of the measured signal does not match the configured burst type.
- **73 - "MIMO signal detected"**
The measurement expects a SISO signal and detected a MIMO signal. Use a MIMO receive mode to measure this signal.
- **74 - "more streams than antennas"**
The measured signal has more streams than expected due to the configured number of antennas. Increase the configured number of antennas to measure this signal.
- **75 - "matrix inversion failed"**

The inversion of the channel matrix failed for a MIMO measurement. Check that the antennas are connected correctly to the instrument.

- **76 - "SIG CRC failed"**
The cyclic redundancy check of a SIGNAL field failed.
- **77 - "parity check failed"**
The parity check of a SIGNAL field failed.
- **78 - "bursts not identical"**
In training mode for composite MIMO measurements, at least some symbols of sequential bursts need to be identical to be used as training data. Setting a fix scrambler initialization can solve this problem.
- **79 - "wrong modulation"**
The modulation scheme of the measured signal does not match the configured modulation scheme.
- **80 - "SISO signal detected"**
Received signal contains only one stream.
- **81 - "fewer streams than antennas"**
Received signal has fewer streams than configured antennas. Reconfigure the number of antennas.
- **82 - "signal fields of different antennas are not identical"**
The signal fields of all captured antennas are compared. In a valid MIMO transmission, all signal fields are identical. If not, some basic parameters such as MCS or data length are not the same and a MIMO measurement is not possible.
- **83 - "bit errors in some of the received symbols"**
Adjust input signal power and repeat training.
- **84 - "first and last symbols need to be skipped"**
Due to a sequence counter in the MAC header, the first symbol and the last symbol change from one burst to another. You can configure the first and the last symbol to be skipped.
- **85 - "received and saved bursts are not identical"**
The signal field of the received burst is not identical to that of the saved burst. Some basic parameters such as MCS or data length are not the same and a composite MIMO measurement is not possible.
- **86 - "no symbols for measurement"**
Too many head and tail symbols have been skipped.
- **93 - "OCXO oven temperature too low"**
The accuracy of measurement results can be impaired because the oven-controlled crystal oscillator has a too low temperature. After switching-on the instrument, the OCXO requires a warm-up phase to reach its operating temperature.
- **96 - "frequency/level change too fast"**
There was a too fast frequency and/or level change. If you use GPRF with list mode, increase the duration of the list entry.
- **101 - "firmware error"**
This value indicates a firmware or software error. If you encounter this error for the first time, restart the instrument.
If the error occurs again, consider the following hints:
 - Firmware errors can often be repaired by restoring the factory default settings.

- If a software package has not been properly installed, this failure is often indicated in the "Setup" dialog > section "SW/HW-Equipment > Installed Software".
- Check for software updates correcting the error. Updates are provided in the customer web on GLORIS (registration required): <https://gloris.rohde-schwarz.com>.

If you get firmware errors even with the properly installed latest software version, send a problem report including log files to Rohde & Schwarz.

- **102 - "unidentified error"**

This value indicates an error not covered by other reliability values. For troubleshooting, follow the steps described for "101 firmware error".

- **103 - "parameter error"**

This value indicates that the measurement could not be performed due to internal conflicting parameter settings.

For 1xEV-DO:

- The 1xEV-DO signaling test application is not enabled.
- Forward link measurement data were requested, but the test application direction is not set to "Forward" or "Forward & Reverse".
- Reverse link measurement data were requested, but the test application direction is not set to "reverse" or "forward & reverse".
- Measurement data were requested, but the corresponding measurement is not assigned.

For CDMA2000: The "1st Service Option" is not set to a loopback service option.

A good approach to localize the conflicting settings is to start with a reset or preset or even restore the factory default settings. Then reconfigure the measurement step by step and check when the error occurs for the first time.

If you need assistance to localize the conflicting parameter settings, contact Rohde & Schwarz (see <https://www.service.rohde-schwarz.com>).

- **104 - "not functional"**

The application could not be started with the configured parameter set.

7.5.2 Common measurement settings

The commands valid for all LTE measurements are described here: [Section 5.5.2, "Common measurement settings"](#), on page 1111

7.5.3 SRS measurement commands

The commands for the "LTE SRS" measurement are divided into the groups listed below.

• Measurement control and states	1402
• Measurement parameters	1404
• Trigger settings	1406
• Limits	1408
• Power dynamics results (diagrams)	1409
• Power dynamics results (single values)	1410

7.5.3.1 Measurement control and states

The following commands control the measurement and return the current measurement state.

<code>INITiate:LTE:MEAS<i>:SRS</code>	1402
<code>STOP:LTE:MEAS<i>:SRS</code>	1402
<code>ABORT:LTE:MEAS<i>:SRS</code>	1402
<code>FETCh:LTE:MEAS<i>:SRS:STATe?</code>	1402
<code>FETCh:LTE:MEAS<i>:SRS:STATe:ALL?</code>	1403

INITiate:LTE:MEAS<i>:SRS

STOP:LTE:MEAS<i>:SRS

ABORT:LTE:MEAS<i>:SRS

Starts, stops or aborts the measurement:

- The `INITiate:...` command starts or restarts the measurement. The measurement enters the "RUN" state.
- The `STOP:...` command halts the measurement immediately. The measurement enters the "RDY" state. Measurement results are kept. The resources remain allocated to the measurement.
- The `ABORT:...` command halts the measurement immediately. The measurement enters the "OFF" state. All measurement values are set to NAV. Allocated resources are released.

To query the current measurement state, use `FETCh...STATe?`.

Example: See [Running a measurement](#)

Usage: Event

Firmware/software: V2.0.20

Manual operation: See ["SRS \(Softkey\)"](#) on page 1383

FETCh:LTE:MEAS<i>:SRS:STATe? [<Timeout>[, <TargetMainState>[, <TargetSyncState>]]]

Queries the main measurement state.

Without query parameters, the state is returned immediately.

With query parameters, the state is returned when the <TargetMainState> and the <TargetSyncState> are reached or when the <Timeout> expires.

Query parameters:

<Timeout>	numeric Default unit: ms
<TargetMainState>	OFF RUN RDY Target <MainState> for the query Default is RUN.

<TargetSyncState> PENDING | ADJUSTED

Target <SyncState> for the query
Default is ADJ.

Return values:

<MainState> OFF | RUN | RDY

Current state or target state of ongoing state transition

OFF: measurement off

RUN: measurement running

RDY: measurement completed

*RST: OFF

Usage: Query only

Firmware/software: V2.0.20
V4.0.140 added optional parameters.

Manual operation: See "SRS (Softkey)" on page 1383

FETCH:LTE:MEAS<i>:SRS:STATE:ALL? [<Timeout>[, <TargetMainState>[,
<TargetSyncState>]]]

Queries the main measurement state and the measurement substates.

Without query parameters, the states are returned immediately.

With query parameters, the states are returned when the <TargetMainState> and the <TargetSyncState> are reached or when the <Timeout> expires.

Query parameters:

<Timeout> numeric

Default unit: ms

<TargetMainState> OFF | RUN | RDY

Target <MainState> for the query
Default is RUN.

<TargetSyncState> PENDING | ADJUSTED

Target <SyncState> for the query
Default is ADJ.

Return values:

<MainState> OFF | RUN | RDY

Current state or target state of ongoing state transition

OFF: measurement off

RUN: measurement running

RDY: measurement completed

*RST: OFF

<SyncState> PEND | ADJ

PEND: transition to <MainState> ongoing

ADJ: <MainState> reached

<ResourceState>	QUE ACT INV QUE: waiting for resource allocation ACT: resources allocated INV: no resources allocated
Usage:	Query only
Firmware/software:	V2.0.20 V4.0.140 added optional parameters.
Manual operation:	See "SRS (Softkey)" on page 1383

7.5.3.2 Measurement parameters

The following commands define measurement control parameters for the SRS measurement.

CONFigure:LTE:MEAS<i>:SRS:TOUT	1404
CONFigure:LTE:MEAS<i>:SRS:REPetition	1405
CONFigure:LTE:MEAS<i>:SRS:SCONdition	1405
CONFigure:LTE:MEAS<i>:SRS:MOEXception	1405
CONFigure:LTE:MEAS<i>:SRS:SCOnt:PDYNamics	1406
CONFigure:LTE:MEAS<i>:SRS:HDMode	1406

CONFigure:LTE:MEAS<i>:SRS:TOUT <Timeout>

Defines a timeout for the measurement. The timer is started when the measurement is initiated via a `READ` or `INIT` command. It is not started if the measurement is initiated manually.

When the measurement has completed the first measurement cycle (first single shot), the statistical depth is reached and the timer is reset.

If the first measurement cycle has not been completed when the timer expires, the measurement is stopped. The measurement state changes to `RDY`. The reliability indicator is set to 1, indicating that a measurement timeout occurred. Still running `READ`, `FETCh` or `CALCulate` commands are completed, returning the available results. At least for some results, there are no values at all or the statistical depth has not been reached.

A timeout of 0 s corresponds to an infinite measurement timeout.

Parameters:

<Timeout>	numeric
	Default unit: s

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

CONFigure:LTE:MEAS<i>:SRS:REPetition <Repetition>

Specifies the repetition mode of the measurement. The repetition mode specifies whether the measurement is stopped after a single shot or repeated continuously. Use the **CONFigure:...:MEAS<i>:...:SCount** commands to specify the number of measurement intervals per single shot.

Parameters:

<Repetition> SINGleshot | CONTInuous
SINGleshot: Single-shot measurement
CONTInuous: Continuous measurement

Firmware/software: V2.0.20

Manual operation: See ["Repetition"](#) on page 1387

CONFigure:LTE:MEAS<i>:SRS:SCONdition <StopCondition>

Qualifies whether the measurement is stopped after a failed limit check or continued. With **SLFail**, the measurement is stopped and reaches the **RDY** state when one of the results exceeds the limits.

Parameters:

<StopCondition> NONE | SLFail
NONE: Continue measurement irrespective of the limit check.
SLFail: Stop measurement on limit failure.
***RST:** NONE

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Manual operation: See ["Stop Condition"](#) on page 1387

CONFigure:LTE:MEAS<i>:SRS:MOEXception <MeasOnException>

Specifies whether measurement results that the R&S CMW identifies as faulty or inaccurate are rejected.

Parameters:

<MeasOnException> OFF | ON
OFF: Faulty results are rejected.
ON: Results are never rejected.
***RST:** OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Manual operation: See ["Measure on Exception"](#) on page 1387

CONFigure:LTE:MEAS<i>:SRS:SCount:PDYNamics <StatisticCount>

Specifies the statistic count of the measurement. The statistic count is equal to the number of measurement intervals per single shot.

Parameters:

<StatisticCount> numeric
 Number of measurement intervals
 Range: 1 to 1000
 *RST: 1

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.0.20

Manual operation: See ["Statistic Count"](#) on page 1388

CONFigure:LTE:MEAS<i>:SRS:HDMMode <HighDynamicMode>

Enables or disables the high dynamic mode for power dynamics measurements.

Parameters:

<HighDynamicMode> OFF | ON
 *RST: OFF

Example: See [Specifying measurement-specific settings](#)

Firmware/software: V2.1.25

Manual operation: See ["High Dynamic Mode"](#) on page 1388

7.5.3.3 Trigger settings

The following commands define the trigger parameters.

TRIGger:LTE:MEAS<i>:SRS:CATalog:SOURce?	1406
TRIGger:LTE:MEAS<i>:SRS:SOURce	1407
TRIGger:LTE:MEAS<i>:SRS:SLOPe	1407
TRIGger:LTE:MEAS<i>:SRS:THReshold	1407
TRIGger:LTE:MEAS<i>:SRS:TOUT	1408
TRIGger:LTE:MEAS<i>:SRS:MGAP	1408

TRIGger:LTE:MEAS<i>:SRS:CATalog:SOURce?

Lists all trigger source values that can be set using [TRIGger:LTE:MEAS<i>:SRS:SOURce](#).

Return values:

<Sourcelist> string
 Comma-separated list of all supported values. Each value is represented as a string.

Usage: Query only

Firmware/software: V2.0.20

Manual operation: See ["Trigger Source"](#) on page 1389

TRIGger:LTE:MEAS<i>:SRS:SOURce <Source>

Selects the source of the trigger events. Some values are always available. They are listed below. Depending on the installed options, additional values are available. You can query a list of all supported values via the TRIGger:...:CATalog:SOURce? command.

Parameters:

<Source> string
 'IF Power': Power trigger (received RF power)
 *RST: 'IF Power'

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.20

Manual operation: See ["Trigger Source"](#) on page 1389

TRIGger:LTE:MEAS<i>:SRS:SLOPe <Slope>

Qualifies whether the trigger event is generated at the rising or at the falling edge of the trigger pulse (valid for external and power trigger sources).

Parameters:

<Slope> REDGe | FEDGe
 REDGe: Rising edge
 FEDGe: Falling edge
 *RST: REDG

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.20

Manual operation: See ["Trigger Slope"](#) on page 1389

TRIGger:LTE:MEAS<i>:SRS:THReshold <TrigThreshold>

Defines the trigger threshold for power trigger sources.

Parameters:

<TrigThreshold> numeric
 Range: -50 dB to 0 dB
 *RST: -20 dB
 Default unit: dB (full scale, i.e. relative to reference level minus external attenuation)

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.20

Manual operation: See ["Trigger Threshold"](#) on page 1389

TRIGger:LTE:MEAS<i>:SRS:TOUT <TriggerTimeOut>

Selects the maximum time that the measurement waits for a trigger event before it stops in remote control mode or indicates a trigger timeout in manual operation mode. This setting has no influence on "Free Run" measurements.

Parameters:

<TriggerTimeOut> numeric | ON | OFF
 Range: 0.01 s to 167772.15 s
 *RST: 0.1 s
 Default unit: s
 ON | OFF enables or disables the timeout.

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.20

Manual operation: See ["Trigger Timeout"](#) on page 1390

TRIGger:LTE:MEAS<i>:SRS:MGAP <MinTrigGap>

Sets a minimum time during which the IF signal must be below the trigger threshold before the trigger is armed so that an IF power trigger event can be generated.

Parameters:

<MinTrigGap> float
 Range: 0 s to 1E-3 s
 *RST: 50E-6 s
 Default unit: s

Example: See [Configuring the trigger system](#)

Firmware/software: V2.0.20

Manual operation: See ["Min Trigger Gap"](#) on page 1390

7.5.3.4 Limits

The following command defines limits for results which characterize the power dynamics of the SRS signal.

CONFigure:LTE:MEAS<i>:SRS:LIMit:PDYNamics <Enable>, <OnPowerUpper>, <OnPowerLower>, <OffPowerUpper>

Defines limits for the ON power and OFF power determined with the power dynamics measurement.

Parameters:

<Enable>	OFF ON OFF: disables the limit check ON: enables the limit check *RST: ON
<OnPowerUpper>	numeric Upper limit for the ON power Range: -256 dBm to 256 dBm *RST: 4.9 dBm Default unit: dBm
<OnPowerLower>	numeric Lower limit for the ON power Range: -256 dBm to 256 dBm *RST: -10.1 dBm Default unit: dBm
<OffPowerUpper>	numeric Upper limit for the OFF power Range: -256 dBm to 256 dBm *RST: -48.5 dBm Default unit: dBm

Example: See [Specifying limits](#)

Firmware/software: V2.0.20

Manual operation: See ["Power dynamics limit settings"](#) on page 1392

7.5.3.5 Power dynamics results (diagrams)

The following commands return the contents of the power dynamics result diagram.

```

FETCh:LTE:MEAS<i>:SRS:TRACe:PDYNamics:CURRent?
FETCh:LTE:MEAS<i>:SRS:TRACe:PDYNamics:AVERAge?
FETCh:LTE:MEAS<i>:SRS:TRACe:PDYNamics:MAXimum?
READ:LTE:MEAS<i>:SRS:TRACe:PDYNamics:CURRent?
READ:LTE:MEAS<i>:SRS:TRACe:PDYNamics:AVERAge?
READ:LTE:MEAS<i>:SRS:TRACe:PDYNamics:MAXimum?

```

Return the values of the power dynamics traces. Each value is sampled with $48 T_s$, corresponding to $1.5625 \mu s$. The results of the current, average and maximum traces can be retrieved.

Note that the GUI shows only the beginning of the trace returned via a remote command. The last $800 \mu s$ cannot be displayed at the GUI.

See also [Section 7.3.2, "Measurement results"](#), on page 1383.

Return values:

<Reliability>	decimal Reliability indicator
<Power>	float 2048 power values, from -1100 µs to +2098.4375 µs relative to the start of the SRS symbol. The values have a spacing of 1.5625 µs. The 705th value is at the start of the SRS symbol (0 µs). Default unit: dBm

Example: See [Running a measurement](#)

Usage: Query only

Firmware/software: V2.0.20

7.5.3.6 Power dynamics results (single values)

The following commands return the statistical results of the power dynamics measurement.

```

CALCulate:LTE:MEAS<i>:SRS:PDYNamics:CURRent?
CALCulate:LTE:MEAS<i>:SRS:PDYNamics:AVERage?
CALCulate:LTE:MEAS<i>:SRS:PDYNamics:MINimum?
CALCulate:LTE:MEAS<i>:SRS:PDYNamics:MAXimum?
FETCh:LTE:MEAS<i>:SRS:PDYNamics:CURRent?
FETCh:LTE:MEAS<i>:SRS:PDYNamics:AVERage?
FETCh:LTE:MEAS<i>:SRS:PDYNamics:MINimum?
FETCh:LTE:MEAS<i>:SRS:PDYNamics:MAXimum?
FETCh:LTE:MEAS<i>:SRS:PDYNamics:SDEViation?
READ:LTE:MEAS<i>:SRS:PDYNamics:CURRent?
READ:LTE:MEAS<i>:SRS:PDYNamics:AVERage?
READ:LTE:MEAS<i>:SRS:PDYNamics:MINimum?
READ:LTE:MEAS<i>:SRS:PDYNamics:MAXimum?
READ:LTE:MEAS<i>:SRS:PDYNamics:SDEViation?

```

Return the current, average, minimum, maximum and standard deviation single-value results of the power dynamics measurement.

The values described below are returned by `FETCh` and `READ` commands. A `CALCulate` command returns limit check results instead, one value for each result listed below.

Return values:

<Reliability>	decimal Reliability indicator
---------------	--

<OutOfTolerance>	decimal Out of tolerance result, i.e. the percentage of measurement intervals of the statistic count for power dynamics measurements exceeding the specified power dynamics limits. Default unit: %
<OffPowerBefore>	float OFF power average value for the time period before the SRS symbol. Default unit: dBm
<OnPowerRMS1>	float ON power average value over the first SRS symbol. Default unit: dBm
<OnPowerPeak1>	float ON power peak value for the first SRS symbol. Default unit: dBm
<OnPowerRMS2>	float ON power average value over the second SRS symbol (NCAP returned for FDD). Default unit: dBm
<OnPowerPeak2>	float ON power peak value for the second SRS symbol (NCAP returned for FDD). Default unit: dBm
<OffPowerAfter>	float OFF power average value for the subframe after the SRS symbol. Default unit: dBm
Example:	See Running a measurement
Usage:	Query only
Firmware/software:	V2.0.20

7.5.4 Combined signal path commands

For some settings, the relevant command depends on the active scenario. While the combined signal path (CSP) scenario is active, these settings are configured via commands of the signaling application. While the standalone (SA) scenario is active, they are configured via measurement commands.

The following table provides the mapping for SRS measurement commands. For general measurement settings, see [Table 5-14](#).

Table 7-1: Mapping for SRS measurement commands

Setting	Commands for SA scenario	Commands for CSP scenario
High dynamic mode	CONFigure:LTE:MEAS<i>:SRS:HDMODE	Fixed value OFF

7.6 List of commands

ABORT:LTE:MEAS<i>:SRS.....	1402
CALCulate:LTE:MEAS<i>:SRS:PDYNamics:AVERage?.....	1410
CALCulate:LTE:MEAS<i>:SRS:PDYNamics:CURREnt?.....	1410
CALCulate:LTE:MEAS<i>:SRS:PDYNamics:MAXimum?.....	1410
CALCulate:LTE:MEAS<i>:SRS:PDYNamics:MINimum?.....	1410
CONFigure:LTE:MEAS<i>:SRS:HDMODE.....	1406
CONFigure:LTE:MEAS<i>:SRS:LIMit:PDYNamics.....	1408
CONFigure:LTE:MEAS<i>:SRS:MOEXception.....	1405
CONFigure:LTE:MEAS<i>:SRS:REPetition.....	1405
CONFigure:LTE:MEAS<i>:SRS:SCONdition.....	1405
CONFigure:LTE:MEAS<i>:SRS:SCOut:PDYNamics.....	1406
CONFigure:LTE:MEAS<i>:SRS:TOUT.....	1404
FETCH:LTE:MEAS<i>:SRS:PDYNamics:AVERage?.....	1410
FETCH:LTE:MEAS<i>:SRS:PDYNamics:CURREnt?.....	1410
FETCH:LTE:MEAS<i>:SRS:PDYNamics:MAXimum?.....	1410
FETCH:LTE:MEAS<i>:SRS:PDYNamics:MINimum?.....	1410
FETCH:LTE:MEAS<i>:SRS:PDYNamics:SDEViation?.....	1410
FETCH:LTE:MEAS<i>:SRS:STATe:ALL?.....	1403
FETCH:LTE:MEAS<i>:SRS:STATe?.....	1402
FETCH:LTE:MEAS<i>:SRS:TRACe:PDYNamics:AVERage?.....	1409
FETCH:LTE:MEAS<i>:SRS:TRACe:PDYNamics:CURREnt?.....	1409
FETCH:LTE:MEAS<i>:SRS:TRACe:PDYNamics:MAXimum?.....	1409
INITiate:LTE:MEAS<i>:SRS.....	1402
READ:LTE:MEAS<i>:SRS:PDYNamics:AVERage?.....	1410
READ:LTE:MEAS<i>:SRS:PDYNamics:CURREnt?.....	1410
READ:LTE:MEAS<i>:SRS:PDYNamics:MAXimum?.....	1410
READ:LTE:MEAS<i>:SRS:PDYNamics:MINimum?.....	1410
READ:LTE:MEAS<i>:SRS:PDYNamics:SDEViation?.....	1410
READ:LTE:MEAS<i>:SRS:TRACe:PDYNamics:AVERage?.....	1409
READ:LTE:MEAS<i>:SRS:TRACe:PDYNamics:CURREnt?.....	1409
READ:LTE:MEAS<i>:SRS:TRACe:PDYNamics:MAXimum?.....	1409
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